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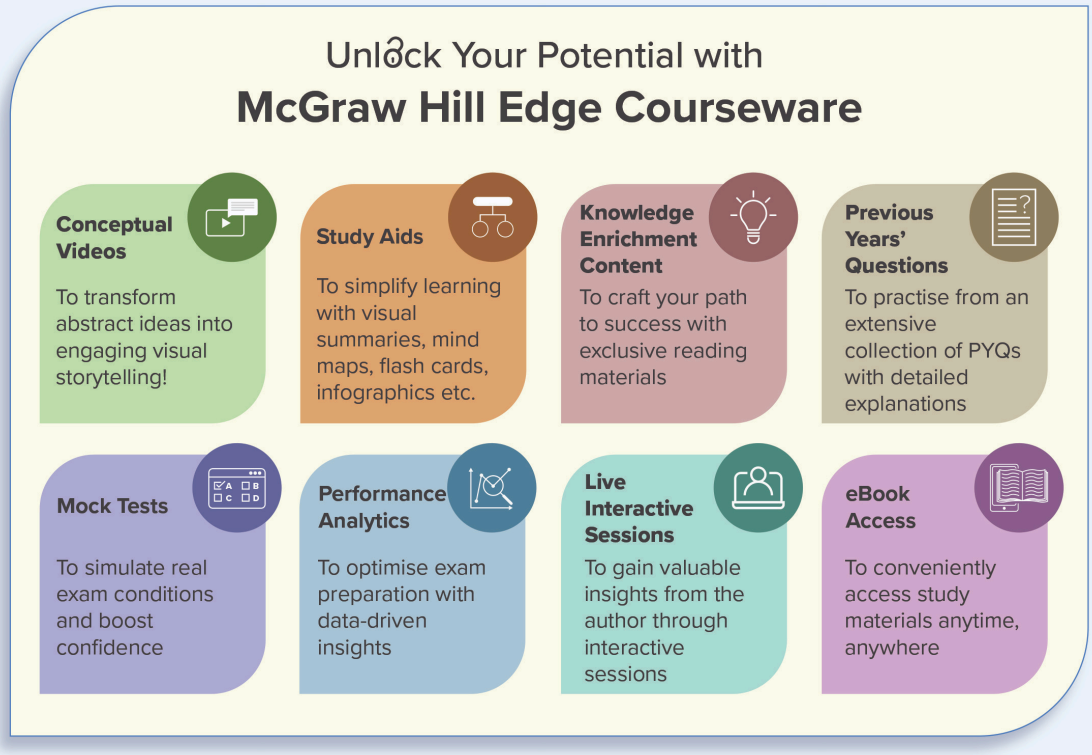

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Environment, Agriculture & Disaster Management, 4e

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Dedicated to

My Parents

Shri Om Prakash Agrahari

and

Late Smt. Meena Agrahari

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About the Author

Dr. Ravi P. Agrahari, Ph.D., **Ex-Scientist** IIT Delhi, is a recipient of Junior Research Fellowship and Senior Research Fellowship from CSIR, Government of India. Dr. Agrahari is presently working as an independent consultant in the field of Environment, Renewable and Sustainable Energy and Biogas issues at the **Bag Energy Research Society, Varanasi**. He is an expert in the field of energy and has published various papers in many reputed international journals. He has attended many international conferences in Sweden and France. Dr Agrahari is well known for his expertise in **'Environment Science'** and **'Science and Technology'** and has authored several books on the same for UPSC examinations. He is also popular as an **'Anthropology Optional Teacher'** among UPSC aspirants. As a teacher of UPSC aspirants for over two decades, Dr. Agrahari has been associated with over 50 reputed coaching institutes in more than 15 states in India. He is also associated with various NGOs and Civil Societies for the welfare of weaker section and minority students.

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To answer subject queries of aspirants, he can be reached at a telegram group **https://t.me/drravipagrahari/Dr. Ravi P. Agrahari (Official)**.

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Preface

This book is a compilation of work from my teaching experience of last two decades. The topics in the book have been chosen with utmost care, keeping in mind the dynamic nature and latest trends of UPSC examinations. The book is hence interdisciplinary with national and international institutions and organisations to connect with all modern dimension of Environmental Ecology, Biodiversity, Climate Change, Agriculture, and Disaster Management. Additionally, a question bank of previous years Prelims (2011 onwards) and Mains (2013 onwards) UPSC exams has been provided with all detailed solutions (both content and Video discussions).

Writing such a comprehensive book would not have been possible without the constant support from my family and friends, regular inputs and suggestions from my students.

I would like to express my sincere appreciation and heartiest thanks to Dr. Poonam Saini (MD) for their hard work at every step of writing this book.

For entire infographics help I would like to thank Gaurav Sharma, without his support it was difficult to simplify many complex concepts. It is also to mention being a visiting faculty at Vision IAS, some of my content are used in this book preparation as well.

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My family member Babeeta Agrahari, Col. Vinod Kumar, Radha Rani Goyal, Namrata Agrahari, Hemant Goyal, Stuti Agrahari, Ankit Agrahari, Anshika Agrahari, Aksh Agrahari (Aksh), Advait Agrahari (Adi) and Aarya Agrahari have been great help in editing and supporting me in this journey.

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Looking forward to receiving suggestions towards the improvement of the book.

God bless you all!!

Let's ace the race together

Dr. Ravi P. Agrahari
PhD & Ex-Scientist IIT Delhi

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ALYSIS OF UPSC - ENVIRONMENT PRELIMS EXAM: NUMBER OF QUESTIONS (2011–2025)

IT	Topics	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	Basics of Ecology & Its related issues	2	1	7	3	2			1	1		6	2	6	5	
	Climate change & Climatic Parameters	2	2	2				1	2		2	2	3	4	3	5
	International Conference & Summits	1	1		6	7	11	3	3			1	3		2	3
	Different Regulatory bodies, laws and Acts in India		3		1	1	1		2	5	1	1	2			
	Environmental Impact Assessment															
	Pollution and Solid waste management	2	2	3	2		1	1		2	2	4		1	2	3
	Biomes, Natural vegetation and Floristic regions of India	2	1	1	2	3	1					2	1		2	1
	Biodiversity, Wildlife conservation & Species Recovery Progranmme in India	6	5	4	3	2	3	5		2	6		2	2	4	1
	Miscellaneous Topics	1			1			1	2		1	1	3			
AL	All Chapters	16	15	17	18	15	17	11	10	10	12	17	16	13	18	13

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PART A

ENVIRONMENTAL ECOLOGY, BIODIVERSITY AND CLIMATE CHANGE

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Part A – Environment

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Upcoming Environment Mains 2026–2027 Model Answers

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SECTION I

Environmental Ecology, Biodiversity and Climate Change

Unit I Basics of Ecology and Its Related Concepts

Introduction 1

IN THIS CHAPTER...

1.1 Ecology
1.2 History of Ecology
1.3 Approaches of Ecology
1.3.1 Scientific Ecology
1.3.2 Deep Ecology
1.4 Sub-divisions of Ecology
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1.5 Hierarchy of Ecology/LevelS of Population
1.5.1 Individual/Organism/Species
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1.5.6 Biosphere
1.6 Habitat and Niche
1.7 Environmentalism

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1.1 ECOLOGY

A German biologist **'Ernst Haeckel'** for the first time in 1869, used the term 'ecology' and defined it as '**the total relations of animal to both its organic and inorganic environment**.' In simple terms, ecology is the study of interaction between living organisms and their environment and the distribution pattern of plants and animals on the earth.

1.2 HISTORY OF ECOLOGY

The word 'ecology' (Oekologie), first proposed by a zoologist named '**H. Reiter'** in 1885, is derived from Greek words, **Oikos** i.e. dwelling place or home and **Logos** i.e. study.

Haeckel (1869) coined the term **'Oekologie'** from which modern ecology has been derived. He defined content of his oekologie as 'comprising the relation of the animal to its organic as well as its inorganic environment, particularly its friendly or hostile relations with the neighbouring plants and animals.'

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The most widely accepted definition of ecology was proposed by '**Ernst Haeckel**'. According to him, **'Ecology is the study of the reciprocal relationship between living organisms and their environments'**.

Prof. R. Mishra (Father of Indian Ecology), defined ecology as 'interactions of forms, functions and factors'. These three interactive aspects i.e. forms, functions, and factors, integrate to construct the triangle of nature.

1.3 APPROACHES OF ECOLOGY

There are two approaches to ecology.

1.3.1 Scientific Ecology

Scientific Ecology is the study of the inter-relationship between species and their environment.

In this approach, the relationship is of a detached observer separate from the object of study. The focus is on measurable data 'out there', collected by the experts who know their 'facts and figures'.

1.3.2 Deep Ecology

Deep Ecology involves seeing ourselves as part of the living earth and finding our role in protecting the planet.

The phrase 'deep ecology' was coined by the Norwegian philosopher **Arnies Naess** in 1973.

Deep ecology's core principle is the belief that the living environment as a whole should be respected and regarded as having certain inalienable legal rights to live and flourish, independent of its practical instrumental benefits for human use.

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A deep ecology is a holistic approach to face world problems that bring together thorough thinking, feeling, spirituality and action. This leads to a deeper connection with life, where ecology is not just seen as something 'out there', but something we are part of and have a role to play in.

1.4 SUB-DIVISIONS OF ECOLOGY

Two major sub-divisions of ecology are **autecology** and **synecology**.

1.4.1 Autecology

A study of the individual species with their environment is known as autecology.

It includes the study of its geographical distribution, taxonomic position, morphological characters, reproduction, life cycle and behaviour concerning ecological factors that might influence these activities.

1.4.2 Synecology

A study of the group of organisms in relation to their environment is called synecology. It comprises community ecology and the study of the ecosystems.

Do You Know?

Bionomics: It is used as the **synonym of ecology**. Bionomics is the comprehensive study of an organism and its relation to the environment.

1.5 HIERARCHY OF ECOLOGY/LEVELS OF POPULATION

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Community, population, organism, organ, cell and gene are widely used terms for significant biotic levels shown in a hierarchical arrangement in a descending order. Hierarchy means **'an arrangement into graded series'** (a system that has many levels from the lowest to the highest) (Figure 1.1).

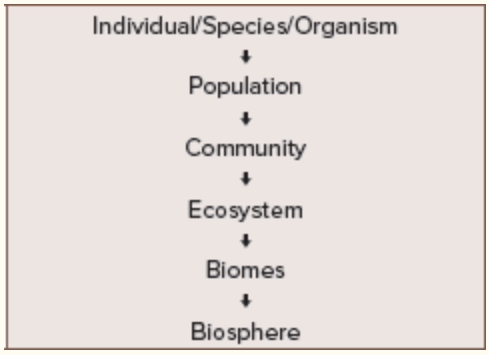

Figure 1.1 Ecological Hierarchy

ENV 01- Ecological Hierarchy

1.5.1 Individual/Organism/Species

Ecology starts with an individual. An individual is a living organism of a particular species. Generally, they do not breed with individuals of other groups.

1.5.2 Population

A group of individuals of a given species that live in a specific geographical area at a certain time.

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Do You Know?

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Do You Know?

Populations have the same species/organisms but may have different genetic makeup such as hair/eye/skin color and size among themselves and other populations.

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1.5.3 Community

Community includes all the populations in a specific area at a given time. It contains populations of organisms of different species. A great community usually has biodiversity.

1.5.4 Ecosystem

Ecosystems include more than a community of living organisms (biotic) interacting with the environment (abiotic). Here we also measure how they depend on other abiotic factors such as rocks, water, air and temperature.

1.5.5 Biome

A biome is a set of ecosystems sharing similar characteristics with their abiotic factors adapted to their environments.

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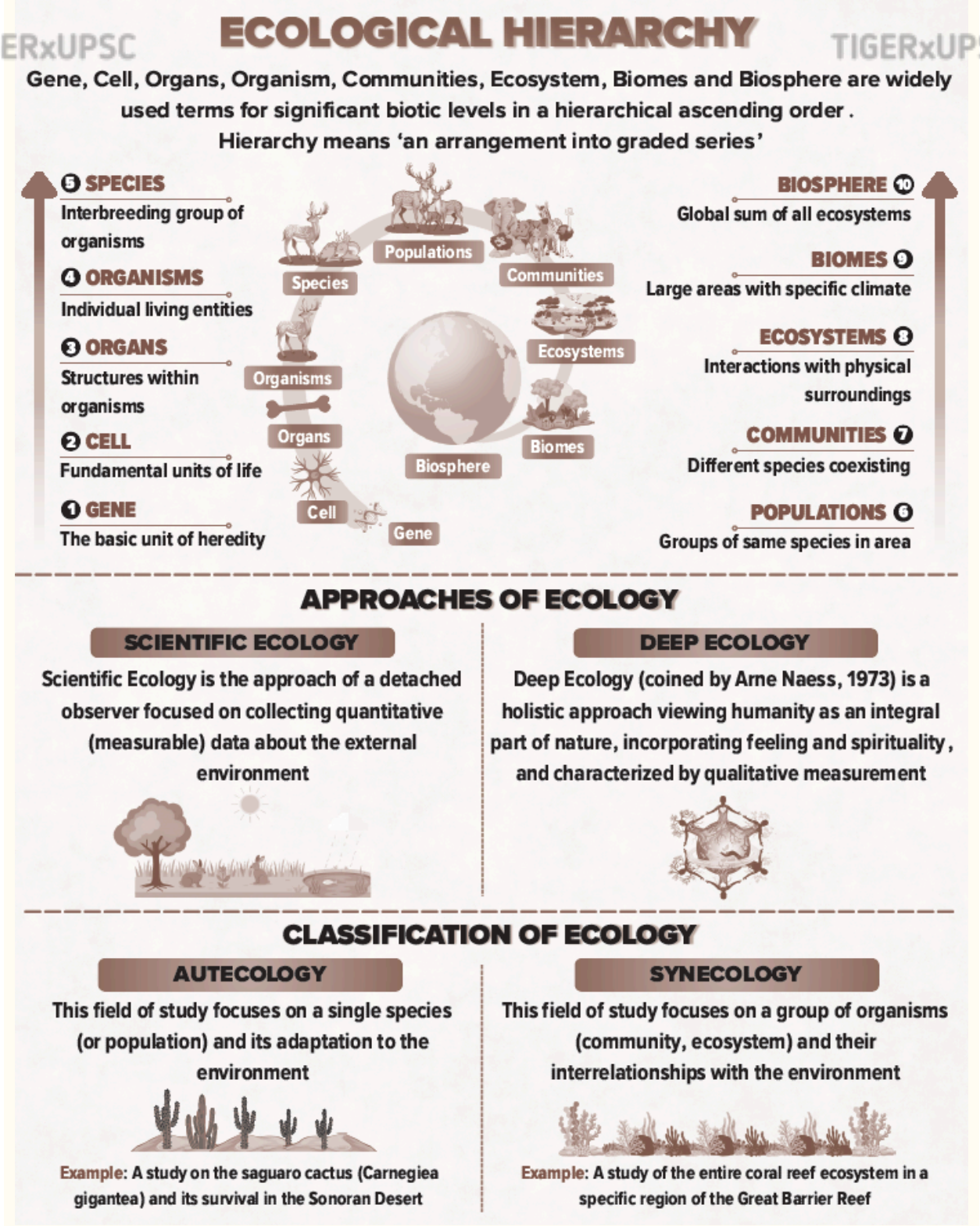

These are extensive ecological areas on the earth's surface, with fauna and flora (animals and plants) adapting to their environment. Biomes are

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often defined with the help of abiotic factors such as climate, relief, geology, soils and vegetation.

A **biome** is **NOT** an ecosystem but looks like a massive ecosystem. If we observe, we will find out that plants or animals in any of the biomes have special adaptations making it possible for them to exist in that area. We can find out many units of ecosystems within one biome.

1.5.6 Biosphere

A biosphere is the sum total of all the ecosystems established on Earth. It is the living (and decaying) component of the earth system. The biosphere is a life-supporting layer that surrounds the earth and makes plant and animal life possible without any protective device. **The average thickness of the biosphere or life supporting layer consisting of air, water, soil and rock is about 30 km.** The upper limit of the biosphere is determined by the availability of oxygen, moisture, temperature and air pressure (Tables 1.1–1.3).

Table 1.1 Important Elements of the Earth's Crust

Elements	Percentage (%)	Elements
Oxygen	46	**Silicon**
Aluminum	8	**Iron**
Magnesium	4	**Calcium**
Potassium	2.3	**Sodium**
Other	Less than 1.0	

Table 1.2 Important Elements of the Earth

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Elements	Percentage (%)	Elements
Iron	35	Oxygen
Silicon	15	Magnesium
Nickel	2.4	Sulphur
Calcium	1.1	Aluminum
Other	Less than 1.0	

Table 1.3 Gases Present in Atmosphere (Dry Air)

Gas	Percentage (%
Nitrogen	
Oxygen	
Argon	
Carbon dioxide	
Neon	
Helium	
Hydrogen	

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ENV 02- Habitat and Niche

1.6 HABITAT AND NICHE

The habitat of an organism is '**the place where it lives**', whereas the definition of ecological niche has changed over time. **Joseph Grinnell,** in 1917, coined the term **niche**, which he used as largely equivalent to a species habitat. In 1958, **George Evelyn Hutchinson** used the term 'niche' to describe the multi-dimensional space of resources available to and used by species. **Joseph Grinnell** probably was the first to use it in a research program in 1917. In his paper 'The niche relationships of the California Thrasher' and as further elaborated by **Charles Elton** in 1927, 'niche' can be termed as **'the position or status of an organism within its community and ecosystem resulting from the organism's structural and functional adaptations'** (Table 1.4). By analogy, habitat is often said to be the organism's address and the niche is its 'profession'.

Table 1.4 Comparison between Habitat and Niche

	Habitat: The Organism's Home		Niche: The Organis
1.	It is a **specific place** or locality where a community resides	1.	It is an **ecological c** delimited by the **func**
2.	A habitat has a number of niches	2.	Niche does **not** have

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3.	Habitat supports a **number of species**	3.	Niche supports a **sin**
4.	A number of environmental variables occur in a habitat	4.	Niche has a specific

Do You Know?

Small niches have maximum chances of disturbances, but broad niches try to compensate the environmental gradient easily, making it difficult to face the extinction of a species.

1.7 ENVIRONMENTALISM

Environmentalism is an activity that **comprises many approaches like any social, political, or religious/ethical movement**. These concepts include environmental education, nature conservation, wildlife protection, environmental management and environmental activism, etc. The **Conservation society of Bishnois** in India practices environmentalism. **Green peace, an international NGO**, practices peace and environmentalism when it speaks **against killing whales**. **Chipko movement** too exhibits environmentalism. It contains political and philosophical inputs. Given the wide range of environmental issues, a person interested in improving the environment can be called as an environmentalist or environmental activist. Today, there are a lot of local organisations in several developed and developing countries concerned about environmental problems and their solutions.

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Ecological Factors 2

IN THIS CHAPTER...

2.1 Ecological Factors
2.2 Edaphic Factors
2.2.1 Soil Formation
2.2.2 Soil Types
2.2.3 Soil Profile
2.2.4 Soil Complex
2.2.5 Soil Water
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2.4 Climatic Factor
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2.4.5 Water
2.5 Biotic Factor (Organismal)
2.5.1 Effect of Interactions
2.5.2 Types of Interaction

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2.1 ECOLOGICAL FACTORS

Those conditions, which affect the existence and metabolic processes of an organism are referred to as the environmental or ecological factors. **The ecological factors are broadly classified into two broad categories, namely abiotic and biotic. They are further divided into sub-categories.** (Refer Fig. 2.1).

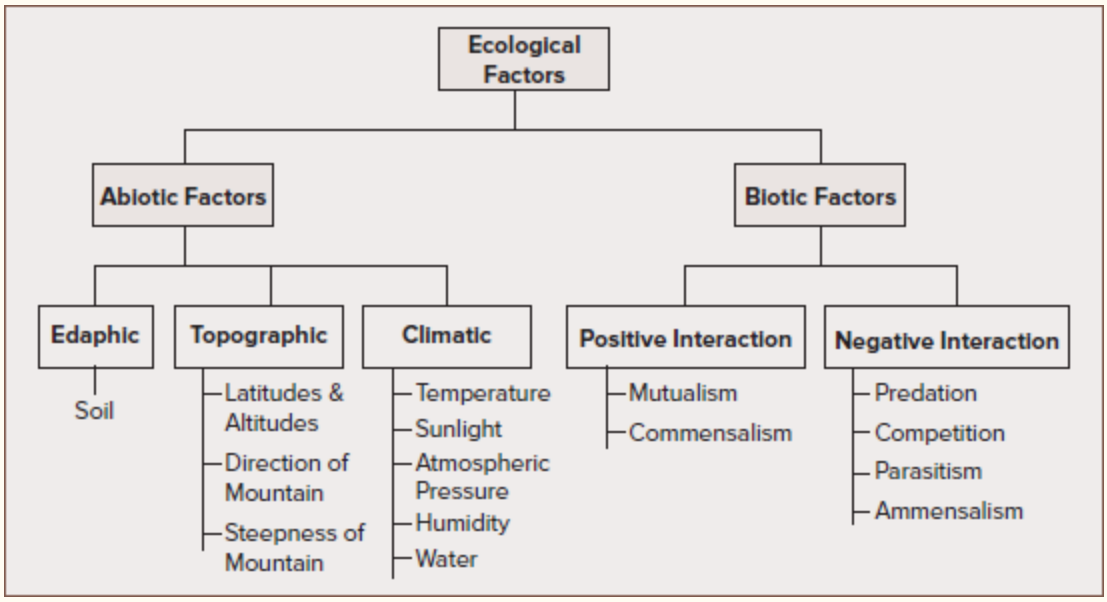

Figure 2.1 Types of Ecological Factors

A. Abiotic Factors

Abiotic elements encompass the non-living components constituting an organism's surroundings. A biotope denotes a habitat with consistent environmental attributes, distinguished by a specific array of abiotic ecological elements. These abiotic components can be categorised into the subsequent three groupings:

1. Edaphic Factors
2. Topographic Factors
3. Climatic Factors

B. Biotic Factors

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Living organism also do various interaction in Ecology in different means, as Positive or negative interaction.

2.2 EDAPHIC FACTORS

It is an abiotic factor related to the physical or chemical composition of the soil found in a particular region.'Pedology' is the science dealing with the **study of soil**. Soil is formed in the intermediate zone between the atmosphere and rock cover of the earth in the lithosphere. It is defined as the **uppermost weathered layer** of the earth's crust in which different types of organisms and their dead and decay products are present.

2.2.1 Soil Formation

There are a lot of theories about the origin of the earth. **Earth originated approximately 4.6 billion years ago but life existed around 3.2 billion years ago**. According to some theories, earlier the earth was part of sun. It cooled down slowly and primitive life forms appeared, perhaps in the form of viruses, bacteria, algae and fungi.

The soil formation occurs in two stages:

(a) Weathering **or breakdown of rocks,**

(b) Pedogenesis**, which involves changes in the weathered rock material into soil.**

The weathering process occurs through the following methods:

(i) **Chemical weathering:** It occurs due to hydration, hydrolysis, carbonation, oxidation-reduction or chelation of rock material.

(ii) **Physical weathering:** It mainly occurs due to rain and heat. It may even occur due to heating and cooling, wetting and drying, freezing, solution formation and glaciations.

(iii) **Biological weathering:** Various bacterial and fungal forms and more commonly the **lichens** are able to cause weathering of rock due to their **acidic action**, thus forming the soil.

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The soil formation is affected by many **passive factors** (nature of parent material, topography and time) and **active factors** (rainfall, temperature, humidity/evaporation, wind and biosphere effects).

2.2.2 Soil Types

On the basis of formation of soil, the soils are classified into two major categories:

(a) **Residual soils:** They are formed by weathering and pedogenesis of the rock at a specific place.

(b) **Transported soils:** After weathering, the soil may be transported by various agencies and thereafter, the pedogenesis occurs. The transported soils are of four types:

(i) **Alluvial:** Transported by **water.**

(ii) **Colluvial:** Transported by **gravity.**

(iii) **Eolian:** Transported by **air.**

(iv) **Glacial:** Transported by slipping of **glaciers.**

2.2.3 Soil Profile

The soil layers distinguished on the basis of color, hardness and other tangible properties are called **Horizons**. They are exposed by excavation from the surface to the parent rock material. The study of these horizons constitutes the soil profile. As many as five horizons are identified which are designated as horizons O, A, B, C and R/D (Figure 2.2).

1. **Horizon O:** It is the uppermost horizon, largely comprised of fresh and/or partly decomposed organic matter. Further two zones are differentiated in this horizon namely O_1 and O_2. **This horizon is well developed in forests but poorly developed in grasslands and are absent in desert.**
2. **Horizon A:** It is the uppermost mineral horizon of the soil. From this layer, minerals like Fe, Al are generally washed downward and hence it is called as **Eluvial zone (Eluviation)**. It is sub-divided into zones

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designated as A_1 and A_2 (and sometimes A_3). While **A_1** is **dark** colored and **rich** in **organic** matter, **A_2** is **light** colored and has **little organic** matter.

3. **Horizon E:** It is a zone of maximum leaching.
4. **Horizon B:** It is the second mineral horizon in which the materials accumulate and hence it is called as **Illuvial Zone (illuviation).** It is made up of **dark** colored, coarse textured particles. Sometimes it is also further differentiated into three zones, named as B_1, B_2 and B_3.

Do You Know?

The two horizons A and B together constitute the '**solum**'. The solum along with the horizon O forms the **'top soil'**.

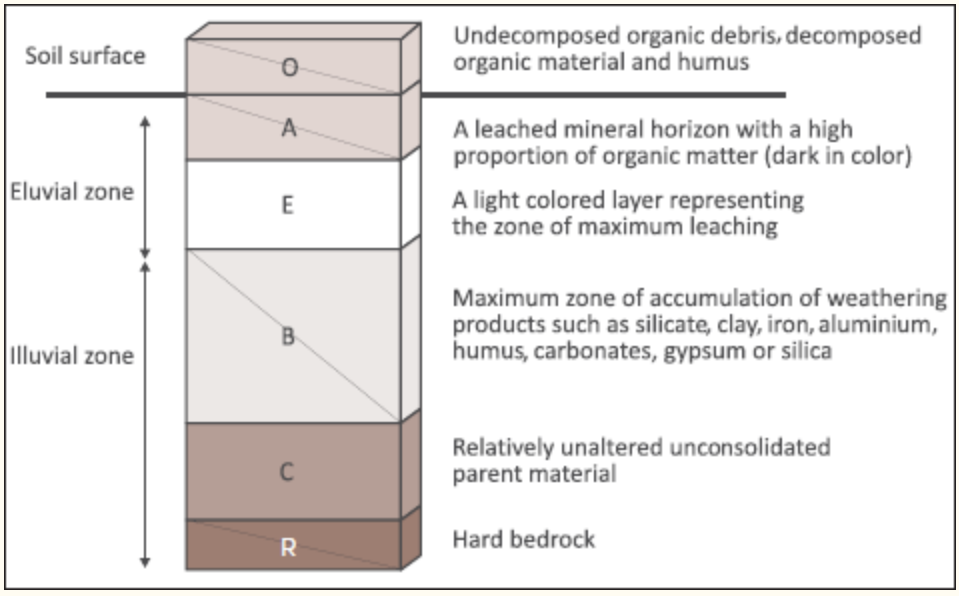

Figure 2.2 Soil Horizon

5. **Horizon C:** It is made up of incompletely weathered, large rock pieces not undergone the process of pedogenesis. It is also called as **sub-soil**. If weathering is fast, this horizon is poorly developed.
6. **Horizon R:** It represents **unweathered parent rock material** or the bed rock.

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Do You Know?

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Do You Know?

Podzol - Podzol are typical soils of **coniferous** or **boreal** forest. Podozols are generally derived from either **quarts rich sands** and **sandstones** or sedimentary debris from **magmatic rocks**. Most podzols are poor soils for agriculture due to the sandy portion, resulting in a low level of moisture and nutrients. Podozolic soils have B horizons that are enriched with amorphous material composed or humified organic matter, aluminium and iron. In this area the mean annual precipitation is above 700 mm.

2.2.4 Soil Complex

Generally soil consists of six constituents namely **inorganic matter, organic matter, soil organisms, soil moisture, soil solution and soil air.** Roughly, the soil contains 50–60% mineral matter, 25–30% water, 15–25% air and a little percentage of organic matter.

■ *Inorganic matter*

The mineral component constitutes some 50–60% of soil. It is present in the form of particles of different sizes. Accordingly, they have been identified as gravel, sand, silt and clay. The Atterberg's system identifies two types of sand particles as coarse and fine. On the other hand, the British standards recognise three types of sand particles namely coarse, medium and fine. We shall follow here the Atterberg's system which designates soil particles in the following manner:

Gravel	Bigger than 2.0 mm
Coarse sand	Between 2.0 and 0.2 mm
Fine sand	Between 0.2 and 0.02 mm
Silt	Between 0.02 and 0.002 mm
Clay	Smaller than 0.002 mm

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Do You Know?

Loam = 40% Sand + 40% Silt + 20% Clay.

A soil having sand, silt and/or clay in a specific proportion is called as **loam**. It is the **best soil** for plant growth. Depending upon the percentage of various components, several textural groups of soil have been recognised. While loam has 50% sand + 50% silt and/or clay, the salt has about 90% silt + 10% sand. **The specific arrangement of soil particles is called as soil structure**. The permanent aggregates of soil are called as **peds** and the disturbances caused by ploughing and digging forms **clods**. The tiny soil units aggregate and collectively impart various shape such as prism like, platy, spheroidal and block like.

■ *Organic Matter*

The organic matter present in the soil is contributed by the **dead and decay of living organisms** i.e., plants and animals. It is present in various forms designated as litter, duff and humus. The undecomposed organic matter is called as **litter**. It decays slowly and when it is partly decomposed, it is called **duff**. Fully decomposed and finely divided, amorphous organic matter which has become mixed with the mineral matter is called as **humus**. **Humus contains carbohydrates, sugar alcohols, lipids (fats, oils and waxes), tannins, resins, lignin, amino acids, flavonoids, alkaloids, carotenoids, etc**. The conversion of litter into humus is called as **humification** whereas **breaking of humus into minerals, CO_2 and water as mineralisation.** The decomposition of litter is brought about the interaction of soil organisms.

Do You Know?

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'High soil biomass is formed' under humid climate, at soil pH ranging between 5–8, intermediate temperature (17°–27° C) and good drainage conditions.

Soil Organisms

Various types of organisms are found in the soil which has been divided into the following three categories on the basis of their size.

1. **The Megafauna/Macrofauna:** It comprises of rodents, rabbits and moles which live partly or wholly inside the soil.
2. **The Mesofauna:** It largely comprises of arthropods (insects), annelids (earthworms) and nematodes.
3. **The Micro-organism:** It represents the soil microflora and microfauna which comprises of algae, fungi, bacteria, protozoans etc.

The soil organisms perform several functions such as:

- Nitrogen fixation in the soil.
- Decomposition of organic matter for humus formation.
- Plays an important role in nutrient cycles.
- Formation of auxin (plant growth hormone) by some bacteria/fungi, thus helping in root growth.
- Brings sub-soil to the surface thus improving soil aeration.
- Performs antibiosis i.e., antagonistic effect to toxins produced in some soils.
- Causes soil-borne diseases.

Soil Solution

Various minerals on being dissolved in soil water forms a solution called as **soil solution.** It contain not only major and minor elements but also a few other elements that are needed for plant growth. **The clay micelle of the soil are negatively charged which bind cations** rather loosely. These cations can be exchanged by other cations. This ability of the soil is described as **cation exchange capacity** (CEC). The CEC of soil organic matter is totally pH dependent. **The humus has a high CEC**. On the basis

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of mineral matter present in the soil solution, the soils have been divided into two categories:

1. **Oligotrophic:** They have **less** concentration of **nutrients** in the soil solution.
2. **Eutrophic:** They have **more** concentration of **nutrients** in the soil solution.

The nature of soil solution determines the soil pH, ranging from 2.0 to 10.5 but more **commonly between 3.0–8.4**. A soil may, therefore, show acidic, alkaline or neutral reaction (**soil reaction**).

The acidity is primarily due to **H+** or **Al+3**. On the basis of the nature of soil solution, we have a specific type of vegetation.

Low soil biomass is formed under arid climate, high or low soil pH, extremes of temperature and poor drainage.

Plants like **Rhododendron spp. Maize, Barley, Potato, Tomato** etc. grow on **acidic soil** whereas some ferns grow in alkaline soil. Species of **Rhizophora, Cassiopeia, Heritiera are common halophytes.**

Do You Know?

The **halophytes** grow on saline soils, **calciphytes** (calcioles) on calcium rich soils and **oxylophytes** (calcifuges) on acidic soils.

2.2.5 Soil Water

The soil retains water in four different forms:

1. **Capillary Water:** Water present in the **capillary spaces** enclosed between soil micelle is called as capillary water. It represents the available form of water that plants absorb.
2. **Hygroscopic Water:** Water present in the form of a thin film around each soil particle by cohesive and adhesive forces is called as hygroscopic water. Usually it is not available to plants.

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3. **Combined Water:** It is present in the soil in chemically combined form e.g., water molecules held by sulphates, silicates etc.
4. **Water Vapour:** It is present in the pore spaces.

■ *Field Capacity*

The amount of water (%) retained by a unit weight of undisturbed soil after complete drainage under controlled conditions is called as **field capacity or holard.**

Thus, 'field capacity' includes all the four forms of soil water i.e., capillary water, hygroscopic water, combined water and water vapour whereas, **'storage capacity'** refers to **capillary water**.

2.3 TOPOGRAPHIC FACTOR

These factors pertain to the terrestrial landscape characteristics. Topographical elements impact the climate in diverse locales. These variables encompass latitude, elevation, mountain orientation, mountain gradient, and similar factors.

2.3.1 Latitudes and Altitudes

Latitudes represent the distance from the equator. Altitudes represent the height above the sea level. These factors influence temperature levels, wind speed, the presence of oxygen, and the intensity of light. Temperatures reach their peak at the equator and hit their lowest points at the poles.

At high altitudes, wind speed escalates, air pressure and temperature decline, while light intensity and humidity rise. These discrepancies lead to diverse vegetation at varying altitudes, resulting in noticeable zonal distinctions.

2.3.2 Direction of Mountains

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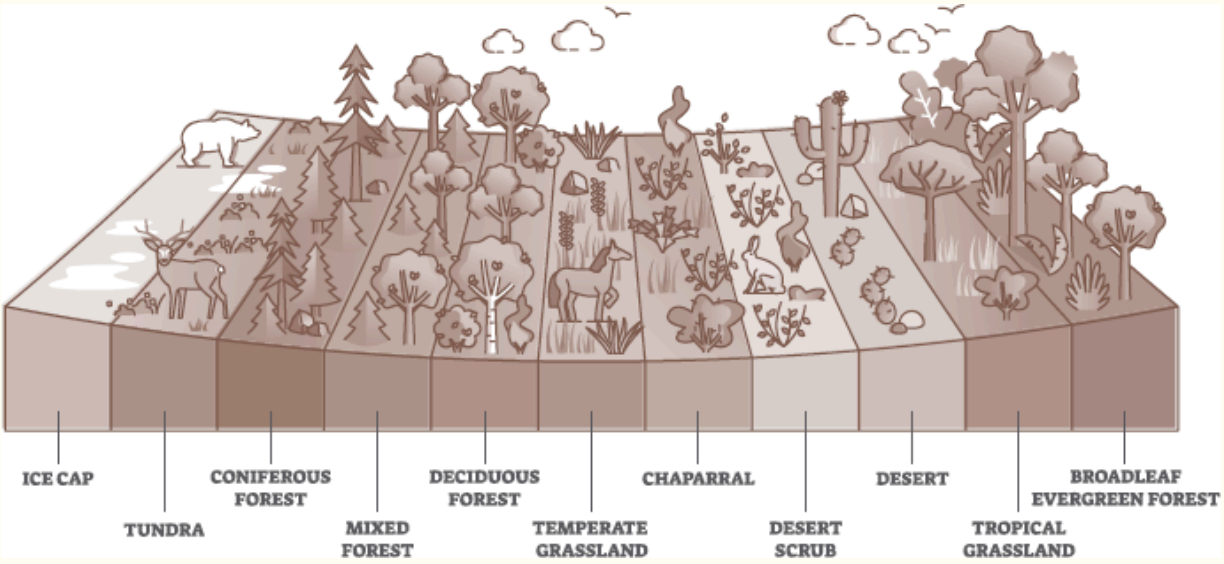
The northern and southern slopes of mountains host distinct flora and fauna due to variations in specific climatic elements, including humidity, precipitation, light intensity, daylight duration, and temperature zones.

2.3.3 Steepness of the Mountains

The steepness of the mountain plays a role in influencing regional rainfall patterns and facilitating the runoff of rainwater. Degradation of hilly terrain contributes to water scarcity and rapid top soil erosion, leading to inadequate vegetation. In contrast, plains exhibit abundant vegetation due to gradual surface water drainage and improved soil water retention.



2.4 CLIMATIC FACTOR

These are abiotic factors that determine the climatic conditions of a particular geographical region. There are the following climatic ecological factors:

2.4.1 Temperature

The temperature is an expression of the quantum of heat or thermal energy. It shows great altitudinal and latitudinal variations.

- **Temperature Based Terrestrial Zonations**

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As many as six temperature based latitudinal zones are differentiated north of equator which are characterised by a specific type of vegetation namely **snow, tundra**, **coniferous forests, deciduous forests, grasslands and desert and tropical rain forests.** A similar type of zonation having a specific type of vegetation has been also differentiated on the basis of altitude.

Do You Know?

The alpine zone has predominance of Rhododendron spp. and Juniperus. The temperate zone has predominance of conifers, oaks and Magnolia. The tropical and temperate zones show abundance of mixed deciduous forests and bamboos. Shorea forms the base line in the eastern Himalayas.

In **Himalayas, as many as four temperature based zones** are differentiated from base to upward. They are identified as **tropical, sub-tropical, temperate** and **alpine.** These zones are characterised by a specific type of vegetation. Due to several other factors also such as soil type, annual rainfall etc., the vegetation of eastern Himalayas differs from the western.

Raunkiær, on the other hand divided the total vegetation into four temperature based groups as mentioned below:

■ *I. Megatherms*

Those plant groups which **require high temperature throughout the year** are called megatherms e.g., **tropical rain forests.**

■ *II. Mesotherms*

Those plant groups which **require high and low temperatures alternately** are called as mesotherms e.g., **deciduous forests**.

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III. Microtherms

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Those plant groups which **need low temperature for their growth** are named as microtherms e.g., **coniferous forests**.

IV. Hekistotherms

These plant groups **require a very low temperature for their growth.** They cannot withstand high temperature e.g., '**alpine forests**.'

- **Aquatic Thermal Zonation**

On the basis of temperature, as many as three aquatic strata have been differentiated where the temperature ranges between 5° to 23°C. This stratification becomes distinct in lakes during summer.

I. Epilimnion

It is the **uppermost zone** where the **temperature decreases gradually**.

II. Thermocline or Metalimnion

It is the **middle zone** where **the temperature falls rapidly**.

III. Hypolimnion

It is the **lower most or the bottom zone** where the **temperature gradients are not distinct. Its temperature may go as low as 5°C.**

- **Thermal Injuries & Its Types**

When a plant is subjected to extremes of temperature, it gets injured and the injuries are of three types:

I. Desiccation

It is the state of **extreme dryness**, or the process of extreme drying. Here **transpiration rate exceed absorption**, due to hot or cold spells of wind, the plant tissue is injured and the injury is called as desiccation.

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II. Chilling

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If a **plant of hot climate is transferred to colder regions**, it gets injured and the injury is called as **chilling.**

III. Freezing

During winters the plants growing on hills show formation of intercellular ice. As a result the plant tissue is injured and the injury is called as **freezing.**

If a plant is exposed to a temperature a few degrees above the maximum temperature, its protoplasm gets killed and this has been referred to as **thermal death point.**

2.4.2 Sunlight

The sole source of energy on our planet is the sun which is responsible for life. The solar energy received in the form of electro-magnetic waves in wave length of visible part of solar spectrum ranges from 390 nm to 760 nm, the rest is invisible. The visible spectrum, distinguishable into seven colors restricted to different wave lengths, is expressed as **VIBGYOR.** The invisible spectrum beyond violet region comprises ultraviolet rays and radio waves.

QUICK INSIGHT

According to Geiger, 1961, from the total solar input, some **86% light is consumed in the evaporation of water** and the remaining **14% heats the atmosphere**. **The plants utilise less than 1% of total incident light**. From the visible spectrum about 83% is absorbed, some 12% is reflected and remaining 5% is transmitted. The ability of a surface to reflect the incoming radiations (from sun) is called as **albedo value.** It is expressed in percentage. It is 80% for fresh snow. 20-30% for sand 5-10% for the forest.

According to Geiger, **from the morning solar radiations**, 42% are reflected (33% by clouds and 9% by dust particles), 10% are absorbed by

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the ozone layer and only 48% reaches the surface of the earth.

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The vegetation of a place is not only affected by the intensity of light but also by its quality and duration. The light intensity of the atmosphere is affected by the following factors:

1. **Atmospheric Components:** This includes atmospheric gases such as nitrogen, oxygen and carbon dioxide (CO_2) etc. and suspended particles such as clouds, smoke, dust etc.

   In industrial areas the light may be reduced due to additional atmospheric gases and smoke.
2. **Vegetation Cover:** The intensity of light is reduced by additional covers of vegetation. In multi-storied forests, as in tropical rain forests, the ground vegetation receives just 2–10% light in comparison to open fields.
3. **Topography:** The intensity of light and duration of exposure varies with the topography of the habitat. The **Chasmophytes** (plants growing in rock) are exposed to a comparatively low intensity of light and for a shorter duration, in comparison to those growing in open fields.
4. **Aquatic Stratification:** On the basis of light penetration, the ocean has been divided into the following three zones:
   - **Euphotic Zone:** It measures upto 80 meters in depth. This is a zone of complete solar radiation in water body.
   - **Disphotic Zone:** It ranges from 80 to 200 meters in depth (moderate solar radiation).
   - **Aphotic Zone:** It lies below 200 meters in depth (no sunlight is received in this zone).

Do You Know?

From the total solar radiations falling on water surface, about 10% are reflected and the remaining 90% penetrate into water.

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2.4.3 Atmospheric Pressure

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The term 'atmosphere' here refers to the gaseous mantle surrounding our planet. However, with reference to a plant it also includes the one surrounding the root system. The aerial mantle shows that gases are present upto the height of 300 km i.e., upto **ionosphere** and not in the **exosphere.**

Do You Know?

There is maximum concentration of gases in the **troposphere** and lower **stratosphere** i.e., upto a height of 20 km.

The atmosphere contain some 78% nitrogen, 21% oxygen, 0.9% argon and 0.03% CO_2 by volume. Besides, it also contains other rare gases like hydrogen, oxides of nitrogen, methane, ozone etc.

The nitrogen of the air is fixed in the soil by bacteria and blue green algae from where it is made available to plants. The atmospheric oxygen is necessary for respiration and weathering of rocks. **If all the autotrophs disappear, the entire oxygen will be consumed in the weathering process and there will be no life left on the planet.**

The soil air is needed for root respiration. It is affected by size of the pore, pore frequency, type of the root system, soil organisms and rate of respiration. Plants growing in conditions of sub-optimum serration show shallow root system, presence of special aerating tissue or organs. Their oxygen requirement is low. Besides, they possess the ability to respire anaerobically.

2.4.4 Humidity

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Humidity signifies the volume of water vapor within the atmosphere. It impacts the transpiration pace in plants and perspiration in animals.

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Consequently, it influences the geographical arrangement of plants and animals across the globe.

2.4.5 Water

Water is extremely essential for sustaining life on the planet. In nature it occurs in several forms such as:

1. **Salt water:** It is present in sea and oceans
2. **Fresh water:** It is present in lakes and rivers
3. **Snow and Ice:** It is formed at higher altitudes and poles
4. **Water vapour:** It is always present in the atmosphere

The chief source of soil water is atmospheric precipitation. It may occur in the form of snow, rain, hail, sleet, frost, dew, fog or mist. A good amount of water is lost by way of run off or run away which goes back to sea and ocean through rivulets and rivers. A certain amount of water percolates into the soil. Water reaching the deep saturated zones of soil is called as **gravitational water** or **ground water**. The top layer of gravitational water is called **water table**.

2.5 BIOTIC FACTOR (ORGANISMAL)

In nature there is interdependence in between plants, in between animals and also between plants and animals. These interactions may be for numerous purposes such as nutrition, habitat, pollination, reproduction, growth, fruit and seed dispersal etc.

The interdependence may be between two individuals of the same species **(intraspecific)** or of different species **(interspecific)**. This may be beneficial or harmful to one or both the components. Thus, any activity of a living being, if influences other organism/s is referred to as biotic **effect**.The inherent power of a population to increase in number is called **biotic potential.**

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De Bary, 1879 coined the term **symbiosis** for all types of interactions between organisms. It literally means 'living together'.

2.5.1 Effect of Interactions

There are different types of effects in interaction among organisms.

■ *Positive Interactions*

They indicate one way or mutually beneficial type of relationship between two partners helping in survival and growth. They include **commensalism**, **protoco-operation** and **mutualism**.

■ *Negative Interactions*

They indicate one way or mutually harmful type of relationship including competition, predation, parasitism and antibiosis. The concepts of **parasitism** and **predation** are combined to put together as **exploitation**.

2.5.2 Types of Interaction

There are different types of interaction among organisms.

Table 2.1 Different types of interaction among organisms

Associations	Species - A
Neutralism	0
Commensalism	0
Proto-combination	+
Mutualism & Symbiosis	+
Amensalism	0

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Parasitism	+
Predation	+

A. Neutralism

It is the most common type of interaction between two or more species where no partner is affected by the other. It occurs mostly between and among unrelated populations of different species.

B. Commensalism

It is a type of relationship where one partner is benefitted but the other remains unaffected. 'Benefit' here implies **mainly for shelter, support or locomotion**. The most common examples of commensalism are as under:

- **Climbers:** Plants with weak stems use other plants as their support. They are rooted in the soil and thus have no nutritional relationship with the commensal, eg. *Piper, Ficus, Pisum*.
- **Epiphytes:** There are many plants which grow as epiphytes on other plants. The most common angiospermic epiphytes are orchids e.g., Vanda. Many of these epiphytes possess a special hygroscopic velamen tissue on their aerial roots. These plants grow on other plant for mechanical support and not for nutrition.
- **Epizoans: Some algal forms grow on the shells of animals**. e.g. *Cladophora* grow on the shell of molluscs.

C. Proto-cooperation

It is a type of relationship, which is beneficial to both the partners, but it is not obligatory for their survival. **Some large birds nest on the lower branches whereas the small ones live much higher. These large birds protect the smaller ones from tree climbing predators.**

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D. Mutualism & Symbiosis

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Mutualism is a type of association where both the partners are benefitted. **This is a permanent and obligatory association involving physiological interdependence**.

ENV 03- Symbiosis

Symbiosis refers to a close and prolonged association between two organisms of different species. Mutualism refers to **mutually beneficial interactions between members** of the same or different species. Mutualistic interactions need not necessarily be symbiotic.

- **Lichens:** It is the classical example of mutualism. Here the relationship between **algal** and **fungal partners** is obligatory and permanent. The algal partner prepares the food by photosynthesis whereas the fungal partner performs the act of reproduction. **Lichens are pioneer in xerosere succession** and also work as an **indicator of air pollution (against SO_2 gas)**.
- **Coral exhibit symbiosis with Algae. CNIDARIA/Coelentrates (Non-chordates) members are of both nature, some of these species live in colonies (corals), while others have a solitary like–span (Hydra).**
- **Certain termites have a symbiotic relationship with protozoa that live in the termite's gut. All protozoans are heterotrophs and live as predators or parasites. If they live as parasites then symbiosis is difficult.**

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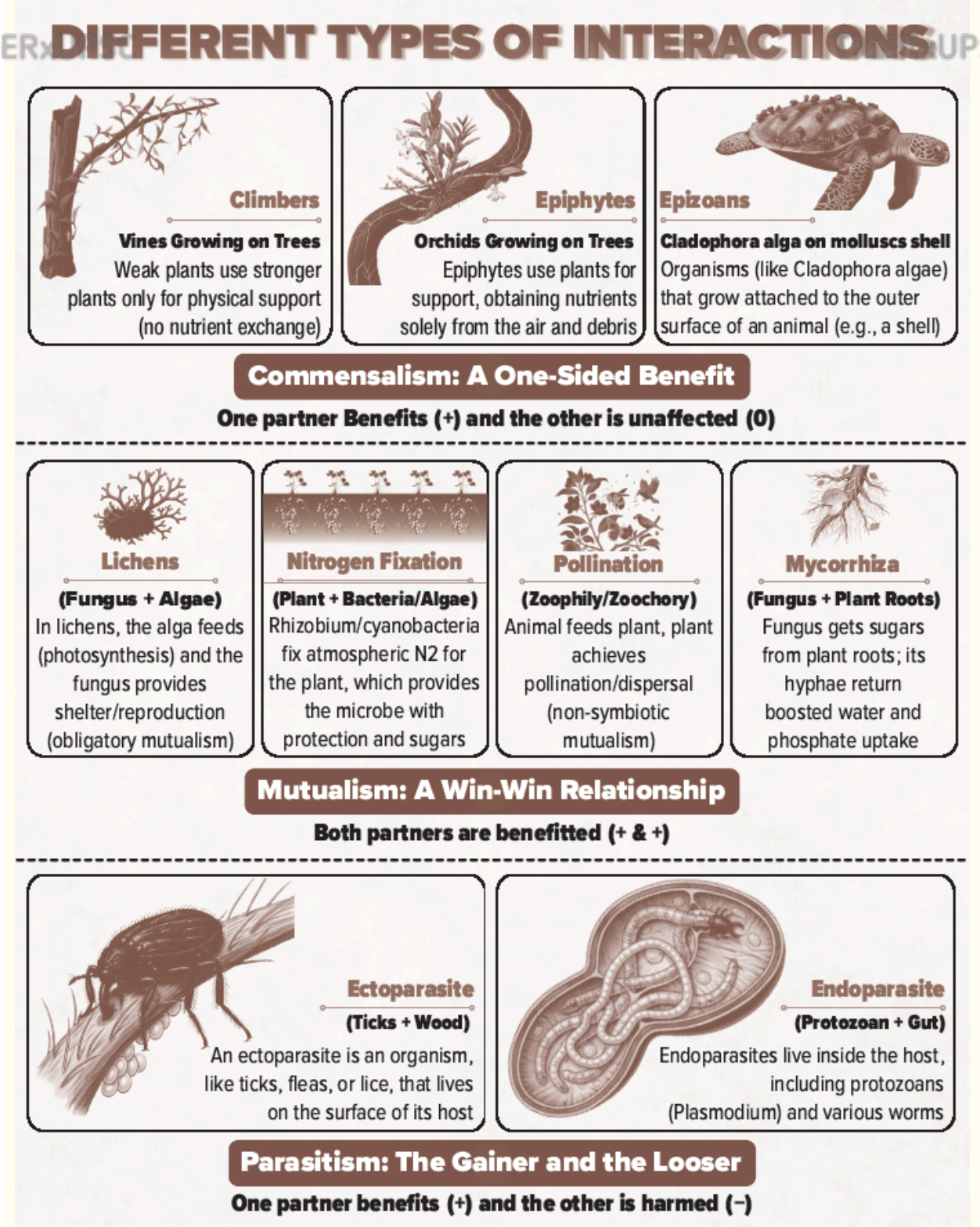

- **Nitrogen fixation:** Nitrogen fixation is a process that implies the transformation of the relatively non-reactive atmospheric N_2 into its more

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reactive compounds (nitrates, nitrites or ammonia). The roots of numerous leguminous plants possess nodules for a nitrogen-fixing bacterium, Rhizobium. **The coralloid roots of Cycas contain species of both Nostoc** and **Anabaena**. These blue-green algae fix atmospheric nitrogen for the partner and receive protection and nutrients in return.

- **Nitrogen-Fixing Plants:** An N-fixing crop is a natural way to provide plant-adjusted N without any industrial harm to nature.

  Nitrogen-fixing plants are:

  - **Clovers, vetches and peas are nitrogen-fixing plants used by farmers worldwide** and in the US Southern Great Plains in particular.
  - **Peas and beans serve a dual purpose:** they can be utilised as a nitrogen-fixing cover crop during the summer or harvested as a food source. In either application, they contribute valuable plant-suitable nitrogen to the soil. Southern peas, which thrive in warmer climates, are best sown when the soil temperature reaches 60°F.
  - **There are various examples of legumes (Pulses crop) - Beans (Kidney bean, black bean,** Fava / Faba beans, Mung beans, **etc.), Peas (Green Pea, Chickpea, pigeon pea, Sweet pea, etc.),** alfalfa, peanuts (groundnuts), soybeans, lupins, lentils, clover, cowpeas, horse gram, etc.
- **Zoophily:** Several types of animals are involved in the pollination mechanism. Many insects such as bees, moths, butterflies (entomophily), birds (ornithophily) are involved in the pollination of flowers when they visit them for food in the form of nectar.
- **Zoochory:** Several animals bring about dispersal of seeds when they visit plants in search of food.
- **Mycorrhiza:** This is the association of growth found with the ‘roots’ of higher plants. It may be ‘ectotrophic (grow outwards) or endotrophic’ (grow inside). The fungal partner gets food and protection from the roots of higher plants and their hyphae substitute for root hair of the host, thus helping in absorption. **Fungi help to provide phosphate for plant as a good fertilizer for plant growth. Mycorrhiza is also responsible to maintain soil pH**.

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- **Fungus Growing Ants: The fungus-growing ants are obligate symbionts** with a fungus they use for food. These ants are found across the American continents and the West Indies. These ants are known as fungus growers because **they maintain an obligate mutualism with fungi cultured inside their nests,** and which is the only food source for the larvae and an important resource for the adult ants as well.
- **Mushroom: Certain mushroom have also mutualistic relationship.** The fungi will combine their mycelium with the tree's roots. As a result, the fungi make it easier for the trees to get more nutrients and moisture from the soil and the fungi get access to sugars from the trees.
- **Figs and Fig-wasps:** Figs have a unique pollination process due to their inverted flower structure, such that both fig-wasps and the fig co-evolve.

  Wasps, attracted by the fig's aroma, enter the fig and lay eggs within. Pollination takes place as the **newborn female wasps leave the fruit loaded with pollen.**

  The **male wasps mate and die within the figs** and are broken down by the fig.

  This represents two ecological processes, namely **obligate mutualism** (by pollinating wasps) **and parasitoidism** (by non-pollinating fig wasps).

  Numerous figs serve as both food sources and habitats for a variety of animals, making them **keystone species** in their surroundings.

E. Amensalism

It is a type of relationship where one component population is harmed and the other remains unaffected. The harmful affect is due to certain chemicals called as allelochemical.

F. Parasitism

It is a kind of relationship where one component population is benefitted and the other is harmed. **The gainer is called as parasite and the looser as host**. The host is always larger than the parasite.

On the basis of its position on the host, the parasites are classified into two

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categories as **ectoparasite** and **endoparasite**. Endoparasite lives inside the host whereas Ectoparasite lives on the surface.

Numerous fungi (*Albugo*, *Puccinia*, *Ustilago*, etc.), bacteria (*Pseudomonas*, *Mycobacterium*, *Vibrio*, *Leptospira*, etc.) and viruses are parasites on plants and animals. In the same way several protozoans (*Plasmodium*, *Entamoeba*, etc.), flat worms, tape worms, round worms live as endoparasites in other animals.

Different Types of Mushroom on the Basis of Habitat & Its Uses

Mushrooms are generally growing on dead and decay matter those are rich with organic compound. This property of mushroom shows a saprophytic nature. Mushrooms grow from fungal spores that thrive in damp, dark conditions.

They require a medium that is high in decaying plant matter. They often spring directly from dead trees. Plants, on the other hand, grow from seeds and require plenty of sun and soil, and don't do well in overly damp environments.

Although mushroom is considered as a vegetable, mushrooms are neither a plant nor animal food. They are a type of fungus that contains a substance called ergosterol, similar in structure to cholesterol in animals. Ergosterol can be transformed into vitamin D with exposure to ultraviolet light.

Various Properties of Mushrooms:

- **Decrease the risk of cancer**.
- Promote **lower cholesterol**.
- **Protect brain health**.
- Provide a source of **Vitamin D & B6**.
- Stimulate a **healthier gut**.
- Support a **healthy immune system**.
- Source of **Selenium** – which helps your body make antioxidant enzymes to prevent cell damage.

Mushrooms also have **psychoactive properties** – they affect brain and cause changes in mood, awareness, thoughts, feelings or behaviour. Alcohol, caffeine, nicotine, marijuana etc. are psychoactive substances.

Certain mushrooms have **insecticidal property** and also common as **hallucinogenic drugs**, such as LSD (lysergic acid diethylamide).

Some mushrooms have **bioluminescence property** (light emitting variety of mushroom). **Panellus stipticus** is one of the brightest-glowing bioluminescent mushrooms on Earth.

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One more bioluminescent variety of mushroom is identified in mushroom i.e. **Roridomyces phyllostachydis**.

ENV 04-Parasitism and Parasitoid

Parasites vs. Parasitoids

The main difference is that parasitoids inevitably kill their host to complete their life cycle, while parasites generally do not. Parasites live on or in a host, benefiting from it without causing its death, such as a flea on a dog. Parasitoids, typically insects, develop on or inside a single host, consuming its resources until the host dies, at which point the parasitoid is free-living.

Parasites

- **Host relationship:** The parasite lives on or in the host's body for a portion of its life cycle.
- **Outcome for host:** The host is harmed, but not killed, as the parasite depends on it for nutrients or shelter.
- **Examples:** Fleas, ticks, mites, and some bacteria or viruses.

Parasitoids

- **Host relationship:** Immature stages (larvae) develop inside or attached to a single host, which is typically another insect.
- **Outcome for host:** The host is eventually killed and consumes its resources to support the parasitoid's development.
- **Life cycle:** Adult parasitoids are free-living and do not require a host.
- **Examples:** Most wasps, a few beetles, flies, moths, etc., are also included in this category. About 68000 recorded species are parasitoids.

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Do You Know?

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Do You Know?

Carabid Beetle: They are **ecto-parasitoids** as larvae. These ground beetles are mostly carnivorous, have defensive secretions and are mostly nocturnal (except tiger beetles)

Parasitoid Wasps and Flies: While not all species belong to the parasitoid family, these are natural enemies of plantations like apples and pears. For example, Ichneumonoid wasps, tachinid flies, bee flies, and thick-headed flies.

G. Predation

It is similar to parasitism in the sense that here also one population is benefitted and the other is harmed. The **looser is called as prey and the gainer as predator.** But, unlike parasitism, the gainer is larger than the looser. The predator kills the prey and feeds upon it. The **insectivorous plants** such as **Drosera, Utricularia/Bladder worm, Dionaea, Aldrovanda, Pinguicula, Nepenthes, Sarracenia, Cephalotus, Darlingtonia, Heliamphora, Pitcher plant, Venus flytrap, etc.** are good examples of predation. **These plants feed upon insects and animalcules to meet their nitrogen deficiency**. Besides, the herbivores feed not only upon herbs but also on shrubs and trees.

Different adaptation in animals also reduced their chance of being captured by predators. E.g. **Hedgehogs have a coat of stiff, sharp spines. If attacked they curl into a prickly and unappetising ball that deters most predators. Marmots live in burrows and hide themselves in holes when there is a predator.**

H. Competition

It occurs as a result of over production, limitations of food supply, light, CO_2 as also for the availability of space for establishment. The competition may be intraspecific or interspecific. The intraspecific competition occurs within members of the same species. It is directly related to density of the population occurring between members of different species. As a result, the two species may co-exist or one species may exist with the same or reduced

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density and the other may be eliminated. The organisms involved in competition may show **allelopathy**. The toxins secreted by one population may interfere with the growth of the other population. The bitter almonds contain HCN which inhibits the germination of poppy seeds.

Do You Know?

Allelopathy is that effect where plants secrete certain chemical substance **(Allelochemical)** those inhibit the growth of surrounding organisms. It plays an important role in future weed control and crop productivity. The allelopathic compounds can be used as natural herbicides and other pesticides. These are less disruptive to the global ecosystem than use of synthetic agrochemicals. **Sunflower**, **Eucalyptus**, **Black walnut,** etc. are certain example of allelopathic effect in nature.

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Adaptations in Plants and Animals

3

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3.1 ADAPTATION

The special features of plants and animals which enable them to survive under specific ecological conditions are called **adaptation**. Different types of adaptation are found in living organisms such as – Photosynthetic pigments, webs of spiders, roots of plants, complex locomotion devices of fishes and horses, which give information about evolutionary changes.

3.2 TYPES OF ADAPTATIONS

Adaptation is generally of three types.

3.2.1 Inherent Adaptation

This type of adaptation is transferred from one generation to another by hereditary means. It is acquired by previous generation through evolution.

3.2.2 Acquired Adaptation

This type of adaptation is acquired by any particular species under specific environmental conditions. For instance, change in skin colour due to sunlight. Sometimes, different adaptations are acquired by the same species because of environmental changes. For example, some fishes are herbivorous while some are carnivorous.

3.2.3 Ecological Adaptation

This type of adaptation is based on the habitat in which plants and animals live. In different habitats, like fresh water, saline water (ocean), pelagic sea organisms developed various kinds of adaptation. Different adaptations are

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shown by different communities like desert adaptation, forest adaptation, etc. Adaptation can be classified as follows:

1. Morphological Adaptation
2. Physiological Adaptation
3. Protective or Behavioural Adaptation.

3.3 ADAPTATIONS IN ANIMALS

3.3.1 Aquatic Adaptation

An animal shows different types of adaptation at various levels.

Primary Aquatic Adaptation

In primary community like fishes, adaptation is studied under primary aquatic adaptation. Their morphological features are as follows:

- **Streamlined Body:** It helps in the locomotion of fishes, because it reduces the water resistance on the body.
- **Fins:** It works as paddle, which controls their velocity.
- **Gills:** It is the main adaptation among all aquatic animals, for gaseous exchange during respiration.
- **Lateral Line Organs:** These are the receptor organs that are present only in fishes. Its function is to identify the mechanical disturbance in water. These organs can identify the objects with a very minor disturbance in water which helps them to identify the food and the prey. This process helps in swimming in very deep and shallow water.
- **Swim Bladder:** It is found in many fishes. In some fishes, it helps in swimming while in others, it functions as a sensory organ. It also works as a sound producing organ.
- **Kidney:** It regulates the process of osmoregulation.

Secondary Aquatic Adaption

Secondary community includes aquatic amphibians, reptiles and mammals. They show different types of adaptations like:

- **Shape of the Body:** It helps aquatic animals in swimming.

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- **Locomotory Organs:** These are the modifications of paddles and webbed feet.
- **Skull:** The cranium of such species is small and conical which helps them to move against the water current.
- **Absence of Neck: The** neck is absent and the head is attached to the trunk.
- **Light Bones:** Bones are light weighted and sponge like.
- **Absence of Hair and Skin Glands:** Skin is naked without hair. A fatty layer is present below the skin, which works as insulator to maintain the body temperature.
- **Respiration:** It is performed by lungs. They are elastic in nature, to store more air; it helps them to respire for longer duration.
- **Absence of External Ear or Pinnae**.
- **Lateral Lines** are present in some species. Sometimes it functions as a receptor organ.

3.3.2 Pelagic Adaptation

Pelagic zone is the upper layer of the water in the oceans, where light is present and substratum is absent. This is rich in food and light. Such environmental conditions make the pelagic animals a more active predator. Pelagic animals comprise many types of organisms, which are called **Plankton** and **Nekton**.

Main adaptations of pelagic animals are:

- Photosensitive
- Well-developed eyes and sensory organs
- Strong swimming organs
- Bioluminescence

3.3.3 Study of Aquatic Animals in Oceans on the Basis of Habitat

On the basis of habitat, aquatic animals are divided into three categories: Nekton, Plankton and Benthos.

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I. Nekton

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It includes aquatic animals, such as vertebrates like fishes, sharks, turtle, etc., and invertebrates like squids. These organisms feed on zoo-planktons to obtain energy so that they can move faster than water current. These are also called **active swimmer**.

II. Plankton

It includes micro-organisms that are free-floating in nature. They comprise zoo-planktons and phytoplanktons. Zooplanktons include small crustaceans, larva of invertebrate. Phytoplankton includes dinoflagellates.

Phytoplanktons are considered extremely vital to the ocean ecosystem and the planet because:

- Phytoplankton acts as a source of oxygen and a carbon sink for the carbon cycle- Biological carbon pump.
- They act as primary producers- 'grasses of the sea' creating organic compounds using sunlight and CO2 through photosynthesis.
- They are also at the first trophic level in the food chains of the oceans, being an essential source of nutrition for zooplankton, small fishes like krill and ultimately for larger marine animals.
- They are also essential for their supporting services, like enabling the smooth functioning of biogeochemical cycles.

However, they have no bearing on the density of ocean water, as it is not dependent on the quantity of phytoplankton, but on factors such as salinity, temperature, and pressure

III. Benthos

These animals live on sea floor. They either live independently at the ocean bed or are attached to the sea surface. Like nekton and planktons, they do not swim in water. Echinoderms, crustaceans, mollusks, poriferans and annelids come under it.

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ENV 05-Bioluminescence

Bioluminescence

It is an important feature of aquatic animals. Animals living in deep sea levels are identified by the presence of different luminescent levels. They are found abundantly in euphotic areas due to the presence of flagellates. Luminescence occurs mainly in oceanic fishes. It also occurs in some cephalopods and crustaceans. These are light-emitting organs, which are generally present on the ventral side. In some animals, they are present on different body parts like in anglerfish, luminous glands are present on tentacles, which are formed of dorsal fin ray.

Bioluminescence Reaction

'Luciferin' is derived from a Latin word 'Lucifer' which means 'Light Bringer'. Hence, the light emitted by living organisms due to certain chemical reactions is called bioluminescence. Two types of substances involved in the process of bioluminescence are:

1. **Luciferin:** It is a light-producing substance.
2. **Luciferase Enzyme:** Luciferin gets oxidised in the presence of luciferase enzyme to produce oxyluciferin and light.

Some examples of bioluminescence organisms are **sea cucumber, dragon fish, hatchet fish,** etc.

Functions of Bioluminescence

The functions of bioluminescence are given below:

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- To lure prey or hunt for prey.
- To protect themselves from other predators.
- To find a mating partner.
- Execute different vital activities.

Do You Know?

Some animals detach their body part to distract predator, e.g. Brittle star.

Echolocation

Aquatic animals use sound waves to get information about their surroundings. They produce a sound that reflects back when they strike an object. This is known as echolocation. It helps the animals to navigate and feed in deep dark water bodies, e.g. **toothed whale, sperm whale, dolphin,** etc.

Deep Sea Adaptation

Benthic community live in deep sea or oceanic floor also known as **'benthos'**. It includes deep sea fishes, crustaceans, squids, echinoderms and a few worms.

Benthos consists of the following ecological features:

- Absence of light
- Competition for food
- Soft bottom
- Stagnant water
- Low temperature
- Absence of phytoplankton
- Their body is partially flattened due to high pressure.
- Some species have large eyes while others have small eyes. The species having small eyes have well-developed olfactory organ.
- Most of the species develop spine, stalk and other associated organs. These appendages are useful in the absence of light and work as contact

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receptors.

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■ *Marine Upwelling Zones*

Upwelling of water refers to the phenomenon when **ocean currents** interact to bring the cold nutrient-rich water to the surface, where the bright sunlight fuels the rapid growth of phytoplankton. It improves marine productivity in the following ways:

- It leads to enrichment of nutrients like **nitrogen and phosphorus** in the surface water.
- The influx of vital nutrients and sunlight fuels the growth of **phytoplankton blooms**, which provide food for several organisms like zooplankton and fish, i.e.. phytoplankton forms the **basis of the marine food web**.
- An increase in food availability creates scope for the **enrichment of marine biodiversity**.
- The abundance of marine life has economic benefits by providing **highly productive fishing grounds**. For example, the coasts of **Peru** (Humboldt current) or **Japan** (Kuroshio and Oyashio currents)

3.3.4 Desert Adaptation

The desert ecosystem is characterised by low rainfall and high pressure. Many biological factors like temperature, light, etc. influence the presence of organisms, their distribution and adaptation.

Herbivorous, **rodents**, **reptiles** and **insects** are commonly found animals in this region, while birds and mammals are rare species.

Features

- Desert animals and plants have high water retention capacity.
- Due to less water consumption, reptiles and insects release dry excreta.
- Rodents feed on dry seeds so that they consume less water.
- Generally, all desert animals are burrowing animals to escape from intense sunlight.

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- Many desert animals go for long summer sleep of a few months. This process is called aestivation. All the metabolic functions of the body get slowed down during this time.
- For defensive purposes, their body is covered with horny scales.
- They can also change their colour according to their surroundings for self-defense.
- Mostly all the desert animals are poisonous in nature.

3.3.5 Flight Adaptation

Flight adaptation is of two types:

1. Active
2. Passive

Example of active flight birds—insects, bats and flying lizards.

Example of passive flight birds—eagle and vultures.

Features of Flight Adaptation

- Bones are hollow and pneumatic which makes their body light weighted.
- Broader wings help them in flying.
- Small tail helps in maintaining their balance.
- Sensory organs and brain are well developed, and eyes are surrounded by a sclerotic layer, to protect the eyes from high pressure during flight.
- Physical features are also in favour of birds, like absence of urinary bladder, rectum is smaller, presence of single ovary in female birds which is situated on the left side.

3.3.6 Subterranean Adaptation

Some animals are adapted to dig and live underground in burrows. This type of adaptation is called subterranean adaptation or fossorial adaptation.

E.g. Talpa-mole, *Sorex*, *Erainaceous*, *Hytrix*, *Tachyglossus*, some nocturnal animals also show this type of adaptation.

Features of Fossorial Adaptation

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- Presence of snout
- Fusiform body which helps the animals to make easy entry and exit from the burrow
- Pinna is small
- Hibernation
- Small eyes
- Small legs

Features of Parasitic Adaptation

- Various types of suckers are present which are attached internally or externally to the host.
- Generally complete their life cycle on primary and secondary hosts. Both host and parasite live in the same habitat. e.g., Parasite of malaria, i.e. *Plasmodium* have two host—First: Female *Anopheles* mosquito and Second: Humans
- Have high reproductive capacity, and reproductive organs are well developed. e.g. On an average, every liver fluke lays 5 lakh eggs, whereas *Ascaris* can lay more than 2 lakh eggs.
- Endoparasites complete a complex life cycle in which they produce different polyembryo.

3.3.7 Poisonous Species

Poisonous species are organisms that are harmful when touched, eaten, or inhaled, with toxins acquired from their environment or diet.

Examples include the Golden Poison Frog from its diet of toxic ants, the Pufferfish with toxins from its diet, and the poisonous Hooded Pitohui bird, which gets toxins from its insect food.

Here are some examples of poisonous species across different groups:

Animals

Frogs: Many Poison Dart Frogs are highly toxic due to chemicals acquired from their diet of ants and beetles, according to the textbook website.

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Fish: The Pufferfish contains a lethal poison called tetrodotoxin, which is more potent than cyanide, and is acquired from its diet. The Stonefish, considered the most venomous fish, has spines that deliver venom under pressure.

Birds: The Hooded Pitohui of Papua New Guinea is one of the few known poisonous birds, containing batrachotoxin in its skin and feathers, which it likely acquires from its insect diet.

Amphibians: The **Spanish Ribbed Newt** can poison predators by pushing its sharp ribs through its skin. **All frogs produce poisonous skin secretions, but not all are lethal for humans.** Colourful exteriors often indicate distasteful and poisonous species. For **example**, dart frogs are considered some of the widely found poisonous frogs, with the golden poison frog being the most poisonous.

Insects & Invertebrates: Blister beetles secrete a toxic substance that can cause blisters.

Butterflies: These species, due to their bright colours, are easy targets for predators. Hence, they are usually **equipped with distasteful or toxic chemicals used for deterrence.** Thus, **brightly coloured butterflies usually indicate poisonous species. Example:** Monarch butterfly, which ingests such chemicals through plants.

Mammals: Some species, like the European mole and shrews, have poisonous saliva that is not deadly to humans but can immobilise prey.

Fungi

Many mushrooms, such as species in the **genera Cantharellus and Morchella**, are poisonous and can cause illness or death if consumed.

3.4 ADAPTATIONS ON THE BASIS OF TEMPERATURE (IN ANIMAL)

Like plants, animals are also classified into three categories.

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3.4.1 Poikilothermic or Ectothermic Animals

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Cold-blooded animals are known as poikilotherms. The body temperature of these animals changes with the environment. e.g. invertebrates, fish and amphibians.

3.4.2 Homeothermic or Endothermic Animals

Warm-blooded animals are known as homeothermic. Their body temperature does not depend upon environmental temperature. e.g. birds and mammals.

3.4.3 Heterotherms

Some animals such as **monotremes** and **marsupials** (Kangaroo, Opossum, etc.) have a limited power of temperature regulation. They respond to extreme temperatures by **hibernating** or **aestivating** mechanisms.

3.5 TERMINOLOGY OF ANIMAL ADAPTATIONS

3.5.1 Mimicry

The concept of mimicry was given by H.W. Bates in 1862. It is the characteristic feature of insects, in which one organism copies another organism's appearance, structure or colour. Some birds and mammals also perform this action to protect themselves from predators.

Mimicry can be classified into two categories:

(i) Protective mimicry

(ii) Aggressive mimicry.

(i) Protective Mimicry: This type of mimicry is done for protection from the predators. Protective mimicry is of two types.

(a) Concealing Mimicry: In this type of mimicry, animals try to look similar to another organism to hide themselves in the surroundings.

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e.g. it is difficult to differentiate between *sargassum* fish and *sargassum* algae.

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(b) Warning Mimicry: When warning is given for security purpose, it is called warning mimicry. e.g. A non-venomous snake is similar to a highly venomous snake. Highly venomous snakes have flattened head and tail which is exactly similar to the nonvenomous snakes.

(ii) Aggressive Mimicry: In this type of mimicry, animals imitate their surrounding environment not only for their protection but also for predation.

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MIMICRY

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The concept of mimicry was given by H. W. Bates in 1862. It is a characteristic feature of insects, though some birds and mammals also exhibit it. Mimicry is a phenomenon where one organism copies the appearance, structure, or color of another organism to gain an advantage, often for protection or predation

TYPES OF MIMICRY

Mimicry can be broadly classified into two categories:

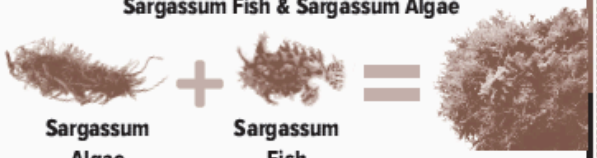
1 Protective Mimicry

This type of mimicry is done for protection from the predators

a Concealing Mimicry

The animal tries to resemble another organism or its surroundings to hide itself

E.g. It is difficult to differentiate between Sargassum Fish & Sargassum Algae



b Warning Mimicry

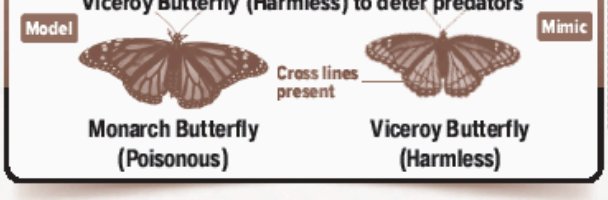
Harmless animals imitate the appearance of harmful ones to warn predators

E.g. A Non-Venomous Kingsnake mimicking the, brightly colored Venomous Coral Snake to deter predators

i Batesian Mimicry

One animal that is harmless looks similar to the one who is harmful

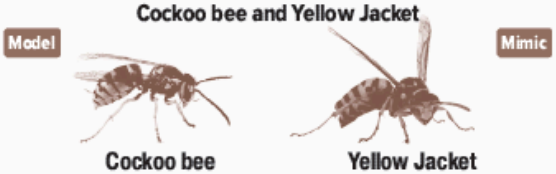
E.g. Monarch Butterfly (Poisonous) is mimicked by Viceroy Butterfly (Harmless) to deter predators



ii Mullerian Mimicry

Multiple dangerous species resemble each other, sharing common predators

E.g. It is difficult to differentiate between Cockoo bee and Yellow Jacket



2 Aggressive Mimicry

Animals imitate their surroundings for protection and predation

Mimicry can also be classified as Batesian mimicry and Mullerian mimicry.

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(i) **Batesian Mimicry:** In this type of mimicry, one animal that is harmless looks similar to the one that is harmful. e.g. Monarch butterfly is poisonous, so the Viceroy butterflies mimic like the Monarch, to protect themselves from the predators.

(ii) **Mullerian Mimicry:** This type of mimicry occurs between two or more inedible and poisonous species, that may resemble each other and share one or more common predators. They mimic each other's warning signals to protect themselves from the predator.

Reasons for Mimicry: Natural selection and mutation is considered as the main reasons for mimicry.

3.5.2 Hibernation

Hibernation is a condition where warm-blooded animals try to maintain their body temperature and protect themselves from cold temperature during winter. So, it is also known as **winter sleep**. (The animals whose body temperature does not change or vary with the change in surrounding temperature, are known as warm-blooded animals). e.g., bat, bear, rodents, etc.

3.5.3 Aestivation

Aestivation is a state of animal dormancy, similar to hibernation, characterised by inactivity and a lowered metabolic rate that is entered in response to high temperatures and arid conditions **(Summer Sleep)**. It takes place during times of heat and dryness, the hot dry season, which are often the summer months.

3.5.4 Diapause

Diapause is the stage of development in which larvae stop their development in unfavourable conditions (like very high/low temperatures, scarcity of food) and when they get favourable conditions, they again start to develop where they stopped.

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3.5.5 Dormancy

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Dormancy is a state of the body in which the metabolism, growth, etc. are kept at a very low level for varying periods of time.

3.5.6 Migration

Migration involves long-distance or short-distance movement of animals from one region to another. Many organisms that fly or swim undertake extensive migrations. Arctic terns are sea birds that make a round trip of thousands of miles between their North Atlantic and the Arctic breeding grounds to the Antarctica every year. In Africa, wild beasts migrate long distances, following a geographical of fresh vegetation. Locust migrates in search of new feeding grounds from the food depleted areas in large numbers to the arid regions.

3.5.7 Camouflage

In some animals, the capacity to blend with surroundings or camouflage is a common adaptation. Some insects, reptiles and mammals have markings on their bodies, which make it difficult to distinguish them from shadows and branches, or from other members of the group.

3.5.8 Warning Colouration

Concealing form and colouration enables a species to avoid its natural predator. The brightly coloured and highly poisonous dart frogs *(Phyllobates bicolour; Dendrobates pumilio)* of the tropical rain forests of South America are easily recognised and avoided by the predators.

3.5.9 Marsupials Adaptations

Marsupials are Metatherian Mammal characterised by premature birth and continued development of the newborn while attached to the nipples on the mother's lower belly i.e. **marsupium pouch**. It is not a universal feature but

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many marsupial species have a marsupium pouch. There are different natures of marsupials worldwide like forest habitat, trees, forest floor and even burrow underground. Australian marsupials generally live in dry scrub or desert areas and South American marsupials live in forests or tropical rainforest region. Marsupials are endemic in Australasia, Wallacea (It is located between Asia and Australia) and the Americas.

List of marsupials worldwide: kangaroos, wallabies, wombats, koala (*Phascolarctos*), Tasmanian devil, opossum (*Didelphis*), Quoll (*Dasyurus maculatus*) and bandicoot (Perameles).

ENV 06- Adaptations in Bees

3.5.10 Adaptations in Bees: Round and Wagging/Waggle Dance

The following types of dances can be seen in honey bees:

Round Dance: This dance indicates that the food source is about less than 75 m from a hive. The direction of the food source can be identified with the help of the smell of flowers which is present on the body of a scout honey bee.

Tail wagging Dance: With the help of this dance, bees give the information that food source is at a very far distance (more than 75 m). The direction and distance of the food source are indicated according to the position of the sun.

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When the honey bee flies in a straight line, it wags its tail and wings to produce the sound. The distance of a food source can be identified with the help of the speed of dance, speed of wagging tail and speed of sound.

- If the motion is on the upper side of a straight line with a wagging tail, then the food source will be in the same direction of the sun.
- If the motion is on the lower side of a straight line, then the food will be in the opposite direction of the sun.

Finally, we can say that the waggle dance tells the watching bees two things about a flower patch's location – 1. the distance and 2. the direction away from the hive.

3.5.11 Miscellaneous Case: Adaptations to Water Scarcity

Two types of adaptations are prominent in animals living in arid regions, viz. lowering of water loss as much as possible, and adapting to arid conditions. For example, the kangaroo rat conserves water by excreting solid urine, and can live their entire life without even drinking water. Camels show unique adjustments to desert condition, being very economical in water use, tolerant to wide fluctuations in body temperature and are able to maintain blood stream moisture even during extreme heat stress.

Adaptations to Cold

Sessile animals, such as barnacles and molluscs, living in very cold inter-tidal zones of northern shores, and several insects and spiders resist the effect of cold spells by a process known as cold hardening. The freeze tolerant organisms have ice nucleating proteins, which induce ice formation in the extracellular spaces at very low sub-zero temperatures. Some freeze – avoiding animals can tolerate environmental temperatures below 0°C by accumulating **glycerol or antifreeze proteins** that lower freezing point of

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their body fluids. The presence of such antifreeze compounds allows the fish in Antarctica region to remain active in sea water.

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3.5.12 Roll up and Protects

To reduce the chance of being captured by predators, some of the organisms roll up and protect their vulnerable parts. e.g. Hedgehog and Pangolin.

3.5.13 Adaptations in Orangutan

A study was conducted on five captive, group-living Sumatran orangutans (Pongo pygmaeus abelii) living at the San Diego Zoo where a man-made mound was provided to examine the use and making of tools by them to obtain foodstuffs in the artificial-mound holes. Three adult orangutans frequently stripped leaves and twigs from a branch provided (tool-making), and then inserted the tool into a hole to obtain foodstuffs (tool-using).

When the orangutans were unable to insert a tool into a hole, they usually modified the tool and/or changed their tool-using technique, such as changing how they grasped the tool.

3.5.14 Nests for Squirrels

Most squirrel nests are called dreys, and consist of clumped-together collections of leaves, twigs, bark, moss, and other compressed materials. They look like small, round bulbs of leaves bunched together. Squirrels usually build their dreys into tree cavities or around tree branches that are 20 feet high or higher. The squirrel is an omnivore. This means it will eat just about anything though it prefers fruit and nuts. In India, palm squirrels prefer crops such as nuts, sugarcane, grapes, mangoes and apples. In addition, Indian palm squirrels will not hesitate to eat crops as well as eggs and even chicks found on poultry farms.

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ENV 07-Adaptation in Hydrophytes and Xerophytes

3.6 ADAPTATIONS IN PLANTS

Adaptation in hydrophytes, mesophytes, xerophytes and halophytes is as follows.

3.6.1 Hydrophytes

It can be classified into the following categories-

- **Free floating:** *Lemna, Wolffia, Spirodela, Echornia,* etc.
- **Submerged floating:** *Utricularia, Ceratophyllum, Najas,* etc.
- **Rooted submerged:** *Hydrilla*, *Vallisneria*, *Potamogeton,* etc.
- **Rooted with floating leaves**: Nymphaea, Victoria, Trapa, etc.

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HYDROPHYTES

Epidermis
Single layed Epidermis

Stomata
Generally absent

Leaves
Ribbon-shaped or dissected; may be swollen or bulbous (e.g., Eichhornia, Trapa)

Stem
Stem are thin, long & flexible

Specialized tissues
Aerenchyma tissues are present which are filled by air. (Roots, stem & leaves)

Roots are submerged in water, but small in size.

Xylem
Less developed

Nature
Plants may be free floating or submerged

Root
Generally absent, less developed

XEROPHYTES

Epidermis
Waxy coating developed on leaves & stem

Cuticle
Thick, present on leaves & stem

Leaves
Small, thick, leathery; often modified into scales; some grasses with oval leaves

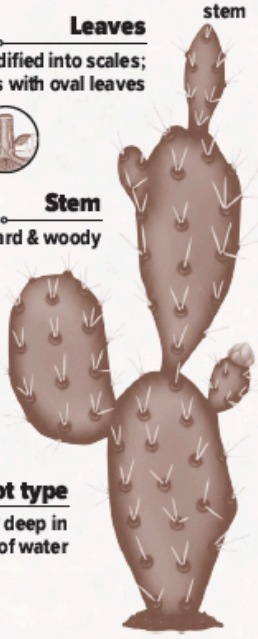
Specialized tissues
Presence of oil glands, mucilage canal in root, stem & leaves

Stomata
Sunken stomata

Stem
Hard & woody

Bark
Thick bark

Xylem
Well developed



Root
Well developed even more elaborate than stem

Root type
Presence of tap roots which grow deep in earth in search of water

Root hairs & caps
Well-developed

Definition

Hydrophytes are plants that are specifically adapted to live in aquatic environments, such as ponds, lakes, and marshes, or in waterlogged soils

Definition

Xerophytes are plants occurring in dry and desert areas. They are also known as plants of dry habitats

Classification

It can be classified into the following categories -

Free Floating
e.g. Eicchornia etc.

Submerged floating
e.g. Utricularia etc.

Rooted with floating leaves
e.g. Victoria plant etc.

Rooted submerged
e.g. Hydrilla etc.

Classification

It can be classified into the following categories -

e.g. Wild Safflower

Ephemeral Annual

They adapt to dry conditions both internally and externally. They are short-lived and are also called drought escapers or drought evaders

e.g. Aleovera etc.

Succulents

They are plants with fleshy, swollen organs due to the active accumulation of water, which they utilize during extreme drought conditions

Adaptation

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1. Roots are generally absent or less developed. e.g. *Ceratophyllum*, *Wolffia.*
2. Root hairs are absent or partially developed.
3. Root cap is absent.
4. Plants may be free floating or submerged.
5. Stems are thin, long and flexible.
6. Leaves may be either ribbon-shaped or dissected.
7. Some leaves can be swollen or Bulbous. e.g. *Echornia*, *Trapa*.
8. Single layered epidermis.
9. Aerenchyma tissues are present which are filled with air. These tissues are found in the roots, stem, or nodes. They increase the submerging capacity of plants.
10. Xylem is less developed.
11. Stomata are generally absent.

3.6.2 Mesophytes

Plants grow in terrestrial habitat under normal conditions and are classified into the following categories:

- **Herbaceous Communities:** The shrubs like vegetation which are found in Arctic and Antarctic grasslands, agricultural and pastoral lands, etc.
- **Community of Woody Plants:** It includes mesophytic bush land, evergreen and deciduous forests. Evergreen forest includes tropical rain forest and sub-tropical forest.

Adaptation

1. Well-developed roots with root cap and root hair.
2. Well-developed stems, which are generally aerial, subaerial or underground.
3. Leaves are green, thin and less broad. They do not have any wax coating.
4. Thin cuticle is present on the outer layer.

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5. Well-developed xylem.
6. Stomata are present on one or both sides of the epidermis.

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3.6.3 Xerophytes

Plants occurring in dry and desert areas are called Xerophytes. They are also known as plants of dry habits. Xerophytes can be classified under two broad categories:

1. **Ephemeral Annual:** These xerophytes are adaptive for dry conditions both externally and internally. During unfavourable conditions, plants are survived in the form of seeds and fruits with hard and resistant seed coat and pericarp. During favourable conditions, they develop into an entire new plant. They are short-lived plants. They are also known as draught escapers or draught evaders.
2. **Succulents:** In this type of adaptation, the plant's organs become **fleshy** and swollen because of active accumulation of water in them, which is utilised during extreme draught conditions.

Adaptation

1. Root system is well developed and more elaborated than shoot system.
2. Presence of tap roots which grow deep in earth in search of water.
3. Well-developed root hair and root cap.
4. Thick cuticle present on leaves and stem.
5. Wax coating develops on the leaves and stem.
6. Presence of thick bark.
7. Stems are hard and woody.
8. Phylloclade and cladodes are present.
9. Generally, leaves are modified into scales *Muchlenbeckia.*
10. In majority of the plants, leaves are small, thick and leathery, e.g. *Ruscus.*
11. In some xerophytic grasses, leaves are oval, e.g. Agropyrom.
12. Xylem is compact and well developed.
13. Presence of sunken stomata.
14. Presence of oil glands, mucilage canal in the root, stem and leaves.

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3.6.4 Halophytes

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The plants which grow in saline soil are called halophytes.

Halophytes are of two types:

(i) Hydrohalophytes – Entirely submerged plants.

(ii) Hygrohalophytes – They have emerged roots, found in coastal areas and are of two types:

- Mangroves or Swamp plants.
- Marsh halophytes.

Adaptation

1. Apart from normal roots, many stilt or prop roots are developed from branches.
2. Presence of aerotrophic respiratory roots. It is called **pneumatophores**.
3. Leaves are thick, small and hairy.
4. Viviparous mode of germination.
5. Presence of thick cuticle.
6. Aerenchymatous tissues are present which are filled with air.
7. Well-developed vascular tissues.
8. Stomata are present on the lower surface of the leaf.
9. Sunken stomata are present on the leaves.

Some common examples of halophytes are:

Crassa cretica, Salicornia herbacea, Salsola, Sensurium portulacastrum, Serpentina alterniflora, Spinifex, Quarrosus, Suaeda fruticosa and some mangroves plants like *Aegiceras majus, Avicennia officinalis, Acanthus mollis, A. Ilicifolius, Bruguiera gymnorrhiza,* etc.

Classification of Vegetation According to Warming

On the basis of requirement of water and composition of soil and soil types, Warming classifies vegetation into 3 large groups and 13 small groups, which are given below:

(i) **Hydrophytes:** Plants which live in water.

(ii) **Helophytes:** Marshy plants.

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(iii) **Oxylophytes:** Plants which grow in acidic soils.
(iv) **Lithophytes:** Plants which grow on stones.
(v) **Halophytes:** Plants which grow in saline soils.
(vi) **Psammophytes:** Plants which grow in sand and pebbles.
(vii) **Chersophytes:** Plants which grow in waste land.
(viii) **Eremophytes:** Plants which grow in desert.
(ix) **Psychrophytes:** Plants which grow in cold soils.
(x) **Psilophytes:** Plants which grow in savannah.
(xi) **Sclerophytes:** Plants which grow in forest and bush land. Sclerenchyma present in the leaves of the plants.
(xii) **Mesophytes:** plants that grow in moderate amount of water
(xiii) **Corniferous:** It includes coniferous forest.

3.7 MICROBIAL ADAPTATIONS/WIDE RANGE OF HABITAT IN MICROORGANISM

Microorganisms can be found in various types of habitats, displaying incredible diversity as they thrive in environments ranging from very cold to extremely hot conditions.

They demonstrate remarkable tolerance to various environmental factors, including limited water availability, high salt content, and low oxygen levels. However, not every microbe can survive in all types of habitats due to their specific adaptations and requirements.

3.7.1 Terrestrial Microbial Habitats

- Only a small fraction (one percent) of the microbes residing in the soil has been identified so far.
- These soil organisms play a crucial role in soil formation and are integral components of their respective ecosystems.
- Bacteria and fungi in the soil primarily derive their nourishment from organic matter, such as decomposing plants and animals.

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- The growth of these microbes is highly influenced by factors like carbon dioxide and oxygen levels, pH, moisture, and temperature in their local environment.

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3.7.2 Aquatic Microbial Habitats

- Microbes inhabit both freshwater and saltwater environments.
- This category includes microscopic plants, animals, bacteria, fungi, and viruses.
- Similar to other microorganisms, aquatic microbes are well-adapted to the specific conditions of their water-based environment, spanning from high-salt ocean water to freshwater lakes and rivers.

3.7.3 Microbial Habitats in other Organisms

- Microbes also reside on various organisms, and their presence can be either beneficial or harmful to the host.
- For instance, bacteria can form nodules on the roots of pea and bean plants, converting atmospheric nitrogen into a usable form for the plants.
- Animals and plants have evolved in many ways to become suitable habitats for the millions of microbes that live in symbiosis with them.

3.7.4 Extreme Microbial Environments

- The microbes living in extreme conditions are called **extremophiles**.
- This means that they love the extreme conditions of their habitat.
- The extremophiles are so well adapted to their own environment.
- Some like the ones in hot springs need extreme temperatures to grow.
- Some microbes, referred to as **thermophiles**, can survive at temperatures higher than the boiling point of water, with some even thriving at temperatures above 100°C (212°F). **Deep-sea hydrothermal vents and volcanic hot springs are two locations where these heat-loving extremophiles can be found**.
- **Psychrophiles**, also known as **cryophiles**, are microorganisms that can thrive and reproduce in temperatures lower than the freezing point of

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water (0°C or 32°F). In order to function and thrive in conditions such as arctic ice, glaciers, permafrost, and deep-sea freezing waters, these cold-loving species have developed unique adaptations, such as cold-adapted enzymes and antifreeze proteins.

- **When the pH is less than 3**, certain microbes can thrive in an extremely acidic environment. In order to keep their internal pH close to neutral, **acidophiles** have active proton export mechanisms, also known as **proton pumps**, that move extra protons out of the cell.

3.7.5 Symbiosis related to Environment Microbiology

Symbiosis

- 'Symbiosis' refers to a relationship between two organisms wherein one organism relies on the other for its survival, and in some cases, they mutually benefit from each other's presence.

Examples

- Photosynthetic plants and microbes play a vital role in providing oxygen which is essential for human survival.
- Trees serve as shelters for various plants and animals, fostering a symbiotic environment.
- Numerous organisms depend on other creatures as their sources of food or nutrients, establishing symbiotic connections in nature.

Symbiotic Microbes

- Microbes frequently engage in symbiotic relationships, wherein one organism lives inside another, known as the host.
- These microbial symbiotic associations can occur between two different types of microbes, or they may form associations with plants and animals.
- Bacteria, with a long history of symbiotic relationships, have co-evolved with their hosts. Similarly, other microbes like fungi and protists also form symbiotic relationships with various organisms.
- **HOST –** usually the LARGER member
- **SYMBIONT –** usually the SMALLER member

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Population Ecology 4

IN THIS CHAPTER...

4.1 Population
4.2 Population Density
4.3 Natality (Birth rate)
4.4 Mortality (Death rate)
4.5 Vital Index
4.6 Survivorship Curve
4.7 Growth Curve
4.6.1 First Type or Highly Convex Type
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4.6.3 Third Type or Highly Concave Type
4.7.1 Exponential Growth Curve/J-Shaped Curve
4.7.2 Logistic/Sigmoid/S-Shaped Growth Curve
4.8 Biotic Potential
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4.1 POPULATION

A group of individuals of a particular species occupying a definite space in which the individual interacts, interbreeds and exchanges genetic information is referred to as 'population'. A population can be divided into small groups called **demes** or **local population**. The gap between human demands for ecological resources, and how much the earth can provide is called **population overshoot**.

4.2 POPULATION DENSITY

Population density refers to the size of any population in relation to some unit of space. It is expressed in terms of the number of individuals or biomass per unit area or volume. Population density changes with time and space.

$$D = \frac{n/a}{t}$$

(where D = density, n = number of individuals, a = area, t = time)

4.3 NATALITY (BIRTH RATE)

It refers to the rate of reproduction or birth per unit time.

$$\text{Birth Rate} = \frac{\text{No. of birth per unit time}}{\text{Average population}}$$

There are two types of natality:

1. **Fundamental/Maximum/Absolute/Potential Natality:** It is the maximum natality under ideal environmental conditions and unlimited resources. It is constant for a given population.
2. **Ecological/Realised Natality:** It is an actual natality under existing environmental conditions and it is not constant.

4.4 MORTALITY (DEATH RATE)

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It refers to number of deaths per unit time.

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$$\text{Mortality rate} = \frac{D}{t}$$

(Where D = number of deaths, t = time)

Mortality is of two types:

1. **Specific/Minimum/Potential Mortality:** It represents the minimum loss of individuals under ideal conditions. It is constant for a population.
2. **Actual/Realised/Ecological Mortality:** It represents loss of individuals under actual environment. It is not constant.

4.5 VITAL INDEX

The representation of birth-death rate in the form of birth-death index is called **vital index.**

$$\text{Vital Index} = \frac{\text{Birth rate}}{\text{Death rate}} \times 100$$

ENV 08- Survivorship Curve

4.6 SURVIVORSHIP CURVE

Survival of individuals is more important than the death of individuals for any population. Survival rate is the difference between total individuals produced and the mortality in a given period of time.

$$x = 100 - qx$$

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(where, *x* = survival rate, 100 = total produce individual, *qx* = mortality rate)

Types of Survivorship Curves: By plotting the number of surviving individuals against time would produce a survivorship curve in the form of a right angle. There are three different types of survivorship curves, which describe the number and behaviour of individuals in their complete life cycle (Figure 4.1).

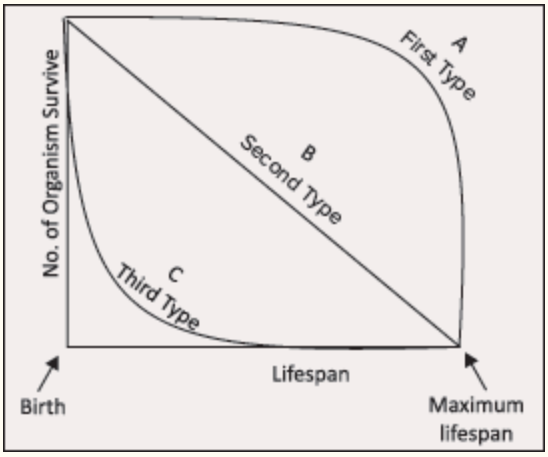

Figure 4.1 Survivorship Curve

4.6.1 First Type or Highly Convex Type

In this type of curve **(curve - A),** population mortality rate is low until the end of lifespan under ideal conditions. In this type of curve, animals take birth and die at about the same time. Examples - deer, sheep, perennial plants, humans, etc.

4.6.2 Second Type or Diagonal Type

Mortality rate in this type of curve **(curve - B)** is constant at all age levels. Example – *Hydra*.

4.6.3 Third Type or Highly Concave Type

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Mortality rate is high during the early stage and then becomes constant in all other ages in this curve **(curve - C)**. Examples – *Ascaris*, annual plants, oyster, etc.

4.7 GROWTH CURVE

When a curve is obtained after plotting number of individuals of a population on the *Y*-axis and the time on the *X*-axis, it indicates the growth of population size in a given time. There are two types of growth curves.

4.7.1 Exponential Growth Curve/J-Shaped Curve

There is no increase in the population size during the early phase because they require time to adjust themselves as per the new environment. After establishing in the new environment, it starts increasing rapidly. This increase in the population continues till the large amount of food is available in the habitat. After some time, the population size starts decreasing because of the increased population size and limited food supply in the habitat. This can be seen as a J-shaped curve (Figure 4.2).

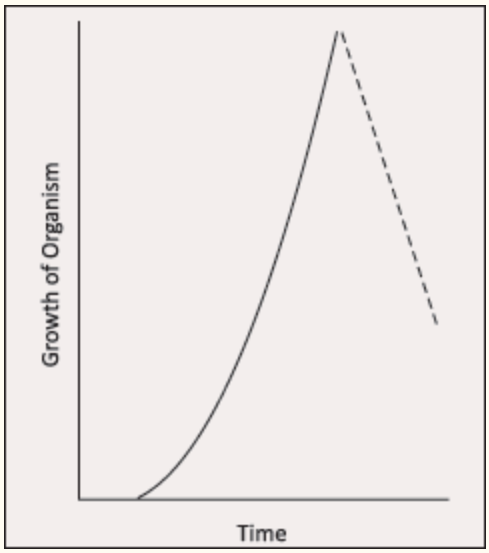

Figure 4.2 J-Shaped Growth Curve

$$\frac{dN}{dt} = rn$$

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where,

$\frac{dN}{dt}$ = Rate of change in population size/Growth rate

r = Biotic potential

N = Population size

In the J-shape curve, density increases rapidly in exponential manner and then stops abruptly as environmental resistance and other factors become more effective.

4.7.2 Logistic/Sigmoid/S-Shaped Growth Curve

When a few organisms are introduced in an area, the population increases slowly in the beginning because it needs time to adjust as per the environment. In the middle phase, population rapidly increases, when it adjusts with the conditions. Finally, in the last phase, the population growth slow down until equilibrium is attained. After which the population size fluctuates according to the variability of environment. The level beyond which no major increase can occur is referred to as 'saturation level' or 'carrying capacity'. In the last phase, the new organisms are almost equal to the number of dying individuals and thus, there is no increase in population size. In this way, one gets sigmoid or S-shaped growth curve (Figure 4.3).

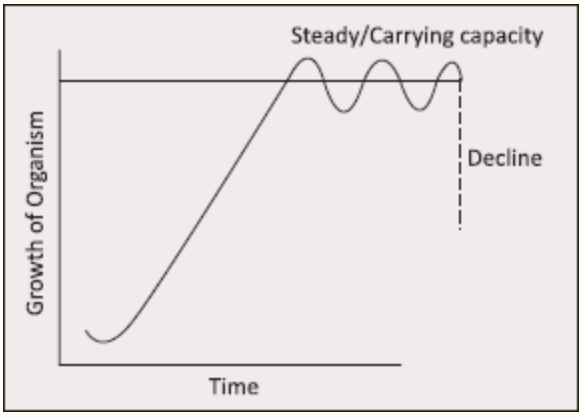

Figure 4.3 S-Shaped Growth Curve

$$\frac{dN}{dt} = rn\left(\frac{K-N}{K}\right)$$

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where,

$\frac{dN}{dt}$ = Rate of change in population size/Growth rate

r = Biotic potential

N = Size of organism

K = Carrying capacity

4.8 BIOTIC POTENTIAL

Maximum reproductive capacity of a population under optimum environmental conditions is called *Biotic potential*. It is generally only reached when environmental conditions are favourable. A species reaching its biotic potential would exhibit exponential population growth and it is said to have a high fertility, that is, how many offspring are produced per mother.

Components of biotic potential include reproductive potential/potential natality (the upper limit to biotic potential in the absence of mortality) and survival potential (because reproductive potential does not account for the number of gametes surviving; survival potential is a necessary component of biotic potential; it is the reciprocal of mortality) (In the absence of mortality, Biotic potential = Reproductive potential).

4.9 CARRYING CAPACITY

A number of factors in the environment, such as food, oxygen supply, diseases, predators and space determine the maximum number of organisms that can be sustained in a given area and is called the carrying capacity or Population carrying capacity of that area.

The carrying capacity of a biological species in an environment is the population size of the species that the environment can sustain indefinitely, given the food, habitat, water and other necessities available in the

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environment. In population biology, carrying capacity is defined as the environment's maximal load, which is different from the concept of population equilibrium.

4.10 POPULATION DISPERSAL

The movement of individuals in-and-out of a population is called population dispersal. It plays a significant role in the geographical distribution of organisms in those areas which was previously unoccupied by the members of the population. Dispersal of organisms occurs for various reasons such as light, temperature, breeding behaviour, psychological reasons or for interchange of genetic material.

Population dispersal occurs in nature in following three ways.

4.10.1 Emigration

It is one-way movement of individuals out from the population. It is a permanent process which causes spread of species to new areas. Emigration under natural conditions occurs mainly because of overcrowding in the population and is generally regarded as an adaptive behaviour that regulates the population on a particular site and prevents overexploitation of that habitat. This type of dispersal offers new opportunity to the individual of a population to interbreed with those of the other population leading to more heterozygosity and adaptability.

4.10.2 Immigration

It is the entry of new individuals which leads to an increase in the density of the population. It may result in decreased mortality among the immigrations or decreased reproductive capacity of the individuals.

4.10.3 Migration

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It is the two-way mass movement of the entire population. It involves periodic returning of the individuals of a population during unfavourable conditions. It is mainly shown by many birds, fishes and certain mammals. Migration of population mainly occurs for food, shelter or reproduction. Migration benefits the population in the process of their dispersal. It also avoids interspecific competition for food and shelter.

4.11 POPULATION DISPERSION

It is the process in which individual tries to fulfil its requirement from resources and factors within the geographical boundary where as in population dispersal the maximum problem occurs with these factors and resources, so organisms move across the geographical boundary.

NOTE: Dispersal always occurs across the boundary but dispersion occurs within the boundary with respect to factors and resources.

ENV 09- r-selection & k-selection

4.12 R- AND K-SELECTION

r-Selected Species: These live in a highly variable population. The fittest individuals in these environment have many offspring and reproduce early, e.g., mosquitos.

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k-Selected Species: These live in populations that are at or near the equilibrium conditions for long period of time. Competition for limited resources is an important feature of k-selected species, e.g., elephant, giraffe, etc.

The term r-selection is used for the species that produce many 'cheap' offspring and live in an unstable environment and k-selection is used for those species that produce few 'expensive' offspring and live in a stable environment (Table 4.1).

Table 4.1 Comparison between r-selection and k-selection

Character	r-Selected Species	k-
Environment	Variable and unpredictable	C
Biotic potential	High	L
Dispersal capacity	High	L
Size	Small	L
Survivorship curve	Concave type	C

Terminology

Ecological Race: It is in the sense that variation is always associated with some ecological factors like Biotic, Edaphic or Climate Conditions. Ecological race is of two types- Ecads or Ecophenes and Ecotypes.

Ecads/Ecophenes: Ecads/ Ecophenes are plants of the same species which appear different due to differing environmental conditions. These variations are not genetically fixed and when transferred to natural conditions, the variations vanish.

For example, in *Euphorbia hirta,* the plant growing in disturbed conditions are prostrate and are erect in undisturbed conditions. The morphological variations are not permanent so when plant is transferred to favourable conditions, the prostrate form may become erect.

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Ecotypes/Genecology: The variations associated with ecological conditions become genetically fixed and such variations are called ecotypes. According to Turresson (1992), an ecotype is '**the product arising as a result of genotypic response of an ecospecies to a particular habitat**'.

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Community Ecology 5

IN THIS CHAPTER...

5.1 Community
5.2 Ecotone and Edge Effect
5.3 Annual Cycle of a Plant
5.4 Community Development
5.5 Community Succession
5.5.1 Causes of Succession
5.5.2 Types of Succession
5.5.3 Plant Succession
5.6 Climax Community
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5.7.5 Ecotone and Ecocline
5.7.6 Ecological Niche
5.7.7 Keystone Spec
5.7.8 Home Range

5.1 COMMUNITY

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A group of different species living in a particular area and interacting with their own environment is called **community.** It is formed by various populations in a particular habitat and their interaction with biotic and abiotic components.

ENV 10- Ecotone and Edge Effect

5.2 ECOTONE AND EDGE EFFECT

The transition zone between two communities/ ecosystem/ biome is known as **ecotone**. For exemple, the ecotone between grassland and forest. Ecotone contains few species from both communities. The total number of species is often greater in the ecotone than in the adjoining communities. The tendency of increased variety and density of some organisms at the community border is due to **edge effect**. The organisms which occur primarily or most abundantly or spend their maximum time in injunction between communities is called **edge species** (Figure 5.1). For example, owl lives on the edge of forest and goes to grassland to feed on rats. **Wetland**, **mangrove**, **estuaries,** etc., are other examples of ecotone.

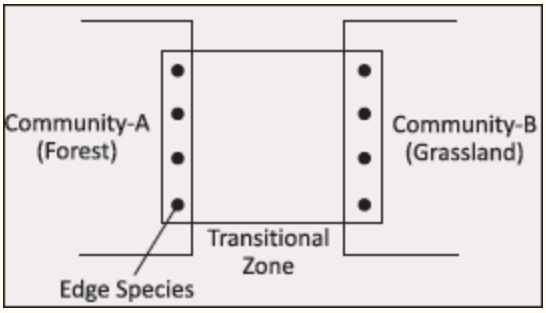

Figure 5.1 Ecotone and Edge Effect

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5.3 ANNUAL CYCLE OF A PLANT

In ecology, annual cycle of a plant is basically subdivided into six phases:

1. **Prevernal Period:** This begins in early March and terminates in mid-April; hibernating mammals emerge and birds migrate to nesting sites.
2. **Vernal Period:** This begins from late April to May. During this period soil temperature increases; new leaves start growing on trees, insects hatch, birds complete their migratory flight and early blooming sets fruits.
3. **Extival Period:** Beginning from June to early July.
4. **Serotinal Period:** Begins from July to early September. This period shows maximum growth activity in plants and animals.
5. **Estivation Period:** It is characterised by very high temperature or extreme dry conditions or both that lower the metabolism of plant and animal communities.
6. **Autumnal Period:** It begins in mid or late September and terminates in early November. There is a gradual defoliation of deciduous vegetation which forces animals to move towards protected habitats. Invertebrates start migrating towards open fields and grasslands from forest. Vertebrates collect food and prepare themselves for adverse conditions.
7. **Heimal Period/Winter Interval:** It begins in late November and organisms which cannot tolerate low temperature either hibernate or die.

5.4 COMMUNITY DEVELOPMENT

The process of succession is generally referred as **community development**. It involves the following stages in a sequential manner:

A. **Nudation:** The initial requirement from succession is formation of bare land. As already described, it is the process through several agencies or natural calamities such as erosion, deposition, wind, water, gravity, slipping of glaciers, fire, lightening, volcanic activities, grazing and by destructive activities of humans.

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B. **Migration:** The introduction of new life-forms in bare land is initiated by the act of migration. It is done by the dispersal of seeds and spores through different agencies. The reproductive structures remain attached to the plants and propagate through soil. The first group of plants those arrive at the bare land is called as **pioneers**.

C. **Ecesis:** The process of establishment of species after migration is called **ecesis**. Under this, the seeds, spores and other vegetative bodies germinate, grow and then reproduce. The mortality rate of seedlings is very high.

D. **Aggregation:** Pioneers are small in number but soon they start increasing. The migrants also begin to multiply and soon they may exceed the pioneers. This situation is called as **aggregation.**

These three phenomenon, i.e. **migration, ecesis and aggregation,** sometimes collectively called as **invasion.**

E. **Competition:** It may arise due to food, space, light or other physical factors. It may occur between individuals of same species (intraspecific) or different species (interspecific). In the end, those species survive which have high reproductive capacity and power adaptability.

F. **Co-action:** The effect of individuals of the same species upon each other is called as **co-action**.

G. **Reaction:** The physical change in the environment brought by an organism living in a particular locality is called as **reaction.**

H. **Stabilisation:** Due to continuous invasion, competition and reaction, the environment keeps on changing. Slowly a complete harmony is established between the organisms and climate of that area. This phenomenon is called as **Stabilisation.** At this stage, a perfect equilibrium is established between abiotic and biotic components of the area. This last stage in which the community is not replaced by any other group of plants is known as climax stage and this last community is called as **climax community**.

5.5 COMMUNITY SUCCESSION

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It is an orderly, progressive, sequential replacement of community over particular area. Clement (1916) while studying plant communities, defined succession as 'a natural process by which the same locality become successively colonised by different group of plant.'

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5.5.1 Causes of Succession

Succession is a universal phenomenon; there are some definite reasons for its occurrence:

A. **Initiation:** It is initiated due to the following factors:

1. **Topographic /Physiogeographical factors:** They produce primary barren areas. The factors responsible for it are:
   (i) **Erosion:** It mainly occurs due to water, air, gravity or by slipping of glaciers.
   (ii) **Deposition:** It also takes place through the agency of water, wind, gravity or by slipping of glaciers.
   (iii) **Submergence:** It occurs due to sinking of a bare land causing formation of a bare land.
   (iv) **Emergence:** It occurs due to emergence of barren area like, from the ocean.
2. **Climatic Factors:** They produce secondary barren area. The factors comprise the following:
   (i) **Wind:** It eliminates the previous vegetation by its high velocity.
   (ii) **Water:** It eliminates the previous vegetation by the impact of rain drops and flood.
   (iii) **Snow:** Snow and storms may eradicate the previous vegetation.
   (iv) **Hail:** Hail storms may also eradicate the previous vegetation.
   (v) **Fire:** It is caused by friction or other volcanic activities.
3. **Biotic factors:** They also produce secondary barren land through different mechanism.
   (i) **Animals:** Barren land was formed due to overgrazing.
   (ii) **Human being:** Due to urbanisation, industrialisation, pollution and other human activities barren lands are formed.

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(iii) **Microorganisms and Insects:** They form barren lands by infectious diseases.

B. **Continuation:** Once started, the process of succession continues through its normal course of stages comprising migration, ecesis, aggregation, competition, co-actions and reaction.

C. **Stabilisation:** When climate is best suitable for community, it stabilises.

QUICK INSIGHT

Grasses are part of ecosystem, not replaced for forests in succession

Grassland ecosystems are an early stage of succession in a region whereas forests are mature ecosystems. However, grass ecosystems are climax ecosystems in grassland regions, where there is not enough rainfall to support a forest.

Two factors are defined for the evolution of temperate grassland biomes:
1. Extreme continental climate with limited supply of water due to low rainfall
2. Human activities mainly due to burning of vegetation

ENV 11- Primary succession vs. secondary succession

5.5.2 Types of Succession

Depending on the nature of barren land on which it develops, succession can be classified under two heads: *Primary* and *Secondary.*

- **Primary Succession vs. Secondary Succession**

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Primary Succession: When a community starts from the primitive substratum into a barren land which was previously unoccupied by any other community, this process is called as primary succession. This first group of organisms established there is called pioneer community or primary colonizers or primary community.

Secondary Succession: When a community establishes itself on a previously built substratum, this process is called as **secondary succession**. The existing community might have disappeared due to sudden environmental changes like climatic factors, fire, and biotic intervention. Thus the area loses its living matter, but the substratum is built again. This succession takes lesser time than primary succession.

- Autogenic Succession vs. Allogenic Succession

  Autogenic Succession: Sometimes, the community modifies its own environment. These changes result into the replacement of one community. This process is called as **autogenic succession.**

  Allogenic Succession: Sometimes the replacement of an existing community is caused largely by external conditions. This is called as **allogenic succession.**

- **Autotrophic Succession vs. Heterotrophic Succession**

  Autotrophic Succession: In this type of succession green plants start dominating the community. This mainly happens in an area rich in organic nutrients. Gradually, the organic matter also increases. This succession is supported by increasing energy flow.

  Heterotrophic Succession: In these communities, heterotrophs dominate the area. It begins in an organic matter rich area. There is a gradual decline in the energy level.

- **Induced Succession**

  It occurs due to an extensive external interference. Here the initial community has high productivity which gradually decreases. The agriculture can be seen as an example of induced succession. Here a steady state is maintained for an ultimate good harvest. Elimination of

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weeds can be deemed as a step towards retrogression as it inhibits the arrival of a stable state.

- **Retrogressive Succession**

  Due to environmental thrust and human interference, the climax vegetation may retrograde into shrub land or savanna. This is referred to as **retrogressive succession.**

- **Deflated Succession**

  When the succession does not proceed through its normal course, it is called **deflated succession**.

- **Cyclic Succession**

  At a certain stage, when the normal succession line is substituted by the repetition of previous stage due to external thrust, it is called **cyclic succession.**

5.5.3 Plant Succession

The process of **succession of plants** is called as **sere.** Sere originating in water is known as **hydrosere**. Those arising under dry conditions are known as **xerosere**. And those developing on exposed rock surface are known as **lithosere** and in sand they are called as **psammosere** (Figure 5.2).

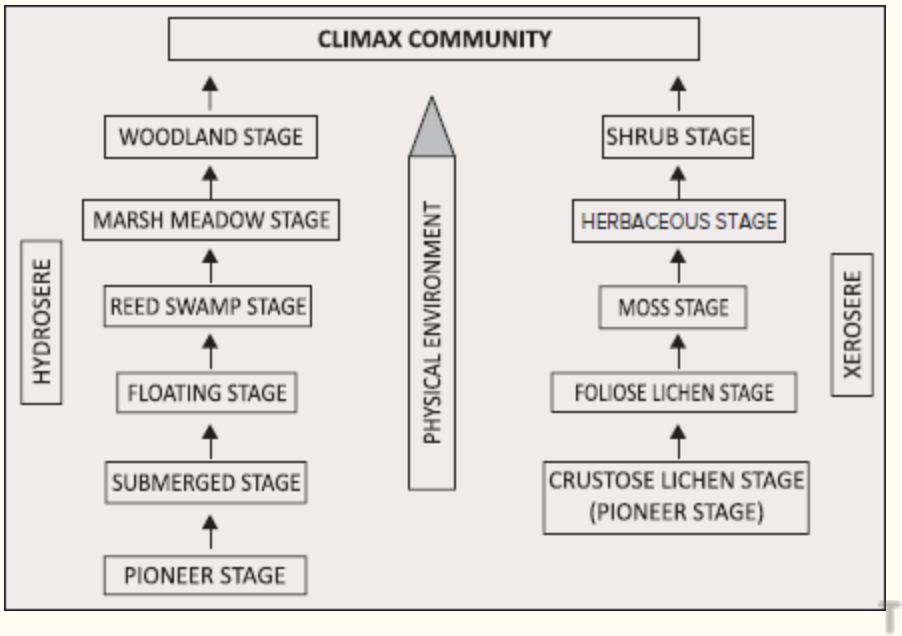

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Figure 5.2 A diagrammatic representation of the development stages of plants succession during hydrosere and xerosere (Lithosere) leading to the formation of mexophytic climax forest community

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- **Hydrosere**

  Ecological succession also occurs in water bodies like ponds and lakes. Water bodies are prone to silting as a result of soil erosion from the surrounding areas. Under this succession, firstly the blockage of river by landslides and construction of dams begins; which leads to the formation of new lakes where hydrosere succession sets in. Later the dead remains of plants and animals settles at the bottom, resulting into the increased succession rate. Succession rate is slower in larger water bodies whereas in small water bodies, it is high.

- **Lithosere and Xerosere**

  This type of succession occurs in dry areas where availability of water is negligible. The succession on rocks is called as lithosere and succession on soil is termed as psammosere.

  The succession of the ecological community in an extremely dry area is called xerophytic succession. This type of succession can generally be seen in sandy deserts, saline deserts (area of Kutch), etc. It is deficient of water and carbonic components.

  Stages of xerarch succession are as follows:

  1. **Lichen Stage:** Due to the high temperature, and normal water availability, very few organisms may develop into lichens, which are the most successful species. They have the ability to survive in dry areas.
  2. **Moss Stage:** The development of second community begins with the process of soil accumulation by trapping dust and humus. It includes mosses. **Mosses are also called as amphibians of the plants** because they can live in both land and water. **Lichens and mosses are the pioneer species in different types of succession**.
  3. **Herb Stage:** The accumulation of soil particles in the lichen-moss carpet provides the suitable substratum for the germination of

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herbaceous plants. Leaves litter and dead herbs add humus to the soil, which result in increasing the number of weeds.

4. **Shrub Stage:** Increased amount of soil provides favourable conditions for the development of shrubs.
5. **Climax Forest:** The shrubs are deeply rooted and hence, can absorb sufficient quantity of water and nutrients. This results into a humid atmosphere which is favourable for the development of trees.

At last, when an equilibrium is established between plant community and their environment, it becomes climax forest community.

5.6 CLIMAX COMMUNITY

The more or less stable community that reaches to the equilibrium with the environment is called climax community.

According to **Clement**, climax community has following features:

1. It is a climate indicator as it flourishes only in a particular environment.
2. It cannot be replaced by any other community in a particular climate. There may develop communities having own ecological domination.
3. A climax community passes through developmental stages like birth, growth and development. These developmental stages may vary for different communities.
4. Phylogenetic relationship exists among different climax communities.

ENV 12- Terminologies

5.7 TERMINOLOGY

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5.7.1 Foundation Species

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Those species which play major role in evolution, development of ecosystem, ecological balance and stability are called foundation species, e.g., some special type of **algae** '**kelp**'.

5.7.2 Flagship Species

Those species of plants and animals which are used for promoting conservation and preservation of natural habitats are called flagship species, e.g. tigers, rhinos, elephants.

5.7.3 Indicator Species

Those species which are easily influenced by geoclimatic and environmental change are described as indicator species. These are used for environmental monitoring, e.g., **lichen**, **touch-me-not plant**, **coral reefs**.

5.7.4 Biotope

An area with uniform environmental conditions in plant and animal population is called biotope, e.g., mangroves.

5.7.5 Ecotone and Ecocline

Ecotone is described as a transitional zone where two communities meet and integrate, e.g., region between two forests where plants integrate. Gradient of change is described as ecocline.

5.7.6 Ecological Niche

It is the functional role describing the position of species or population in an ecosystem.

5.7.7 Keystone Species

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The species which have greater influence on a community, related to their low abundance or biomass are called keystone species. These species play a vital role in controlling the related abundance of other species. The removal of keystone species causes serious disruption in the functioning of the community. For example, in the tropical rain forest different species of figs are the keystone species. During the time of food scarcity, there fruits are eaten by monkeys, birds, bats and other vertebrates. Thus, by protecting the fig trees, the animals dependent on them are also conserved. Only a few species work as keystone species, and others work as critical link species.

Do You Know?

Link Species: Mycorrhizal fungi in soil are a link species as they establish essential links in the absorption of nutrients from the soil and organic residue. Some critical link species may provide food for the network species; others play important role as pollinators of flowers or dispersal agents of seeds and fruits. Tropical rain forests are rich in critical linked species due to high degree of animal dependent pollination and dispersal.

5.7.8 Home Range

In ecology, it is related to the concept of territory, where the organism (animals) moves on a **periodic basis**.

Home range is the **entire area an animal uses for survival and reproduction, including resources like food, water, and shelter**.

Unlike a territory, a **home range is not actively defended from other animals, and different individuals' home ranges may overlap**. The **size and shape of a home range are influenced by factors** such as body size, diet, habitat quality, population density, and reproductive status.

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Ecosystem 6

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6.1 CONCEPT OF ECOSYSTEM

An ecosystem is a cluster of natural resources and living organisms. Ecosystem implies a group of plants, animals and their waste, minerals and nutrients; and flow of food, nutrients and energy from one trophic level to another trophic level in a food chain. **A.G. Tansely** first defined ecosystem as **'the whole system, including not only the organism complex but also the whole complex of physical factors, i.e., environment.'**

■ *Biome*

Similar to the existence of many geographical regions on the earth, there are several ecosystems within a particular geographical region. Many characteristics of these ecosystems have similarities of different kinds.

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These are similarities of having the same climate; many species have similar physical structures, similar climatic adoptions and life cycles. A group (or set) of ecosystems having such similarities is called a 'biome'.

ENV 13- Biotic and Abiotic components of the Ecosystem

6.2 ROLE OF BIOTIC COMPONENTS IN ECOSYSTEM

All ecosystems are full of species of diversified nature. It is always desired to have a long-lasting ecosystem with various functional groups (metabolic groups). The following are the elements of these functional groups:

1. **Producers,** such as green plants, algae, photosynthetic bacteria, etc.
2. **Consumers,** such as animals, protozoa, some bacteria, etc.
3. **Decomposers,** such as fungi, bacteria, etc. (Viruses are not decomposers)

6.2.1 Producers

Producers refer to those organisms which either produce food or convert the compounds from one form to another. To accomplish this, they obtain energy from sunlight or inorganic compounds (e.g., water, phosphates, sulphates, and nitrates) and prepare organic compounds (e.g., carbohydrates, proteins, fat, etc.). Producers such as plants store the chemical energy within themselves and utilise it for their growth and survival.

6.2.2 Consumers

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Consumers refer to those organisms which consume the food (i.e., organic compounds) produced by the 'producers' and gain the energy for their survival. Although they cannot produce food for themselves, they can transform organic compounds from one form to another.

The organic compounds (i.e., foods) are consumed in different stages by different classes of consumers. They are, therefore, grouped under the following categories:

1. **Primary consumers (or herbivores)** are animals such as goats, deer, cattle, grasshoppers, etc., which feed on primary producers.
2. **Secondary consumers** (or **small carnivores**) are animals such as lizards, cats, foxes, owls, hawks, etc., which feed on the primary consumers.
3. **Tertiary consumers** (or **large carnivores**) are animals such as wolves, etc., which feed on secondary consumers.
4. **Quaternary consumers** (or **largest carnivores** or **omnivores**) are animals such as bears, tigers, lions, etc., which feed on tertiary consumers.

The quaternary consumers are not eaten up by any other macro-consumers, but can be consumed by micro-consumers (i.e., decomposers).

6.2.3 Decomposers and Detritivores

Decomposers (or micro-consumers) are organisms that eat dead organisms (animals, plants and other living beings) and transform the organic compounds back into inorganic compounds. These inorganic compounds are then used by the 'producers' to prepare food. In this way, there occurs a cyclic exchange of substances between the living community and the environment.

Decomposers like **bacteria and fungi** are tiny organisms that feed on dead or decaying matter in the soil. Their numbers and activity change with soil depth, with the lowermost soil having the least diversity of microbes.

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Viruses are never decomposers because they take nutrition from only a living host.

Detritivores (e.g., earthworms) break down detritus into smaller particles. This process is called fragmentation. Detritus consists of dead plant remains and dead remains of animals. Woodlice (Terrestrial crustaceans), **Millipedes (a group of arthropods—insects), etc., are other examples of detritivores**. While decomposers break down dead, organic materials, detritivores, such as millipedes, earthworms, and termites, eat dead organisms and wastes.

6.3 STABILITY OF ECOSYSTEM

In the context of conservation ecology, stable populations are often defined as ones that do not become extinct. The larger and more diverse is the ecosystem, the more stable it can be. Marine ecosystems are among the most stable and largest of earth's aquatic ecosystems. They include oceans, salt marsh and intertidal ecology, estuaries and lagoons, mangroves and coral reefs, the deep sea and the sea floor.

6.4 CHARACTERISTIC FEATURES AND FUNCTIONS OF DIFFERENT ECOSYSTEMS

Several kinds of ecosystems exist on the earth. A majority of them include the following.

1. **Forest Ecosystem**
2. **Grassland Ecosystem**
3. **Desert Ecosystem**
4. **Aquatic Ecosystem**
    a. Pond Ecosystem
    b. Steam Ecosystem
    c. Lake Ecosystem

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d. River Ecosystem
e. Ocean Ecosystem
f. Estuaries Ecosystem

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6.4.1 Forest Ecosystem

Depending on the climatic conditions, rainfall, temperature and other factors, there are several kinds of forest ecosystems. A majority of them include the following:

A. **Tropical Rainforest Ecosystem:** Such forests are located in equatorial regions of the earth such as in the Congo River basin of Africa, Central America, etc., where the rainfall is high (annual rainfall is more than 150 cm) and the average annual temperature exceeds 18°C. These forests are characterised by the following features:
   (i) Warm and humid climate
   (ii) High diversity of tree species
   (iii) Broad-leaved and tall plants
   (iv) Highly diverse animal life
   (v) Abundance of insects and invertebrates

   The most commonly found animals in this ecosystem are: **bats, tropical birds, monkeys, carnivorous animals, etc.**

B. **Tropical Seasonal Forest Ecosystem (or Savannas) Ecosystem:** Such forests are located in plains of Africa, South-east Asia, Australia and some parts of India where the rainfall is seasonal but high (annual rainfall is about 100–150 cm). The following features characterise these forests:
   (i) Warm climate; dominantly found species in this ecosystem are: elephants, zebras, giraffes, etc.
   (ii) Dry and wet seasons alternately
   (iii) Coarse grass and scattered trees

C. **Temperate Forest Ecosystem:** Such forests are found in Eastern regions of Asia and the United States, North Central Europe, etc., where the annual rainfall is moderate (about 75–150 cm) and average annual

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temperature is not more than 20°C. The following features characterise these forests:

(i) Tall deciduous trees

(ii) Dominance of hardwood trees which are suitable for quality furniture and building purposes

(iii) Abundance of insects and birds

Common species found in this ecosystem are the following:

- Animals: Rabbits, Squirrels, Frogs, Lizards, Snakes, etc.
- Mammals: Deer, Bears, etc.
- Vegetation: Oaks, Chestnuts, Maples, Elms, Beeches, Hickories, etc.

D. Boreal (or Taiga) Forest Ecosystem: Such forests are found across east–west band of North Europe, North Asia and North America, and in close vicinity below 60°N latitude, where the climate is cold. The annual rainfall varies from 10 cm to 35 cm, and the average annual temperature ranges between 6°C in winter and 20°C in summer. The following features characterise these forests:

(i) Cold climate due to high altitudes and high latitudes

(ii) Dominated by conifers, which are an important source for making paper pulp and lumber

Common species found in this ecosystem are:

- Birds: Owls, eagles, migratory birds, etc.
- Animals: Foxes, rabbits, squirrels, deer, wolves, hyenas, etc.
- Vegetation: Firs, pines, cedars, larches, spruces, hemlocks, etc.

E. Mediterranean Shrub Forest Ecosystem: Such forests are found in South Africa, South Australia along the Mediterranean, Chile and the coast of California, etc., where the rain falls in winter only. The rain is less and the temperature is moderate. The following features characterise these forests:

(i) Dry climate with moist air.

(ii) Existence of '**chaparral**', which are dense shrubs of 2 to 5 m in height.

(iii) Vegetation is broad-leaved and of resinous plants such as rubber.

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Common species found in this ecosystem are: reptiles, small mammals, and large mammals.

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■ *Structures and Functions of Forest Ecosystems*

Different components of a forest ecosystem are as follows:

- **Abiotic components:** These include
  - **(i)** Inorganic compounds present in air and soil
  - **(ii)** Organic compounds present in air and soil
  - **(iii)** Scattered debris of dead bodies
- **Producer organisms include:**
  - **(i)** Trees of diversified species
  - **(ii)** Vegetation of different kinds
  - **(iii)** Shrubs
- **Primary consumers include:**
  - **(i)** Mini animals: Ants, Spiders, Beetles, etc.
  - **(ii)** Small animals: Mice, squirrels, mongooses, etc.
  - **(iii)** Bigger animals: Deer, antelopes, giraffes, elephants, etc.

  The smaller animals eat tree leaves and stems, etc., while the bigger ones graze on shoots and fruits etc.
- **Secondary consumers include:**
  - **(i)** Birds
  - **(ii)** Animals such as frogs, lizards, snakes, foxes, etc.
- **Tertiary consumers** include animals such as tigers and lions.

  Decomposers include fungi, bacteria and actinomycetes.

The rate of decomposition varies from forest to forest. It is faster in tropical forests than the temperate forests.

6.4.2 Grassland Ecosystem

Grassland ecosystem is a terrestrial-type ecosystem where rainfall is usually seasonal (annual rainfall is about 25–75 cm) and temperatures are moderate.

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There are different types of grasslands; the main amongst them are the following:

1. The **'Steppes'** of Asia and Europe
2. The **'Prairies'** of the United States and Canada
3. The **'Veldts'** of Africa
4. The **'Pampas'** of South America

Temperate grasslands are very common. They are found in those regions which are not too moist (neither favourable for the growth of forests nor for evolution of deserts).

Characteristics and Features: The grassland ecosystem is characterised by the following features:

(i) Climate is neither too dry nor too moist
(ii) Soils are composed of deep organic layers
(iii) Vegetation consists of flowering plants, short and tall grasses
(iv) Abundance of large mammals of diversified origins

The animal species generally found in this ecosystem are asses, horses, antelopes, larks, jackals, etc.

■ *Structure and Functions*

Various components of grassland ecosystems and their functions are given as follows:

1. **Abiotic Components:** These are the nutrients present in soil and the surrounding air environment. The nutrients are comprised of:
   (i) Carbon, hydrogen, nitrogen, oxygen, phosphorus, sulfur (CHNOPS), etc., which are present in the atmosphere and supplied by water to the grassland.
   (ii) Nitrates, phosphates, sulphates, etc., that remain present in the soil.
2. **Biotic Components:** These comprise the following metabolic groups:
   (i) Producer organisms which contribute in production of biomass. Examples of such organisms are grasses, shrubs, etc.
   (ii) Consumer organisms which include the following:

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(a) Primary consumers are those which directly feed on grasses. These include numerous species of insects, termites and grazing animals. The main examples are goats, sheep, cows, buffaloes, deer, rabbits, etc.

(b) Secondary consumers are those which feed on primary consumers.

(c) Tertiary consumers are those which feed on secondary consumers. The examples are hawks, tigers and lions.

3. **Decomposers:** These are saprotrophic microorganisms which play an active role in decomposition of dead/decayed organic matter. They consume the dead body and discharge their waste in the same ecosystem. Thus, the nutrients/minerals return to the soil. The nutrients are, thus, again available to 'producers' of the ecosystem.

6.4.3 Desert Ecosystem

Desert refers to an area in which the rainfall is negligible (annual average rainfall is less than 25 cm) and the rate of evaporation is very high. In deserts, the days are hot, but the nights are cold. The seasonal variation of temperature is also too much, which may vary from below 0°C in summer.

Characteristics Features: The following features characterise the desert ecosystems:

(a) Climate is generally dry. It is hot in the day and cold in the night.
(b) There is a scarcity of water.
(c) Solids are generally rich in minerals.
(d) There is a scarcity of organic matters.
(e) Scanty flora and fauna.

The plant and animal species dominantly found in this ecosystem are as follows:

- Plants: Hard Grasses, Acacias, Euphorbias, Cacti, Succulents, etc.
- Non-vertebrates: Reptiles, Snakes, etc.
- Insect-eating birds: Swallows, Swifts, etc.
- Seed-eating birds: Doves, Quails, etc.

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- Other animals: Ants, Wasps, Scorpions, Locusts, Spiders, Lizards, Rats, etc.
- Large mammals: Camels, Jackals, Cats, etc.

■ *Structure and Functions of Desert Ecosystem*

The structure and functions of different components of a desert ecosystem are as follows:

I. **Abiotic components** are the nutrients present in the soil and air. Also, there is a scarcity of water and a lack of organic matter in the soil.

II. **Biotic components are:**

(a) Producer organisms including some grasses, bushes, shrubs, hardy grasses, succulents and low-height plants such as lichens and xerophytic mosses.

(b) Consumer organisms including insects, reptiles, birds, camels, etc.

(c) Decomposers including thermophilic fungi and bacteria.

Since the vegetation in the desert is too low, the quantity of dead organic matter is also substantially lesser. Hence, very few decomposers are found in deserts.

6.4.4 Aquatic Ecosystem

(A) Pond Ecosystem

A pond is a stationary, freshwater aquatic ecosystem. It may be a natural pond or a man-made pond. The ecosystem of a natural pond will generally be different from the ecosystem of an artificial pond. It is because of the arrangements within an artificial pond. The details of abiotic and biotic components in both the above types of ponds are self-explanatory.

(B) Ocean Ecosystem

An ocean is a huge pool of salty water that extends over almost an infinitely large area. It covers about 70% of the earth's surface. Ecosystem of an ocean is very stable and naturally well-balanced.

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Types: There are five oceans on the earth: the Pacific Ocean, Atlantic Ocean, Indian Ocean, Arctic Ocean and Antarctic Ocean.

There are several seas; the main amongst them are: the Caspian Sea, Arabian Sea, Caribbean Sea, South-China Sea, Black Sea and Red Sea.

Characteristic features of an oceanic ecosystem are the following:

(i) They have a high concentration of salts. It is about 3.5%.

(ii) There is an abundance of minerals such as Na, Cl, Ca, S, Mg, etc.

(iii) Salinity is less near the North and South poles, and also on river mouths (i.e., the place where the river falls in the ocean).

(iv) Salinity is higher in deeper regions of the ocean.

Structure and functions of an ocean/sea ecosystem are as follows:

(i) Producers trap the solar energy to produce organic matter.

- Macroscopic seaweeds, e.g., red and brown algae, blue-green algae (cyanobacteria), etc.
- Phytoplanktons, e.g., **diatoms**, **dinoflagellates**, etc.

These are found at different depths under the ocean.

(ii) Primary consumers are fishes, molluscs, crustaceans, etc.

(iii) Secondary consumers are fish such as mackerel, herring, etc.

(iv) Tertiary consumers are fish such as whale, shark, haddock, etc.

(v) Decomposers are fungi, bacteria, etc.

(C) Estuary Ecosystem

Earlier in this chapter, we studied that aquatic ecosystems are mainly of two types, viz., freshwater ecosystems such as those in ponds, lakes, streams and rivers; and marine (salty) water ecosystems such as those in sea/ocean. However, there are some ecosystems in which the freshwater and marine water boundaries overlap in certain regions, such regions are known as estuaries and their ecosystems as estuaries ecosystems. An example of estuaries is the areas/zones/regions at the junction of a river outlet (mouth) into the sea/ocean. Such examples are

- **Ganga delta** in the Bay of Bengal
- **Ratnagiri** in Maharashtra

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- **Pichavaram** near Chennai

In this region, it is difficult to distinguish between the boundaries of river water and marine water. Such ecosystems are characterised by the following features:

(i) Physical condition of water is non-uniform
(ii) Nutrients are rich
(iii) Abundance of fishes
(iv) Good breeding sites for fishes

The river–ocean joining area provides a commercially significant fishing area also.

ENV 14-Energy Flow in Ecosystem

6.5 ENERGY FLOW IN ECOSYSTEM

Ecosystems function through the input of energy, mainly solar radiation which is trapped by green plants (primary producers) and is used to prepare food (chemical organic matter) through the process of photosynthesis. Thus, solar radiation is the main driving force of the ecosystem.

6.5.1 Energy Flow

Energy flow (transfer of organic molecules is unidirectional passing through various trophic levels) is **non-cyclic.** The functioning of an ecosystem depends on the pattern of energy flow because all aspects of living components of an ecosystem depend on it and it also helps in the

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distribution and circulation of organic and inorganic matter within the ecosystems.

Ecosystem energetics or energy flow also helps in the circulation of chemical substances (nutrients) within the ecosystem. There is a progressive loss of energy by the respiration of organisms through increasing tropic levels. **The energy pattern and flow are governed by the first and second laws of thermodynamics.**

6.5.2 First Law of Thermodynamics

It states that in any system of constant mass, energy is neither created nor destroyed, but it can be transformed from one type to another type (e.g., electrical energy can be converted into mechanical energy). In terms of ecosystem, energy in flow or energy input into the system will be balanced by energy outflow from the system.

6.5.3 Second Law of Thermodynamics

It is a state that when work is done, then one form of energy is transformed into another form. In the context of ecosystem, there is a dissipation of energy from each transfer point (trophic level) and, thus, the dissipated or lost energy is not again available to the ecosystem.

Solar radiation is the basic input of energy entering the ecosystem. Only plants make food through the process of **photosynthesis** using a small proportion of radiant solar energy. Thus, green plants transform a part of solar energy into food energy or chemical energy, which is used by the green plants to develop their tissues and, thus, is stored in the primary producers or autotrophs at the bottom of trophic levels. The chemical energy stored at the first trophic level becomes the source of energy to the herbivorous animals at the second trophic level of the food chain. Some portion of the energy is lost from first trophic level through respiration and

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some portion is transferred to plant-eating animals (herbivores) at the second trophic level.

Thus, the chemical energy consumed by herbivores helps in the building of their own tissues and is stored at the second trophic level and becomes the source of energy for carnivores at the third trophic level.

The first and the second laws of thermodynamics govern the energy pattern and flow. Thus, the flow of energy is unidirectional and once used and spent is never available for reuse.

6.5.4 Lindeman's 10% Energy Law

The **10% law** for the transfer of energy from one trophic level to the next was introduced by Raymond Lindeman (1942). According to this law, during the transfer of energy from one trophic level to the next, only about 10% of the energy from organic matter is stored as fresh. The remaining is lost during transfer, broken down during respiration, or lost to incomplete digestion by higher trophic levels.

Example: Plants absorb 1% solar energy for primary production and can store only 10% of the utilised energy as net production available for the herbivores. When the plants are consumed by an animal, about 10% of the energy in the food is fixed into animal flesh which is available for the next trophic level (carnivores or omnivores). When a carnivore or an omnivore consumes that animal, only about 10% of energy is fixed in its flesh for the higher level.

6.6 TROPHIC LEVEL

Feeding (or transfer and assimilation of food energy) takes place in a hierarchical order in the ecosystem through various levels. Thus, the levels through which food energy passes from one group of organisms to the other group are called trophic levels. The chain of transformation and transfer of

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food energy in the ecosystem from one group of organisms to the other group through a series of steps or levels (trophic level) is called the food chain (Figure 6.1). The point where food energy is transferred from one group of organisms to another group is called the **trophic level.** On average, four trophic levels of a food chain are identified.

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FLOW OF ENERGY IN TROPHIC LEVELS

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ENERGY FLOW

Energy Flow is unidirectional & non cyclic.

The functioning of an ecosystem depend on the pattern of energy flow .

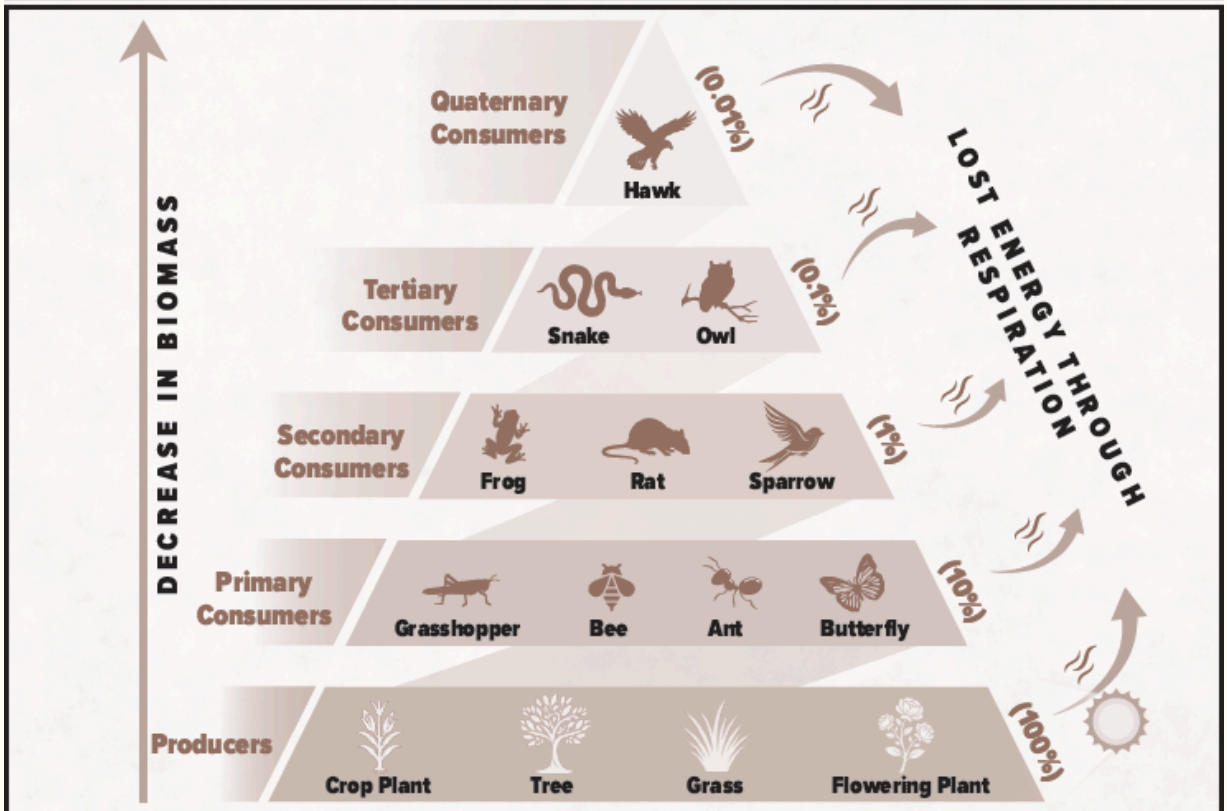
Energy Flow - Solar energy / Radiations are the main driving force in the ecosystem .

There is progressive loss of energy by respiration of organism through increasing trophic levels .

The energy pattern & flow are governed by Law of Thermodynamics and Lindemann 10 % energy law.

2ND LAW OF THERMODYNAMICS

It states that when work is done , then one form of energy is transformed into another form . In the context of ecosystem there is dissipation (loss) of energy from each transfer point (Trophic level) & thus the dissipated or lost energy is not again available to the ecosystem .



LINDEMAN'S 10% ENERGY LAW

According to this law during the transfer of energy from one trophic level to the next , only about 10% of the energy from organic matter is stored as fresh . the remaining is lost during transfer , broken down during respiration or lost to incomplete digestion by higher trophic levels .

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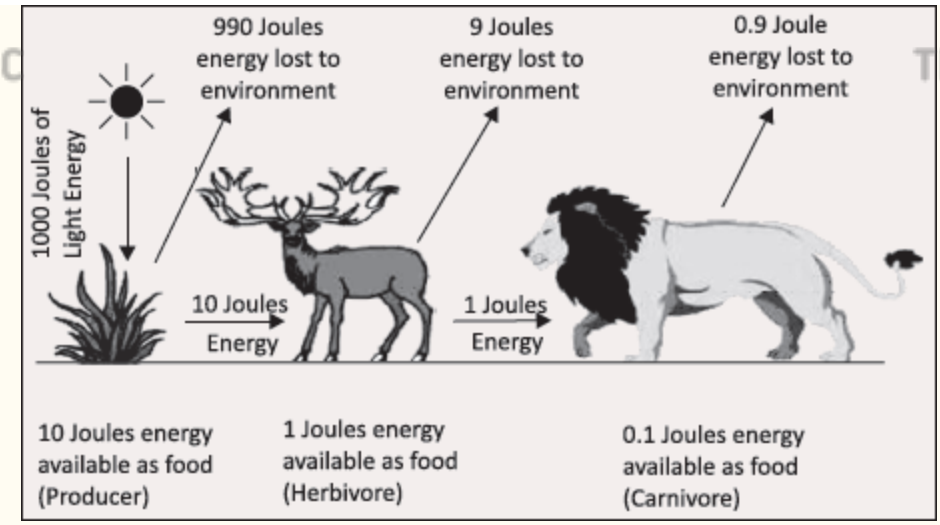

Figure 6.1 Progressive Loss of Energy in a Food Chain

It describes the feeding status of an organism in an ecosystem. It is based on the concept of who feeds on whom, accordingly the following trophic levels (i.e., tiers) can be assigned to different organisms:

(i) **First Trophic Level: Green plants,** which are the producers fall in this category. They get energy from the sun and produce food by the process of photosynthesis for the second trophic level organisms.

(ii) **Second Trophic Level: Herbivores** which are primary consumers, fall in this category. They depend on primary consumers for their food energy.

(iii) **Third Trophic Level: Carnivores** which are secondary consumers fall in this category. They depend on primary consumers for their food energy.

(iv) **Fourth Trophic Level: Omnivores** which are tertiary consumers fall in this category. They depend on secondary consumers for energy.

6.7 FOOD CHAIN

A food chain is in fact the sequence of energy transfer from the lower trophic levels to the upper or higher trophic levels.

In every ecosystem, the food is a source of energy. It is obtained from plants (which are producers) and consumed by different categories of

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consumers (viz., primary, secondary, tertiary consumers, etc.). Thus, the transfer of food energy occurs in repeated stages in which, it is being eaten by a series of organisms. It shows the flow of food energy from a primary producer to different consumers in successive stages.

ENV 15- Types of Food Chain

■ *Types of Food Chains*

In this biosphere, anything that is created (i.e., has taken birth) will perish (i.e., will die) also. In all ecosystems, it is not only the living organisms which are being eaten but the dead matter is also being eaten as food. Hence, there are different food chains for different types of foods. Accordingly, the food chains can be of the following types:

1. **Grazing food chain:** Here, the green plants are being consumed by animals.
2. **Detritus food chain:** Here, the dead organic matter is being consumed by bacteria, worms, fungi, etc.

6.7.1 Grazing Food Chain

In this chain, the green plants/grasses, etc., are being grazed by animals. The forest and grassland ecosystems follow this food chain. A typical grazing food chain may be one as shown below.

(i) Grass Grasshopper → Frog → Snake → Hawk

(ii) Green plants → Goat → Wolf → Lion

Ecosystems having such types of food chains desire energy from the sun.

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TYPES OF FOOD CHAIN

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A food chain is a sequence of energy transfer from the lower trophic level to upper or higher trophic level. It shows the flow of food energy from primary producer to different consumers in successive stages.

GRAZING FOOD CHAIN

Those food chains that start from green plants. In grassland & forest such type of food chain dominates.

DETRITUS FOOD CHAIN

In In this food chain the dead animals and plants & fallen leaves are consumed by detritivores and their predators. Eg. Mangroves and Estuaries ecosystem.

PARASITIC FOOD CHAIN

This food chain start from host-parasitic interaction.

6.7.2 Detritus Food Chain

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In this chain, the dead animals and plants and fallen leaves are consumed by detritivores and their predators. The mangrove leaf and estuarine ecosystems follow this food chain. A typical detritus food chain may be one as shown below.

(i) Dead plants → Soil mites → Insects → Lizard

(ii) Dead organic matter → Bacteria → Protozoa → Rotifers

Ecosystems having such a type of food chain do not depend much on solar energy, rather they depend on the influx of organic matter created in other systems.

6.8 FOOD WEB

When the feeding relationships in a natural ecosystem become more complicated, the food chain does not remain simple and linear, rather it is also complicated by several interconnected overlapping food chains. This happens when a greater number of species feed on many kinds of prey. Such a complicated food chain is called a **food web.**

In different ecosystems, the individual food chains in which the consumers feed on a single species, generally do not exist. It is because most of the consumers feed on multiple kinds of species. Similarly, some species fall prey to a single kind of predator, while many other species are eaten by several kinds of predators. Consequently, individual food chains are interconnected through different types of organisms. This interconnection is called a **'food web'**.

NOTE: The nature of food chains and food webs depends on the richness or poorness of biodiversity of a natural ecosystem. The richer the biodiversity, the longer and more complicated the food chain and food webs. The ecosystem stability also depends on biodiversity, food chains and food webs. As biodiversity increases, the food chain and food webs become longer and more complicated, the ecosystem becomes more stable. A simpler food chain represents an unstable ecosystem and poor biodiversity.

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ENV 16- Types of Ecological Pyramid

6.9 TYPES OF ECOLOGICAL PYRAMIDS

Pyramid is a type of model which depicts the trophic structure of an ecosystem in the form of model. **'Elton'** gave the concept of ecological pyramid. It is obvious that the number of species, the total biomass and energy availability decrease with successive higher trophic levels in the food chain in such a way that the shape becomes like a pyramid. This is called **ecological pyramid.** There are three types of ecological pyramids, e.g., (i) the pyramids of numbers, (ii) biomass pyramid and (iii) energy pyramid.

If a food chain represents primary producers of very large size (trees) at trophic level 1, and parasites and hyperparasites at trophic levels 2 and 3 as primary and secondary consumers, respectively, the resultant pyramid is an inverted pyramid because one single primary producer (tree) can support numerous primary consumers (parasites) and secondary consumers (hyperparasites).

Pyramids of biomass include the biomass (total weight of organic matter) of each trophic level instead of a number of species as is done in the case of a pyramid of numbers. Thus, the total weight of whole organic matter (biomass) at each trophic level represents the **standing crop** at a single point in time.

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The energy pyramid is constructed, thus, on the basis of the total amount of energy used at each trophic level per unit area per unit time. Energy pyramids help in the comparative study of the productivity of different ecosystems. There are three types of pyramid forms, viz., **upright**, **inverted** and **rhamboidal** on the basis of **number**, **biomass** and **energy** level of organism in an ecosystem (Figure 6.2).

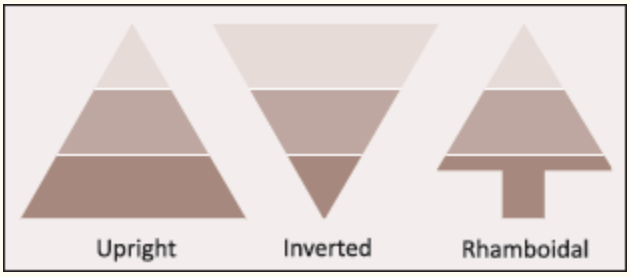

Figure 6.2 Different Shapes of Pyramid

The ecological pyramids thus formed are of three types as discussed below (Figure 6.3):

6.9.1 Pyramid of Numbers

It shows the number of organisms at each trophic level. It is expressed in numbers per unit area. For example, in a grassland, the producers are grasses which are maximum in number. This number decreases towards the higher trophic levels. The herbivores (rabbits) are lesser than grasses. The secondary consumers (wolves) are lesser than the deer and, finally, the tertiary consumers (lions) are still lesser than wolves.

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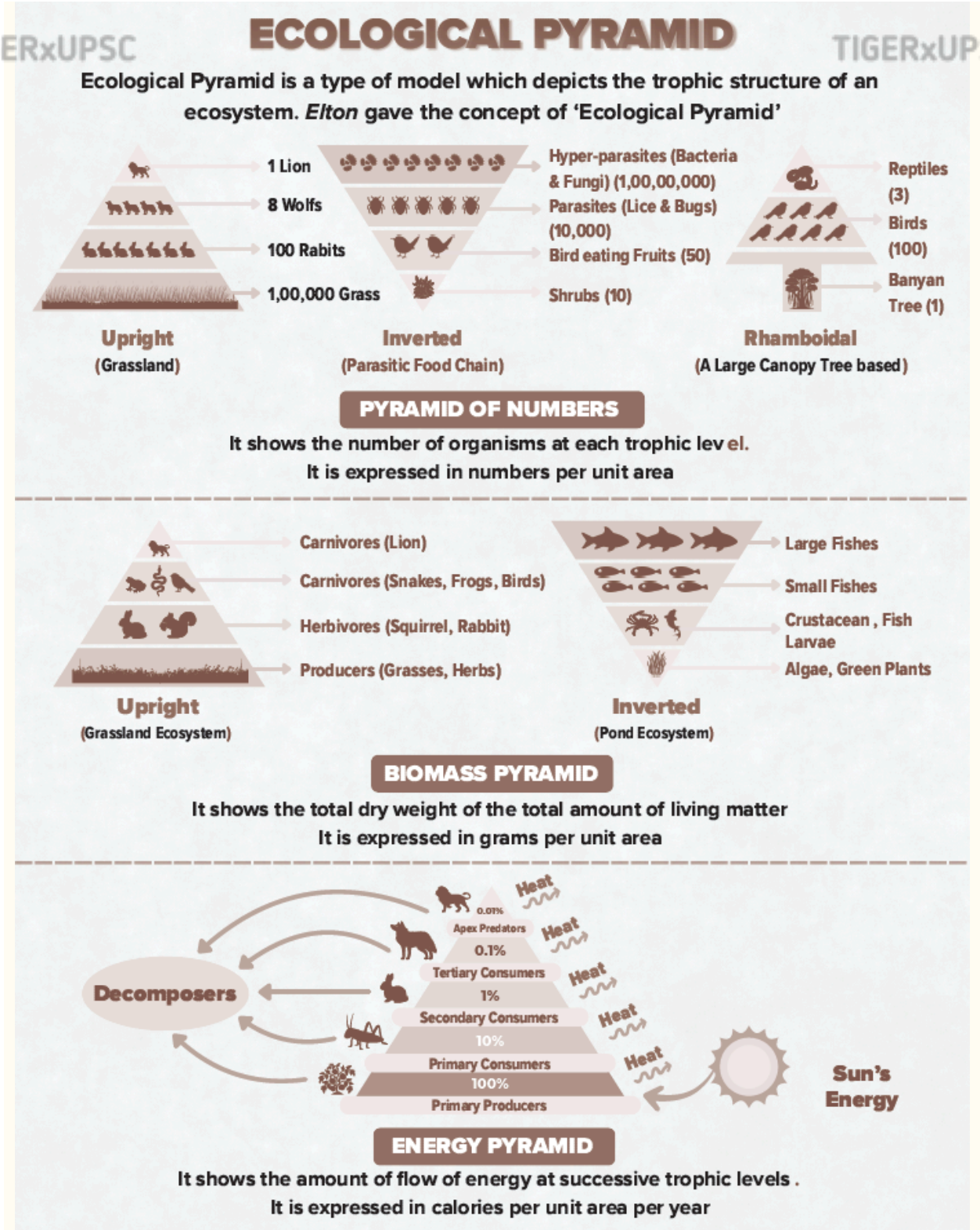

In a forest or lake ecosystem, the pyramid is always in an upright position. However, the **pyramid of numbers in a parasitic food chain is**

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inverted. For example, a single tree supports many fruit-eating birds. These birds support more numbers of parasites such as lice and bugs. These parasites support a large variety of other parasites such as bacteria and fungi. **A big size canopy-based tree (e.g., Banyan tree, etc.) ecosystem forms a rhamboidal pyramid**.

6.9.2 Pyramid of Biomass

It shows the total dry weight of the total amount of living matter. It is expressed in grams per unit area. In many ecosystems, such as in forests, the biomass of producers is much greater than that of the herbivores, which in turn outweighs the carnivores and so on. Thus, we experience a gradual decline in the biomass of organisms at each trophic level. These pyramids are in an upright position. However, in a pond ecosystem, producers are small organisms. So, their biomass is also less. But the primary and secondary consumers are bigger, so their biomass is more. Therefore, the value of biomass shows a gradual increase, making it an **inverted pyramid in the pond ecosystem**.

6.9.3 Pyramid of Energy

It shows the amount of flow of energy at successive trophic levels. It is expressed in calories per unit area per year. It represents the number of calories transferred from one trophic level to the next. Generally, it is about 10% to 15% efficient from herbivores to carnivores. The trophic level at the base denotes the position of the highest amount of energy, whereas the amount of energy at the trophic level situated at the apex is the lowest.

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6.10 PRODUCTIVITY OF ECOSYSTEM

The total accumulated amount of energy stored by the autotrophic primary producers in **per unit area per unit time** is called **ecological productivity.** The production of organic matter or energy by autotrophic primary producers is called primary production and the green plants involved in the production activity are called **primary producers.**

The productivity of the ecosystem depends on two factors, e.g., (i) the availability of the amount of solar radiation to the autotrophic primary producers and (ii) the efficiency of the plants to convert solar energy (light energy) into chemical energy (food energy) which is used by green plants to build up their tissues (Table 6.1).

■ *Types of Productivity*

1. Primary Productivity
   - (a) Gross Primary Productivity
   - (b) Net Primary Productivity
2. Secondary Productivity
3. Net Productivity

6.10.1 Primary Production/Productivity

Primary production/productivity is measured in two ways, e.g.,

1. **Gross Primary Production** (GPP): Gross primary production is the total amount of energy produced by the autotrophic primary producers at trophic level one. In other words, gross primary producers. It refers to the total amount of energy assimilated by autotrophic primary producer green plants.

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2. **Net Primary Production** (NPP): Net primary production excludes the amount of energy which is lost through respiration by autotrophic primary production represents the usable amount of energy at the trophic level. Net primary production is the real growth of an organism (plant or animal).

The ecosystem productivity whether, gross or net, is generally measured in grams/m^2/day or year.

6.10.2 Secondary Productivity

Secondary productivity is the productivity which transfers from one trophic level to another trophic level through the food chain. For example, grasses are eaten up by insects under secondary productivity.

NPP = GPP – R (Respiration/Loss)

Table 6.1 World Productivity Estimate List

Ecosystem Type	NPP (grams organic matter per M^2/year)
Tropical rain forest	2,200
Tropical seasonal forest	1,600
Temperate evergreen forest	1,300
Temperate deciduous forest	1,200
Boreal forest	800
Woodland and shrubland	700

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Source: Whittaker, R.H. 1975. Communities and Ecosystems, Ed. 2. New York. Macmillan Publ. Co., p. 385.

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6.10.3 Net Productivity

Net productivity is the productivity of the substances left after the death of an organism into the soil by decomposers.

6.11 BIO-MAGNIFICATION

A study of ecological pyramids such as the pyramid of energy shows that at each step in the food chain energy tends to decrease. However, a few substances are known to bioaccumulate and concentrate with each link in the food chain. This process is known as **biomagnification**. Different animals and plants show specific preferences for the uptake and accumulation of nutrients, pollutants and elements. Bluegill fish accumulates Sb, As, Be, Se and cadmium. Certain brown seaweeds take up Zn, Cd, Pb, Hg over a long period. Many such chemicals and pesticides are known to persist for long periods in the environment and their concentrations build up geometrically as they are transferred to different steps in the food web.

A study of DDT residues in the food chain presents a suitable example of biomagnification.

Thus, widespread use of many chemicals like DDT has aroused considerable concern in recent years and some countries have legally banned its use. Similarly, the difficulty of dispersing or containing large amounts of radioactive wastes without contaminating man's food chain is one major reason why we could not yet enjoy the peaceful use of atomic energy.

Exam Pointer

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Bioaccumulation and Biomagnification

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Bioaccumulation is the gradual buildup of foreign substances within the body of an organism over time. Bioaccumulation happens when the rate of consumption exceeds the rate of excretion.

Biomagnification refers to the increase in concentration of toxic substances in organisms at higher trophic levels, leading to the highest amounts at the apex predator trophic levels.

6.12 BIOGEOCHEMICAL CYCLES

A biogeochemical cycle is a pathway by which a chemical substance moves through both biotic (biosphere) and abiotic (lithosphere, atmosphere and hydrosphere) compartments of the earth. The term 'biogeochemical' tells us that biological, geological and chemical factors are all involved. The circulation of chemical nutrients such as carbon, oxygen, nitrogen, calcium, water, etc., through the biological and physical world is known as **biogeochemical cycles**.

In an ecosystem, the inorganic elements enter into the biotic system and again return to an abiotic system in a cyclic manner involving some decomposers. During the cyclic flow of elements, a series of chemical reactions occur in the abiotic and biotic environments. Therefore, these cycles of individual elements are called biogeochemical cycles.

Env 18- Carbon Cycle

6.12.1 Carbon Cycle

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Carbon is the basic element of all organic compounds. It occurs in the atmosphere in the form of carbon dioxide gas. Terrestrial plants directly use carbon in the process of photosynthesis. Aquatic plants utilise dissolved form of carbon dioxide. The CO_2 is present in water either in the form of dissolved carbon dioxide gas or combined with water molecules to form carbonic acid. The latter dissociates into hydrogen ions and bicarbonate ions (Figure 6.4). The bicarbonate ions dissolved in water are used by aquatic plants in the process of photosynthesis:

$$CO_2 + H_2O - H_2CO_3 \text{ [Carbonic acid]}$$

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CARBON CYCLE

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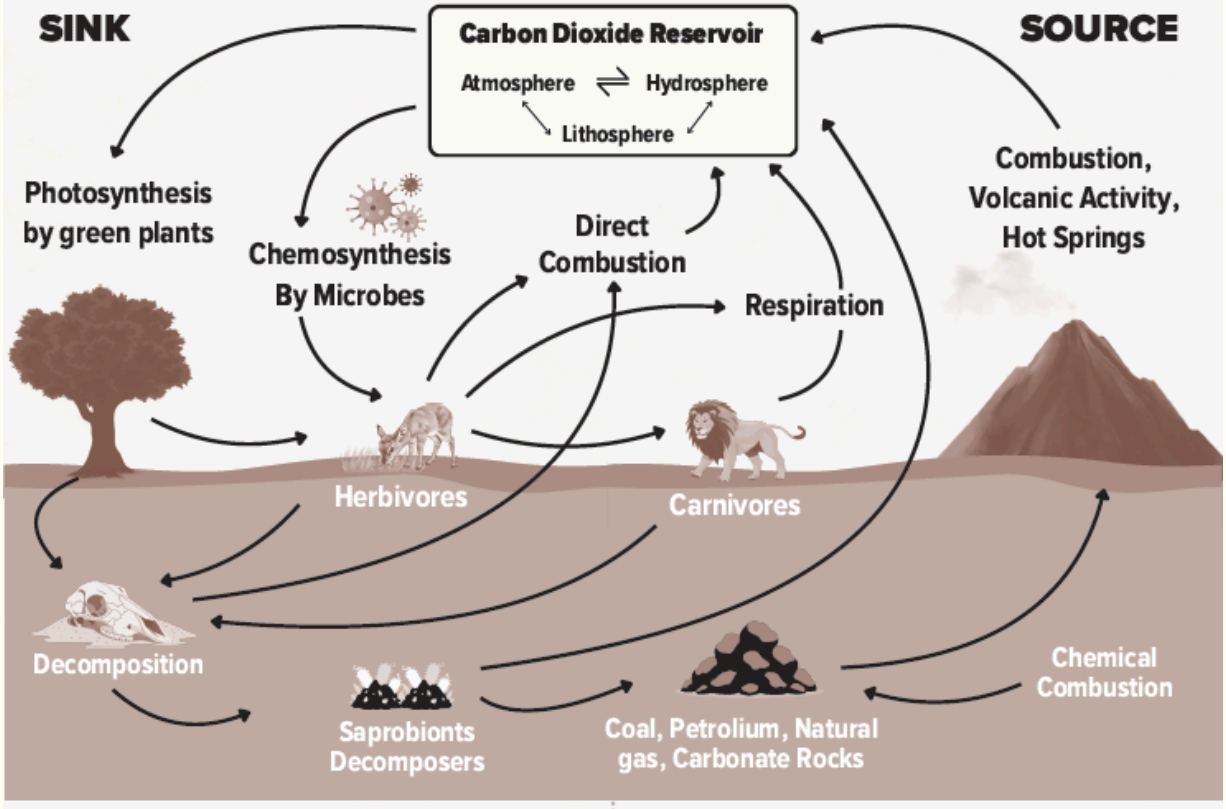

REASONS OF CARBON SINK

- **Oceans**: Absorb a significant amount of atmospheric carbon dioxide
- **Photosynthesis**: Plants utilize sunlight and CO_2 to synthesize food
- **Forests and plants**: Absorb CO_2 via photosynthesis, storing carbon in biomass
- **Soils**: Store carbon as organic matter, serving as both a sink and a source
- **Carbon capture and storage (CCS)**: Industrial process designed to capture carbon emissions
- **Other ecosystems**: Tidal marshes, mangroves, and seagrass meadows are carbon sinks

REASONS OF CARBON SOURCE

- **Fossil Fuel combustion**: Burning fossil fuels releases large amounts of CO_2
- **Respiration**: Animals, humans, and plants release carbon dioxide during respiration
- **Deforestation**: Removing forests releases stored carbon and lowers CO_2 absorption
- **Volcanic eruptions**: Geological processes release carbon from the Earth's interior
- **Decomposition**: Microbes decay organic matter, releasing carbon to the atmosphere
- **Industrial processes**: Certain industrial activities also release carbon emissions

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There is a regular exchange of carbon dioxide between the atmosphere and hydrosphere. The inorganic carbon, in the form of carbon dioxide, is

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assimilated by green plants which fix CO_2 into carbohydrates by the process of photosynthesis. The process of photosynthesis occurs in all green plants which possess chlorophyll. Some photosynthetic bacteria can also fix CO_2 in the presence of bacteriochlorophyll.

Photosynthesis is the basic anabolic process in the biosphere through which inorganic form of carbon (i.e., carbon dioxide) enters into organic form (i.e., carbohydrates). The carbohydrates are then converted into different organic compounds such as proteins, fats, nucleic acids, pigments, growth hormones, vitamins, etc., within the plants. Since energy transfer in ecosystems occurs in the consumption and storage of carbohydrates and fats, the carbon moves from producers to consumers and from both to decomposers.

An enormous amount of inorganic carbon in the form of carbon dioxide is withdrawn from the abiotic environment but the latter does not show any form of carbon depletion. This is because the carbon is continuously being deposited into the abiotic environment by the biotic components. There are two major sources through which the organic carbon is returned to abiotic environment: (i) biological sources and (ii) non-biological sources.

(i) **Biological sources:** The major catabolic process through which the organic compounds (mainly carbohydrates) are oxidised to release an inorganic form of carbon (i.e., CO_2) is respiration that takes place in all living organisms. Besides respiration, the organic matter is decomposed by microorganisms, which is another important biological source of the conversion of organic carbon to inorganic carbon.

(ii) **Non-biological sources:** These include combustion of carbon-containing fuel, burning of fossil, fuels, volcanic eruptions, hot springs and weathering of carbonate rocks.

Carbon Dioxide is the primary form of carbon that circulates the Earth through the Carbon Cycle. **Intake of carbon from the atmosphere takes place through photosynthesis by plants and phytoplankton** and **carbon**

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sinks like oceans. However, **carbon is also released into the atmosphere** through a variety of natural processes like **respiration by animals and plants, decomposition of organic matter, volcanic action, forest fires and weathering of carbonate rocks,** among many others.

ENV 19- Nitrogen Fixation in Pulses crop

6.12.2 Nitrogen Cycle

Nitrogen is the most important element needed for the metabolism, growth, development and reproduction of all living organisms. It is so vital that life is not possible without it. The atmospheric air is the great reservoir of this gas. It constitutes about 78% of the atmosphere. However, the atmospheric nitrogen is chemically inert and cannot be directly used by most of the living organisms. They can utilise the combined nitrogen and organic nitrogen compounds for their metabolism and growth.

The elemental nitrogen is converted to usable forms by the process known as nitrogen fixation and added to the soil. Nitrogen is fixed by the activity of physical and biological agencies. An additional amount of nitrogenous compounds is added to the soil artificially through industrially produced fertilisers. Plants obtain their nitrogen supply from the soil and convert them into essential biomolecules required for their healthy development. Animals and human beings obtain their nitrogen requirements from plants. The nitrogenous compounds then again reach the soil by death and excretion in the form of animal (example- Urea, Creatinine, Uric Acid

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etc.) and plant residues (example- Death of Vegetation etc.). The organic matter in the soil undergoes microbial decomposition by the process or ammonification into ammonia and by nitrification into nitrates. These compounds are again absorbed by the plants or converted into nitrogen gas by the process of denitrification. Thus, there exists a biogeochemical cycle, called the nitrogen cycle, in the biosphere which deals with the conversion of atmospheric nitrogen to fixed inorganic nitrogen, to simple organic compounds, to complex organic compounds in plants and animals and finally returning back to atmosphere in the form of gas.

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NITROGEN CYCLE

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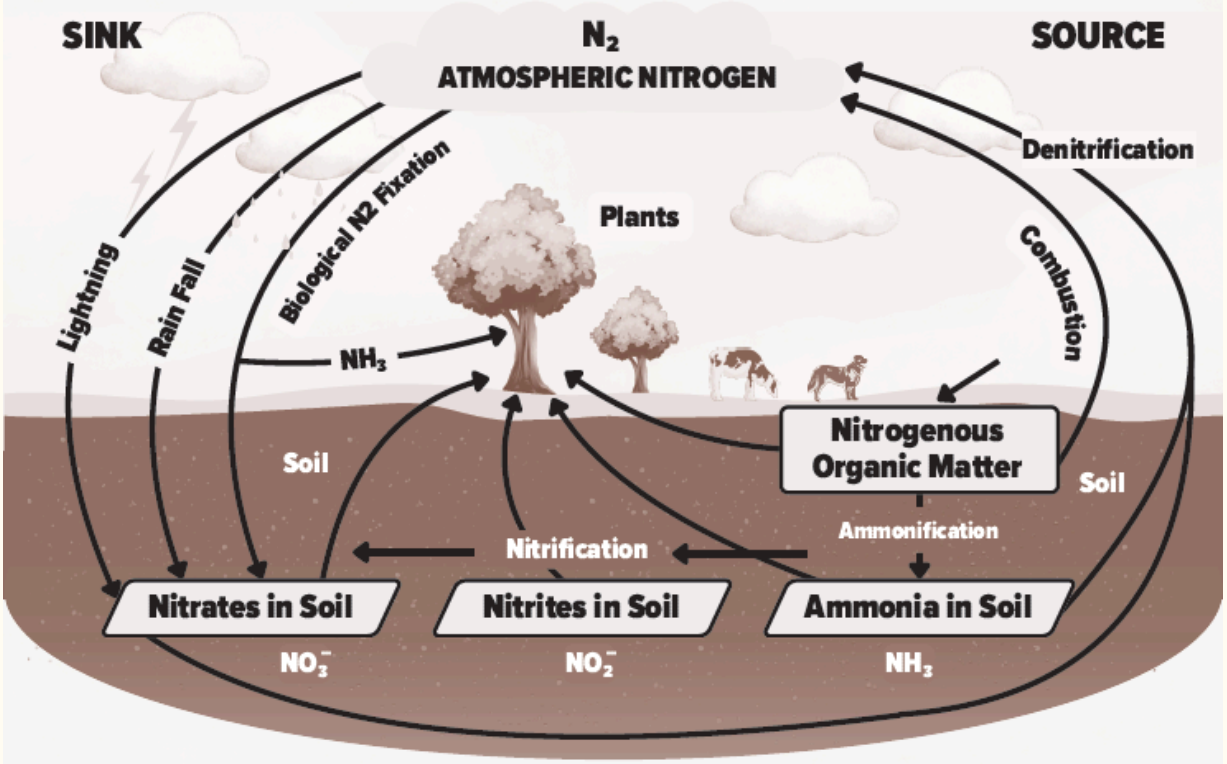

REASONS OF NITROGEN SINK

- **Atmosphere (Nitrogen Fixation):** Atmospheric N_2 is the main sink but requires fixation for biological use
- **Biomass:** Plants and animals assimilate and store nitrogen in their tissues, acting as a sink
- **Soils:** A sink storing nitrogen in organic matter and through nitrification
- **Oceans:** Marine organisms and sediments store nitrate, acting as sinks and sources
- **Riparian Buffers:** Vegetated areas along streams that filter nitrogen from runoff, protecting waterways

REASONS OF NITROGEN SOURCE

- **The main source of nitrogen include:** Atmospheric precipitation, geological sources, agriculture, livestock/ poultry
- **Lightning:** Atmospheric nitrogen is converted to usable compounds by lightning strikes
- **Fertilizer application:** Artificial fertilizers add reactive nitrogen to the environment
- **Animal Waste:** Livestock and poultry manure is a significant nitrogen source
- **Decomposition:** Bacteria and fungi break down organic waste, returning nitrogen to the soil as ammonia

Nitrogen Emissions Going up

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Nitrogen particles make up the largest fraction of PM2.5, the class of pollutants closely linked to cardiovascular and respiratory illness, according to the first-ever quantitative assessment of nitrogen pollution in India.

While the burning of crop residue is said to be a key contributor to winter smog in many parts of North India, it contributes over 240 million kg of nitrogen oxides (NOx: a generic term for the nitrogen oxides that are most relevant for air pollution, namely, nitric oxide and nitrogen dioxide) and about 7 million kg of nitrous oxide (N_2O) per year.

The Indian Nitrogen Assessment assesses the sources, impacts, trends and future scenarios of reactive nitrogen in the Indian environment, said N. Raghuram, Professor and Former Dean, School of Biotechnology, GGS Indraprastha University, who is one of the authors of the report in 2018.

Though agriculture remains the largest contributor to nitrogen emissions, the non-agricultural emissions of nitrogen oxides and nitrous oxide are growing rapidly, with sewage and fossil-fuel burning — for power, transport and industry — leading the trend.

Indian NOx emissions grew by 52% from 1991 to 2001 and 69% from 2001 to 2011.

Annual NOx emissions from coal, diesel and other fuel combustion sources are growing at 6.5% a year currently, the report says.

‘As fertilizer, nitrogen is one of the main inputs for agriculture, but inefficiencies along the food chain mean about 80% of nitrogen is wasted, contributing to air and water pollution plus greenhouse gas emissions, thereby causing threats for human health, ecosystems and livelihoods,’ Dr. Raghuram said at a conference to mark World Environment Day.

Agricultural soils contributed to over 70% of N_2O emissions from India in 2010, followed by wastewater (12%) and residential and commercial activities (6%). Since 2002, N_2O has replaced methane as the second largest greenhouse gas (GHG) from Indian agriculture.

Chemical fertilisers (over 82% of it is urea) account for over 77% of all agricultural N_2O emissions in India, while manure, compost and so on make up the rest. Most of the fertilisers consumed (over 70%) go into the production of cereals, especially rice and wheat, which accounts for the bulk of N_2O emissions from India.\

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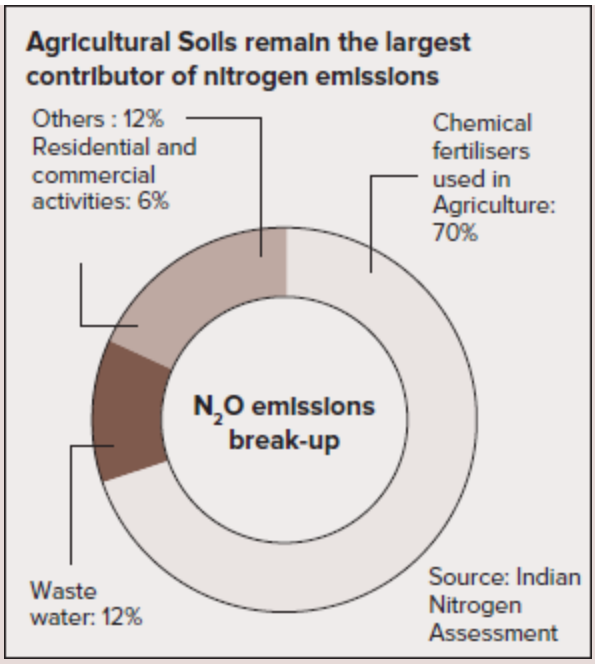

Cattle Emissions

Cattle account for 80% of the ammonia production, though their annual growth rate is 1%, due to a stable population.

India is globally the biggest source of ammonia emission, nearly double that of NOx emissions.

But at the current rate of growth, NOx emissions will exceed ammonia emissions and touch 8.8 tonnes by 2055, the report says.

The poultry industry, on the other hand, with an annual growth rate of 6%, recorded an excretion of reactive nitrogen compounds of 0.415 tonnes in 2016.

That is anticipated to increase to 1.089 tonnes by 2030.

The authors suggest that nutrient recovery/recycling from wastewater for agriculture could cut down N_2O emissions from sewage and wastewater by up to 40%.

A few important events of the nitrogen cycle are briefly discussed below:

- **Nitrogen Fixation:** Both biological and physical mechanisms are involved in the conversion of nitrogen into usable forms (i.e., nitrates and ammonia). The process is termed nitrogen fixation, which has been described earlier in this chapter. Various agencies (such as lightning, rainfall, microflora and **blue-green algae**) are involved in nitrogen fixation, which convert atmospheric nitrogen into ammonia or nitrates. The symbiotic nitrogen fixers convert nitrogen directly to ammonia.

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- **Rate of Fixed Nitrogen:** Plants absorb nitrogen from the nitrogen pool of the soil and ultimately convert it into organic nitrogenous compounds, viz., amino acids, proteins, nucleic acids, etc. Insectivorous plants, such as *Drosera*, *Ultriclaria,* etc., obtain nitrogen from the digestion of trapped insects. The plant proteins and other organic nitrogenous compounds are eaten or grazed by animals.
- **Ammonification:** The nitrogenous compounds (especially proteins and nucleic acids) of dead remains of plants, animals and their excretory products are decomposed into ammonia by a number of micro-organisms. The conversion of nitrogenous organic compounds into ammonia is termed ammonification, and bacteria which carry out this process are called ammonifying bacteria.
- **Rate of Ammonia:** The ammonia of the soil can be lost to the atmosphere, but this can only happen under alkaline conditions. Most of the ammonia is directly absorbed and metabolised by the plants or it can be converted to nitrates by the process known as *nitrification*.
- **Nitrification:** Ammonia is oxidised to nitrite and nitrate ions by a group of nitrifying bacteria, e.g., *Nitrosomonas* and *Nitrobacter*. The nitrifying bacteria obtain their energy from the oxidation of ammonium and nitrate ions. *Nitrosomonas* group oxidises ammonia to nitrite.
- The nitrates, produced by nitrification, can again be absorbed by plants. However, the nitrates and ammonia can be lost to atmosphere by the process known as **denitrification**.
- **Denitrification:** The nitrates and ammonia are converted to nitrous oxide and finally to nitrogen gas by several denitrifying bacteria (e.g., *Pseudomonas fluorescence, P. denitrificans, Bacillus subtilis* and *Thiobacillus denitrificans,* etc.).

The nitrogen gas, released from denitrification, is lost to the atmosphere and completes the complex nitrogen cycle.

As per an All India Coordinated Research Project on '**Long Term Fertilizer Experiments**' (2020-21), **Continuous and excessive use of nitrogenous fertilizers had a detrimental effect on soil health**, such as:

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- Leaching out Nitrate contamination in groundwater
- Eutrophication (nutrient enrichment) of aquatic ecosystems
- Alteration of soil chemistry, such as increased acidity
- Emission of N2O in the atmosphere in the presence of solar heating in soil, thereby causing air pollution.

6.12.3 Phosphorus Cycle

Phosphorus is the second most important element (the first being nitrogen) needed for all vital activities of the organisms. It is a component of the nucleic acids (DNA and RNA) and proteins (which together form nucleoproteins), phospholipids, sugar phosphates, ATP, NADP+, and a number of phosphorylated compounds.

A large amount of phosphorus accumulates in storage tissues of fruits and seeds and during active growth in meristematic tissues. Phosphate plays a key role in energy metabolism. Incorporated into ATP, it is part and parcel of the universal 'energy currency' of all types of living systems (Figure 6.6).

It is actively involved in the synthesis of protein, photosynthesis, respiration and other metabolic processes. It also acts as an activator of some enzymes.

The living organisms obtain phosphorus from the environment. It occurs in the soil as rock phosphate, apatite or calcium phosphate, iron phosphate or aluminium phosphate. However, the usable inorganic fraction of soil phosphorus is found in the form of dihydrogen phosphate (H_2PO_4). It becomes available in the soil for plant use by the natural erosion of rocks. It is usually deficient in the soil and, therefore, added regularly to the soil as fertiliser consisting of rock phosphate, bone meal and fish meal. Mud from lakes and shallow sea beds.

The natural reservoir of phosphorus is rock, which contains phosphorus in the form of phosphates. When rocks are weathered, minute amounts of

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these phosphates dissolve in soil solution and are absorbed by the roots of the plants.

The inorganic phosphate usually occurs in the insoluble form. Plants fail to absorb this form of phosphorus. It is dissolved by microorganisms and the roots. The dissolved phosphorus is absorbed by plants and converted to organic form. From plants, it travels to various trophic levels in the ecosystem. In animals, it is also deposited as part of the skeleton. When the plants and animals die, the organic phosphorus is converted to inorganic form by several decomposers (i.e., some microscopic animals and phosphatising bacteria). The liberated phosphorus is again absorbed by plants and enters the phosphorus cycle.

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PHOSPHORUS CYCLE

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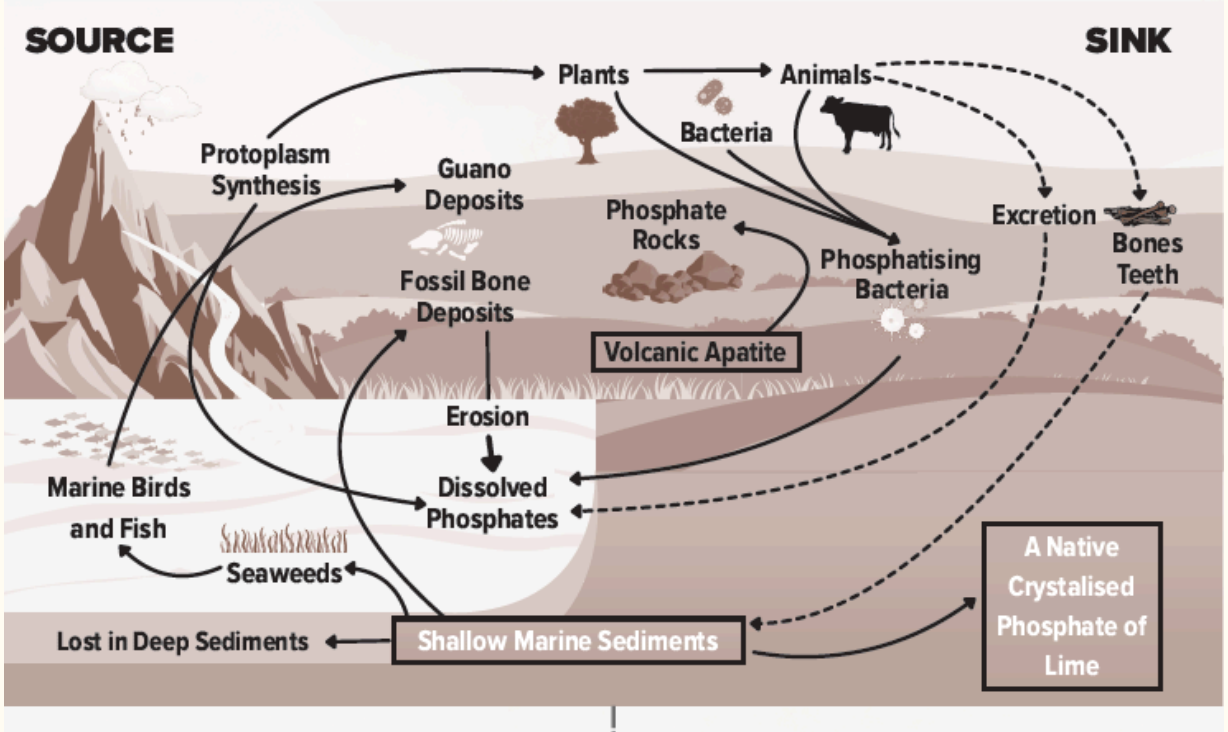

PHOSPHORUS SINKS

- **Ocean and aquatic sediments:** The largest sink for phosphorus, where it accumulates on the ocean floor over long periods
- **Soil organic matter:** Temporarily stores phosphorus in terrestrial ecosystems
- **Plant and animal biomass:** Phosphorus is incorporated into living organisms and is stored in their tissues
- **Deep ocean burial:** Over geological time, deep burial of phosphorus in sediments sequesters it from the active cycle
- **Wetlands:** These areas trap and store phosphorus in their sediments and vegetation
- **Mineral formation:** In certain conditions, such as alkaline soils, phosphorus can become immobilized through mineral formation

PHOSPHORUS SOURCE

- **Weathering of rocks:** This is the primary natural source, where rain and other elements break down phosphate-containing rocks, releasing phosphorus into the soil and water
- **Anthropogenic inputs:** Human activities are significant sources of phosphorus:
  - **Detergents:** Many detergents contain phosphates that end up in water systems
  - **Wastewater:** Sewage and other wastewater discharge phosphorus-rich effluents
  - **Fertilizers:** Phosphorus-rich fertilizers are applied to crops, and excess amounts are washed into waterways
- **Volcanic activity:** Volcanic ash can release phosphorus-rich particles into the atmosphere

Exam Pointer

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Weathering of rocks is an essential first step to release **phosphorus** from rocks into water and soil in the form of phosphate salts.

This is due to the **orthophosphate** (the most abundant form of phosphorus on the earth) existing in solid form within rocks, bound to other structures to form compounds.

When released, it readily mixes with water or other salts in the soil. The **bio-availability of the dissolved phosphorus** is higher than in rocks and is returned to the soil after decay of the consuming organisms.

Hence, the **atmosphere has a lesser role to play** in the phosphorus cycle due to its inherent tendency not to enter the gaseous form readily.

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Questions Asked in UPSC Prelims Exam

UNIT I

Unit - I Prelims PYQs (2011 onwards)

1. Consider the following: **[UPSC (Pre) 2024]**

1. Carabid beetles
2. Centipedes
3. Flies
4. Termites
5. Wasps

Parasitoid species are found in how many of the above kinds of organisms?

(a) Only two
(b) Only three
(c) Only four
(d) All five

2. Consider the following plants: **[UPSC (Pre) 2024]**

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1. Groundnut
2. Horse-gram
3. Soybean

How many of the above belong to the pea family?

(a) Only one
(b) Only two
(c) All three
(d) None

3. Which one of the following shows a unique relationship with an insect that has coevolved with it and that is the only insect that can pollinate this tree? **[UPSC (Pre) 2024]**

(a) Fig
(b) Mahua
(c) Sandalwood
(d) Silk cotton

4. Consider the following: **[UPSC (Pre) 2024]**

1. Butterflies
2. Fish
3. Frogs

How many of the above have poisonous species among them?

(a) Only one
(b) Only two
(c) All three
(d) None

5. The organisms "Cicada, Froghopper and Pond skaters" are? **[UPSC (Pre) 2024]**

(a) Birds

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(b) Fish
(c) Insects
(d) Reptiles

6. Consider the following statement: **[UPSC (Pre) - 2023]**

Statement 1: Marsupials are not naturally found in India.
Statement 2: Marsupials can thrive only in a montane grassland with no predators. Which one of the following is correct in respect of the above statements?

(a) Both Statement 1 and Statement 2 are correct and Statement 2 is the correct explanation of Statement 1.
(b) Both Statement 1 and Statement 2 are correct and Statement 2 is not the correct explanation of Statement 1.
(c) Statement 1 is correct, but Statement 2 is not correct.
(d) Statement 1 is not correct, but Statement 2 is correct.

7. Which of the following organisms perform waggle dance for others of their kin to indicate the direction and the distance to the source of their food? **[UPSC (Pre) - 2023]**

(a) Butterflies
(b) Dragonflies
(c) Honeybees
(d) Wasps

8. Consider the following statements **[UPSC (Pre) - 2023]**

1. Some mushrooms have medicinal properties.
2. Some mushrooms have psycho-active properties.
3. Some mushroom have insecticidal properties.

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4. Some mushrooms have bioluminescent properties. How many of the above statements are correct?

(a) Only one
(b) Only two
(c) Only three
(d) All four

9. Consider the following statements regarding the Indian spaniel: [UPSC (Pre) - 2023]

1. They build nests by making burrows in the ground.
2. They store their food materials, like nuts and seeds, in the ground.
3. They are omnivorous.

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

10. Consider the following statements: [UPSC (Pre) - 2023]

1. Some microorganisms can grow in environments with temperatures above the boiling point of water.
2. Some microorganisms can grow in environments with temperatures below the freezing point of water.
3. Some microorganisms can grow in a highly acidic environment with a pH below

How many of the above statements are correct?

(a) Only one
(b) Only two
(c) All three
(d) None

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11. Which one of the following makes a tool with a stick to scrap insects from a hole in a tree or a log of wood? **[UPSC (Pre) - 2023]**

(a) Fishing cat
(b) Orangutan
(c) Otter
(d) Sloth bear

12. Which of the following are nitrogen-fixing plants? **[UPSC (Pre) - 2022]**

1. Alfalfa
2. Amaranth
3. Chickpea
4. Clover
5. Purslane (Kulfa)
6. Spinach

Select the correct answer using the code given below:

(a) 1, 3 and 4 only
(b) 1, 3, 5 and 6 only
(c) 2, 4, 5 and 6 only
(d) 1, 2, 4, 5 and 6

13. Certain species of which one of the following organisms are well known as cultivators of fungi? **[UPSC (Pre) - 2022]**

(a) Ant
(b) Cockroach
(c) Crab
(d) Spider

14. In case of which one of the following biogeochemical cycles, the weathering of rocks is the main source of release of nutrient to enter

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the cycle? **[UPSC (Pre) - 2021]**

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(a) Carbon cycle

(b) Nitrogen cycle

(c) Phosphorus cycle

(d) Sulphur cycle

15. Which of the following are detritivores? **[UPSC (Pre) - 2021]**

1. Earthworms
2. Jellyfish
3. Millipedes
4. Seahorses
5. Woodlice

Select the correct answer using the code given below.

(a) 1, 2 and 4 only

(b) 2, 3, 4 and 5 only

(c) 1, 3 and 5 only

(d) 1, 2, 3, 4 and 5

16. Which of the following have species that can establish symbiotic relationships with other organisms? **[UPSC (Pre) - 2021]**

1. Cnidarians
2. Fungi
3. Protozoa

Select the correct answer using the code given below.

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

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17. In nature, which of the following is/are most likely to be found surviving on a surface without soil? **[UPSC (Pre) - 2021]**

1. Fern
2. Lichen
3. Mos
4. Mushroom

Select the correct answer using the code given below.

(a) 1 and 4 only

(b) 2 only

(c) 2 and 3

(d) 1, 3 and 4

18. Consider the following kinds of organisms: **[UPSC (Pre) - 2021]**

1. Copepods
2. Cyanobacteria
3. Diatoms
4. Foraminifera

Which of the above are primary producers in the food chains of oceans?

(a) 1 and 2

(b) 2 and 3

(c) 3 and 4

(d) 1 and 4

19. Consider the following animals: **[UPSC (Pre) - 2021]**

1. Hedgehog
2. Marmot
3. Pangolin

To reduce the chance of being captured by predators, which of the above organisms rolls up/ roll up and protects/ protect its/their

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vulnerable parts?

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(a) 1 and 2
(b) 2 only
(c) 3 only
(d) 1 and 3

20. Consider the following statements: **[UPSC (Pre) - 2019]**

1. Agricultural soils release nitrogen oxides into environment.
2. Cattle release ammonia into environment.
3. Poultry industry releases reactive nitrogen compounds into the environment.

Which of the statements given above is/are correct?

(a) 1 and 3 only
(b) 2 and 3 only
(c) 2 only
(d) 1, 2 and 3

21. Which of the following leaf modifications occur(s) in the desert areas to inhabit water loss? **[UPSC (Pre) - 2018]**

1. Hard and waxy leaves
2. Tiny leaves
3. Thorns, instead of leaves

Select the correct answer using the codes given below

(a) 2 and 3 only
(b) 2 only
(c) 3 only
(d) 1, 2 and 3

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22. What can be the impact of excessive/ inappropriate use of nitrogenous fertilisers in agriculture? **[UPSC (Pre) - 2015]**

1. Proliferation of nitrogen-fixing micro-organisms in soil can occur
2. Increase in the acidity of soil can take place
3. Leaching of nitrate to the groundwater can occur.

Select the correct answer using the codes given below:

(a) 1 and 3 only

(b) 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

23. Which one of the following is the best description of the term 'ecosystem'? **[UPSC (Pre) - 2015]**

(a) A community of organisms interacting with one another

(b) Part of the earth which is inhabited by living organisms

(c) A community of organisms together with the environment in which they live

(d) The flora and fauna of a geographical area

24. Which of the following add(s) carbon dioxide to the carbon cycle on the planet Earth? **[UPSC (Pre) - 2014]**

1. Volcanic action
2. Respiration
3. Photosynthesis
4. Decay of organic matter

Select the correct answer using the codes given below:

(a) 1 and 3 only

(b) 2 only

(c) 1, 2 and 4 only

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(d) 1, 2, 3 and 4

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25. Consider the following: **[UPSC (Pre) - 2014]**

1. Bats
2. Bears
3. Rodents

The phenomenon of hibernation can be observed in which of the above kinds of animals?

(a) 1 and 2 only

(b) 2 only

(c) 1, 2 and 3

(d) Hibernation cannot be observed in any of the above

26. Lichens, which are capable of initiating ecological succession even on a bare rock, are actually a symbiotic association of **[UPSC (Pre) - 2014]**

(a) Algae and bacteria

(b) Algae and fungi

(c) Bacteria and fungi

(d) Fungi and mosses

27. Which of the following leaf modifications occur(s) in desert areas to inhibit water loss? **[UPSC (Pre) - 2013]**

1. Hard and waxy leaves
2. Tiny leaves or no leaves
3. Thorns, instead of leaves

Select the correct answer using the codes given below:

(a) 1 and 2 only

(b) 2 only

(c) 1 and 3 only

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(d) 1, 2 and 3

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28. Which one of the following terms describes not only the physical space occupied by an organism, but also its functional role in the community of organisms? **[UPSC (Pre) - 2013]**

(a) Ecotone
(c) Habitat
(b) Ecological niche
(d) Home range

29. In the grasslands, trees do not replace the grasses as a part of an ecological succession because of **[UPSC (Pre) - 2013]**

(a) Insects and fungi
(b) Limited sunlight and paucity of nutrients
(c) Water limits and fire
(d) None of the above

30. Which of the following add(s) nitrogen to the soil? **[UPSC (Pre) - 2013]**

1. Excretion of urea by animals
2. Burning of coal by man
3. Death of vegetation

Select the correct answer using the codes given below
(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

31. With reference to the food chains in ecosystems, which of the following kinds of organism is/ are known as decomposer

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organism(s)? **[UPSC (Pre) - 2013]**

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1. Virus
2. Fungi
3. Bacteria

Select the correct answer using the codes given below:

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

32. With reference to food chains in ecosystems, consider the following statements: **[UPSC (Pre) - 2013]**

1. A food chain illustrates the order in which a chain of organisms feed upon each other.
2. Food chains are found within the populations of a species.
3. A food chain illustrates the numbers of each organism, which are eaten by others.

Which of the statements given above is/are correct?

(a) 1 only

(b) 1 and 2 only

(c) 1, 2 and 3

(d) None of the above

33. Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity? **[UPSC (Pre) - 2013]**

(a) Oceans, lakes, grasslands, mangroves

(b) Mangroves, oceans, grasslands, lakes

(c) Mangroves, grasslands, lakes, oceans

(d) Oceans, mangroves, lakes, grasslands

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34. What would happen if phytoplanktons of an ocean are completely destroyed for some reason? **[UPSC (Pre) - 2012]**

1. The ocean as a carbon sink would be adversely affected
2. The food chains in the ocean would be adversely affected
3. The density of ocean water would drastically decrease

Select the correct answer using the codes given below:

(a) 1 and 2 only

(b) 2 only

(c) 3 only

(d) 1, 2 and 3

35. Consider the following: **[UPSC (Pre) - 2011]**

1. Photosynthesis
2. Respiration
3. Decay of organic matter
4. Volcanic action

Which of the above add carbon dioxide to the carbon cycle on the Earth?

(a) 1 and 4 only

(b) 2 and 3 only

(c) 2, 3 and 4 only

(d) 1 and 4

36. In the context of ecosystem productivity, marine upwelling zones are important as they increase the marine productivity by bringing the **[UPSC (Pre) - 2011]**

1. Decomposer microorganisms to the surface
2. Nutrients to the surface
3. Bottom-dwelling organisms to the surface

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Which of the statements given above is/are correct?

(a) 1 and 2 only

(b) 2 only

(c) 2 and 3 only

(d) 3 only

37. A pesticide, which is a chlorinated hydrocarbon, is sprayed on a food crop. The food chain is: Food crop - Rat - Snake - Hawk. In this food chain, the highest concentration of the pesticide would accumulate in which one of the following? **[UPSC (Pre) - 2010]**

(a) Food crop

(b) Rat

(c) Snake

(d) Hawk

38. Some species of plants are insectivorous. Why? **[UPSC (Pre) - 2010]**

(a) Their growth in shady and dark places does not allow them to undertake sufficient photosynthesis and thus, they depend on insects for nutrition.

(b) They are adapted to grow in nitrogen-deficient soils and thus, depend on insects for sufficient nitrogenous nutrition.

(c) They cannot synthesise certain vitamins themselves and depend on the insects digested by them.

(d) They have remained in that particular stage of evolution as living fossils, a link between autotrophs and heterotrophs.

39. Epiphytes are plants which depend on other plants for **[UPSC (Pre) - 2001]**

(a) Food

(b) Mechanical support

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(c) Shade
(d) Water

40. Within biological communities, some species are important in determining the ability of a large number of other species to persist in the community. Such species are called **[UPSC (Pre) - 2000]**

(a) Keystone species
(b) Allopatric species
(c) Sympatric species
(d) Threatened species

Answers

1. (b) 2. (c) 3. (a) 4. (c) 5. (c) 6. (c) 7. (c) 8. (d) 9. (b) 10. (c)
11. (b) 12. (a) 13. (a) 14. (c) 15. (c) 16. (d) 17. (c) 18. (b) 19. (d) 20. (d)
21. (d) 22. (c) 23. (c) 24. (c) 25. (c) 26. (b) 27. (d) 28. (b) 29. (c) 30. (c)
31. (b) 32. (a) 33. (c) 34. (a) 35. (c) 36. (b) 37. (d) 38. (b) 39. (b) 40. (a)

Explanations

1. Correct answer (b): EXPLANATION

Parasites vs. Parasitoids

The main difference is that parasitoids inevitably kill their host to complete their life cycle, while parasites generally do not. Parasitoids, typically insects, develop on or inside a single host, consuming its resources until the host dies, at which point the parasitoid is free-living.

Carabid Beetle: some species are indeed parasitoids, *particularly in their larval stages*. These parasitoid *larvae feed on a host insect,*

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ultimately causing its death.

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Centipede: *NOT parasitoid*. They are predators that inject venom through pincer-like appendages to hunt and kill their prey.

Parasitoid Wasps and Flies: While not all species belong to the parasitoid family, these are *natural enemies of plantations like apples and pears*. Examples include: Ichneumonoid wasps, tachinid flies, bee flies, thick-headed flies, etc.

Termites: *NOT parasitoid*: They are detritivores that feed on dead organic matter, mostly plants like wood, leaves and soil humus.

All the above 5 species belong to the arthropod group of species. They possess an exoskeleton, a segmented body and paired jointed appendages.

2. **Correct answer (c): EXPLANATION**

Popularly called legumes, the pea family is a diverse group of flowering plants. They are recognized by their fruit, i.e. legume and compound, stipule leaves. There are various example of legumes- **Peas (Pisum sativum), Beans (Phaseolus), Soybean (Glycine max), Peanut (Arachis hypogaea), Lentils (Lens culinaris), Chickpeas (Cicer arietinum), Clover (Trifolium), Sweet Pea (Lathyrus) etc.**

Importance of legumes:

- In their root nodules, the majority of legumes have a symbiotic connection with rhizobia bacteria that allows them to **fix atmospheric nitrogen**.
- The nitrogen-fixing capacity reduces fertiliser cost for farmers, making cultivation economical and beneficial for soil health.

3. **Correct answer (a): EXPLANATION**

Figs have a unique pollination process due to its inverted flower structure, such that **both fig-wasps and the fig co-evolves.**

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- **Wasps, attracted by the fig's aroma, enter the fig and lay eggs** within. Pollination takes place as the **new born female wasps leave the fruit loaded with pollen**.
- The **male wasps mate and die within the figs** and are broken down by the fig.
- This represents two ecological processes namely **obligate mutualism (by pollinating wasps)** and **parasitoidism (by non-pollinating fig wasps).**
- Numerous figs serve as both food sources and habitats for a variety of animals, making them **keystone species** in their surroundings.

4. Correct answer (c): EXPLANATION

1. **Butterflies:** These species due to their bright colours are easy targets for predators. Hence, they are usually **equipped with distasteful or toxic chemicals used for deterrence**. Thus, **bright coloured butterflies usually indicate poisonous species. Example: Monarch butterfly**, which ingests such chemicals through plants.
2. **Fishes:** Unlike venomous fishes which injects the toxin, poisonous fishes rely on ingestion for any species to be affected by the poison. For **example, puffer fish**, **tropical reef fishes** like giant moray etc.
3. **Frogs:** *All frogs produce poisonous skin secretions, but not all are lethal for humans*. Colourful exteriors often indicate distasteful and poisonous species. For **example, dart frogs** are considered some the widely found poisonous frogs, with the golden poison frog the most poisonous.

5. Correct answer (c): EXPLANATION

Insects are the largest group among arthropods. A chitinous exoskeleton, three jointed leg pairs, compound eyes, a pair of antennae,

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and a three-part body **(head, thorax, and abdomen)** are **characteristics of insects**.

1. **Cicada:** They are **arboreal insects** which consume the sap water from trees for their diet.
2. **Froghopper:** like all insects, they have a three-part body (head, thorax, and abdomen), three pairs of jointed legs, and an exoskeleton. Adults are known for **their powerful jumping ability**, which is how they got the name "**froghopper**," capable of leaping many times their height to escape predators.
3. **Pond-skater:** Like other insects, they have six legs and a segmented body. They **use the surface tension of the water** to "skate" or glide across the surface, an ability aided by tiny, water-repellent hairs on their legs.

6. **Correct answer (c): EXPLANATION**

Marsupials are a group of animals which are natively found in the Americas, Oceania and some islands of Indonesia (Wallacea). **Hence Statement 1 is correct.**

Marsupials inhabit the **arid and semi-arid scrublands of Australia, temperate deciduous forests, and tropical rainforests of islands** as well. Hence **Statement 2 is not correct.**

Marsupials vs Placental Mammals

- While mammals are at a comparatively mature stage, marsupials' young are prematurely born and feed on the pouch of the marsupials for post-natal nourishment and growth. **Examples** of Marsupials include **kangaroos, koalas, opossums, Tasmanian devil, possums, wombats, wallabies, and bandicoots**.

7. **Correct answer (c): EXPLANATION**

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- **Waggle Dance by Honeybees:** Honeybees have one of the most intricate nonhuman communication systems. They may physically "waggle dance" to communicate with one another where to find resources like food, water, or nest sites.
- The **direction, distance,** and **quality of a resource** are communicated to the bee's nestmates through this dance.

8. Correct answer (d): EXPLANATION

- **Mushrooms:** A mushroom is the above-ground part of a fungus that looks fleshy and releases spores.
- **Medicinal Effects:** It focuses on **bioactive substances** that are present in fungi, such as **proteins, polysaccharides, and terpenoids**. These substances have ***anti-inflammatory, immunomodulatory, antioxidant, and anticancer activities*.**
- **Psycho-Active properties:** Some mushrooms also called "**magic mushrooms**" contain **hallucinogenic substance** that are activated upon consumption. They are found majorly in **subtropical humid forests.**
- **Insecticidal properties:** Some have pest-repellent properties due to their production of natural compounds and various proteins, which can be **toxic, repellent, or antifeedant to insects.**
- **Bioluminescence:** Scientists have discovered a rare species of bioluminescent mushroom, **Filoboletus manipularis**, in the Ranipuram forests of Kasaragod, Kerala.

9. Correct answer (b): EXPLANATION

There are various squirrels found in India, notable being the **Indian Giant squirrel, Indian Palm Squirrel and the Namdapha Flying Squirrel.**

Habitat: Indian squirrels are **arboreal, and build nests on trees**.

Statement 1 is incorrect

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They bury nuts and seeds in the ground to create food reserves for times when food is scarce, such as during cold weather. **Statement 2 is correct**.

Diet: ***Mostly squirrels feed on nuts and fruits for their diet. But they also rely on insects making them omnivorous***. Hence **Statement 3 is correct.**

Activity: While Indian squirrels are mostly diurnal, the Namdapha Flying squirrel, which is endemic to Arunachal Pradesh is **nocturnal and elusive** in nature. (IUCN status of Namdapha squirrel: **Critically Endangered**)

10. Correct answer (c): EXPLANATION

Microbes or microorganisms, are too small to be seen with the unaided eye. From hot springs to the interior of the human body, this varied group of organisms includes **bacteria, viruses, fungi, algae, archaea, and protozoa.**

- Some microbes, referred to as **thermophiles**, can **survive at temperatures higher than the boiling point of water**, with some even thriving at temperatures above 100°C (212°F). **Deep-sea hydrothermal vents and volcanic hot springs are two locations where these heat-loving extremophiles can be found.** Hence**, Statement 1 is correct.**
- Psychrophiles, also known as **cryophiles**, are microorganisms that can thrive and reproduce **in temperatures lower than the freezing point of water (0°C or 32°F)**. In order to function and thrive in conditions such as arctic ice, glaciers, permafrost, and deep-sea freezing waters, these cold-loving species have developed unique adaptations, such as cold-adapted enzymes and antifreeze proteins. Hence, **Statement 2 is correct**.
- When the **pH is less than 3**, certain microbes can thrive in an extremely acidic environment. Hence **Statement 3 is correct**.

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11. Correct answer (b): EXPLANATION

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Orangutan: They are the **biggest arboreal mammal**, distinguished by their red fur. These big apes are extremely intelligent animals that have 96.4% of human genes.

- **Diet:** Apart from being omnivorous, they also exhibit geophagy (soil consumption)
- **Intelligence:** They use sticks as tools to scrape insects (termites) from tree or log holes and for seed harvesting from hard-husked fruits. Moreover, they use tools for acoustic communication, for example, leaves to amplify sounds and natural remedies like herbs for wound treatment.

12. Correct answer (a): EXPLANATION

Note – Mostly leguminous crops have nitrogen fixation ability to synthesis food material (protein). **Alfalfa, chick pea and clover** are example of leguminous crop so they are right because they have nitrogen fixation ability.

Nitrogen fixation refers to the conversion if nitrogen to ammonia. Biologically, nitrogen fixation is done by legumes and actinorhizal plants like alder and bayberry.

13. Correct answer (a): EXPLANATION

The fungus-growing ants (called **Attine ants**) are **obligate symbionts with a fungus they use for food. These ants are found across the American continents and the West Indies**.

These ants are known as **fungus growers** because they maintain an **obligate mutualism** with **fungi cultured inside their nests**, and which is the only **food source for the larvae** and an important resource for the adult ants as well.

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14. Correct answer (c): EXPLANATION

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Weathering of rocks is an essential first step to **release phosphorus** from rocks into water and soil in the form of phosphate salts. Hence, Option (c) is correct.

15. Correct answer (c): EXPLANATION

Detritivores refer to a group of heterotrophic organisms that **consume dead and decomposing matter (detritus)** to obtain energy. They are usually **arthropods**. Detritivore examples: Earthworms, Millipedes, Woodlice, Dung flies, Starfish, Sea Cucumber, Fiddler Crab etc.

Earthworms: (Annelida) They mostly dwell in the moist compost-rich soil, consuming **detritus**. Earthworm activity loosens and mixes the soil, helping nutrients break down and become available for plants. They are also considered **ecological indicators of soil**. **(1 is correct)**

Jellyfish:(Cnidaria) They are free-swimming marine animals that are armed with stinging cells for predation and defence. jellyfish are **not detritivores**; they are primarily carnivorous, feeding on living organisms like small fish and zooplankton. **(2 is incorrect)**

Millipede: (Arthropoda) millipedes are **detritivores**, meaning they feed on dead and decaying organic matter like dead leaves, wood, and other plant material. Their role as detritivores is crucial for ecosystems because they break down this organic material, recycling nutrients back into the soil to support plant growth. **(3 is correct)**

Seahorse: seahorses are **not detritivores**. They are predators that eat live prey. Detritivores consume dead and decaying organic matter, while seahorses use their snouts to suck up small living crustaceans like shrimp and plankton. **(4 is incorrect)**

Woodlice: (Arthropoda) woodlice are **detritivores**, meaning they feed on dead organic matter like decaying leaves and wood. **(5 is correct)**

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16. Correct answer (d): EXPLANATION

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Cnidarians: Cnidarians are coelenterates (Phylum of Nonchordata) They comprise aquatic organisms found in marine and freshwater areas. **Symbiotic relations: Zooxanthellae and Corals**. The algae provide the corals with nutrients and oxygen, while the cnidarian corals benefit them with shelter and nutrients like carbon dioxide and nitrogen for survival – **Mutualism (1 is correct)**

Fungi: They are eukaryotic organisms comprising yeasts, moulds, or a combination of both. **Symbiotic relations example (1) Mycorrhizae: Mutualism between fungi and higher plant roots**. where the fungi increase the capability of plant roots to absorb water and nutrients while the plant roots provide shelter and sugars produced during photosynthesis. **(2) Lichens:** It is a composite structure formed by the **mutualistic** relationship of fungi and photosynthetic algae or cyanobacteria. **(2 is correct)**

Protozoa: They are a group of free-living or parasitic single-celled eukaryotes. **Symbiotic relations example (1) Termites and protozoa**: The **protozoa digest the cellulose in the gut of termites** from the wood they consume, giving them energy, while deriving food and shelter from the termite – **Mutualism (3 is correct)**

17. Correct answer (c): EXPLANATION

Ferns: Ferns require soil or organic matter to thrive, and they typically have roots that need to penetrate the ground. **(1 is incorrect)**

Lichens and mosses are well-adapted to surviving on surfaces without soil, as they can **directly attach to rocks, bark, or other surfaces and obtain the necessary nutrients from the air and moisture**. **(2 & 3 are correct)**

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Mushroom: Mushrooms are fungi that need decaying organic matter to survive, which is usually found in or around soil. **(4 is incorrect)**

18. Correct answer (b): EXPLANATION

Note – Primary producers are autotrophic in nature those have ability to do photsynthesis and synthesize own food material.

Copepod: It is a type of crustacean found in almost all water bodies, both freshwater and marine. **They mostly feed on phytoplankton or are parasitic in nature**. Being parasitic, they can be used to treat water for mosquitoes, preventing malaria and dengue outbreaks. **(1 is incorrect)**

Cyanobacteria: They are one of the first organisms to have produced oxygen using **photosynthetic pigments** and found almost everywhere. They are commonly also called blue-green algae. **(2 is correct)**

Diatoms: They are a **form of algae** and generate 20-50% of the oxygen produced annually. Algae are **autotrophic** in nature and have **chlorophyll** so act as a primary producer. **(3 is correct)**

Foraminifera: They are unicellular organisms that are mostly marine. Foraminifera are considered **consumers**, often categorized as zooplankton because they are not able to produce their own food. **(4 is incorrect)**

19. Correct answer (d): EXPLANATION

Hedgehog: Hedgehogs are spiny mammals that are primary **nocturnal and omnivorous** in nature. Their spines are not poisonous but they use them as a defense mechanism by **rolling into a tight balls** with the spines exposed. In places without natural predators, hedgehogs become an invasive species as seen in New Zealand and Scotland islands. **(1 is correct)**

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Marmot: They are large ground squirrels that are the **largest and heaviest of the squirrel family**. They typically live in burrows and are found in the mountainous and dry grassland regions, undergoing **hibernation** in winter. Their primary defenses include sophisticated alarm calls, **extensive burrow systems for quick escape**, and communal living that increases vigilance, **against predators**. But they **never roll their own body**. **(2 is incorrect)**

Pangolin: Also called scaly anteaters, they are nocturnal and solitary mammals found in Asia and Sub-Saharan Africa. They are both arboreal and ground-dwelling and can **roll into a tight ball when threatened**. **(3 is correct)**

20. Correct answer (d): EXPLANATION

As per the first ever Indian Nitrogen Assessment in 2018, **nitrogen makes up for the largest portion of PM 2.5 particles** which are linked to **cardiovascular and respiratory illnesses**.

- Of the total emissions, **chemical fertilizers used in agriculture emit 70% of N2O**. It is due to inefficiencies in application leading to wastage. Since 2002, N2O has replaced methane as the second largest GHG from Indian agriculture.
- India is the biggest source of NH3 (ammonia) emissions with **cattle contributing 80%.**
- **Poultry industry** also recorded an **annual growth rate of 6% in excretion of reactive nitrogen compounds.**

21. Correct answer (d): EXPLANATION

Same Question UPSC asked in 2013 Prelim examination

There are many desert plants that have adopted different methods for arresting water loss. Some modifications are as follows:

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- Succulents have fleshy leaves for water storage and a waxy exterior to prevent evaporation.
- xerophytes have long roots to reach deep underground in search of water and tiny leaves to reduce evaporation.
- Cacti have fleshy and thick stems to store water and leaves modified as thorns to reduce the surface area reducing evaporation.

22. Correct answer (c): EXPLANATION

As per an All India Coordinated Research Project on 'Long Term Fertilizer Experiments' (2020-21), continuous and excessive use of nitrogenous fertilizers had a detrimental effect on soil health such as:

- Excessive nitrogen fertilizer use actually **suppresses** nitrogen-fixing microorganisms. Plants with abundant, readily available nitrogen from fertilizers have less need to rely on symbiotic relationships with microbes like Rhizobium or free-living bacteria to fix atmospheric nitrogen. High levels of soil nitrogen can inhibit the activity and population of these beneficial microbes. **(1 is incorrect)**
- Alteration of soil chemistry, such as increased salinity **(2 is correct)**
- Nitrate contamination in groundwater **(3 is correct)**
- Eutrophication of aquatic ecosystems
- Emission of N2O in the atmosphere, thereby causing air pollution.

23. Correct answer (c): EXPLANATION

Ecosystem: It refers to the complex of living organisms, their physical environment, and all their interrelationships in a particular unit of space. (Britannica)

Biodiversity: The collection of all the plants (flora) and animals (fauna) in a specific geographical area is called biodiversity or biota

Habitat: It is a place where an organism resides, including all living and nonliving factors or conditions of the surrounding environment.

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Biome: It is the largest geographical biotic unit where a major community of plants and animals along with similar life forms and environmental conditions live.

24. Correct answer (c): EXPLANATION

Same Question UPSC asked in 2011 Prelim examination

Carbon Dioxide is the primary form of carbon that circulates the Earth through the Carbon Cycle.

- Intake of carbon **(carbon sink)** from the atmosphere takes place through photosynthesis by plants and phytoplankton and carbon sinks like oceans.
- However, carbon is also released **(carbon source / add carbon di oxide in the atmosphere)** into the atmosphere through a variety of natural processes like - Respiration by animals and plants, Decomposition of organic matter, Volcanic Action, Forest Fires and Weathering of Carbonate rocks among many others.

25. Correct answer (c): EXPLANATION

Hibernation is a **winter sleep**. In severe winter condition, **some organism slow down their metabolic activity** and try to save own life. Bats, Bears and rodents, all have hibernation ability.

- **Bats:** Bats undergo hibernation by attaining a state called **torpor**- decreased activity with decreased temperature and metabolism to conserve energy during winters.
- **Bears:** Bears of the northern region hibernate during winter by dropping their body temperature and heartbeat, maintaining energy levels.
- **Rodents:** They undergo seasonal eating for sustained energy levels during winter hibernation.

26. Correct answer (b): EXPLANATION

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Lichens: It is a composite structure formed by the **mutualistic relationship** of fungi and algae.

- Lichens can grow on a variety of substrates in extreme environments and are hence considered an environmental indicator (specially against SO2 gas).
- Lichen can chemically weather rocks and stones, thereby weakening structures and altering soil chemistry.
- They are also considered as pioneer species as they an easily grow on a bare rock as the first organism (Xerosere succession).

27. Correct answer (d): EXPLANATION

There are many desert plants that have adopted different methods for arresting water loss. Some modifications are as follows:

- Succulents have fleshy leaves for water storage and a waxy exterior to prevent evaporation.
- xerophytes have long roots to reach deep underground in search of water and tiny leaves to reduce evaporation.
- Cacti have fleshy and thick stems to store water and leaves modified as thorns to reduce the surface area reducing evaporation.

28. Correct answer (b): EXPLANATION

Ecotone: It is a transitional area between two community or ecosystems or biome etc. They are characterised by overlapping and unique biodiversity. The influence of the bordering ecosystems is called edge effect.

Ecological Niche: It is defined as the particular role of a species encompassing the specific conditions and resources it needs to survive ans interact with the ecosystem.

Habitat: It is a place where an organism resides, including all living and nonliving factors or conditions of the surrounding environment.

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Home Range: A home range is the geographic area where an animal spends its time performing normal activities like foraging, resting, and breeding.

29. Correct answer (c): EXPLANATION

As per the National Geographic, grasslands are characterised by mostly grasses and infrequent trees. The main reasons are as follows:

1. **Low precipitation:** While it is enough for grasses to grow in abundance, it is not enough for all trees to sustain.
2. **Forest Fires:** Grasses are well-adapted to grow back after fires unlike trees.
3. **Animal Grazing:** Large animals like African elephants often trample on the ground, discouraging the growth of trees.

Trees do not replace by grassland means it has an ecological succession of grassland. Grassland are weeds (unwanted plants) so can grow anywhere even in scarcity of water and nutrients. Grassland area also not have effect due to fire.

30. Correct answer (c): EXPLANATION

Excretion of Urea: Urea is a nitrogen-rich compound that is commonly found in animal excreta. It is converted by microorganisms into forms of nitrogen that can be absorbed by plants, like ammonia and nitrates. Hence **1 is correct.**

Burning of coal: When coal is burnt, the flue gas produced may contain nitrogen compounds but it is usually released into the air as a pollutant. It is not absorbed by soil. Hence **2 is not correct**.

Death of vegetation: When plants die, decomposers break down the organic nitrogen into inorganic forms like ammonium (NH_4^+) that is

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released into the soil and allows for plants to absorb. Hence **3 is correct**.

31. Correct answer (b): EXPLANATION

Heterotrophs take nutrition from other organism because they can not synthesise own food material due to absence of chlorophyll (photosynthesis).

There are different types of heterotrophs – **(a) Parasites** – they take nutrition from living host e.g. Virus (1 is wrong) **(b) Saprophytes** – they take nutrition from dead and debris e.g. Bacteria and fungi. So, bacteria and fungi are decomposers. (2 and 3 are rights)

32. Correct answer (a): EXPLANATION

A food chain refers to the **linear sequence** of relationships among organisms in an ecosystem, showing the transfer of energy and nutrients from one organism to another.

- It is **unidirectional**, starting from producers and ending at the apex predator. (**1 is right**)
- No, food chains are not found within the populations of a single species; they are a sequence of different species showing the flow of energy from one to another. A food chain involves **multiple species at different trophic levels,** starting with producers (like plants) and moving to consumers (herbivores, carnivores, etc.). (**2 is wrong**)
- **A food chain illustrates the sequence of organisms** and the flow of energy as one is eaten by another, **not the specific numbers of each organism**. (**3 is wrong**).

33. Correct answer (c): EXPLANATION

Note: kindly follow Whittaker productivity chart, UPSC asked question from this list.

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Explanation- Productivity refers to the rate of biomass production per unit area.

Mangroves are highly productive ecosystems due to abundant sunlight, rich nutrients from both land and sea, and tidal flushing.

Grasslands have moderate to high productivity, generally lower than mangroves but higher on average than lakes or the open ocean.

Lakes have variable productivity depending on nutrient levels, but on average, it is generally lower than grasslands.

Oceans, specifically the vast open ocean, have the lowest primary productivity per unit area due to nutrient limitation in surface waters.

34. Correct answer (a): EXPLANATION

Phytoplanktons are considered extremely vital to the ocean ecosystem and the planet as a whole because:

1. Phytoplankton acts as a source of oxygen and a carbon sink for the carbon cycle- Biological carbon pump.
2. They act as primary producers- "grasses of the sea" creating organic compounds using sunlight and CO2 through photosynthesis
3. They are also at the first trophic level in the food chains of the oceans, being an essential source of nutrition for zooplankton, small fishes like krill and ultimately for larger marine animals.
4. They are also essential for their supporting services, like enabling the smooth functioning of biogeochemical cycles.

However, they have no bearing on the density of ocean water, as it is not dependent on the quantity of phytoplankton, but on factors such as salinity, temperature, and pressure. (3 is wrong)

35. Correct answer (c): EXPLANATION

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Carbon Dioxide is the primary form of carbon that circulates the Earth through the Carbon Cycle.

- **Intake of carbon (carbon sink)** from the atmosphere takes place through **photosynthesis by plants** and **phytoplankton** and **carbon sinks like oceans**.
- However, **carbon is also released (carbon source / add carbon dioxide in atmosphere) into the atmosphere** through a variety of natural processes like - **Respiration by animals and plants, Decomposition of organic matter, Volcanic Action, Forest Fires and Weathering of Carbonate rocks** among many others.

36. Correct answer (b): EXPLANATION

Upwelling of water refers to the phenomenon when **ocean currents interact to bring the cold nutrient-rich water to the surface**, where the bright sunlight fuels the rapid growth of phytoplankton.

It improves the marine productivity in the following ways:

1. It leads to enrichment of nutrients like **nitrogen and phosphorus to** the surface water.
2. The influx of vital nutrients and sunlight fuels the growth of **phytoplankton blooms,** which provide food for a number of organisms like zooplankton and fish, i.e. phytoplankton forms the **basis of the marine food web.**
3. An increase in food availability creates scope for the **enrichment of marine biodiversity.**
4. The abundance of marine life has economic benefits by providing **highly productive fishing grounds.** For example, the coasts of **Peru** (Humboldt current) or **Japan** (Kuroshio and Oyashio currents)

Note: Organism would remain where they present. There have no movement can be observed at the upwelling zone. **(1 and 3 statement wrong)**

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37. Correct answer (d): EXPLANATION

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Bioaccumulation and Biomagnification:

- **Bioaccumulation** is the gradual buildup of foreign substances within the body of an organism over time. Bioaccumulation happens when the rate of consumption exceeds the rate of excretion.
- **Biomagnification** refers to the increase in **concentration of toxic substances in organisms at higher trophic levels**, leading to the **highest amounts at the apex predator trophic levels**.

38. Correct answer (b): EXPLANATION

Insectivorous plants: These plants grow in nitrogen-deficient soils, but their main nutritive elements are nitrogen.

To **fulfill demand of nitrogen they intake insect as a food**. Insect have presence of protein and nitrogen is main element in protein compounds.

For example: Venus Flytrap and Pitcher Plant etc.

Insectivorous plants are found in damp waterlogged areas, such as swamps and bogs, where nutrients are less available but enough sunlight is available for photosynthesis.

39. Correct answer (b): EXPLANATION

Epiphytes: They are plants that use the surrounding plants or other surfaces as mechanical support for growth but do not usually develop any parasitic relation with the host.

It can be considered an example of **commensalism**, where the epiphyte benefits without any impact on the host.

Epiphytes have **hygroscopic root** due to presence of "**Velamen tissue**" those intake moisture from atmosphere.

Due to epiphytic nature, it can collect nutrients directly from the atmosphere or surrounding debris.

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40. Correct answer (a): EXPLANATION

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Keystone Species: Those species members are responsible to maintain entire biodiversity in a particular area are called Keystone species. Insects as a pollinator are important to maintain entire biodiversity.

Allopatric Species: These species are formed due to the creation of a **geographical barrier** that prevents free breeding and then gives rise to a unique kind of species. **For example: Darwin's finches on the Galapagos Islands, etc.**

Sympatric Species: Sympatric species are groups of organisms that **live in the same area with overlapping ranges but stay separate because they cannot interbreed**. For example, species of apple maggot fly within the same area feed on two different hosts- hawthorn and apple trees.

Threatened Species: As per IUCN, threatened or Red-List species refer to those that are **vulnerable to extinction in the near future** due to a variety of reasons, such as climate change, habitat degradation, pollution, etc. Based on the current population analysis, they are categorised into three groups: **Vulnerable, Endangered & Critically Endangered**

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Unit II Climate Change & Climatic Parameters

Climate Change & Its Related Issues 7

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7.1 CLIMATE CHANGE

- Climate is the average weather of a place over many years. Climate change refers to long-term shifts in weather patterns that affect regional, local, and global climates. This includes the rise in global temperatures since the mid-20th century. The rapid climate changes we see today are mainly due to human activities. These activities include using coal, oil, and gas for homes, industries, and transportation.
- Greenhouse gases, mainly carbon dioxide (CO2), come from burning fossil fuels. They trap the sun's heat and raise Earth's temperature. The planet has already warmed by about 1.2°C. Atmospheric CO2 levels are 50% higher than they were in the 19th century. Scientists emphasize that we must slow down temperature increases to avoid the worst impacts of climate change. We need to limit warming to 1.5°C by 2100. Without strong action, warming could exceed 2°C this century. In the worst-case scenario, it could reach 4°C, leading to significant sea-level rise, displacing millions, and causing irreversible loss of plant and animal species.

7.2 REASONS BEHIND CLIMATE CHANGE

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- It is an interesting topic to know how Earth's climate was hundreds and thousands of years back. Scientists made it possible for all of us to know the reasons behind climate change by carrying out several experiments that dealt with studies of rock texture, types of plants that existed, their annual ring pattern, glacier size, fossils, sediments, ocean bed mapping, gradual change in Earth's orbit around the sun and much more.
- When the Earth was created, the climatic change was due to several physical and chemical reactions on the Earth such as volcanic eruption, solar radiation, etc., but it was all natural. After which, it stabilised and cooled down, which favoured the existence of life form. Fast forwarding to the 1700s, when the Industrial Revolution started, everything related to climate started to change gradually due to huge emission of greenhouse gases (gases that trap solar energy and heat Earth's atmosphere) from industries leading to a condition named 'global warming'.

7.2.1 Natural Causes Behind Environmental Changes

- Continental drift theory
- Volcanic eruptions
- Changes that took place in Earth's orbit
- Earth's tilted axis
- Variation in solar radiation
- Ocean currents
- Presence of space dust between the sun and the Earth

Continental Drift Theory

- Millions of years back, **Pangea,** the supercontinent, gradually drifted apart after breaking into several landmasses.
- After continental drift took place, all these landmasses formed continents with different shapes, sizes, land features, mountains, rivers, and deserts, which changed the ocean current direction and wind direction, ultimately changing the climate.

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Volcanic Eruptions

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- Volcanic eruption throws a huge amount of dense gases into the atmosphere, including sulphur dioxide (SO_2), dust, water vapour, and ash. These large volume of dense gases forms a gaseous cloud that contributes to climate change immensely by actually preventing sun rays from reaching landmass, which lowers the overall temperature of that particular area.
- When sulphur dioxide binds with water molecules in the atmosphere, they act as potential reflectors of incoming solar radiations and decrease atmospheric temperature, i.e. global cooling.

Changes Took Place in Earth's Orbit

- The various seasons we see on Earth are due to the varied distance between the Earth and the sun throughout the Earth's revolution. It is surprising to know that Earth's orbit is not fixed even though it seems so as we see the axis pointing towards North Star (Polaris) towards a fixed direction in the north.
- Earth's axis also moves a bit more than 1/2° every 100 years.
- Thus, it is not possible for the Polaris to always point towards the North. It has been found that the gradual alteration in the direction of axis of the Earth, called precession, also contributes to climatic changes.

Earth's Tilted Axis

- One complete orbital cycle around the sun by the Earth takes one year.
- The angle at which Earth's axis is tilted is 23.5° from the perpendicular axis of Earth's orbital path.
- The hemisphere, be it northern or southern, when tilted towards sun experiences summer season while the hemisphere away from the sun experiences the winter season.
- Melting of ice has been proven to contribute to shift in Earth's tilted axis, bringing more severe seasons and climatic changes.

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- More the tilt, warmer the summer and colder the winter whereas lesser the tilt, lesser warm summer and moderate winter variation will be.

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■ Variation in Solar Radiation

- More the radiation from sun reaching the Earth, warmer will be the climate and seasons while lesser the radiation from sun reaching the Earth, colder will be the climate and milder will be the seasons.
- Solar radiation is found to play a role in cloud formation. Since the ice age, the temperature of the Earth atmosphere has been changing and the intensity of solar radiation is one of the major contributors.

■ Ocean Currents

- Direction, strength and temperature of ocean currents vary according to the position of ocean, adjacent landmass or continent, wind direction and overall temperature.
- El Nino is one such event which has a global climatic effect which depends on the Humboldt Ocean current along the coast of Peru, a country situated on the western side of South America.

■ Presence of Space Dust Between the Sun and the Earth

- In the space, thousands of collisions occur between asteroids which create cosmic dusts consisting of metals. On entering Earth's atmosphere with a very high force, they heat up on colliding with water molecules and eventually melt and evaporate. This affects the climate.
- In the Polar region, the cosmic dust combines with noctilucent clouds giving rise to ice crystals, this phenomenon is considered as climate change indicator.
- Cosmic dust particles containing metals also react with the ozone present in the stratosphere.
- When cosmic dusts make it to the Earth's ocean, it increases iron concentration in the ocean affecting phytoplankton, which releases certain

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gases and all these potentially affect climate change.

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7.2.2 Anthropogenic (Man-Made) Causes Behind Environmental Changes

- Burning of fossil fuel
- Industrial processes
- Burning of biomass
- Deforestation
- Agricultural practices
- Change in land-use pattern

■ *Burning of Fossil Fuel*

- Fossil fuel, a non-renewable source of energy, that took millions of years to form, is now being used in coal, petroleum, diesel, natural gas, etc., for industrial purposes, transportation, and producing electricity, domestic purposes.
- Burning of fossil fuel contributes to 3/4th of the total CO_2 emission, around 1/5th of the total methane emission and a huge amount of nitrous and sulphur oxides.
- All these gases are considered greenhouse gases which trap the solar radiation and outgoing thermal radiation, resulting in heating up the Earth's temperature, leading to 'global warming'.

■ *Industrial Processes*

Industrial processes emission can be classified into two main categories:

Direct Emission:

- Combustion of fuel at the site undergoes various chemical and physical changes to create heat energy and power which may accidentally leak into the surrounding atmosphere.
- Iron, plastic, steel and cement industries, which deal with the production of chemicals in huge amounts daily, contribute about 1/3rd to the direct

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emission.

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Indirect Emission:

- At a power plant, when combustion of fossil fuel takes place to produce electricity which is next consumed by various industries for operating their machinery and all come under indirect emission.

Short-lived Climate Pollutants (SLCPs)

- Short-lived climate pollutants (SLCPs) are actually agents with quite short life span in the atmosphere ranging from a few days to a few decades and give a warming influence on climate.
- The most harmful short-lived climate pollutants are **black carbon, methane, hydrofluoro-carbons and tropospheric ozone**, which are the most important contributors to the increasing global greenhouse effect after CO_2.
- They are dangerous air pollutants that have varied harmful effects on human health, agriculture, atmosphere and ecosystems.
- HFCs are currently present in small quantities in the atmosphere yet their contribution to climate change has climbed to as much as 19% of global CO_2 emissions by 2050.

■ *Burning of Biomass*

- Burning forest products, forest fires, ‘jhoom’ cultivation collectively contribute about 1/5th of the total carbon dioxide emissions at a global level.
- This gives rise to short-lived climate pollutant.

Deforestation

- Deforestation is now considered to be the 2nd largest anthropogenic source of CO_2 emission in the atmosphere, combustion of fossil fuel being the 1st. With increasing population, demand for land, agricultural field, industrial spaces, mining has also been on the rise.
- All these can be achieved smoothly only if forests are cleared.
- Thus, deforestation and forest degradation are major greenhouse gas contributors because burning forest biomass releases huge amounts of gases and particulate matters.

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Agricultural Practices

- Reports reveal that the agriculture sector alone contributes about 14% of the global greenhouse gas emission. The advancement of technology has eased agriculture process to a great extent but that didn't help much since it still has a great adverse effect on global warming.
- Using more synthetic fertilizers, toxic or harmful pesticides, monoculture plantation has resulted in huge emissions of CO_2, methane (from paddy cultivation, digestion process in cattle known as enteric fermentation) and nitrous oxide into the atmosphere.

Change in Land-Use Pattern

- As mentioned earlier, an increase in population has led to an increase in demand for agricultural productivity and industrial activity to meet the increasing demand of this growing population.
- Activities like clearing land, slash and burn cultivation, and deforestation have dangerously degraded forest and soil quality.
- Degraded soil quality with low organic compounds has reduced the efficiency of forest and soil to act as **'carbon sink'**.
- Recently it has been proved scientifically that huge concrete buildings in developed and developing countries capture the solar radiation during day time and emit greenhouse gases at night which is popularly known as **'urban heat island effect'**.

7.3 EVIDENCE FOR CLIMATE CHANGE

1. Gradual rise in sea level.
2. Rise in global temperature.
3. Ocean water getting warmer.
4. Greenland and Atlantic ice sheaths are shrinking
5. Arctic sea ice has also thinned out
6. Melting of glaciers
7. Severity in seasons along with the intensity and frequency of climatic hazards has greatly increased (floods, heavy rainfall, dry spell, cyclones,

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etc.)

8. Acidification of oceans
9. Death of corals

7.4 IMPACTS OF CLIMATE CHANGE

- Extreme weather occurrences are intensive in nature, thereby threatening both lives and livelihoods.
- Few regions in the world may become uninhabitable due to further warming, turning farmland into desert.
- In other regions, the opposite is occurring, as the intense rainfall results in huge flooding – as recently witnessed in Germany, China, Belgium and the Netherlands.
- Poorer countries' population suffers the most due to non-availability of money for adapting to climate change.
- Experiencing very hot climates by various farms in developing countries will only get worse.
- There is a persistent threat to our oceans and their habitats.
- Due to climate change, since 1995 the Australian Great Barrier Reef has already lost its half corals due to warmer seas.
- There is a frequent rise in Wildfires due to climate change, thereby increasing the risk of hot, dry weather.
- There will be a worsening climate change on the releasing of greenhouse gases into the atmosphere which are trapped for centuries when the frozen ground melts in areas such as Siberia.
- There will be scarcity of food and water for the animals to survive in a hotter planet.
- For instance, polar bears could struggle to survive as the ice on which they are dependent has started melting, and elephants will find it harder to get nearly 150–300 litres of water a day.
- According to the scientists' belief, planet could lose nearly 550 species in case of no action.

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How will World's various parts be affected?

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- World's areas will get affected due to climate change in different ways.
- Few places will be hotter than others, and others will receive more rainfall and face more droughts respectively, if the temperature rise is not limited to 1.5°C:
  - Vulnerability due to flooding caused by extreme rainfall will be greater in the UK and Europe.
  - Middle East nations will face extreme heatwaves and, finally, the farmland will turn into desert.
  - Pacific oceans Island nation may extinct under rising seas levels.
  - Most of the African countries is to face droughts and food shortages.
  - There will be drought conditions in the western part of the US, while there will be more extreme storms in other areas.
  - Australia will have to face intense heat and drought.

7.4.1 Impact of Climate Change on the Indian Region

Observed Changes in Global Climate

- From the pre-industrial times, the average temperature has risen globally by nearly 1°C.
- The intensity of the warming rate cannot be illustrated alone by natural variations but also take into account the effects of human activities.
- The present climate change is due to greenhouse gases (GHGs), aerosols emissions and **land usage and land cover (LULC)** changes at the time of the industrial period which has significantly altered the atmospheric composition, and subsequently, the planetary energy balance.
- Since the 1950s, there has been a substantial rise in weather and climate extremes globally due to climate warming (for example, droughts, heat waves, heavy precipitation, severe cyclones), precipitation and wind patterns changes (including global monsoon systems shifts), global oceans warming and acidification, melting of sea ice and glaciers, rising sea levels, and marine and terrestrial ecosystems changes.

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■ Projected Changes in Global Climate

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- Global climate models project a continuation of human-induced climate change during the twenty-first century and beyond.
- If the current GHG emission rates are sustained, the global average temperature is likely to rise by nearly 5°C, and possibly more, by the end of the twenty-first century.
- Even if all the commitments (called the 'Nationally Determined Contributions') made under the 2015 Paris Agreement are met, it is projected that global warming will exceed 3°C by the end of the century.
- However, the temperature rise will not be uniform across the planet; some parts will experience greater warming than the global average.
- Such large changes in temperature will greatly accelerate other changes that are already underway in the climate system, such as the changing patterns of rainfall and increasing temperature extremes.

■ Leading Causes for Climatic Change

- The main contributor to climate change is anthropogenic activities which are pushing up concentrations of greenhouse gases and this has led to a rise in temperature and atmospheric moisture content.
- A higher concentration of water vapour, in turn, leads to intense rainfall during monsoon.
- Heating leads to vaporisation, which is directly linked to decreasing soil moisture, resulting in droughts. This can lead to a reduction in food production and the availability of potable water.
- Rising sea levels would make India's big cities vulnerable to erosion and damage to coastal projects.

■ Climate Change in India: Observed and Projected Changes

(i) Temperature Rise Over India

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- India's average temperature has risen by around 0.7°C during 1901–2018.

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- Regions of North India have undergone warming more than the South.
- The GHG-induced warming leads to temperature rise, partially offset by forcing because of anthropogenic aerosols and LULC changes.
- According to RCP8.5 scenario, at the end of the twenty-first century, there will be a projected rise in average temperature over India by nearly 4.4°C, relatively to the recent past (1976–2005 average).
- Over three decades (1986–2015), the warmest day and the coldest night temperature of the year have increased by nearly 0.63°C and 0.4°C respectively.
- By the end of the twenty-first century, these temperatures are projected to rise by approximately 4.7°C and 5.5°C, respectively, relative to the corresponding temperatures in the recent past (1976–2005 average), under the RCP8.5 scenario.
- By the end of the twenty-first century, the frequencies of occurrence of warm days and warm nights are projected to increase by 55% and 70%, respectively, relative to the reference period 1976–2005, under the RCP8.5 scenario.
- Under the RCP8.5 scenario, the summer (April –June) heatwaves frequency over the country at the end of the twenty-first century is projected to be three to four times higher than the 1976–2005 baseline period.
- The mean heatwave duration events are also estimated to be nearly double, but with a considerable spread among models.
- In response to the combined surface temperature and humidity increase across the country, heat stress magnification is expected, specifically over the Indus and Indo-Gangetic river basins.

(ii) Indian Ocean Warming

- During 1951–2015, the Indian Ocean Sea surface temperature (SST) on average has risen by 1°C, noticeably higher than the global 0.7°C average SST warming, over that particular period.
- The tropical Indian Ocean heat content in the upper 700 m has shown a notable rise in trend over the previous six decades (1955–2015), with a

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major noticeable rise over the past two decades (1998–2015).
- The ocean heat content and SST in the Indian Ocean are estimated to rise continuously in the twenty-first century.

(iii) Changes in Rainfall/Monsoon

- The precipitation (June to September) in the summer has decreased by nearly 6% over the country from 1951 to 2015, with a significant decline over the Western Ghats and Indo-Gangetic Plains.
- Annual rainfall over India showed a declining trend during 1951–2015 and the reduction ranged between 1 and 5 mm over central India, Kerala and the far Northeast regions.
- Contrarily, precipitation increased over J&K and Northwest India.
- Based on multiple datasets and climate model simulations, there is an emerging consensus, that the radiative effects of anthropogenic aerosol forcing over the Northern Hemisphere have considerably offset the expected precipitation increase from GHG warming and contributed to the observed decline in summer monsoon precipitation.
- The coming decades are projected to witness a considerable rise in the mean, extreme and inter-annual variability of rainfall associated with the monsoon.
- Recently, during summer monsoon weather, there has been a shift toward a higher number of dry spells and more extreme wet spells (27% higher in the period 1981–2011 relative to 1951–1980).
- Globally, the frequency of localised heavy precipitation occurrences has risen in regard to rising atmospheric moisture content.
- Simply put the frequency and intensity of both heavy rainfall events and dry days have gone up since the 1950s.
- These trends are prominent over Central India and South Peninsular regions during the southwest monsoon (June–September) and northeast monsoon (October–December) respectively.
- Over Central India, the frequency of daily precipitation extremes with rainfall intensities exceeding 150 mm per day increased by about 75% during 1950–2015.

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- With continued global warming and anticipated reductions in anthropogenic aerosol emissions in the future, it projects an increase in the mean and variability of monsoon precipitation by the end of the twenty-first century, together with substantial increases in daily precipitation extremes.

(iv) Droughts

- The overall decrease of seasonal summer monsoon rainfall during the last 6–7 decades has led to an increased propensity for droughts over India.
- Both the frequency and spatial extent of droughts have increased significantly during 1951–2016.
- In particular, areas over Central India, southwest coast, southern peninsula and North-eastern India have experienced more than two droughts per decade, on average, during this period.
- The area affected by drought has also increased by 1.3% per decade over the same period.
- Climate model projections indicate a high likelihood of increase in the frequency (>2 events per decade), intensity and area under drought conditions in India by the end of the twenty-first century under the RCP8.5 scenario, resulting from the increased variability of monsoon precipitation and increased water vapour demand in a warmer atmosphere.

(v) Sea-level Rise

- Sea levels have risen globally because of the continental ice melt and thermal expansion of ocean water in response to global warming.
- Sea-level rise in the **North Indian Ocean (NIO)** occurred at a rate of 1.06–1.75 mm per year during 1874–2004 and has accelerated to 3.3 mm per year in the last two and a half decades (1993–2017), which is comparable to the current rate of global mean sea-level rise.
- At the end of the twenty-first century, steric sea level in the NIO is projected to rise by approximately 300 mm relative to the average over 1986–2005 under the RCP4.5 scenario, with the corresponding projection for the global mean rise being approximately 180 mm.

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(vi) Tropical Cyclones

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- There has been a significant reduction in the annual frequency of tropical cyclones over the NIO basin since the middle of the twentieth century (1951–2018).
- In contrast, the frequency of **very severe cyclonic storms (VSCSs)** during the post-monsoon season has increased significantly (+1 event per decade) during the last two decades (2000–2018).
- However, a clear signal of anthropogenic warming on these trends has not yet emerged.
- Climate models project a rise in the intensity of tropical cyclones in the NIO basin during the twenty-first century.
- Severe cyclonic storms in the Arabian Sea are gaining more strength and the trend is projected to continue.
- The number of extremely severe cyclonic storms formed in the Arabian Sea has increased in 20 years.

(vii) Impact of Climate Change on Indian Agriculture

- Climate change affects all three aspects of food security: availability, access and absorption.
- When production decreases, the availability of food decreases.
- An increase in temperature and CO_2 can prove to be extremely harmful in the long run.
- Changing atmospheric temperature directly and indirectly affects nutrient levels in soil, soil moisture content, drought, depleting groundwater table reducing water availability.
- More frequently occurring droughts and floods could pose threat for farmers and governments.
- On the other hand, changing temperature also changes behavioural pattern in fishes and animals, which would damage the ecological balance and affect the economy since many fishermen are dependent on fishing for their livelihood.
- Climate change has the highest impact on the poor as they don’t have any income for food, which affects their health and absorption.

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- Every year, there is nearly 4–9% climatic impact on agriculture. The contribution of agriculture to the country's GDP is nearly 15% and there is about 1.5% GDP loss due to climate change.
- Rice, wheat, maize and sorghum are the worst hit by this phenomenon.
- By 2030, rice and wheat are likely to see about 6–10% decrease in yields.
- Crops like potatoes, soybean, chickpea and mustard, on which climate change will have a neutral or positive impact.
- Crops like soybean and gram are likely to benefit from higher level of CO_2 in the atmosphere, which helps in CO_2 fertilisation.
- But the positive effects are unlikely to last more than 10–15 years.
- In northern India, especially in Punjab and Haryana, there will be a neutral-to-positive impact on mustard, due to very low winter temperature.
- There will be a very lesser production impact on nearly 1°C rise in temperature. Nonetheless, there will be a negative impact on similar temperature rise in eastern and central India.
- Due to increased CO_2 concentration, there will be a positive impact on potato production which will increase its yield by 11.12% at increased 550 PPM of CO_2 and 1°C rise in temperature. On the other hand, by 2050, a further rise in CO_2 level with a 3°C rise in temperature results in potato production decline by 13.72%.
- Variability in rainfall has more effect on Kharif crops whereas minimum temperature on Rabi crops.
- Due to heat stress with 1°C temperature rise is likely to have a negative impact in Rabi season, resulting in the 4 metric tonnes (MT) loss of wheat. On the contrary, there will be a positive impact on legumes due to the high CO_2 atmospheric level.

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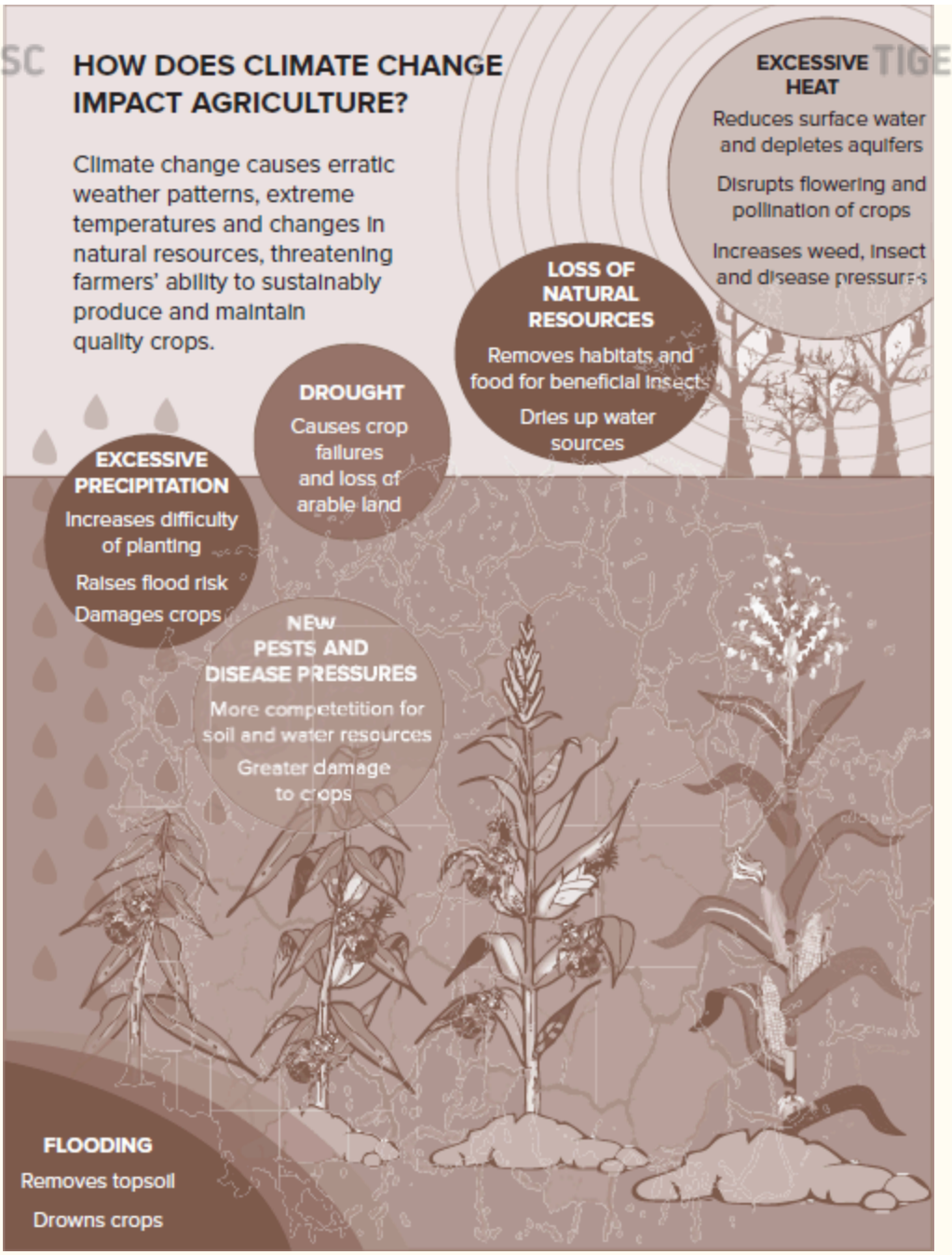

Figure 7.1 Impact of Climate Change on Agriculture

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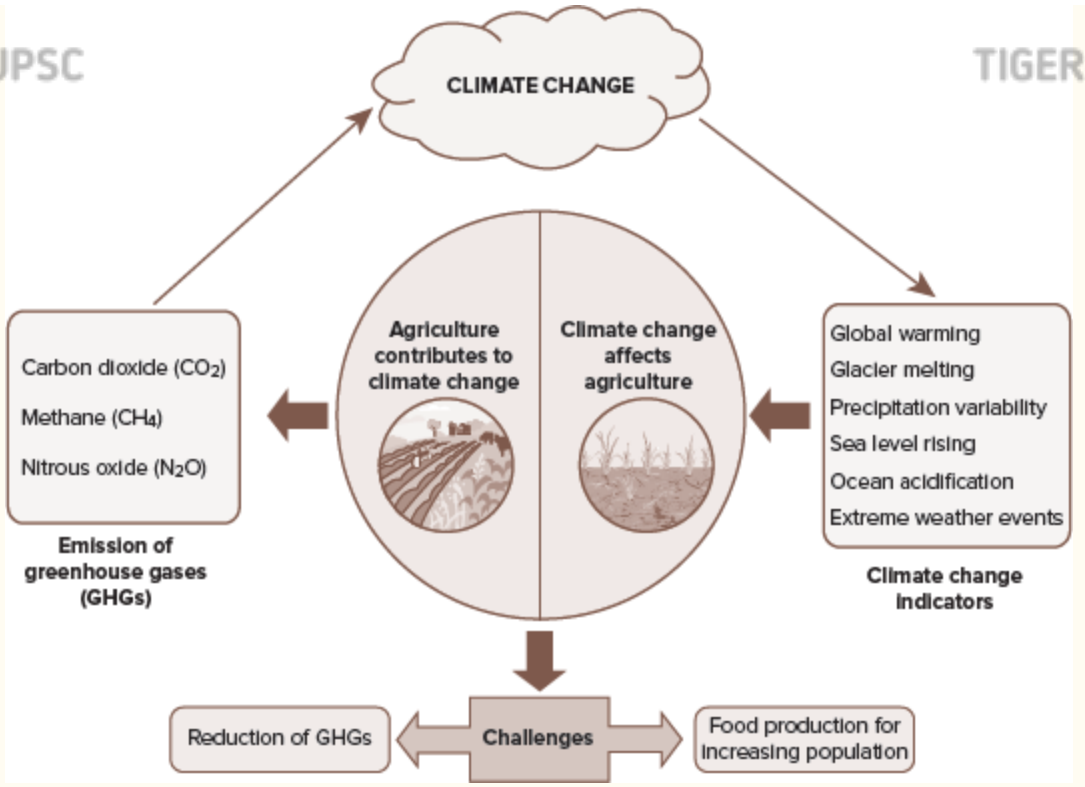

Figure 7.2 Climate Change Indicator & Its Effect on Agriculture

ENV 05-Bioluminescence

Climate-Smart Agriculture (CSA)

Climate-Smart Agriculture (CSA) is a method of managing landscapes like cropland, livestock, forests, and fisheries to tackle the challenges of food security and climate change. It aims to achieve three goals at the same time:

1. **Increased productivity:** Improve nutrition and increase incomes, especially for rural poor who rely on agriculture for their livelihoods. This can be done by producing more and better-quality food.

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2. **Enhanced resilience:** Build the ability to adapt to long-term stresses like shorter growing seasons, unpredictable weather, droughts, pests, diseases, and other climate-related risks.
3. **Reduced emissions:** Decrease greenhouse gas emissions per unit of food produced, prevent deforestation caused by agriculture, and encourage practices that improve carbon storage.

- The World Bank Group (WBG) has steadily increased support for climate-smart agriculture through its Climate Change Action Plan (2021–2025), focusing on the same three goals of productivity, resilience, and emission reduction.
- A significant portion of World Bank agricultural funding now goes toward projects for climate adaptation and mitigation.
- The Maharashtra Project on Climate Resilient Agriculture is one of the largest CSA investments at US$420 million. It aims to strengthen farm resilience and improve climate outcomes.
- CSA practices have been implemented in thousands of projects, benefiting large areas through better irrigation and drainage technologies that promote more efficient and climate-resilient land use.

Climate Smart Village (CSV)

- The 'Climate-Smart Village' (CSV) approach in India is a part of a project led by an international research programme **'Climate Change, Agriculture and Food Security' (CCAFS)**. The project of CCAFS is carried out under Consultative Group on International Agricultural Research (CGIAR) headquartered in **Montpellier, France**.
- **CSV** project was launched in 2011 with 15 climate-smart villages in West Africa, East Africa and South Asia. Other villages are now being chosen in Latin America and Southeast Asia. All the villages are in high-risk areas, likely suffering most from a changing climate. Villages are also locations where partners have already established vital links with local communities.
- CCAFS CSVs have successfully combined global knowledge with local action to help farmers sustainably produce more food, while curbing greenhouse gas emissions and increasing resilience to climate change. In CSVs, farmers and researchers test and implement portfolios of climate-smart agricultural practices, technologies, and services, which can be combined together to make the best out of an increasingly difficult situation.
- In the Indian state of Haryana, farmers have implemented climate-smart practices such as laser-land levelling and alternate wetting and drying of rice, reducing water use, improving soil health and bringing economic rewards. Farmers have also been receiving agro-advisories on their mobile phones, with inputs from met departments, scientists, input dealers and farmers, which allow them to make timely decisions.

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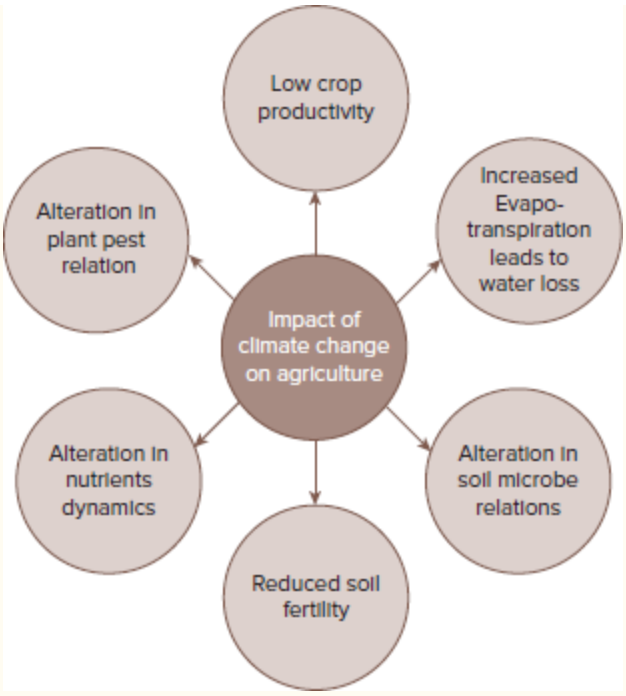

Figure 7.3 Schematic View of Impact of Climate Change on Agriculture

Do You Know?

CCAFS **brings together the world's best researchers in agricultural science, climate science, environmental, and social sciences to identify and address the most important interactions, synergies, and trade-offs between climate change and agriculture**. CCAFS is a partner of Future Earth which is **an interdisciplinary research programme bringing together natural and social sciences**, as well as the humanities, engineering and law, and is focused on designing and producing research together with stakeholders from outside the scientific community.

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Figure 7.4 Climate-Smart Agriculture advocates use of farmers' local knowledge to ensure easy adoption.

7.4.2 Impact of Climate Change on Oceans

- The oceans face serious, long-term, and far-reaching negative consequences due to climate change.
- The three major examples of human activities through which billions of tons of CO_2 and other heat-trapping gases are released every year into the atmosphere are fossil fuel burning, livestock raising, and forest clearing which makes our planet warmer.
- By absorbing 25% and 90% of excess CO_2 and heat, the ocean has shielded us from the climatic change worst impacts. But the same has come at a high cost for marine ecosystems and their harmful impacts such as:
    1. **Increasing ocean temperature: leads to a** decline in ocean wildlife, bleaches coral reefs, and shifts the living of fish.
    2. **Ocean acidification:** results in carbonate ions reduction, important for shell-forming animals such as oysters, crabs, and shrimp.
    3. **Decreasing oxygen:** chokes marine animals, reduces their habitats, and forces them to places which have more vulnerability to predators.
    4. **More intense tropical storms and higher sea level:** destroys the coastal communities and wetland habitats consisting of mangroves and salt marshes.

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- To decrease the climate change effects and protect the ocean, two major actions are required:
    1. Reducing of net carbon dioxide emissions to zero by 2050;
    2. Increasing ocean health through pollution and overfishing reduction and protecting critical marine and coastal habitats.

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■ *What is Ocean Acidification?*

- It refers to the ocean pH reduction over an extended period, as a result of caused carbon dioxide intake (CO_2) from the atmosphere.
- This is due to pollution which is particularly caused by human action.
- The oceanic ecosystems start to get affected more due to the increase in ocean acidification.

What is the Ocean's pH?

- The oceans have a pH similar to other fluids, i.e., a hydrogen potential.
- The pH measures the hydrons activity in the fluid, i.e., the liquid's acidity and basicity degree.
- Lesser pH points towards more acidic fluid and higher pH refers to more alkaline fluid.
- The oceans mean pH is normally 8.25: which allows optimal development of the marine ecosystem.
- The major problem is the decreasing of oceans pH over the years which is currently around 8.14.

■ *There are numerous causes of oceans' acidification. The three major ones are:*

1. the ocean's excessive absorption of CO_2
2. the ocean's absorption of nitrogen compounds
3. the ocean's absorption of sulphur compounds

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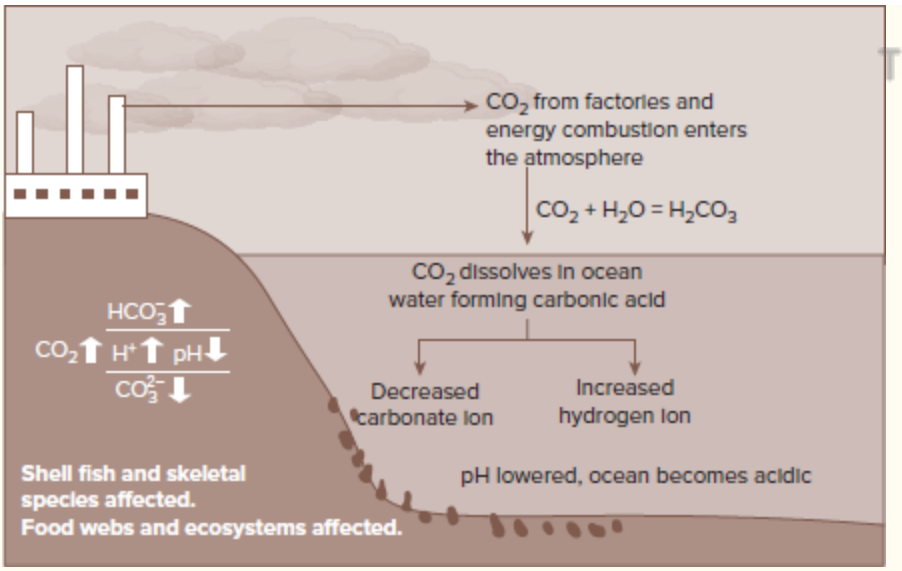

Figure 7.5 Pictorial & Schematic View of Ocean Acidification

Nitrogen Compounds' Role in Ocean Acidification

- Human activities and specifically agricultural activities contribute to the huge quantities emission of nitrogen compounds into the atmosphere, particularly by N_2O (nitrous oxide).
- On dissolving these nitrogenous compounds in the ocean, they have acidifying properties and lead to ocean's global acidification.
- In comparison to the acidification as a result of high CO_2 emissions, that produced by the nitrogen cycle is rather less. Still, the same has become the subject of concern, especially near coastlines where agricultural activities are important.

Consequences

- The atmospheric carbon dioxide (CO_2) concentration has increased due to the burning of fossil fuels and changes in land usage since the industrial revolution.
- The 30% atmospheric CO_2 is absorbed by the ocean, and this results in its increasing level in the ocean.
- There is a chain of chemical reaction of CO_2 being absorbed by seawater, resulting in increased hydrogen ions concentration, thereby increasing acidity in seawater and thus less abundant carbonate ions.

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- Marine structures like sea shells and skeletons are made up of carbonate ions and there decreasing level in the ocean makes building of shells difficult for calcifying organisms such as oysters, clams, sea urchins, shallow water corals, deep sea corals, and calcareous plankton and further effects the behaviour of non-calcifying organisms as well.
- Specific fish's ability in predator detection is declined in more acidic waters, thereby risking the entire food web.
- The world's oceans are getting affected by ocean acidification including coastal estuaries and waterways.
- Fish and shellfish are the source of dependency for many world economies and people as they are staple food from the ocean and is a primary source of protein.

7.4.3 Marine Heatwaves

- There is an occurrence of heatwaves in the atmosphere and our familiarity with the extended periods of excessively hot weather.
- On the other hand, there is also the occurrence of heatwaves in the ocean and termed as marine heatwaves, or MHWs.
- Marine heatwaves (MHWs) are periods of extremely warm sea surface temperature that persist for days to months and can extend up to thousands of kilometres.
- When ocean temperatures are extremely warm for an extended period of time, these marine heatwaves can have significant impacts on marine ecosystems and industries.
- Marine heatwaves can occur in summer or winter – they are defined based on differences with expected temperatures for the location and time of year.
- Annual MHW days have increased substantially since the early twentieth century.
- MHWs have already become longer-lasting and more frequent, extensive and intense in the past few decades, and this trend will accelerate with further global warming.

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- Between 1982 and 2016, we detect a doubling in the number of MHW days, and this number is projected to further increase on average by a factor of 16 for global warming of 1.5°C relative to preindustrial levels and by a factor of 23 for global warming of 2.0°C.
- This number is projected to grow even more as the planet continues to get warm, with MHW days extending for most of the year.
- Warm ocean waters strengthen tropical storms and hurricanes, and this year's warm Atlantic ocean temperatures are fuelling one of the most active starts to a hurricane season on record.
- Sudden environmental extremes can have stronger impacts on species than the slower changes over longer periods, both on land and marine environments.

Marine Heatwave Properties

- On defining the marine heatwaves, each event can be associated with a set of properties including
  - duration (start to end date time, in days),
  - intensity (the temperature anomaly above the climatology, in degree Celsius (°C), defining the intensity of being warmer than expected for that mean time and this being a maximum value or an average over the whole event), and
  - cumulative intensity (the combined intensity over the duration of the event, similar to 'degree heating days' or 'degree heating weeks').
- The annual frequency can be defined over the entire year (count of events in a particular year) and the annual marine heatwave days (the count of days in a year that was part of a marine heatwave).

What Causes Marine Heatwaves?

- A whole range of factors can cause marine heatwaves, and not all factors are important for each event.
- The most common drivers of marine heatwaves include ocean currents which can build up warm water and air-sea heat flux, or warming through the ocean surface from the atmosphere.

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- Winds can enhance or suppress the warming in a marine heatwave, and climate models like El Niño can change the likelihood of events occurring in certain regions.

Impacts and Implications of Marine Heatwaves?

- The marine heat waves affect the oceanic ecosystem structure, thereby supporting certain and suppressing other species.
- For example, in western Austria, the fish communities had a much more ‘tropical’ nature and have switched from kelp forests to seaweed turfs after the 2011 marine heatwave.
- The particular species’ habitat ranges can be changed by the Marine heatwaves like the spiny sea urchin off southeastern Australia which has been expanding southward into Tasmania at the expense of kelp forests which it feeds upon.
- There will be an impact on aquaculture and fisheries by the Marine heatwaves, thereby resulting in economic losses.
- There is a drastic effect on biodiversity by marine heatwaves.
- In the year 2016, there was intense bleaching of the Great Barrier Reef due to marine heatwaves across northern Australia and is theoretically linked to the perishing of mangrove in the Gulf of Carpentaria.
- The theoretical marine heatwaves impact include large-scale fish stock location shifting, kelp forests collapsing, coral bleaching, and unprecedented sea birds perishing.

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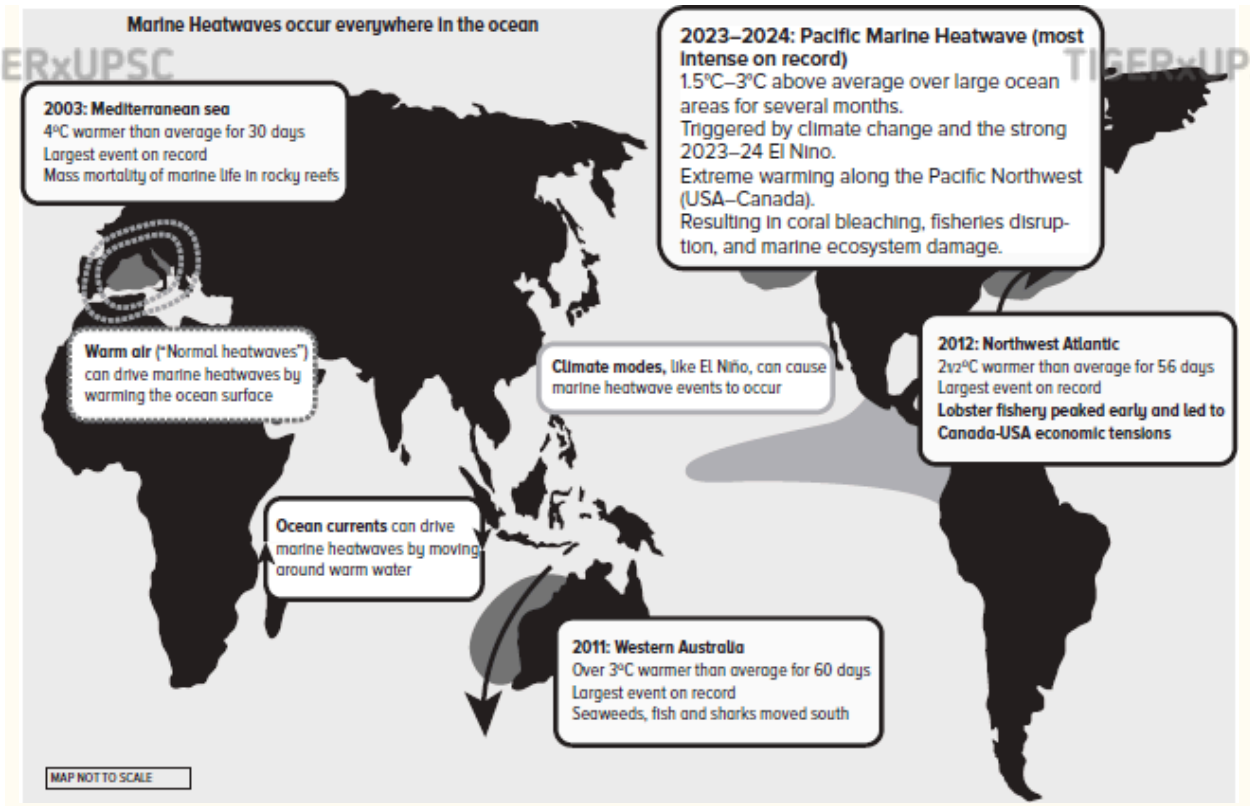

Figure 7.6 Occurrence of Marine Heatwaves in the Ocean

7.5 GREENHOUSE EFFECT

- Accumulation of greenhouse gases in excess like water vapour, CO_2, O_3, CH_4 and chlorofluorocarbons (CFC) is mainly responsible for disrupting the balance of energy and therefore, increasing the Earth's temperature.
- Fossil fuel-based power plants and industries and standards of life are responsible for releasing such gases in the atmosphere.
- These greenhouse gases trap the incoming infrared rays from the sunlight in the atmosphere and help them penetrate into the Earth's surface.
- But they also prevent the sunlight from reflecting back into the space.
- This leads to an increase in the Earth's temperature. This phenomenon is known as the greenhouse effect.

Do You Know?

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The greenhouse effect occurs when greenhouse gases, CO_2, methane and nitrous oxide reflect excess infrared radiation (which is an extra radiation that greenhouse gases reflect) back on the Earth which provides that extra heat on the Earth. Infrared radiation also remains trapped inside the atmosphere and cannot escape it. This finally leads to global warming.

How do Atmospheric conditions affect Global warming?

Without an atmosphere to trap heat, the Earth's average temperature would be around 18°C (0°F), making the planet largely frozen and uninhabitable. The moon, which has no atmosphere, experiences extreme temperature swings between scorching days and freezing nights.

The atmosphere acts as an insulating blanket, slowing the escape of heat radiated from the Earth's surface. This natural process maintains the average temperature at a life-sustaining 15°C (59°F).

Certain trace gases, known as greenhouse gases (including carbon dioxide, water vapour, and methane, etc.), efficiently absorb and re-emit infrared radiation (heat). This property is central to the greenhouse effect that regulates the planet's climate.

7.5.1 Global Warming

- The gradual rise in the Earth's temperature due to excessive emission of greenhouse gases in the atmosphere causes global warming.
- In 1824, Joseph Fourier discovered this greenhouse effect which was first experimented in 1858 by John Tyndall and lastly reported in 1896 by Svante Arrhenius.
- Global warming is nothing but solely the consequence of the greenhouse effect.

7.5.2 Mechanism of Global Warming

The mechanism of global warming involves several steps related to the Earth's energy balance.

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Incoming Solar Radiation: Energy from the Sun reaches Earth as shortwave radiation, mostly visible light. About half of this energy passes through the atmosphere and reaches the surface, while the rest is either reflected or absorbed by clouds and the atmosphere.

Absorption and Re-emission: The Earth's surface, including land and oceans, absorbs this solar energy and warms up. The heat is then radiated back toward the atmosphere as longwave infrared radiation.

The Natural Greenhouse Effect: Naturally occurring greenhouse gases, like water vapor, carbon dioxide (CO_2), methane, and nitrous oxide, absorb some of the outgoing infrared radiation. This process of absorbing and re-emitting heat in all directions, including back toward the surface, slows the escape of heat into space. It helps maintain a habitable average temperature of about 15°C (59°F).

The Enhanced Greenhouse Effect: Since the Industrial Revolution, human activities, mainly the burning of fossil fuels (coal, oil, and natural gas), deforestation, and some agricultural and industrial practices, have greatly increased the concentration of these greenhouse gases in the atmosphere.

Increased Heat Trapping: This rise in greenhouse gases acts like a thicker insulating blanket. It absorbs more outgoing infrared radiation and re-emits even more heat back toward the surface and lower atmosphere.

Energy Imbalance and Warming: The extra heat trapping causes an energy imbalance. More energy enters the Earth's system than escapes into space. To restore this balance, the Earth's surface and lower atmosphere must warm up until the outgoing energy equals the incoming energy. This ongoing warming is what we call global warming.

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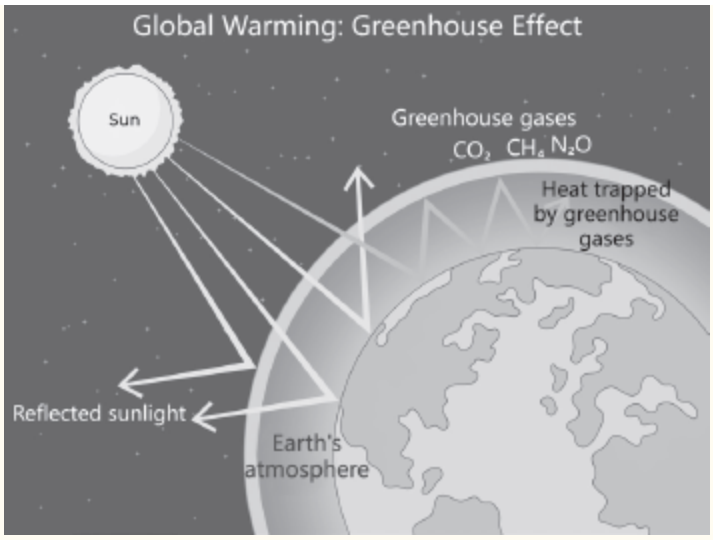

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Do You Know?

The atmosphere gets heated more by terrestrial radiation, which is long-wave radiation from the Earth's surface, than by incoming solar radiation. Most incoming solar radiation goes through the atmosphere and warms the Earth's surface. The surface then radiates heat back as long-wave radiation. Greenhouse gases, such as carbon dioxide, absorb this radiation and warm the atmosphere.

7.5.3 Greenhouse Gases

1. Carbon dioxide (CO_2)
2. Methane (CH_4)
3. Nitrogen oxide (nitrous oxide)
4. Chlorofluorocarbons (CFCs)
5. Water vapour
6. Ozone (O_3)
7. Aerosols

Do You Know?

Many greenhouse gases occur naturally such as water vapour, carbon dioxide, methane, nitrous oxide and ozone. While other gases like hydrofluoro carbons (HFCs),

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perfluoro carbons (PFCs) and sulphur hexafluoride (SF_6) result exclusively from human industrial processes.

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7.5.4 Sources of Greenhouse Gases

Sources of PFCs

- Semiconductors, equipment of refrigerator and aluminium produce.

Sources of SF_6

- These include electricity sector, **circuit breaker and gas-insulated switchgear, transmission lines of high voltage and mini-stations**.
  - **(i)** Magnesium industry-moulding and dye casting in order to prevent molten magnesium oxidation.
  - **(ii)** SF_6 is usually used in **semi-conductor manufacturing processes in electrical industries**.
  - **(iii)** Application of **adiabatic property in tennis balls, shoe soles,** and in **truck tyres**.
  - **(iv)** It is also used in **leak detectors**, for **medical purposes, electronic appliances, sound proofing walls and windows, degassing aluminum specialties** etc.

Sources of HFCs

- Foam insulation manufacturing releases HFCs. It is also used in **residential and commercial refrigerators and air conditioners**.

Sources of Carbon Dioxide

- Combustion of fossil fuels like coal, oil and petroleum, etc., contributes major CO_2 emission along with other sources like oxidation of carbon compounds in marshy areas, and forests. Cement manufacturing industries also release carbon dioxide into the atmosphere.
- Human activities release GHG which contributes about 50% to global warming. Industrial countries contribute around 76% of annual emissions.

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- The chief sources are fossil fuel burning (67%) and deforestation, land clearing and burning (33%).
- Carbon dioxide has persisted in the atmosphere for about 500 years. Carbon dioxide concentration in the atmosphere was 355 ppm in 1990 that has been on the rise ever since and has been increasing at a rate of 1.5 ppm every year.

ENV 21- Confusion with methane in UPSC Exam

Sources of Methane

- Methane traps heat five times more than carbon dioxide does. There are various methane sources including **natural wetlands (21.3%), paddy fields (20.4%), biomass (10.2%), enteric fermentation in ruminant cattle (14.8%), termites (7.4%), landfills (7.4%), drilling (8.3%), coal mines (6.5%), ocean (1.9%) and other sources (1.8%)**.
- Methane is released when bacteria break down organic matter in presence of moisture in anaerobic condition such as paddy fields, swamps, natural wetlands, landfills and ruminant cattle, sheep and termites.
- Production and usage of oil and natural gas, incomplete combustion of organic wastes are major sources of methane which persist in the atmosphere for 7–10 years maximum.
- One molecule of CH_4 has the capability to trap about 25 times more heat than CO_2 molecule. Concentration of methane in atmosphere is about 1.675 ppm and it has been on the rise at a rate of 1% each year.

Do You Know?

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Methane due to oxidation, gives rise to carbon dioxide and water vapour; both are greenhouse gases.

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Chlorofluorocarbons (CFCs)

- 15–20% of global warming has been recorded to be caused by CFCs. CFCs also play a major role in depleting ozone layer. **Leakage in air conditioners and refrigerators, several industries that release several solvents, plastic foams, aerosols, propellants** and **many more,** also contribute to CFCs.
- It takes 10–15 years for CFCs to reach the stratospheric layer of atmosphere and usually trap 1500 to 7000 times more heat per molecule than CO_2 while they pass through tropospheric layer of atmosphere.
- Concentration of CFC in atmosphere is about 0.00225 ppm and it has been on the rise at a rate of 0.5% each year.

Nitrous Oxide (N_2O)

- N_2O is generally 230 times more efficient than CO_2 in creating global warming. **N_2O also plays a major role in depleting ozone layer**. At least 6% of the total global warming is contributed by N_2O which is produced by **combustion of forest, grassland, livestock waste and other biomass or organic wastes, mainly nitrogen-rich fuel like coal including emission from nitrogen contaminated groundwater, ocean, and nitrogenous fertilisers that are heavily used in agriculture**.
- In the troposphere, it lingers on for around 140–190 years and can trap heat 230 times more than CO_2 per molecule. Concentration of N_2O in the atmosphere is around 0.3 ppm and has been on the rise at a rate of 0.2% each year.

Do You Know?

Paddies are a potential **source of anthropogenic nitrous oxide (N_2O) emission as well.** In paddies, both the **soil and the rice plants emit N_2O** into the atmosphere. The

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rice plant in the paddy is considered to act as a channel between the soil and the atmosphere for N_2O emission.

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Water Vapour

- Water vapor is an important greenhouse gas that helps keep the Earth warm by absorbing long-wave radiation. The amount of water vapor in the atmosphere mainly depends on temperature, pressure, and evaporation sources.
- Water vapor is most abundant in warm tropical and subtropical regions and least common in polar areas because warm air holds more moisture and supports higher evaporation. As altitude increases, water vapor decreases since cooler, less-dense air holds less moisture, gas particles disperse more easily, and evaporation rates fall due to lower temperatures and greater distance from moisture sources.

Table 7.1: Atmospheric concentration of GHGs (Most to least) or effectiveness in trapping heat

S.No.	Gas	Formula	Spec
1	Water vapor	H_2O	The n
2	Carbon Dioxide	CO_2	A maj huma
3	Methane	CH_4	-
4	Nitrous Oxide	N_2O	-
5	Ozone	O_3	-
6	Fluorinated	Hydrofluorocarbons (HFCs),	These

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Table 7.2: List of Countries of CO_2 emissions

S.No.	Country	% of Glo
1	China	34%
2	United States	12%
3	India	7.6%
4	European union	6.4%
5	Russia	5.3%
6	Japan	2.4%
7	Iran	2.0%

Table 7.3: Per capita CO_2 emissions by country/ territory

S.No.	Country	Emission per c
1	Palau	62.59
2	Qatar	43.55
3	Kuwait	24.90
4	Brunei	21.12
5	New Caledonia	20.90
6	Bahrain	20.70

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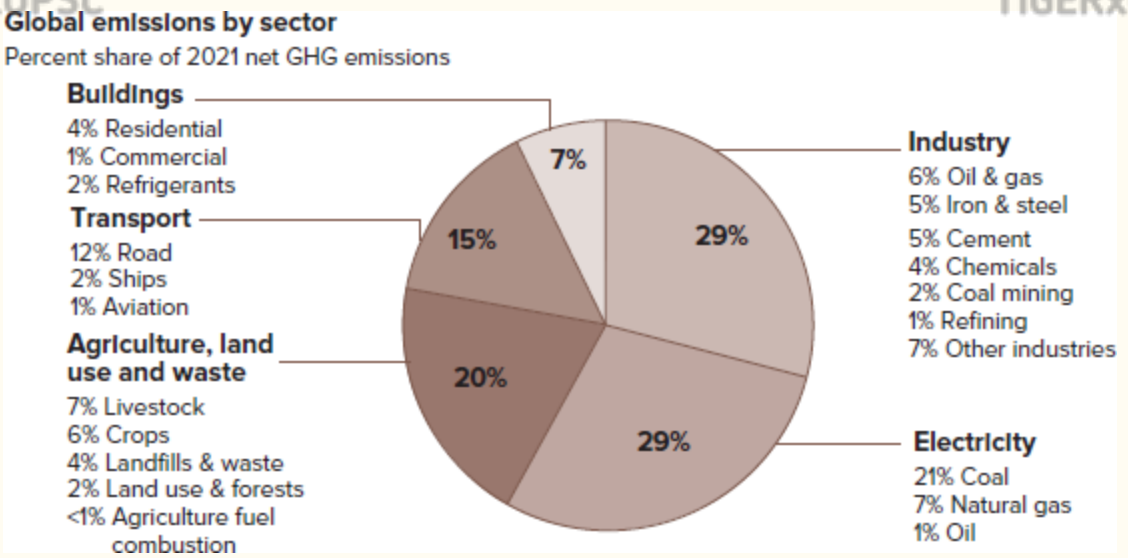

Figure 7.7 Annual greenhouse gas emissions by various sectors

ENV 22- Global Warming Potential (GWP)

7.6 GLOBAL WARMING POTENTIAL (GWP)

- GWP is used to measure heat generation by greenhouse gases trapped in the atmosphere.
- GWP is used to study the comparison of the global warming impacts of different gases.
- It is a measure of the quantity of energy emitted in 1 tonne of a gas absorbed over a given period of time, in comparison to the 1 ton carbon dioxide (CO_2) emissions. Kindly follow table 7.4.

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7.7 IMPACT OF GREENHOUSE EFFECT

7.7.1 Climate Change

- **Melting of glaciers** due to global warming has been of great concern now because rising water level and exposure to darker surface on the Earth would eventually lead to the rise in temperature.
- Rising ocean temperature has also been proven to have ill-effects on corals. In 2022, coral bleaching has been recorded in the Great Barrier Reef for two successive years.

7.7.2 Impact on Forest

- Species distribution of trees in forests has been affected badly due to an increase in temperature.

7.7.3 Impact on Agriculture

- Agro ecosystem has been affected badly through following ways:
  1. Increase in severe evaporation from plants and soil leading to decline in plant survival rate, the higher expense for irrigation, and decline in soil quality.
  2. Decline in soil moisture because of excessive evaporation and transpiration.
  3. There is an increase in insects, diseases and weeds. The activities of microorganisms also increased. The decomposition of organic matter will be promoted.

7.7.4 Human Diseases

- Heatstroke is the major cause of death during summers in North India. Tropical countries are the major victims of malaria, dengue and filariasis because high temperature and rainfall create favourable breeding conditions for vectors like flies and mosquitoes.

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7.8 PROJECTIONS FOR THE FUTURE

7.8.1 Increase in CO_2

- In 1994, the atmospheric concentration of CO_2 was 359 ppm.
- It has been estimated that when CO_2 level reaches up to 400 ppm, the global temperature will go up by 1°C.
- If this trend persists, the level of CO_2 in the atmosphere is expected to grow 600 ppm by 2050.

7.8.2 Increase in Methane

- **Methane atmospheric concentration** is rising at the rate of 1 to 1.3% per year. A rapid increase of methane in the air after 1950 was observed due to industrialisation and commercial rearing of livestocks.
- Livestock and organic waste release methane everywhere.

Table 7.4 Main Greenhouse Gases and their #GWP

Greenhouse Gas	Formula	100-year GWP (AR5)
Carbon Dioxide	CO_2	1 (reference)
Methane	CH_4	28
Nitrous Oxide	N_2O	265
Hydrofluorocarbons	HFCs	Varies (e.g. HFC-134a: 1,300)
Perfluorocarbons	PFCs	Varies (e.g. CF4 : 6,630)
Sulphur Hexafluoride	SF_6	23,500
Nitrogen Trifluoride	NF_3	16,100

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7.8.3 Increase in Nitrous Oxide

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- **Activities of microorganisms in agricultural fields release** N_2O in the atmosphere. Denitrification is a process where nitrates or nitrites are reduced to gaseous nitrogen (N_2) or nitrogen oxides (NO, N_2O) with the help of soil micro-organisms in presence of O_2.

7.9 MAJOR CAUSES OF INCREASEIN GREENHOUSE GASES

1. **Change in land-use pattern:** With the growing industrialisation, change in land-use pattern was unavoidable. To set up industries and carry out developmental projects, forests are cleared, thus, contributing one-third of total CO_2 emission.
2. **Combustion of fossil fuels:** Combustion of fossil fuels to produce heat for industrial activities emits large volumes of CO_2.
3. **Rise of cattle and dairy farming:** Enteric fermentation of carbohydrates in ruminant cattle releases methane. Cattle manure is also a source of methane gas that are used to make biogas.
4. **Smart usage of fertilisers in agriculture:** Limited and smart usage of nitrogenous fertilisers should be carried out to lower atmospheric concentration of nitrous oxide.
5. **Combustion of natural gas and petroleum products:** Combustion of liquid fuels, gasoline and fuel oils contribute to about 36% of the GHG's emissions. Combustion of natural gas contributes about 20% of emissions.

7.10 STEPS TO REDUCE EMISSION OF GREENHOUSE GASES

1. **Alternative sources of energy:** Solar, wind, and tidal energy which are renewable and nonconventional energy sources, produce about eighteen per cent of the total electricity and are being generated globally.

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2. **Generation of nuclear power:** Higher production of nuclear power should be taken into consideration through effective measures. At present, it contributes only fourteen per cent of the total electricity produced and consumed globally.
3. **Afforestation:** Plantation of more trees should be encouraged, instead of deforestation.
4. **Urban planning:** Occupying agricultural land to carry out non-agricultural purposes should be avoided as much as possible. Environmental protective measures in urban areas like multiple transport choices, green areas and green belts should be implemented.
5. **Building designing:** Usage of ecofriendly construction material should be encouraged.
6. **Controlled burning of biomass:** Establishing biomass-fuelled power plants should be encouraged to reduce GHG emission.
7. **Reduction in the growth of population:** Proper implementation of policies and measures to control population growth should be done.
8. **Reduction in consumerism:** Judicious use of goods along with lesser wastage of resources should be checked.
9. **Mass education about ecology and environment:** Awareness and knowledge should be spread through proactive use of media, workshops, conferences and seminars among the general public regarding the importance of conserving resources and protecting the environment.
10. **Proper development and implementation of policies** along with efficient use of technologies to mitigate the adverse effect of GHG emission. For example, fitting filters to diesel vehicles to capture black carbon, replacing inefficient cooking stoves with cleaner alternatives like LPG/CNG and launching electric vehicles.

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ENV 23- Global Dimming

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7.11 WHAT IS GLOBAL DIMMING?

- It is the gradual reduction in the global direct irradiance quantity at the Earth's surface.
- The concept that was observed during the beginning of systematic measurements in the 1950s.
- The Gerald Stanhill, an English scientist in Israel, first spotted these effects and compared the sunlight records of Israel currently with the 50s and discovered a 22% drop in the sunlight.
- He further carried out his investigation and discovered the same almost everywhere, with sunlight declining by about 10% over the US, 16% in British Isles and by 30% in Soviet Union.
- This shows that there was a substantial decline in the sun's energy quantity reaching the Earth's surface, hence global dimming.
- As per the Gerald Stanhill, there is variation of global dimming from place to place, with the overall, the decline falling from 1% and 2% globally per decade between the 1950s and the 1990s.
- Global dimming is assumed to be the result of rise in particulates such as sulphate aerosols in the atmosphere, which are due to human activities.
- Ironically, global dimming has an opposite effect of global warming as it leads to cooling effects, so as a whole, it is environmentally beneficial, even though it brings about elements of literal darkness on Earth.
- There is a variation of global dimming effects by location, with few areas being worst affected than others. For example, there is substantial fall in Northern Hemisphere with 4% to 8% reduction between 1961 and 1990.
- The Global Dimming issue was first highlighted through a documentary called Horizon by BBC on 15 January 2005 as the different regions noticed the different levels of global dimming.

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- Until, there is a lesser amount of global dimming in Southern Hemisphere which is very small, whereas there is a significant amount in Northern Hemisphere to the tune of 4–8%. The Europe and North America regions noticed partial recovery from dimming, whereas the China and India areas have experienced a rise in global dimming.

■ *Various Causes of Global Dimming*

1. **Aerosols**
   - They are the major cause of global dimming, being a colloid of fine solid particles or liquid droplets in the air and other gases.
   - The aerosols in the atmosphere, mostly scatter sun's light, sending it back to space as well as exerting a cooling influence on the Earth's climate.
   - Sulphate aerosols in the atmosphere result from human activities and interfered with the hydrological cycle by reducing evaporation and rainfall in some areas.
2. **Particulate matter**
   - These are the burning fossil fuels as well as ICE by-products consisting of sulphur dioxide, ash and soot.
   - They directly absorb solar energy and reflect it back into the space, before it reaches the Earth surface.
   - Thus causing the energy and sun's light dimming that reaches the Earth different parts.
3. **Water droplets**
   - Water droplets may consist of airborne particulates such as sulphur dioxide, soot and ash, which create polluted clouds.
   - These clouds consist of a heavier and larger number of droplets than normal clouds, thereby changing the cloud properties, resulting in 'brown clouds'.
   - These clouds lead to global dimming, thereby reflecting light and energy from the sun back into space.
4. **Vapours**

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- They are the result of various sources such as water bodies evaporation.
- However, the said vapour is the result of planes flying high in the sky, termed as contrails.
- They led to global dimming by reflecting heat from the sun back into space.

5. **Wildfires**
   - The severity of wildfires has led to the entire cities staying in dim light for days due to smoke in the atmosphere.
   - Even though clearing away of the smoke, it leads to global dimming as particulate matter and fine solid particles remains in the air thereby causing global warming and global dimming.

Observation of Global Dimming

- Scientists have noted that after the 9/11 attack in the USA, the temperature rose by about 1°C within three days.
- More flights have directly resulted in a higher concentration of aerosols which led to global dimming. Because of this condition, solar infrared radiation gets trapped in the atmosphere.
- It is estimated that if there is 3% reduction in solar radiations annually, complete darkness is inevitable in about 330 years.
- The drought in Sub-Saharan Africa during 1970s and 1980s is considered to be caused by global dimming.
- Cloudy regions adversely affect crop ripening; even water bodies are depleted from getting sufficient energy from the sun, which alters oxygen production and even global rainfall pattern.

Exam Pointer

Difference Between Global Dimming and Global Warming

- Surface and near-surface phenomena are usually known as global dimming, while the entirely atmospheric phenomenon is known as global warming.

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- Research revealed that the effect of global warming is double the effect of global dimming.
- A different form of air pollution other than the type that causes global warming is responsible for the global dimming.

Causes

- Global warming is caused by the increase in the concentration of greenhouse gases like carbon dioxide as a result of the burning of fossil fuels.
- On the other hand, global dimming is mainly the result of aerosols produced as a by-product of burning fossil fuels.
- On a comparative front, both global warming and dimming are related to the burning of fossil fuels by industry and internal combustion engines and will continue to impact human societies if they continue unabated.

Effects on Global Temperature

- As the name suggests, global warming results in warmer temperatures on Earth because of the effect of greenhouse gases.
- In contrast, global dimming results in lower temperatures as the sun's heat, rays and energy is reflected away and does not reach the surface of the Earth.

Effects on Rainfall

- Global warming means the oceans evaporate more and as a result, it leads to more rainfall.
- Global dimming does the opposite as it reduces such evaporation, meaning there is less rainfall.

Effects on the Weather

- Global warming results in more severe weather such as prolonged droughts, severe rainfall and floods, melting ice caps, rising ocean levels and strong winds which could bring about more typhoons and hurricanes.
- Global dimming leads to calmer weather as there is less energy in the water cycle, meaning fewer storms.
- However, both phenomena have been documented to ring about similar changes to the weather like droughts.

Tackling the phenomena

- Global dimming was identified several decades ago and has since been slowed, stopped or possibly been reversed in the 1980s.
- Unfortunately, global warming is a continuing problem and no efforts to stall, slow or end it have since been successful.

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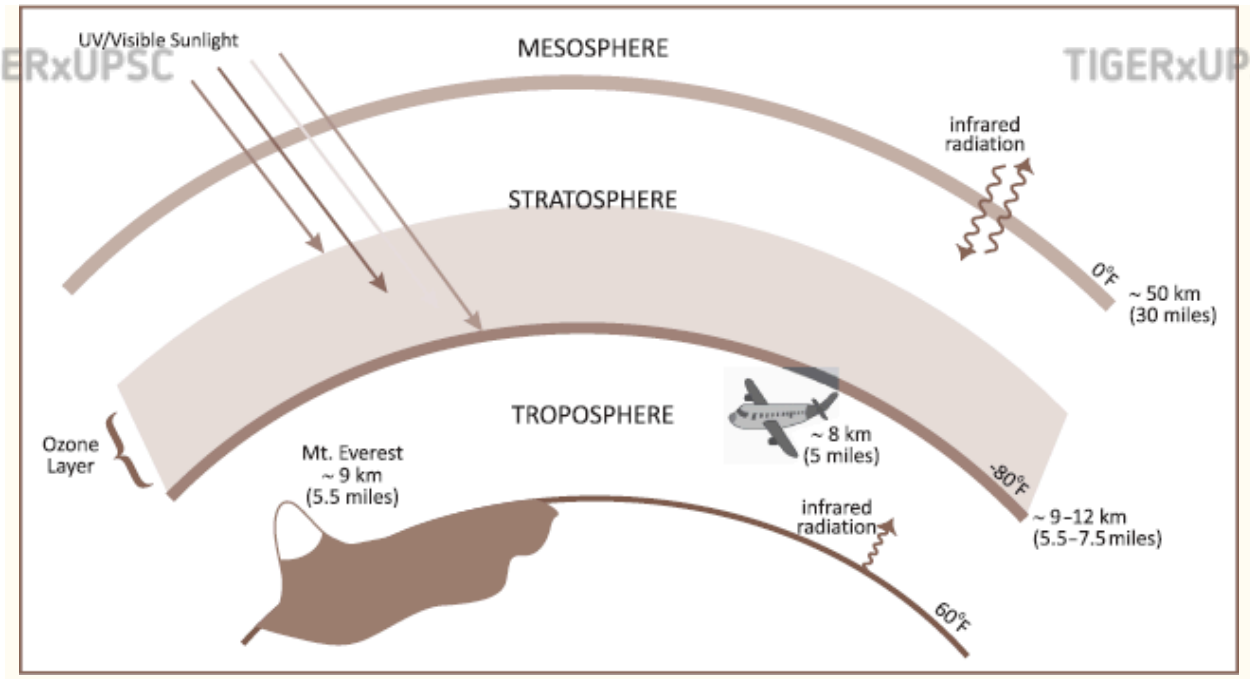

Figure 7.8 Pictorial view of atmosphere, ozone layer and coming radiation effect towards the Earth

7.12 GLOBAL COOLING

- There was a sudden decrease in the atmospheric temperature recorded during the 1970s which resulted in global cooling.
- Scientific community does not support this phenomenon much.
- Two physical mechanisms (aerosol and orbital forcing) were held responsible for the sudden trend in temperature decline from 1940 to the early 1970s.

■ *Aerosol*

- Volcanic eruption injects SO_2 in the stratospheric layer which converts into sulphuric acid and later condenses to form sulphate aerosol.
- We know that aerosols play a big role in reflecting the incoming solar radiation back to the space (that actually increases albedo effect). These sulphate aerosols also mix with rainwater to form acid rain.

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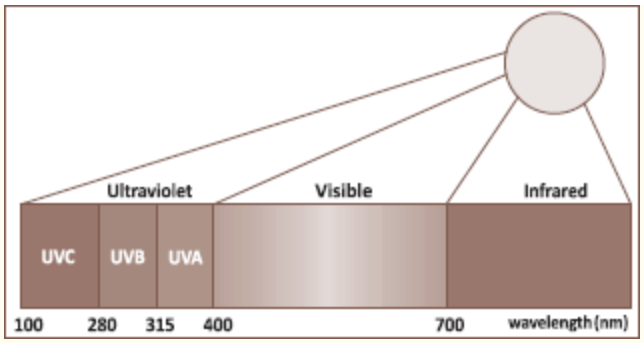

Figure 7.9 Different wavelength of UV radiation

■ *Orbital Forcing*

- A gradual and cyclic alteration in the tilt and shape of the Earth's orbit is believed to be taking place by some scientists. This has a negative impact on the duration and intensity of the global season pattern.

7.13 STUDY OF OZONE LAYER

Atmosphere is composed of different layers:

1. Troposphere
2. Stratosphere
3. Mesosphere
4. Thermosphere
5. Exosphere

The ozone layer is present above the Earth's surface at the height of 15 km to 30 km, in the stratosphere. It has very low density. Ozone layer plays an important role in our life in the following ways:

- Ultraviolet and infrared rays from the sun cannot directly reach the Earth's surface.
- It absorbs the harmful part of the sunlight and protects the life on the Earth from the harmful effects of these rays.
- In 1985, using satellites, balloons and surface stations, a team of researchers discovered a balding patch of ozone in the upper stratosphere, which is equal to the size of the United States, over Antarctica. This is called as ozone hole.

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- It was first discovered by **Dr. J. Foreman, Brain Gardiner** and **J. Shanklin**, a scientist of British Antarctica survey.

Do You Know?

The thickness of ozone layer is measured in 'Dabson unit'. 1 Dabson unit = 0.01 mm thickness of ozone layer.

Ozone

- It is a triatomic form of oxygen (O_3) found in the Earth's upper and lower atmosphere.
- The ozone layer is situated in the stratosphere about 15 to 30 km above the Earth surface.
- It protects living organisms by absorbing harmful ultraviolet radiations (UVB) from the sun.
- The ozone layer is being destroyed by CFCs and other substances.
- Ozone depletion is progressing globally except in the tropical zone.

7.13.1 Stratospheric Ozone and Ultraviolet Radiation (UVR)

- **Ultraviolet radiations are** high-energy electro-magnetic waves emitted from the sun.
- It is made up of wavelength from 100 nm to 400 nm.
- **UV radiations include UV-A the least** dangerous form of UV radiation, with a wavelength ranges from 315 nm to 400 nm.; **UV-B** with a wavelength ranges between 280 nm to 315 nm, and **UV-C** which is the most dangerous with a range between 100 nm to 280 nm. UV-C is unable to reach Earth's surface due to stratospheric ozone's ability to absorb it.

7.13.2 Ozone Cycle

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- The Earth is covered with a thin layer of gases which is known as atmosphere.
- There are several atmospheric layers, out of which one is called stratosphere that extends from 10–50 km above sea level.
- In the stratosphere, the ozone layer is formed that protects the Earth from harmful UV rays known to cause cancer in living organisms on Earth.
- The formation and destruction of ozone in stratosphere is called ozone cycle.

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OZONE FORMATION AND DEPLETION

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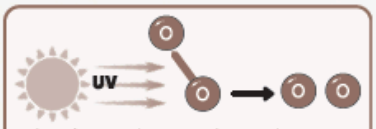
Ozone Formation (The Natural Cycle)

Ozone (O_3) is formed in the stratosphere when oxygen molecules (O_2) interact with high-energy UV radiation.

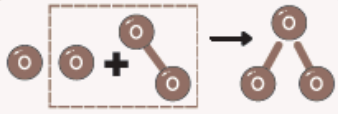


- Stratospheric photolysis (UV-induced O_2 splitting) is the initial step in ozone layer formation (O_3)

Reaction

$O_2 \xrightarrow[\text{(Wavelength <240nm)}]{\text{UV (Photolysis)}} [O] + [O]$

Step 1: Oxygen Dissociation (Photolysis)



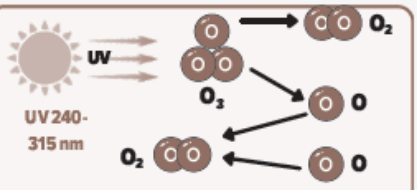
Oxygen atom combine with oxygen gas to form ozone molecule (O_3).

- Oxygen atom (O) collides with an Oxygen molecule (O_2) to form ozone (O_3)

Reaction

$[O] + O_2 \rightarrow O_3$

Step 2: Ozone Creation



- The O atom produced by O_3 breakdown is used to either regenerate O_3 or form molecular oxygen (O_2)

Reaction

Formation: $[O] + [O] \rightarrow O_2$

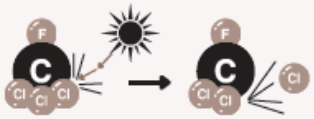
Regeneration: $[O] + O_2 \rightarrow O_3$

(Continuous Regeneration)

Step 3: Natural Breakdown and Regeneration

Ozone Destruction (The CFC Threat)

Ozone is a highly reactive molecule that readily reacts with chlorine atoms . The destruction process is a catalytic reaction caused by Chlorofluorocarbons (CFCs).



- Stratospheric photolysis breaks $CFCl_3$ (ODS) with UV light, releasing highly reactive Chlorine atoms (Cl)

Reaction

$CFCl \xrightarrow{\text{UV (Photolysis)}} Cl$

$CFCl_3 + \text{UV light} \rightarrow CFCl_2 + Cl$

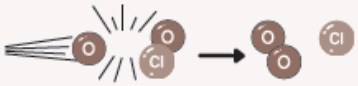
Step 1: Chlorine Generation

- Chlorine atom (Cl) destroys ozone (O_3), forming Chlorine monoxide (ClO) and molecular oxygen (O_2)

Reaction

$Cl + O_3 \rightarrow ClO + O_2$

Step 2: First Ozone Attack



A free [O] atom hits ClO to make O_2 Gas and a free Cl atom.

Reaction : $[O] + ClO \rightarrow O_2 + Cl$

- Chlorine monoxide (ClO) and ozone (O_3) react to form two oxygen molecules (O_2), freeing the chlorine atom (Cl) to continue the cycle

Reaction $ClO + O_3 \longrightarrow Cl + 2O_2$

Step 3: Cycle Continuation

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ENV 24- Ozone Formation and Breakdown

Ozone Formation & Destruction

Step 1:

- It is formed in the stratosphere when oxygen molecules dissociate after absorbing an ultraviolet photon whose wavelength is shorter than 240 nm.
- In stratosphere, short wavelength, high energy radiating UV rays strike an oxygen molecule and breaks it into two oxygen atoms (process is called photolysis).

$$O_2 \xrightarrow{\text{UV (photolysis)}} [O] + [O]$$

Step 2:

- When this oxygen atom collides with an oxygen molecule, an ozone (O) molecule is formed.

$$[O] + O_2 \rightarrow O_3$$

Step 3:

- Now when solar radiation of UV rays strike an O_3 molecule, it breaks down into O_2 and an oxygen atom. This oxygen atom can now again either bind with another oxygen atom to form O_2 or bind with an O_2 to form O_3

$$[O] + [O] \rightarrow O_2$$

$$[O] + O_2 \rightarrow O_3$$

How Does CFC Deplete Ozone

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- Chlorofluorocarbons or CFCs, an anthropogenic gas used for propellants in various sprays (like hair spray); also used as coolant in refrigerators and air conditioners.
- In 1970, **Dr. P. Crutzen** discovered that catalytic reaction is responsible for destroying O_3 (ozone) in the stratosphere.
- Ozone is a highly reactive molecule due to which it readily reacts with hydrogen, nitrogen, bromine and chlorine atoms.
- When CFC molecule reaches up to stratosphere, UV rays interact with CFC molecule and releases chlorine atom in stratospheric atmosphere. Here, chlorine atom collides with an ozone molecule in stratosphere to generate **chlorine monoxide (ClO)** leaving behind an O_2 molecule.
- Now this chlorine monoxide (ClO) collides with a nascent oxygen to form an O_2 molecule and leave behind a chlorine atom, free to bind with another ozone molecule.
- **A single chlorine atom can break more than 1,00,000 ozone molecules in the coming 6–7 years.** The reaction is given below:

$$Cl + O_3 \rightarrow ClO + O_2$$
$$ClO + [O_3] \rightarrow 2O_2 + Cl$$

- This formation and dissociation of ozone molecules is constantly carried out in the stratosphere which releases thermal energy, which heats up the stratosphere and creates a layer of shield that absorbs UV radiations and blocks the entry of UV rays on the Earth's surface.
- In 1987, **Montreal Protocol** was held in Canada when the reason behind ozone depletion was discovered. Twenty-three nations involved promised to reduce the usage of CFC by 50% by the year 1999.

Do You Know?

The ozone layer is on track to recover within four decades, with the global phaseout of ozone-depleting chemicals already benefitting efforts to mitigate climate change.

It has been estimated that if current situation continues, the ozone hole might recover upto 98% by 2060 or maximum 2080.

This is definitely great news and evidence to prove Montreal Protocol's success.

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7.13.3 Causes of Depletion

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- Chlorofluorocarbons (CFCs), nitrogen oxides (from nitrogenous fertilisers) and hydrocarbons are the major sources of ozone depletion.
- **CFCs are widely utilised as coolants in air conditioners and refrigerators; also used as cleaning solvents, as aerosol propellants; in foam insulation and also in fire extinguishing equipment.**
- CFCs escape as aerosols and reach up to the stratosphere.
- The supersonic aircraft that fly high up in the atmosphere's stratospheric layer disturb the ozone layer.
- A number of free radical catalysts can destroy ozone, the most important are the **hydroxyl radical (OH•), nitric oxide radical (NO•), chlorine radical (Cl•) and bromine radical (Br•)**.
- All of these have both natural and man-made sources; at the present time, **most of the OH• and NO• radicals in the stratosphere are naturally occurring, but human activity has drastically increased chlorine and bromine levels**.
- These elements are found in stable organic compounds, especially chlorofluorocarbons, which can travel to the stratosphere without being destroyed in the troposphere due to their low reactivity.

7.13.4 Location of the Ozone Hole

- The temperature at the South Pole is much lower than that of the North Pole.
- During winter season, the temperature at the South Pole drops down to as low as '400°C.
- This low temperature contributes to ice crystals, formation. In the presence of these ice crystals, hydrogen chloride (HCl) gas breaks down into chlorine molecule which later dissociates into a chlorine atom that reacts with the ozone molecule in the stratosphere.
- **Extreme cold temperature at the South Pole and the formation of ice crystals are the major reasons behind ozone depletion over the South Pole.**

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More Ozone Hole Towards Antarctica (South Pole)

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- It takes around 2–7 years for CFCs to reach stratosphere and once they reach stratosphere, the winds play the role of spreading them all around the world.
- **During summers solstice, frozen stratospheric clouds and six month darkness over Antarctica along with extreme cold** and **freeing polar vortex wind** provide ideal conditions for chlorine to dissociate from the CFCs and then react with ozone molecules.
- When two air masses having different physical properties such as temperature, humidity, density, etc., come together and collide, they do not merge readily due to the converging atmospheric circulation.
- The transition zone or the layer of discontinuity is thus created between these two air masses which is a three-dimensional surface.
- This is called a **front** which is much stronger in South Pole than North Pole.
- The chlorine and bromine atoms actually break the ozone molecule mainly during spring time, i.e. between September and December, which is the major cause behind the ozone hole.

Polar Vortex

- A **polar vortex** is a widespread low pressure region in the upper layer of atmosphere found at North and South Poles which gets stronger in winter but weaker during summer at each pole.
- In the South Pole, Antarctic polar vortex exists, which has been observed to be the main reason behind the ozone hole over Antarctica.
- The nitric acid present in the polar stratospheric clouds also react with CFCs and form chlorine atom. These chlorine atoms catalyse the photochemical destruction of ozone molecules.
- During winter season, the concentration of chlorine becomes more, thus leading to a higher number of ozone molecule destruction which is even greatest when the sunlight returns during spring.

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- When the temperature is below 80°C, these polar stratospheric clouds are formed.
- Ozone depletion at the North Pole is less severe than at the South Pole because the exchange of air between the Arctic and the mid-latitudes is much greater.
- When ozone concentration is depleted seasonally over the Arctic, it is known as an **'ozone dent'**.
- But the depletion of ozone is much severe over the Antarctic, which is known as an **'ozone hole'**.

Polar Vortex

What exactly is a polar vortex?

- It is described as a **whirling cone of low pressure over the poles that is strongest in the winter months** due to the increased temperature contrast between the polar regions and the mid-latitudes, such as the US and Europe.
- **The polar vortex spins in the stratosphere,** a layer of the atmosphere 10–48 km above the ground and above the troposphere, where most familiar weather patterns develop.
- Usually, **when the vortex is strongest,** cold air is less likely to plunge deep into North America or Europe. In other words, it forms a wall that protects the mid-latitudes from cold Arctic air.
- But occasionally, **the polar vortex is disrupted and weakens, due to wave energy propagating upward from the lower atmosphere.** When this happens, the stratosphere warms sharply in an event known as sudden stratospheric warming, in just a few days, miles above the Earth's surface.
- **The warming weakens the polar vortex,** shifting its location somewhat South of the pole or, in some instances**, 'splitting' the vortex up into 'sister vortices'.**

Effects:

- **The split higher up in the atmosphere can give rise to both, sudden and delayed effects,** much of which involve declining temperatures and extreme winter weather in the Eastern US along with Northern and Western Europe.
- A sudden stratospheric warming also leads to **a warm Arctic not only in the stratosphere but also in the troposphere as well.** A warmer Arctic, in turn, favours more severe winter weather in the Northern Hemisphere mid-latitudes including the Eastern US.

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What happens in Antarctic during winter?

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- During summer solstice, i.e. winter season in Antarctica, a **strong polar vortex** appears.
- Next, polar stratospheric clouds form which have the potential to concentrate the pollutants that later break down the ozone molecules, thus catalysing the process.
- When spring arrives after winter and the sun comes back after the six long months of polar night duration, the ozone levels get severely depleted around the Antarctic leading to ‘ozone hole’.
- The ozone hole gets even worse when sun comes back because of the availability of UV radiations.
- The ozone concentration in the atmosphere is represented in ‘Dobson units’; the average ozone concentration in the atmosphere is about 300 Dobson units.
- Whenever the concentration drops below 220 Dobson units, it is considered that an ozone hole has been formed.

7.13.5 Tropospheric and Stratospheric Ozone

- Stratospheric ozone (Naturally formed) shields harmful UV rays from reaching the Earth surface, thus preventing fatal diseases like skin cancer.
- But tropospheric ozone (Human made) is not at all helpful. It enters our respiratory system and can cause a variety of health effects from mild-to-severe levels.

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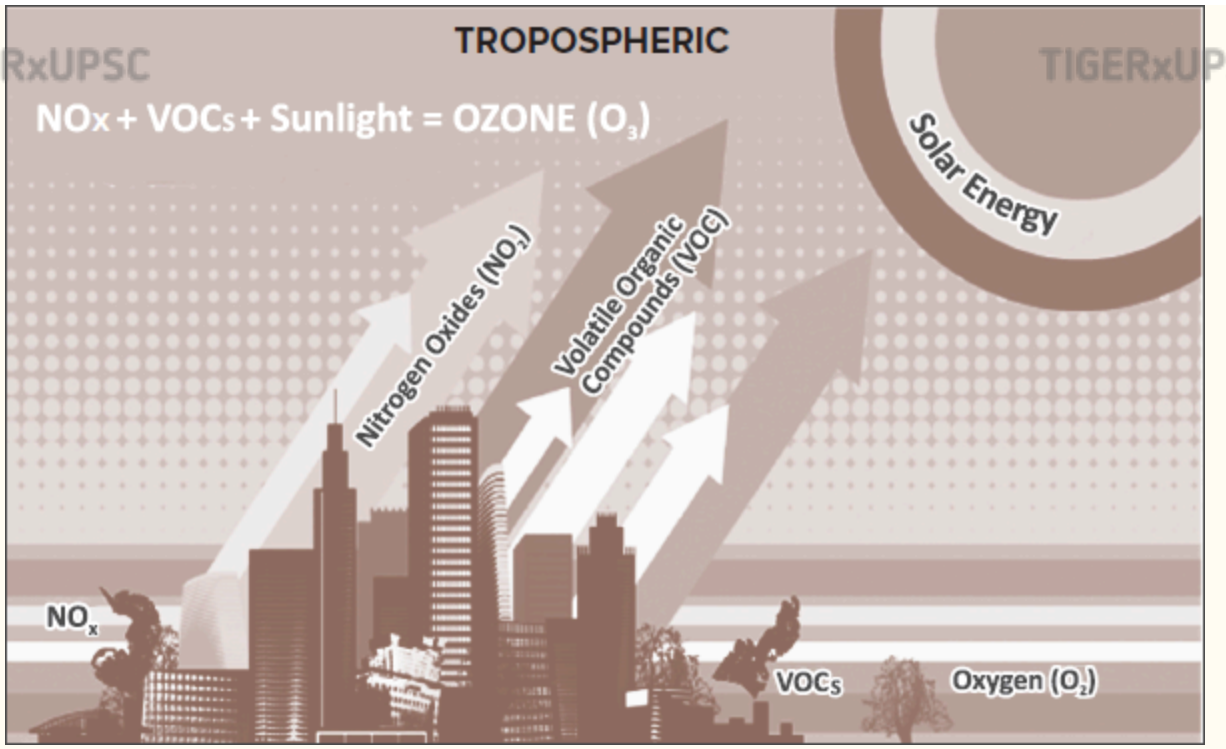

Figure 7.10 Various sources of ozone formation in tropospheric region

7.13.6 Ozone Formation in Troposphere

- Ozone is emitted from various sources which is formed through a series of complex reactions that involve both light and chemical precursors.
- Now efforts are being made to restrict the amount of chemical precursors that generate ozone in the atmosphere.
- **The chemicals include nitrous oxides (NOx) and volatile organic compounds (VOCs) that are produced through various sources and come together to form ozone in the presence of sunlight.**
- **NOx Sources:** Animal manure, combustion of fossil fuels, agricultural activities, sewage treatment, nitrogenous fertilisers, microbial activities in wet tropical forests majorly produce NOx.
- **VOCs Sources:** Combustions of various types of industrial products, refining fossil fuels, burning electrical appliances generate large amount of VOCs. They are also produced in huge quantities by natural sources like vegetation fields.

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7.13.7 EPA and Ozone

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Ozone and other air pollutants were regulated by the **Environmental Protection Agency (EPA)** soon after the passage of the **Clean Air Act in 1971**.

The ozone standard recommended by the EPA is 75 parts per billion or ppb, which is taken after an average of three years of testing carried out annually.

This regulation is the primary standard which **'sets limits to protect public health, including the health of 'sensitive' populations such as asthmatics, children, and the elderly.'**

The Air Quality Index (AQI) was established by EPA, which calculates the presence of toxic air pollutants in the atmosphere and designates colour and level to ranges of pollutant concentrations.

7.13.8 Ozone in the Body

- As mentioned earlier, tropospheric ozone is extremely harmful to living organisms on the Earth.
- Ozone enters the body through the respiratory tract and reacts with the linings of wind pipe and chemical reactions take place within tissues where molecules break down and then recombine with the excess oxygen.
- This recombination destroys the air passage linings, whose function is to keep away bacteria and other microorganisms from entering the respiratory tract. The immune reaction takes place inside airways passage resulting in excess mucus formation and constriction to reduce air flow through both the lungs.
- **The physiological symptoms include shortness of breath, coughing, wheezing, chronic asthma, and often result in severe pulmonary infection and increase risk of heart attack.**

Table 7.5 Different levels of ozone and their health effects

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AQI Levels	Levels of Health Concern	Ozone parts per billion
201+	**Hazardous**	Health warning of emergency co to be affected.
106+	**Very Unhealthy**	Health alert: The risk of health ef
86–105	**Unhealthy**	Some members of the general pu effects; members of sensitive gro serious health effects.
71–85	**Unhealthy for Sensitive Groups**	Members of sensitive groups ma general public is not likely to be a

7.13.9 Effects of Ozone Depletion

Direct Effect

1. Ozone absorbs the heat from the sun which results in an increase in the temperature of stratosphere.
2. Harmful effects caused by an increase in UV radiations are given below:
   - It reduces the immunity power and increases susceptibility towards different types of harmful infections.
   - These rays cause skin cancer and cataracts in the eyes of human beings.
   - The UV rays also damage DNA in the human body and ultimately affect genes.

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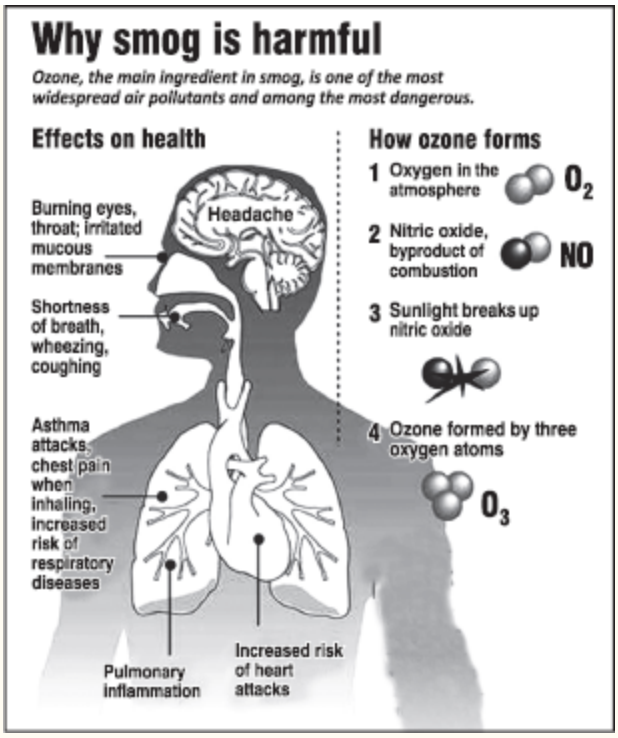

Figure 7.11 Harmful effect of smog in human body

Indirect Effect

- Most fruits and vegetables are very sensitive to the ultraviolet rays
- Reduction in growth
- Increase in harmful mutation
- Young cells and larvae of aquatic ecosystem are also severely affected
- Plastic becomes brittle when it comes in contact with the ultraviolet rays

7.13.10 International Programs to Check Ozone Depletion

- **The Montreal Protocol,** which came into force in 1989, is one of the most effective **international environmental treaties to gradually phase out the emission of Ozone Depleting Substances (ODSs) into the atmosphere.**
- **197 member** parties signed it and thus, it became the **first such international treaty with complete ratification.**

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- Several amendments were carried out and **Kigali amendment was the eighth amendment.**
- **Successful cut-down of 8% of chlorofluorocarbons production and other ODSs has greatly contributed to the repair of the ozone hole.**

Amendments to the Montreal Protocol

- **The London Amendment (1990)** amended the ODS emission schedule. **It made the complete phase-out of CFCs, halons, and carbon tetrachloride mandatory by 2000 in developed countries,** and by 2010 in developing countries. In the list of controlled ODSs, methyl chloroform was also added, which in developed countries phase-out was targeted in 2005, and for developing countries in 2015.
- **The Copenhagen Amendment (1992)** potentially boosted the phase-out of ODSs and **effectively incorporated hydrochlorofluorocarbons (HCFC) which must be phased-out for developed countries,** starting from 2004. CFCs, halons, carbon tetrachloride, and methyl chloroform which were previously targeted for complete phase-out in 2000, in developed countries have been amended and the year was preponed to 1996.
- **The Montreal Amendment (1997)** added the **phase-out of HCFCs along with methyl bromide in developed and developing countries in 2005 and 2015,** respectively.
- **The Beijing Amendment (1999)** added more stringent **control over both the production and trade of HCFCs.** Bromochloromethane was next added to the list of controlled ODS, with 2004 being the target for its phase-out.
- **The Kigali Amendment (2016), the eighth amendment** actually **extended controls to phase down the production and consumption of hydrofluorocarbons (HFCs) which were added previously during Beijing amendment**. This is because these substances were adopted by most of the industries and hence needed more time to phase down.

7.13.11 Alternative Technologies

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Alternatives to hydrofluorocarbons (HFCs) include natural refrigerants like **ammonia (R717)**, **carbon dioxide (CO2 or R744), and hydrocarbons (HCs) such as propane (R290) and isobutane (R600a)**. Other options are newer technologies like **hydrofluoroolefins (HFOs)** and **HFC-HFO blends**, which have a **lower global warming potential (GWP)**. The best alternative depends on the specific application and equipment.

Natural refrigerants

Natural refrigerants are chemicals found in the environment that have a low global warming potential (GWP).

- **Hydrocarbons:** This group includes propane (R290) and isobutane (R600a). They are very energy-efficient and have a low GWP.
- **Applications:** Commonly used in home refrigerators, small air conditioners, and commercial refrigeration systems.
- **Considerations:** Propane and other hydrocarbons are flammable. This requires specific safety designs and standards for their use. India and China have safely used hydrocarbons in room air-conditioning systems for several years.
- **Carbon Dioxide (CO_2 or R744):** With a GWP of 1, CO_2 is an environmentally friendly option.
- **Applications:** It is found in commercial refrigeration, vending machines, and some mobile air-conditioning systems.
- **Considerations:** CO_2 systems operate at higher pressures than traditional systems, which requires specialized equipment.
- **Ammonia (NH3 or R717):** Ammonia is very energy-efficient and has a GWP of zero.
- **Applications:** It has been used historically in large industrial refrigeration facilities, such as food processing and cold storage.
- **Considerations:** It is toxic and requires strict safety measures and trained personnel for handling. This limits its use in homes.

Hydrofluoroolefins (HFOs)

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HFOs are new synthetic refrigerants with low to very low GWP and zero ozone depletion potential (ODP).

- Examples: HFO-1234yf and HFO-1234ze(E).
- Applications: They are used in mobile air conditioning for new vehicles and in some commercial refrigeration and chiller applications.
- Considerations: Some HFOs are mildly flammable, and studies are ongoing regarding the environmental impact of their degradation byproducts.

Low-GWP HFCs and HFO/HFC blends

Certain HFCs and newly formulated blends provide a way to move away from older, higher-GWP HFCs.

- **R32:** This HFC has a GWP of 675, which is significantly lower than HFCs like R410A. This makes it a transitional option for room air conditioners.
- **HFC/HFO Blends:** Combinations such as R454C and R448A are being used to achieve lower GWP than the HFCs they replace. These often require new equipment designs.

Do You Know?

By 2030, developing countries have promised to completely phase out (100%) the emission of HCFs yet 0.5% would still persist because of their usage in air conditioning purposes and refrigerator equipment since it is kind of unavoidable.

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7.14 ACID RAIN

- Acid rain, defined as the type of precipitation with acidic components, like sulphuric or nitric acid which falls in the wet or dry form to the ground from the atmosphere.
- This consists of rain, snow, fog, hail or even dust which is acidic in nature.
- In the year 1872, **Robert Angus Smith** first used the term acid rain.
- Burning of fossil fuel wood, coal or petroleum products releases carbon, sulphur and nitrogen in the atmosphere.
- These three elements, when burned in the presence of atmospheric oxygen, get converted to their respective oxides, which are water-soluble.
- These oxides react with large quantities of vapour present in the atmosphere to form acids like **carbonic acid, sulphuric acid, sulphurous acid and nitrous acid.**

Figure 7.12 demonstrates the acid rain pathways in our environment:

1. SO_2 and NOx emissions are released into the atmosphere.
2. The pollutants are converted into acid particles which are travelled long distances.
3. The acid particles fall to the Earth in the form of wet and dry deposition such as dust, rain, snow, etc.
4. This may cause harmful effects on soil, forests, streams, and lakes.
   - When sulphur dioxide (SO_2) and nitrogen oxides (NOx) are emitted into the atmosphere and transported by wind and air currents., thereby resulting in acid rains.
   - There is reaction of SO_2 and NOx with water, oxygen and other chemicals resulting in the formation of sulphuric and nitric acids and mixing with water and other materials before falling to the ground.

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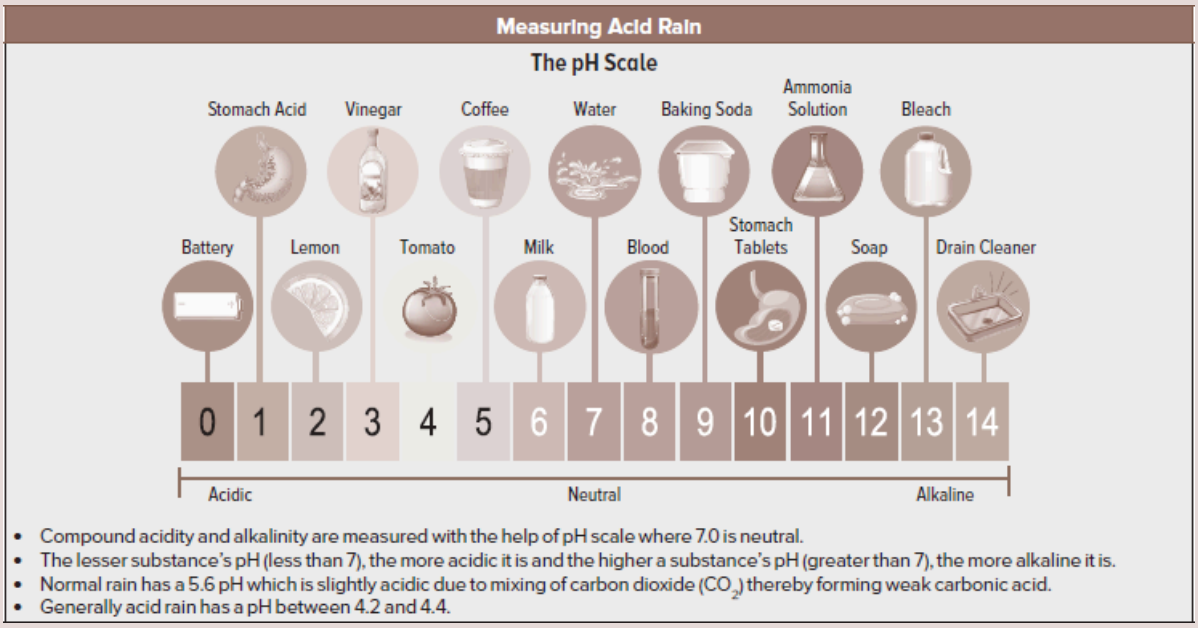
- The small portion of SO_2 and NOx is from natural sources like volcanoes, whereas the major one comes from the fossil fuels burning.

Measuring Acid Rain



- Compound acidity and alkalinity are measured with the help of pH scale where 7.0 is neutral.
- The lesser substance's pH (less than 7), the more acidic it is and the higher a substance's pH (greater than 7), the more alkaline it is.
- Normal rain has a 5.6 pH which is slightly acidic due to mixing of carbon dioxide (CO_2) thereby forming weak carbonic acid.
- Generally acid rain has a pH between 4.2 and 4.4.

Major atmospheric SO_2 and NOx sources are:

- Electricity generation due to fossil fuels burning leading to the two-thirds SO_2 and one-fourth NOx in the atmosphere.
- Vehicles and heavy equipment.
- Oil refineries, manufacturing and other industries.
- Long distances winds blow SO_2 and NOx across borders thereby making it a problem for everyone and not just those who live close to these sources.

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7.14.1 Forms of Acid Deposition

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Wet Deposition

- Wet deposition is what we most commonly think of as acid rain.
- The sulphuric and nitric acids formed in the atmosphere fall to the ground mixed with rain, snow, fog, or hail.

Dry Deposition

- Acidic particles and gases can also deposit from the atmosphere in the absence of moisture as dry deposition.
- The acidic particles and gases may deposit to surfaces (water bodies, vegetation, buildings) quickly or may react during atmospheric transport to form larger particles that can be harmful to human health.
- When the accumulated acids are washed off a surface by the next rain, this acidic water flows over and through the ground, and can harm plants and wildlife, such as insects and fish.

Effects of Acid Rain

- Acid pollution plays havoc with human nervous system, respiratory system and digestive system by making the person an easy prey to neurological diseases.
- This happens because the acids produce highly toxic compounds which contaminate the drinking water and enter human body.
- Acid rain contains air pollutants which contribute to a variety of hazards, associated with reduced visibility due to smog, etc.

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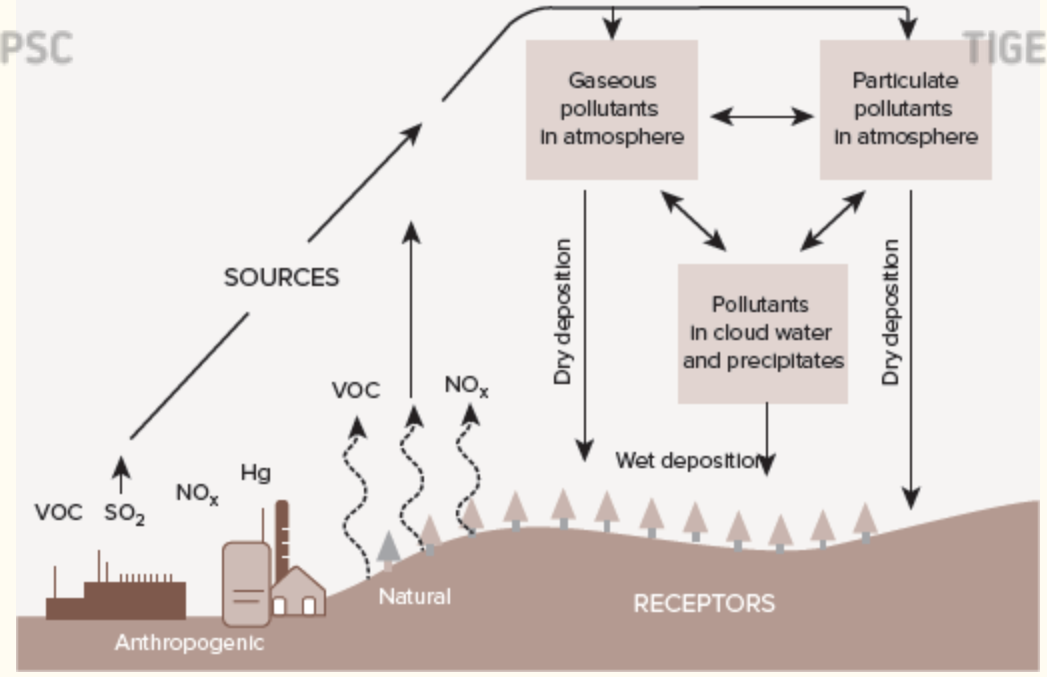

Figure 7.12 Schematic View of Different Ways of Acid Rain

7.14.2 Impact of Acid Rain on Agriculture

- Nutrients like calcium, potassium, iron and magnesium are leached away from soil by acids which are essential for the plant growth.
- Acid precipitation on vegetables reduces the rate of photosynthesis.
- Destroys the soil fertility.
- Retards the growth of the crops such as pea, beans, radish, potato, spinach, broccoli and carrots, etc.
- Acid deposition weakens the trees like pine, spruce, ashes and birch which can be easily attacked by pathogens and drought.

7.14.3 Effects on Building and Monuments

- Acids dissolve limestone and marble, destroying features and structures of historic buildings like monuments.
- The Parthenon in Athens, the Taj Mahal in Agra, the Colosseum in Rome, Medieval Cathedrals in Europe and the Washington Monuments in Washington DC are slowly dissolving and flaking away because of acid rain.

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7.14.4 Effect on Taj Mahal

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- Effect of Mathura refinery can be observed on Taj Mahal in Agra, due to the impact of acid rain.
- It generates lots of SO_2 gas every day which settles on the marble of Taj Mahal and can be called as a stone cancer.
- This sulphuric acid reacts with calcium carbonate ($CaCO_3$) in the marble to form calcium sulphate ($CaSO_4$) which causes pitting on the Taj.
- Discolouring, chipping and breaking off the edge of the marble slabs, and the formation of cracks in the marble are some of the signs of deterioration of the monument.

ENV 26- Artificial Rain

7.15 ARTIFICIAL RAIN/CLOUD SEEDING

Artificial rain is a scientific technique that uses chemicals to create condensation in clouds, leading to rainfall. This process is also called cloud seeding.

Clouds are categorized as warm or cold based on artificial rain:

Warm clouds and cold clouds: Warm clouds, at or above 1 degree Celsius, consist of liquid water droplets. In contrast, cold clouds, below 1 degree Celsius, can contain supercooled water droplets, ice crystals, or a mix of both. A cloud's temperature depends on its altitude; high-level clouds are generally colder and consist of ice, while low-level clouds are warmer and mainly water-based.

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CLOUD SEEDING

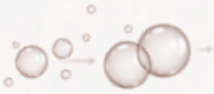
How Cloud Seeding Works?

1 Seeding

Airplane releases Dry Ice (Solid CO_2) or AgI into the sky to form artificial nuclei in clouds

2 Droplet Formation

These nuclei attract supercooled water, rapidly forming larger droplets or ice crystals



3 Growth & Coalescence

Droplets collide and merge via coalescence, significantly increasing their size and mass

4 Precipitation

Droplets reach critical mass, overcoming cloud updrafts to fall as rain or snow

Definition and Classification

- **Artificial rain (Cloud Seeding)** is an advanced scientific technique using chemicals to cause condensation in clouds, leading to rainfall
- Clouds are classified as warm clouds (at or above 1°C) and cold clouds (below 1°C) for artificial rain purposes

Cloud Type	Composition	Altitude	Seeding Materials
Warm Clouds	Primarily liquid water droplet	Generally lower altitude	Salts (NaCl, $CaCl_2$) are common hygroscopic materials
Cold Clouds	Supercooled water/ice crystal mixture, or ice crystals only in low temperatures	Typically higher altitude	Glaciogenic seeding materials include AgI, KI, and dry ice (CO_2)

Importance of Artificial Rain

- Can be used to shift excessive rainfall to another area to reduce disaster risk (e.g., during the China Olympics)
- Can reduce the risk of natural calamities by controlling high-pressure areas to reduce the speed of storms
- Enhances crop yields by providing critical moisture. Crucial for rain-fed agriculture during growth stages

Recent Projects in India

Delhi (October 23, 2025)

Delhi successfully trialed artificial rain to combat severe pollution

Ramgarh Dam, Jaipur, Rajasthan (July 31, 2025)

India conducted its first drone cloud seeding drill

Project Varshadhari (2017)

Karnataka used a Beechcraft to disperse AgI, NaCl, and Potassium Chloride

- ***Warm clouds***

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- Composition: Mainly liquid water droplets.
- Altitude: Usually found at lower altitudes where temperatures are warmer.
- Examples: A summer shower in a warm climate that stays above freezing is a warm rain cloud. Rain forms from the merging of liquid droplets. Common low-level clouds like stratus, stratocumulus, and cumulus are typically warm clouds.

■ Cold clouds

- Composition: Can mix supercooled water droplets and ice crystals. At very low temperatures, they are made entirely of ice crystals.
- Altitude: Typically found at higher altitudes where the air is cooler.
- Examples: A thunderstorm cloud serves as an example of a cold cloud. It contains ice crystals at its top, even if rain is falling to the ground. High-level clouds such as cirrus, cirrostratus, and cirrocumulus are cold clouds formed from ice crystals in extremely cold air.

Do You Know?

Middle-level clouds (altocumulus and altostratus) can contain either water droplets or ice crystals depending on the temperature.

■ Cloud seeding materials in warm rain clouds and cold rain clouds

In warm clouds, hygroscopic materials like salts (sodium chloride, calcium chloride) help water droplets merge and grow into larger raindrops.

In cold clouds, silver iodide (AgI), potassium iodide (KI), or dry ice (frozen carbon dioxide) form ice crystals in temperatures below freezing; this is known as glaciogenic seeding.

■ Importance of Artificial Rain

1. In agriculture, for irrigation purposes.

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2. To shift excessive rainfall from one area to another, reducing the risk of disasters. This occurred during the China Olympics.
3. It can lower the risk of natural disasters by controlling high-pressure areas, which can reduce storm speeds.

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Do You Know?

- Project Varshadhari, a cloud seeding experiment, was conducted in Karnataka in 2017 using BQ100 Beechcraft. Silver iodide, sodium chloride, and potassium chloride were used to condense water vapor in Magadi and Ramanagaram, Karnataka.
- On July 31, 2025, at Ramgarh Dam near Jaipur, Rajasthan, scientists conducted a drone-based drill for artificial rain induction. This marked the first attempt in India to use UAVs for cloud seeding, a significant milestone for the country's agricultural and climate strategies.
- An artificial rain trial recently took place in Delhi to combat severe air pollution. A successful test flight occurred on October 23, 2025, in the Burari area.

ENV 27- Mass Extinction

7.16 MASS EXTINCTION

These events are due to drastic changes to the planet's conditions such as atmospheric changes, volcanic activity, asteroids attack to the surface. Scientists have linked the previous five events of mass extinction in planet history through fossil records.

- ***First***

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- At the **end of Ordovician age**, the first mass extinction took place, which was **nearly 440 million years ago**.
- **During that time, animal species lived mostly in the sea**. As a result of **huge glaciations**, most of the planet's water was locked up on the planet, thus causing sea levels to drop precipitously.
- At the time of the Ordovician mass extinction, **around 60% of animal species perished on Earth**.

Second

- At the **later stages of Devonian age**, about **360 million years ago**, the second mass extinction event took place over a period of around 20 million years.
- Pointing of **successive glaciations** led to the **wiping of approximately 70% of all marine species, specifically in the shallow sea areas**.
- This is where coral reefs lived and over 100 million years later corals didn't return to the ocean on evolving a new species.
- Few formative amphibian animals and insects inhabited the land, and many terrestrial creatures were made extinct during this time.

Third

- This is the **most devastating extinction in the planet's history – as nearly 96% of all animal species perished, all from terrestrial vertebrates to marine animals and insects**.
- Therefore, it is often termed **'The Great Dying'**.
- It occurred at the **end of the Permian age**, around 225 million years ago.
- **Its cause is unknown**, although it was observed that **either the Earth is hit by a comet or unsettling of massive volcanic activity**, **causing an abrupt release of methane from the seafloor**, thereby leading to massive climate change.
- Only the 4% of species on the planet survived this extinction.

Fourth

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- This took place at the **end of the Triassic** age around 18 million years, beginning 200 million years ago in which half of all marine invertebrates as well as most of the land animals, especially amphibians, were perished away.
- Its likeliest causes were **volcanic activity** or **hitting the planet by a comet or asteroid.** As a result, **nearly half of all planet animal species were vanished, including the dinosaurs, marine reptiles and various plant species**.
- In this event, **mammals were least affected**, thereby resulting in the providing of historical niche through which the mammals, and subsequently the humans, came to dominate the planet.

■ *Fifth*

- Most people are aware of the fifth mass extinction event as all the **dinosaurs became extinct**.
- Due to the **hitting of huge asteroid on the planet, about 65 million years ago** (crater found off Mexico's Yucatan Peninsula), resulted in the occurrence of this event. This event led to the huge activating temperature and sea level changes and altering the atmospheric chemical makeup.
- It was **observed the rising of global temperatures by 10 degrees and ocean levels by as much as 300 metres as compared to today's levels** which would lead to the submerging of 40% of the land mass, as per today's sea level.

■ *Sixth*

- The idea of currently being in the midst of a sixth mass extinction event is a grievous as we **humans are its main cause**.
- The humans are responsible for the single species extinction, and the humans killed off animal species are rising exponentially.
- The wiping of native species results from our deliberate or inadvertent actions by transporting the invasive species from one part of the world to another.

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- The spreading of diseases from one continent to another as a traveller and decimated wildlife.
- Deadly chemicals are responsible for poisoning of seas, thereby perishing species of coral reef and declining of fish species dramatically. Due to rise in global human population, the pressure on food's natural resources has become intense, resulting in wiping out of many species, particularly fish, which is close to extinction.
- There is a frequent occurrence of extreme climatic events, increasing sea levels due to the pollution as a result of the burning of fossil fuels which is changing the planet's atmosphere.
- This will eventually lead to the human race's downfall and will have a catastrophic effect on animal species unless there is an occurrence of radical shift in attitude and process.

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MASS EXTINCTION

Mass extinction events are caused by drastic changes in planet conditions due to reasons like alterations in the atmosphere, volcanic activity, or asteroid attacks.
Scientists have identified **five previous mass extinction** events through fossil records.

ORDOVICIAN	DEVONIAN	PERMIAN	TRIASSIC	CRETACEOUS	HOLOCENE
1ST Mass Extinction	2ND Mass Extinction	3RD Mass Extinction	4TH Mass Extinction	5TH Mass Extinction	6TH Mass Extinction
~440 million years ago	~360 million years ago	~225 million years ago	~200 million years ago	~65 million years ago	Ongoing
~60% of all animal species lost	~70% of all marine species lost	~96% of all animal species lost	~76% of all animal species lost	~75% of all animal species lost	??%
Massive glaciations caused sea levels to drop precipitously, leading to the perishing of most animal species, which were primarily marine	Successive glaciations devastated shallow-water marine species, causing coral reefs to vanish for over 100 million years	'The Great Dying' was the most devastating extinction, likely due to a comet or massive volcanism causing rapid climate change	Half of marine invertebrates and many land animals (amphibians) perished, likely due to volcanic activity a comet/asteroid impact	A massive Yucatan impact killed the dinosaurs and caused severe environmental upheaval. Mammals survived and became dominant	The Sixth Mass Extinction, caused by humanity, is driven by pollution/disease, overexploitation, and climate change from fossil fuels

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Climatic Parameters 8

IN THIS CHAPTER...

8.1 Carbon Pricing
8.2 Ecological Footprint
8.3 Carbon Footprints
- 8.3.1 Concept of Carbon Footprint
- 8.3.2 Measures to Reduce Carbon Footprint
- 8.3.3 Common Carbon Metric (CCM)

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8.9 Carbon Space
8.10 Carbon Fertilisation
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8.12 Green Jobs
8.13 Greenwashing and Its Work

8.1 CARBON PRICING

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- Carbon pricing is like putting a price tag on the pollution caused by carbon dioxide (CO_2) and other greenhouse gases. It's a way to make companies and people pay for the pollution they create when they release these gases into the atmosphere. For eg., When companies or individuals emit these gases, they have to pay for it. It's similar to paying a fee for disposing of garbage. The more you pollute, the more you pay.

Carbon Tax

- Carbon tax is a tax that the government levies on fuel that burns to release greenhouse gases (GHGs) in the atmosphere.
- When a fuel is burnt, carbons present in hydrocarbon of that fuel get converted into CO_2, which is considered as a GHG.
- It is a type of environmental tax, also called carbon pricing.
- It is levied on the content of carbon released when 1 tonne of fuel is burnt.
- Hydrocarbon content varies according to the type of fossil fuel, so does the amount of CO_2 released when fossil fuels are burnt.
- This is the main reason for varying carbon tax rates.

Do You Know?

1 tonne carbon = 3.664 CO_2

- Finland was the first country to introduce a carbon tax in 1990. The concept of taxing activities that cause pollution was first articulated by economist **Arthur Pigou** in the 1920s.
- The confusion likely stems from Finland and New Zealand having different approaches to carbon pricing: **Finland uses a national carbon tax, which is a direct price on emissions,** while **New Zealand uses a domestic Emissions Trading Scheme (ETS), which is a cap-and-trade system.**
- Finland's tax applies directly to the carbon content of fuels, and the government sets the price. New Zealand's ETS sets a limit (cap) on total

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emissions and allows companies to buy and sell permits to emit, with the price fluctuating based on market demand.

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The Social Cost of Carbon (SC-CO2)

- SC-CO_2 is nothing but an estimated amount measured in dollar to represent the damage caused by the emission of 1 tonne of CO_2 in the atmosphere in a given year.

Implementation of Carbon tax

- In 2010, India first imposed a carbon tax of `50 per metric tonne on all coal mines in India and all coals imported.
- India's carbon tax rate is currently among the lowest in the world, at just US$1.6 per tonne of CO_2 emissions. By increasing this tax rate, India can not only reduce its carbon emissions but also set an example for other G20 countries.
- All the commitments that were made under United Nations Framework Convention on Climate Change (UNFCCC) conferences by member countries have geared up the implementation of carbon taxes and move towards clean energy.
- Introducing carbon tax in India highlighted India’s responsiveness to reduce fossil fuel usage and boost the economy through clean energy.

8.2 ECOLOGICAL FOOTPRINT

- It is a measurable ecological resource required by a particular population or product for the consumption of natural resources (such as food and fibre plant-based products, livestock and fish products, timber and other forest products, urban infrastructure space) and its waste absorption, especially the carbon emissions.
- This measurement takes into account the resources a population needs to
  (1) produce goods and
  (2) ‘assimilate’, or clean up, its waste.

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- Biologically productive land and water can include arable land, pastures and parts of the sea that contain marine life.
- The ecological footprint is the only metric that measures how much nature we have and how much nature we use.
- The footprint helps:
  - Countries improve sustainability and well-being
  - Local leaders optimise public project investments
  - Individuals understand their impact on the planet

What Is Earth Overshoot Day?

- Overshoot is defined as the ecological deficit of the entire planet.

8.3 CARBON FOOTPRINTS

8.3.1 Concept of Carbon Footprint

- A carbon footprint is a measurement of the total amount of GHGs released by an individual which includes any particular company, organisation, event, product or anything in that matter in a given time.
- Calculating carbon footprint is a complicated method since it requires a huge amount of data and data interpretation must be carried out accurately to achieve the desired result.
- Apart from CO_2 emission, other GHGs are also measured, including hydrofluorocarbons, ozone and methane.
- Carbon footprint is represented in tonnes of CO_2 released.
- It can also be expressed in kg carbon or kg carbon dioxide.
- 1 kg carbon = 3.67 kg CO_2 (carbon dioxide).
- Calculating carbon footprint is a smart way to assess and analyse the amount of pressure being put on natural resources and the vulnerability of exceeding the balance between the demand and supply chain.
- Carbon accounting is actually the activity of analysing all the data and calculating the carbon footprint of the concerned individual.

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- After calculating it, new policies or strategies can be implemented depending on the technology, habitat and natural resources available in that particular place.

8.3.2 Measures to Reduce Carbon Footprint

- Proper timely maintenance of the car should be carried out, including oil and air replacement, checking or filter.
- Keeping the engine off while waiting or in traffic can effectively reduce carbon emissions by following the traffic rules.
- Using clean fuel.
- Keeping the condition of engine healthy and servicing should be done on a regular basis.
- Planting more trees on the roadside will effectively increase carbon absorption, reducing air pollution levels.
- Unnecessary usage of devices such as air conditioning. Use it only when required.

Do You Know?

Coir is a natural fibre and the government under the Pradhan Mantri Gram Sadak Yojana in the year 2020 has allowed the usage of coir-based geotextiles for rural road construction. Whereas in the Hot Mix Asphalt technology, asphalt is heated and poured over the mixture of stone, sand, gravel and, thereafter, heavy roller is driven on it for compacting the road surface. This results in emitting many gases due to heating and does not help decrease the carbon footprint.

During the copper smelting and refining, a by-product termed as copper slag is obtained which is a non-hazardous, non-toxic material. This eco-friendly industrial by-product has been used in government road projects for past many years as per the Sterlite Copper Company reports.

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ENV 28- Common Carbon metric (CCM)

8.3.3 Common Carbon Metric (CCM)

- CCM is a tool for measuring energy use and reporting GHG emissions from any building operations (i.e., building's carbon footprint).
- CCM is the calculation used to define the measurement, reporting and verification of GHG emissions associated with a building operation in reference to climate change.
- The purpose of a CCM for buildings is to give the sector that represents 40% of the world's energy consumption and related 1/3rd of global GHG emissions a tool that does not exist today—a way to measure, report and verify reductions in a consistent and comparable way.
- There is the greater capacity of the building sector in comparison to others in giving swift, deep and cost-effective GHG mitigation. This has a short-term achievement of substantially raising building energy efficiency. With the help of easily available technologies, equipment, management systems and various alternative generation systems, there is a decline in energy consumption by a projected 30%–50% by 2020 in both new and existing buildings.
- Whereas all building's life-cycle (including construction and demolition) stages generate carbon emissions, its operational phase accounts for a majority of 80%–90% of emissions due to the usage for heating, cooling, ventilation, lighting and appliances. Hence, this stage of the building's life cycle is the focus of the CCM.

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- The actual calculation is done in the carbon dioxide weight equivalent ($kgCO_2e$) emitted per metre square per year = kgCO e/m^2/year (by building and climate region).

8.4 CARBON OFFSETTING

- This method needs cooperation among countries to help each other out on reducing GHG emission.
- One country that reduces its GHG emission effectively to compensate for the GHG emission made by some other country is called 'carbon offsetting'.
- Buying and selling carbon offsetting have actually benefited many developed countries.

Carbon Trade

- It is the credits buying and selling, thereby allowing a company or other entity in the certain amount of carbon dioxide emissions.
- There is government authorisation in carbon credits and trade so as to gradually decline the overall carbon emissions and diminish their contribution to climate change.
- This is also termed as carbon emissions trading.
- Carbon trading is a market-based system aimed at reducing GHGs that contribute to global warming, particularly carbon dioxide emitted by burning fossil fuels.
- Emissions trading is a trade to decrease CO_2 and other GHG emission, thereby fighting climate change.
- This has a very simple principle.
  - Establish the quantity of total CO_2 tonnage emitted by a group.
  - Association with the emission group and climatic warming by CO_2 emissions requires an emissions allowance for every tonne of CO_2 emission.
  - Group members can buy allowances, such as government-organised auctions.
  - Paying penalties during the CO_2 emission in excess of allowances.

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- Less CO_2 emission conveys less payment for allowances as the climate action is an investment which pays off.

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Carbon Markets

Carbon markets are likely to be one of the most widespread tools in the fight against climate change. Carbon markets are also known as emissions trading systems or cap-and-trade systems. They create economic incentives for reducing emissions by allowing entities to buy and sell carbon credits. This helps to create a market-based approach to reducing GHG emissions and has been adopted by many countries and regions as a strategy to mitigate climate change.

Carbon markets are a mechanism designed to reduce GHG emissions by putting a price on carbon. They create a market for buying and selling carbon credits, which represent the right to emit a certain amount of GHGs. In carbon markets, companies or entities that emit fewer GHGs than their allotted limit can sell their excess credits to those who exceed their limit. This allows for a more efficient allocation of emission reduction efforts. While carbon markets involve financial transactions and can transfer resources from the private sector to the state. Instead, they facilitate the exchange of emission.

8.5 CARBON SINK

- Carbon sinks can be seen as blotting paper or sponges that absorb the CO_2 and store it for a long period.
- Ocean, plants (forests) and soil are major natural carbon sinks, whereas landfills are artificial carbon sinks.
- Soon after the implementation of Kyoto Protocol, carbon sink became a widely known process or method to reduce excess CO from the atmosphere.

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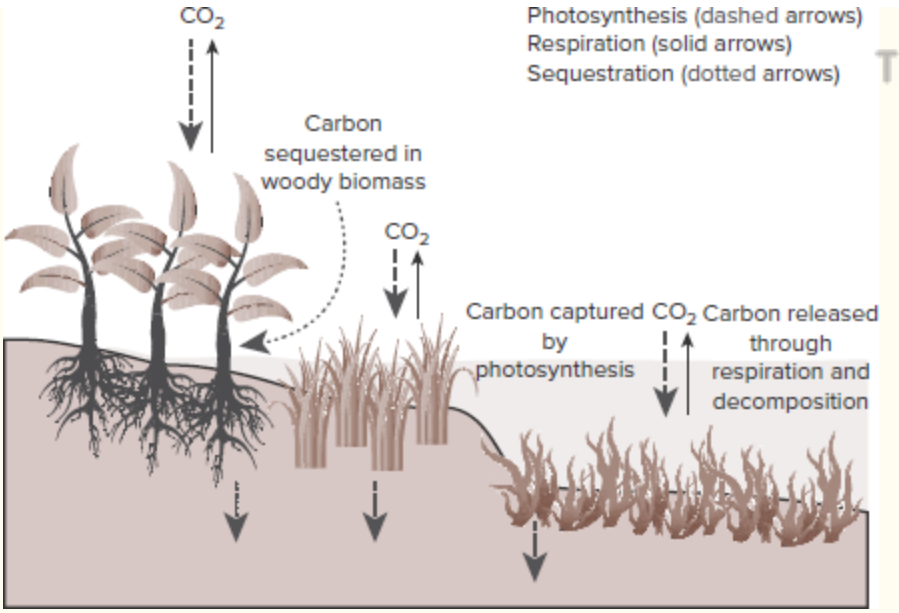

Figure 8.1 Carbon Sequestered in Soil

Blue Carbon

- Blue carbon is simply the term for carbon captured by the world's ocean and coastal ecosystems.
- Simply put, blue carbon is the carbon stored in coastal and marine ecosystems.
- Seagrasses, mangroves and salt marshes along our coast 'capture and hold' carbon act as something called a carbon sink.
- Nowadays, the coastal ecosystems (mangroves, tidal marshes and seagrass meadows) are being appreciated for their role in reducing climatic change as they accumulate and sequester greater carbon per unit area than terrestrial forests.
- These ecosystems give important climatic change adaptation benefits, such as food security and coastal protection for various coastal communities.
- During the degrading or damaging of ecosystems, their capacity as carbon sink is vanished or badly affected. The stored carbon is released, resulting in carbon dioxide (CO_2) emissions, thus contributing to climate change.
- The coastal ecosystems carry on in playing their role as long-term carbon sinks.

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- They are presently being appreciated for their role in climate change mitigation as their ability to remove carbon dioxide (CO_2) from the atmosphere makes them net carbon sinks.
- The restoration and conservation of peatlands and terrestrial forests have been considered as an important climate change mitigation component.
- Various countries are formulating policies and programmes linked with their economic growth for sustainable development through reducing the carbon footprint.
- These incorporate steps in conserving and sustainably managing natural systems in relevance to the UNFCCC, reducing emissions from deforestation and forest degradation in developing countries.
- It is a process of capturing CO_2 from the atmosphere and then storing it underground or in plants for a long period of time.
- Thus, the process by which carbon sink works is called carbon sequestration.

8.6 CARBON SEQUESTRATION

- It is a process of capturing CO_2 from the atmosphere and then storing it underground or in plants for a long period of time.
- Thus, the process by which carbon sink works is called carbon sequestration.

8.6.1 Classification of Carbon Sequestration

■ *Terrestrial Sequestration*

- It is a type of carbon sequestration where carbon dioxide from the atmosphere is captured using plants.
- Naturally, CO_2 is taken by plants for photosynthesis to produce their food which is stored in the form of starch in plant stem, root and soil.
- The same process is enhanced here using advanced technology to keep the maximum amount of carbon stored in plant parts for a long time.

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- Terrestrial sequestration stores only the carbon of carbon dioxide and not the entire carbon dioxide in the form of a gas.
- Thus, if in the future the process is disturbed and the state of soil is interrupted, then the carbon might leak out, which will combine with oxygen present in the atmosphere to form carbon dioxide again.
- Abandoned or uneconomic coal seams are other potential CO_2 storage options.
- Coal contains fractures that impart some permeability to the system.
- It has higher affinity to absorb gaseous CO_2, as compared to methane.
- If CO_2 is injected into coal seams, it can displace CH_4, thereby enhancing coal bed methane recovery.
- Storage of CO_2 underground is now being undertaken at a commercial scale in one deep saline reservoir (aquifer).
- Particularly in Europe and North America, thousands of oil and gas fields are approaching the end of their economically productive life.
- These depleted fields could act as effective storage sites for anthropogenic CO_2 that would otherwise be emitted into the atmosphere.

■ *Geological Sequestration*

- When carbon dioxide is formed due to anthropogenic activities, they are captured at the site where it is formed and stored deep underground called geologic zones for long-term storage. Separation of CO_2 from other gases that are released by several power plants such as coal, electricity, ethanol and many more must be done carefully before they are pumped underground for a long-term storage.

8.6.2 Methods of Capturing Carbon Dioxide

- Biological, chemical or physical processes are used to capture carbon dioxide naturally from the atmosphere.

■ *Biological Processes*

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- **Afforestation:** The first and foremost requirement for carbon sequestration through biological method is the plant. Thus, afforestation must be done. Planting trees along roadside must be taken seriously.
- **Restoration of Wetland:** It has been recorded that wetland holds around 14.5% of the world's soil carbon whereas only 6% of the total land portion of this world is found as wetland. Ramsar convention has, thus, come into the picture to take the required steps to save wetlands from degrading.
- **Modification of Agricultural Practices:** Taking agro-forestry seriously will be helpful in this case. In agricultural land, both crops and trees will act as carbon sink effectively absorbing 20% of total carbon dioxide emitted annually. Using proper fertilisers, disturbing soil less, irrigating crops better and increasing both the yield and ability of both plant and soil to hold carbon for a longer period of time.
- **Fertilisation of Ocean with Iron:** Iron fertilisation promotes phytoplankton growth and helps remove carbon from the atmosphere efficiently even though for a shorter period of time. Releasing nitrogen oxides and disrupting oceans' nutrient balance are a few methods that can increase carbon sequestration.
- **Fertilisation of Ocean with Urea:** Adding urea, a nitrogen-rich fertiliser, encourages phytoplankton growth. This was proposed by Ian Jones.

Do You Know?

1 m deep soil has been proven to hold approximately 1,500 gigatons of organic carbon.

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CARBON SEQUESTRATION

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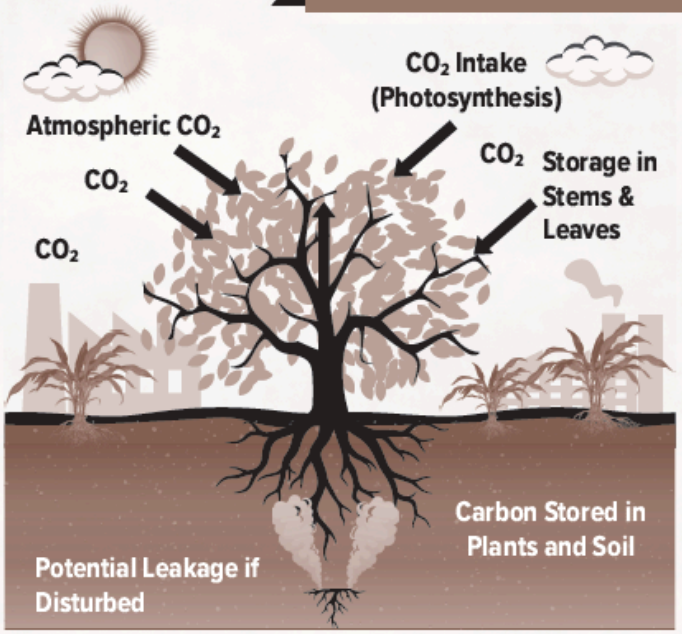
It is a process of capturing CO_2 from the atmosphere and then storing it underground or in plants for a long period of time. Thus, the process by which carbon sink works is called carbon sequestration.

Classification of Carbon Sequestration

TERRESTRIAL SEQUESTRATION



Mechanism

Plants capture atmospheric CO_2 through photosynthesis

Storage

Carbon is chemically converted and stored as starch in plant stems, roots, and soil

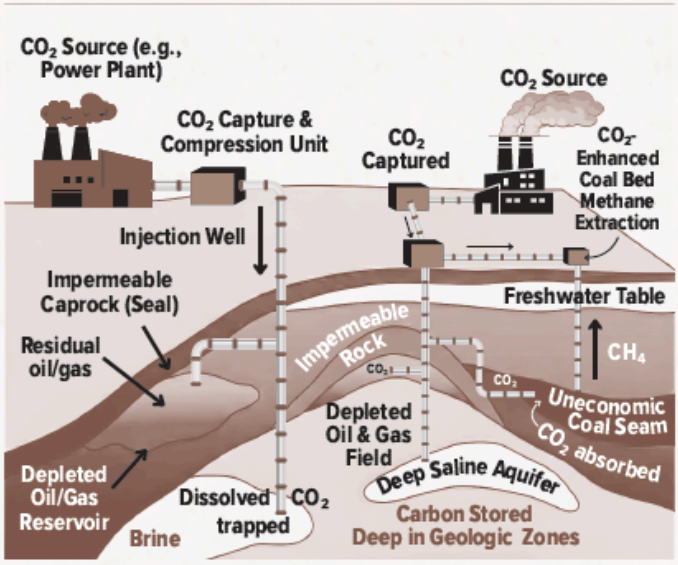
Form of Storage

Unlike geological methods, **this stores elemental Carbon, not CO_2 gas**

Risk Factor

If the soil or plants are disturbed/degraded, the stored carbon can react with oxygen and leak back into the atmosphere as CO_2

GEOLOGICAL SEQUESTRATION



Source

Captures anthropogenic CO_2 from industrial sources (like power plants) at the site of generation

Process

CO_2 is separated from other gases and pumped deep underground for long-term confinement

Storage Sites

- Coal Seams
- Deep Saline Aquifers
- Depleted Oil/Gas Fields

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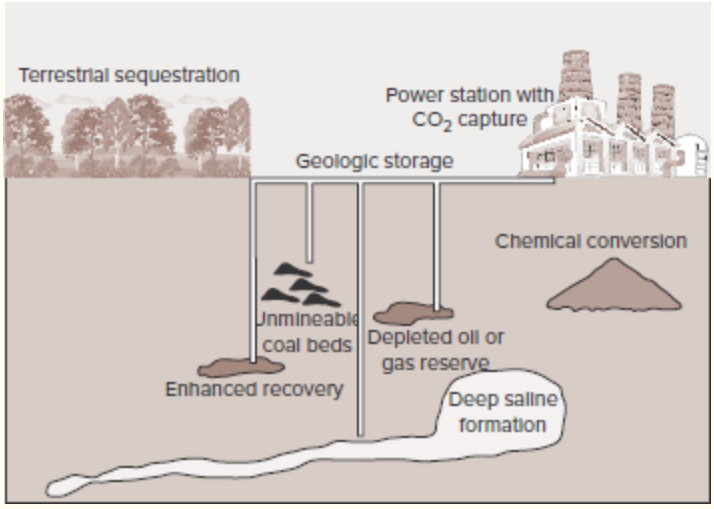

Figure 8.2 Carbon Sequestration Options

Physical Processes

- **Bioenergy with Carbon Capture and Storage (BECCS):** In power plants and industries where carbon dioxides are released, the introduction of biomass that would capture and store the CO_2 is a great way to remove CO_2 from the atmosphere. It is also known as bioenergy with carbon storage, like biochar.
- **Biochar Burial:** Pyrolysis, a chemical reaction that involves breaking down biomass waste using heat, gives rise to charcoal, known as biochar which acts as an effective soil improver when dumped inside a landfill.
- **Biogenic Carbon:** CO_2 emission caused by natural combustion, fermentation, digestion and anything that is related biological substances is recycled naturally in the carbon cycle. If this can be turned into biochar, the carbon might become inert, making it easier to stay sequestered inside the soil.
- **Subterranean Injection:** CO_2 can be injected into depleted oil and gas reservoirs and other geological regions using advanced technology and injected deep into the ocean.

Chemical Processes

- **Mineral Carbonation:** A process where CO_2 reacts with a mineral to form carbonates that are harmless and stable in nature and can be stored, is

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called 'carbon sequestration by mineral carbonation' or mineral sequestration. This is usually done to remove CO_2 from the atmosphere. Naturally, reactions occur between CO_2 and minerals such as Ca or Mg to form calcium oxide (CaO) and magnesium oxide (MgO), respectively.

- **Industrial Use:** Cement that has been used traditionally for a long time actually releases or emits heat and CO_2. With technology advancing, many new types of cement are coming into the market like Novacem which can absorb CO_2 from its surrounding atmosphere while hardening. TecEco is another such type, which has been out in the market since 2002, producing 'Eco Cement'.
- In Estonia and Europe, several power plants release shale ash that has the potential to absorb 60%–65% of the carbonaceous CO_2 and 10%–11% of the total CO_2 emissions from the atmosphere.
- **Chemical Scrubbers:** Chemical scrubbers are devices installed in factory or industry chimney/exhaust that absorb specific pollutants or gas that are harmful and pollute the air if released directly into the atmosphere. Klaus Lackner was the first one to propose the usage of artificial trees to remove carbon dioxide from the atmosphere using chemical scrubbers.
- **Ocean-related**
  1. **Basalt Storage:** When CO_2 is injected deep into the sea, initially it combines with seawater and then reacts with the basalt present there. Ca^{2+} and Mg^{2+} ions are released later after stable carbonate minerals are formed. The measures taken to reduce leakage include hydrate formation since CO_2 in hydrate is denser than CO_2 in seawater. The safe depth of injecting CO_2 is below 8,900 ft.
  2. **Acid Neutralisation:** Carbon dioxide combines with water molecules to form carbonic acid and leads to ocean acidification, affecting the CO_2 absorption rate. Various bases are known to be used to neutralise acidity and boost the CO_2 absorption rate. This includes crushed limestone, NaOH to oceans.

Do You Know?

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Researchers have been successful in finding that Juan de Fuca plate situated in the western coast of the United States has the capacity to store 208 gigatons of CO_2.

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8.6.3 Direct Air Capture (DAC)

Direct Air Capture is a method for removing carbon dioxide (CO_2) directly from the air using engineered mechanical systems. This technology, along with carbon capture and sequestration (DACCS), is a way to reduce carbon dioxide and a potential negative emissions technology. It plays a vital role in reaching net-zero emissions targets to help fight climate change.

■ *How Direct Air Capture Works*

The basic process has two main stages: capture and regeneration.

1. **Air Contact and Capture:** Large fans pull air from the environment into a processing facility or contactor unit. Inside, the air flows over special chemical agents (solvents or sorbents) that attach to the CO_2 molecules. The air, now with less CO_2, is returned to the atmosphere.
2. **Regeneration and Release:** When the capture material is full of CO_2, heat or a vacuum is used to release the pure, concentrated CO_2gas from the solvent or sorbent.
3. **Storage or Utilisation:** The captured CO_2 can be compressed and transported for permanent geological storage underground (sequestration) or turned into feedstock for various applications.

■ *Types of DAC Technologies*

There are two main technology pathways based on the capture material used:

- **Liquid DAC (L-DAC):** Uses a basic chemical solution in water (like potassium hydroxide) that absorbs the CO_2. The process to regenerate the chemical solution and release the CO_2 requires very high temperatures, usually between 300°C and 900°C.

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- **Solid DAC (S-DAC):** Uses solid sorbent materials (like amine-functionalized granules) in filters that chemically attach to the CO_2. These systems work at lower, more adjustable temperatures (about 80°C to 120°C) and often use a temperature or temperature-vacuum swing process for regeneration.

■ *Applications of Captured CO_2*

The captured CO_2 has several possible uses, but permanent sequestration is crucial for climate benefits:

- **Permanent Storage:** Injecting CO_2 deep into appropriate geological formations, such as saline aquifers or basaltic rock, where it turns into stable carbonates.
- **Enhanced Oil Recovery (EOR):** Injecting CO_2 into oil wells to extract more oil. This use is controversial since the oil is eventually burned, releasing CO_2 back into the air.
- **Synthetic Fuels:** Combining captured CO_2 with renewable hydrogen to produce carbon-neutral synthetic fuels for sectors like aviation and shipping that are difficult to electrify.
- **Commercial Products:** Using it in beverages for carbonation, in greenhouses to improve plant growth, or in making construction materials like concrete.

ENV 29- Carbon Capture, Utilization and Storage (CCUS)

8.7 CARBON CAPTURE UTILISATION AND STORAGE (CCUS)

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NITI Aayog launched CCUS policy framework and its deployment mechanism in India.

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Initiative for CCUS in India

- **National Centres of Excellence:** Establishment of two National Centres of Excellence in CCUS at **IIT Bombay and JNCASR (Jawaharlal Nehru Centre for Advanced Scientific Research) Bengaluru**.
- These **centre will facilitate capturing and mapping of current R&D and innovation activities** in the domain and also develop networks of researchers, industries and stakeholders.
- **Mission Innovation Challenge on CCUS** to enable **near-zero CO_2 emissions from power plants and carbon-intensive industries.**
- **Accelerating CCS Technologies (ACT)** to facilitate R&D and innovation that can lead to the development of safe and cost-effective CCUS technologies.

About CCUS

- Encompasses technologies to **remove CO_2 from flue gas and the atmosphere**, followed by **recycling the CO_2 for utilisation** and determining safe and **permanent storage options**.
- **Flue gases are produced when coal, oil, natural gas, wood or any other fuel is combusted in an industrial furnace**, a power plant's steam-generating boiler, or other large combustion device.
- **CO_2 captured using CCU technologies is converted** into fuel (methane and methanol), refrigerants, building materials, etc.
- Captured gas is used **directly in fire extinguishers, pharma, food and beverage industries** as well as the **agricultural sector**.

Carbon Dioxide Uses

- Plants undergo photosynthesis, a process through which they convert carbon dioxide into oxygen while utilising both carbon and oxygen to produce carbohydrates.
- Carbon dioxide serves as a refrigerant and is also employed as a fire extinguisher.

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- In greenhouses, carbon dioxide is used to promote plant growth through a process known as carbon fertilisation. Additionally, it is a key component in carbonated beverages, soft drinks and beers, providing them with their characteristic fizziness.
- The application of solid carbon dioxide in the form of dry ice finds significant use in large-scale refrigeration processes.
- An everyday example of carbon dioxide usage is observed when baking powder or yeast releases it during the process of making cake in which its batter rise.

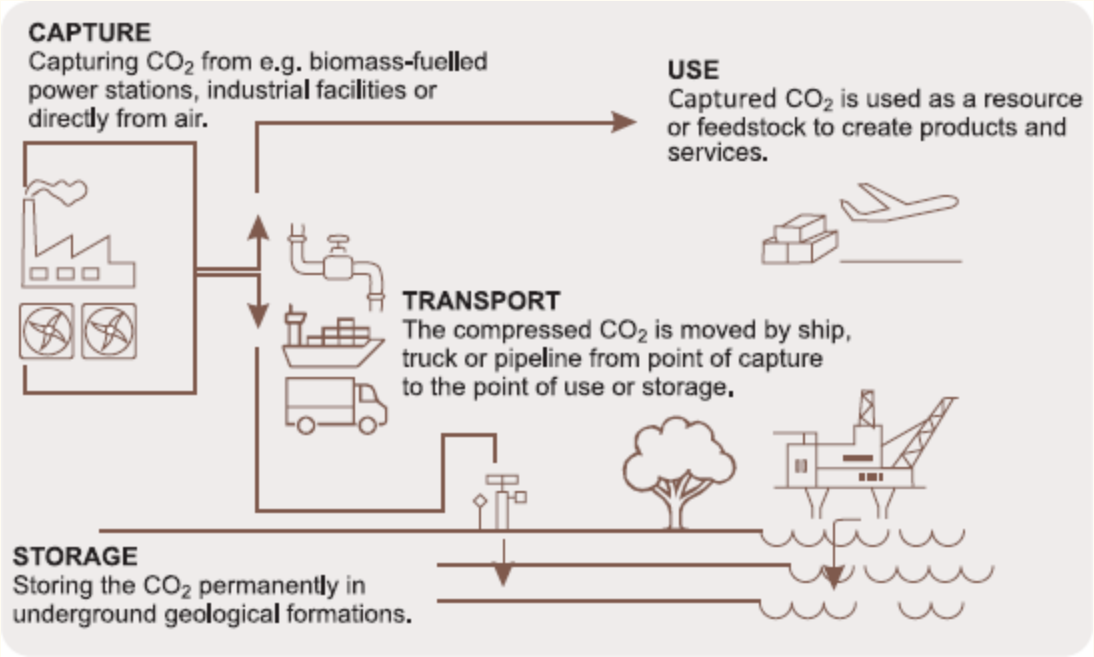

Figure 8.3 Carbon Capture, Utilisation and Storage (CCUS)

8.8 CLIMATE JUSTICE

- It is a phrase used to develop global warming as an ethical and political issue, rather than purely environmental or physical in nature.
- This can be undertaken by linking the climatic change effects to justice concept, specifically environmental and social justice and by exploring issues such as equality, human rights, collective rights and historical climatic responsibilities.

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- As per the statement 'climate justice' associates human rights and its development for achieving a human-centred approach, protecting the most vulnerable rights and sharing the climatic change benefits and burden and its resolution equitably and fairly.
- The Paris Agreement is the first international agreement to exclusively include the 'climate justice' concept.
- Its Preamble depicts: 'the importance of some of the concept of "climate justice", when addressing climate change'.
- It is used to recognise climate change as an ethical, legal and political issue, subsuming the environmental and social justice issues.
- The term 'climate justice' can be utilised to convey actual legal action on climatic change issues, that draws from and aims to achieve these values.
- Fundamentally, climate justice undertakes to ensure that planet and people are treated in the fair ways in which we:
  (a) attempt to decrease climate changes further, such as reducing the amount of burning of fossil fuels for producing energy (mitigation) and
  (b) adjusting to climatic changes such as producing crops resistant to droughts where there is a reduction in rainfall levels due to climate change (adaptation).

8.9 CARBON SPACE

- The amount of CO_2 that is allowed to be released into the atmosphere to maintain a habitable climate and not become a victim of dangerous climate change, is called 'carbon space'.
- In 2009, the Copenhagen Summit (United Nations Climate Change Conference) was held, where most of the countries including India, the United States, China, Japan and European countries have come forward and agreed upon using energy resources sustainably and move towards green fuel so that the global mean temperature does not go beyond 2°C.

Measures that can regulate carbon space are:

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- Using clean fuel to make cleaner emissions and setting a particular standard for CO_2 emissions.
- Proper waste disposal and treating them before releasing into the environment directly.
- Formulation and implementation of policies regarding clean air is an urgent requirement.
- These policies should be made by the members of a committee consisting of representatives from various stakeholders, government servants, scientists, politicians and the public.
- Promotion of afforestation at a large scale.
- Spreading awareness regarding degrading air quality, global warming, threats of climate change and how everybody can effectively regulate carbon space.
- According to Intergovernmental Panel on Climate Change (IPCC), there is a need to decline the global emissions by the year 2030 to 45% over 2010 levels to restrict the rise in temperature to 1.5°C.
- The global CO_2 emissions were 33 Gt in the year 2010. Hence, there is a need to restrict the annual CO_2 emissions level below 18.2 Gt in the year 2030 in order to meet this target. Even though, there would be 37.71 Gt CO_2 emission in 2030 on the attainment of enhanced Nationally Determined Contributions (NDC) which would be double the CO_2 amount which the world would emit in 2030.
- There would be an emission of 409 Gt CO_2 by the years 2020–2030, against the budget availability of 400 Gt CO_2.
- The voluntary mitigation submission of 142 countries of the total 195 signatories countries, as submitted to the United Nations Climate Framework Convention, there would be occupying of enhanced available carbon space by the year 2030 with the enhanced NDCs.
- China is responsible for 31%, and the other seven major countries such as the United States, the United Kingdom, Russia, Australia, Japan, Canada and the European Union are responsible for 34% of the world's total global emissions. Even though the fourth largest emitter in absolute in absolute terms, India is responsible for just 7% of total global emissions.

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- Hence, China and the developed world account for more than 70% of the carbon emissions to date.
- India will utilise just 7%, and China will undertake the remaining one-third of the global carbon space between 2020 and 2030.

ENV 30- Carbon Fertilization

8.10 CARBON FERTILIZATION

- The carbon fertilization effect is the increased rate of photosynthesis in plants, resulting from increased levels of carbon dioxide in the atmosphere.
- Simply, the speeding-up of photosynthesis is known as CO_2 fertilization—a well-known consequence of higher CO_2 concentrations, along with increased water use efficiency.
- It is the artificial enrichment of the greenhouse atmosphere through carbon dioxide, which is an important plant and vegetable nutrient.
- It helps in the production-level enhancement, both qualitatively and quantitatively.
- It is specifically good for cold climates and can be used for growing various types of vegetables (asparagus, celery, lettuce, tomatoes, aubergines, etc.), greenhouse fruit (strawberries) and ornamental plants.
- During high CO_2 conditions, plants lose less water through their leaves, and hence less impacted by dry conditions.
- During high concentration of carbon dioxide, plants maintain narrow openings of leaf surface which protects them from water loss.

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- Elevated CO_2 concentration increases plant growth and leaf photosynthesis, even under water stress conditions.
- This stimulated leaf photosynthesis at higher CO_2 concentration is associated with stomata diffusion processes under water stresses.
- Elevated CO_2 concentration enhances water use efficiency under water stresses.
- Plants grow faster leads to the sequestration of more carbon dioxide and the growth also increases crop yields.
- During high concentrations of carbon dioxide, plants maintain narrow openings of leaf surface which protects them from water loss.

'Arctic greening': Scientists have found that as northern portions of the planet warm up with an increase in atmospheric carbon dioxide, there's been an increase in plant growth in these regions.

Zinc and iron deficiencies: As per the scientists, the expected level of CO_2 in the century's second half will diminish the zinc, iron and protein levels in wheat, rice, peas and soya beans.

In the country of nearly two billion, people receive their 60% zinc or iron requirement from these crops.

Their deficiencies will result in the annual loss of nearly 63 million life-years.

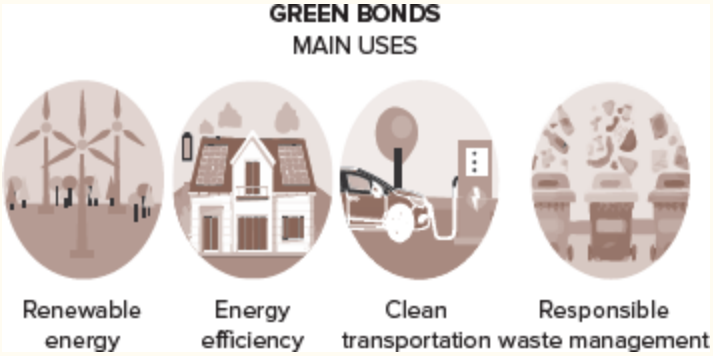

Figure 8.4 Different Uses of Green Bond

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ENV 31- Green Bonds and Green Jobs

8.11 GREEN BOND

In Budget Speech 2022, Finance Minister announced that the government has proposed to issue sovereign green bonds to mobilise resources for green infrastructure and to move towards carbon neutrality.

Climate change affects all of us. But it is expected to hit developing countries the hardest. Its potential effects on temperatures, precipitation patterns, sea levels and frequency of weather-related disasters pose risks for agriculture, food and water supplies. At stake are recent gains in the fight against poverty, hunger and disease and the lives and livelihoods of people in developing countries. The good news is that there are many solutions to address climate change, but they need financing. **Green bond is one of the financing options available to private firms and public entities to support climate and environmental investments**. Investors are attracted to green bonds because they allow a closer connection to positive social and environmental impacts.

What Are Green Bonds?

A green bond is a **debt instrument** which is used to raise **capital to fund 'green' projects (i.e., Climate and Environmental Projects)**. Green projects include projects relating to renewable energy, clean transportation, sustainable water management, etc. These bonds are typically asset-linked and backed by the issuing entity's balance sheet, so they usually carry the

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same credit rating as their issuers' other debt obligations. Green bonds may come with tax incentives to enhance their attractiveness to investors. The World Bank is a major issuer of green bonds. The processes used by the World Bank to issue more than 200 green bonds in 25 currencies are now international best practices, known as the Green Bond Principles, and have been adopted by the financial markets. **As of January 2023, green bonds have raised $2.5 trillion globally to support green and sustainable projects**.

Green bonds come with tax incentives such as tax exemption and tax credits, making them a more attractive investment compared to a comparable taxable bond. These tax advantages provide a monetary incentive to tackle prominent social issues such as climate change and a movement towards renewable sources of energy. To qualify for green bond status, they are often verified by a third party such as the Climate Bond Standard Board, which certifies that the bond will fund projects that include benefits to the environment.

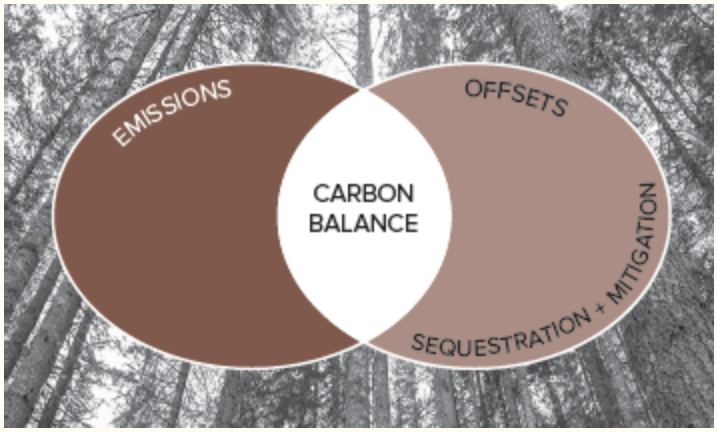

Figure 8.5 Relationship Among Emissions, Offsets, Sequestration and Mitigation with Carbon Balance

Blue Bonds

Blue bonds are sustainability bonds to finance projects that protect the ocean and related ecosystems.

- This can include projects to support sustainable fisheries, protection of coral reefs and other fragile ecosystems, or reducing pollution and acidification.

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All blue bonds are green bonds, but not all green bonds are blue bonds.

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Climate Bonds

'Green bonds' and 'climate bonds' are sometimes used interchangeably, but some authorities use the latter term specifically for projects focusing on reducing carbon emissions or alleviating the effects of climate change.

The climate bonds initiative is an organisation that seeks to establish a standard for certifying climate bonds.

Carbon Neutrality

Carbon neutrality is a state of net-zero carbon dioxide emissions.

This can be achieved by balancing emissions of carbon dioxide with its removal (often through carbon offsetting) or by eliminating emissions from society (the transition to the 'postcarbon economy').

The term is used in the context of carbon dioxide-releasing processes associated with transportation, energy production, agriculture and industry.

8.12 GREEN JOBS

Green growth is one of the seven top priorities of the Union Budget 2023–24 for ushering green industrial and economic transition, environmentally friendly agriculture and sustainable energy in the country. It will also generate a large number of green jobs.

What are Green Jobs?

Green jobs are decent jobs that contribute to preserving or restoring the environment, be they in traditional sectors such as manufacturing and construction, or in new, emerging green sectors such as renewable energy and energy efficiency. Jobs that have a direct, positive impact on the planet traditionally involve renewable energy, electric transport, energy efficiency or nature conservation. But now, as more sectors transition to low-carbon models, every job has the potential to become 'green'.

With unemployment rising due to the pandemic, there's now a chance to reconfigure the jobs landscape while putting the environment at the centre stage. At the enterprise level, green jobs can produce goods or provide

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services that benefit the environment, for example, green buildings or clean transportation.

However, these green outputs (products and services) are not always based on green production processes and technologies. Therefore, green jobs can also be distinguished by their contribution to more environment friendly processes. For example, green jobs can reduce water consumption or improve recycling systems. Green jobs defined through production processes do not necessarily produce environmental goods or services.

Green jobs help:

- Improve energy and raw materials efficiency
- Limit GHG emissions
- Minimise waste and pollution
- Protect and restore ecosystems
- Support adaptation to the effects of climate change

ENV 32- Green Washing

8.13 GREENWASHING AND ITS WORK

Greenwashing is the process of conveying a false impression or providing misleading information about how a company's products are more environmentally sound. Greenwashing is considered an unsubstantiated claim to deceive consumers into believing that a company's products are environment friendly.

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How Greenwashing Works?

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Greenwashing works by leveraging the rising demand for products that are perceived as environmentally sound. This can encompass items that are marketed as more natural, healthier, chemical-free, recyclable or less wasteful of natural resources.

The term ‘greenwashing’, also referred to as ‘green sheen’, was coined in the 1960s, and a prominent early example occurred within the hotel industry. Hotels displayed notices in their rooms urging guests to reuse their towels under the pretence of environmental conservation. However, the primary motivation behind this move was to reduce laundry costs for the hotels, rather than a genuine commitment to sustainability.

Other Types of Greenwashing

One common form of greenwashing is to include misleading labelling or bury environmentally unsound practices in the fine print. This can include use of terminology such as ‘eco-friendly’ or ‘sustainable’, which is vague and not verifiable.

Key Takeaways

- Greenwashing is a tactic used by companies to profit from the increasing interest in environment friendly products.
- It involves creating a misleading perception that a company or its offerings are environmentally conscious or eco-friendly, even if they may not truly adhere to sustainable practices.
- Critics have raised concerns about certain companies resorting to greenwashing in order to take advantage of the growing popularity of socially responsible and environmentally focused investments, known as Environmental, Social and Governance (ESG) investments.

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Questions Asked in UPSC Prelims Exam

UNIT II

Unit - II Prelims PYQs (2011 onwards)

1. Consider the following statements: **[UPSC (Pre) 2025]**

I. Without the atmosphere, temperature would be well below freezing point everywhere on the Earth's surface.

II. Heat absorbed and trapped by the atmosphere maintains our planet's average temperature.

III. Atmosphere's gases, like carbon dioxide, are particularly good at absorbing and trapping radiation.

Which of the statements given above are correct?

(a) I and III only

(b) I and II only

(c) I, II and III

(d) II and III only

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2. With reference to 'Direct Air Capture', an emerging technology, which of the following statements is/are correct? **[UPSC (Pre) 2025]**

I. It can be used as a way of carbon sequestration.

II. It can be a valuable approach for plastic production and in food processing.

III. In aviation, it can be a source of carbon for combining with hydrogen to create synthetic low-carbon fuel.

Select the correct answer using the code given below.

(a) I and II only

(b) III only

(c) I, II and III

(d) None of the above statements is correct

3. Consider the following statements: **[UPSC (Pre) 2025]**

I. Carbon dioxide (CO_2) emissions in India are less than 0.5 t CO_2/ capita.

II. In terms of CO_2 emissions from fuel combustion, India ranks second in the Asia-Pacific region.

III. Electricity and heat producers are the largest sources of CO_2 emissions in India. Which of the statements given above is/are correct?

(a) I and III only

(b) II only

(c) II and III only

(d) I, II and III

4. With reference to the planet Earth, consider the following statements: **[UPSC (Pre) 2025]**

I. Rain forests produce more oxygen than that produced by oceans.

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II. Marine phytoplankton and photosynthetic bacteria produce about 50% of world's oxygen.

III. Well-oxygenated surface water contains several folds higher oxygen than that in atmospheric air.

Which of the statements given above is/are correct?

(a) I and II

(b) II only

(c) I and III

(d) None of the above statements is correct

5. Artificial way of causing rainfall to reduce air pollution makes use of **[UPSC (Pre) 2025]**

(a) Silver iodide and potassium iodide

(b) Silver nitrate and potassium iodide

(c) Silver iodide and potassium nitrate

(d) Silver nitrate and potassium chloride

6. Consider the following statements: **[UPSC (Pre) 2024]**

Statement-I: The atmosphere is heated more by incoming solar radiation than by terrestrial radiation.

Statement-II: Carbon dioxide and other greenhouse gases in the atmosphere are good absorbers of long wave radiation.

Which one of the following is correct in respect of the above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I

(c) Statement-I is correct, but Statement-II is incorrect

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(d) Statement-I is incorrect, but Statement-II is correct

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7. With reference to "water vapour", which of the following statements is/are correct? **[UPSC (Pre) 2024]**

1. It is a gas, the amount of which decreases with altitude.
2. Its percentage is maximum at the poles. Select the answer using the code given below:

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

8. According to the Environmental Protection Agency (EPA), which one of the following is the largest source of sulphur dioxide emissions? **[UPSC (Pre) 2024]**

(a) Locomotives using fossil fuels
(b) Ships using fossil fuels
(c) Extraction of metals from ores
(d) Power plants using fossil fuels

9. Consider the following statements: **[UPSC (Pre) – 2023]**

1. Aerosols
2. Foam agents
3. Fire retardants
4. Lubricants

In the making of how many of the above are hydrofluorocarbons used?

(a) 1 Only
(b) 2 Only
(c) 3 Only

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(d) All four

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10. With the reference to the Earth's atmosphere which one of the following statements is correct? **[UPSC (Pre) – 2023]**

(a) The total amount of insolation received at the equator is roughly about 10 times of that received at the poles.

(b) Infrared rays constitute roughly two-third of insolation.

(c) Infrared waves are largely absorbed by water vapour that is concentrated in the atmosphere.

(d) Infrared waves are a part of visible spectrum of electromagnetic waves of solar radiation.

11. Consider the following activities: **[UPSC (Pre) – 2023]**

1. Spreading finally crowned basalt rock on farmlands extensively.
2. Increasing the alkalinity of oceans by adding lime.
3. Capturing carbon dioxide released by various industries and the pumping it into abandoned subterranean mines in the form of carbonated waters.

How many of the above activities are often considered and discussed for carbon capture and sequestration?

(a) 1 Only

(b) 2 Only

(c) All three

(d) None

12. Consider the following statements: **[UPSC (Pre) – 2023]**

Statement 1: Carbon markets are likely to be one of the most widespread tools in the fight against climate change.

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Statement 2: Carbon markets transfer resources from the private sector to the state. Which one of the following is correct in respect of the above statements?

(a) Both Statement 1 and Statement 2 are correct and Statement 2 is the correct explanation of Statement 1.

(b) Both Statement 1 and Statement 2 are correct and Statement 2 is not the correct explanation of Statement 1.

(c) Statement 1 is correct but Statement 2 is not correct.

(d) Statement 1 is not correct but Statement 2 is correct.

13. Among the following crops, which one is the most important anthropogenic source of both methane and nitrous oxide? [UPSC (Pre) – 2022]

(a) Cotton

(b) Rice

(c) Sugarcane

(d) Wheat

14. Which one of the following best describes the term "greenwashing"? [UPSC (Pre) – 2022]

(a) Conveying a false impression that a company's products are eco-friendly and environmentally sound.

(b) Non-inclusion of ecological/environmental costs in the Annual Financial Statements of a country.

(c) Ignoring the consequences disastrous ecological while infrastructure development undertaking.

(d) Making mandatory provisions for environmental costs in a government project/programme.

15. Consider the following: [UPSC (Pre) – 2022]

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1. Carbon monoxide
2. Nitrogen oxide
3. Ozone
4. Sulphur dioxide

Excess of which of the above in the environment is/are cause(s) of acid rain?

(a) 1, 2 and 3
(b) 2 and 4 only
(c) 4 only
(d) 1, 3 and 4

16. The 'Common Carbon Metric', supported by UNEP, has been developed for [UPSC (Pre) – 2021]

(a) assessing the carbon footprint of building operations around the world
(b) enabling commercial farming entities around the world to enter carbon emission trading
(c) enabling governments to assess the overall carbon footprint caused by their countries
(d) assessing the overall carbon footprint caused by the use of fossil fuels by the world in a unit time

17. In the context of India's preparation for Climate-Smart Agriculture, consider the following statements: [UPSC (Pre) – 2021]

1. The 'Climate-Smart Village' approach in India is a part of a project led by the Climate Change, Agriculture and Food Security (CCAFS), an international research programme.
2. The project of CCAFS is carried out under Consultative Group on International Agricultural Research (CGIAR) headquartered in

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France.

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3. The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) in India is one of the CGIAR's research centres.

Which of the statements given above are correct?

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

18. In rural road construction, the use of which of the following is preferred for ensuring environmental sustainability or to reduce carbon footprint? **[UPSC (Pre) – 2020]**

1. Copper slag
2. Cold mix asphalt technology
3. Geotextiles
4. Hot mix asphalt technology
5. Portland cement

Select the correct answer using the codes given below:

(a) 1, 2 and 3 only

(b) 2, 3 and 4 only

(c) 4 and 5 only

(d) 1 and 5 only

19. Which one of the following statements best describes the term 'Social Cost of Carbon'? It is a measure, in monetary value, of the: **[UPSC (Pre) – 2020]**

(a) long-term damage done by a tonne of CO_2 emissions in a given year.

(b) requirement of fossil fuels for a country to provide goods and services to its citizens, based on the burning of those fuels.

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(c) efforts put in by a climate refugee to adapt to live in a new place.

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(d) contribution of an individual person to the carbon footprint on the planet Earth.

20. The term "sixth mass extinction/sixth extinction" is often mentioned in the news in the context of the discussion of [UPSC (Pre) – 2018]

(a) Widespread monoculture practices in agriculture and large-scale commercial farming with indiscriminate use of chemicals in many parts of the world that may result in the loss of good native ecosystems.

(b) Fears of a possible collision of a meteorite with the Earth in the near future in the manner it happened 65 million years ago that caused the mass extinction of many species including those of dinosaurs.

(c) Large-scale cultivation of genetically modified crops in many parts of the world and promoting their cultivation in other parts of the world which may cause the disappearance of good native crop plants and the loss of food biodiversity.

(d) Mankind's over-exploitation/misuse of natural resources, fragmentation/loss of natural habitats, destruction of ecosystems, pollution and global climate change.

21. Which of the following statements best describes "carbon fertilisation"? [UPSC (Pre) – 2018]

(a) Increased plant growth due to increased concentration of carbon dioxide in the atmosphere.

(b) Increased temperature of Earth due to increased concentration of carbon dioxide in the atmosphere.

(c) Increased acidity of oceans as a result of increased concentration of carbon dioxide in the atmosphere.

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(d) Adaptation of all living beings on Earth to the climate change brought about by the increased concentration of carbon dioxide in the atmosphere.

22. In the context of mitigating the impending global warming due to anthropogenic emissions of carbon dioxide, which of the following can be the potential sites for carbon sequestration? [UPSC (Pre) – 2017]

1. Abandoned and uneconomic coal seams
2. Depleted oil and gas reservoirs
3. Subterranean deep saline formations

Select the correct answer using the codes given below:

(a) 1 and 2 only
(b) 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

23. The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere, because it absorbs [UPSC (Pre) – 2013]

(a) The water vapour of the air and retains its heat
(b) The ultraviolet part of the solar radiation
(c) All the solar radiations
(d) The infrared part of the solar radiation.

24. Acid rain is caused by the pollution of environment by [UPSC (Pre) – 2013]

(a) Carbon dioxide and nitrogen
(b) Carbon monoxide and carbon dioxide
(c) Ozone and carbon dioxide
(d) Nitrous oxide and sulphur dioxide

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25. Which of the following can be threats to the biodiversity of a geographical area? **[UPSC (Pre) – 2012]**

1. Global warming
2. Fragmentation of habitat
3. Invasion of alien species
4. Promotion of vegetarianism

Select the correct answer using the codes given below:

(a) 1, 2 and 3

(b) 2 and 3

(c) 1 and 4

(d) 1, 2, 3 and 4

26. Consider the following statements: **[UPSC (Pre) – 2012]**

Chlorofluorocarbons, known as ozone-depleting substances, are used

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurizing agents in aerosol cans

Which of the statement/s given above is/are correct?

(a) 1, 2 and 3 only

(b) 4 only

(c) 1, 3 and 4 only

(d) 1, 2, 3 and 4

27. Human activities in the recent past have caused the increased concentration of carbon dioxide in the atmosphere, but a lot of it does not remain in the lower atmosphere because of **[UPSC (Pre) – 2011]**

1. Its escape into the outer stratosphere.
2. The photosynthesis by phytoplankton in the oceans.

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3. The trapping of air in the polar ice caps.

Which of the statement/s given above is/are correct?

(a) 1 and 2

(b) 2 only

(c) 2 and 3

(d) 3 only

28. The formation of ozone hole in the Antarctic region has been a cause of concern. What could be the reason for the formation of this hole? **[UPSC (Pre) – 2011]**

(a) Presence of prominent tropospheric turbulence; and inflow of chlorofluorocarbons

(b) Presence of prominent polar front and stratospheric clouds; and inflow of chlorofluorocarbons

(c) Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons

(d) Increased temperature at polar region due to global warming

29. Consider the following, which can be found in the ambient atmosphere: **[UPSC (Pre) – 2010]**

1. Soot
2. Sulphur hexafluoride
3. Water vapour

Which of the above contribute to the warming up of the atmosphere?

(a) 1 and 2 only

(b) 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

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30. Due to their extensive rice cultivation, some regions may be contributing to global warming. To what possible reason/reasons is this attributable? **[UPSC (Pre) – 2010]**

1. The anaerobic conditions associated with rice cultivation cause the emission of methane.
2. When nitrogen-based fertilisers are used, nitrous oxide is emitted from the cultivated soil.

Which of the statement/s given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

31. Consider the following: **[UPSC (Pre) – 2010]**

1. Oxides of Hydrogen
2. Oxides of Nitrogen
3. Oxides of Sulphur

Which of the above causes/cause acid rain?

(a) 1 and 2 only

(b) 3 only

(c) 2 and 3 only

(d) 1, 2 and 3 Explanation

32. Which of the following conditions indicates the impact of global warming? Select the correct answer from the codes given below: **[UPSC (Pre) – 2009]**

1. Melting of glaciers
2. Rise in sea level
3. Changes in weather conditions
4. Rise in global temperature

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Code:

(a) 1 and 2

(b) 1, 2 and 3

(c) 2, 3 and 4

(d) All the above

33. Consider the following: **[UPSC (Pre) – 2008]**

1. Rice fields
2. Coal mining
3. Domestic animals
4. Wetlands

Which of the above are sources of methane, a major greenhouse gas?

(a) 1 and 4 only

(b) 2 and 3 only

(c) 1, 2 and 3 only

(d) 1, 2, 3 and 4

34. Which one of the following countries is the first country in the world to propose a carbon tax for its people to address global warming? **[UPSC (Pre) – 2006]**

(a) Australia

(b) Germany

(c) Japan

(d) New Zealand

Answers

1. (c) 2. (c) 3. (c) 4. (b) 5. (a) 6. (d) 7. (a) 8. (d) 9. (c) 10. (c)
11. (c) 12. (b) 13. (b) 14. (a) 15. (b) 16. (a) 17. (d) 18. (a) 19. (a) 20. (d)

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21. (a) 22. (d) 23. (d) 24. (d) 25. (a) 26. (c) 27. (b) 28. (b) 29. (d) 30. (c) 31. (c) 32. (d) 33. (d) 34. (d)

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Explanations

1. Correct answer (c): EXPLANATION

- The atmosphere is a blanket of gases that regulates the movement of heat from the sun to and from the Earth's surface.
- The atmosphere shields us from the harmful UV radiation that comes from the sun.
- Without the atmosphere, Earth's surface would, on average, be much colder, staying well below freezing. Hence **1 is right**
- It traps heat from the sun, preventing extreme temperature differences between day and night, keeping the day warm by not allowing all the heat to be trapped nor allowing the trapped heat to escape completely. Hence **2 is right**
- Aiding this process are greenhouse gases like carbon dioxide which are effective in trapping the heat in the atmosphere, and also a concern when concentration increases drastically. Hence **3 is right**

2. Correct answer (c): EXPLANATION

- **Direct Air Capture (DAC)** refers to the capture of carbon directly from the atmosphere using chemical processes, without relying on point source emissions. Hence, **I is correct**
- However, since the carbon in the atmosphere is diluted as compared to a point source, significant energy and costs are required in the DAC process.
- The DAC carbon can also be used as a feedstock for plastic production making it a possibly valuable approach, further permitting use of this low-carbon plastic in food packaging. The carbon can also be treated

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and suitably used for carbonation of beverages and for pest control of food items. Hence, **II is correct.**

- As per IEA (International Energy Agency), commercial efforts have started to create synthetic aviation fuel using DAC carbon, making it a low-carbon alternative. Hence, **III is correct.**

3. Correct answer (c): EXPLANATION

- **As of 2023, the World Bank has reported** India's per capita CO2 emissions at 2.1 metric tonnes, significantly higher than 0.5 tonnes. Hence, **I is not correct.**
- As per the **IEA (International Energy Agency),** India ranks second, following China, in terms of CO2 emissions from fuel combustion in the Asia Pacific region. Hence, **II is correct.**
- As per PIB report, the energy sector contributed 75.66% to the overall CO2 emissions. Hence, **III is correct.**

4. Correct answer (b): EXPLANATION

- Rainforests produce a significant amount of oxygen but the oceans produce more. The majority of Earth's oxygen comes from marine photosynthesis, primarily from phytoplankton. Hence, **I is not correct.**
- Marine phytoplankton and photosynthetic bacteria are responsible for a large portion of global oxygen production through photosynthesis, with estimates often centering around 50-70%. Hence, **II is correct**
- The concentration of oxygen in air is much higher than in water. Air is approximately 21% oxygen, while well-oxygenated surface water typically contains around 8-10 mg/L of dissolved oxygen, which is significantly less than in the atmosphere. Hence, **III is not correct.**

5. Correct answer (a): EXPLANATION

- Cloud seeding is a weather modification technique used to alter precipitation, often to increase rain or snow, reduce hail, or disperse fog.

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- It involves dispersing agents like silver iodide, potassium iodide, liquid propane and dry ice, and hygroscopic materials like table salt to stimulate cloud formation and enhance rainfall.
- **Silver iodide (AgI):** Acts as the primary cloud seeding agent because its crystalline structure is similar to ice, allowing it to form ice nuclei around which water vapor can condense to form rain.
- **Potassium iodide (KI):** Is often used as a supplementary seeding agent to improve the efficiency of the process, especially in colder clouds.
- Cloud seeding disperses particles using aircraft, ground-based generators, drones with electric charges, or infrared lasers to trigger rainfall.
- However concerns regarding effectiveness in reducing pollution and accumulation of seeding agents in sensitive ecosystems persist.

6. Correct answer (d): EXPLANATION

- The sun emits short wave radiation to the earth's surface, known as insolation, which is partly absorbed by the surface and rest is reflected, termed as albedo.
- The **atmosphere is heated more by *terrestrial radiation* (long-wave radiation from the Earth's surface) than by incoming solar radiation**. Most incoming solar radiation passes through the atmosphere, heating the Earth's surface, which then radiates heat back as long-wave radiation that is absorbed by greenhouse gases like carbon dioxide, warming the atmosphere. Hence **Statement I is incorrect.**
- The absorbed heat is radiated back as terrestrial long wave radiation. However, it does not entirely escape the Earth. Instead, it is **absorbed by the greenhouse gases present in the atmosphere like carbon dioxide, methane, water vapour** etc. Hence **Statement II is correct.**

7. Correct answer (a): EXPLANATION

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Water vapor is an important greenhouse gas that maintains the temperature of the earth by absorbing long wave terrestrial radiation. The amount of water vapor in the atmosphere is mainly dependent on temperature, pressure and source of evaporation.

- Water vapor is maximum at the tropical and subtropical regions where the temperature of the air is high, so less abundant towards polar regions. Hence it can hold more vapor and aided by higher evaporation rates. Hence **2 is not correct.**
- However, with increasing altitude the amount of water vapor decreases. This is because as the **height increases, air becomes cooler and less dense**. Hence the gas particles are not suspended easily. Moreover, with decreasing temperature and increasing distance from the source, the rate of evaporation at higher altitudes decreases. Hence **1 is correct.**

8. Correct answer (d): EXPLANATION

- As per the EPA, sulphur dioxide is the most concerning due to its highest concentration among the oxides of sulphur. It mentions burning of fossil fuels by power plants as the highest contributor of SO2 in the atmosphere.
- SO2 has both health and environmental effects: Health effects include respiratory issues and the conversion of SO2 into other oxides can lead to the accumulation of particulate matter in the human body. Environmental issues include acid rain and haze.

9. Correct answer (c): EXPLANATION

- Hydrofluorocarbons (HFCs) are primarily produced for use as **aerosol propellants, insulating foams, refrigerants and finds minor usage in solvents and fire retardants.**
- **HFCs are generally not used in the making of lubricants**, which are typically made from mineral oils, synthetic oils, or greases to reduce friction between moving parts.

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- The Kigali Amendment adds HFCs to the Montreal Protocol to phase out by the next three decades, helping to avoid up to 0.5 degree Celsius of global temperature rise by 2100, while continuing to protect the ozone layer.

10. Correct answer (c): EXPLANATION

- The long wave terrestrial radiation emitted by the earth's surface is mainly composed of infrared waves. Water vapour, being a greenhouse gas, absorbs most of the infrared radiation and then radiates back into the atmosphere. Hence option (c) is correct.

11. Correct answer (c): EXPLANATION

- **Enhanced rock weathering (ERW)** is a climate mitigation method where powdered basalt is spread on farmland. **Its magnesium and calcium react with atmospheric CO_2, forming stable carbonates or bicarbonates**, thereby storing carbon in soils and waterways.
- Ocean liming is a climate mitigation approach where hydrated lime is added to surface waters, increasing alkalinity and enhancing the ocean's ability to absorb and store atmospheric CO_2, by **reducing the acidity of the oceans.**
- **Carbon capture and storage (CCS)** involves capturing CO_2 from industrial sources, transporting it, and **injecting it, often as carbonated water**, into abandoned mines or other deep geological formations to safely store it and reduce atmospheric carbon.

12. Correct answer (b): EXPLANATION

- As per UNEP, carbon markets are tradeable credits for governments and the private to control greenhouse emissions. The **aims is to achieve climate targets** and implement climate actions cost effectively. Hence **Statement 1 is correct.**
- The **Article 6 of the Paris Agreement** lays down the standards and mechanism of International Carbon markets with multilateral governance under UNFCCC.

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- Carbon markets enable trading of carbon credits, each representing one tonne of GHG emissions reduced or removed, **involving transfer of resources like money or carbon taxes.** They include compliance markets with government-mandated caps and voluntary markets, incentivizing emission reductions and placing a price on carbon. Hence **Statement II is correct but it is not an explanation of how carbon markets play a role in countering climate change.**

13. Correct answer (b): EXPLANATION

- Methane is released from anaerobically decomposing organic matter. Mostly, rice cultivation undergoes flood irrigation which creates a lack of oxygen in the waterlogged soils. Hence, microbes decompose the organic matter and methane is released as by-product.
- As per IPCC, except for about 15% of upland rice, the rest of rice undergoes flood irrigation. Hence it contributes to 5-20% of anthropogenic methane emissions.
- Nitrous-oxide is released due to application of nitrogenous fertilizers on crops.
- **System of Rice Intensification (SRI)** is a suitable alternative method of cropping in order to reduce emissions. It involves-
- Planting of younger seedlings as they display greater resilience
- Wider plantation of seedlings to reduce competition for nutrients and improved resource use efficiency
- Use of organic fertilizers to improve soil health and reduce nitrogenous emissions.
- Alternate Wetting and Drying method in SRI reduce water consumption and prevents anaerobic decomposition thereby reducing the amount of methane emissions.

14. Correct answer (a): EXPLANATION

As per the UN, **greenwashing refers** to deceptive tactics used by entities to **mislead the public into believing that their actions are**

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environmentally sound, but when in reality, it is otherwise. It can be manifested in a number of ways:

- Claiming to be on track towards environment goals without any credible plan in place.
- Being purposefully vague about the products used or the processes and quality to prevent scrutiny.
- Exaggerating achievements or actions taken to meet environment sustainability criteria without any actual improvement in standards.
- Communicating sustainability of a certain product in isolation while undermining environmental standards for other activities by the entity.

15. Correct answer (b): EXPLANATION

- **Acid Rain** refers to a broad term encompassing any form of precipitation from the atmosphere containing acidic compounds like carbonic acid, sulphuric acid, sulphurous acid and nitric acids etc.
- Acid rain is caused by emissions of sulphur dioxide and nitric oxides, which react with water and oxygen in the presence of sunlight to form acids.
- Emissions can also occur naturally, such as through volcanic eruptions, in small amounts. However, most emissions are due to the burning of fossil fuels.

16. Correct answer (a): EXPLANATION

- The **Common Carbon Metric (CCM),** developed by UNEP, provides a standardised system to measure the climate impact of buildings, which currently contribute a third of global energy use and significant CO_2 emissions.
- Its objective is to offer a uniform method for the building sector to assess, report, and compare energy use and greenhouse gas emissions from building operations.

17. Correct answer (d): EXPLANATION

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- The **Climate-Smart Village (CSV) approach**, part of the Climate Change, Agriculture and Food Security (CCAFS) program, seeks to enhance agricultural practices to better adapt to and mitigate the impacts of climate change.
- **CCAFS is a research program under the CGIAR framework** (formerly the Consultative Group on International Agricultural Research), headquartered in Montpellier, France.
- It **focuses on linking climate change research with agriculture and food security**, developing strategies to help farmers adapt to changing conditions while promoting sustainable and resilient farming practices.
- The **International Crops Research Institute for the Semi-Arid Tropics (ICRISAT), headquartered in Hyderabad**, India, is part of the CGIAR network and ranks among its 15 global research centres, focusing on improving agriculture in semi-arid regions.

18. Correct answer (a): EXPLANATION

- **Copper slag** is a **byproduct of copper smelting** and contains heavy metals, which are harmful as they can leach into soil and water. However, when processed and stabilised, **it can be used as a cost-effective aggregate in rural road construction**, improving strength and durability while recycling industrial waste and reducing environmental pollution.
- **Cold mix asphalt** is more environmentally friendly than hot mix asphalt, as it requires no heating of bitumen, thereby reducing fuel consumption and greenhouse gas emissions. It also reduces the release of volatile organic compounds (VOCs), fumes, and particulate matter, leading to cleaner air and healthier working conditions.
- **Geotextiles** are **permeable synthetic fabrics** that can be used in road construction for drainage and reinforcement. They prevent mixing of soil and aggregates, improve load distribution, reduce rutting, and extend pavement life. Moreover, it reduces the need for thick aggregate layers of material, which **cuts down on emissions** due to logistics and preparation.

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- **Portland cement** and **hot asphalt technology** are major emitters of CO_2, hence not eco-friendly. **Portland cement** generates through two main processes: the chemical reaction of calcination and the high heat required for the kiln, which is primarily produced by burning fossil fuels, those generate CO_2.

19. Correct answer (a): EXPLANATION

- The **Social Cost of Carbon (SCC)** estimates the economic damages caused by emitting one extra tonne of CO_2 annually.
- It tries to represent the current value of future harms like reduced crop yields, health risks, flood-related damages, and lost ecosystem services, helping weigh carbon-reduction benefits against policy costs.

20. Correct answer (d): EXPLANATION

- A **mass extinction** is a relatively brief geological period during which a large share of Earth's biodiversity ranging from microbes to plants and animals becomes extinct. So far, there have been five mass extinctions, and it is believed that we are in the midst of a sixth mass extinction.
- Unlike the previous events, the **sixth mass extinction is driven by anthropogenic causes** primarily due to unsustainable use of natural resources, emissions and climate change.
- Initiatives to prevent or delay the sixth mass extinction include: Paris Agreement, 30 by 2030 initiative and Kunming - Montreal Agreement, among others.

21. Correct answer (a): EXPLANATION

- Increased plant growth due to increased atmospheric carbon dioxide is called **carbon fertilization**.
- This happens because carbon dioxide is essential for photosynthesis, so higher concentrations can increase the rate of this process, leading to faster growth, increased biomass, and greater water-use efficiency in plants.

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22. Correct answer (d): EXPLANATION

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- The process of capturing carbon directly from emission sources and from the atmosphere, and subsequently storing it in natural or artificial reservoirs, is termed **carbon sequestration.**
- **Abandoned and uneconomic coal seams** can be used as a carbon sink to store carbon dioxide through a process called carbon sequestration. These seams can trap as the gas molecules attach to the surface of the coal.
- **Depleted oil and gas reservoirs** can function as carbon sinks through carbon capture and storage (CCS) projects, as they are proven to be secure geological storage sites that have held hydrocarbons for millions of years.
- **Deep saline aquifers** are underground geological formations containing saltwater (brine) that are unsuitable for drinking or agriculture. Deep saline aquifers act as a CO_2 sink by injecting captured carbon dioxide into the saline water, where it is stored through several mechanisms: **dissolution** into the brine, **structural trapping** under a layer of impermeable rock, and eventually, **mineral trapping** as it reacts with minerals to form solid carbonates.

23. Correct answer (d): EXPLANATION

Carbon dioxide traps heat by absorbing infrared radiation emitted from Earth's surface, re-radiating some of it back down and increasing the atmosphere's temperature. This is the "greenhouse effect": incoming solar radiation is mostly visible light, which CO_2 doesn't strongly absorb, but Earth's re-emitted infrared radiation is absorbed by CO_2 and other greenhouse gases, slowing the rate at which heat escapes into space.

- **Absorption:** CO_2 molecules absorb long-wave infrared radiation emitted by the Earth's surface after it has been warmed by the sun. This absorbed energy causes the CO_2 molecules to vibrate.

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- **Re-radiation:** The CO_2 molecules then re-emit this energy as infrared radiation in all directions, including back down towards Earth's surface.
- **Trapping heat:** This process traps heat in the lower atmosphere, preventing it from escaping directly into space, and leading to a gradual increase in the atmosphere›s temperature.

24. Correct answer (d): EXPLANATION

- **Acid Rain** refers to a broad term encompassing any form of precipitation from the atmosphere containing acidic compounds like carbonic acid, sulphuric acid, sulphurous acid and nitric acids etc.
- Acid rain is caused by emissions of sulphur dioxide and nitric oxides, which react with water and oxygen in the presence of sunlight to form acids.
- Emissions can also occur naturally, such as through volcanic eruptions, in small amounts. However, most emissions are due to the burning of fossil fuels.

25. Correct answer (a): EXPLANATION

Biodiversity loss refers to the altered character of the quantity and type of flora and fauna in a particular geographical area attributed to a number of factors. Mostly anthropogenic factors cause threats to biodiversity, such as:

1. Global Warming induces climate change
2. Habitat fragmentation and degradation
3. Invasion of Alien Species
4. Overexploitation of natural resources
5. Pollution

Promotion of vegetarianism refers to the adoption of a plant-based lifestyle that is associated with improved health and the reduction of

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carbon footprint by the meat industry. However, it is not associated directly with loss of Biodiversity.

26. Correct answer (c): EXPLANATION

- Chlorofluorocarbons (CFCs) are synthetic compounds once widely used in *refrigeration, air conditioning, aerosol propellants, and foam-blowing agents etc.*
- *Chlorofluorocarbons (CFCs)*, particularly **CFC-11 and CFC-12**, were used as "blowing agents" in the **production of plastic foams** like *polyurethane* and *polystyrene*, as they helped create the porous structure of the foam by turning into a gas during the manufacturing process.
- **Chlorofluorocarbons (CFCs) were never used in the development or production of tubeless tyres**. The manufacturing process of tubeless tyres primarily relies on various types of rubber, carbon black, sulfur for vulcanization, and other additives; CFCs are not a typical component or processing agent in this industry.
- CFCs, particularly **CFC-113**, were used for **cleaning electronic components** because of their effectiveness as a non-flammable solvent for grease and flux residues.
- *Chlorofluorocarbons (CFCs)* were **widely used as pressurizing agents in aerosol cans** because they are non-toxic, non-flammable, and stable chemicals that effectively propel products like hairspray, deodorant, and insecticides out of the can.

27. Correct answer (b): EXPLANATION

- Carbon dioxide, unlike hydrogen and helium, is a *heavy gas and hence does not have enough kinetic energy to escape the lower atmosphere*. Hence, **1 is incorrect.**
- Oceans and polar ice caps are natural carbon sinks that significantly absorb a lot of carbon dioxide from the atmosphere.
- The phytoplankton in the ocean absorb carbon dioxide from the atmosphere for photosynthesis. Hence, **2 is correct.**

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- CO_2 requires extremely low temperatures to become a solid (sublimation point of −78.5°C at standard atmospheric pressure). While parts of Antarctica can occasionally reach these temperatures, it is not a widespread or consistent enough phenomenon to pull large amounts of CO_2 out of the atmosphere and store it as dry ice. Hence, **3 is incorrect.**

28. Correct answer (b): EXPLANATION

- **Polar front:** In the *winter, a strong polar vortex forms around the South Pole*, creating a powerful vortex of winds that isolates the air inside.
- **Polar stratospheric clouds (PSCs):** The trapped air inside the vortex becomes extremely cold, allowing PSCs to form. These clouds are crucial because they provide a surface for chemical reactions.
    - They *convert inactive chlorine-containing compounds into reactive forms.*

Chlorofluorocarbons (CFCs): These are *long-lasting, human-made chemicals* that rise from the lower atmosphere into the stratosphere. When the sunlight returns after winter, it triggers the breakdown of CFCs into chlorine.

- On the PSCs, inactive chlorine is converted into a highly reactive form, such as ClO (Chlorine Oxide Radical). This reactive chlorine then rapidly destroys ozone molecules in catalytic cycle, causing the formation of the “Ozone hole”

29. Correct answer (d): EXPLANATION

- **Soot,** also known as *black carbon*, is a potent greenhouse gas that absorbs heat from the atmosphere as well as *reduces the reflectivity of polar snow, thereby reducing the albedo effect*. It comes in the *category of PM 2.5* and is a *product of incomplete combustion.*
- **Sulphur Hexafluoride** (SF6) is a *synthetic compound* with a very stable molecular structure and a *global warming potential of about*

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22000 times that of carbon dioxide, making it a potent greenhouse gas. Due to its unique dielectric properties, it is extensively used in electrical fields such as for insulation, current interruption and in semiconductor industries.

- **Water vapor** is the *most abundant greenhouse gas* and a *major contributor to the greenhouse effect*, its concentration is not directly controlled by human activity. The *main source of water vapor in the atmosphere* is **evaporation from oceans, lakes, and rivers**, driven by the Sun's energy.

30. Correct answer (c): EXPLANATION

- Methane is released from anaerobically decomposing organic matter. Mostly, rice cultivation undergoes flood irrigation, which creates a lack of oxygen in the waterlogged soils. Hence, microbes decompose the organic matter, and methane is released as a by-product.
- As per IPCC, except for about 15% of upland rice, the rest of the rice undergoes flood irrigation. Hence, it contributes to 5-20% of anthropogenic methane emissions.
- Whenever farmers applied nitrogenous fertilizer in soil, it emitted Nitrous oxide in atmosphere in the presence of sunlight and also became a cause of global warming effect.

31. Correct answer (c): EXPLANATION

- **Acid Rain** refers to a broad term encompassing any form of precipitation from the atmosphere containing acidic compounds like carbonic acid, sulphuric acid, sulphurous acid and nitric acids etc.
- Acid rain is caused by emissions of sulphur dioxide and nitric oxides, which react with water and oxygen in the presence of sunlight to form acids.
- Emissions can also occur naturally, such as through volcanic eruptions, in small amounts. However, most emissions are due to the burning of fossil fuels.

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32. Correct answer (d): EXPLANATION

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- **Melting of glaciers -** Higher global temperatures cause rapid ice and glacier melt, especially in the Himalayas, Arctic, and Antarctic.
- **Rise in sea level -** Melting glaciers and thermal expansion of seawater from warming contribute to global sea-level rise.
- **Changes in weather conditions -** Global warming increases the frequency and intensity of extreme weather (heatwaves, cyclones, floods, droughts).
- **Rise in global temperature -** Direct effect of greenhouse gas accumulation leading to an overall warming trend in Earth's climate system.

33. Correct answer (d): EXPLANATION

Methane is released from anaerobically decomposing organic matter.

- **Rice fields:** Mostly, rice cultivation undergoes flood irrigation, which creates a lack of oxygen in the waterlogged soils. Hence, microbes decompose the organic matter, and methane is released as a by-product.
- **Coal mining:** Coal mines have naturally trapped methane in the seams. If coalbed extraction is not performed, then the trapped methane can leach out into the atmosphere through ventilation and drainage systems and through ground fissures.
- **Domestic Animals:** Ruminant animals such as cows, sheep, and goats produce methane during *enteric fermentation*, where gut microbes digest fibrous plants in their multi-chambered stomachs. This process generates methane, mostly expelled into the atmosphere through belching.
- **Wetlands:** Similar to rice fields, wetlands release methane when microbes start deposing the organic matter in the waterlogged areas.

34. Correct answer (d): EXPLANATION

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- Finland was the first country to introduce a carbon tax in 1990. The concept of taxing activities that cause pollution was first articulated by

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economist **Arthur Pigou** in the 1920s.

- The confusion likely stems from Finland and New Zealand having different approaches to carbon pricing: **Finland uses a national carbon tax, which is a direct price on emissions**, while **New Zealand uses a domestic Emissions Trading Scheme (ETS), which is a cap-and-trade system**.
- Finland's tax applies directly to the carbon content of fuels, and the government sets the price. New Zealand's ETS sets a limit (cap) on total emissions and allows companies to buy and sell permits to emit, with the price fluctuating based on market demand.

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Unit III International Conference & Summits

Environmental Movements, Organisations, International Cooperation and Laws in the World

9

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9.1 Related Terminology

9.1.1 Summits

9.1.2 Meeting

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9.1.5 Protocol

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9.1.7 Congress

9.2 Principles of International Environmental Law

9.3 Environmental Conventions After UN

9.3.1 Green Issues

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9.3.3 ‘Green Polities’, ‘Eco-greens’ or ‘Green Movement’

9.3.4 Green Party

9.4 Stockholm Declaration/UN Conference on Human Environment (1972)

9.5 United Nations Environmental Programme (UNEP)

9.6 Brundtland Commission (1987) / World Commission on Environment and Development (WCED)

9.7 United Nations Conference on Environment and Development (UNCED)/The Earth Summit (1992)

9.7.1 Rio Declaration

9.7.2 Agenda 21

9.7.3 The Forest Principles/Rio Forest Principles

9.7.4 United Nations Framework Convention on Climate Change (UNFCCC)

9.8 Reduction Emission from Deforestation and Forest Degradation (REDD & REDD+)

9.9 Forest Carbon Partnership Facility (FCPF)

9.10 Bio Carbon Fund, Initiative for Sustainable Forest Landscape (ISFL)

9.11.1 Carbon Fund vs. Biocarbon Fund

9.11 Climate Action Tracker (CAT)

9.12 EP100

9.13 International Code for Ships Operating in Polar Waters (Polar Code)

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9.14 Greenhouse Gas Protocol
9.15 Un Convention on Combat Desertification (UNCCD)
9.16 Un Commission on Sustainable Development (CSD)

19.16.1 RIO +5 | 1997 Newyork, United States

19.16.2 Rio+10 | 2002 World Summit on Sustainable Development (WSSD) | Johannesburg Declaration

19.16.3 RIO+20 | 2012 RIO De Janeiro, Brazil

9.17 Sustainable Development Goals
9.18 Vienna Convention, Montreal Protocol & Kigali Agreement
9.19 Global Environmental Facility (GEF)
9.20 United Nations Educational, Scientific and Cultural Organisation (UNESCO)
9.21 Conservation International (CI)
9.22 World-Wildlife Fund (WWF)
9.23 Trade-Related Analysis of Fauna and Flora in Commerce (TRAFFIC)
9.24 Un Convention on the Law of the Seas (UNCLOS), 1982
9.25 Basel Convention
9.26 Rotterdam Convention
9.27 IPCC (Intergovernmental Panel on Climate Change)
9.28 Global International Water Assessment (GIWA)
9.29 Millennium Ecosystem Assessment (MA)
9.30 Stockholm Convention on Persistent Organic Pollution
9.31 Global Climate Change Alliance (GCCA)
9.32 Climate and Clean Air Coalition to Reduce Short-Lived Climate Pollutants (CCAC)
9.33 Minamata Convention on Mercury
9.34 International Whaling Commission (IWC)
9.35 Food and Agriculture Organisation (FAO)

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9.35.1 Globally Important Agricultural Heritage System (GIAHS)

9.35.2 Global Alliance for Climate-Smart Agriculture (GACSA)

9.36 The World Heritage Convention

9.37 Convention on Biodiversity (CBD)

9.38 European Union Nature Restoration Law

9.39 The Economics of Ecosystems and Biodiversity (TEEB)

9.40 International Treaty on Plant Genetic Resources for Food and Agriculture

9.41 CITES (Convention on International Trade on Endangered Species)

9.42 Convention on Conserving Migratory Species (CMS)

9.43 Ramsar Convention (Convention on Wetlands)

9.43.1 Wetland Conservation in India

9.44 New York Declaration on Forests

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9.46 100 Million Farmers (100MF) – World Economic Forum Initiative

ENV 33- Terminology

9.1 RELATED TERMINOLOGY

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9.1.1 Summits

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It is a productive inter-governmental meeting where states' heads meet to discuss about important topics for which constructive debate is required. They are accompanied by media exposure, tight security, and a pre-planned agenda. One from a respectful post is always invited to a summit because his/her opinion is valued. Summits are generally formal in nature.

9.1.2 Meeting

An event where participants actively attend educational sessions, participate in discussions, social functions.

9.1.3 Convention

It is described as a gathering of individuals, delegates or sometimes even representatives and members of a particular company or country where a discussion is carried on the topic of common interest for all. The minutes or achievements are then published in a written legal document or a treaty.

9.1.4 Treaty

When two international parties or nations come to an agreement after discussing about the topic concerned on a global scale, it can be referred to as a treaty between two or many countries.

9.1.5 Protocol

A legally bound document that is open to corrections and alterations and actually amend the main treaty, is known as a protocol.

9.1.6 Conference

A gathering of participants targeted mainly for discussing, finding solutions for problems, analysing facts and consulting other organisations to share

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and exchange views, ideas, strategies, indulge in a debate to bring out a positive outcome regarding the topic concerned.

9.1.7 Congress

The gathering of a large group of individuals with a pre-planned schedule at a regular interval of time for discussion on the subject concerned is known as congress. It may last for a couple of days and multiple sessions until the target is reached.

9.2 PRINCIPLES OF INTERNATIONAL ENVIRONMENTAL LAW

Environmental Principles	Brief Explanation
1. State Responsibility	A State is liable for viol obligations.
2. Good Neighbourliness	States must use their t neighbouring states.
3. Cooperation	States must work toget problems.
4. Sustainable Development	Development must me future generations.
5. Polluter Pays	The polluter must bear

9.3 ENVIRONMENTAL CONVENTIONS AFTER UN

- The **developmental phase (second phase)** of international environmental laws began with the creation of **UN and its specialised**

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agencies in 1945.

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- The earliest global-level environmental conference was organised in Stockholm in 1972 under the aegis of UNESCO. Popularly referred to as the United Nations Conference on the Human Environment, it emerged from deliberations held by an intergovernmental group of experts on the scientific foundations of resource conservation and rational utilisation.

9.3.1 Green Issues

In the early 1970s, human impact on the environment led to new social, political, and the ecological movements. The so-called 'green issue' grew in importance, fuelling environmental-ecological movements worldwide, including in India. These are classified as 'new social movements' because of public interest in the natural environment.

9.3.2 Greenpeace

Green Peace was Founded in 1971. The term "Greenpeace" was first used by a Canadian environmentalist. Its **headquarters is at** Amsterdam, Netherlands. It has **Global Reach** in **40 countries** worldwide. It emerged particularly for growing environmental consciousness in the late 1960s and early 1970s, particularly concerning nuclear testing, ocean pollution, and deforestation.

Objectives and Mission of Green Peace: It promotes **environmental protection** and sustainable practices globally and create awareness about **Climate change, Deforestation, Overfishing, Genetic engineering (GMOs) & Non-nuclear issues**, that promote policy measures and practical actions to ensure a healthier and safer environment for all

Methods and Activities: Campaigning, Direct Action, Research and Advocacy & Awareness Programs

Achievements and Influence: It is one of the most recognized and influential environmental NGOs globally. It helped in shaping

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international environmental policies & treaties and Inspired the formation of many **national and regional environmental movements.**

Significance in India: It acts as a model for environmental activism and supports awareness campaigns on deforestation, climate change, and marine conservation

Key Characteristics: It is **non-governmental, non-profit and have Global network -** operates in multiple countries but with coordinated campaigns. It is known for **peaceful but bold activism.**

9.3.3 'Green Polities', 'Eco-greens' or 'Green Movement'

As an outcome of these conferences, environmental movements emerged under various names such as 'Green Polities', 'Eco-greens' and the 'Green Movement', particularly in Germany and North America during the 1980s. By 1989, European elections brought environmental issues into the political mainstream, with Green parties across Europe prioritising them in their campaigns.

9.3.4 Green Party

The Green Party is a political movement rooted in ecological awareness and participatory democracy. The idea of a Green Party is not entirely new, as Western countries have witnessed several political parties and alliances formed on an environmental platform. For example, the German Green Party (now Alliance '90/The Greens), established in 1980, is among the oldest and most prominent. Many such parties have entered national parliaments, and some have also gained representation in the European Parliament.

Aim: Green Party aims to create an equitable and sustainable society with the focus of efforts primarily, though not exclusively, through the electoral system.

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- ***Future prospects of Green Party***

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- **To oppose genetically modified organisms and corporate control over agriculture.**
- **To resist unchecked urban expansion by builders that threatens soil and water bodies.**
- **To expand democratic space where it is increasingly constrained by various forces.**

9.4 STOCKHOLM DECLARATION/UN CONFERENCE ON HUMAN ENVIRONMENT (1972)

The United Nations Conference on the Human Environment (UNCHE) or Stockholm Declaration was held in Stockholm (Sweden) in 1972. **June 5th** was declared World Environment Day to mark the first day of the 1972 Stockholm Conference, the **first major UN conference to address environmental issues.**

■ *Total 3 dimensions of Stockholm Conference*

- Countries agreeing **not to harm each other's environment**
  - **Formulation of a structured plan to assess and examine threats to the Earth's environment.**
- Creation of an international institution, the United Nations Environment Programme (UNEP), to promote cooperation and coordination among nations

■ *Outcome of Stockholm Conference*

- The United Nations Environment Programme (UNEP) was created as a result of this conference.
- The conference encouraged nations worldwide to systematically monitor environmental conditions and to establish dedicated environmental ministries and regulatory bodies.
- **Department of Environment (1980) was created in India.**
- **Environment Ministry (1985) was created in India.**

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Exam Pointer

50 YEARS OF STOCKHOLM CONFERENCE

The **Stockholm+50 conference** was held in Stockholm, Sweden, to commemorate 50 years since the landmark 1972 United Nations Conference on the Human Environment, commonly known as the Stockholm Conference. The 1972 event was pivotal in bringing environmental issues to the forefront of global discussions for the first time.

Theme:

- ***"A healthy planet for the prosperity of all – our responsibility, our opportunity."***

Agenda:

- Sharing experiences and best practices to advance efforts in protecting the planet.
- Promoting a sustainable and inclusive recovery in the post-COVID-19 era.

Other outcomes and successes of the Stockholm Conference

Identification of a theme of sustainable development: 20 years later, the 1992 UN Conference on Environment and Development – the Earth Summit – in Rio de Janeiro defined sustainable development.

Establishing Key Principles

Precautionary Principle: The Vienna Convention for the Protection of the Ozone Layer (1985) was the first multilateral environmental agreement to formally adopt precautionary approaches in environmental protection. This principle emphasises taking preventive action in the face of uncertainty to avoid harm to the environment.

Polluter-Pays Principle (PPP): This principle places the responsibility for preventing and controlling pollution on those who generate it. Polluters are required to bear the costs associated with safeguarding human health and protecting the natural environment.

Stockholm+50: Action-Oriented Recommendations

1. **Human Well-Being:** Prioritise human well-being as the cornerstone of a healthy planet and shared prosperity.
2. **Environmental Rights:** Recognise and enforce the right to a clean, healthy, and sustainable environment.
3. **Economic Reforms:** Implement economy-wide changes to tackle existing environmental challenges effectively.
4. **Sectoral Transitions:** Accelerate transitions in industries and sectors with significant environmental impacts.
5. **Support for Developing Countries:** Facilitate access to digital tools and technological assistance for developing nations to address environmental issues.

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ENV 34 - UN Environment Programme (UNEP)

9.5 UNITED NATIONS ENVIRONMENTAL PROGRAMME (UNEP)

- UNEP has been accountable for the founding and application of both local and universal treaties focusing on the depletion of ozone.
- It was established in **1972** in **Nairobi**.
- The United Nations Environment Programme (UNEP) is the leading environmental authority in the United Nations system.
- UNEP strengthens environmental practices and policies with the help of its skill while implementing commitment in the environmental field at the regional, country, and global levels.
- Its aim is, catering leadership and encouraging cooperation to take care of the environment by encouraging, informing, and helping countries and communities in improving the life quality without compromising for future generations.
- UNEP collaborates with all 193 Member States, along with civil society organisations, the private sector, and other key stakeholders, to respond to global environmental challenges through the UN Environment Assembly, which serves as the highest decision-making forum on environmental matters.
- It also provides administrative support by hosting the secretariats of several important multilateral environmental agreements and scientific bodies, including the Convention on Biological Diversity (CBD), the Minamata Convention on Mercury, the Convention on Migratory Species,

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and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).

- The following global conventions were signed under the sponsorships of UNEP:
- Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), adopted in Washington in 1973. Convention on Migratory Species (CMS), concluded in Bonn in 1979. Vienna Convention for the Protection of the Ozone Layer (Vienna, 1985), along with the Montreal Protocol on Substances that Deplete the Ozone Layer (Montreal, 1987). Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal, adopted in Basel in 1989.
- UNEP facilitates the implementation of environmental policies and actions through coordination among different departments and agencies. It collaborates with a wide range of partners, including UN bodies, international institutions, national governments, non-governmental organisations, the private sector, and civil society, and is actively involved in financing and executing environment-related projects.
- United Nations Environmental Programme also hosts several environmental conventions like Ozone Secretariat, Montreal's, CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora), the Convention on Biological Diversity, the Convention on Migratory Species, Basel Convention, Stockholm Convention on Persistent Organic Pollution (POPs) etc.
- **Major Programmes of the UNEP:**
- **Billion Tree Campaign** — The campaign drew inspiration from **Wangari Maathai**, a Nobel Peace Prize laureate and founder of the Green Belt Movement, which began its work in Africa in 1977 and went on to plant over 30 million trees. In response to the growing challenges of climate change and broader sustainability concerns such as water security and biodiversity loss, the **United Nations Environment Programme** launched the Billion Tree Campaign in 2006. The programme initially set a target of planting one billion trees by 2007.

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- **Pain for the Planet**
- **Awareness and Preparedness for Emergencies at Local Level (APELL)**
- **Faith for Earth**–Launched in 2017, to encourage, empower and engage with faith-based organisations as partners, at all levels, towards achieving the Sustainable Development Goals and fulfilling the 2030 Agenda.
- **Clean Up the World** — This is a global environmental campaign conducted annually during the **third week of September**. It originated in **Sydney, Australia, in 1993** and is organised in partnership with the **United Nations Environment Programme**. The campaign aims to unite non-governmental organisations, community groups, educational institutions, governments, businesses, and individuals in collective actions such as improving water quality, cleaning streets, parks, forests, rivers, and canals, while also promoting environmental awareness among children.
- **Seal the Deal**

Do You Know?

The World Meteorological Organisation and UNEP established the Intergovernmental Panel on Climate Change (IPCC) in 1988.

CHAMPIONS OF THE EARTH AWARDS

'Global 500' **awards**:. The **United Nations Environment Programme** conferred the **Global 500** awards for contributions to environmental protection between **1987 and 2003**, after which the programme was discontinued and later succeeded by the **Champions of the Earth** award in **2005**. It was awarded to any individual or organisation that deals with environment protection related issues at political, social and legal levels.

Champions of the Earth — The **United Nations Environment Programme** instituted the *Champions of the Earth* awards in **2004** as an annual initiative to honour exceptional leadership in environmental policy. Each year, **six laureates** from different geographical regions are recognised, along with **one additional special award**.

Six categories of the "Champions of the Earth Awards":

1. Entrepreneur vision

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2. Policy leadership
3. Science and innovations
4. Lifetime achievements
5. Inspiration and action
6. General champion

In **2018**, **India** received **two of the six categories** under the *Champions of the Earth* awards:

1. Kochi International Airport — Entrepreneur category
2. Narendra Modi — Policy Leadership

The award to the Prime Minister recognised his role in advancing the **International Solar Alliance** and his commitment to phase out single-use plastics. Following this pledge, India implemented a nationwide ban on the manufacture, import, stocking, distribution, sale, and use of specified single-use plastic items with low utility and high littering potential, effective **1 July 2022**.

Several other Indians have received the Champions of the Earth award:

Purnima Devi Barman **(2022):** Received the Entrepreneurial Vision award for her work with the "Hargila Army" to save the Greater Adjutant stork.

Jinali Mody **(2025):** A Young Champion of the Earth for her work in sustainable fashion.

Madhav Gadgil **(2024):** Awarded the Lifetime Achievement award for his work on environmental conservation and for chairing the Western Ghats Ecology Expert Panel.

9.6 BRUNDTLAND COMMISSION (1987) / WORLD COMMISSION ON ENVIRONMENT AND DEVELOPMENT (WCED)

The **United Nations General Assembly** set up the **World Commission on Environment and Development (WCED)**, which was chaired by **Gro Harlem Brundtland**. The Commission functioned independently, without direct control from any government or the formal United Nations system.

The **three main mandates** of the commission are:

1. To inspect critically the issues related to environment and development and deal with those through practical solutions.
2. To come up with ideas in which international cooperation can be promoted.
3. To strengthen the understanding and involvement among individuals, non-governmental organisations, institutes, governments, etc.

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A **Brundtland Report/Our Common Future** report was submitted by the commission on global agenda for an alteration.

Brundtland Report

1. **Catalyst for Global Action**: The Brundtland Report (Our Common Future, 1987) transformed international environmental negotiations by highlighting the urgent need to address environmental degradation.
2. **Foundation of Sustainable Development**: It played a major role in shaping the global ideology of **sustainable development**.
3. **Holistic Concept of Sustainability**: Introduced the integrated **physical–biological–social** approach to sustainability.
4. **Definition of Sustainable Development**: "Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."
5. **Two key contributions to global environmental agreements:** a) **Human-centred approach:** Prioritised human welfare within environmental sustainability. b) **Social equity:** Brought social justice and equity into environmental discussions.
6. **Linking Environment with Economy**: Shifted global thinking by addressing environmental issues within the context of economic development.
7. **Introduced Several Major Environmental Principles**: **Polluter Pays Principle**, **Differential standards** for developed vs developing nations & **Precautionary Principle.**

ENV 35- Earth Summit

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9.7 UNITED NATIONS CONFERENCE ON ENVIRONMENT AND DEVELOPMENT (UNCED)/THE EARTH SUMMIT (1992)

- The United Nations Conference on Environment and Development (UNCED), which can be called the **Earth Summit, held on 3rd June, 1992 in Rio de Janeiro**, came into **effect from March 21st, 1994**. Earth Summit is one of the **outcomes of the Brundtland Commission**.

  The primary objective was to promote socio-economic progress while reducing environmental damage through cooperative efforts between developing and developed nations.

It highlighted on

(i) The urgent need for development initiatives at a global level to account for environmental impacts.

(ii) Spotting the need for greater attention towards eco- efficiency by governments and businesses.

(iii) Finding alternative sources of fuel-based energy, reducing the sources of pollution.

- In the Earth summit, leaders focused on ensuring a healthy planet for future generations. The message, 'The change in attitude and behaviour can bring the necessary changes', was spread all across the globe.
- By inspecting the source of production of toxic elements and their illegal disposal systematically, introducing non-conventional sources of energy efficiently will bring down the pollution level. Lesser dependency on fossil fuel and greater awareness at the global level related to water scarcity can make positive changes.

QUICK INSIGHT

Earth Summit Resulted in Following Documents

NON-BINDING

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(I) **Rio declaration** for Environment protection
(II) **Agenda 21:** It is a non-binding, voluntarily implemented action plan by United Nations. It contains total 5 documents, concluded in 300 pages. Agenda 21, recommended for the UN Convention on Combat Desertification (UNCCD), came into force in 1996.
(III) **Forest Principles**

BINDING
(I) **Convention on Biological Diversity (CBD)**
(II) **United Nations Framework for Climate Change (UNFCCC)**
(III) **United Nations Convention to Combat Desertification (UNCCD)**

9.7.1 Rio Declaration

The declaration focused on goals and helped bring the agenda of developing countries to the front seat.

The declaration consisted of 27 principles intended to guide countries in future sustainable development.

This was adopted by **175 countries at the UNCED.**

- The goal of the Rio Declaration was 'new and equitable global partnership.'
- This declaration, unfortunately, ignored the most vital and difficult political issues such as resource transfer, lifestyles and consumption, environment and trade.
- The declaration provided the crucial backbone for all negotiations on global environmental problems since its adoption.

■ *Rio Declaration on Environment and Development*

This agreement supported Agenda 21.

Rio declaration highlighted:

- The importance of adopting sustainable development programmes within states,
- The need for taking proper responsibilities by developed countries,
- The need to put a complete end to poverty, and

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- The need for more women participation in sustainable development practices.

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9.7.2 Agenda 21

"Agenda 21"It is a **non-binding action plan adopted by the United Nations in 1992** to promote sustainable development. **Agenda 21** comprises **forty chapters** addressing a wide range of issues, including poverty-related challenges, public health risks, sustainable agriculture, deforestation, desertification, land degradation, hazardous waste management, air and water pollution, the release of toxic substances, and the loss of biological diversity.

- **Purpose:** A comprehensive, voluntarily implemented blueprint for governments, the UN, and major groups to achieve sustainable development at local, national, and global levels.
- **Structure:**
- It is classified into four broad sections that focus on social and economic aspects, conservation and sustainable management of resources, the involvement of major groups, and the methods required for implementation.
- **Relevance:** Agenda 21 was the main predecessor to the subsequent Sustainable Development Goals (SDGs) of the 2030 Agenda, which reaffirm its core principles.

9.7.3 The Forest Principles/Rio Forest Principles

They are **non-legally binding** principles that deal with management, conservation and sustainable development of all kinds of forest, both natural and human, in all physical and climate-related areas.

It encourages governments to involve local communities in the formulation, implementation, and planning of national forest policies, while integrating environmental protection with the social and economic development of forest-related sectors.

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9.7.4 United Nations Framework Convention on Climate Change (UNFCCC)

On 11th December 1990, the **United Nations General Assembly (UNGA)** established the **Intergovernmental Negotiating Committee (INC) for UNFCCC**. This treaty was produced on 'Earth summit' 1992 and came into effect in 1994.

Presently, it has **198 member nations/Parties** (Countries that sign the treaty are known as "Parties"). UNFCCC makes the framework to tackle with the aim of limiting the global temperature. Since the provision made in emission reduction in 1992 was not sufficient to control the emission, KYOTO protocol was made in which developed countries came forward with reductions in carbon dioxide emission plans. The **UNFCCC has two sister Conventions** also agreed in Rio, the **UN Convention on Biological Diversity (UNCBD)** and the **Convention to Combat Desertification (UNCCD).**

Timeline of UNFCCC

1979 – Climate-related concerns entered the global agenda with the organisation of the First World Climate Conference.

1988 – An international scientific body, the Intergovernmental Panel on Climate Change (IPCC), was created to examine climate variability and its consequences.

1990 – Scientific findings published in the IPCC's initial assessment, along with deliberations at the Second World Climate Conference, highlighted the need for a legal response, leading the United Nations General Assembly to begin discussions on a climate convention.

1991 – Formal negotiations commenced when the Intergovernmental Negotiating Committee (INC) held its first meeting.

1992 – After extensive negotiations, the INC completed the text of the United Nations Framework Convention on Climate Change, which was made available for signature at the Rio Earth Summit together with the CBD and UNCCD.

1994 – The UNFCCC became operational following its formal entry into force.

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1995 – Implementation oversight began with the convening of the first Conference of the Parties (COP-1) in Berlin.

1996 – Institutional support for the Convention was strengthened through the establishment of the UNFCCC Secretariat.

1997 – A legally binding instrument, the Kyoto Protocol, was adopted at the third Conference of the Parties held in December.

(Connecting Link): Kyoto Protocol

Classification of UNFCCC Countries

- The UNFCCC countries (signatories) were initially divided into three groups, viz., Annexe- I, Annexe-II, Annexe-III countries.
- **Annex I Parties:** This group comprises industrialised countries and the European Community that committed to reducing greenhouse gas emissions, particularly carbon dioxide, to an average of **5.2 per cent below 1990 levels**. Countries unable to meet these targets are permitted to purchase emission credits or invest in conservation measures.
- **Annex II Parties:** This category includes developed countries and the European Community that are responsible for providing financial assistance to developing nations. It also covers a subset of Annex I countries belonging to the **Organisation for Economic Co-operation and Development**, established in **1961**
- **Annexe-III countries:** Developing countries.

According to another classification, the world's countries are divided into two groups, viz., Annexe countries and Non-Annexe countries.

1. **Annexe countries:** Developed countries
2. **Non-Annexe countries:** Developing countries

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EARTH SUMMITS

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The term "Earth Summit" usually refers to the series of major UN conferences on environment and sustainable development held under the UN system

1st Earth Summit | Rio de Janeiro | 1992

United Nations Conference on Environment and Development (UNCED)

Theme: Environment & Sustainable Development

Mission: Address global problems & create legal frameworks

The 5 Major Outcomes

Rio Declaration on Environment & Development- 27 principles	Agenda 21	Forest Principles	UNFCCC (United Nations Framework Convention on Climate Change)	CBD (Convention on Biological Diversity)
(Precautionary Principle, Polluter-Pays Principle)	Global Action Plan: Social, Economic, Conservation, & Institutional	Non-binding principles for sustainable forest management	Foundation for Kyoto Protocol, Paris Agreement	First global treaty on biodiversity conservation

Note: **UNCCD (UN Convention to Combat Desertification)** was adopted in 1994 as a follow-up to 1992 decisions

2nd Earth Summit | Johannesburg | 2002

World Summit on Sustainable Development

Theme: Sustainable Development

Mission: Focus on **Implementation** rather than new treaties

Key Documents

Johannesburg Declaration	JPOI (Plan of Implementation)
Commitment to eradicate poverty	Set goals for water, energy, & fisheries

The WEHAB Framework

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EARTH SUMMITS

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3rd Earth Summit — Rio + 20 | 2012

UN Conference on Sustainable Development (UNCSD)

Theme: Green Economy & Institutional Frameworks

Mission: Shift to green economy & strengthen UNEP

The Major Outcomes

The Future We Want	Birth of SDGs	Financing
The official outcome document	Set up the working group that led to Agenda 2030 (SDGs)	New mechanisms for developing nations

4th Earth Summit — Stockholm + 50 | 2022

(Not officially called Earth Summit IV globally , but considered a continuation)

Theme: A Healthy Planet for the Prosperity of All

Outcomes: Stockholm + 50 Leadership Dialogue Messages

Emphasis on

- Circular Economy
- Climate action acceleration
- Protection of vulnerable communities
- Linking environment with post-COVID recovery

Summit	Year	Focus	Major Documents
1st Earth Summit	1992	Environment + Development	Rio Declaration, Agenda 21, Forest Principles, UNFCCC, CBD
2nd Earth Summit	2002	Implementation + Poverty	Johannesburg Declaration, JPOI, WEHAB
3rd Earth Summit	2012	Green Economy + Institutions	The Future We Want, SDG basis
4th Earth Summit	2022	Planetary health	Leadership Dialogue Messages

"Momentum for Change: Climate Neutral Now"

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Climate Neutral Now was an initiative launched by the **UNFCCC Secretariat** during the **UN Climate Change Conference in 2015**. The initiative seeks to motivate and assist individuals, institutions, and governments at all levels to take concrete steps towards achieving a **climate-neutral world by the middle of the century**, in line with the objectives of the **Paris Agreement**, adopted in the same year.

Accordingly, the initiative encourages companies, organisations, governments, and citizens to pursue climate neutrality through a **three-step approach**:

1. **Assessing their greenhouse gas emissions**
2. **Reducing emissions to the maximum extent possible**
3. **Offsetting unavoidable emissions by using UN-certified emission reductions (CERs)**

In **2017**, the **Momentum for Change** initiative identified **five priority focus areas**, namely *Women for Results*, *Financing for Climate-Friendly Investment*, *ICT Solutions, Planetary Health*, and *Climate Neutral Now*.

Chronology

1992 : The UN Framework on Climate Change (UNFCCC), was adopted at the Rio Earth Summit.

1994 : The UNFCCC came into force on 21 March 1994.

1995 : The COP was created to review the implementation of UNFCC and the first meetings (COP1) were held in Berlin.

1997 : The Kyoto protocol was adopted at COP3 in Japan. The protocol, which came into force in February 2005, put an obligation to reduce carbon emissions on developed countries by 2012. In the first phase of the Kyoto Protocol from 2008–2012, 37 Industrial nations and European Union were obligated to reduce the greenhouse gas emission up to 5.2% with the level of 1990.

2005 : The Montreal action plan was produced at COP11 held in the Canadian city. The plan was an agreement to extend the expiry date of the Kyoto protocol beyond 2012 and negotiate ‘deeper cuts in greenhouse-gas emissions’.

2005 : Canada became the first member country that refused to be a part of the protocol.

2015 : The first binding period under the protocol terminates.

Operating Mechanism of UNFCCC

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1. **Conference of the Parties (COP)**
   Supreme decision- making body of the convention.
   All states that are parties to the convention are represented at COP.
   Meets every year, unless the parties decide otherwise.
2. **Conference of the Parties serving as the meeting of the Parties to the Kyoto Protocol (CMP)**
   Oversees the implementation of the Kyoto Protocol All states that are parties to the Kyoto Protocol are represented.
   States that are not parties participated as observers.
3. **Conference of the Parties serving as the Meeting of the Parties to the Paris Agreement (CMA) Monitors and guides the implementation of the Paris Agreement and adopts decisions to ensure its effective functioning. All countries that have ratified the Paris Agreement are represented in the CMA. Countries that are not Parties to the Agreement participate in the proceedings as observers.**

First Commitment Period 2008–12

- A total of **37 industrialised countries**, along with the **European Community**, agreed to cut their greenhouse gas emissions to an **average of about five per cent below 1990 levels**.

Do You Know?

The targets applied to the six greenhouse gases— Carbon dioxide, Methane, Nitrous oxide (N2O), Sulphur Hexafluoride (SF6), Hydrofluorocarbons (HFCs) and Perfluorocarbons (PFCs).

These six GHGs are in CO_2 equivalents in determining reductions in emissions.

- A Party's **assigned amount** refers to the total quantity of emissions it is permitted to release during a commitment period in order to meet its specified emissions target.

COP1: Berlin Meet (1995)

- Parties to the UNFCCC met in Berlin (COP-1) to outline specific targets on emissions, thus laying the groundwork for the **Kyoto Protocol.**

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■ COP 3: Kyoto Protocol (1997)

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Adopted within the broader framework of the **United Nations Framework Convention on Climate Change**, this agreement establishes a system under which participating countries accept legally enforceable limits on their greenhouse gas emissions. The arrangement was finalised in **Kyoto, Japan, in December 1997**, and became legally operational on **16 February 2005**, marking the first instance of binding international commitments on emission reductions.

- The third Conference of the Parties achieved a historical milestone with the adoption of the **Kyoto Protocol,** the world's first greenhouse gas emissions reduction treaty.
- There are now 198 Parties to the Convention and 192 Parties to the Kyoto Protocol.

Provisions under Kyoto Protocol:

- Joint Implementation permits emission-reduction projects to be carried out and traded at the project level among Annex (developed) countries, while the Clean Development Mechanism (CDM) enables project-based emission offsets to be generated in Non-Annex (developing) countries, which can then be used by entities in Annex countries to meet their compliance requirements.
- Under the Kyoto Protocol, several developed nations accepted legally binding limits or reductions on greenhouse gas emissions across two commitment periods: the first spanning 2008–2012, and the second covering emissions from 2013–2020.

■ For reduction in Carbon emission, the following programmes were suggested:

1. **International emissions trading/Carbon Trading:** To reduce the greenhouse emission, a new mechanism was introduced which is known as **carbon trading**.

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- **Carbon Market:** A carbon trading system emerged under **Annex B** of the **Kyoto Protocol**, allowing countries with binding emission reduction commitments to trade surplus allowances. Nations that reduce emissions beyond their assigned targets may sell excess **Assigned Amount Units (AAUs)**, while other countries can meet their obligations by purchasing these units or by using offset credits generated in developing countries. International trading of carbon credits takes place through platforms such as the Chicago Climate Exchange, Nord Pool, the European Climate Exchange, and Powernext.

2. **Clean Development Mechanism (CDM):** The **Clean Development Mechanism**, provided for under the Kyoto Protocol, enables **Annex I countries** with emission reduction commitments to undertake emission-reduction projects in **developing countries**. These projects generate **Certified Emission Reductions (CERs)**, where **one CER represents one tonne of carbon dioxide equivalent**, and can be traded to help meet compliance targets. This mechanism significantly strengthened the practical implementation of the Kyoto Protocol. Although participation is not mandatory for Annex I countries, **India**, as a responsible stakeholder, has adopted the CDM framework.
3. **Joint Implementation: Article 6 of the Kyoto Protocol** has defined it for **Annex B countries**, emission reductions are accounted for in **Emission Reduction Units (ERUs)**, with **each ERU representing one tonne of carbon dioxide (CO_2)** reduced. Joint implementation usually implies on Industrialised or developed nations which are obliged to follow this. A provision of creating Green Climate Fund from these nations that are given to the developing nations for the reduction in carbon emission is a great step taken.

■ *COP 7: Marrakesh Conference (2001)*

- COP-7 resulted in the **Marrakesh Accords,** setting the stage for ratification of the **Kyoto Protocol.**.

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This aimed to establish a formal framework governing the operational rules for **International Emissions Trading**, the **Clean Development Mechanism**, and **Joint Implementation**, along with provisions for compliance and accounting systems.

COP 11: EU's Emissions Trading (2005)

The **European Union Emissions Trading Scheme (EU ETS)**, recognised as the world's **first and largest emissions trading system**, was introduced as a central component of the European Union's climate strategy. The facilities covered under this scheme together account for **nearly half of the EU's carbon dioxide emissions**.

- **Kyoto Protocol – Speed-up through Ratification:** A significant milestone was reached in **2005** when the **Russian Federation** deposited its instrument of ratification to the **Kyoto Protocol**, thereby enabling the Protocol to formally enter into force.

COP 13: Bali Conference (2007)

- At **COP-13**, Parties agreed on the **Bali Road Map**, which incorporated the **Bali Action Plan** and set the direction for a renewed negotiating process on climate change. The Action Plan was structured around **five core areas**: a shared vision, mitigation, adaptation, technology development, and financial support.
- According to the plan, rich countries would cut emissions by 40% by 2020 and bring more funds.
- On the other hand, emerging economies would join the effort in reducing emissions in their countries, enabled by finance and technology from industrialised countries.

Exam Pointer

Nationally Appropriate Mitigation Actions (NAMA)

Nationally Appropriate Mitigation Actions (NAMAs) refer to country-specific policies and measures designed to reduce greenhouse gas emissions. The concept recognizes that

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mitigation efforts vary across nations, guided by the principles of **equity** and **common but differentiated responsibilities and respective capabilities (CBDR–RC)**. NAMAs also emphasize the importance of financial support from developed countries to assist developing nations in achieving emission reductions.

The term "NAMA" was formally introduced under the **Bali Action Plan**, which was part of the **Bali Road Map** adopted during the United Nations Climate Change Conference in Bali in December 2007.

Implementation:

- No NAMAs were implemented in **2010**.
- The **Programme of Activities (PoA)** under the Clean Development Mechanism (CDM) is considered a precursor to future NAMA frameworks and had already been developed, laying the groundwork for their implementation.

NAMAs represent a flexible and inclusive approach to climate mitigation, tailored to the unique circumstances of each country while fostering international collaboration

Status: By **September 2012**, nearly **50 countries** had submitted information on their NAMAs to the **UNFCCC**. These submissions varied widely in scope, covering aspects such as alignment with the **Copenhagen Accord**, target sectors, specific mitigation measures, and intended greenhouse gas reduction goals.

Criticism of NAMAs: NAMAs have faced criticism for moving away from **carbon-pricing mechanisms** and for encouraging large **subsidy-based programmes**, largely financed by developed countries and implemented by developing countries on a voluntary basis.

COP 16: Cancun Conference (2010)

- COP-16 resulted in the **Cancun Agreements,** a comprehensive package by governments to assist developing nations in dealing with climate change.
- The conference was successful in erasing the difference between developed and developing countries.. The **Green Climate Fund**, the **Technology Mechanism**, and the **Cancun Adaptation Framework** were formally put in place.

Green Climate Fund (GCF)

The **Green Climate Fund (GCF)** was established in **2010** as part of the **UNFCCC's** financial framework to facilitate the transfer of climate-related finance from developed countries to developing countries. Its purpose is to support both **climate change mitigation** and **adaptation efforts** in response to the impacts of a changing climate.

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The Fund plays a key role under the **Paris Agreement**, particularly in addressing the responsibilities of countries with historically high emissions.

How does it help?

- **Provides financial support for projects, programmes, policies, and other initiatives in developing countries through designated thematic funding windows.**
- **Designed as a central instrument for mobilising climate finance, with the objective of raising USD 100 billion annually by 2020.**
- **Encourages a transition towards low-emission and climate-resilient development by assisting developing countries in reducing greenhouse gas emissions and adapting to climate impacts, with special attention to those most vulnerable.**
- **Aims to maximise the effectiveness of funding across mitigation and adaptation activities, while ensuring a balanced approach and promoting environmental, social, economic, and developmental co-benefits, including gender responsiveness.**

Who governs the Fund?

The Fund is overseen by a **Board** that holds full authority over funding decisions. The Board operates under the **guidance of the Conference of the Parties (COP)**, to which the Fund remains accountable.

■ *COP 18: Doha Conference (2012)*

During **COP-18/CMP-8**, countries committed to advancing negotiations for a **comprehensive global climate agreement to be finalised by 2015**, while also exploring measures to **strengthen climate action before 2020**, going beyond the emission-reduction pledges already made.

Do You Know?

The modified list of Greenhouse gases includes the following 7 gases:

1. Carbon dioxide (CO_2)
2. Methane (CH_4)
3. Nitrous oxide (N_2O)
4. Hydrofluorocarbon (HFCs)
5. Perfluorocarbon (PFCs)
6. Sulphur hexafluoride (SF_6)
7. Nitrogen trifluoride (NF_3)

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The 'Doha Amendment' established a **second commitment period (2013–20).** It was decided to reduce the emission of greenhouse gases up to 18% as compared to the 1990 level during this period.

■ *COP 21: Paris Agreement (2015)*

About Paris Agreement

1. During **COP-21 in Paris (12 December 2015)**, countries participating under the **UN climate framework** finalised a globally significant accord aimed at addressing climate change and mobilising stronger action and investment towards a **low-emission and sustainable development pathway**.
2. The **Paris Agreement** advanced the principles of the Convention by **bringing all nations—developed and developing—under a single cooperative framework** to undertake climate mitigation and adaptation efforts. It also reinforced mechanisms to **enhance assistance for developing countries**, thereby redefining the direction of international climate cooperation.
3. At the heart of the Agreement lies the objective of **limiting the rise in global average temperature during this century**, by keeping it **well below 2°C compared to pre-industrial levels**, while **pursuing additional efforts to restrict warming to 1.5°C**.
4. The Agreement further focuses on **strengthening national capacities to manage climate-related impacts** and on **reorienting financial flows** so that development pathways are consistent with **low greenhouse-gas emissions and climate resilience**.

The **Paris Agreement became legally effective on 4 November 2016**.

Do You Know?

- It established obligatory commitments by all countries to make countrywide determined contributions (NDCs), and to chase domestic measures aimed at attaining them.

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- Commit all countries to put forward **new NDCs every five years**, with the lucid expectation that they will "represent a progression" ahead of earlier ones.

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- For the first time, developing countries were also encouraged to make voluntary contributions.
- The existing target of mobilising USD 100 billion annually was extended from 2020 up to 2025, with a commitment to set a higher collective finance goal for the period beyond 2025.
- A proposal was made for a new mechanism, comparable to the Clean Development Mechanism under the Kyoto Protocol, allowing emission reductions achieved in one country to be credited towards another country's Nationally Determined Contribution (NDC).
- Paris deal will come into effect from 2020. This means making sure each nation pledges its own Nationally Determined Contribution (NDCs) for post-2020 action.

PARIS Agreement Article 6.0 Article 6 of the Paris Agreement is a key element in discussions about sustainable development and climate change, as it outlines provisions for international collaboration in implementing Nationally Determined Contributions (NDCs).

This article allows nations to voluntarily engage in cooperative efforts, utilizing carbon markets (sections 6.2 and 6.4) and non-market approaches (section 6.8) to meet their climate goals while fostering sustainable development and maintaining environmental integrity.

Additionally, Article 6 establishes guidelines for carbon markets, offering a framework for global cooperation on reducing emissions. It facilitates joint efforts through carbon trading and other mechanisms to support countries in achieving their climate objectives.

Article 6 provides a framework under which countries can **cooperate voluntarily by transferring emission reduction outcomes** among themselves, thereby supporting the achievement of their **Nationally Determined Contributions (NDCs)**.

Under the **Paris Agreement**, Article 6 enables **both market-based and non-market-based forms of international cooperation**. It serves as an important tool for countries to work together in lowering emissions and fulfilling their climate commitments.

Article 6.2: This provision allows countries to apply **internationally transferred mitigation outcomes (ITMOs)**, such as carbon credits, towards their own NDCs, encouraging cooperation and the sharing of emission-reduction efforts.

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Article 6.4: This clause creates a **UN-supervised mechanism** for the generation and exchange of **high-integrity carbon credits**, thereby supporting the development of international carbon markets.

Article 6.8: This section focuses on **non-market approaches**, including cooperation in areas such as **finance, technology transfer, and capacity building**, enabling countries to collaborate on climate action without engaging in emissions trading.

India and Paris agreement

1. **On 22nd April 2016**, India **formally signed the Paris climate agreement, which** was adopted by over 190 countries in December 2015.
2. **India ratified the Paris agreement** on climate change on **2nd October 2016** to become the **62nd nation to join the deal**.
3.. **India** contributes about **6% of global carbon dioxide emissions**, while **China** accounts for nearly **28%**, the **United States** around **16%**, and the **European Union** approximately **10%**. However, when emissions are measured on a **per capita basis**, **India ranks below at least ten other countries**.
4. As part of the initial commitments to the agreement, India also plans to reduce its carbon emission intensity - emission per unit of GDP **by 33–35% from 2005 levels over 15 years**. It aims at **producing 40% of its installed electricity capacity by 2030 from non-fossil fuels, but it increased 50% under Glasgow Convention (COP 26, 2021)**

COP 22: Marrakech Meeting (2016)

Morocco witnessed meetings under the Convention, along with the Kyoto Protocol and the Paris Agreement.

International Solar Alliance (ISA)

The International Solar Alliance (ISA) has over 124 member and signatory countries, with over 120 countries having ratified the Framework Agreement to become full members as of late 2025. Notable members include India, France, Spain, the United States, the United Kingdom, Australia, Germany, Japan, and Brazil. All member states of the UN are eligible to join the ISA.

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Eligibility: The ISA was originally focused on sun-belt countries between the Tropics of Cancer and Capricorn, but after a 2020 amendment**, all member states of the UN are eligible to join**.

It works with governments to improve energy access and security worldwide and promotes solar power as a sustainable transition to a carbon-neutral future. This alliance is a treaty-based inter-governmental organisation. The initiative was proposed by **Narendra Modi** during the **India–Africa Summit** and subsequently discussed in meetings of participating countries ahead of the **2015 United Nations Climate Change Conference in Paris**. The **Framework Agreement of the International Solar Alliance** was opened for signature in **Marrakech, Morocco, in November 2016**.

In January 2016, India and France entered into a significant agreement under the International Solar Alliance (ISA), jointly laying the foundation stone for its headquarters at the National Institute of Solar Energy (NISE) in Gwal Pahari, Gurugram. The Government of India allocated five acres of land at the NISE campus for the ISA headquarters, while France committed `1.75 billion (US$26 million) to support the construction of the campus and operational expenses for its initial five years. The Alliance, also known as the International Agency for Solar Policy and Application (IASPA), reflects the shared vision of both nations to advance solar energy initiatives and promote sustainable energy solutions globally.

Note: The ISA entered into a cooperation arrangement with the **World Bank** to accelerate the mobilisation of finance for solar energy. Banking institutions are expected to play a critical role in mobilising **over US$1 Trillion in investments by 2030** to achieve the ISA's objective of large-scale deployment of **affordable solar energy**.

- Governing members: India and France are the current President and Co-President, respectively.

Advantages for India

Launching of ISA has set up the stage for India's proactive and forward-looking leadership on climate change and the transition to a less carbon-intensive growth trajectory.. It is expected to **reduce the cost of solar technologies**, thereby supporting faster national development. The initiative will also assist **India in achieving its Intended Nationally Determined Contribution (INDC) targets**.

Withdrawal of US From Paris Agreement

Note: U.N.'s Climate Science Panel said that the world would have to reduce greenhouse gas emissions by 40–70% below 2010 levels by 2050 in order to have a fair chance to keep the rise in the global temperature below 2°C and CO_2 down to 430–480 parts per million (ppm) equivalent 2100.

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The **ISA officially came into force** as a treaty-based organization when its Framework Agreement was opened for signature in **Marrakesh, Morocco**, in November 2016, during COP22.

The **major outcomes** were:

- Two years to 2018 were given to nations for agreeing with rules and procedures for the Paris Agreement.
- Five-year plan as agreed by the countries on Loss and Damage.
- **Countries collectively endorsed a call for urgent action through the Marrakech Action Proclamation for Our Climate and Sustainable Development, which reaffirmed the need for immediate measures and expressed the commitment of nations to act in line with the agreed objectives.**

■ *COP 23: Bonn Conference (2017)*

It is actually the extended version of the Paris Agreement (COP 21) that was held in 2015.

Achievements

1. COP23 was successful in initiating the **'rulebook'** framing that is likely to complete in 2018 and reviewed in COP24, in Poland.
2. The **Talano dialogue** was one of the successful outcomes that took place in COP23 where a **decision was taken by all the countries to share their progress report through stories.**
3. The **Gender Action Plan** was finalised here which is a great step ahead to **involve the participation of women in all UNFCCC meetings to effectively boost the gender-responsive policy**.
4. **Indigenous people were allowed to have a greater voice in expressing their knowledge, ideas and strategies** to combat climate change effects.
5. A historic move was made while **introducing a discussion related to agriculture and how it can be made resilient to climate change**.

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6. Collaboration was made among WHO and UNFCCC to launch a special initiative specifically to connect vulnerable island countries to protect them from the health effects of climate change.
7. Very little discussion was made on fund allocation to developing countries by developed countries.

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Do You Know?

Talona Dialogue: A traditional approach used in Fiji and the Pacific to engage in an inclusive, participatory and transparent dialogue. The purpose of Talanoa is to share stories, build empathy and trust. During the process, participants advance their knowledge through common understanding. It creates a platform of dialogue, which results in better decision-making for the collective good. By focusing on the benefits of collective action, this process will inform decision-making and move the global climate agenda forward.

■ *COP 26: Glasgow (2021)*

According to the **UNFCCC**, COP26 focused on the following key objectives:

1. **Secure Net Zero by 2050**
   - Achieve **global net-zero emissions by mid-century** and keep the **1.5°C temperature goal within reach**. Encourage countries to submit **enhanced 2030 emission-reduction targets** consistent with achieving net zero by 2050. To meet these targets, countries were urged to **phase down coal use**, **halt deforestation**, **accelerate the transition to electric vehicles**, and **increase investment in renewable energy**.
2. **Adapt to Protect Communities and Natural Habitats**
   - Strengthen collective efforts to **protect and restore ecosystems** and to build **resilient infrastructure, early-warning systems, and climate-resilient agriculture**, so as to prevent loss of lives, livelihoods, and homes.
3. **Mobilise Climate Finance**

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- Developed countries were reminded of their commitment to **mobilise at least USD 100 billion per year** in climate finance to support developing nations.

4. **Work Together to Deliver**
   - A major task at COP26 was to **finalise the Paris Rulebook**. Countries worked jointly to agree on **detailed operational rules** necessary for the effective implementation of the **Paris Agreement**.

India's Commitment

A **five-point action framework to address climate change**, known as **Panchamrit**, was unveiled by PM **Narendra Modi** during the **COP26** held in **Glasgow**.

1. **500 GW of non-fossil electricity capacity by 2030**
   India will install 500 gigawatts of non-fossil fuel based power capacity (solar, wind, hydro, nuclear, biomass).
2. **50% of energy requirements from renewable sources by 2030**
   India aims to meet 50% of its electricity needs from renewable / non-fossil sources by 2030.
3. **Reduction of carbon emissions by 1 billion tonnes by 2030**
   India will reduce its projected total carbon emissions by 1 billion tonnes between now and 2030.
4. **Reduction of emissions intensity of GDP by 45% by 2030**
   India has committed to **reducing its emissions intensity (emissions per unit of GDP) by 45 per cent relative to 2005 levels**.
5. **Net-zero emissions by 2070**
   India targets to achieve net-zero greenhouse gas emissions by the year 2070.

India in Glasgow

- **India** also articulated the vision of **"One LIFE, One World"** at **Glasgow**.
- The **Prime Minister of India** urged **developed economies** to commit **USD 1 trillion** towards global climate finance.
- The **Glasgow Climate Pact** calls for the **phase-down of unabated coal**, rather than a complete phase-out, and similarly addresses **subsidies for fossil fuels**.
- Several countries criticised **India** for seeking a change from the original language, as the final text refers to a **phase-down** instead of a **phase-out** of coal.

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- India supported the **phase-down approach** on the grounds that it would help improve the **commercial viability of clean technologies and energy storage**, which are essential for scaling up renewable energy capacity.

What is Net-Zero Goal?

- It is also referred to as carbon-neutrality.
- Net zero does not imply that a country's emissions fall to absolute zero.
- Instead, it refers to a condition where the emissions released are balanced by the absorption and removal of greenhouse gases from the atmosphere.
- Absorption can be enhanced by expanding natural carbon sinks, such as forests.
- Removal involves the use of advanced technologies, including carbon capture and storage.
- In some cases, absorption and removal may exceed total emissions, resulting in negative emissions; Bhutan is an example of a carbon-negative nation.

Global Endeavours

- Active campaigning for Net Zero Emissions Efforts over the past two years have focused on persuading all countries to commit to achieving net-zero emissions by 2050.
- It is widely argued that attaining global carbon neutrality by 2050 is essential for meeting the objectives of the Paris Agreement.
- Under existing policies and actions, the world remains on a trajectory that could result in a temperature rise of about 3–4°C by the end of the century.

Actions Taken

- UK, France and several other countries have already **passed legislation committing to the achievement of net-zero emissions by mid-century (2050).**
- The **European Parliament adopted the NetZero Industry Act in April 2024** (in the final plenary session of its 2019–2024 legislature). Officially, the EU's longterm legally binding target is to reach **climate neutrality by 2050.**
- Countries such as **Canada**, **South Korea**, and **Germany** have announced their intent to transition towards a **net-zero emissions future**.
- **China** has committed to achieving **net-zero emissions by 2060**.

■ *Green Grid Initiative*

India and UK have launched the **'green grids' initiative**, formally known as the **One Sun One World One Grid (OSOWOG)**, was highlighted on the sidelines of **COP26**. The initiative seeks to **interlink electricity grids across national boundaries**, enabling the large-scale and rapid adoption of

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renewable energy. **India** first advanced the idea of cross-border solar energy sharing at the **International Solar Alliance** in **2018**, with the objective of allowing regions with surplus renewable power to supply electricity to other countries. The proposal forms part of a wider effort to accelerate the deployment of **affordable low-carbon technologies**, covering over **70 per cent of the global economy**, and represents an important component of **India's net-zero emissions strategy**.

In 2015, at Indo-African meet, Sri Narendra Modi first introduced the concept of Solar Alliance in COP 21 Meeting at the Paris Convention on 30 November 2015.

Main Aims & Objective of the Initiative

- To help in increasing investment in research and development centres by pooling skill, technology and financial resources.
- To speed up the process to decarbonise energy production.
- Create an international network of global interconnected solar power grids, to ensure a 24×7 supply of green power.
- It will bring together a group of governments, legislators and international organisations to accelerate the construction of the new infrastructure needed for a world powered by clean energy.
- The initiative aims to bring together the financial organisations, international coalition of national governments, power system operators.
- Plan to improve connectivity between the world's power grids to accelerate the transition to greener energy.
- It will allow areas with excess renewable power to send it to areas facing a shortage.
- Transport solar power to different countries by creating a three-phase 2600 GW of interconnection by connecting Asia, Africa and the rest of the world by 2050.
- It will also accelerate the construction of electricity interconnectors, charging points and flexible grids.

Challenges Associated

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- It would require large capital investment to set up long transmission lines.
- Cooperation between countries for solar power transfer between neighbouring countries.
- Security risks emanating from terror organisations.

The initiative is a welcome step towards Sustainable Development, aimed at addressing the issue of high cost of energy storage.

■ *India's Updated Nationally Determined Contributions (NDCs)*

India submitted its **revised Nationally Determined Contributions** to the **UNFCCC**.

About India's NDCs

NDC Version
First NDC (Formerly called INDC)
Updated first NDC (reflecting revised targets such as those ann COP 26)

■ *India's updated NDCs:*

- India's updated Nationally Determined Contributions (NDCs) were developed in alignment with the country's national circumstances and the principle of **Common but Differentiated Responsibilities and Respective Capabilities (CBDR–RC)**.

Key Features

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- The updated NDCs outline a roadmap for India's transition to cleaner energy during the **2021–2030 period**.
- Two of the three quantitative targets from India's 2015 NDCs were revised, focusing on:
  - **Emissions Intensity:** Adjusting targets related to emissions intensity reductions.
  - **Non-Fossil Fuel Sources:** Increasing the share of non-fossil fuel sources in installed electricity capacity.
- While India announced two major commitments at COP26 in Glasgow—achieving **500 GW of non-fossil fuel energy capacity** and reducing emissions by **one billion tonnes of CO_2 by 2030**—these were not formally included as official targets in the updated NDCs.

This update reflects India's commitment to balancing its developmental needs with global climate goals.

■ *Nationally Determined Contributions (NDCs)*

- NDCs represent the **post-2020 climate actions** that each Party is required to communicate under the **Paris Agreement**.
- The **Glasgow Climate Pact**, adopted during **COP26 under the UNFCCC**, called upon Parties to **revisit and strengthen their 2030 climate commitments**.

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INDIA'S NATIONALLY DETERMINED CONTRIBUTION (NDC)

		QUANTITATIVE TARGETS			
		Targets for 2030	**Previous NDC, 2015**	**Updated NDC, 2022**	**Progress**
	1.	**Reduce the emissions intensity of its GDP**	By **33 to 35% by 2030** from 2005 level.	**By 45%** by **2030** from 2005 level.	Estimated **reduction of 28%** over 2005 levels
	2.	**Cumulative electric power installed capacity from non-fossil fuel-based energy resources** with the help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF).	About **40%**	About **50%**	**42.3% achieved** by end of November, 2022
	3.	**Create an additional carbon sink** through additional forest and tree cover.	**2.5 to 3 billion tonnes of co2 equivalent**	**Same as earlier**	
		QUALITATIVE TARGETS			
		Previous NDC, 2015		**Updated NDC, 2022**	
	4.	Put forward and further propagate a **healthy and sustainable way of living** based on traditions and values of conservation and moderation.		Put forward and further propagate a healthy and sustainable way of living based on traditions and values of conservation and moderation, including through a mass movement for '**LIFE-Lifestyle for Environ- ment**' as a key to combating climate change.	
	5.	**Adopt a climate friendly and a cleaner path** than the one followed hitherto by others at corresponding level of economic development.			
	6.	**Better adapt to climate change** by enhancing investments in development programmes in sectors vulnerable to climate change, particularly agriculture, water resources, Himalayan region, coastal regions, health and disaster management.			
	7.	**Mobilize domestic and new & additional funds** from developed countries to implement the above mitigation and adaptation actions in view of the resource required and the resource gap.			
	8.	**Build capacities, create domestic framework and inter- national architecture** for quick diffusion of cutting edge climate technology in India and for joint collaborative R&D for such future technologies.			

Figure 9.1 India's Nationally Determined Contribution (NDC)

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■ *Targets in Nationally Determined Contributions (NDCs) by end of 2022*

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- Countries specify measures to lower greenhouse gas (GHG) emissions, aligning with the objectives of the Paris Agreement to limit global warming.
- Nations outline strategies to enhance resilience and adapt to the impacts of rising temperatures, addressing climate-related challenges.
- NDCs are submitted every five years to the UNFCCC Secretariat, ensuring regular updates and strengthened commitments.

These targets reflect each country's efforts to contribute to global climate action while addressing their unique circumstances and priorities

■ *COP 27: Sharm El-Sheikh, Egypt (2022)*

The 27th Conference of the Parties (COP27) under the United Nations Framework Convention on Climate Change was convened in Sharm El-Sheikh, Egypt, from 6 to 20 November 2022, with the theme "**Together for Implementation**."

Key outcomes of the COP27

COP 27 delivered the historic Loss and Damage Fund, modest mitigation progress, and renewed focus on adaptation and finance, but weak action on fossil fuel phase-out.

Areas of Discussion	Important Decisions and Developments
1. Loss and Damage Fund (Historic Achievement)	• Parties agreed to **establish a dedicated fund** for nation • A long-standing demand of developing countries

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Areas of Discussion	Important Decisions and Developments
2. Sharm El-Sheikh Implementation Plan	• Focus on **mitigation, adaptation, finance,** and **just tra** • Calls for urgent global action to meet climate goals.
3. Mitigation Work	• New programme to accelerate **emission reductions fro**

Do You Know?

(1) **CBDR–RC (Common but Differentiated Responsibilities and Respective Capabilities)** is a **core principle** of the **UNFCCC**. It recognises that while **all countries share a collective obligation** to address climate change, their duties are **not uniform**. The level of responsibility varies depending on:

1. **Historical contributions to greenhouse gas emissions**, and
2. **Differences in national capacity**, including economic strength and technological capability.

(2) **Loss and Damage Fund**

The Loss and Damage Fund is a global financial mechanism created under the UNFCCC at COP27 (2022) to help developing and vulnerable countries cope with the losses and damages caused by climate change that cannot be prevented or adapted to.

What it covers?

- **Extreme climate events** such as floods, cyclones, droughts, and heatwaves **Gradual climate processes** including sea-level rise, desertification, and the melting of glaciers **Widespread losses**, affecting human lives, sources of livelihood, physical infrastructure, natural ecosystems, and cultural heritage

Why was it needed?

Because developing countries contribute least to climate change but suffer most from its impacts. They demanded financial support based on climate justice and equity.

When was it created?

- Created: COP27 (Egypt, 2022)
- Operationalised: COP28 (UAE, 2023)

Who pays?

Primarily developed nations (historical emitters). Contributions from other capable countries and institutions are voluntary.

Who receives?

Climate-vulnerable developing countries and small island states.

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(3) **LiFE – Lifestyle for Environment**

LiFE stands for Lifestyle for Environment, an initiative launched by the Government of India at UNFCCC COP26 (Glasgow, 2021).

It seeks to encourage **environmentally responsible living** at the **individual, community, and national levels**, promoting a transition from a **"use and dispose" approach** to a model of **thoughtful and efficient use of resources**.

Core Idea

LiFE encourages behavioural changes that lead to:

- Reduced carbon footprint
- Sustainable consumption
- Conservation of energy, water, and resources

It emphasises people-led climate action along with government policies.

■ *Global Methane Assessment: 2030 Baseline Report*

The **'Global Methane Assessment: 2030 Baseline Report'** was released by the **Climate and Clean Air Coalition** in collaboration with the **United Nations Environment Programme**. The report analyses projected methane emission trends over the next decade and evaluates their implications for achieving the targets under the **Global Methane Pledge (GMP)**.

Key Outcomes

- Amount of methane in the atmosphere is 260% of pre-industrial levels.
- Human-driven methane emissions are responsible for nearly 45% of current net warming.
- 2021 saw the largest annual increase recorded.
- The **Global Methane Pledge (GMP)** was announced at **COP26** in **2021** with the objective of accelerating efforts to curb methane emissions.
- The initiative is jointly led by the **United States** and the **European Union**.
- Its central goal is to achieve a **collective reduction of methane emissions by at least 30 per cent below 2020 levels by 2030**.
- The pledge has **over 100 participating countries**, which together account for **around 45 per cent of global anthropogenic methane emissions**.

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Do You Know?

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Do You Know?

India has not joined the pledge, citing concerns related to its potential impact on **trade**, the **agricultural sector**, and the significance of **livestock in the rural economy**.

COP 28: Dubai, UAE (2023)

It was held in Dubai, UAE from 30 Nov – 12 Dec 2023. Its theme was **"**Delivering the first **Global Stocktake" (GST)** of the Paris Agreement.

COP28 Climate and Health Declaration

At the **28th UN Climate Change Conference (Dubai, 2023)**, countries adopted a declaration highlighting the growing recognition that **climate change poses direct and indirect risks to human health**, and calling for greater integration of health considerations into climate action, including the development of **health systems capable of functioning under climate stress**.

Key Aspects

- The conference featured the **first dedicated Health Day** in the history of the COP process, with **more than 120 countries** expressing support for the declaration. The document draws attention to the **health impacts of climate change**, including rising temperatures, deteriorating air quality, climate-sensitive diseases, food insecurity, and weather-related disasters. It emphasises the need to **strengthen national healthcare systems** so they can remain effective under changing climatic conditions, particularly in **countries facing higher vulnerability**. The declaration encourages **substantial and sustained reductions in emissions**, including those generated within the **health sector**, such as from cooling infrastructure, waste handling, energy consumption, and medical supply chains. It underlines the role of **clean energy adoption, resilient infrastructure, early-warning mechanisms, and improved air quality** in protecting public health. Financial support, including access to **climate-related funding mechanisms**, is highlighted as essential for enabling countries to

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upgrade health systems. Although the declaration does **not create legal obligations**, it carries **considerable political weight** in shaping future climate–health discourse.

Key Outcomes

Areas of Discussion	Important Decisions and Developm
1. First Global Stocktake (GST) Adopted	• Assessed global progress on the Paris Agreem • Found the world off-track to limit warming to 1.5 • Called for urgent, deep, and sustained emissior
2. Historic Call to "Transition Away from Fossil Fuels"	• For the **For the first time in the history of CO explicitly referred to the need for "transition systems."** • **This was viewed as a symbolic and significa calling for a complete phase-out of fossil fue**
3. Loss and Damage Fund Operationalised	**The fund that was established at COP27 becam** • **Several countries also announced initial fina**
4 Renewable Energy Push	**A global objective was set to triple renewable e** • **Countries also agreed to aim for doubling th by 2030.**

India's Position on the Declaration

India chose not to endorse the COP28 Climate and Health Declaration.

- The declaration called for lowering emissions associated with **health-sector operations**, including **cooling requirements for hospitals, vaccines, medicines, and logistics systems**. India argued that such expectations are **not currently feasible**, given that cooling infrastructure is **critical for healthcare delivery**, especially in **high-temperature regions and rural areas**. Authorities expressed concern that rigid commitments could **compromise vaccine storage, drug safety, and hospital services**. As a result, India refrained from joining the declaration, despite its **non-binding character**. India has clarified that it **supports climate-sensitive health planning**, but requires **greater**

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flexibility and timelines consistent with its **development priorities and health infrastructure realities**.

■ *COP 29: Baku, Azerbaijan (2024)*

COP29 was the UN climate conference held in **Baku, Azerbaijan (11–22 November 2024)** under the UNFCCC with the main focus of **Climate finance** — helping developing countries pay for climate action.

Key Outcomes

Areas of Discussion	Important Decisions and Developme
1. New Global Climate Finance Goal	• Countries agreed to mobilize **at least USD 300 b** nations. • With private + public finance combined, the targe
2. Support for Vulnerable Countries	• More resources for **adaptation, National Adapta** least-developed nations. • Launch of the **Baku Adaptation Roadmap**.
3. Gender & Inclusion	• Extended the **Lima Work Programme on Gende** • Renewed commitments to include **Indigenous P**
4. Implementation Focus	COP29 stressed turning past commitments into delivery and transparency.

Gaps / Criticisms

- Many countries felt the **USD 300 billion** goal is *still too low* compared to actual climat
- Limited progress on **fossil-fuel phase-out** and **stronger mitigation measures**

■ *COP 30: Belém, Brazil (2025)*

COP30 is the **30th United Nations Climate Change Conference** under the UNFCCC. It was held in **Belém, Brazil**, in the **Amazon rainforest**, from **10–21 November 2025**. This location symbolised the importance of forests, biodiversity, and Indigenous peoples in climate action. It delivered progress on **finance, NDC updates, and transparency**. It failed to agree on a **fossil-fuel phase-out**, leaving the world off-track for the 1.5°C goal.

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Key Aspects

- Updating and strengthening **Nationally Determined Contributions (NDCs)** for the 2035 target year.
- Increasing **finance** for developing countries.
- Protecting the **Amazon rainforest** and supporting Indigenous communities.
- Increasing **global ambition** to keep global warming below **1.5°C**.

Key Outcomes

Climate Finance	• Countries agreed to **significantly increa**
	• Targets included:
	  • **Tripling adaptation finance**
	  • More funding for **just transition**, rene
	• However, no *fixed* new global finance nu decisions.

Updated National Climate Plans (NDCs)	• COP30 was important because countrie **NDCs aligned with 1.5°C for 2035**. • Many countries submitted stronger plans Paris Agreement goals.
Forests, Amazon Protection & Indigenous Rights	Because COP30 took place in the Amazon: • Strong emphasis on **ending deforestati** • Support for **forest finance mechanisms** • Recognition of **Indigenous peoples as** • New Amazon-related initiatives were dis
Transparency & Accountability	• Progress on the **Enhanced Transparen** Agreement.

Upcoming UNFCCC COP 31-32

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Table 9.1 Brief of UNFCCC COP 21 – COP 30

COP	Year	Location	Key Focus/Outcomes
COP 21	2015	**Paris, France**	Adoption of the Paris Agreement **warming to well below 2°C, pr**
COP 22	2016	**Marrakesh, Morocco**	Focused on implementing the Pa on water, land, and adaptation.
COP 23	2017	**Bonn, Germany**	Presided over by Fiji, the confere **the Paris Agreement's principl** and emphasised the **need to inc**
COP 24	2018	**Katowice, Poland**	Adopted the Paris Agreement W the "Paris Rulebook," which deta would be implemented

Global Financial mechanisms under UNFCCC	
Loss and Damage Fund (LDF)	• Established during **COP27 held in Egypt** and **operation** • Aims to provide **financial assistance to countries most** • "Loss and damage" refers to the negative effects of clima adaptation efforts.
Green Climate Fund (GCF)	• **Established in COP 16, 2010, developed countries ple by 2020 to** support developing countries raise and realiz
Adaptation Fund	• **Established in 2001** to finance concrete adaptation proje Parties to the **Kyoto Protocol**. • **Receives 5% share of proceeds from new market-bas** Article 6.4 of the Paris Agreement.
Special Climate Change Fund (SCCF)	• **Established in** 2001 (COP7) to finance projects relating capacity building; energy, transport, etc. • It is administered by **Global Environment Facility (GEF**

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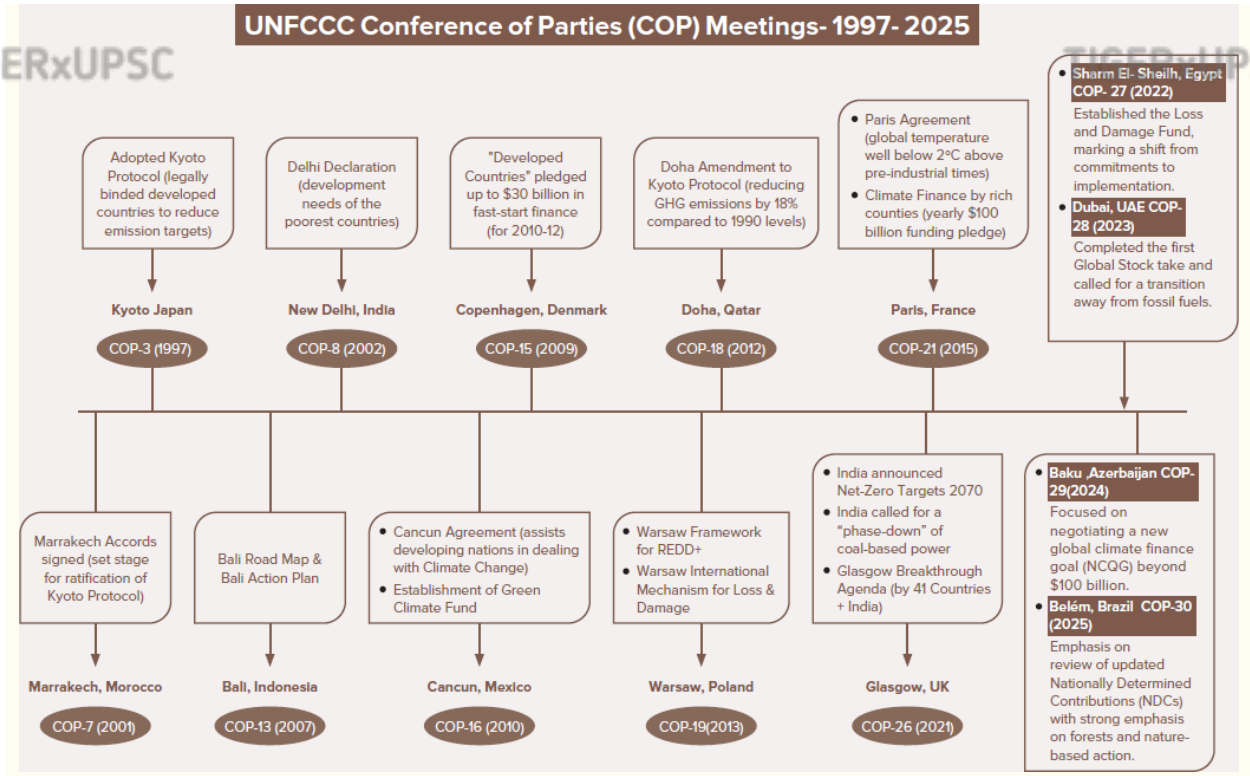

JUST TRANSITION

A Just Transition under UNFCCC means ensuring that climate actions particularly the shift from fossil fuels to clean energy are socially fair, inclusive and equitable. **Recognized first in the Paris Agreement Preamble,** the concept has evolved through the **Katowice Just Transition Declaration (COP24), Glasgow Climate Pact (COP26), and COP27 – 28 outcomes**.

UNFCCC treats Just Transition as a cross-cutting theme combining mitigation, adaptation, finance, and workforce protection. It stresses social dialogue, green job creation, social safety nets, and economic diversification. For developing countries, UNFCCC calls for international finance, technology transfer, and capacity building to support vulnerable workers and coal-dependent regions.

The concept of just transition, broadly defined as ensuring that no one is left behind or pushed behind in the transition to low- carbon and environmentally sustainable economies and societies.

According to the International Labour Organization (ILO), a **just transition** entails transforming economies inclusively and equitably, ensuring that climate action benefits are distributed fairly while minimising costs for vulnerable groups. It focuses on creating decent employment opportunities aligned with low-carbon development pathways, involving affected workers, communities, and industries in planning and decision-making processes.

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By prioritising equity, a just transition aims to share the costs and benefits of climate action fairly, particularly for marginalised and low-income groups. Ultimately, it represents a planned shift away from high-carbon development pathways toward sustainable alternatives, balancing environmental goals with social and economic considerations to safeguard livelihoods and promote social justice.

Need for a Just Transition

- **Impact on Employment:** The International Labour Organization (ILO) estimates that the green transition could create approximately **24 million new jobs by 2030**, but it may also result in the loss of around **6 million jobs** in carbon-intensive industries, highlighting the need for worker protection and reskilling initiatives.
- **Climate Mitigation Imperatives:** Reducing emissions from the energy sector is essential to achieving the **Paris Agreement's goal** of limiting global warming to **1.5°C**. Scientific analyses suggest that global greenhouse gas emissions must **peak by 2025** and decline by **43% by 2030**.
- **Energy Security Considerations:** Expanding renewable energy sources reduces reliance on volatile fossil-fuel markets, which are frequently impacted by **geopolitical tensions** and **price fluctuations**.
- **Preventing Social Unrest:** Poorly designed climate policies can lead to public opposition, as demonstrated by the **Yellow Vest protests in France (2018)**, where climate measures disproportionately affected lower-income groups.

Challenges in Achieving a Just Transition

- **High Transition Costs:** Transitioning coal-mining and thermal power sectors may require significant financial resources, with estimates suggesting that **India alone** could need over **a trillion dollars** in the coming decades (Just Transition, Just Finance report).
- **Regional Inequalities:** Coal-dependent regions, such as **Mpumalanga in South Africa** and **Jharkhand in India**, face limited options for economic diversification, exacerbating disparities in readiness for transition.
- **Energy Access and Affordability:** Coal remains a critical energy source in many developing nations, meeting development and energy-access needs. For instance, coal accounts for **55% of India's commercial energy demand**, and coal-based thermal power plants generate over **70% of electricity**.
- **Economic Risks from Stranded Assets:** The shift to clean energy could render fossil-fuel assets economically unviable, leading to significant financial losses. For example, **Saudi Arabia** derived approximately **68% of its government revenue in 2022** from oil.
- **Additional Constraints:** Challenges such as **inadequate grid infrastructure**, **limited technology transfer**, and **insufficient institutional capacity** further hinder the transition process.

Initiatives for Just Transition in India

- **Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY):** Focuses on ensuring minimum development provisions in mining-affected regions through **District Mineral Foundations (DMFs)**.

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- **District Mineral Foundations (DMFs):** Non-profit trusts established under the **Mines and Minerals (Development and Regulation) Amendment Act, 2015**, by State Governments in all mining-impacted districts.
- **Production-Linked Incentive (PLI) Scheme for High-Efficiency Solar PV Modules:** Aims to create a domestic manufacturing ecosystem for **advanced solar technologies**.
- **International cooperation:** Collaboration with the **Asian Development Bank** to design a **Just Transition Worker Support Facility** for coal-dependent districts and states.
- **National Clean Energy Fund (NCEF):** Finances clean-energy initiatives using proceeds from a **coal cess**.
- **Other measures:** Development of **solar parks and solar cities**, the **National Green Hydrogen Mission**, and **Green Energy Corridors** to support renewable-energy integration

9.8 REDUCTION EMISSION FROM DEFORESTATION AND FOREST DEGRADATION (REDD & REDD+)

REDD and REDD+ were introduced under the UNFCCC **to reduce global greenhouse gas emissions by protecting forests**, especially in developing countries.

- REDD **(Reducing emissions from deforestation and forest degradation):** This initial idea was discussed in 2005 at the UNFCCC Conference of the Parties. It was then officially launched in 2008 with the establishment of the UN-REDD Programme.
- REDD+ **(with the "+" for additional activities):** The framework was formally recognised in 2010 under the Cancun Agreements at COP-16 . Key decisions for the REDD+ framework were completed by 2013, and the final rulebook by 2015.

REDD = Reducing Emissions from Deforestation and Forest Degradation

It is a climate-change mitigation strategy developed under the **UNFCCC** to **reduce greenhouse gas emissions** by:

1. **Stopping deforestation**
2. **Reducing forest degradation**

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Purpose of REDD

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- Cutting down forests releases massive amounts of CO_2.
- Forest loss contributes to around **15% of global greenhouse gas emissions**.
- Protecting forests is one of the **easiest and fastest** ways to fight climate change.

REDD+

REDD+ is an **expanded version** of REDD that includes additional activities to enhance forest carbon storage.

REDD+ includes five key actions

1. **Reducing deforestation**
2. **Reducing forest degradation**
3. **Conservation of forest carbon stocks**
4. **Sustainable management of forests**
5. **Enhancement of forest carbon stocks** (e.g., reforestation, afforestation, restoring degraded forests)

REDD expanded into REDD+

Because many developing countries argued that:

- Merely "reducing" deforestation was insufficient
- They should also be rewarded for **protecting** forests and **increasing** forest cover

Thus, REDD+ recognises **both forest protection and forest improvement**.

Table 9.2 Key Differences: REDD vs REDD+

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Feature	REDD	REDD+
Focus	Reduce deforestation & degradation	REDD + conservation + s enhancement of carbon s
Scope	Narrow	Broader, holistic forest ma
Incentives	Payments for *avoided loss*	Payments for avoided los
Goal	Reduce emissions	Reduce emissions + incre

■ How REDD+ Works

1. A developing country protects or improves its forests.
2. It measures how much **carbon emissions** were avoided or how much **carbon storage** increased.
3. It receives **financial incentives** (payments, carbon credits, funding) from developed countries, donors, or carbon markets.

This encourages countries to **keep forests standing** instead of cutting them down.

■ Benefits of REDD+

- Helps **fight climate change**
- Protects **biodiversity**
- Supports **Indigenous peoples and forest communities**
- Promotes **sustainable forest economies**
- Helps countries meet **NDC (Paris Agreement) targets**

■ Challenges / Criticisms

- Difficult to measure forest carbon accurately
- Risk of corruption or misuse of funds
- May ignore indigenous land rights if not implemented properly

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- Needs long-term finance and political will

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India has prepared National REDD+ Strategy in 2018

India is an active participant in the UNFCCC REDD+ mechanism and has developed a strong national framework to protect and enhance its forests.

- India released its National REDD+ Strategy in 2018, aligned with UNFCCC rules and the Warsaw Framework.
- A three-tier institutional system was created:**The National REDD+ Cell, State REDD+ Cells, and a Technical Committee play key roles in implementation.**
- **India employs robust forest monitoring systems, such as the Forest Survey of India (FSI), to assess forest carbon levels.**
- **REDD+ aligns with India's NDC goals of establishing an additional carbon sink of approx. 2.5–3 billion tonnes of CO_2 by 2030.**
- **Several REDD+ pilot initiatives have been conducted across various states, including Himachal Pradesh, Madhya Pradesh, Assam, and the Terai region.**
- **In international REDD+ discussions, India emphasises equitable financing, active community involvement, and the protection of Indigenous rights.**

Key challenges include limited finance, forest fires, development pressures, and the need for stronger local-level MRV and benefit-sharing. MRV is a system used to track and ensure the effectiveness of climate actions, including REDD+ projects.

9.9 FOREST CARBON PARTNERSHIP FACILITY (FCPF)

FCPF (Forest Carbon Partnership Facility) is a **World Bank-led global partnership**, launched in **2008**, to help developing countries implement **REDD+**. It has two main components:

1. **Readiness Fund** – Helps countries prepare REDD+ strategies, set up MRV systems, and ensure safeguards.
2. **Carbon Fund** – Provides **results-oriented payments** for verified reductions in emissions due to deforestation and forest degradation.

Purpose: Reduce forest-related carbon emissions, strengthen governance, support communities, and promote sustainable forest management.

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Example: India participates in FCPF to enhance forest carbon stocks and improve REDD+ readiness.

9.10 BIO CARBON FUND, INITIATIVE FOR SUSTAINABLE FOREST LANDSCAPE (ISFL)

Bio Carbon Fund, Initiative for Sustainable Forest Landscape (ISFL) is a **multilateral fund** that is solely being **managed** by the **World Bank**. It works on the principle of REDD+ and provides fund to the countries which perform well in dealing with the reduction of greenhouse gas emissions from the land. It can be achieved through better land policy implementation, sustainable agriculture, promoting afforestation and reforestation, involving the tribal sections who protect the rural forest areas and many more strategies. This will, in turn, enhance the livelihood of rural habitats, boost economy, reduce greenhouse gas emissions and help reach the goal of the Paris Agreement of restraining the global average temperature rise below 2°C. The result-based financing pilot is expected to encourage countries to perform better and thus promote better implementation of sustainable use of land at a large scale and conserve carbon emission.

ENV 36 - Climate Action Tracker (CAT)

9.11 CLIMATE ACTION TRACKER (CAT)

9.11.1 Carbon Fund vs. Biocarbon Fund

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Features	Carbon Fund (FCPF) [REDD+, reducing emissions from forest loss]	Bio [Bro sequ
Managed by	World Bank through **the** Forest Carbon Partnership Facility (FCPF)	Wor Fun
Purpose	Provides **results-oriented payments** to developing countries for reducing emissions from deforestation and forest degradation (REDD+)**.**	Sup and sequ
Focus	Primarily on **REDD+ activities** — increased reforestation, reducing forest degradation,	Broa 1. A

The Climate Action Tracker (CAT) is an independent scientific initiative designed to monitor government climate actions and evaluate them against the Paris Agreement's goal of limiting global warming to "well below 2°C" while striving to cap it at 1.5°C.

Developed and maintained by **Climate Analytics and the New Climate Institute**, both based in Germany, CAT operates independently of governments and the UNFCCC, offering impartial and scientifically grounded assessments. It analyses the climate policies of 39 countries and the European Union, collectively accounting for approximately 85% of global emissions.

Purpose / Objectives

1. **Assess National Policies** – Measures if a country's climate action aligns with the Paris Agreement.
2. **Provide Transparency** – Shows progress toward global climate targets.

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3. **Inform Policy & Public** – Helps governments, NGOs, and citizens understand the **effectiveness of climate commitments**.
4. **Highlight Gaps** – Identifies countries falling short of targets and where urgent action is needed.

How It Works

- Tracks **greenhouse gas emissions**, **NDCs**, and climate policies.
- Uses scientific modelling to **estimate future warming based on current pledges**.
- Classifies countries' climate efforts as:
  "Role model" / "Almost sufficient" / "Insufficient" / "Highly insufficient" / "Critically insufficient"
- **Emissions Gap:** The **2030 Emissions Gap** refers to the difference between the greenhouse gas emissions required to limit global warming to below 2°C above pre-industrial levels—and ideally to 1.5°C by 2100—and the emissions expected to result from current Nationally Determined Contributions (NDCs), pledges, and policies. This gap underscores the need for more ambitious climate action to meet global targets.

■ *Funding Sources*

The Climate Action Tracker is made possible through the support of various foundations and governments, which fund different aspects of its work. Key contributors include:

- **Federal Government of Germany** via the International Climate Initiative
- **Climate Emergency Collaboration Group**, facilitated by Rockefeller Philanthropy Advisors
- **ClimateWorks Foundation**
- **European Climate Foundation**

In the past, consortium members such as the Potsdam Institute for Climate Impact Research (2009–2016) and Ecofys, a Navigant company (2009–2019), also played a role in supporting its work.

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ENV 37- EP100

9.12 EP100

EP 100 is a global program led by a climate group, working alongside the alliance to save energy. It brings together over 120 businesses that are committed to improving energy efficiency and sharing their progress. These companies set ambitious energy goals as part of their business strategies, helping them innovate, stay competitive, and reduce greenhouse gas emissions.

Energy efficiency is crucial for tackling climate change, as it can contribute to over 40% of the reductions needed in energy-related emissions. EP100 members are leading the way by cutting emissions, improving their operations, and inspiring others to follow their example. Indian companies like Dalmia Cement, Godrej & Boyce, JSW Cement, Mahindra & Mahindra, Mahindra Heavy Engines, Swaraj Engines, and Ultratech Cement are part of this initiative.

A climate group, also manages the Under2 Coalition, the largest network of regional governments working to achieve net-zero emissions by 2050 or sooner. The coalition helps governments speed up climate action through four main areas:

1. **Pathways –** Providing resources and support to help governments plan for long-term emissions reductions.

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2. **Policy Action** – Sharing successful policies and ideas to make the shift to a net-zero economy faster.
3. **Transparency** – Encouraging governments to openly share their climate goals, actions, and progress.
4. **Diplomacy** – Amplifying the voices of regional governments to push for stronger climate action and deliver real results.

Through EP100 and the Under2 Coalition, the climate group is helping businesses and governments work together to fight climate change and build a sustainable future.

Energy Productivity

Energy productivity refers to the measurement of economic output relative to energy consumption. When organisations adopt efficiency measures, members of the EP100 initiative can enhance their energy productivity, achieving greater economic value for each unit of energy utilised while simultaneously lowering greenhouse gas emissions.

By adopting smarter energy practices, there is a reduction in the overall demand for primary energy, leading to an accelerated adoption of more efficient technologies and energy-efficient practices. The International Energy Agency (IEA) has conducted research indicating that this shift towards improved energy productivity could contribute up to US$18 trillion to the global Gross Domestic Product (GDP) by the year 2035.

Energy Productivity = Economic Output/ Energy Input

9.13 INTERNATIONAL CODE FOR SHIPS OPERATING IN POLAR WATERS (POLAR CODE)

The International Code for Ships Operating in Polar Waters, known as the Polar Code, is a regulatory framework introduced by the International Maritime Organization (IMO). Recognising the unique challenges posed by polar regions, the Code establishes mandatory requirements for ships navigating these demanding environments.

The Polar Code is enforced under both the International Convention for the Safety of Life at Sea (SOLAS) and the International Convention for the

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Prevention of Pollution from Ships (MARPOL). It addresses various aspects essential for safe and environmentally responsible operations in polar waters, including ship design, construction, equipment, operations, crew training, search and rescue procedures, and environmental protection measures.

This comprehensive framework ensures that ships are adequately prepared to operate in the extreme conditions of the Arctic and Antarctic regions. The Polar Code officially came into effect on 1 January 2017.

9.14 GREENHOUSE GAS PROTOCOL

The Greenhouse Gas Protocol (GHG Protocol) is an international accounting mechanism for government and company leaders to recognise, quantify and administer greenhouse gas emissions.

A decade-long partnership between the **World Resources Institute (WRI)** and the **World Business Council for Sustainable Development (WBCSD**), the GHG Protocol is operational with companies, governments and environmental groups around the globe to build a new generation of convincing and effective programmes for tackling climate change.

It serves as the groundwork for nearly every GHG standard and programme in the world, from the International Standards Organisation to The Climate Registry, as well as hundreds of GHG inventories conducted by individual companies.

The GHG Protocol also offers developing countries a globally accepted management device to help their businesses compete in the global marketplace, and their governments make knowledgeable decisions about climate change.

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9.15 UN CONVENTION ON COMBAT DESERTIFICATION (UNCCD)

- It is **an international agreement** which introduces both environment and development for sustainable land management. It was accepted in June 1994 in Paris, which came into force in December 1996 on the recommendation of the Rio conference's Agenda 21. This convention aims to **lessen the effects of drought through nationwide action programmes that incorporate an extended period of strategies, which the International Corporation reinforces.**
- The **year 2006** was proclaimed the "**International Year of Deserts and Desertification**" to highlight the issues affecting drylands, which encompass arid, semi-arid, and dry sub-humid regions. These areas are highly susceptible to desertification, a process that profoundly impacts ecosystems and human livelihoods. The UNCCD serves as the leading international framework to address this concern, with 197 countries having ratified the agreement. Canada became the first nation to withdraw from the UNCCD in 2013 but rejoined in 2016, reaffirming its dedication to combating desertification. The inaugural Conference of Parties (COP1) of the UNCCD took place in Rome in 1997, initiating global efforts to tackle desertification and promote sustainable land management.

India ratified the United Nations Convention to Combat Desertification (UNCCD) **in December 1996**, with the Ministry of Environment, Forest and Climate Change designated as the nodal ministry for its implementation. Desertification poses a major challenge for the country, as highlighted by a **2016 ISRO report**, which revealed that **29% of India's land was degraded**. This underscores the need for sustainable land management practices and proactive measures to address land degradation effectively.

By embracing a grassroots approach, the UNCCD empowers local communities to take an active role in combating desertification,

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fostering a more inclusive and effective path towards sustainable land management. It published a report "Global Land Outlook"

Issue of Land Degradation and Desertification

- **Land degradation** is the loss of soil productivity, reducing its ability to support current and future needs.
- Globally, land degradation impacts **40% of the Earth's land area**, causing the loss of **100 million hectares** of fertile land annually.
- In India, **32% of the land** is degraded, while **25% is undergoing desertification**, emphasising the need for sustainable land management practices.

Table 9.3 List of all UNCCD Conference of Parties (COP) Meeting

COP	Date	Location
COP 1	29 Sep – 10 Oct 1997	Rome, Ita
COP 2	30 Nov – 11 Dec 1998	Dakar, Se
COP 3	15 – 26 Nov 1999	Recife, Br
COP 4	11 – 22 Dec 2000	Bonn, Ge
COP 5	1 – 12 Oct 2001	Geneva, S
COP 6	25 Aug – 5 Sep 2003	Havana, C

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Table 9.4 Major UNCCD COPs and their key outcomes, focusing only on the most important milestones:

COP	Year	Location	Major Decisions / Outcomes
COP 6	2003	Havana, Cuba	Designated GEF as a financial mec support for affected countries.
COP 8	2007	Madrid, Spain	Adopted 10-Year Strategic Plan (20 land degradation, sustainable land r services, and partnerships.
COP 11	2013	Windhoek, Namibia	Emphasis on Land Degradation Ne development plans.
COP 12	2015	Ankara, Turkey	Alignment with 2030 Agenda / SDG improved monitoring & reporting.

■ *IMPORTANT DIMENSIONS UNDER UNCCD*

1. **UNCCD 2018-2030 Strategic Framework:** A Global Commitment to Land Degradation Neutrality
2. **The UNCCD Strategic Framework** represents the most comprehensive global effort to achieve Land Degradation Neutrality (LDN), aligning with the 2030 Agenda for Sustainable Development.

Key Components:

- **Land Degradation Neutrality (LDN):** Refers to maintaining or improving the quality and quantity of land resources to support ecosystem services, food security, and sustainable development.
- **LDN Target Setting Program:** The Global Mechanism and UNCCD Secretariat work with international partners to assist countries in

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establishing voluntary national LDN targets. India has pledged to restore 26 million hectares of degraded land by 2030.

- **LDN Fund:** An innovative impact investment fund that combines public, private, and philanthropic resources to finance sustainable land management and restoration projects. Launched at COP 13 in Ordos, China, the fund uses public investments to attract private capital for land restoration initiatives.

Other Key Initiatives

The Great Green Wall (2007): Combating Desertification in Africa

- **Launch and Vision:** Initiated by the African Union in 2007, the Great Green Wall aims to restore degraded landscapes across Africa and improve the lives of millions in the Sahel, one of the world's poorest regions.
- **Scale:** When completed, the wall will span 8,000 km across the continent, making it the largest living structure on Earth.
- **Sahel Region Focus:** The initiative targets the Sahel, a transitional zone between the Sahara Desert and the fertile lands to the south**.**
- **Participating Countries:** Eleven countries are involved, including Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal, and Sudan.
- **Integrated Approach:** The project employs sustainable land management practices such as tree planting, agroforestry, and natural resource management.
- **International Partnerships:** The initiative receives support from global partners, including the European Union and other organizations.
- **UNCCD Coordination:** The Great Green Wall is coordinated by the United Nations Convention to Combat Desertification (UNCCD).

Exam Pointer

Global Green wall Initiative: Restoring Land Across the Sahel Region

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The Great Green Wall Initiative is an ambitious African-led project focused on combating desertification and restoring degraded land across the Sahel region.

Goals by 2030:

- Land Restoration: Restore 100 million hectares of degraded land.
- Carbon Sequestration: Capture 250 million tonnes of carbon.
- Job Creation: Generate 10 million green jobs, transforming livelihoods in one of the world's most vulnerable regions.

This initiative not only addresses environmental challenges but also aims to improve economic opportunities and resilience for communities in the Sahel.

Figure 9.2 Great Green Wall of Trees

3) COP 14: New Delhi Declaration (2019)

UNCCD COP14: Outcomes of the 2019 Conference in Greater Noida

The 14th Conference of Parties **(COP14) to the United Nations Convention to Combat Desertification (UNCCD) was held from September 2 to September 13, 2019, in Greater Noida, Uttar Pradesh**. The conference decided to adopt the **New Delhi Declaration**, describing the measures for global land restoration of degraded land.

Key outcomes

Funding Needs: The Parties to the Convention have identified the mobilisation of $300 billion for land restoration as their primary financial objective. To sustainably manage the land in a way that is profitable, small

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and medium-sized private enterprises must be actively engaged in the process.

Incentivising Investments: Ibrahim Thiaw, UNCCD Executive Secretary, talked about ways of encouraging investors to invest in land restoration.

Four Takeaways:

Carbon Sequestration: Enhancing nature's ability to capture carbon by putting the land in good condition.

Drought Preparedness: Countries showed their commitment to addressing the issues related to drought.

People-Centric Approach: The restoration of land will succeed if people are given the priority they deserve.

Land Tenure Security: The issue of land tenure security was discussed in detail, as it is a prerequisite for restoration efforts.

India's Commitment: India's then Environment Minister, Prakash Javadekar, committed that India would continue to issue land titles to the rightful forest dwellers. Tribals and forest dwellers should not be considered as mere consumers of forests, but as integral parts of the forest developmental strategies as they reside in and protect forests.

India's Initiatives: The New Delhi Declaration acknowledged India's seven initiatives for sustainable land use, including:

- **National Afforestation Programme**
- **Har Khet Ko Pani** (irrigation for every farm)
- **More Crop Per Drop** (water use efficiency in agriculture)
- **Rural Employment Guarantee Scheme**
- **Soil Health Cards**

Global Impact: The outcomes of COP14 will influence discussions at the UN Climate Action Summit in New York on September 23, 2019. The conference emphasized land-based solutions for climate action, biodiversity

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conservation, and the interconnected implementation of the three Rio Conventions.

Rio Conventions: The three conventions—climate change, land, and biodiversity—originated from the 1992 Earth Summit in Rio de Janeiro, underscoring their mutual support in addressing global environmental challenges.

The New Delhi Declaration serves as a significant step toward global collaboration in combating desertification and promoting sustainable land management practices

Delhi Declaration COP 14: Highlights

Promote initiatives to combat land degradation at local, national, and regional levels.

- Support projects aimed at preventing drought and erosion, while transforming barren land into cultivable areas.
- Promote initiatives to combat land degradation at local, national, and regional levels.
- Support projects aimed at preventing drought and erosion, while transforming barren land into cultivable areas.
- Develop strategies to mitigate drought to prevent land degradation.
- Foster green job creation by encouraging technology adoption and investment through collaboration with stakeholders.
- Advance efforts to prevent land degradation in alignment with the Paris Agreement.
- Implement the UN Ecosystem Restoration program (2021–2030) on behalf of all nations.
- Strengthen programs initiated by African nations to combat land degradation.
- Endorse the benefits of the Peace Forest Initiative and call for collective action to prevent land from becoming barren.
- Advocate for greater accountability of local governments in preventing land degradation and ensuring transparency in land rights.
- Encourage local governments to take responsibility for preserving natural resources and ecosystems.
- COP14 member countries acknowledged India's contributions.
- Prime Minister Narendra Modi's leadership in achieving land restoration goals and promoting South-South cooperation received widespread appreciation.

4) COP 15: Abidjan (2022)

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Fifteenth session of Conference of the Parties (COP15) of United Nations Convention to Combat Desertification (UNCCD), held in Abidjan (Ivory Coast).

- The COP15 concluded with a global pledge to boost drought resilience and invest in land restoration for future prosperity with the theme "Land Life Legacy: From scarcity to prosperity".
- As an outcome of the COP, the member states agreed to establish an Intergovernmental Working Group on Drought for 2022-2024 to **support a shift from reactive to proactive drought management** and **agreed and committed to accelerate the restoration of one billion hectares of degraded land by 2030.**
- Other significant outcomes of the COP 15 included three key declarations:
  1. Abidjan Call issued by the Heads of State and Government to boost long-term environmental sustainability
  2. Abidjan Declaration on achieving gender equality for successful land restoration
  3. COP15 "Land, Life and Legacy" Declaration, which responds to the findings of the UNCCD's flagship report, Global Land Outlook 2

■ *5) COP 16: Riyadh, Saudi Arabia (2024)*

The 16th Conference of the Parties (COP16) to the United Nations Convention to Combat Desertification (UNCCD) concluded in Riyadh, Saudi Arabia, with nearly 200 nations committing to prioritising land restoration and drought resilience. This marked the first time a UNCCD COP was hosted in the Middle East and North Africa (MENA) region.

Despite the progress made, negotiators were unable to finalise a binding protocol or agreement on addressing drought. With droughts projected to impact 75% of the global population by 2050, the urgency for action remains critical.

■ *Key outcomes of COP 15*

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New Commitments	• Accelerate the restoration of **one billion hectares of degraded** collection and monitoring systems. • Establish an **Intergovernmental Working Group on Drought** proactive drought management. Develop and implement strateg other increasing disaster risks. • Strengthen collaboration among the **three Rio Conventions**—t (CBD), UNCCD, and UNFCCC—to ensure greater synergy in a
New Declaration Issued	• **Abidjan Call:** A commitment by Heads of State and Governmer sustainability. • **Abidjan Declaration:** Focused on achieving gender equality as efforts. • **COP15 'Land, Life and Legacy' Declaration:** A response to fir *Land Outlook 2*, emphasising sustainable land management.
Other Initiatives	• **Drought in Numbers, 2022 Report:** Revealed that over **2.3 bil** globally. • **Business for Land Initiative:** A platform to highlight corporate degradation neutrality. • **Sahel Sourcing Challenge:** A program empowering communiti

■ *Key Outcomes of COP16 UNCCD*

- **Global Drought Framework:** Nations advanced discussions on establishing a global drought framework, with plans to finalise it at COP17 in Mongolia in 2026.
- **Financial Commitments:** Over USD 12 billion was pledged to combat desertification, land degradation, and drought.
  - **Riyadh Global Drought Resilience Partnership:** USD 12.15 billion was committed to supporting 80 vulnerable countries, including USD 10 billion from the Arab Coordination Group.
  - **Great Green Wall (GGW) Initiative:** The African-led GGW initiative received Euros 11 million from Italy for Sahel landscape restoration and Euros 3.6 million from Austria to enhance coordination across 22 African nations.
  - **ØVision for Adapted Crops and Soils (VACS):** Nearly USD 70 million was allocated to the VACS initiative, which aims to build resilient food systems through climate-adapted crops and healthy soils.

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- **Indigenous Peoples and Local Communities:** Indigenous Peoples and Local Communities were actively involved to ensure their perspectives and challenges were addressed. The Sacred Lands Declaration, presented at the Indigenous Peoples Forum, emphasised their role in global land and drought governance.
- **Business4Land Initiative:** This initiative underscored the importance of private sector involvement in addressing Desertification, Land Degradation, and Drought (DLDD). It highlighted the need for private sector advocacy, sustainable finance, and environmental, social, and governance (ESG) strategies, as the private sector currently contributes only 6% of financing toward land restoration and drought resilience.
- **UNCCD's Science-Policy Interface (SPI):** Parties agreed to continue the UNCCD's SPI, established at COP11 in Windhoek, Namibia, in 2013, to translate scientific research into actionable recommendations for policymakers.

COP16 reaffirmed the global commitment to combating desertification and drought while emphasizing the need for collaborative efforts, financial investments, and inclusive governance to achieve sustainable land restoration goals.

Committee on Science and Technology (CST)

The United Nations Convention to Combat Desertification (UNCCD) established the Committee on Science and Technology (CST), comprising government representatives with expertise in fields relevant to combating desertification and addressing the impacts of drought. The CST focuses on identifying research priorities and proposing strategies to enhance collaboration among researchers. This committee adopts a multidisciplinary approach and ensures participation from all member Parties.

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Upcoming UNCCD COP 17

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9.16 UN COMMISSION ON SUSTAINABLE DEVELOPMENT (CSD)

It was established in **1992** by the **UN General Assembly**. It was created to **monitor and ensure effective follow-up** of the **1992 Earth Summit** (Rio Conference) and implementation of **Agenda 21**.

Key Functions

- **Monitor progress** on sustainable development commitments (Agenda 21, Rio Declaration, Johannesburg Plan of Implementation).
- **Review implementation** of global agreements on: Climate change, Desertification, Biodiversity, Oceans and forest management, Sustainable consumption & production
- Act as a **policy coordination platform** for environment–development issues.
- Facilitate **dialogue** among governments, NGOs, scientists, women, youth, and business groups.
- Promote **technology transfer**, capacity building, and finance for sustainable development.

Structure

- A **functional commission** under the **UN Economic and Social Council (ECOSOC)**.
- Included **53 member states**, elected for 3-year terms.

Important Milestones

- Played a key role in reviewing progress leading up to:
- **Rio+5** (1997)

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- **World Summit on Sustainable Development (WSSD), Johannesburg (2002)**
- **Rio+20 (2012)**

Transition to HLPF

- In **2013**, CSD was **replaced** by the **High-Level Political Forum on Sustainable Development (HLPF)**.
- HLPF now monitors the implementation of the **2030 Agenda & SDGs**.

Significance

- First dedicated global platform to integrate **environment + economy + social** perspectives.
- Shaped global norms for **sustainable development governance**.
- Strengthened global understanding of sustainability challenges before SDGs were adopted.

19.16.1 RIO +5 | 1997 Newyork, United States

- Rio+5: Special session to appraise the status of Agenda 21
- In 1997, the United Nations General Assembly (UNGA) held a special meeting to evaluate the status of Agenda 21. The focus was to move sustainable growth **From Agenda to Action.** The forum focused on identifying key strategies and management systems for "operationalising" sustainable development both nationwide and global wide.

19.16.2 Rio+10 | 2002 World Summit on Sustainable Development (WSSD) | Johannesburg Declaration

- **Rio+10: World Summit on Sustainable Development**
- The World Summit on Sustainable Development, held in 2002 and informally called "Rio+10," marked the 10th anniversary of the first Earth Summit in Rio de Janeiro. Despite high expectations, the summit ended in disappointment, as it failed to deliver any significant new commitments to address environmental crises. Governments demonstrated limited urgency

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and enthusiasm, even as global environmental conditions continued to deteriorate. The summit concluded with weak, non-binding agreements aimed at promoting sustainable development, highlighting the need for stronger and more impactful global action.

19.16.3 RIO+20 | 2012 RIO De Janeiro, Brazil

- Rio+20: UN Conference on Sustainable Development
- A 20-year follow-up to the 1992 Earth Summit was Rio+20, hosted by Brazil in Rio de Janeiro and the 10th anniversary of 2002, WSSD in Johannesburg.
- The conference centred on Agenda 21, the outcome document from Earth Summit 1992. It was considered 'Revolutionary' in the sense that it essentially created the term sustainable development and created the global environmental agenda for the next 20 years.
- In this conference, some partnership declarations were done.
- Various issues were initiated under this conference:
    1. Requirement of sustainable development
    2. Removal of food crisis
    3. Development of transport system in developing countries with financial and technical support
    4. Development of Green economy
    5. Removal of Poverty
    6. International partnership for sustainable development.

Partnership for Action on Green Economy (PAGE)

- It was founded in 2013 as a direct response to Rio+ 20 declaration – The Future We Want.
- The partnership supports nations and regions in reframing economic policies and practices around sustainability to foster economic growth, create income and jobs, reduce poverty and inequality, and strengthen the ecological foundations of their economies.
- It brings together the expertise of five UN agencies, namely United Nation Environment Programme (UNEP), International Labour Organisation (ILO), United Nations Industrial Development Organization (UNIDO), United Nation Development Programme (UNDP) and United Nations Institute for Training and Research (UNITAR). PAGE assist

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countries in achieving and monitoring the emerging Sustainable Development Goals, especially SDG 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

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Club of Rome

- It is not for profit organisation founded in 1968 in Rome (Italy).
- Since 2008, its headquarters shifted to Switzerland.
- It is an organisation of individuals who share a common concern for the future of humanity and strive to make a difference.
- Its mission is to promote understanding of the global challenges facing humanity and to propose solutions through scientific analysation, communication and advocacy.
- It published its report The Limits to Growth in 1972.

9.17 SUSTAINABLE DEVELOPMENT GOALS

- The Sustainable Development Goals (SDGs) or Global Goals are a collection of 17 interlinked global goals designed to be a blueprint to achieve a better and more sustainable future for all.
- The SDGs were adopted in 2015 by the United Nations General Assembly (UN-GA) and are intended to be achieved by the year 2030 with **17 Goals, 169 Targets & 247 Indicators.**
- They act as a **blueprint for global development** focusing on ending poverty, protecting the planet, and ensuring prosperity for all. **SDGs replaced the earlier MDGs (Millennium Development Goals, 2000–2015).**

■ *Principle: "Leave No One Behind" (LNOB)*

Key Features of SDGs

- **Universal**: Applicable to all countries—developed, developing, least developed.
- **Integrated**: Address economic, social, and environmental dimensions together.

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- **Inclusive**: No one left behind (LNOB principle).
- **Time-bound**: Target year – **2030**.
- **Interconnected**: Achievements in one goal support others.
- **Data-driven**: 247 indicators to monitor progress

Importance of SDGs

- Provide a **holistic approach** to global development.
- Highlight climate changes, inequality, health, poverty, and sustainable consumption.
- Encourage **global partnerships**.
- Link national policies with global commitments.
- Promote **sustainable development + human rights + environmental protection**.

Challenges in Achieving SDGs

- Slow progress in climate, hunger, inequality.
- Funding gaps, conflicts, climate disasters.
- Limited data for indicators in many countries.
- Impact of COVID-19 setback.

Sustainable Development Solutions Network (SDSN)

- It was launched in 2012 by UN Secretary General Ban Ki-moon to mobilise global scientific and technological expertise to promote practical problem solving for sustainable development and implement SDGs.
- Objective: To promote integrated approaches to implement the sustainable development goals (SDGs) and The Paris Agreement on Climate Change, through education, research, policy analysis, and global cooperation.
- Membership: over 1700 members in 15 networks across 144 countries.

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Figure 9.3 Sustainable Development Goals (SDGs)

India & SDGs

- Nodal Institution: NITI Aayog (monitors SDGs).
- States prepare State Indicator Framework (SIF).
- India releases SDG India Index annually.

Key Indian initiatives aligned with SDGs:

- Swachh Bharat Mission – SDG 6
- Ayushman Bharat – SDG 3
- Beti Bachao Beti Padhao – SDG 5
- UJALA / Solar Mission – SDG 7
- PM Awas Yojana – SDG 11
- Jal Jeevan Mission – SDG 6

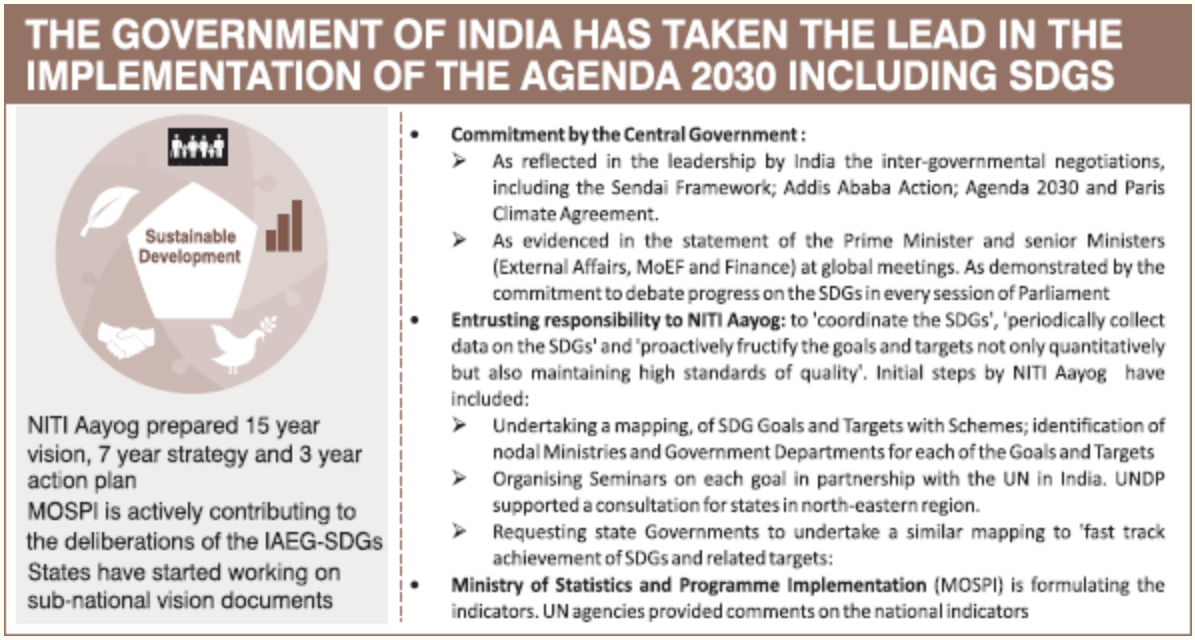

Figure 9.4

Table 9.5 Sustainable Development Goals (SDGs) vs Millenium Development Goals (MDGs)

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MDGs (2000–15)	SDGs (2015–30)
8 goals	17 goals
Only developing countries	Universal
Narrow focus (health, education)	Broad focus (economic, s
Top-down approach	Participatory approach
Limited indicators	247 indicators

Millennium Development Goal (MDGs): MDG was adopted in the millennium summit of the United Nations in 2000. Under MDG, total of eight international development goals were decided to achieve by 2015. During that period, all UN member states and at least 22 international organisations committed to achieve MDG by 2015.

Eight Goals under MDG:

1. Eradication of extreme poverty and hunger
2. Achieve universal primary education
3. Promote gender equality and women empowerment
4. Reduce children mortality
5. Improve maternal health
6. To combat HIV/AIDS, malaria and other diseases
7. Ensure environmental sustainability
8. Develop a global partnership for development

ENV 38- Vienna Convention, Montreal Protocol & Kigali Agreement

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9.18 VIENNA CONVENTION, MONTREAL PROTOCOL & KIGALI AGREEMENT

Vienna Convention (1985)

- It is a multilateral international environmental agreement signed in 1985, which gave structure for reducing the manufacture of chlorofluorocarbons internationally because of their ozone layer destruction contribution. However, it does not include a legally binding reduction goal for using **Ozone-Depleting Substances (ODS).**
- Opened for signature in 1985
- Entered into force in 1988
- Ratified By: 197 (Universal)

Montreal Protocol (1985)

- It is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances that are responsible for ozone depletion.
- **197 member parties** have signed it and thus it became the **first such international treaty with complete ratification.**
- Several amendments were carried out, and the **Kigali amendment was the 8th amendment.**
- Successful cut down of 8% of chlorofluorocarbons production and other ODSs has greatly contributed to the repair of the ozone hole.

Ozone Depleting Gas Index

Ozone-Depleting Substances (ODS) Report by NOAA

Released by: The United States' National Oceanic and Atmospheric Administration (NOAA).

Purpose: The report monitors the overall stratospheric concentration of ozone-depleting chlorine (ODC) and bromine originating from long-lived

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ozone-depleting substances (ODSs), comparing it to its peak levels in the early 1990s.

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Key Finding

- In 2022, the overall concentration of ozone-depleting chlorine in the mid-latitude stratosphere returned to levels observed in 1980, a time when ozone depletion was not a significant concern.
- This indicates substantial progress in reducing ozone-depleting substances and restoring the ozone layer to pre-depletion

■ *Kigali Agreement (2016)*

- A historic agreement was signed by 97 nations with a vision of amending the Montreal protocol in 28th **meeting of the Montreal Protocol,** held **in Kigali,** the capital city of an African country, on 15th October 2016.
- The countries that signed the agreement are expected to cut down the manufacture and use of Hydrofluorocarbons (HFC), a greenhouse gas, by approximately 80–85% from their respective baselines, till 2045.
- This step is expected to halt the average rise in global temperature up to 0.5°C by 2100.
- Kigali agreement is, **thus, basically an amendment of the Montreal Protocol.**

■ *What are Hydrofluorocarbons (HFCs)*

- Hydrofluorocarbons are **organic compounds containing hydrogen, carbon, and fluorine.**
- Commonly used as **substitutes for** ozone-depleting substances like Chlorofluorocarbons (CFCs), such as **in refrigerators and air-conditioners.**

■ *HFCs emission does not escalate ozone layer depletion, but it leads to global warming*

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- Even though HFCs are not as harmful as CFCs, they are a thousand times more potent in causing global warming than commonly known greenhouse gases like carbon dioxide, methane, etc.

Kigali Agreement: Important Points

- It is a **legally binding agreement** among the **signatory parties with non-compliance measures.**
- It has been decided to come into effect from 1st January 2019 provided it is ratified by at least 20 member parties by then.
- It has **shown much-appreciated flexibility** while setting phase-down targets for different countries with different economic capacities based on their developmental aspirations, socio-economic compulsions and scientific and technological progress.
- Provision for a multilateral fund for developing countries was also mentioned in this agreement.

Kigali Amendment and India

India demonstrated a flexible and cooperative approach throughout the negotiation process, aligning its commitments with global climate goals. Despite consuming only 3% of hydrofluorocarbons (HFCs)—significantly lower than nations like the USA (37%) and China (25%)—India agreed to join the third group of signatory parties under the agreement.

This decision reflects India's balanced strategy of adhering to international agreements, such as the Paris Agreement, while pursuing its ambitious **'Make in India'** initiative to boost industrial production. However, the country's tropical climate, coupled with rising middle-class incomes, has led to increased demand for air conditioners, refrigerators, and automobiles, which must be factored into the program›s implementation.

As a responsible global player, India has voluntarily taken proactive measures, including ordering the termination of the production of HFC-23, a byproduct of commonly used refrigerants. This initiative is expected to reduce emissions by 100 million tons of carbon dioxide equivalent over the next 15 years, showcasing India's commitment to sustainable development and climate action.

Significance of the Kigali Agreement

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The Kigali Agreement is a landmark amendment to the Montreal Protocol, aimed at phasing down the use of hydrofluorocarbons (HFCs), potent greenhouse gases that contribute to global warming. It strengthens the ambition of the Paris Agreement by targeting the reduction of HFCs to limit the rise in average global temperature to below 2°C compared to pre-industrial levels.

■ *Implications of the Kigali Agreement on India*

India faces significant challenges due to the widespread use of HFCs in industries such as air conditioning, refrigeration, and automobiles. The agreement will have financial and technological implications for these sectors:

1. ***Financial Implications:***
   - Industries will need to invest heavily in research and development (R&D) to identify safer alternatives to HFCs or purchase patented substances and technologies from multinational corporations (MNCs).
   - This increased cost of production may reduce demand among buyers, shrinking the consumer base for these products.
2. ***Technological Implications:***
   - Developed nations have already begun using HFC substitutes, backed by strong technical expertise.
   - Without significant advancements in technology transfer or domestic research, Indian industries may struggle to compete with advanced products in the global market.
   - The Kigali Agreement represents an opportunity for India to innovate and transition toward sustainable practices, but it also requires strategic investments in R&D, technology transfer, and policy support to mitigate the financial and technical challenges

Table 9.6 List of Signatory Parties and Their Targets Under the Kigali Agreement

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Groups	Countries	Targets
Rich and Developed Economies	USA, UK, EU	Begin phasi reduce them 2036.
Emerging Economies	China, Brazil, Some African Countries	Start phase- HFCs to 20
Developing Economies and Hottest Climatic Countries	India, Pakistan, Iran, Saudi Arabia	Begin phasi reduce them by 2047.

This structured approach ensures equitable timelines for HFC reduction, considering the economic and climatic conditions of each group.

This will actually provide an impetus to the business sector and scientific institutions to invest their financial and intellectual resources in searching for more economic and environmentally safer alternatives.

■ *What next, after HFCs?*

Next comes HFO refrigerants. HFO stands for hydrofluoroolefin, composed of hydrogen, fluorine and carbon atoms connected by single bonds between the atoms and *at least one double bond between the carbon atoms.*

HFO-1234yf has physical properties similar to HFC-134a because of which it can be used in current HFC-134a systems with minimal system modifications.

■ *What are the main characteristics of HFO-1234yf?*

- Less toxic.
- Low GWP (Global Warming Potentials).

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- Zero ozone depletion potential.
- Totally safe contribution to climate change.
- Same operating pressures as the current HFC-134a system.

ENV 39- Global Environmental Facility (GEF)

9.19 GLOBAL ENVIRONMENTAL FACILITY (GEF)

- The Global Environment Facility was founded in October 1991, a **$1 billion pilot programme of the World Bank** to help protect the global environment and encourage environmentally sustainable developments.
- The Global Environment Facility (GEF) was restructured during the 1992 Rio Earth Summit, becoming an independent institution that provides grants and concessional funding to developing countries for environmental projects. It is headquartered in Washington, D.C., and is the only multilateral fund addressing all aspects of environmental health.

The GEF works with 10 agencies, leveraging their expertise in climate change adaptation:

- United Nations Development Programme (UNDP)
- United Nations Environment Programme (UNEP)
- World Bank
- Food and Agriculture Organisation (FAO)
- Inter-American Development Bank (IDB)
- United Nations Industrial Development Organisation (UNIDO)
- Asian Development Bank (ADB)
- African Development Bank (AfDB)
- European Bank for Reconstruction and Development (EBRD)

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  - International Fund for Agricultural Development (IFAD)
  - World Wildlife Fund (WWF) and Conservation International
- The GEF unites **186 countries**, including both developed and developing nations, in partnership with international institutions, civil society organizations (CSOs), and the private sector to address global environmental challenges.
- **Objective:** The GEF aims to assist developing countries in addressing critical environmental issues across five focal areas:
  - Biodiversity loss
  - Chemicals and waste
  - Climate change
  - International waters
  - Land degradation
- **Significance:**
  - The GEF is the **largest public funder** of projects aimed at improving the global environment.
  - It operates independently, providing grants for initiatives related to biodiversity, climate change, international waters, land degradation, the ozone layer, and persistent organic pollutants (POPs).
- **GEF Trust Fund:** The GEF Trust Fund was established to tackle urgent environmental challenges, offering financial support to developing countries and economies in transition to meet the objectives of international environmental conventions and agreements.

This structure highlights the GEF's role, objectives, and collaborative efforts in addressing global environmental issues.

GEF serves as the financial mechanism for five major international environmental conventions:

- Convention on Biological Diversity (CBD)
- United Nations Framework Convention on Climate Change (UNFCCC)
- UN Convention to Combat Desertification (UNCCD)
- Stockholm Convention on Persistent Organic Pollutants (POPs)
- Minamata Convention on Mercury

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The GEF, although not connected formally to the Montreal Protocol on Substances that Deplete the Ozone Layer (MP), helps the implementation of the Protocol in countries with economies in transition.

As part of its restructuring, the Global Environment Facility (GEF) was designated as the financial mechanism for **the UN Convention on Biological Diversity and the UN Framework Convention on Climate Change**. Additionally, in collaboration with the Montreal Protocol under the Vienna Convention on Ozone-Depleting Substances (ODS), the GEF began funding initiatives to assist the Russian Federation, as well as countries in Eastern Europe and Central Asia, in reducing their reliance on ozone-depleting chemicals.

Areas of work

The GEF work aims at **seven main areas:**

1. Biodiversity
2. Change in climate (mitigation and adaptation)
3. Chemicals
4. International water bodies
5. Degradation of Land
6. Sustainable management of forest/REDD+ and
7. Depletion of the Ozone layer

Other key information

- **Funds managed by Global Environment Facility (GEF): Special Climate Change Fund (SCCF)** and **Least Developed Countries Fund (LDCF)**
- Also supports **Montreal protocol,** Global Ship Ballast Water Treaty, **UN Law of the Sea Treaty, MARPOL treaty** for shipping (International Convention for the Prevention of Pollution from Ships) and UN Agreement on conservation and management of straddling fish stocks and highly migratory fish stocks.

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Table 9.7 Special Climate Change Fund (SCCF) vs Least Developed Countries Fund (LDCF) *(Both managed by the Global Environment Facility – GEF)*

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Feature	Special Climate Change Fund (SCCF)	Least Deve
Established Under	UNFCCC (2001, Marrakech Accords)	UNFCCC (200
Objective	Finance climate change activities in **all developing countries**	Support **Least** & implementing
Primary Focus	Adaptation, technology transfer, sectoral resilience	**National Adap** LDC-specific ac
Priority Areas	• Adaptation projects in vulnerable sectors (water, agriculture, health) • Technology transfer • Capacity-building • Energy, industry & transport vulnerability	• Funding for l • Adaptation ir • Climate-resil

9.20 UNITED NATIONS EDUCATIONAL, SCIENTIFIC AND CULTURAL ORGANISATION (UNESCO)

- The United Nations Educational, Scientific and Cultural Organisation (UNESCO) is United Nations (UN) based agency, founded on 16th November 1945. It is headquartered in Paris.
- Its declared purpose is to contribute peace and safety by promoting international collaboration through instructive, precise and cultural reforms to increase universal respect for justice. UNESCO has 195 member states and nine associate members.

UNESCO pursues its objectives through five major programmes:
1. Education
2. Natural sciences

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3. Societal/human sciences
4. Culture
5. Communication Information

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- Projects funded by UNESCO include literacy, technical, and teacher-training programmes, worldwide science programmes, the advertising of independent media and freedom of the press, local and cultural history projects, the support of cultural diversity, translations of world literature, international cooperation agreements to secure the global cultural and natural heritage (World Heritage Sites) and to conserve human rights, and attempts to bridge the global digital divide.
- It is also an associate of the United Nations Development Group.
- UNESCO's aims are to supply the construction of peace, the abolition of poverty, sustainable development and intercultural dialogue via education, the sciences, culture, communication and information.
- UNESCO's strategies and actions also worked earlier to attain Sustainable Development Goals (SDGs), with other international communities.

World Heritage Programme

- A World Heritage Site refers to a landmark or area that is granted legal protection under an international convention managed by the United Nations Educational, Scientific and Cultural Organisation (UNESCO).
- **Designation:** UNESCO designates World Heritage Sites for their cultural, historical, scientific, or other forms of significance.
- **Criteria:** These sites are recognised for their cultural and natural heritage, deemed to hold exceptional value for humanity.
- **Selection Process:** To qualify as a World Heritage Site, a location must be a unique landmark with geographical and historical significance, possessing special cultural or physical importance.
- **Administration:** The list of World Heritage Sites is maintained by the International World Heritage Programme, which is overseen by the UNESCO World Heritage Committee. This committee consists of 21 member states elected by the General Assembly.

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Man and the Biosphere Programme (MAB)

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- UNESCO's Man and the Biosphere Programme (MAB), launched in 1971.
- It is an Intergovernmental Scientific Programme that aims to establish a scientific foundation for the development of relationships between people and their environment.

MAB combines the natural and societal sciences, financial side and education to develop human livelihoods and the even-handed sharing of benefits and to protect natural and managed ecosystems, thus promoting innovative approaches to fiscal development that are publicly and ethnically appropriate, and environmentally sustainable.

The MAB Programme strives to

- Identify and review the changes in the biosphere consequential from human and natural actions and the effects of these alterations on humans and the environment, in the context of climate change;
- Study and contrast the dynamic interrelationships among natural/near-natural ecosystems and socio-economic processes, in the context of the accelerated failure of biological and cultural diversity with unforeseen consequences that impact the capacity of ecosystems to continue to offer services critical for human well-being;
- Ensure essential human welfare and a liveable environment in the background of rapid urbanisation and energy using up as drivers of environmental change;
- Encourage the exchange and raising of knowledge on environmental troubles and solutions, and foster ecological education for sustainable development.

9.21 CONSERVATION INTERNATIONAL (CI)

- It was founded in 1987 with the goal of protecting nature as a source of food, freshwater, our health, our society, our development, everything depends on the ecosystem and natural biodiversity.

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- Conservation International (CI) is an environmental organisation that aims to protect nature.
- CI works by connecting nature to science and technology, policies, intergovernmental partnership, communities, stakeholders, business and much more to reach their goal of building a healthier environment for all living beings to survive prosperously.
- CI also plays an important role in designating Hotspots.
- Hotspots are such places that are rich in biodiversity but are also threatened due to anthropogenic activities.
- To qualify as a biodiversity hotspot, an area must meet two strict criteria:
    1. Contain at least 1,500 species of vascular plants found nowhere else on Earth (known as 'endemic' species).
    2. Have lost at least 70 per cent of its primary native vegetation.
- There are around 35 places all around the globe that have been designated as biodiversity hotspots and constitute about 2.3% of the Earth's land surface.
- They have irreplaceable, endemic flora and fauna of diverse nature and are estimated to provide 35% of 'ecological services' to the Earth.
- There are four major biodiversity hotspots in India:
    1. The Himalayas
    2. Indo-Burma Region
    3. The Western Ghats
    4. Sundarban

Earth's Terrestrial Hot Spots of Biodiversity

Africa

1. Eastern Afro-Montane
2. The Guinean forests of Western Africa
3. The Horn of Africa
4. Madagascar and the Indian Ocean Islands
5. Maputo land, Podo land, Albany hotspot
6. Succulent Karou
7. East Melanesian islands
8. South Africa's Cape floristic hotspot
9. Coastal forests of Eastern Africa

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Asia and Australia

1. Himalayan hotspot
2. The Eastern Himalayas
3. Japan biodiversity hotspot
4. Mountains of South-West China
5. New Caledonia
6. New Zealand biodiversity hotspot
7. Philippine biodiversity hotspot
8. Western Sunda (Indonesia, Malas and Brunei)
9. Wallace (Eastern Indonesia)
10. The Western Ghats of India and Islands of Sri Lanka
11. Polynesia and Micronesian Islands Complex including Hawaii
12. South-Western Australia

North and Central America

1. California Floristic Province
2. Caribbean islands hotspot
3. Modrean pine-oak wood lands of the USA and Mexico border
4. The Mesoamerican forests

South America

1. Brazil's Cerrado
2. Chilean winter rainfall (Valdivian) Forests
3. Tumbes-Choco-Magdalena
4. Tropical Andes
5. Atlantic forest

Europe and Central Asia

1. Caucasus region
2. Iran-Anatolia region
3. The Mediterranean basin and its Eastern Coastal region
4. Mountains of Central Asia

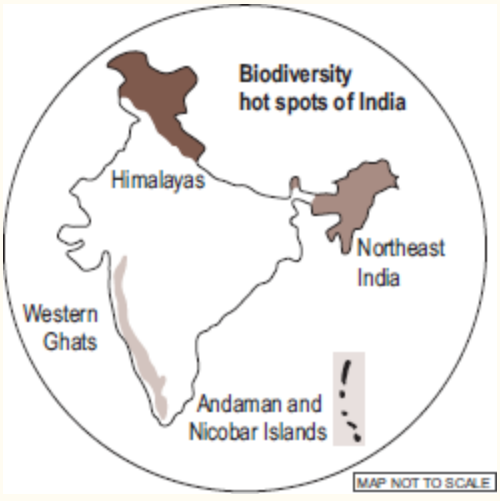

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Figure 9.5 Biodiversity Hot Spots of India

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9.22 WORLD-WILDLIFE FUND (WWF)

- The World-Wide Fund for Nature (WWF) is an international **Non-Governmental Organisation (NGO)** founded in 1961, functioning in the field of wilderness preservation and the cutback of humanity's impact on the environment.
- The **living planet report** has been published every two years by WWF since 1998, it is based on the living planet index and environmental.
- **World's largest conservation organisation** with over five million supporters globally, working in more than 100 countries, sustaining around 1,300 conservation and environmental projects.
- The group's job is to end the degradation of the planet's natural setting and build a prospect in which humans live according to nature.
- At present, much of its work concentrates on the preservation of three bionics, that contain the majority of the world's biodiversity: **oceans and coasts**, **forests** and **freshwater ecosystems**.
- In other issues, it is also worried about endangered species, sustainable production of supplies and climate change.
- Its pictogram is '**Giant Panda**'.

■ *WWF has three global goals*

(i) Saving endangered species
(ii) Protecting endangered habitats
(iii) Addressing global threats such as toxic pollution, over-fishing and climate change.

From functioning to save the **giant panda** and getting back the **Asian rhino** to establishing and serving to manage parks and reserves globally, WWF has been a conservation leader for over 40 years.

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'Earth hour', a movement started by the 'World Wildlife Fund' for the reduction in carbon emission. It began in **Sydney (Australia),** 2007. Under it, every year on last Saturday of March, less important electrical appliances should be shut from 8:30 pm to 9:30 pm. **India participated it in 2009.**

Mission of WWF

- WWF's mission is **to conserve nature** through available scientific knowledge and advance that knowledge to work to preserve the diversity and health of ecological systems.
- **Endangered Species Protection:** Protecting natural areas, and wild populations of plants and animals, including endangered species.
- **Sustainable Development:** Promoting sustainable approaches to the preference of renewable natural resources and using them proficiently to reduce pollution.

WWF is dedicated to reversing the deprivation of our planet's natural environment and constructing a future in which human needs are met in harmony with nature.

9.23 TRADE-RELATED ANALYSIS OF FAUNA AND FLORA IN COMMERCE (TRAFFIC)

- Trade-Related Analysis of Fauna and Flora in Commerce (TRAFFIC), a **non-governmental organisation**, was first established in 1976, which is a wildlife trade monitoring network.
- It is a **programme developed jointly by WWF** (World Wide Fund for Nature) and **IUCN** (International Union for Conservation of Natural and Natural Resources).
- **TRAFFIC was introduced in India in 1991**. It works with **MoEFCC, WCCB (Wildlife Crime Control Bureau) & Forest departments, in focused area of** trade in tigers, pangolins, elephants, turtles, medicinal plants, and timber
- Its Headquarter is at **Cambridge, United Kingdom**

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- It is governed by the TRAFFIC Committee, composed of members of TRAFFIC's partner organisations, WWF and IUCN.
- TRAFFIC works in close co-operation with the Secretariat of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).
- Its staff includes experts from various backgrounds: biologists, conservationists, academics, researchers, communicators or investigators, etc.

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Objective

- TRAFFIC is an organisation that mainly deals with **monitoring wild flora and fauna trade** to ensure the **trade is legal and does not harm the ecological balance** or **threaten** the **existence of species involved in the trade**.
- In recent times, along with being a global network, it also involves **research analysis** related to conservation programmes to deliver **practical and innovative solutions** to the wildlife conservation issues for **sustainable trade of wildlife**.

TRAFFIC in News

1. DETECT-IT Tool (TRAFFIC + WWF + Hewlett Packard Enterprise)
   - TRAFFIC, WWF, and HPE jointly developed **DETECT-IT**, a technology-based tool.
   - It **searches and analyses global trade data** to identify **illegally traded fish products**.
   - Aim: Strengthen monitoring of **illegal, unreported and unregulated (IUU) fishing** and protect marine biodiversity.
   - Helps governments detect suspicious trade flows quickly.
2. Seizure of Indian Star Tortoise in Bangkok
   - TRAFFIC reported seizure of the **Indian Star Tortoise**, a *threatened species* under the **IUCN Red List**.
   - Found mainly in **India and Sri Lanka**.

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- The species is heavily trafficked for the **exotic pet trade**, making it one of the most smuggled tortoises globally.

3. TRAFFIC Report on the Golden Triangle
   - According to a TRAFFIC study, the **Golden Triangle** region (border area of **Thailand–Laos–Myanmar**) is a **major hub of illegal wildlife trade**.
   - High trafficking rates of: **Tigers**, **Bears**, **Elephants** & **Pangolins**
   - Region known for open markets, weak enforcement, and transnational criminal networks.

TRAFFIC Campaigns in News

1. "WANTED ALIVE" Campaign
   - TRAFFIC launched the **WANTED ALIVE series** focusing on conservation of **four Asian big cats**: **Tiger**, **Leopard**, **Snow Leopard** & **Clouded Leopard**
   - Aim: Raise awareness and reduce illegal trade in their body parts.
   - Message: These big cats must be **protected in the wild**, not killed for trade.
2. "Don't Buy Trouble" Campaign (TRAFFIC India)
   - One of TRAFFIC India's earliest consumer-awareness initiatives.
   - Advises **tourists** to avoid purchasing wildlife products as souvenirs.
   - Focus: Prevent demand for illegal wildlife products like shells, skins, claws, exotic pets, etc.
   - Helps reduce unintentional support for illegal wildlife trafficking.

9.24 UN CONVENTION ON THE LAW OF THE SEAS (UNCLOS), 1982

UNCLOS is the **global legal framework** that governs all activities and rights related to the **oceans and seas**. It was adopted in **1982** and came into force in **1994**.

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Objectives

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- Establish rights and responsibilities of nations regarding the **use of oceans**.
- Regulate **marine natural resources**, navigation, and environmental protection.
- Define maritime zones, delimitation rules, navigation rights, and seabed governance.

Table 9.8 Maritime Zones under UNCLOS

Zone	Distance from Baseline	Key Rights
Territorial Sea	Up to **12 nautical miles (NM)**	• Full sovereignty; i
Contiguous Zone	**12–24 NM**	• Limited control to immigration, fisca
Exclusive Economic Zone (EEZ)	Up to **200 NM**	• Sovereign rights f managing natural
Continental Shelf	Up to **200–350 NM**	• Rights over seabe
High Seas	Beyond national jurisdiction	• Open to all states

UNCLOS

- The Coastal States put into effect **sovereignty** over their territorial sea which they have the right to set up its breadth to a limit not to exceed **12 nautical miles;** foreign vessels are allowed ‘innocent passage’ through those waters; ships and aircraft of all countries are permitted ‘transit passage’ through straits used for international navigation; states bordering the straits can adjust navigational and other aspects of passage.

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- Coastal States have sovereign rights in a 200-nautical mile exclusive economic zone (EEZ) with reference to natural resources and certain financial activities and exercise jurisdiction over marine science research and environmental security.
- All other States have **freedom** of navigation and overflight in the EEZ, as well as the freedom to lay **submarine cables and pipelines.**
- The Coastal States have sovereign rights over **the continental shelf** for **exploring and exploiting** it; the shelf can extend at least 200 nautical miles from the shore, and more under specified circumstances.
- The Commission on the **"Limits of the Continental Shelf"** shall make recommendations to States on the shelf' s outer boundaries when it extends beyond 200 miles.
- All States enjoy the traditional freedoms of navigation, overflight, scientific research and fishing on the high seas; they are obliged to adopt or cooperate with other States in adopting measures to manage and conserve living resources.
- **The Land-locked States** have the **right to access** from the sea and enjoy freedom of transit through the territory of transit States.
- States are bound to prevent and control marine pollution and are liable for damage caused by violationof their international obligations to combat such pollution.
- All marine **scientific research** in the EEZ and on the continental shelf is subjected to the consent **of the coastal States,** but in most cases, they are obliged to grant consent to other States when the research is to be conducted for peaceful purposes and fulfils specified criteria.
- States Parties are obliged to **settle the dispute by peaceful means** concerning the interpretation or application of the Convention.
- Disputes can be submitted to the 'INTERNATIONAL TRIBUNAL FOR THE LAW OF THE SEA' established under the Convention, to the International Court of Justice, or to arbitration. Conciliation is also available and, in certain situations, submission to it would be obligatory. The Tribunal has special jurisdiction over deep seabed mining disputes.

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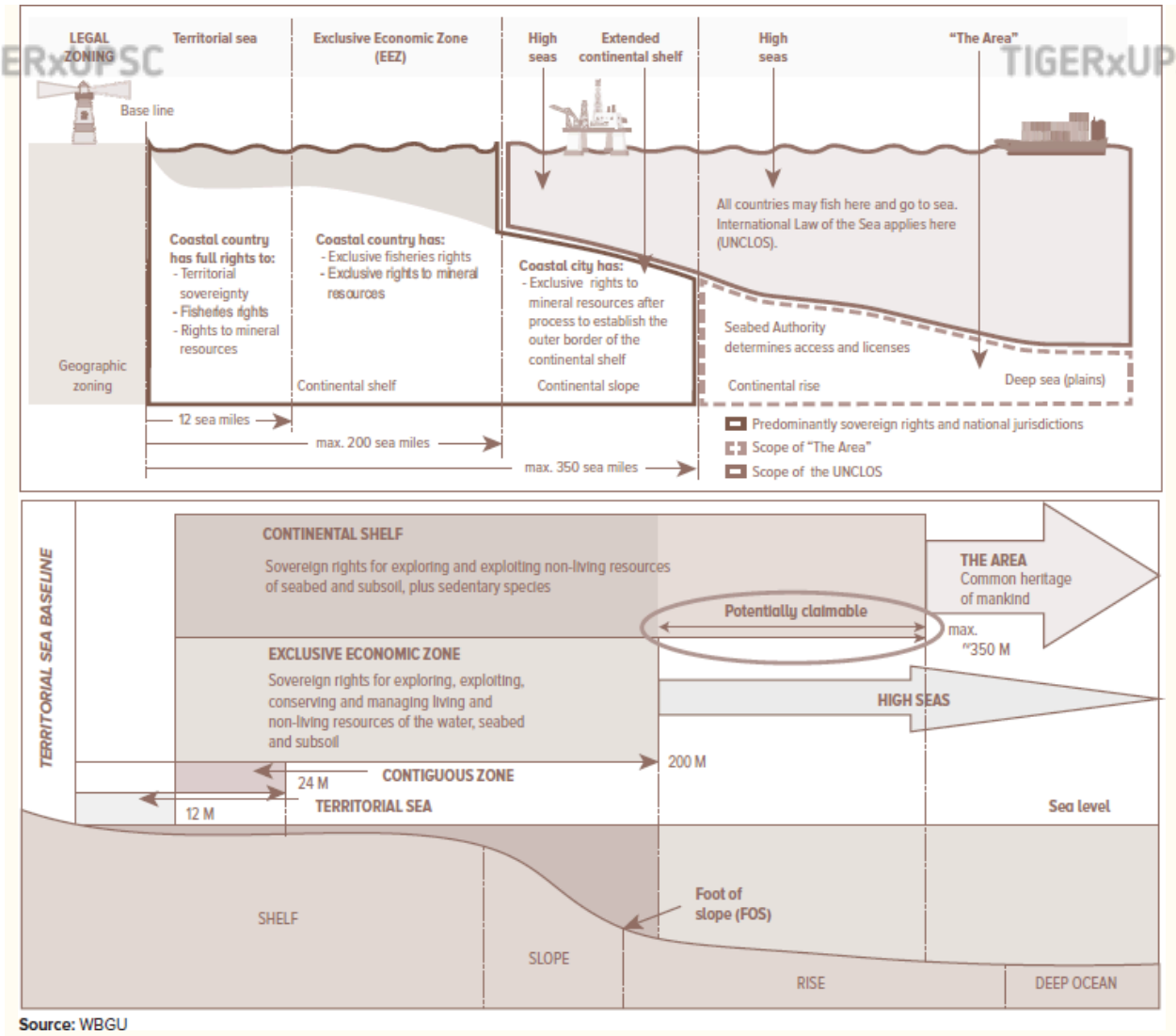

Source: WBGU

Figure 9.6 Different Zonal Distribution in Seas

■ *India and UNCLOS*

- India is a **party to UNCLOS** (ratified in 1995).
- Uses UNCLOS rules to define its **territorial sea**, **EEZ**, and **continental shelf**.
- Engages with ISA for **deep sea mining** in the Indian Ocean.
- Important for India's maritime security, Blue Economy, and SAGAR policy.

9.25 BASEL CONVENTION

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- The Basel Convention on the **Control of Transboundary Movements of Hazardous Wastes and their Disposal** was adopted in 1989 and it came into force in 1992.
- It is the most comprehensive global environmental agreement on hazardous wastes and other wastes.
- With **191 Parties**, it has nearly universal membership.
- The Basel Convention regulates the Transboundary movements of hazardous wastes and other wastes and obliges its Parties to ensure that such wastes are managed and disposed of in an environmentally sound manner.
- Parties also should to minimise the quantities that are transported, to treat and dispose of wastes as close as possible to their place of generation and to prevent or minimise the generation of wastes at source.
- The objective of the Basel Convention is to defend human health and the environment against the unfavourable effects of dangerous wastes.
- The convention is also intended to minimise the rate and toxicity of wastes generated, to ensure their environmentally sound management as closely as possible to the source of generation, and to assist LDCs in the environmentally sound management of the hazardous and other wastes they generate.
- It particularly focuses on preventing the transfer of hazardous waste from developed to less developed countries.
- **Wastes Covered -** Toxic industrial waste, Chemical waste, Biomedical waste, E-waste, Lead-acid batteries, Persistent organic pollutants (POPs) & **Plastic waste (after 2019 amendment)**. In this way many categories of **plastic waste** brought under **PIC procedure** to curb illegal global plastic waste dumping.

Key Features

(a) Prior Informed Consent (PIC)

- No hazardous waste can be exported without prior consent of the importing country.
- Exporting country must ensure safe handling and disposal.

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(b) Prohibition on dumping
- Addresses global dumping of toxic waste in developing nations.

(c) Environmental Sound Management (ESM)
- Countries must ensure environmentally safe disposal.

■ *Aims and Provisions*

- The convention aimed to decrease the generation of dangerous waste, restriction of the Transboundary movement of hazardous waste.
- The Convention aims to protect human health and the environment against the adverse effects resulting from the generation, Transboundary movements and management of hazardous wastes and other wastes.
- Transboundary movement of hazardous waste is possible only when the disposal conditions are environment friendly and making of a regulatory committee to monitor it.

9.26 ROTTERDAM CONVENTION

- The provision of the Rotterdam Convention entered into force on 24 February 2004, with the following objective – Human health and environment is a matter of concern. If any individual or organisation tries to harm it, it can be arrested only by shared responsibility of the concerned parties and government.
- It aims to promote shared responsibility and cooperation among Parties in addressing the international trade of certain hazardous chemicals, to protect human health and the environment from potential harm.
- The agreement establishes a prior informed consent (PIC) procedure for the import of hazardous chemicals.
- It covers pesticides and industrial chemicals that have been banned or severely restricted for health or environmental reasons by Parties and which have been notified by Parties for inclusion in the **Prior Informed Consent (PIC) procedure**.
- The Convention makes **legally binding obligations** for the discharge of the Prior Informed Consent (PIC) procedure.

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Major Provisions

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The Convention encourages the exchange of information on a very broad variety of chemicals. It does so through:

- The necessity for a Party to inform other Parties of each nation about the ban or restriction of a chemical.
- The necessity for a Party that plans to export a chemical that is prohibited or severely restricted for utilisation within its territory must inform the importing Party that such sell abroad will take place, before the first shipment and annually thereafter.
- The necessity for an exporting Party, when exporting chemicals that are to be used for work related purposes, to make sure that an up-to-date security data sheet is sent to the importer.
- Labelling requirements for exports of chemicals incorporated in the PIC procedure, as well as for additional chemicals that are barred or severely restricted in the exporting country.

Recent Updates

- **International trade of 2 hazardous pesticides** was **recommended for 'prior informed consent' (PIC).**
- Recommendations were made by the **Chemical Review Committee (CRC)** for 2 hazardous pesticides:
  - **Iprodione:** A **fungicide used on vines, fruits, trees and vegetables**, classified as carcinogenic and toxic for reproduction.
  - **Terbufos**: A **soil insecticide** used commonly on sorghum, maize, beet and potatoes, found to **pose risk to aquatic organisms** due to its toxicity.
- **In reference of India**
  - In India, the use of these **chemicals was permitted by the 2015 Anupam Verma committee report**.
  - India is **among the largest exporters of Terbufos**.

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ENV 40- IPCC (Intergovernmental Panel on Climate Change)

9.27 IPCC (INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE)

The **Intergovernmental Panel on Climate Change (IPCC)** is an international body established in **1988** by the **United Nations Environment Programme (UNEP)** and the **World Meteorological Organisation (WMO)** to evaluate climate change.

Purpose

- Provides a clear scientific perspective on the current state of knowledge regarding climate change and its potential ecological and socioeconomic impacts.
- Offers scientific information to governments at all levels for developing effective climate policies.
- Role: The IPCC serves as the UN body responsible for assessing science-based observations and research on climate change, helping guide global and national climate action.
- It does **various reviews and evaluates the most recent scientific, technological and socio-economic information** produced globally that are pertinent to understanding climate change.
- It **does not perform any research or scrutinise climate related data or parameters**.

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- In **2007,** the **IPCC was awarded the Nobel Peace Prize**.

Currently, 195 countries are members of IPCC. IPCC is administered in agreement with WMO and UN rules and procedures. It addresses information gaps related to the science, impacts and strategy responses to the climate trouble. It consists of three working groups:

- **Working Group I:** For scientific assessments
- **Working Group II:** For impacts, adaptation and mitigation
- **Working Group III:** For socio-economic dimensions of climate change

It also focuses on adverse impacts of climate change on higher malaria incidences, droughts, coastal flooding, agricultural disruption to become more serious in future.

Do You Know?

There has been an average rise in global temperatures by about 0.8°C over the last century and sea levels have also risen by about 20 cm.

IPCC Assessment Reports

- The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 to evaluate scientific knowledge on climate change.
- **First Report:** The First Assessment Report (AR1) was released in 1990, followed by an update in 1992.
- **Subsequent Reports:** The IPCC has published Assessment Reports approximately every six years:
  - **AR2:** Released in 1995
  - **AR3:** Released in 2001
  - **AR4:** Released in 2007
  - **AR5:** Released in 2014
  - **AR6:** Released between 2021–2023
- **Recognition:** The IPCC received the Nobel Peace Prize for its contributions to global climate science and awareness.

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5th Assessment Report (2014)

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- Temperature rise by 2100 could be 4.8°C from pre-industrial times.
- Atmospheric Concentrations of CO_2, CH_4 and N_2O were at unprecedented levels in the last 800000 years.
- More frequent and longer heat waves "virtually certain".
- A large fraction of species faces extinction.
- This report formed the scientific basis for Paris Agreement 2015.

6th Assessment Cycle (AR6)

The **IPCC** released the third and final part of the **Sixth Assessment Report (AR6),** titled '**Climate Change 2022: Mitigation of Climate Change'**. It provides an **updated global assessment of climate change mitigation progress and pledges** and examines the sources of global emissions.

Reports of AR6 (2021–2023)

Key Releases: WG I: 9 Aug 2021, **WG II & III:** 28 Feb & 4 Apr 2022, **Synthesis Report:** 20 Mar 2023

Key Findings

- 1.5°C warming is now inevitable, even in best-case scenarios.
- Historical emissions are primary drivers of current climate crisis.
- Increased heat waves, longer warm seasons, shorter cold seasons globally.
- Air pollution reduces intensity/frequency of monsoon rains in South Asia.
- Urbanization increases intense rainfall in South Asian cities.
- Human-induced climate change affects all regions.
- Global temperature may exceed 1.5°C between 2021–2040.
- **Mitigation:** Halve GHG emissions by 2030, achieve net-zero by 2050 to limit warming.

Table 9.9 Mitigation targets for achieving Climate goals as per IPCC AR6

To limit global warming to 1.5°C Global GHG emissions peak before 2025,	To limit global warming to 2 must peak before 2025, rec

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reduced by 43% by 2030	achieve net zero CO_2 emis
Reach Global net zero CO_2 emissions in the early 2050s.	Reach Global net zero CO_2
Methane is to be reduced by 34% by 2030.	

9.28 GLOBAL INTERNATIONAL WATER ASSESSMENT (GIWA)

GIWA, led by the United Nations Environment Programme (UNEP), has been working since 1999. 50% of the funding of GIWA is covered by the Global Environment Facility (GEF) while other major donors are the National Oceanic and Atmospheric Administration (NOAA), the Finnish Department for International Development Co-operation and the Swedish International Development Co-operation Agency (SIDA).

■ *Objective*

To construct a comprehensive and incorporated global assessment of international waters. It is to be a methodical assessment of the environmental circumstances and problems in global waters, comprising marine, coastal and freshwater areas, and surface waters and groundwaters.

GIWA has been planned to examine the existing problems and their societal root causes, develop scenarios of the future situation of the world's water resources and evaluate policy reforms. GIWA is focusing on 66 transboundary water areas globally. Therefore, the evaluation includes marine water areas, surface freshwater areas and groundwater.

Five major areas of concern

1. Freshwater shortages

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2. Pollution
3. Habitat and community modification
4. Unsustainable exploitation of fisheries and other living resources
5. Global climatic change

9.29 MILLENNIUM ECOSYSTEMASSESSMENT (MA)

The United Nations Secretary-General, Kofi Annan brought forward the concept of Millennium Ecosystem Assessment in 2000. It was initiated in 2001, with the objectives of assessing the impact of environmental changes on human well-being and the scientific basis for action needed to improve the conservation process and sustainable use of those systems and their involvement in human well-being.

MA included knowledge held by the private sector, practitioners, local communities and native peoples. MA has no aim to make new primary knowledge, but as an alternative, seek to add value to on hand information by collating (order), evaluating, summarising, interpreting, and communicating it in a useful form.

9.30 STOCKHOLM CONVENTION ON PERSISTENT ORGANIC POLLUTION

- The Stockholm Convention is a global treaty to protect human health and the environment from persistent organic pollutants (POPs).
- POPs are chemicals that remain intact in the environment for long periods, become widely distributed geographically, accumulate in the fatty tissue of living organisms and are toxic to humans and wildlife.
- POPs circulate globally and can cause damage wherever they travel.
- It is an international treaty, **legally binding** and presented by UNEP and signed on May 22, 2001.
- It came into effect from May 17, 2004 onwards. India ratified in 2006.
- **Notable non-ratifying states – the United States, Israel and Malaysia.**

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- **186 Parties** (185 states and the European Union)
- The Stockholm Convention focuses on eliminating or reducing releases of POPs.
- It sets up a system for tackling additional chemicals identified as unacceptably hazardous.
- Ultimately, the Convention points the way to a future free of dangerous POPs and promises to reshape our economy's reliance on toxic chemicals.
- The Stockholm Convention is perhaps best understood as having **five essential aims:**
  1. Eliminate dangerous POPs, starting with the 12 worst
  2. Support the transition to safer alternatives
  3. Target additional POPs for action
  4. Cleanup old stockpiles and equipment containing POPs
  5. Work together for a POPs-free future
- **The Global Environmental Facility (GEF)** is the designated interim financial mechanism for the Stockholm Convention.

Do You Know?

key information for India – Ministry of Environment notified the 'Regulation of POP Rules' in 2018, under the Environment (Protection) Act, 1986.

Persistent organic pollutants (POPs): Those **organic matters which do not get decomposed after chemical, biological, light process, etc. are called Persistent Organic Pollutants. It remains in the environment for a longer duration of time.** It accumulates in the tissues of animals and human beings and severely affects their health. Some bromine-containing flame retardants (e.g. Polybrominated diethyl ether) were included in the list of Persistent Organic Pollutants during Stockholm Conference.

9.31 GLOBAL CLIMATE CHANGE ALLIANCE (GCCA)

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The **Global Climate Change Alliance (GCCA)** was launched by the **European Union in 2007** to support **least developed countries, developing nations, and small island states** that are highly vulnerable to climate change.

Since **2017**, the upgraded **GCCA+** also helps countries implement their **Nationally Determined Contributions (NDCs)**, following IPCC findings that current global targets are insufficient to keep temperature rise below 2°C.

GCCA+ works through EU delegations and focuses on building climate resilience by:

1. Integrating climate change into development and poverty-reduction policies.
2. Strengthening resilience to climate-related disasters.
3. Supporting adaptation and mitigation plans and actions.

Its interventions span multiple sectors such as climate-smart agriculture, fisheries, water, sanitation, forestry, coastal management, renewable energy, and livelihood enhancement.

■ *Key activities include*

- Supporting climate policy development, including NDCs.
- Mainstreaming climate change into legislation.
- Disaster risk reduction, protective infrastructure, restoration of forests, and early warning systems.
- Promoting renewable energy and energy efficiency.
- Capacity building, institutional strengthening, and awareness-raising.
- Enhancing carbon sinks through reforestation and vegetation restoration.
- Establishing monitoring, evaluation, and MRV systems (Measurement, Reporting, and Verification) for mitigation and GHG inventories

9.32 CLIMATE AND CLEAN AIR COALITION TO REDUCE SHORT-LIVED CLIMATE POLLUTANTS

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(CCAC)

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The Climate and Clean Air Coalition to Reduce Short-Lived Climate Pollutants (CCAC) was initiated on **16 February 2012** by the United Nations Environment Programme (UNEP) and six founding countries: **Bangladesh, Canada, Ghana, Mexico, Sweden, and the United States**.

The coalition focuses on reducing **short-lived climate pollutants (SLCPs)**, such as methane, black carbon, and hydrofluorocarbons (HFCs), to safeguard **human health**, enhance **agricultural productivity**, and protect the **environment**.

■ *Objectives*

- To **reduce short-lived** harmful **climatic pollutants** that include **black carbon, methane**, **hydrofluorocarbons** and **tropospheric ozone** that **promote global warming** at an alarming rate, leading to climate change.
- To collaborate internationally with various stakeholders, scientists, research institutions, **share technology-based data** and promote **clean fuel energy**.
- To **slow down** the average **rise in global temperature** up to 0.6°C by 2050.
- To **reduce crop loss**, **health issues**, **environmental pollution, premature death** and much more through **cost-effective methods**.
- To provide **financial support** among partner countries and programmes that are working on reducing climatic pollutants.

Short-Lived Climate Pollutants (SLCPs)

Short-lived climate pollutants (SLCPs) are agents with quite a short lifespan in the atmosphere ranging between a few days to a few decades and give a warming influence on climate. The most harmful short-lived climate pollutants are **black carbon, methane, hydrofluoro carbons and tropospheric ozone,** which are the most important contributors to the increasing global greenhouse effect after CO_2. They are dangerous air

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pollutants that have varied harmful effects on human health, agriculture, atmosphere and ecosystems. Although hydrofluorocarbons (HFCs) exist in relatively small concentrations in the atmosphere, their contribution to climate change is projected to rise significantly, potentially accounting for up to **19% of global CO2 emissions** by **2050**.

Climate and clean air coalition (CCAC) is, as the name suggests, a coalition of governments, businesses, intergovernmental organisations, several scientific institutions, multi-stakeholders which also include social organisations. They all **voluntarily** came together to protect the climate from improving **air quality** by **reducing short-lived climatic pollutants** prevalent in the atmosphere.

9.33 MINAMATA CONVENTION ON MERCURY

- The Convention is a global treaty that aims to protect human health and the environment from anthropogenic emissions and releases of **mercury** and **mercury compounds**.
- It derives its name from the **Japanese city Minamata** which had gone through a devastating incident of mercury poisoning.
- It aims to control anthropogenic releases of mercury throughout its lifecycle.
- The Convention was agreed at the **5th session of the Inter-governmental Negotiating Committee on Mercury in Geneva, Switzerland** in January 2013 and was adopted in October 2013 at Kumamoto, Japan.
- It was effective from 16 August 2017.
- The convention aims to phase out use of mercury by 2020.
- This convention has **128 signatories** and **142 parties**.

The **Minamata Convention of Mercury** has held six Conference of Parties (COP) meetings to date, with a seventh schedule for June 2027.

The list of all Minamata Convention COP meetings is as follows:

- **COP 1:** September 24 – 29, 2017, in Geneva, Switzerland.
- **COP 2:** November 19-23, 2018, in Geneva, Switzerland
- **COP 3:** November 25 – 29, 2019, in Geneva, Switzerland
- **COP 4:** This meeting was split into two segments:

  First segment: November 1 – 5, 2021, held virtually.

  Second segment: March 21 – 25, in Bali, Indonesia.

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- **COP 5:** October 30 – November 3, 2023, in Geneva, Switzerland.
- **COP 6:** November 3 – 7, 2025, in Geneva, Switzerland.
- **COP 7:** June 14 – 18, 2027, in Geneva, Switzerland

Key outcomes of COP-6

- **Dental Amalgam:** Parties agreed to adopt new amendments to phase out dental amalgam globally by 2034.
- **Skin-Lightening Products:** The convention will increase global efforts to eliminate mercury-added skin-lightening products.
- **Other Areas of Focus:**
  - **Trade Control:** The meeting discussed controlling the trade of mercury.
  - **Waste Management:** Strategies for managing mercury waste were a key topic.
  - **Artisanal and Small-Scale Gold Mining (ASGM):** The meeting addressed the significant challenge of rising mercury use in ASGM, which is a priority for the convention.
- **Enhanced Cooperation:** The meeting promoted enhanced cooperation with other global agreements and international partners.
- **Participant Engagement:** There was a strong focus on the full and effective participation of Indigenous Peoples and local communities, who are particularly vulnerable to mercury exposure.

Upcoming Minamata Convention COP 7

■ *INDIA: Union Cabinet approves Ratification of Minamata Convention on Mercury*

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The Union Cabinet has approved the proposal for ratification of Minamata Convention on Mercury and depositing instrument of ratification enabling India to become a Party of Convention in February 2018. The approval entails ratification of the convention along with flexibility for continued use of mercury-based products and processes involving mercury compounds up to 2025.

■ *Mercury pollution & its Effects*

Mercury is a global and ubiquitous metal that occurs naturally and has broad uses in everyday objects. It is released in the atmosphere, soil and water from a variety of sources such as burning coal for power plants, waste from industrial and medical products like batteries, measuring devices, such as thermometers and barometers, etc., extraction of minerals (smelting of gold), electric switches and relays in equipment, lamps (including some types of light bulbs), etc.

According to WHO, mercury is considered as one of the top ten hazardous chemicals of major public health concern. Once mercury released into atmosphere, it bioaccumulates and biomagnifies in food chain and easily enters to human body. It has toxic effects on nervous, digestive and immune systems and on lungs, kidneys, skin and eyes. Exposure to even a small amount of mercury may cause serious health problems. It is a threat to the child in early stage of life. It may also cause skin rashes and dermatitis.

Do You Know?

Mercury is the only metallic element that is liquid at standard conditions for temperature and pressure.

9.34 INTERNATIONAL WHALING COMMISSION (IWC)

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- The International Whaling Commission is an **Inter-governmental Organisation** whose purpose is the **protection of whales and the administration of whaling.**
- The legal structure of the IWC is the International Convention for the Regulation of Whaling.
- **This Convention was founded on 2nd December 1946**, making it one of the first pieces of international ecological legislation.
- All member countries of the IWC are signatories.
- India is a member of IWC.
- The IWC has an existing membership of 88 Governments from countries all over the world.

This organisation has been active against commercial whaling. In **1986, it adopted a moratorium on commercial whaling.** This ban still continues. **In 1947, it created the Southern Ocean Whale Sanctuary surrounding the continent of Antarctica.** Here, the IWC has banned all types of commercial whaling. IWC has designated only two such sanctuaries till date. Another is the Indian Ocean Whale Sanctuary by the tiny island nation of Seychelles.

9.35 FOOD AND AGRICULTURE ORGANISATION (FAO)

- FAO is a UN-based organisation founded in 1945.
- The specialised agency of the United Nations leads international efforts to defeat hunger.
- World Food Day is celebrated on 16th October.
- The day is celebrated to mark the anniversary of the founding of the FAO in 1945.
- Its headquarters is in Rome, Italy.
- It has 195 members (194 member nations, 1 member organisation, i.e. European Union). India is part of it.
- There are **three main goals of FAO**:

1. Eradication of hunger, food insecurity and malnutrition.

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2. Eliminate poverty and the driving forward economic and social progress for all.
3. Sustainable management and utilisation of natural resources including land, water, air, climate and genetic resources for the benefit of present and future generations.

- **Codex Alimentarius Commission** or CAC is the body responsible for all matters regarding the implementation of the Joint FAO/WHO Food Standards Programme.

9.35.1 Globally Important Agricultural Heritage System (GIAHS)

- GIAHS works to safeguard and support the world's agricultural heritage system.
- In 2002, FAO (Food and Agricultural Organisation) started an initiative for the identification and dynamic conservation of GIAHS. GIAHS initiative is now GIAHS programme to promote public understanding, awareness, national and international recognition of agricultural heritage system. It provides food and livelihood security for millions of poor and small farmers in a sustainable manner.
- This system resulted in outstanding landscape, maintenance and adaptation of globally significant agricultural biodiversity, indigenous knowledge system and resilient ecosystem. **GIAHS has three sites in India: Koraput (Orissa)- for rich agricultural biodiversity, Saffron heritage (Kashmir) and Kuttanad (Kerala) – below sea level cultivation of Rice.**

9.35.2 Global Alliance for Climate-Smart Agriculture (GACSA)

- The concept of Climate-Smart Agriculture (CSA) was originally developed by FAO and officially presented at the **Hague Conference on Agriculture, Food Security and Climate Change in 2010,** through the

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paper "Climate- Smart Agriculture: Policies, Practices and Financing for Food Security, Adaptation and Mitigation".

- In 2014 an alliance was set up with this issue as its focal point: The GASCA (Global Alliance for Climate-Smart Agriculture).
- GACSA is an inclusive, voluntary and action- oriented multi-stakeholder platform on Climate- Smart Agriculture (CSA).
- Its vision is to improve food security, nutrition and resilience in the face of climate change.
- GACSA aims to catalyse and help create transformational partnerships to encourage actions that reflect an integrated approach to the three pillars of CSA. Membership in the Alliance does not create any binding obligations.
- Each member individually determines the nature of their participation. Today, GACSA has 525 members and India is also one of them.

9.36 THE WORLD HERITAGE CONVENTION

- The World Heritage Convention was **adopted by the United Nations Educational, Scientific and Cultural Organisation (UNESCO) General Conference at its 17**th **session in Paris on 16th November 1972.**
- The Convention came into power in 1975. In August 1974, Australia became one of the first countries to endorse the Convention.
- The World Heritage Convention aims to encourage cooperation among nations to defend heritage worldwide that has exceptional universal value. This conservation is significant for current and future generations.
- States that are parties to the Convention agree to identify, protect, conserve and present World Heritage properties.
- The World Heritage Convention is administered by a **World Heritage Committee**, which meets yearly and comprises 21 members chosen from those States that are parties to the Convention.
- The **Committee's main tasks** are to:

1. Decide on the inscription of new properties on the World Heritage List,

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2. Discuss all matters relating to the discharge of the Convention,
3. Consider requests for international help,
4. Advise State Parties on how they can make sure States assemble their obligations under the Convention to protect World Heritage Properties.

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The World Heritage Fund

- A trust fund, the World Heritage **Fund for the Protection of World Cultural and Natural Heritage of Outstanding Universal Value (the World Heritage Fund) ,**is established under the Convention.
- The Fund is sponsored by contributions from state parties and contributions from private organisations and individuals. They can also apply for long-term loans and, in particular cases, non-repayable grants.

The World Heritage Convention and India

- India adopted this convention on 14 November, 1977.
- Presently, 44 places of India are recognised under UNESCO's World Heritage Sites.
- India's first two sites that were recognised on the list at the 7th Session of the World Heritage held in 1983 were the Agra Fort and the Ajanta Caves.
- Out of these 44 sites, 36 are cultural sites, 7 are natural sites and 1 is a mixed site.

9.37 CONVENTION ON BIODIVERSITY (CBD)

Structural Formate of CBD

- **1988:** The United Nations Environment Programme (UNEP) established an Ad Hoc Working Group to investigate the desirability and possible form of an international convention on biological diversity.
- **May 22, 1992:** The agreed text of the Convention on Biological Diversity was adopted at the Nairobi Conference.
- **June 5, 1992:** The CBD was opened for signature at the United Nations Conference on Environment and Development (Earth Summit) in Rio de Janeiro, Brazil. It received 168 signatures.

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- **December 29, 1993:** The Convention entered into force.
- **November/December 1994:** The first Conference of the Parties (COP 1) was held in Nassau, Bahamas, launching the International Coral Reef Initiative.
- Presently, it is adopted by **196 countries.**
- **International Day of Biodiversity–The United Nations has announced May 22nd as The International Day for Biological Diversity (IDB)** to increase understanding and awareness of biodiversity issues.
- Firstly in 1993, the Second Committee of the UN General Assembly designated "The International Day for Biological Diversity" (IDB) but in December 2000, the **UN General Assembly adopted 22nd May as IDB.**

Table 9.10 Description of the Conference of Parties

Meeting	Location	Date
COP 1	Nassau, Bahamas	November-De
COP 2	Jakarta, Indonesia	November 19
COP 3	Buenos Aires, Argentina	November 19
COP 4	Bratislava, Slovakia	May 1998
COP 5	Nairobi, Kenya	May 2000
COP 6	The Hague, Netherlands	April 2002
COP 7	Kuala Lumpur, Malaysia	February 200

■ *Objectives of convention on Biodiversity*

1. Conservation of biodiversity
2. Sustainable use of the components of biodiversity

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3. Fair and equitable sharing of benefits arising out of the utilisation of genetic resources

Specificity of CBD

- The agreement covers all ecosystems, species and genetic resources.
- The convention is a legally binding international treaty.
- It recognised for the first time in international law that the conservation of biological conservation of biodiversity is "a common concern of humankind" and is an integral part of the development process.
- There are many sub-protocols associated with CBD.

Key Protocols and Strategic Plans

- The CBD has resulted in two supplementary, legally binding protocols: the Cartagena Protocol on Biosafety and the Nagoya Protocol on Access and Benefit-sharing (ABS).
- The Strategic Plan for Biodiversity 2011–2020 included the 20 "Aichi Biodiversity Targets" to address biodiversity loss.
- Recent Developments and the Future COP 15 resulted in the Kunming-Montreal Global Biodiversity Framework (KMGBF), which includes goals like protecting 30% of global land and sea by 2030.
- COP 16 in Cali, Colombia (2024), focused on implementing and financing the KMGBF.
- The CBD is a central international treaty for global biodiversity governance with near-universal participation (196 parties, excluding the United States and the Vatican).

Key Features and Protocols Parties: There are 196 parties to the Convention, with the United States and the Vatican being the only UN member states not to have ratified it. Secretariat: The Secretariat is headquartered in Montreal, Canada, and operates under the United Nations Environment Programme (UNEP). Protocols: **Two supplementary agreements have been adopted to support the CBD's objectives:**

1. **Cartagena Protocol on Biosafety:** Regulates the safe handling, transfer, and use of living modified organisms (LMOs) resulting from modern biotechnology.
   - The Biosafety Protocol makes clear that products from new technologies must be based on the precautionary principle and allow developing nations to balance public health against economic benefits.

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- It will, for example, let countries ban imports of genetically modified organisms if they feel there is not enough scientific evidence that the product is safe and requires exporters to label shipments containing genetically altered commodities such as corn or cotton.

2. **Nagoya Protocol on Access and Benefit-sharing (ABS):** Provides a legal framework for implementing fair and equitable sharing of benefits arising from the use of genetic resources. The Nagoya Protocol is an international agreement that ensures the benefits from using natural genetic resources (like from plants, animals, or microorganisms) are shared fairly and equitably between the countries that provide them and the researchers or companies that use them.

- **It Prevents "Biopiracy**": Before the protocol, companies could take genetic material from a developing country's plants, develop a lucrative product (like a new medicine or cosmetic), and the source country or indigenous community might receive no benefit.
- **It Ensures Fair Compensation:** The Protocol creates a legal framework that requires users (like pharmaceutical companies or researchers) to get "Prior Informed Consent" from the provider country before accessing resources. Once a product is developed and profits are made, the benefits (which could be money, technology transfer, or shared research results) must be shared with the provider country or the local community that held the knowledge.
- **It Adds Transparency and Legal Certainty:** It makes the process of accessing and using genetic resources clear and predictable for both sides, which helps encourage research while respecting the rights of the source countries.
- **It Protects Traditional Knowledge:** It also covers the associated traditional knowledge held by indigenous and local communities about how these resources are used.

Ultimately, the Nagoya Protocol aims to give countries and communities a financial incentive to conserve their biodiversity and use it sustainably.

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Global Biodiversity Framework: At COP 15 in Montreal (2022), parties adopted the Kunming-Montreal Global Biodiversity Framework (KMGBF), which includes the ambitious "30-by-30" target to protect 30% of land and sea areas by 2030.

Aichi Biodiversity Targets

- The Aichi was Target adopted by the Convention on Biological Diversity (CBD) at its Nagoya conference.
- In the COP-10 meeting, the parties agreed that previous biodiversity protection targets were not achieved, so there was a need to develop new plans and targets.
- The short-term plan provides a set of 20 ambitious yet achievable targets, collectively known as the Aichi targets.

These are grouped in five strategic goals. They are as follows:

- **Strategic Goal A:** Address the underlying causes of biodiversity loss by mainstreaming biodiversity across government and society.
- **Strategic Goal B:** Reduce the direct pressure on biodiversity and promote sustainable use.
- **Strategic Goal C:** To improve the status of biodiversity by safeguarding ecosystems, species and genetic diversity
- **Strategic Goal D:** Enhance the benefits to all from biodiversity and ecosystem services.
- **Strategic Goal E:** Enhance implementation through participatory planning, knowledge management and capacity building.

CoP 15: Kunming-Montreal Global Biodiversity Framework

The 15th Conference of Parties (COP15) of the Convention on Biological Diversity recently concluded in Montreal, Canada (7–19 Dec. 2022).

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COP 15 resulted in the Kunming-Montreal Global Biodiversity Framework (KMGBF), which includes goals like protecting 30% of global land and sea by 2030. COP 16 in Cali, Colombia (2024), focused on implementing and financing the KMGBF.

- **Objective:**
  - The Global Biodiversity Framework (GBF) has been developed to replace the Aichi Biodiversity Targets, which expired in 2020. While the GBF and its supporting documents are **not legally binding**, it has been widely recognised as a landmark initiative, often referred to as the **'Paris Agreement for Nature'.**
  - Promoted under the auspices of the Convention on Biological Diversity (CBD), the GBF is considered the most significant agreement to date within this framework. It has been celebrated as a **'huge, historic moment'** and a **'major win for our planet and humanity',** emphasising its importance in advancing global efforts to protect biodiversity and ensure sustainable development.

Upon the launch of Agenda 2030, CBD released a technical note mapping and identifying synergies between the 17 Sustainable Development Goals (SDGs) and the 20 Aichi Biodiversity Targets. This helps to understand the contributions of biodiversity to achieving the SDGs.

■ *Post-2020 Global Biodiversity Framework*

The post-2020 Global Biodiversity Framework (GBF) is a new plan aimed at providing guidance for action until 2030. Its initial draft was made public in July 2021, and subsequent discussions and negotiations took place during the COP 15 meetings.

During the development of the framework, two key contentious issues emerged among the Parties. The first involved disagreements over strategies to reduce agricultural pollution, with varying commitments and approaches to address the environmental impacts of farming practices. The second centred on the equitable sharing of benefits related to digital sequence

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information, raising concerns about access, utilisation, and fairness in the distribution of these resources.

Ultimately, the Convention adopted the final version of the Global Biodiversity Framework on 19 December 2022. The framework sets forth ambitious goals, including the "30 by 30" initiative, which aims to designate at least **30% of the Earth's land and oceans** as protected areas by 2030. This commitment is a vital step toward conserving biodiversity, safeguarding ecosystems, and ensuring their sustainability for future generations.

Do You Know?

30 by 30 (or **30x30**) is a worldwide initiative for governments to designate 30% of Earth's land and ocean area as protected areas by 2030.

Major Outcome: The Kunming–Montreal Global Biodiversity Framework (GBF)

COP-15 concluded with the adoption of the **Kunming–Montreal Global Biodiversity Framework**, one of the most significant global conservation agreements since the Paris Climate Agreement. Its goal is to halt and reverse biodiversity loss by **2030** and achieve full recovery by **2050**.

Key Targets of the GBF

1. 30×30 Target
   - Protect **30% of the world's land, inland waters, coastal areas, and oceans by 2030**.
   - Restore **30% of degraded ecosystems** in terrestrial, freshwater, and marine environments.
2. Reduce Extinction Risk

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- Reduce human-induced extinction rates dramatically and halt extinctions of known threatened species.

3. Cut Harmful Subsidies
   - Reduce subsidies harmful to biodiversity by **$500 billion per year** by 2030.
4. Increase Biodiversity Finance
   - Mobilize **$200 billion per year** from public and private sources for biodiversity.
   - Increase financial flows from developed to developing countries to at least **$20–30 billion per year**.
5. Sustainable Use & Benefit Sharing
   - Strengthen regulation and benefit-sharing of **genetic resources** and **digital sequence information (DSI)**.
6. Reduce Pollution
   - Reduce nutrient loss by **50%**, pesticide risk by **50%**, and eliminate plastic pollution.
7. Mainstream Biodiversity Across All Sectors
   - Integrate biodiversity considerations into agriculture, fisheries, forestry, energy, mining, and infrastructure planning.

Implementation Mechanisms Adopted

- **Monitoring framework** with standardized indicators.
- **National Biodiversity Strategies and Action Plans (NBSAPs)** to be updated to align with the GBF.
- **Ratchet mechanism** for periodic progress reviews, similar to the Paris Agreement process.

Additional Agreements at COP15

- Creation of a **new multilateral fund** for biodiversity within the Global Environment Facility (GEF).
- Agreement on **gender equality**, Indigenous rights, and participation of local communities.

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- Steps toward a **global strategy for invasive alien species**.

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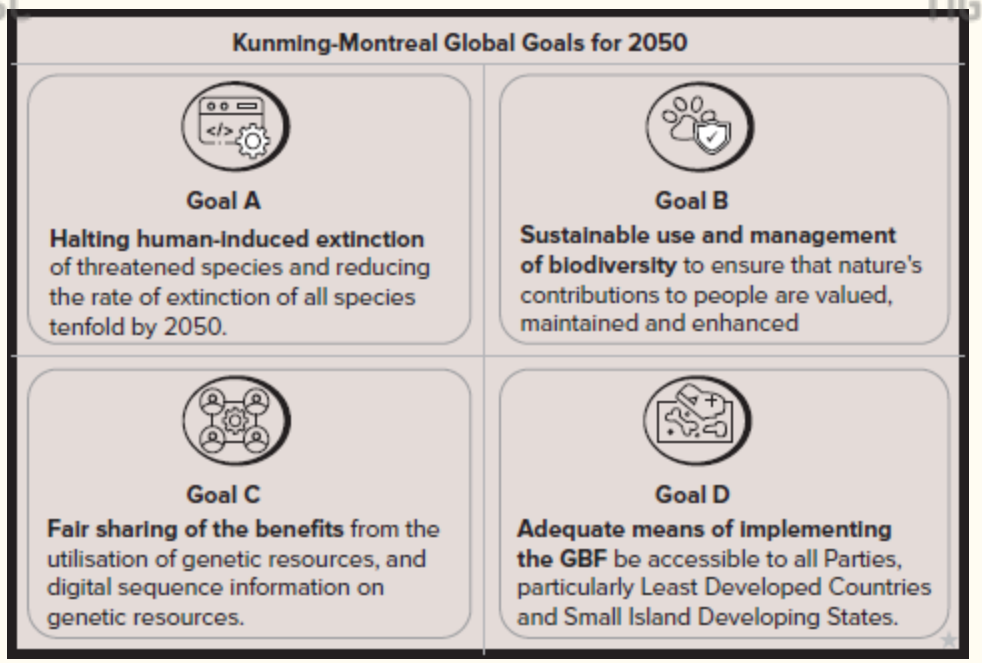

Figure 9.7 Kunming- Montreal Global Goals for 2050

■ *COP16 (16th Conference of Parties)*

When & Where

- Original: 21 Oct – 1 Nov 2024, Cali, Colombia
- Resumed: 25 – 27 Feb 2025, Rome, Italy

Purpose

- Review global progress on biodiversity under the **KunmingMontreal Global Biodiversity Framework (KMGBF)**.
- Make decisions on biodiversity conservation, sustainable use, and benefit-sharing.

Key Decisions & Achievements

1. **Benefit-sharing from Genetic Resources:** Agreed on a mechanism for sharing benefits from **digital sequence information (DSI)**.
2. **Indigenous & Local Communities:** Recognition of their role in biodiversity conservation and inclusion in decision-making.

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3. **Financing Biodiversity:** Commitment to mobilize **US $200 billion per year by 2030** for conservation, especially in developing countries.
4. **Implementation of KMGBF:** Emphasis on achieving targets by 2030 and ecological recovery by 2050.

Challenges

- Original session in Colombia suspended due to quorum loss.
- Some details on finance and benefit-sharing required the resumed session in Rome.
- Actual implementation of commitments remains a key challenge.

Upcoming CBD COP 17

Significance

- First major global “check-in” under KMGBF.
- Moves toward equity, sustainable use of resources, and conservation finance.
- Sets the stage for stronger global biodiversity governance and action.

9.38 EUROPEAN UNION NATURE RESTORATION LAW

The EU Nature Restoration Law is a landmark regulation requiring member states to legally restore degraded ecosystems across the EU. Key goals include restoring at least 20% of land and sea areas by 2030 and all degraded ecosystems by 2050, with specific targets for habitats, agriculture, and urban areas. This law is a crucial part of the EU›s Biodiversity Strategy

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for 2030 and aims to improve ecosystem health, carbon storage, and resilience to climate change.

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Key Requirements

- **Binding targets:** Member states must have effective restoration measures in place for at least 20% of their land and sea areas by 2030, with plans for all degraded ecosystems by 2050.
- **Ecosystem focus:** The law covers a wide range of ecosystems, including terrestrial, coastal, freshwater, forest, agricultural, and urban areas.
- **Agricultural targets:** Member states must implement measures to improve biodiversity, such as increasing grassland butterflies and the stock of organic carbon in cropland mineral soils.
- **Urban targets:** By 2030, there should be no net loss of urban green space and tree canopy cover. An increasing trend in both is required after 2030.
- **National plans:** Each member state is required to submit a national restoration plan by the end of 2026.
- Natura 2000 **priority:** Restoration efforts will prioritize the Natura 2000 network of protected areas until 2030.

How it works

- The law entered into force on August 18, 2024, following its publication in the EU's Official Journal.
- It is the first continent-wide law of its kind, aiming to return nature to a good ecological state.
- The law supports the EU's broader environmental goals, including the EU Biodiversity Strategy for 2030, and aligns with international commitments, such as the Global Biodiversity Framework.

9.39 THE ECONOMICS OF ECOSYSTEMS AND BIODIVERSITY (TEEB)

The initiative on 'The Economics of Ecosystems and Biodiversity' (TEEB) is important in implementing the CBD programme of work on incentive

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measures, and in particular its work on valuation. This major international initiative, funded by the European Commission, Germany, United Kingdom, Norway, the Netherlands and Sweden, and managed by the United Nations Environment Programme as part of its Green Economy Initiative (GEI), seeks to draw attention towards the **economic value of the nature we live in**. People utilise nature freely but if we look at the rate at which we are losing the nature and its resources, if the **value** of the **consequences of the loss** is estimated, it will **highlight** the fact that **proper measures** are urgently needed to focus on and control the rate of loss of biodiversity.

Members of the TEEB Advisory Board include the Executive Director of UNEP, the Executive Secretary of the CBD and the Director-General of IUCN, amongst others.

■ *Objectives*

- TEEB focuses on **recognising** the **benefits** that are provided by nature and its vast **biodiversity**.
- **Demonstrates** the **value** of all those in **economic terms** to **highlight the importance** of such sources.
- It also focuses on **capturing** all those **estimated values** and involvement of various **decision- making bodies** and **stakeholders** to follow appropriately a **structured approach** in implementing the policies so that the loss of biodiversity can be reduced.

■ *Recently in News*

The **bee population has declined** drastically in the last 2 decades in **Germany**. Recently, **TEEB economists** have estimated the **economic loss** one would have to face if bees go extinct. It has highlighted that **bees are extremely helpful** in the **economic field for farmers** and how disastrous it would be for them in the absence of bees since **bees play a major role in pollination**.

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One vital area is economics where many economies are unaware of the **great value** of the diversity of animals, plants and other life-forms and the role they play to keep the ecosystem healthy and functioning starting from forests and freshwaters to soils, oceans and even the atmosphere. **The Economics of Ecosystems and Biodiversity** aims to bridge the gap in understanding and driving action in this area.

In May 2008, the **TEEB Interim Report** released at CBD COP9 highlighted the link between poverty and the loss of ecosystems and biodiversity. It produced how several Millennium Development Goals were at major risk due to the negligence and deterioration of ecosystems and biodiversity.

9.40 INTERNATIONAL TREATY ON PLANT GENETIC RESOURCES FOR FOOD AND AGRICULTURE

- The International Treaty on Plant Genetic Resources for Food and Agriculture (PGRFA) is also universally known as the International Seed Treaty. It is a *wide-ranging international agreement in agreement with the Convention on Biological Diversity, which focuses on guaranteeing food safety through the conservation, swap over and sustainable use of the world's plant genetic resources for food and agriculture (PGRFA), as well as the fair and even-handed benefit sharing arising from its use.* It **also recognises Farmers' Rights.**
- This treaty establishes the Multilateral System of Access and Benefit-sharing to facilitate plant germplasm exchanges and benefit sharing through Standard Material Transfer Agreement (SMTA).
- The treaty was negotiated by the Food and Agriculture Organisation of the United Nations (FAO) and Commission on Genetic Resources for Food and Agriculture (CGRFA) and since 2006 had its own Governing Body under the aegis of the FAO.

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- The meeting addressed various topics, including the implementation of Farmers' Rights, financial regulations, the funding strategy, the partnership with the Global Crop Diversity Trust, and the execution of the Multilateral System (MLS) for access and benefit-sharing, along with other related matters.

9.41 CITES (CONVENTION ON INTERNATIONAL TRADE ON ENDANGERED SPECIES)

- It is an international agreement to which States and regional economic integration organisations adhere voluntarily.
- CITES's major function is to **control the international trade of species that do not threaten the existence of the species concerned.**
- It was **signed on 3**rd **March** 1973 **in Washington** and came into **force in July** 1975.
- CITES is also known as the Washington Convention.
- Presently, it has **184 member nations**.
- CITES's Secretariat **is administered by UNEP.**
- Each Party to the Convention must designate **one or more** Management Authorities (In India Management Authority is **Director of Wildlife Preservation, MoEFCC**) in charge of administering that licensing system and **one or more** Scientific Authorities (In India-Scientific Authorities: Zoological Survey of India, Botanical Survey of India, Wildlife Institute of India) to advise them on the effects of trade on the status of the species.
- Other Management Authority which is competent to grant permits in India is Wildlife Crime Control Bureau (WCCB).
- CITES, with the support of WWF (that provides scientific and technical support), protects endangered species and undertakes follow- up after every three years.
- **CITES is legally binding on all its Parties, requiring each member to adopt domestic legislation to ensure its implementation at the national level. All 184 Parties have the right to participate in meetings, propose considerations, and vote on decisions.**

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The species protected under CITES are classified into three Appendices, based on the degree of protection they require for their conservation.

(a) **Appendix I:** Species **cannot be traded internationally for primarily commercial purposes.** (Example: Tiger, Himalayan brown bear, elephant, and Tibetan antelope)

(b) **Appendix II:** Species included **can be traded** internationally for commercial purposes, **but within strict regulations,** requiring determinations of sustainability and legality. (Example: Hippopotamus, big leaf mahogany, and the grey wolf)

(c) **Appendix III:** A **species integrated at the request of a country with the collaboration of other countries to help in checking illegal exploitation.** (Example: Walrus, Hoffmann's two-toed sloth, and the red-breasted toucan).

MIKE and ETIS are tools developed by CITES to monitor and assess policies related to the trade in elephant products.

- **MIKE:** Stands for '**Monitoring the Illegal Killing of Elephants**', focusing on tracking levels, trends, and causes of elephant mortality.
- **ETIS:** Refers to the '**Elephant Trade Information System**', which monitors illegal trade in elephant products, such as ivory.
- **Supervision:** Both programs operate under the supervision of the CITES Standing Committee.
- **MIKE-ETIS Subgroup:** The Standing Committee has established a dedicated subgroup to oversee the development, modification, and implementation of the MIKE and ETIS programs.

These tools play a critical role in supporting evidence-based policymaking and strengthening global efforts to combat illegal trade and protect elephant populations.

■ *CoP 19 (2022)*

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The **19th Meeting** of Conference of the Parties (CoP 19) to **CITES,** also known as the **World Wildlife Conference,** was held in **Panama.**

Key Outcomes Under CITES Protection

- ***Species Added to Protection:*** Multiple species, including sharks, rays, trees, amphibians, turtles, tortoises, and several songbirds, were brought under CITES protection.
- ***Sea Cucumbers:*** A proposal to include sea cucumbers in Appendix II of the convention was accepted.
- ***Shisham (Dalbergia sissoo):*** On India's initiative, a proposal to clarify the permissible quantity of Shisham items, such as furniture, was considered.
- ***Operation Turtshield:*** Resolution documents on tortoises and freshwater turtles highlighted India's commendable success in combating wildlife crime through Operation Turtshield, led by the Wildlife Crime Control Bureau.
- ***Ivory Trade:***
  - A proposal to allow controlled trade in ivory was defeated.
  - India, for the first time, abstained from voting, marking a shift from its decades-old stance of strongly opposing ivory trade.
  - International ivory trade was banned in 1989, when all African elephant populations were placed in Appendix I of CITES.
  - Later, populations from countries like Namibia and Botswana were moved to Appendix II.

About Shisham (Dalbergia sissoo)

- **Description:** A deciduous tree growing up to 25–30 meters, native to the Indian subcontinent and southern Iran.
- **Distribution:** Found throughout the sub-Himalayan region, from Ravi to Assam.
- **Uses:** Known as Indian rosewood, Shisham is widely used to craft durable, dark-reddish furniture, including tables, beds, bookshelves, and chairs.
- **CITES Listing:** Included in Appendix II of the convention.

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- **Significance:** Declared the state tree of Punjab, reflecting its cultural and ecological importance.

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These updates underscore India's active role in wildlife conservation and its contributions to global efforts under CITES.

CoP20 (2025)

- **Location:** Samarkand, Uzbekistan
- **Dates:** November 24 – December 5, 2025
- **Significance:** This is the first time a CITES CoP is being hosted in Central Asia.
- **Key discussions:** Proposals to re-open international trade in ivory and saiga antelope horn, a CITES Strategic Vision, a language strategy for the Convention, and the role of CITES in preventing zoonotic disease emergence are all on the agenda.
- **Outcomes:** The meeting is advancing decisions on regulating trade in various species, including big cats, sea cucumbers, sharks, and plants.

CoP21 (Future meeting)

- **Scheduled:** 2028
- **Key topics:** The meeting will likely build on the outcomes of CoP20. Discussions could include streamlining the CITES process and focusing on legal and sustainable wildlife trade.
- **Focus:** Parties may focus on the implications of the amendments to the Appendices and work on capacity building for issuing permits.

■ *India and CITES*

- **Appendix III Listings:** Since ratifying CITES in 1976, India has listed 39 taxa under Appendix III to regulate international trade and ensure species conservation.
- **Monitoring the Illegal Killing of Elephants (MIKE):**
  - An international program that tracks levels, trends, and causes of elephant mortality.
  - India has 10 sites participating in the MIKE program.

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- **Regulatory Acts for Wildlife Trade:**
  - Wild Life (Protection) Act, 1972
  - Export–Import Policy (EXIM Policy) under the Foreign Trade (Development and Regulation) Act, 1992
  - Customs Act, 1962
- **CITES Management Authority (CITES-MA):**
  - The Director of Wildlife Preservation, Government of India, serves as the Management Authority for CITES implementation in India.
- **Scientific Authorities Supporting CITES-MA:**
  - The Government of India has designated five scientific authorities to assist the CITES-MA**:**
    (i) Zoological Survey of India (ZSI), Kolkata
    (ii) Botanical Survey of India (BSI), Kolkata
    (iii) Central Marine Fisheries Research Institute (CMFRI), Cochin
    (iv) Wildlife Institute of India (WII), Dehradun
    (v) Institute of Forest Genetics and Tree Breeding (IFGTB), Coimbatore

These measures reflect India's commitment to wildlife conservation and its active role in implementing CITES regulations.

ENV 41 Convention on Conserving Migratory Species (CMS)

9.42 CONVENTION ON CONSERVING MIGRATORY SPECIES (CMS)

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- It is an environmental treaty, which works with UNEP. It is responsible for conserving migratory species and cross border conservation of species.
- CMS works in association with other international institutes, non-governmental organisations, media partners and corporate sectors.
- CMS includes species under two appendixes:
- **Appendix I** includes those migratory species whose safety is greatly focused.
- **Appendix II** includes those species which require international assistance.
- **WWF attends CMS meetings and gives guidelines** to the authority concerned for conserving the migratory species.

Birdlife International

- It was established by T. Gilbert Pearson and Jean Theodore Delacour as an "International committee" for the conservation of birds.
- Birdlife International is the biggest agreement in the world for nature conservation.
- It is the world's largest conservation partnerships for conservation of birds and their habitats.
- It was earlier known as International Committee for Bird Preservation.
- It has **133 participants** all across the globe. It has 6 coordination offices around the world; its headquarters is located in Cambridge, Britain.
- All are collectively called Birdlife International Secretariat.
- This secretariat decides guidelines, programmes and policies of Birdlife International. Birdlife International also **provides an official Red List** for birds, for the **International Union for Conservation of Nature (IUCN).**
- It publishes a quarterly magazine, **World Birdwatch,** which contains recent news and authoritative articles about birds, their habitats, and their conservation around the world.
- It also publishes important bird areas (IBA) inventory.
- It identifies the sites referred to as 'Important Bird and Biodiversity Areas'.

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Recent issues with India

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- India will become the 54th country to sign the Memorandum of Understanding on the Conservation of Migratory Birds of Prey in Africa and Eurasia (Raptors MOU).
- It is an important international agreement to protect migratory birds of prey.
- It is a **binding agreement** concluded under the United Nations Environment Programme's Convention on Migratory Species (CMS).
- Birdlife was also concerned about key issues affecting raptors including agrochemicals, power lines and renewable energy.
- It provides scientific information, which develops guidelines for national strategies for bird of prey conservation.

Bombay Natural History Society established in 1883 is among the oldest conservation organisation in the world. It has carried out pioneer research and conservation on many birds of prey including South Asia's critically endangered vultures, and other migratory raptors such as Amur Falcon (Falco amurensis).

ENV 42 Ramsar Convention (Convention on Wetlands)

9.43 RAMSAR CONVENTION (CONVENTION ON WETLANDS)

- It is an international treaty for the conservation and sustainable utilisation of wetlands.
- It also recognises the fundamental ecological functions of wetlands and their economic, cultural, scientific and recreational value.

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- On 2nd February 1971, in an Iranian town of Ramsar, a conservation treaty was signed solely devoted to the protection of aquatic ecosystems in both marine and freshwater.
- The convention was brought into force with effect from December 21st 1975.
- The Conference of the Contracting Parties (COP) for the Ramsar Convention on Wetlands is held every **three years**.
- The contracting parties are committed under **three pillars:**
  1. Work towards the wise use of all their wetlands;
  2. Designate suitable wetlands for the list of Wetlands of International Importance (the "Ramsar List") and ensure their effective management;
  3. Cooperate internationally on transboundary wetlands, shared wetland systems and shared species.
- Other **International Organisation partners of the Ramsar convention** are: Several organizations, including BirdLife International, the International Union for Conservation of Nature (IUCN), the International Water Management Institute (IWMI), Wetlands International, WWF International, and the Wildfowl and Wetlands Trust, contribute to the convention by serving as technical advisors, conducting research and analysis, and offering financial support. Additionally, these organizations participate as observers in the convention's proceedings.\

■ *World Wetlands Day*

- World Wetlands Day was celebrated for the first time on **2 February**, 1997, i.e. the 16th anniversary of the Ramsar Convention in the Iranian city, Ramsar.

RAMSAR CONVENTION COP 1: The **first Ramsar Convention Meeting (COP 1)** was held in the city of **Ramsar, Iran**, where the treaty was adopted on **February 2, 1971**. The convention is an intergovernmental treaty focused on the conservation and wise use of wetlands.

RAMSAR CONVENTION COP 15: The last **Ramsar Convention (COP15)** was held in **Victoria Falls, Zimbabwe, from July 23 to July 31,**

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2025. The 15th meeting of the Conference of the Contracting Parties concluded with new resolutions on wetland restoration, the protection of migratory species, and equitable governance. **Key outcome**: Adoption of new resolutions on wetland restoration, species protection, and governance, as well as an India-led resolution on promoting sustainable lifestyles for wetland use.

RAMSAR CONVENTION COP 16: The **16th Meeting of the Conference (COP 16)** of the Contracting Parties to the Ramsar Convention on Wetlands is planned to be held in **Panama** in **July 2028**.

FACTS CORNER

- The parties involved in the contract need to designate at least one wetland of its area having international importance.
- Presently, there is over 2550 **places (till January 2026)** designated as Ramsar sites.
- The **United Kingdom has the maximum number of Ramsar sites (wetlands of international importance**) in the world, with 176, **followed by Mexico** with 144.
- **India has the highest number in Asia, with 96 sites as of January 2026 (Note: Students should always check this number whenever they revise this book)**, making it the country with **the third-highest number of Ramsar sites globally.**
- The largest wetland in the world is the **Pantanal**, a vast tropical wetland **located in the upper Paraguay River basin across Brazil, Bolivia, and Paraguay**. Covering an area between 140,000 and 195,000 square kilometers, it is a rich ecosystem of flooded grasslands, savannas, and forests, with 80% of the area submerged during the rainy season. The Pantanal spans parts of Brazil (about 80%), Bolivia, and Paraguay. It is a mosaic of flooded grasslands and dense savanna.
- Today, the Ramsar List is the world's largest network of protected areas.

Montreux Record

Montreux Record (under Ramsar Convention) is a catalogue of wetland sites in the List of Wetlands having international significance where changes in ecological character have occurred or are occurring due to technological developments, pollution or other anthropogenic interference.

It is maintained as a part of the Ramsar List. **In the Conference of the Contracting Parties 1990, Montreux Record was established.**

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Conference of the Contracting Parties (1993) determined that the Montreux Record should imply to the sites which require national and international conservation attention.

List of **Indian wetlands** of international importance included in the Montreux Record is-
(i) Keoladeo National Park, Rajasthan
(ii) Loktak Lake, Manipur.

Do You Know?

Chilika Lake was **removed** from the register in **2002** in light of the **improved conditions** of the lake. In fact, Chilika lake is the **first Ramsar site in Asia** to be **removed** from the Montreux Record.

9.43.1 Wetland Conservation in India

National Wetland Conservation Programme (NWCP)

- The Government of India in association with States started the National Wetlands Conservation programme in 1985–86.
- There are a total of **93 Ramsar Sites (September 2025)** now in India (Lists are given in Section-V).
- The focus of this programme is to conserve and use wetlands smartly, so that their future degradation can be conserved.

Objectives of the Scheme

The main objectives of this scheme are as follows:

- To lay down policy guidelines for conservation and management of wetlands in the country
- To undertake intensive conservation measures in priority wetlands
- Monitoring the implementation of programme
- Monitoring the situation of Indian wetlands

Wetlands (Conservation and Management) Rules, 2017

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- The rules stipulate setting up of a **State Wetlands Authority** in each State and union territories that will be **headed by the State's environment minister** and include a range of government officials. They will also include one expert each in the fields of wetland ecology, hydrology, fisheries, landscape planning and socioeconomics to be nominated by the state government.
- These authorities will need to develop a comprehensive list of activities to be regulated and permitted within the notified wetlands and their zone of influence, recommend additional prohibited activities for specific wetlands, define strategies for conservation and wise use of wetlands, and undertake measures for enhancing awareness within stakeholders and local communities on values and functions of wetlands. Wise use is defined as the principle of sustainable use that is compatible with conservation.
- The State authorities will prepare a list of all wetlands of the State or union territory based on which a comprehensive digital inventory of all wetlands will be created and updated every ten years.
- The rules prohibit activities like conversion of wetland for non-wetland uses including encroachment of any kind, setting up of any industry and expansion of existing industries, manufacture or handling or storage or disposal of hazardous substances and construction and demolition waste, solid waste dumping, discharge of untreated wastes and effluents from industries, cities, towns, villages and other human settlements.
- Under the rules, **powers have been given to the State governments** so that protection and conservation work can be done at the local level. **The central government has mainly retained powers regarding monitoring.**
- To oversee the work carried out by States, the rules stipulate for setting up of the **National Wetlands Committee,** which will be **headed by the MoEF&CC Secretary,** to monitor the implementation of these rules.
- The Committee will also advise the Central Government on appropriate policies and action programmes for conservation and wise use of wetlands, recommend designation of wetlands of international importance

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under the Ramsar Convention, advise on collaboration with international agencies on issues related to wetlands, etc.

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Wetlands International

- It is a nonprofit organisation dedicated to the conservation and restoration of wetlands.
- It has 20 regional, national and project offices in different nations of the world.
- Its headquarters is located in Ede, Netherlands.
- This organisation either executes or supports different programmes and policies for wetlands conservation and restoration.
- It is also associated with different research, community-based projects, collaboration with governments international policies and global conferences.
- It was established in 1954 as an international Wildfowl enquiry, and its main function was the conservation of aquatic birds.
- Later on, it became the International Waterfowl and Wetland Research Bureau.
- And its function was extended to the conservation of wetlands along with aquatic birds.

9.44 NEW YORK DECLARATION ON FORESTS

- It is a **voluntary and non-legally binding** political **declaration that grew out of dialogue among governments, companies and civil society, spurred by the United Nations Secretary-General's Climate Summit held in New York in 2014.**
- It aims to protect and restore forests by implementing **10 Goals**.
  1. At least **halve the rate of loss of natural forests globally by 2020 and strive to end natural forest loss by 2030.**
  2. **Support and help meet the private-sector goal of eliminating deforestation** from the production of agricultural commodities such as palm oil, soy, paper and beef products by no later than 2020, recognising that many companies have even more ambitious targets.
  3. Significantly reduce deforestation derived from other economic sectors by 2020.
  4. Support alternatives to deforestation driven by basic needs (such as subsistence farming and reliance on fuel wood for energy) in ways that alleviate poverty and promote sustainable and equitable development.

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5. **Restore 150 million hectares of degraded landscapes and forestlands by 2020** and significantly increase the rate of global restoration thereafter, which would restore at least an additional 200 million hectares by 2030.
6. Include ambitious, **quantitative forest conservation and restoration targets for 2030** in the post-2015 global development framework, as part of new international sustainable development goals.
7. Agree in 2015 to reduce emissions from deforestation and forest degradation as part of a post-2020 global climate agreement, in accordance with internationally agreed rules and consistent with the goal of not exceeding 2°C warming.
8. Provide support for the development and implementation of strategies to reduce forest emissions.
9. Reward countries and jurisdictions that, by taking action, reduce forest emissions particularly through public policies to scaleup payments for verified emission reductions and private-sector sourcing of commodities.
10. Strengthen forest governance, transparency and the rule of law, while also empowering communities and recognising the rights of indigenous peoples, especially those pertaining to their lands and resources.

- Achieving these outcomes could reduce emissions by 4.5–8.8 billion tons per year by 2030.
- By working in partnership, we can achieve these goals and chart a new course towards conserving, restoring, and managing healthy forests for the benefit of all. We invite others to join us in committing to a world where people and forests grow together.

9.45 UN OCEAN CONFERENCE (UNOC)

First UNOC held in 2017 at UN Headquarters in New York, co-hosted by the Governments of **Fiji and Sweden.**

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The **second UNOC,** co-hosted by **Kenya and Portugal** at **Lisbon,** ended with the **Lisbon Declaration**, a political declaration entitled **'Our Ocean, Our Future: call for action'.**

Key Highlights of second UNOOC

India participated in the **Coastal Clean Seas Campaign** and pledged to work toward banning **single-use plastics** to reduce marine pollution.

- **Global Collective Action:**
  - Over **150 countries** are committed to scaling up **science-based and innovative actions** to address the ocean emergency.
  - These efforts support the implementation of **Sustainable Development Goal 14 (Life Below Water),** which focuses on conserving and sustainably using marine resources.
- **Alignment with Global Frameworks:**
  - Actions align with the **United Nations Decade of Ocean Science for Sustainable Development (2021–2030)**, emphasising the importance of advancing ocean science for sustainable development.
  - Emphasis was placed on implementing the **Paris Agreement (2015)** and the **Glasgow Climate Pact (2021)** to ensure the health, productivity, sustainable use, and resilience of the ocean.
- **Ocean Conservation Targets:**
  - Nations pledged to conserve or protect at least **30% of the global ocean by 2030** through **Marine Protected Areas** and other effective area-based conservation measures.
- **UNESCO Contribution:**
  - UNESCO launched the **State of the Ocean Report (pilot edition)** during the event, providing insights into ocean health and sustainability.

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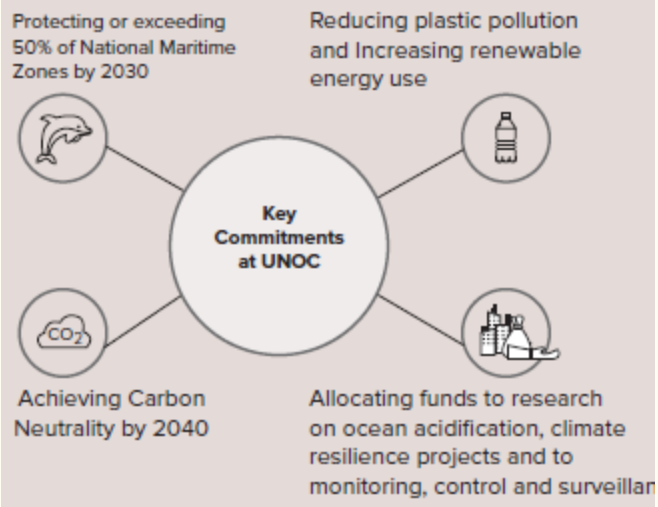

Figure 9.8 Key Commitment at UNOC

These commitments aim to address climate change impacts on marine ecosystems and promote sustainable ocean management practices globally.

9.46 100 MILLION FARMERS (100MF) – WORLD ECONOMIC FORUM INITIATIVE

100 Million Farmers (100MF) is a **World Economic Forum (WEF) initiative** that aims to accelerate the global transition toward **regenerative, sustainable, and climate-resilient agriculture**.

■ *Objective*

To mobilise and support **100 million farmers worldwide** to shift from conventional farming to **nature-positive**, **net-zero**, and **climate-resilient** practices.

Key Features

- Encourages **regenerative agriculture** (soil health, biodiversity, water conservation).
- Promotes **low-carbon farming** and the reduction of emissions from agriculture.

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- Strengthens **farmer resilience** to climate shocks like droughts, floods, and extreme weather.
- Focuses on **local food systems**, farmer livelihoods, and sustainable value chains.
- Works through **multi-stakeholder partnerships**: governments, agribusiness, research institutions, and farmer organisations.

Importance

- Agriculture contributes significantly to **GHG emissions**, soil degradation, and water stress.
- 100MF aims to **transform food systems** to be climate-friendly while improving **farmer income and sustainability**.

International Green Days

- **International Tiger Day:** Also known as **Global Tiger Day,** it is an annual celebration to raise awareness for tiger conservation, held annually on **29 July.**
- **World Environment Day:** World Environment Day (WED) is observed every year on **5 June** to raise global awareness to take positive environmental action to protect nature and the planet Earth. It is run by the **United Nations Environment Programme** (UNEP). WED is the United Nations' most important day for encouraging worldwide awareness and action for the protection of our environment. Since **it began in 1974,** it has grown to become a global platform for public outreach that is widely celebrated in over 100 countries.
- **World Forestry Day (WFD):** WFD or International Day of Forests is celebrated worldwide every year on **21st March** at the international level in order to increase the public awareness among communities about the values, significance and contributions of the forests to balance the life cycle on Earth.
- **World Habitat Day (WHD):** The UN has designated the **first Monday of October** of every year as World Habitat Day. The purpose of WHD is to reflect on the state of our towns and cities, and on the basic right of all to adequate shelter.
- **World Ozone Day (WOD): September 16** is observed as the International Day for the Preservation of the Ozone Layer. This designation was made on December 19, 1994, in commemoration of the date, in 1987, on which nations signed the Montreal Protocol on Substances that Deplete the Ozone Layer.
- **World Water Day, 22 March: In the year 1992, March 22 was first officially added in schedule 21 of the United Nations Conference on Environment and Development** as World Water Day in Rio de

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Janeiro, Brazil. The celebrations to mark world water day started in the year 1993. The day aims to increase awareness among people about the importance, need and conservation of water. **The World Water Development Report** is also released by the UN every year around World Water Day.

- **International Day of Forests, 21 March: The United Nations General Assembly proclaimed 21 March the International Day of Forests (IDF) in 2012.** The Day celebrates and raises awareness of the importance of all types of forests. On each International Day of Forests, countries are encouraged to undertake local, national and international efforts to organise activities involving forests and trees, such as tree-planting campaigns.
- **World Environment Protection Day:** Celebrated on **26th November** every year to raise global awareness of the need to take positive actions to protect the environment. This event is organised by the United Nations Environment Programme (UNEP). The natural resources on Earth are vulnerable to many human impacts.
- **The International Day for Biological Diversity:22 May.** It is a United Nations sanctioned international day for the promotion of biodiversity issues.
- **Earth Day:** It is an annual event on **April 22** to demonstrate support for environmental protection. First held on April 22, 1970.
- **World Oceans Day:8th June.** World Oceans Day is celebrated to remind everyone about the major role that oceans play in everyday life. They are the lungs of our planet, providing most of the oxygen we breathe.
- **International Mangrove Day:26th July.** This day is observed to raise awareness of the importance of mangrove ecosystems as a unique, special and vulnerable ecosystem and to promote solutions for their sustainable management, conservation and uses.

Mind Map 1 on Declarations and Conventions of this Chapter

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Questions Asked in UPSC Prelims Exam

UNIT III

Unit - III Prelims PYQs (2011 onwards)

1. Which organization has enacted the Nature Restoration Law (NRL) to tackle climate change and biodiversity loss? **[UPSC (Pre) 2025]**

(a) The European Union
(b) The World Bank
(c) The Organization for Economic Cooperation and Development
(d) The Food and Agriculture Organization

2. Consider the following statements: **[UPSC (Pre) 2025]**

Statement I: At the 28th United Nations Climate Change Conference (COP28), India refrained from signing the 'Declaration on Climate and Health'.

Statement II: The COP28 Declaration on Climate and Health is a binding declaration; and if signed, it becomes mandatory to decarbonise the health sector.

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Statement III: If India's health sector is decarbonised, the resilience of its health-care system may be compromised.

Which one of the following is correct in respect of the above statements?

(a) Both Statement II and Statement III are correct and both of them explain Statement I

(b) Both Statement II and Statement III are correct but only one of them explains Statement I

(c) Only one of the Statements II and III is correct and that explains Statement I

(d) Neither Statement II nor Statement III is correct

3. Consider the following statements: **[UPSC (Pre) 2025]**

Statement I: Article 6 of the Paris Agreement on climate change is frequently discussed in global discussions on sustainable development and climate change.

Statement II: Article 6 of the Paris Agreement on climate change sets out the principles of carbon markets.

Statement III: Article 6 of the Paris Agreement on climate change intends to promote inter-country non-market strategies to reach their climate targets.

Which one of the following is correct in respect of the above statements?

(a) Both Statement II and Statement III are correct and both of them explain Statement I

(b) Both Statement II and Statement III are correct but only one of them explains Statement I

(c) Only one of the Statements II and III is correct and that explains Statement I

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(d) Neither Statement II nor Statement III is correct

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4. Which one of the following is the correct description of "100 Million Farmers"? **[UPSC (Pre) 2024]**

(a) It is a platform for accelerating the transition towards food and water systems that are net-zero (carbon), nature-positive and that aims to increase farmer resilience.

(b) It is an international alliance and a network of individuals and farming organisations interested in supporting and strengthening the development of the organic animal husbandry.

(c) It is a digital platform fully integrated with service providers and built on blockchain that lets buyers, sellers and third parties trade fertilizers quickly and securely.

(d) It is a platform with the mission of encouraging the farmers to form Farmer Product Organisations or Agribusiness Consortiums, thus facilitating the access to global open markets to sell their products.

5. Consider the following statements: **[UPSC (Pre) 2024]**

Statement-I: The European Parliament approved the Net-Zero Industry Act recently.

Statement-II: The European Union intends to achieve carbon neutrality by 2040 and therefore aims to develop all of its own clean technology by that time.

Which one of the following is correct in respect of the above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

(b) Both Statement-I and Statement-II are correct, but Statement-II does *not* explain Statement-I

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(c) Statement-I is correct, but Statement-II is incorrect
(d) Statement-I is incorrect, but Statement-II is correct

6. "Climate Action Tracker" which monitors the emission reduction pledges of different countries is a: **[UPSC (Pre) 2022]**

(a) Database created by coalition of research organisations
(b) Wing of "International Panel of Climate Change"
(c) Committee under "United Nations Framework Convention on Climate Change"
(d) Agency promoted and financed by United Nations Environment Programme and World Bank

7. Consider the following statements: **[UPSC (Pre) 2022]**

1. The Climate Group is an international non-profit organisation that drives climate action by building large networks and runs them.
2. The International Energy Agency in partnership with the Climate Group launched a global initiative "EP100".
3. EP100 brings together leading companies committed to driving innovation in energy efficiency and increasing competitiveness while delivering on emission reduction goals.
4. Some Indian companies are members of EP100.
5. The International Energy Agency is the Secretariat to the "Under2 Coalition".

Which of the statements given above are correct?

(a) 1, 2, 4 and 5
(b) 1, 3 and 4 only
(c) 2, 3 and 5 only
(d) 1, 2, 3, 4 and 5

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8. Which one of the following statements best describes the 'Polar Code'? [UPSC (Pre) 2022]

(a) It is the international code of safety for ships operating in polar waters.

(b) It is the agreement of the countries around the North Pole regarding the demarcation of their territories in the polar region.

(c) It is a set of norms to be followed by the countries whose scientists undertake research studies in the North Pole and South Pole.

(d) It is a trade and security agreement of the member countries of the Arctic Council.

9. With reference to the New York Declaration on Forests, which of the following statements are correct? [UPSC (Pre) 2021]

1. It was first endorsed at the United Nations Climate Summit in 2014.
2. It endorses a global timeline to end the loss of forests.
3. It is a legally binding international declaration.
4. It is endorsed by governments, big companies and indigenous communities.
5. India was one of the signatories at its inception. Select the correct answer using the codes given below.

(a) 1, 2 and 4

(b) 1, 3 and 5

(c) 3 and 4

(d) 2 and 5

10. Momentum for Change: Climate Neutral Now' is an initiative launched by [UPSC (Pre) 2018]

(a) The Intergovernmental panel on Climate Change

(b) The UNEP Secretariat

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(c) The UNFCCC Secretariat
(d) The World Meteorological Organization

11. With reference to the 'Global Alliance for Climate- Smart Agriculture (GACSA)', which of the following statement/s is/are correct? **[UPSC (Pre) 2018]**

1. GACSA is an outcome of the Climate Summit held in Paris in 2015.
2. Membership of GACSA does not create any binding obligations.
3. India was instrumental in the creation of GACSA.

Select the correct answer using the codes given below:

(a) 1 and 3 only
(b) 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

12. The Partnership for Action on Green Economy (PAGE), a UN mechanism to assist countries transition towards greener and more inclusive economies, emerged at **[UPSC (Pre) 2018]**

(a) The Earth Summit on Sustainable Development 2002, Johannesburg
(b) The United Nations Conference on Sustainable Development 2012, Rio de Janeiro
(c) The United Nations Framework Convention on Climate Change 2015, Paris
(d) The World Sustainable Development Summit 2016, New Delhi

13. Consider the following statements in respect of Trade Related Analysis of Fauna and Flora in Commerce (TRAFFIC) **[UPSC (Pre) 2017]**

1. TRAFFIC is a bureau under United Nations Environment Programme.

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2. The mission of TRAFFIC is to ensure that trade in wild plants and animals is not a treat to the conservation of nature.

Which of the above statement/s is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

14. With reference to 'Global Climate change Alliance', which of the following statement/s is/are correct? **[UPSC (Pre) 2017]**

1. It is an initiative of the European Union.
2. It provides technical and financial support to targeted developing countries to integrate climate change into their development policies and budgets.
3. It is coordinated by World Resources Institute (WRI) and World Business Council for Sustainable Development (WBCSD)

Select the correct answer using the codes given below:

(a) 1 and 2 only

(b) 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

15. Consider the following statements: **[UPSC (Pre) 2017]**

1. Climate and Clean Air Coalition (CCAC) to reduce short- lived climate pollutant is a unique initiative of G20 group of countries.
2. The CCAC focuses on methane, black carbon and hydrofluorocarbons. Which of the statement/s given above is/are correct?

(a) 1 only

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(b) 2 only

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(c) Both 1 and 2

(d) Neither 1 nor 2

16. Consider the following pairs: **[UPSC (Pre) 2016]**

Terms sometimes their seen in the news: Origin

1. Annexe-I Countries: Cartagena Protocol
2. Certified Emissions Reductions: Nagoya Protocol
3. Clean Development Mechanism: Kyoto Protocol Which of the pair/s given above is/are correctly matched?

(a) 1 and 2 only

(b) 2 and 3 only

(c) 3 only

(d) 1, 2 and 3

17. Which of the following best describes/describe the aim of 'Green India Mission' of the Government of India? **[UPSC (Pre) 2016]**

1. Incorporating environmental benefits and costs into the Union and State Budgets thereby implementing the 'green accounting'.
2. Launching the second green revolution to enhance agricultural output to ensure food security to one and all in the future.
3. Restoring and enhancing forest cover and responding to climate change by a combination of adaptation and mitigation measures.

Select the correct answer using the codes given below.

(a) 1 only

(b) 2 and 3 only

(c) 3 only

(d) 1, 2 and 3

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18. With reference to an initiative called 'The Economics of Ecosystems and Biodiversity (TEEB)', which of the following statement/s is/are correct? **[UPSC (Pre) 2016]**

1. It is an initiative hosted by UNEP, IMF and World Economic Forum.
2. It is a global initiative that focuses on drawing attention to the economic benefits of biodiversity.
3. It presents an approach that can help decision-makers recognise, demonstrate and capture the value of ecosystems and biodiversity.

Select the correct answer using the codes given below.

(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

19. Which of the following statement/s is/are correct? **[UPSC (Pre) 2016]**

Proper design and effective implementation of the UN-REDD+ Programme can significantly contribute to

1. protection of biodiversity
2. resilience of forest ecosystems
3. poverty reduction

Select the correct answer using the codes given below.

(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

20. What is 'Greenhouse Gas Protocol'? **[UPSC (Pre) 2016]**

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(a) It is an international accounting tool for government and business leaders to understand, quantify and manage greenhouse gas emissions

(b) It is an initiative of the United Nations to offer financial incentives to developing countries to reduce greenhouse gas emissions and to adopt eco-friendly technologies

(c) It is an inter-governmental agreement ratified by all the member countries of the United Nations to reduce greenhouse gas emissions to specified levels by the year 2022

(d) It is one of the multilateral REDD+ initiatives hosted by the World Bank

21. With reference to 'Agenda 21', sometimes seen in the news, consider the following statements: [UPSC (Pre) 2016]

1. It is a global action plan for sustainable development
2. It originated in the World Summit on Sustainable Development held in Johannesburg in 2002. Which of the statement/s given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

22. With reference to the Agreement at the UNFCCC Meeting in Paris in 2015, which of the following statement/s is/are correct? [UPSC (Pre) 2016]

1. The Agreement was signed by all the member countries of the UN, and it will go into effect in 2017.
2. The Agreement aims to limit greenhouse gas emissions so that the rise in average global temperature by the end of this century does

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not exceed 2°C or even 1.5°C above preindustrial levels.

3. Developed countries acknowledged their historical responsibility in global warming and committed to donate $1000 billion a year from 2020 to help developing countries to cope with climate change.

Select the correct answer using the codes given below.

(a) 1 and 3 only

(b) 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

23. Consider the following statements: **[UPSC (Pre) 2016]**

1. The Sustainable Development Goals were first proposed in 1972 by a global think tank called the 'Club of Rome'.

2. The Sustainable Development Goals have to be achieved by 2030.

Which of the statement/s given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

24. The term 'Intended Nationally Determined Contributions' is sometimes seen in the news in the context of **[UPSC (Pre) 2016]**

(a) pledges made by the European countries to rehabilitate refugees from the war-affected Middle East

(b) plan of action outlined by the countries of the world to combat climate change

(c) capital contributed by the member countries in the establishment of the Asian Infrastructure Investment Bank

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(d) plan of action outlined by the countries of the world regarding Sustainable Development Goals

25. What is/are the importance/importances of the 'United Nations Convention to Combat Desertification'? **[UPSC (Pre) 2016]**

1. It aims to promote effective action through innovative national programmes and supportive international partnerships.
2. It has a special/particular focus on South Asia and North Africa regions and its Secretariat facilitates the allocation of a major portion of financial resources to these regions.
3. It is committed to a bottom-up approach, encouraging the participation of local people in combating desertification.

Select the correct answer using the codes given below.

(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

26. Consider the following statements: **[UPSC (Pre) 2016]**

1. The International Solar Alliance was launched at the United Nations Climate Change Conference in 2015.
2. The Alliance includes all the member countries of the United Nations.

Which of the statement/s given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

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27. With reference to the International Union for Conservation of Nature and Natural Resources (IUCN) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which of the following statement/s is/are correct? [UPSC (Pre) 2015]

1. IUCN is an organ of the United Nations and CITES is an international agreement between governments.
2. IUCN runs thousands of field projects around the world to better manage natural environments.
3. CITES is legally binding on the States that have joined it but this Convention does not take the place of national laws.

Select the correct answer using the codes given below:

(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

28. Which one of the following is associated with the issue of control and phasing out of the use of ozone- depleting substances? [UPSC (Pre) 2015]

(a) Bretton Woods Conference
(b) Montreal Protocol
(c) Kyoto Protocol
(d) Nagoya Protocol

29. What is Rio+20 Conference, often mentioned in the news? [UPSC (Pre) 2015]

(a) It is the United Nations Conference on Sustainable Development
(b) It is a Ministerial Meeting of the World Trade Organization
(c) It is a Conference of the Inter-governmental Panel on Climate Change

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(d) It is a Conference of the Member Countries of the Convention on Biological Diversity

30. Which of the statements regarding 'Green Climate Fund' is/are correct? **[UPSC (Pre) 2015]**

1. It is intended to assist the developing countries in adaption and mitigation practices to counter climate change.
2. It is founded under the aegis of UNEP, OECD, Asian Development Bank and World Bank. Select the correct answer using the codes given below:

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

31. 'Bio Carbon Fund Initiative for Sustainable Forest Landscapes' is managed by the **[UPSC (Pre) 2015]**

(a) Asian Development Bank
(b) International Monetary Fund
(c) United Nations Environment Programme
(d) World Bank

32. With reference to 'Forest Carbon Partnership Facility', which of the following statement/s is/are correct? **[UPSC (Pre) 2015]**

1. It is a global partnership of governments, businesses, civil society and indigenous peoples.
2. It provides financial aid to universities, individual scientists and institutions involved in scientific forestry research to develop

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ecofriendly and climate adaptation technologies for sustainable forest management.

3. It assists the countries in their REDD+ (Reducing Emissions from Deforestation and Forest Degradation+) efforts by providing them with financial and technical assistance.

Select the correct answer using the codes given below:

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

33. With reference to an organization known as 'Birdlife International', which of the following statement/s is/are correct? [UPSC (Pre) 2015]

1. It is a Global Partnership of Conservation Organizations.
2. The concept of 'biodiversity hotspots' originated from this organization.
3. It identifies the sites known/referred to as 'Important Bird and Biodiversity Areas'.

Select the correct answer using the codes given below:

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

34. Consider the following international agreements: [UPSC (Pre) 2014]

1. The International Treaty on Plant Genetic Resources for Food **International Treaty on Plant Genetic Resources for** and Agriculture.
2. The United Nations Convention to Combat Desertification.

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3. The World Heritage Convention.

Which of the above has/have a bearing on biodiversity?

(a) 1 and 2 only

(b) 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

35. Consider the following statements regarding 'Earth Hour': [UPSC (Pre) 2014]

1. It is an initiative of UNEP and UNESCO.
2. It is a movement in which the participants switch off the lights for one hour on a certain day every year.
3. It is a movement to raise awareness about climate change and the need to save the planet. Which of the statement/s given above is/are correct?

(a) 1 and 3 only

(b) 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

36. If a wetland of international importance is brought under the 'Montreux Record', what does it imply? [UPSC (Pre) 2014]

(a) Changes in ecological character have occurred, are occurring or are likely to occur in the wetland as a result of human interference.

(b) The country in which the wetland is located should enact a law to prohibit any human activity within five kilometers from the edge of the wetland.

(c) The survival of the wetland depends on the cultural practices and traditions of certain communities living in its vicinity and therefore the cultural diversity therein should not be destroyed.

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(d) It is given the status of 'World Heritage Site'.

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37. With reference to Bombay Natural History Society (BNHS), consider the following statements: [UPSC (Pre) 2014]

1. It is an autonomous organisation under the Ministry of Environment and Forests.
2. It strives to conserve nature through action-based research, education and public awareness.
3. It organises and conducts nature trails and camps for the general public. Which of the statement/s given above is/are correct?

(a) 1 and 3 only
(b) 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

38. With reference to 'Global Environment Facility', which of the following statement/s is/are correct? [UPSC (Pre) 2014]

(a) It serves as a financial mechanism for 'Convention on Biological Diversity' and 'United Nations Framework Convention on Climate Change'.
(b) It undertakes scientific research on environmental issues at the global level.
(c) It is an agency under OECD to facilitate the transfer of technology and funds to underdeveloped countries with a specific aim to protect their environment.
(d) Both (a) and (b)

39. With reference to a conservation organisation called 'Wetlands International', which of the following statement/s is/are correct? [UPSC (Pre) 2014]

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1. It is an intergovernmental organisation formed by the countries which are signatories to Ramsar Convention.
2. It works at the field level to develop and mobilise knowledge, and use the practical experience to advocate for better policies.

Select the correct answer using the codes given below.

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

40. The Millennium Ecosystem Assessment describes the following major categories of ecosystem services: provisioning, supporting, regulating, preserving and cultural. Which one of the following is a supporting service? **[UPSC (Pre) 2012]**

(a) Production of food and water
(b) Control of climate and disease
(c) Nutrient cycling and crop pollination
(d) Maintenance of diversity

41. Regarding 'carbon credits', which one of the following statements is not correct? **[UPSC (Pre) 2011]**

(a) The carbon credit system was ratified in conjunction with the Kyoto Protocol.
(b) Carbon credits are awarded to countries or groups that have reduced greenhouse gases below their emission quota.
(c) The goal of the carbon credit system is to limit the increase of carbon dioxide emissions.
(d) Carbon credits are traded at a price fixed from time to time by the United Nations Environment Programme.

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Answers

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1. (a) 2. (c) 3. (a) 4. (a) 5. (c) 6. (a) 7. (b) 8. (a) 9. (a) 10. (c)
11. (b) 12. (b) 13. (b) 14. (a) 15. (b) 16. (c) 17. (c) 18. (c) 19. (d) 20. (a)
21. (a) 22. (b) 23. (b) 24. (b) 25. (c) 26. (a) 27. (b) 28. (b) 29. (a) 30. (a)
31. (d) 32. (c) 33. (c) 34. (d) 35. (c) 36. (a) 37. (c) 38. (a) 39. (b) 40. (c)
41. (d)

Explanations

1. Correct answer (a): EXPLANATION

The **EU Nature Restoration Law (NRL)** is the EU's first continent-wide legislation to revive degraded ecosystems, with binding targets to restore **20% of land and sea by 2030** and all ecosystems by 2050. It focuses on improving soil health, enhancing urban green spaces, and restoring biodiversity across terrestrial, freshwater, coastal, and marine systems. Member states must prepare national restoration plans, making it a central tool to tackle **climate change, biodiversity loss, and environmental decline**. **Statement (a) is right**

2. Correct answer (c): EXPLANATION

- **"At the 28th United Nations Climate Change Conference (COP28), India refrained from signing the 'Declaration on Climate and Health'."** — **Correct,** India **did not sign** the Declaration. **Statement (I) is right**
- **"The COP28 Declaration** is described as a **voluntary, non-binding, political declaration**. It does **not impose legal obligations** or enforce mandatory decarbonisation of the health sector. **Statement (II) is wrong**

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- One of the reasons India gave for *not signing* the Declaration was that rapidly curbing greenhouse-gas use (e.g. for cooling, cold-storage of vaccines/medicines) could **impair reliability and resilience of health services**, especially in resource-constrained or remote areas. While decarbonisation is desirable, in India's context this concern was considered serious enough to abstain from signing. **Statement (III) is correct**

Therefore, the correct option is: **Only one of the Statements II and III is correct and that explains Statement I.** (i.e. Statement III explains why India refrained from signing)

3. Correct answer (a): EXPLANATION

- Article 6 is often discussed because it covers **cooperative approaches, carbon markets, and international carbon trading mechanisms**, which are key to achieving Nationally Determined Contributions (NDCs). **Statement (I) is correct**
- Article 6.2 and 6.4 provide **frameworks for voluntary and international carbon markets**, including rules for **trading emission reductions between countries**. **Statement (2) is right**
- Article 6.8 covers **non-market approaches**, supporting **cooperation between countries without involving carbon trading**, e.g., technology transfer or joint mitigation initiatives. **Statement (3) is right**
- **Relation between statements:** Both Statements II and III describe specific mechanisms of Article 6.
- **Statement I is general**, and both II and III explain why Article 6 is frequently discussed internationally.

4. Correct answer (a): EXPLANATION

"100 Million Farmers" (100MF) is a World Economic Forum initiative aiming to empower small-scale farmers worldwide and accelerate the transition to **net-zero, nature-positive food and water**

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systems. By 2030, it seeks to reach 100 million farmers through financing, collaboration, and promotion of **regenerative agriculture** to improve soil health, climate resilience, water conservation, and livelihoods. So **option (a)** is correct.

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5. Correct answer (c): EXPLANATION

- The European Parliament adopted the Net-Zero Industry Act in April 2024 (in the final plenary session of its 2019–2024 legislature). So the statement that the Parliament "approved" the Act "recently" is accurate. **Statement (1) is correct**
- Officially, the EU's long-term legally binding target is to reach **climate neutrality by 2050**. The 2040 target currently under discussion (proposed by the Commission in 2024) is for a **90% reduction in net greenhouse-gas emissions by 2040 (from 1990 levels)** — not full carbon neutrality by 2040.

Thus, saying the EU "intends to achieve carbon neutrality by 2040" is **not correct** under current law/policy. **Statement (2) is wrong**

6. Correct answer (a): EXPLANATION

"Climate Action Tracker" (CAT) is **a database created by a coalition of research organisations.** It is run by independent research groups, including **Climate Analytics** and the **New Climate Institute**. CAT **monitors and assesses climate action pledges** of countries against the goals of the Paris Agreement. It is **not a UN agency, IPCC wing, or UNFCCC committee**. **Statement (a) is correct**

7. Correct answer (b): EXPLANATION

- The Climate Group is an **international NGO** that mobilizes governments and businesses through initiatives and networks. **Statement (1) is correct**

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- **EP100** was launched by **The Climate Group alone**, not in partnership with the IEA. **Statement (2) is wrong**
- This is the **primary aim of EP100**, focusing on energy productivity and efficiency. **Statement (3) is right**
- Indian companies like **Tata Steel** and others are members of EP100**. Statement (4) Is right**
- The **Climate Group** acts as the Secretariat to the "Under2 Coalition", not the **International Energy Agency**. **Statement (5) is wrong**

8. Correct answer (a): EXPLANATION

International code of safety for ships operating in polar waters officially called the **International Code for Ships Operating in Polar Waters**, it was adopted by the **International Maritime Organization (IMO)**. It covers **safety, pollution prevention, and operational guidelines** for ships in **Arctic and Antarctic waters**. **Statement (a) is correct**

9. Correct answer (a): EXPLANATION

- NYDF was **launched at the UN Climate Summit in New York, September 2014**. **Statement (1) is right**
- NYDF sets **voluntary targets**, including halving deforestation by 2020 and ending it by 2030. **Statement (2) is right**
- NYDF is **non-binding** and relies on voluntary commitments by governments and companies. **Statement (3) is wrong**
- Endorsers include **over 190 governments, companies, indigenous groups, and civil society organizations**. **Statement (4) is right**
- India **did not sign** the declaration at its launch in 2014. **Statement (5) is wrong**

10. Correct answer (c): EXPLANATION

"Momentum for Change: Climate Neutral Now" is an initiative launched by **The UNFCCC Secretariat.** It encourages **governments,**

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businesses, and individuals to measure, reduce, and offset their greenhouse gas emissions to support the goals of the Paris Agreement. **Statement (c) is right**

11. Correct answer (b): EXPLANATION

- GACSA was launched in **2014 at the UN Climate Summit in New York**, **not Paris 2015**. **Statement (1) is wrong**
- GACSA is a **voluntary alliance**, and members **do not have legally binding obligations**. **Statement (2) is right**
- India is a signatory and member of GACSA but was not among the founding members or instrumental in its creation. The alliance was primarily initiated by countries such as the Netherlands and New Zealand, with support from the FAO. **Statement (3) is wrong**

12. Correct answer (b): EXPLANATION

The **Partnership for Action on Green Economy (PAGE)** is a **UN initiative** that helps countries transition to **greener, inclusive economies**. It was launched in **2013**, following discussions at the **Rio+20 Conference (United Nations Conference on Sustainable Development 2012, Rio de Janeiro). Statement (b) is right**

13. Correct answer (b): EXPLANATION

- TRAFFIC is a **joint programme of WWF and IUCN**, and works closely with UNEP, but it is **not a bureau under UNEP**. **Statement (1) is wrong**
- TRAFFIC focuses on **monitoring and ensuring sustainable trade in wild flora and fauna**, to prevent threats to biodiversity. **Statement (2) is right**

14. Correct answer (a): EXPLANATION

- The Global Climate Change Alliance (GCCA), launched in 2007, and its successor, GCCA+ (after the 2015 Paris Agreement), are EU

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initiatives. **Statement (1) is right**

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- They provide technical and financial support to vulnerable developing countries, especially LDCs and SIDS, to help them adapt to and mitigate climate change. **Statement (2) is right**
- GCCA is Coordination by the **European Commission**, not WRI or WBCSD. **Statement (3) is wrong**

15. Correct answer (b): EXPLANATION

- The CCAC was **launched in 2012 by the United Nations Environment Programme (UNEP)** and **is a voluntary partnership of countries, not limited to the G20**. **Statement (1) is wrong**
- CCAC specifically targets **short-lived climate pollutants (SLCPs)** such as **methane, black carbon, and HFCs**. **Statement (2) is right**

16. Correct answer (c): EXPLANATION

- **Annex-I countries** are industrialized countries listed under the **UNFCCC**, not the Cartagena Protocol. **Statement (1) is wrong**
- **Certified Emissions Reductions (CERs)** are **carbon credits issued under the Clean Development Mechanism (CDM) of the Kyoto Protocol**, not the Nagoya Protocol. **Statement (2) is wrong**
- The **CDM** is one of the mechanisms under the **Kyoto Protocol** for reducing greenhouse gas emissions. **Statement (3) is right**

17. Correct answer (c): EXPLANATION

- This is related to **green accounting initiatives** under economic policy, not specifically part of the Green India Mission. **Statement (1) is wrong**
- The Green India Mission focuses on **forests, biodiversity, and climate change**, not agricultural productivity. **Statement (2) is wrong**
- This is the **primary aim of the Green India Mission**, part of the **National Action Plan on Climate Change (NAPCC)**. **Statement (3) is right**

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18. Correct answer (c): EXPLANATION

- TEEB is **hosted by UNEP (United Nations Environment Programme)**, **not the IMF or World Economic Forum**. **Statement (1) is wrong**
- TEEB highlights **the economic value of biodiversity and ecosystems** to influence policy and decision-making. **Statement (2) is right**
- TEEB provides **frameworks and tools for valuing ecosystems and biodiversity** in economic terms. **Statement (3) is right**

19. Correct answer (d): EXPLANATION

- The **UN-REDD+ Programme** (Reducing Emissions from Deforestation and Forest Degradation) aims to **reduce forest carbon emissions**, promote **sustainable forest management**, and **enhance forest** Sustainable forest management under REDD+ helps **protect habitats and biodiversity**. **Statement (1) is right**
- REDD+ enhances **forest ecosystem resilience** through conservation and restoration. **Statement (2) is right**
- REDD+ includes **safeguards and benefit-sharing mechanisms** that can support local communities, contributing to poverty reduction. **Statement (3) is right**

20. Correct answer (a): EXPLANATION

The **Greenhouse Gas (GHG) Protocol** provides standards and guidance for organizations and companies to **measure and report GHG emissions** consistently. It is **not a UN financial initiative, treaty, or REDD+ program**. Hence, **Statement (a) is right**

21. Correct answer (a): EXPLANATION

- Agenda 21 is a **comprehensive action plan adopted at the Earth Summit** for **sustainable development at global, national, and local levels**. Hence, **Statement (1) is right**

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- Agenda 21 was **adopted at the UN Conference on Environment and Development (UNCED), Rio de Janeiro, 1992**, also called the **Earth Summit**, **not in Johannesburg 2002**. The 2002 Johannesburg Summit reviewed progress on Agenda 21 but did not originate it. Hence, **Statement (2) is wrong**

22. Correct answer (b): EXPLANATION

- The Paris Agreement was **adopted in December 2015**. It **entered into force on 4 November 2016**, not 2017. Also, it was **signed by UNFCCC parties**, not literally all UN member countries. Hence, **Statement (1) is wrong**
- This is the central goal of the Paris Agreement. Hence, **Statement (2) is right**
- The **Green Climate Fund (GCF)** target was **$100 billion per year**, not $1000 billion. Hence, **Statement (3) is wrong**

23. Correct answer (b): EXPLANATION

- The **Club of Rome** is known for the report *"Limits to Growth"* (1972), which discussed global development and environmental issues. However, the **Sustainable Development Goals (SDGs)** were officially adopted by the **UN in 2015**. They were **not proposed by the Club of Rome**. Hence, **Statement (1) is wrong**
- The SDGs were adopted in **2015** with a target year of **2030**. Hence, **Statement (1) is right**

24. Correct answer (b): EXPLANATION

Intended Nationally Determined Contributions (INDCs) were submitted by countries before the **Paris Agreement (2015)** to indicate their commitments for reducing greenhouse gas emissions and adapting to climate impacts. **Statement (b) is correct**

25. Correct answer (c): EXPLANATION

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- UNCCD promotes national action programmes (NAPs) supported by international cooperation. **Hence, (2) is correct**
- UNCCD does **not** prioritise specific regions for financial allocation.It is a **global convention**, and its Secretariat does **not** allocate major resources specifically to South Asia or North Africa. **Hence, (2) is incorrect**
- One of the defining features of the UNCCD is its **bottom- up, community-driven approach**. **Hence, (3) is correct**

26. Correct answer (a): EXPLANATION

- ISA was launched by India and France at **COP21 (Paris, 2015)**. Hence, **Statement (1) is correct.**
- ISA does **not** include all UN member countries. Originally it was meant for countries between the Tropic of Cancer and Tropic of Capricorn, and although membership was later opened to all UN countries, **not all have joined**. Hence, **Statement (2) is wrong.**

27. Correct answer (b): EXPLANATION

- IUCN is **not** an organ of the UN. It is an **independent international organization**. CITES *is* an international agreement between governments. Since the first part is wrong, the entire **statement (1) is incorrect.**
- IUCN is involved in **many field conservation projects** globally. So better to manage natural environments. Hence, **Statement (2) is correct.**
- CITES is legally binding, but countries must implement it through their **own national laws**. Hence, **Statement (3) is correct.**

28. Correct answer (b): EXPLANATION

- **Bretton Woods Conference:** The Bretton Woods Conference (1944) sought to design a post-World War II international economic framework to ensure global stability and reconstruction.

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- **Montreal Protocol:** The Montreal Protocol (1987) is a landmark international treaty aimed at protecting the ozone layer by phasing out the production and consumption of ozone-depleting substances (ODS) such as chlorofluorocarbons (CFCs). Hence, **(b) is correct**.
- **Kyoto Protocol:** The Kyoto Protocol, adopted in 1997 under the UN Framework Convention on Climate Change (UNFCCC), was the first international treaty legally binding developed nations to reduce greenhouse gas emissions.
- **Nagoya Protocol:** The Nagoya Protocol, adopted in 2010 under the Convention on Biological Diversity (CBD), establishes a legal framework for fair and equitable sharing of benefits arising from the utilisation of genetic resources.

29. Correct answer (a): EXPLANATION

The United Nations Conference on Sustainable Development (UNCSD). It was held in **Rio de Janeiro in 2012**, 20 years after the 1992 Earth Summit. **Statement (a) is right**

30. Correct answer (a): EXPLANATION

- The GCF's core mandate is to help developing countries with **adaptation and mitigation** efforts. So, **Statement 1 is correct.**
- The GCF was **not** founded under these organizations. It was established by the **UNFCCC** at the **Cancún Conference (COP 16) in 2010**. Hence, **Statement 2 is wrong.**

31. Correct answer (d): EXPLANATION

The BioCarbon Fund Initiative for Sustainable Forest Landscapes (ISFL), managed by the World Bank, supports developing countries in reducing greenhouse gas emissions from land use. It promotes sustainable land-use planning, REDD+ activities, and climate-smart agriculture, while enhancing rural livelihoods through jurisdictional

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approaches that deliver large-scale environmental, social, and economic benefits. **Statement (d) is right**

32. Correct answer (c): EXPLANATION

- The Forest Carbon Partnership Facility (FCPF) is a **global partnership** launched by the **World Bank**. It includes **developed and developing country governments**, **businesses**, **civil society**, and **indigenous peoples' organizations**. Hence, **statement 1 is right.**
- FCPF **does not** fund universities or individual research projects. It focuses on **national-level REDD+ readiness**, capacity building, and pilot performance-based payments. Its funding is directed to **countries**, not academic research bodies. Hence, **statement 2 is wrong.**
- This is the **main purpose** of FCPF. It helps countries: Prepare national REDD+ strategies, Build capacity, Establish systems for MRV (Measurement, Reporting, Verification) & Implement pilot REDD+ programs. Hence, **statement 3 is right.**

33. Correct answer (c): EXPLANATION

- BirdLife International is a **global partnership** of NGOs working for bird conservation. It includes over **120 partner organizations** worldwide. So, **statement 1 is right**
- The concept of **biodiversity hotspots** was proposed by **Norman Myers**, a British ecologist, in 1988.BirdLife International did **not** originate this concept. So, **statement 2 is wrong**
- BirdLife International is responsible for identifying and designating **Important Bird and Biodiversity Areas (IBAs)**. So, **statement 3 is right**

34. Correct answer (d): EXPLANATION

- **Food and Agriculture (ITPGRFA) is** Directly related to **conservation and sustainable use** of plant genetic diversity. So, **statement 1 is right.**

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- **Main focus of United Nations Convention to Combat Desertification (UNCCD)** is land degradation, it **protects dryland ecosystems**, which are rich in unique biodiversity. It aims to **prevent loss of biodiversity** due to desertification. So, **statement 2 is right**
- **World Heritage Convention (WHC),** Protects **natural heritage sites** like national parks, wildlife sanctuaries, coral reefs, forests, etc. It is Clearly linked to **biodiversity conservation**. So, **statement 3 is right**

35. Correct answer (c): EXPLANATION

- **Earth Hour is NOT an initiative of UNEP or UNESCO.** It was started by the **World Wide Fund for Nature (WWF)** in Sydney in **2007**. So, **statement 1 is wrong.**
- Earth Hour is observed every year, usually on the **last Saturday of March**. People, businesses, and landmarks worldwide **turn off non-essential lights for one hour** (usually 8:30–9:30 PM local time). So, **statement 2 is right.**
- Earth Hour aims to spread awareness about **climate change**, **energy conservation**, and **environmental protection.** So, **statement 3 is right.**

36. Correct answer (a): EXPLANATION

- This is **exactly** what the Montreux Record signifies. It is a register under the **Ramsar Convention** for sites facing: ecological deterioration, human-induced changes, or vulnerability needing priority conservation attention. So, **option (a) is right.**
- No such fixed-distance prohibition exists under Montreux Record or Ramsar guidelines. So, **option (b) is wrong.**
- Ramsar does consider cultural values, but **Montreux Record is specifically about ecological character changes**, not cultural traditions. So **option (c) is wrong.**
- UNESCO World Heritage status has **no link** with inclusion in Montreux Record. So, **option (d) is wrong.**

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37. Correct answer (c): EXPLANATION

- **BNHS is NOT under the Ministry of Environment & Forests. It** is an **independent, non-governmental (NGO), non-profit organisation**, established in 1883. It collaborates with MoEFCC but is **not an autonomous body of the government**. So, **Statement 1 is wrong.**
- BNHS is known for wildlife research, conservation action, publications, education programmes, and awareness campaigns. So, **statement 2 is right.**
- BNHS regularly conducts **nature trails, bird- watching camps, wildlife tours**, and educational programmes. So, **statement 3 is right**

38. Correct answer (a): EXPLANATION

- The Global Environment Facility (GEF) acts as the **financial mechanism** for multiple international environmental conventions, including: **UNCBD**, **UNFCCC,** UNCCD, Stockholm Convention on POPs, Minamata Convention, etc. So, **statement (a) is right.**
- GEF **does not conduct scientific research itself**. It provides **funding** for projects implemented by agencies such as: UNEP, UNDP, World Bank, FAO But it is **not a scientific research body**. So, **statement (b) is wrong.**
- GEF is **not under OECD**. It is an **independent international partnership** of 183 countries, international institutions, NGOs, and CSOs. It is neither created by OECD nor limited to underdeveloped countries. Hence, **statement (c) is wrong.**

39. Correct answer (b): EXPLANATION

- Wetlands International is **not** an intergovernmental organization. It is an **independent, global non-profit organization**. Hence, **statement 1 is wrong**.
- Wetlands International works on **ground-level wetland conservation**, restoration, and knowledge generation. It uses **field experience** to

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influence **better national and international policies**. Hence, **statement 2 is right.**

40. Correct answer (c): EXPLANATION

According to the **Millennium Ecosystem Assessment (MEA)**, **Supporting Services** are those that are necessary for the production of all other ecosystem services. They include: **Nutrient cycling**, **Soil formation**, **Primary production**, **Photosynthesis & Pollination.**

Let's analyse the options:

1. **Production of food and water → Provisioning service**
2. **Control of climate and disease → Regulating service**
3. **Nutrient cycling and crop pollination → Supporting service, Hence (c) is correct**
4. **Maintenance of diversity** → Sometimes considered a part of supporting or an overarching result, but **not the standard MEA category**. **Statement (c) is right**

41. Correct answer (d): EXPLANATION

- Carbon credit mechanisms like **CDM, JI, and Emission Trading** were established under the Kyoto Protocol (1997). Hence, **option a is right**
- Under Kyoto's cap-and-trade system, entities reducing emissions below their assigned amount can earn tradable credits. Hence, **option b is right**
- The system provides financial incentives to reduce emissions. Hence, **option c is right**
- No UN body (including UNEP) fixes the price. Carbon credit prices are determined by **market forces**. Hence, **option d is wrong**

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UNIT IV Different Regulatory Bodies in India for Environmental Protection

Different Regulatory Bodies in India for Environmental Protection

10

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10.12.5 Institute of Forest Genetics and Tree Breeding (IFGTB), Coimbatore, Tamil Nadu
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10.12.10 Indian Environmental Society (IES), New Delhi
10.12.11 World Wildlife Fund (WWF), India
10.12.12 Environment Protection Training and Research Institute (EPTRI), Hyderabad, Telangana
10.12.13 National Environmental Engineering Research Institute (NEERI), Nagpur, Maharashtra
10.13 Compensatory Afforestation Fund Management and Planning Authority (CAMPA)

10.1 GOVERNMENT ORGANISATIONS: INDIA

The Ministry of Environment, Forests and Climate Change (MoEF) is the nodal organisation of the central government for planning, coordinating and supervising environmental and forestry programmes.

Ministry of Water Resources has a nodal part in all matters concerning India's water resources. **Ministry of New and Renewable Energy Sources** has an obligation to cover the entire renewable energy division, comprising solar, wind, hydro, biomass, geothermal and tidal energy sources.

The Department of Biotechnology, **Geological Survey of India** and the **National Institute of Oceanography** are involved in dealing with the

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subjects related to seismology and understanding of the seas in relation to sustainability of the environment.

10.2 MINISTRY OF ENVIRONMENT AND FORESTS (MOEF)

It is an important organisation in the administrative structure of the central government for the planning, promoting, coordinating and supervising the implementation of India's ecological and forestry policies and programmes. It also serves as the significant agency in the country for the United Nations Environment programme, United Nations Conference on Environment and Development, etc. **(Its chief concern is the implementation of strategies and programme dealing with preservation of the country's natural resources (biodiversity, forests and wildlife, etc.)**

Objectives of the Ministry: Conservation and survey of natural resources and protection and welfare of the environment and animals are the key objectives of MoEF.

The MoEF looks after:

1. Environmental laws and policies
2. Environmental monitoring and control
3. Survey and conservation of natural resources
4. Management of forests and conservation of wildlife
5. Environmental education, awareness and information
6. International cooperation

India is a party to five major international conventions related to wildlife conservation: Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), International Whaling Commission (IWC), UNESCO—World Heritage Convention and the Convention on the Conservation of Migratory Species of Wild Animals. **In 1985, the topics of wildlife and forests were added to the list and a new MoEF was created which remained under the charge of the Prime Minister.**

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10.3 ENVIRONMENTAL INFORMATION SYSTEM (ENVIS)

ENVIS, a Central Sector Scheme of the Ministry, has been implemented since 1982**. The purpose of the scheme is to integrate country-wide efforts in environmental information collection, collation, storage, retrieval and dissemination through ENVIS websites.** The ENVIS Portal (http://envis.nic.in) is a new initiative, **runs parallel with the Digital India Objective, working on improving digital literacy in the environment sector** and delivering services digitally all over the country.

- ENVIS, a Central Sector Scheme implemented in 1982.
- The purpose of the scheme is to integrate country-wide efforts in environmental information collection, collation, storage, retrieval and dissemination through ENVIS websites.
- The ENVIS Portal (http://envis.nic.in) is a new initiative, runs parallel with the Digital India Objective, working on improving digital literacy in the environment sector and delivering services digitally all over the country.
- It provides environmental information to decision-makers, policy planners, scientists and engineers, research workers, etc., all over the country.
- ENVIS has started implementing the World Bank–assisted Environment Management Capacity Building Technical Assistance Project since January 2002, which aims at structuring the ENVIS scheme by extending its reach through the involvement of institutions/organisations in state governments, academic sector, corporate sector, NGO sector, etc.

10.4 SURVEY OF FLORA AND FAUNA

10.4.1 Botanical Survey of India (BSI)

Botanical Survey of India

- It is an apex research organisation under the MoEF. It was **established on 13 February 1890** with the basic target to explore the country's floral resources and recognise the plant species with financial value (taxonomic and floristic studies).
- It has nine regional circles situated in different regions of the country.

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Primary Functions of BSI

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- Publication of National, State and District Floras.
- Exploration, inventorying and documentation of phytodiversity in general and protected areas, hotspots and fragile ecosystems in particular.
- Identification of Threatened and Red List species and conservation of those species and the concerned areas.
- Ex-situ conservation of critically threatened species in botanical gardens.
- Survey and documentation of traditional knowledge (ethnobotany) associated with plants.
- Develop a national database of Indian plants, including herbarium and live specimens, botanical paintings and illustrations, etc.

10.4.2 Zoological Survey of India (ZSI)

- Established in 1916.
- It is a prominent research institution under the MoEF.
- Its headquarters is in Kolkata.
- The current mandate of ZSI is survey, collection, documentation (including the traditional knowledge associated with animals) and ex-situ conservation of animal diversity of the country.
- It is a national centre for faunistic survey and exploration of the resources leading to the advancement of knowledge on the exceptionally rich faunal diversity of the country.
- It has 16 regional stations located in different geographic locations of the country.

10.4.3 Forest Survey of India (FSI), Dehradun

- It was founded in 1981.
- FSI, an organisation under the ministry, is engaged in the assessment of the country's forest resources at a regular interval.
- It is headquartered in Dehradun, Uttarakhand.
- The organisation's precursor was the 'Preinvestment Survey of Forest Resources' (PISFR), a project started in 1965 with aid from UNDP and FAO.

Forest Research Institute

- Located in Dehradun, Uttarakhand, it is one of the premier institutions in the field of forestry research in India. In 1991, the University Grants Commission declared it as a deemed university.
- The Forest Research Institute campus hosts the Indira Gandhi National Forest Academy, the staff college that trains officers selected for the Indian Forest Service.

10.5 BIODIVERSITY

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Biodiversity is the variability between living organisms and the environmental developments of which they are part, including diversity within and between species and ecosystems.

10.5.1 Convention on Biodiversity (CBO/CBD)

The **Convention on Biodiversity (CBO/CBD)** is one of the **key agreements** adopted during the **Earth Summit held in Rio de Janeiro in 1992**. This Convention has been ratified by 196 nations.

The **three objectives** of the CBO are:

1. Conservation of biological diversity.
2. Sustainable use of its components.
3. Fair and equitable sharing of benefits arising out of the use of genetic resources.

10.5.2 Biological Diversity Act

The **Government of India enacted the Biological Diversity Act, 2002**. According to **Section 8 of the Act, the National Biodiversity Authority was established by the Government of India in October 2003 in Chennai.** In 2000, a **Cartagena Protocol on Biosafety (CPB)** was adopted under the guidance of the CBD.

10.5.3 Objective of Cartagena Protocol on Biosafety (CPB)

The primary objective is to safeguard the transfer, handling and use of modified living organisms resulting from modern biotechnology. India is a part of both **CBD** and **CPB. A Nagoya Protocol on Access and Benefit Sharing** was adopted in 2010, after six years of intense discussions under the guidance of CBD.

10.5.4 National Biodiversity Authority

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The Government of India established the National Biodiversity Authority in October 2013, at Chennai, under Section 8 of the Biological Diversity Act. **The State Biodiversity Boards** are to be established by **the state governments**. **Biodiversity Management Committees** are to be organised by **the local bodies.**

Objectives

- To **control access to the country's biological resources** to preserve and **sustainable use of biological diversity.**
- To **respect and defend the knowledge** of local societies related to biodiversity.
- To **protect the sharing of benefits with the local individuals** as conservers of biological resources and holders of knowledge and information related to the use of biological resources.
- **Conservation and improvement of areas of importance from the perspective of biological diversity** by announcing them as biological diversity heritage sites.
- **Protection and restoration of threatened species**, involvement of state government institutions in the broad scheme of implementation of the Biological Diversity Act through the constitution of committees.

10.6 ENVIRONMENT RESEARCH IN INDIA

10.6.1 Ministry of Environment, Forest and Climate Change (MoEF)

The MoEF provides **various grants-in-aid projects to many research institutes,** in various areas under the broad ambition of environment defence and management. In **addition, the ministry has taken** a number of new initiatives to strengthen scientific research in the area of ecological science. Some of these include institution of the National Environmental Sciences Fellow Programme, Mahatma Gandhi Chair in Ecology and Environment, etc. The overall target of the Research and Development Scheme of the Ministry is to encourage basic and practical research in various aspects of ecology and environment.

10.6.2 National Resource Management System

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It is a part of an umbrella scheme of the Planning Commission that came into operation in 1985. The system's main aim is to apply remote sensing technology for the calculation and administration of the country's natural resources.

10.6.3 Forestry Research and Training

It is a nodal division for two major central sector plan schemes namely **Grants-in-Aid to Forestry and Wildlife Sector** and **Capacity Building in Forestry Sector**. This division looks after the administrative, monetary and other matters of three research and training institutions—Indira Gandhi National Forest Academy, Dehradun; Institute of Forest Management, Bhopal; and Indian Council of Forestry Research and Education, Dehradun.

10.6.4 Wildlife Research

Wildlife Institute of India is a leading training and research institution in the field of wildlife and protected area management in South Asia. It covers environmental, biological, socio-economic and managerial aspects of wildlife conservation.

Centres for Excellence: This **scheme was launched in 1983** to **promote institutions for Environmental Sciences and Management.** A few campaigns launched by the Centre for Environment Education in 2009:

- **Pick Right:** To spread consciousness about climate change, its causes and effects.
- **Kaun Banega Bharat ka Paryavaran Ambassador:** To choose a person to represent on environmental issues, who can encourage people to make the right changes in the standard of living.

10.7 ENVIRONMENTAL POLICIES OF INDIA

The government's environmental policy focuses on the following areas:

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1. Conservation of natural resources by straight action such as the declaration of Reserved Forests, Biosphere, Wetlands, Mangroves and defence of endangered species.
2. Check further deprivation of land and water through Wasteland Management and Restoration of river water quality programmes.
3. Monitoring progress through Environmental Impact Assessment and readings of major project proposals.
4. Penalties on industries which violate the **Pollution Control Act**.

10.8 CENTRAL POLLUTION CONTROL BOARD (CPCB)

- The CPCB of India is a statutory organisation **established under the Water (Prevention and Control of Pollution) Act, 1974.**
- It works under the MoEF.
- CPCB is also entrusted with the controls and functions under the Air (Prevention and Control of Pollution) Act, 1981.
- It serves as a field creation and delivers technical services to the MoEF under the provisions of the Environment (Protection) Act, 1986.
- It coordinates the accomplishments of the State Pollution Control Boards by providing technical assistance and supervision and also resolves disputes among them.
- It is the apex organisation in the country in the field of pollution control, as a technical wing of MoEF.
- The board is led by its chairman, who is nominated by the central government.
- The CPCB performs its various functions through the following major project budget heads:
  - Pollution estimation
  - R&D and laboratory governance
  - Formulating standards and guidelines for industry-particular emissions and effluent levels
  - Training
  - Management of library and information database technology and its implementation for pollution control
  - Mass awareness and publication
  - Managing hazard waste

■ *Statistics of Environmental Data*

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- CPCB administers statistics of ecological data consisting of air and water quality.
- CPCB monitors water quality by analysing river and pond water quality levels and air quality by measuring SO_2, NO_2, RSPM and SPM levels in the air.

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10.9 GREEN AWARDS

10.9.1 Rajiv Gandhi Environment Award

This award is given to manufacturing units that make a noteworthy contribution towards the growth of new, or the innovative alteration of existing, technologies or acceptance and use of clean technologies and practices that substantially decrease or prevent ecological pollution.

10.9.2 Indira Gandhi Paryavaran Puraskar (IGPP)

GPP is an environment award **instituted by the MoEF, Government of India, in 1987,** to **encourage public participation in environment**. It is awarded to individuals and organisations that have made significant and measurable contributions in the field of **environment protection and improvement.**

10.9.3 Amrita Devi Bishnoi Award

This award is given for important contributions in the field of **wildlife protection,** which is recognised as having shown exemplary bravery or having done exemplary work for the protection of wildlife.

Do You Know?

G.B. Pant Institute of Himalayan Environment and Development (GBPIHED), Almora, was established in August 1988 by the ministry as an autonomous institute with a command of attaining **sustainable development and environmental conservation in the Indian Himalayan Region.** The identified thematic categories for Institute R&D activities include (i) watershed processes and management; (ii) biodiversity conservation and management; (iii) environmental assessment and management; (iv)

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socio-economic development; (v) biotechnological application; and (vi) knowledge products and capacity building.

10.10 NATIONAL ACTION PLAN ON CLIMATE CHANGE (NAPCC)

The National Action Plan on Climate Change (NAPCC) was launched by the Government of India on June 30, 2008. The then Prime Minister Manmohan Singh released it to outline India's strategy for tackling climate change through sustainable development.

ENV 43- National Action Plan on Climate Change (NAPCC)

The NAPCC presents a national strategy that combines environmental concerns with development planning. It focuses on adapting to climate change and reducing its impacts. The Prime Minister's Council on Climate Change (PMCCC) oversees the plan, while the Ministry of Environment, Forest and Climate Change (MoEFCC) coordinates its efforts.

The plan has eight main components known as **'National Missions'**. These missions represent **long-term, integrated strategies aimed at achieving essential goals in climate action**.

The Eight Core Missions of the NAPCC

Each mission is managed by a specific ministry responsible for its execution and budget. The eight missions are:

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1. **National Solar Mission:** Supports solar energy technologies and boosts solar power generation.
2. **National Mission for Enhanced Energy Efficiency:** Aims to improve energy efficiency in large industries, urban areas, and through demand management.
3. **National Mission on Sustainable Habitat:** Promotes energy efficiency in buildings, urban waste management, and better urban planning.
4. **National Water Mission:** Seeks to conserve water, reduce waste, and ensure fair water distribution, particularly in response to climate change.
5. **National Mission for Sustaining the Himalayan Ecosystem:** Aims to protect the Himalayan ecosystem through research, monitoring, and conservation.
6. **National Mission for a Green India (GIM):** Strives to increase forest cover, restore degraded land, and improve ecosystem services.
7. **National Mission for Sustainable Agriculture:** Aims to make Indian agriculture more resilient to climate change through sustainable methods, better water use, and crop diversification.
8. **National Mission on Strategic Knowledge for Climate Change:** Focuses on building a strong knowledge base and research capabilities to support climate policy.

Exam Pointer

The **National Mission for a Green India (GIM)**, Launched in 2014, is **one of the eight core missions under the NAPCC**. Its main goal is to address climate change by increasing and enhancing the quality of India's forest cover through adaptation and mitigation measures. The mission goes beyond traditional tree planting. It takes a comprehensive, landscape-based approach that encourages community participation and aims to improve multiple ecosystem services like biodiversity, water regulation, and soil conservation, alongside carbon absorption.

The Green India Mission has several main objectives

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- Expand forest and tree cover by 5 million hectares (mha) on forest and non-forest lands.
- Improve the quality of existing forest cover by an additional 5 mha.
- Enhance ecosystem services such as carbon absorption and biodiversity across a total of 10 mha.
- Improve forest-based livelihoods for around 3 million households.
- Increase annual CO_2 absorption.

Key Focus Areas and Implementation

GIM uses a community-centered approach to restore various ecosystems. Key areas of focus include:

- Restoring degraded lands like forests, grasslands, wetlands, and abandoned mining sites.
- Encouraging urban and agro-forestry.
- Highlighting decentralized governance with Gram Sabhas (village councils) in planning and monitoring, along with developing 'community foresters.'
- Coordinating with other government schemes like the Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS) and the Compensatory Afforestation Fund Management and Planning Authority (CAMPA).
- Focusing on vulnerable ecosystems such as the Aravalli ranges, Western Ghats, and the Himalayas, as outlined in a revised roadmap.

Do You Know?

Salim Ali Centre for Ornithology and Natural History (SACON)

The MoEF established the Salim Ali Centre for Ornithology and Natural History. It is a public–NGO partnership between the MoEF and BNHS (the Bombay Natural History Society). The Salim Ali Centre for Ornithology and Natural History (SACON) has been registered in 1990 under the Societies Registration Act. It aims to establish and develop

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a Centre of Excellence to assist, institute, conduct and promote scientific research in ornithology, and of species, habitats and ecosystems with and within which avifauna coexist, and develop scientific solutions to species, habitat and landscape conservation problems which are concerned about socio-economic realities and involve people's aspirations. The minister for MoEF presides over the SACON Society. Since 1992, SACON has undertaken 174 Research and Environment Assessment Projects whose financial value sums approximately 12 crores.

10.11 RIVER PROJECTS IN INDIA

ENV 44- National Ganga River Basin Authority (NGRBA)

10.11.1 National Ganga River Basin Authority (NGRBA)

NGRBA is a planning, sponsoring, implementing, monitoring and handling authority for the River Ganga. It was established on 20 February 2009 by the central government under Section 3(3) of the Environment Protection Act, 1986, which also specified the Ganges as the **National River** of India.

Previously it worked under the Ministry of Environment, but from July 2014, it shifted to the Ministry of Water Resources, River Development and Ganga Rejuvenation.

The Prime Minister is the chairman of the authority.

It works for maintaining the water quality of the River Ganga up to adequate standards. The pollution mitigation activities are conducted through the existing implementation mechanisms in the state and also through special-purpose vehicles at the pollution hotspots.

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In 2016, the central government issued a notice under River Ganga (Rejuvenation, Protection and Management) Authorities Order 2016 to form a new body named National Council for River Ganga (Rejuvenation, Protection and Management), (NCRG) to replace the existing NGRBA.

Exam Pointer

Ganga Action Plan (GAP)

The Ganga Action Plan (GAP) was the first major government-led river cleanup effort in India. It started on January 14, 1986, under then-Prime Minister Rajiv Gandhi. The plan aimed to improve the water quality of the Ganga River by intercepting, diverting, and treating domestic sewage and industrial waste before it entered the river.

The main goals of the Ganga Action Plan included: **Pollution reduction, sewage treatment, industrial waste control, non-point source control, and biodiversity conservation**.

The GAP made a significant impact on some sections of the river and helped raise public awareness about river pollution. However, it ultimately did not meet its long-term goals due to weak infrastructure, poor maintenance, a lack of public involvement, and insufficient enforcement of regulations. Recognizing these issues, the government of India launched the more comprehensive **Namami Gange Programme in 2014 to continue cleaning and revitalizing the Ganga basin**.

Namami Gange (Integrated Ganga Conservation Mission/Programme under National Ganga River Basin Authority): Namami Gange Programme, an Integrated Conservation Mission, is a **Flagship Programme initiated by the Union Government in June 2014 with a budget allocation of ₹20,000 crores** to achieve its twin objectives to effectively mitigate pollution and conserve and rejuvenate our National River Ganga.

The main aim of the programme is mass involvement, encouraging people living on the banks of the river to come forward, help clean the Ganga and finally achieve sustainable results. Involving the States and grassroots level institutions such as Urban Local Bodies and Panchayati Raj

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Institutions in its implementation can bring out great results. It is being implemented by the **National Mission for Clean Ganga**, and **State Programme Management Groups**.

Steps that were considered to be taken for controlling pollution are the efficient treatment of wastewater through bioremediation before releasing into the river, in-situ treatment wherever necessary, using innovative technologies, encouraging people for afforestation, setting up of sewage treatment plants (STPs) along with effluent treatment plant. Immediate short-term measures for restricting pollution at exit points to stop the inflow of sewage into the river, etc.

■ *Key Achievements*

1. **Setting up STPs:** A total of 63 sewage management projects were implemented in the states of Uttarakhand, Uttar Pradesh, Bihar, Jharkhand and West Bengal.
2. **Coming up with River-Front Development:** A total of 28 river-front development projects, including 33 entry-level projects for building, modernising and renovating 182 ghats, 118 crematoria, have already been initiated.
3. **Cleaning River Surface:** To collect floating solid waste from the surface and dispose of them at proper places.
4. **Biodiversity Conservation:** Several biodiversity conservation projects under the Namami Gange Programme for restoring and rejuvenating marine life in Ganga have been undertaken. The **Biodiversity Conservation** and **Ganga Rejuvenation** projects have been implemented by the Wildlife Institute of India, Fish and Fishery Conservation in Ganga River by the Central Inland Fisheries Research Institute and the Ganges River Dolphin Conservation Education Programme by the Centre for Environment Education.
5. **Afforestation:** A Detailed Project Report (DPR) has been decided to be prepared by the Forest Research Institute for Forestry Interventions for Ganga, over a period of five years (2016–2021). The aim is **to contribute towards a clean Ganga, particularly Anant Van and Aviral Dhara besides the maintenance of ecological and geological integrities of the River scape**. According to the DPR, a total of 134,105.8 ha of land will be treated, which includes Uttarakhand, Uttar Pradesh, Bihar, Jharkhand and West Bengal, that lies at the river bank.
6. **Public Awareness:** Various events, workshops, seminars, conferences, rallies, campaigns, exhibitions, shramdaan, cleanliness drives, competitions, plantation drives and numerous IEC activities were organised to promote community participation. Gange Theme song was released widely, which was also played on digital media.
7. **Monitoring Industrial Effluent:** Real-time **Effluent Monitoring Stations** have been established in a total of 572 Grossly Polluting Industries (GPIs) out of 760. 135 GPIs so

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far have been notified of their closure while others have been given deadlines.

8. **Ganga Gram: The** Ministry of Drinking Water and Sanitation has identified 1,674 Gram Panchayats that are located on the river bank in five states (Uttarakhand, Uttar Pradesh, Bihar, Jharkhand and West Bengal). UNDP acted as an executing agency for the rural sanitation programme and played a major role in developing Jharkhand as a model state.

Arth Ganga

- A **concept** espoused by the Prime Minister during the **National Ganga Council meeting in Kanpur in 2019**.
- Focuses on **creating economic livelihood opportunities to sustain the activities under the Namami Gange Programme**, the flagship programme of the government to clean the Ganga and its tributaries.
  Aim: To contribute about **3% to the GDP from the Ganga** Basin.
- Expected to **generate economic benefit of more than ₹1,000 crores over the next 5 years**.

District Ganga Committees (DGCs)

- Minister for Jal Shakti launched **Digital Dashboard for DGCs Performance Monitoring System.**
- **DGCs:** Constituted in districts on Ganga River basin to ensure **people's participation** in management and pollution abatement in Ganga and its tributaries.
  DGCs are **mandated to:**
- Ensure proper **utilisation of assets created under Namami Gange**
- **Monitor drains/sewage** falling in Ganga and its tributaries
- **Create strong connect of people** with Ganga rejuvenation
- **District Collector is the Chairperson** of DGC

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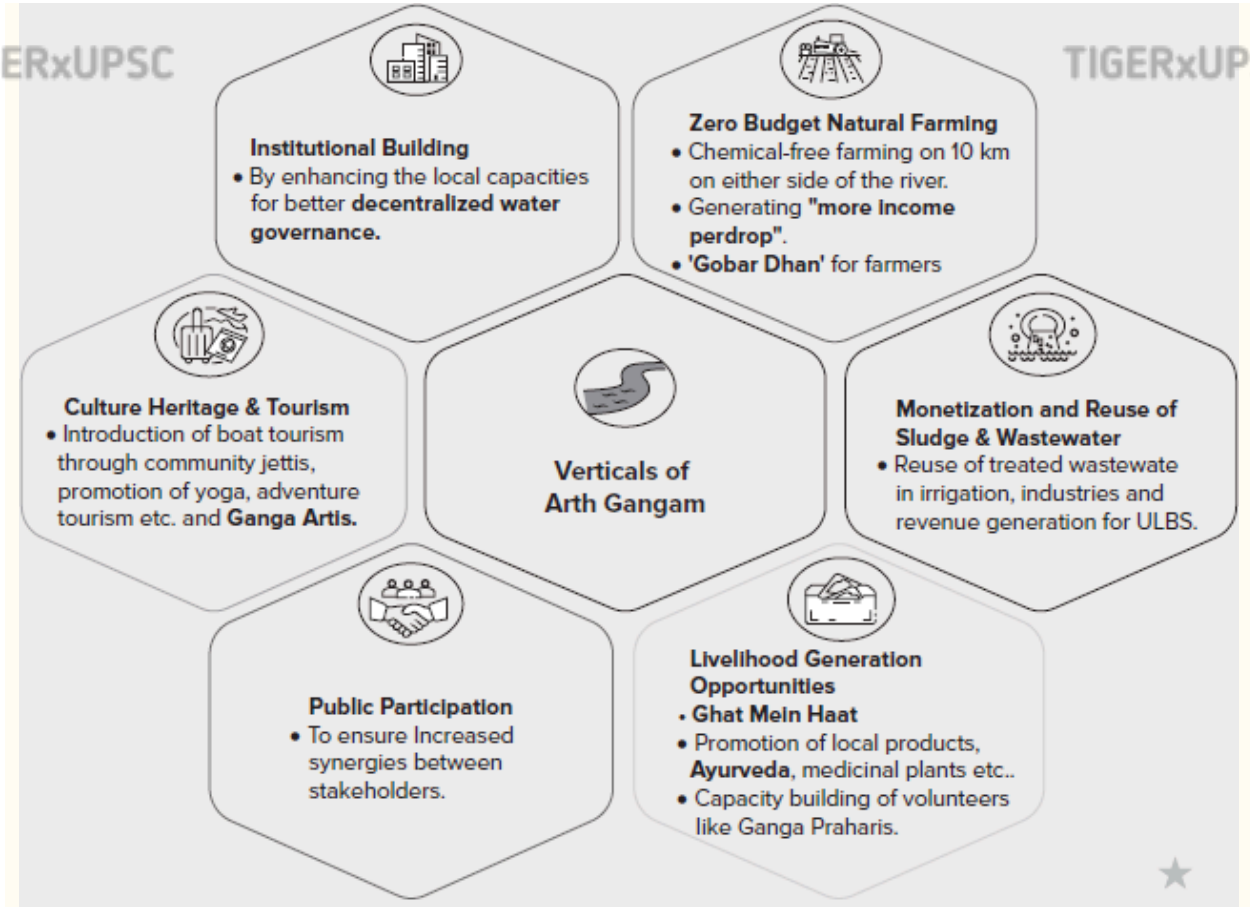

Figure 1 Verticals of Arth Ganga

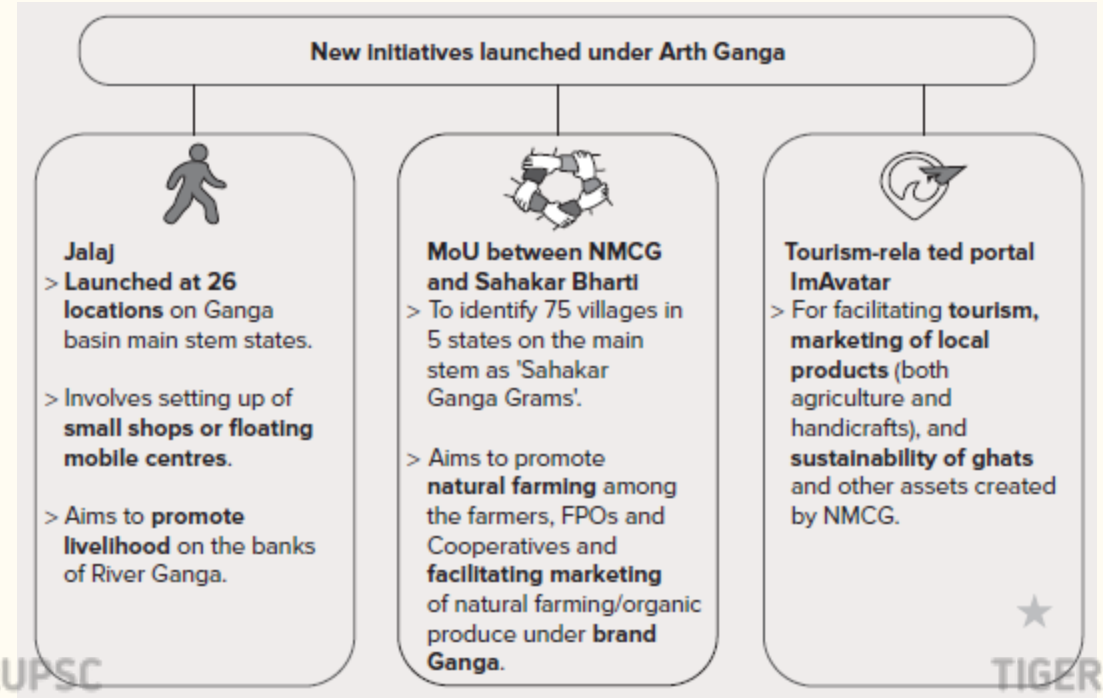

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Figure 2 New initiatives launched under Arth Ganga

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10.11.2 Yamuna Action Plan (YAP)

■ *Revitalisation Action Plan*

The **Yamuna Action Plan (YAP)** is a multi-phase project that started in 1993 to reduce serious pollution in the Yamuna River. This initiative is a partnership between the Governments of India and Japan and it focuses mainly on increasing sewage treatment capacity in towns along the river.

■ *Objectives and Key Features*

The main goal of the YAP is to improve the river's water quality by tackling major pollution sources which include domestic and industrial wastewater, agricultural runoff, and solid waste.

Key features include

- **Sewage Treatment Infrastructure:** A key part has been building and upgrading numerous Sewage Treatment Plants (STPs) and sewer networks in cities and towns along the Yamuna.
- **Waste Management:** Efforts also include better solid waste management to stop garbage and plastics from being dumped into the river.
- **Public Awareness:** The plan has outreach programs to encourage community participation and raise awareness about keeping the river clean.
- **Industrial Waste Control:** The Central Pollution Control Board (CPCB) inspects industrial units regularly to enforce rules and prevent untreated waste from being discharged.
- **Water Quality Monitoring:** The plan involves tracking key water quality indicators like Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), and Fecal Coliform (FC) levels to measure progress.

■ *Phases of the Yamuna Action Plan*

The YAP has been carried out in several phases with funding mainly from the Japan International Cooperation Agency (JICA).

- **YAP Phase I (1993-2003):** This phase covered 15 towns in Uttar Pradesh, Haryana, and Delhi, focusing on building 29 STPs. While it had some successes, the overall river

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quality in Delhi remained a major concern due to the gap between sewage generation and treatment capacity.

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- **YAP Phase II (2004-onwards):** This phase focused more on the critically polluted 22-km stretch of the river in Delhi.
- **YAP Phase III (Under Execution):** Approved in 2011, this phase aims to rehabilitate and upgrade existing STPs and trunk sewers, especially in Delhi's main drainage areas (Kondli, Rithala, and Okhla).

■ *Current Status and Challenges*

Despite significant investment and efforts over the past thirty years, the Yamuna River remains heavily polluted, particularly the segment in Delhi.

Key challenges include

- A persistent gap between sewage generation from growing urban populations and the available treatment capacity.
- Insufficient freshwater flow, especially downstream of the Wazirabad barrage, limiting the river's natural ability to dilute pollution.
- Operational inefficiencies in existing treatment plants.

In response, governments continue to launch new initiatives. For instance, the Delhi government announced a detailed 45-point action plan in 2025 aimed at cleaning the river within two years, focusing on specific goals and coordinated efforts among different agencies.

10.12 GREEN INSTITUTES/INITIATIVES

10.12.1 Ecomark Scheme

Ecomark is a **certification mark issued by the Bureau of Indian Standards** (the National Standards Organisation of India) to **products conforming to a set of standards aimed at the least impact on the ecosystem**. The marking **scheme was started in 1991**.

10.12.2 Haryali Yojana

It means to include village communities in the **application of watershed projects under all the area development programmes**, namely, the

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Integrated Wasteland Development Programme, Drought Prone Area Programme and Desert Development Programme. The **programme involves the Panchayat Raj Institutions more meaningfully in the preparation, application and supervision of financial development activities in rural areas.**

10.12.3 National Bureau of Plant Genetic Resources (NBPGR)

It was established in 1976 as the nodal **agency at the national level to manage plant genetic resources for food and agriculture and carry out related research and human resource development**.

10.12.4 National Botanical Research Institute (NBRI)

Engaged in the field of taxonomy and modern biology, it is a CSIR research institute situated in Lucknow.

10.12.5 Institute of Forest Genetics and Tree Breeding (IFGTB), Coimbatore, Tamil Nadu

It is a forestry research institute situated in Coimbatore district of Tamil Nadu. It works under the Indian Council of Forestry Research and Education (ICFRE) of the Ministry of Environment, Forest and Climate Change, Government of India. IFGTB works to recognise and evolve diversities of species used in afforestation and societal forestry that will contribute to the national aim of achieving a growth of 3 to 4 cubic metres of biomass per ha per year, within the environmental considerations applicable to the area.

10.12.6 Tropical Forest Research Institute (TFRI), Jabalpur, MP

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It works under ICFRE, Ministry of Environment, Forest and Climate Change, Government of India.

10.12.7 Indian Institute of Forest Management, Bhopal, MP

It is an escorial management institute which constantly endeavours to evolve information useful for the managers in the area of forest, environment and natural resources management and allied sectors.

10.12.8 Wildlife Institute of India (WII), Dehradun, Uttarakhand

The prime concern of the institute is to build capability and grow skills for providing solutions to problems of wildlife conservation in general and protected area management in specific.

10.12.9 Wildlife Trust of India (WTI), Noida, UP

It is a non-profit conservation organisation committed to urgent action to avoid damage to India's wildlife.

10.12.10 Indian Environmental Society (IES), New Delhi

It has been set up to upsurge consciousness and knowledge about the environment, including the major ecological problems fronting the world today.

10.12.11 World Wildlife Fund (WWF), India

Its mission is to promote nature conservation and environmental protection as a basis for sustainable and equitable development.

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10.12.12 Environment Protection Training and Research Institute (EPTRI), Hyderabad, Telangana

The main objective of EPTRI is to provide training, consultancy, applied research services and advocacy in the area of environmental protection to industry, regulatory bodies, government and NGOs.

10.12.13 National Environmental Engineering Research Institute (NEERI), Nagpur, Maharashtra

The mission of the NEERI is to bestow itself to the service of manhood by giving innovative and effective solutions to environmental and natural resource problems. It strives to allow individuals and organisations to attain productive and sustainable use of natural resources on which life and human activity depend.

10.13 COMPENSATORY AFFORESTATION FUND MANAGEMENT AND PLANNING AUTHORITY (CAMPA)

CAMPA **promotes afforestation and regeneration activities to make up for forest land that is diverted to non-forest uses** or for purposes other than reforestation. According to the **Forest (Conservation) Act 1980**, when forest land is to be used for non-forestry purposes, an equivalent area of non-forest land must be identified for Compensatory Afforestation (CA), and funds must be allocated for CA.

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ENV 45- Compensatory Afforestation Fund Management and Planning Authority (CAMPA)

The Central Government passed the **Compensatory Afforestation Fund Act 2016** to provide **detailed guidelines and activities for using the CAMPA Fund. CAMPA funds are used to offset the loss of forest land and ecosystem services by creating compensatory afforestation (CA)**, enhancing the quality of forests through assisted natural regeneration, boosting biodiversity, improving wildlife habitats, controlling forest fires, protecting forests, and implementing soil and water conservation measures.

CAMPA activities involve local communities in afforestation, nursery development, soil and water conservation, and forest protection. A **National Compensatory Afforestation Fund** exists under the **Public Account of India**, and each state has its own **State Compensatory Afforestation Fund**.

These funds will receive payments for

1. **Compensatory Afforestation**
2. **Net Present Value (NPV) of Forest**
3. **Other project-specific payments**

The **NPV** will be **determined by an expert committee** formed by the central government. This committee will assess both the material and non-material benefits of forests. NPV reflects the ecological cost of forests.

Do You Know?

The C&AG audits the Annual Accounts of the National Authority in general. The C&AG conducts the audit of each state's CAMPA through the respective Accountant General's Office.

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Salient Features

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- The Act aims to **improve efficiency and transparency in managing CAMPA funds**.
- It seeks to **establish compensatory afforestation fund management and planning authorities** to manage funds at national and state levels.
- The **sharing ratio between the centre and states will be 10%**, i.e., the National Fund will receive 10% of fund collected and the remaining 90% will go to the respective state fund.

The National CAMPA Advisory Council will have the following roles

- Set general guidelines for state CAMPA.
- Provide scientific, technological, and other assistance needed by state CAMPA.
- Recommend actions to state CAMPA based on plan reviews.
- Offer a way for state CAMPA to address inter-state or center-state issues.

The fund can be used for

- Artificial regeneration
- Assisted natural regeneration
- Forest management
- Forest protection
- Infrastructure development
- Wildlife protection and management
- Supply of goods and other forest products
- Conservation devices

Other benefits of CAMPA

This policy will **enhance forest resource management and create direct jobs in tribal and backward areas of the country**. It will also **increase the availability of timber and various non-timber forest products, improving the living standards of forest-dependent communities**. This progressive CAMPA policy will be vital for **achieving national goals related to INDCs, REDD+, and UN Sustainable Development Goals**.

Challenges

- **Finding non-forest land for afforestation** is a significant challenge.
- The law states that selected **land should ideally be close to existing forests to make management easier**.

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- **In case of non-availability of land** near the forest being diverted, **land in any other part of the state can be used**.
- If suitable non-forest land cannot be found, **degraded forests may be chosen for afforestation, but in such cases, the area designated for afforestation must be twice that of the diverted forest**.
- It's **challenging to assess the net ecological loss**.
- Compensating for the loss of natural species **with non-native species in artificial plantations threatens existing ecosystems**.
- Biodiversity loss cannot be measured in monetary terms.
- The policy **does not adequately address** the Forest Rights Act (FRA) 2006, which emphasizes the role of the Gram Sabha; under CAMPA, community participation is not required in compensatory afforestation programs.
- This **neglects the rights of tribal and forest dwellers**.
- The **Kanchan Chopra Committee and the Indian Institute of Forest Management Committee on NPV recommend sharing compensation with forest-dwelling communities**.
- The Kanchan Chopra Committee **also suggests revising the NPV every five years**.

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Environmental Laws and Acts in India

11

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11.14.1 Waste Management Rules

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11.1 THE WILDLIFE (PROTECTION) ACT, 1972

It is a statutory Act targeted at the protection of plants, birds and animal species. [Statutory law or statute law is a written rule set down by a body of legislature or by a singular legislator (in the situation of an absolute monarchy).]

Salient features of the Act

- It covers the whole of India.
- It has six schedules, which give varying degrees of protection.
- The Act provides for the protection of wild animals, birds and plants; and for matters connected with or incidental to.
- National Park, Wildlife Sanctuaries, Tiger Reserves and Conservation Reserves are the protected areas registered under this Act.
- Its provisions paved the way for the formation of the Central Zoo Authority.
- The Act created six schedules that gave varying degrees of protection to classes of flora and fauna.
- The National Board for Wildlife was constituted as a statutory organisation under the provisions of this Act.
- The Act also provided for the establishment of the National Tiger Conservation Authority.
- It gives statutory authority to Project Tiger.

ENV 46- The Wildlife (Protection) Act, 1972

It initially had six schedules, which were reduced to four schedules after the coming into force of the Amendment Act 2022 w.e.f. 1st April 2023, which gives varying degrees of protection. It reduces the number of schedules from six to four, eliminating Schedule V (Vermin) and merging some content into Schedule II.

Exam Pointer

OLD Structure	NEW structure

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Schedule I and part II of Schedule II provide absolute protection – offences under these are prescribed the highest penalties. Species listed in **Schedule III and Schedule IV** are also protected, but the penalties are much lower. **Schedule V** includes the animals, which may be hunted. The plants in **Schedule VI** are prohibited from cultivation and planting.	**Schedule I** (Highest protection) **Schedule II** (Lower protection for animals) **Schedule III** (Protected plant species) **Schedule IV** (CITES-listed specimens)

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Do You Know?

The original Schedule VI, which regulated the cultivation of restricted plants, was removed, and now the new Schedule III and IV are used instead.

11.1.1 Schedules under the Wildlife Protection Act 1972

Schedule I

- This Schedule covers endangered species need to be protected.
- High penalties for violation of the law are under this Schedule.
- Species under this Schedule are prohibited to be hunted throughout India, except under threat to human life.
- Absolute protection is accorded to species on the list.
- The trade of these animals is prohibited.
- ***Examples*** – Bengal Tiger, Red Panda, Great Indian Bustard, Blackbuck, Himalayan Brown Bear, Brow-Antlered Dear, Blue whale, Common Dolphin, Cheetah, Clouded Leopard, Hornbills, Indian Gazelle etc.

Original Schedule	New Schedule	What happened
Schedule I: *Endangered species*	***Schedule I***: *Endangered species with the highest level of protection.*	*Remains, with a new structure.*
Schedule II: *Protected animals*	***Schedule II***: *Animals with a lower level of protection.*	*Combined with content from the original Schedule II.*
Schedule III: *Protected animals with lower penalties*	***Schedule III***: *Protected plant species.*	*Content has been moved and the schedule redefined.*
Schedule IV: Protected animals with lower penalties	***Schedule IV:*** Species listed under CITES.	Content has been moved, and the schedule redefined, with CITES-listed species added.
Schedule V: Vermin	***Schedule V:*** (Vermin)	Eliminated entirely.

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Original Schedule	New Schedule	What happened
Schedule VI: Restricted plants	***Schedule III:*** Restricted plants.	Merged into the new Schedule III, along with the content of the new Schedules II and IV.

Exam Pointer

Some of the animals granted protection under the Schedule I include:

The Black Buck, Clouded Leopard, Snow Leopard, Swamp Deer, Himalayan Bear, Asiatic Cheetah, Kashmiri Stag, Fishing Cat, Lion-tailed Macaque, Musk Deer, Rhinoceros, Brow Antlered Deer, Chinkara (Indian Gazelle), Capped Langur, Golden Langur, Hoolock Gibbon.

Schedule II

- Animals under this list are also accorded high protection.
- Their trade is prohibited.
- They cannot be hunted except under threat to human life.
- ***Examples*** – Kohinoor (insect), Assamese Macaque, Bengal Hanuman Langur, Large Indian Civet, Indian Fox, Larger Kashmir Flying Squirrel, Kashmir Fox etc

Exam Pointer

Some of the animals listed under Schedule II include:

Pig Tailed, Assamese Macaque, Stump Tailed Macaque, Himalayan Black Bear, Himalayan Newt/Salamander, Jackal, Flying Squirrel, Giant Squirrel, Sperm Whale, Indian Cobra, King Cobra.

Do You Know?

Schedule I and Schedule II deliver absolute protection. Offences committed under these are considered for highest penalties. Schedule II includes a lesser degree of protection than Schedule I.

Schedule III and Schedule IV

Species registered in **Schedule III** and **Schedule IV** are also protected, but the penalties are much lesser.

- This list is for species that are **not endangered**.
- This **includes protected species, but the penalty for any violation is a less** compared to the first two schedules.
- **Schedule III:** Protected plant species
- **Schedule IV**: Species listed under CITES

Exam Pointer

Animals protected under Schedule III include: Chital (spotted deer), Bharal (blue sheep), Hyena, Nilgai, Sambhar (deer), Sponges.
Animals protected under Schedule IV include: Flamingo, Hares, Falcons, Kingfishers, Magpie, Horseshoes Crabs.

Schedule V

Schedule V (now repealed) contains the animals which may be hunted.

Exam Pointer

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It includes **only four species of wild animals**: *Common Crows, Fruit Bats, Rats, Mice*.

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Schedule VI

The plants in **Schedule VI** (now repealed) are prohibited from cultivation and planting. For hunting, the Enforcement authorities have the power to compound offences under this Schedule (i.e., to impose penalties on offenders).

- This **list contains plants that are forbidden (not allowed) from cultivation**.
- Examples, pitcher plant, blue vanda, red vanda, kuth etc.
- **Schedule VI of the Act lists all 6 plants of Indian origin included in CITES appendices. Beddomes cycad, Blue Vanda, Kuth, Ladies slipper orchids, Pitcher plant, Red Vanda.**

WPA, 1972 is against taxidermy (preservation of dead wild animals as a trophy or in the form of skins, antlers, horns, eggs, teeth or nails).

11.1.2 Wild life Protection Act 1972 - Several Amendments

- The 2002 Amendment Act, which came into force in January 2003, has made punishment and penalty for offences under the Act more stringent.
- WPA amendment 2006 provided for the National Tiger Conservation Authority and Wildlife Crime Control Bureau.
- WPA amendment 2013 provided the provision of imprisonment up to 7 years and protects hunting rights of the Scheduled Tribes in the Andaman and Nicobar Islands.

Recently, the Wild Life (Protection) Amendment Act, 2022 came into force from 1st April 2023, which seeks to give effect to India's obligations under the Convention on International Trade in Endangered Species of Wild Fauna and Flora ('CITES').

■ *Changes in the Wildlife (Protection) Amendment Act (referred to as the Amendment Act) 2022*

1. **Changes in the Preamble**: Instead of words, protection of wild animals, birds and plants', the words '**conservation, protection and management of wildlife**' were substituted in the Preamble.
2. **Reduction in the number of schedules**: The Amendment Act reduces the **number of schedules from six to four**.
3. **Implementation of CITES: The addition of Chapter VB** provides regulation of international wildlife trade as per **provisions laid by CITES**. The Central Government shall form a Managing Authority and a Scientific Authority to regulate the import and export of specimens of species.
4. **Management of Wildlife Sanctuaries: Under Section 33B**, the Amendment Act empowers State governments to constitute an Advisory Committee comprising the Chief Wildlife Warden, members of the legislature, wildlife NGOs, and representatives from Panchayati Raj institutions to manage wildlife sanctuaries. Together with the Gram Sabha, this Committee is responsible for preparing management plans for wildlife sanctuaries and conservation reserves. This marks the first step toward shifting the decision-making process to a more people-centric approach.
5. **Criteria to declare animals 'Vermin':** The Principal Act (Under Schedule V) and the amendment both use the term 'Vermin' to describe species which can be hunted under certain conditions, to keep their numbers in check. For example, Mice, rats, crows.
6. **References to Forest Rights Act, 2006 (FRA):** According to **Clause 13, 15 and 17,** which seek to amend Sections **33, 35 and 38** of the principal Act, the management of Sanctuaries should be done according to the provisions of the FRA wherever applicable.

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The Amendment Act empowers the Gram Sabha and other local village **institutions to collectively protect** the forests, wild animals and biodiversity and take action against any activity posing a threat to wildlife.

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7. **Permission to carry out subsistence fishing in and around Protected Areas:** Under Sections 29 and 33, the Amendment Act allows low-intensity, small-scale fishing in rivers and other water bodies in and around protected areas. This **lifts the blanket ban,** which prohibits all types of fishing activity in and around Protected Areas, which had caused many people to **lose their livelihoods**.
8. **Invasive species:** The **amendment to Section 62A** of the main Act acknowledges the threat to ecosystems from invasive alien species. It also allows for the creation of a Scientific Authority, as per CITES, to provide advice on matters concerning the trade of these species. The amendment gives the Central Government the authority to regulate or ban the import, trade, possession, or spread of such species.
9. **Use of Protected Areas:** The amendment to **Section 29** eases the rules regarding cattle grazing and water use by local communities inside protected areas until they are relocated.
10. **Trade of Live Elephants:** The amendment introduces a **new sub-section (4) under Section 43**. It states that **transferring or transporting a captive elephant for religious or other purposes by someone with a valid ownership certificate will follow the terms and conditions set by the Central Government**. This change could allow for the legalisation of elephant trade in India, which raises concerns about the terrible conditions faced by the country's captive elephants.
11. **Ownership of captive animals and trophies:** According to **Section 42A**, anyone who owns a captive animal or trophy can voluntarily give it to the government without expecting compensation. These live animals will be cared for by rescue centres if they cannot be released into the wild.
12. **Prohibition of renewal of firearm licenses:** As per the amendment to Section 34, no renewal of any firearms license under the Arms Act, 1959, will be granted to anyone living within ten kilometers of a sanctuary unless the Chief Wildlife Warden or an authorized officer is informed.
13. **Increased penalties:** The Amendment Act also raises the penalties for breaking the rules of the Act. For general violations, the maximum fine increases from **₹25,000 to ₹1 lakh**. For specially protected animals, the minimum fine rises from **₹10,000 to ₹25,000**.
14. **Change in the definition of 'Zoos':** Zoos will now serve as areas for ex-situ conservation, rescue centres, and breeding centres, in addition to their role in displaying animals.

11.2 THE WATER (PREVENTION AND CONTROL OF POLLUTION) ACT, 1974

The Act aims to stop and control water pollution and to maintain/restore the freshness of water by establishing central and state pollution control boards to monitor and enforce the regulations.

The Central Government shall appoint and constitute a central board to be called the Central Pollution Control Board to exercise the powers conferred on and perform the functions assigned to that board under this Act.

The central board shall consist of the following members, namely:

- A full-time chairman, being a person having special knowledge or practical experience in respect of environmental protection, to be nominated by the Central Government.
- A maximum of five such officials are nominated by the Central Government to represent that government.
- Such many persons, not exceeding five, to be nominated by the Central Government, from amongst the members of the state boards.
- Not more than three Non-officials, to be nominated by the Central Government, for representing the agriculture, fishery, industry, trade or any other field which, in the government's opinion, has to be represented.
- Nomination of two persons from the Central Government-owned, controlled or managed companies or corporations.
- Central Government to appoint full-time member-secretary, having scientific, engineering and pollution control management aspects experience, qualifications and knowledge.

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■ *Objectives of this Act*

Join Channel

- To avoid and control water pollution.
- To maintain or restore the wholesomeness of water.
- To establish boards for the inhibition and control of water pollution.
- To confer on and allocate to the boards, the power and purposes relating to the Act.
- Establishment of central and state boards for pollution control.
- Delivery of joint boards for two or more states.
- Exclusion of the use of streams and wells for the disposal of pollutants.
- Consent of pollution control board to open fresh outlets and discharges into streams and wells.
- Provisions for funds, budgets accounts and audits of the central and state pollution control boards.
- The Act defines various penalties for the defaulters and procedures for the same, the above stated.

11.3 THE FOREST CONSERVATION ACT, 1980

It is for the conservation of forests to keep a check on the diversion of forestland for non-forest purposes and for matters connected there. The Act extends to the whole of India.

Salient Features of the Act

- Approval from the Central Government (restrictions on the power of the state government) is mandatory before any reserved forest is declared as de-reserved or if forestland is diverted to non-forest purposes.
- No forestland may be cleared of trees, which have grown naturally in that land for reforestation without prior approval of the Central Government.
- No state government or any other authority may direct any forest land or any portion may be assigned by way of lease or any other method to any corporation, organisation or agency without approval of the Central Government.
- Admissible punishment to the offender of the provision of Section 2 of the Forest Conservation Act.

Amendment

(a) Regional chief conservator of the forest has the power to choose a proposal involving forest land up to 5 hectares.

(b) Regional chief conservator in discussion with state advisory group containing the representative of the concerned state govt. has the power to approve land between 5 and 20 hectares.

(c) More than 20 hectares of forest land is vital to be placed before the advisory committee constituted by the MOEF.

- In 1992, some amendment was made in the Act which made provisions for allowing some non-forest activities in forests, without cutting trees or limited cutting with prior approval of Central Government.

11.4 THE AIR (PREVENTION AND CONTROL OF POLLUTION) ACT, 1981

ENV 47- The Air Prevention and Control of Pollution Act, 1981

The Air (Prevention and Control of Pollution) Act, 1981, was passed by the Central Government to stop the deterioration of air quality.

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Objectives of the Act

- Avoidance, control and reduction of air pollution
- Preserving the quality of air
- Establishing boards for avoidance and control of air pollution

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Salient Features of Act

The **Air Act, 1981** was enacted to implement decisions of the **1972 Stockholm Conference**. Its purpose is to **prevent, control and reduce air pollution** and maintain air quality across India.

Key Provisions of the Act

1. **Institutional Structure:**
   Central Pollution Control Board (CPCB): It coordinates pollution control and sets national standards.
   State Pollution Control Boards (SPCBs): It grant consent, monitor industries and enforce state-level norms.
2. **Declaring Air Pollution Control Areas:** States can **declare any area as "Air Pollution Control Area."**Industrial units in these areas must take special permissions.
3. **Standards & Enforcement:** CPCB sets **National Ambient Air Quality Standards (NAAQS)** and **States implement** emission standards for: Industries, Vehicles & Power plants
4. **Penalties:** In the case of violation, have imprisonment up to **6 years a fine or both.** Continuous violations lead to additional daily penalties
5. **1987 Amendment -** Strengthened enforcement: Under this amendment, SPCBs are empowered to **close industries & cut power/water**. It introduced **citizen suits** (any person may complain) and Added **Air Quality Monitoring Network**

QUICK INSIGHT

Different Procedures adopted to solve the Air Pollution Problem

1980 – 1990's

- **Environment (Protection) Act, 1986** - Enabled stricter air emission rules.
- **National Ambient Air Quality Standards** — 1982 & 1994 versions
- **Introduction of Lead-Free Petrol** (1990–2000)

2000 – 2010

- **CNG Conversion (Delhi, 2001–04)** Supreme Court mandated CNG for **a**utos**, t**axis**, b**uses, drastically reduced SO_2 and PM in Delhi.
- **National Auto Fuel Policy (2003) -** Timeline for Bharat Stage (BS) emission standards.
- **Bharat Stage Emission Norms (2000 onwards)** BS-I → BS-IV across India by 2017.

2010 – 2020 (Strong Regulatory Reforms)

- **National Ambient Air Quality Standards (NAAQS 2009–10 Updated) -** It added PM2.5 standards for the first time and aligned with WHO guidelines (partial)
- **National Air Quality Monitoring Programme (NAMP) -** 804 stations expanded to over **900+ stations**
- **Air Quality Index (AQI) – 2014,** It launched under Swachh Bharat **and provides** real-time data for 400+ cities
- **Graded Response Action Plan (GRAP), 2016 -** Action escalation based on AQI levels (Moderate → Severe+)

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- **Bharat Stage VI (BS-VI) Fuel, 2020-** Direct jump from BS-IV to BS-VI **&** 80% reduction in NOx from vehicles

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2020 – Present(National, Legal, and Technological Measures)

- **National Clean Air Programme (NCAP), 2019**
- **Commission for Air Quality Management (CAQM) Act, 2021**
- **Scrapping Policy (2021–22) -** Phased removal of old vehicles**,** Automated testing stations
- **Electric Mobility Initiatives - FAME-I (2015), FAME-II (2019), FAME-III (proposed),** State EV policies launched (Delhi leads), **Biomass Management (Stubble Burning Control) -** Happy seeder, super seeder incentives**,** PUSA decomposer**,** CAQM Action Plan 2023–24.
- Bio-decomposer capsule: Developed by the Indian Agricultural Research Institute from microbial consortium (seven **fungal strains) in capsule form. It** accelerates the decomposition process of stubble by turning it into manure over a period of 15–20 days, since 2020. Farmers mix the capsules with jaggery and gram flour to create a liquid solution, which is then sprayed on crop stubble.

11.4.1 National Ambient Air Quality Standard (NAAQS)

Prior to **November 2009 values**, India had set air quality standards on 11 April 1994, and this was later revised on 14 October 1998. The 2009 values further lowered the maximum allowable limits for pollutants and made the standards even across the nation. Earlier, less severe standards were given for industrial zones as compared to residential areas.

Ambient air quality refers to the condition or quality of air around us. National Ambient Air Quality Standards are the values for ambient air quality set by the **CPCB** that is relevant nationwide. The CPCB has discussed this power with the Air (Prevention and Control of Pollution) Act, 1981.

The current NAAQSs were notified on 18 November 2009. The 12 pollutants considered under the NAAQS: SO_2, NO_2, Ozone (O_3), PM10, PM 2.5, Lead, CO, Arsenic, Nickel, Benzene, Ammonia and Benzopyrene.

Do You Know?

The government will also seek an additional support system of implementation from the **National Environment Protection Authority** and the **National Green Tribunal** to safeguard effective application of standards.

11.4.2 National Air Quality Monitoring Programme (NAMP)

The CPCB is performing a nationwide programme of ambient air quality monitoring programme known as the National Air Quality Monitoring Programme (NAMP). The network consists of 804 operating stations covering 344 cities/towns in 28 states and 6 Union Territories of the country.

The objectives of the NAMP are

1. To control the status and developments of ambient air quality.
2. To determine whether the given ambient air quality standards are dishonoured.
3. To recognise non-attainment cities.
4. To attain the information and understanding essential for developing defensive and remedial measures.
5. To appreciate the natural cleansing process occurring in the atmosphere through pollution thinning, spreading, wind-based measures, dry deposition, precipitation and chemical alteration of pollutants generated.

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Under NAMP, four air pollutants, viz., SO_2, oxides of nitrogen as NO_2, suspended particulate matter, and respirable suspended particulate matter (RSPM/PM10), have been recognised for regular checking at all the sites. The checking of meteorological parameters such as wind speed and wind direction, relative humidity and temperature was also combined with the monitoring of air quality.

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Do You Know?

The checking of pollutants is done for 24 hours with a frequency of twice a week, to have 104 observations in a year.

11.4.3 National Air Quality Index (NAQI)

In April 2015, India launched its first-ever National Air quality Index and acknowledged the increasing level of air pollution. This Index was launched by Shri Narendra Modi, Prime Minister of India in New Delhi. The first phase of the National Air Quality Index includes ten cities such as Delhi, Agra, Lucknow, Bangalore, etc. The government is also planning to extend it to 22 state capitals and other 44 cities with a population of more than one million.

Exam Pointer

Air Index Quality categorises based on

Colour-coded warnings. It has been classified into 6 bands. Each band is a cut point of concentration with a colour code to visually represent the level of harshness that people can recognise easily.

The 6 bands are:

1. **Good:** 0–50% of the maximum permissible limit.
2. **Satisfactory:** 50–100% of the maximum permissible limit.
3. **Moderately polluted:** 101–200% of the maximum permissible limit.
4. **Poor:** 201–300% of the maximum permissible limit.
5. **Very poor:** 301–400% of the maximum permissible limit.
6. **Severe:** 401–500% of the maximum permissible limit.

Do You Know?

For calculating the Air Quality Index, 8 pollutants are considered: CO, O_3, NO_2, SO_2, NH_3, Pb, PM10, PM 2.5.

11.4.4 Graded Response Action Plan (GRAP)

The Graded Response Action Plan (GRAP) is an emergency framework designed to control air pollution in the Delhi-National Capital Region (NCR). It is run by the **Commission for Air Quality Management (CAQM)**. GRAP outlines specific actions that become progressively tougher as air quality worsens. **Its goal is to stop the Air Quality Index (AQI) from reaching 'Severe Plus' or 'Emergency' levels.**

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ENV 48- Graded Response Action Plan

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Key Features of GRAP

- **Statutory Framework**: GRAP received approval from the Supreme Court in 2016 as part of the *M. C. Mehta vs. Union of India* case. It was officially announced by the **Ministry of Environment, Forest and Climate Change (MoEFCC) in 2017. The CAQM implements GRAP in NCR and nearby areas.**
- **Pre-emptive & Coordinated**: The plan activates based on AQI forecasts from agencies like the India Meteorological Department (IMD). This helps to take preventive actions before pollution reaches peak levels. It requires cooperation among 13 agencies across Delhi, Uttar Pradesh, Haryana, and Rajasthan.
- **Incremental Approach**: The plan follows an incremental model. When a higher stage is triggered, actions from all earlier stages remain in effect.

Stage	Category	AQI Range (Delhi)	General Measures
Stage I	Poor	201-300	Enhanced mechanical sweeping and water sprinkling of roads, strict enforcement of PUC (Pollution Under Control) norms, and enforcement of NGT/SC orders on overaged vehicles.
Stage II	Very Poor	301-400	Banning the use of diesel generators (except for essential services), intensifying actions at identified pollution hotspots, and stopping firecracker usage.
Stage III	Severe	401-450	Banning non-essential construction and demolition activities (excluding specific linear public projects), strict restrictions on plying BS-III petrol and BS-IV diesel four-wheelers, and potential shift to hybrid learning for junior school classes.
Stage IV	Severe +	> 450	Stopping the entry of most trucks into Delhi (except essentials), potentially implementing an odd-even vehicle rationing scheme, and potential closure of schools/colleges and non-emergency commercial activities.

- **Accountability:** GRAP specifies the actions that must be taken, the trigger levels, and the agencies responsible. This helps ensure accountability.**GRAP Stages and General Measures**

GRAP categorises the air quality into **four stages**, each with a corresponding set of mandatory actions:

11.4.5 National Clean Air Programme 2019

"NCAP 2019" most likely refers to India's **National Clean Air Programme (NCAP)**, launched in January 2019 to improve air quality in 131 of its most polluted cities. The program aims to reduce particulate matter pollution by 40% by 2026, with an initial target of a 20-30% reduction by 2024 compared to 2017 levels. Alternatively, it could refer to a vehicle safety rating from **Global NCAP**, which conducted a car-to-car crash test in 2019.

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■ *National Clean Air Programme (NCAP)*

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- **Goal:** To improve air quality in 131 non-attainment cities by reducing particulate matter (PM) pollution by 20-30% by 2024 (later revised to 40% by 2026) compared to 2017 levels.
- **Funding:** The program leverages funds from various government schemes and has provided financial support to cities for implementing action plans.
- **Implementation:** Cities are required to create and implement City Action Plans (CAPs) that include measures for reducing pollution from sources like dust, vehicles, and waste burning.
- **Challenges:** Despite significant funding, the program has faced criticism for slow progress and uneven implementation, with some studies suggesting pollution levels worsened in certain cities.

Global NCAP

- **Purpose:** An independent organisation that tests the safety of vehicles by conducting crash tests and providing a star rating.
- **2019 Activity:** In 2019, Global NCAP performed a car-to-car crash test to demonstrate the safety differences between Africa-spec and Euro-spec versions of the Nissan NP300 pickup truck.

11.4.6 Commission for Air Quality Management (CAQM)

The **Commission for Air Quality Management (CAQM)** is a **statutory body** created under the Commission for **Air Quality Management in the National Capital Region and Adjoining Areas Act, 2021**. It was initially formed in 2020 through an Ordinance. The CAQM is led by a chairperson who should have at least 15 years of experience in environmental and pollution fields or 25 years of administrative experience. People can appeal orders and directions issued by the CAQM at the National Green Tribunal (NGT)..

Functions of CAQM

- Coordinate actions on the **monitoring of air quality.**
- **Planning and executing plans** to prevent and control air pollution in the National Capital Region (NCR).
- **Conduct research and development** through networking with technical institutions.

CAQM bans the use of coal in Delhi and nearby cities from January 2023.

11.4.7 New Targets for National Clean Air Programme (NCAP)

The **National Clean Air Programme (NCAP)**, launched in 2019, aims to put in place measures to prevent, control, and reduce air pollution. It initially set a nationwide goal of cutting particulate matter (PM10 and PM2.5) concentrations by 20 to 30 per cent by 2024, using 2017 as the base year**.**

Do You Know?

The centre has set a **new target of a 40% reduction in PM concentration** in cities covered under the **NCAP by 2026**, updating the earlier goal of 20%–30% reduction by 2024.

Key Features

- **Preparation of city-specific action plans,** which include measures for strengthening the air quality monitoring network, reducing vehicular and industrial emissions, increasing public awareness, etc.
- **Implemented in 131 cities**:
  - **123 Non-Attainment cities** (NACs) identified under **NCAP based on non-conforming to National Ambient Air Quality Standards (NAAQS)** consecutively for five years.
  - **8 million Plus Cities identified by the 15th Finance Commission (XV-FC),** for receiving performance-based grants for air quality improvement.

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- **PRANA (Portal for Regulation of Air Pollution in Non-Attainment cities)** monitors the implementation of NCAP.

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■ *Swachh Vayu Sarvekshan*

- Ministry of Environment, Forest and Climate Change will launch **Swachh Vayu Sarvekshan** under NCAP.
- 131 cities to be ranked in the country for implementing City Action Plans prepared NCAP for **reducing air pollution up to 40% by 2025–26**.
- **Cities categorised into 3 groups based on population** (population > 10 lakh, population between 3 and 10 lakh and population < 3 lakh).

■ *For the survekshan (survey), cities are required to*

1. Conduct an annual self-assessment using the framework on the PRANA online portal. This assessment looks at different parts of city management and services to measure their performance in various areas.
2. Report on the activities and measures taken to solve specific environmental issues. These include solid waste management, road dust management, handling construction and demolition waste, controlling vehicular emissions, and managing industrial pollution. Cities must provide information on the steps taken to improve and manage these areas effectively.
3. By following these steps, cities can engage in the **survekshan** and show their commitment to sustainable and responsible urban management. This effort promotes better living conditions for residents and a healthier environment. Based on the **self-assessment and third-party assessment, three best-performing cities in each group will receive a cash award**.

11.4.8 WHO's Air Quality Guidelines 2005 & Updated Guidelines 2021 (AQGs 2021)

Exam Pointer

International Standards

World Health Organization (WHO) Air Quality Guidelines, 2005, offer global leadership on verges and bounds for 4 key air pollutants that show health risks - particulate matter (PM), ozone (O_3), nitrogen dioxide (NO_2) and sulphur dioxide (SO_2). Indian standards are slightly less severe than WHO guidelines.

■ *The WHO's new updated Global Air Quality Guidelines (AQGs)*

The WHO released revised air quality the first major update to the standards in 15 years. The revised recommendations, produced from evidence-based data and information and a growing awareness that no amount of air pollution is safe for human beings, resulted in nearly all pollutant standards being strengthened in comparison to the AQGs established in 2005. If target levels of pollution are implemented by governments, these new guidelines will save lives by reducing preventable airborne pollutant deaths.

- They provide recommendations on air quality guideline levels as well as interim targets for six key air pollutants.
- They also offer qualitative statements on good practices for the management of certain types of PM, for example, black carbon/elemental carbon, ultrafine particles and particles originating from sand and dust storms, for which there is insufficient quantitative evidence to derive AQG levels.
- The guidelines identify the levels of air quality necessary to protect public health worldwide.
- **WHO's** new guidelines **(published in 2021)** recommend **air quality levels for six pollutants: PM 2.5 and PM 10, ozone (O_3), nitrogen dioxide (NO_2), sulphur dioxide (SO_2)** and **carbon monoxide (CO).**
- The annual average for **PM2.5** should not exceed 5 micrograms per cubic metre of air, while the 24-hour average should not exceed 15 micrograms per cubic metre.

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- The annual average for **PM 10** should not exceed 15 micrograms per cubic metre of air, while the 24-hour average should not exceed 45 micrograms per cubic metre.
- The **ozone levels** average should not exceed 100 micrograms per cubic metre over 24 hours.
- The **nitrogen oxide** levels should not exceed 25 micrograms per cubic metre over 24 hours.
- The **sulphur dioxide** levels should not exceed 40 micrograms per cubic metre over 24 hours.
- The **carbon monoxide** levels should not exceed 4 micrograms per cubic metre over 24 hours.

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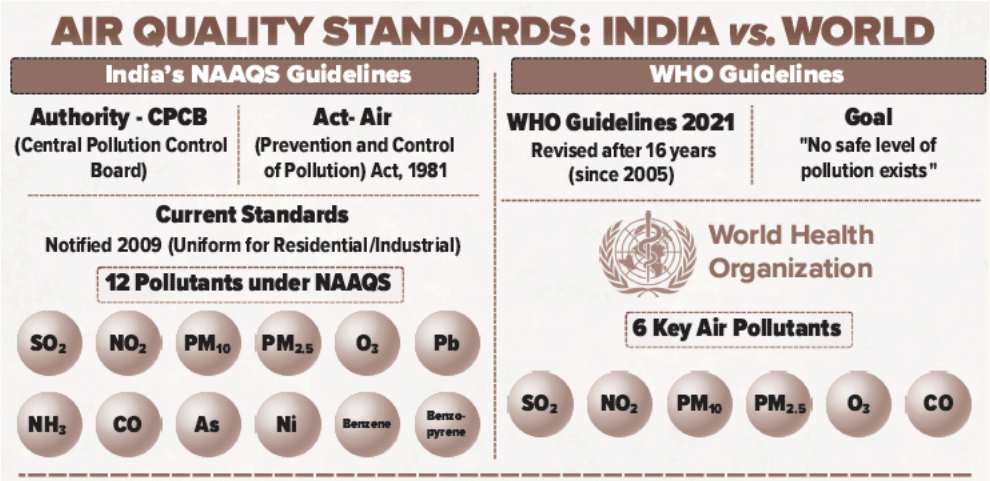
Difference between 2005 and 2021 AQG levels

The new WHO Global **AQGs** from 2021 demonstrate the harmful effects of air pollution on human health at considerably lower concentrations than previously thought. The guidelines propose new air quality standards to safeguard people's health by lowering levels of important air contaminants, some of which are also linked to climate change.



How Relaxed are Indian Standards? (WHO 2021 vs. India NAAQS)

Pollutant	Metric	WHO 2005	WHO 2021	India NAAQS (Current)
$PM_{2.5}$	Annual	10 µg/m³	5 µg/m³	40 µg/m³ (India allows **8x** more pollution)
	24-hour	25 µg/m³	15 µg/m³	60 µg/m³ (India allows **4x** more pollution)
PM_{10}	Annual	20 µg/m³	15 µg/m³	60 µg/m³ (India allows **4x** more pollution)
	24-hour	50 µg/m³	45 µg/m³	100 µg/m³ (India allows **~2x** more pollution)
O_3 Ozone	Peak	NS**	60 µg/m³	—
	8-hour	100 µg/m³	100 µg/m³	100 µg/m³
NO_2	Annual	40 µg/m³	10 µg/m³	40 µg/m³ (India allows **~4x** more pollution)
	24-hour	NS**	25 µg/m³	80 µg/m³ (India allows ~3x more pollution)
SO_2	Annual	—	—	50 µg/m³
	24-hour	20 µg/m³	40 µg/m³	80 µg/m³ (India allows **2x** more pollution)
CO	Annual	NS**	4 mg/m³	—
	8-hour	—	—	4 mg/m³

µg/m³ = Micrograms per Cubic Meter | **NS**-** Not Set

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The Centre has launched the National Clean Air Programme (NCAP) to tackle air pollution in 122 cities by aiming for a 20%–30% reduction in PM10 and PM2.5 concentrations by 2024, compared to 2017.

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Changes required in India's NCAP

- India's air quality rules are more relaxed than WHO standards. Stricter targets are necessary under the proposed Clean Air for All Mission.
- We need an airshed approach to plan actions based on shared geographic and weather factors.

WHO's 2021 Air Quality Guidelines set much tighter limits than the guidelines from 2005.

Exam Pointer

Keeping 2017 as the base year for the comparison of particulate matter concentration, the Indian government, through its dedicated NCAP, aims to reduce the PM concentrations to 20%–30% by 2024 in 122 cities.

Do You Know?

- **Ozone at ground level** is formed by the reaction with sunlight (photochemical reaction) of pollutants such as nitrogen oxides (NOx) from vehicle and industry emissions and volatile organic compounds emitted by vehicles, solvents and industry. **As a result, the highest levels of ozone pollution occur during periods of sunny weather and not during inclement weather**.
- **PM** is a common proxy indicator for air pollution. It affects more people than any other pollutant. The major components of PM are sulfate, nitrates, ammonia, sodium chloride, black carbon, mineral dust and water. It consists of a complex mixture of solid and liquid particles of organic and inorganic substances suspended in the air. **While particles with a diameter of 10 microns or less, (PM10) can penetrate and lodge deep inside the lungs**, the even more health-damaging particles are those with a diameter of 2.5 microns or less, (PM2.5). **PM2.5 can penetrate the lung barrier and enter the blood system**. Chronic exposure to particles contributes to the risk of developing cardiovascular and respiratory diseases, as well as of lung cancer.
- **Excessive ozone in the air** can have a marked effect on human health. It causes breathing problems, trigger asthma, reduce lung function and cause lung diseases. Ozone triggers asthma because it is very irritating to the lungs and airways.

11.5 THE ENVIRONMENT (PROTECTION) ACT, 1986

The Environment (Protection) Act, 1986 is a key legislation for environmental regulation in India. It has three main goals:

1. To protect and restore air, water, and land.
2. To prevent harm to all living beings.
3. To promote a positive relationship between people and their environment.

11.5.1 The 1986 Act was general legislation created under Article 253.

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ENV 49- The Environment (Protection) Act, 1986

Key Features of the Act:

- **Government Authority:** The Act gives the Central Government the power to take necessary steps to protect and improve the environment. This includes coordinating actions, planning programs, setting pollution standards, and regulating industrial activities.
- **Regulatory Power:** The government can issue binding written orders to individuals or authorities. This can include closing industries or regulating essential services like electricity and water if they do not comply.
- **Penalties:** Anyone violating the Act can face up to five years in prison, a fine of up to 1 lakh rupees, or both. Ongoing violations can result in an additional fine of up to ₹5,000 per day.
- **Inspection and Sampling:** The Act allows Central Government officers to enter premises and collect samples of air, water, soil, or other materials for testing to ensure compliance with the Act's standards.
- **Citizen Suits:** Section 19 of the Act permits anyone, not just government officials, to file a complaint in court for offences under the Act. They must give the Central Government a notice of at least 60 days before doing so.
- **Associated Bodies:** The Act sets up a framework for creating statutory bodies like the **Genetic Engineering Appraisal Committee (GEAC**), which reviews activities involving genetically modified organisms, and the **Central Ground Water Authority (CGWA)**, established to manage groundwater resources.

Do You Know?

1. Ecological sensitive zone (ESZ)are the areas declared under the Environment (Protection) Act, 1986.
2. There are no provisions for people participating under the Environment (Protection) Act, 1986. Under Section 6 of this Act, the Central Government decides all criteria for pollution control.

Eco-Sensitive Zones

Eco-Sensitive Zones (ESZs) in India are designated **under the Environment (Protection) Act, 1986** by the **Ministry of Environment, Forest and Climate Change (MoEFCC).** These zones act as '**shock absorbers' around National Parks and Sanctuaries**. They regulate activities like mining and have strict rules against polluting industries. Hotel construction and tree cutting are regulated, while organic farming and renewable energy are allowed. This is done through State-led Zonal Master Plans (ZMPs) to safeguard biodiversity and reduce human-wildlife conflict. They serve as transition areas.

Declaration Process

1. **Identification:** State Governments survey and identify ecologically important areas, usually within 10 km of Protected Areas, but the distance can vary.
2. **Proposal:** States submit proposals with their recommendations to the Central Government (MoEFCC).
3. **Notification:** The MoEFCC formally declares the ESZ based on these proposals, as outlined in Section 3 of the EPA, 1986.

What are Eco-sensitive zones?

How eco-sensitive zones are managed and declared under India›s Environment (Protection) Act, 1986?

- **Legal Basis:** The Act does not specifically mention "Eco-sensitive Zones." Instead, **Sections 3(2)(v) and 5(1)** give the Central Government the authority to limit industrial activities or ban certain actions in designated

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areas.

- **Criteria for Declaration:** The government considers factors such as biological diversity, pollution levels, suitable land use, and closeness to protected areas when declaring these zones. They are also known as 'No Development Zones' or 'Ecologically Fragile Areas.'
- **Management Model:** The zones follow a core and buffer structure
- **Core Area**: Has the legal designation of a national park
- **Buffer Area**: Does not have national park status and may consist of reserved forests, wildlife sanctuaries, or tiger reserves

Criteria

- The MoEF has approved a comprehensive set of guidelines laying down parameters and criteria for declaring ESAs.
- A committee constituted by MoEF put this together. The guidelines lay out the criteria based on which areas can be declared as ESAs.
- These include species-based (endemism, rarity, etc.), ecosystem-based (sacred groves, frontier forests, etc.) and geomorphologic feature-based (uninhabited islands, origins of rivers, etc.).
- The National Wildlife Action Plan (2002–2016) of MoEFCC stipulated that state governments should declare land falling within 10 km of the boundaries of national parks and wildlife sanctuaries as eco-fragile zones or ESZs under the Environmental (Protection) Act, 1986.

Activities in ESZ

- **Prohibited:** Commercial mining, setting of sawmills and industries causing pollution, commercial use of firewood and major hydropower projects and tourism activities.
- **Regulated:** Felling of trees, drastic change in agricultural systems and commercial use of natural water resources.
- **Permitted:** Agriculture and horticulture practices by local communities, rainwater harvesting, organic farming, adoption of green technology and use of renewable energy sources.

11.6 HAZARDOUS WASTE MANAGEMENT RULES 1989 TO 2016

Hazardous Waste Management Rules, 1989, were formed to safeguard the handling, generation, processing, treatment, packaging, storage, transference, reprocessing, gathering, conversion, and offering for sale, devastation and disposal of hazardous waste.

These rules were modified later in 2000, 2003 and the Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2008, was released in place of the former notification.

Later in 2016, in supersession of the Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008, the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 came into force.

Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016

Key Features of the Rules

Objective: Ensure safe and environmentally friendly handling, retrieval, and disposal of hazardous waste.

Balanced Approach: Designed to support both environmental protection and industrial convenience.

Govt Alignment: Supports 'Ease of Doing Business' and 'Make in India' while promoting sustainable practices.

Waste Classification: Introduces a clear distinction between hazardous waste and other types of waste.

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Resource Recovery: 'Other wastes,' such as tyres, paper waste, metal scrap, and used electronics, are recognised for recycling and reuse.

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11.6.1 2022 Amendments to Hazardous and Other Wastes Rules, 2016 (Crisp Summary)

Expanded Scope: Coverage now includes 'other wastes' in addition to hazardous waste.

Waste-Management Hierarchy: Prevention, minimization, reuse, recycling, recovery, co-processing, safe disposal.

Simplified Procedures: Tighter and clearer processes for authorization, transport, and import/export.

Mandatory SOPs: SPCBs must check infrastructure and enforce waste-specific SOPs.

Efficiency Measures: A single-window consent system was introduced; co-processing is prioritised over disposal.

Import Exemptions: Some imports, like metal scrap, paper waste, and reusable EEE, are exempt from ministry approval.

State Responsibilities: Designate recycling zones, register workers, form labour groups, and improve safety and skill systems.

Technical Updates: Updated waste lists and concentration limits to align with global standards. Import Bans: Bans on household waste, waste oils and fats, critical medical equipment, tyres for direct reuse, plastic waste, and EEE scrap.

Annual Reporting: SPCBs must send waste inventories to CPCB by September 30.

Extended Producer Responsibility: New Schedule IX requires EPR for tyre producers, recyclers, and retreaters.

11.7 THE PUBLIC LIABILITY INSURANCE (PLI) ACT, 1991

This Act is about compensation to victims of an accident, which happens as a result of handling any hazardous substance. The MoEF administer PLI Act.

Salient features of the Act

- The Act relates to all owners associated with the manufacture or handling of any hazardous chemicals.
- This Act was ratified after the Bhopal Gas leak disaster, where **MlC (Methyl Iso Cyanide)** leaked from the plant of Union Carbide India Ltd. and caused the demise of over 3,000 persons and severe injuries to a large number of others. (Owners have failed to subscribe to PLI policies because of ignorance.)
- The MoEF has issued directions under Section 18(1)(a) of the Water Act and the Air Act to CPCB to ensure better implementation of the Public Liability Insurance (PLI) Act, 1991.

This Act establishes an **Environmental Relief Fund (ERF),** in which all such user industries promise to deposit an equal quantity to the yearly premium amount of such insurance policies. The instructions have been issued and will confirm that all the liable handling user industries promise to the PLI insurance policy and deposit the agreed amount in ERF.

- The payable compensation under this Act is also regardless of the company's negligence. The victims can file a claim within the incident 5 years with the Collector when exposed to the hazardous substances used by a concerned industry.
- On the receipt of the application, the Collector shall, after giving notice of the application to the owner and after giving the parties an opportunity of being heard, hold an inquiry into the claim and award the compensation by determining the fair amount of relief.

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Criticisms

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- Even though, under this Act, the claim amount was specified nearly two decades ago. As specified in the Schedule, the amounts under this Act were stipulated nearly two decades ago. Hence, under this Act, the compensation is very low. The victims' families who have died due to or have suffered permanent disability due to gas leakage are entitled to a maximum compensation of ₹25,000, besides a maximum of ₹12,500, for medical expenses.

11.8 THE NOISE POLLUTION (REGULATION AND CONTROL) RULES, 2000

The objective of Noise Pollution (Regulation and Control) Rules is to preserve the ambient air quality standards in respect of noise. The state government shall classify the areas into industrial, commercial, residential or silence areas/zones for the resolution of application of noise standards for unlike areas.

An area including not less than 100 metres around hospitals, educational institutions and courts may be stated as a silence area-I zone.

A loudspeaker or a community address system shall not be utilised at night (between 10.00 p.m. to 6.00 a.m.) except in locked premises for community within, e.g., auditoria, conference community halls and banquet halls.

11.9 PROTECTION OF PLANT VARIETIES AND FARMERS' RIGHTS ACT, 2001

The Protection of Plant Varieties and Farmers' Rights Act, 2001 (PPVFR Act) is a statutory Act ratified to establish an effective scheme for protecting plant varieties, the rights of farmers and plant breeders and inspiring the development and cultivation of new varieties of plants.

Purpose of the PPVFR Act, 2001

- Protect new plant varieties and the rights of farmers and breeders.
- Encourage the development of new plant varieties and meet TRIPS obligations.

Salient Features

- A **sui generis** system is used for plant variety protection.
- Breeders, farmers, and researchers share concurrent rights.
- Farmers are defined as cultivators, conservers, and breeders.
- The Act provides important rights for farmers, including the right to seed and compensation for crop failure.
- There is no penalty for farmers who unknowingly violate breeders' rights.

Breeders' Rights: Breeders have **exclusive rights** to produce, sell, market, distribute, import, or export a protected variety. They can appoint agents or licensees and seek civil remedies for infringement.

Researchers' Rights: Researchers can freely use registered varieties for research and experiments. They can use these varieties to develop new ones, but repeated use requires the breeder's consent.

Farmers' Rights: Farmers can register new varieties they develop. They are free to save, use, exchange, or sell their produce or seed, except when it comes to branded seed. They may receive recognition, rewards, and compensation if the variety fails. They are also exempt from legal fees under the Act.

11.10 BIOLOGICAL DIVERSITY ACT, 2002

The **Biological Diversity Act, 2002**, regulates access to **biological resources and related traditional knowledge** while ensuring fair and equal sharing of benefits.

Core Objectives and Scope

- **Conservation and Use:** The Act supports the conservation and sustainable use of India's biological resources for commercial, research, bio-survey, and bio-use purposes**.**

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- **Benefit Sharing:** It establishes a framework for accessing biological resources and requires the sharing of benefits gained from their use.
- **Intellectual Property Rights (IPRs):** The Act includes the sharing of research findings and the submission for IPRs related to Indian biological resources.
- **Compliance:** It was passed to fulfill obligations under the Convention on Biological Diversity (CBD).

Salient Features: The Three-Tiered Structure

A three-tiered structure operates at national, state, and local levels to manage implementation:

Table 11.1 The Three-Tiered Structure

Body	Level	Primary Responsibilities
National Biodiversity Authority (NBA)	National	Deals with access requests from foreign entities/institutions/companies and the transfer of research results to outsiders.
State Biodiversity Boards (SBB)	State	Manages all matters related to access by Indian entities for commercial activities. The industry must provide prior notice of resource use.
Biodiversity Management Committees (BMCs)	Local	Established by local self-governments for conservation, sustainable use, and documentation of biodiversity within their areas. NBA/SBB must consult BMCs.

Key Exemptions and Rights

- **Researchers:** Indian researchers do not need prior agreement or notice to SBB for attaining biological resources for research within India.
- **Local Users:** Local people, the public, growers, cultivators, and traditional practitioners (Vaids and Hakims) are exempt from free usage of biological resources within India.

Benefit-Sharing Mechanisms: It includes joint ownership of intellectual property rights, technology transfer, and setting up production or research units in the source area. They can also involve monetary or non-monetary compensation, along with creating venture capital funds to support those claiming benefits.

11.11 THE SCHEDULED TRIBES AND OTHER TRADITIONAL FOREST DWELLERS (RECOGNITION OF FOREST RIGHTS) ACT, 2006

ENV 50- The Scheduled Tribes and Other Traditional Forest Dwellers Act, 2006

Salient Features

- The Act grants land rights to forest dwellers based on their current holdings, **limited to four hectares**.

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- It considers claims from **Scheduled Tribes and Other Traditional Forest Dwellers (OTFDs)** who have lived on and relied on forest land for three generations, or 75 years.

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Do You Know?

India's forests are governed by two main laws:

- Indian Forest Act, 1927: It empowers the government to declare any area to be a reserved forest, protected forest or village forest.
- Wild Life (Protection) Act, 1972: It permits any area to be established as a 'protected area', namely a national park, wildlife sanctuary, tiger reserve or community conservation area.

QUICK INSIGHT

Types of Rights: *Title rights, use rights, relief and development rights, and forest management rights. The Act also allows forest land to be set aside for essential community infrastructure.*

11.11.1 Criteria to Claim Rights Under FRA For Forest-Dwelling Scheduled Tribes (FDST) -

Must belong to a listed Scheduled Tribe in that area.

- Must have primarily lived on forest land before December 2005.
- Must depend on the forest for livelihood.

For Other Traditional Forest Dwellers (OTFD)

- Must have lived in forests for three generations, or 75 years, before December 2005.
- Must depend on the forest for livelihood.

Do You Know?

The Ministry of Tribal Affairs (MoTA) serves as the main agency. Rights can be inherited but not transferred.

Process of Recognition of Rights

- The Gram Sabha starts the process by passing a resolution that identifies rightful claimants.
- Claims are then reviewed at the sub-division or taluka level, and subsequently at the district level.
- Each committee consists of three government officials (from Forest, Revenue, and Tribal Welfare) and three elected local members.

The Act allows for ownership and sale of minor forest produce (MFP) and protects existing rights in national parks and sanctuaries without creating new ones.

11.11.2 Bamboo as Minor Forest Produce

- Under the FRA, bamboo is classified as MFP, giving forest dwellers the right to collect, use, and sell it.
- MFP includes all non-timber forest products such as **cane, honey, wax, lac, medicinal plants, and brushwood.**
- The FRA recognises forest dwellers' rights to live, cultivate, and sustainably use forest resources**.**

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Exam Pointer

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Critical Wildlife Habitat

The phrase 'critical wildlife habitat' is **defined only in the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006.**

Union **MoTA (Ministry of Tribal Affairs) officially decides and declares Habitat Rights** for Primitive and Vulnerable Tribal Groups in any part of India.

Baigas are considered a **particularly vulnerable tribal group (PVTG)** in the Indian Constitution and **rely mostly on shifting cultivation, forest produce and fishing for sustenance, spread over forested areas of Madhya Pradesh and Chhattisgarh**.

In 2016, they became India's first community to get habitat rights. The government of Madhya Pradesh/district collector has recognised the habitat rights, so district-level committees have to adjudicate the rights, whereas the Tribal Ministry only issues guidelines

11.12 NATIONAL GREEN TRIBUNAL ACT, 2010

ENV 51- National Green Tribunal Act, 2010

Key Information

The National Green Tribunal (NGT), established on October 18, 2010 under the NGT Act, serves as a specialised body for the swift and effective resolution of cases involving environmental protection, forest conservation, and compensation for environmental damage. Guided by principles of natural justice rather than the Code of Civil Procedure, the Tribunal aims to dispose of cases within six months of filing. With its principal bench in New Delhi and additional benches in Bhopal, Pune, Kolkata, and Chennai, the NGT is founded on Article 21 of the Constitution, which upholds the right to a healthy environment.Jurisdiction and Powers

The NGT has appellate authority over orders made under several key environmental acts:

- Water (Prevention and Control of Pollution) Act, 1974
- Water (Prevention and Control of Pollution) Cess Act, 1977
- Forest (Conservation) Act, 1980
- Air (Prevention and Control of Pollution) Act, 1981
- Environment (Protection) Act, 1986
- Biological Diversity Act, 2002

Exclusions: The NGT cannot hear matters related to the Wildlife (Protection) Act, 1972, the Indian Forest Act, 1927, or various state-level forest/tree preservation laws. The Act applies only to civil cases and excludes criminal offenses.

Composition

- **Chairperson:** A retired Judge of the Supreme Court.
- **Members:** Retired High Court Judges (Judicial members) and experts with a minimum of 15 years of experience in environment/forest conservation.
- **Structure:** Each bench must comprise at least one Judicial Member and one Expert Member.

Appeals

Any person aggrieved by an NGT decision may appeal to the **Supreme Court** within 90 days.

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11.12.1 Different Directions by NGT

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The NGT is a specialised judicial body established to handle environmental cases efficiently and effectively.The 'different directions' cover a wide range of environmental issues, including:

- **Waste Management:** States and local bodies must strictly implement Solid Waste Management Rules, ban open burning of waste, and ensure proper disposal of biomedical waste.
- **Pollution Control:** Authorities will issue orders against air and noise pollution. They will act against industries that do not comply and impose penalties for environmental damage based on the 'polluter pays' principle.
- **Conservation and Resources:** Authorities will suspend environmental clearances for projects that threaten critical habitats, like a hydro project near a bird's wintering site. They will take action against illegal mining and regulate groundwater extraction.
- **Infrastructure and Development:** Agencies will direct the use of treated wastewater for public parks, regulate construction activities in sensitive areas, and ensure environmental compliance for large projects like the Delhi-Mumbai Expressway.
- **Suo Motu Cognizance:** Authorities will act on newspaper articles or letter petitions about important environmental concerns, such as mangrove depletion or illegal construction on lakes.

The NGT's decisions are based on the principles of **sustainable development, the precautionary principle, and the polluter pays principle**. Failure to follow NGT directions can result in **imprisonment and heavy fines**.

11.13 WETLANDS (CONSERVATION AND MANAGEMENT) RULES

11.13.1 The Wetlands (Conservation and Management) Rules, 2010

The Wetlands (Conservation and Management) Rules were notified by the Ministry of Environment, Forest and Climate Change (MoEF) to strengthen the conservation, management, and protection of India's wetlands and prevent their degradation. As the first legally enforceable framework for safeguarding eco-sensitive wetlands in the country, these rules support India's commitments under the Ramsar Convention and align with the objectives of the National Wetlands Conservation Programme**.**

11.13.2 Wetlands (Conservation and Management) Rules, 2017

These updated rules from 2017 replaced the 2010 rules and prohibit various activities in wetlands.

Key Provisions and Functions

- **Constitution of State Wetland Authority:** This mandates the establishment of a State Wetland Authority in every state and union territory, led by the environment minister. This body decides on the 'wise use principle' (sustainable use that aligns with conservation) and has led to the delegation of powers.
- **National Wetland Committee (NWC):** This replaces the Central Wetlands Regulatory Authority and is led by the Secretary to the Ministry of Environment, Forest and Climate Change. Its functions include monitoring implementation, advising the Central Government on policies, recommending Ramsar site designations, and providing guidance on international collaboration.
- **Digital Inventory:** State authorities must create a list of all wetlands, which will serve as a digital inventory updated every 10 years.
- **Regulated Activities:** The rules require the creation of a clear list of activities that are either regulated, permitted, or prohibited within notified wetlands and their zones of influence.
- **Prohibited Activities:** The rules specifically ban the release of unwanted waste from villages, towns, cities, and industries, as well as the dumping of solid waste into wetlands. They also prohibit converting wetland areas for non-wetland uses.

11.14 WASTE MANAGEMENT AND HANDLING

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11.14.1 Waste Management Rules

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Construction and Demolition (C&D) Waste Management Rules, 2016

India's **Construction and Demolition (C&D) Waste Management Rules, 2016**, notified by the Ministry of Environment and Forests (MoEF).

The C&D Waste Rule refers to India's **Construction and Demolition Waste Management Rules**, **primarily the 2016 notification (updated/revamped in 2025),** which mandates responsible handling (segregation, collection, recycling, treatment) of waste from building construction, renovation, and demolition to promote circular economy, reduce landfill burden, and control pollution, placing duties on waste generators (like large construction firms) and service providers for proper waste flow, with requirements for management plans, authorization for processors, and increasing mandatory use of recycled materials in infrastructure projects.

Key Aspects of the C&D Waste Rules

- **Definition:**Waste from construction, remodelling, repair, and demolition (debris, rubble, building materials).
- **Applicability:** To all generators and entities involved in civil construction activities.
- **Responsibilities of Waste Generators:**
  - **Segregation:** Separate concrete, soil, steel, wood, plastic, bricks, etc.
  - **Management Plan:** Large generators (over 20 tons/day) need an approved plan before starting work.
  - **Channelisation:** Hand over waste to authorised processing facilities or collection centres.
  - **Charges:** Pay for collection, transport, and processing.
- **Extended Producer Responsibility (EPR):** Service providers are responsible for managing their waste.
- **Processing & Recycling:** Rules for site selection, authorisation (Form-I) for facilities by State Pollution Control Boards (SPCBs).
- **Mandatory Use of Recycled Materials:** Road projects must use increasing percentages of C&D-derived materials (For example 5 per centby 2026-28, 10 per cent by 2028-30, 15 per cent from 2030-31).
- **Enforcement:** Centralised digital system, audits, and clear roles of agencies.

Goals: Prevent unscientific disposal, promote reuse, recover recyclable value, and address pollution.

Waste Management and Handling

According to the Environment (Protection) Act, 1986, the Central Government has made the following rules relating to the disposal of solid wastes. Radioactive wastes are covered under the provisions of the Atomic Energy Act. 1962.

The Municipal Solid Wastes (Management and Handling) Rules, 2000

It came into effect in January 2004 to apply to all municipal authorities responsible for collection, segregation, storage, transportation, processing and disposal of municipal solid wastes.

These rules lay down the responsibility of the management of solid waste disposal, 'anaerobic digestion', 'biodegradable digestion', 'composting', 'vermicomposting', etc.

The management of solid waste has been made the responsibility of the municipal authority.

The District Magistrate, Deputy Commissioner, shall have the overall responsibility for the enforcement of the provisions under these rules.

■ *Solid Wastes (Management and Handling) Rules, 2016*

India's Solid Waste Management Rules, 2016, replaced the Municipal Solid Wastes (Management and Handling) Rules, 2000.

Salient Features of the 2016 Rules

- **Expanded Scope:** These rules apply to municipal areas and urban agglomerations, census towns, industrial townships, railway areas, airports, SEZs, pilgrimage centres, and government establishments.
- **Mandatory Segregation:** Waste must be sorted at the source to allow for recovery, reuse, and recycling.

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- **Three Waste Streams:** ***Wet*** (biodegradable) ***Dry*** (plastic, paper, metal, wood, etc.) ***Domestic hazardous waste*** (diapers, sanitary waste, mosquito repellents, etc.)
- **Handling Segregated Waste:** Waste must be handed over to authorised collectors, rag pickers, or local bodies. States must include rag pickers in formal systems.
- **Generator Responsibilities & Fees:** Payment of a user fee for waste collection. Spot fines for littering and not segregating waste. Bulk generators, such as hotels, restaurants, and event organisers, must manage their own waste, often through composting or biomethanation.
- **Processing & Disposal:** Composting or biomethanation is preferred for biodegradable waste, ideally on-site. Clear criteria exist for locating landfills and waste-processing facilities.
- **Specific Inclusions:** Street vendors must use waste containers, and horticulture waste is also included under these rules.

11.14.2 e-waste Management Rules, 2016

e-waste Management Rules, 2016, notified by the Union Ministry of Environment have introduced major changes in comparison to the e-Waste (Management & Handling) Rules, 2011.

■ *Highlights*

- Establishing 'reverse chain' – Extended Producer Responsibility (EPR) has been put in place to ensure electronic products are recovered for safe recycling. A 'buy back' policy for electronics has also been suggested under which producers can buy the products back for safe recycling.
- The informal sector workers will be trained to handle e-waste properly rather than burning it after extracting the precious metals. The informal sector will be formalised and brought into the mainstream as they handle over 90% of e-waste generated in India.
- New rules also have Compact Fluorescent Lamp and other mercury-containing lamps and equipment in their ambit.
- It does not elaborate on penal clauses.

Figure 11.1 e-waste Type

■ *e-WASTE (MANAGEMENT) RULES, 2022*

The Ministry of Environment, Forest and Climate Change (MoEF&CC) has notified e-waste (Management) Rules, 2022, in exercise of the powers conferred by the Environment (Protection) Act, 1986.

■ *Key Provisions of e-waste (Management) Rules, 2022*

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- **'e-waste' Definition:** Electrical and electronic equipment, whole or in part discarded as waste, and rejects from manufacturing, refurbishment and repair processes.
- **Coverage:** Every manufacturer, producer, refurbisher, dismantler and recycler involved processing of e-waste with certain exceptions.
- Compulsory registration of manufacturer, producer, refurbisher or recycler of the e-products with CPCB.
- Introduction of **Extended Producer Responsibility (EPR) Certificates** to incentivise registered electronic waste recyclers by introducing (not part of the 2016 Rules).

Do You Know?

Extended Producer Responsibility: Responsibility of any producer of electrical or electronic equipment for meeting recycling targets to ensure environmentally sound management of e-waste.

- CPCB to conduct random sampling of electrical and electronic equipment placed on the market to monitor and verify the compliance of reduction of hazardous substances provisions.
- Imposition and collection of environmental compensation on the producer in case of non-fulfilment of obligations as per CPCB guidelines.

Status of e-waste generation in India

- India is the third largest electronic waste producer in the world after China and the United States.
- Produces about 4mMT (million metric tonnes) of e-waste. This is expected to increase 40-fold by 2050.
- Computer devices account for nearly 70% of annual e-waste production, followed by the telecom sector, medical equipment and electric equipment.
- Only 1.5% of electronic waste generated is recycled through institutional processes.
- Uttar Pradesh, Uttarakhand, Tamil Nadu and Haryana are among the states that have a bigger capacity to dismantle and recycle e-waste.

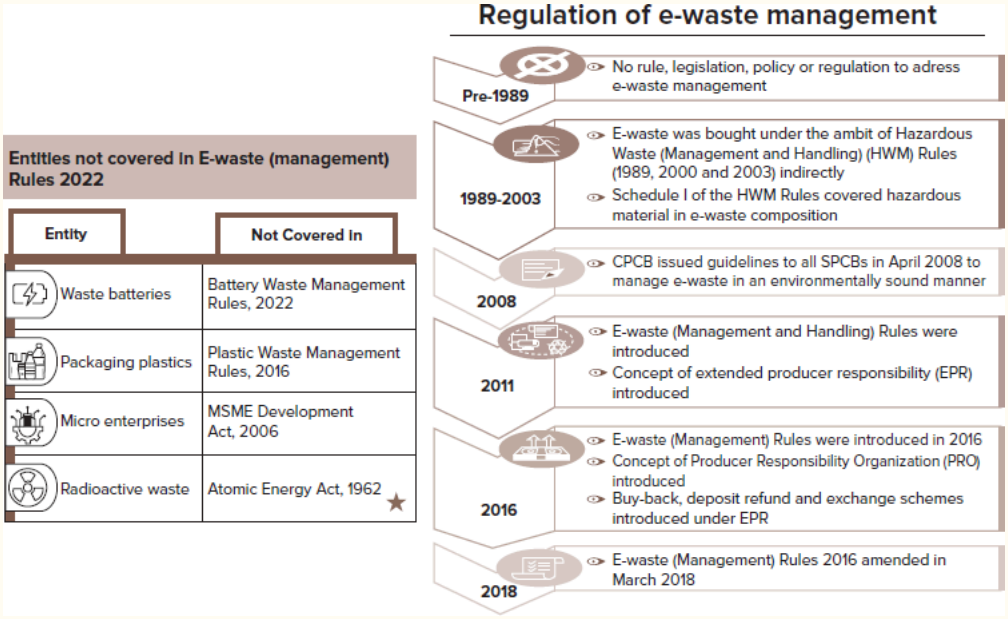

Figure 11.2 e-Waste Management

11.14.3 Biomedical Waste (Management and Handling) Rules, 1998 and 2016

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The Biomedical Waste (Management and Handling) Rules, 1998

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The Biomedical Waste Management and Handling) Rules that came into force in 1998 notified the rules for managing and handling biomedical wastes generated from hospitals, clinics and other institutions. Biomedical is categorised as any waste, which is generated during the diagnosis, treatment or immunisation of human beings/animals or research activities etc.

According to the Rules, the institution/body generating biomedical waste which includes a hospital, nursing home, clinic, dispensary, veterinary institution, animal house, pathological laboratory, blood bank by whatever name is called to take all steps to ensure that such waste is handled without any adverse effect to human health and the environment.

Biomedical Waste Management Rules, 2016

The new Rules are more comprehensive as they define the roles of stakeholders, waste generators and waste treatment facility operators, local authorities, state governments and pollution control boards.

Highlights

- It has expanded the scope to include vaccination camps, blood donation camps, surgical camps and other Healthcare activities.
- **Barcode** has been made mandatory for bags and containers containing biomedical waste.
- The new rule has also made things simple for hospitals by providing a reduction in the waste categorisation number to lessen the confusion faced by waste handlers.
- Health-care workers at my school through training and immunisation regularly.
- Following WHO and NACOs (National AIDS Control Organisation) prescription pretreatment of lab waste, blood samples, laboratory waste, microbiological waste and blood bags through disinfection or on-site sterilisation is now mandatory.
- It aims to phase out the use of chlorinated plastic bags, gloves and blood bags within 2 years.
- Biomedical waste has been classified into 4 categories instead of 10 to improve segregation of waste at source.
- State Government to provide land for setting up common biomedical waste treatment and disposal facility.
- No occupier shall establish an on-site treatment and disposal facility, if a service of it is available at a distance of 75 km.

11.14.4 Plastic Waste Management (PWM) Rules, 2011, 2016, 2018, 2022 & 2024

India's first major plastic waste law was the **Plastic Waste (Management and Handling) Rules, 2011**. Before this, the 1999 **Plastics Manufacture, Sale and Usage Rules** mainly set a minimum carry-bag thickness of 20 microns and did not cover larger waste management issues. The 2011 rules introduced the first organized framework for managing plastic waste, but they mostly applied to municipal areas. They were later replaced by the more detailed Plastic Waste Management Rules, 2016.

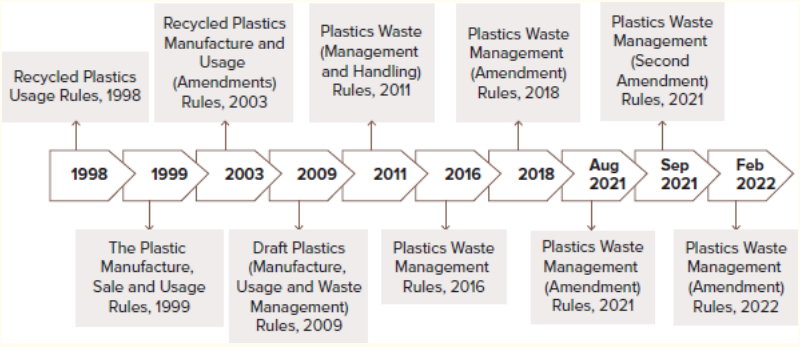

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Figure 11.3 Timeline of the PWM Rules in India.

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11.14.5 Plastic Waste Management (Amendment) Rules, 2022

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Summary of the 2022 Amendments

Carry Bag Thickness: The minimum thickness is now 75 microns until December 31, 2022, and 120 microns after that. Recycled Plastic in Food Packaging: Allowed if it meets FSSAI standards.

Sachet Ban: Plastic sachets cannot be used for gutkha, tobacco, or pan masala.

Plastic Sheet Thickness: The minimum is 50 microns, unless a thinner sheet is necessary for how the product works.

Extended Producer Responsibility (EPR) Producers of branded goods (PIBOs) must collect and manage plastic waste equal to the amount they sell in the market. PIBOs and Plastic Waste Producers (PWPs) must register on the Central Pollution Control Board (CPCB) online portal, or on the State Pollution Control Board (SPCB) for single-state operations. If they do not register, their operating consent will not be renewed.

Single-Use Plastic (SUP) Ban: Starting July 2022, items like plastic sticks, earbuds, flags, cutlery, plates, cups, and straws are banned.

Disposal & Processing: Non-recyclable or non-recoverable plastic packaging will be phased out. Multi-layered plastics can be used for co-processing, such as waste-to-energy, cement kilns, roadmaking, pyrolysis, and gasification. Recycling must follow **IS 14534:1998 standards.**

Supply Chain Regulation: Plastic manufacturers cannot supply raw materials to unregistered units.

Local Body Responsibility: Urban local bodies must create infrastructure for managing plastic waste**.**

11.14.6 Plastic Waste Management (Amendment) Rules, 2024

From March 14, 2024, new rules will tighten standards for biodegradable plastics. They will require full degradation into non-microplastic substances and require CPCB certification.

The definitions of Producers and Importers will be expanded. Raw-material manufacturers will need to file quarterly and annual reports that include pre-consumer waste data.

Local bodies must assess waste generation and infrastructure every year. Wider EPR responsibilities will now apply across the plastic value chain.Comparative overview of key Plastic Waste Management (PWM) Rules and their subsequent Amendments in India

Table 11.2 Plastic Waste Management Rules and Amendments (2016-2024)

Feature	PWM Rules, 2016	Amendment Rules, 2018	Amendment Rules, 2022	Amendment Rules,
Applicability Scope	Expanded to rural areas, industrial townships, etc.	Clarified rules for multi-state entities.	No major change from 2016 scope.	Local bodies (ULBs/ assigned specific duties.
Minimum Carry Bag Thickness	**50 microns**	50 microns	Increased to **75 microns** (Sep 2021) and **120 microns** (Dec 2022).	120 microns

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Single-Use Plastics (SUP) Ban	Ban on plastic in *gutkha/pan masala* sachets.	No significant change.	Nationwide ban on **19 identified SUP items** (effective July 1, 2022).	No new bans; compliance with exis
EPR (Extended Producer Responsibility)	Introduced the concept of EPR for PIBOs.	Centralized registration portal established (CPCB).	Detailed, legally binding EPR guidelines with mandatory targets, credit mechanisms, and environmental compensation for non-compliance.	Expanded defin "Producer/Importer/M to cover more entities
Biodegradable/Compostable	General rules for disposal.	No specific detailed regulation.	Recycling standard (IS14534:1998) emphasised.	Strict definiti "biodegradable" microplastics) and **CPCB certificatio** before marketing.
Reporting	Annual reports (Form 4).	Online CPCB registration for multi-state entities.	Mandates online reporting via CPCB portal for all PIBOs and processors.	Mandates quarterly online reports for all manufacturers/impor

11.15 LEGAL PROVISION IN INDIA FOR ENVIRONMENTAL PROTECTION

Legal provisions to control environmental pollution existed in India even before independence but did not play a significant role in protecting the environment.

11.15.1 Constitutional Provisions

During the 1970s, through the Stockholm Conference, every nation understood the importance of the environment, so they started various constitutional provisions too apart from various other initiatives. **In India under Article 48A (Directive Principles of State Policy) and Article 51A(g).**

Article 48A 'The State shall endeavour to protect and improve the environment and to safeguard forests and wildlife of the country.' *(Directive Principles of State Policy)*.

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Article 51A(g) provides: 'It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures' *(Fundamental Duties of a citizen)*.

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11.15.2 Directive Principles

The Directive Principles under **Articles 49** and **51A (f)** also recognise the importance of protecting the sites of cultural heritage as a part of the total environment. Our Constitution provides necessary safeguards for framing and enforcing environmental legislation.

11.15.3 Fundamental Rights

The Parliament has passed several laws to control pollution and to protect the environment; for instance, Man's fundamental right to life includes the right to a wholesome environment. If this right is obstructed, he can move the court for a constitutional remedy. A Petition may be filed before the Supreme Court and the High Courts, under Articles 32 and 226 respectively of the Constitution of India, to issue orders or directions or writs wherever there is violation of environmental laws or other such offences.

In the case of violation of fundamental right

- **Art. 32:** Remedies for enforcement of fundamental rights including writs (Supreme Court)
- **Art. 226:** Power of high courts to issue certain writs.

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Environmental Movements and Environmentalists of India

12

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12.2 Environmentalists of India

12.2.1 Sunderlal Bahuguna

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12.2.5 Baba Ampte
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12.1 ENVIRONMENTAL MOVEMENTS IN INDIA

- An environmental movement can be defined as a social or political movement, for the conservation of environment or for the improvement of the state of the environment.
- At present Environmental movements in India that focus on dams, displacement and resettlement, effectively articulated their agenda on the human consequences of tampering with the courses of natural resources. They have initiated protest action against the forces and agencies responsible for environmental degradation.
- In India, it is also seen that the ethnic practices of worshipping plants, trees, forests, and rivers reflect the natural and social domains and the wisdom of seeing unity in the living and the non-living world in the Indian tradition.
- The environmental movements in India encapsulate all categories of caste, class, race, religion, nations and also categories of species divisions and the divisions of the organic and inorganic world.

12.1.1 Bishnoi Movement

- **Bishnoi is a religious sect found in the Western Thar Desert and northern states of India.**
- **It was founded by Guru Maharaj Jambaji in 1485 AD in the Marwar (Jodhpur) desert region of western Rajasthan, India.**

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- Jambaji witnessed persistent cutting of trees during drought and their eventual death drought continued.
- He also recognised the importance of trees within his local ecosystem (keeping animals alive) and banned cutting down green trees and killing birds or animals.
- **It is a non-violent community of nature worshippers. They are highly committed towards environment conservation.**
- In 1730, the Maharajah (king) of Jodhpur sent some soldiers to gather wood from the forest near the village of Khejarli.
- When solders began to cut trees, the Bishnoi's protested in anguish, but were ignored by the soldiers, so they decided to literally hug the trees, proclaiming, "A chopped head is cheaper than a felled tree." They were beheaded by the soldiers.
- This voluntary martyrdom continued until 363 Bishnoi villagers were killed in the name of the sacred Khejarli forest. Amrita Devi was one of them who sacrificed her life along with her three daughters.
- **This movement started around 1700 AD against deforestation.**
- The Chipko movement initiated in 1973 was an inspiration by the story of Amrita Devi.
- When the king of this region came to know the protest and killing, he rushed to the village and apologised, and declared the region a protected area.
- It is noteworthy that this legislation still exists today.

12.1.2 Narmada Bachao Movement

- The most popular movement in the environmental history of India is the movement against the Narmada River Valley Project.
- The Narmada is the largest west flowing river on the Indian peninsula.
- The Narmada covers its 1,312 km long course to the Arabian Sea through lovely forested hills, rich agricultural plains and narrow rocky gorges in a series of falls.

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- **Narmada Bachao Andolan was initiated by Medha Patkar** along with other colleagues.
- She visited the dam site in 1985.
- Due to environmental and rehabilitation issues, she obtained the Supreme Court stay order in the year 1996.
- According to their argument, the dam did not follow the necessary environmental and social conditions as per the Ministry of Environment and Forests guidelines.
- So, they desired to cut the World Bank finance for the project.
- NBA arranged numerous mass protests against the project which gathered international attention.
- **This movement was centered around the issue of human rights.**
- The main leaders of the movement, like Medha Patkar, were working towards proper rehabilitation programme for the dam displaced.
- **Due to improper implementation of the rehabilitation programmes by the state, the human rights activists have become the articulators of antidam protests.**
- Their demands included complete stopping of the dam, resettlement and rehabilitation benefits to the oustees.
- The movement, however, gained wider public attention with mobilisation and organisation of oustees (mostly tribals) and the joining of eminent social workers like Baba Amte, Sunderlal Bahuguna and Medha Patkar.
- Though its wider public attention is due to its coverage (impact) in three states, the most notable feature of this movement is the international support it has received.
- **The mode of campaign under NBA includes court actions, hunger strikes, rallies and gathering of notable film and art personalities.**
- The NBA, with its leading spokespersons Medha Patkar and Baba Amte, received the Right Livelihood Award in 1991.
- The issues of land for the displaced, the rehabilitation policy at national level and development planning without displacement have become national issues with NBA interventions, influencing policy-making and mass movements.

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- NBA has been effective in its multiple strategies at the executive, legislative and judicial level, campaigning against the destruction and displacement caused by large dams and for the rights of the affected people farmers, labourers, fishermen and others.

12.1.3 Chipko Movement

- **It was started in 1973 in Chamoli district of Uttarakhand.**
- **Sunderlal Bahuguna, Gaura Devi and Chandi Prasad Bhatt were prominent leaders of this movement.**
- **Chipko movement was against deforestation.**
- Chipko is one of the globally known environmental movements in India.
- Chipko movement attracted the attention of the world on the environmental problems of the Alaknanda catchment area in the mid Western Himalayas.
- Chipko movement, launched to protect the Himalayan forests from destruction, has its roots in the pre-independence days.
- Many struggles were organised to protest against the colonial forest policy during the early decades of the twentieth century.
- **The main demand of the people in these protests was that the benefits of the forests, especially the right to fodder, should go to local people.**
- In early 1973, the forest department allotted ash trees to a private company.
- This incident provoked the Dasholi Gram Swarajya Sangha (DGSS), a local cooperative organisation to fight against this injustice by lying down in front of timber trucks and burning resin and timber depots as was done in the Quit India Movement.
- When these methods were found unsatisfactory, Chandi Prasad Bhatt, one of the leaders, suggested embracing the trees to prevent them from being cut in a meeting in the Mandal on April 1, 1973.
- With its success, the movement has spread to other neighboring areas, and then onwards the movement is internationally known as Chipko

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movement.

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- The name of the movement, ‘Chipko’, comes from the word ‘embrace’, in Hindi.
- It is said that the villagers hugged or embraced or stuck to the trees in the forest to prevent them from being cut by the contractors.
- The name ‘Chipko’ was derived from the consensual strategy of clinging on to the trees as a non-violent direct action.
- The other demands included 1) on the basis of minimum needs of the people, a reorganisation of traditional rights should take place; 2) arid forest should be made green with people’s participation and increased tree cultivation; 3) village committees should be formed to manage forests; 4) forest related home-based industries should be developed and the raw materials, money and technique for it should be made available; and 5) based on local conditions and requirements, local varieties should be given priority in afforestation.
- The most notable characteristics of this movement were the involvement of women.
- Chandi Prasad Bhatt was bestowed the Ramon Magsaysay Award in 1982, and Sundarlal Bahuguna was awarded the Padma Vibhushan in 2009.

12.1.4 Appiko Movement

- Appiko Movement is India’s one of the forest- based environmental movements,often looked at as a continuation of the Chipko Movement.
- **The movement took place in the Uttara Kannada district of Karnataka in the Western Ghats.**
- It can be said that Appiko movement is the southern version of the Chipko movement.
- The Appiko Movement was locally known as ‘Appiko Chaluvali’.
- The locals embraced the trees which were to be cut by contractors of the forest department.

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- The Appiko movement used various techniques to raise awareness such as foot marches in the interior forest, slide shows, folk dances, street plays etc.
- The second area of the movement's work was to promote afforestation on denuded lands.
- The movement later focused on the rational use of ecosphere by introducing alternative energy resources to reduce pressure on the forest.
- The Appiko Movement has forced the government to amend its forest policy.
- Some specific amendments include a ban on clear felling, no further dispensing of discounts to logging companies, and suspension on felling of green trees in the tropical rainforest of the Western Ghats.
- It has produced awareness among villagers throughout the Western Ghats about the environmental destruction of forest prosperity.
- People now closely observe the misuse of forests by the forest department and have been able to display the inconsistency between the professed and actual practice of forest management.

12.1.5 Silent Valley Movement

- The Silent Valley Movement is one of India's most important ecological movements.
- Silent Valley is a narrow valley of the Kunthipuzha River in the state of Kerala in the Malabar region at the southern end of the Western Ghats.
- It stands at an elevation between 2400 and 1000 meters.
- It has 8950 hectares of rainforest, one of the few remaining rainforests in India, with valuable flora and fauna.
- In 1973, the then state government of Kerala decided to build a dam across a gorge in the Kunthipuzha River, which flows through the Silent Valley.
- The proposed project would generate 200 MW of electricity and form the basis for regional economic development.

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- However, the proposed project was not ecologically viable, as it would drown a chunk of the valuable rainforest of the valley and threaten the life of a host of endangered species of both flora and fauna.
- Hence, by 1979, students, rural school teachers, science forums, journalists, citizens, and voluntary organisations made a well-reasoned case against the project.
- As a result, an organisation called Save Silent Valley Committee emerged which spearheaded the protest against the construction of the dam across the Kunthi River in the Silent Valley.
- The central issues of the Silent Valley protests included the protection of the tropical rainforest, maintenance of the ecological balance, and an opposition destructive development.
- The campaigns and petitions were the main strategies adopted by the activists in the movement, basing it on the non-violent, Gandhian ideological orientation.
- The movement also portrays how the coalition was formed of the people from different professions who came together to fight against the project.
- It has also enmeshed several themes within it such as the protest against the destruction of forest, an opposition to ecologically unsustainable development, and above all, maintenance of the ecological balance.
- This diversity of the theme attracted people belonging to the different walks of life, thereby giving energy and vigor to the movement to move and to transform.
- In 1980, the M.G.K. Menon Committee set up to review the plan, came out with suggestion to scrap it. This grassroot programme became the substratum of Indian environmental activism.

12.1.6 Jungle Bachao Movement

- The tribal community of **Singhbhum district of Jharkhand** (during the British Raj, it was part of the Chota Nagpur Division of the Bengal Presidency) agitated against the forest policy of the Government in 1982.

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- The Government wanted to replace the natural soil, forests with the high-priced teak.
- **Many environmentalists refer to this movement as Greed Game Political Populism.**

12.1.7 Baliapal Movement

- The popularly known Baliapal Movement is a **struggle against the establishment of the National** Testing Range (NTR), India's first missile testing project in the Baliapal and Bhograi blocks of Odisha's Balasore district.
- This movement was against testing of missiles on the land of Baliypal village which may result in infertile soil.
- On the east coast of India, along the Bay of Bengal, a small part of land has become the place for a life or death fight waged by laborer farmers and fisher folk against India's central government and military formation.
- In the Baliapal and Bhograi village area of Baleshwar district, the populations of 54 villages (approximately 45,000 people, according to government estimate) were facing compulsory expulsion and repositioning from their homes and lands to allow governrnent's National Testing Range-site for the testing and launching of satellites, rockets and long-range missiles.
- Like most other environmental movements in India, the Baliapal movement also followed a non-violent path of resistance.
- The ideology of the movement draws upon the cultural and economic dimensions of the peasants' everyday reality, and hence, more Gandhian in its appearance.

12.1.8 Tehri Dam Movement

- One of the most protracted and a celebrated environmental movement in recent years is the movement against the Tehri Dam, which is being built on the **Bhagirathi River in Garhwal, Uttaranchal.**

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- **The Tehri Dam is the tallest dam in India, 2nd highest in Asia (in all dam types) and the 12th highest in world.**
- The Tehri Dam, **partly funded by Russia,** is a major hydroelectric project that was conceived long back in 1949 by the Geological Survey of India.
- However, it was only in 1963 that detailed investigations to this end were made.
- After the visit of the then Minister for Power and Irrigation, Shri K. L. Rao, in 1965, the site was finally confirmed.
- By 1967, several experts, both of India and abroad, visited the site and recorded their opinions.
- The cost of the project as initially prepared in 1967 was estimated at Rs. 126.8 crores.
- However, the figure moved up to Rs. 3000 crores in 1989 and in March 1990, it went up to Rs. 4142 crores.
- The projected dam height is 260.5 meters, impounding 3.22-million cubic metre of water.
- The construction of the Tehri Dam is opposed on the ground of seismic data projecting earthquake disaster and displacement of the people of old Tehri town and the neighbouring villages.
- **The Tehri dam is situated in the Central Himalayan Seismic Gap (Zone V), a major geologic fault zone.**
- This project has dishonored concerns about the ecological consequences of localising a large darn in the delicate ecosystem of the Himalayan foothills.
- The anti-Tehri Dam movement is spearheaded by Tehri Baandh Virodhi Sangharsh Samiti (Committee for the Struggle against the Tehri Dam) founded by prominent leaders like V. D. Saklani, Sunderlal Bahuguna and other leaders active in the movement.
- The movement continued from 1980s to 2004.
- The core of the conflict was against the displacement of town inhabitants and environmental consequences of the weak ecosystem.

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12.1.9 Navdanya Movement, 1982

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- It's about authorising women or anti globalisation campaigns.
- Ecological activist Vandana Shiva has always had an advantage in her fights against the authorities.
- Her eco-feminist drivers stored a farming system positioned on engaging women, altering the present system.
- She originated Navdanya in 1982, an organisation encouraging biodiversity conservation and organic farming.
- It has helped create markets for farmers, stimulated quality food for consumers, and linked the seed to cooked food.
- In 2000, Navdanya launched the Jal Swaraj Movement to defend our water from denationalisation and to promote old-style water harvesting systems and reasonable access to water.
- Navdanya is dedicated to reducing the cost of agricultural inputs by helping seed banks and supporting farmer re-education and teaching in low cost eco friendly, native and ancestral farming methods.
- Low cost organic agriculture permits farmers to feed their families and avoid indebtedness, poverty and hunger.

12.2 ENVIRONMENTALISTS OF INDIA

12.2.1 Sunderlal Bahuguna

- Born on January 9, 1927 Sunderlal is a renowned environmentalist, Chipko Movement leader and a follower of Mahatma Gandhi's philosophy of Non-violence and Satyagraha. He has struggled for the conservation of forests in the Himalayas, as a first member of the Chipko movement in 1970s, and later directed the Anti-Tehri Dam movement starting on 1980s to early 2004. He was awarded the Padma Vibhushan, India's second highest civilian honour, on January 26, 2009.

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12.2.2 Gaura Devi

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- Born in 1925 in a tribal Marchha family of Lata village in Neeti valley of Chamoli district, she was actively involved in the panchayat and other community work.
- She was elected as the president of the Mahila Mangal Dal, having responsibilities of ensuring cleanliness in the village and the protection of community forests.
- On the historic day of March 26th, 1974, she moved forward to stop cutting of trees by the forest officials.
- 27 women and girls from Reni village led by Gaura Devi saved their shared forest, initiating a decade of Chipko direct movements by women throughout the Uttarakhand region.

12.2.3 Chandi Prasad Bhatt

- Born in 1934, he is an Indian Gandhian environmentalist and societal activist, who started Dasholi Gram Swarajya Sangh (DGSS) in Gopeshwar in 1964. DGSS was later developed as the mother-organisation to the Chipko Movement, in which he was one of the forerunners, and for which he was awarded Ramon Magsaysay Award for Community Leadership in 1982, followed by the Padma Bhushan in 2005.
- Now he is acknowledged for his work on subaltern social ecology and is considered as one of the recent environmentalists of India.Gandhi Peace Prize of 2013 is given to Chandi Prasad Bhatt.

12.2.4 Medha Patekar

- She was born on 1st December 1954, in Mumbai.
- Medha Patkar was keen on public service from a very young age.
- She is known for her active role in Narmada Bachao Andolan. Violence was never in her agitation, those were a complete reconstruction of Gandhiji's Non-Violence Movement.

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12.2.5 Baba Ampte

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- Born on December 26th, 1914 in Wardha district of Maharashtra, Ampte was a lawyer and contributed to the freedom struggle against the British Empire.
- Baba Ampte will also be remembered for his peace and justice walks across India, his unity with the aborigine tribes, and his disapproval of big dams like the Sardar Sarovar project on Narmada River.

12.2.6 Arundhati Roy

- She was born on November 24th, 1961 in Shillong.
- Arundhati Roy is an Indian author, best known for the 1997 Man Booker Prize winning novel 'The God of Small Things'.
- Along with a group of activists known for the Narmada Bachao Andolan, she protested against the construction of dam on the Narmada River. Many people got displaced from this area without any compensation. State governments involved claimed that they had no land to give them compensation.

12.2.7 Vandana Shiva

- Vandana Shiva was born on 5 November 1952, in Dehradun, Uttarakhand.
- Shiva has donated ideas and carried out activist campaigns. She has assisted grassroots institutions of the Green movement in Africa, Asia, Latin America, Ireland, Switzerland and Austria with campaigns against genetic engineering.
- In 1987, she established the Research Foundation for Science, Technology and Ecology striking outsourced claims. This led to the formation of Navdanya in 1991, a national drive to safeguard the diversity and integrity of living resources, particularly native seed, the advancement of organic farming and fair trade.

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Do You Know?

Navdanya was founded on 1987, but was registered as a trust on 1991.

- In the field of intellectual property rights and biodiversity, Shiva and her team at the Research Foundation for Science, Technology and Ecology dared the biopiracy of Neem, Basmati and Wheat. She has also attended expert groups of government on Biodiversity and IPR legislation.

12.2.8 Kallen Pokkudan

- Born on 1930 in Kerala, he is an environmental activist in Kerala. He has been working for the protection of mangrove forests since 1989.
- He has won 7th P S Gopinathan Nair Environmental Award for 2014.

12.2.9 John C. Jacob

- Born on 11th October 1936, at Nattakam in Kottayam, he was one of the pioneers of the environmental movement in Kerala, India.
- At Payyanur, in 1972 he started a zoology club that would form the genesis of campus-based nature conservation activities. The club also became involved in protesting threats to the environment.
- He has also set up the Society for Environment Education, Kerala (SEEK), which published an environment-focused magazine Soochimukhi. Additionally, Jacob led the movement against the Silent Valley project in South Kerala.
- In recognition to his contributions to the environment, he was awarded the Kerala Biodiversity Board's Haritham Award and the state government's Vana Mitra Award.

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Questions Asked in UPSC Prelims Exam

UNIT IV

Unit - IV Prelims PYQs (2011 onwards)

1. In the context of WHO Air Quality Guidelines, consider the following statements: **[UPSC (Pre) – 2022]**

1. The 24-hour mean of PM2.5 should not exceed 15 ug/m³ and annual mean of PM 2.5 should not exceed 5 µg/m³.
2. In a year, the highest levels of ozone pollution occur during the periods of inclement weather.
3. PM10 can penetrate the lung barrier and enter the bloodstream.
4. Excessive ozone in the air can trigger asthma.

Which of the statements given above are correct?

(a) 1, 3 and 4

(b)1 and 4 only

(c) 2, 3 and 4

(d)1 and 2 only

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2. Which one of the following has been constituted under the Environment (Protection) Act, 1986? **[UPSC (Pre) – 2022]**

(a) Central Water Commission
(b) Central Ground Water Board
(c) Central Ground Water Authority
(d) National Water Development Agency

3. At the national level, which ministry is the nodal agency to ensure effective implementation of the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006? **[UPSC (Pre) – 2021]**

(a) Ministry of Environment, Forest and Climate Change
(b) Ministry of Panchayati Raj
(c) Ministry of Rural Development
(d) Ministry of Tribal Affairs

4. Consider the following statements: **[UPSC (Pre) – 2020]**

1. 36% of India's districts are classified as 'overexploited' or 'critical' by the Central Ground Water Authority (CGWA).
2. CGWA was formed under the Environment (Protection) Act.
3. India has the largest area under groundwater irrigation in the world.

Which of the statement/s given above is/are correct?

(a) 1 only
(b)2 and 3 only
(c) 2 only
(d)1 and 3 only

5. Consider the following statements: **[UPSC (Pre) – 2019]**

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1. As per the recent amendment to the Indian Forest Act, 1927, forest dwellers have the right to fell the bamboos grown on forest areas.
2. As per the Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, bamboo is a minor forest produce.
3. The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 allows ownership of minor forest produce to forest dwellers.

Which of the statement/s given above is/are correct?

(a) 1 and 2 only
(b)2 and 3 only
(c) 3 only
(d)1, 2 and 3

6. As per the Solid Waste Management Rules, 2016 in India, which one of the following statements is correct? [UPSC (Pre) – 2019]

(a) The waste generator has to segregate waste into five categories.
(b) The rules are applicable to notified urban local bodies, notified towns and all industrial townships only.
(c) The rules provide for exact and elaborate criteria for the identification of sites for landfills and waste processing facilities.
(d) It is mandatory on the part of waste generator that the waste generated in one district cannot be moved to another district.

7. Consider the following statements: The Environment Protection Act, 1986 empowers the Government of India to[UPSC (Pre) – 2019]

1. State the requirement of public participation in the process of environmental protection, and the procedure and manner in which it is sought.

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2. Lay down the standards for emission or discharge of environmental pollutants from various sources.

Which of the statement/s given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

8. Consider the following statements: **[UPSC (Pre) – 2019]**

1. As per law, the Compensatory Afforestation Fund Management and Planning Authority exists at both National and State levels.
2. People's participation is mandatory in the compensatory afforestation programmes carried out under the Compensatory Afforestation Fund Act, 2016.

Which of the statement/s given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

9. Consider the following statements: **[UPSC (Pre) – 2019]**

1. Under Ramsar Convention, it is mandatory on the part of the Government of India to protect and conserve all the wetlands in the territory of India.
2. The Wetlands (Conservation and Management) Rules, 2010 were framed by the Government of India based on the recommendations of the Ramsar Convention.
3. The Wetlands (Conservation and Management) Rules, 2010 also encompass the drainage area or catchment regions of the wetlands as

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determined by the authority.

Which of the statement/s given above is/are correct?

(a) 1 and 2 only

(b)2 and 3 only

(c) 3 only

(d)1, 2 and 3

10. How is the National Green Tribunal (NGT) different from the Central Pollution Control Board (CPCB)? **[UPSC (Pre) – 2018]**

1. The NGT has been established by an Act whereas the CPCB has been created by an executive order of the Government.
2. The NGT provides environmental justice and helps reduce the burden of litigation in the higher courts whereas the CPCB promotes cleanliness of streams and wells, and aims to improve the quality of air in the country.

Which of the statement/s given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

11. Consider the following statements: **[UPSC (Pre) – 2018]**

1. The definition of 'Critical Wildlife Habitat' is incorporated in the Forest Rights Act, 2006.
2. For the first time in India, Baigas have been given Habitat Rights.
3. Union Ministry of Environment, Forest and Climate Change officially decides and declares Habitat Rights for Primitive and Vulnerable Tribal Groups in any part of India.

Which of the statement/s given above is/are correct?

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(a) 1 and 2 only
(b) 2 and 3 only
(c) 3 only
(d) 1, 2 and 3

12. Which of the following best describe the aim of 'Green India Mission' of the Government of India? **[UPSC (Pre) – 2016]**

1. Incorporating environmental benefits and costs into the Union and State Budgets, and thereby implementing the 'green accounting'.
2. Launching the second green revolution to enhance agricultural output so as to ensure food security to one and all in future.
3. Restoring and enhancing forest cover and responsibility to climate change by a combination of adaption and mitigation measures.

Select the correct answer using the codes given below:
(a) 1 only
(b)2 and 3 only
(c) 3 only
(d)1, 2 and 3

13. The Genetic Engineering Appraisal Committee is constituted under the **[UPSC (Pre) – 2015]**

(a) Food Safety and Standards Act, 2006
(b) Geographical Indications of Goods (Registration and Protection) Act, 1999
(c) Environment (Protection) Act, 1986
(d) Wildlife (Protection) Act, 1972

14. With reference to 'Eco-Sensitive Zones', which of the following statement/s is/are correct? **[UPSC (Pre) – 2014]**

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1. Eco-Sensitive Zones are the areas that are declared under the Wildlife (Protection) Act, 1972.
2. The purpose of the declaration of Eco- Sensitive Zones is to prohibit all kinds of human activities in those zones except agriculture.

Select the correct answer using the codes given below.

(a) 1 only

(b)2 only

(c) Both 1 and 2

(d)Neither 1 nor 2

15. The National Green Tribunal Act, 2010 was enacted in consonance with which of the following provisions of the Constitution of India? [UPSC (Pre) – 2012]

1. Right to a healthy environment, construed as a part of Right to Life under Article 21.
2. Provision of grants for raising the level of administration in the Scheduled Areas for the welfare of Scheduled Tribes under Article 275(1).
3. Powers and functions of Gram Sabha as mentioned under Article 243(A).

Select the correct answer using the codes given below:

(a) 1 Only

(b)2 and 3

(c) 1 and 3

(d)1, 2 and 3

16. How does National Biodiversity Authority (NBA) help in protecting the Indian agriculture? [UPSC (Pre) – 2012]

1. NBA checks the biopiracy and protects the indigenous and traditional genetic resources.

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2. NBA directly monitors and supervises the scientific research on the genetic modification of crop plants.
3. Application for Intellectual Property Rights related to genetic/biological resources cannot be made without the approval of NBA.

Which of the statement/s given above is/are correct?

(a) 1 Only
(b)2 and 3
(c) 1 and 3
(d)1, 2 and 3

17. If the National Water Mission is properly and completely implemented, how will it impact the country? [UPSC (Pre) – 2012]

1. Part of the water needs of urban areas will be met through the recycling of wastewater.
2. The water requirements of coastal cities with inadequate alternative sources of water will be met by adopting appropriate technologies that allow the use of ocean water.
3. All the rivers of Himalayan origin will be linked to the rivers of peninsular India.
4. The expenses incurred by the farmers for digging borewells for installing motors, and pump-sets to draw groundwater will be completely reimbursed by the Government.

Select the correct answer using the codes given below:

(a) 1 Only
(b) 1 and 2
(c). 3 and 4
(d) 1, 2, 3 and 4

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Answers

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1. (b) 2. (c) 3. (d) 4. (b) 5. (b) 6. (c) 7. (b) 8. (a) 9. (c) 10. (b)
11. (a) 12. (c) 13. (c) 14. (d) 15. (a) 16. (c) 17. (b)

Explanations

1. Correct answer (b): EXPLANATION

The 24-hour mean of PM2.5 should not exceed 15 µg/m³ and the annual mean should not exceed 5 µg/m³ according to the 2021 WHO Global Air Quality Guidelines. **Statement 1 is right**

Ground-level ozone (O3) forms from reactions between pollutants in sunlight, so it peaks on hot, sunny, and stagnant days, not during rain or storms (inclement weather). **Statement 2 is wrong**

Particles of 10 micrometers or less (PM10) can lodge deep in the lungs, but it's the even smaller particles (PM2.5) that are fine enough to cross into the bloodstream. **Statement 3 is wrong**

High concentrations of ground-level ozone are known to cause respiratory problems, including triggering asthma attacks. **Statement 4 is right**

2. Correct answer (c): EXPLANATION

Central Ground Water Authority (CGWA): Established in January 1997 under Section 3(3) of the Environment (Protection) Act, 1986, to manage groundwater resources.

Central Water Commission (CWC): Established under the Water (Prevention and Control of Pollution) Act, 1974, not the EPA.

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Central Ground Water Board (CGWB): A subordinate office under the Ministry of Water Resources, established by renaming the Exploratory Tube wells Organization in 1970. **Option (c) is right**
National Water Development Agency (NWDA): A registered society under the Societies Registration Act, 1860.

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3. **Correct answer (d): EXPLANATION**
The nodal ministry at the national level for the Forest Rights Act (FRA), 2006, is the **(d) Ministry of Tribal Affairs**, as it primarily deals with empowering tribal communities and forest dwellers by recognizing their traditional rights over forest lands, though the Ministry of Environment, Forest and Climate Change (MoEFCC) and state governments also play crucial roles in implementation.
   - ***Ministry of Tribal Affairs*** **(MoTA):** Acts as the nodal agency for overall coordination and policy at the central level.
   - ***State Governments*****:** Responsible for on-the-ground implementation, processing claims, and issuing titles.
   - ***Gram Sabhas*****:** Crucial local bodies that play a vital role in the claim and verification process.

4. **Correct answer (b): EXPLANATION**
   - The Central Ground Water Authority (CGWA) classifies ‘assessment units’ (blocks, mandals, talukas), not entire districts. The percentage of overexploited/critical units is around 17-20%, not 36% of districts. **Statement 1 is wrong**.
   - The CGWA was constituted under Section 3(3) of the **Environment (Protection) Act**, 1986, to regulate and control the development and management of groundwater resources. **Statement 2 is right**.

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- India has the largest area under groundwater irrigation in the world, utilizing more groundwater for agriculture than any other country. **Statement 3 is right**.

5. Correct answer (b): EXPLANATION

- The 2017 amendment to the Indian Forest Act, 1927, removed bamboo from the definition of "tree," easing restrictions on bamboo in *non-forest areas*, but it did not grant forest dwellers the right to fell bamboo in forest areas; those remain subject to existing laws. **Statement (1) Incorrect.**
- The FRA, 2006, explicitly lists bamboo as a Minor Forest Produce (MFP). **Statement (2) Correct.**
- The FRA, 2006, grants forest dwellers the right of ownership, access, collection, use, and disposal of Minor Forest Produce, including bamboo, traditionally collected by them. **Statement (3) Correct.**

6. Correct answer (c): EXPLANATION

- The Solid Waste Management Rules, 2016, place significant emphasis on scientific waste disposal and management. Schedule I of the rules outlines detailed and specific criteria for the identification of sites for landfills and waste processing facilities to prevent environmental pollution and health hazards. **Statement (c) is right**

7. Correct answer (b): EXPLANATION

- The Environment (Protection) Act, 1986 empowers the Central Government to lay down standards for the emission or discharge of environmental pollutants from various sources (Section 3(2)(iv)).

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Statement (1) is wrong

- However, it does not mandate public participation in the process of environmental protection. Public consultation requirements exist under specific rules and notifications (like the EIA Notification, 2006), not directly under the Act itself. **Statement (2) is right**

8. Correct answer (a): EXPLANATION

The Compensatory Afforestation Fund Act **(CAMPA Act), 2016** establishes institutional mechanisms at the Central and State/UT levels to ensure the **efficient, transparent, and timely use of funds** collected for compensatory afforestation when forest land is diverted for non-forest purposes, mitigating the environmental impact of such diversion.

Statement (1) right

The Act **does not** make people's participation mandatory in compensatory afforestation programmes; it allows for consultation and involvement but it is not a legal requirement.

Statement (2) wrong

9. Correct answer (c): EXPLANATION

- The Ramsar Convention encourages conservation and "wise use" but doesn't legally bind India to protect *every single* wetland; it focuses on wetlands of international importance and promoting general conservation.

Statement (1) wrong

- The Wetlands (Conservation and Management) Rules, 2010, were framed by India to regulate activities, not as a direct mandate or application of Ramsar principles, although both aim for conservation. **Statement (2) wrong**

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- The 2010 Rules specifically define "wetland" to include the direct drainage area or catchment region, recognizing the link between the surrounding land and the wetland's health. **Statement (3) right**

10. Correct answer (b): EXPLANATION

- The CPCB was established as a statutory body under the **Water (Prevention and Control of Pollution) Act, 1974**, not by an executive order of the government. Both the NGT (established by the National Green Tribunal Act, 2010) and the CPCB are statutory bodies created by Acts of Parliament.

Statement (1) wrong

- It accurately describes the fundamental difference in the roles of the two organizations.
- The **NGT** is a specialized judicial body that provides environmental justice and handles environmental disputes, which helps in reducing the burden of litigation on the High Courts and the Supreme Court.
- The **CPCB** is a regulatory and technical body with functions including promoting cleanliness of streams and wells, and improving the quality of air in the country, as outlined in its founding acts. **Statement (2) right**

11. Correct answer (a): EXPLANATION

- The definition of 'Critical Wildlife Habitat' (CWH) is explicitly incorporated in Section 2(b) of the **Forest Rights Act, 2006** (Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006). CWHs are areas within national parks and sanctuaries required to be kept as inviolate for wildlife conservation based on scientific evidence. **Statement 1 is right**

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- The Baigas tribal community in Madhya Pradesh became the first community in India to be officially granted Habitat Rights under the Forest Rights Act in 2015. This was a landmark step in recognizing the traditional rights of Particularly Vulnerable Tribal Groups (PVTGs). **Statement 2 is right**
- Habitat Rights for PVTGs are recognized and vested by the respective **State Governments**, through a decentralized process involving the Gram Sabha (village assembly) and district-level committees. The Union Ministry of Tribal Affairs is the nodal ministry for the implementation of the FRA, but the Ministry of Environment, Forest and Climate Change does not officially decide and declare these rights.

Statement 3 is wrong

12. Correct answer (c): EXPLANATION

- The Green India Mission (GIM) is part of the National Action Plan on Climate Change (NAPCC) and focuses on forests and ecosystems. It is not directly related to implementing 'green accounting' in Union and State budgets, which is a broader economic concept. **Statement (1) wrong**
- The mission focuses on the forestry sector and climate change adaptation/mitigation. It is a distinct initiative from the "second green revolution," which focuses specifically on enhancing agricultural output and food security. **Statement (2) wrong**
- A primary aim of the GIM is to protect, restore, and enhance India's diminishing forest cover and respond to climate change through a combination of adaptation and mitigation measures, such as enhancing carbon sinks and improving the resilience of vulnerable ecosystems. **Statement (3) right**

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13. Correct answer (c): EXPLANATION

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The Genetic Engineering Appraisal Committee (GEAC) is a statutory body constituted under the "Rules for the Manufacture, Use/Import/Export and Storage of Hazardous Micro Organisms/Genetically Engineered Organisms or Cells, 1989" which were framed under the **Environment (Protection) Act, 1986.**

Option (c) is right

14. Correct answer (d): EXPLANATION

- Eco-Sensitive Zones (ESZs) are not declared under the Wildlife (Protection) Act, 1972. They are notified by the Ministry of Environment, Forest and Climate Change (MoEFCC) under the **Environment (Protection) Act, 1986**. **Statement (1) wrong**
- The purpose of declaring ESZs is not to prohibit all kinds of human activities. Instead, the aim is to *regulate* and manage certain activities to minimize their negative impact on the nearby protected areas. While many polluting activities like commercial mining and sawmills are prohibited, various activities are permitted or regulated, including ongoing agricultural and horticultural practices, rainwater harvesting, and organic farming. **Statement (2) wrong**

15. Correct answer (a): EXPLANATION

- The Supreme Court of India has interpreted the fundamental Right to Life under Article 21 of the Constitution to include the right to a clean and healthy environment. The establishment of a specialized body like the NGT was a legislative step to enforce this right effectively and provide an accessible forum for environmental justice. **Statement (1) right**

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- Article 275(1) deals with financial grants from the central government to certain states for specific welfare schemes, particularly for Scheduled Tribes and administration in Scheduled Areas. This has no direct relation to the constitutional basis for a national environmental judicial tribunal. **Statement (2) wrong**
- Article 243(A) is part of the provisions related to Panchayats (local self-governance) and defines the powers and functions of a Gram Sabha at the village level, as determined by state legislatures. This provision is not the constitutional basis for the national NGT. **Statement (3) wrong**

16. Correct answer (c): EXPLANATION

- The National Biodiversity Authority (NBA) implements the Biological Diversity Act, 2002, one of the key objectives of which is to prevent biopiracy and protect India's indigenous biological resources and associated traditional knowledge. **Statement (1) right**
- This function is primarily handled by the **Genetic Engineering Appraisal Committee (GEAC)**, which is a separate statutory body under the Ministry of Environment, Forest and Climate Change (MoEF&CC). While the NBA has an oversight role regarding the access to genetic resources used in such research, it does not directly monitor and supervise the scientific research itself. **Statement (2) wrong**
- Section 6 of the Biological Diversity Act, 2002, mandates that prior approval from the NBA must be obtained before applying for any intellectual property rights (IPR) in or outside India for an invention based on biological resources or associated knowledge obtained from India. This measure ensures the fair and equitable sharing of benefits arising from the use of such resources. **Statement (3) right**

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17. Correct answer (b): EXPLANATION

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- A key objective of the National Water Mission is to ensure that a considerable share of the water needs of urban areas is met through the **recycling of wastewater**. This is a core strategy for efficient urban water management. **Statement (1) right**
- The mission explicitly aims to meet the water requirements of coastal cities that lack adequate alternative freshwater sources by adopting appropriate technologies, such as **low-temperature desalination**, to allow for the use of ocean water. **Statement (2) right**
- This statement refers to the large-scale inter-linking of rivers project, which is a massive infrastructure plan distinct from the core objectives and implementation strategy of the National Water Mission. The mission focuses more on demand-side management, efficiency, conservation, and integrated management within existing river basins. **Statement (3) wrong**
- The National Water Mission promotes the efficient use of groundwater and participatory management, but it does not include a provision for completely reimbursing farmers' expenses for digging borewells and installing pumps to draw groundwater. In fact, the mission aims to regulate groundwater extraction and encourage sustainable practices. **Statement (4) wrong**

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UNIT V Environment Impact Assessment

Environmental Impact Assessment (EIA) 13

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13.1 ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

- Environment Impact Assessment (EIA) is an **assessment of possible environmental effects of a coming project or its development.**
- As per the **United Nations Environment Programme (UNEP),** EIA is defined as **an instrument to recognise the projects' social, economic and environmental effects before any decision-making.**
- **In India, EIA has been made mandatory under the Environment Protection Act, 1986.**
- The **assessment is carried out by an Expert Appraisal Committee (EAC)**, which consists of scientists and project management experts.

Meaning of EIA

- **EIA is an important tool for integrating the objectives of environmental management into the decision-making process to ensure environmentally grounded and sustainable developments.**

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Objectives of EIA

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- The main objective of EIA is to **maximise the positive environmental impacts and minimise the adverse impacts.**

13.2 TYPES OF EIA

- The EIA types are as follows
  Comprehensive EIA: It consists of an annual data estimation of all four seasons.
  Rapid EIA: It consists of data estimation for one season of the year.

Do You Know?

History of EIA

- **EIA was formalised and established in the world at the United Nations Conference on the Environment at Stockholm in the year 1972.**
- The USA decided to establish the National Environmental Policy Act in 1970 for environmental protection, and hence, the **USA became the first country to enact legislation on EIA, and after that, Australia adopted EIA in 1970, Canada and New Zealand in 1974**.
- **Presently, all developed countries have enacted legislation on EIA, whereas a few evolving countries are in the process of adopting and making laws on EIA.**

13.3 EIA IN INDIA

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In India, **EIA was started in 1976-77 to examine the river valley projects** from an environmental angle. But during that time lacked legislative support.

1986: After the 1984 Bhopal gas tragedy, the Environment Protection Act was passed to protect the environment and punish those harming it.

1994: The Environmental Impact Assessment notification of 1994 is the first subordinate legislation that makes environmental clearance mandatory for building, expanding and modernising projects. 1994 rules were subsequently amended 70 times.

Do You Know?

Environmental Impact Assessment (EIA) is a critical process in India, governed by the **Environmental Protection Act, 1986**, and the **EIA Notification, 2006**, aimed at evaluating the potential environmental impacts of major projects. Environmental NGOs and activists play a pivotal role in influencing EIA outcomes by raising public awareness, contesting flawed data, offering legal assistance, and amplifying the concerns of affected communities.

2006: The EIA notification issued in September becomes the main tool to minimise the adverse impact of rapid industrialisation on the environment. Nearly 50 amendments and 230 changes are introduced subsequently.

March 2020: The Ministry of Environment, Forests and Climate Change puts out a draft EIA to replace the 2006 notification. The government says this is a streamlined notification but environmentalists claim it further dilutes the process.

On Jan, 1994 EIA notification [under the Environment of (Projection) act (1986)] was issued by which the EIA process became a **Statutory requirement**, rather than an administrative requirement for

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several projects/activities likely to **have significant environmental impacts & health implications.**

- Since then, the EIA notification has undergone several amendments in incorporating provisions for 'Public Hearing' & bringing in several important projects/ activities into the purview of EIA, requiring 'Environmental clearance' by the Ministry of Environment & Forest (MoEF).
- **According to this notification, different projects/development activities have been divided into majority requiring 'Environmental Clearance' (EC) either from Central Govt. i.e., MoEF (Category A) and at the state level from State Environmental Impact Assessment Authority (SEIAA).**

13.4 STEPS IN ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

EIA is a systematic process in which different types of decisions are taken to reduce the harmful effects of the environment and human health.

This process has different steps:

1. **Clarity of objective: In this step, we should follow the government and other environmental quality standards.** What is to be achieved and at what level, and at what costs, has to be ascertained.
2. **Different survey analysis: To check** existing environmental conditions by using checklists. The suitability of land area, availability of water, provision for waste disposal, labour, capital, should be in mind while conducting the survey.
3. **Sample and different data storage.**

13.5 PROCEDURE OF EIA

- Once an application for environmental clearance is submitted,

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- The **proposal is examined by the Ministry and sent to expert committees** to assess the environmental impacts of the project.
- **They may visit sites** and then **recommend for approval or rejection** of the proposal.
- In case of projects, which are submitted with the required documentation, a **decision is taken within 90 days.**
- **If the application is rejected, a new application may be submitted with** the suggested modification in the project.

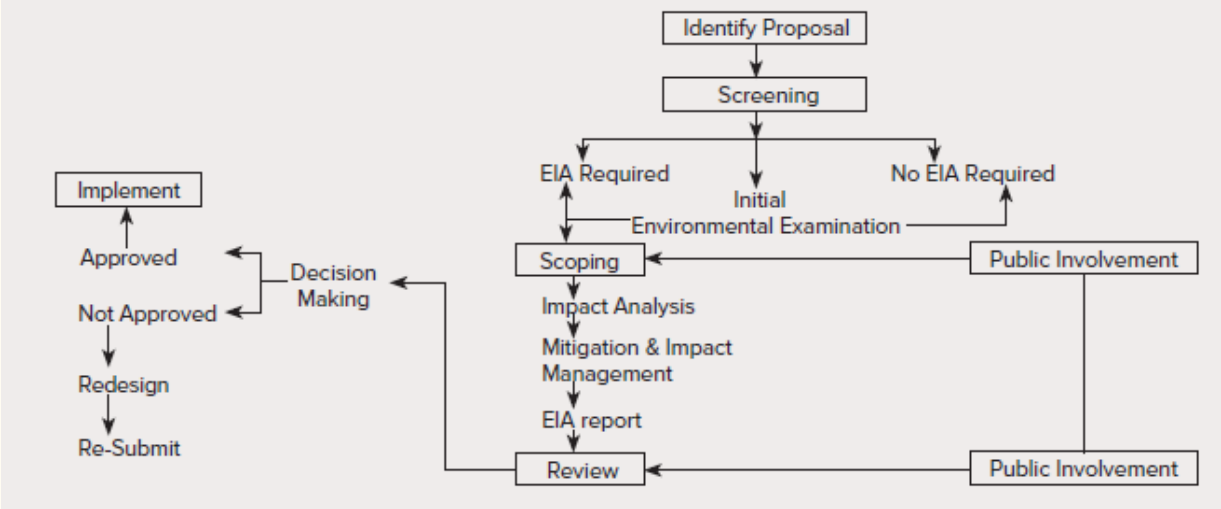

Figure 13.1 Procedure of EIA

13.6 THE EIA PROCESS IS HIGHLIGHTED IN EIGHT STEPS

The first step is to define a project and study all the likely activities involved in its process, so as to understand the range and reach of the project. This helps in deciding the possible zones of environmental impacts.

1. **Screening:** Screening is the **first and simplest process in project evaluation. It decides if the project needs EIA or not.**
2. **Scoping:** The **EIA study team**, which was organised after preliminary assessment, gets engaged in discussions with developers, investors, and

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regulatory agencies. It studies and addresses all issues of importance and the concerns raised by various groups.

3. **Impact Analysis:** The **key impacts on the environment,** such as changes in air quality, noise levels, impacts on wildlife, impact on biodiversity, impact on local communities, are identified. Predication of impacts is both qualitative and quantitative.
4. **Mitigation: This stage includes recommended actions** that can offset the adverse impacts of the project. This is done with the idea of lessening the negative effects and improving the scope for project benefits. **Mitigation measures include preventive, corrective and compensatory actions.**
5. **Reporting:** It represents **the outcome of the EIA report to be available to the decision-making body and other involved agencies.**
6. **Review of EIA**: It **inspects the EIA report's efficacy and appropriateness** and gives the required information necessary for decision-making.
7. **Decision-Making:** It determines the **project's decision whether rejected, approved or requires further change.**
8. **Post Monitoring:** This stage is applicable after the commissioning of the project.

   It ensures the project impacts do not exceed legal standards and the mitigation measures implementation is as per the prescribed EIA report.

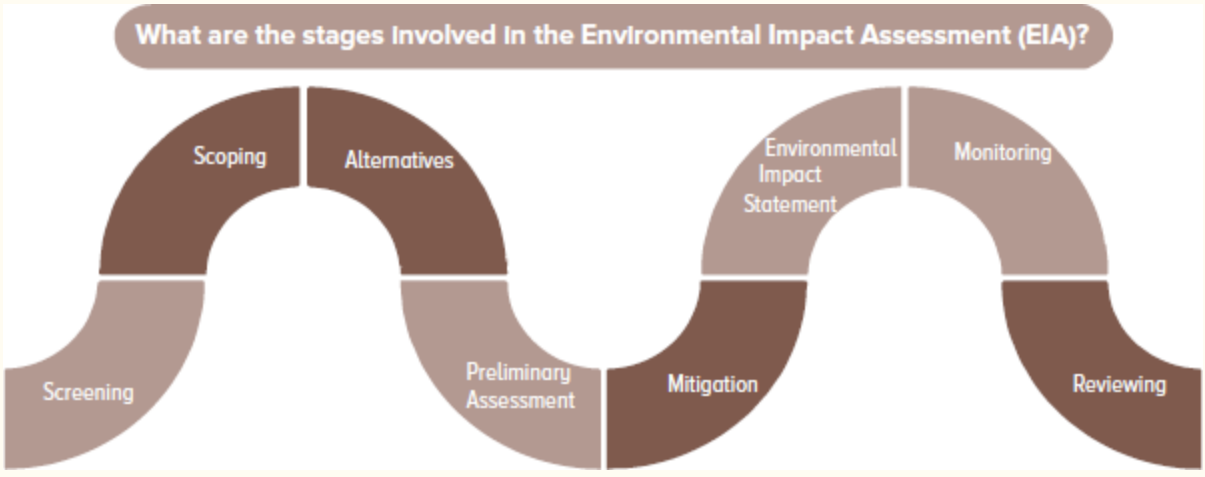

Figure 13.2 Stages Involved in the Environmental Impact Assessment (EIA)

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STAGES OF EIA (ENVIRONMENTAL IMPACT ASSESSMENT)

(11 KEY STAGES OF EIA: FROM SCREENING TO MONITORING OF EIA IN INDIA)

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1. **Screening:** Determines EIA necessity based on project size , type, and potential impact. In India, EIA is mandatory for Category A/B1 projects, and exempt for B2
2. **Scoping:** Identifies key environmental issues to be studied . The Terms of Reference (ToR) are prepared and approved, specifying the parameters for assessment
3. **Baseline Data Collection:** Collection of data on the project area's existing environmental (air, water, soil, biodiversity) and socio-economic conditions
4. **Impact Prediction & Assessment:** Predicts likely direct , indirect, and cumulative impacts during construction and operation using scientific models
5. **Evaluation of Alternatives:** Examines project options (sites, technologies, designs) and the "no-project" scenario to minimize environmental damage
6. **Mitigation Measures:** Strategies developed to avoid , minimize, or compensate for adverse project impacts, which forms the basis for the EMP
7. **Preparation of EIA Report:** Compiles all data, assessments, mitigation plans, and the EMP into the formal EIA document for submission
8. **Public Consultation:** Involves community participation through public hearings and written comments to ensure transparency and address local concerns
9. **Appraisal:** Expert committees (EAC/SEAC) review the EIA Report and public feedback to recommend approval , rejection, or further study
10. **Decision-making:** Authorities grant or deny Environmental Clearance (EC), conditions may be imposed for environmental protection
11. **Post-Clearance Monitoring:** Ensures compliance with EC conditions through periodic monitoring , audits, and inspections

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13.7 DETAILED PROCEDURE OF ENVIRONMENTAL IMPACT ASSESSMENT

- There is a detailed procedure of Environmental Impact Assessment in the country.
- It consists of the following steps to form the EIA report:

13.7.1 Screening

- This step is used to determine the project's EIA requirement and remove the projects that does not need EIA.
- In this step, the small projects are kept from complicated assessment as they too require exhaustive scrutinising, because they too can harm the environment.
- The process applies only for 'B' Category projects and for category 'A' projects.
- The Category 'B' projects are further categorised into category 'B1' and 'B2' projects by State Level expert appraisal Committee (SEAC) for further determination. Projects which require Environment Impact assessment report are termed under Category 'B1'. The remaining projects which do not require EIA report are termed under Category 'B2'.

13.7.2 Preliminary Assessment

- At this stage, there is extensive available data research pertaining to the projects and further its impact evaluation on the environment by the experts.
- Thereafter, the experts have to predict the intensity of the project impact.
- Thereafter, the project's requirement for comprehensive EIA or rapid EIA is based on evaluation of initial assessment results.
- In accordance with the authorities' decision, the preparation of EIA report is undertaken by the experts in cognisance with the other team members.

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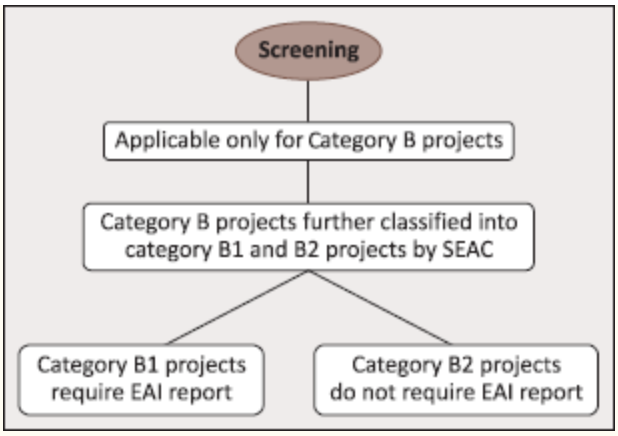

Figure 13.3 Screening Procedure

13.7.3 Scoping

- The EIA specialist team remain involved into the detailed discussion with the investors, developers, institutions, stakeholders, local people etc. to get their feedback and further addressing the issues raised by them.
- On the basis of their feedback, the team chooses the projects' critical impacts for inclusion in the comprehensive EIA report.
- The elaborated terms of reference (TORs) for the inclusion in the project's EIA report for the permission is required are prepared by the Expert Appraisal Committee regarding Category 'A' projects and State level Expert Appraisal Committee pertaining to Category 'B1' projects.
- There shall be an intimation of TORs to the applicant within sixty days (60) of the form receipt by EAC/SEAC.
- In case of application rejection by the regulating authority on the EAC/SEAC recommendations, the decision, along with the reasons within sixty days have to be communicated to the candidate from the receiving application date.

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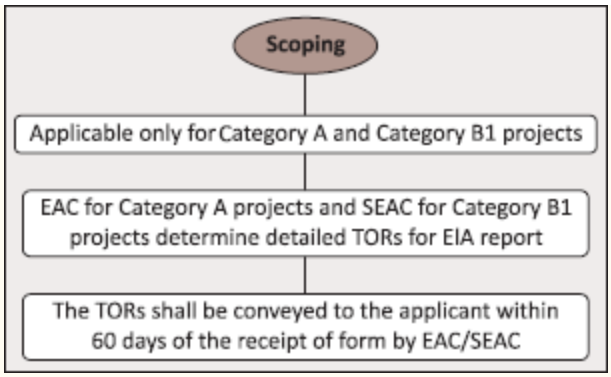

Figure 13.4 Scoping Procedure

13.7.4 Main EIA

- The main process of Environmental Impact Assessment consists of directing and finding a solution for the key notable impacts.
- Till the reaching of possible solutions, the process consists of questions series asked by experts from different agencies.
- The assessment of mitigation procedures is being undertaken after forecasting the evaluation of the projects' critical impacts and their effect on the environment.
- The different mitigation stages consist of a change in operating methods, disposal treatment system, project site, raw materials, engineering designs, waste treatment, social service, etc.
- The compensation is given in case, any persons are found to be affected.
- The EIA report is formulated on the basis of mitigation costs and measures identification.
- The project report consists of the following stages:
  - Recommended development
  - Environment Impacts
  - Involvement of major environmental issues
  - Forecasting
  - Mitigation procedures and measures
  - Composed Information Gaps and uncertainties
  - EIA process summary for the general public

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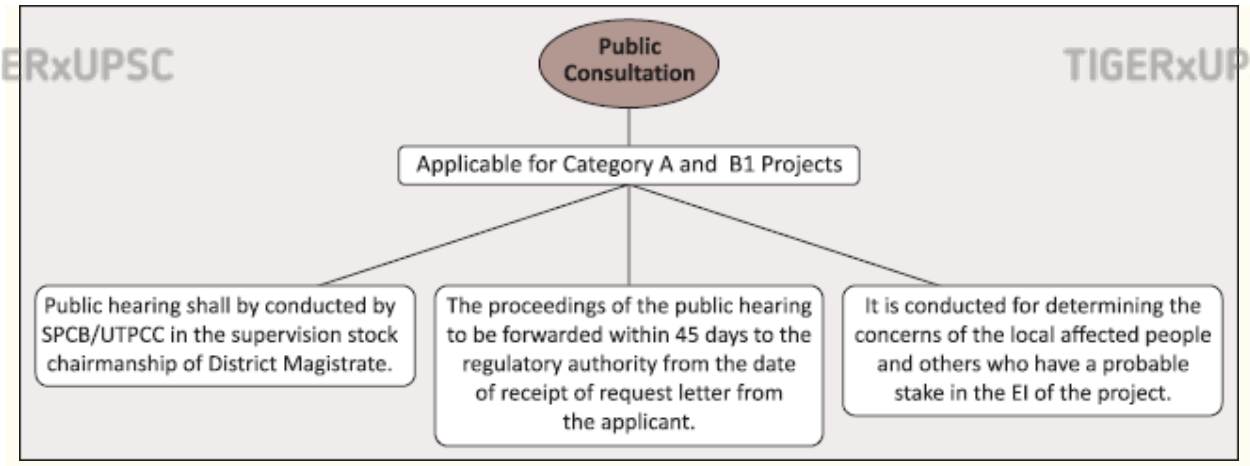

Figure 13.5 Public Consultation Procedure

13.7.5 Public Hearing

- The State Pollution Control Board conducts a public hearing at the project site or in its close proximity to discover the affected local person problems.
- The concerned persons also give written responses on the website by announcing the project within 7 days of accepting the application.
- The following projects in the country don't require any public hearing:
  - Small scale industries situated in notified industrial areas or under the industrial development authorities jurisdiction.
  - Units situated in the special economic zones and export processing zones.
- The project developer gets the NOC from the SPCB to complete the public hearing.
- In order to get environmental clearance, the application to MoEF Secretary along with the NOC has to be made.

13.7.6 Appraisal

- Appraisal means the inspection of documents like the Final EIA report, outcome of the public consultations, including public hearing proceedings by the Expert Appraisal Committee or State Level Expert Appraisal Committee for grant of ecological clearance.

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- Appraisal Committee shall make clear-cut suggestions to the concerned regulatory authority either for grant of previous ecological clearance on stipulated terms and conditions, or rejection of the application together with reasons for the same.
- The evaluation of an application should be finished by the concerned Expert Appraisal Committee or State Level Expert Appraisal Committee within 60 days from the receipt of the final EIA report and other documents and the recommendations of the Expert Appraisal Committee or State Level Expert Appraisal Committee shall be located before the competent authority for a concluding decision within the next 15 days.

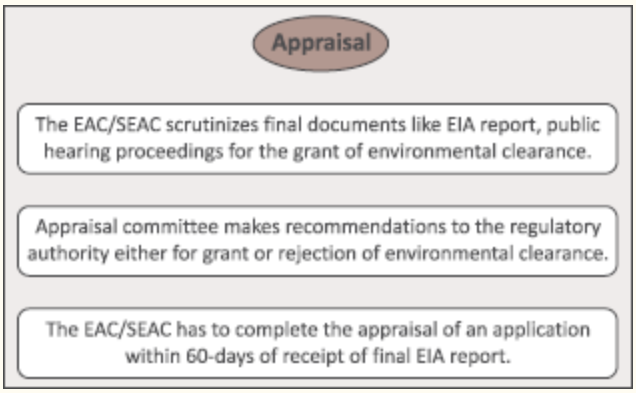

Figure 13.6 Appraisal Procedure

13.8 GUIDING PRINCIPLES OF EIA

The following eight guiding principles are used to govern the entire EIA process.

(i) **Participation:** The procedure should provide appropriate chances for adequate participation and involvement of all the affected public and other probable stakeholders and their contribution in decision-making.

(ii) **Transparency:** EIA processes, consequences and conclusions should be open and easily reachable.

(iii) **Certainty:** The assessment process should have clear objectives and should be conducted in an agreed time frame.

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(iv) **Accountability:** In the assessment process, the decision-makers and project supporters are accountable to all parties for their actions and choices and should provide clear and detailed reasons for their decisions.

(v) **Credibility:** The assessment process and the preparation of EIA report should be commenced with professionalism / objectivity.

(vi) **Cost Effectiveness:** The assessment procedure and its results should guarantee environmental security at a minimal price to society.

(vii) **Flexibility:** The procedure and the various stages involved in the environment clearance procedure should be adaptable to deal competently with any suggestion and should easily accommodate any decision.

(viii) **Practicality:** The process should focus more on practical problems and information/outputs freely usable in decision building and scheduling and further focus on practical output that the project proponent can easily implement.

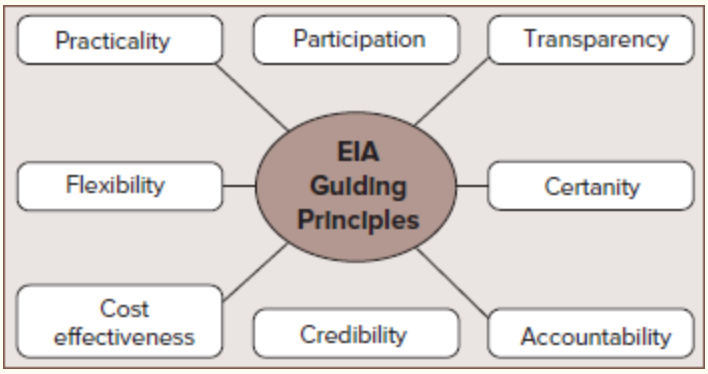

Figure 13.7

13.9 KEY ROLES OF NGOS AND ACTIVISTS IN EIA

NGOs and activists are crucial watchdogs in Environmental Impact Assessments (EIAs), acting as bridges for public awareness, conducting independent research, facilitating community participation, challenging flawed EIAs through legal action (like PILs), monitoring compliance, and lobbying for stronger environmental policies to ensure accountability and

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environmentally sound development. They hold governments and corporations accountable, ensuring transparency and often influencing project outcomes, sometimes leading to cancellations or significant modifications.

Key Roles of NGOs and Activists in EIA

1. **Public Awareness & Participation:** Educate local communities about project impacts and their rights. Facilitate informed participation in public hearings, bridging gaps with authorities.
2. **Research & Technical Expertise:** Conduct independent studies to supplement or challenge official EIAs. Provide scientific data and alternative, sustainable development suggestions.
3. **Legal Intervention:** File Public Interest Litigations (PILs) to challenge non-compliant or biased EIAs in court. Ensure adherence to environmental laws and norms.
4. **Monitoring & Transparency:** Act as watchdogs, monitoring project sites and compliance with EIA conditions post-approval. Expose violations and demand accountability from developers and regulators.
5. **Advocacy & Policy Reform:** Lobby for stricter EIA guidelines and policy changes. Engage with policymakers to promote sustainable practices.
6. **Community Mobilization:** Organize grassroots movements and protests, as seen in cases like the POSCO project, to delay or halt destructive projects.

■ *Examples of Impact*

- **Narmada Bachao Andolan (NBA):** Challenged the **Sardar Sarovar Dam's EIA**, leading to project modifications and World Bank withdrawal.
- **Vedanta's Niyamgiri Mining:** The proposed Vedanta bauxite mine in the Niyamgiri Hills was ultimately cancelled due to significant opposition from **Niyamgiri Suraksha Samiti (NSS) a Non-Governmental Organizations (NGOs)** and the local **Dongria Kondh**

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tribe , who successfully challenged the project's flawed **Environmental Impact Assessments (EIAs)** and asserted their indigenous rights.

- **Kudankulam Nuclear Power Plant:** Protests questioned EIA adequacy, leading to more safety measures and delays.
- **Tata Mundra Power Plant**: **NGO**: **Machimar Adhikar Sangharsh Sangathan (MASS)** filed complaints, leading to tighter environmental controls by the **IFC (International Finance Corporation)**.
- **Posco Steel Project**: **NGO**: **POSCO Pratirodh Sangram Samiti** and **Abhay Sahoo** led protests, halting the project over environmental concerns.

Criticism of NGO Activity

- In some cases, NGOs have been criticized for delaying development projects unnecessarily. For instance, the **Sterlite Copper Plant** in Tamil Nadu was shut down after environmental protests, leading to significant economic losses, affecting copper production and jobs.
- Such delays or closures can have negative impacts on industrial development and economic growth, raising concerns about a balance between environmental activism and development needs.

Conclusion

NGOs and activists ensure greater transparency and accountability in the EIA process, influencing policy changes such as improved public consultation and data transparency. However, it is important to balance environmental concerns with development to avoid undue harm to the economy and livelihoods.

13.10 ENVIRONMENTAL IMPACT ASSESSMENT (EIA) 2006 AND DRAFT EIA 2020 – COMPARATIVE OVERVIEW

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The EIA Notification 2006 has been the governing framework for environmental clearances for various projects. In 2020, the Ministry of Environment, Forest and Climate Change proposed a draft EIA Notification 2020, which aims to replace the 2006 notification.

13.10.1 Salient Features of Environmental Impact Assessment 2006

EIA notification 2006 was introduced under the **Environment Protection Act 1986 (EPA)** to govern the process of granting **Environmental Clearance (EC)** for projects like dams, mining, thermal power plants, infrastructure developments such as highways, ports, airports, and large-scale construction initiatives.

The notification streamlined the environmental clearance process by decentralising it into two categories:**Category A** projects, which are **evaluated at the national level**, and **Category B** projects, **reviewed at the State level.**

Category A projects: Evaluations for these projects are conducted by the **Impact Assessment Agency** in conjunction with the **Expert Appraisal Committee**.

Category B projects: Appraisals are performed by the **State Level Environmental Impact Assessment Authority (SEIAA)** with the support of the **State Level Appraisal Committee,** specifically formed for this purpose.

For **Category A projects**, conducting an **EIA** is mandatory, but **they bypass the screening phase**. Conversely, **Category B projects go through a screening process** and are further divided into two subcategories:

- **B1 projects**, which **necessitate a mandatory EIA.**
- **B2 projects**, which are **exempt from requiring an EIA.**

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13.10.2 Major Differences Between Draft (2020) and Current Notification (2006)

1. **Post-Facto Approval**
   The draft notification introduces a significant change by allowing **post-facto approvals**. This implies that **projects can receive clearance even if they have commenced construction or are operational without having previously secured environmental clearances.**
2. **Public Consultation Process**
   The draft notification proposes to **reduce the time allocated for public responses during public hearings for environmental clearance applications from 30 days to 20 days.** This reduction may compromise the ability of affected stakeholders to adequately prepare and share their views, comments, or suggestions, undermining the purpose of meaningful public hearings.
3. **Compliance Report Issue**
   Under the 2006 notification, project proponents are required to submit compliance reports every **six months** to demonstrate adherence to the approved terms and conditions. **The draft notification, however, revises this requirement, mandating compliance reports only once a year, potentially reducing oversight.**
4. **Diluting the EIA Process**
   The draft notification significantly alters the EIA process by **granting the central government authority to label certain projects as "strategic." Once categorised as strategic, no information regarding these projects will be disclosed to the public**.
   Additionally, **the draft increases the size threshold for new construction projects exempted from detailed scrutiny**, EIA studies, and public consultation, **raising the limit from 20,000 square meters to 1,50,000 square meters.**

Conclusion

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The draft EIA Notification 2020 proposes several significant changes to the existing 2006 framework. While it aims to streamline processes and reduce regulatory burden, critics argue that it may dilute environmental safeguards. The proposed changes include shorter public consultation periods, post-facto approvals, extended validity for clearances, and more exemptions. These modifications could have far-reaching implications for environmental governance in India, potentially altering the balance between development and environmental protection.

13.11 STRATEGIC ENVIRONMENTAL ASSESSMENT (SEA)

- SEA is defined as the standardised, structured and extensive process to assess and identify the environmental impacts of proposed plans or programmes for their inclusion at the **initial decision-making stage**, at par with economic and social deliberations.
- SEA, in comparison to EIA, includes a wider range of activities and long-duration projects.
- SEA can be enforced on the entire sector (like national energy policy) or to a particular geographical area (like regional development scheme context).
- Even though SEA does not restore or lessen the project-level EIA requirement (however, it can happen in a few cases), it can assist in simplifying and concentrating on the inclusion of the environmental concerns (considering biodiversity) for the decision-making process, thereby creating a project-level EIA a larger successful process. **SEA is characterised as proactive and 'sustainability driven', whereas EIA is mostly reactive**.

■ *STRATEGIC ENVIRONMENTAL ASSESSMENT (SEA) Vs. EIA*

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- **SEA is a proactive process and is used before decision making, while EIA is a reactive process** and is used at the end of decision making.
- **SEA is conducted by a public authority,** whereas EIA is conducted by the project proponent or consultants.
- **SEA is used at the policy, planning and programme level** while EIA is used to ascertain the likely environmental impacts of the proposed project prior to its execution.
- **SEA considers the wide range of potential alternatives,** whereas EIA is concerned with a limited number of feasible alternatives.
- **SEA stressed the attainment of objectives and maintaining systems,** whereas EIA emphasised mitigating and reducing impacts.
- **SEA focused on sources of environmental deterioration, and** EIA focused on **symptoms** of environmental deterioration.
- **SEA provides early warning of collective impacts,** whereas EIA provides a limited review of collective impacts.
- **In SEA, the uncertainty is high as compared to EIA** where there is more certainty.

Table 13.1 Difference between EIA and SEA

Environmental Impact Assessment (EIA)	Strategic Environmental Assessment (SEA)
It is usually reactive to a development proposal.	Is pro-active and informs development proposals.
Assesses the effect of a proposed development on the environment.	Assesses the effect of a policy, plan or programme on the environment, or the effect of the environment on development needs and opportunities.

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Environmental Impact Assessment (EIA)	Strategic Environmental Assessment (SEA)
Addresses a specific project.	Addresses areas, regions or sectors of development.
Has a well-defined beginning and end.	Is a continuing process aimed at providing information at the right time.
Assesses direct impacts and benefits.	Assesses cumulative impacts and identifies implications and issues for sustainable development
Focuses on the mitigation of impacts.	Focuses on maintaining a chosen level of environmental quality.
Has a narrow perspective and a high level of detail.	Has a wide perspective and a low level of detail.

13.12 ADVANTAGES AND DISADVANTAGES OF EIA

Environmental Impact Assessment (EIA) has significant **advantages** as a planning and decision-making tool, but it also faces several **disadvantages** in practice.

13.12.1 Advantages of EIA

The primary benefits of the EIA process include:

- **Environmental Protection:** EIAs identify potential negative environmental impacts in advance, allowing for mitigation measures or

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project redesigns to avoid or minimize damage to ecosystems, air, water, and soil quality.

- **Informed Decision-Making:** The process provides decision-makers (government authorities and project proponents) with a clear understanding of a project's full environmental, social, and economic consequences before approval, leading to better outcomes.
- **Cost-Effectiveness:** Addressing environmental issues early in the planning stage can prevent expensive modifications, clean-up costs, or legal penalties later in the project life cycle.
- **Sustainable Development:** EIA links environmental considerations with development planning, helping to ensure that projects are environmentally sound and promote long-term sustainability rather than short-term gains.
- **Public Participation and Transparency:** The process mandates public consultation, allowing affected communities and other stakeholders to voice their concerns and contribute local knowledge, which increases project acceptance and accountability.
- **Regulatory Compliance:** EIA helps ensure that projects adhere to national and international environmental laws and regulations, avoiding legal issues and fines.

13.12.2 Disadvantages and Limitations of EIA

Despite its potential, the EIA process faces several challenges and shortcomings:

- **Time-Consuming and Costly:** The in-depth research and consultation required can be a lengthy process, which project proponents may view as a cause of delays and increased costs.
- **Lack of Data Credibility:** The quality of EIA reports can be compromised by a lack of reliable baseline data, or by the use of erroneous, single-season, or biased data.
- **Potential for Bias:** Reports are often commissioned by the project proponent, creating an inherent conflict of interest that can lead to reports

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downplaying negative impacts in favour of project approval.

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- **Inadequate Public Participation:** Public hearings can sometimes be a mere formality with insufficient notice periods, making it difficult for local communities to participate meaningfully or for their indigenous knowledge to be properly considered.
- **Insufficient Monitoring and Enforcement:** A major weakness is the lack of effective post-approval monitoring to ensure compliance with the proposed mitigation measures and clearance conditions.
- **Limited Scope:** EIAs often focus only on direct, biophysical impacts of a specific project, potentially ignoring cumulative, indirect, or long-term impacts (e.g., climate change or regional development effects).

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UNIT VI Pollution and Solid Waste Management

Pollution and Solid Waste Management 14

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14.1 POLLUTION

Pollution is an unwanted change in physical and chemical characteristics of air, water and soil, which causes a potential health hazard to living organisms. **E.P. Odum**, a famous ecologist, defined pollution as an unwanted change in the chemical, physical and biological characteristics of air, water and soil, adversely affecting human life. **John Mclaughlin** defines pollution as the incorporation of waste material or excess energy into the environment by man, which immediately or accidentally destroys man and his surroundings.

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14.2 POLLUTANTS

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Pollutants are the constituents of pollution, which can be either invasive substances or naturally arising contaminants. The earliest pollutants found in the atmosphere were perhaps of natural origin. Smoke, fumes, ash and gases from volcanoes and forest fires, sand and dust from windstorms in arid or dry regions, fog in humid, low-lying regions, were part of our atmosphere since long before anthropogenic complications came to the scene.

14.3 DIFFERENT WAYS OF THE STUDY OF POLLUTANTS

14.3.1 Natural vs. Anthropogenic

Natural sources: Smoke and gases produced from forest fires, volcanic eruptions, dust from a dust storm, pollens, spores and volcanic ashes etc.

Man-made sources: Smoke and trace gases produced from forest fires, thermal power plants, automobiles, mining, deforestation, construction etc.

ENV 52- Primary vs. Secondary Pollutants

14.3.2 Primary vs. Secondary Pollutants

Primary pollutants: Come into the atmosphere directly from a procedure such as combustion of fuel in an engine discharging CO and CO_2 out of the

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exhaust and SO_2 is released in a power plant or factory. A volcanic eruption discharges smoke and ash which are regarded as primary pollutants. NOx is produced by fossil fuels combustion at high temperature or even by thunderstorms.

- These pollutants enter the environment exactly the way they are produced.
- They are related to different categories such as particulate, aerosol and acidic.
- They are poisonous.
- They do not show synergy.

Secondary pollutants: These are produced when primary pollutants react with something else in the air and produce something else completely. For example, when NOx reacts under high temperatures, with volatile organic compounds, the formation of ground-level ozone takes place. The secondary pollutants that can lead to the production of smog and poor air quality are of this nature. For example, smog is a mixture of smoke and fog that can be the consequence of burning coal which releases smoke and SO_2. It can be the outcome of vehicles and industries releasing emissions that react with UV radiation from the sun to form secondary pollutants.

- These pollutants emerge as a result of their contact with primary pollutants and other substances of the atmosphere.
- These are generally oxidising in nature.
- They are highly poisonous.
- They show synergy
- They are present in nature in altered form.

Exam Pointer

Volcanic eruptions are a major natural source of environmental pollution. They release gases, ash, and debris that impact air, water, and land at local, regional, and sometimes global levels.

Key Pollutants and Environmental Effects

1. **Air Pollution (Gases and Ash)**

Volcanic emissions contain toxic gases and abrasive ash. This creates hazards near the volcano and hundreds of kilometers downwind.

- **Sulfur Dioxide (SO_2):** This gas irritates the respiratory system, eyes, and skin. It reacts with water vapor and sunlight

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in the atmosphere to form a haze of fine sulfate aerosol droplets. This is known as volcanic smog or 'vog' in Hawaii. During significant eruptions, SO_2 released into the stratosphere can create a global aerosol blanket. This blanket reflects sunlight and causes short-term cooling at the surface, either globally or regionally.

- **Carbon Dioxide (CO_2):** Although global volcanic CO_2 emissions are much lower than those from human activities, CO_2 can gather in low-lying areas. This displaces oxygen and poses a serious asphyxiation risk to humans and animals.
- **Hydrogen Halides (HCl, HF):** Hydrogen chloride and hydrogen fluoride dissolve in water droplets within the volcanic plume. This can result in acid rain. Ash particles coated in these acids can poison grazing land, harming livestock and agriculture.
- **Volcanic Ash:** Made of fine, sharp fragments of glass and rock, ash is a significant nuisance and danger. Inhaling it can cause respiratory issues, while ash fall can damage crops, collapse roofs, and disrupt aviation by harming aircraft engines.

2. **Water Pollution**

Ash and acidic gases directly affect water quality

- **Contamination of Water Supplies:** Ash fall can pollute drinking water sources, clog treatment facilities, and change the pH balance of lakes and rivers.
- **Acidification:** Acid rain from volcanic gases lowers the pH of soil and water bodies, harming plants and aquatic life.
- **Leaching of Metals:** Rainwater interacting with volcanic ash can cause toxic elements and heavy metals, like fluoride and arsenic, to seep into surface and groundwater. This creates poisoning risks.

3. **Land Pollution**

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- **Ash Blanketing:** Thick layers of ash can cover agricultural fields, stopping crop production for years and damaging infrastructure.
- **Lava Flows and Pyroclastic Flows:** These flows destroy everything in their path through heat and burial, causing severe localized damage to land and property.
- **Lahars:** Volcanic mudflows called Lahars, occur when ash and debris mix with water from rain or melted snow and ice. These fast-moving flows can devastate valleys far from the volcano, burying communities and landscapes.

14.3.3 Biodegradable and Non-biodegradable Pollutant

Biodegradable pollutant: These pollutants decompose into simple products by bacteria. Raw materials of nature that are reutilised in the ecosystem are these simple products. Their excessive presence threatens the environment, e.g., paper, garden waste, domestic sewage, agro-based residues, fertilisers etc.

Non-biodegradable pollutants: These substances are not decomposed by bacteria. They stay in the ecosystem for a long time, get gathered and often biomagnifies to a dangerous level when they go up to the next trophic level in the food chain; e.g., plastics, pesticides, heavy metals, rubber, nuclear wastes etc.

14.3.4 Source and Non-source Pollutants

Point sources: They are discrete and limited sources that discharge the pollutants/effluents through a pipe or chimneys from industries or municipal areas. Pollution originating from the point sources can be organised through on-site treatment.

Non-point sources are diffused sources that release pollutants over a large area, e.g., run-off from construction spots and agricultural fields. It is

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challenging to treat such pollution because of its widespread sources over a large part.

Examples of environmental pollutants: Gases (NO, NO_2, CO and CO_2), **deposited matter** (soot, smoke), metals (mercury, lead, zinc etc.), **fluoride, agrochemicals** (pesticides, herbicides, fertilisers etc.), **radioactive waste, noise** etc.

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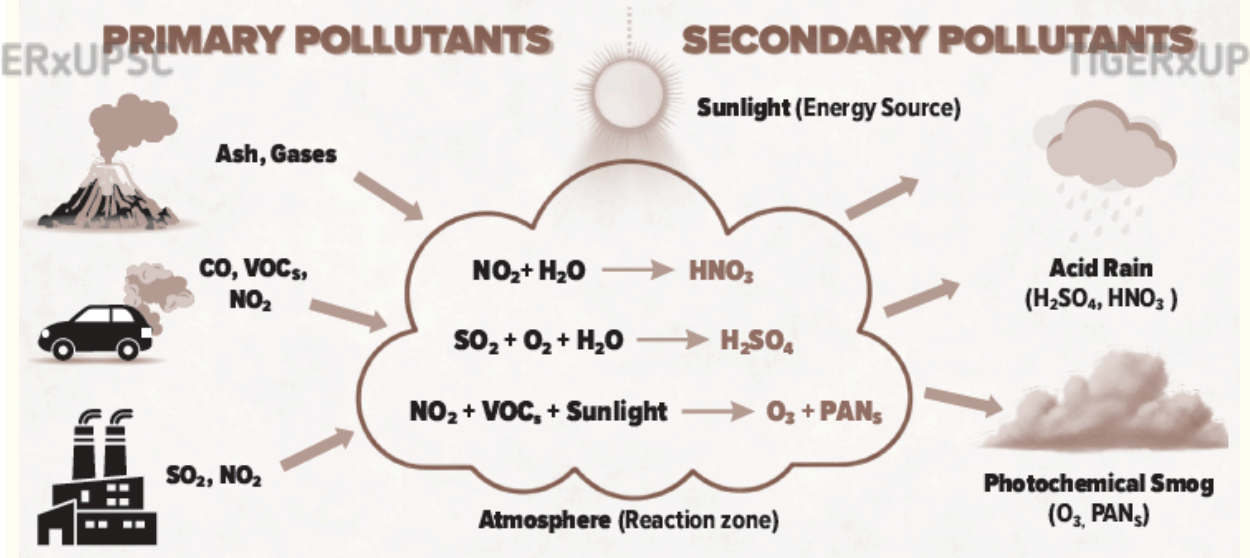
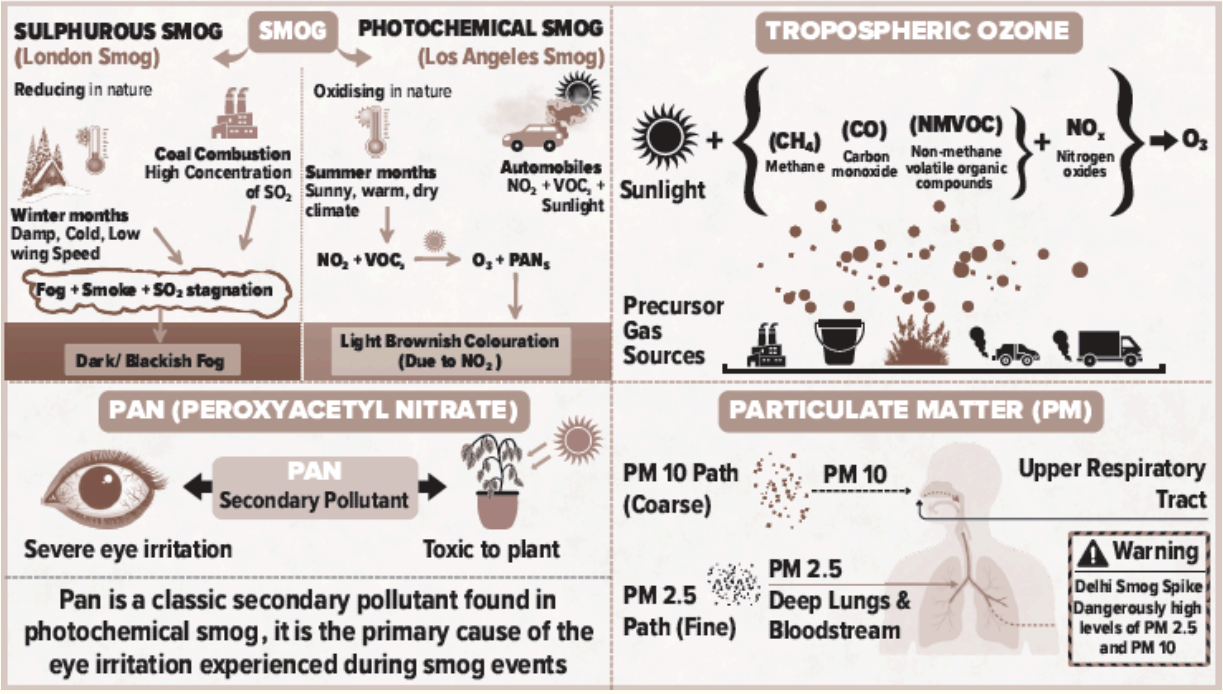


CHARACTERISTICS OF PRIMARY POLLUTANTS	CHARACTERISTICS OF SECONDARY POLLUTANTS
Enters environment in original form	Formed after chemical changes
Found as aerosols or particulates	Found generally as oxidisers
Generally poisonous in nature	Highly poisonous in nature
Does not show synergistic effects	Often shows synergistic effects



14.4 AIR POLLUTION

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Air pollution generally refers to pollutants that directly harm living beings (animals, plants and people) and their artefacts on confined and regional scales.

14.4.1 Sources of Air Pollution

- **Burning of fossil fuels** releases SO_2, H_2SO_4, H_2S and CO_2 in the atmosphere. **(Thermal Power Plants emit CO_2, SO_2 and NOx)**
- NO, NO_2, CO (poisonous gas) and HNO_3 are released from automobile exhausts and thermal power plants.
- **Agriculture and biomass waste burning** generate many greenhouse gases (CO_2, N_2O, CH_4) and other air polluting gases (CO, NH_3, NOx, SO_2, Volatile Organic Compounds) etc. in the atmosphere.

Table 14.1 Difference between Primary & Secondary Pollutants

Primary Pollutants	Secondary Pollutants
They are pollutants which are passed into the environment in the form they are produced	The pollutants develop as a result of the interaction of primary pollutants and environmental constituents
They can be particulate, aerosol, reduced and oxidised	They are generally oxidising
They are less toxic	They are more toxic
They do not show synergism	Secondary pollutants show synergism
They persist in the state in which they are released in the environment directly	They are modified product

- From industries and automobiles, metals (nickel, arsenic and lead etc.), hydrocarbons, fluoride and photochemical smog are released. Chemical substances are released from spray, paints etc.

1. **Natural Pollutants:** Pollutants discharged from natural sources or as an outcome of natural activity. E.g. **pollens** and **volatile organic compounds** from plants; gases such as SO_2, H_2S etc. from **volcanoes and decay of dead organic materials, particles from wildfires and sea; natural radioactivity** etc.
2. **Primary Pollutants:** These pollutants are **released directly into the air as a result of natural or anthropogenic activity,** e.g., CO_2, CO, SO_2 and NOx, hydrocarbons and particulate matters released from burning fuel.

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3. **Secondary Pollutants:** These are produced as an **outcome of chemical interaction between primary pollutants and normal atmospheric compounds** in the presence of electromagnetic radiation from the sun e.g. the primary pollutant SO_2 reacts with O_2 in the atmosphere to form **sulphur trioxide (SO_3)** which is a secondary pollutant. **Smog**, a secondary air pollutant, is a combination of smoke and fog produced by a **complex reaction between** NOx **and a wide range of hydrocarbons activated by sunlight. It is produced mostly in the stagnant air over urban areas.**

Do You Know?

Carbon monoxide poisoning: Carbon monoxide poisonous gas pollution occurs primarily from emissions produced by fossil fuel-powered engines. The incomplete reaction of air with fuel produces colourless and odourless carbon monoxide. It binds to the chemicals in our blood, called haemoglobin. It reacts 250 times more effectively than oxygen and also stays bound to the haemoglobin for longer than oxygen does. The effect is that the blood is starved of oxygen, affecting rest of the body.

ENV 53- Carbon Monoxide Poisoning

14.4.2 Smog

Smog is a type of air pollution, initially named for the **combination of smoke and fog**. Typical smog results from large amounts of coal combustion in an area and is produced by a combination of smoke and SO_2. In the 1950s, a different type of smog, called **photochemical smog**, was first designated.

The burning of fossil fuels can create an atmospheric pollution problem known as photochemical smog. Photochemical smog is a condition that

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develops when **primary pollutants** such as oxides of nitrogen, volatile organic compounds created from fossil fuel combustion interact under the influence of sunlight to produce **secondary pollutants**. The major chemical pollutants in photochemical smog are NO, NO_2, VOCs (volatile organic compounds), ozone (O_3) and PAN (peroxyacetyl nitrate). NO_2 decreases visibility due to its yellowish colour. It also causes heart and lung problems. Ozone (O_3) contributes to bronchial constriction, coughing and wheezing. PAN causes eye irritation, high toxicity to plants. **SO_2 and NO_2 are the primary pollutants while ozone, PAN and smog are secondary pollutants.**

QUICK INSIGHT

India Is the Largest Emitter of Sulphur Dioxide in the World

India is the largest emitter of anthropogenic sulphur dioxide in the world, as per the data released by environmental NGO Greenpeace on 19 August, 2019. Anthropogenic sulphur dioxide is produced from the burning of coal and it is known to largely contribute to air pollution.

Key findings:

- SO_2 hotspots across the world have been mapped.
- The SO_2 hotspots were detected by the **OMI (Ozone Monitoring Instrument) satellite.**
- India has over 15% of all anthropogenic sulphur dioxide (SO_2) hotspots in the world.
- **The main SO_2 hotspots in India include** Singrauli in Madhya Pradesh, Korba in Chhattisgarh, Talcher and Jharsuguda in Odisha, Neyveli and Chennai in Tamil Nadu, Kutch in Gujarat, Ramagundam in Telangana and Chandrapur and Koradi in Maharashtra.

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- Norilsk smelter complex in Russia is the largest SO_2 emission hotspot globally.

Smog Pollutants and Their Sources

The emissions from motor vehicles, industry, paints, solvents and gasoline fumes react with the heat during summertime and form the **ground-level ozone**, of which smog is its main component. So, in short, **smog is nothing but a type of air pollution. Smog is very common in cities with a sunny, warm, dry climate with a large number of vehicles.**

While O_3 in the stratospheric layer protects the earth from harmful UV rays, O_3 **on the ground level is extremely toxic to human health. During the summer season, the higher air is warm and prevents vertical circulation causing the smog to worsen**. It can build up to extremely dangerous levels by remaining for extended periods over the heavily populated areas. It can also travel with the wind and disturb the sparsely populated areas.

- **Smog is basically derived from the merging of two words; smoke** and **fog**. Smog is also used to describe the type of fog which has smoke or soon in it.
- Smog is yellowish or blackish fog formed mainly by a mixture of pollutants in the atmosphere which consists of fine particles and ground-level ozone.
- Smog which occurs mainly because of air pollution can also be defined as a mixture of various gases with dust and water vapour. Smog also refers to hazy air that makes breathing difficult.

Smog Formation

- The atmospheric pollutants or gases that form smog are released into the air when fuels are burnt. When sunlight and its heat react with these gases and fine participles in the atmosphere, smog is formed.
- It is purely caused by air pollution. **Ground-level ozone and fine particles are released in the air due to complex photochemical reactions between VOC, sulphur dioxide (SO_2) and nitrogen oxides (NOx).**
- These VOC, SO_2 and NOx are called precursors. The main sources of these precursors are pollutants released directly into the air by gasoline and diesel-run vehicles, industrial plants and activities, and other human activities.

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- Smog is often caused by heavy traffic, high temperatures, sunshine and calm winds. These are a few of the factors behind the increasing level of air pollution in the atmosphere.
- During the winter months, when the wind speeds are low, it helps the smoke and fog to become stagnate at a place forming smog and increasing pollution levels near the ground closer to where people are respiring. It hampers visibility and disturbs the environment.
- The time that smog takes to form depends directly on the temperature. Temperature inversions are situations when warm air does not rise instead stays near the ground.
- During situations of temperature inversions, if the wind is calm, smog may get trapped and remain over a place for days.
- But it is also true that smog is more severe when it occurs farther away from the sources of release of pollutants.
- This is because the released pollutants from heavy traffic get drifted due to the wind and photochemical reactions result in the formation of smog. Smog can thus affect and prove to be dangerous for suburbs, rural areas as well as urban areas or large cities.

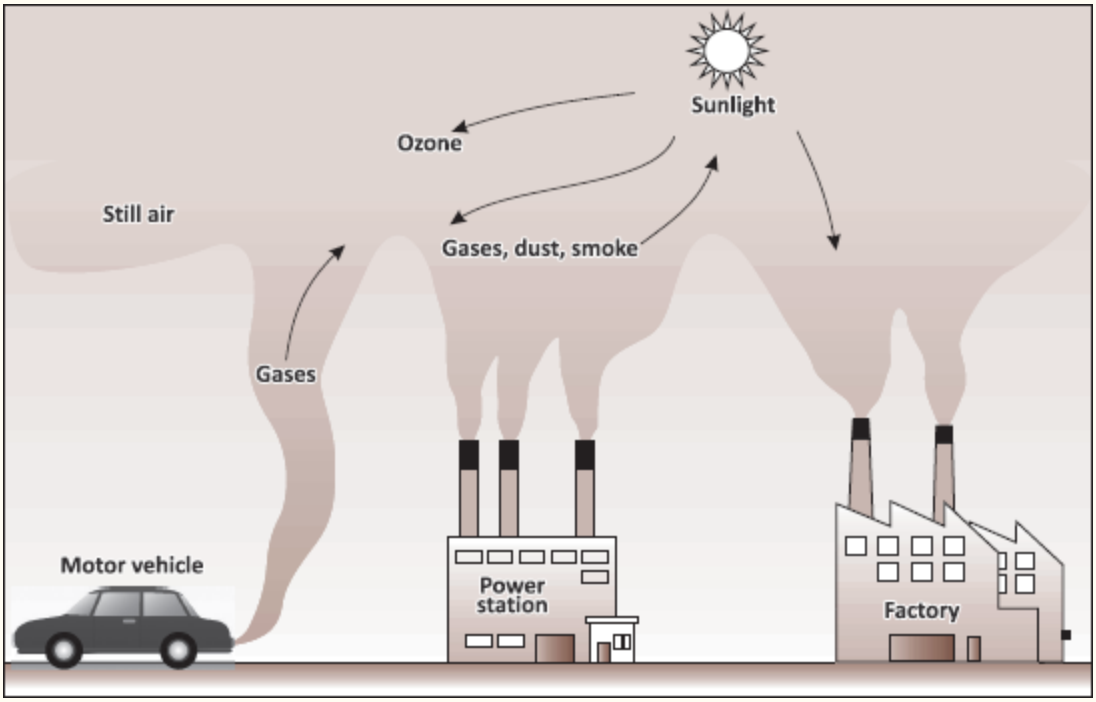

Figure 14.1 Smog Formation

14.4.3 Types of Smog

At least two distinct types of smog are recognised: sulphurous smog and photochemical smog.

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Sulphurous Smog

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- Sulphurous smog is also called **London smog** (first formed in London)
- Sulphurous smog results from a high concentration of SULPHUR OXIDES in the air and is caused by the use of sulphur-bearing fossil fuels, particularly coal (Coal was the main source of power in London during the nineteenth century. The effects of coal-burning were observed in the early twentieth century).
- This type of smog is aggravated by dampness and a high concentration of suspended particulate matter in the air.

Photochemical Smog

- Photochemical smog is also known as **Los Angeles smog**.
- Photochemical smog forms most prominently in the urban areas where there are large numbers of automobiles in use. (Nitrogen oxides are the primary emission)
- Photochemical smog (summer smog) forms when pollutants such as nitrogen oxides (primary pollutant) and organic compounds (primary pollutant) react together in the presence of sunlight. A gas called Ozone (secondary pollutant) is formed.
- **Nitrogen Dioxide + Sunlight + Hydrocarbons = Ozone** (Ozone in stratosphere is beneficial, but near the earth's surface it results in global warming as it is a greenhouse gas).
- The resulting smog causes a **light brownish colouration of the atmosphere**, **reduced visibility**, **plant damage**, **irrigation of the eyes** and **respiratory distress**.

Smog in India

New Delhi, the capital of India, has become the choking city as the city develops thick smog in November and December. This dense smog contains dangerously high levels of particulate matter (PM) 2.5 and PM 10 which puts millions of life at threat. In 2017, the Government of India has come out with Graded Response Action Plan to tackle the problem of Smog in Delhi and NCR.

14.4.4 Effects of Smog

- Smog is harmful and it is evident from the components that form it and the effects that can happen from it.
- It is harmful to humans, animals, plants and nature as a whole. Many deaths were recorded, notably, those relating to bronchial diseases. Heavy smog is responsible for **decreasing the UV radiation greatly**.
- Heavy smog results in a low production of the crucial natural element vitamin D leading to cases of rickets among people.
- When a city or town gets covered in smog, the effects are felt immediately. Smog can be responsible for any ailment from minor pains to deadly pulmonary disease such as lung cancer. Smog is well known for causing irritation in the eye.

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- It may also result in inflammation in the tissue of lungs; giving rise to pain in the chest. Other issues or illnesses such as cold and pneumonia are also related to smog.
- The human body faces great difficulty in defending itself against the harmful effects of smog.
- Minor exposure to smog can lead to greater threats of asthma attacks; people suffering from asthma problems must avoid exposure.
- Smog also causes premature deaths and affects densely populated areas. The highly affected people include old people, kids and those with cardiac and respiratory complications as they have an easy tendency to be at a disadvantage of asthma.
- The ground-level ozone present in the smog also inhibits plant growth and causes immense damage to crops and forests. Crops, vegetables such as soybeans, wheat, tomatoes, peanuts and cotton are subjected to infection when they are exposed to smog.
- The smog results in mortifying impacts on the environment by killing innumerable animal species and green life as these take time to adapt to breathing and surviving in such toxic environments.
- Smog is a devastating problem especially due to the fast modernisation or industrialisation as the hazardous chemicals involved in smog formation are highly reactive is spread around in the atmosphere.
- Smoke and sulphur dioxide pollution in urban areas is at much lower levels than in the past, as a result of the law passed to control emission and in favour of cleaner emission technology.

14.4.5 Tropospheric Ozone

The maximum tropospheric O_3 formation occurs when **NOx**, **CO** and **VOCs**, **interact up in the atmosphere in the presence of sunlight. NO_x, CO and VOCs are known as ozone precursors**. Vehicle exhaust, industrial releases and chemical solvents are the major anthropogenic sources.

Do You Know?

VOCs are organic compounds that effortlessly become vapours or gases. Carbon along with elements such as H, O, Fl, Cl, Br, S or N.

How does it affect human beings?

- The O_3 **deters plant growth** and can cause extensive damage to crops and forests and the **fog decreases visibility**.

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- It has also been assessed that **more than 100 million people worldwide live in areas with O_3 levels above the recognised standards for health security.**
- Ground-level O_3, SO_2, NO_2 and CO are especially harmful to senior citizens, children and people with heart and lung disorders such as emphysema, bronchitis and asthma.
- It results in inflammation of breathing passages, declining the lungs capacity and triggering shortness of breath, discomfort when inhaling deeply, wheezing and coughing.
- It can result in eye and nose irritation, dries out the protective layer of the nose and throat and restrict the body's ability to fight infection, increasing susceptibility to illness.
- Hospital admittances and respiratory deaths often upsurge during periods when O_3 levels are high.

14.4.6 Effects of Air Pollution

(a) **Effects on human health:** Irritation of eye, nose and throat, respiratory system, nickel particulates in tobacco smoke affect respiratory tract, lead results in convulsions, delirium, coma and even death etc., cadmium (from cigarette) cause heart disease, kidney and liver damages, radioactive substances effect on genes of human beings.

(b) **Effects on plants:** NO_2 causes premature leaf fall (i.e., **abscission**) and inhibits plant growth leading to lesser yields of crop plants, SO_2 bleaches the leaf surfaces and causes **chlorosis** (i.e., absence of chlorophyll and yellowing of the leaf), especially in leafy vegetables, **PAN (peroxyacetyl nitrate) harms leafy vegetables resulting in premature fall, discolouration and curling of sepals**.

(c) **On animals:** When animals feed on the plants coated with particulate (especially with Fl, Pb, As) they get affected with Arsenic poisoning (cattle and sheep are main victims). Lead poisoning leads to bronchitis and loss of appetite in pet animals.

(d) **Effect on aesthetic beauty:** Industries, automobiles, sewage and trash heaps emit foul smells causing further damage of aesthetic beauty.

(e) **On climate:** Due to deforestation and fuel burning in industries and automobiles, the CO_2 concentration of the atmosphere is likely to double (by the year 2020). This upsurge in CO_2 will enhance the atmospheric temperature of earth, resulting in the melting of polar ice, glaciers etc. Consequently, it will lead to flooding of coastal areas. Rainfall patterns, if changed, will affect the agricultural output and also causes changes in the composition of gases. The ozone layer is thinning due to an increase in aerosols.

Table 14.2 Major air pollutants, sources and their effects

S. No.	Pollutants	Sources	Effects
1.	**Oxides of carbon** (CO_x)	• Burning of coal, oil and other fuels for energy production	• CO_2 has a chief role in greenhouse effect, forms weak carbonic

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S. No.	Pollutants	Sources	Effects
	- Carbon dioxide (CO_2) - Carbon monoxide (CO)	• Transport • Burning of biomass	acid leading to acid rains • CO affects human health by combining to haemoglobin, which may lead to asphyxia • Haemoglobin has 250 times more attraction to CO as compared to that of CO_2
2.	**Oxides of sulphur** (SO_x) - Sulphur dioxide (SO_2) - Sulphur trioxide (SO_3) - Sulphate (SO_4)	• Burning of sulphur containing fuel e.g. coal, petroleum extraction and refining • Paper manufacturing; • Municipal burning • Ore smelting for metal extraction	• SO_2 has maximum damaging effects as it can cause severe injury to human and other animal lungs and is an important precursor to acid rain • Ill effects comprise corrosion of paints, metals and injury or death to animals and plants
3.	**Oxides of Nitrogen** (NO_x) - Nitrogen oxide (NO) - Nitrogen dioxide (NO_2) - Nitrous oxide (N_2O) - Nitrate (NO_3)	• Burning of fuels; • Biomass burning • By-product in the manufacturing of fertilizers.	• Form the secondary pollutants: peroxyacetyl nitrate (PAN) and nitric acid (HNO_3) • Inhibition of plant growth and tissue damage; • Results in irritation to eyes, viral infection like influenza; • Nitrate made in the atmosphere impairs the visibility, whereas in soil encourages plant growth.

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S. No.	Pollutants	Sources	Effects
4.	**Hydrocarbons (HCs)** also called **Volatile Organic Compound (VOCs)** Methane (CH_4) Butane (C_4H_{10}) Ethylene (C_2H_4) Benzene (C_6H_6) Benzopyrine ($C_{20}H_{12}$) Propane (C_3H_6)	• Evaporation from gasoline tanks, carburettors; • Burning of fuels, biomass; • Municipal landfills; • Microbial activity of sewage; • Industrial process involving solvents	• Can have a carcinogenic influence on humans; • Higher levels are toxic to plants and animals; • Can convert into damaging compounds through complex chemical variations that occur in the atmosphere; • Some are extra reactive with sunlight and create **photochemical smog**
5.	**Other organic compounds** - Cholorofluoro carbon (CFCs), - Formaldehyde (CH_2O) - Methylene chloride (CH_2Cl_2) - Tricholoro Ethylene (C_2HCl_3) - Vinyl chloride (C_2H_3Cl) - Carbon Tetra Chloride (CCl_4) - Ethylene oxide (C_2H_4O)	• Aerosol sprays; • Foam and plastics for making disposable fast food containers; • Refrigeration	• CFCs cause decrease in strato- spheric ozone that permits greater penetration of UV light at earth's surface; strengthened UV radiation cause skin cancer and lethal effects on various life forms
6.	**Metals and other inorganic compounds** - Lead (Pb), Mercury (Hg)	• Oil wells and refineries; • Transport vehicles; • Municipal landfills; fertilizer,	• Cause respiratory complications, toxicity and even death to humans and other animals; • Injury to crops;

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S. No.	Pollutants	Sources	Effects
	- Hydrogen sulphide (H_2S) - Hydrogen Fluoride (HF) - Manganese	• Ceramic, paper, chemical and paint industries; • Pesticides, fungicides; • Aluminum production; • Coal gasification	• Prove to be carcinogenic
7.	**Liquid droplet** - Sulphuric acid (H_2SO_4) - Nitric acid (HNO_3) - Oil - Pesticides e.g. DDT and malathion	• Agricultural pesticides; fumigation; oil refineries; reactions of pollutants in the atmosphere	• Contribute to acid rains; • Corrosion; • Damage to various life forms
8.	**Suspended particulate matter (SPM)** - Dust, soil, sulphate, salts, heavy metal salts, fire particles of carbon (soot), silica, asbestos, liquid sprays, mist etc. - Pollen, volcanic dust range from 0.01 to 100 micrometres	• Fuel combustion; • Building construction; • Mining; • Thermal power station; • Stone crushing; • Industrial processes; • Forest fires; • Refuse incineration	• Have chronic effects on the respiratory system • Deposition on the surface of green leaves thus interfering with the absorption of CO_2 and release of O_2 • Obstruction of sunlight; • Particles size that range between 0.1 to 10 PM, cause the greatest lung damage.

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S. No.	Pollutants	Sources	Effects
9.	**Photochemical oxidants** - Ozone (O_3), peroxyacetyl nitrates (PANs), - Formaldehyde (CH_2O) - Acetaldehyde (C_2H_4O) - Hydrogen peroxide (H_2O_2) - Hydroxyl radical (HO)	• Fireproofing, thermal and electrical insulation, acoustic • Photochemical reactions in the atmosphere that involve sunlight, oxides of nitrogen and hydrocarbons • Foam insulation, particleboard, tobacco, ceiling tiles, furnishing • Building materials: concrete, stone; water and soil	• Create haze; • Irritation to eyes, nose and throat; • Respiratory problems; • Blocking of sunlight
10.	**Radon**		• Causes lung cancer

14.4.7 Control Measure of Air Pollution

Air pollution can be controlled using the following equipments:

Cyclone separators, Gravitational settling, Scrubbers/wet collectors and **Fabric filters** are devices to control particulate contaminates.

Dry absorption methods and **wet absorption methods** are devices to control gaseous contaminants.

- **Electrostatic Precipitators:** This works on the principle of electrostatic precipitation, in which electrically charged particulates existing in polluted gas are isolated from the gas stream under the influence of an electrical field.
- **Cyclone separator:** This process takes place by utilisation of a centrifugal force to separate the particulates from the polluted gas. The centrifugal force offers a much greater force than the gravitational force. The centrifugal force is generated by spinning

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(a kind of rotation) a gas stream. Consequently, the cyclone separator removes smaller particulates more effectively than removed by gravitation such as in settling chambers.

- **Spray tower:** In this case, the clean water enters into a spray tower on its top and is then sprayed through the spray nozzles. Simultaneously, the polluted gas is allowed to flow upwards. Consequently, an intermingling takes place between the sprayed water and polluted gases. Thus, liquid droplets containing the particulates settle down under the influence of gravitation and exit from the bottom of the spray tower.

QUICK INSIGHT

WAYU - Air Pollution Mitigation Device

In 2018, the pollution control device WAYU (Wind Augmentation Purifying Unit) for traffic junctions was inaugurated in Delhi.

- WAYU is developed by the Council of Scientific and Industrial Research–National Environmental Engineering Research Institute (CSIR-NEERI) as a part of the Technology Development Project funded by the Department of Science and Technology.
- The device can purify the air in an area of 500 metre square. The device consumes only half a unit of electricity for 10 hours of running and has a maintenance cost of only ₹1,500 per month.

How does it work?

- The device works on two principles: wind generation for dilution of air pollutants and active pollutants removal.
- The device has filters for PM removal and activated carbon (charcoal) and UV lamps for removing poisonous gases such as VOCs and carbon monoxide.
- The device has one fan and filter for sucking and removing PM. There are two UV lamps and half kg of activated carbon charcoal coated with special chemical Titanium Dioxide.

■ *Fabric Filters (Bughouse Filters)*

In this system, a stream of the polluted gas is prepared to pass through a fabric that screens out the particulate pollutant and permits the clear gas to pass through.

14.4.8 International Laws on Air Pollution

- **The Convention on Long-Range Transboundary Air Pollution (CLRTAP)** is planned to defend the human environment against air pollution and to gradually decrease and

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prevent air pollution, including long-range transboundary air pollution.

- **Helsinki Protocol on Reduction of Sulphur Emission (1985)** aimed for a 30% decrease in sulphur emissions or transboundary fluxes by 1993.
- **Nitrogen oxide protocol** delivers the control or decrease of nitrogen oxides and their transboundary instabilities. It came into force in 1991.
- **Kyoto protocol** is an international contract linked to the UNFCCC, which pledges its parties to internationally obligatory emission-reduction targets of greenhouse gases.

14.5 WATER POLLUTION

Water is an essential component of life. Freshwater is one of the most important substances for sustaining human life. But in the past two decades, unsustainable development, improper waste management and mismanagement of water resources have resulted in heavy pollution. Water is a universal solvent, with chemicals of various kinds dissolved in it, water gets contaminated easily.

14.5.1 Sources of Water Pollution

Based on their origin, the sources of water pollutants are broadly categorised into:

- **Point sources,** where the release of **effluent is from the exact site**; for example, sewage exits of a municipal area or effluent opening of a factory.
- **Non-point sources**, **where entry of pollutants occurs over a widespread area**; for example, city stormwater flow, agriculture run-off, etc. Point source pollution can be efficiently checked with suitable technology. Non-point source pollution is problematic to control and needs the application of control measures on a big scale.

14.5.2 Types of Water Pollutants

1. **Physical pollutants:** They are **insoluble matter such as suspended solids, insoluble particles of soil, silt, garbage, plastic etc.**

Thermal pollution of water occurs when hot waste from industries is discharged into lakes, ponds, rivers etc. The temperature change can kill the life forms thriving in natural conditions.

2. **Chemical pollutants:** Pollutants that dissolve in water and hence change their characteristics come under this category. These can be organic or inorganic.

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(i) **Inorganic matter:** Include nitrates, phosphates, acids, alkaline effluents, salts and heavy metals.

(ii) **Organic matter:** Include oil, gas, petroleum, dyes, paints, detergents, cleaning solvents, domestic sewage and animal waste. The biodegradable organic waste is an oxygen-depleting waste.

(iii) **Radioactive pollution** occurs when radioactive substances are released by nuclear research labs after processing the uranium ore.

3. **Biological pollutants:** Disease-causing bacteria, virus, protozoa, parasites and worms pose a serious threat to human health.

(i) **Water-borne diseases** such as gastroenteritis, dysentery, typhoid, polio, viral hepatitis, worm infections and influenza are caused due to biological pollutants.

(ii) **Domestic sewage, human and animal waste, solid waste** from slaughterhouses and wastewater from tanneries bring biological changes in water.

Table 14.3 Different Polluting Agents, their sources and effects

Pollutants	Sources	Effects
• **Biological agents** Bacteria, parasitic fungi, and protozoa	• Human sewage • Animal and plant wastes • Decaying organic matter • Industrial wastes (oil refineries, paper mill, food processing units) • Natural land and urban runoffs	• Oxygen–requiring bacteria feed on this biological garbage and deplete O_2 in the water body, life is destroyed due to lack of oxygen; • Foul stench, poisoned livestock
• **Chemical agents** Inorganic chemicals and minerals, acids,	• Natural run-off from land • Urban storm • Industrial wastes • Acid deposition	• Toxic to various life forms and humans through the food chain, can result in genetic and birth defects

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Pollutants	Sources	Effects
salts, metals like lead and mercury, plant nutrients like phosphates and nitrates.	• Leaded gasoline • Lead smelting • Irrigation • Pesticides • Agricultural run-offs • Mining • Oil fields • Domestic sewage inadequate wastewater treatment • Food processing industries • Detergents containing phosphates	• Increased solubility of harmful minerals in the water • Make water unfit for domestic, agricultural and industrial uses • Salinity build-up in soil • Upsets ecosystem of water bodies and causes eutrophication
• **Organic Chemicals** Pesticides, herbicides, detergents, chlorine compounds, oil, grease and plastics	• Agriculture, forestry • Pest control industries • Home and industrial wastes • Water disinfection processes • Paper industry • Bleaching process • Machine and pipeline wastes	• Toxic to aquatic life forms as well as organisms that depend on such water bodies • Eutrophication of water bodies
• **Radioactive substances**	• Oil spills (home, community and industrial use)	• Radionuclides enter the food chain and causes birth and genetic defects
• **Physical agents** Particulates and heat	• Nuclear waste from research	• Causative agent for cancer

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Pollutants	Sources	Effects
	laboratories and hospitals • Processing of uranium ore • Nuclear plants • Soil erosion, run-offs from the agriculture • Mining • Forestry and construction activities • Power plants, industrial cooling	• Filling of waterways, harbours and reservoirs • Upsurge in temperature lowers the solubility of oxygen in the water • Lessening biotic life in the water bodies

QUICK INSIGHT

Chlorination in Water Treatment

Water chlorination is the process of adding chlorine to water. This method is used to kill certain bacteria and other microbes in tap water as chlorine is highly toxic. **When dissolved in water, chlorine converts to an equilibrium mixture of hypochlorous acid (HOCl) and hydrochloric acid (HCl)**

(a) $Cl_2 + H_2O \rightarrow HOCl/HClO + HCl$

(b) $HClO/HOCl \rightarrow H^+ + ClO–$

That is why chlorine is assigned to clean water because of its oxidising nature. In **wastewater treatment**, **chlorination** is used for elimination of ammonia, which is converted to an oxidised volatile form.

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14.5.3 Municipal Wastewater

Liquid wastes from domestic actions such as kitchen, toilet and other household wastewater are discharged straight into a river or into a big water body nearby. Many rivers in India, including the river Ganga, are polluted by unselective discharge of wastewater. **The famous Dal Lake in Kashmir is also seriously polluted by domestic waste**. Domestic wastes mostly carry biodegradable organic wastes. Surplus input of nutrients comes from detergent residues (e.g., phosphates) and organic remains (e.g. nitrates).

14.5.4 Industrial Wastewater

Both small-scale and large-scale industrial activities produce wastewater contaminated by a variety of organic and inorganic wastes. Almost all the rivers of India, at least in definite stretches, are seriously polluted by the release of industrial sewage. Even the marine ecology is not spared, and most of the coastal water bodies are threatened by pollution from the discharges of coastal prawn-culture farms and fish processing industries. Most constituents of industrial discharges are poisonous to ecological systems even at low content, and many are non-biodegradable.

Hot water is another prominent pollutant from factories and industries. Many industries, including power plants and oil refineries, use water as coolant for the machines.

QUICK INSIGHT

Arsenic in Groundwater

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Arsenic is toxic to almost all life forms, but now researchers at the University of Washington have discovered that some microbes in the Pacific Ocean not only tolerate the stuff, but actively breathe it. The discovery has implications for how life may adapt to a changing climate, as well as where we might find it on other planets.

Arsenic in groundwater

- Arsenic in groundwater is a **geogenic contaminant** i.e., caused by natural geologic processes.
- Concerns: **Incidence of high arsenic in groundwater reported from various parts of the country, particularly in the Ganga-plains** is a serious threat to the health of the human being.
- Arsenic is naturally present at high levels in the groundwater of a number of countries.

Difference between organic arsenic and inorganic arsenic: Atoms of arsenic bond with other elements to form molecules – if carbon is one of these elements, then the arsenic compound is an organic compound. If there is no carbon present, then the arsenic compound is in an inorganic compound.

Inorganic arsenic is a known human carcinogen: This form of arsenic is linked with increased risks of cancer and other health effects.

14.5.5 Surface Run-Off from Land

Pollutants in terrestrial run-off (and stormwater) vary accordingly with the nature of land over which it runs. The run-off from agricultural land is contaminated with pesticide residues and residues of inorganic fertilisers. The run-off from urban areas chiefly comprises biodegradable organic pollutants. Industrial locations may contribute to varied types of pollutants, such as heavy metals, acids and various inorganic compounds. All these pollutants in the run-off profoundly contaminate our surface water and groundwater resources.

14.5.6 Oil Spills and Oil Zapper

An oil spill occurs when liquid petroleum is discharged into the environment by vehicle, vessel or pipeline. It happens on a big scale and is mostly seen in water bodies. It occurs due to social carelessness and is considered as chief form of pollution.

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ENV 55- Oil Spills and Oil Zapper

The sources of the spill include crude oil released by tankers on land. In water bodies, the spill happens due to drilling rigs, offshore oil platforms and wells. Oil spills and their effects can also be seen with refined petroleum and waste oil from large-scale industries.

■ *How do oil spills occur?*

Oil Spills may occur because of several reasons:

1. **When oil tankers have equipment faults:** When oil tankers break down, they get trapped on a narrow land. When the tanker is tried to shift from narrow land, a scratch may cause a hole in the tanker that will result in large amounts of oil being discharged into the oceanic bodies. However, although this type of oil spill is most frequently known and has the maximum media attention, only 2% of oil in water bodies is a consequence of oil spill.
2. **From nature and human activities on land:** The majority of oil is spilled from natural seeps, geographical seeps from the ocean bed along with leaks that occur when products using petroleum or various types of oil are used on continents, and later the oil is washed off into various water bodies.
3. **Water Sports:** Other reasons for oil spills are spills by petroleum operators of released oil. This occurs when various water sports or water vehicles such as motorboats and jet skis leak fuel.
4. **Drilling works carried out in sea:** For drilling works in the sea, the oil and petroleum are used but when released into the sea, causes an oil spill.

■ *How Can Oil Spills Cause Damage to the Environment?*

The effects of an oil spill depend on various factors counting the quantity and nature of oil spilled and how it acts with the aquatic environment.

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Prevalent weather conditions will also affect the oil's physical features and its behaviour. Other key factors comprise the biological and ecological attributes of the area; the ecological importance of key species and their sensitivity to oil pollution along with the time of year.

Oil spills may influence the environment in the following ways:

- **Physical smothering of organisms:** Mainly caused by oils with great viscosity, i.e., heavy oils. Suffocating organisms' physical ability to inhibit critical functions such as respiration, feeding and thermoregulation.
- **Chemical toxicity:** This is a feature of lighter chemical constituents that are easily bio-available, i.e., absorbed into organs, tissues and cells, and can have sub-lethal or lethal toxic effects.
- **Ecological changes:** Caused by the damage of key organisms with a particular function in an ecological community.
- Can be substituted by diverse species undertaking alike functions in which case the consequences for the ecosystem as a whole may not be deep.
- More harmful is the niche in the community being substituted with organisms acting on completely different tasks thereby changing the ecosystem dynamics.
- **Indirect effects:** Damage of shelter or habitat through lubricating or clean-up operations.

QUICK INSIGHT

Deep-water Horizon incident/BP Oil Spill

Deep-water Horizon oil spill of 2010, also called the **Gulf of Mexico oil spill of 2010**, the largest marine oil spill in history, produced by April 20th, 2010, an explosion on the Deep-water Horizon oil rig – located in the Gulf of Mexico, approximately 41 miles (66 km) off the coast of Louisiana – and its subsequent sinking on April 22.

The Deep-water Horizon rig, owned and functioned by offshore-oil-drilling company Transocean and leased by oil company BP, was situated in the Macondo oil prospect in the Mississippi Canyon, a valley in the continental shelf. The oil well over which it was located on the seabed 4,993 feet (1,522 metres) below the surface and extended approximately 18,000 feet (5,486 metres) into the rock. On the night of April 20, a surge of natural gas blasted through a concrete core lately installed by contractor Halliburton in order to seal the well for later use.

Once released by the fracture of the core, the natural gas moved up the Deep-water rig's riser to the platform, where it burnt, killing 11 workers and injuring 17. The rig overturned and sank on the morning of April 22, 2010, breaking the riser, through which drilling mud had been injected to counter the upward pressure of oil and natural gas. Without any conflicting force, the oil began to discharge into the gulf.

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OIL SPILLS and OIL ZAPPER

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WHAT IS AN OIL SPILL?

Discharge of liquid petroleum into the environment (water bodies) by vehicles, vessels, or pipelines.
Oil Spill is a chief form of pollution caused largely by anthropogenic causes .

SOURCES OF OIL SPILLS

Shipping & Transport: Tanker accidents, operational discharges, or hull breaches

Offshore Drilling: Accidental releases during exploration/production

Land Sources: Industrial runoff and natural seeps entering the ocean

ENVIRONMENTAL IMPACT

Physical Smothering (Heavy Oils)	Chemical Toxicity (Lighter Oils)	Ecological Disruption
Oil suffocates organisms and impairs feeding/thermoregulation	Toxic chemicals damage/kill cells by absorbing into tissues	Key species loss changes ecosystems

CASE STUDY: Mumbai Shoreline (2010)	LOCATION: Arabian Sea	SCALE: 800 tonnes of oil spilled	CAUSE: Collision between two vessels

WHAT IS OIL ZAPPER?

Technique	Bioremediation (using bacteria to degrade crude oil)
Developed By	TERI (The Energy and Resources Institute)
Supported By	Dept. of Biotechnology & Ministry of Science and Technology

Composition

5 Bacterial Strains

+

(Oil zapper) Packed in sterile polyethylene bags

HOW IT WORKS?

APPLICATION
Applied directly to the spill or sludge

DIGESTION
Bacteria feed on the hydrocarbons

Hydrocarbons → CO_2 + Water

CONVERSION
Harmful hydrocarbons are converted into harmless

KEY ADVANTAGES

Eco-friendly:	Versatile:	Resilient:	Proven Success:
Non-pathogenic and residue-free	Works in both water and land environments	Operates effectively at high temperatures (up to 70°C)	Restored Mumbai shoreline (2010) by treating 130,000+ tonnes of sludge

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- *Oil Zapper*

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The new method of using bacteria to dispose of the oil spills is called 'Oil Zapping'. **Oil Zapping is a bio-remediation method including the usage of 'oil zapping' bacteria.** It was used to clean up the Mumbai shoreline damaged by the oil spill that occurred in August 2010. The Maharashtra Pollution Control Board used **The Energy and Resources Institute (TERI)** services, which has established the oil zapping bacteria. TERI had developed this technique for seven years. **The Oil Zapping project was reinforced by the Department of Biotechnology (Government of India) and the Ministry of Science and Technology.**

How does Oilzapper Work?

5 different bacterial strains are arrested and combined with a carrier material like a powdered corncob. **This mixture of 5 bacteria is called Oilzapper**. Oilzapper nourishes on hydrocarbon compounds existing in crude oil and the hazardous hydrocarbon waste produced by oil refineries, known as **Oil Sludge** and alters them into harmless CO_2 and water. The Oilzapper is precisely filled into sterile polyethylene bags and closed aseptically for safe transference. The shelf life is 3 months at ambient temperature. The method was successfully used and 130,000 tonnes of oily sludge/oil-contaminated soil were cured as per the leading news sources.

Qualities of Oilzapper

- Economical, easy to use, and ecologically sound
- Naturally happening and non-pathogenic
- Appropriate for both aquatic and terrestrial environments
- Cleans up and reinstates the site to its original form, leaving behind no damaging residue
- Persists and stays operative under high temperature (up to 70°C)
- Cleans up the oil-contaminated location in 4 months

14.5.7 Water Quality Parameters

The water pollutants from various sources seriously impact living beings directly or indirectly. Water quality parameters indicate the level of pollutants in the water. Various parameters measuring water pollution are dissolved oxygen (DO), biochemical oxygen demand (BOD) and chemical

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oxygen demand (COD), eutrophication, bio magnification, algal bloom, MPN, TDS etc.

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ENV 56- Water Quality Parameters

Biological Oxygen Demand (BOD)

Biochemical Oxygen Demand measures the degree of water pollution from oxygen demanding wastes and plant nutrients. It is the quantity of dissolved O_2 required by aerobic organisms to decompose organic material existing in a given water sample like sewage at a specific temperature in a particular period. BOD is calculated in parts per million (ppm). BOD value is proportional to the quantity of organic waste existing in water, **with the upsurge in organic material in the sewage, the BOD rises**, BOD levels of industrial waste may be several times higher than that of domestic waste.

Table 14.4 BOD level and water quality

BOD Level in mg/litre	Water Quality
1-2	**Very Good:** Not much organic waste exists in the water supply.
3-5	**Fair:** Reasonably clean.
6-9	**Poor:** Slightly polluted, usually designates that organic matter is existing and microorganisms are decaying that waste.
10 or more	**Very Poor:** Highly polluted, contains organic matter.

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Do You Know?

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Do You Know?

Biochemical Oxygen Demand is the amount of dissolved oxygen needed by aerobic biological organisms in a body of water to break down organic material present in a given water sample at a certain temperature over a specific time period. It is a measure of, the amount of oxygen that requires for the bacteria to degrade the organic components present in water/wastewater. The BOD value is most commonly expressed in **milligrams of oxygen consumed per litre of sample** and is often used as a robust surrogate of the degree of organic pollution of water; BOD can be used as a gauge of the effectiveness of wastewater treatment plants. **Algal blooms** may occur in freshwater as well as marine environments. **Algal bloom changes the colour of the water**. These blooms led to the **killing of fishes** and **numerous reports of skin rashes**, **unappealing odours** and **accumulations of foam** and **shoreline scrums**.

Chemical Oxygen Demand (COD)

COD measures the amount **of O_2 essential to oxidise soluble and particulate organic matter in the given sample water.** COD measures the capacity of water to consume O_2 during the decomposition and nitrite available in the contaminated water or sewage.

The BOD is limited to organic waste, and it is not a reliable method for measuring water pollution. Hence **COD is used to measure all the oxygen-consuming pollutants (organic and inorganic) present in water.**

COD values and BOD values may not match, waste of textile industry and paper mills etc. **High cellulose content has COD values higher than BOD values, as cellulose is not readily attacked by dissolved oxygen (DO).** Hence, the oxygen-consuming capacity of the cellulose is measured by the COD test and not by the BOD test.

Table 14.5 Basic Difference between BOD and COD

BOD	COD
BOD is only a calculation of consumed O_2 by marine microorganisms to decompose and oxidize organic matter.	COD is the **measure of dissolved oxygen for the oxidation of both organic and inorganic constituents.**

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BOD	COD
Although some of the organic compounds, which can be broken down by microorganisms, are countable for the biological oxygen demand, they may not be encountered in measuring chemical oxygen demand.	

Dissolved Oxygen (DO)

- It is the quantity of O_2 available in the water. It refers to the level of free non-compound oxygen (oxygen not bonded to any other element or free oxygen molecules within the water) present in the water or sewage. It is a tool used to evaluate the quality of water.
- The oxygen slowly diffuses across the water surface from the atmosphere and it is also produced as an end product of photosynthesis of aquatic plants including phytoplankton and algae.
- The dissolved oxygen in higher concentration means the quality of water is better. When sewage enters a water body, microorganisms start to decompose the organic materials and microbes depleting the available oxygen in the water consume the dissolved oxygen.
- Depleted O_2 levels affect the survival of aquatic species and when organic compounds decompose under anaerobic conditions, it gives an undesirable foul smell associated with septic conditions due to the release of methane and hydrogen sulphide.
- These gases are harmful to other aquatic life forms that die a slow death. The resource value of the water body is decimated and it becomes unfit for fishing, hunting or any form of recreation. The inorganic matter acts as a food for algae, which proliferate rapidly leading to algal bloom.

14.5.8 Most Probable Number (MPN)

- The water polluted with organic trash such as sewage/sludge will have a high population of bacteria such as *E. coli* and *coliforms*.
- MPN test helps in the detection of both *E. coli* and coliforms.
- MPN method statistically envisages the number of organisms existing in the water body. Polluted water shows a high value of MPN.
- Coliforms existing in human intestines are not certainly harmful to us. But its existence specifies the presence of human waste in the water.

14.5.9 Total Dissolved Solids (TDS)

- The volume of salts and solids dissolved in water is assessed by testing the TDS and salinity content. Excessive quantities of TDS damage water quality.

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- Some of the dissolved materials make water quality poor such as calcium, phosphorus, iron sulphates, carbonates, nitrates, chlorides and other salts and heavy metals.
- Elevated TDS can result in water giving a bitter or salty taste; causing incrustations, films or precipitates on fixtures; corrosion of fixtures, and lessened efficiency of water filter and equipment.
- A raised TDS indicates:
  (i) The content of the dissolved ions turns water to be corrosive, salty or brackish in taste, leading to scale formation and interfering and decreasing the efficiency of water heaters.
  (ii) Many contain higher levels of ions which are above the Primary of Secondary Drinking Water Standards, such as raised levels of nitrate, arsenic, aluminium, copper, lead, etc.

14.5.10 Biological Magnification

The event through which certain pollutants get gathered in tissues in growing concentrations along the food chain is called biological magnification. Pollutants (e.g., DDT and polychlorinated biphenyls (PCB)) are non-biodegradable, i.e., when an organism absorbs, they fail to metabolise and break down or get excreted out. These pollutants usually get accumulated in fat cell-containing tissues of an organism. The typical example of biological magnification is that of **PCB**. A factory toxic waste which contaminated water in Great Lakes in the USA during the 1960s, resulting in deterioration of fishes and birds. It was exposed that the PCB concentration, which was 0.000002 ppm in water, amplified along the food chain to as much as 4.83 ppm in fishes and 124 ppm in birds. Many other stubborn pesticides (e.g. DDT) display similar magnification. Pollutants such as **PCB and DDT** are known to cause disturbances in animal reproduction.

14.5.11 Eutrophication

Further inorganic nutrient input with the influx of wastewater, putrefaction of organic wastes also upsurge water bodies' nutrient content. Accessibility of extra nutrients causes profuse development of algae (algal bloom), especially the blue-green algae. Such algal blooms may completely cover the water surface, often discharge toxins in water, and sometimes lead to a

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deficiency of O_2 in the water. This leads to **BOD**. Thus, in bloom-infested water, the development of other algae may be subdued due to toxins, and **aquatic animals (e.g., fish) may die due to toxicity or lack of O_2** (i.e. effect of BOD). The procedure of nutrient enhancement of water and consequent damage of species diversity is referred to as eutrophication.

ENV 57- Eutrophication

Eutrophication is the phenomenon in which high amounts of nutrients from sewage, fertilisers and detergents enter the streams and lakes leading to the enrichment of the water bodies and triggering excessive growth of phytoplankton and other aquatic weeds. Eutrophication is of two types, they are:

1. **Natural Eutrophication:** It is a normal process of increasing nutrient levels and biological productivity over time. It occurs gradually at a rate that may not be noticeable in a human lifetime.

The lakes become more productive and fill up with plants and other vegetation ultimately becoming marshy and then are metamorphosed into a terrestrial ecosystem in the process of succession.

2. **Cultural or accelerated eutrophication:** Anthropogenic activities act as catalysts and accelerate the process of eutrophication and increase biological productivity, which is known as cultural eutrophication.

It is known mainly by domestic sewage and fertiliser run-off from croplands.

14.5.12 Algal Bloom

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Eutrophication is followed by algal bloom. An algal bloom is a fast rise or gathering in the population of algae (microscopic) in a water system.

- Algal blooms may happen in freshwater and also in marine environments. The excessive growth of algae can be observed as green, yellowish-brown or red dispersion in the water. Cyanobacteria blooms are often called **blue-green algae**.
- The concentration of algae in the bloom is as high as millions of cells per millilitre.
- Some species of algae produce **neurotoxins**, so the water becomes poisonous and kills fish and other life forms. A **harmful algal bloom (HAB)** causes negative effects on other organisms via the creation of natural toxins, mechanical injury to other organisms or other means.

Algal bloom can enter the food chain, make its way up and disrupt the links in the aquatic food web and cause mortality.

1. The formation of neurotoxins causes mass deaths of fish, seabirds, sea turtles and marine mammals.
2. Human illness or demise via ingestion of seafood contaminated by toxic algae.
3. Mechanical harm to other organisms, such as disturbance of epithelial gill tissues in fish, leading to asphyxiation.
4. Oxygen reduction of the water column (hypoxia or anoxia) from cellular breathing and bacterial degradation.

14.5.13 Effects on Human Health

Domestic sewage comprises pathogens such as viruses, bacteria, parasitic protozoa and worms. Contaminated water, therefore, can transmit the germs of water-borne diseases such as jaundice, cholera, typhoid, amoebiasis, etc. Such contamination may create the water unfit for drinking, bathing, swimming and even for irrigation. Heavy metal contamination of water can lead to serious health problems. Mercury poisoning (Minamata disease) due to ingestion of fish caught from Hg-contaminated Minamata Bay in Japan, was discovered in 1952. Mercury compounds in wastewater are transformed by bacterial action into tremendously toxic methyl mercury, which can result in numbness of limbs, lips and tongue, deafness, blurring of vision, mental derangement. Cadmium pollution results in itai-itai disease (ouchouch disease, a painful disease of bones and joints) and cancer of the liver and lung.

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EUTROPHICATION

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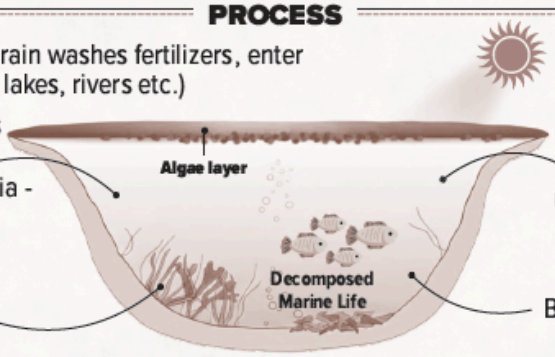
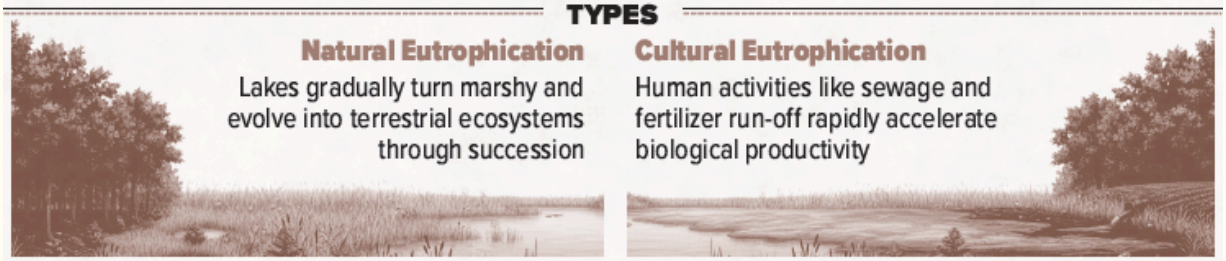
The process of nutrient enrichment in water bodies and the consequent loss of species diversity

PROCESS

1. **Nutrient Runoff:** Heavy rain washes fertilizers, enter into water bodies (Ponds, lakes, rivers etc.)
2. **Rapid Growth:** Nutrients supply increases productivity (Cyanobacteria - algal bloom etc.) in water bodies
3. **Sunlight Blocked:** Overgrowth of bloom stop penetration of solar radiation in water bodies so submerged plant die
4. **Oxygen Depletion:** Bacterial decay consumes remaining oxygen
5. **Ecosystem Death:** Oxygen starvation kills most aquatic life



TYPES



Natural Eutrophication

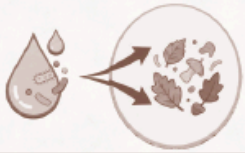
Lakes gradually turn marshy and evolve into terrestrial ecosystems through succession

Cultural Eutrophication

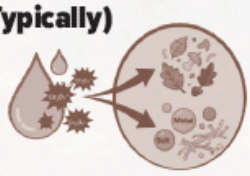
Human activities like sewage and fertilizer run-off rapidly accelerate biological productivity

Biochemical Oxygen Demand (BOD) Vs Chemical Oxygen Demand (COD)

BOD < COD (Typically)



Consumed O_2 by Aerobic Microorganisms



O_2 Essential to Oxidise

Calculations of consumed O2 for organic matter only	Measure of O_2 for oxidation of organic + inorganic matter
1. It is the quantity of dissolved O_2 required by aerobic organisms to decompose organic material existing in a given water sample	1. It measures the amount of O_2 essential to oxidise soluble and particulate organic and inorganic matter
2. **Limited to organic waste** and is proportional to the quantity of organic material existing in the water	2. Measures dissolved O2 required for the oxidation of **organic and inorganic constituents**
3. Considered **not a reliable method** for measuring total water pollution as it excludes inorganic waste	3. Used as a **more reliable method** to measure the total capacity of water to consume O2
4. Shows lower values for waste with high cellulose content (e.g., textile/paper mills) as cellulose is not readily decomposed by bacteria	4. Shows values higher than BOD for cellulose-rich waste as it measures the full oxygen-consuming capacity of cellulose
5. Relies on aerobic biological organisms to break down organic material	5. Relies on the oxidation process to measure the capacity of water to consume O_2

14.5.14 Groundwater Pollution

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In India, at many locations the groundwater is susceptible to contamination due to seepage from industrial and municipal wastes and effluents, sewerage channels and agricultural run-off. For example, additional nitrate in drinking water is hazardous for human health and may be deadly for infants. It reacts with haemoglobin and forms nonfunctional methaemoglobin that damages oxygen transport. This is called methemoglobinemia or blue baby syndrome. Excess fluoride in drinking water leads to teeth deformity, hardened bones and stiff and painful joints (skeletal fluorosis). At many locations in India, groundwater is contaminated with arsenic, mainly from certainly occurring arsenic in bedrocks. Uranium found in groundwater is also a problem in Punjab and leads to the delivery of mentally retarded children.

Overexploitation of groundwater may perhaps enhance the leaching of arsenic from soil and rock sources and pollute groundwater. Continuing exposure to arsenic leads to black-foot disease. Arsenic results in diarrhoea, peripheral neuritis, hyperkeratosis and also lung & skin cancers.

Do You Know?

Nanoparticles are particles between 1 and 100 nanometres in size. As they are very tiny particles, they can accumulate in the environment and contaminate water and soil and they can enter the food chains. Zinc oxide and Titanium oxides nanoparticles trigger the production of free radicals in presence of ultraviolet light.

14.5.15 Improving Water Quality

The industrial and municipal sewages are treated in **Effluent Treatment Plant (ETP)** before disposal in water bodies. Generally, the following treatments are given in ETP

(i) **Primary treatment:** Process involves the parting of large debris, followed by sedimentation in tanks or clarifiers.

(ii) **Secondary treatment:** A biological process, carried out by microorganisms. Here, the sewage is pumped in narrow stabilisation or oxidation ponds, where the microbes oxidise only the organic matter. The procedure results in the discharge of CO_2 and the

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creation of sludge or biosolid. The sludge is endlessly aerated to further its oxidation. Algae grown in the highlighted region of the wastewater provide aeration by generating O_2.

(iii) **Tertiary treatment:** This physio-chemical procedure eliminates turbidity in sewage resulting from the existence of nutrients (nitrogen, phosphorus, etc.), dissolved organic matter, metals or pathogens. This step includes chemical oxidation of sewage by resilient oxidising agents, Such as Cl gas, perchlorate salts, O_3 gas and UV rays. After tertiary treatment, the wastewater can be released into natural waters or recycled for irrigation.

14.5.16 Marine Pollution

About 71% of the earth's surface is covered with oceans. Oceans are the ultimate sink of natural and man-made pollutants. In addition to the pollutants entering the ocean through rivers, oceanic pollution consists of navigational discharge of oil, detergents, sewage, garbage and radioactive waste, offshore oil mining, oil spills and volcanic eruption. Marine pollution happens when damaging, or potentially destructive chemicals, particles, industrial, agricultural and domestic waste, noise or the spread of invasive organisms enter into the marine ecosystem.

■ *Sources of Marine Pollution*

Both physical and anthropogenic sources contribute to marine pollution. While the main cause of marine pollution may be similar to that of general water pollution, it includes some unexpected natural events and anthropogenic sources which can be accidental or deliberate. Some of the sources are

1. **Plastics:** Represents the latest contaminant in the marine environment. The plastic discarded is well known to be caused primarily from fishing-related activities, and non-point source invasion from beaches.
2. **Crude oil and petroleum:** Naturally occurring seeps have been dripping large quantities of oil into the sea for millions of years.

The quantity of oil entering the marine ecosystem has increased significantly in recent years due to **marine** transportation, offshore drilling, nearshore refining and street run offloaded with waste oil from automobiles.

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3. **Intentional release of petroleum products:** Some petroleum and petroleum products are released in the oceans intentionally, quietly and routinely, during the loading, unloading, discharging and flushing of tanker ships. This is harmful to seabirds, zooplanktons and microorganisms.
4. **Sewage disposal into the seas and oceans: One of the biggest** marine pollute is the volume of sewage waste **discharged into the sea.**
5. **Volcanic eruption:** It adds enormous quantities of carbon dioxide, methane, compounds, oxide of nitrogen, etc. into the oceans.
6. **Cosmic dust:** Another source of pollution is meteors and cosmic dust from space falling into the oceans

■ *Effects of Pollutants of Marine Life*

1. **Impact of industrial effluents:** Leads to a spike in heavy metals in water.

Bioaccumulation of mercury in fish and shellfish are passed on to humans who consume the same. Industrial effluents also deplete oxygen levels, which suffocates and kills aquatic life.

2. **Impact of oil spills:** Leads to massive killing of fish. Shore-based life forms such as crabs, lobsters, starfish, barnacles and seaweeds are also badly affected.

Seabirds face high causality due to oil spills. Their plumage gets drenched in oil, making it impossible to fly and eventually die.

3. **Impact on coral reefs:** Pollution of ocean water leads to the disproportionate amount of algal growth and underwater plant growth.

This reduces the dissolved oxygen level in the water and the corals get a diminished supply of oxygen, affecting their growth.

4. **Impact of plastics:** Plastic beads about 20 microns or so in size pose a challenge to the Antarctic krill and other zooplankton.

The coral reefs, which get chocked by micro plastic, get severely affected, eventually leading to their death.

■ *Marine Pollution Control*

Some of the important preventive and technological steps are:

1. **Treatment of sewage:** Installation of treatment plants along the drains which discharge liquid and solid wastes into the bays and seas.
2. **Sludge Processing:** Sludge containing pollutants and harmful bacteria should be treated before it is discharged into the seas, gulfs and oceans.
3. **Seepage oil-proof tankers:** The loading and unloading of crude oil and petroleum in the tankers should be through the ‘top load system’ including special arrangements for

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the washing and cleaning of the tankers.
- There should be dry-docking for servicing, repairs, cleaning the hull, etc. to keep the residual oil from entering into the sea.
- To prevent oil tanker accidents by improving the navigation standards
- In case of a tanker accident, the responsibility of cleaning the polluted marine area should be of the country to which the tanker belongs.

4. **Effective international law:** There is a need to legislate more effective laws at the international level with severe financial compensation/ punishment to those countries that do not abide by the international laws, rules and regulations.

■ *Beach Pollution in India*

- The National Centre of Coastal Research (NCCR) has released a report on beach pollution in India.

Highlights of the report:

- The NCCR conducted a qualitative analysis of the litter on six different beaches on the eastern and western coasts. The study notes that beach pollution is on the rise in the country.
- Tourism and fishing are the biggest culprits, contributing most of the plastic litter on beaches.
- The study found that plastic litter from tourism alone accounted for 40%–96% of all beach litter.
- At Chennai's Elliot's Beach, for instance, plastics left by tourists accounted for 40% of all the litter, while at Gopalpur in Odisha, it was as high as 96%. As for the other four beaches, plastics formed 66% of the overall litter on Fort Kochi Beach, 60% at Karnataka's Karwar beach, 87% at Visakhapatnam's R.K. Beach and 81% at Andaman Island's Rangachang beach.
- After tourism, fishing was the next biggest source of litter. While fishing nets were a major contributor, the processing of fish on the beach also produced a lot of litter.
- Also, the proportion of biomedical litter was high in urban areas, such as Elliot's Beach and Fort Kochi Beach.
- Other than the plastic litter dropped by tourists, similar waste from creeks and inlets made its way into the sea in the monsoon.
- Most of the litter consisted of plastic bottles, cutlery and thermocol.

■ *Way ahead- need of the hour*

- India needs a national marine litter policy to control and manage waste on land and prevent its entry into the marine environment.
- Experts suggest the installation of debris booms and fin deflectors upstream as measures to reduce the quantity of floating solid waste entering coastal waters.

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- India also needs to start blue-flagging its beaches. The 'blue flag' is a globally recognised eco-label awarded to beaches and marinas that adhere to strict environmental and safety norms.

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More about 'Blue Flag' project

- Launched in December 2017 by the Environment Ministry, the prime objective of the project is to enhance standards of cleanliness, upkeep and basic amenities at beaches.
- Under the project, each state or union territory has been asked to nominate a beach which will be funded through the ongoing Integrated Coastal Management Programme.

Criteria for certification:

- To achieve the Blue Flag standards, a beach has to strictly comply with 33 environment and tourism-related conditions.
- The standards were established by the Copenhagen-based Foundation for Environmental Education (FEE) in 1985.
- For example, a beach must be plastic-free and equipped with a waste management system. Clean water should be available for tourists, apart from international amenities.
- The beach should have facilities for studying the environmental impact around the area.

14.5.17 Ocean Acidification

Ocean acidification is the continuing reduction in the earth's oceans' pH, resulting from the uptake of CO_2 from the atmosphere. An estimated 30–40% of the CO_2 from anthropogenic activity released into the atmosphere dissolves into oceans, rivers and lakes. Ocean acidification has been called the 'evil twin of global warming' and 'the other CO_2 problem'.

Causes of Ocean acidification

- Emissions have amplified over the last 200 years, chiefly **due to increased industrialisation and agriculture leading to more burning of fossil fuels, cement manufacturing and land use change**.
- Anthropogenic CO_2

Chemical Change

- To achieve chemical equilibrium, some of it interacts with the water to produce carbonic acid. Some of these additional carbonic acid molecules combine with a water molecule to give a bicarbonate ion and a hydronium ion, thus snow balling ocean acidity (H+ ion concentration).

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- As the oceans absorb CO_2, it mixes with water (H_2O) to form carbonic acid (H_2CO_3). The carbonic acid then detaches, producing a hydrogen ion (H+) and a bicarbonate ion (HCO_3-)
- Ocean Acidification is a calculation of the concentration of hydrogen ions– the more hydrogen ions, the lower the pH higher is the acidity. Hence, growing levels of hydrogen ions from the absorption of CO_2 lowers the ocean's pH and raises its acidity.

Biological change

- There has been a 30% reduction in pH and a 16% reduction in carbonate ion content. Carbonate ions are taken up by organisms to make shells and reef systems thus it may be more problematic for organisms to do this in the forthcoming time as carbonate ions decline further.
- Organisms are a social food source (e.g. molluscs), food for fish (e.g. pteropods are eaten by salmon) and form ecosystems and shelters for fish (e.g. coral reefs).

■ *Human consequences*

Oceans acidification is expected to threaten food safety, damage fishing industries and raise the risk of inundation and erosion in low-lying areas by waning natural shoreline protection. In terms of adaptation, ocean acidification impends marine and coastal ecosystem resilience and hence declines the capability of both humans and natural systems to adapt to the current change.

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OCEAN ACIDIFICATION

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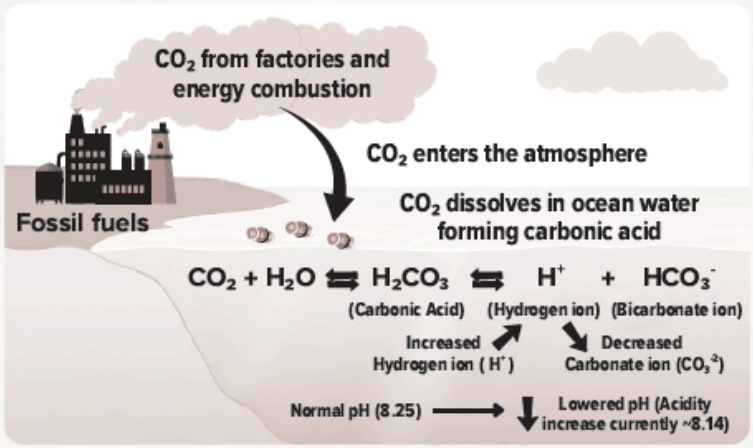
Ocean acidification is the long-term decrease in the pH of Earth's oceans caused by the absorption of atmospheric CO_2, with about 30–40% of human-generated CO_2 dissolving into oceans, rivers, and lakes.



Primary Causes of CO_2 (Anthropogenic Emissions)

Amplified emissions over the last 200 years due to:

- **Deforestation**
- **Burning of fossil fuels**
- **Increased industrialization and cement manufacturing**
- **Agriculture and land use change**

BIOLOGICAL CONSEQUENCES

Carbonate Decline

16% drop, reducing material needed for shells and reefs

Calcifiers Affected

Shell-building becomes difficult for oysters, corals, and plankton

Ecosystem impact

Affects food webs and ecosystems i.e. coral reefs

Behavioural Effects

Impairs predator detection in some fish, harming the food web

HUMAN & SOCIO-ECONOMIC CONSEQUENCES

Food Security Threat

Damages fishing industries, a primary source of protein and dependency for many economies

Coastal Risk

Increases flooding and erosion risk in low-lying areas by reducing natural shoreline protection

Reduced Resilience

Reduces the ability of humans and ecosystems to adapt by weakening marine and coastal resilience

GLOBAL MEASURES & RESPONSE

IAEA Initiative:
Set up the Ocean Acidification Centre in Monaco to support science and policy

UNFCCC Mandate:
Aims to stabilize greenhouse gases and prevent harmful climate impacts

National Actions:
Australia, China, the EU, and the USA research or include ocean acidification in adaptation plans

Within UNFCCC:
Use indicators, targets, and geo-engineering

- ***Measures***

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- In reply to the international scientific community's call for more synchronised research and activity, the IAEA has formed the Ocean Acidification International Coordination Centre, which will be launched soon. The Centre is located at the Environment Laboratories of the IAEA in Monaco and is reinforced by several IAEA Member States. It will be recognised under the IAEA Peaceful uses Initiative project.
- The Ocean Acidification International Coordination Centre will aid the scientific community, policymakers, schools, media and the general public.

Ocean Acidification and the UNFCCC Mandate

- The UNFCCC aims is to steady the amount of greenhouse gas concentrations in the atmosphere at a level that would stop hazardous intrusion with the climate system.
- **National Actions on Ocean Acidification:** National actions and lessons learned on ocean acidification can be used to inform international policy development. Several Parties including Australia, China, Germany, Japan, the Republic of Korea, the United Kingdom, the United States of America and the European Union, have ongoing or planned research programmes looking at various aspects of ocean acidification or research coordination thereof. Some countries, including Australia, Chile, Indonesia, Ireland, Japan, Portugal and the U.K., make reference to ocean acidification and its implications for marine ecosystems and fisheries in their national climate change plans and strategies, or more specifically, in their climate change adaptation plans.

Ways to Address Ocean Acidification within the UNFCCC:

- Effective indicator
- Effective Targets
- Geo-engineering

Policy Actions

- **Clean Development Mechanism**
  The CDM permits emission-reduction projects in emerging countries to receive certified emission reduction (CER) credits, each equal to one tonne of CO_2. These CERs can be transacted, sold and used by developed countries to achieve a part of their emission lessening targets under the Kyoto Protocol.
  The instrument encourages sustainable development and emission reductions while giving developed countries some flexibility in how they attain their emission-reduction restriction targets.
  It is the chief source of income for the UNFCCC Adaptation Fund, which was recognised to fund adaptation projects and programmes in emerging countries. Parties to the Kyoto Protocol are principally vulnerable to the adverse effects of climate change. The Adaptation Fund is funded by a 2% levy on CERs issued by the CDM.
- **Geo-engineering**
  Geo-engineering is a method that aims to change the earth's albedo that will have little to no effect on ocean chemistry and does not offer a solution to alleviate the threats posed by ocean acidification. The risks associated with large-scale manipulation of

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earth systems, which are great, need to be a driving factor in deciding whether or not geo-engineering schemes are a viable option for combating climate change.

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14.6 SOIL POLLUTION

Soil pollution generally occurs from different human activities, such as dumping wastes, using agrochemicals, mining operations and urbanisation. The depreciation in soil quality or soil degradation may occur due to the following reasons: high rate of soil erosion, lack of microorganisms living in soil, absence or abundance of required moisture in the soil, highly fluctuating temperature, less amount of humus in the soil, presence of different types of pollutants in the soil.

14.6.1 Natural Pollutants

1. **Plant Origin**

- The life cycle of plants carries pathogens and pests, which become soil contaminants.
- Nitrogen oxide combines with soil water and forms nitrous and nitric acids, increasing the soil acidity and lowering the pH. This creates unfavourable conditions for plants and soil microfauna and flora.

2. **Animal Origin**: Animal droppings such as urine and blood are pollutants from animal sources. When animals are left to decompose, there is a spike in the organic matter in the soil. This overloading of organic matter creates unhealthy conditions for plant growth. The soil becomes hypertonic, resulting in wilting and stunting plant growth. Some other common natural pollutants are: Accidental spills, acid rain, industrial accidents, deforestation, genetically modified plants, nuclear wastes, intensive farming, landfill and illegal dumping, mining and other industries, oil and fuel dumping, buried wastes, disposal of coal ash, electronic waste, disposal of ammunition and war agents, agricultural practices such as the application of pesticides, herbicides, and fertilisers, and drainage of contaminated surface water into the soil.

14.6.2 Man-Made Pollutants

1. **Agrochemical**

- Chemical fertilisers and pesticides, when added to crops, improve yield. These remain in the soil even after harvesting, resulting in increased soil salt content.
- Crop waste is usually burnt in the field as stubble burning. This yields hydrocarbons and PM, apart from carbon monoxide and nitrogen dioxide. Carbon monoxide is very toxic and detrimental to root growth and soil microbial substance.

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2. **Pesticides**

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- Commonly used pesticides such as DDT, Aldrin which are organo-chlorides dissolve more easily in the bloodstream than in water affecting the nervous and endocrine systems.
- They cause the eggshells of birds to lack calcium, making them easily breakable and are accountable for the deterioration of the numbers of birds of prey and falcons in the 1970s.

3. **Industrial waste**

- Fly ash from thermal power plants, effluents from dye and paint factories, are released in the open. They contaminate the soil and disturb the composition which balances the healthy presence of microflora and fauna.

4. **Seepage from landfills**

- Landfills created to dump solid waste often have leachates formed by the decomposing content that contaminate the soil.

5. **Acid Rain**

- Acid rain affects the soil pH value, which in turn hinders plant growth. It strips the soil of its nutrient components, making the trees and plants susceptible to attack from viruses, fungi and bacteria.

14.6.3 Impact of Soil Pollution

■ *Agriculture*

(i) Reduced soil fertility
(ii) Reduced crop yield
(iii) Increased erosion
(iv) Increased salinity

■ *Health*

Heavy metals present in paints, dyes and chemicals, pesticides and insecticides cause contamination of soils, which is toxic to human and animal life. E.g. **Arsenic:** Damages skin, kidney and central nervous system.

Mercury: Damages brain and nervous system and leads to pulmonary fibrosis, lung cancer and kidney failure.

■ *Environment*

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(i) Reduced vegetation
(ii) Ecological imbalance

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■ *Urban areas*

(i) Clogging of drains
(ii) Inundation of areas
(iii) Foul smell and release of gases
(iv) Waste management problem

14.6.4 Control Measures

1. **Organic farming:** Adoption of organic farming practices such as using organic manure **'panchagavya'** to improve crop yield as well as pest management.

Many herbal decoctions are prepared using plant material to kill pests and weeds. These must be widely advocated.

Table 14.6 Various activities of noise pollution & its effect on human beings

Activities	Sound Pressure (dbA)	Perceived Loudness	Effect
Rocket engine	180	Painful	Eardrum rupture
Jet takeoff (25 metres away)	150	Painful	Eardrum rupture
Jet takeoff (100 metres away), earphones at loud level	130	Painful	
Thunderclap, textile loom, live rock music, jet takeoff (161 metres	120	Uncomfortably loud	Human pain threshold

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Activities	Sound Pressure (dbA)	Perceived Loudness	Effect
away), siren (close range), chain saw			
Steel mill, riveting, automobile horn at 1 metre, stereo held dose to ear	110	Uncomfortably loud	
Busy urban street, diesel truck, food blender, cotton spinning machine	90	Very loud	Hearing damage (8 hrs), speech interference
Conversation in the restaurant, average office, background music	60	Moderately loud	Intrusive
Whisper, rustling levels	20	Very quiet	
Breathing	10		
Silence zone is a zone comprising not less than 100 metres around hospitals, educational institutions, courts, religious locations or any other area acknowledged as such by a competent authority.			

2. **Natural farming:** it is the practice of no-tilling and no-agrichemical usage and depends on natural methods and mulching.

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3. **Application of biological system:** certain plants remove metal contamination of soil. **Phytoextraction, rhizofiltration and phytostabilisation** are the three processes by which this takes place. In Chernobyl, sunflower crops were used to remove radioactive contaminants from groundwater.
4. **Bio-augmentation:** in this method, **bacterial cultures are added to the soil** to speed up the process of breakdown of contaminants.
5. **Bioremediation and phytoremediation:** it is the use of microorganisms (bacteria and fungi) to damage the environment into a less toxic form. The microorganism may be native to a polluted area or they may be invasive.

Recently researchers have identified the microorganism, which can decompose plastic waste. **Ideonella sakaiensis** is a bacterium from the genus ideonella and family Comamonadaceae capable of breaking down the **PET (polyethylene terephthalate)** plastic was isolated from outside a plastic bottle recycling facility.

14.7 NOISE POLLUTION

- Sound is the intermediate for communication. It is almost unattainable to lead our day-to-day life without sound.
- Sound is irritating and damaging if it becomes noise. Noise is any undesirable or exceptionally high levels of sound that can irritate, cause stress or damage the hearing ability.
- Noise pollution was first considered as a problem in **the United Nations Environment Conference in Stockholm in 1972.**
- According to the **Environment Protection Agency (EPA), USA,** sound becomes unwanted when it either inhibits normal activities such as sleeping, conservation or disrupts or reduces one's quality of life.
- Since sound is invisible, tasteless and without any smell, it has not become the subject of study or debate such as air or water pollution.
- The persistent and escalating source of sound can often be considered an irritation, which can have major consequences to one's overall health.

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ENV 58- Noise Pollution

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14.7.1 Types of Sound

Depending on the nature of the continuity or periodicity of sound, the following types may be mentioned.

1. **Continuous noise:** The incessant sound level is less than 5 dB continuously. E.g. running fan.
2. **Intermittent noise:** The noise that is not produced continuously but with breaks. E.g. drilling machine.
3. **Impulse noise:** It is the instinct noise characterised by a noise level of more than 40 dB within half-second with the duration of less than one second. E.g. firing of the weapon.

14.7.2 Impact of Noise Pollution

1. **Annoyance:** Creates irritation to the receptors because of sound level variation. Irregular occurrence causes anger and frustration.
2. **Physiological effects:** Activities such as breathing rate, blood pressure, heart rate, pulse rate amd blood cholesterol are affected.
3. **Loss of hearing:** Prolong contact to high sound levels may lead to hearing loss. This is mostly ignored but has a negative influence on hearing function.
4. **Human performance:** The working efficiency of the workers/ humans will be affected due to distraction of concentration.
5. **Nervous system:** Pain, ringing in the ears, feeling of tiredness, affecting the functioning of the human system.
6. **Sleeplessness:** It affects the sleep rhythm, causing restlessness and lack of concentration in their activities.
7. **Damage to building material:** Buildings and building material may be damaged from exposure to infrasonic/ ultrasonic waves. They can even collapse.

14.7.3 Noise Control Techniques

The techniques employed are broadly classified as:

■ *Control at the source*

1. Reducing the noise level from the domestic sector
2. Maintenance of automobile
3. Control over vibration

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4. Prohibition on the usage of loudspeakers
5. Selection and maintenance of machinery

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■ *Control in the transmission path/medium*

(i) Installation of barriers
(ii) Design of buildings
(iii) Green belt development

■ *Using protective equipment at receivers end*

(i) Job rotation
(ii) Reduced exposure time
(iii) Hearing protection
(iv) Documentation of noise measurements, continuous monitoring are the need of the hour.

Table 14.7 Noise Levels in Various Zones

Category of area/zone	Limits in dB(A)Leq	
	Day time (6 am to 10 pm)	Night time (10 pm to 6 pm)
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silence zone	50	40

14.7.4 Legislative Measures to Control Noise Pollution

Air (Prevention and Control of Pollution) Act 1981

It was amended in 1987 to include noise as an air pollutant.

Ambient noise levels for various areas are as follows: Real-Time Ambient Noise Monitoring Network was launched by the Government of

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India in 2011.

- **Phase-1:** five remote noise-monitoring terminals each have been installed in *different* noise zones in seven metros (Delhi, Hyderabad, Kolkata, Mumbai, Bangalore, *Chennai* and Lucknow).
- **Phase-2:** another 35 monitoring stations will be installed in the same seven cities.
- **Phase-3:** it will cover installing 90 stations in 18 other cities.

Bureau of Indian Standard

- Bureau of Indian Standard has set the standards, 125 dB is the maximum decibel limit for horns used in the commercial vehicle, 125 dB for two-wheelers.
- A violation can lead to confiscation of driver's licence or a fine under the Motor Vehicle Act

Ministry of Environment, Forest and Climate Change

- Ministry of Environment, Forest and Climate Change recommends a code of practice for controlling noise from sources other than industries and automobiles.

QUICK INSIGHT

Decibel: Also known as a dB, it is a unit for measuring the relative intensity of sounds, equal to one-tenth of a bell. A bell is used in physics to measure the difference in the intensity level of sounds to normal human ears, equal to ten decibels. It is estimated that 0 dB corresponds roughly to the quietest sound that a healthy young adult can hear. The normal conversation has a level of 60–70 dB, sound from the noisy street, 90–95 dB while sounds above about 100 dB tend to be uncomfortably loud and can damage our ears if heard for a long time. Sounds with a level above 120 dB can damage peoples' ears within a very short time, perhaps only a few minutes. The noise level of 100 decibels would correspond to noise from a machine shop.

QUICK INSIGHT

Green Mufflers: Green mufflers are barriers grown near noisy places to reduce the impact of noise pollution. Normally 4 to 5 rows of green plants are grown near noisy places such as a highway or industrial areas to obstruct the sound-reaching citizens. The trees even block the sharp lights of the vehicles at the time of night. There is a scheme named '**Green Muffler**' **which is now mandatory to construct highways in many parts of the world**.

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14.8 NUCLEAR HAZARDS

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Nuclear pollution may be defined as the physiological pollution of air, water and soil caused by radioactive elements. The radioactive wastes are mainly released from thermo-nuclear explosions in which disintegration of atomic nuclei of certain elements emits electromagnetic waves of very short wavelength. The radioactive elements are uranium, thorium, radium etc. Nuclear pollution poses a serious threat to all living organisms.

The electromagnetic waves are b-rays, g-rays and a-rays. The rays are normally a health hazard in the form of internal radiation through ingestion, inhalation or open wounds. These do not cause external hazards since they cannot penetrate other layers of skin. b-rays can be a health hazard either as internal or external radiation due to ionisation in the tissues. X-rays and g-rays are very penetrating compared to a and b. They constitute the chief health hazard of external radiation although g-rays and can be a hazard also as internal radiation.

14.8.1 Causes of Nuclear Hazards

The main sources of radioactive pollution are both: natural and man-made.

■ *Natural Sources*

(i) Cosmic rays which are fast-moving highly energetic radiations, reach the earth from outer space. The intensity of cosmic rays in the biosphere is low, therefore they are not a health hazard. But they are a major hazard in space.

(ii) The terrestrial radiations from nuclides of radioactive elements are present in the rocks, soil and water. These elements could be Uranium 235, Uranium 238, Radon 222, Radium 224 and Thorium 232 etc.

■ *Man-made Sources*

(i) Testing of nuclear weapons

(ii) The use of Uranium 235 and Plutonium 239 for fission

(iii) Hydrogen or helium as fusion material

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■ Atomic Explosion

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Atomic explosions are uncontrolled chain reactions. They give rise to large neutron-flux conditions that cause other materials in the surroundings to become radioactive. Huge clouds of fine radioactive particles and gases are thrown up in the environment and are carried away to distant areas by the winds. Gradually, they settle down on earth as fall out or are brought down by rain. Strontium-90, Iodine-131, Caesium-137, unused explosive and activation products are the radioactive substances present in the fallout. When raindrops containing these radioactive particles fall on earth, radioactivity is transferred to soil particles causing soil pollution. From the soil, these radioactive materials are washed into different water sources where the aquatic organisms absorb and accumulate them through the food chains and many pass them to human beings.

■ Radioactive Isotopes

A large number of radioactive isotopes such as 125I, 14C, 32P and their compounds find wide usage in scientific research. The wastewater from these research institutions contains varying amounts of radioactive materials. When this wastewater reaches different water sources such as rivers, streams, lakes etc., through the sewers, they cause water pollution. Radioactive iodine and phosphorus also enter the food chain through water and may finally reach man through fish etc.

■ Other Sources

Varying concentrations of radiation enter the human body during different medical treatments. For instance, X-rays are common for detecting skeletal disorders; therapy for cancer patients often includes radium and other isotope radiations.

14.8.2 Effects of Nuclear Hazards

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Various effects of nuclear hazards are as follows:

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1. In high doses, radiation can cause instant death; whereas, in lower doses, it can impair the functioning of body organs.
2. Tumours, cancer and developmental changes are other long-range effects of radiation pollution.
3. When nuclear radiations and X-rays pass through genetic cells, they affect the chromosomes. When the genetic cells are affected, then the consequences are serious because the genes are affected i.e. mutation occurs. Mutations are generally transmitted from generation to generation and so the damage is inherited by progenies of the affected generation e.g. sufferers of the Hiroshima blast exist even today.
4. Some cells are more vulnerable to injury by radiations, for example

(a) Actively dividing cells (That is why cancer cells can be destroyed by radiations as they also belong to the actively growing category)

(b) Bone marrow cells which produce blood corpuscles

(c) Cells of the skin

(d) Reproductive cells

(e) Foetus or embryo

Iodine 131 damages the white blood cells, bone marrow, lymph nodes and spleen and causes tumours, skin cancer, sterility and defective eyesight.

The pollutants which happen to fall in the fields find their way into cattle grazing on the plants growing in these fields. Children who drink the milk of such cattle get leukaemia because the radioactive pollutants such as **strontium 90** get deposited in the bones in the same manner as calcium does and causes bone cancer.

14.8.3 Control Measures of Nuclear Hazards

■ *Radiation Exposure Protection*

During medical examinations, the following care should be taken.

(i) Photofluorography X-ray equipment should not be used for chest radiography.

(ii) Adequate protection should be given to limit the exposure to the desired part of the body.

(iii) Avoid routine use of X-rays such as

(a) Chest and lower back X-ray examinations

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(b) Chest radiography

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(c) Tuberculosis screening by chest radiography

Radiation Contamination Protection

Radioactive Particles are particularly hazardous, if inhaled. They radiate in the body continuously, until eliminated. This can be minimised by using laboratory hoods, air filters and exhaust systems, eliminating dry sweeping, wearing protective clothing and radiation indicators (to see the total amount of radiation to which one has been exposed), using respirators where indicated, and prohibiting smoking and eating where radioactive material is used.

Controlled area that causes exposure to radiation are required to have controlled accessibility and should be marked 'Restricted Area' or 'Radiation Zone' etc. The level of radiation pollutants should be regularly checked in the high-risk areas. Radiation resistant cases or walls should be constructed for screening workers from radioactive materials.

14.9 IRON AND STEEL INDUSTRY POLLUTION

The iron and steel industry has noteworthy effects on air, water and soil. In the sinter plants the leading emissions produced from material handling, exhaust from windbox, discharge end and cold screen are particulate emissions, mostly iron oxides, sulphur oxides, calcium oxides, hydrocarbons, carbonaceous compounds and chlorides.

The release of PM, CO, SO_2, organic compounds and other pollutants originate from several processes in the coking power plants. During the moulding operation, iron oxides, magnesium oxide and carbonaceous compounds are produced.

The most important emissions from the basic O_2 process are released during the O_2 blow period: Iron oxides, mainly heavy metals and fluorides.

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The processes which produce emissions during electric steel-making are:

- melting: iron oxide
- refining: calcium oxide from the slag
- charging: iron and metallic oxides from the alloy in the scrap
- dumping slag and tapping steel: iron oxides and oxides from the fluxes.

The environmental impact of steel production

Steel production has many influences on the environment, including air releases (CO, SOx, NOx, PM2), wastewater contaminants, dangerous wastes and solid wastes. The major ecological impacts of integrated steel mills are from coking and iron-making.

Climate change

Nearly all of the GHG releases related to steel production are from the CO_2 emissions connected to energy consumption.

Emissions to air

Coke production is the main pollution source from steel production. Air productions such as coke oven gas, naphthalene, ammonium compounds, crude light oil, sulphur and coke dust are discharged from coke ovens.

Emissions to water

Water emissions are generated from the water used to cool coke after it has ended baking. Satisfying water becomes polluted with coke breezes and other compounds. While the quantity of polluted water can be great, quenching water is fairly easy to reuse. Most pollutants can be detached by filtration.

Waste

Slag, limestone and iron ore impurities gathered at the top of the molten iron make up the main portion of iron-making by-products. SO_2 and H2S are volatised and arrested in air emissions regulator equipment and the

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remaining slag is sold to the construction industry. While this is not a pollution inhibition technique, solid waste does not reach landfills. Gaseous emissions and metal dust are the most prominent sources of waste from electric arc furnaces.

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Measures industries take to minimise emissions:

- Alteration in technology is a significant measure.
- The contaminating emissions in the iron and steel industry are regulated by a variety of approaches such as dry or wet electrostatic precipitators, scrubbers, bag houses, hoods, furnace enclosures; chemical, catalytic and biological methods.
- The Protocol agreed at the Conference in Kyoto (1997) set out promises to bind GHG emissions for industrial countries for the period 2008–2012.

This document is very significant for engineers, scientists and operators to implement technology policies and public strategies that Control of Pollution in the Iron and Steel Industry stop, reduce or eliminate opposing environmental costs in the iron and steel industry.

■ *Steel Slag*

- Steel slag, a by-product of steel making, is produced during the separation of the molten steel from impurities in steel-making furnaces.
- The slag occurs as a molten liquid melt and is a complex solution of silicates and oxides that solidifies upon cooling.
- The major mineral components of steel slag are made of minerals such as silica, alumina and titanium from iron sand, and combinations of calcium and magnesium oxides.
- Many studies have shown that steel slag can be applied in ceramics, road pavement materials and other materials.
- Steel slag powder has hydraulic properties, and its hydration process is similar to that of cement.
- The iron and steel slag used as aggregates for asphalt roads are iron and steel by-products.
- Asphalt roads are made of a mixture of aggregates, binders and fillers.
- The aggregates are typically iron and/or steel slag, sand, gravel or crushed rock, and they are bound together with asphalt itself, which is a bitumen.

14.10 COPPER SMELTING PLANTS

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Copper smelting plants separate elemental copper from copper concentrates through multiple sulphide oxidising stages.

- The copper concentrates are fed through the flash smelting furnace with oxygen-enriched air.
- In the furnace, the concentrates are instantly oxidised, after which they melt and separate by their reaction heat into copper matte with a grade of 65% and slag consisting of iron oxide, silica and other compounds.
- In copper smelting process, first, the carbon (C) combusts with oxygen (O_2) in the air to produce carbon monoxide (CO).
- Second, the carbon monoxide reacts with the ore and removes one of its oxygen atoms, releasing carbon dioxide.
- Hence, it does not release a lethal quantity of CO into the environment.
- The principal air pollutants emitted from the processes are sulphur dioxide and PM.
- Wastewater from primary copper production contains dissolved and suspended solids that may include concentrations of copper, lead, cadmium, zinc, arsenic and mercury and residues from mould release agents (lime or aluminium oxides). Fluoride may also be present, and the effluent may have a low pH.

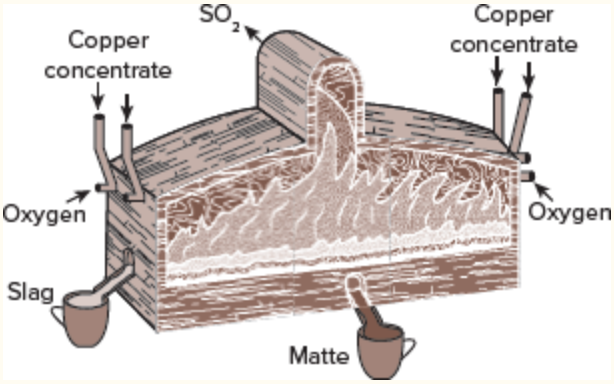

Figure 14.2 Pictorial View of Copper Smelting Plants

14.11 LEAD POISONING

Lead poisoning happens when one absorbs too much lead by consuming, inhaling or swallowing a substance containing lead in it. Lead can harm almost every organ system.

In children, too much lead in the body can result in enduring problems with growth and development. **These can affect conduct, hearing in capacities, hyperactivity, low IQ and education and can slow the child's growth**.

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In adults, **lead poisoning can harm brain and nervous system, stomach and kidneys**. It can also lead to high blood pressure and other health problems.

14.11.1 Reasons for Lead Poisoning

Worldwide, seven sources appear to account for most lead exposure:

- Gasoline additives
- Food can soldering
- Lead-based paints
- Ceramic glazes
- Drinking water systems
- Cosmetic and folk remedies
- In air and soil

14.11.2 Who Is at Highest Risk of Lead Poisoning?

Lead poisoning can occur at any age but maximum cases can be perceived in children. They often place their hands and objects in their mouths, sometimes consume non-food items, their bodies absorb Pb at a higher rate since their brains develop rapidly. The greatest common source of Pb exposure for children is **lead-based paint**, **dust** and **soil** that are polluted by it.

Others at risk for lead poisoning include individuals who

- Drink water which flows through pipes soldered with lead & lipstick.
- Work with lead (for example, metal smelters, pottery makers and stained glass artists).
- Consumption of food from cans made with lead solder.
- Cook or store food in ceramic containers. Some ceramic glaze comprises lead that may not have been properly fired or cured.
- Consume or inhale traditional or folk medicines containing lead, such as herbs and vitamins from other countries.
- Live in groups with a lot of industrial pollution.

Contact of pregnant women to high levels of lead can cause miscarriage, premature birth and low birth weight, as well as minor malformations. **Females with high blood lead content grow early osteoporosis, lower backache, joint pain and pernicious anaemia**.

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Statement of World Health Organization (WHO): Lead poisoning is common in developing countries due to emissions from industrial units. The World Health Organization estimates that 15-18 million children in developing countries are suffering from permanent brain damage due to lead poisoning.

14.12 THERMAL POLLUTION

Thermal pollution may be defined as 'the sudden rise or fall in the temperature of water bodies (river, lake, ponds, spring, ocean etc.) due to man-made or natural activities. It occurs when the water used by any industry or plant is returned to the water body at a higher temperature. The burning of coal releases CO_2, SO_2, NOx, etc. gases into the atmosphere, which can be seen as good pollutants.

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14.12.1 Causes (or Sources) of Thermal Pollution

Water as Cooling Agent in Power, Manufacturing and Industrial plants: Manufacturing industries are the largest source of thermal pollution. These plants pull water from nearby sources to keep machines cool and then discharge back to the source having higher temperatures. When heated water goes back to the water body its temperature increases. This procedure negatively affects the marine animals and can also wipe away streamside vegetation.

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Soil Erosion: Constant soil erosion results in the rise of water bodies, making them additionally exposed to sunlight. It can be a major cause of the increasing temperature of water bodies.

Deforestation: Trees and plants inhibit sunlight from coming directly on lakes, ponds or rivers. Deforestation results in the direct exposure of water bodies to sunlight, therefore absorbing additional heat and increasing its temperature.

Run-off from Paved Surfaces: Urban run-off released to surface waters from paved surfaces such as roads and parking lots can make the water warmer. During summer seasons, the pavement turns hot forming warm run-off that enters the sewer systems and water bodies.

Natural Causes: Natural causes like volcanoes and geothermal activity plays a significant role in increasing the temperature of water sources.

14.12.2 Effects of Thermal Pollution

Decrease in Oxygen Levels: The warm temperature decreases the levels of Dissolved Oxygen in water. The warm water holds comparatively less O_2 than cold water. The lessening in DO can be fatal for plants and animals.

Increase in Toxins: The continuous flow of high temperature released from the industries, upsurges the amount of toxins into the natural water bodies. These toxins may have strict impact on the local environment and make them vulnerable to numerous diseases.

Loss of Biodiversity: Sudden increase in the temperature of water body may cause significant loss of its biodiversity.

Affects Reproductive Systems: Extreme temperature can result in the release of immature eggs or can inhibit normal development of certain eggs.

Increases Metabolic Rate: Increased metabolic rate causes organisms to consume more food than what is normally required. It disrupts the stability of food chain.

Migration: The warm water can also result in particular species of organisms migrating to the appropriate environment.

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14.12.3 Control Measures of Thermal Pollution

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It is necessary to control thermal pollution in order to protect the aquatic ecosystem. It can be controlled by the following measures:

- **Cooling ponds/reservoirs** constitute the simplest method of controlling thermal discharges.
- **Cool towers** by this medium heat is immersed in the atmosphere to lower the temperature of water.
- **Artificial lake:** Artificial lakes are man-made bodies of water that offer likely alternatives to once-through cooling.

14.13 DIFFERENT TYPES OF CARBONS

14.13.1 Black Carbon

It is a PM or Aerosol (it is not a gas), which contributes to global warming. Black carbon is generally called **Soot**. It is a type of air pollutant, which occurs due to incomplete combustion.

ENV 60 Black carbon, Brown carbon, Blue carbon, and Green carbon

Source

- Burning of biomass
- Preparing food from solid fuels
- Expulsion of diesel

Functions of Black Carbon

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Influence on Earth's atmosphere

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- By absorbing the heat from the atmosphere, it warms the earth
- It decreases the Albedo of snow and ice by depositing on it that further results in warming earth's surface.
- Black carbon is a powerful absorber of heat from sunlight and air. It makes the snow patches and glaciers black by depositing on it.
- It also impacts the local clouds and Monsoon. It also increases the melting rate of Glaciers such as Hindukush–Himalayan Glaciers.

■ *Lifetime*

- Black carbon can occur from a few days to weeks in the atmosphere.
- By reducing its emission, its impact on the atmosphere and glaciers can be ended in a few months.

■ *Contribution of India at Global Level*

According to an estimate, China and India contributed to 35% and 25% of the black carbon in the global atmosphere respectively.

■ *Government Measures*

'Project Surya' is a government initiative to reduce the emission of Black Carbon. It can be done through successive use of stove technology, solar cooker, solar lamp and biogas plants.

14.13.2 Brown Carbon

Recent atmospheric studies have shown brown carbon as a ubiquitous factor of Organic Aerosol. Atmospheric aerosol contains different originating sources of light absorbing organic matters (except Soot) such as: Humic substances **(HULS)**, coal tar substances, which originates from combustion and Bio Aerosol.

■ *Possible Sources of Brown Carbon*

- The main source of brown carbon is the burning of biomass, which mainly includes the burning of wood for domestic purposes.
- Smoke originating from agricultural fire.

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- Brown carbon is generally counted in greenhouse gases and black carbon is used for the particles which are the outcome of impure combustion such as soot and dust.

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14.13.3 Green Carbon

Green Carbon is carbon detached with the help of photosynthesis and deposited in the plants and soil of natural ecosystems and is an integral part of the global carbon cycle.

Afforestation and reforestation are measures that can be taken to enhance biological carbon sequestration.

14.13.4 Blue Carbon

Blue Carbon denotes coastal, aquatic and marine carbon sinks detained by the indicative vegetation, aquatic organisms and sediments. Coastal ecosystems such as tidal marshes, mangroves and sea grasses eliminate carbon from the atmosphere and ocean, loading it in plants and putting it in the sediment below them through natural procedures.

■ *The Blue Carbon Initiative*

The Blue carbon initiative is the first combined programme with a complete and synchronised global agenda fixated on eliminating climate change **through the conservation and restoration of coastal marine ecosystems.**

14.13.5 Activated Carbon

Activated carbon is made from various carbon-rich materials through a high-temperature process called activation.

1. Renewable Sources: Agricultural byproducts are common and more sustainable sources. These include coconut shells, wood, rice husks, and nutshells.
2. Non-Renewable Sources: Fossil fuel-based materials like coal (bituminous, lignite, anthracite), peat, and petroleum coke are also commonly used.

Activated carbon is generally a **useful tool for protecting the environment and human health when used properly**. Its main purpose is to **remove pollutants from air and water**. However, its production and disposal can

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create certain environmental challenges, making it a complex product with both positive and negative sides.

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■ *Properties of Activated Carbon*

Activated carbon is a highly porous form of carbon. Its most important properties include a large internal surface area, a complex pore structure, and high chemical and thermal stability. These features allow it to adsorb a wide range of substances.

Physical Properties

- **High Surface Area:** Its defining characteristic is its exceptionally large surface area, created during the activation process. This vast internal surface is where adsorption occurs.
- **Pore Structure:** It has a complex network of pores classified by size:
  - Micropores (<2 nm): These are the main sites for adsorbing small molecules and gases, accounting for most of the surface area.
  - Mesopores (2 to 50 nm): These serve as paths for molecules to reach the micropores and are important for adsorbing medium-sized molecules.
  - Macropores (>50 nm): These act as main transport channels into the internal structure.
- Appearance and Form: It is a black, odorless, and tasteless material available in different forms, such as powder, granular, extruded pellets, and beads, each suited for specific applications.
- Density: The apparent density is relatively low (typically 0.4 to 0.5 g/cm^3) due to its high porosity.
- Hardness/Mechanical Strength: Depending on the raw material and manufacturing process, it can be quite resistant to wear and physical stress, which is important for fixed-bed industrial applications.

Chemical Properties

- **Adsorption Capacity:** It can hold molecules on its surface mainly through Van der Waals forces (physical adsorption) and sometimes through chemical bonding (chemisorption). This capacity is often measured by the iodine number (for small molecules) and molasses number (for large molecules).
- **Chemical Inertness:** Activated carbon is generally stable and chemically inert, meaning it does not readily react with the substances it adsorbs. This allows for safe use with a wide range of chemicals.
- **Surface Chemistry:** Its surface can be changed with different functional groups (like oxygen, nitrogen, or sulfur compounds) to increase its attraction to certain polar or non-polar contaminants.

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- **Hydrophobicity:** Most activated carbons are non-polar and hydrophobic, which means they have a low affinity for water but can effectively adsorb non-polar organic substances from water.
- **Reactivatability:** A key property is its ability to be regenerated, usually by using heat or chemicals, which restores its adsorption capacity and allows for reuse.

Advantages of Activated Carbon

- **Excellent pollutant removal:** It effectively captures a wide range of organic compounds, heavy metals (like mercury), and toxins from air and water, making these sources safer for human use and the environment.
- **Medical uses:** Medically, it is the most effective treatment for certain types of oral poisoning and drug overdoses. It traps toxins in the gut before the body can absorb them.

Drawbacks of Activated carbon

- **Production impact:** The manufacturing process is energy-intensive and can release significant greenhouse gases (CO_2, CH_4) and other emissions (SO_2, NO_x), especially when it comes from coal or uses inefficient methods.
- **Disposal challenges:** Once saturated with pollutants, "spent" carbon can become hazardous waste. Improper disposal (like in regular landfills) can lead to soil and water contamination with concentrated toxins.

Environmental Pollution and Effects - Environmental Impact

- **Production Emissions:** High energy use during activation causes greenhouse gas emissions and air pollutants like SO_2 and NO_x.
- **Chemical Contamination:** The chemical activation process uses strong acids or bases that can risk water and soil contamination if not managed properly.
- **Waste Disposal:** The main environmental issue is managing spent carbon, which is non-biodegradable and requires special handling or regeneration to prevent releasing captured pollutants back into the environment.

Effect on Humans

- **Health Benefits:** When used in filtration or under medical supervision, it protects human health by removing harmful chemicals and pollutants from air, water, and the body during poisoning situations.
- **Health Risks:** Direct exposure to activated carbon dust should be avoided to prevent respiratory and eye irritation. In a medical setting, the main risk is pulmonary aspiration (inhaling it into the lungs), which can be fatal if the patient is not fully conscious. It can also interfere with absorbing essential nutrients and oral medications.

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How to Solve the Problem

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Solutions focus on improving sustainability throughout the product life cycle

- **Sustainable Sourcing:** Prioritize renewable, waste biomass materials (like coconut shells or wood waste) instead of fossil fuels to reduce the carbon footprint.
- **Efficient Production:** Use energy-efficient manufacturing processes, renewable energy sources, and recover waste heat and gases produced during manufacturing.
- **Regeneration and Reuse:** This is crucial. Spent activated carbon, especially granular types (GAC), can be regenerated with methods like thermal, chemical, or biological processes. Regeneration restores much of its adsorption capacity and significantly reduces waste volume and disposal costs.
- **Responsible Waste Management:** For highly toxic contaminants (like high mercury levels) that cannot be safely regenerated, disposal should occur at approved hazardous waste facilities with appropriate safety measures to prevent leaching.

14.14 ECOMARK

It is a governmental scheme to recognise environment friendly products. The scheme will provide official recognition and marking for domestic and other customer products which come across certain ecological standards decided by the Bureau of Indian Standards.

Objectives

- To improve the production in units and to recognise the eco-friendly products.
- Any product whose production, manufacturing, use and settlement are done in such a way that it rarely affects the environment, then it is called ‘environment friendly product’.
- This project will help in capacity building by organising workshops, seminars, conferences related to the industries and environment.
- This will help in changing views of stakeholders regarding the need for active industries.

14.15 ENVIRONMENT ISSUES AND HEALTH EFFECTS

14.15.1 Iodine – 131

- Iodine – 131 (I-131) is produced by nuclear tests and then it is accumulated by plants. The milk of the cattle’s, which feeds on those plants, then produces contaminated milk, which is later consumed by humans.

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- It causes damage to the Thyroid Gland, especially in children.

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14.15.2 Mercury

- It is generally found in water in the form of monomethyl mercury and is highly poisonous. It occurs in all kinds of industrial residues.
- The maximum amount of mercury in fishes are found in Coastal regions. Mumbai, Kolkata, Karvar (Karnataka), South Koyal (Bihar) are some severely affected areas.
- Compact fluorescent lamp (CFL) as an energy efficient product gives rise to new controversies.
- Mercury is highly poisonous as compared to other substances; it is 1,000 times more effective than colchicines (use in genetic research).

14.15.3 Fluorine

- It is found in nature (in air and water) as fluoride.
- **Fluoroisis** is a common problem in many states; its main reason is consuming highly Fluoride contaminated water.
- Fluoride is the main cause of dental fluorosis and stiffness of joints (esp. spinal cord). Humped back is the main reason for stiffness of spinal cord.
- Pain on bending of knees, bones and joints to the outer side is called knock knee syndrome.
- In animals, fluoride causes erosion in teeth, blurriness and spots in the skin. It also causes scarcity of milk and lameness.

14.15.4 DDT

- Harmful pesticides such as BHC, PCB, DDT etc. do not degrade easily and remain for a longer duration in the environment. Their concentration is constantly increased by successive applications in soil and water. DDT was earlier used in marshy areas to check mosquitoes.

14.15.5 Lead in Paints

- Modern houses are filled with damaging chemicals, lead is one of them. Lead is also used in paints & lipstick.
- In many countries including India, using lead in paints is prohibited. Still it is mixed by the manufacturers.
- Lead dust is the main reason for lead poisoning.
- Human body is unable to dissociate lead that accumulate in infants, and damages the central nervous system.
- Lead is used even after being poisonous because other options are expensive and they reduce the quality of paint.

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14.15.6 Magnetite Particles and Its Pollution

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- Magnetite is a mineral and one of the main iron ores, with the chemical formula Fe_3O_4.
- It is one of the oxides of iron, and is ferromagnetic; it is attracted to a magnet and can be magnetised to become a permanent magnet itself.
- It is the most magnetic of all the naturally occurring minerals on earth.
- Small grains of magnetite are very common in igneous and metamorphic rocks.
- Magnetite reacts with oxygen to produce hematite.
- Magnetite nanoparticles are becoming increasingly important for the development of novel biomedical and nanotechnology applications.

Sources of Magnetite Pollution

- Vehicle brake wear is by far the most dominant source of airborne magnetite in the roadside environment.
- The automotive industry uses ceramic or ferrite magnets to make vehicles safer. One of the most impressive uses is in the Anti-lock Braking System (ABS) of vehicles. This use of magnets in motor vehicles can generate magnetic particles.
- Engines of motor vehicles can generate magnetic particles as they use high powered magnets which generate magnetic flux.

Impacts of Magnetite Particles

- Magnetite can have potentially large impacts on the brain due to its unique combination of redox activity, surface charge and strongly magnetic behaviour.
- Magnetite particles can act as pollutants and lead to neurodegenerative problems.
- Magnetic particles can penetrate and damage the blood–brain barrier and led to the early development of Alzheimer's disease.

14.15.7 Benzene Pollution

- Benzene is a chemical that is a colourless or light-yellow liquid at room temperature. It has a sweet odour and is highly flammable.
- Benzene evaporates into the air very quickly. Its vapour is heavier than air and may sink into low-lying areas.
- Benzene dissolves only slightly in water and will float on top of water.

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ENV 61- Benzene Pollution

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Presence of Benzene

- Benzene is formed from both natural processes and human activities.
- Natural sources of benzene include volcanoes and forest fires. Benzene is also a natural part of crude oil, gasoline, and cigarette smoke.
- Benzene is widely used in the United States. It ranks in the top 20 chemicals for production volume.
- Some industries use benzene to make other chemicals that are used to make plastics, resins and nylon and synthetic fibres. Benzene is also used to make some types of lubricants, rubbers, dyes, detergents, drugs and pesticides.

Exposure of Benzene

- Outdoor air contains low levels of benzene from tobacco smoke, gas stations, motor vehicle exhaust and industrial emissions.
- Indoor air generally contains levels of benzene higher than those in outdoor air. The benzene in indoor air comes from products that contain benzene such as glues, paints, furniture wax and detergents.
- The air around hazardous waste sites or gas stations can contain higher levels of benzene than in other areas.
- Benzene leaks from underground storage tanks or from hazardous waste sites containing benzene can contaminate well water.
- People working in industries that make or use benzene may be exposed to the highest levels of it.
- A major source of benzene exposure is tobacco smoke.

Effect of Benzene on Human Body

- Benzene works by causing cells not to work correctly.
- For example, it can cause bone marrow not to produce enough red blood cells, which can lead to anaemia. Also, it can damage the immune system by changing blood levels of antibodies and causing the loss of white blood cells.
- People who breathe in high levels of benzene may develop the following signs and symptoms within minutes to several hours:
- Drowsiness
- Dizziness
- Rapid or irregular heartbeat
- Headaches
- Tremors
- Confusion

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- Unconsciousness
- Death (at very high levels)
- Eating foods or drinking beverages containing high levels of benzene can
- Direct exposure of the eyes, skin or lungs to benzene can cause tissue injury and irritation.

14.15.8 Furnace Oil

- Furnace Oil is a fraction obtained from petroleum distillation, either as a distillate or a residue.
- Broadly speaking, fuel oil is any liquid petroleum product that is burnt in a furnace or boiler for the generation of heat or used in an engine for the generation of power.
- Fuel oil, also called furnace oil, fuel consisting mainly of residues from crude-oil distillation.
- It is used primarily for steam boilers in power plants.
- Fuel oil is made of long hydrocarbon chains, particularly alkaline, cyclo-alkaline andaromatics.
- The term fuel oil is also used in a stricter sense to refer only to the heaviest commercial fuel that can be obtained from crude oil i.e. heavier than gasoline and naphtha.
- It is a dark, viscous liquid at room temperature.

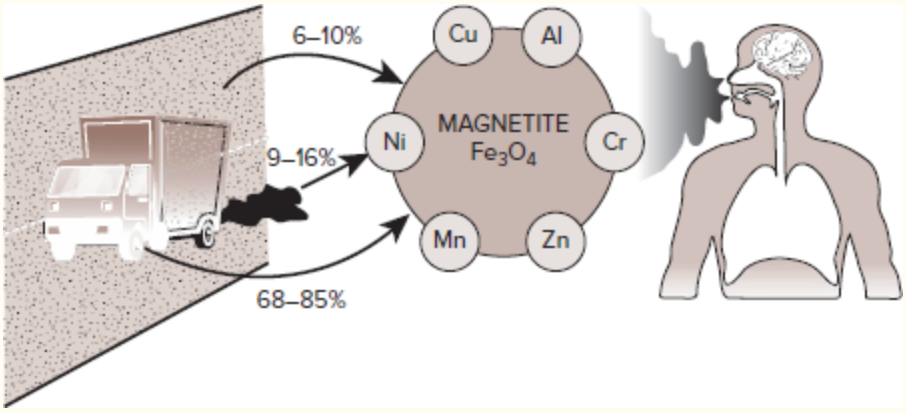

Figure 14.3 Sources of Magnetite Pollution

Pollution

- Petroleum coke is a carbon-rich solid derived as a by-product from oil-refining and has high Sulphur content.
- Furnace oil is the residue fraction obtained from petroleum distillation, also containing sulphur.
- Using petroleum coke and furnace oil contributes to high SO_2 emission and secondary sulphate formation as PM.

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14.16 SOLID WASTE MANAGEMENT

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Solid wastes are those solids that are discarded and rendered useless by human beings including animal waste.

14.16.1 Sources of Solid Waste

Waste material from construction, industries and thermal power plant, radioactive waste, garbage, dead animals, sludge and rubbish which includes nonconsumable combustible and non-combustible solids. (Combustible rubbish such as rubber, plastic, paper, wood, textiles, cardboards etc., Non-combustible rubbish such as metal scraps, metal containers, glass, ceramic crockery's, tin-cans, aluminium articles).

14.16.2 Causes of Urban Solid Waste

Growth in technology, growth in consumption, increase in urbanisation which includes certain things such as construction of buildings, markets, shopping malls etc. construction of roads, railways, airways, construction of bridges, dams etc. construction of water supply and sewage disposal systems etc. over population.

14.16.3 Effects of Urban Solid Waste

Increase in the number of diseases (bacterial and viral) such as typhoid fever, cholera, paratyphoid fever, bacillary dysentery and infections hepatitis (i.e. infectious jaundice), poliomyelitis etc. Solid wastes may also obstruct the sanitations and gully pits leading to waterlogging. This may form a breeding ground for mosquitoes and thus the danger of malaria and dengue is also present. Pollution of soil and underground water due to dumping.

14.16.4 Control Measures of Urban Solid Waste

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By minimising the generation of domestic garbage and other wastes. By throwing out the rubbish and other unwanted materials at a proper place from where they may be easily taken for disposal. By enforcing laws and by incorporating various methods of solid waste disposal.

14.16.5 Methods of Solid Waste Disposal

■ *Pulverisation*

In this method, the solid waste is converted to powder form by grinding. It thus becomes almost odourless and of reduced volume. After this treatment, the waste may be used as manure or for landfilling.

■ *Land Filling*

Solid wastes are discarded into low-lying regions. The waste is discarded in layers of 1.5 m or so on and each layer is concealed by at least 20 cm thick soil. Each layer is then kept for at least 7 days and compaction is carried out for its settlement, before starting filling the next layer. Insecticides such as DDT are sprayed on top to prevent the breeding of mosquitoes and flies. Generally, the waste gets stabilised within 2 to 12 months and settles down by 20 to 40% of its original height.

Advantage of this method: It is artless and cheap. Skilled labour is not necessary. Low-lying regions can be retrieved and put to better use. No remainder or by-product; hence no further discarding. No costly plant and equipment are necessary. Separation of different kinds of solid wastes is not necessary.

Disadvantage of this method: Continuous emission of foul odour near the site of disposal, use of insecticides are necessary, the requirement of large land area.

■ *Composting*

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In this technique, the organic substances in solid waste is processed anaerobically and transformed into humus and steady mineral compounds. Its quantity is also considerably lowered and is made free of most of the pathogenic creatures so that it can be effortlessly and securely handled.

■ Incineration

This technique is usually used when appropriate dumping land areas are unavailable. When solid waste is accumulated, the non-combustible and inert ingredients such as earth, broken glass chinaware; metals etc. are isolated to decrease the weight on the hearth. The flammable garbage, rubbish and dead organisms are burnt. The least temperature in the burning chamber is kept at more than 6°C to incinerate all the organic matters and oxidise the filthy smelling gases. The final products turn to ashes and clinkers, the ashes are recycled as aggregate for low-grade solid or as road material.

Advantages of this method: No odour and dust nuisance, all the pathogens and insects are destroyed.

Disadvantage of this method: Nuisance of smoke, odour and ash during improper functioning of incinerators and relatively high initial cost

■ Pyrolysis

In this technique, the chemical elements and chemical energy of some organic trashes is recovered by destructive distillation of solid remaining. For that, the flammable constituents of solid waste are heated in a specifically designed chamber, known as a **pyrolysis reactor** at 650 to 1,000°C in an oxygen-free (or low-oxygen) environment.

■ Disposal into Sea

The solid waste is disposed into the sea, which is approx. 30 m deep, and at a reasonable distance (approx. 16 to 20 km) from the coastal area and with strong forward currents.

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Advantages of this method: Cheap and simple

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Disadvantages of this method: Some portions of solid waste may return and spoil the beaches. The lighter components of solid waste float, spread and tend to return to the shores during high tides. This method is not feasible during monsoons or stormy weather.

14.16.6 Causes, Effects and Treatment of Industrial Wastes

A wide variety of inorganic and organic pollutants such as oils, greases, plastics, plasticisers, metallic wastes, suspended solids, phenols, toxins, acids, salts, dyes, cyanides, DDT etc. and heavy metals such as Na, Cu, Cr, Cd, Hg, Pb etc. remain present in industrial effluents. They are not easily degradable and cause serious pollution issues.

■ *Causes and Effects*

- These toxins enter into the food chain and become extremely toxic to living beings.
- They poison the biological purification mechanism of sewage treatment when discharged through the sewage system.
- Metallic contaminants such as Hg, Zn, Cd etc. destroy bacteria and beneficial microorganisms in the soil and thus decrease soil fertility.
- When food containing radionuclides is consumed by a man, they concentrate in specific body organs and create diseases.
- Excessive use of nitrogenous fertilisers inland leads to the accumulation of nitrate in the soil, causing diarrhoea and cyanosis in children.

■ *Treatment of Industrial Waste*

Industrial waste treatment processes are

- Anaerobic digestion
- Chlorination
- Deionisation
- Filtration
- Screening
- Incineration
- Lagooning
- Spray irrigation
- Vacuum filtration

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- Precipitation

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Primary treatment: In this, suspended organic solids of large size are removed through sedimentation tanks.

Secondary treatment: Fine suspended and also organic matters, which are dissolved, are removed by the process of flocculation or filtering or sedimentation.

Final treatment: Extra fine organic matter along with pathogen bacteria are removed through calorimeter.

- **Shredding or comminuting:** Breaks large-sized sludge solids into smaller sizes. Then these sizes pass through screens of course, medium or fine size as per the suitability.
- **Incineration:** It is the process of burning screened sewage bad odour is avoided when the incineration

Sedimentation: Sludge is retarded in the sediment doubled tank for the detention period.

Coagulation and Floc: The size of very fine suspended particles in sludge is increased into the flocculated size and the product obtained by mixing of sewage and coagulant is called floc.

Grim State of Pollution Level

- In metropolitan cities, the air we breathe is so harmful that it is equivalent to the smoking of 10 to 20 cigarettes a day.
- In India, more than 40,000 premature deaths occur due to air pollution every year as reported by World Bank.
- In 23 Indian cities with a population of more than one million, auto exhausts and industrial emissions dangerously cross the safer limit.
- Current studies disclose that the number of patients with respiratory sicknesses and allergies has approximately doubled since 1990.

Table 14.8 Different types of E-waste, their sources and health effects

Health Consequences of E-waste			
S.No	**Pollutants**	**Source**	**Health effects**
1.	**Lead**	1. Glass panels	1. Harm to the central and

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Health Consequences of E-waste			
S.No	**Pollutants**	**Source**	**Health effects**
		2. Gaskets in computer monitors 3. Solderer in PCB board	peripheral nervous system, Blood system, reproductive system 2. Inhibits brain development among children. 3. Lead tends to gather in the environment and has great acute and chronic effects on plants, animals and microorganisms
2.	**Cadmium**	1. Infra-red detectors 2. Semiconductor chips 3. Some older cathode ray tubes	1. It gets collected in the human body especially in kidney
3.	**Mercury**	1. It is estimated that 22% of the yearly world consumption of	1. Cause harm to body organs including the brain and kidney, as well

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Health Consequences of E-waste			
S.No	**Pollutants**	**Source**	**Health effects**
		mercury is used in electrical and electronic equipment 2. Used in relays, switches, sensors, lamps, mobile phones and batteries	as the foetus. The developing foetus is highly susceptible to mercury exposure 2. It enters the food chain from the aquatic life forms and gets biomagnified, which cause harmful effects to humans consuming
4.	**Toners**	1. Found in the plastic printers cartridge containing black and colour toners.	1. Inhalation is the main exposure pathway, and acute exposure may result in respiratory tract irritation. 2. Carbon black is categorised as 2B (possible) carcinogen
5.	**Phosphor and additive**	1. Phosphor is an inorganic	1. The phosphor coating on the

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Health Consequences of E-waste			
S.No	**Pollutants**	**Source**	**Health effects**
		chemical compound that is applied as a coat on the **interior** *of* the CRT (Cathode Ray Tube) faceplate.	CRT tube contain heavy metals, such as cadmium and other rare earth metals, example, zinc, vanadium.
6.	**Barium**	1. Barium is a soft silvery-white metal that is used in computers in the front panel of a CRT, to protect user from radiation	1. Short-term exposure to barium causes brain swelling, muscle weakness, damage to the heart, liver and spleen
7.	**Beryllium**	1. It is commonly found in mother boards and finger clips	1. Exposure to it cause lung cancer 2. Also cause a skin disease that is characterised by poor wound healing and wart-like bumps

■ *Role of Individual*

- People should be persuaded not to wash, bathe or ease in rivers, lakes and ponds etc.

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- Burning of dead bodies and thereafter throwing the ashes into the river must be avoided.
- Effluents from thermal power plants should not be discharged directly into rivers, lakes etc. they should be chemically treated to bring down the pollutants within the safety limits and then only be discharged away from the living area.
- Smoke to be discharged into the air must be treated before exhausting it to mix with air.
- Religious offerings and other wastes should not be thrown into river.

14.17 E-WASTE POLLUTION

- Technology has not only made life easy for everyone, but it has also cluttered our lives with a lot of hi-tech trash that is proving to be a hazard to the environment and health of living beings.
- E-waste/Electronic waste is loosely used for the discarded electrical and electronic equipment along with the accessories at the end of functioning life and is fated for recovery, recycling or disposal. Computers and cell phones are the two major sources.

ENV 62- e-Waste Pollution

14.17.1 Types of E-Waste

Sources of E-waste can be broadly divided into:

- **White goods:** The large electrical goods used domestically such as air conditioners, refrigerators, washing machine etc.
- **Brown goods:** These are relatively light electronic consumer durables such as televisions, cameras etc.
- **Grey goods:** Products that are bought in a foreign country and sold more cheaply than in the manufacturer's usual distribution system. These goods include computers, printers, scanners, mobile phones etc. Importantly, waste from the white and brown goods is less toxic as compared with grey goods.

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Do You Know?

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Do You Know?

Electronic waste or E-waste had ferrous and non-ferrous metals both. Non-ferrous metals such as copper, aluminium, silver, gold, platinum, palladium etc. The presence of elements such as lead, mercury, arsenic, cadmium, selenium and hexavalent chromium is classified as hazardous waste. Plutonium is a radioactive element therefore it is not used in computer or electronic goods. Heptachlor is used as a pesticide chemical.

14.17.2 e-Waste Indian Scenario

- The Indian electronics industry dates back to the 1960s. The size of the industry is expected to grow at 25% during 2012–2020 to reach the US $400 Bn in 2022.
- In India, among the top ten cities, **Mumbai ranks first in generating e-waste** followed by **Delhi**, **Bangalore**, **Chennai**, **Kolkata**, **Ahmadabad**, **Hyderabad**, **Pune**, **Surat** and **Nagpur**.
- Recovery of metals by primitive means is a hazardous process. The recycling process, if not carried out properly, can cause damage to human beings through the inhalation of gases during **recovery** and **recycling**.

14.17.3 Environmental Assessment of Electronic Goods

- **Electronic product environment assessment tool (EPEAT)** is a method for consumers to evaluate the effects of a product on the environment. It measures lifecycle, ecological standards and ranks products as gold, silver or bronze based on a set of environmental performance *criteria.*
- It is managed by the **Green Electronics Council -**a programme of the International **Sustainable Development Foundation (ISDF)** which foresees a world where commerce, communities and nature thrives in harmony.
- Since 2007, companies have taken seriously to greening their products and many have made giant strides. Example: Apple's **iPhone 6** and **onwards series is launched mercury-free**, **arsenic-free** and **beryllium-free**.

QUICK INSIGHT

- R2 certification is generally regarded as responsible recycling certification, a company-level certification that is fully reliant on R2 standards and based on it.
- There has been the establishment of certain requirements by R2 practices that an organisation needs to meet for the handling of electronic waste in a proper manner.
- Thus environmentally responsible practices in the electronics recycling industry are compiled under the R2 Code of Practices.
- An R2-certified company will be able to assure its upstream clients and customers that it takes appropriate measures at its facility to protect the environment, worker

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and public health and data security.
- Furthermore, an R2-certified company will be able to make the same assurances about each of its 'Focus Material' downstream vendors throughout the recycling chain to final disposition.

14.18 CEMENT INDUSTRY POLLUTION

14.18.1 Greenhouse Gas (GHG) / CO_2 Emissions

- The global cement industry emits roughly **2.4 gigatonnes CO2 equivalent (GtCO2e)** annually — about **6–8% of all global human-made CO2 emissions**.
- For each ton of cement produced, roughly **0.9 to 1 ton of CO2** is emitted.
- The emissions come from two main sources:
  - **"Process emissions"**: chemical breakdown of limestone (calcium carbonate) during clinker formation — this accounts for a large share (~50–60%) of cement CO2 emissions.
  - **Energy emissions**: burning fossil fuels to heat the kiln (over 1,400 °C), plus energy used for grinding, materials handling, etc.
- Over the decades, cement's emissions have grown dramatically: global cement CO_2 emissions have more than doubled since the early 2000s.

14.18.2 Air Pollution & Other Emissions

Beyond CO_2 / climate change, cement and concrete production contribute to other air pollutants and environmental harms:

- Emissions of **nitrogen oxides (NO2)**, **sulphur oxides (SO2)**, and **particulate matter (PM)** (including fine PM2.5 and PM10) — from kiln combustion, raw material processing, dust from grinding/storage.
- Health impacts: Exposure to cement dust and PM affects workers and communities nearby — risks include respiratory diseases, cardiovascular issues, and other pollution-related health problems.
- High energy consumption and resource depletion: Cement manufacture consumes large amounts of fuel and raw materials (limestone, clay, aggregates), contributing to resource extraction impacts.

14.18.3 'Hard to Abate' Industry

- Cement is considered one of the **hardest sectors to decarbonise**, because CO_2 emissions come not only from energy use, but from the **chemical process** of making clinker, which is fundamental to cement's properties.

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- High kiln temperature (≈ 1400 –1500 °C) demands lots of fossil fuel, making energy-related emissions persistent.
- As global demand for infrastructure and buildings rises (especially in developing countries), cement consumption and hence emissions are expected to **grow further** unless significant changes are made.

14.18.4 Mitigation Efforts and Trends in 'Cleaner' Cement

- Some portion of CO_2 from cement is naturally reabsorbed over time via a process called **carbonation** (when cement/concrete carbonates over its life) — global cement materials absorb a significant amount of CO_2 cumulatively.
- The industry is exploring **lower-carbon cement** (For example, using supplementary cementitious materials like fly ash, slag, calcined clays) to reduce the clinker-to-cement ratio and thereby cut emissions.
- Energy-efficient technologies, alternative fuels, waste-derived fuels, and — in advanced cases — **carbon capture and storage (CCS)** are being trailed to curb emissions from kilns and plants

Do You Know?

- In the cement manufacturing process, **crushed limestone is mixed with other materials like clay or shale** to **achieve the desired balance of calcium, silicon, aluminum, and iron oxides needed to form cement clinker**. This step describes the raw materials blend but doesn't directly explain the source of the **CO_2** emissions.
- During a **high-temperature process** called **calcination**, calcium carbonate ($CaCO_3$) in the limestone decomposes into calcium oxide (lime, CaO) and releases carbon dioxide (CO_2) gas:
  $CaCO_3$ (Heat) → CaO +CO_2 : This chemical reaction (process emissions) accounts for about 60-65% of the total CO_2 emissions from cement production. The remaining emissions come from burning fossil fuels for the heat required in the kiln.

Table 14.9 Cement Formation Process & Pollution

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Stage	Process	Major Pollution
1. Raw Material Extraction	Mining of limestone, clay, shale; drilling, blasting, excavation; transported to crushers.	• Dust (PM10, PM2.5) • Noise from blasting • Land degradation • Habitat loss
2. Crushing	Large rocks crushed using jaw/hammer/impact crushers to 15–75 mm size for easier grinding.	• High dust emissions • Noise & vibration • Fugitive particulate from conveyors
3. Raw Grinding & Mixing	Grinding into fine raw meal; blending in silos (Storage container)	• Fine dust (PM) • High electricity use → indirect CO_2 • Material spillages
4. Preheater & Pre-calciner	Raw meal heated; partial calcination	• CO_2 from $CaCO_3$ breakdown • Dust in gas streams • NO_x formation
5. Rotary Kiln (Clinker Formation)	High-temperature burning (1400–1500°C) to form clinker	• **Maximum CO_2 emissions** (calcination + fuel) • SO_2, NO_x • CO, VOCs from fuels • Heat pollution

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Stage	Process	Major Pollution
6. Clinker Cooling	Rapid air cooling of hot clinker	• Dust release • Hot air emissions
7. Finish Grinding (Cement Milling)	Grinding clinker + gypsum; add fly ash/slag	• PM emissions • High energy use → indirect CO_2
8. Cement Storage	Stored in silos	• Dust leakage • Silo vent emissions
9. Packing & Dispatch	Bagging (50 kg bags) or bulk loading	• Dust around packing machines • Plastic waste (bags)
10. Transportation	Trucks, belt conveyors, bulk carriers	• Diesel exhaust: CO_2, NO_x, PM • Road dust • Noise

14.19 CIGARETTE AND ITS POLLUTANTS

A cigarette is a highly engineered product composed of a few primary components that, upon combustion, create a complex mix of over 7,000 chemicals, many of which are hazardous to human health and the environment.

14.19.1 Cigarette Components & Ingredients

The basic physical components of most commercially manufactured cigarettes are:

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- **Tobacco:** A blend of different types of cured tobacco leaves (e.g., bright/Virginia, Burley) which naturally contain nicotine (the addictive agent) and other compounds.
- **Chemical Additives/ Flavourings:** Hundreds of additives (like sugars, humectants such as propylene glycol, cocoa, and menthol) are used to enhance flavour, control moisture, prolong shelf life, and make the smoke smoother and easier to inhale.
- **Paper Wrapping:** Porous paper that encases the tobacco blend and can influence burning rate and smoke dilution.
- **Filter (Butt):** Typically made of **cellulose acetate, a type of plastic**. Filters trap some of the tar and cool the smoke, but do not make cigarettes safer.

14.19.2 Pollutants and Ingredients

The burning process is what creates most of the highly toxic pollutants found in cigarette smoke. These pollutants are divided into **gas** and **particulate (tar) phases**.

■ *Key Pollutants and Their Levels*

Levels of specific pollutants vary by brand, tobacco type, and how the cigarette is smoked, but many are present in microgram (μg) or nanogram (ng) quantities per cigarette:

Table 14.10 Key Pollutants and their Levels

Pollutant	Chemical Class	Primary Effect(s)
Tar	Complex Mixture/Particulate Matter	Contains most carcinogens; paralyses lung cilia
Carbon Monoxide (CO)	Gas	Reduces the blood's oxygen-carrying capacity

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Pollutant	Chemical Class	Primary Effect(s)
Nicotine	Alkaloid/Drug	Addictive agent, causes increased heart rate/BP
Benzene	Aromatic Hydrocarbon (Carcinogen)	Linked to leukemia and bone marrow harm
Formaldehyde	Aldehyde (Carcinogen)	Respiratory irritant, linked to various cancers
Hydrogen Cyanide	Nitrile/Toxic Gas	Highly toxic, damages cilia
Arsenic, Cadmium, Lead	Heavy Metals (Carcinogens)	Accumulate in organs, linked to cancer/heart disease
Polycyclic Aromatic Hydrocarbons (PAHs)	Carcinogens	DNA damage, numerous cancers (e.g., Benzopyrene)
Tobacco-Specific Nitrosamines (TSNAs)	Carcinogens	Potent carcinogens formed during curing/smoking

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14.19.3 Different Types of Smoke

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Cigarettes produce different types of smoke, varying in pollutant levels:

- **Mainstream Smoke (MS):** Inhaled directly by the smoker through the filter.
- **Sidestream Smoke (SS):** The smoke that emanates from the burning tip of the cigarette between puffs. SS is often more concentrated in some toxins than MS (e.g., higher levels of ammonia, benzene, and certain nitrosamines) due to less complete combustion.
- **Secondhand Smoke (SHS):** A mixture of SS and exhaled MS. There is no safe level of exposure to SHS.
- **Thirdhand Smoke (THS):** The persistent toxic residue that clings to surfaces and dust long after a cigarette is extinguished, posing long-term exposure risks

14.20 CHEWING GUM COMPONENTS AND THEIR POLLUTION

Modern chewing gum is composed primarily of a **non-biodegradable, synthetic rubber "gum base"** that poses significant environmental problems when littered. While generally considered safe for human consumption, concerns exist regarding the ingestion of microplastics released during chewing.

14.20.1 Chewing Gum Components

A typical piece of chewing gum is made of five main types of ingredients:

- **Gum Base (20−30%):** The insoluble, non-chewable part that gives gum its texture and elasticity. This is typically a mix of synthetic polymers (plastics/synthetic rubber) such as polyethylene, polyvinyl acetate, and butyl rubber. Natural alternatives made from tree sap (chicle) are less common today.
- **Sweeteners (*upto* 60%):** Both traditional sugar (sucrose, corn syrup) and sugar-free alternatives (xylitol, sorbitol, mannitol) are used.
- **Softeners (1−3%):** Ingredients like glycerine or vegetable oils help retain moisture and keep the gum soft and flexible.
- **Flavourings (<2%):** Natural and artificial flavours, such as peppermint, spearmint, or fruit flavours, provide the taste.
- **Additives:** This can include colourings, fillers (like calcium carbonate), and antioxidants (like BHT).

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14.20.2 Environmental Pollution and Problems

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The main problem with modern chewing gum is that the plastic gum base is not biodegradable, making it a major source of pollution.

- **Litter and Cleanup Costs:** Improperly discarded gum (chewing gum litter) is a pervasive problem, highly visible on sidewalks and public spaces. It is expensive and labour - intensive to clean up, costing cities millions annually.
- **Plastic Pollution:** Because the base is plastic, discarded gum remains in the environment for years and eventually breaks down into **microplastics**. These microplastics contaminate soil and water and can be ingested by marine animals, potentially disrupting the food chain and ecosystem balance.
- **Drain Clogging:** In some cases, large accumulations of discarded gum can contribute to clogged drains and local flooding problems.

14.20.3 Effect on Human Health

In general, chewing gum ingredients are considered safe for consumption. In fact, sugar-free gum can have oral health benefits by increasing saliva production, which helps neutralise plaque acids and remove food particles, reducing the risk of cavities.

However, there are potential problems associated with excessive or specific types of gum consumption:

- **Microplastic Ingestion:** Recent studies have shown that the act of chewing gum releases hundreds of microplastic particles into the saliva, which can then be swallowed. While the long-term health effects of this microplastic ingestion require further study, concerns exist about potential cell damage, inflammation, or hormonal imbalance.
- **Jaw Problems:** Constant, excessive chewing can overwork jaw muscles, potentially leading to **temporomandibular joint disorder (TMJ)**, which causes jaw pain, headaches, and muscle imbalance.
- **Digestive Issues:** Chewing signals the body to prepare for food, releasing stomach acids and enzymes. Swallowing excess air while chewing can also lead to bloating and abdominal pain. Sugar alcohols in sugar-free gum (like sorbitol and mannitol) can have a laxative effect if consumed in large quantities, causing diarrhoea and cramps.
- **Tooth Decay:** Gums containing sugar can contribute to tooth decay and cavities, as the sugar feeds harmful oral bacteria.
- **Mercury Release:** For individuals with older amalgam (silver) dental fillings, the abrasive action of chewing gum may accelerate the release of trace amounts of mercury vapor from the fillings.

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14.21 FOOD CONTAMINATION AND HEALTH

Food can get contaminated with biological or chemical contaminants at any point during its journey from the producer to the consumer i.e. during production, handling and distribution, cooking and serving.

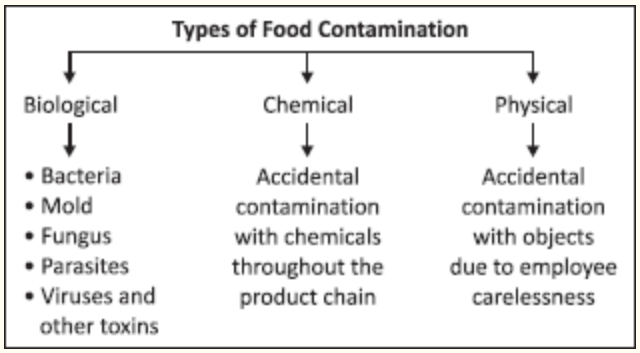

Figure 14.4 Various means of food contamination

14.21.1 Biological Contaminants

- Bacterial diseases such as typhoid, cholera and ova of some worms invade human body through food.
- Cooked food, if left at room temperature for long gets spoiled because of invasion of the spores of bacteria and fungi.
- Over some time they secrete toxic substances. Some of the secreted toxins are heat-sensitive, i.e., they become harmless if food is sufficiently heated while some toxins are heat resistant. Certain infections such as botulism are heat resistant and are often fatal.

14.21.2 Chemicals in Food

1. Presently, more than **3,000 synthetic and natural chemicals** are being used in food items. Many of the food items sold in the market contain food additives that are used to improve appearance, flavour, texture or storage. In India, generally the new food additives are introduced from the west. Like certain medicines even if their ill effects on health are reported, their use in our country continues.
2. **Adulteration of food, another menace is common practice in our country**. Although many government organisations such as **Agricultural Produce Grading and Marketing (Agmark) Act**, **Food Produce Order (FPO)**, Meat Product Order, Vegetable Oil Control Order, Control the Quality of food stuff, its production and distribution, yet many adulterated foods are sold in the market.
3. **Injection of hormones** such as steroids and oestrogen to increase the growth and size of poultry animals rapidly, is a global practice. The hormones enter the body on the

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consumption of eggs or flesh of the treated animals. This affects the normal balance and consequently leads to excessive/abnormal growth in children and causes diseases.

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14.21.3 Metal Contamination

- The metals antimony, zinc and tin get into food from cheap cooking utensils.
- The metal contamination of canned food is common and can be identified by the change in colour or metallic taste. Toxic metals such as mercury, lead, cadmium and arsenic may also get into the body through contaminated food, besides air and water.
- Regardless of their mode of entry, their effects are severe. The toxic effects of metal fumes inhaled are much more harmful than ingestion of metal.

Do You Know?

The Food Safety and Standards Authority of India (FSSAI)

It has been recognised under Food Safety and Standards Act, 2006 which strengthens numerous acts and instructions that have previously controlled food-related issues in numerous Ministries and Departments. FSSAI has been shaped for putting forward science-based standards for objects of food and to control their production, storage, circulation, sale and import to safeguard the availability of safe and wholesome food for human ingestion.

14.22 POLLUTER-PAY PRINCIPLE

In the polluter-pay principle, the polluter has to pay for the damages that have been done. This principle has been accepted as a part of the sustainable development discourse. Many countries around the world have applied this principle to fix the accountability for the damaging environment.

India's context

In India, this polluter-pays principle has been interpreted by the Supreme Court to not only pay for the actions one has done, but also to pay for reclaiming and reconstructing the damage done to environment and

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ecosystems. Though this principle appears ideal for the government to follow, there are many **flaws in this principle that have to be rectified:**

1. **Identifying who is a polluter:** This is the biggest challenge for any government as it is difficult to define a polluter whether the industries that produce goods or the citizens that consume these goods.
2. **Rational disinterest:** Common man has a fear to sue a company, as he is of the opinion that he gets entangled in legal disputes for many years and of course this fear is a proven fact. Example - Bhopal gas tragedy.
3. **Compensation is very meagre:** Compensation is claimed by certain groups who are directly affected.
4. **Problems in the assessment of damages:** This often leads to poor compensation.
5. **The injurer may not be able to pay a high amount of damages:** This is the biggest challenge to impose hefty fines on the culprits. One way to remedy this is to have an insurance cover for the damages caused by the companies or individuals.

However, as a mature democracy, we need to incorporate this principle on a sound basis by fixing the above problems cited. We need to have sustainable development and the time is ripe to walk the talk.

14.23 PLASTIC POLLUTION

Plastic pollution is the accumulation of plastic *objects and particles* (*e.g.* plastic bottles, bags *and* microbeads) in the earth's environment *that adversely affects* wildlife, wildlife habitat and humans. Plastics that act as pollutants *are categorised into micro-,* meso- or macro debris, based on size. Plastics are inexpensive and durable, and as a result levels of plastic production by humans are high.

Plastic pollution can afflict land, waterways and oceans. It is estimated that 1.1 to 8.8 million metric tonnes (MT) of plastic waste enters the ocean from coastal communities each year.

Approximately, 380 million tonnes of plastic is produced worldwide each year. From the 1950s up to 2018, an estimated 6.3 billion tonnes of plastic has been produced worldwide, of which an estimated 9% has been recycled and another 12% has been incinerated. This large amount of plastic waste enters the environment, with studies suggesting that the bodies of

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90% of seabirds contain plastic debris. In some areas, there have been significant efforts to reduce the prominence of free-range plastic pollution, through reducing plastic consumption, litter clean-up, and promoting plastic recycling.

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14.23.1 Types of plastic debris

Three major forms of plastic contribute to plastic pollution: Microplastics as well as mega- and macroplastics. **Mega and microplastics have accumulated in the highest densities in the Northern Hemisphere,** concentrated around urban centres and water fronts. Both mega- and macroplastics are found in packaging, footwear and other domestic items that have been washed off of ships or discarded in landfills.

Plastic debris is categorised as either primary or secondary. Primary plastics are in their original form when collected. Examples of these would be bottle caps, cigarette butts and microbeads. Secondary plastics, on the other hand, account for smaller plastics that have resulted from the degradation of primary plastics.

■ *Microdebris*

Microdebris are plastic pieces between **2 mm and 5 mm in size**. Plastic debris that starts off as mesoor macrodebris can become microdebris through degradation and collisions that break it down to smaller pieces. **Microdebris is more commonly referred to as nurdles.** Nurdles are recycled to make new plastic items, but they easily end up released into the environment during production because of their small size. They often end up in ocean waters through rivers and streams. **Microdebris that come from cleaning and cosmetic products are also referred to as scrubbers.** Because microdebris and scrubbers are so small in size, filter-feeding organisms often consume them.

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Macrodebris

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Plastic debris is categorised as macrodebris when it is **larger than 20 mm**. These include items such as plastic grocery bags. Macrodebris are often found in ocean waters and can have a serious impact on native organisms. Fishing nets have been prime pollutants. Even after they have been abandoned, they continue to trap marine organisms and other plastic debris.

Persistent organic pollutants

It was estimated that the global production of plastics is approximately 500 million metric tonnes (mt)/2025. Their abundance has been found to transport persistent organic pollutants, also known as POPs. These pollutants have been linked to increased distribution of algae associated with red tides.

Exam Pointer

Major plastic polluter countries (Top 12 mismanaged plastic waste polluters)

China (27.7%), Indonesia (10.1%), the Philippines (5.9%), Vietnam (5.8%), Sri Lanka (5.0%), Thailand (3.2%), Egypt (3.0%), Malaysia (2.9%), Nigeria (2.7%), Bangladesh (2.5%), South Africa (2.0%), India (1.9%), Rest of the world (27.3%)

Total plastic waste polluters

Around 400 million tonnes of plastic waste is generated each year around the world. About 60% of the plastic waste in the ocean comes from the following top 5 countries: China, Indonesia, the Philippines, Thailand and Vietnam.

14.23.2 Effects of Plastic

Effects of plastic on land

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Plastic pollution on land poses a threat to plants and animals – including humans who are based on the land. Estimates of the amount of plastic concentration on land are between four and twenty-three times that of the ocean. The amount of plastic poised on the land is greater and more concentrated than that in the water. Chlorinated plastic can release harmful chemicals into the surrounding soil, which can then seep into groundwater or other surrounding water sources and also the ecosystem of the world. This can cause serious harm to the species that drink the water.

■ *Effects of plastic on oceans*

In 2012, it was estimated that there were approximately 165 million tonnes of plastic pollution in the world's oceans. The Ocean Conservancy reported that China, Indonesia, the Philippines, Thailand and Vietnam dump more plastic in the sea than all other countries combined.

One study estimated that there are more than 5 trillion plastic pieces (defined into the four classes of small microplastics, large microplastics, meso and macroplastics) afloat at sea.

The litter that is being delivered into the oceans is toxic to marine life and humans. The toxins that are components of plastic include diethylhexyl phthalate, which is a toxic carcinogen, as well as lead, cadmium and mercury.

Plankton, fish and ultimately the human race, through the food chain, ingest these highly toxic carcinogens and chemicals. Consuming the fish that contain these toxins can cause an increase in cancer, immune disorders and birth defects.

■ *Effects on animals*

Plastic pollution has the potential to poison animals, which can then adversely affect human food supplies. Some marine species, such as sea turtles, have been found to contain large proportions of plastics in their

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stomach. When this occurs, the animal typically starves, because the plastic blocks the animal's digestive tract. Sometimes Marine mammals are entangled in plastic products such as nets, which can harm or kill them.

Sea turtles are affected by plastic pollution. Some species are consumers of jelly fish, but often mistake plastic bags for their natural prey. This plastic debris can kill the sea turtle by obstructing the oesophagus. Baby sea turtles are particularly vulnerable according to a 2018 study by Australian scientists. Large amounts of plastics have been found in the stomachs of beached whales.

■ *Effects on humans*

Compounds that are used in manufacturing pollute the environment by releasing chemicals into the air and water. Some compounds that are used in plastics, such as phthalates, bisphenol A (BRA), polybrominated diphenyl ether (PBDE), are under close statute and might be very hurtful. Even though these compounds are unsafe, they have been used in the manufacturing of food packaging, medical devices, flooring materials, bottles, perfumes, cosmetics and much more. The large dosages of these compounds are hazardous to humans, destroying the endocrine system. BRA imitates the female's hormone called oestrogen. PBD destroys and causes damage to thyroid hormones, which are vital hormone glands that play a major role in the metabolism, growth and development of the human body.

14.23.3 Microbeads

Microbeads are manufactured solid plastic particles of less than 1 mm in their largest dimension. They are most frequently made of polyethylene but can be of other petrochemical plastics such as polypropylene and polystyrene. They are used in exfoliating personal care products, toothpastes and in biomedical and health science research.

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Microbeads can cause plastic particle water pollution and pose an environmental hazard for aquatic animals in freshwater and ocean water.

Use of Microbeads

Microbeads are added as an exfoliating agent to cosmetics and personal care products, such as soap, facial scrub and toothpastes. They may be added to over-the-counter drugs to make them easier to swallow. In biomedical and health science research microbeads are used in microscopy techniques, fluid visualisation, fluid flow analysis and process troubleshooting.

Exam Pointer

Microbeads in Fishes

Microbeads are tiny plastic particles smaller than 5 mm. They appear in cosmetics, toothpaste, face scrubs, detergents, and industrial cleaners. These microbeads slip through wastewater filters and enter rivers and oceans.

How Microbeads Enter Fishes

Wastewater discharge carries microbeads into water bodies. Zooplankton ingest them, thinking they are food. Small fish eat zooplankton, which introduces microbeads into the food chain. Larger fish then consume smaller fish, leading to bioaccumulation and biomagnification. Humans who eat fish also ingest microplastics.

Impacts on Fish

- **Physical:** Digestive blockage, reduced appetite, and poor growth.
- **Chemical:** Microbeads attract toxins like PCBs, PAHs, pesticides, and heavy metals, which transfer to fish tissues.
- **Biological:** Gut inflammation, reduced immunity, reproductive issues, and organ damage.

Impacts on Humans

Microplastics from fish may lead to inflammation, hormonal disruption, and oxidative stress. The long-term health effects are still being studied.

Environmental Impacts

Microbeads disrupt aquatic food chains, contribute to a decline in fish populations, and last a long time because they do not break down.

Global Response

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Several countries including the USA, UK, EU, and India have banned microbeads in personal care products. UNEP supports eliminating microplastics in cosmetics.

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Environmental effects

When microbeads are washed down the drain, they subsequently pass unfiltered through sewage treatment plants and make their way into rivers and canals, resulting in plastic particle water pollution. One of the various animals affected by microbeads was perch, a freshwater fish. The beads can absorb and concentrate pollutants such as pesticides and polycyclic hydrocarbons.

A variety of wildlife, from insect larvae, small fish, amphibians and turtles to birds and larger mammals, mistake microbeads for their food source. This ingestion of plastics introduces the potential for toxicity not only to these animals but to other species higher in the food chain. Harmful chemicals thus transferred can include hydrophobic pollutants that collect on the surface of the water such as PCBs, DDT and polycyclic aromatic hydrocarbons (PAHs).

Do You Know?

Banning production and sale in cosmetics

There are many natural and biodegradable alternatives to microbeads that have no environmental impact when washed down the drain, as they will either decompose or get filtered out before being released into the natural environment. Some examples to use as natural exfoliates include ground up almonds, oatmeal, sea salt and coconut husks. Microbead is banned in India from 2020 onwards.

QUICK INSIGHT

Single-use Plastic

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Several single-use plastic (SUP) items identified by the Ministry of Environment, Forest and Climate Change (MoEFCC) were **banned across India from 1st July 2022.**

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About the SUP Ban

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SUP items which have low utility and high littering potential **identified under the Plastic Waste Management (PWM) Amendment Rules, 2021.**

- Their manufacture, import, stocking, distribution, sale and use are prohibited.
- **Ban not applicable to compostable plastic products**.
  - **Penalties for violation:** Under the **Environment (Protection) Act 1986,** imprisonment up to 5 years, or a penalty up to Rs 1 lakh, or both.

Compostable Plastics

Compostable Plastics are **derived from renewable materials** such as corn, potato and tapioca starches, cellulose and soy protein. Compostable plastics are **non-toxic and decompose back into carbon dioxide, water and biomass when composted**.

Important provisions of PWM Amendment Rules, 2021

- Aimed to curb pollution caused by littered and unmanaged plastic and phase out SUP items by 2022.
- **SUP** is defined as a **plastic item intended to be used once for the same purpose** before being disposed of or recycled.
- **Thickness of plastic carry bags increased** from 50 microns to 75 microns with effect from **30th September 2021** and to 120 microns with effect from the 31st December 2022.
- **Complete ban on use of sachets** using plastic material for storing, packing or selling **gutkha, tobacco and pan masala.**

Extended Producer Responsibility (EPR) Guidelines given legal force: Plastic packaging waste not covered under identified SUP items shall be collected and managed in an environmentally sustainable way through **EPR of the producer, importer and brand owner (PIBO)** as per PWM Rules, 2016.

Related news:

- **Maharashtra, Tamil Nadu and Gujarat among top Indian plastic waste producers**
- Findings are **based on Central Pollution Control Board (CPCB) data**.

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Other Key findings

- India **ranks 5th in the generation of plastic waste** with an annual discharge of 3.5 million tonnes in 2020.
- Almost **83% of this comes from just 10 states/ UTs**.
- On a per capita basis, **India's plastic waste generation has almost doubled** over **FY2016-20**; however, **remains lower than most of the developed economies.**

Puneet Sagar Abhiyan (PSA)

- **UNEP** has signed MoU with **National Cadet Corps (NCC)** to **tackle issue of plastic pollution through PSA and Tide Turners Plastic Challenge programme**.
- **Aim:** To **increase awareness amongst local population** about importance of keeping the beaches and river fronts clean.

Global Plastics Outlook: Policy Scenarios to 2060

- The 'Global Plastics Outlook: Policy Scenarios to 2060' report **was recently released by Organisation for Economic Co-operation and Development (OECD).**
- **Projections of the report for the year 2060**:
  - **Use of plastics and plastic waste will almost triple globally,** driven by economic and population growth.
  - **Largest increases are expected in emerging economies** in Sub-Saharan Africa and Asia.
  - Following impacts from plastic lifecycle are projected to be more than double: **GHG emissions, Ozone formation, acidification and human toxicity.**

14.24 OIL POLLUTION

Oil pollution refers to the release of oil or oil-based substances into the environment, causing harm to ecosystems and wildlife. It can occur in various forms, including oil spills from ships or pipelines, oil discharge from industrial processes, and the improper disposal of used oil.

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Sources of Oil Pollution

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Marine Transportation: Oil tankers and ships can accidentally release oil into the oceans through collisions, accidents, or leaks during transportation.

Industrial Activities: Industries like oil refineries and chemical plants may discharge oil or oil-contaminated wastewater into rivers and coastal areas.

Oil Extraction: Oil drilling and extraction operations, both onshore and offshore, can result in oil spills and leaks.

Urban Runoff: Oil can enter water bodies through runoff from roads and urban areas, carrying oil from vehicles and machinery.

Oil Spills: Deliberate or accidental oil spills can occur during the storage, transport, or transfer of oil.

Magnitude of the Problem

Oil pollution is a significant global issue. According to estimates, millions of tons of oil enter the oceans each year from various sources, including spills, routine discharges, and urban runoff. These incidents harm marine life, contaminate water, and disrupt ecosystems. The exact extent of the problem is challenging to quantify due to underreporting and the widespread nature of smaller, chronic oil pollution events.

Solving the Issue

Prevention: Implement and enforce strict regulations and safety measures to prevent oil spills and leaks.

Cleanup and Remediation: Develop and use effective oil spill response technologies to mitigate the damage caused by oil pollution.

Alternative Energy: Transition to cleaner and more sustainable energy sources to reduce the demand for oil.

Oil Pollution in India

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India faces oil pollution challenges, especially along its vast coastline. The Indian government has regulations and agencies in place to address oil pollution, including the Oil Pollution Act of 1974 and the National Oil Spill Disaster Contingency Plan.

■ International and National Regulations

Internationally, the International Maritime Organization (IMO) sets standards and regulations to prevent oil pollution from ships. The United Nations Convention on the Law of the Sea (UNCLOS) also addresses oil pollution liability. Nationally, countries have their own regulations and agencies to manage oil pollution.

Addressing oil pollution requires a coordinated effort at the global, national, and local levels, emphasizing prevention, preparedness, and response measures to protect the environment and human health.

14.25 LIGHT POLLUTION / PHOTO POLLUTION

Astronomers and sky enthusiasts marked the **2022 International Dark Sky Week** from April 22 to 30 with the motive to raise awareness about the negative impact of light pollution and celebrate the night sky.

■ About International Dark Sky Week

- An **annual event** hosted by **International Dark-Sky Association (IDA).**
- **Aim:** To raise awareness about the **negative impact of light pollution** and celebrate the night sky.

■ About Light Pollution or Photo Pollution

- Presence of **excessive, misdirected, or obtrusive artificial (usually outdoor) light**.

■ Components of light pollution

- **Glare:** excessive brightness that causes visual discomfort.
- **Skyglow:** brightening of the night sky over inhabited areas.
- **Light trespass:** light falling where it is not intended or needed.
- **Clutter:** bright, confusing and excessive groupings of light sources.

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Ways to tackle light pollution

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- **Promoting the use of LED technology and green energy** for displays and illuminations.
- **Avoid illuminated advertisements** (ads) that obscure or diminish the effectiveness of traffic signals.

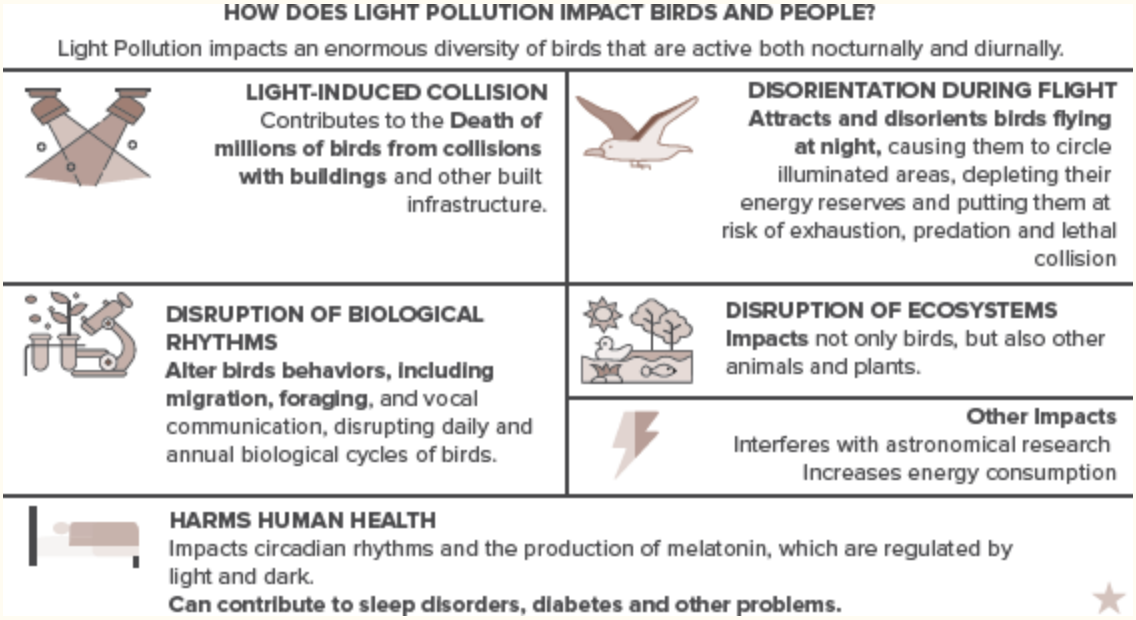

Figure 14.5 Harmful Impact of Light Pollution

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Questions Asked in UPSC Prelims Exam

UNIT VI

Unit - VI Prelims PYQs (2011 onwards)

1. Consider the following statements: **[UPSC (Pre) 2025]**

Statement I: Studies indicate that carbon dioxide emissions from the cement industry account for more than 5% of global carbon emissions.

Statement II: Silica-bearing clay is mixed with limestone while manufacturing cement.

Statement III: Limestone is converted into lime during clinker production for cement manufacturing.

Which one of the following is correct with respect to the above statements?

(a) Both Statement II and Statement III are correct and both of them explain Statement I

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(b) Both Statement II and Statement III are correct but only one of them explains Statement I

(c) Only one of the Statements II and III is correct and that explains Statement I

(d) Neither Statement II nor Statement III is correct

2. Consider the following statements: **[UPSC (Pre) 2025]**

Statement I: Activated carbon is a good and attractive tool to remove pollutants from effluent streams and to remediate contaminants from various industries.

Statement II: Activated carbon exhibits a large surface area and a strong potential for adsorbing heavy metals.

Statement III: Activated carbon can be easily synthesized from environmental wastes with high carbon content.

Which one of the following is correct in respect of the above statements?

(a) Both Statement II and Statement III are correct and both of them explain Statement I

(b) Both Statement II and Statement III are correct but only one of them explains Statement I

(c) Only one of the Statements II and III is correct and that explains Statement I

(d) Neither Statement II nor Statement III is correct

3. Consider the following **[UPSC (Pre) 2025]**

I. Cigarette butts

II. Eyeglass lenses

III. Car tyres

How many of them contain plastic?

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(a) Only one (b) Only two
(c) All the three (d) None

4. Consider the following statemz **[UPSC (Pre) 2024]**

Statement-I: Many chewing gums found in the market are considered a source of environmental pollution.

Statement-II: Many chewing gums contain plastic as gum base.

Which one of the following is correct in respect of the above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I

c. Statement-I is correct, but Statement-II is incorrect

d. Statement-I is incorrect, but Statement-II is correct

5. Consider the following **[UPSC (Pre) 2024]**

1. Pyroclastic debris
2. Ash and dust
3. Nitrogen compounds
4. Sulphur compounds

How many of the above are products of volcanic eruptions?

(a) Only one (b) Only two
(c) Only three (d) All four

6. Consider the following statements regarding mercury pollution **[UPSC (Pre) – 2023]**

1. Gold mining activity is a source of mercury pollution in the world.
2. Coal-based thermal power plants cost mercury pollution.

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3. There is no known safe level of exposure to mercury. How many of the above statements are correct?

(a) 1 only (b) 2 only

(c) All three (d) None

7. Magnetite particles, suspected to cause neurodegenerative problems, are generated as environmental pollutants from which of the following? **[UPSC (Pre) – 2021]**

1. Brakes of motor vehicles
2. Engines of motor vehicles
3. Microwave stoves within homes
4. Power plants
5. Telephone lines

Select the correct answer using the codes given below.

(a) 1, 2, 3 and 5 only

(b) 1, 2 and 4 only

(c) 3, 4 and 5 only

(d) 1, 2, 3, 4 and 5

8. 'R2 Code of Practices' constitutes a tool available for promoting the adoption of **[UPSC (Pre) – 2021]**

(a) environmentally responsible practices in the electronics recycling industry

(b) ecological management of 'Wetlands of International Importance' under the Convention

(c) sustainable practices in the cultivation of agricultural crops in degraded lands

(d) 'Environmental Impact Assessment' in the exploitation of natural resources

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9. With reference to furnace oil, consider the following statements: [UPSC (Pre) – 2021]

1. It is a product of oil refineries.
2. Some industries use it to generate power.
3. Its use causes sulphur emissions into environment. Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

10. Why is there a concern about copper smelting plants? [UPSC (Pre) – 2021]

1. They may release lethal quantities of carbon monoxide into environment.
2. The copper slag can cause the leaching of some heavy metals into environment.
3. They may release sulphur dioxide as a pollutant.

Select the correct answer using the codes given below.
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

11. Steel slag can be the material for which of the following? [UPSC (Pre) – 2020]

1. Construction of base road
2. Improvement of agricultural soil
3. Production of cement

Select the correct answer using the codes given below:
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only **(d) 1, 2 and 3**

12. Which of the following are the reasons/factors for exposure to benzene pollution? [UPSC (Pre) – 2020]

1. Automobile exhaust

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2. Tobacco smoke
3. Wood burning
4. Using varnished wooden furniture
5. Using products made of polyurethane

Select the correct answer using the codes given below:

(a) 1, 2 and 3 only
(b) 2 and 4 only
(c) 1, 3 and 4 only
(d) 1, 2, 3, 4 and 5

13. Why is there a great concern about the 'microbeads' that are released into the environment? **[UPSC (Pre) – 2019]**

(a) They are considered harmful to marine ecosystems.
(b) They are considered to cause skin cancer in children.
(c) They are small enough to be absorbed by crop plants in irrigated fields.
(d) They are often found to be used as food adulterants.

14. Consider the following **[UPSC (Pre) – 2019]**

1. Carbon monoxide
2. Methane
3. Ozone
4. Sulphur dioxide

Which of the above are released into the atmosphere due to the burning of crop/biomass residue?

(a) 1 and 2 only (b) 2, 3 and 4 only
(c) 1 and 4 only (d) 1, 2, 3 and 4

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15. In the context of solving pollution problems, what is/are the advantage/advantages of the bioremediation technique? [UPSC (Pre) – 2017]

1. It is a technique for cleaning up pollution by enhancing the same biodegradation process that occurs in nature.
2. Any contaminant with heavy metals such as cadmium and lead can be readily and completely treated by bioremediation using microorganisms.
3. Genetic engineering can be used to create microorganisms specifically designed for bioremediation.

Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

16. In the cities of our country, which among the following atmospheric gases are normally considered in calculating the value of Air Quality Index? [UPSC (Pre) – 2016]

1. Carbon dioxide
2. Carbon monoxide
3. Nitrogen dioxide
4. Sulfur dioxide
5. Methane

Select the correct answer using the codes given below:

(a) 1, 2 and 3 only

(b) 2, 3 and 4 only

(c) 1, 4 and 5 only

(d) 1, 2, 3, 4 and 5

17. There is some concern regarding the nanoparticles of some chemical elements that are used by the industry in the manufacture

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of various products. Why? **[UPSC (Pre) – 2014]**

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1. They can accumulate in the environment, and contaminate water and soil.
2. They can enter the food chains.
3. They can trigger the production of free radicals. Select the correct answer using the codes given below

(a) 1 and 2 only (b) 3 only

(c) 1 and 3 only (d) 1, 2 and 3

18. Which of the following are some important pollutants released by the steel industry in India? **[UPSC (Pre) – 2014]**

1. Oxides of sulphur
2. Oxides of nitrogen
3. Carbon monoxide
4. Carbon dioxide

Select the correct answer using the codes given below:

(a) 1, 3 and 4 only

(b) 2 and 3 only

(c) 1 and 4 only

(d) 1, 2, 3 and 4

19. Which of the following can be found as pollutants in the drinking water in some parts of India? **[UPSC (Pre) – 2013]**

1. Arsenic
2. Sorbitol
3. Fluoride
4. Formaldehyde
5. Uranium

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Select the correct answer using the codes given below:

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(a) 1 and 3
(b) 2, 4 and 5
(c) 1, 3 and 5
(d) All of these

20. Due to improper/indiscriminate disposal of old and used computers or their parts, which of the following are released into the environment as e-waste? **[UPSC (Pre) – 2013]**

1. Beryllium 2. Cadmium
3. Chromium 4. Heptachlor
5. Mercury 6. Lead
7. Plutonium

Select the correct answer using the codes given below:
(a) 1, 3, 4, 6 and 7 only
(b) 1, 2, 3, 5 and 6 only
(c) 2, 4, 5 and 7 only
(d) 1, 2, 3, 4, 5, 6 and 7

21. Photochemical smog is a resultant of the reaction of? **[UPSC (Pre) – 2013]**

(a) NO2, O3 and peroxyacetyl nitrate in the presence of sunlight
(b) CO, O2 and peroxyacetyl nitrate in the presence of sunlight
(c) CO, CO2 and NO2 at low temperature
(d) High concentration of NO2, O3 and CO in the evening

22. Lead, ingested or inhaled, is a health hazard. After the addition of lead to petrol has been banned, what still are the sources of lead poisoning? **[UPSC (Pre) – 2012]**

1. Smelting units
2. Pens and pencils

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3. Paints
4. Hair oils and cosmetics

Select the correct answer using the codes given below:

(a) 1, 2 and 3

(b) 1 and 3

(c) 2 and 4

(d) 1, 2, 3 and 4

23. The acidification of oceans is increasing. Why is this phenomenon a cause of concern? **[UPSC (Pre) – 2012]**

1. The growth and survival of calcareous phytoplankton will be adversely affected.
2. The growth and survival of coral reefs will be adversely affected.
3. The survival of some animals that have phytoplankton larvae will be adversely affected.
4. The cloud seeding and formation of clouds will be adversely affected.

Which of the statement/s given above is/are correct?

(a) 1, 2 and 3 (b) 2 only

(c) 1 and 3 (d) 1, 2, 3 and 4

24. Consider the following **[UPSC (Pre) – 2011]**

1. Carbon dioxide
2. Oxides of Nitrogen
3. Oxides of Sulphur

Which of the given above is/are the emission/ emissions from coal combustion at thermal power plants?

(a) 1 only

(b) 2 and 3

(c) 1 and 3

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(d) 1, 2 and 3

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25. Recently, 'oil zapper' was in the news. What is it? [UPSC (Pre) – 2011]

(a) It is eco-friendly technology for the remediation of oily sludge and oil spills.

(b) It is the latest technology developed for undersea oil exploration.

(c) It is a genetically engineered high biofuelyielding Maize variety.

(d) It is the latest technology to control the accidentally caused flames from oil wells.

26. Chlorination is a process used for water purification. The disinfecting action of chlorine is mainly due to [UPSC (Pre) – 2010]

(a) The formation of hydrochloric acid when chlorine is added to water.

(b) The formation of hypochlorous acid when chlorine is added to water.

(c) The formation of nascent oxygen when chlorine is added to water.

(d) The formation of hydrogen when chlorine is added to water.

27. Excessive release of the pollutant carbon monoxide (CO) into the air may produce a condition in which oxygen supply in the human body decreases. What causes this condition? [UPSC (Pre) – 2010]

(a) When inhaled into the human body, CO is converted into CO2.

(b) The inhaled CO has a much higher affinity for haemoglobin as compared to oxygen.

(c) The inhaled CO destroys the chemical structure of haemoglobin.

(d) The inhaled CO adversely affects the respiratory centre in the brain.

Answers

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1. (b) 2. (a) 3. (c) 4. (a) 5. (d) 6. (c) 7. (b) 8. (a) 9. (d) 10. (b)
11. (d) 12. (d) 13. (b) 14. (d) 15. (c) 16. (b) 17. (d) 18. (d) 19. (c) 20. (b)
21. (a) 22. (b) 23. (a) 24. (d) 25. (b) 26. (b) 27. (b)

Explanations

1. Correct answer (b): EXPLANATION

The cement industry is a major industrial source, responsible for approximately 5-8% of global anthropogenic emissions.

Statement (I) right

In the cement manufacturing process, crushed limestone is mixed with other materials like clay or shale to achieve the desired balance of calcium, silicon, aluminum, and iron oxides needed to form cement clinker. This step describes the raw materials blend but doesn't directly explain the **source** of the CO2 emissions. **Statement (II) right**

During a high-temperature process called calcination, calcium carbonate (CaCO3) in the limestone decomposes into calcium oxide (lime, CaO) and releases carbon dioxide (CO2) gas:

$CaCO3 (Heat) \rightarrow CaO+CO2$

This chemical reaction (process emissions) accounts for about 60-65% of the total emissions from cement production, making it the key explanation for Statement I. The remaining emissions come from burning fossil fuels for the heat required in the kiln. **Statement (III) right**

Therefore, both Statement II and Statement III are factually right, but only Statement III provides the direct explanation for the high level of carbon emissions mentioned in Statement I.

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2. Correct answer (a): EXPLANATION

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Activated carbon, or activated charcoal, is a type of carbon widely used to remove impurities from water, air, and other substances. Hence, Statement I right.

It is specially treated to develop tiny pores that create a very large surface area, making it highly effective for adsorption and chemical reactions. (Adsorption means atoms or molecules stick to a surface, which is different from absorption.) These pores work like a microscopic sponge.

Hence Statement II is right and explains Statement I as to how activate carbon can be used to remove pollutants from effluent streams.

Activated carbon is often produced from waste materials such as coconut husks, along with other agricultural byproducts, industrial effluents and even municipal waste i.e. waste with high carbon footprint.

Hence Statement III is right and explains Statement I as to how remediation of contaminants can be done to produce activated carbon.

The process usually includes two steps: carbonization, where the material is heated without air, and activation, where oxidizing agents like steam or CO_2 at high temperatures develop a porous structure. This makes the method both cost-effective and sustainable, turning waste into a valuable adsorbent.

3. Correct answer (c): EXPLANATION

- **Cigarette Butts** form one of the most abundant plastic waste as it contains cellulose acetate, a man-made plastic that takes over 10 years to degrade and leaches out harmful chemicals like arsenic, lead and nicotine.

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• **Eyeglass lenses** made of plastics like polycarbonate are increasingly preferred due to their light weight, optical clarity and impact resistance.

• **Car tyres** account for about 28% of microplastics, as per the IUCN. They are primarily made of a plastic called synthetic rubber, and the internal structure includes cords made of nylon or polyester.

4. Correct answer (a): EXPLANATION

• Many chewing gums found in the market are a significant source of environmental pollution. Improperly discarded gum is a common form of litter worldwide, which is costly to clean up and persists in the environment for a long time. **Statement-I right**

• Many modern chewing gums use synthetic polymers (which are essentially plastics or synthetic rubber) as their gum base. Examples of these synthetic materials can include polyethylene and polyvinyl acetate, which are also used in various plastic products. **Statement-II right**

• The fact that the gum base is made of non-biodegradable synthetic polymers is the primary reason why discarded chewing gum is a persistent environmental pollutant. These plastic components do not break down easily in nature, leading to long-lasting litter and contributing to microplastic contamination as they degrade into smaller pieces over time. **Statement-II explains Statement-I**

Therefore, both statements are right, and the second statement provides the direct explanation for the phenomenon described in the first statement.

5. Correct answer (d): EXPLANATION

Pyroclastic debris refers to the fragmented solid material ejected during explosive eruptions, which includes a range of sizes from large bombs

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and blocks to smaller particles.

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Statement (1) right

Volcanic ash and dust are the finest particles (less than 2 mm in diameter) of pyroclastic debris, formed when magma violently fragments upon eruption. Ash plumes can travel vast distances.

Statement (2) right

Nitrogen gas (N2) is a less abundant as component of volcanic emissions. Volcanic lightning during super-eruptions can “fix” atmospheric nitrogen, forming bioavailable nitrogen compounds like nitrates that accumulate in volcanic deposits.

Statement (3) right

Sulphur dioxide (SO2) and hydrogen sulphide (H2S) are common and significant volcanic gases released during eruptions. These compounds are a major contributor to acid rain and can form atmospheric aerosols that influence climate. **Statement (4) right**

6. **Correct answer (c): EXPLANATION**

Artisanal and small-scale gold mining (ASGM) is the single largest source of anthropogenic mercury emissions globally. Miners use liquid mercury to extract gold from ore, and the subsequent heating of the mercury-gold amalgam releases highly toxic mercury vapor into the atmosphere and aquatic environments.

Statement (1) right

Coal naturally contains trace amounts of mercury. When burned in power plants to generate electricity, this mercury is released into the atmosphere as emissions, which then settles onto land and water bodies, contributing to widespread contamination. **Statement (2) right**

Mercury is a potent neurotoxin, and both the World Health Organization (WHO) and the National Center for Biotechnology Information (NCBI)

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state that there is no known safe level of exposure for humans. Even low levels of exposure can cause serious, often irreversible, health effects, particularly in developing fetuses and children. **Statement (3) right**

7. **Correct answer (b): EXPLANATION**
   Magnetite particles, a form of iron oxide, are generated in the environment through high-temperature processes or friction involving iron and steel components.
   The friction and wear between brake pads and discs generate fine particles, including magnetite, due to the high heat and iron content of the materials. **Statement (1) right**
   The high-temperature combustion of fossil fuels (especially diesel and coal) and general engine wear produce magnetite particles as a component of exhaust and particulate matter. **Statement (2) right**
   Microwave ovens use electromagnetic radiation to heat food and do not involve combustion or significant physical wear of iron/steel components that would release airborne particulate magnetite as a pollutant.
   Statement (3) wrong
   Fossil fuel power plants, particularly coal-fired ones, are a major source of airborne magnetite particles due to high-temperature combustion processes. **Statement (4) right**
   Telephone lines are static infrastructure (copper or fiber optic) and do not have processes involving high heat, combustion, or friction that would generate and release airborne magnetite particles as an environmental pollutant. **Statement (5) wrong**

8. **Correct answer (a): EXPLANATION**

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The R2 Code of Practices is a standard for promoting **environmentally responsible practices in the electronics recycling industry**, guiding safe handling and refurbishment of e-waste, not wetlands, agriculture, or general EIA for resources.

9. Correct answer (d): EXPLANATION

- Furnace oil, also known as fuel oil or heavy fuel oil, is a residual fuel obtained during the distillation and refining process of crude petroleum. It is one of the heavier components that remains after lighter fractions like gasoline and diesel have been extracted. **Statement (1) right**
- Due to its high energy density and relatively low cost, furnace oil is widely utilized as a fuel in various industrial sectors, including cement, steel, textiles, and in thermal power plants (both captive and utility scale) for generating heat, steam, and electricity.

Statement (2) right

- As a residual product, furnace oil typically contains a significant amount of sulfur. When this fuel is burned for energy, the sulfur compounds react to form sulfur dioxide ( ), a major air pollutant that contributes to acid rain and respiratory problems. Environmental regulations often focus on limiting the sulfur content in furnace oil to mitigate these emissions. **Statement (3) right**

10. Correct answer (b): EXPLANATION

Copper smelting is the process of extracting pure copper from its ores through heating and chemical reduction

Environmental concerns include:

1. Sulfur dioxide emissions: Smelting sulfide ores releases large amounts of SO_2, which can form acid rain, damaging vegetation,

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soil, and water bodies.

2. Heavy metal contamination: Trace metals like arsenic, lead, and cadmium may be released into the atmosphere.
3. Particulate Matter: Dust from furnaces and slag handling can cause respiratory issues in nearby populations.
4. Incomplete combustion of carbon-based fuels (coke, coal) used in furnaces produces carbon monoxide (CO). However, modern furnaces are designed to minimise CO emissions, hence the released amount is not lethal.

11. Correct answer (d): EXPLANATION

Steel slag is a valuable industrial byproduct that is increasingly used in various applications as part of a circular economy approach. All the mentioned uses are technically feasible and actively practiced in various parts of the world, including India.

- Steel slag is widely used as a superior aggregate material for the base and sub-base layers of roads due to its high strength, durability, and load-bearing capacity. Projects in India have successfully demonstrated its effectiveness in building heavy-duty roads, offering advantages over conventional materials.

Statement (1) right

- Steel slag contains beneficial mineral components like calcium, magnesium, and silicon oxides, and has alkaline properties. It is used as a liming agent to neutralize acidic soils and as a silicate fertilizer to improve soil quality and crop yields, particularly for crops like rice. **Statement (2) right**
- Processed steel slag can be used as a raw material component in the manufacture of cement clinker or as a supplementary cementitious material in the production of blended cements like Portland slag

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cement. This practice helps reduce the environmental impact of cement production by minimizing the need for natural raw materials and lowering emissions. Therefore, steel slag can be used as a material for all three of the listed purposes. **Statement (3) right**

12. Correct answer (d): EXPLANATION

Benzene is a volatile organic compound and a known human carcinogen. Exposure occurs from various sources, but primarily from combustion processes.

1. **Automobile exhaust:** This is a significant source of benzene in outdoor air, as it is a component of gasoline that is released during combustion in vehicle engines.
2. **Tobacco smoke:** Tobacco smoke is a major source of benzene exposure, accounting for a large percentage of total exposure in some populations. Benzene is formed during the burning of tobacco, leading to exposure for both active and passive smokers.
3. **Wood burning:** The incomplete combustion of wood in fireplaces, stoves, or open burning releases various pollutants, including benzene, into the air.

Statements 4 and 5 are generally not considered major or direct sources of significant benzene exposure:

4. **Using varnished wooden furniture:** While varnishes and paints can contain volatile organic compounds (VOCs), benzene is typically not a significant component of emissions, especially once the product has fully cured.
5. **Using products made of polyurethane:** Polyurethane products and the process of making them primarily involve other chemicals like isocyanates and VOCs, but benzene itself is generally not a significant emission source from the final cured product.

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Therefore, the main reasons for exposure to benzene pollution among the given options are automobile exhaust, tobacco smoke, and wood burning.

13. Correct answer (b): EXPLANATION

Microbeads are tiny plastic particles, typically less than 5 mm in size, used in personal care products like exfoliants, toothpaste, and cosmetics. Their presence in the marine ecosystem is harmful in several ways:

1. **Ingestion:** They can be consumed by marine animals, causing physical blockages, malnutrition, and internal injuries. Moreover, toxic chemicals like pesticides, mercury, etc. can accumulate inside their bodies and pass down the food chain.
2. **Chemical pollution**: Microbeads having large surface areas can adsorb pollutants, which when consumed, can cause hormonal and reproductive issues, toxicity in marine life.
3. **Ecosystem Disruption:** Associated issues like accumulation and reduction of dissolved oxygen can alter the biodiversity of the region.

14. Correct answer (d): EXPLANATION

All four of the listed substances are released into the atmosphere, either directly as primary pollutants or indirectly as secondary pollutants formed from precursors released during the burning process:

1. **Carbon monoxide (CO):** This is a direct product of incomplete combustion of organic matter and is a major air pollutant from biomass burning.
2. **Methane (CH_4):** As another product of incomplete combustion, methane is released during the burning of organic material and is a

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potent greenhouse gas.

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3. **Ozone (O3):** Ground-level ozone is not directly emitted. However, biomass burning releases significant amounts of precursor gases such as nitrogen oxides (NOx), carbon monoxide, and volatile organic compounds (VOCs). These precursors react in the presence of sunlight to form ozone in the lower atmosphere, contributing significantly to regional ozone pollution.
4. **Sulphur dioxide (SO2):** Biomass contains a certain amount of sulfur, and when it is burned, the sulfur is oxidized, resulting in the emission of sulfur dioxide. Therefore, all four substances are associated with the burning of crop/biomass residue and are released into the atmosphere.

15. Correct answer (c): EXPLANATION

- Bioremediation uses microorganisms or plants to naturally degrade or detoxify pollutants, **accelerating processes that already occur in ecosystems.**
- **Heavy metals cannot be degraded** by microorganisms. They can only be transformed, immobilised, or accumulated, not completely eliminated.
- Genetic engineering enhances the natural pollutant - degrading abilities of microorganisms and plants using recombinant DNA technology.
- By modifying organisms, scientists can **improve pollutant uptake, degradation rates, enzyme production, tolerance to contaminants, and create specialised metabolic pathways,** making bioremediation more efficient, cost-effective, and environmentally friendly.

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16. Correct answer (b): EXPLANATION

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The Air Quality Index (AQI) is a tool used by governments to convey air pollution levels and associated health risks in a simple format. It converts complex pollutant data into a single number, color, and rating, with higher values indicating worse pollution.

AQI measures key pollutants such as PM2.5, PM10, ground-level ozone (O_3), carbon monoxide (CO), sulfur dioxide (SO_2), and nitrogen dioxide (NO_2), with country-specific scales in use.

17. Correct answer (d): EXPLANATION

- Nanoparticles are highly **stable and persistent**, and can accumulate in soil and water, potentially causing long-term environmental contamination
- Once nanoparticles enter the environment, especially easily spreadable particles like **microplastics, aerosols, etc.**, they can be absorbed by plants and microorganisms, moving up the food chain and affecting animals and humans.
- Certain nanoparticles can induce **oxidative stress** in cells by generating free radicals, which can damage proteins, DNA, and cell membranes, leading to health risks. Examples include:

a. Silver nanoparticles (AgNPs): Used in antimicrobial coatings, they can produce reactive oxygen species (ROS) in human cells.

b. Titanium dioxide nanoparticles (TiO2): Common in sunscreens and paints, they can trigger free radical formation under UV exposure.

c. Carbon nanotubes: Used in electronics and composites, they can cause oxidative damage and inflammation in lung tissue.

d. Zinc oxide nanoparticles (ZnO): Found in cosmetics and sunscreens, they can induce ROS, leading to cytotoxicity.

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18. Correct answer (d): EXPLANATION

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The steel industry is one of the 17 most polluting industrial sectors identified by the Central Pollution Control Board (CPCB) in India. The main pollutants released from integrated steel plants, particularly from processes involving high-temperature combustion of coal and coke in blast furnaces and coke ovens, include all four options listed:

1. **Oxides of sulphur (SOx):** These are released primarily from the combustion of sulfur-containing fossil fuels like coal and coke. They contribute to acid rain and respiratory problems.
2. **Oxides of nitrogen (NOx):** These are formed during high-temperature combustion processes in the furnaces and power generation units within steel plants. They contribute to smog and acid rain.
3. **Carbon monoxide (CO):** This toxic gas is produced due to incomplete combustion of carbon-based fuels and is a byproduct of the chemical reactions during the reduction of iron ore in blast furnaces. While some is reused as fuel, significant amounts can be released.
4. **Carbon dioxide (CO2):** The steel industry is a major emitter of this greenhouse gas, primarily from the combustion of fossil fuels for energy and from the chemical process where carbon reacts with iron oxides.

Therefore, all four substances are important pollutants released by the steel industry.

19. Correct answer (c): EXPLANATION

Groundwater pollution in India is mainly due to two causes: Geogenic and Anthropogenic.

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- Geogenic pollution refers to the **naturally occurring elevated levels** of certain elements in groundwater that pose health risks. It arises from the geochemical properties of aquifer materials, where contaminants in rocks dissolve during rock-water interactions, or from environmental factors.
- Geogenic pollutants include: Arsenic (West Bengal and Bihar), Fluoride (Rajasthan, MP, Punjab), Uranium and Iron (Assam).
- Uranium contamination in groundwater is a significant, emerging issue in India, affecting 16 states with levels often exceeding the WHO guideline of 30 µg/L.
- Anthropogenic groundwater pollution is caused by **human activities**, including agricultural runoff (fertilisers, pesticides), industrial and mining discharges (untreated wastewater, chemical spills), and urban waste (sewage, dumping).
- These activities introduce contaminants like **nitrates, heavy metals, and organic chemicals** into groundwater, posing serious health risks such as blue baby syndrome (due to nitrates) and kidney disorders, and often requiring complex and costly remediation.

20. Correct answer (b): EXPLANATION

Improper disposal of e-waste releases a host of heavy metals as well as organic chemicals into the atmosphere.

Heavy Metals:

- Lead (Pb): Found in old CRT monitors and electronics; affects the nervous system and kidneys.
- Mercury (Hg): A potent neurotoxin causing sensory impairment, muscle weakness, and memory loss.
- Cadmium (Cd): Highly toxic; damages lungs and kidneys and impairs brain development in children.

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- Chromium (Cr): A carcinogen that can damage DNA and cause multiple health problems.
- Beryllium (Be): Causes chronic and acute beryllium disease, including lung cancer.
- Arsenic (As): Poisonous; contaminates soil and water, leading to severe health effects.

Organic Pollutants:

- Brominated Flame Retardants (BFRs): Used in plastics; they can leach into water and disrupt endocrine functions.
- Dioxins and Furans: Highly toxic byproducts released

21. Correct answer (a): EXPLANATION

- Photochemical smog is a brownish-gray haze formed when solar ultraviolet radiation reacts with hydrocarbons and nitrogen oxides in the atmosphere.
- It consists of anthropogenic pollutants, including ozone, nitric acid, and various organic compounds, which are often trapped near the ground by temperature inversion, worsening air quality and health impacts.
- **Peroxyacetyl nitrate (PAN)** is a secondary atmospheric pollutant formed by photochemical reactions between volatile organic compounds (VOCs) and nitrogen oxides (NOx).
- It is a strong irritant to eyes and respiratory systems, more phytotoxic than ozone, and serves as an indicator of photochemical pollution.

22. Correct answer (b): EXPLANATION

Lead despite being banned from addition to petrol can be released into the environment through a number of ways:

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1. Smelting Units

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- Airborne Emissions: During high-temperature processing, lead fumes and dust are released into the air.
- Wastewater & Slag: Effluents and solid residues from smelting can leach lead into soil and water bodies.

2. Lead-Based Paints
   - Paint Chips & Dust: Flaking, sanding, or scraping produces lead dust or chips.
   - Indoor Pollution: Inhalation of dust or ingestion (common in children) leads to systemic absorption of lead.
3. Other processes include:
   - Contaminated Soil: Lead persists in soils near industrial sites, battery recycling units, or highways.
   - Water Pipes: Lead plumbing pipes or solder can leach lead into drinking water.
   - Industrial Emissions: Mining, smelting, and battery manufacturing release lead into air and soil.
   - Consumer Products: Toys, ceramics, cosmetics, and imported items may contain lead.
   - E-Waste: Improper recycling of electronics exposes workers and the environment to lead.
   - Food Contamination: Vegetables grown in contaminated soil or seafood from polluted waters can contain lead.

23. Correct answer (a): EXPLANATION

Calcareous phytoplankton, use calcium carbonate ($CaCO_3$) to build their shells. Increased acidity reduces the concentration of available carbonate ions, making it difficult for these organisms to form and maintain their shells. **Statement (1) is right**

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Corals also build their skeletons from calcium carbonate. Lower pH levels hinder the calcification process, leading to weakened coral structures, slower growth, and increased vulnerability to bleaching and degradation.

Statement (2) is right

Many marine animals, including shellfish, sea urchins, and some fish, have larval stages that are planktonic and often calcify during development or rely on phytoplankton for food. Acidification impairs the ability of these larvae to form shells, reducing their survival rates and impacting the entire food web. **Statement (3) is right**

While ocean life (phytoplankton) does influence the atmospheric sulphur cycle and thus cloud formation, the *acidification process itself* is not considered to directly or primarily affect "cloud seeding" or cloud formation in the way described in the options. **Statement (4) is wrong**

24. Correct answer (d): EXPLANATION

Coal-fired thermal power plants are major sources of air pollution and greenhouse gases. When coal is burned to generate electricity, several pollutants are released.

- Greenhouse Gases (GHGs)
  - Carbon dioxide (CO_2): Primary GHG, contributing to global warming.
  - Methane (CH_4) and Nitrous oxide (N_2O) are released in smaller amounts during coal handling and combustion.
- Particulate Matter like fly ash and bottom ash: Fine particles containing heavy metals like mercury, arsenic, and lead. Cause respiratory and cardiovascular diseases when inhaled.
- Sulfur dioxide (SO_2): Forms acid rain; irritates the respiratory system. Comes from sulfur present naturally in coal.

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- Nitrogen Oxides (NOx): Includes NO and NO_2, contributing to smog formation, acid rain, and ozone formation.
- Heavy Metals: Mercury (Hg), Lead (Pb), Cadmium (Cd), Arsenic (As): Toxic elements that can bioaccumulate in ecosystems.
- Trace Organic Pollutants: Polycyclic aromatic hydrocarbons (PAHs) and other organics can form during incomplete combustion.

25. Correct answer (b): EXPLANATION

- Oilzapper Technology is a bioremediation method developed by TERI that employs a consortium of natural oil-degrading bacteria to break down crude oil, oily sludge, and other hydrocarbon wastes into harmless carbon dioxide and water.
- The process involves immobilising five bacterial strains on a carrier, such as powdered corncob, which is then applied to contaminated sites—like oil spills or drill cuttings—to restore soil and reduce environmental pollution. **Option (a) is right**

26. Correct answer (b): EXPLANATION

Formation of Active Chlorine

- When chlorine (Cl_2) is added to water, it reacts with water to form hypochlorous acid (HOCl) and hydrochloric acid (HCl):
- Hypochlorous acid partially dissociates into hypochlorite ions (OCl^-), depending on the pH.

Mechanism of Disinfection

- Oxidation of Cellular Components: HOCl and OCl^- oxidise essential components of microbial cells, including proteins, enzymes, and nucleic acids, disrupting cell metabolism.
- Penetration of Cell Membranes: Hypochlorous acid, being uncharged and small, penetrates microbial cell walls easily, leading to

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irreversible damage to cell structures.

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- Inactivation of Microorganisms: This results in the death or inactivation of bacteria, viruses, and protozoa, making water safe for consumption. **Option (b) right**

27. Correct answer (b): EXPLANATION

The condition caused by excessive carbon monoxide (CO) exposure is hypoxia, specifically carbon monoxide poisoning.

- CO binds to hemoglobin in red blood cells with an affinity about 200–250 times greater than oxygen.
- This forms carboxyhemoglobin (COHb), which prevents hemoglobin from carrying oxygen to body tissues.
- As a result, the oxygen supply to vital organs decreases, leading to symptoms like headache, dizziness, shortness of breath, and in severe cases, unconsciousness or death. **Statement (b) is right**

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UNIT VII Biomes, Natural Vegetation and Floristic Regions of India

Biomes of the World

15

IN THIS CHAPTER...

15.1 Biomes
15.2 Difference between Biome and Ecosystem
15.3 Factors affecting Biome Location
15.4 Classification of the Biomes
15.4.1 Walter's classification of Biomes
15.4.2 Whittaker's classification of Biomes
15.5 Importance of Biomes
15.6 Important Biomes of the World
15.6.1 Aquatic Biomes
15.6.2 Terrestrial Biomes
15.6.3 Forests
15.6.4 Deserts

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15.1 BIOMES

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Biomes are large geographical areas on the Earth that can be best distinguished by the prevailing climate and the flora (plant species) and fauna (animal species) that have adapted to the climatic conditions.

15.2 DIFFERENCE BETWEEN BIOME AND ECOSYSTEM

Biomes and Ecosystem have overlapping features and are often confused with each other. However, there are fundamental differences between the two:

- Ecosystem forms the sum total of interaction of the organisms with their abiotic environment in an area and is characterised by the flow of energy and nutrient cycling. An Ecosystem can range from a small area of a few square meters to very large areas.
- **Biome is a vast geographical area and has a characteristic climate.** The plant and animal species residing in a biome have adaptive features that help them cope with the climatic conditions.
- A biome may consist of many ecosystems.

15.3 FACTORS AFFECTING BIOME LOCATION

Climate, topography, soil in the terrestrial environment and equivalent influences in the aquatic environment are the significant elements that demarcate biomes.

The climate of an area is one of the important **determinants** for the kind of flora and fauna found. **Due to the influence of the climatic conditions, the plants and animals make adaptations to survive in the physical environment.** Since biomes are characterised by the kind of vegetation and

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animals found in them, they correspond with the different climatic zones of the world.

For example, the vegetation found in tropical evergreen forests consists of dense canopy of tall trees whereas the arid and hot deserts consist of sparsely distributed shrubs and cactus. There is abundance of precipitation in the tropical areas which supports dense growth of the trees whereas in hot deserts, the shrubs or trees grow at wide distances from one another to avoid the competition for limited groundwater resources.

Climate is in turn influenced by the latitude (distance from the equator), altitude, prevailing winds, distance from the sea and the ocean currents.

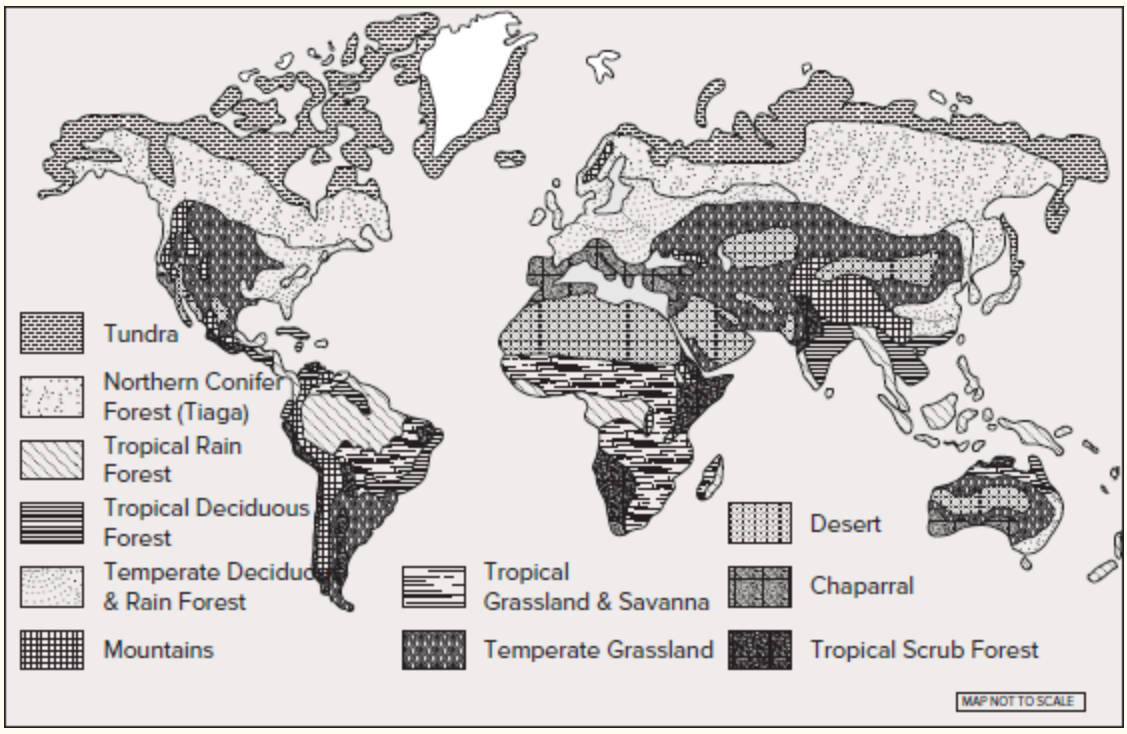

Figure 15.1 Major Biomes of the World

15.4 CLASSIFICATION OF THE BIOMES

There is no single classification of biomes. Broadly, biomes can be divided into terrestrial and aquatic biomes.

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Two most common classifications used are **Walter's biome** classification and **Whittaker's biome** classification.

15.4.1 Walter's classification of Biomes

Heinrich Walter classified the biomes according to the annual cycle of temperature and precipitation. Walter classified biomes into units called zonobiomes that correspond to the climate zones. (Refer to Table 15.1 for list of 9 zonobiomes).

Table 15.1 Zonobiomes Classified by Walter

Zonobiome	Climate
I	Equatorial with diurnal climate
II	Tropical with summer rains
III	Sub-tropical arid climate
IV	Mediterranean summer drought and winter rains
V	Warm temperature
VI	Temperate with a short cold period in the winter (nemoral)
VII	Arid-Temperate (continental)
VIII	Cold-Temperate (boreal)
IX	Arctic (including Antarctic)

15.4.2 Whittaker's classification of Biomes

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R.H Whittaker classified biomes according to the average temperature and precipitation. He produced a graph by plotting the observed vegetation types in the different regions with respect to the average precipitation and temperature received there.

15.5 IMPORTANCE OF BIOMES

Biomes are the home to the various plant and animal species and demonstrate the interdependence among them. The dense forests provide shelter to the various species of insects, wild animals etc. Many medicinal plants are found in the forests and are a basis for the traditional medicines. Biomes offer ecological services.

Aquatic biomes support life on Earth. The first signs of life on Earth evolved in water. Biomes we see today have been formed over a vast period of time.

It is necessary to conserve and preserve them. Anthropogenic activities have resulted in the exploitation of the biomes and are causing serious damage to them. Pollution and global warming are the main culprits that have disturbed the delicate balance between the flora and fauna that exists in the biomes.

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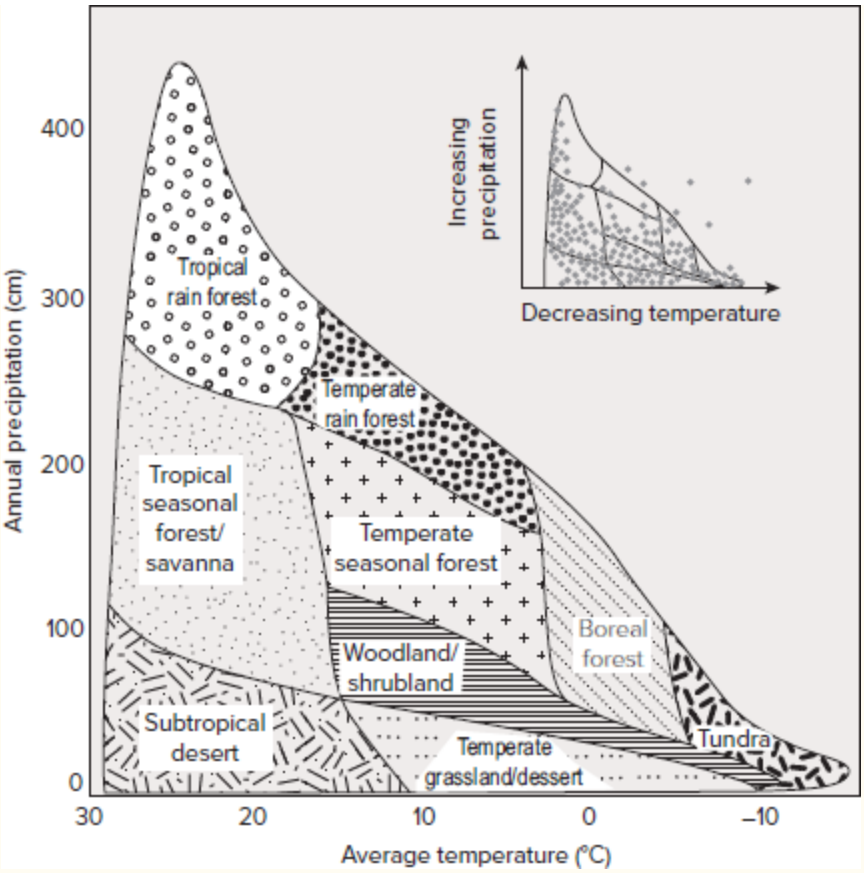

Figure 15.2 Whittaker Parameter

15.6 IMPORTANT BIOMES OF THE WORLD

15.6.1 Aquatic Biomes

■ *Freshwater Regions*

The water in these regions has a low salt concentration. The important freshwater biomes are:

Ponds and Lakes

The size and depth of lakes and ponds vary from a few to thousands of Km^2. Many ponds are seasonal. Lakes and ponds often have limited species diversity. They are divided into three zones on the basis of distance from shoreline and depth.

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(a) **Littoral zone: It is the topmost zone, near-shore area, shallow and receives the most amount of sunlight and has abundant oxygen and nutrients.** It supports rooted and floating aquatic plants, algal species like diatoms, grazing snails, insects (egg and larvae of insects like dragonfly and midges), crustaceans, fish and amphibians.

(b) **Limnetic zone:** It is an open surface water area that has the penetration of light similar to littoral zone. **It supports planktons i.e. floating microscopic organisms phytoplanktons and zooplanktons. Nektons i.e. actively swimming organisms like various fish are also found here.**

(c) **Profundal zone:** It is the deepest part of pond where sunlight do not penetrate. Heterotrophs, i.e. organisms; who cannot manufacture their own food, live here.

Streams and Rivers

- The temperature of streams and rivers at their source is much less than that at their mouth.
- Water is clear at the source of the river and is much cleaner and have high dissolved oxygen and support fishes.
- At the middle part of the river, the diversity of fish species increases and algal and aquatic plants appears.
- Near the mouth, sediments brought with the river flow turn the water murky. Also, due to less oxygen, fish that survive on less oxygen, survive.

Wetlands

- These areas consist of marshes, swamps and bogs i.e. areas of standing water.
- Hydrophytes, which are plant species adapted to humid conditions, are found here. Examples are pond lilies, black spruce, tamarack etc.
- Wetlands are the richest of the ecosystems in terms of species diversity.
- Wetlands are not considered as freshwater ecosystem exclusively and there is presence of salt marshes and contains salts.

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- Fauna associated with wetlands includes waterfowl like ducks, geese, swans, and other birds like pelicans. Herons, ibis etc.; Amphibians like green tree frog, red-eyed frog; mammals like common water rat, bats; Reptiles like turtles, water snakes.

■ *Marine*

Oceans

- Oceans dominate almost 75% of the Earth. They support rich species diversity.
- The four zones found in oceans are:
  (a) **Inter-tidal zone:** This is the junction of ocean and land, the area between high tide and low tide and is divided into four zones: Spray zone, high intertidal zone, mid-intertidal zone and low intertidal zone. The stratification is clearer in rocky areas than the sandier areas as the sand keeps moving when waves strike; thus, the number of algae and plants found is less. Examples of flora found here include red algae, kelps or seaweed; fauna include predatory crustaceans, shoreline crabs, sea stars, anemones, shorebirds etc.
  (b) **Pelagic zone:** This zone is the area where the open sea is found, further away from the land. It is divided into:
  (i) **Epipelagic zone** where surface receives illumination but gradually light penetration decreases and photosynthesis is limited at the depth of 200 m (ii) **Mesopelagic zone** where photosynthesis does not occur due to lack of adequate sunlight. Extends from 200–1000 m. (iii) **Bathypelagic or the dark zone** where light is absent and the organisms here make special adaptations to survive in the great pressure. (iv) **Abyssopelagic zone** (v) **Hadopelagic zones** which are the deep ocean trenches.
  (c) **Benthic zone:** This zone is below the pelagic zone and it represents the ocean bottom and consists of sediment surface and some sub-surface layers. **Benthos is the term used for organisms found in the benthic zone. Epibenthos are the organisms living on the**

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sediment surface and hyberbenthos are found just above the sediment surface. Invertebrate organisms such as oysters, clams, sea cucumbers, sea stars, brittle stars, anemones and microbenthos (size less than 32 micrometer) like diatoms, bacteria and ciliates are found here. These organisms survive are scavengers or detrivores.

(d) **Abyssal zone:** It is the same as abyssopelagic zone. Environmental conditions are almost uniform here. The water is very cold 3°–4°C, the pressure is great. Abyssal waters contain high amount of oxygen dissolved per liter but the amount near the sea floor is minimal.

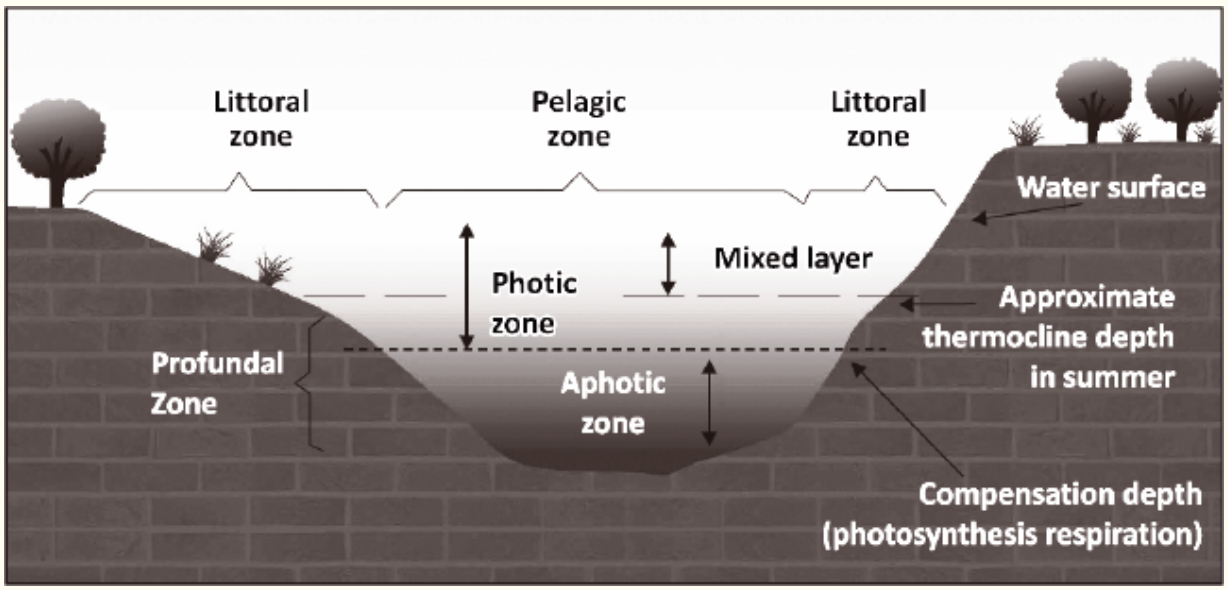

Figure 15.3 Figure showing the zones in the lake/pond. The photic zone is the zone where sunlight penetrates and aphotic zone is where light cannot penetrate.

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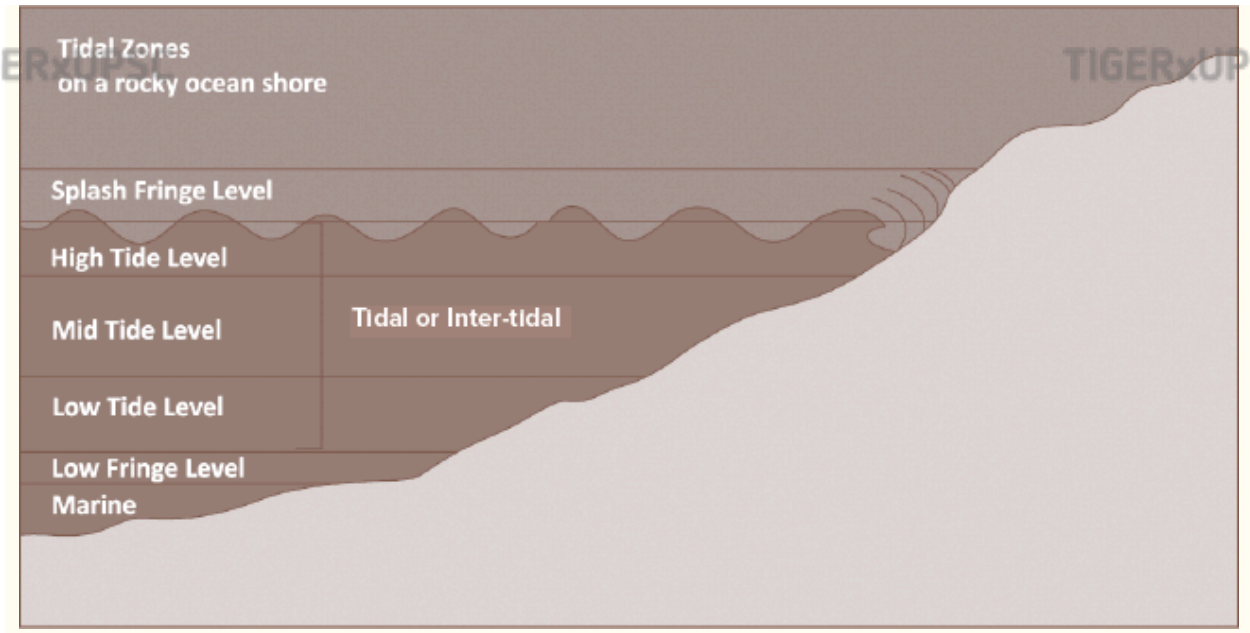

Figure 15.4 A figure showing the stratification of the inter-tidal zone

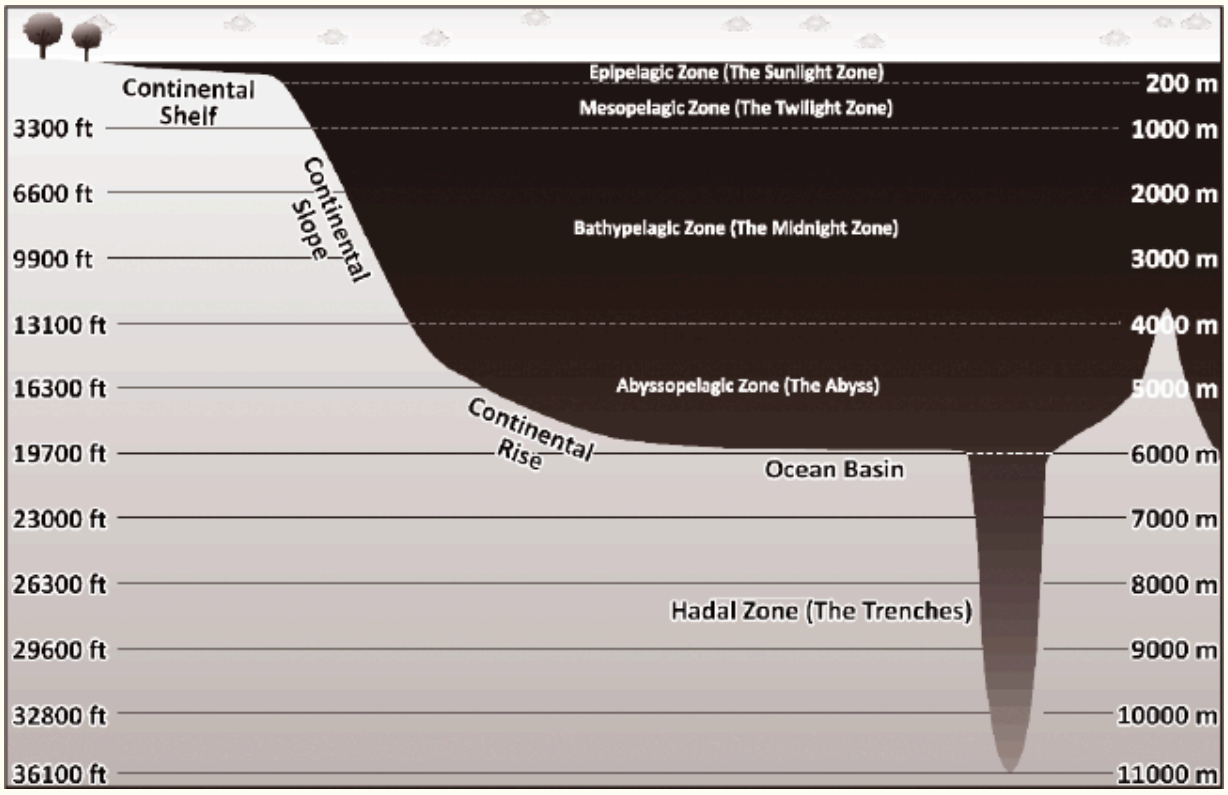

Figure 15.5 Showing different pelagic zones

15.6.2 Terrestrial Biomes

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Tundra

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Some features of Tundra Biomes are—

- Very cold climate
- Little biodiversity
- Lacks tree with little vegetation adapted to low temperature
- Very little precipitation
- Scanty Drainage
- Short season of reproduction and growth
- High oscillation of population due to continuous immigration and emigration.

Arctic Tundra

- It lies in the Northern Hemisphere
- Growing season is 50–60 days
- Average of temperature in winter season is –34°C (–30°F)
- Average of temperature in summer season is 3–12°C (37–54°F)
- Annual Precipitation is 15–25 cm.
- Permanently frozen sub-soil consisting of gravel and finer material, exists. It is called **permafrost**.
- Vegetation
  - Does not consist of deep roots; short plants exist in group to fight cold harsh climate
  - 1700 plants species exist in arctic and sub-arctic zones.
  - It includes sedges, reindeer mosses, liverworts, low shrubs, and grasses, 400 different varieties of flowers, crustose and foliose lichen
- Fauna
  - Mammals and birds have extra insulation layer of fat.
  - Reptiles and Amphibians generally absent
  - Animals either hibernate or migrate to the South (Birds)
  - **Herbivorous mammalian:** Voles, caribou, arctic hares, lemmings and squirrels
  - **Carnivorous mammalian:** Arctic foxes, polar bears, wolves

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- **Migratory birds:** Falcons, loons, ravens, sandpipers, terns, snow birds, ravens, snow buntings and different species of gulls
- **Insects:** Flies, moths, grasshoppers, blackflies, mosquitoes and arctic bumble bees
- **Fishes:** Flat fish, salmon, cod and trout

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Alpine Tundra

- Located at high altitude in the mountains.
- Growing season- Approx. 180 days
- Well-drained soil unlike Arctic tundra.
- Vegetation: Dwarf trees, small-leafed shrubs, tussock grasses and heaths
- Fauna:
  - Mammals: Pikas, Marmots, Mountain Goats, Sheep, Elk
  - Birds: Grouse like birds
  - Insects: Springtails, Butterflies, Beetles, Grasshoppers

Grasslands

Savanna

- Found in Africa, Australia, South America and India
- **Summer temperature: 25°–30°C (78° to 86°F)**
- **Winter temperature: 20°–25°C (68° to 78°F)**
- **Annual Precipitation:** 100 to 150 cm (10 to 30 inches). Rainfall occurs in 6–8 months and the rest of the months are dry and fires occur. Thunderstorms appearing in October mark the starting of the dry spell. January is the driest month.
- **Climatic Savanna:** Results from the climatic conditions
- **Edaphic Savanna:** Results from soil conditions
- **Derived Savanna:** Results from anthropogenic activities like forest clearing.
- Soil is porous and drains rapidly.
- **Vegetation**

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  - Number of species compete for limited area, few species of grasses dominates a particular limited area
  - For example, in drier Savannas, in Serengeti plains or Kenya's Laikipia plateau, Rhodes grass and red oat grass are dominant; Star grasses dominate East African Savannas and Lemon grasses are found in Ugandan Savannas.
  - Deciduous trees and shrubs are scattered.
- **Fauna:** giraffes, zebras, snakes, worms, termites, beetles, lions, leopards, hyenas, buffaloes, kangaroos, mice, moles, gophers, ground squirrels and African elephants.

- Temperate Grasslands

- These grasslands lies in north of the Tropic of Cancer and south of the Tropic of Capricorn.
- **Major temperate grasslands:** Pampas of Argentina and Uruguay, Veldts of South Africa, Puszta of Hungary, Great Plains and Prairies of central North America and Steppes of the Russia
- **Summer Temperature:** They have hot summers where temperature can reach more than 38ºC.
- **Winter temperature:** Can go as low as –40ºC.
- **Precipitation:** 50.8 to 88.9 cm rainfall. Height of grasses depend on the amount of rainfall.
- Soil is deep and dark, nutrient-rich due to the decay of grass-roots.
- **Vegetation**
  - Temperature, rainfall and soil determine the species of grass growing in a particular grassland
  - Trees and shrubs usually don't appear as the precipitation is just enough to support grasses but not the tree-growth. Seasonal drought and grazing by animals also contribute to less tree-growth.
  - Hundreds of flower species are found in the grasses.
  - A few species of grasses include purple needle grass, galleta, blue grama, buffalo grass

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- Some flowers found in the grasslands are asters, clovers, blazing stars, goldenrods, sunflowers, wild indigos.

- **Fauna**
  Different animal species live in different grasslands. Some common species found are gazelles, hawks, owls, snakes, grasshoppers, leafhoppers, zebras, deer, mice, coyotes, foxes, skunks, badgers, blackbirds, grouses, meadowlarks, quails, rhinoceroses, wild horses, lions, wolves, prairie dogs, jack rabbits, sparrows and spiders.

15.6.3 Forests

Tropical Forests

- They are found between the 23.5 degrees North latitude and 23.5 degrees South latitudes.
- Winter is absent in tropical forests.
- **Annual Temperature:** 20–25ºC. There is little variation in temperature all over the year.
- **Precipitation:** >200 cm annually
- **Soil:** Nutrient poor due to torrential rains. Leaching of soil appears at rapid pace.
- The four layers of rainforest are:
  - **Emergent layer:** Consists of very tall trees above the dense canopy that receive highest amount of sunlight and can tolerate humidity and strong winds. The crown of the trees is usually mushroom-shaped.
  - **Canopy:** This layer is found 60–90 feet above the ground. 90% of the organisms live in this layer. Epiphytes grow on the trees.
  - **Understory:** It just receives 2–15% of the sunlight that falls on the canopy. Young trees and herbs are found in this layer.
  - **Forest Floor:** It receives less than 2% of the sunlight that falls on the canopy. It consists of fallen leaves, fruits, seeds and branches which decompose quickly.

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- **Flora:** These tropical forests have high species diversity and trees are about 25–40 m tall trees with shallow roots and buttressed trunks. They are mostly evergreen forests with large dark green leaves. Other plant species found are orchids, palms, bromeliads, vines (lianas), ferns, mosses found in tropical forests.
- **Fauna:** Numerous birds, insects, bats, small mammals
- The sub-division of tropical forests is based on distribution of seasonal rainfall.
  - **Evergreen rainforest:** Rainfall is abundant and there is no dry season.
  - **Seasonal rainforest:** Very wet tropical region with limited dry season
  - **Semi-evergreen rainforest:** Longer dry season
  - **Moist/dry deciduous forest (monsoon):** Trees are deciduous in nature and dry season increases as rainfall decreases.

Temperate Forests

- They are found in North America, western and central Europe, northeastern Asia
- The seasons are characterised with 4–6 frost-free months that support the growing season that lasts from 140–200 days.
- **Temperature range:** 30ºC to –30ºC. Average temperature annually is 10ºC; summers are hot and winters are cold.
- **Precipitation:** 75–150 cm annually.
- The soil is usually fertile as the organic decaying matter is not leached rapidly like in the tropical forests.
- Canopy of forest is moderately dense where penetration of sunlight results in diverse and well defined understory.
- **Flora:** Broad-leaved trees are found. Examples include oak, hickory, cottonwood, elm, willow, beech, hemlock, maple, basswood and spring-flowering herbs. The leaves turn red, auburn or yellow in the autumn season and are shed gradually. This happens every year.
- **Fauna:** squirrels, rabbits, bobcat, timber wolf, skunks, birds, deer, mountain lion, fox and black bear.

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- The seasonal rainfall distribution determines the sub-division of temperate forests:
  - **Moist coniferous and evergreen forests:** They are found in regions of dry summers and wet winters
  - **Dry coniferous forests:** Found prominently in the areas of low precipitation and higher altitudes.
  - **Mediterranean forests:** Rainfall is less than 100 cm per year and distributed in winters
  - **Temperate coniferous:** The winters are mild and annual precipitation is greater than 200 cm.
  - **Temperate rainforests:** Found in areas of mild and frost-free winters; rainfall of more than 1500 mm distributed throughout the year.

Taiga or Boreal Forests

- They are found in Alaska, Siberia, Canada and Scandinavia i.e. in the Northern Hemisphere in the sub-arctic zone.
- The world's largest taiga forest spreads across 5,800 kilometers from the Ural Mountains to Pacific Ocean
- Annual temperature: –7ºC to 21ºC in summers and –54ºC to –1ºC in winters.
- Annual precipitation: 30–85 cm. The majority of precipitation occurs in warm, humid summer months.
- **Soil:** The soil is less fertile due to fewer nutrients. Soil beneath taiga consists of permafrost. Sometimes, bedrock is found beneath the soil. **Permafrost and bedrock prevent the drainage of water which gets collected in the form of shallow bogs called muskegs, on which small grasses and moss grows. The ground looks solid but is wet and spongy.**
- **Flora: Conifers, which have needles instead of broad leaves, are found.** Common examples are spruce, pine and fir. Conifers do not shed their needles and are, therefore, evergreen. No shrubs and flowers are found. Mosses, lichens and mushrooms grow on the soil.

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- **Fauna: Siberian tiger,** bear, lynx, birds that migrate south during winters.

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15.6.4 Deserts

Hot and Dry Desert

- Temperature is usually warm throughout the year. The diurnal temperature range differs greatly due to little humidity in the areas to trap the sun rays.
- Sub-tropical deserts like the Sahara in Northern Africa, Kalahari Desert in Southern Africa, Tanami desert in Australia are examples.
- **Temperature:** Mean annual temperature: 20–25ºC; Maximum range 44–54°C; minimum range could go up to –18ºC.
- **Precipitation:** Does not rise above 25 cm annually. E.g. in Atacama and Sahara desert, less than 1.5 cm is received annually.
- **Soil:** Coarse-textured, sand blows leaving heavier particles behind.
- **Flora:** Prickly pears, false mesquite, yuccas, ocotillo, turpentine bush, sotol, ephedras, agaves, cacti, woody shorttrees and brittlebush.
- **Fauna:** Insects, burrowers, reptiles, kangaroo rats etc.

Semi-Arid desert

- The average summer temperature range from 21ºC–27ºC evening temperature is around 10ºC.
- Annual precipitation is around 2–4 cm.
- **Soil:** Sandy and fine-textured to small rocky fragments. It has a low salt concentration.
- **Flora:** Plants have glossy leaves to radiate sunlight, usually have unfavourable odour and taste. Common species include Creosote bush, bur sage (*Franseria dumosa* or *F. deltoidea*), whitethorn, catclaw, mesquite, brittle bushes (*Encelia farinosa*), lyciums, and jujube.
- **Fauna:** Grasshoppers, ants, kangaroo rats, rabbits, lizards and snakes, birds like California thrasher and burrowing owls

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Coastal desert

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These occur where cold currents are found near the shore. The Atacama Desert in Chile is an example.

- Temperature: Average summer temperature: 13–24°C; average temperatures of winter is 5°C or below. The maximum temperature of the year is about 35°C and the minimum temperature is about –4°C.
- **Precipitation:** 8–13 cm in many areas. Atacama is one of the driest deserts and some weather stations in the desert have not recorded evidence of rain.
- **Soil:** It has a fine texture with good drainage and moderate soil content.
- **Flora:** The plants with thick and fleshy stems absorb rainwater and store it for future use. Common species found are saltbush, buckwheat bush, black bush, rice grass, little leaf horsebrush, black sage, and chrysothamnus.
- **Fauna:** Insects, mammals (coyote and badger), amphibians (toads), birds (great horned owl, golden eagle and the bald eagle), and reptiles (lizards and snakes).

Cold Desert

- Occur in the Antarctic, Greenland and Nearctic regions.
- **Temperature:** Mean summer temperature is between 21–26°C; mean winter temperature is between –2 to 4°C.
- High precipitation occurs during the winters in the form of snowfall and rainfall. Mean annual precipitation is 15–26 cm.
- Soil is heavy and silty. Good drainage leaches out most of the salt out of the soil.
- **Flora:** Plant heights vary from 15–122 cm. Ground is covered by lichens. Plant species are deciduous and have spines.
- **Fauna:** Antelope ground squirrels, badger, kit fox and coyote, gazelles, jackrabbits, gerbils, saiga antelope, kangaroo rats, kangaroo mice, pocket mice, grasshopper mice, desert tortoises, lizards, wolves, sidewinder and viper snakes. Deers with thick fur coats are found during the winters.

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Biogeographical Zones of India

16

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16.1.4 Semi-arid

16.1.5 Western Ghats

16.1.6 Deccan plateau

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16.1 BIOGEOGRAPHIC ZONES

Biogeographic is referred to as the study of the ecosystem, various species and organisms in a given region throughout geological time. India is divided into ten zones according to biogeographic characteristics.

India is blessed with mega diversity due to its different climatic conditions and topography. **India is placed at 10th rank in richness of**

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plant diversity in the world. Ecosystem and biodiversity have their own economic and natural functions, therefore, it is important to have knowledge of species richness in the country.

India has 10 biogeographic zones which are distributed on the basis of climate, soil and biodiversity. Following are the zones described below:

16.1.1 Trans-Himalayas

The cold desert of Ladakh and Lahaul Spiti of Himachal Pradesh comes under this region. It is considered as an extension of Tibetan plateau and is about 5–6% of the country's total landmass. This biogeographic zone is habitat to various endemic species, along with the largest population of wild sheep and goat in the world. Rare fauna like **Snow Leopard** and the migratory Black-necked Crane is found here. It also has an ecosystem of cold dry desert.

16.1.2 Himalayas

The Himalayas are the northernmost boundaries of India and act as barriers between India and Tibet. It lies from Kashmir to Assam. The Himalayas has diverse biotic provinces and biomes. The vast span of the Himalayas comprise 7.2% of the total landmass of India and has vegetation like grassy meadows, moist mixed deciduous forests, alpine and sub-alpine forests and endangered species like **Himalayan Tahr, Musk Deer, Bharal, Ibex, Markhor.**

16.1.3 Desert

Aravalli hills is an extremely dry region and salty desert of Gujarat and Rajasthan. Deserts of India comprise about 6.9% of the country's landmass. Following are the desert zones in India:

1. Western Rajasthan desert
2. Gujarat desert

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3. Cold desert of Jammu and Kashmir and Himachal Pradesh.

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Deserts of India has diversified flora and fauna. They give shelter to endangered species like **Wolf**, **Caracal**, **Desert Cat** and **birds like Houbara Bustard** and **Great Indian Bustard**.

16.1.4 Semi-arid

These are the areas that lies between the desert of western India and the Deccan plateau and includes the Aravalli hill range. It comprises 15.6% of the country's total landmass. They are considered as a transition zone between the desert and forests of Western Ghats.

- Peninsular India has two regions that are considered semi-arid. These semi-arid regions have natural and artificial lakes and marshy lands.
- The grass and shrub layer in this zone helps to grow the highest wildlife biomass. Species like **Chital** and **Sambar** have habitat in moister valley areas and wooded hills respectively.
- Endangered species like **lion**, **caracal**, **jackal** and **wolf** are the features of this semi-arid region.

16.1.5 Western Ghats

They run along the western coast of India and they range from south Gujarat to south Kanyakumari. This mountain range covers 160,000 sq. km and **comprise 5.8% of total landmass of India**. Western ghats are blessed with biotic provinces and biomes.

Western ghat host tropical evergreen forest and represent one of the biodiversity hotspot of the world.

- Western ghat is endemic to many species of animal and they are also habitat of many vertebrate species of peninsular India.
- Endemic species to western ghats are **Grizzled Giant Squirrel**, **Malabar Civet**, **Nilgiri Tahr**, **Nilgiri Langur**, **Lion Tailed Macaque** and **Malabar Grey Hornbill. Cane turtle** and **Travancore Tortoise** are restricted in the central part of the western ghat.

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16.1.6 Deccan plateau

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It is covered by mountain ranges on three sides and lies south of the narmada valley. Deccan plateau is triangular in shape. In the north, lies the Satpura ranges, western ghats lies in the western part of the deccan plateau while the eastern ghats lies in the eastern side are found in the central and south-central part of the Deccan Plateau.

It comprises of 42% of the total landmass of India. It lies in the rain shadow effect region of western ghat and is mostly semi-arid in nature.

- Deccan plateau is one of the biogeographic zones of India and covers India's forests in states like **Maharashtra**, **Odisha and Madhya Pradesh**.
- Though **most of the forests are deciduous** in nature, various parts of the deccan plateau shows richness in biodiversity. These regions comprise **deciduous forests**, **thorny forests**, **scrubland** and supports animal diversity.
- Species that have habitat in the deccan plateau are **Barking deer, Gaur Elephant**, **Chital**, **Sambar, Nilgai** and **Chausingha** in Bihar, Odisha and Karnataka-Tamil Nadu belts. **Wild buffaloes** are found in parts of Odisha, Madhya Pradesh and Maharashtra. **Swamp Deer** are limited to parts of Madhya Pradesh.

16.1.7 Gangetic plain

These are considered as one of the most fertile plains of the world. These plains lie between the Himalayas and the tropic of cancer. These plains are formed as part of the river Ganges and are mostly homogenous in nature. They receive annual 60 cm rainfall. **It covers about 11% of the total landmass of the country.** Sundarbans are located in these gangetic plains.

Species diversity of the gangetic plains includes **Buffalo**, **Swamp Deer**, **Hog-Deer**, **Rhino**, **Elephant** and **Hispid Hare**.

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16.1.8 North-east India

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North-east India is considered as the biogeographical gateway for much of India's flora and fauna. **Eastern Himalayan biodiversity hotspot lies in the North-east India.**

North-east India shows richness in biodiversity and thus, many species are endemic to the region and many are restricted to Khasi hills.

This region is a transition zone between the Indian, Indo-Malayan and Indo-Chinese bio-geographical zones. It is a meeting point of the Himalayan mountains and peninsular India.

16.1.9 Islands

Andaman and Nicobar Islands lies along the Bay of Bengal and houses almost 300 islands, out of which only five are inhabited. Andaman and Nicobar constitutes 0.3% of the total landmass of India. It is one of the three tropical moist evergreen forest zones in India.

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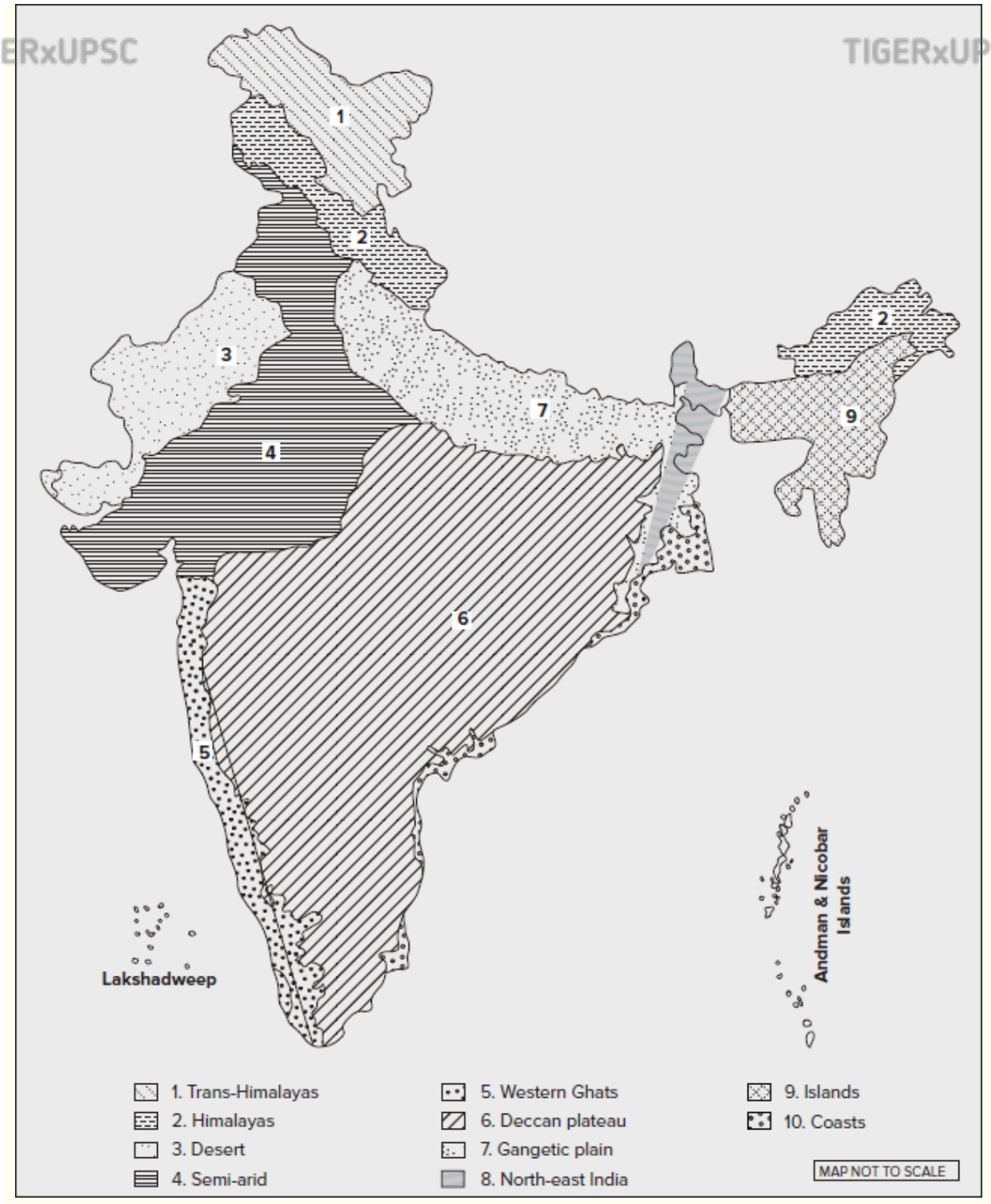

Figure 16.1 Biographical Classification of India: Zones

The islands house an array of flora and fauna that is not found elsewhere. These islands support endemism and a wide diversity of corals. In India,

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endemic island biodiversity is found only in the Andaman and Nicobar Islands. Some of the endemic fauna of Andaman & Nicobar islands include **South Andaman krait, Narcondam hornbill** etc.

16.1.10 Coasts

India has vast coastline of 7500 Km spreading from the west to eastern coasts.

Coastal region of India comprise about 5.2% of the total landmass of India. Coastal region has mangroves, mud flats, coral reefs, sandy beaches, marine angiosperm pastures.

Coastline from Gujrat to Sundarban are estimated to 5423 Km and islands of Lakshadweep makes it of 7500 Km. 25 islands of Lakshadweep are of coral origin and have rich reef lagoon system. Though the islands of Lakshadweep are populated, they lack natural vegetation.

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Natural Vegetation of India/Forests in India

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17.1 VARIOUS TYPES OF FORESTS IN INDIA

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The following types of forests are found in India which cover nearly 17% of the total area of Indian Territory.

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17.1.1 Tropical Forest

A great majority of the forests found in India are of this type. Tropical forests are of two types:

A. Tropical Moist Forests
B. Tropical Dry Forests

■ *Tropical Moist Forests*

These are further classified into the following types based on amount of rainfall they receive

- Tropical moist evergreen forests
- Tropical moist semi-evergreen forests
- Tropical moist deciduous forests
- Littoral and swamp forests

Tropical Moist Evergreen Forests

- These are also known as **tropical rain forests**. In India, such forests are found in wet regions receiving more than 250 cm average rainfall.
- These are climatic forests having luxuriantly growing lofty tress which are more than 45 metres in height. The shrubs, lianas (woody climber) and epiphytes are abundant because of high rainfall.
- These forests are found in Andaman & Nicobar Islands, Western coasts and parts of Karnataka (N. Canara), Annamalai hills (Koorj), Assam and Bengal. Mesophytes also belong here.

Tropical Moist Semi- Evergreen Forests

- These forests grow well where the annual rainfall is between 200 and 250 cm. These forest are found in western coasts of India, upper Assam, Eastern Odisha.

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- These forests are characterised by intermixing of deciduous and evergreen trees and shrubs. Various plants found in these forests are **Orchids**, **ferns**, **grasses** and **herbs**.

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Tropical Moist Deciduous Forests

- These type of forests receive annual rainfall of 100 to 200 cm. Dry season are short in nature. Trees of such type of forests shed leaves in hot summer season.
- These forests are found in western parts of the deccan plateau like North eastern Andhra Pradesh, Mumbai, Gangetic Plains, Punjab and Assam valley.
- The forests of Southern India are dominated by **Teak**. In the north, they are dominated by shal (Shorea robusta). These (shal) forest is considered as one of the valuable timbers of India. Grasses become important both in seral stages and in the areas under fire.

Littoral and Swamp Forests

Littoral and Swampy forests include the following types:

- Beach forests
- Tidal forests or Mangrove forests
- Fresh water swamp forests

Beach Forests

- These forests are found along the sea beaches and deltas of the river. Therefore, the soil is sandy in nature, poor in nitrogen and other nutrient while rich in lime and salts.
- Ground water found in these parts is brackish in nature and only have few meter depth and rainfall varies from 50–75 cm depending on the areas.

Tidal or Mangrove Forests

- Tidal forests grow near the estuaries or the deltas of rivers, swampy margins of islands and along the sea coasts. The soil is formed of silt,

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silt-loam or silt-clay and sand.

- These trees are halophytes in nature and have features like prop roots with advanced knees for support and pneumatophores and viviparous germination of seeds.

Tidal forests can be distinguished into the following four types with overlapping constituent species

- Tree Mangrove Forests
- Low Mangrove Forests
- Salt Water Forests
- Brackish Water Mangrove Forests

■ *Tree Mangrove Forests*

They occur on both east and west sea coasts. The best development occurs in Sundarbans. The forests floor is flooded with salt water daily. Plants may attain a height of 10–15 metres and form an almost closed evergreen forest.

■ *Low Mangrove Forests*

These forests grow on soft tidal mud near estuaries, which is flooded by salt water. Forest is dense but the trees with leathery leaves attain maximum height of 3–6 m. The vegetation consists of a few species which show gregarious growth habit. Besides, a common shrub ***Acanthus elicifolius*** and some grasses are also found at places. Low mangrove forests are more developed on east sea coast than on west coast.

■ *Salt Water Mangrove Forests*

These forests grow beyond mangrove forests in big river deltas where the ground is flooded with tidal water. Silt deposition and salt content in soil are low in these areas. Tree height is upto 20 m. Forests are dense, pneumatophores are common. The common plants are ***Heritiera minor*** and ***Nipa*** at places.

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Brackish Water Mangrove Forests

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They grow near the river deltas where forests floor is flooded with water at least for some time daily. Water is brackish (salty) but during rain, it is nearly fresh. Tree height may reach up to 30 m. Forest is dense. Common species of the forests are two palms; ***Nipa*** and ***Phoenix paludosa***.

Fresh Water Swamp Forests

These forests grow in low lying areas where rain or swollen river water is collected for some time. Ground water table is near surface.

Tropical Dry Forests

These are classified into the following types

- Tropical Dry Evergreen Forests
- Tropical Dry Deciduous Forests
- Tropical Thorn Forests

Tropical Dry Evergreen Forests

These forests are located in areas of abundant rainfall. However, dry season in these areas are relatively longer. Trees found are evergreen in nature and are dense and short in height (10–15 m). They are found in **western ghats**, **eastern ghats** and **Tamil Nadu**.

Tropical Dry Deciduous Forests

These forests grow in areas where annual rainfall ranges from 75 to 100 cm like in Punjab, Bihar, Odisha, MP, UP and parts of peninsular India. Tropical dry deciduous forests cover larger part of Indian forests. Dry season in these forests have leafless trees.

Height of the trees in these forests are 10–15 m and they are not dense. Northern part is dominated by **Sal** species of trees, while in south, it is **Teak** trees that are in abundance.

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Tropical Thorn Forests

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These forest lies in areas where rainfall is in between 20–70 cm annually and dry season is long and hot. These forests are found in Rajasthan, South Punjab and Gujarat. Vegetation grows along the river in these regions.

Areas away from the rivers and lacking irrigation have no trees and are mostly sandy in nature. Vegetation comprise of small trees (8–10 m), thorny bushes and shrubs. The trees in such forests have no leaves for most part of the year and therefore, called thorn scrubs or scrub jungles.

During rainy season, there is luxuriant growth of ephemeral herbs and grasses. Towards the desert region the vegetation diminishes and in arid parts there is almost no vegetation.

The Rajasthan desert can be divided into the following vegetation zones:

- Area of shifting sand dunes at and around Jaisalmer and Bikaner
- Area of established sand dunes near Jodhpur
- Sand stones rocks covered by xerophytic plants like Euphorbia
- Area of halophytic vegetation
- Sandy-loam soil vegetation

17.1.2 Subtropical Montane Forests

These forests are found in region of fairly high rainfall but where temperature difference between winter and summer are less marked. Winter generally goes without rains.

They are found up to the latitude of about 1500 metres in south and up to 188 metres in the north. In composition, subtropical forests are almost intermediate between tropical forests and temperate forests and a sharp demarcation can seldom be made between tropical and subtropical or subtropical and temperate forests.

These forests have been grouped into the following three types:

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- Wet Hill Board Leaved Forests

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- Dry Evergreen Forests
- Pine Forests

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■ *Wet Hill Board Leaved Forests*

They are found in Mahabaleshwar, Coorg, Karnataka, Parts of Assam, Panchmarhi and other Parts of M.P.

■ *Dry Evergreen Forests*

They occupy the foot-hills of the Himalayas. The common constituents of vegetation are **Acacia modesta**, **Olea cuspidate** etc.

■ *Pine Forests*

They are found mostly in western and central Himalayas and in Assam hills. The forests are dominated by species of **Pinus**, Species of **Quercus**, **Berberies**, **Carissa**, **Bauhinia** may also occur rarely in pine forests.

17.1.3 Temperate Montane Forests

These forests grow in the Himalayas at the altitude of 1800 to 3800 metres where humidity and temperature are comparatively low. Montane forests have been classified into the following three types on the basis of moisture regime:

I. Montane Wet Temperature Forests
II. Himalayan Moist Temperature Forest
III. Himalayan Dry Temperate Forests

■ *Montane Wet Temperature Forests*

- These are found in the Himalayas extending from Nepal to Assam at the altitude of 1800 to 3000 m, as well as in some parts of South India (Nilgiris). The forests in south are evergreen and are called **Sholas**. The forests are dense with closed canopy and the trees may be 18 to 20 m high.

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- **Epiphytes** are in abundance. Important plants constituting the vegetation in the Eastern Himalayas are species of Confiners. The members of family Compositae, Rubiaceae, Acanthaceae and Leguminosae form the under growth.

Himalayan Moist Temperature Forest

- These forests grow in the areas of lesser rainfall. The trees are high, sometimes up to 45 metres tall. The dominant elements of vegetation are oak and conifers. Undergrowth is shrubby and consists of deciduous species of **Barberis**, **Spiraea**, **Cotaneaster** etc.

Himalayan Dry Temperate Forests

- These forests, dominated by Rhododendrons, Oaks and Conifers from a narrow belt at the altitude from 3000 to 4000 m in the western Himalayas extending from a part of Uttarakhand through Himachal Pradesh and Punjab to Kashmir.
- The other commonly found species belong to genera **Daphne**, **Desmodium**, **Indigofera**, **Artemisia**, **Cannabis**, **Plectranthus**, **Fraxinus**, **several epiphytic mosses**, **Lichens** etc.

Exam Pointer

Taxus (Himalayan Yew)

Taxus (Himalayan Yew) is found in the Indian Himalayan region, including Himachal Pradesh, Uttarakhand, Sikkim, Jammu and Kashmir, Arunachal Pradesh, and parts of Northeast India. It thrives in temperate to sub-alpine forests, usually at elevations above 2,000 meters. It is highly valued for Taxol (paclitaxel), an important anti-cancer compound. Over-exploitation has seriously threatened its survival.

Key Regions & States

Western Himalayas: Himachal Pradesh, Uttarakhand (Jageshwar area), Jammu and Kashmir, and parts of West Bengal.

Eastern Himalayas: Arunachal Pradesh, Sikkim, Manipur, Meghalaya, and Nagaland.

Habitat: It occurs in temperate to high-altitude forests, particularly in mixed coniferous and oak-fir zones. It is mainly found at elevations above 2,000 meters.

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Significance

Medicinal: The bark, needles, and roots produce Taxol (paclitaxel), which is widely used in cancer treatment.

Traditional Use: It is used in Ayurvedic and Tibetan medicine; local communities use it for herbal tea and as fuel.

Conservation Status

Taxus wallichiana is classified as **endangered** due to unsustainable harvesting for medicinal purposes. This situation requires strict conservation measures.

17.1.4 Sub-alpine Forests

- The Sub-alpine forests are found throughout Himalayas from Ladakh to Arunachal Pradesh at the altitude from 2800 m to 3800 m. Annual rainfall is less than 65 cm, but snowfall occurs for several weeks in a year.
- Strong winds and below 0°C temperature prevails for greater part of the year. Trees are like those of temperate zone. **Epiphytic mosses** and **lichens** are in **abundance**.
  Two types of forests in sub-alpine zone:
  - Silver **Fir-Birch forests** that are found in glacial moraines.
  - **Birch-Rhododendron forests** that grow on rocky substrate. The common trees are ***Betula utilis, Quercus semecarpifolia,*** **many species of *Rhododendron pyrus* spp. Acer spp, Salix, Juniperus spp** etc.

17.1.5 Alpine Forests

Alpine vegetation has been classified into the following three types:

- Alpine Forests
- Moist Alpine Scrubs
- Dry Alpine Scrubs

Alpine Forests

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Plants growing at the altitude from 2900 to 6000 m are called alpine plants. In India, Alpine flora occurs in the Himalayas between 4500 and 6000 m. At lower level, alpine forests consists of dwarf tree with or without conifers and at higher level scrubs and only scattered xerophytic shrubs are left to merge with alpine meadows. The common plants of alpine forests are **Abies**, **Pinus**, **Juniperus**, **Betula**, **shrubby Rhododendrons**, **Quercus**, **Pyrus**, **Salix** etc.

■ *Moist Alpine Scrubs*

This type of vegetation is distributed extensively throughout the Himalayas above 3000 m. It is most often dense and composed of evergreen dwarf Rhododendron species, some birch and other deciduous trees. Ferns and mosses cover the ground with varying amounts of flowering herbs, alpine shrubs and ferns. Alpine pastures include mostly mesophytic herbs with very little grasses.

■ *Dry Alpine Scrubs*

These are open xerophytic formations spread in the UP, Himachal Pradesh, Punjab and Kashmir. Species of this develops general on lime stone rock.

17.2 GLOBAL FOREST RESOURCES ASSESSMENT (GFRA)

The Global Forest Resources Assessment is the only global assessment based on official national data. FRA classifies forests into two main types: **naturally regenerating and planted**. Within naturally regenerating forests, it identifies primary forests, which contain only ***native species***. In the planted forests category, it distinguishes plantation forests, such as ***rubber, from other planted forests*** that don't meet the criteria of a plantation.

India has made a notable leap in global forest statistics, moving up to 9th place worldwide for total forest area, according to the Food and Agriculture

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Organisation's Global Forest Resources Assessment 2025, released on October 22, 2025. In the last assessment, India held the 10th position. Additionally, India remains 3rd globally for the annual net gain in forest area. FAO is a United Nations agency focused on ending hunger and improving the management of natural resources, including forests. The GFRA is FAO's periodic evaluation of the world's forests, offering detailed information on forest area, changes, management, and use.

■ *GFRA 2025: India in Global Context*

- **Global Forest Cover:** The latest GFRA 2025 from the FAO states that the world's total forest area is about 4.14 billion hectares, which represents roughly 32% of the total land area or about 0.5 hectares of forest per person.
- India has **around 72,739 thousand hectares of forest**, making up approximately 2% of the global total.
- Europe has the largest forest area, accounting for 25% of the world's total. More than half of all forests globally (54%) are found in just five countries: ***Russia, Brazil, Canada, the USA, and China***.

India's Success in Expanding Planted Forests

- **Bamboo Plantations:** The worldwide bamboo resource is estimated at 30.1 million hectares, with 21.2 million hectares (70%) located in Asia; India has 11.8 million hectares. The area of bamboo forests worldwide grew by 8.05 million hectares between 1990 and 2025, mainly due to increases in China and India.
- **Rubber Plantations**: India ranks **5th globally, with 831 thousand** hectares of rubber plantations, contributing to a global total of 10.9 million hectares.

Agroforestry in India

- **Agroforestry Area**: India and Indonesia account for nearly 100% of Asia's agroforestry area, totalling about 39.3 million hectares.

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- **Global Contribution**: Together, India and Indonesia contribute around 70% of the global agroforestry area, roughly 55.4 million hectares.

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Deforestation and Net Changes

- India reported a **net gain in forest area from 1990 to 2025**, as expansion exceeded losses due to afforestation efforts. This resulted from decreased deforestation in some countries and increased forest area in others.
- India **contributes 9% of global wood removals, ranking 2nd globally as of 2023**.

India's Achievements

- India ranks **5th among the top carbon sinks worldwide**, with its forests removing **150 million tonnes of CO2 per year from 2021 to 2025.**
- Forest carbon removals in Asia, including India, reached **0.9 billion tons of CO_2 per year during 2021 to 2025**, with significant reductions in deforestation emissions.

India's Forest Status and Changes

- **Total Forest Cover:** According to the India State of Forest Report 2023, India's total forest cover is **715,343 square kilometres, which is 21.76%** of the country's geographical area.
- **Top States with Largest Forest Cover:** The three states with the largest forest areas are ***Madhya Pradesh (77,073 sq km), Arunachal Pradesh (65,882 sq km), and Chhattisgarh (55,812 sq km).***
- **Mangrove Cover:** India's mangrove cover is around 4,992 sq km, mainly concentrated in the Andaman & Nicobar Islands, Gujarat, Maharashtra, and West Bengal.
- **Biodiversity & Protected Areas:** India has 106 national parks, 573 wildlife sanctuaries, 115 conservation reserves, and 220 community reserves, protecting a wide variety of plants and animals.
- **Key Initiatives by the Government of India to Increase Forest Cover**

1. **Budgetary Allocations**

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- 2025–26 Budget: The MoEFCC allocated ₹3,412.82 crore, a 9% increase from the revised estimate of ₹3,125.96 crore in 2024–25.
- Revenue Expenditure: ₹3,276.82 crore (96% of the total allocation), representing an 8% increase.

2. **National Mission for a Green India (GIM)**

- **Launch and Objective:** Launched in February 2014 under the National Action Plan on Climate Change, GIM aims to increase forest and tree cover, restore ecosystems, and boost biodiversity and carbon sinks.
- **Coverage Targets:** The goal is to expand forest and tree cover by 5 million hectares and improve the quality of another 5 million hectares of forest and non-forest land.
- **Ecosystem and Livelihood Enhancement:** The mission also focuses on improving ecosystem services like biodiversity, water, and carbon storage while increasing the incomes of around 3 million families who depend on forests.

3. **National Afforestation Programme**

- **Objective:** To regenerate degraded forests and surrounding areas in the country.
- **Implementation**: This program operates through a three-tier structure involving State Forest Development Agencies, Forest Development Agencies at the division level, and Joint Forest Management Committees at the village level.

4. **Mission LiFE (Lifestyle for Environment)**

- **United Nations Environment Assembly Resolution:** A resolution was adopted on Sustainable Lifestyles based on the principles of Mission LiFE.
- **MeriLiFE Portal:** This was launched to promote individual and collective actions for sustainable living.
- **Ek Ped Ma Ke Naam Initiative:** This initiative aims to encourage tree planting by connecting it to affection for one’s mother or homeland.

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The forest cover data indicates that the world is making tangible strides in reducing deforestation in some countries while expanding forest areas in others. India's rise to 9th place globally for total forest area and its steady 3rd place in net annual gain show what a strong national commitment can achieve. India's ongoing efforts to expand forest cover, support sustainable forestry practices, and carry out initiatives like GIM highlight its dedication to environmental conservation and global climate action.

17.3 INDIA STATE OF FOREST REPORT 2023 - 2024 (ISFR 2023 - 2024)

The **Indian State of Forest Report (ISFR) is a biennial assessment released by the Forest Survey of India (FSI) under the Ministry of Environment, Forest and Climate Change (MoEFCC).** It provides a **detailed overview of forest cover and tree cover in India**. The report uses a **scientific approach that combines satellite-based remote sensing data**, supplied by NSRC, with ground-based field checks through the inventory of Forests and Trees Outside Forests (TOF) under the National Forest Inventory (NFI). **Satellite data indicate forest cover, while the NFI provides information on Growing Stock and Carbon Stock**.

In 2023, the analysis included 751 districts, an increase from 636 in ISFR 2021. Additionally, new data are presented for better analysis:

- Trees between 5-10 cm are now included in the analysis for Tree Cover and Growing Stock, as they significantly contribute to the paper and pulp industry.
- For the first time, the bamboo cover has been estimated and included in Tree Cover.
- Agroforestry has been analysed separately.
- Changes in Forest Cover in the Western Ghats are noted.
- Information on forest fires in 705 Protected areas is included.
- A national assessment of burnt areas is provided.

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- A separate chapter discusses changes over the last decade.

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Data collected are examined and compiled to offer a national view of forest resources. This information is also used to report country data to FAO for the Global Forest Resource Assessment (GFRA).

Do You Know?

The **ISFR 2023-2024** is the **18th edition** in this series.

17.3.1 Highlights of the Report

According to ISFR 2023, India's total forest and tree cover is 25.17% of the country's geographical area.

- Forest cover accounts for 21.76%, while tree cover makes up 3.41%.
- Compared to the 2021 assessment, total forest and tree cover has increased by 1,445 sq km.
- This includes an addition of 156 sq km in forest cover and a significant rise of 1,289 sq km in tree cover, showing stronger growth in trees outside recorded forest lands.

Table India's Total Forest and Tree Cover Geographical Area

Geographical Class	Percentage Area
Forest Cover	21.76
Tree Cover	3.41
Total Forest and Tree Cover	**25.17**
Scrub	1.33

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Geographical Class	Percentage Area
Non Forest	73.50
Geographical Area of the country	**100.00**

To understand this better:

1. **Forest Cover** refers to all lands larger than one hectare with a tree-canopy density of more than 10%, irrespective of ownership or legal status.

- Very Dense Forest (VDF) having canopy density equal to or more than 70%
- Moderately Dense Forest (MDF) having canopy density equal to or more than 40% but less than 70%, and
- Open Forest (OF) having canopy density equal to or more than 10% but less than 40%.

2. **Tree Cover** includes all tree patches outside recorded forest areas that are less than one hectare in size, such as scattered trees in rural landscapes, urban areas, farmlands, roadsides, and homesteads.
3. **Forest Area**: Also known as the Recorded Forest Area (RFA), it is defined as "all such lands which have been notified as forest under any Government Act or Rules or recorded as 'forests' in the Government Records".
4. **Scrub:** Forest lands having canopy density less than 10%, generally with Shrubs interspersed with trees.
5. **Trees Outside Forests:** It refers to all trees growing outside recorded forest areas, irrespective of patch size.

Do You Know?

The term 'forest area' denotes the legal status of the land as per the government records, whereas the term 'forest cover' indicates the presence of trees over any land.

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17.3.2 State Rankings in terms of forest and tree cover

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Increase in area

1. States Showing Maximum Increase in **Forest and Tree Cover** in descending order (Combined)

- Chhattisgarh
- Uttar Pradesh
- Odisha
- Rajasthan

2. Maximum Decrease in **Forest and Tree Cover** in descending order

- Madhya Pradesh
- Karnataka
- Ladakh
- Nagaland

3. States Showing Maximum Increase in **Forest Cover inside Recorded Forest Area (RFA)**

- Mizoram
- Odisha
- Karnataka
- West Bengal

4. Maximum decrease in Forest Cover inside RFA

- Tripura
- Telangana
- Assam
- Andhra Pradesh
- Gujarat

5. States showing Maximum Increase in **Forest cover outside the Recorded Forest Area**

- Gujarat
- Bihar
- Kerala
- Uttar Pradesh
- Assam

6. Maximum decrease in Forest Cover outside RFA

- Madhya Pradesh

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- Rajasthan
- Andhra Pradesh
- Arunachal Pradesh
- Maharashtra

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■ *Area Coverage*

1. States with the Largest Forest and Tree Cover (absolute area)

- Madhya Pradesh
- Arunachal Pradesh
- Maharashtra

2. States with the Largest Forest Cover (absolute area)

- Madhya Pradesh
- Arunachal Pradesh
- Chhattisgarh

3. States/UTs with Highest Forest Cover as a Percentage of Geographical Area

- Lakshadweep – 91.33%
- Mizoram – 85.34%
- Andaman & Nicobar Islands – 81.62%

4. Western Ghats Eco-Sensitive Area

- Forest cover - 73% of the geographical area
- Decadal change in forest cover- decreased by 58 sq. km

5. North-Eastern Region

- Forest and Tree cover- 67% of the geographical area
- Net Decrease in forest cover since 2021- 327.30 sq. km

6. Hills districts

- Forest cover- 40% of the geographical area
- Net increase in forest cover since 2021- 234.14 sq. km

17.3.3 Forest Cover Targets and National Policy Linkage

The report notes that 19 States and Union Territories have achieved **more than 33%** of their geographical area under forest cover.

This is significant because the National Forest Policy, 1988, sets a national ecological target of:

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- At least 33% forest/tree cover for the country, and
- At least 66% forest cover in hilly and mountainous regions

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Among these, eight States/UTs have more than 75% of their geographical area under forest cover, reflecting exceptional forest dominance:

- Mizoram
- Lakshadweep
- Andaman & Nicobar Islands
- Arunachal Pradesh
- Nagaland
- Meghalaya
- Tripura
- Manipur

■ *Mangrove Cover*

- Total cover: 4,991.68 km² (0.15% of India's geographical area).
- **Composition:**
  - Very Dense: 1,463.97 km² (29.33%)
  - Moderately Dense: 1,500.84 km² (30.07%)
  - Open Mangroves: 2,026.87 km² (40.60%)
- Net decline since 2021- 7.43 km².
- Major Decline- **Gujarat**
- Major increase- **Andhra Pradesh and Maharashtra**

■ *Forest Fires (2023-24)*

- Total fire hotspots declined as compared to 2021-22 and 2022-23
- Most affected states: **Uttarakhand, Odisha, Chhattisgarh**

■ *Growing Wood Stock*

- Highest forest growing stock: **Arunachal Pradesh, Uttarakhand, Chhattisgarh, Madhya Pradesh**

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- Dominant species
  - **Forests:** Sal, Teak, Chir Pine, Terminalia
  - **Trees outside Forests:** Mango, Neem, Mahua, Coconut
- Net increase since 2021: 4.25%

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Bamboo Resources

- **Largest bamboo areas**: Madhya Pradesh, Arunachal Pradesh, Maharashtra, Odisha
- **Highest increase:** Arunachal Pradesh
- **Highest decrease:** Karnataka

Agroforestry

- **Increase since 2013:** 20.02%
- **Highest stock:** Maharashtra, Karnataka, Odisha, Rajasthan
- **Shift in dominant species**: Increase in Neem, Prosopis, Eucalyptus, along with Mango.

Carbon Stock

- **Increase since last assessment**: 81.5 Mt
- Annual carbon sink addition: 40.75 Mt (≈ 149.42 Mt CO_2 equivalent)
- **Largest carbon pool: Soil Organic Carbon** (55.06%) followed by biomass, litter and dead wood.
- **Highest state carbon stock:** Arunachal Pradesh, Madhya Pradesh, Chhattisgarh, and Maharashtra.
- **Highest per-hectare carbon stock:** Jammu & Kashmir, Sikkim, Himachal Pradesh, and Andaman and Nicobar Islands.
- **India's total additional carbon sink achieved so far:** 2.29 billion tonnes CO_2 equivalent (towards NDC target since 2005)

As per **decadal trends,** except for the Western Ghats Eco-Sensitive area, there has been a net **increase in:**

- Forest Cover

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- Tree Cover
- Mangrove Cover
- Agroforestry tree cover
- Growing stock- forest, tree outside forest and agroforestry

17.3.4 Key changes in Forest Health and Ecosystem Conditions (2013–2023)

1. Soil health improved

- Shallow to deep soil increased from 83.53% to 87.16%
- Medium to deep humus increased from 11.43% to 18.04%
- Soil organic carbon increased **marginally** from 55.85 t/ha to 56.08 t/ha

2. Vegetative cover improved

- Dense to very dense grass cover increased from 15.64% to 17.21%
- Dense to very dense undergrowth increased from 20.32% to 25.58%

3. Grazing pressure reduced

- Moderate to heavy grazing declined from 41.04% to 35.79%
- Biotic pressure on forests reduced from 31.28% to 26.66%

4. **Mixed-size class forest areas increased** from 21.12% to 38.11%, indicating

- Improved floral diversity
- Better habitat conditions for faunal biodiversity

17.3.5 Disturbances in Forest Areas

1. Soil Erosion:

- The extent of soil erosion is assessed within a 60-metre radius around each sample plot.
- **As per the assessment of Recorded Forest Area (RFA):**
  - 16.84% of the forest area shows no soil erosion
  - 54.42% experiences mild erosion
  - 23.25% is under moderate to heavy erosion
- Moderate and heavy soil erosion have **increased** over the last decade, indicating rising pressure on forest ecosystems and the need for stronger conservation measures.

2. Invasive Species:

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- Forest Survey of India (FSI) has identified 45 major invasive species in a dedicated reference album
- Invasive species presence is observed within a 60 m radius around each plot
- Extent of invasive species occurrence in forest areas
  - 2.51% – Very dense infestation
  - 33.59% – Moderate to dense infestation
  - 18.92% – Scanty presence
  - 23.61% – Free from invasive species

Do You Know?

Comparable data for invasive species were not available for 2013, so long-term trend comparison is not possible.

3. **Illicit Felling:** It is revealed that 6.17% of the forest area is heavily influenced (effect of >25%) by illicit felling, 25% area experiences no illicit felling, and 21.42% of the forest area is affected by moderate (effect of 5-25%) illicit felling, highlighting continued anthropogenic pressure on forest resources.

4. **Girdling:**

- Girdling refers to the complete removal of a strip of bark around the entire circumference of a tree's branch or trunk, affecting vital tissues like the cambium and phloem.
- This practice reduces both the timber volume and timber quality available for harvest.
- During the National Forest Inventory (NFI), girdling was visually assessed within a 60 m radius around each plot centre.
  - 1.79% of the total forest area is **heavily affected** by girdling (more than 25% effect).
  - 30.48% of the forest area shows moderate to occasional girdling impact (effect of 5-25%).
  - 56.02% of forest area shows no injury due to girdling.

Do You Know?

No comparable data is available for 2013 for girdling impact to understand decadal change.

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5. **Lopping:**

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- Lopping refers to the cutting of branches, leaves, or parts of the trunk of living trees, mainly for fodder, fuel, or livelihood needs
- It supports livestock rearing and supplementary rural income, but requires careful regulation to avoid ecological damage.
- Proper forestry practices aim to balance local needs with forest sustainability.
- Forest area affected by lopping (as per assessment):
  - Heavily affected: 3.69% (>25% area affected)
  - Moderate to occasional lopping: 54.99% (0-25% area affected)
  - No lopping incidence: 29.61%

Do You Know?

No comparable data is available for 2013 for girdling impact to understand decadal change.

6. Biotic Influence:

- Biotic influence includes pressure from overpopulation of herbivores, invasive species, pathogens, grazing, fire, pollarding, illicit felling, and lopping.
- Excessive biotic pressure negatively affects forest health and biodiversity.
  - It leads to changes in forest composition and structure and allows invasive species to outcompete native vegetation.
  - Carbon sequestration capacity of forests is reduced due to degradation.
  - Increased runoff and sedimentation adversely impact rivers and streams, lowering water quality.
- Forest degradation status (% of forest area):
  - Highly degraded: 5.93% (>50% affected)
  - Moderately degraded: 20.73% (10-50% affected)
  - Mildly degraded: 53.40% (<10% affected)
  - Not degraded: 12.26%

The India State of Forest Report (ISFR) 2023 shows the country's strong commitment to systematic forest monitoring through remote sensing and field-based inventories. However, the report's broad definition of 'forest cover' includes plantations, orchards, and urban green spaces. This can result in an overestimation of actual natural forest ecosystems. Issues like **forest degradation, biodiversity loss, and canopy quality are not**

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fully addressed. This raises concerns about the accuracy and importance of the reported forest cover figures.

Plantation expansion, particularly monoculture plantations, is often counted as forest gain, even though it has limited ecological value compared to natural forests. The report does well in carbon stock accounting but is less effective in tracking wildlife habitat quality and ecological connectivity. Separate reporting for natural forests and plantations is needed to better guide conservation policies and climate actions.

19th ISFR (2025 - 2026)

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Floristic Regions of India

18

IN THIS CHAPTER...

18.1 Eastern Himalaya
18.2 Western Himalaya
18.3 Assam Region
18.4 Ganga Plain
18.5 Indus Plain
18.6 Deccan Region
18.7 Malabar Region
18.8 Andaman – Nicobar Region

The whole credit of highlighting the regional characteristics of the country's natural vegetation goes to Hooker and Thomson (1855).

Later on C.B. Clarke (1898) divided this part of Asia into six floristic regions:

1. Himalaya region
   (a) Eastern Himalaya
   (b) Western Himalaya
2. Arid region
3. Malabar region
4. Coromandel region
5. Ganga plain
6. Assam region

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In 1907, J.D. Hooker identified eight floristic regions in the British India. These are:

1. Eastern Himalaya
2. Western Himalaya
3. Indus plain
4. Ganga plain
5. Malabar region
6. Deccan region
7. Maldives and Sri Lanka
8. Burma (not in India)

In 1937, C.C. Calder identified six floristic regions in the country: (1) North-western Himalaya (2) Eastern Himalaya (3) Indus plain (4) Ganga plain (5) Deccan region (6) Malabar region.

D. Chatterji (1939), on the basis of certain indigenous plant species in various parts of India, divided India into eight main floristic regions.

18.1 EASTERN HIMALAYA

This region includes the hilly areas of West Bengal, Sikkim and Arunachal Pradesh that has heavy rainfall, low snow fall, high temperature and humidity. This area of undulating topography nurtures about 4000 species of plants which vary from tropical to temperate and Alpine with increasing altitude. Among the subtropical forests extending from Tarai plains to a height of 1524 m **Sal**, **Oak**, **Chestnut** are the main trees. Between 1524 m and 3657 m temperate broad leafed trees like **Quercus**, **Castanopsis**, **Magnolia**, **Pyrus** and **Symplous** dominate. Between 743 m and 3657 m lies a belt of conical forest with trees like **Abies**, **Picea**, **Symplocos**, **Eurya** etc. Beyond 3657 m of height lies a zone of Alpine forests with main trees like **Fir**, **Juniper**, **Pine**, **Birch** and **Rhododendrons**.

A brief comparison between Western and Eastern Himalayas is given below:

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1. Western Himalayas have higher peaks than Eastern Himalayas.
2. Eastern Himalayas are more earthquake prone than Western Himalayas.
3. The Eastern Himalayas receive more rainfall than Western Himalayas. Combined with high temperature than Western Himalayas, Eastern Himalays are much greener.
4. Range of temperature both annual and diurnal are higher in Western Himalays due to distance from sea.
5. It's much colder in Western Himalays owing to high lattitudes and altitudes. Therefore, snowline is at lower altitude in Western Himalayas.
6. Eastern Himalays are more abrupt than Western Himalayas from north to south while Western Himalayas are broader from south to north than Eastern Himalays.

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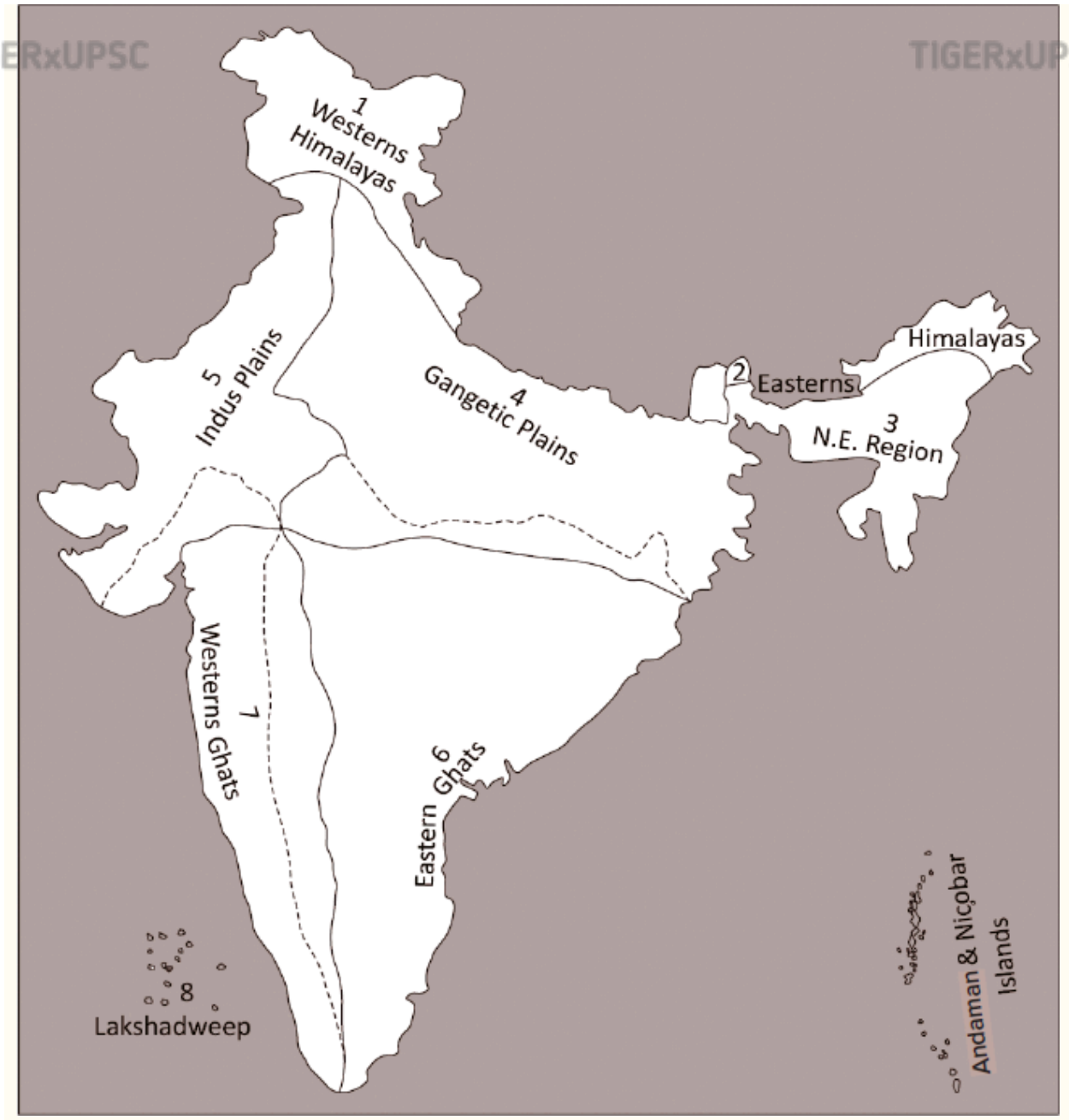

Figure 18.1 Floristic Regions of India

18.2 WESTERN HIMALAYA

The region covering Uttaranchal, Himachal Pradesh, J & K enjoys less rainfall and temperature than its eastern counterpart but the snow fall is heavier. Here again sub-tropical (upto 1524 m), temperate (1524 m to 3657 m) and Alpine (3657 m to 4572 m) vegetation types are noticed with the

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increasing height along the hill slopes. In the sub-montane region the main vegetation is sal, semul, and savanna type. Amongst temperate vegetation chirr, oak, deodar, alder, birch and conifers are important. Similarly Alpine vegetation between the altitude zones of 3657 to 4572 m is dominated by **Juniper**, **Silver**, **Fir**, **Birch** etc.

18.3 ASSAM REGION

The region includes the whole of the northeast including Assam, Meghalaya, Nagaland, Manipur, Tripura and Mizoram. Here vegetation depicts south-east Asian impact which maintains the Himalayan heterogeneity except the Alpine vegetation. The region is rich in various types of **Bamboos** and **Palms** with nilgiri type of **grasslands** at higher altitudes.

18.4 GANGA PLAIN

Being a distinct physiographic unit of country the region has typical floristic charactertics of its own. The vegetation type ranges between semi-arid shrubs (the Aravalli region) to evergreen mangroves (the Ganga delta). Although due to the population and growing agricultural and economic activities natural vegetation of the region has been wiped out but historical evidences sup dense vegetal cover with rich wild life. Sal and Terminalia are the representative species of the primordial vegetation.

The vegetation of Uttar Pradesh belongs to deciduous type which changes to moist deciduous type in eastern Bihar and West Bengal and torn grove type in the Ganga delta. Besides, there different types of grasses all over the region depending upon the local physical conditions.

18.5 INDUS PLAIN

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The floristic region comprises the plains of Haryana, Punjab, Rajasthan (west of the Aravallis kachh and northern-parts of Gujarat). In this region of scanty rainfall (less than 75 cm annually), shrubs and trees with thick bark predominate. Important trees include **Acacias**, **Euphorbias**, **Cacti**, **Phoenix**, **Khejra**, **Kanju** and **Pal** etc. During rains **short grasses** also spring up.

18.6 DECCAN REGION

This region covers major part of the peninsular India. Here **Palm trees (Camamus sps.)** are dominant. Areas characterised by granites and gniesses, support teak, sal trees while semi arid regions contain **thorny shrub**.

18.7 MALABAR REGION

This region stretches all along the west coast from Gulf of Cambay to Cape Comorin. Here vegetation type ranges from moist tropical evergreen to broadleaf mixed and monsoon deciduous type. The Nilgiri hills show temperate forests. The region also contains several species of plants with Malay origin.

18.8 ANDAMAN – NICOBAR REGION

The vegetation of this region has been greatly influenced by the Malayan and Burmese impact. It is of mainly tropical evergreen type with different varieties of palms and canes etc. In Lakshadweep islands, the vegetation is dominated by different varieties of shrubs and **coconut trees**.

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Questions Asked in UPSC Prelims Exam

UNIT VII

Unit - VII Prelims PYQs (2011 onwards)

1. Consider the following statements: **[UPSC (Pre) 2025]**

Statement I: The amount of dust particles in the atmosphere is more in subtropical and temperate areas than in equatorial and polar regions.

Statement II: Subtropical and temperate areas have less dry winds.

Which one of the following is correct in respect of the above statements?

(a) Both Statement I and Statement II are correct and Statement II explains Statement I

(b) Both Statement I and Statement II are correct but Statement II does not explain Statement I

(c) Statement I is correct but Statement II is not correct

(d) Statement I is not correct but Statement II is correct

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2. One of the following regions has the world's largest tropical peatland, which holds about three years' worth of global carbon emissions from fossil fuels, and the possible destruction of which can exert a detrimental effect on the global climate. Which one of the following denotes that region? **[UPSC (Pre) 2024]**

(a) Amazon Basin
(b) Congo Basin
(c) Kikori Basin
(d) Rio de la Plata Basin

3. Consider the following description: **[UPSC (Pre) 2024]**

1. Annual and daily range of temperatures is low.
2. Precipitation occurs throughout the year.
3. Precipitation varies between 50 cm - 250 cm.

What is this type of climate?

(a) Equatorial climate
(b) China-type climate
(c) Humid subtropical climate
(d) Marine West Coast climate

4. With reference to "Gucchi" sometimes mentioned in the news, consider the following statements: **[UPSC (Pre) – 2022]**

1. It is a fungus.
2. It grows in some Himalayan forest areas.
3. It is commercially cultivated in the Himalayan foothills of north-eastern India.

Which of the statement/s given above is/are correct?

(a) 1 only
(b) 3 only

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(c) 1 and 2
(d) 2 and 3

5. The vegetation of savannah consists of grassland with scattered small trees, but extensive areas have no trees. The forest development in such areas is generally kept in check by one or more or a combination of some conditions. Which of the following are such conditions? [UPSC (Pre) – 2021]

1. Burrowing animals and termites
2. Fire
3. Grazing herbivores
4. Seasonal rainfall
5. Soil properties

Select the correct answer using the codes given below.
(a) 1 and 2 (b) 4 and 5
(c) 2, 3 and 4 (d) 1, 3 and 5

6. "Leaf litter decomposes faster than in any other biome, and as a result, the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up into the canopy vicariously, by climbing the trees or growing as epiphytes, rooted on the upper branches of trees." This is most likely the description of: [UPSC (Pre) – 2021]

(a) Coniferous forest
(b) Dry deciduous forest
(c) Mangrove forest
(d) Tropical rain forest

7. The greatest diversity in plants and animals is characteristic of: [UPSC (Pre) – 2016, 1994]

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(a) Temperate deciduous forests
(b) Tropical moist forests
(c) Savanna
(d) Temperate forest with grasslands

8. In India, in which one of the following types of forests is teak a dominant tree species? **[UPSC (Pre) – 2015]**

(a) Tropical moist deciduous forest
(b) Tropical rain forest
(c) Tropical thorn scrub forest
(d) Temperate forest with grasslands

9. Consider the following states: **[UPSC (Pre) – 2015]**

1. Arunachal Pradesh
2. Himachal Pradesh
3. Mizoram

In which of the above states do 'Tropical Wet Evergreen Forests' occur?
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

10. Which one of the following regions of India has a combination of mangrove forest, evergreen forest and deciduous forest? **[UPSC (Pre) – 2015]**

(a) North Coastal Andhra Pradesh
(b) South-West Bengal
(c) Southern Saurashtra
(d) Andaman and Nicobar Islands

11. If you travel through the Himalayas, you are likely to see which of the following plants naturally growing there? **[UPSC (Pre) – 2014]**

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1. Oak
2. Rhododendron
3. Sandalwood

Select the correct answer using the codes given below:

(a) 1 and 2 only (b) 3 only

(c) 1 and 3 only (d) 1, 2 and 3

12. Which of the following is not temperate grass land? [UPSC (Pre) – 2014]

(a) Pampas (b) Veldt

(c) Downs (d) Savannah

13. Which of the following/s is/are unique characteristic/characteristics of equatorial forests? [UPSC (Pre) – 2013]

1. The presence of tall, closely set trees with crowns forming a continuous canopy.
2. Co-existence of a large number of species.
3. The presence of numerous varieties of epiphytes. Select the correct answer using the codes given below:

(a) 1 Only (b) 2 and 3

(c) 1 and 3 (d) All of these

14. Which of the following temperate grasslands are found in Eurasia?[UPPSC (Pre) – 2012]

(a) Downs (b) Veldt

(c) Steppes (d) Prairies

15. The Himalayan Range is very rich in species diversity. Which one among the following is the most appropriate reason for this phenomenon? [UPSC (Pre) – 2011]

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(a) It has a high rainfall that supports luxuriant vegetative growth.
(b) It is a confluence of different biogeographical zones.
(c) Exotic and invasive species have not been introduced in this region.
(d) It has less human interference.

16. Consider the following statements: **[UPSC (Pre) – 2011]**

1. Biodiversity is normally greater in the lower latitudes as compared to the higher latitudes.
2. Along the mountain gradients, biodiversity is normally greater in the lower latitudes as compared to the higher latitudes.

Which of the statement/s given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

17. Consider the following statements: **[UPSC (Pre) – 2010]**

1. The Taxus tree is naturally found in the Himalayas.
2. The Taxus tree is listed in the Red Data Book.
3. A drug called 'Taxol' is obtained from the Taxus tree, which is effective against Parkinson's disease.

Which of the statement/s given above is/are correct?
(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 1, 2 and 3

Answers

1. (c) 2. (b) 3. (a) 4. (c) 5. (c) 6. (d) 7. (b) 8. (a) 9. (c) 10. (d)
11. (a) 12. (d) 13. (d) 14. (c) 15. (b) 16. (a) 17. (b)

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Explanations

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1. Correct answer (c): EXPLANATION

Statement I: The amount of dust particles in the atmosphere is more in subtropical and temperate areas than in equatorial and polar regions. It is right

- Subtropical regions (like the Sahara, Thar) have arid and semi-arid areas with high dust.
- Temperate regions also have dust from dry soils and agriculture.
- Equatorial regions have heavy vegetation and rainfall, which reduces dust. Polar regions have limited dust sources.

Statement II: Subtropical and temperate areas have less dry winds. It is wrong

- Subtropical regions have dry winds (like the trade winds and continental winds) contributing to dust transport.
- Temperate regions also have dry winds in some areas.

Since Statement II is not correct, it cannot explain Statement I.

Answer: (c) Statement I is correct but Statement II is not correct

2. Correct answer (b): EXPLANATION

- **Tropical peatlands** are carbon-rich wetlands in the tropics, storing huge amounts of carbon.
- **The world's largest tropical peatland** is in **the Congo Basin** in Africa. It spans about 145,500 km². It stores an enormous amount of carbon—roughly equivalent to several years of global fossil fuel emissions.

Other options:

- **Amazon Basin:** Contains some peat, but not the largest.

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- **Kikori Basin:** Located in Papua New Guinea, smaller peat areas.
- **Rio de la Plata Basin:** Temperate region; not known for tropical peatlands.

3. Correct answer (a): EXPLANATION

- **Annual and daily range of temperatures is low:** This means temperatures are fairly constant throughout the year and even between day and night.
- **Precipitation occurs throughout the year:** No distinct dry season.
- **Precipitation varies between 50 cm – 250 cm:** Moderate to high rainfall.

Now, compa ring with the options:

a. Equatorial climate: High temperatures throughout the year, low temperature range, and rainfall throughout the year. **It is right**

b. China-type climate (Monsoon): Has hot summers, cold winters, and distinct wet/dry seasons. **It is wrong**

c. Humid subtropical climate: Hot summers, mild winters, rainfall mostly in summer. **It is wrong**

d. Marine West Coast climate: Mild temperatures, low annual temperature range, rainfall throughout the year, also partially matches, but rainfall is usually higher (150–300 cm) and temperature is milder due to oceanic influence. **It is partially right**

4. Correct answer (c): EXPLANATION

Gucchi, also called **Morel mushroom**, is a highly prized edible fungus.

- **It is a fungus. Statement (1) is right.**
- It naturally occurs in Himalayan regions of India. **Statement (2) is right**

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- Gucchi cannot be commercially cultivated; it grows **wild only** and is collected from forests. **Statement (3) is wrong**

5. Correct answer (c): EXPLANATION

1. Savannas are characterized by **grasslands with scattered trees**, and the factors that prevent dense forest growth are well known in ecology:
2. **Fire -** Frequent natural or human-induced fires kill young tree saplings but grasses regenerate quickly.
3. **Grazing herbivores -** Large herbivores (e.g., antelopes, buffaloes, elephants) continuously graze and prevent tree seedlings from growing.
4. **Seasonal rainfall -** Savannas occur in regions with **distinct wet and dry seasons**. The prolonged dry period limits tree growth but grasses survive due to deep root systems.

Why the others are NOT primary controlling factors?

1. **Burrowing animals and termites**
   - Termites and burrowers influence soil structure but are **not major agents** preventing forest formation.
5. **Soil properties**
   - Soil may influence vegetation type, but **savannas are primarily maintained by fire, grazing, and seasonal rainfall**, not just soil.

Therefore, the correct answer is: (c) 2, 3 and 4

6. Correct answer (d): EXPLANATION

Very fast decomposition of leaf litter - Warm and wet conditions cause **rapid microbial activity**, so the forest floor is almost bare.

Dense canopy - Multi-layered canopy with little light reaching the ground.

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Abundant climbers and epiphytes - Lianas (climbers) and epiphytes (e.g., orchids, bromeliads) are **characteristic of tropical rainforests**.

Why the other options do NOT fit:

- **Coniferous forest** → cold climate, slow decomposition, thick litter layer.
- **Dry deciduous forest** → long dry season, slower decay, fewer climbers and epiphytes.
- **Mangrove forest** → coastal wetlands, saline soils, not characterized by epiphytes and climbers.

Therefore, the biome described is: **Tropical rain forest**

7. Correct answer (b): EXPLANATION

Among all major biomes of the world, **tropical moist forests (tropical rainforests)** have the **highest plant and animal diversity** because they offer: Warm temperatures throughout the year, High and uniform rainfall, Dense, multilayered vegetation, Stable climate over long evolutionary periods This makes them the most species-rich ecosystems on Earth.

Other options (why not):

- **Temperate deciduous forests** → moderate diversity
- **Savanna** → grassland with scattered trees, lower diversity
- **Temperate forest with grasslands** → low to moderate diversity

Thus, the greatest biodiversity occurs in: **Tropical moist forests**

8. Correct answer (a): EXPLANATION

Correct Answer: (a) Tropical moist deciduous forest

Explanation:

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Teak (Tectona grandis) naturally grows in: **Tropical Moist Deciduous Forests -** Found in central and southern India (MP, Maharashtra, Odisha, Karnataka)

These forests have: Annual rainfall between **100–200 cm,** A dry season, during which **teak sheds its leaves**

Why not the others?

- **Tropical rain forest** → too wet; teak does not grow here.
- **Tropical thorn scrub forest** → too dry; acacia-type species dominate.
- **Temperate forest with grasslands** → teak is a tropical species; not found here.

Hence, teak is dominant in: **Tropical moist deciduous forests**

9. Correct answer (c): EXPLANATION

Correct Answer: (c) 1 and 3 only

Tropical Wet Evergreen Forests are found in regions with: Very high rainfall (> 250 cm), Warm and humid climate and Dense, multi-layered vegetation

1. Arunachal Pradesh — is right

- The **Eastern Himalayas** receive very high rainfall.
- Tropical wet evergreen forests occur in the foothills and lower elevations.

2. Himachal Pradesh — is wrong

- Himachal Pradesh mainly has **subtropical pine**, **moist temperate**, **dry temperate**, and **alpine** forests.
- It **does NOT** have tropical wet evergreen forests.

3. Mizoram — is right

- Receives high rainfall and has **lush tropical wet evergreen forests**, especially in southern and western parts.

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Therefore, Tropical Wet Evergreen Forests occur in: Arunachal Pradesh & Mizoram

10. Correct answer (d): EXPLANATION

Among the given options, **Andaman and Nicobar Islands** is the only region that clearly contains all three types of forests:

1. Mangrove Forests

Very well developed along the coast (e.g., Baratang, Middle Andaman). One of the **largest mangrove areas** outside mainland India.

2. Evergreen Forests

Tropical evergreen forests dominate the higher rainfall zones. Home to species like *Dipterocarpus*.

3. Deciduous Forests

Moist and dry deciduous forests are present in interior regions.

Why not others?

- **North Coastal Andhra Pradesh** → mangroves + deciduous, but **no evergreen**.
- **South-West Bengal** → mangroves (Sundarbans) + some deciduous, but **no evergreen**.
- **Southern Saurashtra** → dry forests only; **no mangroves or evergreen**.

11. Correct answer (a): EXPLANATION

Several oak species (e.g., *Quercus semecarpifolia*, *Quercus leucotrichophora*) grow in the **Western and Central Himalayas**. **Oak is right**

Rhododendron is Found abundantly in the **Eastern and Western Himalayas** (e.g., *Rhododendron arboreum* is the state flower of Uttarakhand). **Rhododendron is right**

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Sandalwood (*Santalum album*) is native to **peninsular India**, especially **Karnataka, Tamil Nadu, Kerala** — **not** the Himalayan region.

Therefore, the plants naturally growing in the Himalayas are Oak and Rhododendron only.

12. Correct answer (d): EXPLANATION

Grasslands of the world can be categorized as:

- **Temperate Grasslands (Mid-latitudes): Pampas** → South America, **Veld/Veldt** → South Africa, **Downs** → Australia, **Steppes** → Eurasia, **Prairies** → North America
- **Tropical Grasslands (Low latitudes): Savannah** → Africa, Australia, South America

Since **Savannah is a tropical grassland**, it is **not** a temperate grassland. Hence, the correct answer is **Savannah (d)**.

13. Correct answer (d): EXPLANATION

These forests have **multi-layered, dense canopies** blocking sunlight from reaching the ground. **Statement (1) is right** Equatorial forests have the **highest biodiversity on Earth**, with numerous plant and animal species. **Statement (2) is right** High humidity supports a large number of **epiphytes** like orchids, ferns, mosses. **Statement (3) is right**

14. Correct answer (c): EXPLANATION

Temperate grasslands of the world include:

- **Steppes → Eurasia (Europe + Central Asia)**
- Prairies → North America
- Pampas → South America
- Downs → Australia

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• Veld/Veldt → South Africa

Thus, **only Steppes are found in Eurasia**. **Option (c) is right**

15. Correct answer (b): EXPLANATION

It lies at the **junction of multiple biogeographical regions** — the **Palearctic**, **Indo-Malayan**, and **Afrotropical** realms. This confluence results in species from different evolutionary backgrounds coming together, increasing diversity. Hence, the confluence of different biogeographical zones is the most appropriate explanation. **Option (b) is correct**

Why others are not the best reason?

- **High rainfall** — applies mostly to Eastern Himalayas; not to entire range.
- **No exotic/invasive species** — not true; invasives like *Lantana* and *Eupatorium* are found.
- **Less human interference** — human activity exists; also not the main reason for high diversity.

16. Correct answer (a): EXPLANATION

- Lower latitude means near to equator and higher altitudes means towards polar regions. Biodiversity is highest near the equator and decreases toward the poles. **Statement (1) is right**
- A mountain gradient is the **measure of a slope's steepness.** Along **mountain slopes**, biodiversity typically **increases from the base to mid-elevations**, then declines at the top. Note: Mountain gradients refer to **altitude**, not latitude. **Statement (2) is wrong**

17. Correct answer (b): EXPLANATION

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- Taxus is also known as **Himalayan Yew**, the Taxus tree grows naturally in the temperate and sub-alpine regions of the Indian Himalayas, specifically in states like Himachal Pradesh, Uttarakhand, and Arunachal Pradesh. **Statement (1) is right**
- It also grows in the hills of the northeastern states of Meghalaya, Manipur, and Nagaland.
- It is endangered due to overexploitation for medicinal purposes and slow regeneration, hence listed in the Red Data Book. **Statement (2) is right**
- T**axol** is a potent anti-cancer drug obtained from the bark and needles of the Taxus tree, but it is used to treat cancers such as breast and ovarian cancer, not Parkinson's disease.

Statement (3) is wrong

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Unit III Biodiversity, Wildlife Conservation and Species Recovery Programme in India

Biodiversity and Conservation 19

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19.1 BIODIVERSITY

The term 'biodiversity' is used to express biological diversity. Biodiversity is defined as genetic variation within species and different types of ecosystem in an area. Biodiversity provides ecological, social and economic benefits to society. Human civilisation requires biological diversity and natural resources from diverse ecosystems to fulfil essential needs like water, air, food, fodder, etc.

What is Biodiversity?

- Scientists believe that about 50 million different species of flora and fauna exist in the world. However, only 8.7 million have been identified and known to us.
- Different species survive in different environments to which they are adapted. Life is found from mountains to sea trenches, from North Pole to the rain forests of Amazon. Life and its diversity exist in the biosphere which is basically a sum of the lithosphere, hydrosphere and atmosphere.
- It was **Walter Rosen** who coined the term 'Biodiversity' as a title in a seminar in 1986.
- Biodiversity is defined as genetic variation within species and different types of ecosystem in an area. In other words, biodiversity is the summation of various forms of life.

19.2 BIODIVERSITY OF INDIA

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Classification of Groups	Number	
Mammals	350	
Birds	1224	
Amphibians	197	
Reptiles	408	
Fishes	2546	
Flowering plants	15000	

India is gifted with a huge number of species and many of these species are endemic to the mountain ranges in India.

Many species in India trace their origin to Gondwana land, of which present India is a part. The movement of Gondwana land to Laurasia and the subsequent collision of the two land masses set an interchange of species.

However, natural calamities like volcanic eruptions and climate change that happened around 20 million years ago became the reason for the end of many species endemic to India. After that, mammals from Asia found their passage to India from the sides of the newly formed and emerging ranges of the Himalayas. Therefore, Indian species has 12.6% of mammals and 4.5% of birds which are endemic, in contrast with 45.8% of reptiles and 55.8% of amphibians.

Some of the endemic species of India are Lion-tailed Macaque and Nilgiri Tahr, that are found in Western Ghats.

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IUCN stated that India had 2.9% of globally threatened species. These included **Asiatic Lion, Nilgiri Tahr, Lion tailed Macque**. **Vulture was and is still categorised as critically endangered** in India because of the diclofenac drug, which enters the vulture digestive system and causes organ failure in vulture.

In the past few decades habitat fragmentation has posed a threat to the wildlife of India. Therefore, as a part of conservation efforts, a network of national parks and protected areas have been established and expanded over the years. **The first national park in India was established in 1936; and it was Jim Corbett National Park**.

India provided legal backing to conservation efforts of wildlife and their habitat by enacting **Wildlife Protection Act, 1972**. Specie-specific programmes like Project Tiger, Project Crocodile were also started. India, today, has over 573 wildlife sanctuaries, 18 biosphere reserves; 96 wetlands of Ramsar sites.

Constitutional Provisions

- Priority is given to conservation efforts in India, but sometimes this priority is questioned as India has problems like poverty and unemployment.
- However, constitutional provision under **Article**-48 states that, the State shall endeavour to protect and improve the environment and to safeguard the forests and wildlife of the country and **Article 51-A** provides that it shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures.

19.3 TYPES OF BIODIVERSITY

Biodiversity is classified under three categories: 1. Species Diversity 2. Genetic Diversity 3. Ecological Diversity

19.3.1 Species Diversity

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- Species is defined as a group of the same organisms that breed among each other and their offspring have same common origin. It is considered as a basic unit of classification among different organisms.
- Species diversity, on the other hand, is defined as the occurrence of different species and variety of organisms within the same species in any given area.
- Species diversity ranges from micro bacteria to giant animals, from algae to trees of Amazon. Plants species include bryophytes, pteridophytes, gymnosperms, angiosperms while animal species include micro-organisms to mammals like elephants and blue whales.
- Species diversity depends on climatic factors like moderate temperature, adequate rainfall, availability of natural resources like nutrients and sunlight. Therefore, some regions of the world have high species diversity than other areas. Also, the population within particular species will be high if the available resources are abundant.
- Polar region shows less diversity in species and less population within specie because climatic conditions are not favourable. While, on the other hand, tropical regions have adequate rainfall, sunlight and therefore, contain huge species diversity in terms of plant diversity in the tropical rainforest to different animal species.
- Hotspots of biodiversity are those parts of the world which have high species diversity.

19.3.2 Genetic Diversity

- Variation in the genetic material of organisms within a species is termed as genetic diversity.
- Each member of the specie has different genetic composition than other members.
- Every member of the specie has particular characteristics that are individualistic in nature due to its specific individual genetic composition. Simply, each individual has unique genes which make him different than others.

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- Therefore, although belonging to the same homo sapiens species, every human being is still different from each other.
- In a similar way, plants have the same varieties like rice has 1 lakh different varieties in the world. There are different varieties of mangoes, which are of different size, shape and taste due to differences in genetic material.
- The term, gene pool represents a collection of different gene diversity and the occurrence of different species in an area. Gene pool includes a variety of genes within species and the different species in an area. However, there has been intermixing within species that give rise to different species over millions of years.
- Domesticated animals like dogs and domesticated plants like wheat crops have traced their origin to the wild gene pool.
- Importance of genetic variability is being debated. Inbreeding within species having the same genetic material can lead to extinction and therefore, genetic variability is considered important for healthy offsprings. Although, this theory has not been conclusively proven.
- *Biotechnology* : It is defined as the manipulation of the genetic material of different species in order to evolve better organisms with desirable characteristics like genetically modified plants and animals. It is also called Recombinant DNA technology.

19.3.3 Ecological/Ecosystem Diversity

- Ecological Diversity is defined as different plant and animal species cohabitating an area and is interlinked by nutrient exchange cycles and energy flow in various ecosystems.
- It contains variations within the same species as well as variations in different plant, animal and microorganism species.
- Therefore, ecological diversity contributes to abundance of flora, fauna and microorganism in an ecosystem.
- Biosphere, in terms of its richness of different life forms, depends on variation in ecosystem.

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- There are various ecosystems in Earth like forests, deserts, grasslands, marine, freshwater, swamp, marshlands, wetland, etc. which differ in terms of flora, fauna and microbial togetherness.
- Ecological diversity indicates a complex network of different species, their interaction and interconnectedness among each other and with the environment.
- Ecological diversity developed over millions of years through the interaction and interconnectedness among different species.

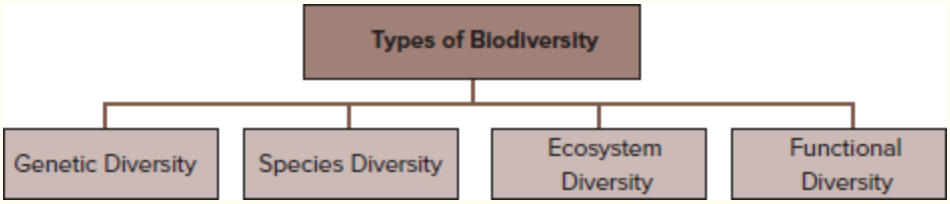

Figure 19.1 Types of Biodiversity

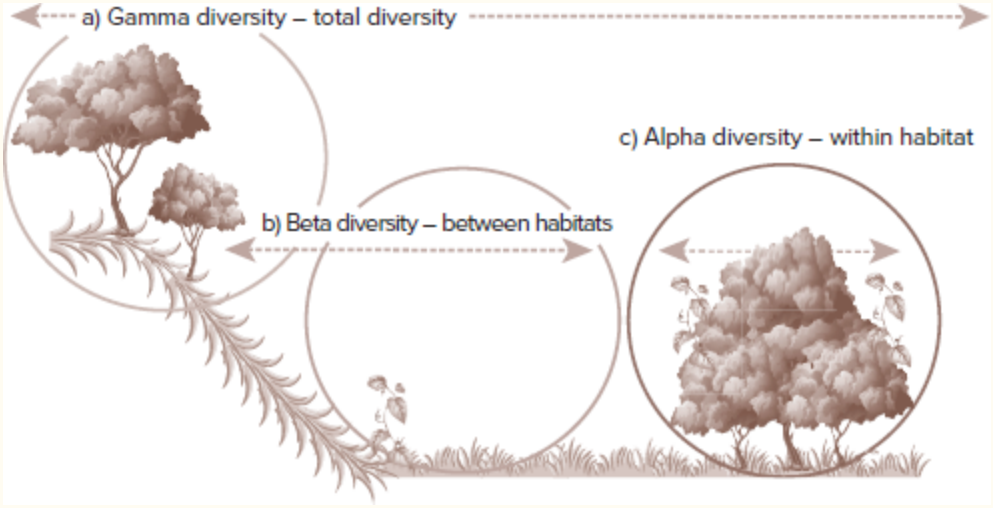

Figure 19.2 Pictorial View of Alpha, Beta and Gamma Diversity

Alpha diversity implies '**species richness**', beta diversity refers to '**susceptibility to changes**' and gamma diversity means '**mobility over larger areas**'.

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Do You Know?

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Do You Know?

Genetic diversity: It is assessed at the level of species.

Species diversity: Assessed at the level of a particular ecosystem.

Ecosystem diversity: It is assessed at continents and global levels.

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The community of living organisms in an ecosystem depends on the availability of biotic resources; like lake ecosystem is different from an ocean ecosystem depending on nutrients and availability of sunlight.

19.4 MEASURING BIODIVERSITY

Species number and their richness are assessed on the basis of geographical factors. These are the following:

19.4.1 Alpha Diversity

Assess the species number in a community and compare it with the number of species of another community.

19.4.2 Beta Diversity

Change in species diversity is assessed based on the changes in the environment.

19.4.3 Gamma Diversity

Estimate the degree to which species with the same habitat expand from one geographical region to other.

19.5 VALUE OF BIODIVERSITY

19.5.1 Social Value

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Activities like owning a garden, developing flower parks, fishing, cattle and animal domesticating, agriculture etc. have social values for human beings. For example owning elephants is a status symbol in many societies, whereas having too many agricultural fields may be a status symbol in other societies. Owning such activities add to an enhanced social value of the owner.

19.5.2 Ethical Value

Many people use species for their monetary benefits; there are many others who try to protect them. This is so on account of the ethical values in society, after all, we, the human beings, being the most intelligent species on Earth, cannot ignore our duty to keep all forms of life on Earth.

Similar to religion, ethical values are also part of our culture. Therefore to preserve these values, conservation of biodiversity is essential. This can be achieved by boycotting the misuse of animals, plants, flowers, birds etc.

19.5.3 Aesthetic Value

Nature has provided different varieties of species; many are versatile in different aspects. Some are beautiful, some are most beautiful. Their uniqueness imparts their aesthetic value. Common examples of such species are:

- Birds like peacocks and butterflies.
- Flowers like roses and dehaliya.

19.5.4 Option Value

Some species have great potential to exhibit their utility in the future. They are superior to many existing species in different aspects such as nutrition value. Such utility value is called option value. The following examples are illustrious in this regard:

(i) **Soyabean whose nutrition value is quite high.**

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(ii) **Sugar tulsi** (*stevia ribaudiana*) which is about 25 times sweeter than sugar.

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19.5.5 Spiritual Value

Plants like tulsi, peepal, sandal and anwala which besides having medicinal values also have spiritual value.

19.5.6 National Value

Species of national importance like:

- Peacock: National Bird
- Lotus: National Flower
- Tiger: National Animal
- Gangetic Dolphin: National Aquatic Animal
- Elephant: National Heritage Animal

19.6 BIODIVERSITY AT GLOBAL, NATIONAL AND LOCAL LEVELS

19.6.1 Global level

The latest estimated total species on Earth is 8 million as per UNEP. Species on land are 6.5 million and 2.5 million in the ocean depths.

19.6.2 National and Local Level

India has identified and described over 108,276 species of bacteria, fungi, plants and animals out of these, 21.2% constitutes fungi, 13.9% flowering plants and 49.3% insects. These species occur on land, fresh and marine waters, or occur as symbionts in a mutualistic or parasitic state with other organisms.

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19.6.3 Comparison of Biodiversity in Different Regions

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The biodiversity in different geographical regions varies widely. A comparison in this regard is given below: Tropical regions are the most (species) rich environments on Earth. Although they cover much less area on Earth surface (about 7%) they hold about 90% of species, including insects.

19.6.4 India as a Mega-Diversity Nation

India has very rich biodiversity at all three levels viz. genes, species and ecosystem.

The salient features of India's biodiversity may be enumerated as follows:

In terms of plant richness, India comes at 10th place in the world and 4th in Asia, 11th in terms of the number of endemic species of higher vertebrates like amphibians, birds and mammals and 10th in the world as far as richness in mammals is concerned. India hosts four biodiversity hotspots of the world. These are **Eastern Himalaya, North East India, Western Ghats and Andaman and Nicobar Islands, Sundaland.**

The crops which first grew in India and spread throughout the world include **rice, sugarcane, asiatic vignas, jute, mango, citrus, banana, several species of millets, spices, medicinal, aromatic and ornamental**. India ranks sixth among the centres of diversity and origin in terms of agro-biodiversity.

India is blessed with huge species diversity and most of the species are endemic to mountains like the Himalayas and the Western Ghats.

19.7 PROTECTED AREA NETWORK OF INDIA

The National Wildlife Database Cell of Wildlife Institute of India (WII) has been developing a National Wildlife Information System (NWIS) on the Protected Areas of the country. India has over 1000 Protected Areas including 106 National Parks, 573 Wildlife Sanctuaries, 115 Conservation

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Reserves and 220 Community Reserves covering a total of 175169.42 km^2 of geographical area of the country which is approximately 5.32% (till January 2026).

Following areas in India is considered as part of Protected areas as per standards of IUCN:

1. **National Parks:** There are 106 existing national parks in India covering an area of 44,402.95 km^2, which is 1.35% of the geographical area of the country (National Wildlife Database, Jan. 2026).
2. **Wildlife Sanctuaries:** There are 573 existing wildlife sanctuaries in India covering an area of 122,564.86 km^2, which is 3.73% of the geographical area of the country (National Wildlife Database, Jan. 2026).
3. **Biosphere Reserves:** Accorded with IUCN Category V. Biosphere reserves are sites established by countries and recognised under UNESCO's Man and the Biosphere (MAB) Programme to promote sustainable development based on local community efforts and sound science.
4. **Reserved and Protected Forests:** Reserved forests and protected forests accorded with IUCN category V and VI as per the protection level.
5. **Conservation and Community Reserves:** Conservation reserve: There are 115 conservation reserves in India.

Community reserve: There are 220 community reserves in India. These reserves acted as buffer zone around protected areas.

6. **Village** and Panchayat Forests: These are forests land that is supervised by village panchayat.
7. **Private Protected Areas:** These lands are administered and owned by private players. These lands are not owned by government.
8. **Conservation areas:** These are geographical areas where conservation activities are undergoing.

Facts Corner

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- There are 10 bio-geographic regions.
- India has 18 Biosphere reserves, of which 13 are part of UNESCO Man and Biosphere Programme (Till Jan 2026).
- There are 96 Ramsar Wetland Sites (Till Jan 2026).
- There are 3 biomes viz. **Tropical dry deciduous forests, tropical humid forest, semi-deserts/warm forest.**
- There are 4 hotspots

Threats to Biodiversity and Ecosystem

Threats to biodiversity and ecosystem can be broadly grouped as Nature and Anthropogenic (man-made).

Natural threats include calamities like cyclone, earthquake, landslide, tsunami etc.

Anthropogenic threats include:

- Global sea levels are rising as a result of human-caused global warming, with recent rates being unprecedented over the past 2,500-plus years.
- Habitat destruction-on land; in streams, rivers, and lakes; and in the oceans. Human activities such as deforestation, bottom trawling in the oceans, the damming and dredging of streams, rivers, lakes and the draining and degradation of wetlands, estuaries, and mangroves are responsible.
- Over-harvesting of plant and animal species as well as overhunting. Lots of fish species are known to have reached their maximum exploitation level, and others will soon be at the same level.
- Introduction of non-native species by humans can affect the ecosystem/biodiversity if these species prey on vulnerable and native members of that ecosystem. One devastating impact of introducing alien species, Nile Perch and Nile Tilapia into Lake Victoria in the 1970s was the extinction of almost half of the 350+ endemic species of fish in the cichlid family.
- Many types of human-caused pollution are a threat- the release of excessive amounts of nitrates and phosphates from sewage and agricultural run-off: Persistent organic pollutants that can concentrate in food webs (and in our own tissues) and adversely affect hormonal and reproductive function: Pharmaceuticals used by people and in livestock production that are toxic to wildlife; acid rain, heavy metals, herbicides, pesticides and plastics.
- Excessive ultraviolet radiation from depletion of the stratospheric ozone layer that can damage the DNA and proteins of land-based, freshwater, and marine organisms.
- War and conflict that can result in habitat destruction. For example- conflict in Kashmir is causing loss of biodiversity.
- Climate change: Climate Change may significantly affect the temperature, rainfalls, water table, extreme events like cyclones, heat waves etc. that pose a threat to the ecosystem and biodiversity.

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Some examples of biodiversity and ecosystem destruction in India:

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In recent years, India has witnessed intensified weather patterns, with the monsoon season becoming more erratic and less dependable over time. Recently, while various factors contributed to flooding in various parts of India, experts point to climate change as a significant driver of increased occurrences of heavy rainfall.

- Construction of roads, buildings in Chennai and Bangalore led to the destruction of lakes, wetlands etc. leading to habitat destruction of many organism as well as distorted ecology.
- Illegal sand mining in Maharashtra, Madhya Pradesh, and Goa is destroying beaches and riverbeds.
- Sardar Sarovar Dam on Narmada River: Recent decision to raise the height of the dam from 122 m to 138.5 m led to the submergence of around 10 lakh trees.
- Use of diclofenac for veterinary purposes led to the killing of vultures who fed on carcasses of animals having diclofenac.
- Pollution due to industrial waste has led to the creation of toxic hotspots in many parts of India. For example- Vapi and Bharuch in Gujarat, Udyogamandal near Kochi etc. Human society is becoming increasingly aware of the importance of biodiversity and its role in sustaining life on Earth. As a result, efforts are being made to conserve the biodiversity at various levels- legal, economic, political, international, technological as well as individual.

19.8 IUCN CATEGORIES

IUCN was founded in October 1948 as the International Union for the Protection of Nature (or IUPN) following an international conference in Fontainebleau, France. It was renamed as International Union for Conservation of Nature and Natural Resources in 1956 with the acronym IUCN.

- IUCN is the world's first global environmental organisation. Today it is the largest professional global conservation network.
- The Union's HQ is located in Gland, near Geneva, in Switzerland.
- It demonstrates how biodiversity is fundamental to addressing some of the world's greatest challenges such as climate change, sustainable development and food security.
- The IUCN Red List is set upon precise criteria to evaluate the extinction risk of thousands of species and subspecies. These criteria are relevant to all species and all regions of the world. The aim is to convey the urgency

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of conservation issues to the public and policymakers, and help the international community to reduce species extinction.

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Species are classified by the IUCN Red List into nine groups, set through criteria such as rate of decline, population size, area of geographic distribution, and degree of population and distribution fragmentation.

- **Extinct (EX)** – Last individual of a species has died.
- **Extinct in wild** – Due to habitat loss, members of species have been known to survive in captivity and outside their natural habitat.
- **Critically endangered (CR)** – There is a very high risk of extinction for these categories of species.
- **Endangered (EN)** – There is a high risk of extinction but relatively less than critically endangered species.
- **Vulnerable (VU)** – These category species are not in endangered categories but they will fall into endangered categories if necessary steps are not taken.
- **Near threatened (NT)** – These categories of species are not threatened yet but still require conservation efforts to check population decline.
- **Least concern (LC)** – These categories of species are least concerned.
- **Data deficient (DD)** – Adequate data is unavailable for such species.
- **Not evaluated (NE)** – Evaluation of such species has not been done.

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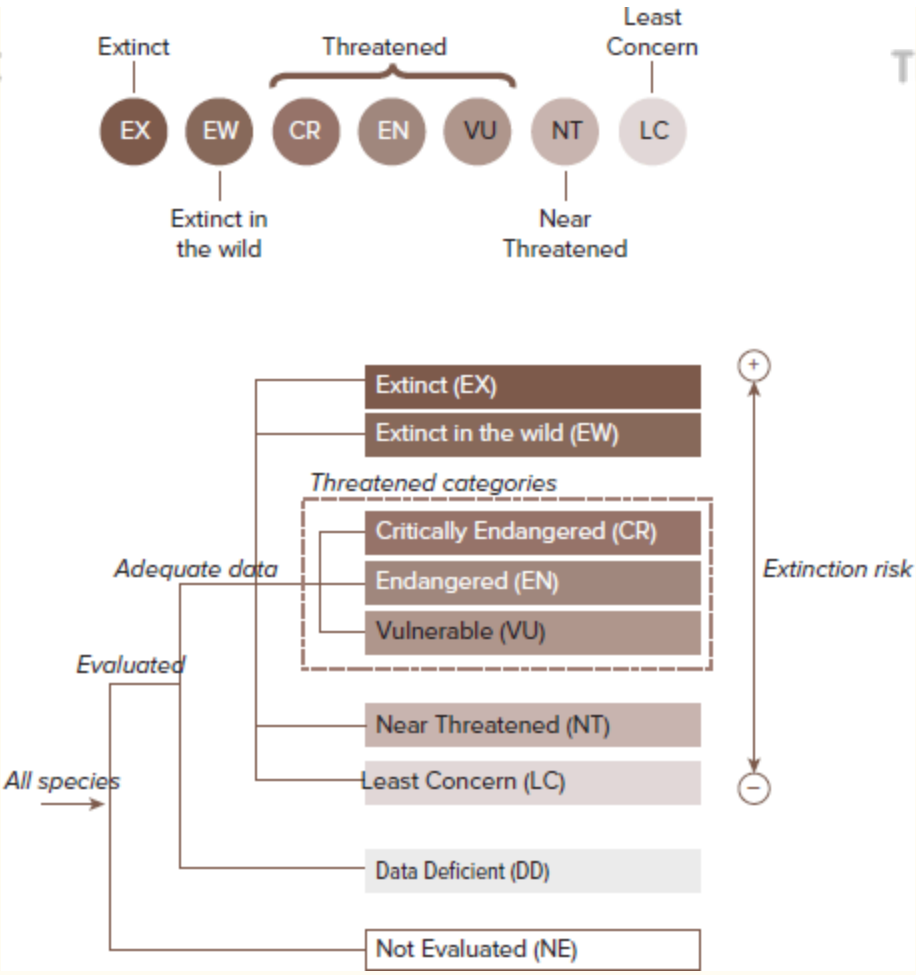

Figure 19.3 IUCN Categories

The term "**threatened**" is used for three categories, that is, critically endangered, endangered and vulnerable in IUCN Red List.

Apart from Red List of IUCN, many more specialised lists exist like blue list and yellow list.

Red List: It contains species which are evaluated and their existence is threatened unless the factors responsible for their depletion are not restricted and controlled.

Blue List: Species which are not threatened yet, but their existence is in a vulnerable category and will become endangered if they are not protected.

Yellow List: These species are not at risk.

IUCN Green Status of Species

- The IUCN Green Status classifies species into nine species recovery categories, indicating the extent to which species are depleted or

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recovered compared to their historical population levels.

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- It provides a fuller picture of species' conservation status including both their extinction risk and recovery progress.
- Each Green Status Assessment measures:
  1. Impact of past conservation on a species
  2. A species' dependence on continuing support
  3. How much a species stands to gain from conservation action within the next ten years.
  4. The potential for it to recover over the next century.
- The IUCN Green Status of species will be integrated into the IUCN Red List of Threatened Species.

19.9 CONSERVATION OF BIODIVERSITY

Conservation means "the management of natural resources, ecosystems and biodiversity etc. of the biosphere; so that the maximum benefits may be derived by the present generation, and substantial potential may be preserved for the future generation."

ENV 28- Common Carbon metric (CCM)

Objective: The specific objectives to conserve our natural resources are the following:

1. To maintain ecological balance.
2. To allow the continuance of natural processes.
3. To safeguard the life-supporting systems.

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4. To preserve the biodiversity and minimise/avoid the extinction of species.
5. To utilise the natural resources in a sustainable way.

■ *Biodiversity Treaty*

This is one of the strategies to protect species and ecosystems, which was introduced in June 1992. A total of 196 countries have ratified the treaty. India was one among them.

Some of its features are given below:

1. Every state has a sovereign right to its biological resources.
2. Transfer of biotechnology, genetic and other related technologies from developed countries to developing countries should be done at favourable terms.
3. Benefits out of utilisation of traditional practices, skills, knowledge should be shared by all the stakeholders.

■ *In-Situ and Ex-Situ Conservation Methods*

Wildlife protection is the most important activity in conservation efforts and therefore conservation efforts are concerned about conserving animals and plants species in a controlled environment. These controlled environments are national parks, wildlife sanctuaries and botanical gardens. Following two approaches/methods are being used for conserving the species and they are:

1. In-situ conservation
2. Ex-situ conservation

In *in-situ approach*, only wild flora and fauna are conserved while in ex-situ conservation approach endangered or rare species are conserved.

19.9.1 In-Situ Conservation

In-situ conservation is the on-site conservation of genetic resources in natural populations of plant or animal species.

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In **In-situ conservation**, protection is given to wild fauna and flora. In-situ conservation emphasises the protection of the ecosystems of the original habitats or natural environment. The priority in this approach is to save the species and their original habitat in the protected areas. These protected areas are managed through state or other effective agencies. The national parks, wild life sanctuaries, and biosphere reserves are included in the protected areas.

Advantages: There are various advantages of in-situ conservation:

1. Advantageous for long-term protection. It provides for a wide coverage of areas as well as species.
2. A number of unknown species that are endangered are also under care.
3. It relatively costs less to conserve species in their original habitat.
4. By conserving specific species of biodiversity, it can also utilise for future economic benefits.
5. It also ensures conservation of fruits and vegetables, medicinal plants etc.

Limitations: In-situ conservation also has some limitations:

1. No demographic certainty in this method.
2. On the basis of various random events, this conservation have a lot of survival and reproduction problems.
3. Environmental uncertainty.
4. Mining, road construction, pouching, livestock grazing etc. are also dangerous in this area.

■ *National Parks*

- They are specially designated areas for conserving the wildlife. Human activities like mining, grazing, agriculture, etc. are restricted. These national parks are regulated by state laws.
- It is associated with the habitat of wild animal species like rhino, tiger, lion etc.
- This boundary is under legislation.

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- Disturbance only in a limited area (only in buffer zone).
- Tourism is allowed.
- No scientific management so genetic research is lacking.

ENV 28- Common Carbon metric (CCM)

■ *Wildlife Sanctuaries*

- These are protected areas concerned with protecting particular species like tiger reserves, crocodile sanctuaries, bird sanctuaries, etc.
- Limited human activities are allowed, like the collection of minor forest products, harvesting, etc. Private ownership is allowed.
- It is species oriented as pitcher plant, Great Indian Bustard etc.
- This boundary is not under legislation.
- Limited disturbances.
- Tourism is allowed here.
- No scientific management, so genetic research is lacking.

Many projects/programmes projects are being carried out in India to prevent the extinction of some endangered wild animal species like Project Tiger, Vulture recovery Programme, Project Crocodile, Project Elephant etc.

■ *Biosphere Reserves*

Biosphere reserves are protected areas in which protection is granted not only to flora and fauna but also to the human communities inhabiting these protected areas. Conservation of animals and plants is being done in these biosphere reserves.

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India has 18 biospheres reserves, out of which 12 are designated under UNESCOs, Man and Biosphere (MAB) Reserve Programme. These biospheres are regulated by state governments.

- It takes into consideration the entire ecosystem.
- Its boundary is defined under legislation.
- Disturbance is limited only in buffer zone.
- Tourism is generally not allowed.
- Scientific management and genetic research is allowed.

Various countries have biosphere reserves within their boundaries and most of them are connected through World Network of Biosphere Reserves (WNBR). WNBR also acts as a platform to share experiences, exchange best practices and also provide policy inputs.

■ Biosphere Reserve

The concept of biosphere reserves originated from the Biosphere conference held in 1968 and organised by UNESCO. This conference was probably the first step towards integrating conservation efforts and proper utilisation of natural resources.

The result of this conference was the launch of **Man and Biosphere (MAB) Programme of UNESCO in 1970**.

Initial projects of MAB included establishing a well-coordinated network of sites representing regions where flora and fauna are protected, research activities are undergoing and training work is carried out. These regions/sites are termed as “biosphere reserves” under MAB Programme.

■ Objectives of Biosphere Reserves (BRs)

Each has three basic objectives which are mutually enforcing and complementary to each other. These activities are:

1. **Conservation:** BRs have a role in conserving the species, their habitat, and overall ecosystem.

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2. **Development:** BRs are also meant to enhance economic and human development in order to ensure sustainable development. Humans residing in these BRs are also protected.
3. **Research and Development:** BRs have a role in contributing to research, monitoring, education, exchange of information and best practices between countries. Training of locals and experts is also done in these BRs.

■ *International Status of Biosphere Reserves (BRs)*

The term "Biosphere Reserve" is used by UNESCO for natural areas where growth/development and conservation activities go hand in hand. The Central Government of a country nominates biosphere reserves that meet certain international criteria and standards to the world network of Biosphere Reserve under Man and Biosphere Programme. Worldwide as of Jan 2026, there are 784 biosphere reserves in 142 countries, including 23 transboundary sites.

Table 19.1 Biosphere Reserves in India

S.No.	Name	Sta
1.	**Nilgiri Biosphere Reserve**	T.
2.	**Nanda Devi National Park and Biosphere Reserve**	
3.	**Gulf of Mannar Biodiversity Reserve**	
4.	**Nokrek Biosphere Reserve**	
5.	**Sundarbans Biosphere Reserve**	
6.	Manas Biosphere Reserve	
7.	**Simlipal Biosphere Reserve**	

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Note: Out of these 18 BRs, **13** Biosphere Reserve are part of MAB Programme of UNESCO. These are Nilgiri, Gulf of Mannar, Sundarbans, Nanda Devi, Nokrek, Parchmarhi, Simlipal, Great Nicobar, Achanakmar-Amarkantak, Cold Desert, Agasthyamalai, Khangchendzonga, and Panna.

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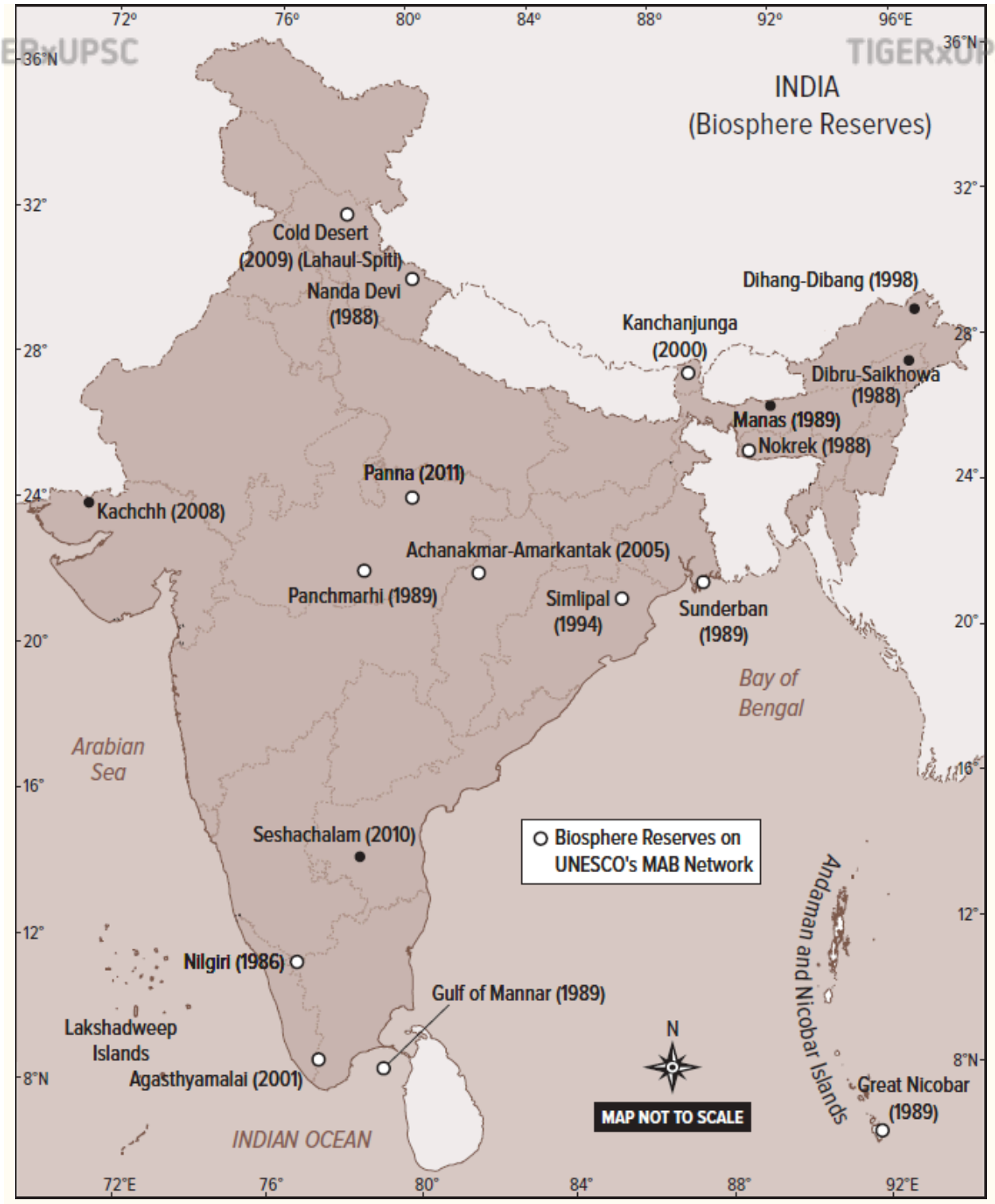

Figure 19.4 Biosphere Reserves

Source: Wildlife Institute of India

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Structure and Function of Biosphere Reserves

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Structures of BRs is demarcated into three zones:

Core Zones: It is a very important and sensitive part of BR. It has a habitat of various plants and animal endemic species. Core areas have an important function of conserving the threatened, economically important species and also their habitat. Special species having scientist interests are also conserved in this part of BRs. In India, Wildlife Protection Zone 1972 forms the basis for the regulation of BRs. Human activities are limited and restricted to training and learning activities.

Buffer Zone: This zone acts as a buffer between the transition zone and core area. This area surrounds the core area and is therefore managed in such ways that pressure on the core area is reduced and preservation of species is not hampered. Human activities like demonstrations, tourist activities, recreations, collection of minor products, fishing, grazing, etc. are allowed. Research activities are promoted in this zone of BRs. Although, human activities are only allowed to a level, that does not disturb ecological stability.

Transition Zone: This zone is the least regulated among three zones and is the outermost part and most accessible to the local community. Conservation activities are planned with the participation of local people and local knowledge. Human activities are allowed like collection of minor forest products, habitat for locals, farming, tourism, recreation, etc.

Three main functions of BR (Conservation, Development and Research)

- To protect the diversity of flora and fauna within their natural habitat.
- To protect the genetic diversity of species in an ecosystem.
- Socio-Economic development of local people through benefit sharing arising out of the use of natural resources and traditional knowledge.
- Research and development activities in these areas.
- Provide training in these areas.

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Management of BRs

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Central Government provides full 100% grants under Biosphere Reserve Scheme to State governments and UTs to implement Management Action Plans.

The Scheme broadly has the following activities:

- Value addition.
- Effective utilisation of threatened species.
- Rehabilitation of endangered species for their protection.
- Socio-economic development of local people.
- Properly maintain corridor areas.
- Establishing proper communication network.
- Promotion of Eco-tourism.

Biosphere Reserve Scheme not only focuses on endangered flora and flora but also local people by providing alternate livelihood and supplementing local people's rights in buffer and transition zones. This makes BR scheme different than other conservation schemes.

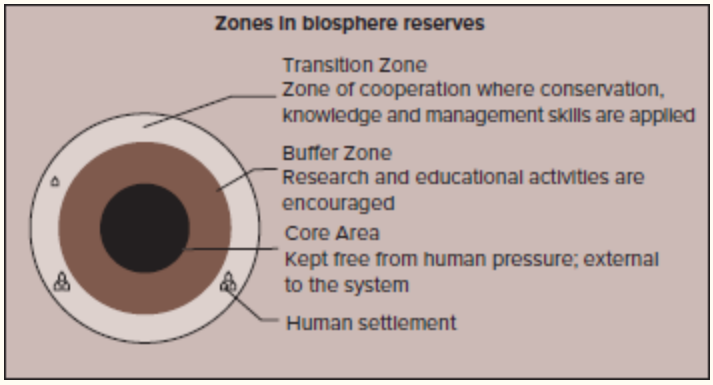

Figure 19.5 Biosphere Reserve Zone

19.9.2 Ex-Situ Conservation

In this approach of conservation, protection to endangered species is given outside their natural habitat in human-made habitats especially made for the purpose of conservation. To execute the process of protection, the

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endangered plant species are collected together and bred under desired conditions in seed banks or botanical gardens.

It is literally an 'off site' conservation policy that involves a couple of techniques linking the transfer of an objective species, experiencing various threats, away from its native habitat to a much safer abode, like in a Botanical Garden, Zoological Garden, Seed Bank or Gene Bank etc.

■ Advantages

1. It is a long term conservation.
2. Due to controlled supervision, assured food, better shelter and security, the species can survive longer and may breed more offspring than usual.
3. The captive breeding (it is a means of conserving species that may not survive in the wild) can be over exploited to yield an increased number of offspring which may later be sent back to their natural ecosystems.

■ Limitations/Disadvantages

1. This technology is costly and requires a high level of technology.
2. Not a viable option for the protection of rare species due to human interference.
3. The species is susceptible to pests and diseases.
4. It is done only for limited species.
5. This may give an undue advantage of overprotection to a few species.
6. Possible extinction of species is also possible due to breeding in a changing environment and demographic variation.

■ Botanical Garden

- Ex-situ Conservation is one of the primary objectives of Botanical Survey of India (BSI).
- A botanical garden or botanic garden is a garden dedicated to collecting, cultivating, preserving and displaying a wide range of plants labelled with their botanical names.

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- It may contain specialist plant collections such as cacti and other succulent plants, herb gardens, plants from particular parts of the world, and so on. There may be greenhouses, shadehouses, again with special collections such as tropical plants, alpine plants, or other exotic plants.

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Zoo

- Zoos are primarily dry facilities where animals are confined within enclosures and displayed to the public, and in which they may also be bred.
- The oldest zoo in India is Alipore Zoological Gardens, Kolkata. The Central Zoo Authority of India (CZA) is the Governing Authority of all zoos in India.

Central Zoo Authority

- It is an associate member of the World Association of Zoos and Aquariums (WAZA).
- The World Association of Zoos and Aquariums (WAZA) is the “umbrella” organisation for the world zoo and aquarium community. Its mission is to provide leadership and support for zoos, aquariums, and partner organisations of the world in animal care and welfare, conservation of biodiversity, environmental education and global sustainability.
- The Central Zoo Authority has been constituted under the section 38A of Wild Life (Protection) Act 1972.
- The Authority consists of a chairman, ten members and a Member Secretary.
- The main objective of the Authority is to complement the national effort in conservation of wildlife. Standards and norms for housing, upkeep, health care and overall animal management in zoos have been laid down under the Recognition of Zoo Rules, 1992.
- Every zoo in the country is required to obtain recognition from the Authority for its operation.
- The Authority evaluates the zoos with reference to the parameters prescribed under the rules and grants recognition accordingly.

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- Zoos which have no potential to come up to the prescribed standards and norms may be refused recognition and asked to close down.
- The Authority's role is more of a facilitator than a regulator. Therefore, it provides technical and financial assistance to such zoos which have the potential to attain the desired standard in animal management.
- Central Zoo Authority also regulates the exchange of animals of endangered category listed under Schedule-I and II of the Wild Life Protection Act among zoos. Exchange of animals between Indian and foreign zoos is also approved by the Authority before the requisite clearances under EXIM Policy and the CITES permits are issued by the competent authority.

Seed Bank

- A seed bank is a kind of gene bank that stores seeds of food crops or any rare species of plants.
- Seed banks protect against the extinction of plant species due to natural catastrophes, outbreak of diseases and even deliberate non-use for a long time.
- Seed banks make our diminishing biodiversity available to future generations.
- Seeds in a seed bank are dried to suitably low moisture content and stored at 18°C or below.
- The bank also documents the plan's identity, sampling location, seed quantity, farming systems in which the crop was grown etc.
- Svalbard Global Seed Vault is a facility on a remote island in the Arctic Ocean and it houses the world's largest collection of seeds. The seeds can be used in the event of global catastrophe or when some species are lost due to natural disasters and hence it is called the doomsday vault.
- India's seed vault is located at Chang La, Ladakh in the Himalayas which stores over 5000 seed accessions (One accession consists of a set of seeds of one species collected from different locations or different populations).
- The vault is a joint venture of National Bureau of Plant Genetic Resources (which comes under Indian Council of Agriculture Research) and the

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Defense Institute of High-Altitude Research (under Defense Research and Development Organisation)

Advantages of Seed Banks

- Ease of storage.
- Economy of Space: Can store a very large amount of plant genetic material in a small space.
- Relatively low labour demands.

19.10 BIODIVERSITY HOTSPOTS

These are areas where species diversity is very high and most of the species are endemic. Also, these regions of the world are threatened due to habitat destruction and other human activities.

ENV 28- Common Carbon metric (CCM)

Species are the basic unit of Earth's life form and every life form is interdependent on each other. Species diversity provides ecological stability in the ecosystem. However, biodiversity is threatened by human activities like urbanisation, pollution, global warming, etc.

Table 19.2 Comparative study among National Park, Sanctuary and Biosphere Reserve

National Park	Sanctuary	

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Habitat for particular wild animal species	Generally species-oriented such as citrus, pitcher	Ec life
The general size range is 0.04 to 3162 sq. km.	The general size range is 0.61 to 7818 sq. km.	Th 56
Boundaries fixed by legislation	Boundaries are not sacrosanct	Bo
Except the buffer zone, no biotic interference	Limited biotic interference	Ex int
Tourism Permissible	Tourism Permissible	To

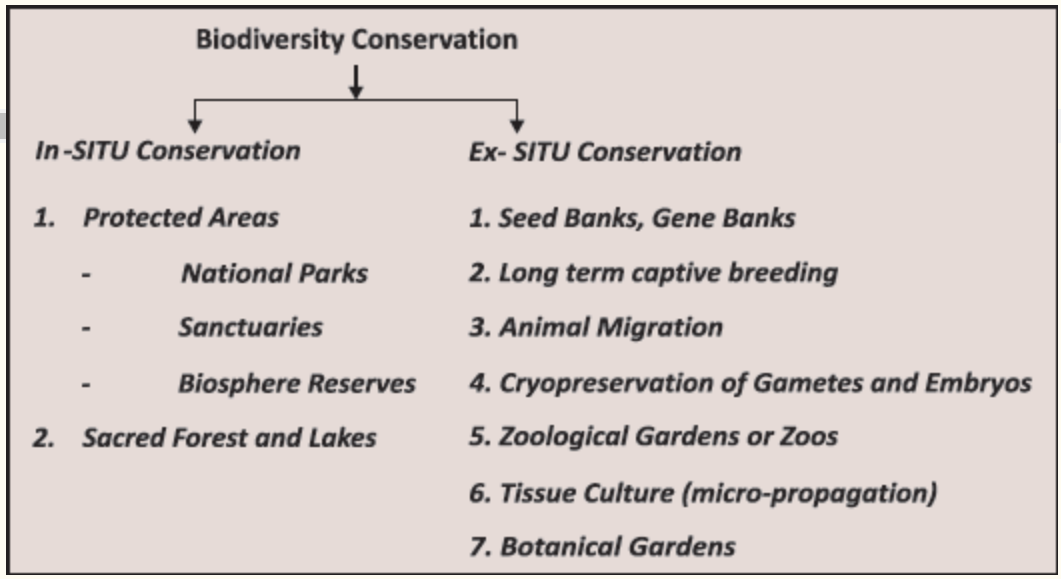

Figure 19.6 Different examples under Biodiversity Conservation

The extinction rate of species in our time is more than the mass extinction rate in the dinosaur era.

Mass extinction of species in our era is actually more than the dinosaur era. In order to avert this crisis, habitat protection of biodiversity is needed. However, biodiversity is not evenly distributed but certain areas have endemic species which are not anywhere else.

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Most of these endemic species are highly endangered due to habitat fragmentation and human activities. These areas which have endemic species are called biodiversity hotspots. Forests and other habitats represent 2.5 of Earth's land area.

19.10.1 Criteria for Becoming Biodiversity Hotspot

There are two criteria for becoming biodiversity hotspots:

- 1500 vascular plants must be endemic to the region. It means such plants are only found in these regions.
- 70% of natural vegetation must be lost.

19.10.2 List of Biodiversity Hotspots

- **North and Central America:** California Floristic Provinces, Madrean pine-oak woodlands, Mesoamerica.
- **The Caribbean:** Caribbean Islands
- **South America:** Atlantic Forest, Cerrado, Chilean Winter Rainfall-Valdivian Forests, Tumbes-Choco-Magdalena, Tropical Andes.
- **Europe:** Mediterranean Basin
- **Africa:** Cape Floristic Region, Coastal Forests of Eastern Africa, Eastern Aformontane, Guinean Forest of West Africa, Horn of Africa, Madagascar and the Indian Ocean Islands, Maputaland-Ponteland-Albany, Succulent Karoo
- **Central Asia:** Mountains of Central Asia
- **South Asia:** Eastern Himalayas, Nepal, Indo- Burma, Myanmar, Western Ghats, India, Sri Lanka
- **South East Asia and Asia-Pacific:** East Melanesian Islands; New Zealand, Philippines, Polynesia-Micronesia, Southwest Australia, Sundarland, Wallacean
- **East Asia:** Japan; Mountains of southwest China
- **West Asia:** Caucasus; Iran-Antolian Biodiversity hotspots are spread over 2.4% of land surface of Earth. They inhabit 50% of endemic plants and

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43% of endemic mammals, birds, reptiles and amphibian species.

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19.10.3 Criticism of Biodiversity Hotspot

Biodiversity Hotspots has following criticism:

- Species other than endemic species are not adequately represented.
- Other than vascular plants, plant taxonomy is not adequately represented.
- Small scale richness of hotspots is not adequately protected.
- Biodiversity hotspots do not allow alteration in land use patterns. Some hotspots have experienced high habitat loss but may not necessarily experience a continuation of such loss in the future. However, some hotspots like Amazon Basin which have experienced less habitat loss yet, but the rate of habitat loss is high. These factors are not considered while determining hotspots.
- They do not protect services that are provided by the ecosystem.
- Phylogenetic diversity has not been considered.

Phylogenetic diversity: It supports the objective of biodiversity conservation to maintain variation in living beings.

■ *Why Biodiversity Hotspots Matter*

- Biodiversity represents all life forms on Earth; species diversity is the reason for the availability of food, air and water. Without biodiversity, there would be no human existence possible and therefore, in regions where biodiversity is threatened, biodiversity hotspots are critical for human survival.
- Biodiversity regions are known for being the most beneficial to the local communities residing there because they are the richest ecosystems and also habitat to most vulnerable human populations of the world. These people are dependent on nature for their survival. Biodiversity hotspots cover 2.3% of the land of the world, forest, wetlands and other ecosystems, however, they represent 35% of ecosystem services on which vulnerable people depend upon.

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19.10.4 Biodiversity Hotspot in India

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India has four regions which satisfy the criterion for being classified as biodiversity hotspots. They are:

■ *The Western Ghats and Sri Lanka*

- Western Ghats are the ranges in western peninsular India. They are close to the Arabian Sea and therefore receive high rainfall due to the orographic effect. The Western Ghats have moist deciduous and rain forests. The region has high endemic species and species diversity.
- 62% reptiles and 77% amphibians are endemic to WGs.
- Sri Lanka also shows species diversity and it has a connection with India through past glacial events. Although the cause of occurrence of species richness of Western Ghats is still unknown but Sri Lanka has bio-geographic similarities with the WG region.

Biodiversity

- Western Ghat has over 6000 vascular plants, of which 2500 are endemic.
- World spices like black pepper and cardamom owe their origin to this region.
- Agsthyamali hills in the southern Western Ghat show the highest richness in species.
- Western Ghats are the habitat of 450 birds species, 140 mammals, 260 reptiles and 175 amphibians.
- About 60% of amphibians and reptiles are purely endemic to the region. **Western Ghat is a natural habitat for Great Indian Hornbill**.
- Although Western Ghat has species richness, but today, they are threatened. The vegetation cover has been reduced from 1.9 lakh km to 43 thousand km in Sri Lanka of the original forest cover; only 1.5% cover remains.

19.10.4.2 The Eastern Himalayas

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- Eastern Himalayas spread in Bhutan, North East India, and South Nepal.

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- The region is young in geographical scale and has high altitude differences. The Himalayas is home to the world's highest peaks, like Mount
  Everest.
- About 40 million years ago, the Indian tectonic plate moving towards the Eurasian Plate forms the basis for the formation of the Himalayas.
- The hypothesis is that the sea bed of that time uplifted due to the collision of tectonic plates and it is shown by the fact that the summit of the Himalayas is made of Marine Limestone.

Biodiversity

- Eastern Himalayas is one of the biodiversity hotspots and is habitat to nearly 163 endangered species, including one-horned rhinoceros, wild asian water buffalo. The list of threatened species includes 50 birds, 17 reptiles, 12 amphibians, 3 invertebrate, 45 mammals and 36 plant species.
- Relict Dragonfly found in the eastern Himalayas is an endangered species and other species of this genus found in Japan.
- They provide habitat to Himalayan Newt, the unique salamander species found in India.
- Overall, Himalayas are home to 10 thousand species of plants, of which one third is endemic to the eastern Himalayas.
- The Himalayas has 300 species of mammals, of which about 12 are endemic to the eastern Himalayas.
- Mammals of Himalayas include langurs, asiatic wild dogs, black bear, blue sheep, takin, golden langur, the Himalayan tahr, the pygmy hog, gangetic dolphin, wild water buffalo, swamp deer call, sloth bears, gaurs, muntjac, sambar, snow leopard.
- The only critically endangered species is namdapha flying squirrel.

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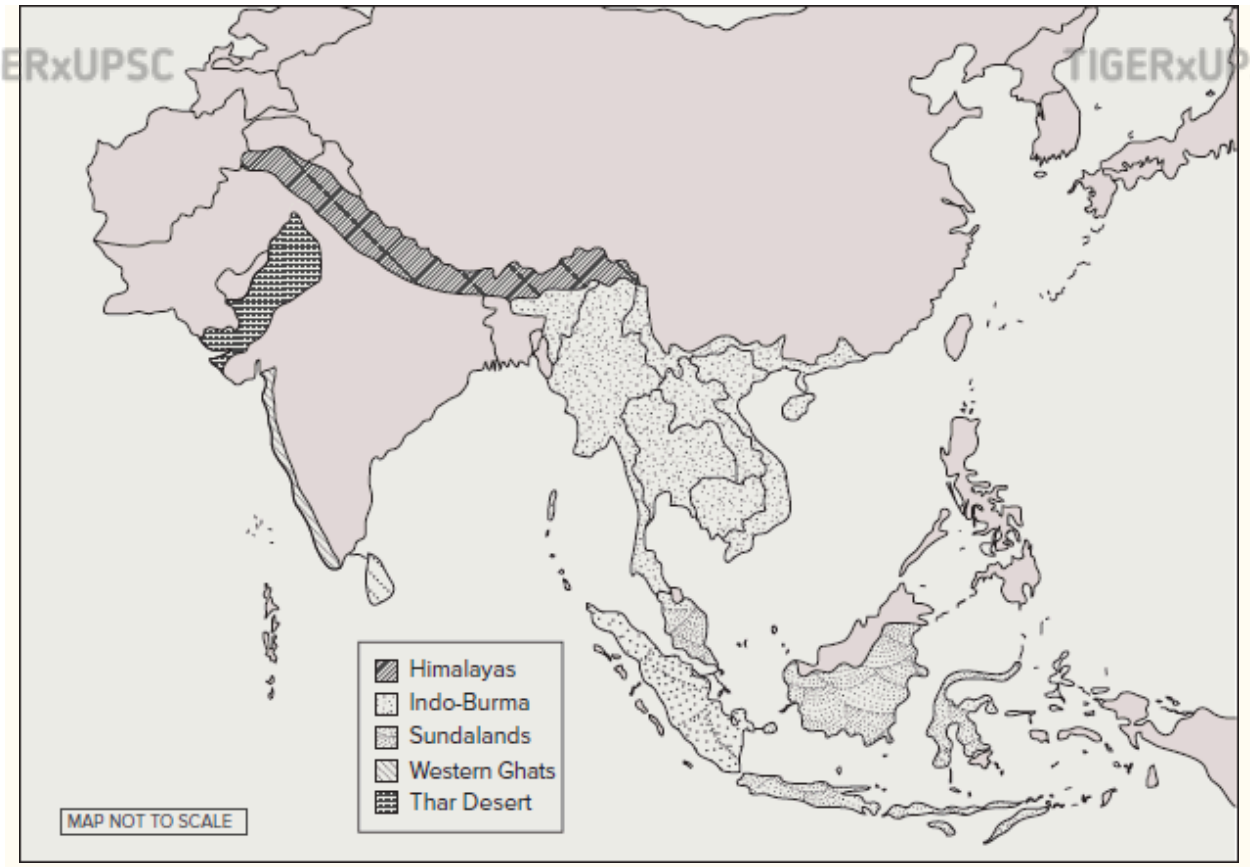

Figure 19.7 Biodiversity Hotspot in India

Indo-Burma Biodiversity Hotspot

- This hotspot is spread in many countries that includes Malaysia, North East India, Bangladesh, China, Cambodia, Vietnam, Thailand and Lao.
- Indo-Burma hotspots cover 2 million km of Asia. It shows varied diversity of climate as it passes through various landforms.

Biodiversity

- In the past few years, Indo Burma biodiversity has been declining. Although, much of the region still exist in the wilderness.
- Recently, six species of mammals have been discovered. These are large-antlered muntjac, leaf deer, saola, annamite muntjac, grey-shanked douc, annamite striped rabbit.
- Indo Burma homes various primate species like monkeys, langurs and gibbons which have population in hundreds only. Freshwater turtles are

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endemic to this region.

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- About 1300 birds species live in this region which includes the endangered white-eared night-heron, the grey-crowned crocias, and the orange-necked partridge.
- Indo Burma has 13,500 flora species in this hotspot, of which half are endemic like Ginger.

■ *Sundaland*

This region exists in SE Asia that covers part of the Indo-Malayan archipelago. Sundaland spreads in Thailand, Malaysia, Singapore, Brunei, Indonesia, India (Nicobar Islands).

In 2013, UNESCO declared the islands as one of the World Biosphere Reserve.

These islands have rich ecosystems, including terrestrial and marine. Mangrove, coral reefs and seagrass beds form the resource treasure of these islands. Marine resources include whales, dolphins, prawns, lobsters, seashells, dugong, turtles, crocodiles and fishes.

Over exploitation of marine resources present a primary threat to the biodiversity of sundaland.

19.10.5 Threats to Biodiversity Hotspots

Over exploitation of natural resources and population pressure causing habitat fragmentation of species are major leading threats to biodiversity hotspots. The threats include the change in land use patterns, pollution, global warming, acid rain, climate change and invasion of exotic species.

However, for rainforests, the primary threat is land conversion. For the tropical regions, it is said that climate will change least. Also, phosphorus is more important for rainforest growth than nitrogen. Hence, change in the nitrogen cycle will not impact the rainforest much. Invasion of exotic

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species is also not much of a problem as the biodiversity is very rich in rainforests.

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Some of the major causes and issues related to threats to biodiversity are:

1. **Habitat Destruction:** Interiors of biodiversity hotspots were earlier inaccessible to mankind but today, with technological advancement, the reach of mankind has increased. This also became the reason for habitat fragmentation and habitat destruction for many species. Activities such as deforestation, pollution, population increment and human settlements are factors that are causing species habitat destruction and thus decreasing their population and making their existence threatened.
2. **Resource Mismanagement:** Increased tourism is one of the causes of environmental degradation and pollution in places like Rishikesh and Dehradun. These destinations have become polluted due to increased population pressure and other human activities. Construction work in Dehradun due to illegal immigrants from Bangladesh is also reducing the natural beauty of Dehradun.
3. **Poaching:** It is one of the reasons for the species to become endangered like tiger, rhinoceros and elephant that once faced extinction due to poaching. Conservation efforts like Project Tiger, Project Elephant have helped the population increment of these threatened species. However, trading tiger hides, elephant tusks, tiger teeth and rhinoceros horn still exists and run on profit.

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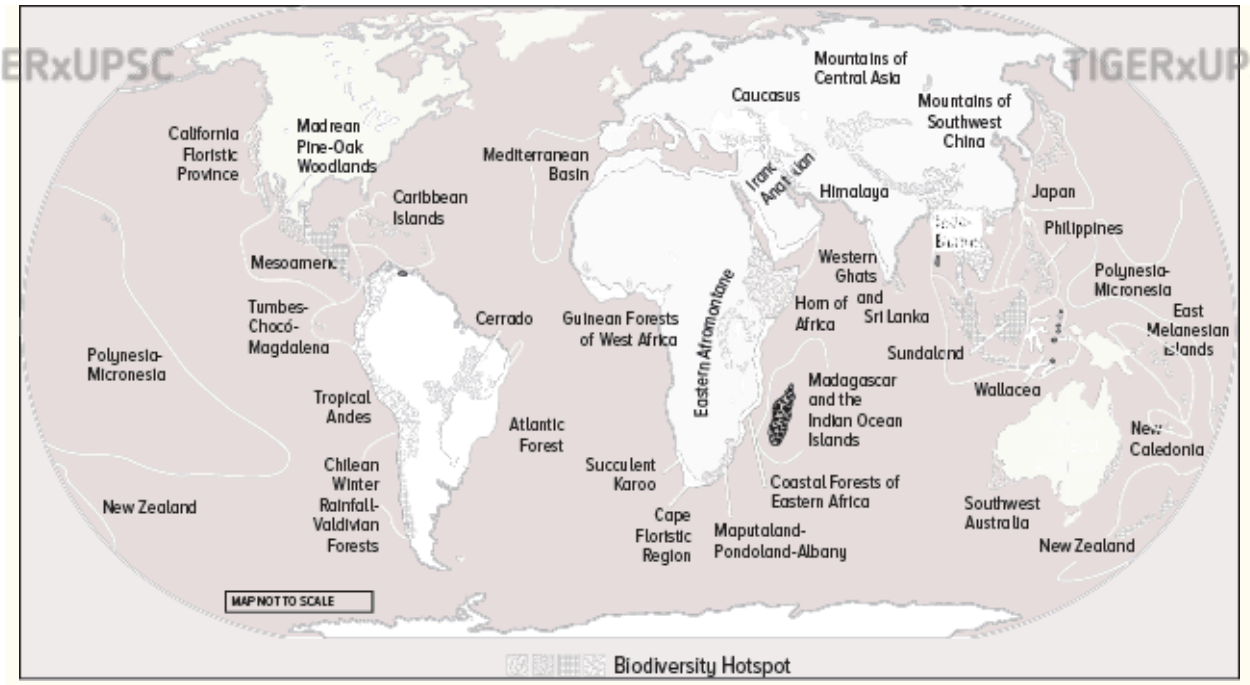

Figure 19.8 Location of Biodiversity Hotspot

19.10.6 Report on the Western Ghats

Western Ghats covers six states and is home to many endangered species. There are thirteen national parks and many sactuaries in the Western Ghats. UNESCO has declared Western Ghat as the most important hotspot. The Western Ghats are sources of many rivers like Godavari, Krishna and Cauvery. Still, the water shortage exists in the region due to leakages of river and when water is available, it is polluted. Western Ghats needed a strategy to maintain ecological stability and sustainability. Therefore, Ministry of Environment and Forest of India set up an expert panel, headed by Madhav Gadgil to recommend means to conserve Western Ghat.

■ *Gadgil Committee Report*

Officially called as 'Western Ghats Ecology Expert Panel' (WGEEP), the committee submitted the report to the Central Government in 2011.The report was more tilted towards environmental conservation and was criticised on the development front.

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The major recommendations of Gadgil Committee Report were:

- It designated the whole ranges of Western Ghats as an **Ecologically Sensitive Area (ESA).**
- The report had recommended 64% of the area to come under Ecologically Sensitive Area (ESA).
- Report classified the various talukas of Western Ghats into Ecologically Sensitive Zone (ESZ) I, II, III.
- In ESZ-I, all development activities were restricted like construction, mining, etc. This zone is the most sensitive of all.
- Recentralisation of power in governance was recommended by the report. It advocates bottom-up approach right from gram sabha level rather than top down approach in governance of Western Ghats.

ENV 28- Common Carbon metric (CCM)

No new dams based on large-scale storage should be permitted in ESZ I.

- **No new polluting industries, including thermal power plants, should be allowed in ESZ I and II.**
- The existing red and orange category industries should be asked to switch to zero pollution by 2016.
- Panel recommended for setting up of a statutory body called Western Ghats Ecology Authority under MoEF under Environment Protection Act 1986.

Gadgil report was criticised as anti-development, and farmers, out of fear, opposed it on some fronts. The major reasons for objection were:

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1. Panel did not take into account ground realities and was overly environment friendly.
2. Most of the recommendations were argued impractical to implement.
3. Providing complete ecological protection cover to the Western Ghats hampers different states in terms of energy and growth.
4. States argued over the need for a new body, that is, WGEA and insisted that enough laws exist for providing protection.
5. Panel does not specify the solution for revenue loss after implementation of the recommendations to states.
6. Panel said no to dam building in Western Ghat and this is a deal breaker for many states as power need is rising.

People with vested interest created fear among people that the Gadgil panel report is contrary to the interest of farmers and that they will lose their lands and agricultural right if the report is implemented. The mining mafia took advantage and misled the people. Therefore, another committee called Kasturirangan Committee was formed. The mandate was to suggest changes in the approach.

■ *Kasturirangan Committee Report*

A high-level Working Group was constituted to examine the Gadgil report. Major recommendations of the Kasturirangan Committee Report were:

- Report recommended **only 37%** of the total area of Western Ghat to be brought under Ecological Sensitive Area (ESA).
- In ESA, there will be a complete ban on quarrying and sand mining.
- Report classified the Western Ghats into a cultural landscape (58% in WGs occupied by human settlement like agriculture, plantations) and natural landscape (90% of it will come under ESA).
- Mining areas from ESA to be phased out for the next 5 years.
- Thermal power plants to be banned in the region and hydropower projects to be allowed only after proper study.

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- Industries as red listed, that is, highly polluting industries to be completely banned in this area.

Building and construction projects up to 20,000 sq. m were to be allowed but townships were to be banned.

- Forest diversion could be allowed with extra safeguards.

Report made recommendations that were considered favourable to farmers, including the exclusion of inhabitant regions and agricultural areas from ESA.

Report assessed that 123 villages fall under ESA. ***Flaws in Kasturirangan Committee Report:***

1. For demarcation of zones in the Western Ghats, the panel had used remote sensing satellites and aerial survey methods. Therefore, ground realities were not taken into account, causing mistakes in reports.
2. Bottom approach was not given adequate attention and bureaucrats were the deciding authorities and not Gram Sabha.
3. Under the report, the mining and quarrying industries were not adequately regulated, causing the destructive effects on ecosystem of Western Ghats.
4. Report identified many non-sensitive ecological areas as ESA and left many areas which were ecologically sensitive areas.

Draft 2018 Notification on the Western Ghats

- After several years of discussions, the government finally notified 56,825 square km area in the Western Ghats (WG) region as an ecologically sensitive area (ESA).
- The notified land spreads over Gujarat, Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu.
- The draft notification, if it gets final clearance, will ban activities such as the operation of:
  a. red category industries,
  b. construction of thermal power plants,
  c. Large scale construction, mining, quarrying and sand mining.

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- However, hydropower projects, orange category industries and other existing activities will be allowed.

19.11 HOPE SPOTS

- Hope Spot Network is a joint initiative of Mission Blue and International Union of Conservation of Nature (IUCN).
- A hope spot is an area of an ocean that needs special protection because of its wildlife and significant underwater habitats.
- They are chosen for their contributions to biodiversity, the carbon sink, and important habitat.
- They can be Marine Protected Area's (MPA) that need attention or any new sites.

Hope Spot Sites from India

- Lakshadweep Islands
- Andaman Nicobar Islands

Marine Protected Areas

- These are designated areas in the ocean within which human activities are regulated more stringently and given special protection than elsewhere in the marine environment.
- These areas are defined by IUCN.

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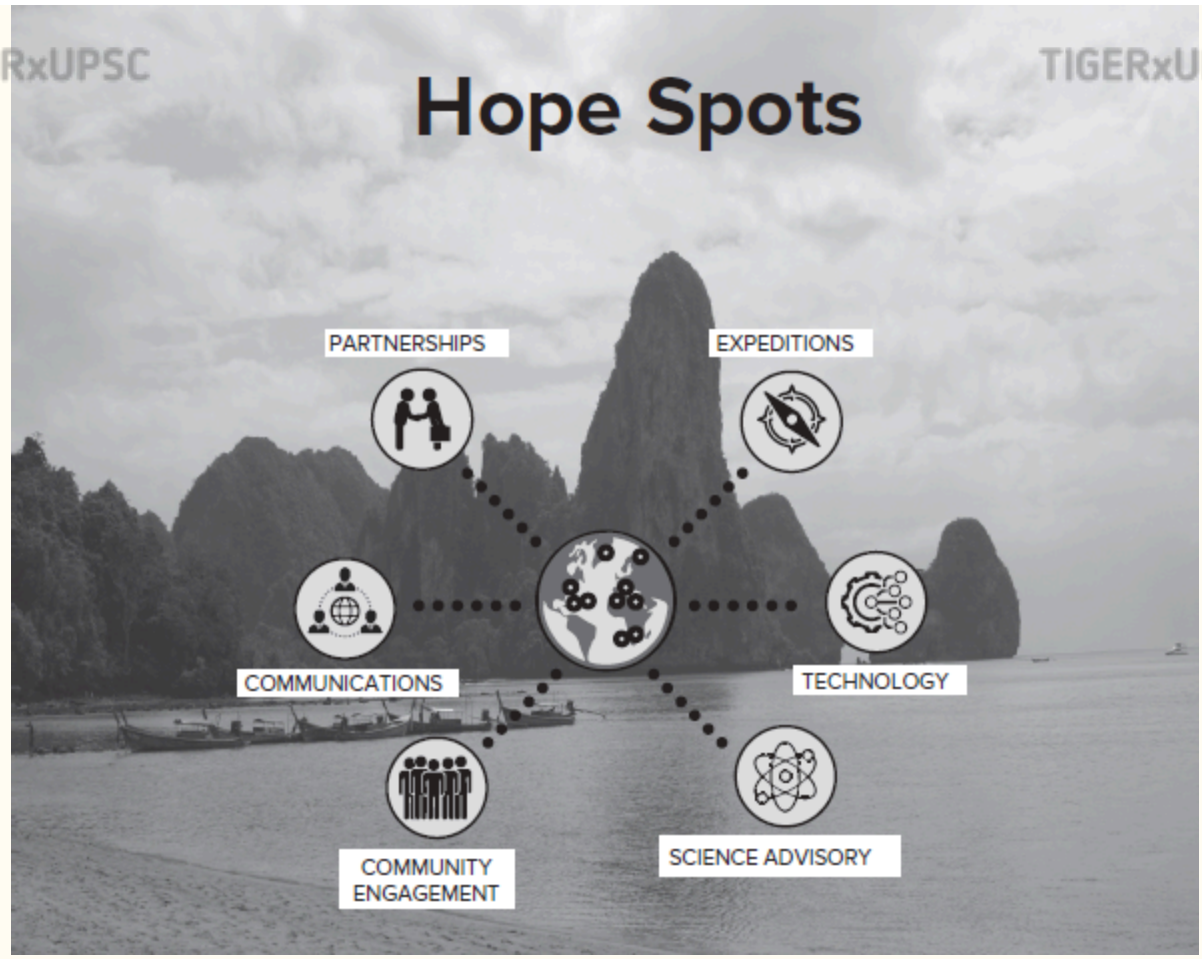

Figure 19.9 Hope Spots

19.11.1 BioWeb

- The BioWeb is the connotation for a network of web-enabled biological devices (e.g. trees, plants, and flowers) which extends an internet of things to the Internet of Living Things of natural sensory devices.
- The BioWeb devices give insights to real-time ecological data and feedback to changes in the environment.
- The biodiversity of today is one giant ecological mesh network of information exchange, and resource humanity should be able to access for a better understanding of the state of our global ecology.
- The BioWeb information technologies emerge from the interdisciplinary fields of biotechnology and nanotechnology.

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- The devices for reading individual ecological systems can be either wireless transmitters implemented into the organic structure of seeds or external inserted network nodes with the ability to read information and wirelessly transmit the information to the internet (or network).

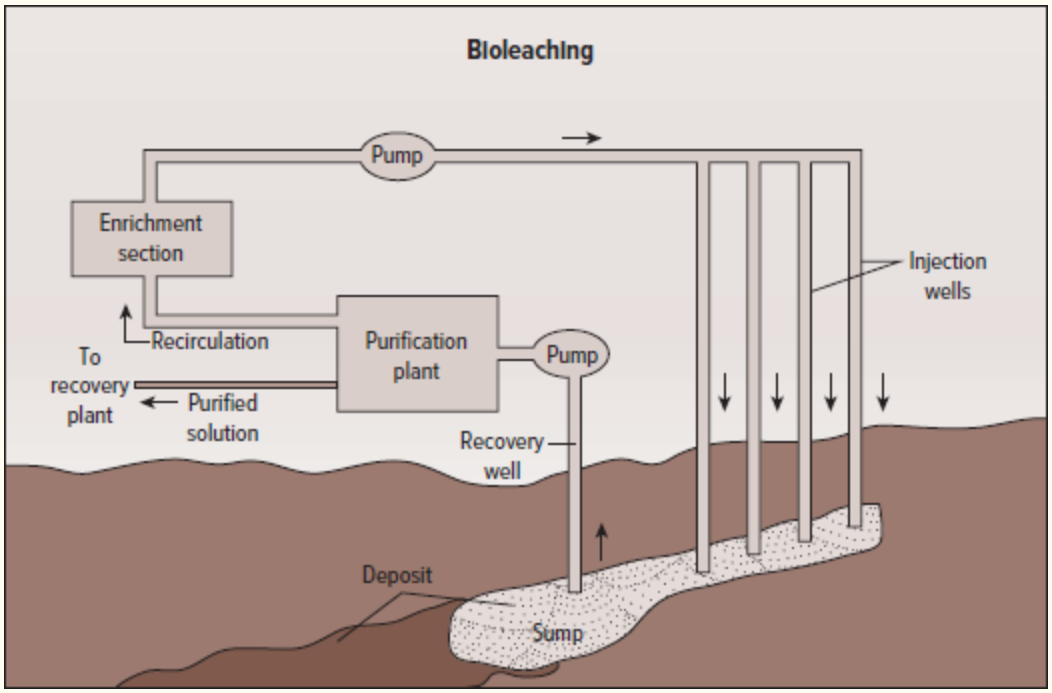

Figure 19.10 Bioleaching

19.11.2 Bioleaching

- Bioleaching is the extraction of metals from their ores through the use of living organisms.
- This is much cleaner than the traditional heap leaching using cyanide. Bioleaching is one of several applications within biohydrometallurgy and several methods are used to recover copper, zinc, lead, arsenic, antimony, nickel, molybdenum, gold, silver, and cobalt.

19.11.3 Biobank

A biobank is a type of biorepository that stores biological samples (usually human) for use in research.

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Biobanks have become an important resource in medical research, supporting many types of contemporary research like genomics and personalised medicine.

Biobanks can give researchers access to data representing a large number of people.

Samples in biobanks and the data derived from those samples can often be used by multiple researchers for cross-purpose research studies.

Biobanks usually incorporate cryogenic storage facilities for the samples.

In August 2021, the first National Heart Failure Biobank (NHFB) in the country that would collect blood, biopsies, and clinical data as a guide to future therapies was inaugurated at the Sree Chitra Tirunal Institute for Medical Sciences and Technology (SCTIMST). The facility will be useful for the research and treatment of post-covid heart failure.

19.11.4 Biolink Zones

- Biolink zones are a land use category developed for biodiversity conservation and landscape adaptation under changing climates.
- Biolink zone was first coined in 1992 to encapsulate a potential new land-use resulting from research on vertebrate fauna and their prospective responses to climate change.
- Biolink zones are identified parts of the landscape where the functional ecological connectivity for biodiversity is enhanced and/or restored to provide space for species (and consequently ecological communities) to self-adapt their distributions and abundances under changing climates through natural processes including: dispersal; re-colonisation; regeneration and restoration of ecological function.

19.11.5 Biopiracy

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It is a phenomenon in which theft of natural resources or traditional knowledge has taken place. To put it simply, whenever the knowledge of local communities is used by individuals or organisation for their profits without taking consent of locals and without sharing the benefits arising out of traditional knowledge with locals it is called as 'biopiracy'.

19.11.6 Bioprospecting

Bioprospecting is the process of making new discoveries and commercialisation of new natural resources based products in underdeveloped countries. This includes the use of traditional knowledge with the locals regarding plants and animals. When bioprospecting is exploitative in nature wherein benefits arising out of it is not shared with locals, it is called biopiracy. It includes the search for natural products like traditional medicine.

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Wildlife Conservation Initiative

20

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Rudyard Kipling in 1894 published his book called ***Jungle Book***, which got the attention of children and also the people. The book, in a way, has shown the animal diversity in the country. Today, we know that India is one of the 17 mega diversities in the world and is home to 7.6% of all

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mammals, 12.6% of birds, 6.2% of reptiles and 6.0% of flowering plant species.

India has four biodiversity hotspots out of 36 in the world. These are the **Eastern Himalaya, North East India, Western Ghats and Andaman and Nicobar Islands, Sundaland.** The country has 18 biosphere reserves, around 573 Wildlife sanctuaries, 106 national parks and 96 wetlands (till Jan 2026). However, this biodiversity is threatened and requires protection and conservation efforts.

20.1 WILD LIFE (PROTECTION) ACT, 1972

It is a Parliamentary Act enacted for the protection of flora and fauna. The Act initially contained six Schedules which later got reduced to four. Species are categorised on the basis of the amount of protection given. Schedule I is the most protected category of species; the hunting and harvesting of such species are restricted.

Highlights

- The Act provides for the protection of wild animals, birds and plants and their habitat to maintain ecological stability.
- The Act is applicable to the whole of India.
- Traps for capturing animals are prohibited except in certain situations.
- Tribal hunting rights in Andaman Nicobar Islands are protected.
- It includes provision for the Convention on International Trade in Endangered Species (CITES) for wild fauna and flora.

Wild Life (Protection) Act, 1972, **has four schedules after the passing of Wild Life (Protection) Amendment Act, 2022 which give varying degrees of protection**

- Schedule I provides for animal species enjoying the highest level of protection.
- Schedule II provides for animal species subject to a lesser degree of protection.

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- Schedule III provides for protected plant species.
- Schedule IV provides for scheduled specimens under CITES.

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Amendments

- In January 2003, the punishment was made more stringent after amending the Act.
- Various Crime Control Bureau and conservation institutions were set up under the Wild Life (Protection) Amendment Act, 2006, such as the National Tiger Conservation Authority (NTCA).
- In 2022, Wild Life (Protection) Amendment Act, 2022 expanded the range of species receiving legal protection and made the provisions of CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora) effective. The number of Schedules were also reduced to four from six.
- Additionally, the Act provided for the utilisation of elephants for 'religious or other lawful purposes.' Furthermore, it introduced enhanced penalties for violations related to both general and specially protected animals. It came into effect from 1st April 2023.

20.2 NATIONAL BOARD FOR WILDLIFE

- Providing policy inputs to the government aid and advice to the governments and promoting and conservation of wildlife by controlling poaching and illegal trade of wild life and its products was the mandate of the Board.
- Board suggests ways to set up and manage national parks, wildlife sanctuaries and other protected areas and on matters relating to restriction of activities in those areas.
- Board needs to recommend the improvement of wildlife conservation methods.
- Board needs to prepare and publish a status report at least once in two years on wild life in the country.

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20.3 PROTECTED AREAS

Protected areas that can be notified under the Wild Life (Protection) Act, 1972, are:

20.3.1 Wildlife Sanctuaries

- Central government or state government can notify constituting a sanctuary for the protection of wildlife.
- Government can decide for the rights of the people residing in these areas.
- Limits of boundaries of such sanctuaries are decided by the state government through a notification which declares that area shall be the sanctuary.
- Changes in boundaries of sanctuaries shall be approved by the resolution passed in the legislature of the state.
- India has approximately 573 sanctuaries (Till Jan 2026).

20.3.2 National Parks

- Central government or state government can notify constituting a national park for the protection of wildlife and its habitat.
- State government cannot change the boundaries of national parks except on the recommendation of the National Board for Wildlife.
- Grazing activities inside national parks are not permitted.
- Provisions applicable to sanctuaries are also applicable to national parks.
- Change in boundaries of national parks shall be approved by the resolution passed in the state legislature.
- There are around 106 National Parks (Till Jan 2026) in India.

20.3.3 Conservation Reserves

- These are areas that act as a buffer zone between the people and the protected areas, such as national parks, sanctuaries, protected forests, etc.
- These areas are owned by the government but used by the community.

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- Local people and agencies, such as gram panchayats, govern these conservation reserves.
- State governments can declare any area as conservation reserves, particularly those adjacent to the national parks and wildlife sanctuaries, after consultation with local governments.
- Conservation Reserve Management Committee has been constituted by the government to manage the conservation reserve.
- These are the first kind in which private lands are protected under legislature.
- There is the scope of community-owned private wildlife resorts and private trusts to come up under community reserves.
- At present, 115 Conservation Reserves (Till Jan 2026) are there in India.

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Tiger Reserves

- These reserves are meant for protecting the tigers.
- Based on the recommendation of the NTCA, the state government created tiger reserves for which they have tiger conservation plans. Under Project Tiger, 58 Tiger Reserves (Till Jan 2026) have been set up by the government, which are governed by NTCA. It was declared in July 2021 that India has achieved the St. Petersburg aim when the number of Royal Bengal Tigers doubled from 1,411 in 2006 to 2,967. According to the census' estimated 2,967 tigers, India is home to 75% of the world's tigers.

Some of the wildlife conservation projects taken by the government of India are:

Project Tiger

It is one of the most successful conservation projects of wildlife which has contributed to not only the increase in the number of tigers but also the conservation of their habitat.

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ENV 28- Common Carbon metric (CCM)

Key Facts

- Project Tiger is funded by the Ministry of Environment, Forest and Climate Change (MoEF&CC).
- India has 58 Tiger Reserves (Till Jan 2026), including Corbett National Park and Ranthambore National Park. NTCA under Project Tiger makes an assessment of tigers, their habitat, their prey and hunting patterns.
- The strategy of Core-Buffer was used for creating tiger reserves in the country.

Tiger reserves functioned under certain core principles mentioned below:

- Human exploitative activities are banned from the core zone of tiger reserves; however, it is allowed in the buffer zone provided that the activities do not hamper the habitat of tigers.

M-STrIPES (Monitoring System for Tiger-Intensive Protection and Ecological Status)

- In 2010, the **NTCA,** along with the **Wildlife Institute of India (WII)**, launched **M-STrIPES**.
- This is a patrol-based wildlife monitoring based on a **Geographic Information System (GIS)** database, designed to assist wildlife protection, monitoring and management of protected areas.
- Under **M-STrIPES** protocols, forest guards are expected to patrol their beats and record their tracks using a GPS, in addition to recording observations in site-specific datasheets.

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- Habitat management should be subjected to fixing the damage caused by human activities in order to maintain ecosystem stability.
- For research purposes, the changes in flora and fauna need to be documented.

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Success of the Project Tiger

Initially, only nine tiger reserves were created by joint efforts of central and state governments, namely:

- Manas Tiger Reserve (Assam)
- Palamau Tiger Reserve (Jharkhand)
- Similipal Tiger Reserve (Orissa)
- Corbett Tiger Reserve (Uttar Pradesh)
- Kanha Tiger Reserve (Madhya Pradesh)
- Melghat Tiger Reserve (Maharashtra)
- Bandipur Tiger Reserve (Karnataka)
- Ranthambore Tiger Reserve (Rajasthan)
- Sundarbans Tiger Reserve (West Bengal)

Tigers are carnivorous and thus at the top of the food chain; therefore, a healthy tiger population is necessary to maintain the ecosystem and sustain species at various levels. Project Tiger has accomplished the goal of increasing the tiger population.

Project Tiger targets increasing not only the tiger population but also their prey and habitat. Therefore, all other wildlife populations have also increased in the tiger reserve.

After the launch of Project Tiger, the focus has been on enlarging the core and buffer zones in the tiger reserve, intensifying protection efforts and development of ecology in the buffer zone and creating more tiger reserves and doing research activities.

Concerns and Challenges

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- India has one of the lowest per capita forest areas in the world. Forests as carbon sinks are deemed to be a major means of controlling climate change. **Depletion of forests is responsible for the reduction of tiger habitats**.
- **Reduced food base:** As forestlands fall to development projects, habitable lands for animals that make for the tiger's food base are also reduced.
- **Poaching:** Another issue that has hindered tiger conservation in India and globally is poaching, which will persist as long as there is an illegal market for tiger body parts.
- **Climate change:** Rising sea level as a result of climate change is on the verge of wiping out Sundarbans, one of the last remaining habitats of the Bengal Tigers.

Conservation Efforts

- **The NTCA has launched the M-STrIPES (Monitoring System for Tigers –** Intensive Protection and Ecological Status), a mobile monitoring system for forest guards.
- At the **St. Petersburg Tiger Summit in 2010**, leaders of 13 Tiger Range Countries resolved to do more for the tiger and embarked on efforts to double its number in the wild, with a popular slogan **'T X 2'**.
- The **Global Tiger Initiative (GTI) programme** of the **World Bank**, using its presence and convening ability, brought global partners together to strengthen the tiger agenda.
- Over the years, the initiative has institutionalised itself as a separate entity in the form of the **Global Tiger Initiative Council, with its two arms:** the **Global Tiger Forum** and the **Global Snow Leopard Ecosystem Protection Programme**.

Some Controversial Projects

- **Highways and railway lines** are being expanded in the corridor connecting Kanha and Pench tiger reserves.
- A **railway line is being constructed** through the Melghat Tiger Reserve.

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The Ken–Betwa river linking project will submerge over 100 sq km of the Panna Tiger Reserve in MP.

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National Tiger Conservation Authority (NTCA)

The NTCA is a statutory body under the Ministry of Environment, Forest and Climate Change constituted under enabling provisions of the Wild Life (Protection) Act, 1972, for strengthening tiger conservation in the country by retaining an oversight through advisories/normative guidelines, based on an appraisal of tiger status, ongoing conservation initiatives and recommendations of specially constituted committees.

The functions of NTCA are as follows:

- Ensuring normative standards in tiger reserve management
- Preparation of reserve-specific tiger conservation plan
- Laying down annual/audit report before Parliament
- Instituting state-level steering committees under the Chairmanship of the Chief Minister and establishment of a Tiger Conservation Foundation.
- According approval for declaring new Tiger Reserves.

International Tiger Day: 29th July

- Global Tiger Day, often called International Tiger Day, is an annual celebration to raise awareness for tiger conservation, held annually on July 29th.
- Global Tiger Day was created in 2010 at the Saint Petersburg Tiger Summit.
- The goal of the day is to promote a global system for protecting the natural habitats of tigers and to raise public awareness and support for tiger conservation issues.

■ *Global Tiger Recovery Programme (GTRP)*

The **GTRP** seeks to empower **Tiger Range Countries** to address the entire spectrum of threats, domestic as well as those that are transboundary in nature, and work towards increased financial sustainability through the integration of conservation objectives into development.

The GTRP was developed to double the number of wild tigers globally by 2022 through actions to:

(i) Effectively manage, preserve, protect and enhance tiger habitats.

(ii) Eradicate poaching, smuggling and illegal trade of tigers, their parts and derivatives.

(iii) Cooperate in transboundary landscape management and in combating illegal trade.

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(iv) Engage with indigenous and local communities.
(v) Increase the effectiveness of tiger and habitat management.
(vi) Restore tigers to their former range.

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Tiger vs. Cheetah

Tigers are massive, striped, solitary ambush predators of Asian forests, known for power, while cheetahs are smaller, spotted, daytime hunters from African savannas, famous for being the world's fastest land animal. Key differences lie in their **appearance** (stripes vs. spots), **size** (tiger is huge, cheetah is smaller), **speed** (cheetah is fastest), **habitat** (Asia vs. Africa), and **hunting style** (power ambush vs. speed chase).

Cheetah Reintroduction Project

India's **Project Cheetah** is a pioneering reintroduction program to bring cheetahs back to the country after they were declared extinct in 1952. It is the world's first inter-continental large wild carnivore trans-location project, primarily centered in Madhya Pradesh's **Kuno National Park**.

■ *Project Status (as of Jan 2026)*

- **Total Population:** India currently has **30 cheetahs**. This includes 12 adults, 9 sub-adults, and 9 cubs, with 19 individuals born on Indian soil.
- **Trans-location:** 20 adult cheetahs were initially brought from Namibia (8 in Sept 2022) and South Africa (12 in Feb 2023). In November 2025, eight more were gifted by Botswana and are awaiting final translocation.
- **Breeding Success:** A major milestone was achieved when the first Indian-born cub, Mukhi, gave birth to a second-generation litter of five cubs in November 2025, signaling good adaptation to the environment.
- **Expansion:** With Kuno National Park nearing capacity, the **Gandhi Sagar Wildlife Sanctuary** is being developed as a second home for cheetahs, and the **Nauradehi Wildlife Sanctuary** as a third site, aiming for a self-sustaining meta-population of 60-70 cheetahs by 2032 across a 17,000 km² landscape.

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- **Mortality:** The project has faced challenges, with several deaths recorded among trans-located adults and cubs due to various causes, including kidney ailments, infections, and interactions with other predators.

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Objectives

The primary objectives of Project Cheetah are:

- To establish a viable, self-sustaining cheetah meta-population in safe habitats across the species' historical range.
- To restore the cheetah's role as a flagship and umbrella species, which helps to conserve the neglected grassland and savanna ecosystems and their associated biodiversity.
- To promote eco-development and ecotourism, enhancing local livelihoods and community engagement in conservation efforts through initiatives like **'Cheetah Mitras'**.

The project is implemented by the National Tiger Conservation Authority (NTCA) in collaboration with the Madhya Pradesh Forest Department and the Wildlife Institute of India (WII).

Project Elephant

In 1992, Project Elephant was launched by the government of India for conserving elephants, their habitats and migratory corridors through scientific management techniques.

Key Facts

- Welfare of domesticated elephants and reducing the conflict between man and elephant is also one of the targets.
- Project Elephant provides protection to elephants from poachers and unnatural deaths.
- States that have a significant population of elephants are provided with technical and financial support. The project is implemented in 16 states, viz., Andhra Pradesh, Karnataka, Kerala, Maharashtra, Arunachal Pradesh,

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Assam, Chhattisgarh, Jharkhand, Meghalaya, Tamil Nadu, Tripura, Uttarakhand, Nagaland, Orissa, Uttar Pradesh and West Bengal.

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Main Activities of Project Elephant

- Restoration of natural habitat and migratory routes of elephants.
- Development of scientific management strategies for the conservation of elephants, habitat and their corridors and increasing population of Asiatic elephant.
- Reduction of man–elephant conflict in habitats of elephants and moderation of the pressure of human activities in elephant habitats.
- Promotion of measures to protect wild habitats from poachers.
- Research and development on the management of elephants.
- Creation of public awareness.
- Ecological development.
- Veterinary care of domestic elephants.
- Setting up of elephant rescue centres.

Haathi Mere Saathi Campaign

Reaching out to the people to improve the conservation and welfare prospects of the elephant, **India's National Heritage Animal**, the **Ministry of Environment and Forests (MoEF),** in partnership with the **Wildlife Trust of India (WTI),** launched the nationwide **Haathi Mere Saathi campaign** in **2011**.

Under this campaign, the public initiative was aimed to increase awareness among people and to develop not just friendship but also companionship between people and elephants.

Elephant Corridors

Asian Elephant Alliance, an umbrella initiative by five NGOs, has come together to secure 96 out of the 101 existing corridors used by elephants across 12 states in India.

- The joint venture aims to secure the 96 remaining elephant corridors, old and new, in the next 10 years.
- The alliance joined hands to raise the mammoth sum as money was the main constraint in securing the land.

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- NGOs Elephant Family, International Fund for Animal Welfare, IUCN Netherlands and World Land Trust have teamed up with **WTI** in the alliance.

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What Are Elephant Corridors?

- Elephant corridors are narrow strips of land that connect two large habitats of elephants. Elephant corridors are crucial to reducing animal fatalities due to accidents and other reasons.

Monitoring of Illegal Killing of Elephants (MIKE)

- The Conference of Parties of CITES mandated the MIKE programme in South Asia in 2003.
- MIKE programme was established by the UN CITES. It is the site-based monitoring of elephant mortality.
- MIKE aims to help Elephant range states improve their ability to monitor elephant populations, detect changes in levels of illegal killing and use this information to provide more effective law enforcement and strengthen any regulatory measures required to support such enforcement.

The main objectives of the MIKE are:

- To assess the pattern of poaching and illegal hunting.
- To determine the changes in the pattern of poaching activities over time.
- To correlate the decision of COP with the changes in the patterns of the poaching to see if the decisions are yielding results or not.

MIKE Sites in India-Total 10

- Chirang Ripu (Assam)
- Dhang Patki (Assam)
- Eastern Dooars (WB)
- Deomali (Arunachal Pradesh)
- Garo Hills (Meghalaya)
- Mayurbhanj (Orissa)
- Mysore (Karnataka)
- Nilgiri (TN)

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- Shivalik (Uttarakhand)
- Wayanad (Kerala)

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Elephant Trade Information System (ETIS)

- ETIS is a comprehensive information system to track illegal trade in ivory and other elephant products.
- It shares the same objectives as those set out for MIKE, with the difference that it aims to record and analyse levels and trends in illegal trade, rather than the illegal killing of elephants.

TRAFFIC

- TRAFFIC is a leading non-governmental organisation working globally on trade in wild animals and plants in the context of both biodiversity conservation and sustainable development.
- Fragmentation of forests makes it all the more important to preserve migratory corridors.

Why Protect Elephant Corridors?

- The movement of elephants is essential to ensure that their populations are genetically viable. It also helps to regenerate forests on which other species, including tigers, depend.
- Nearly 40% of elephant reserves are vulnerable, as they are not within protected parks and sanctuaries. Also, the migration corridors have no specific legal protection.
- Forests that have turned into farms and unchecked tourism are blocking animals' paths. Animals are thus forced to seek alternative routes, resulting in increased elephant–human conflict.
- Weak regulation of ecotourism is severely impacting important habitats. It particularly affects animals that have large home ranges, like elephants.

Need of the Hour

- Efforts should be made to expand elephant corridors, using the successful models within the country. This includes the acquisition of lands using private funds and their transfer to the government.

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- Ending human interference in the pathways of elephants is more of a conservation imperative.

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Gaj Yatra

- 'Gaj Yatra', a nationwide campaign to protect elephants, was launched on the occasion of World Elephant Day in 2017. The campaign is planned to cover 12 elephant range states. The elephant is part of India's animal heritage, and the government celebrates this day to spread awareness about the conservation of the species.
- The 15-month campaign will be led by the WTI. The campaign aims to create awareness about elephant corridors to encourage free movement in their habitat.

■ *Crocodile Conservation Project*

IUCN Red List has listed gharial as a critically endangered species.

Project Crocodile

- In 1975, the Government of India started Project Crocodile with the support of the United Nations Development Fund and the United Nations Food and Agriculture Organisation.
- Project aims to increase the number of gharials in their habitat through captive breeding and rearing programme.
- Project Crocodile has been successful in increasing the number of crocodiles, which were once on the verge of extinction.
- The project also contributed to related fields like conservation of habitat and other species.

Main Objectives

The focus of the programme is to conserve the remaining population of crocodiles and their natural habitat by making crocodile sanctuaries, promoting captive breeding and rearing, and improving governance of the programme with the participation of local people. With the initiation of the

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project, 4,000 gharials/alligators, 1,800 muggers/crocodiles and 1,500 saltwater crocodiles will increase.

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QUICK INSIGHT

UNDP Sea Turtle Project

WII initiated the UNDP Sea Turtle Project with the objective of conserving olive ridley turtles.

- 10 Indian states are implementing this project. Mapping of breeding sites of sea turtles along with the identification of migratory routes along the seacoast has taken place under this project.
- Several thousands of sea turtles migrate to the Indian Eastern Coast for breeding and nesting in winter, especially olive ridley turtles.
- Project has yielded scientific guidelines to safeguard the olive ridley turtles' mortality and precautions to be taken for tourism purposes. Another achievement of the project is the use of Satellite Telemetry for the identification of migratory routes of sea turtles in the sea.

■ *Vulture Conservation*

- Indian vulture owes its origin to the Old World vulture which has its original habitat in Nepal, India and Pakistan.
- Indian vulture has been listed as a critically endangered species since 2002 due to a significant decline in population.
- Diclofenac drug poisoning has become the reason for the deaths of Indian vulture, which causes renal failure in vultures. Diclofenac found its way to the digestive system of the vultures through the dead bodies of their prey, like dead cattle, which are food to the vultures.
- Slender-billed vulture is a bird which once considered a subspecies of vulture and is now treated as a separate species. Earlier, they were listed under long-billed vulture category.

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ENV 28- Common Carbon metric (CCM)

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There has been a decline of 97%–99% of the population in India vulture and white-rumped vulture in Pakistan, Bangladesh and India from 1993 to 2002.

Diclofenac

It is a common anti-inflammatory drug administered to livestock.

It is used to treat the symptoms of inflammation, fever and/or pain associated with disease or wounds.

It leads to renal failure in vultures, damaging their excretory system (direct inhibition of uric acid secretion in vultures).

Diclofenac is used to reduce the joint pain of animals and keep them working for longer hours. It is a **non-steroidal anti-inflammatory drug (NSAID)**.

In 2006, the Indian government decided to ban the use of diclofenac in India.

Conservation Efforts by the Government to Save Vultures

- Vulture recovery plan was introduced by the government in 2006.
- The plan has recommended that diclofenac should be banned and a safe alternative of it should be found. Conservation of vultures should be done through an intensive breeding programme.
- In 2006, in line with the recovery plan, the government banned the use of diclofenac.
- Movement for vulture conservation was led by the Bombay Natural History Society (BNHS).
- The strategy of vulture conservation was to increase their numbers through captive breeding, research and development work on vulture and their habitat, monitoring the risks for vultures in their habitats,

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sensitisation programme for vulture conservation among the people and their release in identified safe zones.

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Steps Taken by Government to Protect Vultures in India Vulture Conservation Breeding Centre

- Vulture Conservation Breeding Centre (VCBC) is a joint project of the Haryana Forest Department and the BNHS. It is a collaborative initiative to save the three species of vultures, the white-backed, long-billed and slender-billed, from looming extinction.
- Jatayu and Sparrow Conservation Breeding Centre is the world's largest facility for the breeding and conservation of Indian vultures and the house sparrow. It is located within the Bir Shikargah Wildlife Sanctuary in the town of Pinjore in the state of Haryana, India. It is run by the Forest Department, Haryana, and BNHS, with the help of the British nature conservation charity Royal Society for the Protection of Birds.

Vulture Action Plan

- Ministry of Environment, Forests and Climate Change launched a Vulture Action Plan 2020–25 for the conservation of vultures in the country. Objectives of the Action Plan for Vulture Conservation 2020–2025 are:
- Drug control
- To ensure minimum use of diclofenac.
- To prevent the poisoning of the principal food of vultures, the cattle carcasses, with veterinary NSAIDs, by ensuring that sale of veterinary NSAIDs is regulated and is disbursed only on prescription and by ensuring that treatment of livestock is done only by qualified veterinarians.
- To carry out safety testing of available NSAIDs on vultures and to develop new ones which do not affect vultures.
- To ensure that DGCI must institute a system that automatically removes a drug from veterinary use if it is found to be toxic to vultures.
- Such a system would ensure that drugs other than diclofenac that are toxic to vultures like aceclofenac and ketoprofen are banned for veterinary use.

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- Upscaling conservation.
- To establish Additional Conservation Breeding Centres along with Vulture Conservation Centres with samples and information collected from the wild are analysed and stored at these centres.
- To implement the Vulture Safe Zone programme at eight different places in the country where there are existing populations of vultures.
- To launch conservation plans for the Redheaded and Egyptian vultures, with breeding programmes for both.
- To declare a Vulture Safe Zone only when no toxic drugs are found in undercover pharmacy and cattle carcass surveys, for two consecutive years, and the vulture populations are stable and not declining.
- To build four rescue centres for different geographical areas like Pinjore in the North, Bhopal in Central India, Guwahati in Northeast and Hyderabad in South India.

Saving Asia Vulture from Extinction (SAVE) Initiative

- It is a consortium of 24 partners working together to implement agreed priority actions listed in the annually reviewed 'blueprint for the recovery of Asia's globally threatened vultures'.

Vulture Census

- Organisation involved in the census is the Bombay Natural History Society (BNHS), a wildlife analysis organisation.

Vulture Restaurants

A vulture restaurant is an **undisturbed area** where non-toxic, poison-free meat and carcasses are provided for vultures and other scavengers. This supplementary feeding supports the vultures in times of food scarcity and when young birds fledge.

Vulture restaurant is a safe food source and provides landowners with a **clean and cost-effective way of disposing of waste** and **unwanted carcasses**. Using vulture restaurants, **farmers are contributing to conservation efforts** and **disposing of their waste in an environment-friendly way**.

Vulture-feeding projects like these restaurants have been implemented successfully internationally, with a region in Nepal that started to provide clean carcasses reporting a

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doubling of nesting pairs of vultures in just two years, according to **Bird Conservation Nepal**.

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Conservation Breeding Centres: Ex-situ Conservation Initiative

- Asia's first VCBC in India in Haryana called a vulture research facility in 2005.
- India, today, has nine vulture breeding centres located in Rani, Guwahati, Pinjore, Buxa (WB) and Bhopal (MP) etc.
- Jatayu Conservation and Breeding Centre (JCBC) in UP's Maharajganj district, the world's first conservation and breeding centre built and designed exclusively for the conservation of Asian king vulture.
- In addition to these four, Central Zoo Authority (CZA) manages four more centres in Junagarh in Gujarat, Nandakanan in Orissa, Hyderabad in Telangana and Muta in Ranchi.

Vulture Safe Zones: In-situ Conservation Initiative

- It is an area of 100 km radius designated as a natural habitat of vultures. This area has been made free of diclofenac drug in the carcasses of animals. That is why it is considered a safe zone for vultures.
- These zones are meant to conserve and increase the number of vultures and act as sites for the release of vultures in the future.
- Vulture safe zones are declared in Gujarat, Jharkhand, Uttarakhand, Uttar Pradesh and Assam with the help of BNHS.
- Vulture restaurants have come up in a few states which provide diclofenac-free carcasses.

■ *India Rhino Vision (IRV) 2020*

- India, Pakistan, Burma Nepal, Bangladesh and Bhutan were once used to be the home of great one-horned or Indian rhinoceros.
- However, by the early 1900s, poaching and habitat fragmentation had caused a decline in the population of great one-horned or Indian rhinoceros to just 200 individuals in Nepal and Bangladesh.
- Poaching and habitat destruction threats for Indian rhinos still exist.

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- 70% of the world population of Indian rhinos resides in Kaziranga National Park. It is worrying that the carrying capacity of the Kaziranga National Park has reached its limit and no more rhinos can be accommodated. Also, the threat of entire species of Indian rhino dying in case of outbreak of some disease or natural disaster in Kaziranga National Park looms.
- India Rhino Vision 2020 was considered a partnership at national, state, local and international levels. The partners include the Assam Forest Department and the Bodoland Territorial Council, WWF, International Rhino Foundation (IRF) and the US Fish and Wildlife Service.
- The goal of the vision was to increase the population of great one-horned rhinos to 3,000 spread over seven protected areas by the end of 2020. IRV is supported by **Save the Rhino International (SRI)** through IRF.
- Relocation of Indian rhinos took place in 2008, and in the next few years, 18 Indian rhinos moved from Pabitora Wildlife Sanctuary and Kaziranga National Park to Manas National Park. Manas National Park is a UNESCO world heritage site and it was the habitat of Indian rhinos before the poaching crisis.

The great one-horned rhinoceros is the largest of the three Asian rhinos and, together with African white rhinos, is the largest of all rhino species. It is listed as vulnerable on the IUCN Red List. With at least half of the total population, India's Kaziranga National Park remains the key reserve for this species. Two species of rhino in Asia, Javan and Sumatran Rhino, are critically endangered.

The New Delhi Declaration on Asian Rhinos 2019

India and four rhino range nations have signed a declaration **'The New Delhi Declaration on Asian Rhinos 2019'** for the conservation and protection of the species at the Second Asian Rhino Range Countries meeting.

The Second Asian Rhino Range Countries meeting was held in New Delhi and organised by the Ministry of Environment, Forest and Climate

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Change of the Central government in collaboration with the IUCN Asian Rhino Specialist Group, WWF-India and Aaranyak.

- India will collaborate with Bhutan, Nepal, Indonesia and Malaysia to increase the population of three species of Asian rhinos, including the greater one-horned rhinoceros found in the Indian sub-continent.
- The declaration was signed to conserve and review the population of the greater one-horned, Javan and Sumatran rhinos every four years to reassess the need for joint actions to secure their future.
- The declaration includes undertaking studies on health issues of the rhinos and their potential diseases and taking necessary steps; collaborating and strengthening wildlife forensics for the purpose of investigation and strengthening transboundary collaboration among India, Nepal and Bhutan for the conservation and protection of the greater one-horned rhino.

Project Lion

- Indian government initiated the Asiatic Lion Reintroduction Project to conserve the Asiatic lion from extinction in the wild through reintroduction of lions.
- Gir Forest in Gujarat contains the last wild population of Asiatic lions. However, there is also a threat of possible epidemic, fall of natural disaster and other causes.
- Project Lion aims to have another region to have an autonomous population of Asiatic lions at Kuno Wildlife Sanctuary, Madhya Pradesh. However, Gujarat is against the translocation of the Asiatic lion and has protested.
- A workshop was held in 1993 on the Population and Habitat Viability Assessment of the Asiatic lion, and a report was presented to the state government which contains basic data on ecology.
- Several teams of the workshop focused on different aspects of the conservation of Asiatic lions such as census of population, habitat

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conservation, monitoring, man–lion interaction, translocation, captive lions, public awareness, reproduction, veterinary, etc.

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Suitability of sites is as follows:

Kuno Wildlife Sanctuary lacked an adequate prey population and heavy human interferences.

The reason for the rejection of **Darrah–Jawahar Sagar Wildlife Sanctuary** was unsuitable habitat and geography.

Kumbhalgarh Wildlife Sanctuary lacked adequate area, suitable terrain and enough water and prey population.

Barda Wildlife Sanctuary has problems like scarcity of water, prey, forage and human interface.

■ *Project Dolphin*

Gangetic River Dolphin

The Ganges River dolphin, or susu, inhabits the Ganges–Brahmaputra–Meghna and Karnaphuli–Sangu river systems of Nepal, India and Bangladesh. It is classified as endangered by the IUCN.

- This dolphin is among the four ‘obligate’ freshwater dolphins; the other three are the baiji, now likely extinct from the Yangtze River in China, the bhulan of the Indus in Pakistan and the boto of the Amazon River in Latin America. Although there are several species of marine dolphins whose ranges include some freshwater habitats, these four species live only in rivers and lakes.
- Being a mammal, the Ganges River dolphin cannot breathe in the water and must come to the surface every 30–120 seconds. Because of the sound it produces when breathing, the animal is popularly referred to as the ‘Susu’.

■ *Threats*

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- The survival of the Ganges River dolphin is threatened by unintentional killing through entanglement in fishing gear; directed harvest for dolphin oil, which is used as a fish attractant and for medicinal purposes; water development projects (e.g., water extraction and the construction of barrages, high dams and embankments); industrial waste and pesticides; municipal sewage discharge and noise from vessel traffic; and over-exploitation of prey, mainly due to the widespread use of non-selective fishing gear.

Dolphins have been included in Schedule I of the **Indian Wildlife (Protection) Act, 1972**, in Appendix I of the **CITES**, in Appendix II of the **Convention on Migratory Species (CMS)** and categorised as **'Endangered' on the IUCN Red List**.

National Dolphin Research Centre

India's and Asia's first Dolphin Research Centre will be set up on the banks of the Ganga river in Patna University campus in Patna, Bihar. It will be named the National Dolphin Research Centre (NDRC). It is expected to be ready by October 2023.

The announcement for the centre was made on the occasion of Dolphin Day (October 5), observed in Bihar for the protection and conservation of Gangetic river dolphins to create awareness to save endangered species.

Role:

- NDRC will play an important role in strengthening conservation efforts and research to save endangered mammals whose population is decreasing.
- Bihar is home to around half of the country's estimated 3,000 dolphin population.

■ *Project Hangul*

- It is the state animal of Jammu and Kashmir. It belongs to the subspecies of European red deer of Asia.
- Kashmir stag is different than its female counterpart. It has antlers with 11–16 points and has long hair on the neck while their females have no such feature.
- Hangul today is restricted to **Dachigam National Park** in Srinagar.

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- The number of hangul has been declining from 5,000 in the late 20th century to 900 in the 1980s. A World Wildlife Fund aided the project.
- Under the Save Hangul Project, a survey of hangul's natural habitat and habitat of leopard and black bear has taken place.

Factors Responsible for the Failure of this Project

- There was no local people participation in the project. It was carried out without the involvement of local communities such as Gujjars, Bakerwals, Nambardars, Chowkidars and Patwaris.
- The government departments allowed the establishment of cement factories around Dachingam National Park. These factories disturbed the wild areas.
- There was illegal and reckless unscientific extraction of limestone stretching over miles after miles was carried under its nose. Those areas created death traps for animals.
- The onset of militancy dealt a blow to conservation efforts.

20.4 IMPORTANT INSTITUTES FROM THE POINT OF VIEW OF WILDLIFE

20.4.1 Wildlife Crime Control Bureau (WCCB)

- To combat organised crime against wildlife, WCCB was established by the government of India in 2006 under MoEF. It is a statutory body.
- WCCB has its headquarters in New Delhi. It has five regional offices at Delhi, Jabalpur, Kolkata, Mumbai and Chennai; three sub-regional offices at Guwahati, Amritsar and Cochin; and five border units at Ramanathapuram, Gorakhpur, Motihari Nathula and Moreh.

Under Wild Life (Protection) Act, 1972, the Mandate is

- Collection of intelligence regarding wildlife crime activities and share it with state government and other institutions involved in curbing organised crime against wildlife.

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- Coordination among different agencies and establishing a centralised data bank of wildlife crime activities is mandated.
- Coordination with international institutions and agencies for controlling wildlife crimes.
- Capacity building of people involved in wildlife crime control agencies for objective investigation of the crimes and assists the state governments in the prosecution of the criminals.
- Advising the government of India and providing policy inputs to the issues related to wildlife crime having national and international ramifications.
- Assist and advise the authorities of the customs department to have a thorough and scientific investigation of the goods as per provisions of the Wild Life (Protection) Act, CITES and EXIM policy.

20.4.2 Wildlife Institute of India (WII)

- WII is an independent institution under MoEF.
- WII has conducted research in the areas like biodiversity, wildlife policies, wildlife management, wildlife forensics, endangered species, spatial modelling, ecological development, understanding habitat ecology and climate change.
- Research facility that has forensics, laboratory, herbarium and electronic library, remote sensing and GIS was founded by V.B. Saharia. WII has contributed in terms of trained personnel who conducted studies in the protection of wildlife and popularised wildlife studies.
- WII is based in Dehradun. It is close to the forests of Dehradun.

20.4.3 Bombay Natural History Society (BNHS)

BNHS was founded in 1883 and is the largest NGO in India that is involved in the conservation of biodiversity and conducting research. It publishes the journal of BNHS and provides grants for research purposes. Prominent environmentalists like Salim Ali and S. Dillon Ripley have been a part of BNHS. BNHS is the partner of Birdlife International in India. The

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Department of Science and Technology has designated it as a 'Scientific and Industrial Research Organisation'.

20.4.4 National Tiger Conservation Authority (NTCA)

- The Wild Life (Protection) Act, 1972, was amended in 2006 for constituting the National Tiger Conservation Authority responsible for the implementation of the Project Tiger plan to protect endangered tigers.
- Chairman of NTCA is the Minister of Environment and Forests.
- NTCA is responsible for developing guidelines and standards for conservation initiatives in tiger reserves other than the national parks and sanctuaries.
- NTCA is responsible for providing information, tiger census, patrolling, disease surveillance, mortality survey and other aspects.
- NTCA will assist states in tiger reserve management through eco-development and ensuring participation of people in the governance of tiger reserves according to the pre-approved plans.
- NTCA is to prepare an annual report along with an audit that is laid down before parliament.
- To ensure coordination and monitoring, Tiger Task Force has recommended state-level steering committees to be headed by respective CMs.
- NTCA has a provision for the role of state government which includes capacity building of staff and their deployment while ensuring forest operation in the forest areas.
- There are safeguards in the provision for securing the rights of the people living in the areas of tiger reserves. These rights include agricultural, livelihood and development. The buffer zone of tiger reserve takes care of such activities.

20.4.5 Zoological Survey of India (ZSI)

ZSI was established in 1916 under MoEF as a national centre for conducting survey of animal species and conducting research for finding

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new animal species in the country. ZSI has its headquarters in Kolkata, and 16 regional offices are located in different geographic locations of the country. Activities carried out by ZSI are:

■ Activities

- To study animals in the country
- To study animals in conservation areas
- To study animals in important ecosystems
- To study the status of endangered species
- To conduct ecological studies and environmental impact assessments

■ ZSI Further Provide

- Advice to the government
- Training of personnel for conducting survey of Indian fauna.
- Library facilities.
- Presentation and publication of research work in journals and books.

20.4.6 Botanical Survey of India (BSI)

- BSI was founded in 1890 under MoEF for the survey of flora, conservation of plants, research and maintaining germplasm and gene bank of endangered species.
- BSI has several regional centres spread across India located at Dehradun, Jodhpur, Pune, Coimbatore, Hyderabad, Port Blair, Shillong, Itanagar, Gangtok and Allahabad, with another Botanic Garden of Indian Republic (BGIR) being developed at Noida. There are five organisational units located in Kolkata/Howrah.
- BSI was created in 1890 under the direction of Sir George King, who became its ex-officio Director.
- The Calcutta Garden became the headquarters of the flora survey and was given regional charge for Bengal, Andaman and Nicobar Islands, North East and Burma.

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- Before 1890, EIC has set up botanical gardens at Sibpur, Poona, Saharanpur and Madras for enhancing botanical knowledge and conducting research under local governments. For instance, the Saharanpur Botanical Garden was acquired by EIC in 1817 for growing medicinal plants.
- Many botanical gardens set up by EIC were for the commercial cultivation of plants for trade purposes.

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Forests for Life Partnership

It is an initiative launched by:

1. Global Wildlife Conservation
2. Rainforest Foundation Norway
3. UNDP
4. Wildlife Conservation Society
5. World Resources Institute.

It aims to:

- Promote forests as a nature-based solution to climate change and biodiversity protection.
- Halt and reverse forest degradation across 1 billion hectares of the most intact forests worldwide.

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IUCN & Threatened Species Categories

21

21.1 International Union Forconservation of Nature (IUCN)
21.2 Work of IUCN
21.3 IUCN Red Data Book: Threatened Species Classification
21.4 Other Divisions: Extinct, Functionally Extinct and Extinct in the Wild
21.5 Critically Endangered, Endangeredand Vulnerable Species in India
21.5.1 Critically Endangered
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21.6.1 Fishes
21.6.2 Amphibians
21.6.3 Reptiles
21.6.4 Birds
21.7 IUCN List of Indian Mammals (Critically Endangered / Endangered / Vulnerable)

21.1 INTERNATIONAL UNION FORCONSERVATION OF NATURE (IUCN)

IUCN is one of the oldest institutions for environmental conservation. It was founded in 1948. It has a network of professionals working in the conservation field. Members of IUCN include 200 governments and 900 NGOs. The headquarters of IUCN are in Gland, near Geneva, Switzerland.

More about IUCN

- The Council of IUCN governs the institution and its members are elected every 4 years in the IUCN World Conservation Congress.
- It is funded by corporations, governments, NGOs, etc.
- It has Observer Status at the United Nations General Assembly.

Biodiversity conservation is the main focus of IUCN. The main areas of function are:

1. Red List of Threatened Species published by IUCN.
2. Various conservation projects in the world.
3. Influence the policies through the collective strength of governments and NGOs.

QUICK INSIGHT

Invasive Species Specialist Group (ISSG)

The **GISD** is a **free, online searchable source** of information **about alien and invasive species** thats **negatively impact biodiversity**.

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The GISD aims to **increase public awareness** about invasive species and to **facilitate effective prevention and management activities**. It focuses on invasive alien species that threaten native biodiversity and natural areas and covers **all taxonomic groups from micro-organisms to animals and plants**.

The **GISD** is **managed by the ISSG of the IUCN Species Survival Commission**. It was developed between 1998 and 2000 as part of the global initiative on invasive species led by the erstwhile **Global Invasive Species Programme.**

21.2 WORK OF IUCN

A global programme is formed by IUCN members and their organisations every 4 years.

21.3 IUCN RED DATA BOOK: THREATENED SPECIES CLASSIFICATION

Any species that is feared to be extinct in the future is considered a threatened species. Depending on the level of threat, IUCN has divided the threatened species into categories such as critically endangered, endangered and vulnerable.

Critically Endangered Species

These species are at the highest threat of extinction; their population has decreased by 80% within a generation.

Endangered Species

These species are declining in population and risk the fear of extinction due to environmental changes, habitat fragmentation and lack of prey/food and water. Risk is lower than the Critically Endangered category.

Vulnerable Species

These species are likely to become endangered if necessary steps are not taken. Threat includes reproductive issues.

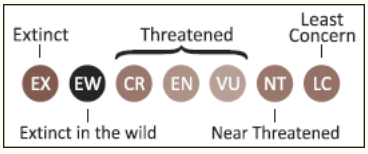

Figure 21.1 IUCN Categories

21.4 OTHER DIVISIONS: EXTINCT, FUNCTIONALLY EXTINCT AND EXTINCT IN THE WILD

When the last member of species is dead and there is no surviving individual remaining that can reproduce and further propagate, the generation is considered an extinct species.

The category 'Extinct in the wild' includes the species extinct in the forests and their native habitat and only known to exist in the human-controlled environment like zoos, and it is highly unlikely that they will ever be restored to their original habitat.

According to the IUCN Red Data List, there are approximately 1,000 threatened species in India.

21.5 CRITICALLY ENDANGERED, ENDANGEREDAND VULNERABLE SPECIES IN INDIA

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21.5.1 Critically Endangered

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Arthopoda

- Rameshwaram Parachute Spider
- Peacock tarantula

Birds

- White-bellied hereon
- Great Indian bustard
- Forest owlet
- Baer's pochard
- Spoon-billed sandpiper
- Siberian crane
- White-rumped vulture
- Indian vulture
- Slender-billed vulture
- Bengal florican
- Bugun liocichla
- Himalayan quail
- Jerdon's courser
- Pink-headed vulture
- Sociable lapwing

Fish

- *Wayanad mahseer*
- Pondicherry shark
- Ganges shark
- *Glyptothorax kashmirensis*
- *Kudermukh glyptothorax*
- *Nilgiri mystus*
- *Horalabiosa arunachalami*
- *Hypselobarbus pulchellus*
- Red canarese barb
- Deccan labeo
- *Mesonoemacheilus herrei*
- Bovany barb
- *Deolali minnow*
- Pookode lake barb
- Common sawfish
- Large-tooth sawfish
- Long-comb sawfish
- *Psilorhynchus tenure*
- Deccan barb
- *Schistura papulifera*

Insects

- Pygmy hog-sucking louse

Reptiles and Amphibians

- Madras spotted shrink
- Northern river terrapin

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- Red-crowned roofed turtle
- Cnemaspis anaikattinesis
- Hawksbill sea turtle
- Gharial
- Ghats wart frog
- Jeypore ground gecko
- Gundia Indian Frog
- Rao’s torrent frog
- Dattatreya night frog
- Sacred grove bush frog
- Amboli bush frog
- White-spotted bush frog
- Green-eyed bush frog
- Griet bush frog
- Kaikatti bush frog
- Mark’s bush frog
- Munnar bush frog
- Ponmudin bush Frog
- Resplendent shrub frog
- Shillong bubble-nest frog
- Anaimalai flying frog
- Sushil’s bush frog
- Amboli toad
- Ghats wart frog

Mammals

- Namdapha flying squirrel
- Himalayan wolf
- Elvira rat
- Andaman shrew
- Jenkins shrew
- Nicobar shrew
- Northern Sumatran rhinoceros
- Chinese pangolin
- Kondana soft-furred rat
- Pygmy hog
- Indian Javan rhinoceros
- Malabar large-spotted civet
- Kashmir stag/Hangul

21.5.2 Endangered

Fish

- Knife-tooth sawfish
- Asian arowana
- Red line torpedo barb
- Golden mahseer

Birds

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- Steppe Eagle
- Great Knot
- Masked Finfoot
- Greater Adjutant
- White-bellied Blue Robin
- Nilgiri Blue Robin
- White Headed Duck
- Green Peafowl
- Narcondam Hornbill
- Spotted Greenshank
- Banasura Laughingthrush

Reptiles

- Perrotet's Vine Snake
- Three-stripped Roofed Turtle
- Green Turtle
- Indian Narrow-headed softshell Turtle
- Goan Day Gecko
- Wyanad Day Gecko
- Keeled Box Turtle
- Boulenger's Dasia
- Poona Skink
- Inger's Mabuya
- Yellow-headed Tortoise
- Asian Forest Tortoise
- Indian Kangaroo Lizard
- Assam Roofed Turtle
- Asian Giant Softshell Turtle
- Travancore Hills Thorntail Snake
- Travancore Earth Snake
- Cochin Forest Cane Turtle

Mammals

- Red Panda
- Hispid Hare
- Dhole
- Lion-tailed macaque
- White-bellied musk deer
- Nilgiri tahr
- Gee's golden Langur
- Wild Ass
- Nilgiri Langur
- Chita/Tibetian Antelope
- Hog Deer
- Eid's Deer/brow-antlered deer

Marine Mammals

- Ganges river dolphin
- Irrawaddy river dolphin

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21.5.3 Vulnerable

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Mammals

- Gaur
- Yak
- Takin
- Barasingha
- Clouded leopard
- Dugong
- Sun bear
- Stump-tailed macaque
- Marbled Cat
- Sperm Whale
- Rusty Spotted Cat
- Indian Rhinoceros
- Snow Leopard
- Asian Elephant
- Asiatic Lion
- Bengal Tiger
- Sambar Deer (Nocturnal)

Birds

- Sarus Crane
- Nicobar Megapode
- Dalmatian Pelican

Reptiles and Amphibians

- Olive ridley sea turtle

21.6 CRITICALLY ENDANGERED (CE) SPECIES IN INDIA

CE species are those which are at the highest risk of extinction.

A taxon is critically endangered when the following conditions are met:

1. 80% decline in population over last 10 years or 3 generations.
2. Having a limited geographical habitat.
3. Having a population of less than 250 individuals and a decline of 25% in population in last 3 years or 1 generation.
4. Miniscule population of 50 mature individuals of species.
5. There exists a high probability of extinction in the wild.

21.6.1 Fishes

India has several fish species classified as **critically endangered** by the International Union for Conservation of Nature (IUCN), many of which are sharks, sawfish, and specific freshwater species endemic to regions like the Western Ghats.

Critically Endangered Marine and Estuarine Species

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Many of India's critically endangered fish are cartilaginous species (sharks and sawfish) that inhabit marine and brackish environments, making them highly vulnerable to overfishing.

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- **Pondicherry Shark** (*Carcharhinus hemiodon*): A very rare, little-known shark species, possibly extinct, last recorded in 1979. It is threatened by unregulated commercial fisheries in inshore habitats.
- **Ganges Shark** (*Glyphis gangeticus*): An exclusively fish-eating shark found in the muddy waters of the Ganges river system and the Bay of Bengal. It is threatened by targeted fisheries, increasing river use, pollution, and the construction of dams and barrages.
- **Large-tooth Sawfish** (*Pristis microdon*): This heavy-bodied sawfish, known for entering large rivers like the Mahanadi and the deltas of the Ganga and Brahmaputra, is highly vulnerable to entanglement in fishing gear due to its prominent, "toothed" snout.
- **Long-comb Sawfish/Narrow-snout Sawfish** (*Pristis zijsron*).
- **Knife-tooth Sawfish** (*Anoxypristis cuspidata*): Found in deltas and shallow coastal waters across the Indo-Pacific, it faces the same threats as other sawfish species.

■ *Critically Endangered Freshwater Species*

Several freshwater species, particularly in the biodiversity-rich Western Ghats, are also at extremely high risk due to habitat degradation, pollution, and dam construction.

- **Humpbacked Mahseer** (*Tor remadevii*): A large, iconic freshwater fish endemic to the Cauvery river system in the Western Ghats. It is endangered by a combination of factors including deforestation, drought, dam construction, and pollution.
- **Nilgiri Mystus** (*Hemibagrus punctatus*): A catfish species found in the Nilgiris and other parts of the Western Ghats.
- **Pookode Lake Barb** (*Pethia pookodensis*): Endemic to the single Pookode Lake in Kerala, it is highly vulnerable to any changes in its confined habitat.
- **Red Canarese Barb** (*Hypselobarbus thomassi*): An endemic species in the Western Ghats facing severe threats.
- **Deccan Barb** (*Puntius deccanensis*): A species from the Deccan region (possibly extinct from some locations).

Conservation efforts, including protection under India's Wildlife (Protection) Act, 1972, and the work of institutions like the National Bureau of Fish Genetic Resources, aim to prevent the extinction of these vital species

21.6.2 Amphibians

India has a significant number of critically endangered amphibian species, with reports from the Zoological Survey of India (ZSI) and IUCN assessments identifying around **20 species** in this category. The majority of these are frogs endemic to the Western Ghats and parts of Northeast India.

■ *Key Critically Endangered Amphibians*

Most of these species belong to the genera *Raorchestes* (bush frogs) and *Indirana* (Indian frogs/leaping frogs).

- ***Raorchestes resplendens* (Resplendent Shrubfrog):** Found exclusively in the high-altitude regions around the Anamudi peak in the Eravikulam National Park, Kerala. It is characterized by short limbs and bright coloration.
- ***Indirana gundia* (Gundia Indian Frog):** This species is found only in the Gundia river basin in the Western Ghats of Karnataka.
- ***Indirana phrynoderma* (Kerala Indian Frog/Toad-skinned Frog):** Endemic to a specific restricted area in the Kerala part of the Western Ghats, threatened by habitat degradation and pesticide use in surrounding plantations.
- ***Micrixalus kottigeharensis* (Kottigehar Bubble-nest Frog):** Part of the "dancing frog" genus, endemic to the Western Ghats, known for their unique "foot-flagging" behavior during mating.
- ***Ingerana charlesdarwini* (Charles Darwin's Frog):** Named after the famous naturalist, this species is highly endangered and found in restricted habitats in Northeast India.
- ***Xanthophryne tigerina* (Tiger Toad):** Found in the Amboli hills in Maharashtra.

Conservation efforts, such as the Amphibian Recovery Project by the Wildlife Trust of India, are actively working to mitigate these threats in biodiversity hotspots like the Western Ghats

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21.6.3 Reptiles

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Several reptile species in India are classified as **critically endangered** by the International Union for Conservation of Nature (IUCN). Many of these are freshwater turtles and marine turtles threatened by habitat loss, pollution, fishing activities, and illegal trade.

■ *Prominent Critically Endangered Reptiles in India*

- **Gharial** (*Gavialis gangeticus*): This unique, fish-eating crocodile with a long, thin snout is one of India's most iconic critically endangered species. Its population has declined drastically due to habitat loss from dams and barrages, sand-mining, and entanglement in fishing nets. The Chambal River Sanctuary is home to the largest breeding population.
- **Red-crowned Roofed Turtle** (*Batagur kachuga*): Also known as the Bengal Roof Turtle, this species is endemic to the Ganges river system. It is threatened by hunting, pollution, and the destruction of its sandbar nesting sites due to sand mining and agriculture. A reintroduction program has been started in the Haiderpur wetland complex.
- **Northern River Terrapin** (*Batagur baska*): This large river turtle is among the most severely threatened freshwater turtles in the world, with populations drastically reduced by hunting for meat and eggs, and accidental drowning in fishing gear.
- **Hawksbill Turtle** (*Eretmochelys imbricata*): A critically endangered sea turtle found in India's coastal areas. Its beautiful, patterned shell makes it a target for the illegal "tortoiseshell" trade.
- **Leatherback Sea Turtle** (*Dermochelys coriacea*): The largest of all living turtles, this marine giant faces threats from habitat loss on nesting beaches, climate change, and bycatch in fishing operations.
- **Sispara Day Gecko** (*Cnemaspis sisparensis*): This species of gecko is critically endangered due to localized habitat loss from the expansion of tea plantations in the Western Ghats.

Conservation efforts for these species often involve legal protection under India's Wildlife (Protection) Act, 1972 (Schedule I) and international treaties like CITES Appendix I.

■ *Indian Crocodilians*

A **crocodilian** is any predatory, semi-aquatic reptile belonging to the order *Crocodilia* (or *Crocodylia*). This order includes all true **crocodiles**, **alligators**, **caimans**, and the **gharial**.

India has three species of crocodilians, all protected under **Schedule I of the Wildlife (Protection) Act, 1972**. They were covered by the Indian Crocodile Conservation Project, launched in 1975 to revive populations through "rear and release" programs.

Table 21.1

Species	Scientific Name	IUCN Status	Habitat Preference
Gharial	Gavialis gangeticus	Critically Endangered	Pure freshwater rivers (Himalayan foothills)
Mugger Crocodile	Crocodylus palustris	Vulnerable	Freshwater (lakes, marshes, rivers) and coastal lagoons
Saltwater Crocodile	Crocodylus porosus	Least Concern	Brackish water, estuaries, and coastal areas

Table 21.2 Key Distinctions: Gharial vs. Crocodiles

Feature	Gharial (Gavialis gangeticus)	Mugger/Saltwater Crocodiles

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Snout	Very long, thin, and narrow, adapted for fishing	Broad, robust, V-shaped (Mugger) or V-shaped/pointed (Saltwater)
Diet	Exclusively fish-eaters (piscivorous). Jaws physically incapable of large prey	Generalist predators; feed on fish, mammals, birds, etc.
Male Feature	Adult males have a bulbous growth ('ghara' or pot) on the snout tip for vocalization	No such structure (ghara) is present in male crocodiles
Movement	Highly aquatic, move poorly on land (belly-sliding)	Agile on land, capable of "high walking"
Threat to Humans	Generally shy and not considered man-eaters	Territorial and can be aggressive towards humans

■ *Conservation Specifics*

Gharial

- **Significance:** An indicator of clean river water.
- **Primary Habitat in India:** National Chambal Sanctuary (tri-junction of UP, MP, and Rajasthan). Also found in Girwa, Gandak, and Son rivers.
- **Threats:** Dam construction, illegal sand mining, river pollution, and fishing nets.

Mugger Crocodile

- **Status:** Vulnerable. Population is stable or recovering in many areas.
- **Habitat:** Widely distributed across India in various freshwater bodies.
- **Threats:** Habitat loss and fragmentation, fishing activities.

Saltwater Crocodile

- **Status:** Least Concern.
- **Primary Habitat in India:** Bhitarkanika National Park (Odisha), Sundarbans (West Bengal), and Andaman & Nicobar Islands.
- **Conservation Success:** The "Baula Project" in Bhitarkanika helped the population rebound significantly from near extinction in the 1970s.

21.6.4 Birds

India is home to several bird species classified as **critically endangered** by the International Union for Conservation of Nature (IUCN). The primary threats include habitat loss, environmental contaminants, and hunting.

■ *Prominent Critically Endangered Bird Species*

Here are some of the most prominent critically endangered bird species found in India:

- **Great Indian Bustard** (*Ardeotis nigriceps*): Considered India's most threatened bird, with only about 150 individuals remaining, mostly in Rajasthan's Desert National Park. They are large, heavy birds highly vulnerable to collisions with power lines and habitat loss in grasslands.
- **Bengal Florican** (*Houbaropsis bengalensis*): A rare bustard species found in the grasslands of Uttar Pradesh, Assam, and Arunachal Pradesh. Fewer than 1,000 mature birds are estimated to exist globally, threatened by the conversion of grasslands to agriculture.
- **Jerdon's Courser** (*Rhinoptilus bitorquatus*): A nocturnal bird endemic to the Eastern Ghats of Andhra Pradesh, once thought extinct until its rediscovery in 1986. Its remaining habitat is threatened by scrub jungle clearing and

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construction projects.

- **Spoon-billed Sandpiper** (*Eurynorhynchus pygmeus*): A small migratory wader that breeds in Russia and winters in Southeast Asia, including coastal areas in West Bengal and Odisha. It faces extreme habitat loss in its specialized breeding and wintering grounds.
- **Forest Owlet** (*Heteroglaux blewitti*): Endemic to the forests of Central India, this small owl was considered extinct until 1997. Its population remains very low due to deforestation.
- **White-bellied Heron** (*Ardea insignis*): A large, dark grey heron found in the foothills of the Eastern Himalayas in Assam and Arunachal Pradesh. It is inherently rare and threatened by habitat degradation of undisturbed rivers and wetlands.

21.7 IUCN LIST OF INDIAN MAMMALS (CRITICALLY ENDANGERED / ENDANGERED / VULNERABLE)

Here is a comprehensive list of all critically endangered, endangered, and vulnerable mammals found in India, along with information on their location, food habits, main threats, IUCN category, and WPA 1972 Schedule.

Table 21.3 Critically Endangered Mammals

These species face an extremely high risk of extinction in the wild.

Species	Location/Distribution	Food	Reasons Behind Threat	IUCN Category	WPA 1972 Sche
Pygmy Hog	Assam (primarily Manas NP)	**Omnivorous:** Roots, tubers, insects, eggs, grass	Habitat loss/degradation (fires, livestock grazing), agriculture expansion	Critically Endangered	Sche I
Malabar Civet	Western Ghats (endemic to south Malabar coast)	**Omnivorous** (poorly studied wild diet); likely small animals, fruits, insects	Habitat destruction, hunting outside protected areas	Critically Endangered	Sche I
Namdapha Flying Squirrel COURSEWARE ON ENVIRONMENT, AGRICULTURE & DISASTER MANAGEMENT Click for a conceptual video	Arunachal Pradesh (known for one specimen in Namdapha NP)	**Arboreal Omnivore:** Fruits, leaves, seeds, nuts, possibly insects/fungi	Small population size, habitat loss due to logging	Critically Endangered	Sche I

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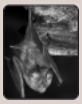
Species	Location/Distribution	Food	Reasons Behind Threat	IUCN Category	WPA 1972 Sche
Kashmir Stag (Hangul) COURSEWARE ON ENVIRONMENT, AGRICULTURE & DISASTER MANAGEMENT Click for a conceptual video	Kashmir Valley (Dachigam NP), parts of Himachal Pradesh	**Herbivorous:** Grasses, leaves, bark, twigs, fruits	Habitat degradation (livestock grazing), poaching, low breeding rates	Critically Endangered	Sche I
Andaman White-toothed Shrew, Jenkin's Shrew, Nicobar Shrew	Andaman and Nicobar Islands (endemic)	**Insectivorous** (insects and small invertebrates)	Habitat loss, limited distribution, natural disasters	Critically Endangered	Sche I
Kondana Rat & Large Rock Rat (Elvira Rat)	Maharashtra (Kondana Rat), Tamil Nadu (Large Rock Rat)	**Herbivorous** (seeds, grains, vegetation); nocturnal foragers	Very restricted habitat (type locality only), habitat modification	Critically Endangered	Sche I
Kolar Leaf-nosed Bat 	Karnataka (Kolar district caves)	**Insectivorous**	Habitat disturbance (granite mining), small population size	Critically Endangered	Sche I

Table 21.4 Endangered Mammals

These species face a very high risk of extinction in the wild.

Species	Location/Distribution	Food	Reasons Behind Threat	IUCN Category	WPA 1972 Schedule
Asian Elephant	Across India (Assam, Kerala, Karnataka,	**Herbivorous:** Grasses,	Habitat fragmentation,	Endangered	Schedule I

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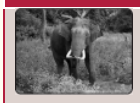
Species	Location/Distribution	Food	Reasons Behind Threat	IUCN Category	WPA 1972 Schedule
	etc.)	leaves, bark, fruits, crops	human-wildlife conflict, poaching for ivory		
Bengal Tiger	Various protected reserves across India	**Carnivorous** (large ungulates like deer, wild boar)	Poaching for body parts, habitat loss, human conflict	Endangered	Schedule I
Asiatic Lion	Gir National Park and surrounding areas in Gujarat	**Carnivorous** (primarily large ungulates)	Restricted gene pool (single location), accidental deaths (open wells, fences)	Endangered	Schedule I
Red Panda	Eastern Himalayas (Sikkim, W. Arunachal Pradesh)	**Herbivorous** (mainly bamboo shoots/leaves, berries, fruits)	Habitat loss/fragmentation, loss of nesting trees	Endangered	Schedule I
Lion-tailed Macaque	Western Ghats rainforests	**Omnivorous:** Fruits, nuts, seeds, insects, small mammals	Habitat loss (logging, roadkill), hunting, access to human food	Endangered	Schedule I
Indian Pangolin & Chinese Pangolin	Across India (Indian Pangolin), Northeast (Chinese)	**Insectivorous:** Ants and termites	Heavy poaching/trafficking for scales and meat	Endangered (Indian) / Critically Endangered (Chinese)	Schedule I

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Species	Location/Distribution	Food	Reasons Behind Threat	IUCN Category	WPA 1972 Schedule
Hoolock Gibbon (Western & Eastern)	Northeast India	**Frugivorous** (primarily fruits, leaves, flowers, insects)	Habitat destruction (deforestation), hunting	Endangered	Schedule I
Hispid Hare (Bristly Rabbit)	Foothills of the Himalayas (Assam, W. Bengal)	**Herbivorous:** Bark, shoots, roots of grasses	Habitat destruction for agriculture and human settlement	Endangered	Schedule I
Endangered Marine Mammals					
Ganges River Dolphin	Ganges-Brahmaputra-Meghna river systems (India, Nepal, Bangladesh, and Pakistan)	**Carnivorous:** Fish and crustaceans	Pollution, entanglement in fishing nets, dam/barrage construction	Endangered	Schedule I

Table 21.5 Vulnerable Mammals

These species face a high risk of extinction in the wild.

Species	Location/Distribution	Food	Reasons Behind Threat	IUCN Category	WPA 1972 Schedule
Greater One-horned Rhinoceros	Assam (Kaziranga NP, Orang NP, etc.), North Bengal, UP	**Herbivorous:** Grasses, leaves, aquatic plants	Poaching for horn, annual floods, human conflict	Vulnerable	Schedule I

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Species	Location/Distribution	Food	Reasons Behind Threat	IUCN Category	WPA 1972 Schedule
Indian Bison (Gaur)	South and Southeast Asia; largest populations in India	**Herbivorous:** Grasses, bark, leaves, fruits	Poaching for meat/horns, habitat loss, food scarcity	Vulnerable	Schedule I
Snow Leopard	Himalayas (J&K, Himachal Pradesh, Uttarakhand, Sikkim)	**Carnivorous**: Blue sheep, marmots, pikas, livestock	Decline in prey base, human conflict (retaliatory killings), illegal trade	Vulnerable	Schedule I
Nilgiri Tahr	Western Ghats (Kerala, Tamil Nadu)	**Herbivorous:** Grasses, leaves, shoots	Poaching, habitat loss/fragmentation, climate change impacts	Vulnerable	Schedule I
Dugong (Sea Cow)	Coastal waters (Gulf of Mannar, Gulf of Kachchh, A&N Islands)	**Herbivorous:** Seagrass and aquatic plants	Hunting (meat/oil), habitat degradation, fishing-related fatalities	Vulnerable	Schedule I
Clouded Leopard	Himalayan foothills, Northeast India	**Carnivorous**	Habitat loss and fragmentation due to deforestation and logging	Vulnerable	Schedule I

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Species Recovery Programme in India

22

IN THIS CHAPTER...

22.1 Species Recovery Programme
22.2 Species Under Idwh Scheme

22.1 SPECIES RECOVERY PROGRAMME

- The Species Recovery Programme of the Union Ministry is implemented under the Integrated Development of Wildlife Habitats – a centrally sponsored umbrella scheme for the management and conservation of parks, wildlife habitats and conservation.
- Started in 2008–09, Integrated Development of Wildlife Habitats (IDWH) is meant to provide support to protected areas (national parks, wildlife sanctuaries, conservation reserves and community reserves except for tiger reserves), protection of wildlife outside protected areas and recovery programmes for saving critically endangered species and habitats.
- The Cabinet Committee on Economic Affairs approved the continuation of the **Centrally Sponsored Umbrella Scheme of Integrated Development of Wildlife Habitats (CSS-IDWH)** beyond the 12th Plan period from 2017–18 to 2019–20.

Integrated Development of Wildlife Habitats (IDWH)

It is a Centrally Sponsored Scheme (CSS) launched by the Ministry of Environment, Forest and Climate Change (MoEF&CC), Government of India, to provide financial and technical assistance to states for wildlife conservation.

Key Objectives

The scheme aims to enhance the protection and management of wildlife and their habitats across the country. Its objectives include:

Habitat Management: Improving and restoring habitats within and outside Protected Areas (PAs).

Species Recovery: Supporting specific recovery programs for critically endangered species.

Human-Wildlife Conflict Mitigation: Addressing conflicts between local communities and wildlife to protect both populations.

Community Involvement: Engaging local communities in conservation efforts and providing alternative livelihood opportunities.

Components of the IDWH Scheme

The umbrella scheme has three main sub-components:

Support to Protected Areas: Provides financial assistance for National Parks, Wildlife Sanctuaries, Conservation Reserves, and Community Reserves.

Protection of Wildlife Outside PAs: Focuses on conservation efforts in landscapes beyond traditional protected areas.

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Recovery Programmes for Critically Endangered Species: Includes species-specific conservation plans for 22 identified species, such as the Snow Leopard, Asiatic Lion, Great Indian Bustard, Gangetic River Dolphin, and Gharial.

The IDWH scheme also serves as the umbrella under which major national conservation initiatives operate, including Project Tiger, Project Elephant, Project Lion, and Project Dolphin

22.2 SPECIES UNDER IDWH SCHEME

The **Integrated Development of Wildlife Habitats (IDWH) scheme** includes a specific component that funds recovery programs for **22 critically endangered species** across India.

These species are prioritised based on their threatened status and include:

Mammals:

1. Snow Leopard
2. Asiatic Lion (part of Project Lion)
3. One-horned Rhinoceros (part of Project Rhino)
4. Swamp Deer (Hard Ground & Eastern Swamp Deer)
5. Wild Buffalo
6. Manipur Brow-antlered deer (Eld's deer)
7. Pygmy Hog
8. Asiatic Wild Dog (Dhole)
9. Ganges River Dolphin (part of Project Dolphin)
10. Finless Porpoise
11. Malabar Civet
12. Indian Wolf

Birds:

13. Great Indian Bustard
14. Bengal Florican
15. Lesser Florican
16. Vultures (multiple species like the White-rumped Vulture)
17. Jerdon's Courser
18. Forest Owlet

Reptiles:

19. Gharial
20. Marine Turtles (multiple species, including Leatherback and Hawksbill)
21. Northern River Terrapin

Fish:

22. Pondicherry Shark

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Below is a detailed overview of the 22 species under India's Species Recovery Programme (part of the IDWH scheme), providing details on their habitat, distribution, threats, conservation status (IUCN, WPA 1972, CITES), and unique facts.

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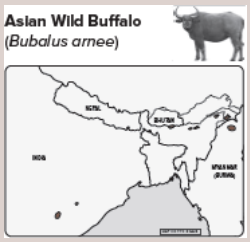
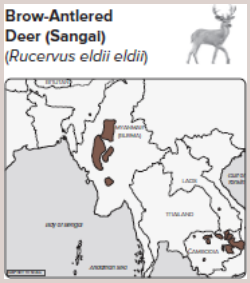
All listed species are included in Schedule I of the Wildlife (Protection) Act, 1972, granting them the highest level of protection, and most are listed on CITES Appendix I, which bans international commercial trade.

Species Name	Habitat & Distribution	Threats	IUCN Category	CITES Status	Uni
Asian Wild Buffalo (*Bubalus arnee*) 	Tall wet grasslands, swamps, and river valleys in Assam, Arunachal Pradesh, & Chhattisgarh.	Hybridization with domestic buffalo, habitat loss, diseases from livestock.	Endangered	Appendix I/III	Pro anc don buf ani Chh
Asiatic Lion(*Panthera leo persica*)	Dry deciduous forest ecosystem (Gir NP & surrounding areas) in Gujarat.	Single population location (vulnerable to disease/calamity), accidental deaths (open wells, trains)	Endangered	Appendix I	Onl occ pop lion Afri
Brow-Antlered Deer (Sangai)(*Rucervus eldii eldii*) 	Floating grasslands ("Phumdis") in Keibul Lamjao NP, Manipur	Habitat destruction (flooding/sinking of phumdis), poaching	Endangered	Appendix I	Kno "da the **nat** the hab
Dugong (Sea Cow) (*Dugong dugon*)	Shallow coastal marine habitats with seagrass beds (Gulf of	Hunting for meat/oil, habitat degradation of seagrass beds,	Vulnerable	Appendix I	The her ma reli

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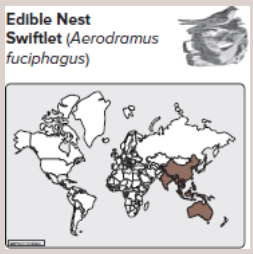
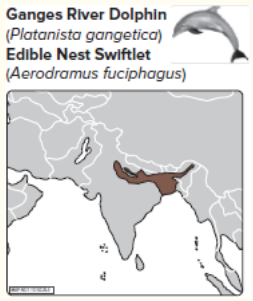
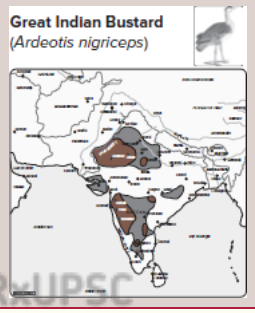
Species Name	Habitat & Distribution	Threats	IUCN Category	CITES Status	Uni
	Mannar, Kachchh, A&N).	entanglement in fishing gear.			sea sur
Edible Nest Swiftlet (*Aerodramus fuciphagus*) 	Coastal areas, nesting in dark sea caves in the Andaman & Nicobar Islands.	Poaching of nests for human consumption (used in bird's nest soup); habitat disturbance.	Least Concern (Globally)	Appendix II	Bui enti sali priz cuis
Ganges River Dolphin(*Platanista gangetica*) **Edible Nest Swiftlet**(*Aerodramus fuciphagus*) 	Ganges-Brahmaputra-Meghna river systems in India, Nepal, Bangladesh.	Pollution, dams/barrages (isolating populations), entanglement in fishing nets.	Endangered	Appendix I	Indi Aqu fun use ech nav
Great Indian Bustard (*Ardeotis nigriceps*) 	Arid grasslands & scrublands in Rajasthan and Gujarat.	Habitat loss, predation by feral dogs, collision with power lines.	Critically Endangered	Appendix I	One hea birc spe gra

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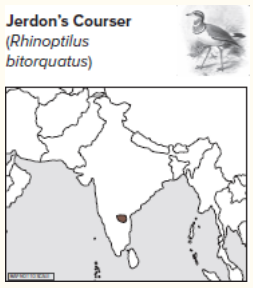
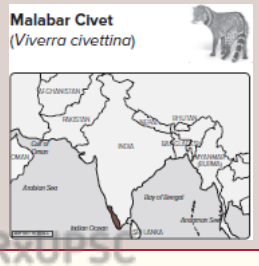
Species Name	Habitat & Distribution	Threats	IUCN Category	CITES Status	Uni
Hangul (Kashmir Stag) (*Cervus hanglu*)	Temperate forests & high-altitude valleys in Dachigam NP, J&K.	Habitat degradation (livestock grazing), poaching, low calf survival rate.	Critically Endangered	Appendix I	The of r in I
Indian Rhinoceros (*Rhinoceros unicornis*)	Floodplains, tall grasslands, and swampy areas in Assam, West Bengal, UP.	Poaching for its horn, habitat loss due to flooding and human settlement.	Vulnerable	Appendix I	The rhin only
Jerdon's Courser (*Rhinoptilus bitorquatus*) 	Scrub jungles with open patches, endemic to the Eastern Ghats in Andhra Pradesh.	Habitat clearance for development, logging, illegal hunting.	Critically Endangered	Appendix I	A r red 198 thou for
Malabar Civet(*Viverra civettina*) 	Wooded plains and hill slopes of the Western Ghats (South Malabar).	Habitat destruction and fragmentation, hunting outside protected areas.	Critically Endangered	Appendix I	A ra and elus cive

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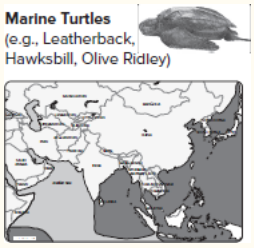
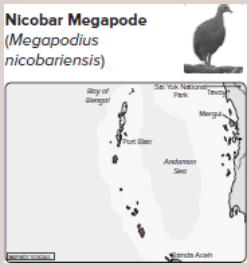
Species Name	Habitat & Distribution	Threats	IUCN Category	CITES Status	Uni
Marine Turtles(e.g., Leatherback, Hawksbill, Olive Ridley) 	Indian coastal waters, nesting beaches across East & West coasts.	Fishing net entanglement, plastic pollution ingestion, coastal development, egg poaching.	CE, EN, VU	Appendix I	Anc rep Rid kno nes (Arr
Nicobar Megapode(*Megapodius nicobariensis*) 	Coastal forests & scrublands of the Nicobar Islands.	Habitat destruction (tsunami effects, human settlement), egg collection, predation by invasive species.	Vulnerable	Appendix I	Gro birc larg nes veg soil egg
Nilgiri Tahr (*Nilgiritragus hylocrius*)	Open montane grasslands & rocky cliffs (“sholas”) of the Western Ghats (Kerala, TN).	Poaching, habitat loss/fragmentation, invasive plant species.	Endangered	Appendix I	Indi goa stat Tan
Snow Leopard (*Panthera uncia*)	High mountains of the Himalayas (J&K, HP, Uttarakhand, Sikkim, Arunachal).	Poaching for fur/bones, retaliatory killings by herders, climate change impact on habitat.	Vulnerable	Appendix I	Kno “Gh Mou to natu

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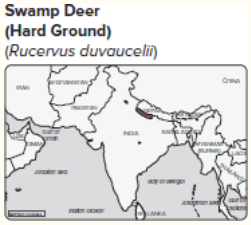
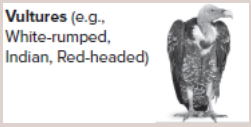
Species Name	Habitat & Distribution	Threats	IUCN Category	CITES Status	Uni
Swamp Deer (Hard Ground) (*Rucervus duvaucelii*) 	Alluvial grasslands & moist deciduous forests in Central India (Kanha NP, MP) and UP.	Habitat destruction/fragmentation, poaching, conversion of grasslands to agriculture.	Vulnerable	Appendix I	Onl wor sex (tho hav one
Vultures (e.g., White-rumped, Indian, Red-headed) 	Across India, favoring open areas near human habitation for scavenging.	Poisoning from the veterinary drug Diclofenac, habitat loss.	Critically Endangered	Appendix II	Nat sca to hea con car pre dis
Northern River Terrapin (*Batagur baska*)	Brackish coastal mangrove estuaries (Sundarbans in India/Bangladesh).	Poaching for meat & eggs, entanglement in fishing nets, habitat degradation.	Critically Endangered	Appendix I	One wor end turt hist the fres turt
Clouded Leopard (*Neofelis nebulosa*)	Himalayan foothills, Northeast India (Meghalaya, Assam, Sikkim).	Habitat loss/fragmentation (logging), poaching for skin/bones.	Vulnerable	Appendix I	Has sha and disp lon size toot rela

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Species Name	Habitat & Distribution	Threats	IUCN Category	CITES Status	Uni
Arabian Sea Humpback Whale (*Megaptera novaeangliae*)	Arabian Sea coastal waters off India's west coast.	Ship strikes, noise pollution, entanglement in fishing gear, small isolated population size.	Endangered	Appendix I	Beli one isol gen dist hun pop glol
Red Panda (*Ailurus fulgens*)	Montane forests with dense bamboo understory in Eastern Himalayas (Sikkim, W. Arunachal, Darjeeling).	Habitat loss/fragmentation (logging), poaching, livestock grazing.	Endangered	Appendix I	Sol its taxo (Ail prin ban
Caracal (*Caracal caracal*)	Arid/semi-arid regions, scrub forests in Central and West India (Rajasthan, Gujarat, MP).	Poaching/hunting, habitat loss, conflict with humans.	Near Threatened (Globally); Critically Endangered in India	Appendix I	Dist poi tuft it loca pre

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Questions Asked in UPSC Prelims Exam

UNIT VIII

Unit - VIII Prelims PYQs (2011 onwards)

1. Regarding Peacock tarantula (Gooty tarantula), consider the following statements:

I. It is an omnivorous crustacean.

II. Its natural habitat in India is only limited to some forest areas.

III. In its natural habitat, it is an arboreal species. Which of the statements given above is/are correct?

(a) I only (b) I and III

(c) II only (d) II and III

2. Consider the following statements: **[UPSC (Pre) 2024]**

Statement-I: The Indian Flying Fox is placed under the "vermin" category in the Wildlife (Protection) Act, 1972.

Statement-II: The Indian Flying Fox feeds on the blood of other animals. Which one of the following is correct in respect of the

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above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I

(c) Statement-I is correct, but Statement-II is incorrect

(d) Statement-I is incorrect, but Statement-II is correct

3. Consider the following pairs: **[UPSC (Pre) 2024]**

Country ***Animal found in its natural habitat***

1. Brazil Indri
2. Indonesia Elk
3. Madagaskar Bonobo

How many of the pairs given above are correctly matched?

(a) Only one (b) Only two

(c) All three (d) None

4. Consider the following statements:[UPSC (Pre) 2024]

1. Lions do not have a particular breeding season.
2. Unlike most other big cats, cheetahs do not roar.
3. Unlike male lions, male leopards do not proclaim their territory by scent marking.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

5. Consider the following: **[UPSC (Pre) 2024]**

1. Cashew
2. Papaya

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3. Red sanders

How many of the above trees are actually native to India?

(a) Only one (b) Only two

(c) All three (d) None

6. Invasive Species Specialist Group (that develops the Global Invasive Species Database) belongs to which one of the following organisations? **[UPSC (Pre) 2023]**

(a) The International Union for Conservation of Nature

(b) The United Nations Environment Programme

(c) The United Nations Commission for Environment and Development

(d) The World Wide Fund for Nature

7. Consider the following fauna: **[UPSC (Pre) 2023]**

1. Lion tailed Macaque
2. Malabar Civet
3. Sambar Deer

How many of the above are generally nocturnal or most active after sunset?

(a) 1 only (b) 2 only

(c) All three (d) None

8. Which of the following is not a bird? **[UPSC (Pre) 2022]**

(a) Golden Mahseer

(b) Indian Nightjar

(c) Spoonbill

(d) White Ibis

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9. With reference to Indian laws about wildlife protection, consider the following statements: [UPSC (Pre) 2022]

1. Wild animals are the sole property of the government.
2. When a wild animal is declared protected, such an animal is entitled to equal protection whether it is found in protected areas or outside.
3. Apprehension of a protected wild animal becoming a danger to human life is sufficient ground for its capture or killing.

Which of the statement/s given above is/are correct?

(a) 1 and 2 (b) 2 only

(c) 1 and 3 (d) 3 only

10. With reference to India's Desert National Park, which of the following statements are correct? [UPSC (Pre) 2020]

1. It is spread over two districts.
2. There is no human habitation inside the Park
3. It is one of the natural habitats of the Great Indian Bustard.

Select the correct answer using the codes given below:

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

11. Among the following Tiger Reserves, which one has the largest area under 'Critical Tiger Habitat'? [UPSC (Pre) 2020]

(a) Corbett

(b) Ranthambore

(c) Nagarjunsagar-Srisailam

(d) Sunderbans

12. If a particular plant species is placed under Schedule VI of The Wildlife Protection Act, 1972, what is the implication?[UPSC (Pre)

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(a) A licence is required to cultivate that plant.

(b) Such a plant cannot be cultivated under any circumstances.

(c) It is a genetically modified crop plant.

(d) Such a plant is invasive and harmful to the ecosystem.

13. With reference to India's biodiversity, Ceylon frogmouth, Coppersmith Barbet, Graychinned minivet and White-throated redstart are: [UPSC (Pre) 2020]

(a) Birds (b) Primates

(c) Reptiles (d) Amphibians

14. Which one of the following protected areas is well- known for the conservation of a subspecies of the Indian swamp deer (Barasingha) that thrives well on hard ground and is exclusively grass in diet? [UPSC (Pre) 2020]

(a) Kanha National Park

(b) Manas National Park

(c) Mudumalai Wildlife Sanctuary

(d) Tal Chhapar Wildlife Sanctuary

15. Which of the following are the most likely places to find the musk deer in its natural habitat? [UPSC (Pre) 2020]

1. Askot Wildlife Sanctuary
2. Gangotri National Park
3. Kishanpur Wildlife Sanctuary
4. Manas National Park

Select the correct answer using the codes given below:

(a) 1 and 2 only (b) 2 and 3 only

(c) 3 and 4 only (d) 1 and 4 only

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16. Consider the following statements: **[UPSC (Pre) 2019]**

1. Asiatic lion is naturally found in India only.
2. A double-humped camel is naturally found in India only.
3. One-horned rhinoceros is naturally found in India only. Which of the statement/s given above is/are correct?

(a) 1 only (b) 2 only

(c) 1 and 3 only (d) 1, 2 and 3

18. In India, if a species of tortoise is declared protected under Schedule 1 of the Wildlife (Protection) Act, 1972, what does it imply? **[UPSC (Pre) 2017]**

(a) It enjoys the same level of protection as the tiger.

(b) It no longer exists in the wild, a few individuals are under captive protection; and now it is impossible to prevent its extinction.

(c) It is endemic to a particular region of India.

(d) Both (b) and (c) stated above are correct in this context.

19. According to the Wildlife (Protection) Act, 1972, which of the following animals cannot be hunted by any person except under some provisions provided by law? **[UPSC (Pre) 2017]**

1. Gharial
2. Indian wild ass
3. Wild buffalo

Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

20. The term 'M-STrIPES' is sometimes seen in the news in the context of **[UPSC (Pre) 2017]**

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(a) Captive breeding of Wild Fauna
(b) Maintenance of Tiger Reserves
(c) Indigenous Satellite Navigation System
(d) Security of National Highways

21. If you want to see gharials in their natural habitats, which one of the following is the best place to visit? **[UPSC (Pre) 2017]**

(a) Bhiterkanika Mangroves
(b) Chambal River
(c) Pulicat Lake
(d) Deepor Beel

22. Due to some reasons, if there is a huge fall in the population of species of butterflies, what could be its likely consequence/consequences? **[UPSC (Pre) 2017]**

1. Pollination of some plants could be adversely affected.
2. There could be a drastic increase in the fungal infections of some cultivated plants.
3. It could lead to a fall in the population of some species of wasps, spiders and birds.

Select the correct answer using the codes given below:
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

23. What is/are unique about the 'Kharai camel', a breed found in India? **[UPSC (Pre) 2016]**

.i. It is capable of swimming up to three kilometres in seawater.
ii. It survives by grazing on mangroves.

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iii. It lives in the wild and cannot be domesticated. Select the correct answer using the codes given below:

(a) 1 and 2 only (b) 3 only

(c) 1 and 3 only (d) 1, 2 and 3

24. With reference to 'Red Sanders' sometimes seen in the news, consider the following statements: [UPSC (Pre) 2016]

1. It is a tree species found in a part of South India.
2. It is one of the most important trees in the tropical rainforest areas of South India. Which of the statement/s given above is/are correct:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

25. In which of the following regions of India do you most likely to come across the 'Great Indian Hornbill' in its natural habitat? [UPSC (Pre) 2016]

(a) Sand deserts of northwest India

(b) Higher Himalaya of Jammu and Kashmir

(c) Salt marshes of western Gujarat

(d) Western Ghats

26. Which of the following is the national aquatic animal of India? [UPSC (Pre) 2015]

(a) Salt-water crocodile

(b) Olive ridley turtle

(c) Gangetic Dolphin

(d) Gharial

27. Concerning 'dugong', a mammal found in India, which of the following statement/s is/ are correct? [UPSC (Pre) 2015]

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1. It is a herbivorous marine animal.
2. It is found along the entire coast of India.
3. It is given legal protection under Schedule 1 of the Wildlife (Protection) Act, 1972. Select the correct answer using the codes given below:

(a) 1 and 2 (b) 2 only
(c) 1 and 3 (d) 3 only

28. Consider the following statements:[UPSC (Pre) 2014]

1. Animal Welfare Board of India is established under the Environment (Protection) Act, 1986.
2. National Tiger Conservation Authority is a statutory body.
3. National Ganga River Basin Authority is chaired by the Prime Minister. Which of the statement/s given above is/are correct?

(a) 1 only (b) 2 and 3 only
(c) 2 only (d) 1, 2 and 3

29. Other than poaching, what are the possible reasons for the decline in the population of Ganga River Dolphins? [UPSC (Pre) 2014]

1. Construction of dams and barrages on rivers
2. Increase in the population of crocodiles in rivers
3. Getting trapped in fishing nets accidental
4. Use of synthetic fertilizers and other agricultural chemicals in crop fields in the vicinity of rivers

Select the correct answer using the codes given below:

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1, 3 and 4 only
(d) 1, 2, 3 and 4

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30. The most important strategy for the conservation of biodiversity together with traditional human life is the establishment of **[UPSC (Pre) 2014]**

(a) Biosphere Reserves
(b) Botanical Gardens
(c) National Parks
(d) Wildlife Sanctuaries

31. In which of the following states is lion-tailed macaque found in its natural habitat? **[UPSC (Pre) 2013]**

i. Tamil Nadu ii. Kerala
iii. Karnataka iv. Andhra Pradesh
Select the correct answer using the codes given below:
(a) 1, 2 and 3 (b) 2 only
(c) 1, 3 and 4 (d) All of these

32. Consider the following: **[UPSC (Pre) 2013]**

i. Star tortoise
ii. Monitor lizard
iii. Pygmy hog
iv. Spider monkey
Which of the above are naturally found in India?
(a) 1, 2 and 3 (b) 2 and 3
(c) 1 and 4 (d) All of these

33. Consider the following fauna of India: **[UPSC (Pre) 2013]**

i. Gharial
ii. Leatherback turtle
iii. Swamp deer
Which of the above is/are endangered?

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(a) 1 and 2 (b) 3 only
(c) All of these (d) None of these

34. Out of all the Biosphere Reserve in India, four have been recognized on the world Network by UNESCO. Which one of the following is not one of them? **[UPSC (Pre) 2013]**

(a) Gulf of Mannar
(b) Kanchenjunga (declared in 2018)
(c) Nanda Devi
(d) Sunderbans

35. Match List-I (Indian Wild Life species) with List-II (Scientific names) and select the correct answer using the codes given below: **[UPSC (Pre) 2012]**

List-I (Indian Wild Life Species)	List-II (Scientific Names)
A. Asiatic Wild Ass	1. Boselaphus tragocamelus
B. Barasingha	2. Cervus duvauceli
C. Chinkara	3. Equus hemionus
D. Nilgai	4. Gazella benetty

(a) 2 3 1 4 (b) 3 2 4 1
(c) 2 3 4 1 (d) 3 2 1 4

36. Consider the following: **[UPSC (Pre) 2012]**

i. Black-necked crane
ii. Cheetah
iii. Flying squirrel
iv. Snow leopard

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Which of the above are naturally found in India?

(a) 1, 2 and 3 (b) 1, 3 and 4

(c) 2 and 4 (d) 1, 2, 3 and 4

37. Vultures which used to be very common in the Indian countryside some years ago are rarely seen nowadays. This is attributed to: [UPSC (Pre) 2012]

(a) The destruction of their nesting sites by new invasive species

(b) A drug used by cattle owners for treating their diseased cattle

(c) Scarcity of food available to them

(d) A widespread, persistent and fatal disease among them

38. Which one of the followings groups of animals belongs to the category of endangered species? [UPSC (Pre) 2012]

(a) Great Indian Bustard, Musk Deer, Red Panda and Asiatic Wild Ass

(b) Kashmir Stag, Cheetal, Blue Bull and Great Indian Bustard

(c) Snow Leopard, Swamp Deer, Rhesus Monkey and Saras (Crane)

(d) Lion-tailed Macaque, Blue Bull, Hanuman Langur and Cheetal

39. In which one among the following categories of protected areas in India are local people not allowed to collect and use the biomass? [UPSC (Pre) 2012]

(a) Biosphere Reserve

(b) National Parks

(c) Wetlands declared under Ramsar Convention

(d) Wildlife Sanctuaries

40. Which one of the following is not a site for an in-situ method of conservation of flora? [UPSC (Pre) 2011]

(a) Biosphere Reserve

(b) Botanical Garden

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(c) National Park
(d) Wildlife Sanctuary

41. Biodiversity forms the basis for human existence in the following ways: **[UPSC (Pre) 2011]**

i. Soil formation
ii. Prevention of soil erosion
iii. Recycling of waste
iv. Pollination of crops

Select the correct answer using the codes given below: Codes:
(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1 and 4 (d) 1, 2, 3 and 4

42. The 'Red Data Books' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of: **[UPSC (Pre) 2011]**

i. Endemic plant and animal species present in the biodiversity hotspots.
ii. Threatened plant and animal species.
iii. Protected sites for conservation of nature and natural resources in various countries.

Select the correct answer using the codes given below?
(a) 1 and 3 (b) 2 only
(c) 2 and 3 (d) 3 only

43. Three of the following criteria have contributed to the recognition of Western Ghats, Sri Lanka and lndo- Burma regions as hotspots of biodiversity: **[UPSC (Pre) 2011]**

i. Species richness
ii. Vegetation density

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iii. Endemism
iv. Ethno-botanical importance
v. Threat perception
vi. Adaptation of flora and fauna to warm and humid conditions

Which three of the above are correct criteria in this context:

(a) 1, 2 and 6 (b) 2, 4 and 6
(c) 1, 3 and 5 (d) 3, 4 and 6

44. A sandy and saline area is the natural habitat of an Indian animal species. The animal has no predators in that area but its existence is threatened due to the destruction of its habitat. Which one of the following could be that animal? **[UPSC (Pre) 2011]**

(a) Indian wild buffalo
(b) Indian wild ass
(c) Indian wild boar
(d) Indian gazelle

45. Two important rivers-one with its source in Jharkhand (and known by a different name in Odisha) and another with its source in Odisha – merge at a place only a short distance from the coast of the Bay of Bengal before flowing into the sea. This is an important site of wildlife and biodiversity and a protected area. Which one of the following could be this? **[UPSC (Pre) 2011]**

(a) Bhitarkanika
(b) Chandipur-on-sea
(c) Gopalpur-on-sea
(d) Simlipal

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46. Consider the following statements: **[UPSC (Pre) 2010]**

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(i) Biodiversity hotspots are located only in tropical regions.

ii. India has four biodiversity hotspots in Eastern Himalayas, Western Himalayas, Western Ghats and Andaman and Nicobar Islands.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both l and 2 (d) Neither 1 nor 2

47. Consider the following statements: **[UPSC (Pre) 2010]**

i. The boundaries of a National Park are defined by legislation.

ii. A Biosphere Reserve is declared to conserve a few specific species of flora and fauna.

iii. In a Wildlife Sanctuary, limited biotic interference is permitted.

Which of the statement/s given above is/are correct?

(a) 1 only (b) 2 and 3

(c) 1 and 3 (d) 1, 2 and 3

Answers

1. (d) 2. (c) 3. (d) 4. (a) 5. (a) 6. (a) 7. (b) 8. (a) 9. (a) 10. (c)
11. (c) 12. (a) 13. (a) 14. (a) 15. (a) 16. (a) 17. (c) 18. (a) 19. (d) 20. (b)
21. (b) 22. (c) 23. (a) 24. (a) 25. (d) 26. (c) 27. (c) 28. (b) 29. (c) 30. (a)
31. (a) 32. (a) 33. (a) 34. (b) 35. (b) 36. (b) 37. (b) 38. (a) 39. (b) 40. (b)
41. (d) 42. (b) 43. (c) 44. (b) 45. (a) 46. (d) 47. (c)

Explanations

1. Correct answer (d): EXPLANATION

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It is a **spider**, not a crustacean, and it is **carnivorous**, feeding on insects and small invertebrates. **Statement (1) wrong** Endemic to a **tiny area (~3 km²) of deciduous forest in Andhra Pradesh**, India. **Statement (2) right**

It lives **in trees and crevices**, making it an **arboreal spider**. **Statement (3) right**

2. **Correct answer (c): EXPLANATION**

Statement-I: The Indian Flying Fox is placed under the "vermin" category in the Wildlife (Protection) Act, 1972. This statement is **correct**. The Indian Flying Fox (*Pteropus giganteus*), or large fruit bat, is listed in Schedule V (Vermin) of the Wildlife (Protection) Act, 1972. This classification is primarily due to the damage they cause to fruit orchards and crops, which permits their culling or hunting under specific conditions to control populations.

Statement-II: The Indian Flying Fox feeds on the blood of other animals. This statement is **incorrect**. The Indian Flying Fox is a frugivorous bat, meaning its diet consists mainly of fruits, nectar, and flowers. It plays an important ecological role as a pollinator and seed disperser. Vampire bats, which feed on blood (hematophagy), belong to a different family found exclusively in the Americas and are not present in India

3. **Correct answer (d): EXPLANATION**

1. Brazil – Indri - Incorrect

• The **Indri** is a lemur species **endemic to Madagascar**, not Brazil.

2. Indonesia – Elk - Incorrect

• **Elk** (Cervus canadensis) are native to **North America and parts of Asia**, but **not Indonesia**.

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3. Madagascar – Bonobo - Incorrect

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• **Bonobos** are found **only in the Democratic Republic of Congo**, not Madagascar.

4. Correct answer (a): EXPLANATION

Lions are **polyestrous**, meaning they can breed **any time of the year**. **Statement (1) right**

Cheetahs (**Acinonyx jubatus**) cannot roar; they **chirp, purr, or hiss**, unlike lions, tigers, leopards, and jaguars. **Statement (2) right**

Male leopards **do scent mark** their territory using urine, feces, and scratch marks. Only the form of territorial marking may differ from lions. **Statement (3) wrong**

5. Correct answer (a): EXPLANATION

Let's examine each species:

1. Cashew (*Anacardium occidentale*) – Not native; originally from **Brazil**, introduced to India by the Portuguese.

2. Papaya (*Carica papaya*) – Not native; native to **Central America**, introduced to India.

3. Red Sanders (*Pterocarpus santalinus*) – ü Native; **endemic to southern Eastern Ghats of India** (Andhra Pradesh).

So, only **Red Sanders** is native to India.

6. Correct answer (a): EXPLANATION

The **Invasive Species Specialist Group (ISSG)**, which develops and maintains the **Global Invasive Species Database**, is part of: **The International Union for Conservation of Nature (IUCN)**

7. Correct answer (b): EXPLANATION

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Let's analyze the activity patterns of the given animals:

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1. Lion-tailed Macaque – Diurnal - Active during the day, not nocturnal

2. Malabar Civet – Nocturnal - Mostly active at night

3. Sambar Deer – Crepuscular/nocturnal - Active during dusk and night

So, **Malabar Civet** and **Sambar Deer** are generally **active after sunset**.

8. Correct answer (a): EXPLANATION

Among the options:

- **Golden Mahseer** – Not a bird; it is a **freshwater fish** found in rivers of India.
- **Indian Nightjar** – Bird
- **Spoonbill** – Bird
- **White Ibis** – Bird

9. Correct answer (a): EXPLANATION

Section 9 of the Wildlife (Protection) Act states that **all wild animals are the property of the government**, irrespective of where they are found. **Statement (1) right**

The Act provides protection **both inside and outside protected areas**, i.e., hunting or harming a protected species is illegal everywhere. **Statement (2) right**

Capture or killing is **not automatically permitted**. Only authorized officials under specific provisions of the Act can act to prevent danger, after following due procedure. Mere apprehension is **not sufficient**. **Statement (3) wrong**

10. Correct answer (c): EXPLANATION

DNP covers parts of **Jaisalmer and Barmer districts**. **Statement (1) right**

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Some **human settlements and villages exist within the buffer zones and fringes** of the park. Complete absence of humans is not true. **Statement (2) wrong**

DNP is an important habitat for the **Critically Endangered Great Indian Bustard**. **Statement (3) right**

11. Correct answer (c): EXPLANATION

The **Critical Tiger Habitat (CTH)** is the core area within a tiger reserve, legally notified under the **Wildlife (Protection) Act** as essential for tiger conservation.

Among the given tiger reserves:

- **Corbett** – CTH ~ 500 km²
- **Ranthambore** – CTH ~ 350 km²
- **Nagarjunsagar-Srisailam** – CTH ~ **1,400 km²** Largest
- **Sunderbans** – CTH ~ 2,585 km², but it is a **mangrove reserve**, area of tiger habitat is smaller compared to NSS in forested terms

12. Correct answer (a): EXPLANATION

- **Schedule VI of The Wildlife Protection Act, 1972**, lists specific plant species that are rare and threatened, such as Beddomes' cycad, various Vanda orchids, and the Pitcher plant.
- The primary implication of a plant being placed on this schedule is that its collection, possession, sale, transport, and cultivation are strictly regulated.
- **Section 17C of the Act** stipulates that no one shall cultivate such a plant except under and in accordance with a **licence** granted by the Chief Wild Life Warden of the state.
- Therefore, cultivation is not completely prohibited but is permitted under a special license, making option (a) the correct answer.

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13. Correct answer (a): EXPLANATION

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All the species mentioned—**Ceylon frogmouth, Coppersmith Barbet, Gray - chinned minivet, and White-throated redstart**—are **birds**.

14. Correct answer (a): EXPLANATION

- The question specifically asks for the protected area well- known for the **"hard ground"** subspecies of the Indian swamp deer (Barasingha), which is exclusively **graminivorous (grass-eating).**
- **Kanha National Park** in Madhya Pradesh is the *exclusive* natural habitat and conservation success story for this particular subspecies (*Rucervus duvaucelii branderi*). The park's management successfully revived its population from the brink of extinction.
- The other subspecies of **Barasingha (wetland/eastern)** are found in swampy grassland areas of North and North-Eastern India, such as parts of **Dudhwa** and **Kaziranga National Park**s, which does not fit the "hard ground" description.
- **Manas National Park (Assam)** is known for other species like the Pygmy Hog and Assam Roofed Turtle, and while it has a swamp deer population, it's the eastern subspecies, not the hard-ground one.
- **Mudumalai Wildlife Sanctuary (Tamil Nadu) and Tal Chhapar Wildlife Sanctuary (Rajasthan)** are not known habitats for any subspecies of the Indian swamp deer.

15. Correct answer (a): EXPLANATION

The **musk deer** (*Moschus spp.*) is a shy, mountain-dwelling species found in the **Himalayan region** of India, preferring alpine forests and high-altitude areas.

- **Askot Wildlife Sanctuary (Uttarakhand)** – Suitable habitat, musk deer present. **Statement (1) right**

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- **Gangotri National Park (Uttarakhand)** – High-altitude Himalayan park, musk deer present. **Statement (2) right**
- **Kishanpur Wildlife Sanctuary (Uttar Pradesh)** –Terai grassland, not natural musk deer habitat. **Statement (3) wrong**
- **Manas National Park (Assam)** – Lowland tropical forests, not musk deer habitat. **Statement (4) wrong**

16. Correct answer (a): EXPLANATION

The Asiatic lion (*Panthera leo persica*) now survives **only in the Gir Forest, Gujarat**, India. **Statement (1) right**

The **Bactrian camel** (double-humped) is naturally found in **Central Asia** (Mongolia, China), **not in India**. India has only the **single-humped camel** (dromedary). **Statement (2) wrong** The **Indian one-horned rhinoceros** (*Rhinoceros unicornis*) is found in **India (Assam, West Bengal, etc.) and Nepal**. **Statement (3) wrong**

17. Correct answer (c): EXPLANATION

The **Blue-finned Mahseer (*Tor khudree*),** a species of freshwater fish, is naturally found in the river systems of the Western Ghats, including the Cauvery River basin. **It is correctly matched**

Irrawaddy Dolphins (*Orcaella brevirostris*) are primarily found in coastal areas and large rivers in South and Southeast Asia (such as the Mekong and Ayeyarwady rivers), as well as in India's Chilika Lake. The Chambal River is a known habitat for the critically endangered **Ganges River Dolphin**, not the Irrawaddy Dolphin. **It is incorrectly matched**

The **Rusty-spotted Cat (*Prionailurus rubiginosus*)**, one of the world's smallest wild cat species, is native to India and Sri Lanka and is known to inhabit various regions across India, including parts of the Eastern Ghats. **It is correctly matched**

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18. Correct answer (a): EXPLANATION

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If a species of tortoise is declared **protected under Schedule I of the Wildlife (Protection) Act, 1972**, it means:

- **It enjoys the highest level of legal protection**, the same as tigers, lions, elephants, etc.
- Hunting, trade, or any exploitation is **strictly prohibited**, except under special provisions of the law.

It **does not imply** that the species is extinct, endemic, or impossible to save.

19. Correct answer (d): EXPLANATION

Under the **Wildlife (Protection) Act, 1972**:

- **Schedule I species** are given the **highest protection**, and hunting them is prohibited except under very special provisions.

Species listed:

1. Gharial – Schedule I - Hunting prohibited

2. Indian wild ass – Schedule I - Hunting prohibited

3. Wild buffalo – Schedule I - Hunting prohibited

All three fall under **Schedule I** and cannot be hunted except under specific legal provisions.

20. Correct answer (b): EXPLANATION

The term **'M-STrIPES'** stands for **Monitoring System for Tigers – Intensive Protection and Ecological Status**.

It is a **software-based monitoring system** used for:

- Patrolling and protection in **tiger reserves**
- Recording data on tiger sightings, poaching, and other threats

21. Correct answer (b): EXPLANATION

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The **gharial (Gavialis gangeticus)** is a freshwater crocodilian that prefers **deep, fast-flowing rivers** with sandbanks for nesting.

Among the options:

- **Bhiterkanika Mangroves** – saltwater/brackish habitat, not suitable for gharials.
- **Chambal River** – ideal habitat; gharials are found here in the wild.
- **Pulicat Lake** – brackish water, not natural gharial habitat.
- **Deepor Beel** – freshwater wetland, but not known for gharials.

22. Correct answer (c): EXPLANATION

Butterflies are important **pollinators**. A decline in their population would reduce pollination in plants that depend on them. **Statement (1) right**

Butterflies **do not** play any known role in controlling fungal infections. Their decline does **not** have a direct link to fungal outbreaks in crops. **Statement (2) wrong**

Butterflies (and their larvae) serve as **prey** for birds, spiders, and parasitic wasps. A fall in butterfly numbers can reduce food availability, causing declines in these predator species. **Statement (3) right**

23. Correct answer (a): EXPLANATION

Kharai camels are known for their ability to **swim in the sea** to reach mangrove islands. **Statement (1) right**

They feed mainly on **mangrove vegetation**, especially in coastal areas and islands. **Statement (1) right**

Kharai camels **are domesticated** and used by local communities, particularly the Kharai camel herders (Maldharis). **Statement (3) wrong**

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24. Correct answer (a): EXPLANATION

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Red Sanders is **endemic to the southern Eastern Ghats**, mainly in **Andhra Pradesh** (particularly in the Seshachalam Hills). **Statement (1) right**

Red Sanders does **not** grow in tropical rainforests.

It grows in **dry deciduous forests** with rocky, hilly terrain — not in rainforest ecosystems. **Statement (2) wrong**

25. Correct answer (d): EXPLANATION

The **Great Indian Hornbill** is most commonly found in:

The Western Ghats and parts of Northeast India.

Among the given options, the correct habitat is: **Western Ghats**

26. Correct answer (c): EXPLANATION

The **national aquatic animal of India** is the: **Gangetic Dolphin**

27. Correct answer (c): EXPLANATION

Dugongs feed mainly on seagrass and are fully herbivorous. **Statement (1) right**

In India, dugongs are found **only in specific regions**: Gulf of Mannar, Palk Bay, Andaman & Nicobar Islands, Gulf of Kachchh (small population). They are **not** found along the *entire* Indian coastline. **Statement (2) wrong**

Dugong (*Dugong dugon*) is listed under **Schedule I**, giving it the highest level of legal protection. **Statement (3) right**

28. Correct answer (b): EXPLANATION

The Animal Welfare Board of India (AWBI) was **established under the Prevention of Cruelty to Animals Act, 1960**, not the Environment Protection Act. **Statement (1) wrong** NTCA was made a statutory body

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through the **Wildlife (Protection) Amendment Act, 2006**. **Statement (2) right** NGRBA (created in 2009) was indeed **chaired by the Prime Minister**. **Statement (3) right**

29. Correct answer (c): EXPLANATION

The major reasons (other than poaching) for the decline of **Ganga River Dolphins** include:

1. Construction of dams and barrages – disrupts migration, alters flow, reduces prey availability.

2. Getting trapped in fishing nets accidentally – a major cause of mortality.

3. Pollution from synthetic fertilizers and agricultural chemicals – contaminates water and affects dolphin health. **Increase in crocodile population (2)** is *not* considered a significant factor in their decline.

30. Correct answer (a): EXPLANATION

Biosphere Reserves - under UNESCO's Man and Biosphere (MAB) Programme - are specifically designed with a **zonation system** (core, buffer, transition) that allows:

• **Strict protection** of biodiversity in the core

• **Sustainable use** in the buffer

• **Human habitation and traditional activities** in the transition area.

Thus, they integrate **conservation + traditional human life**, unlike national parks and wildlife sanctuaries where human activities are highly restricted.

31. Correct answer (a): EXPLANATION

The lion-tailed macaque is endemic to the **Western Ghats**, and occurs naturally in: **Tamil Nadu**, **Kerala**, **Karnataka**

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It is **not** naturally found in **Andhra Pradesh**.

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32. Correct answer (a): EXPLANATION

1. Star tortoise - Naturally found in India

Indian Star Tortoise (*Geochelone elegans*) occurs in: Rajasthan, Gujarat, Tamil Nadu, Karnataka

2. Monitor lizard - Naturally found in India

India has several species: Bengal monitor, Desert monitor, Water monitor

3. Pygmy hog - Naturally found in India

Found only in the **Assam** region (Manas landscape). One of the rarest mammals of the world.

4. Spider monkey - Not found in India

Found in **Central and South America**, not Asia.

33. Correct answer (a): EXPLANATION

Let's check each species **as per the status around 2013**, which UPSC used for this question:

1. Gharial (Gavialis gangeticus) Critically Endangered (CR) — *IUCN*

So it is definitely **endangered**. **It is right**

2. Leatherback turtle (Dermochelys coriacea)

- **Globally:** *Vulnerable (VU)* under IUCN
- **However, in India**, it is listed in **Schedule I of the Wildlife Protection Act**, and was treated by government sources as a **highly threatened / endangered species** in the Indian context. UPSC often accepts a species as "endangered" if it is listed as **highly threatened in India**, even when its IUCN status is VU. **It is right**

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3. Swamp deer (Barasingha)

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Around 2013: **IUCN:** *Vulnerable (VU)*

• Some subpopulations (e.g., Hard-ground Barasingha of Kanha) were severely threatened, but **not classified as endangered**. **It is wrong**

34. Correct answer (b): EXPLANATION

The question refers to the situation **as of 2013**, when **only four Indian Biosphere Reserves** were included in the **UNESCO World Network of Biosphere Reserves (WNBR)**.

As of 2013, the four UNESCO-recognized Biosphere Reserves were:

1. Nilgiri (2000)

2. Gulf of Mannar (2001)

3. Sunderbans (2001)

4. Nanda Devi (2004)

Kanchenjunga Biosphere Reserve

• It was **NOT recognized by UNESCO in 2013**.

• It was added to the WNBR **later in 2018**.

35. Correct answer (b): EXPLANATION

Let's match each species with its correct scientific name:

A. Asiatic Wild Ass - Equus hemionus - 3

B. Barasingha (Swamp Deer) - Cervus duvauceli - 2

C. Chinkara (Indian Gazelle) - Gazella bennettii - 4

D. Nilgai (Blue Bull) - Boselaphus tragocamelus - 1

So the correct sequence **A–B–C–D** is: **3 – 2 – 4 – 1**

36. Correct answer (b): EXPLANATION

Explanation: Let's check each species:

1. Black-necked Crane – *Naturally found in India*, mainly in **Ladakh**. **It is right**

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2. Cheetah – The **Asiatic cheetah is extinct in India** (last seen in 1952). Reintroduced cheetahs (2022) are **not counted** as naturally occurring. **It is wrong**

3. Flying Squirrel – Several species of flying squirrels are **naturally found in India** (e.g., Indian giant flying squirrel). **It is right**

4. Snow Leopard – Naturally found in the **Himalayan region** of India. **It is right**

37. Correct answer (b): EXPLANATION

Vulture populations in India collapsed drastically (over **95% decline**) mainly due to the use of the veterinary drug **diclofenac**.

- When cattle treated with diclofenac died, vultures feeding on the carcasses suffered **kidney failure**, leading to mass mortality.
- This is a well-documented and globally cited conservation issue.

38. Correct answer (a): EXPLANATION

To identify the correct set, recall that **endangered species** in India include animals that face a high risk of extinction in the wild.

Let's evaluate Option A:

Great Indian Bustard – Endangered (critically endangered)
Musk Deer – Endangered Red Panda – Endangered
Asiatic Wild Ass – Near Threatened / Endangered in some assessments (IUCN: Near Threatened, but listed as Endangered in older Indian sources)

39. Correct answer (b): EXPLANATION

National Parks have the highest level of protection after wildlife sanctuaries. No grazing, no firewood collection, no forest produce collection is permitted.

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Others:

Biosphere Reserves - Allow sustainable use in buffer & transition zones.

Ramsar Wetlands - No uniform rule; local use is generally allowed as per national laws.

Wildlife Sanctuaries - Certain regulated activities (like grazing, fuelwood collection) may be allowed.

40. Correct answer (b): EXPLANATION

In-situ conservation means *conserving species in their natural habitats*. Examples: **Biosphere Reserves, National Parks, Wildlife Sanctuaries**.

Ex-situ conservation means *conserving species outside their natural habitats*. Examples: **Botanical Gardens, Zoos, Seed Banks**.

41. Correct answer (d): EXPLANATION

All statements are right

1. Soil formation - microorganisms, decomposers, and plant roots help form soil.

2. Prevention of soil erosion — vegetation cover and root systems hold soil together.

3. Recycling of waste — decomposers (bacteria, fungi, detritivores) break down organic waste and recycle nutrients.

4. Pollination of crops — insects, birds, bats, etc. provide essential pollination services.

42. Correct answer (b): EXPLANATION

• The IUCN Red Data Book does **not** list species specifically because they are endemic or because they occur in hotspots.

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It lists **threatened** species globally. **Statement (1) is wrong**

- IUCN Red List (Red Data Book) includes species that are: Critically Endangered, Endangered, Vulnerable etc. **Statement (2) is right**
- Protected sites are listed under other frameworks (e.g., UNESCO World Heritage, Ramsar sites), not IUCN Red List. **Statement (3) is wrong**

43. Correct answer (c): EXPLANATION

To be classified as a **biodiversity hotspot**, a region must meet **two scientific criteria** (as defined by Norman Myers and Conservation International):

1. Exceptionally high species richness & endemism

2. High degree of threat (i.e., major loss of habitat)

Now let's evaluate the options:

Correct Criteria:

1. Species richness: Hotspots must have a large number of species.

3. Endemism: Hotspots must contain at least **1,500 endemic vascular plant species**.

5. Threat perception: Hotspots must have lost ≥**70% of their primary habitat**, meaning they face extremely high threat.

Incorrect Criteria (Not used for hotspot identification):

2. Vegetation density - Not a hotspot criterion

4. Ethno-botanical importance - Not relevant to hotspot criteria

6. Adaptation to warm and humid conditions – Not required; hotspots can be cold (e.g., Himalaya)

44. Correct answer (b): EXPLANATION

Indian Wild Ass (Equus hemionus khur)

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Found mainly in the **Little Rann of Kutch**, Gujarat, which is: A **sandy, saline desert area,** Known for **salt pans, this** wild ass has **very few or no natural predators** there

- Its main threat is **habitat degradation**, especially due to: Expansion of salt industries, Prosopis juliflora invasion & Increasing human activities

Other options do not fit:

- **Indian wild buffalo** → found in marshy/wet grasslands (Assam/Chhattisgarh), not saline deserts.
- **Indian wild boar** → found in forests/grasslands across India, not sandy-saline habitat.
- **Indian gazelle (Chinkara)** → lives in arid areas but **not specifically saline**, and does have predators.

Correct Answer: Indian wild ass

45. Correct answer (a): EXPLANATION

- **One river originating in Jharkhand** → *known by a different name in Odisha*

This is the **Brahmani River-** Source: **Ranchi Plateau, Jharkhand and** Known in parts of Odisha as **South Koel–Brahmani system**

- **Another river originating in Odisha -** This is the **Baitarani River, its** Source: **Gonasika Hills, Odisha**

These two rivers meet close to the Bay of Bengal and form a rich biodiversity site.

The Brahmani and Baitarani rivers meet near Bhitarkanika, forming the vast mangrove ecosystem of: **Bhitarkanika National Park & Wildlife Sanctuary-** Famous for saltwater crocodiles, Large mangrove forests, Ramsar wetland site & Rich biodiversity

Hence, the **only correct option**:**Bhitarkanika**

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46. Correct answer (d): EXPLANATION

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- Biodiversity hotspots occur **both in tropical and non- tropical regions**.Example: **Mediterranean Basin** (a hotspot) is not tropical. **Statement (1) is wrong**
- India lies within **four global hotspots**, but their names are:

1. Himalaya (includes Eastern Himalayas; *Western Himalayas are NOT recognised as a hotspot*)

2. Western Ghats–Sri Lanka

3. Indo-Burma (includes NE India + Andaman & Nicobar Islands)

4. Sundaland (only **Nicobar Islands**, not Andaman)

So the statement's list is **factually incorrect**. **Statement (2) is wrong**

47. Correct answer (c): EXPLANATION

- National Parks in India are declared under the **Wildlife (Protection) Act, 1972** and their boundaries **can be altered only by the State Legislature**. Thus, boundaries are indeed defined by legislation. **Statement (1) is right**
- A **Biosphere Reserve** aims to conserve **entire ecosystems**, including biodiversity, landscapes, and cultural heritage. It is **not created for a few specific species** (that is the role of a Wildlife Sanctuary or National Park). **Statement (2) is wrong**
- Wildlife Sanctuaries allow **certain human activities** such as grazing, collection of firewood, etc., if regulated. Thus, limited biotic interference is indeed permitted. **Statement (3) is right**

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UNIT IX Miscellaneous Topics

Miscellaneous Topics

23

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23.2 Coral Reefs and Coral Bleaching
23.3 Biorock Technique
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23.20 Issues Related to Stubble Burning
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23.1 BHOPAL GAS TRAGEDY

Bhopal gas tragedy was a gas leak incident in India, considered the world's worst industrial disaster. It occurred on the night of 2–3 December 1984 at the **Union Carbide India Limited (UCIL)** pesticide plant in Bhopal, Madhya Pradesh.

Over 500,000 people were exposed to **Methyl isocyanate (MIC)** gas and other chemicals. The death toll was 3,787 related to the gas release. Others estimate 8,000 died within two weeks.

■ *Who is Responsible?*

UCIL was the Indian subsidiary of Union Carbide Corporation (UCC), with Indian Government-controlled banks and the Indian public holding a 49.1% stake. Warren M. Anderson was a former chairman and chief executive officer of Union Carbide belonging to the United States of America and sought by the Indian government for the Bhopal Gas Tragedy. It has been said that the causes of the leakage were the design of the plant and the economic pressure.

■ *Reasons*

- Poor management systems.
- Lack of early warning system.
- Ignoring warnings.

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- Storing the MIC in large tanks more than the advised level.
- Poor maintenance after the plant stopped MIC production in 1984.
- Safety systems were in a standby mode to save money.
- No emergency plans for the evacuation of people or medical treatment for any catastrophic gas release.

Medical Consequences

- Gas circulated through the bloodstreams of victims, carrying toxins and causing damage to the eyes, lungs, kidneys, liver, intestines, muscles, brain and reproductive and immune systems.
- 43% of the women from the severely affected communities who were pregnant at the time of the disaster aborted.
- Growth and development of children whose mothers were exposed to the gases during pregnancy revealed that the majority of children had delayed gross motor and language sector development.
- Evidence of chromosomal damage.
- Those exposed suffered from post-traumatic stress disorder.

23.2 CORAL REEFS AND CORAL BLEACHING

Corals are sessile animals (not mobile) and stay fixed in one place. They feed by reaching out with tentacles to catch prey, such as small fish and planktonic animals. Corals live in colonies consisting of many individuals, each of which is called **polyp**. They secrete a hard calcium carbonate skeleton, which serves as a uniform base or substrate for the colony. The skeleton also provides protection, as the polyps can contract into the structure if predators approach. It is these hard skeletal structures that build up coral reefs over time. The calcium carbonate is secreted at the base of the polyps, so the living coral colony occurs at the surface of the skeletal structure, completely covering it. Calcium carbonate is continuously deposited by the living colony, adding to the size of the structure. The growth of these structures varies greatly, depending on the species of coral and environmental conditions.

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Coral reefs in India are mainly restricted to the Andaman and Nicobar Islands, Gulf of Mannar, Gulf of Kutch, Palk Strait and the Lakshadweep islands. All of these reefs are Fringing reefs, except Lakshadweep which are Atolls.

Condition Required

Reef building coral communities also require tropical or sub-tropical temperatures, and exist globally in between 30 degrees north and 30 degrees south of the equator.

Uses

- Coral reefs provide habitats for a large variety of organisms. These organisms rely on corals as a source of food and shelter.

- Coral reefs cover less than 0.2% of our oceans, but they contain 25% of the world's marine fish species.
- Coral reefs are second only to rainforests in biodiversity of species.

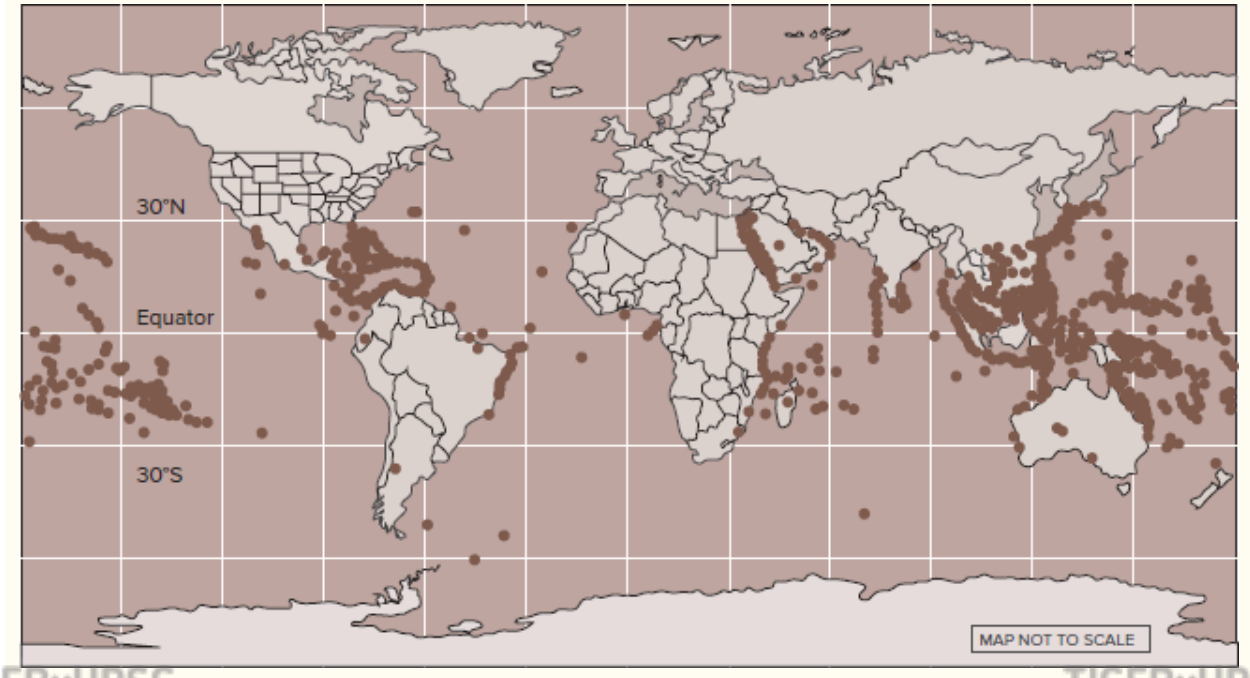

Figure 23.1 Worldwide Distribution of Coral Reefs

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CORAL REEFS & CORAL BLEACHING

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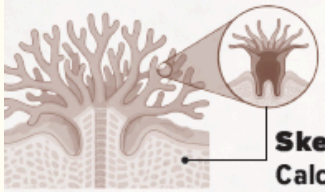
Corals are sessile (fixed) colony animals made of individual polyps

CORAL FUNDAMENTALS



Polyps
Sessile animal, catches prey with tentacles

Skeleton
Calcium carbonate reefs for protection

Feeding
Polyps use tentacles to catch prey (small fish, plankton)

Polyps secrete a hard calcium carbonate skeleton for protection and as a base. This continuous deposition builds reefs over time

DISTRIBUTION

Global Conditions
Tropical/Sub-tropical waters (exist between **30°N and 30°S latitudes**)

India's Reefs

- **Andaman & Nicobar**
- **Gulf of Mannar**
- **Gulf of Kutch**
- **Palk Strait (Fringing)**
- **Lakshadweep (Atolls)**

IMPORTANCE

1. Supports **25%** of marine life in **<0.2%** ocean area, rivaling rainforest biodiversity
2. Acts as a natural barrier against coastal erosion, storms, and tsunamis
3. Vital nursery for **~25% of ocean fish**, supporting global fisheries
4. Provides **major tourism** revenue & income for coastal communities
5. **Source of medicines** for diseases like Cancer and HIV

THREATS

Agricultural, industrial, and sewage runoff combined with sedimentation from land clearing

Warming waters and ocean acidification dissolve skeletons, hindering growth

Overfishing, cyanide or dynamite fishing, and trading for aquarium pets

Outbreaks of predatory starfish, invasive species, and infections

CORAL BLEACHING

1 HEALTHY CORAL
Symbiotic relationship with **zooxanthellae (algae),** algae provide primary food & color

2 STRESSED CORAL
Due to stress factors, algae leave the coral tissue

3 BLEACHED CORAL
Loses major food source, turns white/pale, becomes susceptible to disease

Causes of Bleaching

- **Elevated Sea Temperature (When water is too warm - Primary cause)**
- Decreased sea water temperature
- Bacterial and other infections
- Solar radiation
- Reduced salinity

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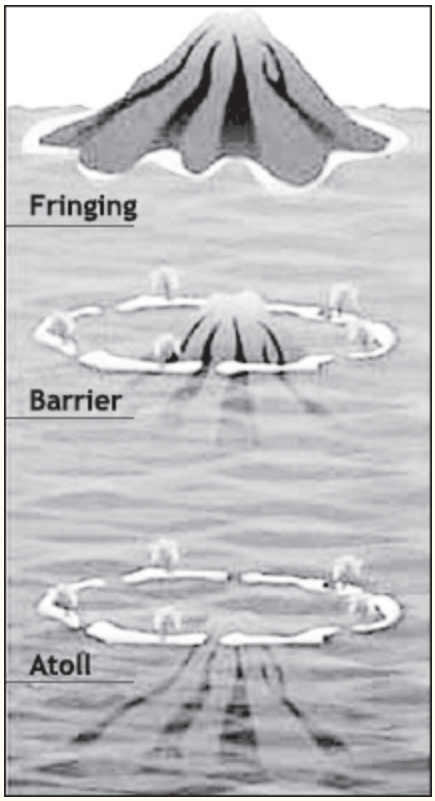

Figure 23.2 Darwin's Three Stage of Atoll formation

- They support the lives of not only their symbiotic algae but also other creatures like sponges, mollusks, crustaceans like crabs and shrimp, star fish, sea urchins, jellyfish, sea turtles and many species of fish.
- They act as large wave-resistant structures.

Threats to Coral Reef

- **Agricultural and industrial runoff:** Agriculturalactivities can introduce herbicides, pesticides, fertilisers and runoff from animal feed. Sewage discharges can introduce nitrogen and phosphate compounds along with pathogens and mixtures of toxics.
- **Increased sedimentation** from land clearing, human sewage and toxic discharges. Many land-based activities have important implications for reefs.

Warming Waters

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- **Ocean Acidification:** The acidic conditions dissolve coral skeletons, which make up the structure of the reef, and make it more difficult for corals to grow.
- Overfishing, fishing using cyanide and dynamite, massive outbreaks of predatory starfish, invasive species.
- Trading of reef habitats to serve as aquarium pets or decorative items.

Why Are Coral Communities Important?

- **Fisheries:** Coral reefs are vital to the world's fisheries. They form the nurseries for about a quarter of the ocean's fish, and thus provide revenue for local communities as well as national and international fishing fleets.
- **Tourism:** Tourism revenues generated by coral reefs are significant. Sustainably managed coral reef-based tourism can also provide significant alternative or additional sources of income to poorer coastal communities in developing countries.
- **Coastal protection:** Coral reefs break the power of the waves during storms, hurricanes, typhoons, and even tsumanis. Thus, they help in preventing coastal erosion, flooding, and loss of property on the shore.
- **Source of medical advances:** Coral reef organisms are being used in treatments for diseases like cancer and HIV.
- **Intrinsic value:** For many coastal societies around the world, coral reefs and their inhabitants are intricately woven into cultural traditions.

Coral Bleaching

It refers to the phenomenon of coral bleaching or whitening of corals due to the **loss of symbiotic algae**.

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ENV 74 Coral Bleaching

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Causes

- **Elevated Sea Temperature**: When water is too warm, corals expel the algae (zooxanthellae) living in their tissues, causing the coral to turn completely white.
- Decreased sea water temperature
- Solar radiation
- Reduced salinity
- Bacterial and other infections

Government of India (GoI) has taken steps to protect and conserve barrier reefs, fringing reefs and atolls in the country through various regulatory and promotional measures, which include:

1. Coral species are listed under **schedule -I** under **Indian Wildlife (Protection) Act, 1972**, which provides highest protection to the coral reefs.
2. The **Coastal Regulation Zone (CRZ) Notification, 2019**, promulgated under Environment (Protection) Act 1986, has specific focus on conservation and management plans of Ecologically Sensitive Areas (ESAs), like corals and coral reefs and prohibit developmental activities and disposal of wastes in the fragile coastal ecosystems.
3. **Ministry of Environment, Forest Climate Change** has initiated a long-term programme on Coral restoration and Marine Biological Station, Zoological Survey of India is involved in monitoring and restoration of coral ecosystem in Lakshadweep since last five years.
4. **Central Marines Fisheries Research Institute (CMFRI)** has undertaken studies to understand ecological changes affecting coral reefs. CMFRI has initiated a comprehensive national project aimed at investigating the resilience potential of various coral reefs in India by integrating advanced climatic modelling, deep-learning, and ecological research and develop resilience-based management actions ensuring the long-term sustainability of coral reef ecosystems.

23.3 BIOROCK TECHNIQUE

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The technology works by passing a small amount of electrical current through electrodes in the water. When a positively charged anode and a negatively charged cathode are placed on the sea floor, with an electric current flowing between them, calcium ions combine with carbonate ions and adhere to the structure (cathode). This results in calcium carbonate formation. Coral larvae adhere to the $CaCO_3$ and grow quickly. **Biorock** refers to the substance formed by electro-accumulation of minerals dissolved in seawater.

ENV 75- Biorock Technique

The formed material has a strength similar to concrete. It can be used to make robust **artificial reefs** on which corals grow at very rapid rates.

In this process, they are able to grow at least four to six times faster than their actual growth.

The technology also helps corals, including the highly sensitive branching corals, to counter the **threats posed by global warming.**

- Biorock methods speed up coral growth in damaged areas and restore authentic coral reef habitat and species.
- This becomes rapidly pioneered by a whole coral reef organism variety such as fish, crabs, clams, octopus, lobster, and sea urchins.
- Species typically found in healthy reef environments are given an electrical advantage over the weedy organisms which often overgrow them in reefs stressed by humans.
- The advantages corals gain from mineral accretion are cancelled if they no longer receive current, at which point weeds will overgrow the corals.

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If the current is maintained, coral reefs can often be restored even in areas where water quality would prevent their recovery by any other method.

- Biorock structures cement themselves to the hard bottom providing a physical wave barrier which, over time, grows larger and stronger. Biorock materials are, to an extent, structurally self-healing. If a section is damaged, the cracks will fill, making them ideal for breakwater shore protection.

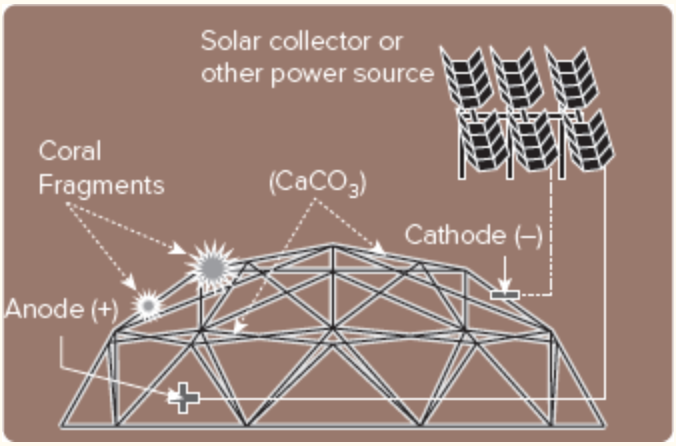

Figure 23.3 Procedure of Biorock Technique

Use of Technology in India

- A biorock structure was **installed one nautical mile off the Mithapur coast in the Gulf of Kachchh** on 19th January, 2020. The scientists have **used solar panels** for power in this case.
- The calcification (conversion into calcium carbonate) has started, but the real success will come when attached and new organisms (corals) start to grow faster.
- Often called the **"rainforests of the sea,"** coral reefs are home to a spectacular variety of organisms.
- **India has four major coral reef areas:** Andaman and Nicobar Islands, Lakshadweep, Gulf of Mannar and the Gulf of Kachchh.
- The coral reefs are **threatened by** climate change induced acidification as well as by anthropogenic factors such as overfishing and pollution.

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- The **Zoological Survey of India (ZSI),** with help from Gujarat's forest department, is attempting for **the first time a process to restore coral reefs using biorock or mineral accretion technology.**

23.4 EURO NORMS AND BHARAT STAGE

Europe has established some emission standards that set specific limits to the amount of pollutants that can be released into the environment. These norms are known as **Euro Norms**. However, the Government of India has utilised these Euro norms for available fuel quality and the method of testing. In India the manufacturers are required to reduce the existing polluting emission levels efficiently by making certain technical changes in their vehicles.

Bharat Stage Emission Standards

The Government of India has initiated with its own emission standard to regulate the output of air pollutants from internal combustion engine equipment, including motor vehicles. The standards are based on EURO Norms and termed **Bharat Stage**. The Central Pollution Control Board under the Ministry of Environment & Forests sets the standards and the timeline for implementation.

Why India felt the need to Shoot ahead from BS IV to BS VI

With 30 most polluted cities in the Top 100 Most Polluted Global Cities (in terms of particulate matter PM10) as per the data published by World Health Organisation in May 2016, India is facing a huge air quality degradation problem.

Vehicular emission is a major contributor to the worsening air quality of Indian cities. Emission of NOx, SO_2, CO_2 and particulate matter is taking a toll on people's health. In cities like Delhi, the PM2.5 level is more than 6 times the prescribed levels by WHO.

In October 2016, India ratified the Paris Climate Agreement. Being a signatory to the agreement, India is obligated to bring down the carbon footprint by 33–55% from the levels recorded in 2005 in the next 12 years.

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Ideally, BS V would have been rolled out by 2021 and BS VI in 2024, but leapfrog to Bharat Stage VI norms by 2020 had to be planned because of the carbon footprint obligations.

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Advantages of BS VI over BS IV

BS VI is in line with Euro VI norm already adopted in European countries. In fact, this new emission norm will also address one major drawback in the Euro VI norm that allows emission of higher PM (particulate matter) in diesel engines.

Some major benefits of Bharat Stage VI norms

- **OBD (On Board Diagnostic)** will be mandatory for every vehicle and it will help monitor the pollution caused by the vehicle in real-time.
- **RDE (Real Driving Emission)** will be introduced for the first time that will measure the emission in real-world conditions and not just under test conditions.
- **Bharat Stage VI** norms will also change the way particulate matter is measured. Number standard instead of mass standard will now measure it, thereby regulating the fine particulate matter as well.
- NOx emission will come down by approximately 25% for the petrol engine and 68% for the diesel engines.
- The PM emission will see a substantial decrease of 80% in diesel engines. The reason behind making OBD mandatory is to make sure that the emission control component work at its optimum efficiency at all times.

Technological Upgrade Challenges

Transitioning to BS-VI norms will require significant engine technology changes, including improvements in engine combustion and calibration, increased injection and cylinder pressures, NOx and PM aftertreatment solutions and transitioning to electronic controls.

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Although BS VI is based on Euro 6, two engine fitments will be required to reduce emission levels which will increase bonnet size as well.

1. **Diesel Particulate Filter (DPF):** For reduction of PM in diesel vehicles
2. **Selective Catalytic Reduction (SCR) Module:** For reduction in NOx emissions

■ Concerns over Fuel Grade

Every pollution emission norm is partially based on the type of fuel used in the vehicle, based on which it limits the amount of permissible emission.

The BS III and BS IV compliant vehicles were able to run on the fuel that was in circulation for the past 10–15 years. The requisite BS IV compliant fuel had to be 50 ppm ultra-low Sulphur.

According to some of the reports, many oil refineries are now equipped to produce BS VI compliant fuel and will soon be able to give out samples for testing.

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EMISSION STANDARDS IN INDIA

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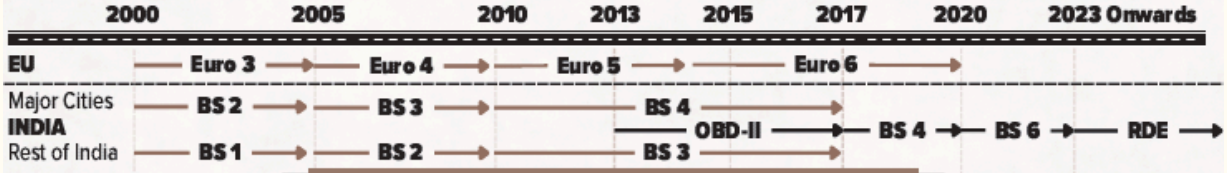
EVOLUTION OF EMISSION STANDARDS IN INDIA

Euro Norms: European standards setting limits for exhaust emissions (NOx, CO, Hydrocarbons, PM)

Bharat Stages (BS): These emission standards are set and implemented by the **Central Pollution Control Board (CPCB)**, under the administrative control of the **MoEF&CC**



HISTORY & EVOLUTION

Catalytic converters made mandatory
1991-92

Mashelkar Committee Report accepted
2002

BS3 nationwide; BS4 in 13 major cities.
2010

Leapfrogging from BS4 to BS6, skipping BS5
2020

1999
Supreme Court enforces **Euro I norms**

2003
National Auto Fuel Policy announced

2016
India ratifies **Paris Climate Agreement**

THE LEAPFROG (WHY BS4 TO BS6?)

- **Health Emergency:** 30 Indian cities in the Top 100 most polluted globally (WHO, 2016)
- **PM 2.5 Levels:** Delhi levels reached **6x** the WHO limits

- **Paris Agreement (2016):** India committed to emissions intensity of GDP by **33–35%** (from 2005 levels) by **2030**

BS6 implemented
in April 2020 to meet the deadline

BS-IV VS. BS-VI VS. BS-VI (PHASE-II)

Component	BS4 (Pre 2020)	BS6 Ph-1 (Apr 2020)	BS6 Ph-2 (Apr 2023)
Diesel NO_x Limit	**250** mg/km	**80** mg/km	**Real Driving Conditions** required
Diesel PM Limit	**25** mg/km	**4.5** mg/km (Mass/Number)	Mandatory **Particulate Number (PN) monitoring**
Fuel Sulphur	**50** PPM	**10** ppm(Ultra Sulphur Fuel)	**10 ppm**
Monitoring	Lab Tests Only	Lab / OBD / RDE	Requires Mandatory **OBD-II** (Advanced) + Mandatory **RDE**
Technology	Catalytic Converter	**DPF / SCR (Mandatory)**	**Sophisticated SCR systems** and upgraded fuel injectors

TECHNOLOGICAL UPGRADES

DPF (Diesel Particulate Filter): Removes soot/Particulate Matter

SCR (Selective Catalytic Reduction): Reduces NO_x using AdBlue

Advanced Fuel: Refineries upgraded to produce ultra-low sulphur fuel

23.5 FLEX FUELS

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A flexible-fuel vehicle (FFV) or dual-fuel vehicle is an alternative fuel vehicle in which the designed vehicle is made to run on more than one fuel, generally gasoline mixed with either ethanol or methanol fuel, and the same tank is used to store both the fuels.

- Modern flex-fuel engines are capable of burning any proportion of the resulting blend in the combustion chamber as fuel injection and spark timing are adjusted automatically according to the actual blend detected by a fuel composition sensor. Flex-fuel vehicles are distinguished from bi-fuel vehicles, where two fuels are stored in separate tanks and the engine runs on one fuel at a time, for example, compressed natural gas (CNG), liquefied petroleum gas (LPG), or hydrogen.
- In a good flex-fuel vehicle, up to 83% ethanol can be mixed with petrol, which is a global standard. Most of such vehicle components are the same as those found in a conventional gasoline-only car. However, some of the components, especially engine and fuel line, are tuned, tweaked and modified to work with blended fuel, and not just petrol.

ENV 76-Flex Fuel Vehicle

Ethanol

- It is a kind of alcohol with more than 99% purity.
- It is a by-product of sugarcane but can also be made from grains. In short, ethanol is a renewable fuel made from various plant materials collectively known as biomass.
- Given the size of the country's sugarcane and grain production, India can meet most of its ethanol requirements indigenously. It will drastically cut the import bill, help boost the farming sector and generate lakhs of jobs. When compared to petrol, whose prices have been hovering above Rs.100 per litre for quite some time, ethanol is

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much cheaper. After a recent hike, ethanol produced from sugarcane juice is now Rs. 63.45 litre.

- In India, the blending percentage of ethanol with petrol has gone up from 1.53% in 2013–14 to 8.5% in 2020–21. The country has set a target of 20% ethanol blending with petrol by 2023–24. And the ultimate goal is to have 100% ethanol-run vehicles.

23.6 ISSUE OF ENDOSULFAN

An organochlorine insecticide which can be used on a variety of vegetables, fruits, cotton and on ornamental plants.

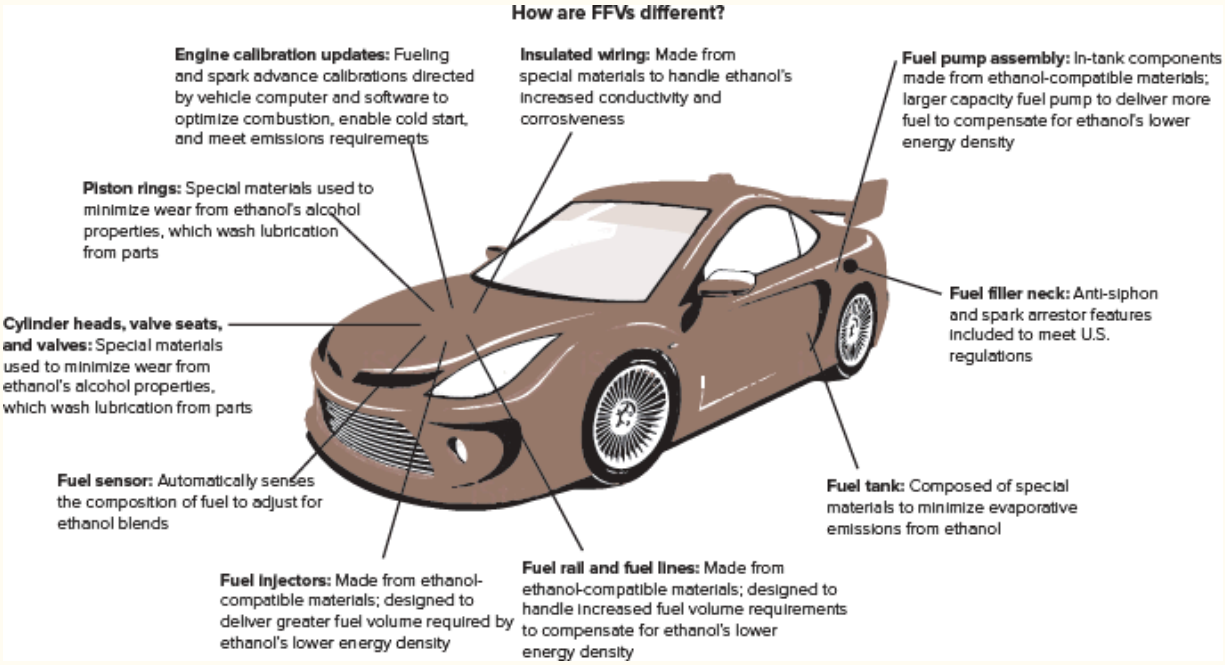

Figure 23.4 How are FFVs Different?

Endosulfan is volatile, persistent, and has the potential to bio-accumulate in aquatic and terrestrial organisms.

Endosulfan is a highly toxic, ubiquitous environmental pollutant that causes long-term harm to humans and wildlife. It is widely considered to be a **Persistent Organic Pollutant (POP)** and is readily absorbed by stomach, lungs and through the skin.

How did the Disaster Happen?

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The endosulfan tragedy happened in Kasaragod district of Kerala in India. The Plantation Corporation of Kerala (PCK), a public sector undertaking under the State Government, which owns three cashew plantations covering 4600 hectares in Kasargod, sprayed endosulfan aerially in these plantations for 24 years (1976 to 2000), three times a year.

These uninterrupted spraying, in spite of several warnings about its disastrous impact on health and environment, has resulted in several chronic, critical and life-threatening ailments in the areas surrounding the plantations.

This tragedy occurred because a toxic chemical was used continuously in a populated area with several natural drinking water sources.

What are its Impacts?

Eleven Gram Panchayats covering twelve villages were severely affected. In such villages there is a very high concentration of chronic health complications like Mental, Physical and behavioural disorders, Sensory loss, Neurological ailments, Cardiovascular diseases, Congenital Anomalies, Dermatological and Musculoskeletal disorders, etc.

Impact on Biodiversity

- The area showed a decline in plant diversity between 40 and 70%, particularly for native species, compared to the natural habitat. Fishes were the worst hit.
- Abnormalities and deformities were recorded in cattle. Animals such as jackal, porcupine, wild boar, civet cats, and bats, which were once common in the area, completely disappeared during the spray. Birds and fishes also showed a decline in their numbers.
- India has signed the Stockholm Convention aimed at phasing out the production and emission of one dozen persistent organic pollutants (POPs), including nine pesticides (Endosulphan) and three industrial chemical emissions.

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- The country would be granted $50,000 from the Global Environment Fund (GEF) to make a preliminary assessment and chart out a plan to tackle POPs. This would be carried out by United Nations Industrial Development Organisation (UNIDO).

23.7 PERMACULTURE

Permaculture is a system of **sustainable agriculture** coined by Australian biologist **Bill Mollison**. It is a combination of permanent agriculture and permanent culture which includes conscious design and maintenance of agriculturally productive natural ecosystem. It is the harmonious integration of landscape and people.

Env 77-Permaculture

The philosophy behind permaculture is one of working with, rather than against, nature; of protracted and thoughtful observation rather than protracted and thoughtless action.

Permaculture makes agriculture sustainable:

1. It is applicable for both rural and urban areas.
2. It included multidisciplinary aspects such as agriculture waste management, hydrological solution, energy, natural building, forestry, waste management, animal systems and aquaculture.
3. It has better productivity and better nutritional quality as compared to normal agriculture, so can be used to fulfil the gap of under nourishment and malnourishment.

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4. Permaculture not only focuses on food security but also ensures food security and nurturing ecosystem in a sustainable way.
5. In permaculture, land will be cleared through mulching, not burning, so no harm to the environment.
6. Permaculture is an effective method to increase carbon sequestration capabilities of plants.
7. Permaculture activated the practice of chickens, bullocks and cows along with traditional agriculture which allows farmers to get rid of distress.

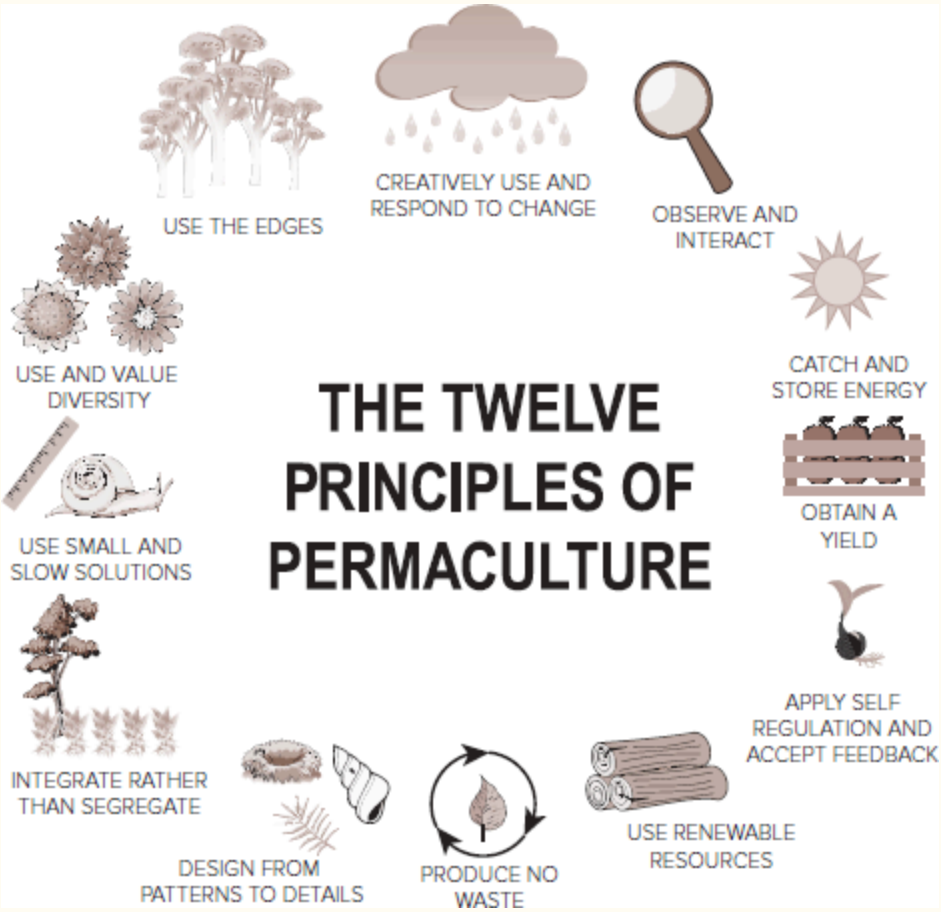

Figure 23.5 Pictorial View of Twelve Principles of Permaculture

■ *Challenges associated with permaculture*

1. Lack of research facility and any particular government support.
2. Displacing green revolution and practising permaculture is really a cumbersome task.

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3. Lack of awareness about permaculture among Indian farmers.
4. It faces the same problems as normal agriculture like irrigation and land requirement.

So, by keeping an eye on the changing global climate scenario and farmers' stress in India, GOI should promote awareness among farmers to practise permaculture.

23.8 XENOBIOTICS

The term Xenobiotic comes from the Greek for xeno (foreign) and biotics (of or pertaining to life). Xenobiotics are compounds that are foreign to an organism or are not part of its normal nutrition. Examples of Xenobiotics are compounds that include drugs, food additives and environmental pollutants. These agents are normally eliminated from the body after metabolism to compounds that are excreted through the bile, kidney, lung or dermis.

■ *Effects*

1. Immunological Responses
2. Allergenic Effects
3. Toxicity
4. A range of effects on the reproductive system including sperm quality and the incidence of male genital abnormalities
5. Spontaneous abortion, premature deliveries and low birth weight
6. Alterations in the ratio of male to female offspring
7. Some chemicals may play a role in certain diseases such as breast and testicular cancer.
8. Our bodies build up over time very persistent and bioaccumulative xenobiotics - most of these will last many years and probably be one of the main causes of our mortality.

■ *Types of Xenobiotics*

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- **Exogenous:** The foreign molecules which are not normally ingested or utilised by the organism but gain entry through dietary foodstuffs, or in the form of certain medicines/drugs used for therapeutic causes or are inhaled through environment.
  Ex: Drugs, food, additives, pollutants, insecticides, chemical carcinogens, etc.
- **Endogenous:** They are not foreign substances but have effects similar to exogenous xenobiotics. These are synthesised in the body or are produced as matabolites of various processes in the body.
  Ex: Bilirubin, Bile, Acids, Steroids.

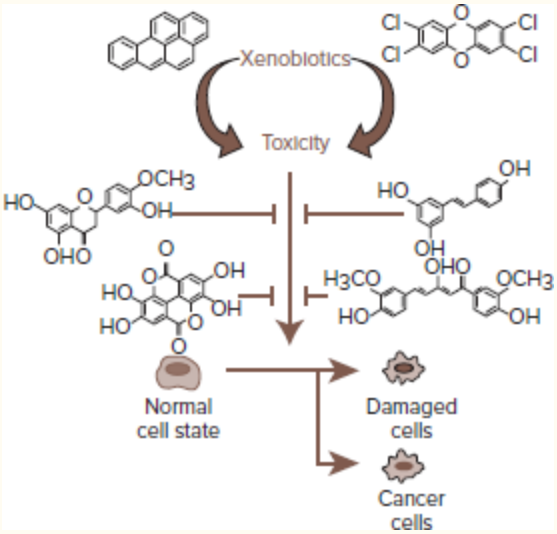

Figure 23.6 Xenobiotics

23.9 BIOREMEDIATION AND PHYTOREMEDIATION

Bioremediation is the use of microbes to clean up contaminated soil and groundwater. Microbes are very small organisms, such as bacteria, that live naturally in the environment. Bioremediation stimulates the growth of certain microbes that use contaminants as a source of food and energy. Contaminants treated using bioremediation include oil and other petroleum products, solvents and pesticides.

How Does It Work?

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- Some types of microbes eat and digest contaminants, usually changing them into small amounts of water and harmless gases like carbon dioxide and ethene.
- For bioremediation to be effective, the right temperature, nutrients, and food also must be present. Proper conditions allow the right microbes to grow and multiply and eat more contaminants.
- If conditions are not right, microbes grow too slowly or die, and contaminants are not cleaned up. Conditions may be improved by adding "amendments." Amendments range from household items like molasses and vegetable oil to air and chemicals that produce oxygen. Amendments are often pumped underground through wells to treat soil and groundwater in situ (in place).
- It may take a few months or even several years for microbes to clean up a site, depending on several factors.

■ Applications

- Bioremediation has the advantage of using natural processes to clean up sites.
- It may not require as much equipment, labour, or energy as some clean-up methods, it can be cheaper.
- The contaminated soil and groundwater are treated onsite without having to dig, pump, and transport them elsewhere for treatment.
- Microbes change the harmful chemicals into small amounts of water and gases, few if any waste by-products are created.

■ Phytoremediation

Phytoremediation is a bioremediation process that uses various types of plants to remove, transfer, stabilise, and/or destroy contaminants in the soil and groundwater. There are several different types of phytoremediation mechanisms. These are:

1. **Rhizosphere biodegradation:** In this process, the plant releases natural substances through its roots, supplying nutrients to microorganisms in

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the soil. The microorganisms enhance biological degradation.

2. **Phyto-stabilisation:** In this process, chemical compounds produced by the plant immobilise contaminants, rather than degrade them.
3. **Phyto-accumulation (phyto-extraction):** In this process, plant roots sorb the contaminants along with other nutrients and water. The contaminant mass is not destroyed but ends up in the plant shoots and leaves. This method is used primarily for wastes containing metals.
4. **Hydroponic Systems for Treating Water Streams (Rhizofiltration):** Rhizofiltration is similar to phyto-accumulation, but the plants used for cleanup are raised in greenhouses with their roots in water. This system can be used for ex- situ groundwater treatment. That is, groundwateris pumped to the surface to irrigate these plants.
5. **Phyto-volatilisation:** In this process, plants take up water containing organic contaminants and release the contaminants into the air through their leaves.
6. **Phyto-degradation:** In this process, plants actually metabolise and destroy contaminants within plant tissues.
7. **Hydraulic Control:** In this process, trees indirectly remediate by controlling groundwater movement. Trees act as natural pumps when their roots reach down towards the water table and establish a dense root mass that takes up large quantities of water.

The growth of mustard and populus plant can remove lead from the soil. In Chernobyl disaster area, the growth of sunflower plant frees soil against uranium isotopes. Certain microorganisms are also used as microremediation but it is not possible to destroy heavy metals or inorganic compounds.

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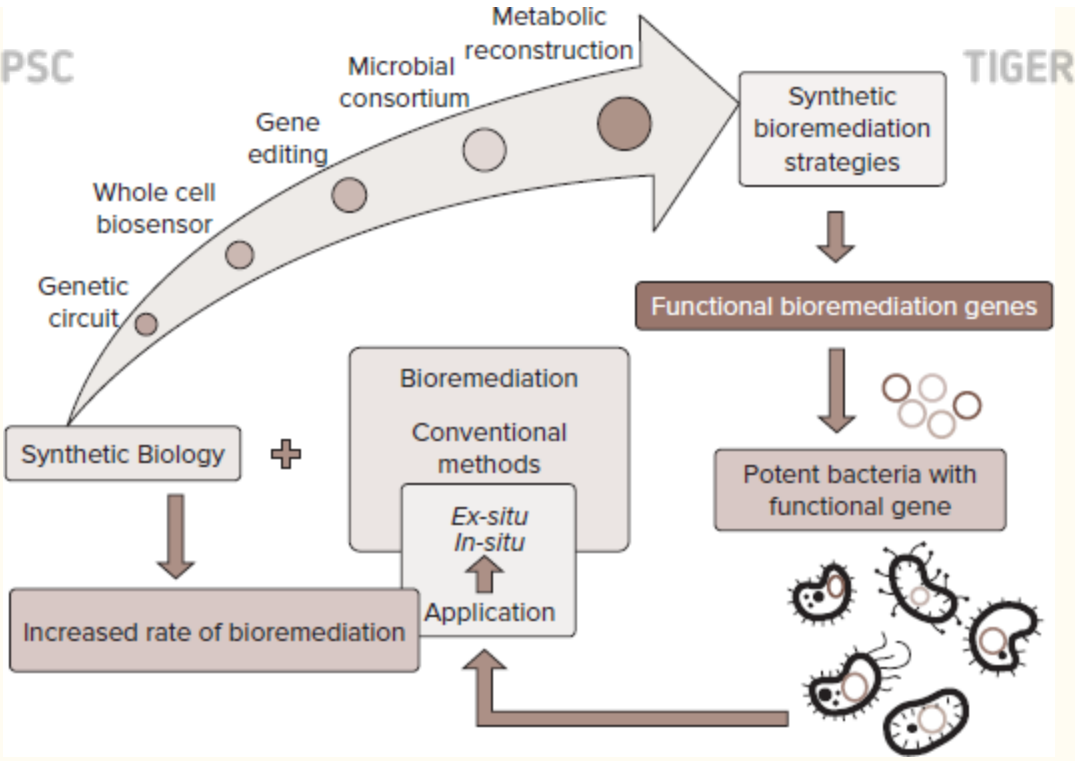

Figure 23.7 Steps of Bioremediation Process

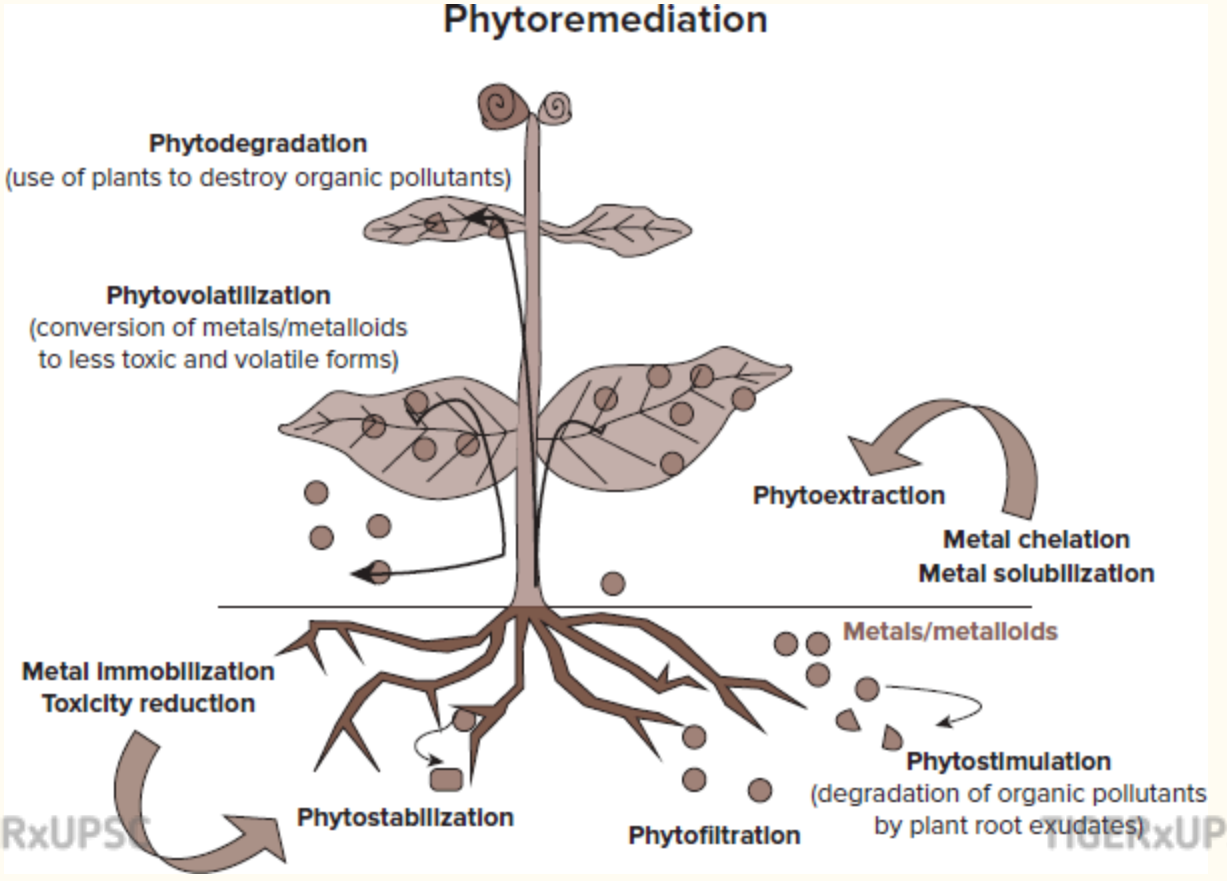

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Figure 23.8 Phytoremediation

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Limitations and Concerns

- The toxicity and bioavailability of biodegradation products are not always known.
- Degradation by-products may be mobilised in groundwater or bio-accumulated in animals.
- Scientists need to establish whether contaminants that collect in the leaves and wood of trees are released when the leaves fall in the autumn or when firewood or mulch from the trees is used.
- Disposal of harvested plants can be a problem if they contain high levels of heavy metals.
- The depth of the contaminants limits treatment. The treatment zone is determined by plant root depth. In most cases, it is limited to shallow soils, streams, and groundwater. Pumping the water out of the ground and using it to irrigate plantations of trees may treat contaminated groundwater that is too deep to be reached by plant roots.
- Generally, the use of phytoremediation is limited to sites with lower contaminant concentrations and contamination in shallow soils, streams, and groundwater. However, researchers are finding that the use of trees (rather than smaller plants) allows them to treat deeper contamination because tree roots penetrate more deeply into the ground.
- The success of phytoremediation may be seasonal, depending on location. Other climatic factors will also influence its effectiveness.
- The success of remediation depends on establishing a selected plant community. Introducing new plant species can have widespread ecological ramifications. It should be studied beforehand and monitored.
- The establishment of the plants may require several seasons of irrigation. It is important to consider the extra mobilisation of contaminants in the soil and groundwater during this start-up period.
- If contaminant concentrations are too high, plants may die.
- Some phytoremediation transfers contamination across media (e.g., from soil to air).

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- Phytoremediation is not effective for strongly sorbed contaminants such as polychlorinated biphenyls.
- Phytoremediation requires a large surface area of land for remediation.

23.10 VERMICOMPOSTING

Vermicomposting is the breaking down of organic material through the use of worms, bacteria, and fungi. In nature, organic matter is decomposed through these organisms. The end product of vermicomposting is a substance called **vermicompost** or "**worm castings**". This is a nutrient-rich organic substance that can be added to soil to increase its organic matter content and available nutrients. Vermicomposting is basically a managed process of worms digesting organic matter to transform the material into a beneficial soil amendment.

Vermicomposts are defined as organic matter of plant and/or animal origin consisting mainly of finely-divided earthworm castings, material, due to interactions between aerobic microorganism and earthworms, as the materials pass through the earthworm gut.

■ *Advantages of Vermicompost*

1. Vermicompost is rich in all essential plant nutrients.
2. Provides excellent effect on overall plant growth, encourages the growth of new shoots/leaves and improves the quality and shelf life of the produce.
3. Vermicompost is free-flowing, easy to apply, handle and store and does not have a bad odour.
4. It improves soil structure, texture, aeration, and water-holding capacity and prevents soil erosion.
5. Vermicompost is rich in beneficial micro flora such as a fixers, P-solubilises, cellulose decomposing micro-flora, etc. in addition to improving the soil environment.

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6. Vermicompost contains earthworm cocoons and increases the population and activity of earthworm in the soil.
7. It neutralises soil protection.
8. It prevents nutrient losses and increases the use efficiency of chemical fertilisers.
9. Vermicompost is free from pathogens, toxic elements, weed seeds, etc.
10. Vermicompost minimises the incidence of pest and diseases.
11. It enhances the decomposition of organic matter in soil.
12. It contains valuable vitamins, enzymes and hormones like auxins, gibberellins, etc.

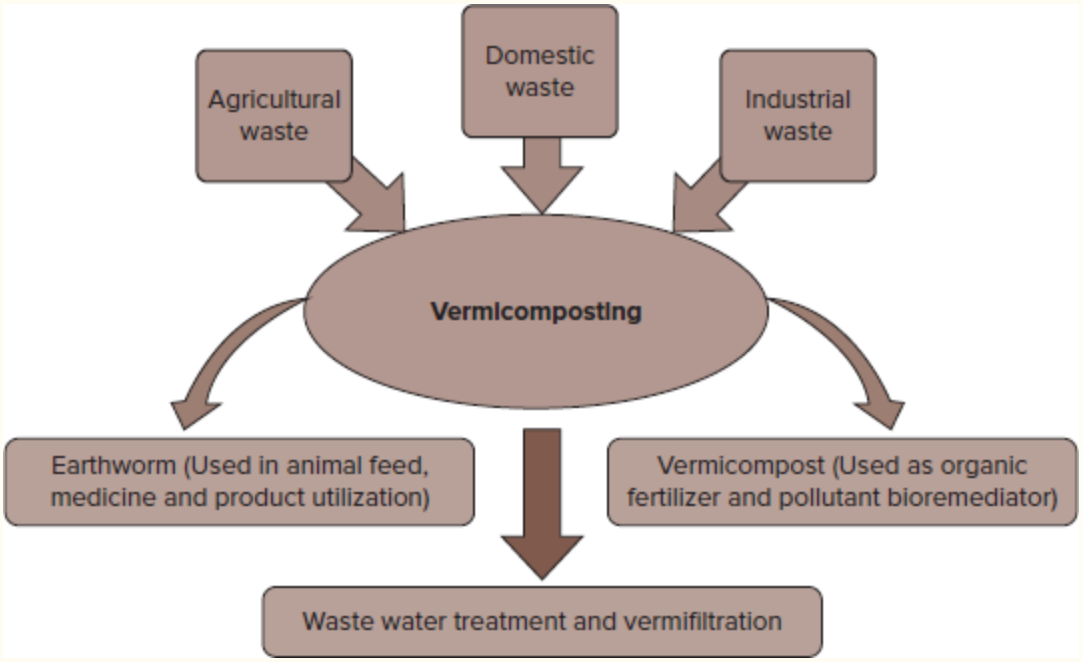

Figure 23.9 Vermicomposting

23.11 INDIAN VULTURES AND THEIR RECENT ISSUES

■ *Distribution and population*

Gyps indicus breeds in south-east Pakistan and peninsular India south of the Gangetic plain, north to Delhi, east through Madhya Pradesh, south to the Nilgiris, and occasionally further south. It was common until very recently,

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but since the mid-1990s has suffered a catastrophic decline (over 97%) throughout its range. All three species were declared Critically Endangered in 2000.

It is found in cities, towns and villages near cultivated areas, and in open and wooded areas. This species feeds almost entirely on carrion. It nests almost exclusively in colonies on cliffs and ruins, although it can be found nesting in trees.

Types

- White-rumped Vulture - *Gyps bengalensis*
- The Indian vulture - *G. indicus*
- Slender-billed vulture - *G. tenuirostris*

Reasons for Decline

- Extensive research has identified the nonsteroidal anti-inflammatory drug (NSAID) diclofenac to be the cause behind this rapid population collapse.
- This was caused because of feeding on carcasses of animals treated with the veterinary drug diclofenac.
- This drug, used to treat domestic livestock, is ingested by vultures feeding on their carcasses, leading to renal failure causing visceral gout.

Why do we need Vultures?

- Vultures provide a crucial ecosystem service through the disposal of livestock carcasses and their loss has huge socio-economic impacts across the Indian Subcontinent.
- Without vultures, hundreds of thousands of animal carcasses are uneaten —left to rot in the sun; these pose a serious risk to human health.
- Livestock carcasses provide a potential breeding ground for numerous infectious diseases, including anthrax, and encourage the proliferation of pest species, such as rats.

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Scavenger birds like vultures ensure that the dead and decomposed organisms are consumed, keeping the living healthy and epidemic free.

Conservation Actions Underway

- The Indian government had passed a bill banning the manufacture of the veterinary drug diclofenac that has caused the rapid population decline across the Indian subcontinent; their aim was to phase out its use.
- Efforts to replace diclofenac with a suitable alternative are ongoing; drug companies have now developed meloxicam, an alternative to diclofenac, which has been tested on Gyps vultures with no ill-effects.
- Captive breeding efforts are ongoing and captive birds laying eggs has raised hopes that they will successfully breed in captivity in the near future.
- In January 2012, the State Government of Karnataka declared 346.14 hectares in and around the hill as Ramadevarabetta Vulture Sanctuary, which is also India's first Vulture Wild Life Sanctuary.
- Ramadevarabetta is one of the world's oldest granite outcrops and is also blown as South India's earliest rock climbing hub.

Without Vultures

- Feral dogs have moved into carcass dumps, increasing the spread of diseases such as rabies.
- Researchers say that the increased number of rabies victims may have cost the Indian economy $34 billion.
- Traditional sky burials of some Himalayan and Parsi communities cannot be carried out.
- Cattle owners will have to pay to have livestock carcasses buried or burnt.

Conservation Actions Required

- Identify the location and number of remaining individuals and identify action required to prevent extinction. Measure the frequency of

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diclofenac treated carcasses available to vultures.

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- Support the ban on the veterinary use of **diclofenac** and support species management or restoration, as needed.
- Initiate public awareness and public support programmes.
- Monitor remaining populations.
- Support captive breeding efforts at a number of separate centres. Promote the immediate adoption of **meloxicam** as an alternative to diclofenac.
- Test other non-steroidal anti-inflammatory drugs (NSAIDs) to identify additional safe alternative drugs to diclofenac and also other toxic ones.

23.12 WETLAND

Wetlands are defined as lands transitional between terrestrial and aquatic eco-systems where the water table is usually at or near the surface or the land is covered by shallow water.

Distribution of Wetlands in India

India, with its annual rainfall of over 130 cm, varied topography and climatic regimes, supports and sustains diverse and unique wetland habitats. Natural wetlands in India consist of the high-altitude Himalayan lakes, followed by wetlands situated in the flood plains of the major river systems, saline and temporary wetlands of the arid and semi-arid regions, coastal wetlands such as lagoons, backwaters and estuaries, mangrove swamps, coral reefs and marine wetlands, and so on. In fact, with the exception of bogs, fens and typical salt marshes, Indian wetlands cover the whole range of the ecosystem types found. In addition to the various types of natural wetlands, a large number of man-made wetlands also contribute to faunal and floral diversity. These man-made wetlands, which have resulted from the needs of irrigation, water supply, electricity, fisheries and flood control, are substantial in number. The various reservoirs, shallow ponds and numerous tanks support wetland biodiversity and add to the country's

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wetland wealth. It is estimated that freshwater wetlands alone support 20% of the known range of biodiversity in India.

According to the National Wetland Inventory and Assessment compiled by the Indian Space Research Organisation (ISRO), wetlands are spread over 1,52,600 square kilometres (sq. km) which is 4.63% of the total geographical area of the country. The majority of the inland wetlands are directly or indirectly dependent on the major rivers like **Ganga**, **Brahmaputra**, **Narmada**, **Godavari**, **Krishna**, **Kaveri** and **Tapti**. They occur in the hot arid regions of Gujarat and Rajasthan, the deltaic regions of the east and west coasts, highlands of central India, wet humid zones of south peninsular India and the Andaman and Nicobar and Lakshadweep Islands.

Indian wetlands are grouped as:

I. **Himalayan wetlands Ladakh and Zanskar:** Pangong Tso, Tso Morad, Chantau, Noorichan, Chushul and Hanlay marshes
Kashmir Valley: Dal, Anchar, Wular, Haigam, Malgam, Haukersar and Kranchu lakes
Central Himalayas: Nainital, Bhimtal and Naukuchital
Eastern Himalayas: Numerous wetlands in Sikkim, Assam, Arunachal Pradesh, Meghalaya, Nagaland and Manipur, Beels in the Brahmaputra and Barak valley.

II. **Indo-Gangetic Wetlands:** The Indo-Gangetic flood plain is the largest wetland system in India, extending from the river Indus in the west to Brahmaputra in the east. This includes the wetlands of the Himalayan terai and the Indo- Gangetic plains.

III. **Coastal wetlands:** The vast intertidal areas, mangroves and lagoons along the 7500 km long coastline in West Bengal, Orissa, Andhra Pradesh, Tamil Nadu, Kerala, Karnataka, Goa, Maharashtra and Gujarat. Mangrove forests of Sundarbans, West Bengal and Andaman and Nicobar Islands. Offshore coral reefs of Gulf of Kutch, Gulf of Mannar, Lakshadweep and Andaman and Nicobar Islands.

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IV. **Deccan:** A few natural wetlands, but innumerable small and large reservoirs and several water storage tanks in almost every village in the region.

■ Benefits

1. Provide habitat for a wide variety and number of wildlife and plants.
2. Filter, clean and store water— in other words, acting like kidneys for other ecosystems.
3. Collect and hold flood waters.
4. Absorb wind and tidal forces.
5. Provide places of beauty and many recreational activities.

■ Legal Framework

Wetlands conservation in India is indirectly influenced by an array of policy and legislative measures (Parikh and Parikli,1999). Some of the key legislations are given below:

- The Indian Fisheries Act, 1857
- The Indian Forest Act, 1927
- Wild Life (Protection) Act, 1972
- Water (Prevention and Control of Pollution) Act, 1974
- Territorial Water, Continental Shelf, Exclusive Economic Zone and other Marine Zones Act, 1976
- Water (Prevention and Control of Pollution) Act, 1977
- Maritime Zone of India (Regulation and fishing by foreign vessels) Act, 1980
- Forest (Conservation Act), 1980
- Environmental (Protection) Act, 1986
- Coastal Zone Regulation Notification, 1991
- Wild Life (Protection) Amendment Act, 1991 and 2022
- National Conservation Strategy and Policy Statement on Environment and Development, 1992
- National Policy And Macro level Action Strategy on Biodiversity, 1999

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India is also a signatory to the **Ramsar Convention on Wetlands** and the **Convention of Biological Diversity.** Apart from government regulation, the development of better monitoring methods is needed to increase the knowledge of the physical and biological characteristics of each wetland resource, and to gain, from this knowledge, a better understanding of wetland dynamics and their controlling processes. India, being one of the mega diverse nations of the world, should strive to conserve the ecological character of these ecosystems along with the biodiversity of the flora and fauna associated with these ecosystems.

23.13 SOCIAL FORESTRY & AGRO FORESTRY

Social forestry may be defined as the science and art of growing trees in and outside traditional forest areas and managing like an existing forest with intimate involvement of the people and more or less integrated with other operation resulting in balanced and complementary land used with a view to providing a wide range of goods and services to the individual as well as to the society.

Key Facts about Social Forestry

It is also called Community Forestry:

- The term social forestry was first used by a forest scientist named Westoby in 1976. Socialforestry is the greatest instrument of landtransformation.
- Consider the number of trees, which we will have if each farmer raises even 10 trees on his farmland. The figure and yield will be colossal and its effect on the economy will be very impressive. If each tree is harvested, say at the 10th year, the farmer having 10 earns Rs. 400 at Rs. 40 only. This increases the area under trees for the benefit of the community as a whole and rural community in particular.

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- Development of trees on agricultural and other wastelands has tremendous effects. Trees control sheet, rill and gully erosion, retain moisture in soil, provide the farmer with fuel and timber for agricultural implements, improve the climate, provide recreation to people, save cow dung for manure and wood required for cremation, which is scarce sometime.
- Fuel alone to the extent of 90 million tons is consumed annually and it is estimated that more than 180 million tons of fuel will be needed. Again, it is said that about 400 million tons of cow dung equivalent to about 60 million tons of fuel wood are burnt annually in our country.

■ Social Forestry Programme

- Government is trying to increase forest areas that are close to human settlement and have been degraded over the years due to human activities needed to be afforested. Trees were to be planted in and around agricultural fields.
- Plantation of trees along railway lines and roadsides, and river and canal banks was carried out.
- They were planted in village common land, government wasteland, and Panchayat land. Social forestry scheme was initiated in India to increase fuel availability in rural areas and to prevent soil erosion.
- This programme was a failure because of the lack of governance. It is important to know that social forestry includes maximum utilisation of land for several purposes.

■ Types of Social Forestry

Social forestry schemes can be categorised into the following groups:

1. Farm Forestry
2. Community Forestry
3. Extension Forestry
4. Agro Forestry

1. **Farm Forestry**

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- Individual farmers are being encouraged to plant trees on their own farmland to meet the domestic needs of the family.
- In many areas, this tradition of growing trees on the farmland already exists. Noncommercial farm forestry is the main thrust of most of the social forestry projects in the country today.
- It is not always necessary that the farmer grows trees for fuel wood, but very often, they are interested in growing trees without any economic motive.
- They may want it to provide shade to the agricultural crop; as wind shelters; soil conservation or to use wasteland. Farm Forestry is another name for Agro-forestry, a part of Social Forestry.

2. Community Forestry

- Another scheme taken up under the social forestry programme is the raising of trees on community land and not on private land that is in farm forestry.
- All these programmes aim to provide for the entire community, and not for any individual.
- The government has the responsibility of providing seedlings, fertiliser but the community has to take responsibility for protecting the trees.
- Some others took advantage and sold the timber for a short- term individual profit. Common land being everyone's land is very easy to exploit.
- Over the last 19 years, large-scale planting of Eucalyptus, as a fast-growing exotic, has occurred in India, making it a part of the drive to reforest the subcontinent, and create anadequate supply of timber for rural communities upon the augur of 'social forestry'.

3. Extension Forestry

- Planting of trees on the sides of roads, canals and railways, along with planting on wasteland, is known as 'extension' forestry, increasing the boundaries of forests.

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- Under this project, there has been the creation of wood lots in the common village lands, government wastelands and Panchayat lands.
- Schemes for afforesting the degraded government forests that are close to villages are being carried out all over the country.

4. **Agro-Forestry**

- Agro-forestry or agro-sylviculture is a land use management system in which trees or shrubs are grown around or among crops or pastureland.
- It combines shrubs and trees in agriculture and forestry technologies to create more diverse, productive, profitable, healthy, ecologically sound and sustainable land-use systems.
- Agro-forestry has a lot in common with intercropping. Both have two or more plant species (such as nitrogen-fixing plants) in close interaction; both provide multiple outputs, and, as a consequence, higher overall yields and, because a single application or input is shared, costs are reduced.
- Beyond these, there are advantages specific to agro-forestry.
- Agro-forestry systems can be advantageous over conventional agricultural and forest production methods. They can offer increased productivity, economic benefits, and more diversity in the ecological goods and services provided.

■ *Topics Covered under Agro-forestry*

Depending upon the application, the positive impacts of agro-forestry comprise different topics.

Biodiversity

- Biodiversity in agro-forestry systems is typically higher than in conventional agricultural systems.
- With two or more interacting plant species in a given land area, it creates a more complex habitat that can support a wider variety of birds, insects, and other animals.

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- agro-forestry is important for biodiversity for different reasons. It provides a more diverse habitat than a conventional agricultural system.
- A further contribution to biodiversity is that the germplasm of sensitive species can be preserved. And as there is no regular cleaning like in conventional agriculture systems, habitats are more consistent.
- Furthermore, agro-forestry systems can serve as corridors between habitats, for instance forests. Agro-forestry also helps to conserve biodiversity by having positive influence on other ecosystem services. Some are mentioned below.

Soil and Plant Growth

- Depleted soils can be prevented from soil erosion due to ground cover plants such as naturally growing grasses of agro-forestry systems helping to stabilise the soil as they increase soil cover compared to short cycle cropping systems.
- Soil cover is a crucial factor in preventing erosion. Cleaner water through reduced nutrient and soil surface runoff can be a further advantage of agro-forestry systems.
- The runoff can be reduced by decreasing its velocity and increasing the infiltration into the soil in an established agro-forestry system. Compared to row cropped fields, the uptake of nutrients can be higher and thus result in a reduction of nutrients loss into streams.

Further advantages concerning plant growth

- Bioremediation (Bioremediation is a process used to treat contaminated media, including water, soil and subsurface material, by altering environmental conditions to stimulate growth of microorganisms and degrade the target pollutants).
- Typically, more droughts resistant
- Increased crop stability

Contribution to Sustainable Agricultural Systems

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- Reducing poverty through increased production of wood and other tree products for home consumption and sale.
- Contributing to food security by restoring the soil fertility for food crops.
- Multifunctional site use, i.e. crops production and animal grazing.
- Measuring global warming and the risk of hunger by increasing the number of drought-resistant trees and the subsequent production of fruits, nuts and edible oils.
- Reducing deforestation and pressure on woodlands by providing farm-grown fuel wood.
- Reducing or eliminating the need for toxic chemicals (insecticides, herbicides, etc.).
- Through more diverse farm outputs, it has improved human nutrition.
- In situations where people have limited access to mainstream medicines, it provides growing space for medicinal plants.

■ *Other Environmental Goals*

- The potential of carbon sequestration is discussed as an important ecosystem service.
- Trees in agro-forestry systems, like in new forests, can regain some of the carbon that was lost by cutting existing forests but not completely replacing them. However, they provide additional food and products to people.
- The rotation age and the use of the resulting products are important factors contributing to the amount and duration of sequestered carbon.
- Agro-forestry systems can offset deforestation of primary forests as they reduce pressure on the latter by providing some of the products needed from forests.

Agro-forestry practices may also realise a number of other associated environmental goals, such as

- Odour, dust, and noise reduction
- Green space and visual aesthetics
- Enhancement or maintenance of wildlife habitat

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National Agro-forestry Policy, 2014

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- The maiden policy comes following the **World Congress on Agro forestry** held in New Delhi from February 10–14th, 2014. It aims to include agro-forestry in the mainstream of agricultural policies and strategies by promoting it in integrated farming systems.

The Policy has linkage with several ongoing mega campaigns, like the 'Green India Mission' under which the state plans to plant 100 crores trees to improve forest cover.

The Major Policy Goals

- Setting up a National Agro-forestry Mission or an Agro-forestry Board to implement the National Policy by bringing coordination, convergence and synergy among various elements of agro-forestry scattered in various existing missions, programmes, schemes and agencies pertaining to agriculture, environment, forestry, and rural development sectors of the Government.
- Improving the productivity, employment, income and livelihood opportunities of rural households, especially of the smallholder farmers through agro-forestry.
- Meeting the ever-increasing demand for timber, food, fuel fodder, fertiliser, fibre and other agro-forestry products; conserving the natural resources and forest; protecting the environment and providing environmental security; and increasing the forest/tree cover. There is a need to increase the availability of these from outside the natural forests.

Basic Objectives

- Encourage and expand tree plantation in a complementarily and integrated manner with crops and livestock to improve productivity, employment, income and livelihoods of rural households, especially the smallholder farmers.

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- Protect and stabilise ecosystems, and promote resilient cropping and farming systems to minimise the risk during extreme climatic events.
- Meet the raw material requirements of wood- based industries and reduce import of wild and wood products to save foreign exchange.
- Supplement the availability of agro-forestry products (AFPs), such as the fuel-wood, fodder, non-timber forest produce and small timber for the rural and tribal populations, thereby reducing the pressure on existing forests.
- Complement achieving the target of increasing forest/tree cover to promote ecological stability, especially in the vulnerable regions.

Develop capacity, strengthen research in agro-forestry and create a massive people's movement to achieve these objectives and minimise pressure on existing forests.

■ *Strategy*

- Establishment of institutional set-up at the national level to promote agro-forestry.
- Simple regulatory mechanism.
- Development of a sound database and information system.
- Investing in research, extension and capacity building material.
- Providing institutional credit and insurance cover for agro-forestry.
- Facilitating increased participation of industries dealing with agro-forestry produce.
- Strengthening farmer's access to markets for tree products.
- Incentives to farmers for adopting agro-forestry.
- Promoting sustainable agro-forestry for renewable biomass- based energy.

23.14 MANGROVES

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The word “Mangrove” is considered to be a combination of the Portuguese word “Mangue” and the English word “grove”. Mangroves are salt-tolerant plants of tropical and subtropical intertidal regions of the world.

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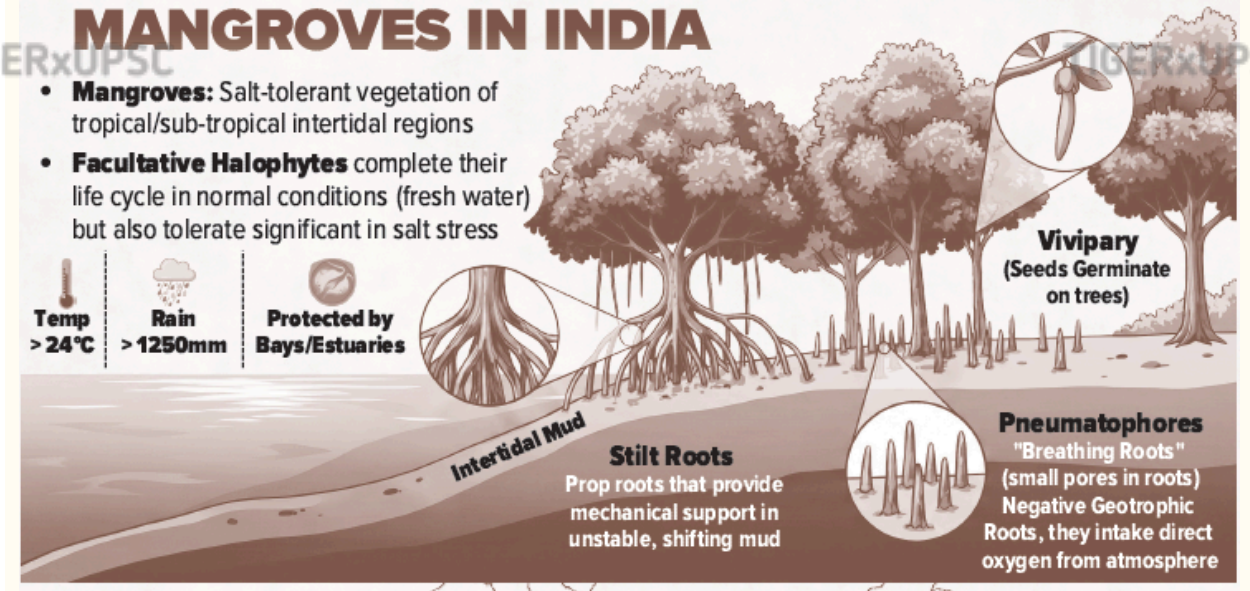
MANGROVES IN INDIA

- **Mangroves:** Salt-tolerant vegetation of tropical/sub-tropical intertidal regions
- **Facultative Halophytes** complete their life cycle in normal conditions (fresh water) but also tolerate significant in salt stress



DISTRIBUTION

- **Total Area (ISFR 2023)- 4998.61 KM²** (**0.15%** of India's total area)
- **Mangrove Cover increased by 293 KM²**
- **State Performance (ISFR 2023)**
  - Loser: Gujarat (-36.4 km²)
  - Gainers: Andhra Pradesh (+13 km²), Maharashtra (+12 km²)

West Coast- 31%
Sparse due to Steep Gradients

East Coast- 57%
Dominant due to large river deltas and gentle slope

West Bengal (Sundarbans)
Largest Single Block (UNESCO Site)

Odisha (Bhitarkanika)
Home to Saltwater Crocs & Olive Ridleys

Pichavaram (Tamil Nadu)
Complex backwater channels

A&N Islands- 12%

Baratang Island (Andaman & Nicobar)- 13% Dense, evergreen mangrove swamps

IMPORTANCE	THREATS
Bio-Shield: Protects against Cyclones/Tsunamis	**Aquaculture:** Conversion for shrimp farming
Blue Carbon: 4x more carbon than rainforests	**Agriculture:** Reclamation of land
Nursery Ground: Breeding site for fish/crabs	**River Diversion:** Increases Salinity
Nature's Kidney: Filters heavy metals	**Pollution:** Industrial runoff, oil, ports

POLICY & GOVERNANCE		CONSERVATION INITIATIVES			
Art 51 A (g) Protect environment	**CRZ-I Status** under EPA, 1986	**ICZM** Integrated Coastal Zone Management	**Ecotourism** Sustainable income source	**Mangrove Planting & Regeneration**	**MISHTI** Aiming to cover ~540 km² across 11 States & 2 UTs

Mangroves in India

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- West Coast
- East Coast
- Andaman and Nicobar Islands

The total area of mangroves in India is about 6,740 sq km, which is about **7% of the world's total area of mangroves. 80% are present along the east coast**, mostly forming the Sunderbans, Bhitarkanika and the Andaman & Nicobar mangroves. Besides, large rivers like Mahanadi, Krishna, Cauveri, and Godavari also harbour major mangroves in their estuarine regions. The remaining 20% mangroves are scattered on the west coast from Kutch to Kerala.

The reason for such a restricted mangrove cover is the peculiar coastal structure and the nature of estuaries formed by the relatively small and nonperennial rivers except Narmada and Tapi.

■ *Where do Mangroves occur?*

The richest mangrove communities occur in

- Tropical and sub-tropical areas, i.e., between the 30°N and 30°S latitudes.
- Water temperature is greater than 24°C in the warmest month.
- The annual rainfall exceeds 1250 mm.
- Where mountain ranges greater than 700 m high are found close to the coast.

■ *Conditions Required*

Mangroves are found extensively in the estuarine regions where mud-flats are wide and gently sloping. Besides estuaries, they also inhabit the intertidal regions of shallow bays and creeks where the environment is conducive for the growth of mangroves.

■ *Do mangroves need Salt?*

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Mangroves are **facultative halophytes**, i.e., the presence of salt in the environment is not necessary for the growth of mangroves and they can grow very well in freshwater.

■ *Importance of Mangroves*

(a) Buffer zone between the land and the sea.
(b) Protect the land from erosion.
(c) Play an invaluable role as nature's shield against cyclones, ecological disasters and as protector of shorelines.
(d) Breeding and nursery grounds for a variety of marine animals.
(e) Harbour a variety of life forms like invertebrates, fish, amphibians, reptiles, birds and even mammals like tigers.
(f) Good source of timber, fuel and fodder.
(g) Main source of income generation for shoreline communities like fisher folk.
(h) Purify the water by absorbing impurities and harmful heavy metals and help us to breathe a clean air by absorbing pollutants in the air.
(i) Potential source for recreation and tourism.
(j) Mangroves are flood buffers. They also help to stabilise the climate by moderating temperature, humidity, wind and even waves. They are specially adapted to withstand salinity, wave action, and can grow in poor soils. They actually protect the land from the impact of the sea.

■ *Adaptations*

Mangrove environment is highly dynamic and harsh and mangrove species are variously adapted to cope with these environmental conditions.

- **Breathing Roots (Respiratory Roots):** Underground tissue of any plant requires oxygen for respiration and in mangrove environment, oxygen in soil is very limited or nil. This necessitates mangrove root system to take up oxygen from the atmosphere. For this purpose, mangrove species have specialised above ground roots called breathing roots or **pneumatophores**. These roots have numerous pores through which

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oxygen enters into the underground tissue. In some plants, buttress roots function as breathing roots and also provide mechanical support to the tree.

- **Stilt roots:** In some mangrove species, roots diverge from stems and branches and penetrate the soil some distance away from the main stem, as in the case of banyan trees. Because of their appearance and because they provide the main physical support to these, they are called Stilt roots. These roots also have many pores through which atmospheric oxygen enters into the roots.
- **Vivipary:** Saline water, unconsolidated saline soil with little or no oxygen is not a conductive environment for seeds to germinate and establish. To overcome this, mangrove species have a unique way of reproduction, which is generally known as vivipary. In this method of reproduction, seeds are still attached to the parent tree. These seedlings are normally called propagules and they photosynthesis while the seeds are in mother tree. The parent tree supplies water and necessary nutrients. They are buoyant and float in the water for some time before rooting themselves in suitable soil.

Uses

- Mangrove wetland is a multiple ecosystem.
- It is considered as the best form of bio-shield since it plays a critical role in reducing the impact of cyclonic storms, hurricanes and tsunami in human lives and properties. For example: Mangrove vegetation in Andhra Pradesh saved lives by reducing the impact of tsunami in 2004.
- It reduces soil erosion. It enhances fishery productivity of the adjacent coastal waters by acting as a nursery ground for commercially important fish, prawn and crabs and supplying organic and inorganic nutrients.
- They are rich in biodiversity and act as habitats for wildlife. For example: Sundarbans—homes to Royal Bengal Tiger, fishing cats, flying foxes, spotted dears, etc.

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Management of Mangroves in India

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India has a long tradition of mangrove forest management. The Sundarbans mangroves, located in the Bay of Bengal (partly in India and partly in Bangladesh), were the first mangroves in the world to be put under scientific management.

1. The area's first management plan was implemented in 1892.
2. 1976 amendment to the Indian Constitution, which states that it shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife.
3. The Government of India set up the **National Mangrove Committee** in the **Ministry of Environment and Forests** (MoEF) in 1976 to advise the government about mangrove conservation and development.
4. The National Forest Policy, 1988 lists effective conservation and management of natural forest ecosystems (including the mangrove ecosystem) as a priority area for forestry research.

Legislative Framework

- The Indian Forest Act, 1927 and the Wildlife (Protection) Act, 1972 provide protection to flora and fauna. Although they do not specifically mention mangroves, these Acts can also apply to the conservation of the flora and fauna of mangrove ecosystems.
- The Forest Conservation Act, 1980 states that no forest area shall be diverted for any non-forestry purpose without prior approval of the Government of India.
- The Environment (Protection) Act, 1986 has had a crucial role in the conservation and management of mangrove ecosystems. It declares a Coastal Regulation Zone in which industrial and other activities such as discharge of untreated water and effluents, dumping of waste, land reclamation and bunding are restricted in order to protect the coastal environment. Coastal stretches are classified into four categories, and mangroves are included in the most ecologically sensitive category.

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Causes for Destruction of Mangroves

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- Natural hazards like cyclones, typhoons and strong wave action destroy mangroves.
- Land reclamations for construction activity, aquaculture, agriculture, tourism.
- Industrial and domestic pollution.
- Port development.
- Dumping of all kinds of waste and debris.
- Deforestation for fuel wood.
- Overharvesting of marine resources.
- Illegal large-scale collection of mangrove fruits, which hinders their natural regeneration.
- Movement of barges which give rise to strong waves that sometimes damage the young mangrove seedlings.
- Obstruction and diversion of water for culvert construction.
- Lack of interest of private landowners in conserving and developing the mangroves on their lands.

Important Mangrove Sites in India

1. **Sundarbans**

   The **Great Sundarbans**, the **largest mangrove region in the world** and **UNESCO World Heritage Site**, is the largest single block of tidal halophytic mangrove forest in the world, covering parts of Bangladesh's Khulna Division and Indian state of West Bengal.

Key Facts

- In the Ganga-Brahmaputra delta, Sundari trees are found, which provide durable hard timber. Palm, coconut, Keora, and agar also grow in some parts of delta.
- India's mangrove forests are known to serve as a habitat for **turtles, crocodiles, gharials** and **snakes**. This region is part of the Great

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Sundarbans and covers a National Park, Tiger Reserve and a Biosphere Reserve.

- The **Sundarbans National Park** is a **National Park**, a **Tiger Reserve** and a **Biosphere Reserve**. It is one of the largest reserves for the **Bengal Tiger**.
- A third of this area is covered by water and marshes. Since 1966, it has been considered a sanctuary for wildlife with an estimated 400 Royal Bengal tigers and more than 30,000 deer. The forest lies at the foot of the Ganges. The seasonally flooded Sundarbans freshwater swamp forest lies inland from mangrove forests.
- The forest covers 10,000 square kilometres, of which about 6,000 square kilometres (2300 sq m) are in Bangladesh.
- It was described as a UNESCO world heritage site in 1997, they are separately listed in the UNESCO world heritage list as the Sundarbans and Sundarbans National Park, respectively.
- The Sundarbans is intersected by a complex network of tidal waterways, mudflats and small islands of salt-tolerant mangrove forests.
- The area is known for the eponymous Royal Bengal Tiger *(Panthera tigris)*, as well as numerous fauna including species of birds, spotted deer, crocodiles and snakes.
- Sundarbans was designated as a Ramsar site on May 21, 1992.
- The fertile soils of the delta have been subject to the intensive human use for centuries, and the eco-region has been mostly converted to intensive agriculture, with few enclaves of forest remaining. Together with the Sundarbans, mangroves are an important habitat for the endangered tiger.

2. *Bhitarkanika Mangroves*

- Bhitarkanika Mangroves is India's second largest forest, located in the state of Odisha.
- Bhitarkanika is formed by two river deltas of Brahmani and Baitarani River and one of the important Ramsar Wetlands in India.
- It is also home to salt-water crocodile and nesting beach olive ridley sea turtle.

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3. *Godavari-Krishna Mangroves*

- The Godavari-Krishna mangrove lies in the delta of the Godavari and Krishna rivers in the state of Andhra Pradesh.
- Mangrove ecoregion is under protection for Calimere Wildlife and Pulicat Lake Bird Sanctuary.

4. *Pichvaram Mangroves*

- Pichavaram mangrove is the world's second largest mangrove forest, situated at Pichavaram near Chidambaram in the state of Tamil Nadu.
- Pichavaram ranks among one of the most exquisite scenic spots in Tamil Nadu and is home to many species of aquatic bird.

Protecting the Sundarban Wetlands

On January 30, 2019, the Indian Sundarban was accorded the status of 'Wetland of International Importance' under the Ramsar Convention.

Key Facts

- The Sundarbans comprises hundreds of islands and a network of rivers, tributaries and creeks in the delta of the Ganga and the Brahmaputra at the mouth of the Bay of Bengal in India and Bangladesh.
- Located on the southwestern part of the delta, the Indian Sundarban constitutes over 60% of the country's total mangrove forest area.
- It is the 27th Ramsar Site in India, and with an area of 4,23,000 hectares is now the largest protected wetland in the country.
- The Indian Sundarban, also a UNESCO world heritage site, is home to the Royal Bengal Tiger. It is also home to a large number of "rare and globally threatened species, such as the critically endangered northern river terrapin (Batagurbaska), the endangered Irrawaddy dolphin (Orcaellabrevirostris), and the vulnerable fishing cat (Prionailurusviverrinus)."
- Two of the world's four horseshoe crab species and eight of India's 12 species of kingfisher are also found here. Recent studies claim that the Indian Sundarban is home to 2,626 faunal species and 90% of the country's mangrove varieties.

5. *Baratang Insland Mangroves*

- Baratang Island Mangrove is a swamp, located at Great Andaman and Nicobar Islands.

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- Mangrove Swamps of Baratang Island are situated between Middle and South Andaman, capital city Port Blair.

23.15 BIOCHAR

Biochar is charcoal that is produced by pyrolysis of biomass in the absence of oxygen (pyrolysis or gasification). It is a stable solid that is rich in carbon and can endure in soil for thousands of years.

on ENV 78- Biochar

- This primitive practice helps in the conversion of agricultural waste into a soil enhancer which holds carbon, enhances food security, and expands soil biodiversity, and decreases deforestation. The process assists in the retaining of soils water and nutrients through the formation of a fine-grained, highly porous charcoal.
- Biochar is found in soils around the world as a result of vegetation fires and historic soil management practices. Intensive study of biochar-rich dark earth in the Amazon (terra preta) has led to a wider appreciation of biochar's unique properties as a soil enhancer.
- Biochar can be an important tool to increase food security and cropland diversity in areas with severely depleted soils, scarce organic resources, and inadequate water and chemical fertiliser supplies.
- Biochar also improves water quality and quantity by increasing soil retention of nutrients and agrochemicals for plant and crop utilisation. More nutrients stay in the soil instead of leaching into groundwater and causing pollution.

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- The application of biochar to agricultural soils reduces N_2O emissions, decreases N leaching, improves soil nitrogen availability, increases crop productivity, and promotes the activity of soil microbes.

Difference between Charcoal and Biochar:

- There are differences between biochar and charcoal. The first difference is just a difference in application. Charcoal usually refers to carbonised biomass used for cooking, whereas biochar is applied to soil and used in agriculture or gardening. The second difference is that biochar can be made from a wide variety of different substances, from wood to grains, bones, chaff, or manure. Charcoal is more commonly made from wood.

23.16 VERTICAL FARMING

It is the method of food production on vertically inclined areas or surfaces. This method helps in the food production in the vertically piled layers usually integrated into structures such as a skyscraper, shipping container or repurposed warehouse instead of growing vegetables and other foods on a single level.

- Using Controlled Environment Agriculture (CEA) technology, this modern idea uses indoor farming techniques. The artificial control of temperature, light, humidity, and gases makes producing foods and medicine indoor possible. In many ways, vertical farming is similar to greenhouses, where metal reflectors and artificial lighting augment natural sunlight. The primary goal of vertical farming is maximising crop output in a limited space.

How Vertical Farming Works?

- There are four critical areas in understanding how vertical farming works: 1. Physical layout, 2. Lighting, 3. Growing medium, and 4. Sustainability features.
- Firstly, the primary goal of vertical farming is producing more foods per square meter. To accomplish this goal, crops are cultivated in stacked

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layers in a tower life structure. Secondly, rotating beds technologies are used to enhance lighting efficiency and to maintain a right proportion of natural and artificial lights.

- Thirdly, instead of soil, aeroponic, aquaponic or hydroponic growing mediums are used. Peat moss or coconut husks and similar non-soil mediums are very common in vertical farming. Finally, the vertical farming method uses various sustainability features to offset the energy cost of farming. In fact, vertical farming uses 95% less water.

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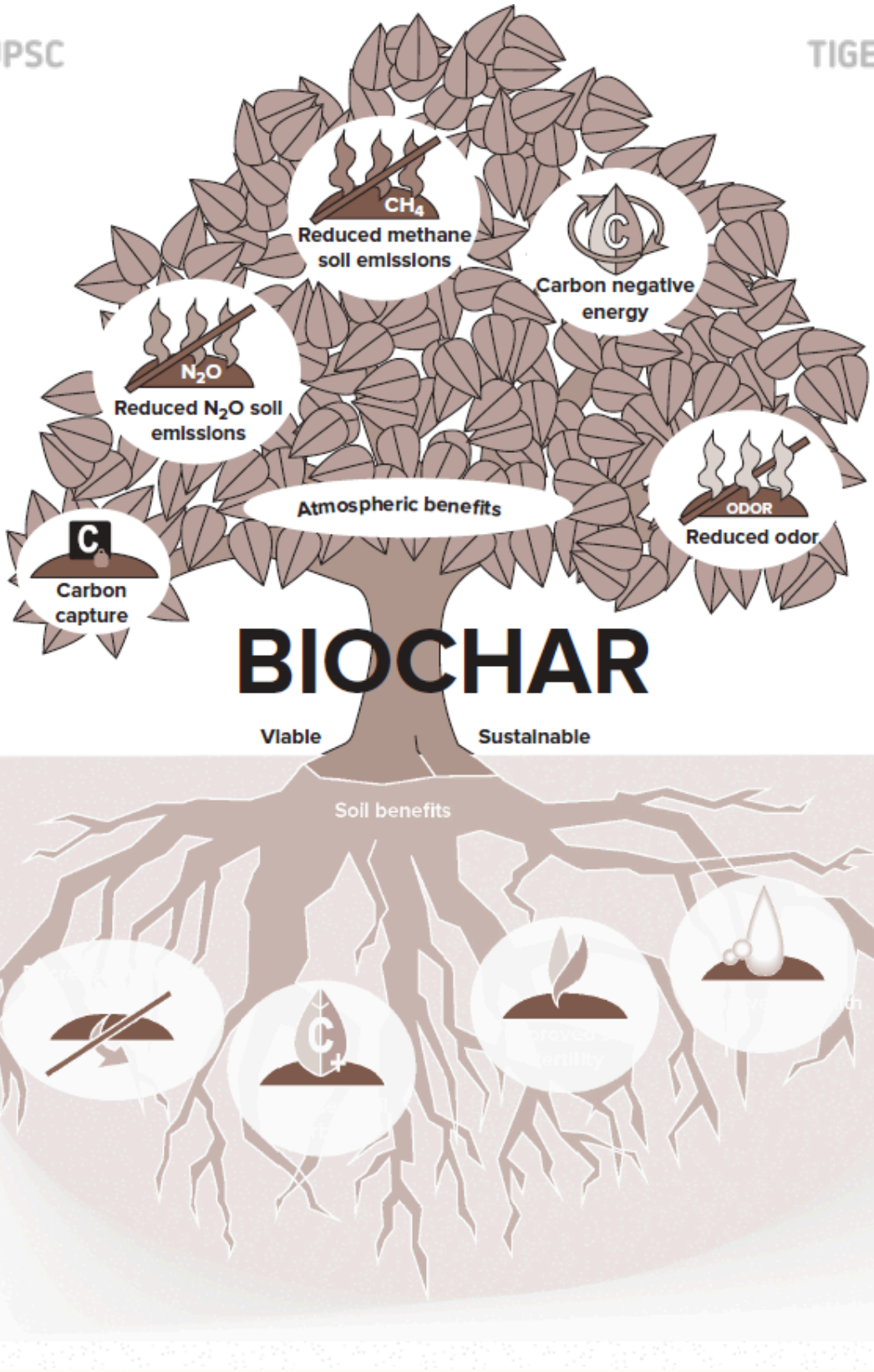

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Figure 23.10 Biochar

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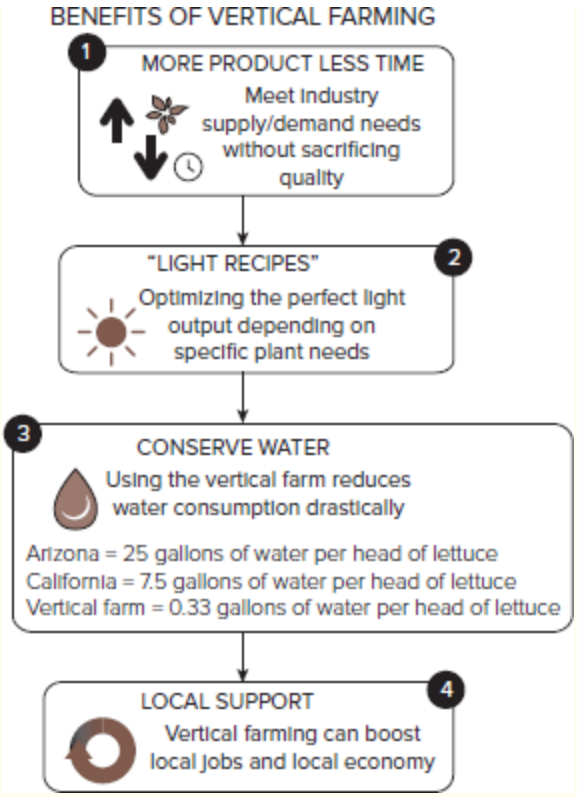

Figure 23.11 Benefits of Vertical Farming

23.17 ORGANIC FARMING

Organic farming system in India is not new and has been followed from the ancient times. It is a method of farming system which primarily aimed at cultivating the land and raising crops in such a way so as to keep the soil alive and in good health by use of organic wastes (crop, animal and farm wastes, aquatic wastes) and other biological materials along with beneficial microbes (biofertilisers) to release nutrients to crops for increased sustainable production in an ecofriendly pollution free environment.

As per the definition of the United States Department of Agriculture (USDA) study team on organic farming, "organic farming is a system which avoids or largely excludes the use of synthetic inputs (such as fertilisers, pesticides, hormones, feed additives, etc.) and to the maximum extent feasible rely upon crop rotations, crop residues, animal manures, off-

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farm organic waste, mineral grade rock additives and biological system of nutrient mobilisation and plant protection."

FAO suggested that "Organic agriculture is a unique production management system which promotes and enhances agro-ecosystem health, including biodiversity, biological cycles and soil biological activity, and this is accomplished by using on-farm agronomic, biological and mechanical methods in exclusion of all synthetic off-farm inputs."

Need of organic farming

With the increase in population, our compulsion would be not only to stabilise agricultural production but to also increase it further in a sustainable manner. The scientists have realised that the 'Green Revolution' with high input use has reached a plateau and is now sustained with diminishing return of falling dividends. Thus, a natural balance needs to be maintained at all costs for the existence of life and property. The obvious choice for that would be more relevant in the present era when these agrochemicals which are produced from fossil fuel and are not renewable are diminishing in availability. It may also cost heavily on our foreign exchange in the future.

The key characteristics of organic farming include

- Protecting the long- term fertility of soils by maintaining organic matter levels, encouraging soil biological activity, and careful mechanical intervention
- Providing crop nutrients, indirectly using relatively insoluble nutrient sources which are made available to the plant by the action of soil micro-organisms
- Nitrogen self-sufficiency through the use of legumes and biological nitrogen fixation, as well as effective recycling of organic materials including crop residues and livestock manures

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- Weed, disease and pest control relying primarily on crop rotations, natural predators, diversity, organic manuring, resistant varieties and limited (preferably minimal) thermal, biological and chemical intervention
- The extensive management of livestock, paying full regard to their evolutionary adaptations, behavioural needs and animal welfare issues with respect to nutrition, housing, health, breeding and rearing
- Careful attention to the impact of the farming system on the wider environment and the conservation of wildlife and natural habitats

■ National Programme for Organic Production (NPOP)

National Programme for Organic Production (NPOP) is managed by Agricultural and Processed Food Products Export Development Authority (APEDA) under the Ministry of Commerce and Industry.

APEDA is assisting the exporters of organic products under various components of its export promotion scheme.

The Ministry of Commerce has implemented the National Programme for Organic Production (NPOP) since 2001. The objectives of NPOP are as under:

1. To provide the means of evaluation of certification programme for organic agriculture and products (including wild harvest, aquaculture, livestock products) as per the approved criteria.
2. To accredit certification programmes of Certification Bodies seeking accreditation.
3. To facilitate certification of organic products in conformity with the prescribed standards.
4. To facilitate certification of organic products in conformity with the importing countries' organic standards as per the equivalence agreement between the two countries or as per importing country requirements.
5. To encourage the development of organic farming and organic processing.

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Do You Know?

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Do You Know?

Sikkim has formally become India's first fully organic state after it successfully implemented organic farming practices on around 75,000 hectares of agricultural land. Formal declaration in this regard was made by Prime Minister Narendra Modi after inaugurating the Sikkim Organic Festival 2016 in Gangtok, Sikkim.

APEDA

- The Agricultural and Processed Food Products Export Development Authority (APEDA) is an Indian Apex- Export Trade Promotion Active Government Body under Ministry of Commerce & Industry.
- According to the citizens' Charter Released by APEDA, its vision is to "Create a World Class Organisation Dedicated to Promote India as the 'Supplier of Choice' of Agro and Value Added Food Products in the Global Markets".
- APEDA is the premier body of export promotion of fresh vegetables and fruits. It provides the crucial interface between farmers, storehouses, packers, exporters, surface transport, ports, Railways, Airways, and all others engaged in export trade to the international market.

23.18 CSIR DEVELOPS LESS POLLUTING FIRECRACKERS

CSIR develops Less Polluting Firecrackers named - **Safe Water Releaser (SWAS), Safe Minimal Aluminium (SAFAL)** and **Safe Thermite Cracker (STAR)**.

Technical Details

- These crackers have a unique property of **releasing water vapour and/or air as dust suppressant** and **diluent for gaseous emissions and matching performance in sound** with conventional crackers.
- **SWAS crackers eliminate usage of (KNO_3) Potassium nitrate and Sulphur with the consequent reduction in particulate matter (30–35%) SO_2 and NOx**. It has matching sound intensity with commercial crackers in the range of 105–110 dBA. SWAS has been tested for shelf life upto 3 weeks with consistent performance.

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- **STAR eliminates usage of KNO3 and S with consequent reduction in particulate matter (35–40%), SO_2 and NOx**. It has matching sound intensity with commercial crackers in the range of 105–110 dBA.
- **SAFAL has minimal usage of aluminium (only in flash powder for initiation) with consequent significant reduction in particulate matter (35–40%)** compared to commercial crackers. It has matching sound intensity with commercial crackers in the range of 110–115 dBA.

Benefits

- Indian Fireworks industry is over 6000-crores worth of annual turnover and provides employment opportunities to over 5 lakh families directly or indirectly.
- This endeavour of CSIR aims at addressing the pollution concerns, at the same time protecting the livelihoods of those involved in this trade.
- These Firecrackers are not only environment- friendly but 15–20% cheaper than the conventional ones.

23.19 NITROGEN POLLUTION

The **annual Frontiers Report 2019, published by the United Nations (UN)**, has included a chapter on nitrogen pollution in its latest edition. The report was released by the **United Nations Environmental Assembly (UNEA) in Nairobi**.

ENV 79- Nitrogen Pollution

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Highlights of the Frontiers Report 2019

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- **Pollution caused by the reactive forms of nitrogen** is now being recognised as a grave environmental concern on a global level.
- It highlights that the growing demand on the livestock, agriculture, transport, industry and energy sectors has led to a **sharp growth in the levels of reactive nitrogen - ammonia, nitrate, nitric oxide (NO), and nitrous oxide (N_2O)** in our ecosystems.
- The report claims that the total annual cost of nitrogen pollution to ecosystem and healthcare services in the world is around $340 billion.
- The report also warns that the scale of the problem remains largely unknown and unacknowledged outside scientific circles.

Nitrogen as an Essential Nutrient

- Nitrogen, which is a **vital macronutrient for most plants**, is the most abundant element in the atmosphere.
- **A little over 78% of dry air on Earth is nitrogen**. But atmospheric nitrogen, or dinitrogen, is unreactive and cannot be utilised by plants directly.
- Until the beginning of the 20th century, farmers depended on a natural process called nitrogen fixation for the conversion of atmospheric nitrogen into reactive nitrogen in the soil: **nitrogen-fixing bacteria like rhizobia live symbiotically with leguminous plants**, providing nitrogen to the plant and soil in the form of reactive compounds like ammonia and nitrate.
- But the **natural nitrogen cycle was inadequate to feed the growing population**. Scientists Fritz Haber and Carl Bosch solved this problem by producing ammonia by combining atmospheric nitrogen with hydrogen gas at high temperature and pressure,known as the Haber-Bosch process. The Green Revolution, which was instrumental in establishing food security in developing countries in the 1960s, was driven by artificial nitrogen-fixation. Today, about half of the world's population depends on this process for its nutrition.

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How Nitrogen Turned into Pollutant from Nutrient? How it is Affecting Health and Environment?

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- **Nitrogen is an inert gas that is necessary for life**, but we're changing it into forms that are harmful, overloading the environment with it, and throwing the natural nitrogen cycle out of whack.
- **Nitrogen compounds running off farmland have led to water pollution problems around the world**, while nitrogen emissions from industry, agriculture and vehicles make a big contribution to air pollution.
- **Over 80% of the nitrogen in soil is not utilised by humans**, while over four-fifths of the nitrogen is used to feed livestock, only about six percent reaches humans in case of non-vegetarian diet, as compared to the 20% that reaches the plate of a vegetarian.
- **Nitrogen becomes a pollutant when it escapes into the environment and reacts with other organic compounds.** It is either released into the atmosphere, gets dissolved in water sources such as rivers, lakes or groundwater, or remains in the soil. While it might lead to favourable growth of species that can utilise this nutrient, nitrogen as a pollutant is often detrimental to the environment and health.

Effects on Health

- According to the World Health Organisation, nitrate-contaminated drinking water can cause reduced blood function, cancer and endemic goitres. Surplus inputs of nitrogen compounds have been found to cause soil acidification. The lowering pH, as a result of the acidification, can lead to nutrient disorders and increased toxicity in plants. It may also affect natural soil decomposition.

Nitrogen Pollution Has a Significant Impact on The Environment

- **It creates harmful algal blooms and dead zones in our waterways and oceans;** the algae produce toxins which are harmful to human and aquatic organisms (and indirectly affect fisheries and biodiversity in coastal areas).

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- **Contamination of drinking water**, 10 million people in Europe are potentially exposed to drinking water with nitrate concentrations above recommended levels. This can have an adverse effect on human health.
- **Food Security:** Excessive nitrogen fertiliser application contributes to soil nutrient depletion. As the world needs to feed an ever-growing population, loss of arable land is a major global problem.
- **The release of Nitrous Oxide is essentially a greenhouse gas** which is harmful to the environment.

International Nitrogen Initiative (INI)

- It is an international programme, set up in 2003 under the sponsorship of the Scientific Committee on Problems of the Environment (SCOPE) & International Geosphere Biosphere Programme (IGBP).
- The key aims of the project are:
- Optimise nitrogen's beneficial role in sustainable food production.
- Minimise nitrogen's negative effects on human health and the environment resulting from food and energy production.
- The programme is currently a sustained partner of Future Earth.
- Future Earth was launched in 2012 at the UN Conference on Sustainable Development (Rio+20).
- It aims to build knowledge about the environmental and human aspects of global change and to find solutions for sustainable development.
- INI holds a conference every 3 years (Last conference held in Melbourne in 2016 which led to Melbourne Declaration).
- It has called nitrogen the "new carbon".

South Asian Nitrogen Hub (SANH)

- 18 research institutions in India are among 50 institutions called South Asian Nitrogen Initiative Hub (SANH) to assess, study quantum and impact of Nitrogen Pollution in South Asia.
- SANH will look nitrogen in 8 countries:
  1. India
  2. Afghanistan
  3. Pakistan
  4. Bhutan
  5. Bangladesh
  6. Maldives
  7. Nepal
  8. Sri Lanka

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23.20 ISSUES RELATED TO STUBBLE BURNING

Only educating farmers about the monetary costs of burning stubble can address the environmental crisis triggered every year in Punjab, says a team of Swiss and Indian researchers who interviewed 600 farmers over two years.

Key Observations

- According to the team, the government's efforts - earmarking funds for specialised farming equipment (for straw management) or enforcing the state-led ban on the practice - are unlikely to solve the problem.
- Farmer cooperative groups - a key link between government and farmers - ought to be playing a more active role in educating farmers.

What is Stubble Burning?

- Stubble burning is a common practice followed by farmers in the neighbouring states Haryana and Punjab to prepare fields for sowing of wheat in November as there is little time left between the harvesting of paddy and sowing of wheat.
- **Stubble burning results in emission of harmful gases** such carbon dioxide, sulphur dioxide, nitrogen dioxide along with particulate matter.

What needs to be done, Supreme Court's Observations?

- The problem is required to be resolved by taking all such measures as are possible in the interest of public health and environment protection.
- Incentives could be provided to those who are not burning the stubble and disincentives for those who continue the practice.
- The existing Minimum Support Price (MSP) Scheme must be interpreted so as to enable the States concerned to wholly or partly deny the benefit of MSP to those who continue to burn the crop residue.

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- Secretary, Union Ministry of Agriculture and Farmers' Welfare, has also been directed to be present to "find a lasting solution."
- The Central government should convene a meeting with the States.

■ *Advantages of Stubble Burning:*

- It quickly clears the field and is the cheapest alternative.
- Kills weeds, including those resistant to herbicide.
- Kills slugs and other pests.
- Can reduce nitrogen tie-up.

■ *What's the Issue?*

Stubble burning is adversely affecting the environment and public health. The problem has not been fully tackled and the adverse impacts on the air quality and consequent impacts on the citizens' health and lives are undisputed.

23.21 COASTAL SAND MINING AND IMPACT ON ENVIRONMENT

Sand Mining is an activity referring to the process of the actual removal of sand from the foreshore including rivers, streams and lakes. Sand is mined from beaches and inland dunes and dredged from ocean beds and river beds. A related process is the mining of mineral sands, such as mineral deposits like diamond, gold and silver. These minerals typically occur combined with ordinary sand. The sand is dug up, the valuable minerals are separated in water by using their different density, and the remaining ordinary sand is re-deposited.

Environmental Impacts of Coastal Sand Mining

Coastal sand mining has a wide range of negative impacts:

- **Coastal Erosion:** Sand acts as a natural buffer against waves and storms. Its removal leads to accelerated erosion of beaches and coastlines,

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making the land more vulnerable to storm surges and tsunamis.

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- **Saltwater Intrusion:** Mining deepens estuaries and river mouths, allowing saline water from the sea to intrude into coastal aquifers, which contaminates freshwater sources for drinking and agriculture.
- **Habitat Destruction:** The activity destroys crucial habitats and nesting grounds for marine and coastal species, including various fish populations, crabs, and endangered species like the Olive Ridley sea turtle.
- **Biodiversity Loss:** Increased turbidity (cloudiness) in the water from disturbed sediment blocks sunlight, which is harmful to marine life like corals and aquatic vegetation that require light to survive.
- **Livelihood Loss:** The degradation of beaches, reduction in fish populations, and salinization of water sources can severely impact the livelihoods of local communities dependent on fishing, tourism, and agriculture.

Examples in India

Rampant sand mining along the Indian coasts has led to severe consequences:

- In Tamil Nadu, erosion has caused the coastline to recede significantly in some districts, displacing families.
- In Kerala, mining has damaged mangrove forests and reduced fish catch in areas like Alappuzha.

Regulations and Policies in India

Sand is classified as a “minor mineral” under the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act), giving state governments the authority to frame specific rules for its management. Key national guidelines and interventions include:

- **Environmental Clearance (EC)**
- **Coastal Regulation Zone (CRZ) Notification, 2011**

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- **Sustainable Sand Mining Management Guidelines (SSMMG), 2016**

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- **Enforcement & Monitoring Guidelines for Sand Mining, 2020**
- **National Green Tribunal (NGT) Interventions**

Sustainable Alternatives

To reduce dependence on natural coastal and river sand, several sustainable alternatives are being explored and adopted.

- **Manufactured Sand (M-Sand):** This is produced by crushing hard rocks into fine, uniform particles. M-Sand offers consistent quality, enhances concrete strength, and significantly reduces the environmental impact of natural sand mining. It is widely considered the most viable alternative for construction.
- **Industrial By-products:** Waste materials from other industries can be repurposed as sand substitutes.
  - **Copper Slag:** A by-product of copper refineries, it can replace a significant percentage of natural sand in concrete.
  - **Quarry Dust:** A fine aggregate produced during the stone crushing process, which can be a valuable substitute for river sand.
  - **Foundry Sand:** A by-product from metal casting operations that can also be used in construction mixes.
  - **Granulated Blast Furnace Slag and Bottom Ash:** Waste products from steel production and coal combustion, respectively.
- **Recycled Waste Materials:**
  - **Crushed Waste Glass:** This can replace natural sand in geotechnical applications and construction, reducing pressure on landfills and conserving natural resources.
  - **Recycled Plastics:** Being tested for use in road construction, offering potential durability benefits.
  - **Construction and Demolition Waste:** Recycled concrete and other building waste can be processed into aggregates

23.22 MIYAWAKI METHOD

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Named after Japanese botanist **Akira Miyawaki**, this method involves **planting two to four different types of indigenous trees within every square metre**. In this method, the trees become **self-sustaining** and **they grow to their full length within three years** and do not require regular maintenance like manuring and watering.

ENV 80 - Miyawaki Method

Objectives

- To **densify green cover within a small fragment of land**.
- Being used by the government and nature lovers to increase green cover in urban and semi-urban areas.
- The micro-forests are expected to enhance the environment and nature-friendly surroundings of the destinations and turn an added attraction for visitors.

The dense green cover of indigenous trees plays a key role in absorbing the dust particles of the area where the garden has been set up. The plants also help in regulating surface temperature. Some of the common indigenous plants that are used for these forests include Anjan, Amala, Bel, Arjun and Gunj.

Kerala Tourism will embrace the Miyawaki method of afforestation to create 22 micro-forests in tourist destinations in 12 districts.

Prime Minister Shri Narendra Modi, during his one episode of ‘Mann ki baat’, spoke about Miyawaki plantation, the Japanese method of creating

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dense urban forests in a small area. The PM also cited the example of a Kerala-based teacher, **Raafi Ramnath**, who used the Miyawaki technique to **transform a barren land into a mini forest** called **Vidyavanam** by planting 115 varieties of trees.

Meanwhile, to fight climate change, curb pollution levels, and increase the green cover of the financial capital, the **Brihanmumbai Municipal Corporation (BMC)** has been creating Miyawaki forests in several open land parcels of Mumbai.

■ *How it is useful?*

With several infrastructure projects like real estate metro rail construction in progress in Mumbai over the past few years, it was recorded that the surface temperature in certain pockets of Mumbai has increased. Therefore, to fight this challenge, such forests are being created.

At the industrial neighbourhood in Marol (Andheri East), the BMC is creating an urban forest with the Miyawaki method. "These green patches play a major role in regulating the carbon levels of a given area, which may, in return help in maintaining a clean air. Also, these forests encourage new biodiversity and an ecosystem is developed around it, which in turn increases the fertility of the soil and regulates surface temperature."

23.23 ENVIRONMENTAL EMERGENCY

- Environmental emergency declared in damaged coastal territories, after an oil spill that saw 6,000 barrels of crude oil pour into the sea.
- The spill occurred at the La Pampilla refinery. It was caused by freak waves caused by a volcano eruption in Tonga.
- Government said it plans "sustainable management" of 21 beaches tarred by 6,000 barrels of oil that spilled from a tanker ship unloading at a refinery.

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Environmental Emergency

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Environmental Emergency is defined as a "sudden onset disaster or accident resulting from natural, technological or human-induced factors, or a combination of these, that causes or threatens to cause severe environmental damage as well as loss of human lives and property."

The Joint United Nations Environment Programme (UNEP)/Office for the Coordination of Humanitarian Affairs (OCHA) Environment Unit (JEU): By pairing the UNEP's technical expertise with OCHA's humanitarian response network, the Joint UNEP/OCHA Environment Unit (JEU) mobilises and coordinates a comprehensive response to environmental emergencies to protect lives, livelihoods, ecosystems and future generations.

UK is the first nation in the world to declare a climate emergency in April 2019 with an objective to eliminate net greenhouse gas emissions completely by 2050 (compared to the current objective of 80%) and is followed by Brazil.

Tonga Volcano

In December 2021, an eruption began on Hunga Tonga–Hunga Ha apai, a submarine volcano in the Tongan archipelago in the southern Pacific Ocean. The eruption reached a very large and powerful climax nearly four weeks later, on 15 January 2022.

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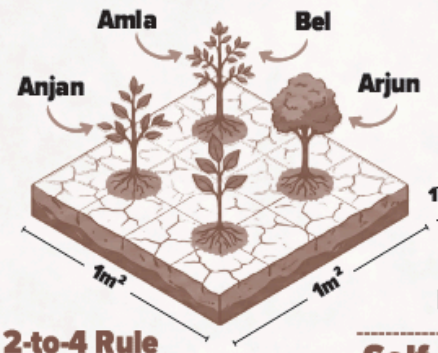
MIYAWAKI TECHNIQUE

A technique for creating dense, native urban forests in small land parcels. Developed by Japanese botanist Akira Miyawaki in the 1970s

Phase 1: The Method (Barren Land)



2-to-4 Rule
Planting 2-to-4 **native species** (e.g. Anjan, Amla, etc. per m^2)

Also known as **"Pot Plantation Method"**

Growth Process & Timeframe **(3 yrs)**

10X Faster Growth | **30X** Denser Greenery

Self-Sustaining after 3 years
(Zero Maintenance)

Phase 2: The Result (Micro Forest)

Dense urban micro- forest

No Regular Maintenance (No manuring or watering needed)

Ecosystem & Soil Health
Absorbs Dust Particles
Increases Soil Fertility
New Biodiversity & Ecosystem

ECOLOGICAL BENEFITS

Urban Climate Control
Mitigates the **Urban Heat Island effect** by regulating surface temperatures (crucial for **concrete jungles** like **Mumbai**)

Pollution Barrier
Shields against noise and dust; absorbs particulates and regulates carbon

Biodiversity
Restores ecosystems, enhancing soil fertility and water conservation

CHALLENGES & LIMITATIONS

Cost Factor
High initial investment required for soil remediation and infrastructure

Scale Limits
Strictly an urban intervention; cannot replace natural afforestation

Labor
Labor-intensive initial phase requiring a skilled workforce

IMPLEMENTATION IN INDIA

Uttar Pradesh
Mahakumbh 2025 (Adopted technique)

Kerala
Raafi Ramnath's 'Vidyavanam' (PM's Mention) & 22 Tourism micro-forests in 12 districts

Maharashtra
BMC Projects (Marol) to counter infrastructure heat

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■ *Freak Waves*

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These “freak waves” are not limited to the Atlantic Ocean or the North Sea.

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One of the most popular locations for rogue waves appears to be off the southeast coast of South Africa.

A freak wave, also known as a rogue wave, is typically characterised as a wave that is double the area’s significant wave height as the average of the highest one-third of waves that occur within a given time period.

Even the largest ships and oil rigs can be disabled and sunk by rogue waves.

■ Reasons behind most Oil Spills

An oil spill is defined as any uncontrollable release of crude oil, gasoline, fuels, or other oil byproducts into the environment. These have become a major environmental issue, owing mostly to increased petroleum exploration and production on continental shelves, as well as the transportation of large amounts of oil in vessels.

■ Impacts

- Oil pollution endangers the health of indigenous people who rely on seafood.
- Oil on ocean surfaces is harmful to many types of aquatic life because it inhibits sufficient amounts of sunlight from accessing the surface and reduces the level of dissolved oxygen.
- Hypothermia (decrease in body temperature to below-normal levels). Crude oil destroys the insulating and waterproofing characteristics of bird feathers and fur.
- This oil can be toxic to the affected animals, causing habitat destruction and a decrease in reproduction rate.
- Mangroves are threatened by oil spills, which occur often in saltwater marshes and mangroves.
- Tourism and commerce may get affected.

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- Fishing activities: When there is a major oil leak, commercial fishing is often suspended immediately.
- Alters water temperature which may lead to lesser oxygen supply at certain depths of water.

■ *Technological Solutions*

Oil Zapper is a cocktail of bacteria that feed on the oil and degrade the hydrocarbons. Newer methods and technologies like Nanosheets that could revolutionise oil spill clean ups and water purification needs to be deployed.

■ *International Convention*

Section 356A-356O in the Part-XI of the Merchant Shipping Act, 1958 gives effects to the international conventions regarding the prevention of pollution of the seas by oil.

■ *Indian Laws*

- India has National Oil Spill Disaster Contingency Plan of 1996 (NOS-DCP) to deal with such situations.

23.24 ENVIRONMENTAL JUSTICE

One look around the world we inhabit, and it is clear that the distribution of environmental hazards is in no way equal. Environmental justice refers to the fair treatment and meaningful involvement of all people with regard to the development, implementation, and enforcement of laws, regulations, policies and practices related to the environment. By bringing all people into the purview, environmental justice tries to bring in the voices of people regardless of their race, colour, income, or country of origin.

While fair treatment aims at no particular population bearing the brunt of negative environmental effects from certain actions, meaningful involvement seeks to provide everyone the opportunity to make well-

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informed decisions. Those who work towards environmental justice thus dream of a world where all people live in a clean and safe environment that is free from waste and pollution and doesn't threaten their well-being.

23.25 ZERO WASTE

Zero waste refers to a way 0f life where we try to reduce the waste we generate. It is reflected in every single aspect of our living - from what we buy, what we eat, how we dress, and how we travel to the way we run our homes and what they are filled with. It is about being mindful of not just what we produce and consume but also now to reuse in a way we conserve all available resources. Zero waste is achievable when we consciously strive to ensure that all materials and products are continued to be used (and reused/ repurposed) for the longest possible time, giving them a longer lifespan. The adoption of a zero-waste lifestyle will help us cut down on actions that will directly or indirectly affect us and our environment including land, water, air, or any living being.

23.26 CARBON ACCOUNTING

We have all heard about financial accounting. But what is carbon accounting? Carbon accounting is just like financial accounting, only that it calculates how much greenhouse gas (GHG) an organisation emits. Instead of the financial impact in the case of financial accounting, we calculate the climate impact here. It helps in estimating the carbon footprints of businesses, governments, and individuals. This is very crucial for tackling climate change as it will help in knowing high GHG emitting companies, make them stay compliant to emission norms and also come up with emission reduction strategies. It is a great tool in fighting climate change as we get to measure the direct and indirect emissions of greenhouse gases into the atmosphere.

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Questions Asked in UPSC Prelims Exam

UNIT IX

Unit - IX Prelims PYQs (2011 onwards)

1. "If rainforests and tropical forests are the lungs of the Earth, then surely wetlands function as its kidneys." Which one of the following functions of wetlands best reflects the above statement? [UPSC (Pre) 2022]

(a) The water cycle in wetlands involves surface runoff, subsoil percolation and evaporation.

(b) Algae form the nutrient base upon which fish, crustaceans, molluscs, birds, reptiles and mammals thrive.

(c) Wetlands play a vital role in maintaining sedimentation balance and soil stabilization.

(d) Aquatic plants absorb heavy metals and excess nutrients.

2. "Biorock technology" is talked about in which one of the following situations? [UPSC (Pre) 2022]

(a) Restoration of damaged coral reefs

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(b) Development of building materials using plant residues
(c) Identification of areas for exploration/extraction of shale gas
(d) Providing salt licks for wild animals in forests/protected areas

3. The "Miyawaki method" is well known for the: [UPSC (Pre) 2022]

(a) Promotion of commercial farming in arid and semi-arid areas
(b) Harvesting wind energy on coastal areas and on sea surfaces
(c) Development of gardens using genetically modified flora
(d) Creation of mini forests in urban areas

4. How is permaculture farming different from conventional chemical farming? [UPSC (Pre) 2021]

1. Permaculture farming discourages mono¬cultural practices but in conventional chemical farming, monoculture practices are predominant.
2. Conventional chemical farming can cause increase in soil salinity but the occurrence of such phenomenon is not observed in permaculture farming.
3. Conventional chemical farming is easily possible in semi-arid regions but permaculture farming is not so easily possible in such regions.
4. Practice of mulching is very important in permaculture farming but not necessarily so in conventional chemical farming.

Select the correct answer using the codes given below:
(a) 1 and 3 (b) 1, 2 and 4
(c) 4 only (d) 2 and 3

5. What is the use of biochar in farming? [UPSC (Pre) 2020]

1. Biochar can be used as a part of the growing medium in vertical farming.

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2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for a long time.

Which of the statement/s given above is/are correct?

(a) 1 and 2 only (b) 2 only

(c) 1 and 3 only (d) 1, 2 and 3

6. With reference to organic farming in India, consider the following statements: **[UPSC (Pre) 2018]**

1. The National Programme for Organic Production (NPOP) is operated under the guidelines and directions of the Union Ministry of Rural Development.
2. The Agricultural and Processed Food Products Export Development Authority (APEDA) functions as the Secretariat for the implementation of NPOP.
3. Sikkim has become India's first fully organic State. Which of the statement/s given above is/are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 3 only (d) 1, 2 and 3

7. Consider the following statements: **[UPSC (Pre) 2018]**

1. Most of the world's coral reefs are in tropical waters.
2. More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and the Philippines.
3. Coral reefs host far more number of animal phyla than those hosted by tropical rainforests.

Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 3 only

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(c) 1 and 3 only (d) 1, 2 and 3

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8. In the context of solving pollution problems what is/are the advantage/advantages of bioremediation technique? [UPSC (Pre) 2017]

1. It is a technique for cleaning up pollution by enhancing the same biodegradation process that occurs in nature.
2. Any contaminant with heavy metals such as cadmium and lead can be readily and completely treated by bioremediation using micro-organisms.
3. Genetic-engineering can be used to create micro- organisms specifically designed for bioremediation.

Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3

(c) 1 and 3 (d) 1, 2 and 3

9. Which of the following have coral reefs? [UPSC (Pre) 2014]

1. Andaman and Nicobar Islands
2. Gulf of Kachchh
3. Gulf of Mannar
4. Sunderbans

Select the correct answer using the codes given below:

(a) 1, 2 and 3 only

(b) 2 and 4 only

(c) 1 and 3 only

(d) 1, 2, 3 and 4

10. There is a concern over the increase in harmful algal blooms in the seawaters of India. What could be the causative factors for this phenomenon? [UPSC (Pre) 2011]

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1. Discharge of nutrients from the estuaries.
2. Run-off from the land during the monsoon.
3. Upwelling in the seas.

Select the correct answer from the codes given below:

(a) 1 only (b) 1 and 2 only

(c) 2 and 3 only (d) 1, 2 and 3

Answers

1. (d) 2. (a) 3. (d) 4. (b) 5. (d) 6. (b) 7. (d) 8. (c) 9. (a) 10. (d)

Explanations

1. Correct answer (d): EXPLANATION

The statement compares **wetlands to kidneys**, highlighting their role in **filtering and cleaning**.

Among the options:

- **Aquatic plants absorb heavy metals and excess nutrients - Statement (d) is right**
- This reflects the **detoxification and purification function** of wetlands, just like kidneys filter waste from blood.

Other options describe water cycling, food webs, or soil stabilization, which are important but **do not capture the "kidney-like" filtering function**.

2. Correct answer (a): EXPLANATION

Biorock technology is used for the **restoration and growth of coral reefs**. It involves **low-voltage electrical currents** to stimulate the

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deposition of minerals like calcium carbonate, helping **coral growth and reef restoration**.

3. Correct answer (d): EXPLANATION

The **Miyawaki method** is a technique developed by Japanese botanist Akira Miyawaki for **creating dense, native forests in small areas**. It is widely used for **restoring urban green cover and biodiversity**.

4. Correct answer (b): EXPLANATION

- Permaculture promotes **polyculture, crop diversity, and ecological balance**. Conventional chemical farming often relies on **monocultures** for high yields. **Statement (1) is right**
- Overuse of chemical fertilizers and improper irrigation can **increase salinity** in conventional farming. Permaculture uses **organic methods and water-efficient practices**, preventing salinity buildup. **Statement (2) is right**
- Permaculture **can be adapted to semi-arid regions** using techniques like rainwater harvesting, mulching, and drought-resistant crops. **Statement (3) is wrong**
- Mulching is **crucial in permaculture** to conserve soil moisture, suppress weeds, and maintain soil health. Conventional chemical farming may or may not use mulching. **Statement (4) is right**

5. Correct answer (d): EXPLANATION

- Biochar is used in **soilless and vertical farming systems** as a component of the growing medium. **Statement (1) is right**
- Biochar improves **soil microbiome**, supporting beneficial microbes including nitrogen-fixers. **Statement (2) is right**

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- Biochar has a **porous structure**, which increases **water retention and nutrient holding capacity**. **Statement (3) is right**

6. Correct answer (b): EXPLANATION

- NPOP is implemented under the **Ministry of Commerce & Industry**, not the Ministry of Rural Development. **Statement (1) is wrong**
- APEDA administers NPOP, including certification and export promotion of organic products. **Statement (2) is right**
- Sikkim achieved **100% organic farming** across the state. **Statement (3) is right**

7. Correct answer (d): EXPLANATION

- Coral reefs thrive in **warm, shallow, sunlit tropical waters** between the Tropic of Cancer and Tropic of Capricorn. **Statement (1) is right**
- The **Great Barrier Reef (Australia)** and the reefs in **Indonesia and the Philippines** together account for a significant portion (>1/3) of global coral reefs. **Statement (2) is right**
- Coral reefs are extremely biodiverse, hosting **more phyla than tropical rainforests** despite being a smaller area. **Statement (3) is right**

8. Correct answer (c): EXPLANATION

- Bioremediation uses **microorganisms or plants** to break down pollutants naturally. **Statement (1) is right**
- Heavy metals **cannot be degraded** by microorganisms; they can only be **immobilized or accumulated**, not completely broken down. **Statement (2) wrong**

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- Scientists can develop **genetically engineered microbes** with enhanced ability to degrade specific pollutants. Oil zapper is one of example. **Statement (3) is right**

9. Correct answer (a): EXPLANATION

Coral reefs are found in **warm, shallow, sunlit waters**, typically along tropical coastlines.

Andaman and Nicobar Islands – extensive coral reefs

Gulf of Kachchh – patchy reefs along Gujarat coast

Gulf of Mannar – coral reefs along Tamil Nadu–Sri Lanka coast

Sunderbans – it is primarily a **mangrove ecosystem**, not coral reefs

10. Correct answer (d): EXPLANATION

Harmful algal blooms (HABs) occur due to rapid growth of algae in water, often producing toxins harmful to marine life and humans.

Causative factors:

Nutrients like nitrogen and phosphorus from rivers and estuaries **promote algal growth**. **Statement (1) is right** Monsoon runoff carries fertilizers, sewage, and other nutrients into the sea, enhancing blooms. **Statement (2) is right**

Upwelling brings **nutrient-rich deep waters** to the surface, promoting algal proliferation. **Statement (3) is right**

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SECTION II Important Lists

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IN THIS CHAPTER...

1. Project Tiger
2. Project Elephant
3. Ramsar Wetland Sites in India
4. World Heritage Sites in India
5. List of Biosphere Reserves in India
6. Extinct Wild Animals in India
7. National Park
8. Wildlife Sanctuaries (WLS)
9. List of Indian Wild Life (Protection) Act, 1972: Schedule-I

1. PROJECT TIGER

Project Tiger was launched by the Government of India in the year 1973 to save the endangered species of tiger in the country. Starting from nine (9) reserves in 1973–2026 the number is grown up to fifty-eight (58).

India's **58 tiger reserves** cover a combined area of **approximately 84,500 square kilometres**, safeguarding critical tiger habitats across **18 states**, with **Madhya Pradesh having the most reserves (9),** and the **largest being Nagarjunsagar-Srisailam Tiger Reserves**.

Table 1 Tiger Reserves of India (updated as on January 2026)

S. No.	Tiger Reserve (Year of Creation)	State	Population of Tigers
1	Bandipur (1973–74)	Karnataka	158
2	Corbett (1973–74)	Uttarakhand	252
3	Kanha (1973–74)	Madhya Pradesh	140
4	Manas (1973–74)	Assam	48
5	Melghat (1973–74)	Maharashtra	35
6	Palamau (1973–74)	Jharkhand	3
7	Ranthambore (1973–74)	Rajasthan	79
8	Similipal (1973–74)	Odisha	28
9	Sunderbans (1973–74)	West Bengal	100
10	Periyar (1978–79)	Kerala	40

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S. No.	Tiger Reserve (Year of Creation)	State	Population of Tigers
11	Sariska (1978–79)	Rajasthan	23
12	Buxa (1982–83)	West Bengal	96
13	Indravati (1982–83)	Chhattisgarh	23
14	Namdapha (1982–83)	Arunachal Pradesh	29
15	Dudhwa (1987–88)	Uttar Pradesh	107
16	Kalakad-Mundanthurai (1988–89)	Tamil Nadu	16
17	Valmiki (1989–90)	Bihar	32
18	Pench (1992–93)	Madhya Pradesh	53
19	Tadoba-Andhari (1993–94)	Maharashtra	115
20	Bandhavgarh (1993–94)	Madhya Pradesh	70
21	Panna (1994–95)	Madhya Pradesh	56
22	Dampa (1994–95)	Mizoram	1
23	Bhadra (1998–99)	Karnataka	33
24	Pench (1998–99)	Maharashtra	44
25	Pakke (1999–2000)	Arunachal Pradesh	9
26	Nameri (1999–2000)	Assam	3
27	Satpura (1999–2000)	Madhya Pradesh	26
28	Anamalai (2008–09)	Tamil Nadu	20
29	Sitanadi (2008–09)	Chhattisgarh	4
30	Satkosia (2008–09)	Odisha	18
31	Kaziranga (2008–09)	Assam	200
32	Achanakmar (2008–09)	Chhattisgarh	19
33	Dandeli-Anshi Tiger Reserve (Kali) (2008–09)	Karnataka	22
34	Sanjay Dubri (2008–09)	Madhya Pradesh	150

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S. No.	Tiger Reserve (Year of Creation)	State	Population of Tigers
35	Mudumalai (2007)	Tamil Nadu	162
36	Nagarhole (2008–09)	Karnataka	125
37	Parambikulam (2008–09)	Kerala	35
38	Sahyadri (2009–10)	Maharashtra	8
39	Biligiri Ranganatha Temple (2010–11)	Karnataka	62
40	Kawal (2012–13)	Telangana	10
41	Sathyamangalam (2013–14)	Tamil Nadu	80
42	Mukundra Hills (2013–14)	Rajasthan	1
43	Nawegaon (2013–14)	Maharashtra	20
44	Nagarjunsagar-Srisailam (1982–83)	Andhra Pradesh	67
45	Amrabad (2014)	Telangana	16
46	Pilibhit (2014)	Uttar Pradesh	65
47	Bor (2014)	Maharashtra	6
48	Rajaji (2015)	Uttarakhand	16
49	Orang (2016)	Assam	28
50	Kamlang (2016)	Arunachal Pradesh	-
51	Srivilliputhur-Megamalai (2021)	Tamil Nadu	14
52	Ramgarh Vishdhari (2022)	Rajasthan	35
53	Ranipur Wildlife Sanctury (2022)	Uttar Pradesh	-
54	Dholpur-karauli Tiger Reserve (2023)	Rajasthan	-
55	Veerangana Durgavati Tiger Reserve (2023)	Madhya Pradesh	15
56	Guru Ghasidas – Tamor Pingla Tiger Resereve (2024)	Chhattisgarh	5
57	Ratapani Tiger Resereve (2024)	Madhya Pradesh	90
58	Madhav (2025)	Madhya Pradesh	5

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Figure 1.1 Tiger Reserves

Source: ENVIS Center on Wildlife & Restricted Area.

Key Figures (as of Jan 2026):

- **Number of Reserves:** 58
- **Total Area:** approx. 84,500 sq km
- **States Covered:** 18
- **Largest:** Nagarjunsagar - Srisailam Tiger Reserve (Andhra Pradesh)
- **Smallest:** Orang Tiger Reserve (Assam)
- These areas are crucial for Project Tiger, aimed at conserving the endangered Bengal tiger and its habitat.

2. PROJECT ELEPHANT

Project Elephant was launched by the Government of India in the year 1992 as a Centrally Sponsored Scheme with the following objectives:

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1. To protect elephants, their habitat and corridors
2. To address issues of man–animal conflict
3. Welfare of captive elephants

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The Project is being mainly implemented in 14 States/UTs, viz. Andhra Pradesh, Arunachal Pradesh, Assam, Chhattisgarh, Jharkhand, Karnataka, Kerala, Maharashtra, Meghalaya, Nagaland, Odisha, Tamil Nadu, Uttarakhand, Uttar Pradesh and West Bengal. Singhbhum Elephant Reserves (Jharkhand) is the largest (13440 Sq. km) in India.

The Ministry of Environment, Forest and Climate Change provides the financial and technical support to major elephant range states in the country through Project Elephant.

The Indian elephant *Elephas maximus* occurs in the central and southern Western Ghats, Northeast India, eastern India and northern India and in some parts of southern peninsular India. It is included in **Schedule I of the Indian Wild Life (Protection) Act, 1972** and in **Appendix I of the Convention on International Trade in Endangered Species of Flora and Fauna (CITES).** There are 33 Elephant reserves in India till Jan 2026, covering about 65000 sq. km of forest across India.

Table 2 Elephant Reserves of India (as on January 2026)

S. No.	Elephant Reserve	State	Date of Notification	Total Area (sq. km)
1	Rayala ER	Andhra Pradesh	09.12.2003	766
2	Kameng ER	Arunachal Pradesh	19.06.2002	1892
3	South Arunachal ER	Arunachal Pradesh	29.02.2008	1957.50
4	Sonitpur ER	Assam	06.03.2003	1420
5	Dihing-Patkai ER	Assam	17.04.2003	937
6	Kaziranga-Karbi Anglong ER	Assam	17.04.2003	3270
7	Dhansiri-Lungding ER	Assam	19.04.2003	2740
8	Chirang-Ripu ER	Assam	07.03.2003	2600
9	Badalkhol-Tamorpingla	Chhattisgarh	15.09.2011	1048.30
10	Lemru ER	Chhattisgarh	2022	450
11	Singhbhum ER	Jharkhand	26.09.2001	4530
12	Mysore ER	Karnataka	25.11.2002	6724
13	Dandeli ER	Karnataka	26.03.2015	2,321
14	Wayanad ER	Kerala	02.04.2002	1200

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S. No.	Elephant Reserve	State	Date of Notification	Total Area (sq. km)
15	Nilambur ER	Kerala	02.04.2002	1419
16	Anamudi ER	Kerala	02.04.2002	3728
17	Periyar	Kerala	02.04.2002	3742
18	Garo Hills ER	Meghalaya	31.10.2001	3,500
19	Intanki ER	Nagaland	28.02.2005	202
20	Singphan ER	Nagaland	16.08.2018	23.57
21	Mayurbhanj ER	Odisha	29.09.2001	3214
22	Mahanadi ER	Odisha	20.07.2002	1038
23	Sambalpur ER	Odisha	27.03.2002	427
24	Nilgiri ER	Tamil Nadu	19.09.2003	4663
25	Coimbatore ER	Tamil Nadu	19.09.2003	566
26	Anamalai ER	Tamil Nadu	19.09.2003	1457
27	Srivilliputtur ER	Tamil Nadu	19.09.2003	1249
28	Agsthyamalai ER	Tamil Nadu	12.08.2022	1,197.48
29	Uttar Pradesh ER	Uttar Pradesh	09.09.2009	744
30	Terai ER	Uttar Pradesh	2022	3049
31	Shivalik ER	Uttarakhand	28.10.2002	5405
32	Mayurjharna ER	West Bengal	24.10.2002	414
33	Eastern Dooars ER	West Bengal	28.8.2002	978

Source: Ministry of Environment

3. RAMSAR WETLAND SITES IN INDIA (AS ON JANUARY 2026)

Sr. No	Ramsar Wetland Sites	States/UTs	Date	Area
1	Kolleru Lake	Andhra Pradesh	2002-08-19	90100
2	Deepor Beel	Assam	2002-08-19	4000

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Sr. No	Ramsar Wetland Sites	States/UTs	Date	Area
3	Gogabil Lake	Bihar	2025-05-13	86.63
4	Kabartal Wetland	Bihar	2020-07-21	2620
5	Nagi Bird Sanctuary	Bihar	2023-10-11	205.817
6	Nakti Bird Sanctuary	Bihar	2023-10-11	332.608
7	Kopra Jalashay	Chhattisgarh	2025-08-08	210
8	Nanda Lake	Goa	2022-06-08	42.01
9	Khijadia Wildlife Sanctuary	Gujarat	2021-04-13	511.745
10	Nalsarovar	Gujarat	2012-09-24	12000
11	Thol Lake Wildlife Sanctuary	Gujarat	2021-04-05	699
12	Wadhvana Wetland	Gujarat	2021-04-05	630
13	Bhindawas Wildlife Sanctuary	Haryana	2021-05-25	412
14	Sultanpur National Park	Haryana	2021-05-25	142.52
15	Chandertal Wetland	Himachal Pradesh	2005-11-08	49
16	Pong Dam Lake	Himachal Pradesh	2002-08-19	15662
17	Renuka Wetland	Himachal Pradesh	2005-11-08	20
18	Hokera Wetland	Jammu & Kashmir	2005-11-08	1375
19	Hygam Wetland Conservation Reserve	Jammu & Kashmir	2022-06-08	801.82
20	Shallbugh Wetland Conservation Reserve	Jammu & Kashmir	2022-06-08	1675
21	Surinsar–Mansar Lakes	Jammu & Kashmir	2005-11-08	350
22	Wular Lake	Jammu & Kashmir	1990-03-23	18900
23	Udhwa Lake Bird Sanctuary	Jharkhand	2024-01-08	935.5
24	Aghanashini Estuary	Karnataka	2023-02-14	4801
25	Ankasamudra Bird Conservation Reserve	Karnataka	2023-03-10	98.76
26	Magadi Kere Conservation Reserve	Karnataka	2023-02-14	54.38
27	Ranganathittu Bird Sanctuary	Karnataka	2022-02-15	517.7
28	Ashtamudi Wetland	Kerala	2002-08-19	6140

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Sr. No	Ramsar Wetland Sites	States/UTs	Date	Area
29	Sasthamkotta Lake	Kerala	2002-08-19	373
30	Vembanad–Kol Wetland	Kerala	2002-08-19	151250
31	Tso Kar Wetland Complex	Ladakh	2020-11-17	9577
32	Tso Moriri	Ladakh	2002-08-19	12000
33	Bhoj Wetland	Madhya Pradesh	19-08-2002	3201
34	Gokul Jalashay	Madhya Pradesh	2025-05-13	448
35	Sakhya Sagar	Madhya Pradesh	2022-01-07	248
36	Sirpur Wetland	Madhya Pradesh	2022-01-07	161
37	Tawa Reservoir	Madhya Pradesh	2024-01-08	20050
38	Yashwant Sagar	Madhya Pradesh	2022-01-07	822.9
39	Lonar Lake	Maharashtra	2020-07-22	427
40	Nandur Madhameshwar	Maharashtra	2019-06-21	1437
41	Thane Creek	Maharashtra	2022-04-13	6521.08
42	Loktak Lake	Manipur	1990-03-23	26600
43	Pala Wetland	Mizoram	2021-08-31	1850
44	Ansupa Lake	Odisha	2021-10-12	231
45	Bhitarkanika Mangroves	Odisha	2002-08-19	65000
46	Chilika Lake	Odisha	1981-10-01	116500
47	Hirakud Reservoir	Odisha	2021-10-12	65400
48	Satkosia Gorge	Odisha	2021-10-12	98196.72
49	Tampara Lake	Odisha	2021-10-12	300
50	Beas Conservation Reserve	Punjab	2019-09-26	6428.92
51	Harike Lake	Punjab	1990-03-23	4100
52	Kanjli	Punjab	2002-01-22	183
53	Keshopur-Miani Community Reserve	Punjab	2019-09-26	343.9
54	Nangal Wildlife Sanctuary	Punjab	2019-09-26	116

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Sr. No	Ramsar Wetland Sites	States/UTs	Date	Area
55	Ropar (Rupnagar Wetland)	Punjab	2002-01-22	1365
56	Keoladeo National Park	Rajasthan	1981-10-01	2873
57	Khichan Wetland	Rajasthan	2025-02-19	54.187
58	Menar Wetland Complex	Rajasthan	2025-02-19	463.414
59	Sambhar Lake	Rajasthan	1990-03-23	24000
60	Siliserh Lake	Rajasthan	2025-07-07	315.97
61	Khachoedpalri Wetland	Sikkim	2024-07-15	172
62	Chitrangudi Bird Sanctuary	Tamil Nadu	2021-11-08	260.47
63	Gulf of Mannar Marine Biosphere Reserve	Tamil Nadu	2022-04-08	52671.876
64	Kanjirankulam Bird Sanctuary	Tamil Nadu	2022-04-08	96.891
65	Karaivetti Bird Sanctuary	Tamil Nadu	2023-05-24	453.715
66	Karikili Bird Sanctuary	Tamil Nadu	2022-04-08	58.442
67	Kazhuveli Bird Sanctuary	Tamil Nadu	2024-01-16	5151.6
68	Koonthankulam Bird Sanctuary	Tamil Nadu	2021-11-08	72.04
69	Longwood Shola Reserve Forest	Tamil Nadu	2023-05-24	116.007
70	Nanjarayan Bird Sanctuary	Tamil Nadu	2024-01-16	125.865
71	Pallikaranai Marsh Reserve Forest	Tamil Nadu	2022-04-08	1247.537
72	Pichavaram Mangrove	Tamil Nadu	2022-04-08	1478.642
73	Point Calimere Wildlife and Bird Sanctuary	Tamil Nadu	2002-08-19	38500
74	Sakkarakottai Bird Sanctuary	Tamil Nadu	2024-07-15	230.49
75	Suchindram Theroor Wetland Complex	Tamil Nadu	2022-04-08	94.229
76	Therthangal Bird Sanctuary	Tamil Nadu	2024-07-15	29.295
77	Udhayamarthandapuram Bird Sanctuary	Tamil Nadu	2022-04-08	43.767
78	Vaduvur Bird Sanctuary	Tamil Nadu	2022-04-08	112.638
79	Vedanthangal Bird Sanctuary	Tamil Nadu	2022-04-08	40.348
80	Vellode Bird Sanctuary	Tamil Nadu	2022-04-08	77.185

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Sr. No	Ramsar Wetland Sites	States/UTs	Date	Area
81	Vembannur Wetland Complex	Tamil Nadu	2022-04-08	19.746
82	Rudrasagar Lake	Tripura	2005-11-08	240
83	Udaipur Jheel	Tripura	2025-05-13	319
84	Bakhira Wildlife Sanctuary	Uttar Pradesh	2021-06-29	2894
85	Haiderpur Wetland	Uttar Pradesh	2021-04-13	6908
86	Nawabganj Bird Sanctuary	Uttar Pradesh	2019-09-19	224.6
87	Parvati Arga Bird Sanctuary	Uttar Pradesh	2019-12-02	722
88	Saman Bird Sanctuary	Uttar Pradesh	2019-12-02	526.3
89	Samaspur Bird Sanctuary	Uttar Pradesh	2019-10-03	799.371
90	Sandi Bird Sanctuary	Uttar Pradesh	2019-09-26	308.54
91	Sarsai Nawar Jheel	Uttar Pradesh	2019-09-19	161.27
92	Sur Sarovar	Uttar Pradesh	2020-08-21	431
93	Upper Ganga River	Uttar Pradesh	2005-11-08	26590
94	Asan Conservation Reserve	Uttarakhand	2020-07-21	444.4
95	East Calcutta Wetlands	West Bengal	2002-08-19	12500
96	Sundarban Wetland	West Bengal	2019-01-30	423000

Source: Ministry of Environment & Forests, Government of India

As of **January 2026**, India has **96 Ramsar Sites**, covering a **total area of over 1.53 million hectares (approximately 15,300 sq. km)**.

These sites are recognised as Wetlands of International Importance under the Ramsar Convention, an Intergovernmental treaty for the Conservation and Sustainable use of Wetlands.

Key Factual Data

Total Sites: India has **96 Ramsar sites**, the highest number in South Asia and the third highest globally after the United Kingdom and Mexico.

First Sites: Chilika Lake in Odisha and Keoladeo National Park in Rajasthan were India's first Ramsar sites, designated in 1981.

Largest Site: The Sundarban Wetland in West Bengal is the largest Ramsar site in India, spanning 4,230 sq. km and encompassing the world's largest mangrove forest.

Smallest Site: Renuka Lake in Himachal Pradesh is the smallest Ramsar site, with an area of just 0.2 sq. km (20 hectares).

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State with the Most Sites: Tamil Nadu has the highest number of Ramsar sites (20), followed by Uttar Pradesh (10).

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Montreux Record: Two Indian wetlands are currently on the Montreux Record (a register of sites facing ecological changes): **Loktak Lake in Manipur** and **Keoladeo National Park in Rajasthan**.

Recent Additions (2025): The two most recent additions in December 2025 are **Siliserh Lake (Rajasthan) and Kopra Reservoir (Chhattisgarh)**.

4. WORLD HERITAGE SITES IN INDIA

India has **44 UNESCO World Heritage Sites** as of **January 2026**, comprising **36 Cultural, 7 Natural, and 1 Mixed Site**, with the 'Maratha Military Landscapes' added most recently in July 2025, showcasing India's rich history, nature, and architecture globally.

India ranks among the top countries for World Heritage Sites, demonstrating its vast heritage.

Breakdown of Sites:

- **Cultural (36):** Examples include the Taj Mahal, Ajanta & Ellora Caves, Agra Fort, and the latest Maratha Military Landscapes.
- **Natural (7):** Examples include Kaziranga National Park and Manas Wildlife Sanctuary.
- **Mixed (1):** Khangchendzonga National Park, recognised for both natural and cultural significance.

Table 3 Natural World Heritage Sites in India (as on January 2026)

S. No.	Name of WH Site	State Location	Year of Notification	Area (sq. km)
1	Great Himalayan National Park Conservation Area	Himachal Pradesh	2014	905.4
2	Western Ghats	Maharashtra, Goa, Karnataka, Tamil Nadu and Kerala	2012	7,953.15
3	Nanda Devi and Valley of Flowers National Parks	Uttarakhand	1988	630.00 87.50
4	Sundarbans National Park	West Bengal	1987	1,330.10
5	Kaziranga National Park	Assam	1985	429.96
6	Keoladeo National Park	Rajasthan	1985	28.73
7	Manas Wildlife Sanctuary	Assam	1985	391.00

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Table 4 Cultural World Heritage Sites in India (as on January 2026)

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S. No.	Name of WH Site	State Location	Year of Notification
1	Maratha Military Landscapes of India 11 in Maharashtra and 01 in	Tamil Nadu	2025
2	Moidams – The MoundBurial System of the Ahom Dynasty	Assam	2024
3	Sacred Ensembles of the Hoysalas	Karnataka	2023
4	Santiniketan	West Bengal	2023
5	Kakatiya Rudreshwara (Ramappa) Temple	Telangana	2021
6	Dholavira: A Harrapan City	Gujarat	2021
7	The Pink City, Jaipur	Rajasthan	2019
8	Victorian Gothic and Art Deco Ensembles of Mumbai	Maharashtra	2018
9	Historic City of Ahmadabad	Gujarat	2017
10	Archaeological Site of Nalanda Mahavihara (Nalanda University) at Nalanda	Bihar	2016
11	The Architectural Work of Le Corbusier, an Outstanding Contribution to the Modern Movement	Chandigarh	2016
12	Rani-ki-Vav (the Queen's Stepwell) at Patan	Gujarat	2014
13	Hill Forts of Rajasthan	Rajasthan	2013
14	The Jantar Mantar, Jaipur	Rajasthan	2010
15	Red Fort Complex	Delhi	2007
16	Champaner-Pavagadh Archaeological Park	Gujarat	2004
17	Chhatrapati Shivaji Terminus (formerly Victoria Terminus)	Maharashtra	2004
18	Rock Shelters of Bhimbetka	Madhya Pradesh	2003
19	Mahabodhi Temple Complex at Bodh Gaya	Bihar	2002
20	Mountain Railways of India	West Bengal	1999
21	Humayun's Tomb	Delhi	1993

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S. No.	Name of WH Site	State Location	Year of Notification
22	Qutb Minar and its Monuments	Delhi	1993
23	Buddhist Monuments at Sanchi	Madhya Pradesh	1989
24	Elephanta Caves	Maharashtra	1987
25	Great Living Chola Temples	Tamil Nadu	1987
26	Group of Monuments at Pattadakal	Karnataka	1987
27	Churches and Convents of Goa	Goa	1986
28	Fatehpur Sikri	Uttar Pradesh	1986
29	Group of Monuments at Hampi	Karnataka	1986
30	Khajuraho Group of Monuments	Madhya Pradesh	1986
31	Group of Monuments at Mahabalipuram	Tamil Nadu	1984
32	Sun Temple, Konârak	Odisha	1984
33	Agra Fort	Uttar Pradesh	1983
34	Ajanta Caves	Maharashtra	1983
35	Ellora Caves	Maharashtra	1983
36	Taj Mahal	Uttar Pradesh	1983

Table 5 Mixed World Heritage Sites in India (as on January 2026)

S. No.	Name of WH Site	State Location	Year of Notification
1	Khangchendzonga National Park	Sikkim	2016

(Source: UNESCO World Heritage Convention: http://whc.unesco.org/en/statesparties/in-title=World)

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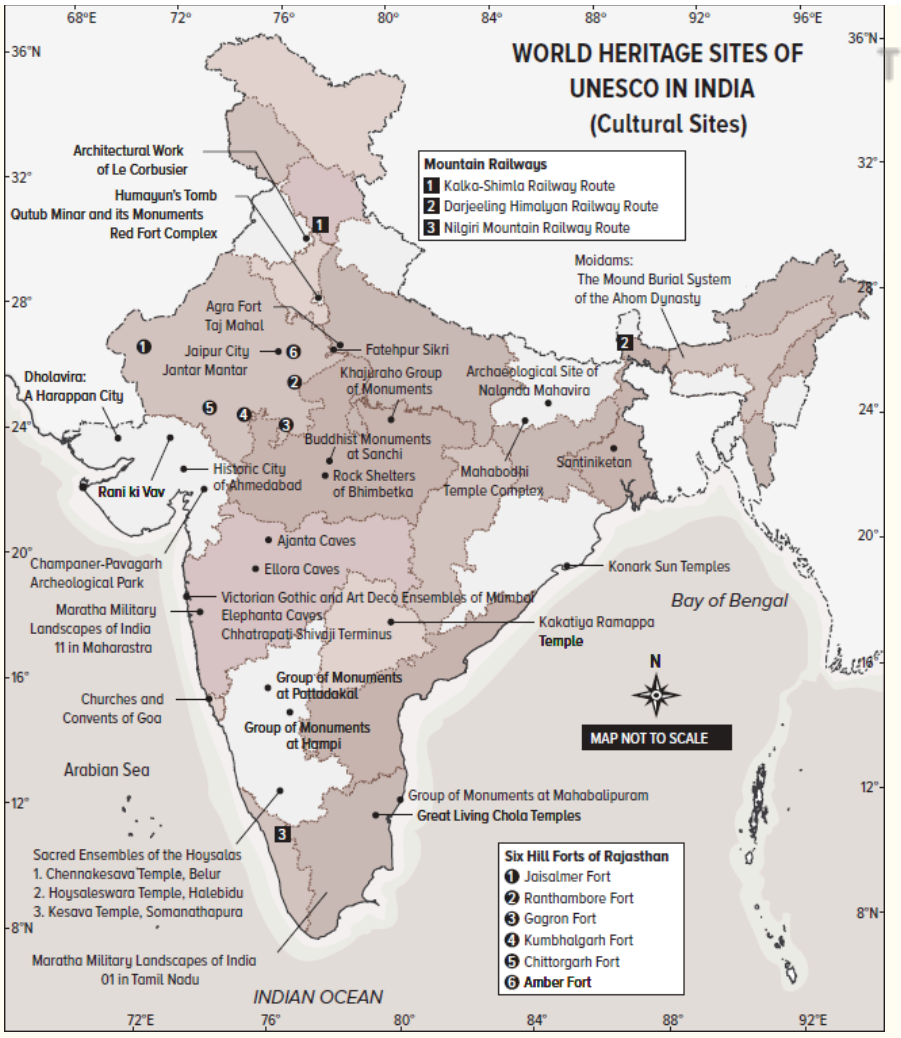

Figure 1.2 UNESCO World Heritage Sites in India (Cultural Sites)

5. LIST OF BIOSPHERE RESERVES IN INDIA

Biosphere Reserves are large areas of biodiversity where flora and fauna are protected. These regions of environmental protection roughly correspond to IUCN Category V Protected areas. The Indian government has established 18 Biosphere Reserves of India, which protect larger areas of natural habitat (than a National Park or Wildlife Sanctuary). Biosphere Reserves of India often include one or more National Parks or sanctuaries, along with buffer zones that are open to some economic uses. Protection is granted not only to the flora and fauna of the protected region, but also to the human communities who inhabit these regions, and their ways of life.

Table 6 List of Biosphere Reserves of India

Biosphere Reserves of India[7]							
	Year	Name	Location	State	Type	Key fauna	Area (km^2)
1	1986	Nilgiri Biosphere Reserve	Part of Waynad,	Karnataka, Tamil Nadu	Western Ghats	Nilgiri Tahr, tiger,	5520

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Biosphere Reserves of India[7]							
	Year	Name	Location	State	Type	Key fauna	Area (km^2)
			Nagarhole, Bandipur and Mudumalai, Nilambur, Silent Valley	and Kerala		lion-tailed macaque	
2	1988	Nanda Devi Biosphere Reserve	Parts of Chamoli district, Pithoragarh district & Bageshwar district	Uttarakhand	Western Himalayas	Snow leopard, Himalayan black bear	5860
3	1988	Nokrek	In West Garo Hills	Meghalaya	Eastern hills	Red panda	820.00
4	1989	Gulf of Mannar	Indian part of Gulf of Mannar extending from Rameswaram island in the north to Kanyakumari in the south of Tamil Nadu and Sri Lanka	Tamil Nadu	Coasts	Dugong	10500
5	1989	Sundarbans	Part of delta of Ganges and Brahmaputra river system	West Bengal	Gangetic Delta	Royal Bengal tiger	9630
6	1989	Manas	Part of Kokrajhar, Bongaigaon, Barpeta, Nalbari, Kamrup and	Assam	Eastern Hills	Asiatic elephant, tiger, Assam roofed turtle, hispid	2837

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Biosphere Reserves of India[7]							
	Year	Name	Location	State	Type	Key fauna	Area (km^2)
			Darrang districts			hare, golden langur, pygmy hog	
7	1989	Great Nicobar Biosphere Reserve	Southernmost of the Andaman and Nicobar Islands	Andaman and Nicobar Islands	Islands	Saltwater crocodile	885
8	1994	Simlipal	Part of Mayurbhanj district	Odisha	Deccan Peninsula	Gaur, royal Bengal tiger, Asian elephant	4374
9	1998	Dihang-Dibang	Part of Siang and Dibang Valley	Arunachal Pradesh	Eastern Himalaya	Mishmi takin, musk deer	5112
10	1997	Dibru-Saikhowa	Part of Dibrugarh and Tinsukia districts	Assam	Eastern Hills	White-winged wood duck, water buffalo, black-breasted parrotbill, tiger, capped langur	765
11	1999	Pachmarhi Biosphere Reserve	Parts of Betul district, Hoshangabad district and	Madhya Pradesh	Semi-Arid	Giant squirrel, flying squirrel	4981.72

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Biosphere Reserves of India[7]							
	Year	Name	Location	State	Type	Key fauna	Area (km^2)
			Chhindwara district				
12	2000	Khangchendzonga National Park	Parts of Kangchenjunga	Sikkim	East Himalayas	Snow leopard, red panda	2620
13	2001	Agasthyamalai Biosphere Reserve	Neyyar, Peppara and Shenduruny Wildlife Sanctuary and their adjoining areas	Kerala, Tamil Nadu	Western Ghats	Nilgiri tahr, Asian elephant	3500.08
14	2005	Achanakmar-Amarkantak Biosphere Reserve	Part of Annupur, Dindori and Bilaspur districts	Madhya Pradesh, Chhattisgarh	Maikala Hills	Four-horned antelope, Indian wild dog, sarus crane, white-rumped vulture, sacred grove bush frog	3835
15	2008	Great Rann of Kutch	Part of Kutch, Morbi, Surendranagar and Patan districts; the largest biosphere reserve in India.	Gujarat	Desert	Indian wild ass	12454

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Biosphere Reserves of India[7]							
	Year	Name	Location	State	Type	Key fauna	Area (km^2)
16	2009	Cold Desert	Pin Valley National Park and surroundings; Chandratal and Sarchu & Kibber Wildlife Sanctuary	Himachal Pradesh	Western Himalayas	Snow leopard	7770
17	2010	Seshachalam Hills	Seshachalam Hill Ranges covering parts of Chittoor and Kadapa districts	Andhra Pradesh	Eastern Hills	Slender loris	4755.997[8]
18	2011	Panna	Part of Panna and Chhattarpur districts in Madhya Pradesh	Madhya Pradesh	Moist deciduous forest	Bengal tiger, Chinkara, Nilgai, Sambhar deer, and Sloth bear	2998.98[9]

Note:

Panna reserve is the third Biosphere reserve of Madhya Pradesh to be included in the UNESCO's World Network of Biosphere Reserves. Before this, Pachmarhi and Amarkantak have been included in the list.

Table 7 Key Faunas in Biosphere Reserves of India

Name	Key Fauna
Nilgiri Biosphere Reserve	Lion-tailed Macaque (EN), Nilgiri Tahr (EN), Malabar Giant Squirrel (LC), Nilgiri Langur (VU)
Nanda Devi	Snow Leopard (VU), Musk Deer (EN), Bharal or Blue Sheep (LC)
Nokrek	Red Panda (EN), Hoolock Gibbons (EN), Red Giant Flying Squirrel (LC)
Great Nicobar	Dugong (VU), Saltwater Crocodile (LC)

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Name	Key Fauna
Gulf of Mannar	Dugong (VU), Olive Ridley turtles (VU)
Manas	Assam Roofed Turtle (EN), Hispid Hare (EN), Golden Langur (EN), Pygmy Hog (EN), Wild Water Buffalo (EN), Bengal florican (CR)
Sunderbans	Royal Bengal Tiger (EN)
Simlipal	Royal Bengal Tigers, Wild Elephants (EN), Gaurs (VU – Indian Bison), Chausingha (VU)
Dibru-Saikhowa	Bengal Tiger, Clouded Leopard (VU), Gangetic Dolphin (EN)
Dehang-Dibang	Takin (VU), Red Panda (EN)
Pachmarhi	Tiger, Gaur, Indian Giant Flying Squirrels (LC)
Khangchendzonga	Red Panda (EN), Snow Leopard (VU), Musk Deer (EN), Great Tibetan Sheep (Argali – NT)
Agasthyamalai	Nilgiri Tahr (EN)
Achanakamar-Amarkantak	Four-horned Antelope (Chausingha (VU)), Indian Wild Dog (VU)
Great Rann of Kutch (Kachchh)	Great Indian Bustard (CR), Indian Wild Ass (NT)
Cold Desert	Snow Leopard (VU), Himalayan Ibex (also referred to as Siberian Ibex – LC)
Seshachalam Hills	Red Sanders (NT), Golden Gecko (LC – Endemic To Tirumala Hills)
Panna	Tiger (EN), Chital (LC), Chinkara (LC), Sambar (VU)

Note: IUCN Categories – LC, EN, VU, NT, CR

India has 13 Biosphere Reserves recognised under UNESCO's Man and the Biosphere (MAB) Programme.

Table 8: List of Biosphere Reserves in India Included under UNESCO Man and the Biosphere (MAB) Programme

S.No.	Name	States/ UT	Year
1	Nilgiri Biosphere Reserve	Tamil Nadu, Kerala and Karnataka	2000
2	Gulf of Mannar Biosphere Reserve	Tamil Nadu	2001
3	Sundarbans Biosphere Reserve	West Bengal	2001
4	Nanda Devi Biosphere Reserve	Uttarakhand	2004

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S.No.	Name	States/ UT	Year
5	Nokrek Biosphere Reserve	Meghalaya	2009
6	Pachmarhi Biosphere Reserve	Madhya Pradesh	2009
7	Simlipal Biosphere Reserve	Odisha	2009
8	Cold Desert	Himachal Pradesh	2009
9	Great Nicobar Biosphere Reserve	Andaman & Nicobar Islands	2013
10	Achanakmar-Amarkantak Biosphere Reserve	Chhattisgarh, Madhya Pradesh	2012
11	Agasthyamalai Biosphere Reserve	Kerala and Tamil Nadu	2016
12	Khangchendzonga National Park	Sikkim	2018
13	Panna Biosphere Reserve	Madhya Pradesh	2020

QUICK INSIGHT

India has 18 Biosphere Reserves in total, but only 13 are in the UNESCO–MAB Network.

Nilgiri was the first Indian biosphere reserve included in MAB.

Cold Desert is the only high-altitude trans-Himalayan biosphere reserve in the MAB list.

Panna is the latest inclusion (2020).

Here are a few details about the Thirteen major bio-reserves in India:

1. **Nilgiri Biosphere Reserve:**
   Location: Western Ghats and Nilgiri Hills in South India
   Area: 5,520 square kilometres
   Fauna: 100 species of mammals, 350 species of birds, 80 species of reptiles including Tiger, Asian elephant Lion-tailed macaque and Nilgiri tahr. Nilgiri has the largest population of Lion-tailed macaque and Nilgiri Tahr, which are two endangered species.
   Flora: 3,300 species of flowering plants grow in Nilgiri and out of them, 132 are endemic to this region. Sholas or the local tropical forests hosts most rare plant species.
2. **Gulf of Mannar:**
   Location: Between India and Sri Lanka, forming part of the Laccadive Sea
   Area: 32,400 square kilometres
   Fauna: Sea turtles, Sharks, Dugongs, Dolphins, Whales. Pearl Oysters, Gorgorian Corals are also frequent in the reserve.
   Flora: Sea Grass, Sea Weed, Sea Cucumber.
3. **Sundarbans:**
   Location: Estuaries of River Ganges and River Brahmaputra. The reserve is also a UNESCO World Heritage site, as it is the largest mangrove forest area in the world.
   Area: 139,500 hectares
   Fauna: The reserve is famous as the breeding ground of the Royal Bengal Tiger. Spotted deer, Rhesus Macaque, Salt Water Crocodile and Giant Adjutant Storks are some other attractions.

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Flora: World's largest mangrove forest hosts several endemic species of plants such as Garjan, Gewa, Goran, Keora and Sundari trees. The name Sundarbans refers to the abundance of Sundari trees in the area.

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4. **Nanda Devi Biosphere:**
   Location: Northeast of Uttarakhand. The Nanda Devi bio-reserve is also designated as a UNESCO World Heritage Site
   Area: 630 square kilometres
   Fauna: Himalayan musk deer, Mainland serow and Himalayan tahr. The reserve is also known to host Snow Leopards, Himalayan Black Bear and Brown Bear, Langurs and Rhesus Macaque.
   Flora: Fir, birch, rhododendron and juniper are abundant here. Vegetation is scarce due to the dry, frosty environment.
5. **Nokrek National Park:**
   Location: West Garo Hills, Meghalaya
   Area: 47.5 square kilometres
   Fauna: A large population of Red Panda can be found in the reserve. Other mammals include Hoolocks, Stump-tailed Macaque, Flying Squirrel, Leopard and Pig-tailed Macaque.
   Flora: Grand rasamala, White meranti, Lali, Chempaka, Wild lemon are some of the endemic plants. The reserve is mainly famous for its wide vegetation of citrus plants such as lemons, oranges and lime.
6. **Pachmari Biosphere Reserve:**
   Location: Satpura Range, Madhya Pradesh
   Area: 4,926 square kilometres
   Fauna: Tigers, Leopard, Wild Boar, Indian Giant Squirrel, Indian Wolf, Chinkara, Nilgai, Chital Deer, Muntjac Deers, Sambar Deer, Bison and Rhesus Macaques.
   Flora: Two commercial trees Sal and Teak can be found in abundance. Other endemic flora include Wild Mango, Jamun, Silver Fern and Arjun.
7. **Similipal Biosphere Reserve:**
   Location: Northeast Odisha
   Area: 2,750 square kilometres
   Fauna: Around 42 species of mammals, 242 types of birds and 30 species of reptiles have been found in the reserve. Tiger, Leopard, Sambar, Asian Elephant, Barking Deer, Gaur, Chausingha, Jungle Fowl, Hill Mynah and Crested Serpent Eagles comprise a large variety of animals and birds here.
   Flora: The tropical moist deciduous forest is famous for its wide range of orchids. As many as 96 species of orchids are found here. Other prominent vegetation include Eucalyptus and large grasslands.
8. **Achanakmar-Amarkantak Biosphere Reserve:**
   Location: Chhattisgarh and Madhya Pradesh
   Area: 383,550 hectares
   Fauna: Tigers, Panthers, Bears, Chital Deer, Sambar Deer, Barking Deer, Foxes, Wild Boar, Jackals and Bisons can be found in the biosphere. The area is also famous for hosting different species of frogs including Cricket Frog, Burrowing Frog, Tree Frog, Ornate Narrow Mouthed Frog, Toad and Bullfrog.
   Flora: Pine, Juniper, Cedar and Cypress can be found here. Some of the species of Pine and Cedar are endemic to the reserve.
9. **Great Nicobar Island:**
   Location: Nicobar Islands
   Area: 1,045 square kilometres
   Fauna: Crab Eating Macaque, Nicobar Megapode, Giant Robber Crab, Nicobar Serpent Eagle, Giant Robber Crab, Palm Civet and Water Monitor Lizards are common in the reserve.
   Flora: Screw Pine, Nipa Palm, Ceylon Iron Wood can be found in abundance.
10. **Panna Biosphere Reserve:**
    Location: Madhya Pradesh
    Area: 542.67 km^2

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Fauna: Bengal tiger, Indian leopard, chital, chinkara, nilgai, Sambar deer and sloth bear, rusty-spotted cat, Asian palm civet. The park is home to more than 200 species of birds including the bar-headed goose, crested honey buzzard, red-headed vulture, blossom-headed parakeet, changeable hawk-eagle and Indian vulture.

11. **Agasthyamala Biosphere Reserve:**

Location: Southernmost end of Western Ghats, Tamil Nadu and Kerala

Area: 3500.36 square kilometres

Fauna: Bengal Tiger, Asian Elephant, Nilgiri tahr.

Flora: 2000 varieties of medicinal plants.

12. **Khangchendzonga Biosphere Reserve:**

Location: Sikkim

Area: 849.5 square kilometres

Fauna: Musk deer, Snow leopard, Himalaya tahr, Dhole, Sloth bear, Viverrids, Himalaya black bear, Red panda.

Flora: Oaks, Fir, Birch, Maple, Villow.

13. **Cold Desert Biosphere Reserve (India):**

Location: Trans-Himalayan region of Himachal Pradesh, covering parts of Lahaul & Spiti and Kinnaur districts.

Area: 7,770 square kilometres

Fauna: Supports ~40 mammals and 200 bird species, including Snow Leopard, Bharal, Ibex, Tibetan Wolf, Kiang, Himalayan Marmot, Snow Partridge, Golden Eagle, and Lammergeier.

Flora: Sparse cold-arid vegetation dominated by alpine grasses, Caragana, Artemisia, Ephedra, Sea buckthorn, with willow and poplar along river valleys.

6. EXTINCT WILD ANIMALS IN INDIA

There was a time when India was home for some of the most beautiful wild animals. Due to excessive hunting and poaching for sport and body parts by man, these wild animals have become extinct. Some of the wild species are on the brink of extinction and some of them are already extinct such as Dangs Giant Squirrel and Aldabra banded snail. Critically endangered species in India are the Great cats, one-horned rhinoceros, Ganges river dolphins, Purple frog, birds' species such as Himalayan Quail, Great Indian Bustard and Indian Horn-bill and many small mammals. The Indian government is planning a re-wilding project for Cheetahs.

Indian Cheetah: Indian Cheetah also known as Asiatic Cheetah have been known to exist in India for a very long time. Due to access hunting, deforestation and habitat loss, the fastest land animal on earth become extinct in India. The Asiatic Cheetah is a rare and critically endangered species of Big Cats family, now found today only in Iran so also known as the Iranian Cheetah, world's last few are occasional sightings in neighbour countries of India. Indian Cheetah was found in semi-desert areas of Rajasthan and Gujarat and other open habitats where prey is available.

Indian Aurochs: The Indian Aurochs once lived in the hot and dry areas of India. The huge wild cattle was larger than most modern domestic cattle with height of 6.6 ft and weight of 1,000 kilograms. The Indian bison or the gaur is the largest species of wild cattle found in India. Zebu and gaur are the Indian cattle, who are quite similar to the extinct wild Indian Aurochs. Extinct type of cattle was the most challenging hunting wild animal like the wild water buffalo and wild boar.

Sivatherium: Sivatherium was a very large approximately 7 ft tall extinct animal found in India. The giant beasts was first discovered in India, where it was called as "Sivatherium" due to the Hindu God Lord Shiva. The Sivatheres were the tallest and fastest monsters found in India ever.

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Sumatran Rhinoceros: The Sumatran Rhinoceros has been declared as one of the extinct animals in India. The Sumatran rhinoceros is the smallest rhinoceros with two horns and only extant species of the genus Dicerorhinus. The extinct Sumatran rhinoceros, once roamed and inhabited rainforests of India and neighbour north-east Asian countries. They are now critically endangered species and estimated to number fewer than 275 only and found in neighbour countries of India.

Pink-Headed Duck: Pink-headed duck was a large diving blackish-brown duck. It was one of the most beautiful birds in India. The long-necked duck once found in throughout India, but its beautiful appearance made it most hunted birds in India. Due to excess of hunting this beautiful bird is falling in the category of extinct birds in India. There is hope that the critically endangered species could still survive in some part of India and our neighbour countries.

Extinct species of India

- Alligaticeras
- Apateodus
- Babiacetus
- Bharattherium
- Deccanodon
- Deccanolestes
- Discoceras
- Eoluvarus
- Eoserranus
- Exaeretodon
- Gigantopithecus
- Godavarisaurus
- Himalayacetus
- Hyperodapedon
- Indian aurochs
- Indohyaenodon
- Indotrigonodon
- Javan rhinoceros
- Kankatodus
- Kharmerungulatum
- Kurmademys
- Kutchisiren
- List of extinct animals of India
- Miocaretta
- Pink-headed duck
- Rebbanosaurus
- Sahnitherium
- Sankuchemys

7. NATIONAL PARK

An area, whether within a sanctuary or not, can be notified by the state government to be constituted as a National Park, by reason of its ecological, faunal, floral, geomorphological, or zoological association or importance, needed to for the purpose of protecting and propagating or developing wildlife therein or its environment. No human activity is permitted inside the national park except for the ones permitted by the Chief Wildlife Warden of the state under the conditions given in CHAPTER IV, WPA 1972.

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National parks in India are IUCN category II protected areas. **India's first national park** was established in 1936 as **Hailey National Park**, now known as **Ram Ganga National Park**. **Hemis National Park** is the **largest national park** with an area of 4,400 km² while **South Button Island National Park** is the **smallest** with an area of just 5.19 km².

There are 106 existing national parks in India covering as area of around 44403 km^2, which is 1.35% of the geographical area of the country (National Wildlife Database, **January 2026**).

Table 9 Complete List of National Parks in India

National Park Area	State/Union Territory	Establishment Year	(sq. km)
Campbell Bay National Park	Andaman & Nicobar Islands	1992	426.23
Galathea Bay National Park	Andaman & Nicobar Islands	1992	110
Mahatama Gandhi Marine (Wandoor) National Park	Andaman & Nicobar Islands	1983	281.5
Middle Button Island National Park	Andaman & Nicobar Islands	1987	0.44
Mount Harriett National Park	Andaman & Nicobar Islands	1987	46.62
North Button Island National Park	Andaman & Nicobar Islands	1987	0.44
Rani Jhansi Marine National Park	Andaman & Nicobar Islands	1996	256.14
Saddle Peak National Park	Andaman & Nicobar Islands	1987	32.54
South Button Island National Park	Andaman & Nicobar Islands	1987	0.03
Papikonda National Park	Andhra Pradesh	2008	1012.86
Rajiv Gandhi (Rameswaram) National Park	Andhra Pradesh	2005	2.4
Sri Venkateswara National Park	Andhra Pradesh	1989	353.62
Mouling National Park	Arunachal Pradesh	1986	483
Namdapha National Park	Arunachal Pradesh	1983	1807.82

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National Park Area	State/Union Territory	Establishment Year	(sq. km)
Dibru-Saikhowa National Park	Assam	1999	340
Kaziranga National Park	Assam	1974	858.98
Manas National Park	Assam	1990	500
Nameri National Park	Assam	1998	200
Rajiv Gandhi Orang National Park	Assam	1999	78.81
Valmiki National Park	Bihar	1989	335.65
Guru Ghasidas (Sanjay) National Park	Chhattisgarh	1981	1440.71
Indravati (Kutru) National Park	Chhattisgarh	1982	1258.37
Kanger Valley National Park	Chhattisgarh	1982	200
Mollem National Park	Goa	1992	107
Vansda National Park	Gujarat	1979	23.99
Blackbuck (Velavadar) National Park	Gujarat	1976	34.53
Gir National Park	Gujarat	1975	258.71
Marine (Gulf of Kachchh) National Park	Gujarat	1982	162.89
Kalesar National Park	Haryana	2003	46.82
Sultanpur National Park	Haryana	1989	1.43
Great Himalayan National Park	Himachal Pradesh	1984	754.4
Inderkilla National Park	Himachal Pradesh	2010	104
Khirganga National Park	Himachal Pradesh	2010	710
Pin Valley National Park	Himachal Pradesh	1987	675
Simbalbara National Park	Himachal Pradesh	2010	27.88
City Forest (Salim Ali) National Park	Jammu & Kashmir	1992	9
Dachigam National Park	Jammu & Kashmir	1981	141
Hemis National Park	Jammu & Kashmir	1981	3350
Kishtwar National Park	Jammu & Kashmir	1981	425

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National Park Area	State/Union Territory	Establishment Year	(sq. km)
Betla National Park	Jharkhand	1986	226.33
Anshi National Park	Karnataka	1987	417.34
Bandipur National Park	Karnataka	1974	874.2
Bannerghatta National Park	Karnataka	1974	260.51
Kudremukh National Park	Karnataka	1987	600.32
Nagarahole (Rajiv Gandhi) National Park	Karnataka	1988	643.39
Anamudi Shola National Park	Kerala	2003	7.5
Eravikulam National Park	Kerala	1978	97
Mathikettan Shola National Park	Kerala	2003	12.82
Pambadum Shola National Park	Kerala	2003	1.32
Periyar National Park	Kerala	1982	350
Silent Valley National Park	Kerala	1984	89.52
Bandhavgarh National Park	Madhya Pradesh	1968	448.85
Fossil National Park	Madhya Pradesh	1983	0.27
Indira Priyadarshini Pench National Park	Madhya Pradesh	1975	292.85
Kanha National Park	Madhya Pradesh	1955	940
Madhav National Park	Madhya Pradesh	1959	375.22
Panna National Park	Madhya Pradesh	1981	542.67
Sanjay National Park	Madhya Pradesh	1981	466.88
Satpura National Park	Madhya Pradesh	1981	585.17
Van Vihar National Park	Madhya Pradesh	1979	4.45
Dinosaur National Park	Madhya Pradesh	2011	0.8974
Chandoli National Park	Maharashtra	2004	317.67
Gugamal National Park	Maharashtra	1975	361.28
Nawegaon National Park	Maharashtra	1975	133.88

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National Park Area	State/Union Territory	Establishment Year	(sq. km)
Pench (Jawaharlal Nehru) National Park	Maharashtra	1975	257.26
Sanjay Gandhi (Borivali) National Park	Maharashtra	1983	86.96
Tadoba National Park	Maharashtra	1955	116.55
Keibul-Lamjao National Park	Manipur	1977	40
Balphakram National Park	Meghalaya	1985	220
Nokrek Ridge National Park	Meghalaya	1986	47.48
Murlen National Park	Mizoram	1991	100
Phawngpui Blue Mountain National Park	Mizoram	1992	50
Intanki National Park	Nagaland	1993	202.02
Bhitarkanika National Park	Odisha	1988	145
Simlipal National Park	Odisha	1980	845.7
Mukundra Hills National Park	Rajasthan	2006	200.54
Desert National Park	Rajasthan	1992	3162
Keoladeo Ghana National Park	Rajasthan	1981	28.73
Ranthambhore National Park	Rajasthan	1980	282
Sariska National Park	Rajasthan	1992	273.8
Khangchendzonga National Park	Sikkim	1977	1784
Guindy National Park	Tamil Nadu	1976	2.82
Gulf of Mannar Marine National Park	Tamil Nadu	1980	6.23
Indira Gandhi (Annamalai) National Park	Tamil Nadu	1989	117.1
Mudumalai National Park	Tamil Nadu	1990	103.23
Mukurthi National Park	Tamil Nadu	1990	78.46
Kasu Brahmananda Reddy National Park	Telangana	1994	1.43
Mahaveer Harina Vanasthali National Park	Telangana	1994	14.59
Mrugavani National Park	Telangana	1994	3.6

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National Park Area	State/Union Territory	Establishment Year	(sq. km)
Clouded Leopard National Park	Tripura	2007	5.08
Bison (Rajbari) National Park	Tripura	2007	31.63
Dudhwa National Park	Uttar Pradesh	1977	490
Corbett National Park	Uttarakhand	1936	520.82
Gangotri National Park	Uttarakhand	1989	2390.02
Govind National Park	Uttarakhand	1990	472.08
Nanda Devi National Park	Uttarakhand	1982	624.6
Rajaji National Park	Uttarakhand	1983	820
Valley of Flowers National Park	Uttarakhand	1982	87.5
Buxa National Park	West Bengal	1992	117.1
Gorumara National Park	West Bengal	1992	79.45
Jaldapara National Park	West Bengal	2014	216.51
Neora Valley National Park	West Bengal	1986	159.89
Singalila National Park	West Bengal	1986	78.6
Sunderban National Park	West Bengal	1984	1330.1

Source: Wildlife Institute of India

8. WILDLIFE SANCTUARIES (WLS)

Any area other than area comprised with any reserve forest or the territorial waters can be notified by the State Government to constitute as a sanctuary if such area is of adequate ecological, faunal, floral, geomorphological, natural, or zoological significance, for the purpose of protecting, propagating or developing wildlife or its environment. Some restricted human activities are allowed inside the Sanctuary area details of which are given in CHAPTER IV, WPA 1972.

Wildlife sanctuaries of India are classified as IUCN Category IV protected areas. There are 573 existing wildlife sanctuaries in India covering an area of 125564.86 km^2, which is 3.73% of the geographical area of the country (National Wildlife Database, Jan 2026).

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Wildlife Sanctuaries in India

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Table 10 Complete List of Wildlife Sanctuaries in India (as on January 2026)

Some important wildlife Sanctuaries in India

State	Wildlife Sanctuary (WLS)	Estd. in	Wildlife Attractions
Andhra Pradesh	Srisailam WLS		N.A.
	Nagarjunasagar WLS	1978	Tiger, Blackbuck, Bustard
Assam	Pabitora WLS	1971	Rhinoceros
Gujarat	Sasangir WLS	1965	Asiatic Lion, Chinkara, Wild Boar, Striped Hyena,Jackal, Common Langur, Hare, Blackbuck, Marsh Crocodile
	Wild Ass WLS	1973	Indian Wild Ass, Ghudkhur, Blue-Bull, Chinkara, Hedgohog, Wolf, Jackal, Fox, Jungle and Desert Cats
Kerala	Idukki WLS	1976	Elephant, Langur, Leopard, Wild Boar, Jungle Fowl, Laughing Thrush, Black Bulbul, Woodpecker, King Fisher, Peacock.
	Periyar WLS	1950	Tiger, Indian Elephant, Nilgiri, Langur, Lion-Tailed Macaque, Nilgiri Tahr, Indian Python, King Cobra
	Wayanad WLS	1973	N.A.
	Aralam WLS	1984	Different Deers, Boar, Elephant, Bison, Leopard, Jungle Cat.
	Neyyar WLS	1958	Elephant, Gaur, Sloth Bear, Nilgiri Tahr, Jungle Cat, Wild Boar, Nilgiri, Langur
	Pappara WLS	1983	Elephant, Gaur, Sambar, Barking Deer, Mouse Deer, Wild Boar, Tiger, Panther, Wild Dog, Lion-Tailed Macaque. Nilgiri Langur, Malabar Squirre
	Parambikulam	1973	Gaur, Elephant, Sloth Bear, Wild Boar, Sambar, Cheetal, Tiger, Panther, Crocodile, Bonet Macaque
	Thattekad Bird Sancturay	1981	Bird Spp. > 250; Elephant, Leopard, Sloth Bear, Porcupine

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State	Wildlife Sanctuary (WLS)	Estd. in	Wildlife Attractions
Karnataka	B.R. Hills WLS	1987	Gaur, Cheetal, Sambar, Bear, Elephant, Panther, Tiger; 200 Spp. of Birds: Racket-tailed Drongo, Crested Eagle
	Dandeli WLS		Barking, Spotted, and Mouse Deer, Sloth Bear, Panther, Tiger, Gaur, Elephant Wild Dog, Civet Cat, Bison, Langur, Giant Flying Squirrel
	Ranganathittu WLS	1984	Paradise of Birds: Spoonbill, Open BillStork, White Ibis, Egret, Cormorant, Heron, Partridge, River Tern, Stone Plougher, Snake Bird
	Jogimatti WLS	2015	Jackals, wolves, sloth bears and leopards
	Yadahalli chinkara WLS	2015	Indian Gazelle, wolves, jackals, wild cats, striped hyenas and civet cats
	Thimlapura WLS	2016	Bears, wolfs, leopards and black bucks
Madhya Pradesh	Karera WLS	1981	245 Bird Sp: Great Indian Bustard, Bearded Bustard, Coloured Bustard, Pintail, Teal, Gadwall, Black-bellied River Tern, Egret, Spoonbill, Heron, Indian Robin, Blackbuck.
Orissa	Chilka Lake Bird Sanctuary Bhitarkanika WLS	1987 1985	Asia's largest inland salt water lagoon, World's largest breeding colonies of Flamingos, White-bellied Sea Eagles, Greylag Goose, Purple Moorhen, Jacana. Heron; Blackbuck, Spotted Deer, Golden Jackal and Hyena. 215 Bird spp. Giant Salt Water Crocodile, King Cobra, Indian Python,Water Monitor Lizard.
Tamil Nadu	Mudumalai WLS	1978	Langur, Bonnet Macaque, Tiger, Leopard, Wild Dog, Hyena, Sloth Bear, Elephant, Gaur, Sambar, Spotted Deer, Indian Muntjac, Mouse Deer, Wild Boar, Flying Squirrel, Birds: Malabar Trogan, Grey Hornbill, Crested Hawk. Eagle, Serpent Eagle.
Telangana	Manjira WLS	1978	Monitor Lizard, Fresh Water Turtle, Cobra, Marsh Crocodile, Painted Stork, Heron, Coot, Teal, Cormorant, Pochard, Spoon Bill, Open Billed Stork
Uttar Pradesh	Hastinapur WLS	1986	Swamp Deer, Cheetal, Nilgai, Wolf, Leopard, Hyena, Wild Boar, Gharial

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State	Wildlife Sanctuary (WLS)	Estd. in	Wildlife Attractions
	Kishanpur WLS	1972	Tiger, Leopard, Swamp Deer
	National Chambal WLS	1992	Rare Gangetic Dolphin, Crocodile (Magar), Gharial (Alligator), Chinkara,Sambar, Nilgai, Wolf, Wild Boar

Source: Wildlife Institute of India

9. LIST OF INDIAN WILDLIFE (PROTECTION) ACT, 1972: SCHEDULE-I

It is a statutory Act targeted for **protection of plants, birds and animal species**. (**Statutory law** or **statute law** is written rule set down by a body of legislature or by a singular legislator (in the situation of an absolute monarchy). This is as contrasting to oral or customary law; or regulatory law transmitted by the executive or common law of the judiciary.) It established schedules for protected plant and animal species; **hunting or collection of the species was banned**.

Salient features of the Act

1. It covers to the whole of India.
2. It has 4 Schedules now, which gave varying degrees of protection.

- **Schedule I** Animal species that will enjoy the highest level of protection including those which are critically endangered.
- **Schedule II** Animal species that will be subject to a lesser degree of protection
- **Schedule III** Protected Plant species
- **Schedule IV Specimens listed in the Appendices under CITES**

A number of wild animal and bird species have been listed in various Schedules of the Indian Wild Life (Protection) Act, 1972. Similarly, these species are also categorised under IUCN Red List, CITES and Convention on Migratory Species (CMS). This database provides information on categorisation of species under these legal instruments.

Table 11 Schedule Birds Species under Indian Wild Life (Protection) Act, 1972

Common Name	Scientific Name	WPA, 1972	IUCN	CITES	CMS
Andaman teal	*Anas gibberifrons albogularis*	Sch I			
Assam bamboo partridge	*Bambusicola fytchii*	Sch I			
Bazas	*Aviceda jerdoni and Aviceda leuphotes*	Sch I			
Jerdon's Baza	*Aviceda jerdoni*	Sch I			
Bengal florican	*Eupodotis bengalensis*	Sch I			
Black-necked crane	*Grus nigricollis*	Sch I	VU	I	I/II
Blood pheasant	*Ithaginis cruentus kuseri*	Sch I			

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Common Name	Scientific Name	WPA, 1972	IUCN	CITES	CMS
Cheer pheasant	*Catreus wallichii*	Sch I	VU	I	
Great Indian bustard	*Ardeotis nigriceps*	Sch I	EN	I	
Great Indian hornbill	*Buceros bicornis*	Sch I	NT	II	
Hawks	*Fam. Accipitridae*	Sch I			
Hooded Crane	*Grus monacha*	Sch I	VU	I	I/II
Hornbills	*Ptiloaemus tickelli austeni, Aceros nipalensis, Rhyticeros undulates ticehursti*	Sch I			
Wreathed Hornbill	*Aceros undulatus*	Sch I			
Rufous-necked Hornbill	*Aceros nipalensis*	Sch I	VU	II	
Houbara bustard	*Chlamydotis undulata*	Sch I	VU	I	I/II
Humes bar-backed pheasant	*Syrmaticus humiae*	Sch I	NT	I	
Indian pied hornbill	*Anthracoceros malabaricus*	Sch I			
Jerdon's Courser	*Cursorius bitorquatus*	Sch I	CR		
Lammergeier	*Gypaetus barbatus*	Sch I			
Large falcons	*Falco peregrinus, F. biarmicus, F. chicquera chicquera*	Sch I			
Lesser Adjutant	*Leptoptilos javanicus*	Sch I			
Peregrine Falcon	*Falco peregrinus*	Sch I			
Red-headed Merlin	*Falco chiquera*	Sch I			

Schedule Birds Species under Indian Wild Life (Protection) Act, 1972

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Schedule Fishes Species under Indian Wild Life (Protection) Act, 1972

Schedule Insects Species under Indian Wild Life (Protection) Act, 1972

Schedule Mammals Species under Indian Wild Life (Protection) Act, 1972

Schedule Reptiles Species under Indian Wild Life (Protection) Act, 1972

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SECTION III

UPSC Mains Model Answers: Environment (2025 – 2023)

UPSC MAINS 2025

1. What is Carbon Capture, Utilization and Storage (CCUS)? What is the potential role of CCUS in tackling climate change?(10 marks; 150 words)

Ans As per IEA, CCUS involves the capture of CO2, generally from large point sources like power generation or industrial facilities that use either fossil fuels or biomass as fuel. The captured CO2 is either used onsite or compressed and stored for future use. It can also be injected into deep geological formations such as depleted oil and gas reservoirs or saline aquifers.

The technology of CCUS is inspired from natural "carbon sinks" like wetlands and rainforests which protect the earth from warming by acting as natural storage sites for carbon dioxide.

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Since CCUS operates on the principle of redirecting carbon emissions to prevent warming, it has huge potential in tackling climate change.

Potential to tackle climate change:

1. **Decarbonisation of high emission sectors:** Power, industry and automobile sector contributes to more than 70% of emissions. Carbon capture through exhaust capture, mobile carbon capture and direct air capture is a possible solution to reducing emissions before switching to cleaner technologies.
2. **Enabling Negative Emissions:** Technologies like BECCS (Bioenergy Carbon Capture and Storage), capture the CO2 released from combustion of biomass based renewable energy, and further stores it permanently. Since plants absorb CO2 during photosynthesis, this ultimately results in net negative emissions of carbon.
3. **Circular Carbon Economy:** The captured CO2 can be used to make synthetic carbon based fuels (e-fuels), fertilisers like urea, and in plastic manufacturing as feedstock. This effectively ensures removal of emissions from atmosphere and recycles it into usable products.
4. **Blue Hydrogen:** Since green hydrogen is not yet cost competitive, blue hydrogen can be produced from synthetic natural gas.

Despite the feasibility of CCUS to tacke climate change, challenges remain such as:

- High cost of capture and storage technologies
- Storage risks - leakage into atmosphere and increased saline stress in soil
- Diversion from decarbonisation as CCUS provides an alternative for greenwashing.
- Lack of legal frameworks can lead to exploitation of natural resources for sequestration of carbon dioxide.

CCUS can play a vital role in climate mitigation, however its success depends upon effective scaling up of technologies in a cost-effective

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manner, ensuring safe storage and aligning with countrywise decarbonisation policies for effective implementation.

2. Examine the factors responsible for depleting groundwater in India. What are the steps taken by the government to mitigate such depletion of groundwater?(10 marks; 150 words)

Ans Groundwater in India is a vital resource crucial for sustaining various sectors, particularly agriculture. Being the world's largest consumer of groundwater, India with its large population specifically faces a dire situation in case of water scarcity due to groundwater depletion.

Some of the factors responsible for groundwater depletion can be summrised as follows:

1. Anthropogenic Factors:

- **Irrigation:** As per Economic Survey 2022, agriculture uses more than 60% of groundwater in India. The main reasons for over-dependence on groundwater are as follows:
  - MSP driven water intensive crops - rice and wheat
  - Unscientific irrigation methods- flood and furrow methods
  - Subsidized electricity and water pumps.
- **Demographic Pressure:** Rapid growth of population has placed a burden on India's drinking water availability. Moreover rapid urbanisation has led to overextraction of groundwater through municipal brewell water supply.
- **Industrial usage:** As per CWGB Report 2022, 5–6% of total annual groundwater extraction in India is for industrial uses. While the proportion is less, localized extraction leads to water stress in affected areas. Moreover, mining of coal and iron requires dewatering of aquifers affecting local groundwater levels.
- **Thermal electricity production:** 70% of India's electricity comes from coal-based thermal power (CEA, 2023). According to a Center for Science and Environment (2018) study, thermal power plants alone account for about 70% of India's industrial water consumption, much of it groundwater-based in arid/semi-arid belts.
- **Sand mining:** Illegal sand mining can lead to riverbed erosion and altered course which can consequently disturb the replenishment of the natural aquifers leading to groundwater depletion

2. Environmental factors:

- **Erratic Monsoons:** Due to climate change and sea surface circulation patterns, monsoons are delayed and irregular. As a result summer peak demand affects groundwater levels especially in densely populated urban areas.

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- **Reduced Natural Recharge:** Encroachment of water bodies and poor drainage have led to poor recharge capacities of ground aquifers.
- **Exceeded carrying capacity:** Particularly in NW India, the extraction rate far exceeds the rate of replenishment of groundwater.

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Case studies reveal that access to groundwater plays a pivotal role in poverty reduction and ensuring food security.

Government Initiatives to address groundwater depletion:

Since water is a state subject, the Central Gvernment augments the efforts of the states through model policies, financial and techncal support:

1. **Jal Shakti Abhiyan: Catch the Rain (2020):**
   - It nudged the states and local stakeholders to create appropriate Rain Water Harvesting Structures (RWHS) to reduce dependence on groundwater for irrigation and household purposes.
   - RWH Cells were set up to monitor the extent of groundwater extraction and recharge.
   - It promotes water reuse and groundwater recharge through bore wells.
   - Moreover, intensive afforestation is done to improve groundwater recharge and reduce soil erosion.

2. **Atal Bhujal Yojana (2020):** It is implemented at Gram Panchayat Level across 7 water stressed states to improve community led sustainable ground water management through:
   - Formation/strengthening of Water User Associations, monitoring and disseminating ground water data
   - Water budgeting - behavorial change
   - Preparation of Gram Panchayat-wise Water Security Plans (WSPs) and their implementation.

3. **Irrigation schemes:** These schemes aim to reduce the usage of groundwater for irrigation and adoption of new scientific methods for effective water usage:
   - **PM-Krishi Sinchayee Yojana-** It encourages adoption of precision-irrigation under **Per Drop More Crop** and reuse of treated municipal wastewater for peri-urban agriculture to enhance groundwater recharge.

4. **Groundwater Estimation Reports:** They are released by Central Ground Water Board every 2 years to track aquifer stress.

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5. **Bhujal Portal:** ISRO Remote Sensing Centre created a database of all groundwater reservoirs across the country, providing information for ground water quality and prospect.
6. **Bhu-Neer Portal:** It is a licensing,, NOC and monitoring portal to track extraction of ground water by industries, infrastructure projects and mining projects as per the guidelines under Ministry of Jal Shakti.

Despite the number of schemes, the sheer scale of extraction makes uniform regulation difficult. The way forward lies in saving water through crop shifts, better irrigation, stricter rules for industries, recharge efforts, and active community participation to ensure long-term water security.

3. Mineral resources are fundamental to the country's economy and these are exploited by mining. Why is mining considered an environmental hazard? Explain the remedial measures required to reduce the environmental hazard due to mining.(10 marks; 150 words)

Ans The importance of minerals can be traced back to the Harappan civilization where extraction of gold, copper, silver, lead and tin were done to supplement the Bronze Age culture. In modern times, critical minerals are indispensable to develop the semiconductor and clean energy industries.

Minerals are valuable natural resources being finite and non-renewable. The economic importance of minerals can be realised as follows:

- Raw materials for many industries including steel, cement, automobiles etc.
- Expansion of the mining sector drives infrastructure and employment growth.
- India is the 15th largest global exporter of minerals.

Despite their economic importance, minerals mining is often considered an environmental hazard due to the following reasons:

1. **Deforestation and Land Degradation**- Large tracts of land are cleared for open-cast and underground mines, especially in mineral-rich states

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like Odisha, Chhattisgarh, Jharkhand, Madhya Pradesh, and Karnataka.

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2. **Biodiversity loss-** It primarily leads to habitat displacement of native species. Moreover, dust on leaves reduce their photosynthetic ability gradually reducing the species coverage.
3. **Pollution**-
   - **Air pollution:** Methane released from coal mines significantly leads to GHG emissions which has a global warming potential of about 30x.
   - **Water Pollution:** Acid Mine Drainage (AMD) contaminates rivers and groundwater with sulphuric acid and heavy metals (arsenic, lead, mercury).
   - **Noise Pollution:** Blasting operations affect local settlements and wildlife
   - **Groundwater pollution:** Fly Ash ponds can leak out and cause pollution of groundwater.
4. **Soil contamination and groundwater depletion:** Deposition of toxic tailings and heavy metals makes land infertile. Moreover sand minings cause hydrological changes which can affect recharge of natural aquifers.
5. **Waste Generation:**
   - Overburden and Mine Spoils occupy vast areas, renders soil unft for agriculture and causes dust pollution when left untreated.
   - Uranium and rare-earth mineral mining generate radioactive wastes (Jaduguda, Jharkhand case)
   - Leaching processes in copper and zinc mines release sulphates and toxic salts into soil and water.

Remedial Measures to reduce environmental hazards of mining:

1. Regulatory and Policy Measures:

- Environmental Impact Assessment (EIA) before mining projects.
- Strict enforcement of the Environment Protection Act (1986), Forest Conservation Act (1980), and MMDR Act (1957).
- Identification of “Go and No-Go zones” to protect ecologically fragile areas.

2. Sustainable Mining Practices:

- Adoption of eco-friendly technologies like underground mining, in-situ leaching, remote sensing for exploration.
- Progressive mine closure plans to ensure rehabilitation and prevent hazards like ground collapse and continued pollution.
- Usage of mine sites for clean energy production like solar parks and wind power plants.

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3. Pollution Control Measures:

- Dust suppression using water sprays and green belts.
- Proper treatment of Acid Mine Drainage and effluents before discharge to prevent water contamination.
- Monitoring and capping methane emissions. Moreover capture methods can enable use of methane as a clean fuel.
- Noise reduction using moise barriers and sound-dampening materials

4. Environmental Restoration:

- Reclamation of abandoned mines with backfilling, soil covering, and re-vegetation.
- Afforestation with native species to restore ecological balance.

5. Waste Management:

- Scientific disposal of tailings through properly lined ponds
- Reuse of fly ash as a partial replacement to cement in construction activities.
- It can also be used after treatment in agriculture to improve water use efficiency.
- Tata Steel, Jamshedpur uses steel slag in road construction, achieving almost 100% utilization of blast furnace slag. Moreover, Indian Railways has also used slag for track laying.

Mining, though vital for economic growth, poses serious ecological and social challenges. A balance must be struck between resource extraction and environmental protection through preventive and remedial measures ensuring mining contributes to development without compromising ecological integrity.

4. Write a review on India's climate commitments under the Paris Agreement (2015) and mention how these have been further strengthened in COP26 (2021). In this direction, how has the first Nationally Determined Contribution intended by India been updated in 2022?(10 marks; 150 words)

Ans The Paris Agreement is a legally binding international treaty on climate change. While its overarching goal targets to limit global warming to 2 deg C above pre-industrial levels, efforts are underway to limit the rise to 1.5 deg C.

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Participating countries submit personalised goals in order to achieve the target in the form of Nationally Determined Contributions (NDCs). Each successive NDC is meant to reflect an increasingly higher degree of ambition compared to the previous version.

India submitted ts first NDC in the year 2015, which are as follows:

1. Reduce emissions intensity of GDP by 33–35% by 2030 (base year 2005)
2. Achieve ~40% cumulative electric-power installed capacity from non-fossil sources by 2030.
3. Create an additional carbon sink of 2.5–3.0 billion tonnes CO_2 equivalent by 2030 through additional forest/tree cover.

India has already achieved and surpassed the first two targets and has since updated its

NDCs by **announcing the "Panchamrit"** Package in COP26 along with other additions:

1. **Panchamrit Package:**
   a. India aims to achieve net-zero emissions by 2070
   b. Increase non-fossil energy capacity to 500 GW by 2030.
   c. Meet 50% of India's energy requirements from renewables by 2030.
   d. Reduce the carbon intensity of GDP by >45% by 2030 (from 2005 levels). e. Reduce projected cumulative carbon emissions by 1 billion tonnes from now through 2030.
2. **LiFE Movement -** to propagate **healthy and sustainable** way of living based on **traditions** and values of **conservation and moderation** as a way to combat climate change.

   While these additions and updated goals were announced in the 2021 COP26, they were formalised by the Government of India in 2022 with a focus on "Working towards Climate Justice". It formalized:

   1. The **LiFE movement** announced in 2021 to combat climate change
   2. To reduce emissions Intensity of its GDP by **45 percent by 2030**, from 2005 level.

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3. To achieve **50%** of India's energy requirements from **non-fossil sources by 2030** through transfer of technology and low-cost international financing including from Green Climate Fund.
4. To mobilize **funding from developed countries** to achieve targets.
5. It also introduced government actions and behavorial additions like domestic capacity building, higher R&D investment and adopt a climate friendly approach to achieving its goals.
6. It also **retained** the **3rd goal of the 2015** NDCs which is yet to be fulfilled.

On paper, India's COP26 announcements and the August 2022 formal NDC update represent a clear upward revision of India's 2030 ambition. In practice, credibility will rest on:

- detailed sectoral roadmaps
- concrete coal transition plans and timelines,
- scaled finance and technology flows, and
- transparent near-term trajectories and reporting (so 2030 promises can be independently tracked).

UPSC MAINS 2024

1. Industrial pollution of river water is a significant environmental issue in India. Discuss the various mitigation measures to deal with this problem, and also the government's initiatives in this regard.(10 marks; 150 words)

Ans As per the World Economic Forum, around 70% of surface water in India is unfit for consumption. Every day, almost 40 million litres of wastewater enter rivers and other water bodies, with only a tiny fraction adequately treated.

Industrial pollution of rivers refers to the contamination of river water by pollutants released from industries in the form of wastewater, thermal pollution, dumping of solid waste, etc. Contaminants have the potential to

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leach into groundwater and pollute drinking water sources and agricultural fields. This can have serious health consequences.

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Mitigation measures:

1. *Exponential increase in the capacity of wastewater facilities* - As per CPCB report, 2021, the wastewater released into water bodies is greater than 200% of the available wastewater treatment capacity.
2. *High-resolution monitoring of water pollution* - Automated, geotagged, and time-stamped real-time sensors can be used to generate heat maps for water pollution for targeted treatment.
3. *Stringent regulations to control effluent standards* - Effluent standards specify limits for various pollutants like BOD, COD, TSS, oil and grease, heavy metals, and pH levels.
4. *Wastewater treatment at source* - As per the provisions of the Environment (Protection) Act, 1986 and the Water (Prevention & Control of Pollution), Act 1974, industrial units and local bodies are required to install effluent treatment plants to treat their effluent/sewage to comply with stipulated environmental standards before discharging into rivers and water bodies.
5. *Re-use of wastewater by industries* - Reusing wastewater in industry can occur within a company or between companies, and it reduces expenditure on treatment as well as the amount of polluted water from being released into rivers.

India's largest river Ganges, has been declared "not fit" for drinking by CPCB. Kanpur, the most polluted city on the river, dumps up to 9 million litres of industrial wastewater illegally from mostly tanneries.

Government initiatives:

As per the Water (Prevention and Control of Pollution) Act, 1974, CPCB is the statutory body for monitoring of water quality. As per its reports, the government has taken up several initiatives:

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1. **Namami Gange-** It is an integrated conservation programme launched by the GoI in 2014 to arrest the pollution of Ganga river and revive it through term activities. Medium term activities are aimed at arresting industrial pollution by buiding wasterwater treatment facilities in PPP mode.
2. **National Water Quality Monitoring Programme (NWMP)-** It is a collaborative effort between CPCB and State PCBs to assess the extent of maintenance and restoration in terms of pollution of water bodies**.**
3. **Online Continuous Effluent Monitoring Systems (OCEMS):** The CPCB has directed 17 categories of highly polluting industries like tanneries, oil refineries, fertilizer plants etc. to install OCEMS to help track the discharges of pollutants from these units.
4. **Draft New National Water Policy (NWP), 2020:** It aims to lower the industrial water footprint by recycling industrial wastewater and reducing the reliance on freshwater for industries.

   The impact of industrial pollution on rivers can be life-threatening. The DuPont pollution case of 2001 caused PFAS to enter the bloodstream of 97% American population. These kinds of cases should serve as a warning about the extent of the impact of industrial pollution of rivers.

   In India, urgent corrective measures like increasing the capacity of wastewater treatment plants and rigorous penalisation of illegal dumping of effluents can be a step in the right direction.

2. What role do environmental NGOs and activists play in influencing Environmental Impact Assessment (EIA) outcomes for major projects in India? Cite four examples with all important details.(10 marks; 150 words)

Ans EIA refers to analytical processes that systematically examine the potential environmental consequences of a project before its approval. It gives an opportunity to:

- Assess key issues effectively and in a transparent manner
- Achieve wider sustainable development objectives.

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The role of environmental activists and NGOs is considered especially crucial in an EIA, as they represent an unbiased and non-political view of an environmental project's sustainability.

Role of NGOs and Activists in influencing EIA outcomes:

1. They can generate extensive public awareness regarding the implications of any project and mobilise community support due to on-ground presence.
2. The government can appoint NGOs to monitor the implementation of EIAs for environmental projects for unbiased feedback.
3. NGOs and activists have been known to challenge falsified and non-compliant EIAs through PIL.
4. NGOs can research on behalf of competent authorities to supply accurate technical data for EIAs due to their field expertise.
5. They can also mobilise people to pressurise the government for stricter environmental regulations and their enforcement during the preparation of EIAs.

Important examples citing the role of NGOs and activists:

1. **Narmada Bachao Andolan:**
   a. NGO/ Activists: Narmada Bachao Andolan, Medha Patkar, Baba Amte
   b. Project: Sardar Sarovar Dam
   c. Impact: The protest led to them securing fresh EIAs, directed by the SC, focusing on the rehabilitation of the displaced masses.
2. **Save Western Ghats Movement**
   a. NGO/ Activists: Peaceful Society, WWF, Soliga and Irula tribes
   b. Project: Kaiga nuclear power plant and Sharavathi underground pumped storage plant.
   c. Impact: The issues raised by the movement on degradation of Western Ghats influenced the Gadgil report, with it declaring 64% of the area as ecologically sensitive.

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3. Pratirodh Sangram Samiti

a. NGO/ Activists: Multiple, including the Civil Society Forum on Human Rights, Greenpeace India and Amnesty International.
b. Project: POSCO 12 MTPA steel plant
c. Impact: Protests and legal challenges over inadequate EIAs and community displacement led to the withdrawal of clearance and POSCO's withdrawal from the project.

4. Mumbai Coastal Road Project (Maharashtra)

a. NGOs/Activists: Vanashakti and the Conservation Action Trust
b. Project: Coastal road threatening marine ecosystems and fisherfolk.
c. Impact: PILs filed against faulty EIAs led to the Bombay HC halting construction and ordering a revision of the EIA.

With increasing development, there will be a tug-of-war between environmental preservation and economic development. As challenges increase, continued engagement of NGOs will drive rigorous assessments and allow incorporation of sustainable alternatives, enhancing environmental governance for a livable Earth for generations to come.

3. The world is facing an acute shortage of clean and safe freshwater. What are the alternative technologies that can solve this crisis? Briefly discuss any three such technologies, citing their key merits and demerits.

(10 marks; 150 words)

Ans Although water covers 70% of the Earth's surface, only about 3% is freshwater, and less than 1% is accessible for human use. However, due to unsustainable usage, nearly 2 billion people lack access to safe drinking water (WHO, 2022).

As per UN Water 2023, 52% of global population will be living in water stressed areas by 2050. Hence, UN has included Clean Water and Sanitation in the list of SDGs (SDG-6).

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However, with the advent of technological advancements, there have been efforts to solve this crisis in the following ways:

1. **Rainwater harvesting-** One of the most traditional methods that collects rainwater for usage reducing reliance on freshwater sources. **E.g:** Rooftop harvesting, bunds, percolation tanks etc.
2. **Desalination:** It refers to the process of removing salts and other minerals from saline water, typically seawater or brackish water, to produce freshwater. **E.g.:** Reverse Osmosis, Thermal Distillation, electrodialysis etc.
3. **Greywater Recycling :** Also termed as water reclamation, it is a process of purifying wastewater and reusing it for non-potable purposes like irrigation, industrial processes, or even groundwater recharge.
4. **Nano-filtration:** It uses nanoparticles to remove contaminants from water at molecular level. It can help in targeted filtration like removal of heavy metals from industrial wastewater.
5. **Atmospheric water harvesting (AWH):** It is a method of extracting potable water from the air for water-scarce regions by tapping into water stored in the atmosphere. **E.g:** fog collection, dew harvesting, and sorption-based methods.

While the list is not exhaustive, it provides an insight into the alternative methods of freshwater collection as most existing water sources are severely polluted making it unsafe for consumption, thereby creating an artificial scarcity.

Technologies in detail:

1. **Desalination:** It removes salt and impurities from water, making it a source of usable fresh water, especially in arid regions. For example, 50% of the municipal water in Israel and Saudi Arabia comes from desalination plants.
   - **Merits:**
     i. Promotes **water security**, fulfilling SDG-6 of UN.

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ii. Promotes **diversification** of water sources, reducing reliance on freshwater bodies in water-stressed areas.

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- **Demerits:**
  i. Since the process is **energy-intensive**, the use of fossil fuels can increase GHG emissions.
  ii. **Brine disposal** is a major concern as the high salinity can disrupt the ecological balance of marine ecosystems.
  iii. The development process is **capital-intensive** as the setup requires complex infrastructure and safety standards since its inception.

2. Greywater Recycling: Greywater is gently used water, but not water that has come into contact with faeces, either from the toilet or from washing diapers.

Significant examples include the **Houseful project of Europe**, which aims to achieve a **circular economy** by recycling over 90% of rainwater and greywater for producing reclaimed water and biogas.

- **Merits:**
  a. **Reduces reliance on freshwater** sources by using greywater from domestic outputs for non-potable purposes.
  b. The **nutrient-rich** greywater after recycling can benefit soil health and reduce reliance on chemical inputs like fertilisers and pesticides.
- **Demerits:**
  a. Improper recycling can lead to the **leaching of harmful pathogens** into agricultural produce through greywater used in irrigation.
  b. Adherence to **safety standards** is difficult to regulate due to a lack of technical expertise.

3. Nano filtration:

- **Merits:**
  i. Nanofiltration can be used for the **selective removal** of contaminants, enabling retention of beneficial minerals in potable water.
  ii. Certain nano-membranes can be used to remove contaminants in a **chemical-free** manner, making them an attractive alternative to traditional ion exchange systems..
- **Demerits:**
  i. Nanofiltration typically requires **higher pressure** than ultrafiltration, leading to increased energy consumption and operational costs.
  ii. **Membrane fouling** (accumulation of impurities) can reduce the efficiency of nano-filtration over time.
  iii. Nano-filtration membranes and particles can be costly to produce, making it a highly capital-intensive setup.

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Despite having 17% of the global population, India possesses only 4% of the freshwater resources globally. Hence, technological interventions through water treatment and recycling is the need of the hour as natural resources deplete rapidly. Effective implementation of schemes like Sujalam 2.0 can be a step towards water sufficiency in India.

UPSC MAINS 2023

1. What is oil pollution? What are its impacts on the marine ecosystem? In what way is oil pollution particularly harmful for a country like India?(10 marks; 150 words)

Ans Oil pollution refers to the resultant contamination of the environment due to the dispersion of oil in excessive quantities. Particularly affecting marine environments, 12% of oil pollution is a result of oil spills into marine bodies. The main sources of oil pollution include:

- Leakage due to **human error** during maintenance of petroleum refineries, barrages, tankers, pipelines and containers.
- **Accidental** breakdown or failure of drilling machinery and pipelines.
- **Natural disasters** like cyclones derail ships carrying colossal quantities of oil
- **Deliberate attacks** by terror groups to harm the economic situation of the target country.
- **Illegal dumping** by industries through improper waste management and weak regulatory oversight.

Impact on marine ecosystem:

Ans Oil spills have decreased significantly over the last two decades, despite an increase in tanker movement. However, recent incidents like the Wakashio MV off the Mauritius coast highlight the need to understand the impacts on marine ecosystems:

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1. **Physical Smothering:** In incidents like the Alaska Exxon Valdez Spill (1989), heavy oils coat the bodies of marine organisms and birds, which:
   a. Reduces their ability to maintain temperature leading to hypothermia
   b. Reduces their mobility which leads to en masse drowning of sea mammals.
2. **Toxicity:** The Deepwater Horizon explosion and blowout of the Gulf of Mexico and subsequent photooxidation of PAHs led to formation of toxic compounds. Consumption of this degraded oil has led to cardiac arrests, mutations, and larval mortality in many marine animals, according to NOAA.
3. **Biomagnification:** Oil spills also tend to move up the food chain through the accumulation of persistent toxic chemicals like PAHs and VOCs.
4. **Water acidification:** Introduction of gases by oil spills in ocean water changes its chemical composition, which in turn makes water more acidic and leads to the degradation of fragile ocean habitats like the coral reefs.
5. **Tar-mats:** Accumulation of oils near shores and coasts can form sticky residues of oil, which decrease the nutrient absorption capacity of sensitive mangroves.

Harmful impact on India:

Ans The recent detection of an oil spill by EOS-4 from the capsized MSC ELSA 3 near the coast of Kerala has raised debates on India's preparedness regarding oil pollution**.**

1. **Environmental Impact**
   a. **Biodiversity hotspots:** India's coastline is dotted with ecologically sensitive zones like the Western Ghats, A&N Islands and Lakshadweep coral reefs, which are particularly at risk of oil pollution.
   b. **Climate resilience:** Coastal ecosystems like mangroves and salt marshes are likely to be affected, which are crucial for carbon

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sequestration and buffers against cyclonic disasters.

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c. **Ecological balance:** Death of sea mammals, fishes, marsh grasses and marine birds due to smothering and toxicity can disrupt the food chain and consequently the ecological balance.

2. **Economic Impact:**
   a. **Fisheries:** The fisheries sector contributes 7% value to Indian agriculture. This can be severely impacted due to a reduction in catch following oil spills, as seen in the aftermath of the MV Rak spill (2011).
   b. **Tourism:** The destruction of the marine ecosystem deters tourists from visiting affected areas, reducing economic growth aided by tourism.
   c. **Trade:** The geostrategic location of India makes it a transit point in the IOR. However, oil spills can block such points, reducing trade and revenue for India.

Despite being a party to multiple conventions like MARPOL and SOLAS, India does not have any dedicated legislation to address oil pollution especially in marine ecosystems. To counter the unpreparedness, India can do the following:

1. Stringent regulations for adherence to safety standards for industries and commercial ships.
2. Use AI and VR to do a predictive analysis of potential oil spill incidents based on weather patterns and shipping routes.
3. Implementation of nanotechnology and bioremediation measures for effective oil cleanup.

The Indian coastline is an ecologically diverse and sensitive zone that is crucial for preserving the ecological balance of the IOR, making ts protection a socio-economic and environmental imperative.

2. The adoption of electric vehicles is rapidly growing worldwide. How do electric vehicles contribute to reducing carbon emissions and what

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are the key benefits they offer compared to traditional combustion engine vehicles?

(10 marks; 150 words)

Ans The higher allocation towards electric mobility schemes in the 2025-26 Union Budget highlights the sustainability of the electric vehicle (EV) sector. EVs occupied 4% of the automobile market share as of 2025 in India, and are projected to account for 40% globally by 2030.

This wide popularity of EVs is driven by nations seeking an alternative to conventional fossil fuel-powered vehicles to meet their Nationally Determined Contributions as per the Paris Agreement.

For example, one electric bus can save ~50 tonnes of CO_2 annually compared to a diesel bus.

Role of EVs in reducing carbon emissions:

In 2024-25, EVs alone contributed to reduction in carbon emissions by 700 lakh tonnes augmenting India's COP 26 target to reduce 1 billion tonnes by 2030.

The main difference between EVs and conventional vehicles lies in the power source and drivetrain - EVs run on electricity stored in rechargeable batteries, while ICE vehicles run on fossil fuels (petrol, diesel, CNG) through combustion.

While EVs start with a carbon debt due to the energy-intensive procedure of manufacturing Li-ion batteries, they can offset it in the following manner:

1. **Tailpipe Emissions -** Unlike conventional ICEs, EVs have zero tailpipe emissions, eliminating the release of CO and CO_2.
2. **Energy efficiency -** EVs are 3–4 times more efficient in converting stored energy into motion. Moreover, EVs are made using lighter fibre-reinforced polymers, which further improves energy efficiency, thereby reducing the need for excess power.

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3. **Regenerative Braking -** Additional energy conservation is done by converting the kinetic energy of the EV into electric energy. This can increase the range of battery capacity by 10%.
4. **Renewable energy sources-** Up to ~90% of carbon emissions can be offset by EVs, as compared to ICE, if power sources for recharging batteries are renewable energy sources like solar and wind.
5. **Vehicle to Grid-** EVs can also be used as alternative power sources by the grid supply for domestic use, which can reduce reliance on diesel generators during power cuts.

Benefits of EV over ICE:

1. **Reduces import dependency on fossil fuels:** India imports ~85% of its crude oil. Especially with tariff wars, EVs can help India reduce dependency on globally volatile markets.
2. **Economical alternative:** With government support through the FAME scheme and tax subsidies at the state level, EVs provide a better alternative to conventional cars due to decreasing battery costs.
3. **Safety features:** With the integration of software to detect the status of car components, EVs provide a safer alternative than conventional vehicles, where servicing is to be done periodically.
4. **Compliance:** All EVs are compliant with BS6 standards, which reduces the possibility of them getting phased out anytime soon.
5. **Reduction in noise pollution:** Since EV powertrains have fewer moving parts to convert electrical energy to mechanical energy, they are very silent, especially as compared to diesel vehicles.

Despite the benefits, EV adoption in the Indian market is met with reluctance due to a lack of charging infrastructure as compared to fossil fuel availability across the country. Moreover, hybrid vehicles, which are a more convenient alternative, have high upfront costs that do not come under the ambit of subsidies under FAME India.

However, with government interventions through the PM-eDrive and FAME India schemes, the budgetary outlay for the development of public

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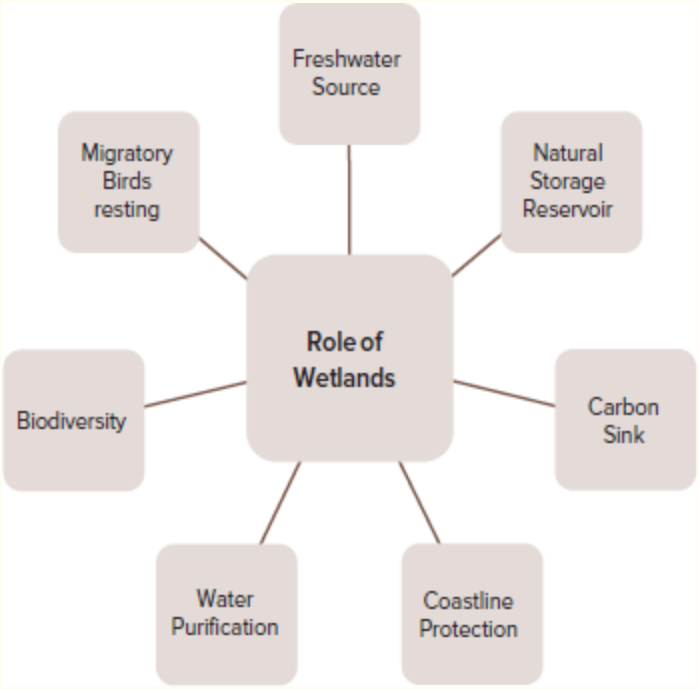
charging infrastructure has increased rapidly. Higher subsidies on purchase and retrofitting existing fuel depots with charging infrastructure can help in the faster adoption of EVs in India.

3. Comment on the National Wetland Conservation Programme initiated by the Government of India and name a few of India's wetlands of international importance included in the Ramsar Sites.(10 marks; 150 words)

Ans Due to high geographical diversity, India has a high number of wetlands, which, as per the National Wetlands Atlas, contribute to approximately 5% of India's geographical area.

Wetlands play an important role in nature conservation by:



National Wetland Conservation Programme (NWCP)

Currently subsumed under the integrated National Plan for Conservation of Aquatic ecosystems (NPCA), NWCP identified 115 wetlands in 1985-86, as per the Ramsar convention, for conservation and management interventions.

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NWCP aimed to prevent further degradation of identified wetlands and ensure their wise utilisation for the benefit of local communities as well as for conservation of biodiversity.

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Key Features of the programme include:

1. **Financial assistance:** Both central and state governments jointly funded the programme for conservation activities. For example: Wular Lake, J&K.
2. **Capacity building** of stakeholders through training, workshops and general awareness. For example: Fishermen of Pulicat Lake
3. **Monitoring**: Technology like remote sensing has been used to monitor the health of wetlands to ensure efficient conservation efforts
4. **Policy framework:** Wetlands received protection under numerous existing frameworks like the Forest (Conservation) Act, the Environmental Protection Act and the Indian Wildlife (Protection) Act.

Critique:

1. The program has played a pivotal role in preserving biodiversity in wetlands such as migratory birds and aquatic species.
2. It successfully raised awareness among local communities on the importance of wetland conservation, hence developing the concepts of eco-tourism and sustainable fishing.
3. Despite the successes, the threat of climate change leading to degradation of wetlands has led to reassessment of NWCP, and bringing it under an integrated program, NPCA.

Ramsar Sites in India:

India ratified the Convention on Wetlands, known as the Ramsar Convention, in February 1982. The Convention provides a framework for national action and international cooperation for the conservation and wise use of wetlands and their Resources.

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India has 75 designated Ramsar sites, which is the largest in South Asia and covers ~8% the known wetland regime. Some notable ones are as follows:

1. Sundarbans wetland, West Bengal:

- It's the largest continuous mangrove forest in the world, a UNESCO World Heritage Site, and a Wetland of International Importance.
- It is home to the Royal Bengal Tiger, saltwater crocodile and northern river terrapin, among many others.

2. Vembanad Kol, Kerala:

- It accounts for 30% of the surface water available for use in Kerala and is the largest brackish lake in south-west India.
- It is home to 90 domestic and 50 migratory bird species, and accounts for the third largest waterfowl population in India during the winter months.

3. Keoladeo National Park, Rajasthan:

- It is a UNESCO World Heritage Site and renowned for its rich birdlife, particularly its migratory waterfowl, and is a major wintering area for birds from Afghanistan, Turkmenistan, China, and Siberia.
- Placed on the Montreux Record in 1990 due to "water shortage and an unbalanced grazing regime".
- Additionally, the invasive growth of the grass Paspalum distichum has changed the ecological character of large areas of the site, reducing its suitability for certain waterbird species, notably the Siberian crane.

To further improve the conservation of Ramsar sites in India, the MoEFCC launched the **Amrit Dharohar** (2023) Initiative. It aims to create, demonstrate, replicate and upscale the effect on wetlands of national and international importance.

4. The Intergovernmental Panel on Climate Change(IPCC) has predicted a global sea level rise of about one metre by AD 2100. What would be its impact in India and the other countries in the Indian Ocean region?(10 marks; 150 words)

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Ans The IPCC's ~1m sea-level rise projection is a synthesis of thermal expansion, glacier melt, Greenland and Antarctic ice sheet loss, and land-water changes, modulated by regional factors like subsidence and ocean dynamics.

Impact on India:

1. **Environmental Impacts**
   a. **Land Submergence:** Low-lying areas, especially the crucial Mahanadi, Krishna-Godavari and Sundarbans deltas, are likely to be submerged.
   b. **Urban flooding:** Major cities like Mumbai, Visakhapatnam, Kolkata, Chennai and Kochi are likely to experience flooding due to storm-surge intensification.
   c. **Agricultural Impact:** Saltwater leaching into soil can increase salinity, leading to osmotic stress, which is detrimental to major crops like rice and wheat.
   d. **Ecosystem Imbalance:** Mangroves and salt marshes like Sundarbans and Rann of Kutchh will lose their ecological importance as brackish breeding grounds due to excessive influx of saltwater.
   e. **Islands:** The Andaman and Nicobar Islands and Lakshadweep are at great risk of submergence, threatening shallow reefs and habitat loss for terrestrial flora and fauna.
2. **Economic Impacts**
   a. **Fisheries:** Change in water composition in brackish and islandic areas will reduce fish breeding, leading to loss of livelihoods for the fishing community.
   b. **Damage to Infrastructure:** Submergence of infrastructure in coastal cities and ports will disrupt trade and cause huge losses to the exchequer.
   c. **Tourism:** Tourist spots like Goa, Puducherry and Kerala beaches will become vulnerable, leading to loss of livelihood.
   d. **Stagflation:** There will be a shortage of goods exacerbated by reduced production in coastal areas, leading to inflation and slowing

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economic growth.

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3. **Security Impact**
   a. **Climate Refugee crisis:** There will be a high influx of climate refugees from Bangladesh, Sri Lanka and the Maldives, which can threaten border security.
   b. **Border Infrastructure**: Destruction due to submergence can make border areas vulnerable to the entry of terrorist elements from Pakistan via the western coast.
4. **Social Impact**
   a. **Health issues:** Flooding can lead to the outbreak of numerous vector-borne diseases like typhoid, malaria and dengue in coastal areas.
   b. **Habitat Security:** Destruction of coastal habitat can render a huge population homeless.
   c. **Income Security:** Loss of livelihoods can threaten the affordability of basic amenities for a majority of agriculture and coastal workers.

Impact on other countries in IOR-

1. Bangladesh-

- It is projected to be most vulnerable in South Asia due to the submergence of the low-lying Ganga-Brahmaputra-Meghna delta leading to displacement of over 20 million people as per ADB.
- Intensified cyclones, storm surges, and salinity intrusion will lead to food insecurity.

2. Sri Lanka

- Risks to coastal cities like Colombo.
- Salinisation of rice paddies and reduction in fish output causing food and income insecurity.
- Damage to coastal tourism (Bentota, Galle, Trincomalee).

3. Maldives

- It faces an existential threat as 80% of the land is below 1m of sea level (UNDP)
- It will lead to the complete displacement of the population and create a refugee crisis.

4. Myanmar & Thailand:

- The Irrawaddy delta is at high risk of inundation and agricultural loss.
- Coral reef degradation would affect fisheries and tourism.

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5. East African coastal countries:

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- Submergence of the economically important tourist islands of Zanzibar, Kilwa, Lamu and Mafia islands
- Degradation of coral reefs, fisheries, mangroves and estuaries will exacerbate poverty due to agricultural livelihood.

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Despite the threat, combined efforts through multilateral forums like the Indian Ocean Rim Association (IORA) and SAARC could provide platforms for joint climate adaptation, disaster relief, and technology sharing.

Strategies include:

- Coastal zone management and climate-resilient agriculture.
- Construction of protective embankments and artificial islands
- Relocation planning for the Maldives and delta populations.
- Strengthening of early warning and disaster-response systems.

Model Answer Archive (2013-2022): Environment

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PART B AGRICULTURE

Part B – Agriculture

Section - I **Agriculture: A Complete Overview**

Chapter - 1 Foundation of Indian Agriculture

Chapter - 2 Cropping Patterns and Irrigation

Chapter - 3 Agriculture and Allied Sectors

Chapter - 4 Food Processing

Chapter - 5 Agriculture Inputs and Technology

Chapter - 6 Institutional and Market Reforms

Chapter - 7 Farmer Welfare and Government Schemes

Chapter - 8 Challenges and Way Forward

Chapter - 9 Miscellaneous Topics

Section - II (2025 -2023) **UPSC Mains Model Answers: Agriculture**

(2013–2022 onwards Agriculture: Mains Model Answers available on McGraw Hill Edge platform)

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Upcoming Agriculture Main 2026–2027 Model Answers

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SECTION I

Agriculture: A Complete Overview

Foundation of Indian Agriculture

1

IN THIS CHAPTER...

1.1 Introduction of Indian Agriculture

1.1.1 Importance of Agriculture in the Indian Economy

1.1.2 Contribution to GDP and Employment

1.1.3 Agriculture in Post-Independence India

1.1.4 Historical Evolution and Trends

1.2 Agro-Climatic Zones of India

1.2.2 Objectives of Agro-Climatic Zonation

1.2.3 Significance of Agro-Climatic Zoning

1.2.3 Government and Policy Relevance

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Agriculture has always occupied a central position in India's civilisational, economic, and socio-cultural landscape.

From the Indus Valley's sophisticated irrigation systems to the colonial emphasis on cash crops, and finally to post-independence self-sufficiency movements, Indian agriculture has witnessed a complex and dynamic journey.

Today, despite rapid industrialisation and service sector growth, agriculture remains a foundational pillar of the Indian economy — not just as a source of food and employment, but as the lifeblood of rural India.

This chapter delves into the multi-dimensional role of agriculture in India's development trajectory — its economic relevance, historical evolution, policy milestones, and key challenges. By understanding these aspects, we also appreciate how agriculture intersects with issues of food security, climate resilience, technology, and inclusive growth.

1.1 INTRODUCTION OF INDIAN AGRICULTURE

1.1.1 Importance of Agriculture in the Indian Economy

- **Foundational sector:** Agriculture is the backbone of the Indian economy, historically and structurally.
- **Supports livelihoods:** As per Periodic Labour Force Survey (PLFS 2023), 45.8% of India's workforce is employed in agriculture and allied sectors.
- **Food security:** With a population of over 1.4 billion, India relies heavily on domestic agriculture for food and nutritional security.
- **Raw material base:** Critical for industries like textiles (cotton), sugar (sugarcane), jute, edible oil, and dairy.
- **Foreign exchange:** India is one of the top global exporters of rice, spices, marine products, cotton, etc.

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CASE STUDY: During the COVID-19 pandemic, while most sectors shrank, agriculture registered a 3.4% growth in 2020-21, showing resilience and acting as a stabilizing force.

1.1.2 Contribution to GDP and Employment

- In 1951, agriculture contributed >50% of India's GDP.
- As of 2023-24, its contribution to GDP has declined to 18%, reflecting economic diversification.
- However, employment in agriculture remains disproportionately high (46%), indicating low productivity per worker and disguised unemployment.

QUICK INSIGHT

The declining share in GDP, but high share in employment, signals the need for rural diversification and Agri-reforms to enhance productivity.

1.1.3 Agriculture in Post-Independence India

1947–1960s

- Characterized by food shortages, low productivity, and frequent famines.
- Heavy reliance on imports (PL-480 agreement with the USA) for wheat in the 1950s and early 1960s.

1960s–1980s: Green Revolution

Introduced HYV seeds, chemical fertilizers, and irrigation in regions like Punjab, Haryana, and western UP.

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- Achieved self-sufficiency in food grains, especially wheat and rice.
- But it created regional imbalance, soil degradation, and water overuse.

1980s–2000s: Diversification & Market Orientation

- Rise of dairy (Operation Flood), horticulture, poultry, and commercial crops.
- Emergence of agri-marketing and private sector involvement.

2000s–Present: Sustainability & Technology

- Focus on climate-smart agriculture, digital agriculture, organic farming, and value chains.

Do You Know?

The shift from food scarcity to food surplus allowed India to launch schemes like PM Garib Kalyan Anna Yojana (free food grain distribution during COVID).

Table 1.1 Agriculture in Five-Year Planning

Plan Period	Agricultural Focus
1st Plan (1951–50)	Focus on land reforms and irrigation
1st Plan (1951–50)	Focus on land reforms and irrigation
3rd Plan (1951–60)	Emphasized self-sufficiency in food produ
4th & 5th Plans	Green Revolution support
6th–8th Plans	Rural employment & poverty alleviation th
10th–12th Plans	Doubling farm incomes, agri-infrastructure

Exam Pointer

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Exam Pointer

The Second Five-Year Plan (195G–G1) prioritised heavy industries, leading to neglect of agriculture initially, contributing to food crises.

1.1.4 Historical Evolution and Trends

■ *Key Trends in Indian Agriculture (1947–2025)*

- Food grain production
  - 1950–51: 50 million tonnes
  - 2023–24 (3rd AE): 330 million tonnes
- Irrigation coverage
  - 1950s: 17%
  - 2020s: 50% of net sown area
- Landholding fragmentation
  - Average holding size declined from 2.3 ha (1970) to 1.08 ha (2019–20)
- Shift in cropping patterns
  - From coarse cereals → rice and wheat → now diversification into horticulture, pulses, and oilseeds

■ *Rise of Allied Sectors*

(a) Dairy, poultry, fisheries contribute 30% of agri-GVA (Gross Value Added in Agriculture)

Do You Know?

India is the world's largest producer of milk and the second-largest producer of fruits and vegetables.

Table 1.2 Agriculture Policies, Programmes and Initiatives (1947 Onwards)

Period	Key Policies / Events

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1950s–60s	Land reforms, Community Development Programm Programme (IADP)
1960s–70s	Green Revolution, FCI creation, MSP introduction, Hi (HYVP)
1980s	Operation Flood, National Agricultural Extension Proje
2000s	National Food Security Mission (2007), Rashtriya Krisl
2010s	Soil Health Card (2015), PMFBY (201G), e-NAM (201

Exam Pointer

PM-KISAN launched in 2019 provides `6,000/year to small and marginal farmers. As of 2024, over 11 crore farmers have benefited.

Conclusion: Agriculture as the Heart of Rural India

Agriculture in India is not just about crops and yield — it is about rural dignity, food sovereignty, and economic justice. From the anxieties of food scarcity in the 1950s to becoming a global leader in food production today, the sector has come a long way. Yet, challenges persist — from land fragmentation and low incomes to climate risks and market access.

The way forward lies in technology adoption, sustainable practices, institutional reform, and holistic support for farmers. As India moves toward a $5 trillion economy, agriculture must not be left behind — instead, it should be reinvented as a driver of inclusive and sustainable growth.

1.2 AGRO-CLIMATIC ZONES OF INDIA

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India's vast geographical diversity, ranging from the Himalayan heights to coastal plains and arid deserts, creates a wide spectrum of climatic and soil conditions. This natural variation profoundly influences what crops can be grown where, how efficiently they grow, and how farming systems are structured.

Recognising this, the Planning Commission (now NITI Aayog) identified 15 agro-climatic zones in the country based on Soil type, Rainfall, Temperature regime, Topography, and Water Resources.

Agro-climatic planning is especially critical for a country like India, where:

- Agriculture is mostly rainfed (about 52% of net sown area)
- Farming is vulnerable to climate variability
- Regional disparities in productivity persist

Do You Know?

These zones were designed for planning and developmental strategies — especially during the 7th and 8th Five-Year Plans.

CASE STUDY (Punjab): The Trans-Gangetic Zone, once a diversified agrarian region, is now dominated by rice-wheat monoculture, leading to groundwater depletion. Diversification is now being incentivized by both state and central schemes.

1.2.1 Objectives of Agro-Climatic Zonation

- Scientific identification of region-specific crop potential
- Promoting sustainable land and water use
- Tailored input planning (seeds, fertilizers, irrigation)
- Better agricultural extension services
- Ensuring balanced regional development

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- Boosting productivity and income through localized strategies

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1.2.2 Significance of Agro-Climatic Zoning

- Customized Crop Planning: Encourages cultivation of crops best suited to local conditions.
- Resource Efficiency: Optimizes use of water, soil, and other inputs.
- Better Yield & Resilience: Enhances productivity while reducing vulnerability to climate risks.
- Planning Interventions: Helps design region-specific schemes (e.g., PM-AASHA, RKVY).
- Foundation of Agro-Ecological Zoning: Forms the basis for sustainable agricultural planning.

1.2.3 Government and Policy Relevance

Agro-climatic classification is used to:

- Design area-based crop insurance (e.g., PMFBY)
- Allocate funds under RKVY, NFSM, and MIDH
- Develop region-specific agri-export clusters
- Formulate climate-resilient agriculture plans

Do You Know?

National Mission for Sustainable Agriculture (NMSA) recommends location-specific cropping strategies based on agro-climatic risk assessments.

Table 1.3 Classification of Agroclimatic Zones

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Agro-Climatic Zone	States Covered	Rainfall & Temperature	Major Soil Types	Major Crops
Western Himalayan Region	Jammu and Kashmir, Himachal Pradesh, and Uttarakhand	Rainfall: 1000–2500 mm Temp: Cold winters, mild summers	Mountain soils, Brown forest soils	Apple, walnut, saffron, barley, wheat, maize, and temperate vegetables
Eastern Himalayan Region	Arunachal Pradesh, Sikkim, and parts of Assam	Rainfall: 2000–4000 mm Temp: Cool to warm, humid	Red loamy, lateritic soils	Rice, maize, tea, ginger, cardamom, and orange
Lower Gangetic Plains	West Bengal, and parts of Assam	Rainfall: 1200–2000 mm Temp: Hot & humid	Alluvial soils	Rice, jute, sugarcane, mustard, and pulses
Middle Gangetic Plains	Eastern UP, and Bihar	Rainfall: 1000–1400 mm Temp: Warm	Alluvial soils	Rice, wheat, maize,

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Agro-Climatic Zone	States Covered	Rainfall & Temperature	Major Soil Types	Major Crops
		summer, cool winter		sugarcane, and lentil
Upper Gangetic Plains	Western UP	Rainfall: 800–1000 mm Temp: Extreme summer & winter	Alluvial soils	Wheat, rice, sugarcane, and mustard
Trans-Gangetic Plains	Punjab, Haryana, Delhi, and Chandigarh	Rainfall: 500–800 mm Temp: Hot summer, cold winter	Alluvial, loamy soils	Wheat, rice, cotton, sugarcane, and pulses
Eastern Plateau and Hills	Jharkhand, Chhattisgarh, Odisha	Rainfall: 1000–1600 mm Temp: Hot summers	Red & yellow soils, lateritic soils	Rice, pulses, oilseeds, and maize
Central Plateau and Hills	Madhya Pradesh, parts of UP	Rainfall: 800–1200 mm Temp: Hot summers	Black soil, red soil	Wheat, soybean, and pulses

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Agro-Climatic Zone	States Covered	Rainfall & Temperature	Major Soil Types	Major Crops
Western Plateau and Hills	Maharashtra, parts of MP	Rainfall: G00–1000 mm Temp: Hot and dry	Black soil (regur), laterite	Cotton, jowar, soybean, and pulses
Southern Plateau and Hills	Karnataka, AP, Tamil Nadu	Rainfall: 700–1200 mm Temp: Hot summers	Red loamy, black soil	Millets, pulses, groundnut, and cotton
East Coast Plains and Hills	Odisha, coastal AP, and Tamil Nadu	Rainfall: 1000–1500 mm Temp: Warm and humid	Alluvial, coastal sandy soils	Rice, coconut, groundnut, and sugarcane
West Coast Plains and Ghats	Kerala, coastal Karnataka, and Goa	Rainfall: 2000–3500 mm Temp: Humid tropical	Lateritic, coastal sandy soils	Coconut, arecanut, spices, paddy, rubber
Gujarat Plains and Hills	Gujarat	Rainfall: 400–1200 mm Temp:	Black soil, saline soil	Groundnut, cotton, bajra, wheat

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Agro-Climatic Zone	States Covered	Rainfall & Temperature	Major Soil Types	Major Crops
		Hot and semi-arid		
Western Dry Region	Rajasthan (Thar)	Rainfall: <400 mm Temp: Extreme heat, dry winds	Sandy desert soils	Bajra, moth bean, guar, mustard
The Islands Region	Andaman and Nicobar, and Lakshadweep	Rainfall: 2500–3000 mm Temp: Tropical & humid	Coastal alluvium, coral soils	Coconut, rice, tropical fruits, and spices

Conclusion

Agro-climatic zoning serves as the scientific foundation for regional agricultural planning in India. In a country as diverse as India, a "one-size-fits-all" approach to agriculture is neither effective nor sustainable. These zones help optimize the choice of crops, agricultural practices, and resource allocation — ensuring that both economic efficiency and environmental sustainability are achieved.

As climate change intensifies, the importance of climate-smart, zone-specific agriculture will only grow. India's future food and livelihood

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security will depend on how well it leverages its agro-climatic diversity to promote inclusive and resilient agricultural development.

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Cropping Patterns and Irrigation

2

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2.1 CROPPING PATTERNS IN INDIA

A cropping pattern refers to the spatial and temporal arrangement of crops grown in a region over a period of time. It captures not only what crops are grown, but also how frequently, in what combinations, and in which seasons. It reflects the interplay between environmental suitability, economic feasibility, institutional support, and social preferences.

In India, with its diverse climatic zones, soil types, and socio-economic contexts, cropping patterns exhibit substantial variation across regions and time. These patterns are dynamic, constantly shaped by changing environmental conditions, technology, market forces, and state interventions.

Table 2.1 Determinants of Cropping Patterns

Cropping Patterns	Details
Climatic Conditions	• Temperature, rainfall, and humidity determine what crops can grow. • E.g., Rice requires a hot and humid climate; wheat prefers cool, dry conditions.
Soil Characteristics	• Soil type, fertility, drainage, and water-retention capacity influence crop suitability. • E.g., Black soil is ideal for cotton; alluvial soil supports wheat and rice.
Topography	• Slopes and elevation restrict or enable certain crops. • E.g., Tea and coffee thrive on hilly terrains under shade.
Economic Considerations	• Crop profitability, market access, cost of cultivation, and price assurances (like MSP) influence decisions. • Farmers tend to grow crops with higher returns and lower risks.

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Cropping Patterns	Details
Social and Cultural Factors	• Food habits and cultural preferences shape cropping decisions. • E.g., Millets were dominant earlier, but their decline reflects changing food habits.
Availability of Infrastructure:	• Irrigation facilities, storage, transport, and processing units facilitate certain crops over others. • Punjab's rice-wheat dominance is aided by irrigation and procurement systems.
Government Policies and Incentives:	• MSP regime, subsidies, crop insurance, procurement policies, etc., heavily influence cropping trends. • BT cotton saw an increase due to policy support and input availability.
Scientific and Technological Inputs:	• HYV seeds, fertilisers, mechanisation, and pest-resistant varieties have enabled farmers to adopt new crops.

2.1.1 Cropping Seasons in India

India has three major cropping seasons, aligned with monsoonal rhythms:

Table 2.2 Cropping Seasons in India

Season	Period	Climatic Conditions	Major Crops
Kharif	June – September	Warm, humid, and monsoonal rainfall	Rice, maize, cotton, soybean, bajra, and tur (Arhar)

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Season	Period	Climatic Conditions	Major Crops
Rabi	October – March	Cool and dry	Wheat, barley, mustard, peas, and gram
Zaid	April – June	Hot and dry	Watermelon, cucumber, moong, and fodder crops

2.1.2 Regional Cropping Patterns

India's regional diversity results in distinct cropping patterns:

- Punjab & Haryana: Rice–Wheat system is dominant due to assured irrigation and procurement.
- Eastern India: Rice, jute, and pulses are aided by rainfall and fertile alluvial soil.
- Western India: Bajra, cotton, and pulses - adapted to arid conditions.
- Southern States: Multiple crops including paddy, millets, groundnut, sugarcane, and coffee.
- Northeast India: Rice, maize, pulses, and horticultural crops due to high rainfall and diverse topography.

2.1.3 Changing Trends in Cropping Pattern

Over the decades, cropping patterns in India have shifted significantly due to policy, economic, and ecological pressures:

- **Green Revolution Phase:** Rise of wheat and rice in northwestern India.
- **Post-Liberalisation Era:** Commercialisation with more horticulture, floriculture, and oilseeds.
- **Recent Trends:**

  Decline of Coarse Cereals: Millets, once staples, saw a decline due to urbanisation and dietary shifts.

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The resurgence of millets after 2018 was driven by their nutritional value and government promotion.

Horticulture Boom: Fruits and vegetables expand rapidly with cold-chain and processing development.

Cash Crops Rise: Cotton, sugarcane, and oilseeds expanded due to market demand and policy support.

Renewed focus on pulses and oilseeds due to import dependency and government missions.

Table 2.3 Major Crops Cultivated in India

Crop Type	Examples
Food Crops	Rice, wheat, millets, maize, pulses
Cash Crops	Sugarcane, cotton, tobacco, oilseeds
Plantation Crops	Tea, coffee, rubber, coconut, and areca nut
Horticultural	Mango, banana, onion, potato, tomato, floriculture

2.1.4 Allelopathy and Its Role in Cropping Systems

Allelopathy: A biological phenomenon where one plant species releases biochemicals (allelochemicals) that influence the growth, germination, or survival of another plant.

Role in Irrigated Agriculture

- Used to suppress weeds (allelopathic properties of wheat or sorghum).
- In crop rotation systems, allelopathic crops reduce pest buildup.
- Green manuring with crops like dhaincha improves soil health and reduces nematodes.

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- Intercropping (e.g., mustard with chickpea) uses allelopathic synergy to improve yields.

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2.1.5 Factors Behind the Success of the Rice - Wheat Cropping System

The Rice-Wheat Cropping System (RWCS), prevalent primarily in Punjab, Haryana, and western Uttar Pradesh, is one of the most productive agricultural systems in India. Its success is driven by a mix of agro-climatic suitability, institutional support, and technological advances.

Agro-Climatic Suitability

Alluvial soils of the Indo-Gangetic plains are fertile and deep, suitable for both rice and wheat.

Climate: Hot and humid monsoon favours rice, while cool and dry winters support wheat.

Technological Factors

High-Yielding Varieties (HYVs) of both rice and wheat introduced during the Green Revolution increased productivity manifold.

Mechanisation (tractors, combine harvesters) enabled timely sowing/harvesting.

Institutional and Policy Support

MSP and Open-Ended Procurement: Assured returns on rice and wheat encouraged their continuous cultivation.

Subsidised Inputs: Fertilizers, electricity (for irrigation), and seeds reduced the cost of production.

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Irrigation Infrastructure: Development of canal systems and widespread use of tube wells ensured year-round water availability.

Socio-Economic Factors

Risk Aversion: Farmers prefer crops with guaranteed procurement over experimenting with new ones.

Knowledge Ecosystem: Extensive agri-extension services and farmer training in RWCS regions.

QUICK INSIGHT

Ensured food security for the country and led to India's transformation from a food-deficient to a food-surplus nation.

2.1.6 Issues with the Rice - Wheat Cropping System (RWCS) and the Need for Diversification

Despite its past success, the RWCS is now showing signs of agro-ecological unsustainability and economic stagnation.

Major Issues

- **Groundwater Depletion**: Water-intensive rice cultivation is causing critical depletion in Punjab & Haryana.
- **Soil Degradation:** Repeated mono-cropping, excessive fertiliser and pesticide use reduce soil fertility.
- **Stubble Burning**: Lack of post-harvest alternatives leads to air pollution and loss of organic matter.
- **Pest Resistance**: Continuous cropping promotes the evolution of resistant pests (e.g. stem borers, bollworm).
- **Methane Emissions**: Paddy fields emit large amounts of methane, contributing to climate change.

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- **Nutritional Imbalance**: Dominance of wheat-rice reduces dietary diversity; neglects protein- and nutrient-rich crops.
- **Disguised Unemployment:** Over-reliance on limited crop cycles reduces job diversity in the rural economy.
- **Economic Vulnerability**: Heavy dependence on MSP and government procurement makes the system financially unsustainable.

2.1.7 Crop Diversification

Crop diversification means growing a variety of different crops on a piece of land. It helps reduce the risk of crop failures due to pests, diseases, or adverse weather. It can improve soil health by reducing the depletion of specific nutrients.

■ *Need for Crop Diversification*

- Reduce ecological pressure on land and water resources.
- Enhance farmers' incomes through high-value crops like horticulture, spices, and pulses.
- Promote nutritional security through millets and coarse grains.
- Strengthening resilience to climate change and price shocks.

■ *Examples of Successful Diversification*

- Sikkim's organic horticulture.
- Millets resurgence in Karnataka and Odisha.
- Agro-forestry in Maharashtra (e.g., bamboo with pulses).

■ *Present Challenges in Crop Diversification and Opportunities from Emerging Technologies*

Crop diversification refers to the strategic shift from mono-cropping systems (like rice-wheat) to multi-cropping systems, including less water-intensive, nutrient-rich, and climate-resilient crops. It improves ecological balance, enhances farmer incomes, and promotes nutritional security.

■ *Present Challenges to Crop Diversification*

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- **Policy-Induced Bias**: MSP, subsidies, and procurement mainly favour wheat and rice; pulses/oilseeds are often neglected.
- **Market Uncertainty**: Lack of assured markets for alternative crops; absence of robust price discovery and procurement.
- **Input and Infrastructure Gaps:** Poor access to quality seeds, cold chains, storage facilities, and processing units for perishable crops.
- **Credit and Insurance Access**: Farmers face difficulty getting loans or insurance for non-traditional crops.
- **Knowledge and Extension Gaps**: Limited awareness and training for cultivating high-value or non-conventional crops.
- **Perception of Risk**: Farmers view diversification as risky due to a lack of historical yield data or familiarity.
- **Initial Investment Needs**: Horticulture or plantation crops require new tools, knowledge, and capital.

Opportunities for Emerging Technologies

Emerging agri-tech tools can bridge knowledge, risk, and input gaps impeding diversification.

Table 2.4 Emerging Agri-Tech Tools in India

Technology	How it Aids Crop Diversification
Precision Agriculture	Real-time monitoring of soil moisture, nutrients, and pests using IoT sensors helps select optimal crops.
Mobile & Digital Platforms	Government & private apps (e.g., Kisan Suvidha, IFFCO Kisan) offer weather forecasts, crop advisories.
Crop Modelling and Analytics	Predictive tools help farmers select climate-appropriate, soil-suitable crops.

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Technology	How it Aids Crop Diversification
Genetic Engineering	Development of drought/heat-tolerant, pest-resistant varieties (e.g. GM mustard, biofortified millets).
Vertical and Indoor Farming	Enables production of vegetables/fruits in urban areas or water-scarce zones, reducing land dependency.
Hydroponics and Aeroponics	Soil-less systems promote high-yielding crops with efficient nutrient use in controlled environments.
Agri-Robotics and Drones	Mechanisation of labour-intensive crops (e.g., tea plucking, fruit spraying) supports diversification.

Government Initiatives Supporting Diversification

- **Crop Diversification Programme (CDP) under RKVY (Rashtriya Krishi Vikas Yojana):** Targets original green revolution states to shift from paddy to pulses/oilseeds.
- **PMFBY (Pradhan Mantri Fasal Bima Yojana):** Expands insurance coverage, encouraging risk-taking in crop choices.
- **Increased MSP (Minimum Support Price):** For pulses, millets, and oilseeds to make them economically viable.
- **Paramparagat Krishi Vikas Yojana**: Promotes organic farming and diversification.
- **National Mission on Horticulture**: Supports infrastructure, cold chains, and skill development.

Way Forward

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- **Policy Reforms:** Rationalize MSP and procurement; incentivize crop rotation & mixed farming.
- **Institutional Support:** Strengthen extension services and agri-clinics to train farmers in new crop practices.
- **Infrastructure Investment:** Build cold chains, warehouses, and value-addition units for perishable crops.
- **Market Access:** Expand e-NAM (National Agriculture Market), FPOs (Farmer Producer Organisation), and export infrastructure for diversified produce.
- **Awareness Campaigns:** Promote the nutritional, economic, and ecological value of crops like millets, pulses.
- **Technology Integration:** Foster public-private partnerships to democratize access to advanced tools and data.

Conclusion

Cropping patterns in India are not merely shaped by nature—they are deeply intertwined with policy, market, and social behaviour. The dominance of the rice-wheat system, while once critical for food security, now presents ecological and economic challenges.

- Going forward, India needs to promote diversified, sustainable, and resilient cropping systems. This will require:
- Shifting consumer preferences,
- Realigning support systems (like MSP),
- Integrating new technologies, and
- Empowering farmers to make climate-smart, market-smart decisions.

Only then can Indian agriculture achieve nutritional, economic, and ecological security together.

2.2 TYPES OF FARMING IN INDIA

Farming in India is not a monolithic practice. It varies drastically based on geography, economic status, technological accessibility, and ecological constraints. The diversity of agro-climatic zones and socio-economic backgrounds of Indian farmers has led to the development of different types of farming systems. These types reflect the purpose of production, scale of operation, nature of inputs, and market orientation.

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2.2.1 Subsistence Farming

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- **Definition**: Farming in which the main objective is to grow crops primarily for the consumption of the farmer's family, rather than for sale.
- **Key Features:**
  - Small and scattered landholdings with low productivity.
  - Traditional tools and indigenous knowledge systems are widely used.
  - Labour-intensive, often depending on family members.
  - Minimal use of modern inputs like fertilisers, HYV seeds, or machinery.
  - Vulnerable to climate fluctuations and pest outbreaks.
  - Practised mostly in regions with low access to markets, infrastructure, and credit.
- **Examples of Crops:**
  - Rice, wheat, millets, maize, pulses, and vegetables.
- **Geographical Distribution:**
  - Tribal belts of Jharkhand, Odisha, Chhattisgarh, Madhya Pradesh, Northeast India, and the Himalayan regions.

2.2.2 Commercial Farming

Definition: Market-oriented Farming, where crops are cultivated for sale in local, national, or international markets.

Key Features:

- Larger landholdings and higher investment.
- Mechanisation and modern inputs (chemical fertilizers, pesticides, irrigation, etc.) are commonly used.
- Market-sensitive crop choices.
- Often linked with contract farming and agri-businesses.
- Greater use of crop insurance, cold storage, and transport networks.

Examples of Crops:

- Sugarcane, cotton, oilseeds, wheat, rice (in Punjab-Haryana belt), vegetables, and flowers.

Geographical Distribution:

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- Punjab, Haryana, Western Uttar Pradesh, Maharashtra, parts of Andhra Pradesh and Karnataka.

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2.2.3 Plantation Farming

Definition: A specialised commercial farming type where a single cash crop is grown over a large area for export or agro-industry.

Key Features

- Monoculture system—Entire estates are dedicated to one crop.
- Requires high capital investment and a large labour force.
- Plantation estates are usually owned by corporate entities or colonial-era landholding structures.
- Export-oriented and industrially processed products.
- High dependency on rainfall or managed irrigation.
- Located in regions with humid, tropical climates.

Examples of Crops

- Tea, coffee, rubber, coconut, oil palm, cocoa, banana, spices.

Geographical Distribution

- **Tea:** Assam, West Bengal (Darjeeling), Tamil Nadu.
- **Coffee**: Karnataka, Kerala, Tamil Nadu.
- **Rubber**: Kerala.
- **Coconut and spices**: Kerala and coastal Andhra.

2.2.4 Mixed Farming

Definition: A system in which both crop production and animal husbandry are practiced on the same farm to ensure diversified income and sustainable resource use.

Key Features

- Balanced integration of livestock (dairy, poultry, sheep, goats) with food and cash crops.
- Promotes manure recycling, risk distribution, and nutrient sustainability.
- Enhances employment generation and reduces farmer vulnerability.

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- Often supported by government extension services and cooperatives.

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Examples

- Wheat + dairy farming in Punjab.
- Millets + goats in Rajasthan.
- Pulses + poultry in Madhya Pradesh.

Geographical Distribution

- Practised widely across Punjab, Haryana, Karnataka, Maharashtra, UP, and Tamil Nadu.

2.2.5 Organic Farming

Definition: A holistic farming approach that eschews synthetic inputs in favour of ecological balance, soil health, and sustainable productivity.

Key Features

- Uses organic manures, compost, biofertilizers, and natural pest control methods.
- Emphasises soil biodiversity, crop rotation, and minimum tillage.
- Certified organic products fetch premium market prices, especially in exports.
- Help combat soil degradation, pesticide residues, and climate change.

Examples of Practices

- Use of Neem-based pesticides, vermicompost, cow dung slurry, and companion cropping.

Geographical Focus

- Sikkim (declared India's first fully organic state).
- Uttarakhand, Kerala, Himachal Pradesh, parts of North-East and Madhya Pradesh.

2.2.6 Precision Farming

Definition: A modern farming approach that uses digital technology, data analytics, and automation to optimise input use and maximise output.

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Key Features

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- Incorporates GPS mapping, IoT devices, drones, soil sensors, and AI models.
- Enables site-specific management of crops (micro-irrigation, nutrient delivery).
- Helps reduce input cost, increase yield, and minimise environmental damage.
- Still at a nascent stage in India but gaining ground rapidly under smart agriculture schemes.

Applications

- Real-time irrigation control via soil moisture sensors.
- Drone-based aerial crop surveillance.
- AI-based pest prediction.

Regions Adopting It

Maharashtra, Andhra Pradesh, Karnataka, Telangana, and pilot projects in Punjab and Tamil Nadu.

Conclusion

The diverse types of farming in India reveal the multi-dimensional nature of its agrarian economy. From traditional subsistence farming rooted in indigenous knowledge to data-driven precision agriculture, Indian farmers navigate a wide spectrum of challenges and opportunities.

Policy reforms such as Paramparagat Krishi Vikas Yojana (for organic farming), Digital Agriculture Mission, and Plantation Crop Schemes aim to support various farming types. Going forward, a balanced integration of technological modernization, climate-resilient practices, and farmer-centric policy design will be critical for sustainable and inclusive agricultural development.

2.3 MAJOR CROPS IN INDIA – SUMMARY TABLE

India grows a wide variety of crops due to its diverse climate and different agricultural zones. Major food crops include rice, wheat, maize, and pulses. Important cash crops are sugarcane, cotton, jute, and oilseeds. The country also produces many spices, fruits, vegetables, and tea. These crops are crucial for

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India's food security and agricultural economy. A table on the following page discusses India's major crops in more detail.

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Table 2.5 Major Crops & Producing States – Conditions, Top Producing States and Challenges

Crop	Type	Ideal Conditions	Top Producing States	Key Challenges
Rice	Food Grain	Hot and humid; rainfall 100–200 cm; clayey and alluvial soils; grown in kharif season.	West Bengal, Punjab, Uttar Pradesh	Groundwater overuse, yield stagnation, heavy MSP dependence.
Wheat	Food Grain	Cool and dry climate; well-drained loamy soil; grown in rabi season.	Uttar Pradesh, Punjab, Haryana	Declining soil fertility, unsustainable input use, water stress.
Maize	Food Grain	Moderate temperature; well-drained fertile soil; adaptable to	Karnataka, Madhya Pradesh, Bihar	Low productivity, poor irrigation support, market linkage gaps.

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Crop	Type	Ideal Conditions	Top Producing States	Key Challenges
		both kharif and rabi.		
Sugarcane	Cash Crop	Hot and humid climate; rainfall 100–150 cm; alluvial/black soil; long growing period.	Uttar Pradesh, Maharashtra, Karnataka	High water consumption, delayed payments, climate sensitivity.
Cotton	Cash Crop	High temperature, low to moderate rainfall; black soil; kharif crop.	Gujarat, Maharashtra, Telangana	Pest outbreaks (pink bollworm), high input cost, market price fluctuations.
Tea	Plantation Crop	Cool climate, high humidity, acidic well-drained soil; hilly slopes.	Assam, West Bengal (Darjeeling)	Labour issues, monoculture risks, declining export competitiveness.

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Crop	Type	Ideal Conditions	Top Producing States	Key Challenges
Coffee	Plantation Crop	Warm and humid; well-distributed rainfall; shaded areas; loamy soil.	Karnataka, Kerala, Tamil Nadu	Price volatility, pest diseases (coffee rust), export dependency.
Arhar (Tur)	Pulse	Semi-arid areas; light soils; grown as kharif crop.	Madhya Pradesh, Maharashtra, Karnataka	Low MSP procurement, poor yield per hectare.
Gram (Chana)	Pulse	Cool and dry climate; loamy soil; grown in rabi season.	Madhya Pradesh, Rajasthan, Maharashtra	Lack of irrigation, low productivity.
Moong & Urad	Pulses	Warm climate, light loamy soil; typically grown in	Rajasthan, Maharashtra, Andhra Pradesh	Highly rain-dependent, poor storage infrastructure.

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Crop	Type	Ideal Conditions	Top Producing States	Key Challenges
		kharif/zaid seasons.		
Soybean	Oilseed	Moderate temperature, moderate rainfall; well-drained black soil.	Madhya Pradesh, Maharashtra, Rajasthan	Volatile prices, pest infestations, poor processing facilities.
Mustard	Oilseed	Cool temperature; dry climate during ripening; loamy soil.	Rajasthan, Uttar Pradesh, Haryana	Lack of irrigation, climate variability.
Groundnut	Oilseed	Warm climate; sandy loam to loamy soil; needs well-distributed rainfall.	Gujarat, Tamil Nadu, Andhra Pradesh	Susceptible to fungal diseases; price instability.
Sunflower	Oilseed	Cool to warm	Karnataka, Andhra	Low acreage, limited

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Crop	Type	Ideal Conditions	Top Producing States	Key Challenges
		climate; deep, well-drained soil; adaptable.	Pradesh, Maharashtra	processing industries.
Jowar	Millet	Drought-tolerant; requires moderate rainfall and well-drained soil.	Maharashtra, Karnataka, Madhya Pradesh	Less support in PDS and MSP, low market demand.
Bajra	Millet	Extremely drought-tolerant; poor sandy soil; high temperature.	Rajasthan, Uttar Pradesh, Gujarat	Poor procurement and awareness, shifting dietary patterns.
Ragi	Millet	Requires cool climate with moderate rainfall; grows in	Karnataka, Tamil Nadu, Uttarakhand	Low market incentives, neglected in food policy.

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Crop	Type	Ideal Conditions	Top Producing States	Key Challenges
		laterite or red soil.		
Small Millets	Millet	Resilient to poor soils and low rainfall; grown in tribal and rainfed areas.	Odisha, Chhattisgarh, Jharkhand	Minimal R&D focus, declining cultivation.

2.3.1 Irrigation in India

India's agriculture is heavily dependent on the monsoon, making irrigation a vital component of agricultural productivity and food security. With over half of India's net sown area still rainfed, developing efficient and equitable irrigation systems is central to achieving sustainable agriculture. The diverse climatic zones and cropping patterns across regions demand an equally diverse set of irrigation solutions—from traditional wells to modern drip systems. However, challenges like groundwater depletion, poor maintenance, and low water-use efficiency continue to hinder progress. This section explores the types of irrigation, assesses efficiency, examines challenges, and highlights innovative interventions like micro-irrigation and the PMKSY initiative.

Table 2.6 Types of Irrigation Systems in India

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Type	Description	Regions/Uses	Advantages	Limitations
Canal Irrigation	Water is diverted from rivers via canals and distributed to fields.	Punjab, Haryana, UP, Andhra Pradesh	Suitable for large areas; regulated supply	Waterlogging, seepage losses, high initial cost
Well Irrigation	Water drawn from wells—dug wells or tube wells—for field irrigation.	Indo-Gangetic plains, Peninsular plateau	Accessible and cheap for small farmers	Groundwater depletion, limited to shallow aquifers
Tank Irrigation	Rainwater stored in tanks and used for irrigation.	Tamil Nadu, Karnataka, Telangana	Useful in low-rain areas, decentralized	Silting, seasonal reliability
Drip Irrigation	Water drips directly to root zones through pipes and emitters.	Maharashtra, Gujarat, Karnataka (horticulture)	High water-use efficiency, reduces weed growth	High cost of setup, maintenance issues

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Type	Description	Regions/Uses	Advantages	Limitations
Sprinkler Irrigation	Water is sprayed like rain using nozzles and pumps.	Rajasthan, Haryana, semi-arid regions	Uniform distribution, water-saving	High energy requirement, evaporation losses

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Irrigation Efficiency in India

- Current efficiency of traditional surface irrigation (like canals) ranges from 30–40%.
- Major water losses occur during conveyance, distribution, and field application.
- Efficiency can be improved through:
  - Lining of canals to prevent seepage.
  - Adopting micro-irrigation systems.
  - Better field leveling and scientific scheduling of irrigation.

Major Challenges in Indian Irrigation System

- **Over-reliance on groundwater:** Over G0% of irrigation depends on groundwater, leading to aquifer depletion.
- **Regional disparity:** Canal systems are better developed in northern India; southern and eastern regions lack extensive infrastructure.
- **Inefficient systems:** Poor maintenance, leakage, and outdated irrigation techniques.
- **Waterlogging and salinity:** Especially in canal-irrigated areas due to unlined canals and poor drainage.
- **Climate variability:** Unpredictable rainfall patterns affect recharge and water availability.
- **Institutional issues:** Delayed canal water supply, inequitable distribution, and weak water user associations.

Water-Use Efficiency & Micro-Irrigation

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What is Water-Use Efficiency (WUE)?

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- WUE is the ratio of agricultural output to water consumed.
- A high WUE implies more crop per drop.

Micro-Irrigation: A Key to Enhancing WUE

- Includes drip and sprinkler systems.
- Benefits:
  - Saves 30–60% of water.
  - Increases yields by up to 50%.
  - Reduces energy and fertilizer usage.
  - Minimizes weed growth and soil erosion.
- Micro-irrigation is essential in water-stressed states like Rajasthan, Maharashtra, and Tamil Nadu.
  - Promoted under ‘Per Drop More Crop’ sub-scheme of PMKSY.

■ *PM Krishi Sinchayee Yojana (PMKSY)*

- Launched: 2015
- Objective: To expand irrigation coverage and improve water-use efficiency.
- Key Components:

■ *Accelerated Irrigation Benefit Programme (AIBP): Completion of long-pending irrigation projects.*

- **Har Khet Ko Pani:** Water for every field, ensuring equitable access.
- **Per Drop More Crop:** Focus on micro-irrigation and efficient water management.
- **Watershed Development:** Recharge groundwater and manage rain-fed areas.
- **Impact:**
  - Promotes integrated water resource management.
  - Encourages convergence of schemes and investments.
  - Enhances resilience of Indian agriculture to climate variability.

Conclusion

Irrigation is not just a technical intervention but a lifeline for Indian agriculture. While India has made significant strides in expanding irrigation coverage, it still faces serious challenges in terms of efficiency, sustainability, and equity. With

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mounting pressure on groundwater and uncertain rainfall due to climate change, the focus must now shift from expanding irrigation to making it smart, efficient, and resilient. Technological solutions like micro-irrigation, institutional reforms, and farmer awareness are crucial to ensure “more crop per drop” and to secure the future of Indian agriculture.

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Agriculture and Allied Sectors

3

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3.1 HORTICULTURE IN INDIA

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Horticulture, which includes the cultivation of fruits, vegetables, flowers, medicinal and aromatic plants, spices, and plantation crops, plays a critical role in diversifying agriculture and ensuring nutritional security. Though it occupies only about 13% of gross cropped areas, it contributes over 33% to the agricultural GDP. India's agro-climatic diversity supports a wide range of horticultural crops, making it the second-largest producer of fruits and vegetables globally. With changing dietary patterns and rising incomes, horticulture is now central to achieving doubling of farmers' income, yet it continues to face systemic challenges.

3.1.1 Features of Horticulture in India

- **High Value Output:** Provides greater return per unit area compared to cereals.
- **Labour Intensive:** Supports rural employment, especially for women and smallholders.
- **Diverse Crop Base:** India leads in mango, banana, papaya, lemon, and okra production.
- **Export-Oriented:** Mango, grapes, pomegranate, and spices contribute significantly to agri-exports.
- **Climate Resilient:** Many horticultural crops are more adaptable to climate change than field crops.
- **Nutrition Sensitive:** Key to combating hidden hunger and micronutrient deficiencies.
- **Year-Round Cultivation:** Enables continuous income generation, unlike seasonal field crops.
- **Government Push**: Mission for Integrated Development of Horticulture (MIDH), NHM, and NHB are major initiatives.

3.1.2 Challenges Faced by the Horticulture Sector

- **Post-Harvest Losses:** Estimated at 20–30% due to lack of cold chains, storage, and processing units.

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- **Fragmented Land Holdings:** Difficulties in adopting mechanisation and modern practices.
- **Limited Processing Capacity:** India processes less than 10% of fruits and vegetables.
- **Inadequate Marketing Infrastructure:** Absence of market linkages, price realisation remains poor.
- **Pest and Disease Management:** Especially challenging for export-quality produce.
- **Water Stress:** Many horticultural crops are water-intensive (e.g., banana, sugarcane).
- Lack of Skilled Manpower**: Poor extension services specific to horticultural needs.**

3.1.3 Operation Greens

Launched in 2018 by the Ministry of Food Processing Industries (MoFPI), the initiative aims to stabilise the supply of **tomato, onion, and potato (TOP) crops** while ensuring that farmers receive fair and remunerative prices for their produce.

- **Key Features:**
  - Price Stabilisation Fund to prevent sales of distress.
  - Development of Value Chains for TOP crops.
  - **Infrastructure Development**: Cold chains, pack houses, ripening chambers.
  - **Farmer-Producer Organisations (FPOs):** Promotion for direct linkages with processors and retailers.
  - Extended in 2020 to cover all fruits and vegetables under Aatma Nirbhar Bharat.
- **Challenges:**
  - Limited awareness among farmers.
  - Operational bottlenecks in fund utilisation.
  - Weak FPO penetration.

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Conclusion

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India's horticultural potential is vast and can play a defining role in food security, employment generation, nutrition improvement, and export earnings. However, to realise this potential, policy attention must shift from production to infrastructure, value addition, and market integration. Initiatives like MIDH, Operation Greens, and PM-FME (Formalisation of Micro Food Processing Enterprises) need better coordination and ground-level penetration. A farmer-centric approach, backed by technology, cold chain networks, and price assurance, can unlock the true promise of India's horticultural sector.

3.2 ANIMAL HUSBANDRY IN INDIA

Animal Husbandry is the backbone of India's rural economy, supplementing agriculture by providing livelihood, food security, and additional income to millions. It comprises dairy, poultry, meat, fisheries, and sericulture sectors.

Contributing over 30% to the agricultural GDP, it is especially crucial for marginal and landless farmers, offering resilience during agrarian distress. India ranks first in milk production, and among the top producers of eggs, fish, and meat globally. Despite its significance, the sector faces several structural and logistical challenges.

3.2.1 Importance of Animal Husbandry in India

- **Income and Employment**: It supports around 20.5 million people; 70% of them are women.
- **Dairy Dominance**: Contributes 60% of the livestock sector's value.
- **Nutritional Support**: Supplies protein-rich food (milk, eggs, meat, fish).
- **Complement to Crop Farming**: Livestock manure enhances soil fertility.
- **Export Potential**: India exports buffalo meat, dairy products, honey, etc.
- **Insurance against Risk**: Provides a financial cushion during crop failure.

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3.2.2 White Revolution in India

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Overview

- It was launched in 1970 by Dr. Verghese Kurien.
- Its flagship program was called Operation Flood (by NDDB)

Impact

- Transformed India from a milk-deficient to the world's largest milk producer.
- Empowered rural women through cooperative dairies.
- Boosted rural economy, improved nutrition.

Key Institutions

- Amul (Gujarat Cooperative Milk Marketing Federation)
- NDDB (National Dairy Development Board)

3.2.3 Challenges in Animal Husbandry Sector

- **Low Productivity:** Despite large livestock numbers, yield per animal is low.
- **Fodder Deficit:** Estimated shortfall of 30–40% in green fodder.
- **Veterinary Services Gap**: Lack of infrastructure and trained personnel.
- **Frequent Disease Outbreaks:** FMD, brucellosis, and bird flu affect livestock health.
- **Unorganised Markets:** Especially in meat and poultry processing.
- **Poor Cold Chain Infrastructure:** Hinders milk and meat preservation.
- **Environmental Concerns:** Methane emissions and overgrazing issues.

Table 3.1 Key Animal Husbandry Sectors in India

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Sector	Key Features	Challenges
Dairy	World's largest milk producer; smallholder-based; White Revolution was a huge success	Low yield, fodder shortage, cold chain gaps
Poultry	Second in eggs, Third in broiler meat; private sector-led industry	Disease outbreaks, feed costs, market volatility
Sheep & Goat	Crucial for arid zones; income from meat and wool	Poor breeds, limited veterinary care
Meat Processing	Major buffalo meat exporter; potential for packaged meat growth	Unregulated slaughter, export restrictions
Fisheries	Third globally; vital for rural nutrition and employment	Overfishing, post-harvest loss, and weak inland aquaculture
Sericulture	Second in silk production; empowers rural women and tribal communities	Low productivity, climate risk, and Chinese competition

Conclusion

Animal husbandry and allied activities are no longer secondary to crop agriculture but form an integral part of India's agri-economy. Diversification into livestock, dairy, fisheries, and sericulture not only enhances rural incomes but builds resilience against climate and price shocks. With focused investments in infrastructure, R&D, disease control,

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and market access, India can unlock the full potential of its animal-based sectors. The future lies in integration, sustainability, and value addition across all verticals of animal husbandry.

3.3 AGRICULTURAL REVOLUTIONS AND SUSTAINABLE AGRICULTURE

The journey of Indian agriculture since independence has been marked by a series of sectoral revolutions that reshaped the landscape of food production, nutrition, and rural livelihoods. From the Green Revolution that ensured food security to the White and Blue Revolutions that diversified agricultural income sources, each wave of transformation addressed a crucial gap in the agrarian economy. These revolutions were not merely technological shifts, but also socio-economic catalysts that aimed at poverty alleviation, nutritional enhancement, and rural empowerment. Together, they reflect India's evolving strategy toward achieving self-sufficiency and sustainability in its agri-food systems.

Table 3.2 Overview Table: Major Agricultural Revolutions in India

Revolution	Focus Area	Key Personality	Impact
Green	Food Grains (Wheat, Rice)	M.S. Swaminathan	Boosted food grain output, especially in Punjab and Haryana. India became self-sufficient in food.
White	Milk & Dairy	Dr. Verghese Kurien	Made India the largest milk

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Revolution	Focus Area	Key Personality	Impact
			producer. Strengthened the rural economy, especially women's dairy cooperatives.
Blue	Fisheries	Dr. Arun Krishnan and Dr. Hiralal Chaudhuri	Enhanced inland and marine fish production. Supported nutrition and export.
Pink	Meat, Poultry	Durgesh Patel	Improved quality and processing of meat. Promoted exports and focused on animal protein.
Silver	Eggs, Poultry	Indira Gandhi	Boosted poultry farming. Addressed protein deficiency through eggs.
Golden	Horticulture (fruits, honey)	Nirpakh Tutej	Helped diversify agriculture. Enhanced farmer

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Revolution	Focus Area	Key Personality	Impact
			income and exports.
Purple	Lavender Cultivation (esp. Jammu and Kashmir)	Dr. Jitendra Singh	Promoted aromatic crops. Helped farmers shift from traditional crops to high-value farming.
Brown	Leather / Cocoa / Non-Conventional Products	Hiralal Chaudri	Focused on sustainable agri-energy sources like biofuels.
Yellow Revolution	Chemical fertilizers, pesticides, and herbicides	M.S. Swaminathan	To overcome the shortcomings of the green revolution and increase fertilizer production.
Grey Revolution	Oilseeds	Sam Pitroda	It significantly increased India's oilseed production, leading to self-sufficiency and a

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Revolution	Focus Area	Key Personality	Impact
			reduction in edible oil imports.

3.3.1 Role of Agricultural Revolutions in Poverty Alleviation and Food Security

- **Food Security:**
  - The Green Revolution was pivotal in reducing India's dependency on food imports in the 1960s.
  - Increased food grain production stabilised prices and improved public distribution systems (PDS).
  - Horticulture, dairy, poultry, and fisheries ensured year-round food availability and nutritional security.
- **Poverty Alleviation**:
  - White and Silver Revolutions provided regular income streams through milk and eggs.
  - Women's participation in dairy cooperatives increased rural empowerment.
  - Horticulture and high-value crops under the Golden Revolution ensured better price realisation and export value.

3.3.2 Pink Revolution – The Need for Policy Action Now

- **Context:**
  - India still faces high protein malnutrition despite being food surplus.
  - Animal protein (meat, poultry, fish) is crucial for balanced nutrition.
- **Opportunities of Pink Revolution:**
  - Enhances nutrition through quality meat and poultry.
  - Opens agro-based exports (Example: Buffalo meat, processed poultry).
  - Promotes employment in food processing and cold chain logistics.

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- **Challenges:**
  - Lack of veterinary infrastructure and disease control.
  - Weak cold storage and transport infrastructure.
  - Cultural taboos and resistance toward meat consumption in certain communities.
- **Way Forward:**
  - Expand meat processing parks under PM-SAMPADA.
  - Strengthen APEDA support for meat exports.
  - Promote awareness of balanced diets, including animal protein, in government nutrition schemes.

Conclusion

Agricultural revolutions in India have played a vital role in lifting millions out of poverty, reducing food insecurity, and creating a diversified agricultural economy. However, the future calls for a more inclusive and nutrition-focused approach, especially with the rising concerns over health, environment, and resource sustainability. Strengthening movements like the Pink Revolution can help bridge nutritional gaps, while integrating ecological balance and farmer-centric policies will be key to India's next agri-transformation.

3.4 SUSTAINABLE AGRICULTURE IN INDIA

In the face of growing ecological degradation, climate change, and land stress, the need for sustainable agricultural practices has never been more urgent in India. Sustainable agriculture aims to harmonise productivity with environmental integrity and social equity. It includes an array of time-tested and innovative methods such as organic and natural farming, crop diversification, agroforestry, and integrated pest management. Supported by government interventions like the National Horticulture Mission and PMKSY, these practices aim to make Indian agriculture more resilient, especially for small and marginal farmers who are most vulnerable to climate and market fluctuations.

Do You Know?

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Do You Know?

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What is Sustainable Agriculture?

An agricultural system that ensures long-term soil health, biodiversity, low carbon emissions, and economic viability for small farmers without overexploiting natural resources.

3.4.1 Government Support & Missions

- **Paramparagat Krishi Vikas Yojana (PKVY**): Promotes organic farming through farmer clusters and certification.
- **National Mission on Sustainable Agriculture (NMSA):** Focuses on water use efficiency, soil health, and climate adaptation.
- **National Horticulture Mission (NHM):** Supports high-value horticultural crops, organic produce, and farmer-market linkages.
- **PM-Krishi Sinchayee Yojana (PMKSY):** Encourages water-use efficiency via micro-irrigation (drip/sprinkler).

Table 3.3 Key Sustainable Practices and Interventions

Practice	Description	Benefits for Small and Marginal Farmers
Organic Farming	No synthetic inputs, uses compost and bio-fertilizers.	Low input cost, premium pricing, and improves soil quality.
Natural Farming (e.g., ZBNF)	Zero-Budget Natural Farming with cow dung and urine.	Drastically reduces input cost, local inputs, and climate-resilient.

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Practice	Description	Benefits for Small and Marginal Farmers
Crop Rotation and Intercropping	Alternating crops to preserve soil fertility and pest control.	Reduces dependency on chemical inputs, ensures income diversification.
Cover Cropping and Mulching	Use of cover crops and mulch to retain moisture and prevent erosion.	Conserves water, enhances soil structure, and lowers irrigation needs.
Agroforestry	Integration of trees with crops and livestock.	Provides fuelwood, fodder, and enhances income during lean agricultural periods.
Precision Farming	Tech-enabled irrigation, sowing, nutrient delivery.	Optimises resource use, reduces cost, and increases productivity.
Integrated Farming System (IFS)	Crop-livestock-poultry-fish combinations for circular economy.	Maximises land use, diversifies income, and ensures food security.

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Practice	Description	Benefits for Small and Marginal Farmers
Integrated Pest Management (IPM)	Combines bio and minimal chemical control of pests.	Reduces pesticide cost, safer food, and improves ecological balance.
System of Rice Intensification (SRI)	Low-seed, high-yield method for rice cultivation.	Water-saving, yield-increasing – especially helpful in water-scarce areas.
Vertical Farming (Hydroponics, Aeroponics, Aquaponics)	Soil-less farming techniques use less space.	High yield per unit area, urban agriculture scope, water-efficient but costly setup.

3.4.2 How These Help Small & Marginal Farmers?

- Low-cost practices like natural farming, crop rotation, and agroforestry require minimal investment.
- IFS and IPM ensure year-round income and reduce input dependence.
- Government support ensures access to credit, training, market, and certification.

Conclusion

Sustainable agriculture is not just an environmental imperative but also a socio-economic necessity for India. By embracing agroecological principles and promoting low-input, high-efficiency techniques, India can secure long-term food security while safeguarding natural resources. The integration of

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vertical farming and precision agriculture further opens pathways for modernisation with sustainability. Ultimately, empowering farmers through education, incentives, and infrastructure is key to mainstreaming sustainability as the cornerstone of India's agricultural future.

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Food Processing 4

IN THIS CHAPTER...

4.1 Food Processing Industry in India
4.1.1 Key areas of significance
4.1.2 Upstream and Downstream Bottlenecks in Food Processing
4.2 Challenges in Marketing and Supply Chain Management
4.3 Why Small Processing Units Struggle
4.4 Can E-commerce Solve the Bottleneck?
4.5 Government Initiatives for Food Processing

4.1 FOOD PROCESSING INDUSTRY IN INDIA

The Food Processing Industry (FPI) in India stands at the confluence of agriculture and industry, holding the potential to transform India's rural economy and ensure food security. It has evolved into a key sector contributing to GDP, employment, export potential, and poverty reduction. With India being among the top producers of cereals, fruits, vegetables, milk, and meat, food processing becomes essential to harness this agricultural abundance. Yet, despite this vast production base, India ranks low in the global food processing value chain, highlighting both the challenges and immense scope for growth.

Exam Pointer

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India's food processing industry has shown steady growth, registering an average annual growth rate of 5.35% (2015–2023). Yet only a small percentage of perishable agricultural produce is processed. For example, 2% of fruits and vegetables, 6% of poultry, and 8% of marine products. This gap offers tremendous opportunities.

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4.1.1 Key areas of significance

- **Economic Contribution**: FPI contributed 7.66% of the GVA of the manufacturing sector in 2022–23 (₹1.92 lakh crore).
- **Employment Generation**: It is among the top employment generators in the manufacturing sector, with more than 20 lakh workers, accounting for 12.2% of the workforce.
- **Export Potential**: India's agri-exports are primarily raw products. Processing can help move up the global value chain and enhance foreign exchange earnings.
- **Reducing Post-Harvest Losses**: UN estimates suggest India wastes nearly 30% of its food production annually. FPI can reduce this wastage significantly.
- **Increasing Farmer Income**: By adding value to produce and minimising losses, food processing can boost incomes and offer price stability.
- **Promoting Crop Diversification**: Different types of processed food require a variety of crops, encouraging diversification in agriculture.
- **Nutritional Security**: Fortified processed food can tackle micronutrient deficiencies effectively.
- **Rural Industrialisation**: FPI can develop entrepreneurship and employment in rural areas, helping curb rural-urban migration.

4.1.2 Upstream and Downstream Bottlenecks in Food Processing

The food processing value chain is divided into upstream and downstream components. Bottlenecks in both stages undermine the efficiency and profitability of the sector.

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■ *Upstream Bottlenecks (Pre-processing)*

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- Weak linkages between farmers and processors
- Lack of modern raw material extraction methods
- Inadequate cold storage near farms
- Inconsistent quality and lack of grading/sorting
- Poor market reforms (e.g., APMC constraints)
- Lack of skilled workforce in farming techniques
- Inadequate rural infrastructure (roads, electricity)

■ *Downstream Bottlenecks (Processing to Consumption)*

- Limited use of modern machinery and processing technologies
- Shortage of food testing labs to ensure quality and standards
- Inefficient logistics and cold chain infrastructure
- Scarcity of skilled human resources
- Entry barriers due to the capital-intensive nature
- Inadequate branding and marketing support, especially for MSMEs

4.2 CHALLENGES IN MARKETING AND SUPPLY CHAIN MANAGEMENT

India's FPI suffers from poor integration across the supply chain. The disconnect between production, processing, and marketing results in wastage and inefficiencies.

Key impediments

- Lack of organised marketing networks
- Fragmented cold chain and storage systems
- Poor transport connectivity to markets
- Regulatory constraints (e.g., frequent export bans)
- Difficulty in maintaining consistent quality
- Weak branding and limited access to export markets

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- Lack of digitisation and real-time tracking systems

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4.3 WHY SMALL PROCESSING UNITS STRUGGLE

Despite being cost-effective and closer to the source of production, small processing units face significant hurdles:

- Limited access to affordable finance
- High cost of packaging, certification, and quality control
- Low awareness of hygiene and food standards
- Lack of training in modern processing techniques
- Difficulty competing with large-scale branded products
- Limited access to formal markets and retail networks. However, promoting micro-units can uplift small farmers by:
  - Adding value to local produce
  - Creating rural employment
  - Empowering SHGs and FPOs
  - Reducing transportation costs and post-harvest losses

4.4 CAN E-COMMERCE SOLVE THE BOTTLENECK?

E-commerce platforms offer a promising solution to bypass traditional bottlenecks in the supply chain and marketing:

- Direct-to-Consumer Sales: Small processors can reach consumers without middlemen.
- Efficient Logistics: Platforms like Amazon, BigBasket provide warehousing and last-mile delivery.
- Branding & Visibility: Allows micro-enterprises to showcase local specialities nationwide.
- Digital Payments: Ensures faster cash flows and reduces dependency on credit cycles. However, digital literacy, infrastructure (like the internet in rural areas), and training remain challenges for full-scale adoption.

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4.5 GOVERNMENT INITIATIVES FOR FOOD PROCESSING

- **Production Linked Incentive (PLI) Scheme for FPI**
  - `10,900 crore outlay (2021–27)
  - Aims to boost global champions in food manufacturing and improve branding
- **PM Formalisation of Micro Food Processing Enterprises (PM-FME)**
  - `10,000 crore outlay
  - ODOP-based support for SHGs, FPOs, and entrepreneurs with credit-linked subsidy and capacity building
- **PM Kisan SAMPADA Yojana**
  - Infrastructural development, value addition, cold chains, and Mega Food Parks
- **Mega Food Parks Scheme**
  - Cluster-based infra with processing, warehousing, logistics, and testing facilities
- **Operation Greens**
  - Focused support for 22 perishable fruits and vegetables
  - Price stabilisation and value chain development
- **Skill Development Schemes (SHRISD)**
  - Training youth for roles in the food industry
- 100% FDI allowed in food processing (automatic route) and in e-commerce for locally produced food (approval route)

Financial Support Measures

- NABARD fund of `2,000 crore
- Priority sector lending status for food & agro units
- Integration with Trade Receivables Electronic Discounting System (TReDS) platform to ease working capital access

Conclusion

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The Food Processing Industry holds transformative potential for India's agri-economy. From reducing food waste to generating rural employment and enhancing exports, the sector is vital for inclusive growth. While the government has rolled out multiple schemes and incentives, the key now lies in effective last-mile implementation, building resilient infrastructure, skilling rural youth, and strengthening the digital ecosystem. A robust food processing sector can serve as a pillar of India's food security and rural prosperity.

A comprehensive, structured, and exam-oriented set of notes on "Soil Health and Fertility" — covering all key subtopics under Chapter 5: Agricultural Inputs and Technology of this PART. The content is tailored for UPSC/BPSC and similar exams.

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Agriculture Inputs and Technology

5

IN THIS CHAPTER...

5.1 Soil Health and Fertility

5.1.1 Importance of Soil Health

5.1.2 Water Use Efficiency (WUE)

5.1.3 Micro-Irrigation

5.1.4 Soil Health Card Scheme (SHC)

5.1.5 Organic and Natural Farming

5.2 Agricultural Technology

5.2.1 Science-Based Technological Interventions in Agriculture

5.2.2 Role of Biotechnology in Agriculture

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5.3 Artificial Intelligence (AI) in Agriculture

5.4 Internet of Things (IoT) in Agriculture

5.5 Precision Farming and Drones

5.6 Genetically Modified (GM) Crops

5.7 Nanotechnology in Agriculture

5.8 Space Technology

5.9 ICT in Agriculture

5.10 Nuclear Technology in Agriculture

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5.1 SOIL HEALTH AND FERTILITY

Soil is the foundational resource of agriculture, directly influencing crop productivity, nutritional security, and environmental sustainability. In a country like India—characterised by diverse agro-climatic zones and smallholder-dominated farming—maintaining soil health is vital for sustainable agriculture and rural livelihoods.

However, decades of chemical-intensive practices, mono-cropping, and poor soil management have led to widespread nutrient depletion, erosion, and acidification. Recognising this, India has initiated numerous scientific and policy measures such as the Soil Health Card Scheme, organic farming promotion, and micro-irrigation to restore soil fertility, ensure efficient resource use, and enhance long-term productivity.

5.1.1 Importance of Soil Health

- Soil is the foundation of agricultural productivity and food security.
- Healthy soil ensures:
  - Optimal plant growth.
  - Sustainable crop yield.
  - Resistance to pests and diseases.
  - Enhanced water retention and nutrient cycling.
- Soil degradation (due to erosion, salinisation, and overuse of chemical fertilizers) threatens agricultural sustainability in India.

Table 5.1 Soil Fertility vs Soil Productivity

Aspect	Soil Fertility	Soil Productivity
Definition	The capacity of soil to supply essential nutrients	The ability of soil to produce crops sustainably

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Aspect	Soil Fertility	Soil Productivity
Indicator	Nutrient availability	Yield level under specific cropping systems
Relation	Fertile soil may not always be productive	Productive soil is generally fertile

Table 5.2 Types of Soils in India

Soil Type	Characteristics	Regions	Suitability
Alluvial Soil	Fertile, loamy, rich in potash and lime	Indo-Gangetic Plains, Coastal India	Rice, wheat, sugarcane, pulses
Black Soil (Regur)	Clayey, moisture-retentive, rich in lime and iron	Deccan Plateau – Maharashtra, MP, Gujarat	Cotton (main), oilseeds, pulses
Red Soil	Rich in iron, poor in nitrogen and humus	Tamil Nadu, Karnataka, Odisha	Millets, pulses, groundnut
Laterite Soil	Rich in iron/aluminium, acidic, low fertility	Western Ghats, Northeastern Hills	Tea, coffee, cashew (with treatment)

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Soil Type	Characteristics	Regions	Suitability
Desert Soil	Sandy, low moisture retention, saline	Rajasthan, Gujarat	Barley, millets (with irrigation)
Mountain Soil	Loamy/silty, acidic, rich in humus	Himalayan region	Tea, fruits, maize
Peaty and Marshy	High organic matter, waterlogged	Kerala, Sundarbans	Rice, jute (with drainage)

5.1.2 Water Use Efficiency (WUE)

- Ratio of crop yield to the amount of water used.
- **India's challenge** More than 80% of water is used in agriculture, with high wastage due to flood irrigation.
- Measures to improve WUE:
  - Micro-irrigation systems: Drip and sprinkler
  - Laser land levelling and mulching
  - Crop selection based on agro-climatic conditions
  - Rainwater harvesting and farm ponds
- Government Schemes:
  - Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)
  - Per Drop More Crop initiative (focuses on micro-irrigation)

5.1.3 Micro-Irrigation

- Includes drip and sprinkler systems
- **Advantages**
  - Saves 30–70% water
  - Increases crop productivity by 20–40%

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- Reduces fertilizer use and weed growth

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- **Barriers**
  - High initial cost
  - Poor awareness among marginal farmers
  - Maintenance challenges
- **Progress**: India has the largest area under micro-irrigation globally, but still <20% of potential coverage.

5.1.4 Soil Health Card Scheme (SHC)

- Launched in 2015 by the Ministry of Agriculture and Farmers Welfare, the scheme aims to assess the nutrient status of soil and provide crop-specific fertilizer recommendations to the farmers..
- **Features**
  - Soil tested every 2 years.
  - **Parameters**: pH, NPK, micro-nutrients, organic carbon.
- **Achievements**
  - Over 22 crore soil health cards distributed up to 2024.Reduction in excess fertilizer use.
- **Limitations**
  - Delays in testing and distribution.
  - Need for better farmer awareness and advisory services.

5.1.5 Organic and Natural Farming

■ *Organic Farming*

Organic farming is a cultivation method that uses natural inputs such as manure, compost, and bio-pesticides, and avoids synthetic chemicals.

Benefits

- Improves soil structure and biodiversity.
- Enhances soil carbon, reduces chemical residue.
- Sikkim: First certified 100% Organic State (201G) under Paramparagat Krishi Vikas Yojana (PKVY).

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■ *Natural Farming / ZBNF (Zero Budget Natural Farming)*

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Natural farming focuses on low-cost, ecological crop production using natural resources, enhancing soil health, biodiversity, and sustainable, chemical-free food systems

Components

- Jeevamrit, Bijamrit, Acchadana (mulching), and Whapasa (soil aeration)
- **Eco-benefits**
  - Restores soil biota
  - Reduces dependence on external inputs
- **Economic benefits**:
  - Lower input costs
  - Higher net returns, especially for marginal farmers

Do You Know?

Subhash Palekar is regarded as the pioneer of natural farming.

- **Ecological and Economic Benefits of Organic State (Sikkim Model)**
  - **Ecological**:
    - Soil rejuvenation and microbial activity improved.
    - No chemical leaching, preserving water quality.
    - Enhanced biodiversity and agro-ecological resilience.
  - **Economical**:
    - Boost in organic tourism and branding of produce.
    - Higher price realisation due to premium markets.
    - Reduced input cost for better income stability ofsmall farmers.
- **SVAMITVA Scheme**
  - Survey of Villages Abadi and Mapping with Improvised Technology in Village Areas.
  - It was launched in 2020 by the Ministry of Panchayati Raj.
  - **Relevance to Agriculture**

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- Provides accurate land ownership records.
- Enables better credit access for the farmers.
- Supports infrastructure planning and dispute resolution.

Conclusion: Soil Health and Fertility

Soil health is not merely an agricultural issue; it is central to food security, climate stability, and the economic well-being of rural communities. The future of Indian agriculture hinges on adopting integrated nutrient management, eco-friendly systems such as organic and natural farming, and technology-enabled soil analysis. A shift from yield-focused agriculture to sustainability-driven farming is imperative, with soil fertility as its core priority. By equipping farmers with knowledge, incentives, and innovative tools, India can ensure that its soils remain productive, resilient, and capable of sustaining future generations.

5.2 AGRICULTURAL TECHNOLOGY

Agricultural technology has emerged as a transformative force in modern Indian agriculture, bridging the gap between traditional practices and scientific innovations. From biotechnology and precision farming to digital tools like Artificial Intelligence (AI), drones, and remote sensing, the landscape of agriculture is rapidly evolving. These technologies not only enhance productivity and sustainability but also empower farmers by providing real-time information, improving resource efficiency, and opening new market linkages. In the era of climate change and resource stress, the adoption of cutting-edge technologies has become essential to ensure food security, reduce ecological footprints, and enhance the socio-economic well-being of farmers.

5.2.1 Science-Based Technological Interventions in Agriculture

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- Agriculture in India is transitioning from traditional knowledge to science-based practices, leveraging genetics, information technology, engineering, and material sciences.
- The Green Revolution marked the beginning of this shift with high-yielding varieties (HYVs), irrigation, fertilizers, and mechanisation.
- **Contemporary interventions include**
  - Molecular breeding and genomics
  - Real-time data analytics
  - Weather forecasting systems
  - Automated machinery and digital platforms

5.2.2 Role of Biotechnology in Agriculture

Biotechnology has significantly contributed to crop improvement, disease resistance, and productivity enhancement.

- **Key Applications**
  - Genetic Engineering: Introduction of traits such as drought tolerance, pest resistance (e.g., Bt cotton).
  - Biofertilizers and Biopesticides: Reduce chemical use and enhance soil microflora.
  - Tissue Culture: Rapid multiplication of disease-free plants (used in banana, sugarcane, etc.)
  - Marker-Assisted Breeding: Accelerates the selection of desirable traits.
- **Socio-economic Impact**
  - Enhances farmer incomes by reducing input costs and increasing yield.
  - Helps mitigate climate-induced risks via stress-tolerant crops.
  - Improves nutritional content of food (e.g., Golden Rice, biofortified millets).

5.2.3 e-Technology in Agriculture

e-Technology refers to the use of digital platforms, mobile apps, and ICT tools to deliver farm-related services and knowledge.

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Table 5.3 Major e-Technology Initiatives in Agriculture

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Initiative	Description
mKisan Portal	Provides SMS-based advisories to farmers in regional languages.
e-NAM (National Agriculture Market)	Integrates APMCs across India for transparent price discovery.
Kisan Suvidha App	Provides information on weather, market prices, and crop advisories.
AgriStack (proposed)	Centralised digital database of farmers for targeted interventions.

Benefits

- Bridges the information gap in rural areas.
- Help farmers make informed decisions about sowing, pest control, and selling.
- Enhances access to credit, insurance, and inputs.

5.3 ARTIFICIAL INTELLIGENCE (AI) IN AGRICULTURE

AI is increasingly used for predictive analytics, crop disease detection, yield estimation, and automation.

- **Applications**
  - Image-based pest/disease diagnosis via smartphone apps.
  - AI-based yield prediction models using satellite and weather data.
  - Chatbots (e.g., Digital Krishi Sakhis) to answer farmer queries.
  - Smart irrigation systems using AI-based scheduling.
- **Benefits**
  - Reduces dependency on traditional extension workers.

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- Improves input use efficiency and productivity.
- Enables real-time decision-making and climate-smart practices.

5.4 INTERNET OF THINGS (IOT) IN AGRICULTURE

IoT involves interconnected devices that collect and share data for intelligent farm management.

- **Key Uses**
  - Soil moisture sensors, weather stations, and drone-based sensors.
  - Automated irrigation systems based on real-time moisture data.
  - Monitoring livestock health through wearable devices.
- **Advantages**
  - Reduces water and fertilizer wastage.
  - Facilitates precision agriculture and real-time monitoring.
  - Enables early warning systems for pest outbreaks.

5.5 PRECISION FARMING AND DRONES

Precision Farming refers to site-specific crop management using GPS, GIS, sensors, and big data.

Drones are used for aerial surveys, pesticide spraying, and crop health imaging.

- **Impact**
  - Maximises productivity per unit input.
  - Minimises ecological damage.
  - Helps in disaster response (For example-post-flood crop damage assessment).

Mechanisation and its Challenges

- **Advantages**

  Improves timeliness and efficiency of operations.

  Reduces labour dependency and drudgery.

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Enables cultivation of larger areas with minimal human input.

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- **Challenges**

  High capital cost of machinery.

  Fragmented land holdings limit utility.

  Inadequate training on machine usage and maintenance.

  Low custom hiring services in remote regions.
- **Solutions**

  Promoting Custom Hiring Centres (CHCs).

  Subsidies under SMAM (Sub-Mission on Agricultural Mechanisation).

  Development of small and affordable implements for marginal farmers.

5.6 GENETICALLY MODIFIED (GM) CROPS

GM crops are developed by inserting foreign genes to obtain desired traits like pest resistance or herbicide tolerance.

- **Examples in India**
  - Bt Cotton (only commercially approved GM crop).
  - Trials for GM mustard, brinjal, and rice is ongoing.
- **Concerns**
  - Biosafety and long-term ecological impacts.
  - Resistance buildup in pests (Exmple- pink bollworm).
  - Fear of corporate control over seeds.
- **Way Forward**
  - Transparent regulatory mechanisms (GEAC, FSSAI).
  - Promotion of indigenous R&D to reduce dependency.
  - Engagement with farmer communities for risk awareness.

5.7 NANOTECHNOLOGY IN AGRICULTURE

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Nanotechnology uses nano-scale particles and materials for improved input efficiency.

- **Applications**
  - Nano-fertilizers for controlled nutrient release.
  - Nano-pesticides for targeted pest control.
  - Improved soil remediation and seed germination.
- **Advantages**
  - Reduces chemical run-off and soil degradation.
  - Improves crop health and sustainability.

5.8 SPACE TECHNOLOGY

ISRO has developed several remote sensing satellites (Example, Cartosat, RISAT) for agriculture.

- **Uses**
  - Crop acreage and yield estimation.
  - Drought and flood monitoring.
  - Mapping of soil moisture, irrigation infrastructure, and land-use changes.

5.9 ICT IN AGRICULTURE

Information and Communication Technologies (ICT) empower farmers with tools for communication, awareness, and decision-making.

- **Examples**
  - Mobile apps, IVRS, community radio.
  - Kisan Call Centres and Common Service Centres (CSCs).
  - Village Knowledge Centres and Krishi Vigyan Kendras.

5.10 NUCLEAR TECHNOLOGY IN AGRICULTURE

Nuclear techniques are used in mutation breeding, food preservation, and insect pest management.

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Applications

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- Gamma radiation for developing high-yield, disease-resistant crop varieties (e.g., Trombay groundnut).
- Irradiation for food preservation.
- Sterile Insect Technique (SIT) for controlling pests like fruit flies.

Conclusion

The future of Indian agriculture is deeply intertwined with the adoption and diffusion of cutting-edge technologies. These innovations offer solutions to complex problems such as declining soil fertility, fragmented land holdings, climate change vulnerabilities, and stagnant productivity. However, the success of technological interventions hinges on affordability, accessibility, farmer awareness, and robust institutional support. With proactive policy frameworks, public-private partnerships, and inclusive digital infrastructure, agricultural technology can pave the way for a more resilient, sustainable, and equitable agrarian economy in India.

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Institutional and Market Reforms

6

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6.1 AGRICULTURAL MARKETING IN INDIA

Agricultural marketing in India plays a crucial role in ensuring that the efforts of millions of farmers reach remunerative and sustainable outcomes. It encompasses all activities involved in the movement of agricultural produce from the farm to the final consumer, including harvesting, grading, packaging, storage, transport, and sale. Although India has made significant progress in agricultural marketing since Independence, the sector still faces issues like fragmented markets, dominance of middlemen, poor infrastructure, and limited access to formal finance. To tackle these problems, the government has rolled out various reforms and policies, including changes to APMC, digital platforms like e-NAM, and new models like contract farming. These efforts aim to create a more efficient, inclusive, and farmer-focused marketing system.

6.1.1 Agricultural Credit and Regulation of Moneylending

- **Importance of agricultural credit**
  - Essential for purchasing seeds, fertilizers, machinery, and irrigation facilities.

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- Acts as a buffer during crop failures or seasonal income shortages.

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- **Sources of credit**
  - Formal sector: Includes commercial banks, cooperative banks, and regional rural banks.
  - Informal sector: Includes moneylenders, traders, relatives—often marked by high-interest rates and exploitative practices.
- **Issues with moneylending**
  - High rates of interest.
  - Usurious practices and debt traps.
  - Absence of proper documentation or grievance redressal.
- **Regulation of moneylending**
  - The Usurious Loans Act and State Moneylenders Acts aim to curb predatory lending.
  - Efforts to enhance formal credit access through:
    - Kisan Credit Card (KCC) Scheme.
    - Financial Inclusion initiatives via Jan Dhan-Aadhaar-Mobile (JAM) trinity.
    - Interest subvention schemes.
- **Challenges**
  - High dependence on informal credit in states like Bihar and UP.
  - Procedural delays and collateral requirements in formal credit institutions.
  - Weak credit penetration among tenants and marginal farmers.

6.1.2 APMC Acts and Reforms

- **Overview**
  - Agricultural Produce Market Committees (APMCs) are state-regulated markets established to ensure fair prices for farmers and prevent exploitation by intermediaries.
  - Farmers are often mandated to sell their produce only through APMC mandis.
- **Major issues**
  - Restricted market access and limited competition.

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  - Monopolistic practices by licensed commission agents.
  - High mandi fees, market cess, and multiple levies.
- **Reform initiatives**
  - **Model APMC Act, 2003:** Suggested to liberalise trade and promote private markets and direct procurement.
  - **Model Agricultural Produce and Livestock Marketing Act (APLM), 2017:** More comprehensive reform proposal encouraging single licenses and unified markets.
  - **2020 Farm Laws (repealed):** Intended to promote barrier-free trade outside mandis and legalise contract farming.
- **Progress in reforms**
  - Some states like Maharashtra, Karnataka, and Gujarat have partially adopted reform measures.
  - Others continue to retain restrictive provisions.
- **Way forward**
  - Need for consensus-driven reforms respecting federalism.
  - Strengthen transparency and farmer participation in APMC operations.
  - Encourage establishment of private markets and Farmer Producer Organizations (FPOs).

6.1.3 E-NAM and Digital Agricultural Markets

- **E-NAM (National Agriculture Market)**
  - Launched in 2016 to integrate APMC mandis across the country via a digital trading platform.
  - Aims to create a unified national market for agricultural commodities.
- **Features**
  - Real-time price discovery and transparency.
  - Enables inter-mandi and inter-state trading.
  - Reduces dependency on intermediaries.
- **Achievements**
  - Over 1300+ mandis integrated across 27 states and UTs.
  - Supported by quality assaying, digital payments, and logistic modules.
- **Limitations**

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- Poor digital literacy among farmers.
- Low penetration in remote and backward regions.
- Infrastructure gaps (grading labs, internet connectivity).
- Mandis often continue to dominate local trade despite e-NAM's presence.

- **Suggested improvements**
  - Capacity building for farmers.
  - Expanding infrastructure for quality testing and storage.
  - Integrating e-NAM with FPOs, SHGs, and private players.

6.1.4 Contract Farming

- **Definition**
  - A pre-agreed contract between farmers and buyers (agri-business firms, food processing companies) regarding production and marketing of agricultural produce.
- **Advantages**
  - Ensures assured markets and fair prices.
  - Promotes better technology adoption and input support.
  - Reduces market risk and post-harvest uncertainty.
  - Enhances quality and standardisation.
- **Challenges**
  - Farmers are often at a weaker bargaining position.
  - Disputes related to pricing and procurement.
  - Lack of legal safeguards and enforcement mechanisms.
  - Limited awareness and trust issues.
- **Government Initiatives**
  - Model Contract Farming Act, 2018, proposed legal framework for fair and enforceable agreements.
  - Some states like Punjab, Maharashtra, and Tamil Nadu have implemented state-specific policies.
- **Way Forward**
  - Strengthen legal backing with dispute resolution mechanisms.
  - Create enabling institutions like Contract Farming Promotion Boards.

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- Link contract farming to food processing and export markets.

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Conclusion

Agricultural marketing remains the weakest link in India's agrarian economy, often restricting farmers from realising fair value for their produce. The dominance of intermediaries, poor market integration, and inadequate infrastructure have historically constrained farmers' income potential. Reforms in APMC Acts, promotion of digital platforms like e-NAM, strengthening of institutional credit, and regulated expansion of contract farming represent key pillars of a modernized, transparent, and farmer-friendly marketing system. However, true transformation requires a multi-dimensional approach—empowering farmers with access to markets, information, finance, and technology. Strengthening FPOs, encouraging private investment in agri-markets, and ensuring digital and logistical inclusiveness will be central to creating a vibrant and inclusive agricultural marketing ecosystem in India.

6.2 MINIMUM SUPPORT PRICE (MSP) AND PROCUREMENT

The Minimum Support Price (MSP) system is a cornerstone of India's agricultural pricing policy, designed to safeguard farmers from the uncertainties of the open market and to ensure remunerative prices for their produce. Introduced during the mid-19G0s in response to the food crisis and the push for self-sufficiency, the MSP mechanism has evolved into a powerful instrument for price support and food grain procurement. It ensures that farmers are guaranteed a minimum price for certain crops, irrespective of market fluctuations. The MSP system, backed by public procurement agencies like the Food Corporation of India (FCI), not only plays a crucial role in providing income security to farmers but also supports the nation's food security apparatus by maintaining buffer stocks. However, over the years, concerns have emerged about its limited crop and regional coverage, sustainability, and compatibility with broader market

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reforms. Understanding the current structure, achievements, limitations, and potential reforms of the MSP regime is key to shaping a more equitable and effective agricultural policy.

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6.2.1 Objectives of MSP

- **Income Protection:** Shields farmers from distress sales during market price crashes.
- **Price Stability:** Helps stabilise food prices through procurement and buffer stocking.
- **Crop Diversification:** Encourages production of specific crops in deficit years.
- **Support Food Security Programs:** Ensures the availability of food grains for PDS.
- **Rural Economic Stability:** Especially for wheat and paddy in northern states.

6.2.2 Current Mechanisms of MSP Procurement

- Open-ended Procurement by FCI – Mainly for wheat and rice.
- Price Support Scheme (PSS) – Pulses, oilseeds, cotton (via NAFED).
- Market Intervention Scheme (MIS) – For perishable horticultural produce.
- PM-AASHA Scheme (2018):
  - PSS
  - Price Deficiency Payment Scheme (PDPS) – Compensates for the difference between MSP and market price.
  - Private Procurement & Stockist Scheme (PPSS)

6.2.3 Achievements of MSP Regime

- Contributed to Green Revolution success.
- Ensured buffer stock maintenance for national food security.
- Provided price assurance, especially to farmers in Punjab, Haryana, and parts of Andhra Pradesh.

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6.2.4 Challenges and Criticism

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- **Limited Coverage**
  - Only 6% farmers benefit directly from MSP procurement.
  - Focus largely on rice and wheat; pulses, oilseeds, and coarse grains poorly covered.
- **Geographical Bias**
  - Skewed procurement from a few states; marginal and tribal farmers excluded.
- **Inadequate Infrastructure**
  - Lack of procurement centers and storage in many rural/tribal areas.
- **Price Distortion**
  - Encourages cultivation of water-intensive crops.
  - Disincentivizes market-led cropping decisions.
- **Fiscal Burden**
  - Heavy subsidy for FCI operations; increasing storage and handling costs.
- **Awareness Gap**
  - Many farmers unaware of MSP and procurement timelines.
- **Calculation Concerns**
  - MSP based on A2 + FL rather than C2 (recommended by Swaminathan Committee).
- **WTO Compatibility**
  - Issues under WTO Agreement on Agriculture regarding market price support.

6.2.5 Way Forward

- **Widen the Procurement Base**
  - Increase coverage of crops and states.
  - Strengthen state-level procurement mechanisms for pulses/oilseeds.
- **Reform MSP Calculation**
  - Consider C2 costs, land rentals, and capital depreciation for better price discovery.
- **Revamp Infrastructure**

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- Invest in decentralised storage, transport, and procurement centres in remote areas.

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- **Promote Alternatives**
  - Expand Price Deficiency Schemes and FPO-led market models.
- **Private Sector Participation**
  - Enable agri-startups and private players in procurement under strict safeguards.
- **Enhance Farmer Awareness**
  - Use digital platforms, panchayat campaigns, and agri-extension services.
- **Comprehensive Strategy**
  - MSP alone can't solve income distress; broader reforms in production, processing, and marketing are needed.

Conclusion

The MSP and procurement framework has undoubtedly played a significant role in transforming India's agricultural landscape, particularly by ensuring stable farmer incomes and supporting food grain availability through public distribution. However, its overdependence on a few crops and regions, rising fiscal costs, and limited reach to small and marginal farmers signal the urgent need for reform. As India moves toward more inclusive and market-oriented agricultural growth, the MSP system must be restructured to align with sustainable practices, regional diversity, and modern supply chain mechanisms. A well-calibrated approach—blending MSP with enhanced infrastructure, diversified procurement, better price discovery, and increased private sector engagement—can create a more balanced and resilient agri-economy. Ultimately, empowering farmers with multiple avenues of fair pricing and market access will be central to achieving both equity and efficiency in India's farm sector.

6.3 STORAGE OF FOOD GRAINS IN INDIA: CURRENT SITUATION

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The storage of food grains forms a crucial link in India's agricultural supply chain and food security framework. Efficient storage not only prevents post-harvest losses and maintains the quality of grains but also enables the government to manage buffer stocks and run public welfare schemes like the Targeted Public Distribution System (TPDS). In a country like India, which experiences both seasonal gluts and regional disparities in production and demand, robust storage infrastructure becomes indispensable. The Food Corporation of India (FCI), established under the Food Corporations Act, 19G4, is the nodal agency tasked with procurement, storage, and distribution of food grains. Over the years, various schemes and policies have been implemented to bridge the storage gap, especially in vulnerable and geographically challenging regions. As India aspires to modernize its agri-logistics, the current storage status and institutional strategies hold vital implications for long-term food security and price stability.

6.3.1 Role of the Food Corporation of India (FCI)

- FCI is the primary agency responsible for:
  - Procurement of food grains from farmers at MSP.
  - Storage of procured food grains for buffer stock maintenance.
  - Distribution of grains under TPDS and welfare schemes.
- Storage requirement is determined by:
  - Volume of procurement.
  - Buffer stock norms.
  - PDS operations—mainly for rice and wheat.

6.3.2 Key Objectives of Storage Infrastructure

- To feed the Targeted Public Distribution System (TPDS) and other welfare programs like mid-day meals, Annapurna Yojana, etc.
- To ensure food security during years of production shortfall, natural calamities, or market disruptions.
- To stabilise prices by releasing stocks during inflationary phases and procuring during bumper harvests.

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6.3.3 Modes of Creating and Managing Storage Capacity

- **Hiring of Godowns**
  - From government-linked warehousing bodies:
    - Central Warehousing Corporation (CWC)
    - State Warehousing Corporations (SWCs)
    - Other state government agencies.
- **Private Warehousing Scheme (PWS)**
  - FCI hires storage space from private warehouse owners, under regulated terms and conditions.
- **Central Sector Scheme (CSS)**
  - Aimed at augmenting storage capacity, especially in:
    - North-Eastern States and other difficult regions like Jammu and Kashmir.
  - **Salient Features**
    - Funds are released as equity to FCI for land acquisition and infrastructure, like:
      - Godowns
      - Railway sidings
- Electrification
- Weighbridges
  - Grants-in-aid are given to state governments in geographically difficult areas to create intermediate storage points.
- Construction of SILOs under PPP Mode
  - Modern storage solution through Public-Private Partnership.
  - SILOs offer:
    - Scientific preservation.
    - Better quality control.
    - Automation and faster handling.
- Private Entrepreneurs Guarantee (PEG) Scheme
  - Introduced to encourage private investment in storage infrastructure.
  - Private investors build godowns, which are then hired by FCI under guaranteed returns for specified periods.

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Table 6.1 Current Storage Capacity in India
(As of February 2024)

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Storage Type	Capacity (in Lakh Tonnes)
FCI – Owned	146.86
FCI – Hired (via PWS, CWC, SWCs, etc.)	214.76
Total with FCI	361.62
State Government Agencies	400.74
Total (FCI + State)	762.36 (≈76 million tonnes)
Additional via PEG Scheme (Private)	≈ 15 million tonnes

Conclusion

India's vast food grain production, driven by schemes like MSP procurement, demands a robust and adaptive storage infrastructure to minimise post-harvest losses, ensure uninterrupted supply, and support public welfare mechanisms. The Food Corporation of India (FCI) plays a central role in managing the nation's food reserves through a combination of owned, hired, and privately built facilities. While significant capacity has been added in recent years—especially through schemes like PEG, CSS, and PPP SILOs, there remain challenges such as regional imbalances, inadequate cold storage, and maintenance of quality. Going forward, India must invest in scientific, decentralised, and tech-driven storage solutions, particularly in high-production and remote areas. Strengthening storage is

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not just a logistical concern, it is a strategic necessity for national food security and farmers' income resilience.

6.4 TRANSPORT OF AGRICULTURAL PRODUCE

6.4.1 Importance of an Effective Transport System in Agriculture

- **Vital for farm-to-market linkage:** Ensures timely delivery of perishables, reducing post-harvest losses.
- **Enhances farmer income:** Reduces cost and dependency on middlemen, enabling better price realisation.
- **Facilitates regional trade:** Allows surplus produce in one region to meet demand in another.
- **Improves accessibility:** Brings remote/rural farmers into the national agricultural market system.
- **Supports input supply:** Ensures timely delivery of seeds, fertilizers, and other inputs to farmers.

6.4.2 Constraints and Challenges in Transport

- **Poor rural road connectivity:** Nearly 30% villages still lack all-weather road access.
- **Lack of multimodal transport:** Weak integration between road, rail, and waterways.
- **High freight cost:** Impacts the competitiveness of agri-produce.
- **Inadequate cold-chain transport:** Affects perishables like fruits, vegetables,and dairy.
  - **Storage-transit mismatch:** Delays due to lack of synchrony between storage hubs and transport systems.
- **Unorganised transport sector:** Fragmented truckers and outdated infrastructure reduce efficiency.

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6.4.3 Government Initiatives

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- **PM Gram Sadak Yojana (PMGSY):** Boosted rural road connectivity significantly.
- **Bharatmala Pariyojana:** Includes rural-farm road linkages to national highways.
- **Kisan Rail Scheme:**
  - Introduced in 2020 for time-sensitive perishable transport.
  - Provides subsidy and special freight corridors.
- Agri-Express Services by Indian Railways for fast movement of farm produce.
- **PM Gati Shakti:** Unified logistics platform to integrate transport and agri-logistics.
- **E-NAM logistics module:** Enables demand-based online booking of transport vehicles.

6.4.4 Storage of Agricultural Produce

- **Importance of Storage**
  - Ensures year-round food availability: Helps tackle seasonal production and consumption mismatches.
  - Supports price stabilisation: Regulates market supply, preventing price crashes during harvest.
  - Enables MSP procurement: Facilitates government purchase and stockpiling.
  - Protects against spoilage and wastage: Especially critical for grains, pulses, and perishables.
  - Improves farmer bargaining power: Allows holding produce till favourable market conditions.
- **Types of Storage Facilities**
  - **Traditional Storage**
    - Kuthla, mud bins, underground pits — low-cost but poor efficiency.
  - **Conventional Godowns**
    - Used by FCI, CWC, SWCs; generally for long-term grain storage.

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- **SILOs**
  - Mechanised, vertical steel/concrete structures.
  - Modern alternative to traditional godowns with better preservation.
- **Cold Storage**
  - For fruits, vegetables, meat, dairy, flowers, etc.
  - Maintains optimal temperature/humidity to reduce spoilage.
- **Controlled Atmosphere Storage**
  - Used for apples, pears, etc., adjusting oxygen/CO_2 levels.
- **On-Farm Storage**
  - Small-scale low-cost structures built by farmers using government aid or NGO support.
- **Constraints in Transport and Marketing of Agricultural Produce**
  - **Fragmented supply chain:** Long chain of intermediaries and informal logistics systems.
  - **Lack of scientific storage:** Leads to 10-15% post-harvest losses annually.
  - **Limited awareness among farmers:** On storage, preservation, and logistics systems.
  - **High perishability:** Fruits and vegetables are most vulnerable due to inadequate cold-chain.
  - **Infrastructure bottlenecks:** Power supply, road access, and storage technology.
  - **Inadequate aggregation facilities:** Small landholding farmers lack the capacity to store in bulk.

6.5 COLD CHAINS IN AGRICULTURE

- **Cold chains are integrated temperature-controlled systems for storing, transporting, and distributing perishable agricultural products Benefits**
  - Helps cut post-harvest losses (India loses nearly `90,000 crore yearly due to poor cold storage).
  - Extends shelf life of produce.

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- Enables export of high-value perishable commodities.

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- **Constraints**
  - High capital and operational costs.
  - Irregular power supply in rural areas.
  - Lack of trained workforce and maintenance.
  - Fragmented ownership of cold chain nodes.
- **Government Support**
  - MIDH (Mission for Integrated Development of Horticulture).
  - PM Kisan SAMPADA Yojana: Grants for setting up cold chains.
  - NABARD & NHB support for cold storage infra.

6.6 FOOD CORPORATION OF INDIA (FCI) AND BUFFER STOCKS

- **FCI Role**
  - Procures foodgrains at MSP.
  - Maintains buffer stocks.
  - Supplies through TPDS and welfare schemes.
- **Buffer Stocks**
  - Minimum stock of food grains is maintained to ensure food security and price stability.
  - Released during scarcity, price rise or disaster.
- **Issues**
  - Overstocking and rising carrying costs (`90,000 crore/year).
  - Leakage, spoilage, and inefficiencies in distribution.
- **Recent Reforms**
  - Digitization of stock tracking.
  - Silo-based storage to reduce wastage.
  - Decentralized procurement policy.

6.7 AGRI-LOGISTICS AND ROLE OF SUPERMARKETS IN SUPPLY CHAIN

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- Agri-logistics refers to the complete movement of farm produce. This includes aggregation, storage, handling, packaging, processing, transportation, and retail. It ensures efficient delivery from farms to markets. It also reduces post-harvest losses and helps farmers get better value for their products.
- **Current challenges**
  - Lack of integrated supply chain.
  - Poor infrastructure and coordination.
- **Supermarkets' Role**
  - Direct sourcing from farmers under contract farming or FPO linkages.
  - Reduce middlemen, ensuring better price to producers and lower prices to consumers.
  - Provide sorting, grading, quality testing, and packaging at source.
- **Emerging Models**
  - BigBasket, Reliance Fresh, Amazon Fresh are adopting farm-to-fork models.
  - Blockchain and QR-based tracking for traceability and quality control.

Conclusion

India's agricultural supply chain—from production to final sale—is undergoing a slow but essential transformation. The focus on better transport, scientific storage, efficient cold chain, and modern agri-logistics is vital for minimising losses and ensuring farmer welfare. With increased private participation, PPP models, government incentives, and tech integration, the future of the agri-supply chain promises improved income for farmers, better food quality for consumers, and enhanced national food security.

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Farmer Welfare and Government Schemes

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7.1 AGRICULTURAL CREDIT

Agricultural credit is a foundational element of India's rural economy. Given the seasonal nature of agriculture, uncertain returns, and the need for timely inputs like seeds, fertilizers, and machinery, access to credit is vital for improving agricultural productivity and rural livelihoods. Over the decades, the Indian government and RBI have developed a multi-tiered system of institutional credit to support farmers, yet challenges in outreach, inclusivity, and sustainability persist.

7.1.1 Institutional Sources of Agricultural Credit

India's agricultural credit system is dominated by institutional players such as:

- **Cooperative Banks**
  - Comprises a three-tier structure:
    - State Cooperative Banks (SCBs) at the apex,
    - District Central Cooperative Banks (DCCBs) at the middle,
    - Primary Agricultural Credit Societies (PACS) at the village level.
  - Primarily cater to short- and medium-term credit needs of small and marginal farmers.
  - Challenges: Political interference, high NPAs, weak financial infrastructure, and lack of modernisation.
- **Regional Rural Banks (RRBs)**

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- Established in 1975 to bridge the gap between cooperative banks and commercial banks in rural areas.
- Jointly owned by the Central Government, State Government, and Sponsor Bank.
- Offer credit for agriculture, dairy, poultry, fisheries, and rural development.
- Increasingly being merged and consolidated for better viability.

- **Commercial Banks**
  - Became active in agricultural credit post the nationalisation of banks in 19G9.
  - Account for the largest share of agricultural credit today.
  - Provide both direct (to farmers) and indirect (to agri-businesses, input dealers) loans.
  - Expanded reach through Priority Sector Lending (PSL) norms.
- **Microfinance Institutions (MFIs)**
  - Include Self-Help Groups (SHGs) and Joint Liability Groups (JLGs).
  - Extend small loans without collateral to landless labourers, women, and marginal farmers.
  - Have helped increase credit penetration in remote areas, but are often criticised for high interest rates.
- **NABARD (National Bank for Agriculture and Rural Development)**
  - Apex refinancing and policy institution for rural credit, established in 1982.
  - **Benefits**
    - Refinance support to RRBs, Cooperative Banks, and Commercial Banks.
    - Developmental funding for infrastructure like irrigation, cold storage, warehouses, etc.
    - Policy planning, research, and innovation support for rural credit mechanisms.
    - Financial Inclusion and capacity-building of rural institutions.

7.1.2 Kisan Credit Card (KCC) Scheme

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- It was launched in 1998, following the recommendations of the R.V. Gupta Committee. It got implemented in cooperative banks, RRBs, and commercial banks..
- **Objective**
  - To provide timely, adequate, and hassle-free credit to farmers for crop production, allied activities, and other short-term requirements.
- **Key Features**
  - Credit limits are fixed based on landholding, cropping pattern, and scale of finance.
  - Validity up to 5 years with annual reviews.
  - Linked to RuPay debit cards for ATM withdrawals and point-of-sale transactions.
  - Includes accident insurance cover for cardholders.
  - Covers both crop loans and working capital for allied activities like dairy, poultry, and fisheries.
- **Benefits**
  - Reduces dependency on informal moneylenders.
  - Simplifies loan disbursal and withdrawal process.
  - Ensures credit is available during critical crop stages.
  - Encourages financial discipline and record-keeping among farmers.

Conclusion

Agricultural credit, particularly from institutional sources, remains the backbone of India's farm economy. While schemes like the Kisan Credit Card and institutions like NABARD have strengthened the system, significant gaps remain in outreach, financial literacy, and timely disbursal —especially for tenant farmers, landless labourers, and smallholders. A comprehensive credit framework, integrating technology, cooperative reforms, and targeted subsidies, is vital for making Indian agriculture truly inclusive and sustainable.

7.2 FARMER INCOME AND INSURANCE

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Agriculture remains the principal livelihood for more than half of India's population. However, Indian farmers, especially small and marginal ones, face persistent challenges—such as volatile incomes, climate-related uncertainties, input cost inflation, and market price fluctuations. In response, the Government of India has implemented a suite of schemes aimed at income support, risk mitigation, and financial inclusion, alongside technological and institutional reforms. Schemes like PM-KISAN, PMFBY, e-NAM, Soil Health Card, and the Doubling Farmers' Income Mission collectively represent a holistic approach to ensure economic stability, risk coverage, and long-term resilience in Indian agriculture.

7.2.1 Major Schemes & Initiatives for Farmer Income and Risk Mitigation

- PM-KISAN (Pradhan Mantri Kisan Samman Nidhi)
- Launched in 2019, this program uses the Direct Benefit Transfer (DBT) model to send financial assistance straight into beneficiaries' bank accounts. This approach ensures transparency and cuts down on waste.
  - **Support:** `6,000 per year in 3 equal instalments to all landholding farm families.
  - **Coverage**: Over 11 crore farmers (as of 2024).
  - **Impact:** Provides liquidity and addresses consumption and input needs, especially for small farmers.
- **Soil Health Card Scheme**
  - It was launched in 2015.
  - **Objective**: Provide individualised soil nutrient status to farmers every 2 years.
  - **Benefits**
    - Encourages balanced Fertilizer use.
    - Reduces input costs.
    - Improves long-term productivity and environmental sustainability.
- **Pradhan Mantri Fasal Bima Yojana (PMFBY)**
  - it was launched in 2016. |

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- **Objective**: Comprehensive risk coverage against crop failure.
- **Features**
  - Coverage for pre-sowing to post-harvest losses.
  - Low farmer premium: 2% (kharif), 1.5% (rabi), 5% (commercial crops).
  - Uses technology (drones, remote sensing) for yield estimation.
- **Issues:** Delayed claim settlement, low awareness, exit of some insurers in past.
- **Reforms (post-2020):** Voluntary participation, improved grievance redressal.

- **E-NAM (Electronic National Agricultural Market)**
  - it was launched in 2016.
  - **Objective**: Pan-India digital platform for transparent agricultural trade.
  - **Features:**
    - A total of 1,360+ mandis have been integrated.Enables inter-state trade, online bidding, real-time price discovery.
    - Encourages fair pricing and reduces middlemen exploitation.
- **Paramparagat Krishi Vikas Yojana (PKVY)**
  - It was launched in 2015 under NMSA.
  - **Objective**: Promote organic farming through a cluster-based approach (50 acres/cluster).
  - **Support**: `50,000 per hectare for 3 years (including certification).
  - **Benefits:**
    - Eco-friendly farming.
    - Creates niche markets with higher returns.
- **Kisan Credit Card (KCC)**
  - Launched in 1998, the scheme was later expanded to include livestock and fisheries in 2019.**Objective**: Provide timely and affordable credit.
  - **Credit Limit**: Based on landholding and cropping pattern.
  - **Interest Subvention**: Effective interest rate as low as 4% for timely repayment.
  - **Relevance**: Ensures credit flow and prevents dependency on moneylenders.

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- **National Mission on Sustainable Agriculture (NMSA)**
  - **Focus:** Climate-resilient farming practices, water use efficiency, soil management.
  - **Components:**
    - Rainfed Area Development.
    - On-farm water management.
    - Climate change adaptation.
- **Doubling Farmers' Income (DFI) Mission**
  - Announced in 2016, the scheme aimed for completion by 2022, but the target was missed.
  - **Strategies**
    - Diversification (horticulture, livestock).
    - Improving productivity and irrigation.
    - Reducing input cost.
    - Post-harvest value addition.
    - Institutional credit & crop insurance.
  - **Status:** Though not formally achieved, key initiatives continue under PM-AASHA, PM-KISAN, PMFBY, and digital agri-stack.

7.2.2 Significance of Crop Insurance, MSP, and Food Processing for Small and Marginal Farmers

- **Crop Insurance**
  - Mitigates risk from drought, floods, pest attacks, hailstorms.
  - Especially vital for small/marginal farmers with little savings.
  - PMFBY offers affordable premiums but needs better execution.
- **Minimum Support Price (MSP)**
  - Acts as price floor for key crops (23 crops currently).
  - Prevents distress sale and ensures income security.
- **Challenges**
  - Limited procurement outside Punjab-Haryana.
  - Does not benefit non-MSP crops.
  - Market distortion and environmental harm (e.g., rice-wheat bias).
- **Food Processing**

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- Help in value addition and rreducepost-harvest losses (~20% of production).
- Expands farmer's income base beyond raw produce.
- Key schemes: PM FME (Formalisation of Micro Food Enterprises), Agri Export Policy.

7.2.3 Agricultural Subsidies – Types and Issues

- **Types of Agricultural Subsidies**
  - Input subsidies:
    - Fertilizer subsidy (~`2 lakh crore)
    - Power subsidy (free electricity in some states)
    - Irrigation subsidy
  - **Price subsidies**
    - MSP procurement operations
  - **Credit subsidies**
    - Interest subvention on KCC and crop loans
  - **Subsidy on insurance premiums (PMFBY)**
  - **Criticism and Distortions**
    - Fertilizer overuse due to price distortion (urea bias).
    - Environmental damage: Rice-wheat monoculture, depleting water table.
    - Skewed regional benefit: Punjab, Haryana, and Western UP benefit more.
    - Unsustainable fiscal burden.
    - Subsidies often regressively benefit large farmers more.
  - **Way Forward**
    - Shift towards Direct Benefit Transfers (DBT).
    - Encourage crop diversification and sustainable practices.
    - Promote targeted subsidies over universal ones.

7.2.4 Digital India and Farm Productivity

- **Potential of Digital India in Agriculture**
  - Precision agriculture via AI, IoT, GIS.

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- Real-time weather, pest alerts.
- Market intelligence and price forecasting.
- Access to credit, insurance, and subsidies through apps.
- Blockchain for transparent land records and supply chains.

- **Government Initiatives**
  - AgriStack: Unified digital infrastructure for personalised advisory.
  - Kisan Sarathi: Digital extension and expert support.
  - Digital Kisan Portal, M-Kisan, KCC mobile app.
  - Drone deployment in agriculture for spraying, mapping.

Conclusion

Enhancing farmer income and providing effective insurance cover are critical to achieving the vision of a sustainable, inclusive, and profitable agricultural economy. While initiatives such as PM-KISAN, crop insurance through PMFBY, and digital platforms like e-NAM have yielded measurable benefits, challenges such as regional disparities, subsidy distortions, limited crop diversification, and infrastructural gaps remain. Moving forward, there is a need to reform subsidy structures, promote sustainable farming practices, integrate digital innovations, and ensure the timely implementation and monitoring of welfare schemes. Only then can the goal of doubling farmers' income and ensuring holistic rural prosperity be realized.

7.3 LAND REFORMS AND TENANCY IN INDIA

Land reforms in India have historically aimed to eliminate feudal structures, ensure equitable land distribution, and enhance agricultural productivity. Given that a majority of Indian farmers are marginal or landless, and land remains a key socio-economic asset, reforms in land ownership, tenancy, and land-use rights are crucial for inclusive rural development, poverty alleviation, and agricultural economic sustainability.

7.3.1 Status of Land Ceiling and Tenancy Laws

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- **Land Ceiling Laws**
  - Introduced post-Independence to fix maximum landholding limits and redistribute surplus land to the landless.
  - Implementation was weak and uneven across states.
  - Exemptions (e.g., for plantations, religious trusts) and benami transactions diluted the impact.
  - As per the latest data, around 72 lakh acres of surplus land were declared; only 56 lakh acres were distributed (mostly of poor quality).
- **Tenancy Laws**
  - Varied across states; many laws were pro-tenant in theory but poorly enforced.
  - Often resulted in concealed tenancy and informal sharecropping (e.g., batai system).
  - Lack of legal recognition for sharecroppers restricted access to credit, insurance, and government schemes.
  - Some states (e.g., Kerala, West Bengal) saw success (Operation Barga), while others saw stagnation.

7.3.2 Digitisation of Land Records:

- **Implemented under Digital India Land Records Modernisation Programme (DILRMP)**
  - Integration of land records, survey/resurvey, and GIS mapping.
  - Aim: Ensure clear land titles, reduce disputes, and facilitate agri-credit and infrastructure development.
- **Progress**
  - Mixed across states – better in Karnataka, Andhra Pradesh, Maharashtra.
  - Challenges include data mismatches, lack of capacity, and resistance from vested interests.

7.3.3 Land Reforms and Socio-Economic Upliftment of Small and Marginal Farmers:

- Redistribution of land has reduced dependency on landlords.

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- Enhanced access to institutional credit, crop insurance, and government subsidies.
- Improved bargaining power and dignity of land-owning farmers.
- In some states, land reforms led to higher productivity and rural industrialisation (e.g., West Bengal in 1980s).

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7.3.4 Role of Land Reforms in Agricultural Development and Success Factors

- **Positive Role in Agriculture**
  - Promoted egalitarian agrarian structure.
  - Incentivized investment in land due to secure tenure.
  - Supported agri-diversification and adoption of new technologies.
- **Factors Behind Success of Land Reforms in Some States**
  - Political will and left-oriented leadership (e.g., Kerala, West Bengal).
  - Strong bureaucratic machinery.
  - People's movements (e.g., Telangana peasant movement, Bhoodan Movement).
  - Effective implementation and monitoring.

7.3.5 Declining Land Holdings and Contract Farming Debate

- **Issue:**
  - Average landholding size in India is now 1.08 hectares (as per Agriculture Census 2015-16).
  - Fragmentation makes capital-intensive and modern farming non-viable.
- **Should Contract Farming and Land Leasing Be Promoted?**
  - **Pros**
    - Encourages private investment in agriculture.
    - Farmers retain land ownership while leasing operational control.
    - Facilitates technology transfer, better inputs, and assured buy-back.
    - Reduces risk for small farmers and ensures price stability.
  - **Cons**

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- Risk of corporate exploitation and monopsony pricing.
- In absence of regulatory safeguards, it may alienate small farmers.
- Environmental exploitation due to profit-centric practices.

- **Way Forward**
  - Legalize and regulate land leasing with proper model leasing laws, ensure fair contracts, and protect land rights of smallholders.

7.3.6 Link Between Land Reforms, Agricultural Productivity and Poverty Alleviation

- Secure land ownership boosts long-term investment in soil, irrigation, and infrastructure.
- Redistributed land fosters inclusive growth and reduces income inequality.
- Productivity improves due to enhanced incentives for land improvement.
- Poverty reduction happened via asset creation and an empowered rural workforce.
- **Challenges in Designing & Implementing Reform**
  - Political resistance from landed elites.
  - Administrative inertia and lack of updated land records.
  - Judicial delays in settling land disputes.
  - Lack of coordination between Centre and States.
  - Weak implementation of land ceiling laws and resurvey initiatives.

Conclusion

Land reforms remain a foundational pillar for agrarian development and rural empowerment in India. While early reform waves yielded limited results due to implementation flaws, the emerging focus on digitisation, contract farming, and regulated land leasing presents a second opportunity to address structural agrarian inequalities. For long-term sustainability, reforms must be farmer-centric, rights-based, and technology-driven, ensuring both economic viability and social justice in Indian agriculture.

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Challenges and Way Forward

8

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8.1 SUSTAINABILITY AND CLIMATE CHANGE IN INDIAN AGRICULTURE

Sustainability and climate resilience have become central concerns in Indian agriculture, a sector inherently vulnerable to ecological stress. Erratic monsoons, increasing temperatures, shifting rainfall patterns, and extreme weather events threaten crop productivity and farmer livelihoods. Ensuring food security while preserving environmental resources demands a transition towards sustainable, climate-smart agricultural practices.

8.1.1 Impact of Climate Change on Indian Agriculture

- **Erratic Rainfall and Droughts**
  - Increased frequency of dry spells and unseasonal rains disrupt crop cycles and sowing patterns.
  - Rainfed regions (60% of cultivated area) are especially vulnerable.
- **Rising Temperatures**
  - Reduces yields of major crops like wheat and rice.
  - A 1°C rise in temperature may reduce wheat yield by 4-5 million tonnes.
- **Floods & Cyclones**
  - Coastal and low-lying areas experience frequent crop damage.
  - Increases salinisation of soil and damages infrastructure.
- **Shift in Cropping Patterns**
  - Decline in productivity of traditional crops in specific regions (e.g., apple cultivation in Himachal shifting to higher altitudes).
- **Increased Pest and Disease Incidence**
  - Warmer climates lead to a rise in crop pests and diseases, especially in rice, cotton, and horticultural crops.

8.1.2 Climate-Resilient Agriculture: Concept and Components

Climate-resilient agriculture (CRA) refers to farming systems that are adaptive, sustainable, and risk-proof against climatic stress.

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Key components include

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- **Drought-Resistant/Heat-Tolerant Crop Varieties**
  - E.g., Drought-tolerant rice and millet varieties promoted by ICAR and CGIAR.
- **Integrated Farming Systems**
  - Combining crops with livestock, fishery, agroforestry for risk minimization.
- **Efficient Water Management**
  - Drip/sprinkler irrigation, micro-irrigation under PMKSY.
  - Rainwater harvesting and watershed development.
- **Soil Health Management**
  - Use of Soil Health Cards, organic manure, and cover cropping to improve fertility and reduce dependency on chemicals.
- **Agroforestry & Tree-based Farming**
  - Trees sequester carbon, reduce erosion, and provide income diversification.
- **Early Warning Systems & ICT**
  - AI-based advisories (like 'Kisan e-Mitra') for weather prediction and pest outbreaks.

8.1.3 National Mission for Sustainable Agriculture (NMSA)

- A key sub-mission under the National Action Plan on Climate Change (NAPCC).
- **Objectives**
  - Promote sustainable agriculture through resource conservation.
  - Enhance productivity in a climate-resilient manner.
  - Integrate traditional knowledge with modern techniques.
- **Key Components**
  - Rainfed Area Development (RAD):
    - Integrated farming systems in vulnerable rainfed areas.
  - Soil Health Management (SHM):
    - Balanced Fertilization, organic farming, bio-fertilizer use.

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- Climate Change & Sustainable Agriculture Monitoring:
  - Data collection, research, and modeling climate impacts.
- Paramparagat Krishi Vikas Yojana (PKVY):
  - Promotion of organic farming practices.
- On-Farm Water Management (OFWM):
  - Water-use efficiency and micro-irrigation techniques.
- **Achievements and Challenges**
  - Over 4 lakh hectares brought under sustainable practices.
  - However, limited financial allocation and fragmented implementation across states remain challenges.

8.1.4 Climate-Smart Practices & Innovations

- Zero Budget Natural Farming (ZBNF) – reducing input costs, improving soil health.
- Millet Promotion (Shree Anna) – climate-resilient and nutritionally superior.
- Solar-Powered Irrigation Pumps – clean energy for water use.
- Crop Diversification Schemes – reducing mono-cropping risks.
- Digital Climate Services – apps, satellite data, precision agriculture.

Conclusion

The dual challenge of feeding a growing population and combating climate change demands a decisive shift toward climate-resilient and sustainable agricultural models. While initiatives like NMSA and PKVY mark crucial steps, ensuring their success requires integrated efforts—ranging from technological innovation and financial incentives to institutional reforms and farmer awareness. Only by embedding sustainability at the core of agricultural planning can India ensure long-term food and livelihood security for millions.

8.2 STRUCTURAL ISSUES IN AGRICULTURE

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Despite being the backbone of the Indian economy, agriculture in India continues to suffer from deep-rooted structural weaknesses. These issues have inhibited productivity, profitability, and long-term sustainability. Fragmented landholdings, low productivity, and poor supply chain infrastructure remain major bottlenecks to realizing the full potential of Indian agriculture in the 21st century.

8.2.1 Fragmentation of Landholdings

- Declining average landholding size: From 2.28 ha (1970-71) to ~1.08 ha (2015-16); expected to decline further.
- **Causes**
  - Inheritance laws and family divisions.
  - Lack of land consolidation policies in several states.
- **Consequences**
  - Non-viability of mechanisation.
  - Low economies of scale.
  - Difficulty in adopting modern farming technologies.
- **Policy response**
  - Model Land Leasing Act (201G) to promote land leasing.
  - Need for land consolidation, cooperative/group farming, and digitisation of land records.

8.2.2 Low Agricultural Productivity

- **Current scenario**
  - Productivity of rice, wheat, pulses is lower than global averages (e.g., rice in India: 2.7 tonnes/ha vs China: 6.7 tonnes/ha).
- **Factors responsible**
  - Fragmented and rain-fed farming.
  - Poor seed replacement rate and unscientific cropping practices.
  - Inadequate use of irrigation, fertilizers, and mechanisation.
  - Extension services are weak and unresponsive.
- **Way forward**
  - Adoption of High-Yielding Varieties (HYVs) and climate-resilient seeds.

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- Investment in micro-irrigation (PMKSY), precision farming, and extension systems (example- Kisan Call Centers, Kisan Sarathi).
- Focus on agro-ecological zones and region-specific best practices.

8.2.3 Supply Chain and Post-Harvest Issues

- **Post-harvest losses**
  - 15–20% of total produce is lost due to poor storage, transport, and market linkages.
- **Marketing bottlenecks**
  - Small farmers have limited access to mandis, and middlemen distort prices.
  - Underdeveloped value chains and poor cold storage facilities.
- **Inadequate processing**
  - Only 10% of India's agricultural produce is processed.
- **Government initiatives**
  - E-NAM for digital marketing and integration of mandis.
  - Operation Greens and PM Formalisation of Micro Food Processing Enterprises (PMFME) for storage and value addition.
  - Mega Food Parks and cold chain infrastructure under MIDH and MoFPI.
  - Promotion of FPOs for aggregation, marketing, and collective bargaining.

8.2.4 Interlinkages: Fragmentation–Productivity–Supply Chain

- Small landholdings → Difficult mechanization → Low productivity → Low surplus → Weak market power → Post-harvest losses.
- Thus, structural problems form a vicious cycle in Indian agriculture, compounding farmer distress.

Conclusion

Structural issues such as fragmented holdings, low productivity, and inefficient supply chains are interlinked and perpetuate agrarian distress. India must focus on comprehensive land reforms, promotion of

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collectivized farming models, and supply chain modernization to unlock the full potential of agriculture. Addressing these deep-rooted issues with a systems-based approach is essential for achieving inclusive, resilient, and sustainable agricultural growth.

8.3 CURRENT AFFAIRS INTEGRATION

A dynamic understanding of agricultural challenges must be supported by real-time data, recent government initiatives, and contemporary developments. Current affairs provide crucial insight into policy performance, emerging trends, and innovative responses to longstanding structural issues in Indian agriculture**. Integrating these examples is key for informed analysis in Mains answers and interview discussions.**

8.3.1 Examples of Structural Issues with Current Developments

- **Fragmentation of Landholdings**
  - **Latest Data (2021-22)**
    - 86.2% of operational landholdings are marginal (<1 ha), as per the Agriculture Census.
  - **Government Step**
    - Digital India Land Records Modernization Programme (DILRMP): Ensures land title clarity and encourages leasing.
    - SVAMITVA Scheme (2020): Drone-based land mapping in rural areas to formalise property ownership and encourage investment.
  - **NITI Aayog Recommendation**
    - Push for Group Farming Models and SHG-based farming to overcome fragmentation barriers.
- **Low Agricultural Productivity**
  - Example: India's cereal yield in 2023 stood at 3.1 tonnes/ha vs. global average of 4.2 tonnes/ha (World Bank data).
  - **Initiatives**

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- Natural Farming Mission and Zero Budget Natural Farming (ZBNF) pilot in Andhra Pradesh and Himachal Pradesh.
- PM-PRANAM (2023): To incentivise states promoting balanced fertilizer use.
- Millet Mission and declaration of 2023 as International Year of Millets: Boosting climate-resilient and nutrient-rich crops.
- Nano Urea promotion by IFFCO to reduce fertilizer dependence and improve soil health.

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- **Supply Chain and Marketing Constraints**
  - **E-NAM Expansion**
    - As of 2025, over 1,260 mandis are integrated under e-NAM; yet many farmers still sell outside it due to poor internet and mandi access.
  - **Mega Food Parks**
    - 41 sanctioned; 23 operational as of mid-2025 (MoFPI report), helping in value addition.
  - **FPO Scheme**
    - Target to promote 10,000 FPOs by 2027 under Central Sector Scheme. Over 4,300 FPOs registered as of April 2025.
  - **PM-FME Scheme (2020–25)**
    - Formalizing unorganized micro food processors. Over 80,000 enterprises assisted till March 2025.
- **Dependence on Monsoon**
  - Over 50% of India's net sown area is still rain-fed.
  - Erratic rainfall patterns due to climate change increase risks of crop failure and farmer distress.
  - Inadequate irrigation infrastructure further aggravates this dependence.
- **Soil Degradation**
  - Overuse of chemical fertilizers and intensive monocropping have led to reduced soil fertility.
  - Practices like over-irrigation and poor drainage result in salinization and waterlogging.
  - According to the ICAR, nearly 30% of India's land is degraded.
- **Farmer Distress**

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- Rising input costs, stagnant farm incomes, and low returns on produce (even with MSP) contribute to indebtedness.
- Over 50% of agricultural households are indebted (NSSO data).
- Mental health issues and suicides among farmers are a grim symptom of this broader economic distress.

8.3.2 New-Age Interventions and Future Outlook

- **Digital Agriculture Mission (2021–26)**
  - Use of AI, IoT, and drones for crop health, insurance, and market data (pilot projects in UP, MP).
- **AgriStack**
  - A unified farmer database integrating land records, crop data, and welfare scheme benefits.
- **AgriTech Startups**
  - Over 2,500 startups as of 2025 working on solutions like precision farming, real-time price discovery, and logistics.

8.3.3 Current Policy Debates and Reforms

- **Minimum Support Price (MSP)**
  - Ongoing debate on legal guarantee of MSP post-2020–21 farm law protests.
- **Private Sector Participation**
  - Increasing role in warehousing, logistics (e.g., Adani Agri Logistics, Ninjacart) amidst pushback from small farmers.
- **Agri Export Trends**
  - Export restrictions on wheat, rice, onion (2023–24) affecting farmer income and global food diplomacy.

Conclusion

The landscape of Indian agriculture is undergoing rapid transformation, driven by a mix of long-standing challenges and fresh interventions. From fragmented holdings to weak supply chains, real-time policy responses such as the promotion of FPOs, e-NAM, digital land records, and AgriStack are

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crucial. Integrating such current examples not only enriches academic understanding but also reflects the vibrancy and complexity of India's agrarian reality.

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Miscellaneous Topics

9

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9.1 WORLD TRADE ORGANIZATION (WTO)

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Global Framework of Trade and Agricultural Subsidies in India

The World Trade Organisation (WTO), established in 1995, functions as the premier international institution governing the rules of global trade. As the successor to the General Agreement on Tariffs and Trade (GATT), the WTO plays a central role in ensuring that trade flows smoothly, predictably, and freely across nations. Among its many objectives, one key area of engagement has been the regulation of agricultural trade, which is often highly distorted due to state interventions, subsidies, and protectionist policies.

India, being an agrarian economy with a vast population dependent on agriculture, faces the dual challenge of protecting its farmers through domestic support measures while also adhering to the obligations and disciplines of the WTO. This has led to complex debates around agricultural subsidies, particularly concerning Minimum Support Prices (MSP), input subsidies, and income support schemes like PM-KISAN.

9.1.1 WTO and the Global Agricultural Trade Framework

The WTO governs agricultural trade primarily through the Agreement on Agriculture (AoA), which aims to bring greater transparency and fairness in global agricultural markets. The AoA is structured around three main pillars:

- Market Access – Reducing tariffs and removing non-tariff barriers.
- Domestic Support – Classifying and disciplining subsidies.
- Export Subsidies – Reducing and eliminating unfair trade advantages through export support. The overarching goal is to ensure a level playing field where international prices reflect genuine market forces rather than state-driven distortions.

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9.1.2 India's Agricultural Subsidies: A Classification

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India provides both direct and indirect subsidies to farmers. These support mechanisms are essential for ensuring food security, enhancing productivity, and alleviating rural distress. However, they have often come under WTO scrutiny for potentially distorting trade.

- **Direct Subsidies**

  These involve direct financial transfers or price support mechanisms offered to farmers.

  - **Minimum Support Price (MSP)**
    - A pre-announced price at which the government procures agricultural commodities.
    - Although India argues it is a safety-net policy, WTO classifies this under Amber Box, which is trade-distorting.
  - **Direct Income Support (e.g., PM-KISAN)**
    - Annual cash transfer to farmers.
    - Often considered as part of Green Box support, provided it meets criteria of minimal distortion.
- **Indirect Subsidies**

  These subsidies reduce the cost of cultivation or input usage without involving direct financial transfers.

  - **Fertilizer Subsidie**s: The Government provides nutrients at below-market prices.
  - **Electricity Subsidies**: Free or highly subsidized electricity for agricultural pumps.
  - **Irrigation and Credit Support**: Canal water, crop loans at subsidized interest.
  - **Investment in Infrastructure**: Storage, cold chains, rural roads, and markets.
- These are more nuanced in WTO treatment—some may qualify as Green Box, while others (like power subsidies) may fall into Amber Box.

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9.1.3 WTO's Classification of Subsidies (Domestic Support Boxes)

WTO classifies domestic support measures into three boxes, based on the level of market distortion:

Table 9.1 WTOs Domestic Support Measures

Box	Nature of Support	WTO Status	Indian Examples
Green Box	Minimal or no trade distortion. Must not influence production or prices.	Allowed without limit	Agricultural R&D, extension services, crop insurance (if not price-linked), PM-KISAN
Blue Box	Production-limiting subsidies tied to specific conditions.	Allowed with conditions	India currently does not use Blue Box subsidies
Amber Box	Trade-distorting subsidies that affect prices and production.	Capped and subject to reduction commitments	MSP procurement, fertilizer, electricity, irrigation subsidies

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De Minimis Limit: WTO allows developing countries like India to provide Amber Box subsidies up to 10% of the total value of agricultural production. Breaching this limit may invite WTO challenges.

9.1.4 Issues Raised by WTO Regarding India's Subsidies

- **Breach of De Minimis Limit**
  - WTO members, particularly the USA and EU, have accused India of exceeding its 10% subsidy limit, especially in rice procurement under MSP.
  - India argues that current calculations use outdated reference prices (198G–88), which unfairly inflate subsidy levels.
- **Export from Public Stockholding**
  - WTO rules prohibit countries from exporting foodgrains procured under subsidized domestic support.
  - India has been questioned over export of rice procured under MSP through the Food Corporation of India (FCI).
- **Lack of Transparency**
  - WTO members have raised concerns over India's delay in notifying its domestic support data.
- **Peace Clause**
  - India invoked the WTO Peace Clause (2013 Bali Agreement) to protect its food security programs from legal challenges even if subsidy limits are breached.
  - However, the clause is temporary and conditional, and does not provide a permanent exemption.

9.1.5 India's Standpoint and Policy Imperatives

- Right to Food Security: India maintains that its support system is crucial for ensuring food security to millions of poor and vulnerable citizens.

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- Need for WTO Reform: India, along with other developing nations, demands revision of outdated subsidy calculation norms and a permanent solution to the public stockholding issue.
- Balanced Trade Rules: India advocates for fairer trade rules that accommodate the unique needs of low-income and agrarian economies.

Conclusion

The interface between WTO rules and Indian agricultural policy reflects a classic tension between global trade liberalization and domestic developmental priorities. While India has made significant strides in aligning its agricultural subsidies with WTO norms, challenges remain in terms of compliance, transparency, and flexibility. Going forward, India must balance its international obligations with its constitutional responsibility to support farmers and ensure national food security. A reformed WTO framework, with greater sensitivity towards the concerns of developing economies, remains vital to achieving equitable and inclusive trade outcomes in agriculture.

9.2 TRIPS AND INDIAN AGRICULTURE

The Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement is one of the key components of the WTO regime, aimed at standardizing intellectual property (IP) protection globally. Enforced in 1995, TRIPS significantly influences agriculture by setting minimum standards for IP rights, particularly regarding plant varieties, seeds, and biotechnology. For a country like India—with a large agrarian base and deep traditional knowledge systems—TRIPS presents both challenges and opportunities in balancing farmers' rights with IP protections.

9.2.1 Key Provisions of TRIPS Relevant to Agriculture

- **Article 27.3(b)**
  - Allows members to exclude plants and animals from patentability, but requires protection of plant varieties by either:

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  - Patents, or
  - An effective sui generis (unique) system, or
  - A combination of both.
- This provision is particularly relevant for seed sovereignty and protection of indigenous varieties.

- **Patents on Biotechnological Innovations**
  - TRIPS mandates patents on micro-organisms and microbiological processes, which affects GM crops, agricultural chemicals, and agri-biotech companies.
- **Protection of Traditional Knowledge**
  - Although not directly covered under TRIPS, its implementation can threaten farmers' traditional rights, especially when corporate patents are based on indigenous knowledge without adequate benefit sharing.

9.2.2 India's Response to TRIPS in Agriculture

- **The Protection of Plant Varieties and Farmers' Rights Act (PPV&FR), 2001**
  - India adopted a sui generis system to protect plant varieties while ensuring farmers' rights.
  - **Key features**
    - Farmers can save, use, sow, resow, exchange, share or sell farm produce including seeds.
    - Recognition and reward for conserving traditional varieties.
    - Breeders are granted exclusive rights for commercialization of new varieties.
- **Biodiversity Act, 2002**
  - Safeguards biological resources and associated traditional knowledge.
  - Establishes mechanisms for Access and Benefit Sharing (ABS).
- **Patent Act Amendments (2002, 2005)**
  - Aligns with TRIPS but excludes methods of agriculture, plants, animals, and seeds from patenting in India.
  - Ensures that biotech patents do not override basic farming rights.

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9.2.3 Impacts and Challenges of TRIPS on Indian Agriculture

- **Positive Impacts**
  - Encourages research and innovation in seed technology and agri-biotech.
  - Helps Indian agri-companies gain IP protection globally.
  - Fosters public-private partnerships in plant breeding.
- **Challenges**
  - Risk of corporate monopoly over seeds and agrochemicals through patents.
  - High seed costs due to private ownership may marginalize small farmers.
  - Biopiracy concerns – patents based on indigenous plants or knowledge without consent or benefit sharing (e.g., neem, turmeric).
  - Diminishing community rights and informal seed systems.
  - Limited capacity among farmers to understand or challenge IP claims.

9.2.4 Way Forward

- Strengthen farmers' education and awareness about their rights under PPV&FR.
- Promote open-source seed systems and community seed banks.
- Enhance R&D in public sector institutions to counterbalance private dominance.
- Ensure effective implementation of biodiversity and IP laws to prevent biopiracy.
- Push for TRIPS reform at global forums to include better protection of traditional knowledge and farmers' rights.

Conclusion

While TRIPS brings global standards and fosters innovation, its implications on Indian agriculture are complex. India's proactive adoption of farmer-centric legislation like the PPV&FR Act showcases a balanced approach between compliance and protection. Moving forward, a careful policy mix is essential to preserve the rights of Indian farmers, prevent

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monopolies, and ensure food and seed sovereignty in the face of expanding global IPR regimes.

9.3 AGREEMENT ON AGRICULTURE (AOA)

The Agreement on Agriculture (AoA) is a key international treaty under the World Trade Organization (WTO) that governs global trade in agricultural products. Signed in 1995 as part of the Uruguay Round agreements, the AoA aims to create a fair and market-oriented agricultural trading system by reducing trade-distorting subsidies, lowering tariffs, and eliminating non-tariff barriers. While developed nations pushed for liberalisation, developing countries—like India—have had to balance trade openness with the imperative of food security, farmer livelihoods, and rural development.

9.3.1 Salient Features of the Agreement on Agriculture (AoA)

- The AoA is built on three key pillars:
  - **Market Access**
    - Seeks to reduce trade barriers like tariffs, quotas, and import bans.
    - Countries were required to convert all non-tariff barriers into tariffs (tariffication) and reduce them over time.
    - Developed countries committed to an average 3G% reduction in tariffs over G years; developing countries to a 24% reduction over 10 years.
    - Least Developed Countries (LDCs) were exempted from tariff reductions.
  - **Domestic Support**
    - Focuses on reducing trade-distorting domestic subsidies provided to farmers.
    - Subsidies are categorized into three boxes based on their impact on trade (explained in detail in next section):
- Amber Box (most trade-distorting): Subject to reduction commitments.
- Blue Box (production-limiting): Exempt from reduction.

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- Green Box (non-trade-distorting): Allowed without limits (e.g., research, environmental subsidies).
  - Developed countries agreed to reduce their Aggregate Measure of Support (AMS) by 20% over six years, while developing countries had to reduce it by 13.3% over ten years.
  - India maintains its AMS within de minimis limits (below 10% of total agricultural production for developing countries).
  - **Export Subsidies**
    - Aims to reduce or eliminate subsidies given to promote exports.
    - Developed countries committed to 3G% reduction in export subsidy expenditure and 21% reduction in volume over G years.
    - Developing countries committed to 24% and 14% reductions respectively over 10 years.
    - India does not provide direct export subsidies but is affected by subsidized agricultural exports from developed countries that depress international prices.

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9.3.2 India's Position on the AoA

- Food security concerns: India has consistently advocated for flexibility for developing countries to maintain public stockholding for food security purposes.
- India maintains that its Minimum Support Price (MSP) and Public Distribution System (PDS) are essential for rural welfare and do not distort international trade.
- The AoA does not adequately account for inflation, currency fluctuation, and rising procurement prices, putting Indian policies at risk of violating subsidy limits.
- India and the G-33 group have demanded a permanent solution for public stockholding, which remains unresolved.

9.3.3 Criticism of the AoA

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- Unequal commitments: Developed countries had already high subsidies and tariffs and made marginal cuts, whereas developing nations had to liberalise more deeply.
- Loopholes: The AoA allowed developed nations to shift subsidies from Amber to Green Box, continuing distortion under a different label.
- Inadequate protection for small farmers: The Agreement fails to safeguard the subsistence and marginal farmers in developing countries from global price fluctuations and import competition.

9.3.4 Recent Developments

- Bali Package (2013): Allowed interim protection for public stockholding programmes for food security (Peace Clause), but without a permanent solution.
- Nairobi Ministerial (2015): Some developed countries agreed to phase out export subsidies, but domestic support remains contentious.
- Ongoing discussions continue under the WTO Committee on Agriculture, where India has been vocal on the need for equity and development-oriented reforms.

Conclusion

The Agreement on Agriculture was a landmark attempt to liberalise global agricultural trade. However, it has largely favoured developed countries, allowing them to maintain their large subsidies under non-actionable categories. For India, the AoA poses significant challenges in preserving its food security systems, farmer welfare mechanisms, and rural employment. The need for a reformed, equitable, and development-sensitive framework remains urgent to protect the interests of agrarian economies in the global trading system.

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SECTION II

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UPSC Mains Model Answers: Agriculture (2025 – 2023)

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1. Explain the factors influencing the decision of the farmers on the selection of high value crops in India.

(10 marks; 150 words)

Ans. High value crops refer to horticultural produce like fruits, vegetables, spices and flowers, which generate higher returns due to market demand and specialized cultivation processes. India is the 2nd largest producer of fruits and vegetables highlighting the growing demand of high value crops among farmers.

The factors which influence the decision of farmers for the selection of high value crops (HVCs) in India can be summarised as follows:

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1. Market demand: Despite the perishable nature of HVCs, high market demand has increased the risk appetite of farmers due to higher return on investment.

2. Policy Support: Under the MIDH programme, the government encourages farmers to cultivate HVCs through distribution of high quality planting materials, financial and technical support for raw inputs and farm mechanization and aids in supply chain connectivity to rural areas. Moreover, initiatives lik e-NAM has further boosted farmer confidence in adoption of HVCs.

3. Higher Income: As compared to traditional food crops, HVCs are profitable due to their demand and market prices. Moreover, contract farming and digital markets like e-NAM has helped in better price realisation by farmers.

4. Poverty Alleviation: Higher price realisation leads to breaking the cycle of seasonal poverty that farmers face due to lower prices and cyclical nature of food crops.

5. Soil Health: While food crops like rice and wheat are know to degrade the soil quality over time, HVC cultivation provides manure which increases soil fertility through nutrient cycling, improves water retention and carbon sequestration capacity of the soil. This in turn reduces the input cost for further cultivation which encourages the farmers to cultivate HVCs.

6. Nutritional Security: Self-consumption of HVCs like fruits and vegetables reduce the instances of malnourishment among the rural agrarian population. This reduces their need to incur expenditure on market-based products reinforcing their decision to pivot to cultivation of HVCs.

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HVCs contributes 33% to the agricultural GDPPP with only 13.1% of cropped area. The way forward includes capacity building among small and marginal farmers to integrate HVC cultivation in their cropping pattern, R&D for increasing yield and shift to modern farming methods like aeroponics and use of nanotechnology in cropping methods for realisation of the full potential of HVC cultivation in India.

2. Elaborate the scope and significance of supply chain management of agricultural commodities in India.

(10 marks; 150 words)

Ans. Supply chain management (SCM) in agriculture encompasses the planning, execution, and control of the flow of agricultural produce from farm to fork, including production, storage, transportation, processing, distribution, and marketing. In India, SCM is critical for food security, farmer income enhancement, and integration with global agri-markets.

Scope of SCM for agri-commodities in India:

1. Pre-harvest Stage:

- Ensure on-time availability and accessibility through government support and Direct Benefit Transfer for agricultural inputs. For example, seeds, fertilizers, farm machinery, pesticides etc.
- Production scheduling and demand forecasting using MIS and AI helps align market demand with production cycle enable informed decision making in crop selection.

2. Post-Harvest Stage:

- Mechanization of harvesting and primary processing can improve efficiency and timely delivery of agri-commodities to markets.
- Storage and cold chain infrastructure reduces wastage due to perishability and rodent infestations. For example, PM-Kisan Sampada Yojana focuses on developing SCM infrastructure for better availability of produce.

3. Marketing and Distribution:

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- Direct Farmer to market linkages through e-NAM, FPOs and contract farming reduce losses and enable better price realisation, by avoiding exploitative intermediaries.
- APEDAs Agri-Export Zones enable farmers access to high-end storage infrastructure, quality standards of processing and direct linkage with global markets.

4. Food Processing (FP): Several initiatives like PM-KSY, PLI schemes for FP and Mega Food Parks reduce post harvest losses and generate rural employment through setting up of FP industries, creating a resilient supply chain.

5. Digital Integration: IoT for cold storage infrastructure, blockchain for logistical traceability and AI for demand prediction can bolster the supply chain for agri-produce by preventing wastage, leakage and demand-supply mapping.

Significance of SCM for agri-commodities in India:

1. Reducing post-harvest losses: Effective SCM provides cold chains, warehouses, pack-houses, and reefer logistics, ensuring produce reaches markets fresh.

Example: Operation Greens under MoFPI reduces wastage of tomato, onion, and potato by creating integrated cold chains.

2. Augmenting farmer's income: By cutting down intermediaries through FPOs, e-NAM and Contract Farming as well as higher demand augments farmer's income. It reduces distress sales due to storage facilities and farmers transform from "price-takers" to "price-negotiators".

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3. Ensuring food security and price stabilisation: Agriculture faces seasonal gluts and shortages, leading to price volatility. SCM helps by:

• Buffer stocking (through FCI, NAFED)

• Efficient logistics for Public Distribution System (PDS)

• Maintaining nutritional security through timely supply of perishable food.

Example: During COVID-19 lockdown, SCM ensured continuity of food supply chains even under mobility restrictions

4. Export competitiveness: SCM ensures compliance with global quality and traceability standards. For example: HACCP, Organic Certification etc. Moreover, efficient SCM reduces the logistics costs (for developed countries it is at 7-8% whereas for India it stands at 14% of GDP)

5. Rural employment: Agri-logistics, warehousing, packaging, and processing generate non-farm rural employment. According to MoFPI, the food processing sector employs 1.8 million people directly and supports 5.1 million indirectly.

6. Women Empowerment: Women constitute about 1/3rd of the post-harvest agricultural workforce. Opportunities in cooperatives, SHG participation in value chains and food processing units can uplift the women in rural areas by providing financial independence. E.g.: Kudumbashree Model in Kerala.

An efficient SCM is the **backbone of food security, farmer prosperity, and global trade competitiveness**. Despite challenges like regulatory bottlenecks and fragmented supply chains, recent initiatives are transforming agriculture from a subsistence activity to a remunerative

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enterprise. Future progress lies in digitally enabled, climate-smart, and inclusive supply chains to achieve the goals of Atmanirbhar Bharat and Doubling Farmer's incomes.

3. Examine the scope of the food processing industries in India. Elaborate the measures taken by the government in the food processing industries for generating employment opportunities.

(15 marks; 250 words)

Ans. The food processing (FP) sector is considered a Sunrise industry with huge scope for growth. As of 2023, the food processing sector grew at an average annual rate of roughly 5.35% over eight years.

Scope of food processing industry

- **Availibilty of raw material:** India ranks first globally in dairy production and is a food-surplus country in almost all sectors except edible oils and pulses. The high production volume, is beneficial for providing raw inputs to the FP industry.
- **Demand:** With rising middle class, the amount of disposable income increases providing opportunities for FP in the domestic market. Globally, the FP industry s projected to reach $300 Bn by 2031 providing significant scope for development.
- **Global competitiveness:** With enablement of 100% FDI through the automatic route, and low cost of produce, value-added products will be more competitive.
- **Formalization of sector:** More than 70% FP industries are spread in the unorganised sector. Thus there is a huge scope of untapped growth that can be realised by formalizing the sector.

The food processing sector is a **priority under the Make in India initiative,** with the Ministry of Food Processing Industries implementing schemes to attract investment and develop infrastructure. The government has undertaken several initiatives to generate employment in the food processing industry:

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1. PM-Kisan SAMPADA Yojana: It aims to create all-round infrastructure like food processing units, quality units and cold chain infrastructure for Agro-Marine and Agro-processing in India. Infrastructure creation and utility creates huge employment opportunities often in rural areas during the lean season.

2. PLI Scheme for FP industry: The scheme provides support to three categories of FP industries:

- I- Incentivises manufacturing of Ready-to-cook/eat foods
- II- Production of innovative/organic products of SMEs
- III- Support for global branding and marketing

These initiatives will provide employment in the non-farm sector and will employ both skilled and semi-skilled labours in the branding and innovation, and manufacturing sectors respectively.

3. PM-Formalisation of Micro FP Enterprises Scheme: This initiative aims to formalise and grow the micro FP enterprises through financial assistance and capacity building bridging the gap between formal and seasonal employment in the food processing sector.

Moreover, the ODOP program enhances employment at the district level by employing local people in creating value added products from agricultural produce.

4. World Food India: It is an annual event aimed at showcasing India's rich and diverse food culture while fostering global investment. Initiatives like B2B/B2G meeting enable industry connections and growth opportunities for small businesses, which in turn has the potential to generate significant employment in the sector.

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Despite being a major producer, the level of food processing in India, particularly for fruits and vegetables, remains low compared to global standards. Focusing on innovation and sustainability as well as in scaling up of FP industries, supported by government policies can be the driving force for geenrating large-scale employment in the FP industry.

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4. What were the factors responsible for the successful implementation of land reforms in some parts of the country? Elaborate.

(10 marks; 150 words)

Ans.Land reforms in India have been a field of continuous effort in which policymakers have tried to address underlying issues like exploitation of tenant rights, over-extraction of land revenue, low agrarian productivity and social inequalities.

However, since Agriculture is a constituent of the state list in the Seventh Schedule of the Indian Constitution, the success of land reforms varies across states, with certain states like West Bengal and Kerala showing the highest extent of efficient implementation of all forms of land reforms.

The factors associated with such success include:

- **Strong political will** ensured effective policy implementation and its sustenance.
  - Operation Barga by the West Bengal government to protect sharecroppers' rights.
  - Decisive action by leaders like EMS Nambroodipad in the Kerala government.
- **Social movements and socio-cultural shifts** initiated participation at the grassroots levels.

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- The Tebhaga movement in Bengal, the Deccan Rebellion in Telangana, the Bhoodan Movement in Maharashtra
- Anti-caste movements in Kerala as part of a broader societal reorganisation.

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- **Bureaucratic Efficiency** in states like Tamil Nadu through simplified land record systems expedited implementation.
- **Decentralised governance** limited the influence of landed elites and ensured community participation through Panchayati Raj Institutions in West Bengal & Gujarat.
- **Modernisation of land records** through GIS mapping and digitisation of ownership reduced disputes and facilitated the timely extraction of records.

Northern States like UP, Bihar and Assam saw extensive collusion between the landed elites and the state government, leading to demotivation of public officials in effectively implementing land reforms. Thus, model states like Kerala, West Bengal and Tamil Nadu, coupled with recommendations of the **Kumarappa Committee**, can serve as examples for future implementation of land reforms in the remaining states.

5. Explain the role of millets for ensuring health and nutritional security in India.

(10 marks; 150 words)

Ans. The UN and FAO declared **2023** as the **International Year of Millets** following India's proposal. This step underscores the need to understand the nutritional value and climate resilience of millets compared to the more widely consumed cereals.

- Millets are **coarse cereals** that provide protein, dietary fibre, iron and calcium along with carbohydrates, making them a much **nutrient nutrient-dense** option for the malnourished population of India.
- As per research, millets are also known to be a good **defence against diabetes** by lowering blood sugar spikes after consumption. India is called the 'diabetes capital' of the world, and hence, millets could be helpful in

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managing the health of a huge number of affected patients as an **alternative to rice and wheat**.

- Millets use **70% less water** than rice, grow in **half the time** of wheat and need **40% less energy in processing**, making them a more sustainable crop as compared to water intensive wheat and rice.
- Millets can also **withstand extreme heat and drought** conditions, hence are labelled as hardy crops that are adaptable to climate change and global warming.

Millets, once cultivated widely, lost significance after the Green Revolution's focus on rice and wheat, reducing millets to a fodder crop. However, with government support, millets are making a comeback.

- Their hardy nature **empowers farmers to adapt** to changing climatic conditions, promoting biodiversity and achieving sustainable development.

Even after 78 years of independence, malnutrition stands at greater than 30% among children and around 20% among women. Integration of millets as a nutrition source under Mission Poshan and awareness programmes can help ensure **health and nutritional security** in India.

6. What are the major challenges faced by the Indian irrigation system in recent times? State the measures taken by the government for efficient irrigation management.

(15 marks; 250 words)

Ans. As per the Economic Survey 2024-25, the GoI has prioritised irrigation development and water conservation practices to enhance access to irrigation facilities. However, irrigation in India, despite sustained progress in percentage of irrigated area has faced a number of **challenges:**

1. **Climate change-** Warming of the Indian Ocean has led to unpredictable monsoons. Since most of the **agriculture is rainfed,** it reduces the yield

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quality of Rabi crops like wheat and reduces germination rate for Kharif crops.

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2. **Water stress** - As per 2017 CGWB report, 17% of water resource assessment area across 17 states are **over-exploited**. This is mainly due to **subsidized electricity** that allows farmers to over-extract water using water pumps.

3. **Soil degradation-** Over-irrigation and excessive fertilizer usage has increased osmotic and alkaline stress in soil reducing crop quality.

4. **Water pollution-** Surface runoff due to over-irrigation and unscientific methods like canal well and surface irrigation methods have led to pollution of water bodies.

5. River water sharing disputes - Disputes between states like Cauvery dispute between Karnataka and Tamil Nadu, Narmada dispute between Rajastha, Gujarat, Maharashtra and MP have also led to delays in release of water from upstream states for irrigation into downstream states leading to water stress.

6. Ineffective policy implementation: Earlier policies were mostly distributed across departments which created coordination issues and irrigation being a state subject leads to non-uniform implementation of schemes.

7. Lack of water conservation infrastructure: Improper implementation of schemes and capital intensive nature of water conservation structures like watersheds and dams makes individual farmer contribution difficult.

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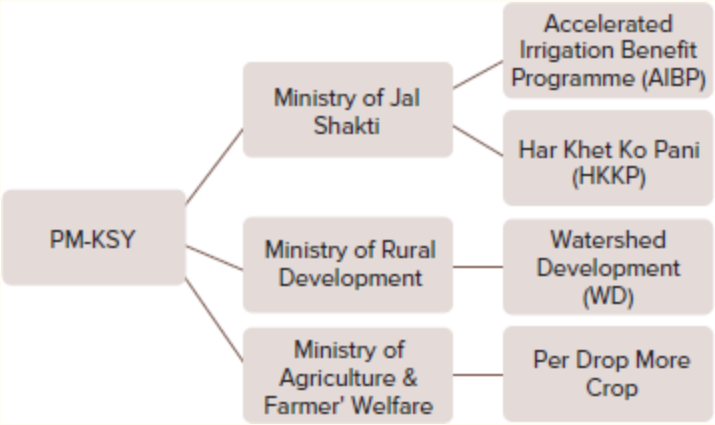
Despite the challenges. The GoI has integrated the irrigation schemes under an umbrella schemes and formulated schemes to address structural issues for premanent redressal of irrigation needs in India.

Initiatives:

1. PM-Krishi Sinchayee Yojana (KSY)- It is an **umbrella scheme** under which various existing and new sub-schemes are implemented. The objective of this scheme includes **enhanced physical access of water** on farm and **expand cultivable area** under assured irrigation, **improve on-farm water use efficiency**, introduce **sustainable water conservation** practices, etc.

It has the following components:



2. PM-KUSUM - To address water shortage, the scheme provides financial assistance to farmers for standalone solar pumps' installation for sustainable extraction of groundwater. This will ensure **water security for small farmers.**

3. Damodar Valley Corporation Irrigation Project- The irrigation component of DVC aims to enhance agricultural productivity and support farmers in West Bengal through-

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- Development of irrigation infrastructure like dams to simultaneously control floods and provide water for irrigation.
- It also uses solar energy plants in order to provide power for extraction of water for irrigation.

4. River interlinking projects - These projects aim to mitigate irrigation issues by transferring water from surplus to deficit areas. Example: Ken-Betwa project.

Due to unpredictability of monsoons and India's rainfed agriculture, irrigation systems need to be strengthened according to the need of different ago-climatic regions and availability of water across different regions.

7. Elucidate the importance of buffer stocks for stabilizing agricultural prices in India. What are the challenges associated with the storage of buffer stock? Discuss. (15 marks; 250 words)

Ans. Buffer stocks refer to a mechanism of price stabilisation through the controlled release of agricultural produce during fluctuations in production. Introduced during the fourth 5-year plan (1969-74), buffer stocks in India are managed by FCI (Food Corporation of India) on behalf of the GoI, facilitating monthly releases of food grains for supply through PDS and during emergencies like crop failure, natural disaster, war, famine etc.

Buffer stocks are crucial for maintaining the stability of agricultural prices in India as:

1. It acts as a **safeguard against price volatilities** during times of scarcity or surplus, by controlling the availability of food crops in the open market. **Eg:** In 2023, GoI released **Tur Dal** in a targeted manner to meet demand due to a shortage caused by import delays.

2. It ensures **food security** by releasing food grains to the public through PDS during crises or natural disasters. **Eg:** Following the **Wayanad**

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landslides, the Kerala government released free ration to the affected population.

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3. Buffer stocks also facilitate **income stability** for farmers by procuring food grains at MSP, providing an assured income.

4. Controlled release of buffer stocks also helps maintain **economic stability** by preventing inflation.

5. Buffer stocks also aid in policy implementation by supplying to consumers at subsidized rates through PDS under NFSA, PM-GKAY etc.

In India, buffer stocks stand at more than 700 lakh MT which is almost 80% greater than the buffer norms. While this is beneficial for ensuring domestic food security, there are certain challenges associated with it:

1. Economic costs: The operational costs associated with buffer stocks are always increasing due to:

- Higher acquisition costs due to rising MSPs (rise by >100% for most crops since 2013-14)
- Inadequate storage infrastructure (~30% owned) has led to FCI hiring storage from private sector which has contributed to substantial increase in hire charges as per CAG 2023 report.
- Delays in land acquisition for augmentation of storage capacity incurs extra hiring costs of storage silos.

2. Open ended procurement: GoI in order to fulfil MSP policy has become the largest procurer of grains (almost 1/3rd of grains). This large stockholding has been flagged by WTO as an Amber Box measure.

3. Inefficient inventory management: Despite being a balancing measure, the release of stocks through Open Market Sale Schmes (OMSS) is very ad-

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hoc in nature. With no policy in place for OMSS, release of inventory instead destabilises markets through sudden releases.

4. Increasing farmer dependency: More than being a protective measure, MSP has become the procurement price for any quantity of food grains sold by farmers. As a result:

- Farmers stick to "safe crops" increasing their dependency on government support potentially losing open market benefits.
- Subsidy bill of government has increased exponentially from ~ INR 7 lakh crore (2004-13) to INR 18 lakh crore (2013-24) as per PIB.

While buffer stock implementation faces challenges due to both domestic and international pressures, a shift towards structured agri-marketing employing digital innovations, can be a step towards reducing reliance on buffer stocks for food security and economic stability of farmers. Initiatives like **e-NAM** is a step towards this direction.

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8. How does e-Technology help farmers in production and marketing of agricultural produce? Explain it.(10 marks; 150 words)

Ans. As per Budget 2025-26, agriculture sector in India contributes around 18.2% to the country's GDP with an annual growth of 5% (2017 to 2023), which is less compared to service and manufacturing sectors.

E-technology in this regards can be a catalyst to speed u the growth of agriculture by promoting awareness and government interventions to formalise the sector. It can ultimately help in enhancing the production and marketing of agriculture enabling better market relaisation for farmers.

GoI has taken up development of DPI for agriculture as an open source, open standard and inter-operable public good to enable inclusive farmer centric solutions through relevant information services under the National e-Govrnance Plan for Agriculture (NeGP-A).

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e-Technology in Production:

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1. Awareness of modern technologies: Digital platforms like Kisan Sarathi and Kisan Suvidha provides a plethora of services and knowledge base to enable farmers to adopt modern scientific technologies of cropping, irrigation and animal rearing.

2. Soil Health: Agencies like ISRO leverages remote sensing and geospatial techniques to map soil fertility and provide farmers with tailored nutrient recommendations.

3. Precision Agriculture Schemes like Drone Didis leverage IoT to apply fertilizers and pesticides to crops in efficient quantities compared to unscientific methods used earlier. It prevents soil degradation, reduces GHG emissions and water pollution.

4. Procurement of genetic material: To improve breed quality of indigenous animals and improve productivity, GoI has set up e-Pashuhaat, an online platform for procurement of germplasm of improved quality for artificial insemination.

e-Technology in Marketing

1. e-NAM: Under the NeGP-A framework, the government has set up a digital marketing platform to enable agri-marketing across the country without depending on APMCs or local infrastructure. This helps with better market realization and prevents exploitation by middlemen.

2. Integrated Scheme for Agricultural Marketing schemes **(AGMARKNET):** It is formulated to promote creation of agricultural

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marketing infrastructure by providing backend subsidy support to State, cooperative and private sector investments

3. e-Choupal: An initiative by ITC, it provides a market platform for agri-produce and also provides cold-storage services for farmers for effective price realisation by extending shelf life of perishable products.

Rapid developments in IT has enabled farmers to come out of the obscurity of rural areas and participate in a more competitive market through access to a knowledge base and set of services provided by both the government and private sector. Further developments would require expediting the Agri-stack for open access to beneficial data.

9. Explain the changes in cropping pattern in India in the context of changes in consumption pattern and marketing conditions.

(15 marks; 250 words)

Ans. India spans across various agro-climaic regions resulting in varied production and consumption patterns. This has however been particular influenced by food availability and cropping patterns adopted thereby changing the supply of crops based on demand and other prevailing conditions.

Evolution of cropping pattern in India

1. Wheat and rice bias- In the aftermath of independence, there was widespread poverty and famine due to import dependency of food grains in India. Spearheaded by Dr. M.S. Swaminathan, the Green Revolution focused on HYV rice and wheat cultivation transforming India into a food surplus country.

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As a result, the widespread availability of rice and wheat altered consumption patterns. So farmers started cultivating rice and wheat even in semi-arid regions like Punjab and HAryana in a widespread manner.

2. Poultry and Dairy- Post the successful Green revolution, the White revolution was started in the 1970s. Completed in 3 phases, Operation Flood made India the largest producer of dairy globally. With increasing incomes, households also included dairy in their diet. This was especially made possible by the establishment of cooperatives like Amul and Mother dairy which has a wide marketing reach.

3. Horticulture - In the early 21st centure, monocropping practices put a strain on the soil quality and also scarcity became a major problem by the second decade (around 2015). So farmers were encouraged to diversify with high value crops. Schemes like National Horticlture Missin helped in this transition.

After the 2020s, the increasing income of the rural population due to MSP policy and dairy income, there is diversification of diet from cereals to including fruits and vegetables along with pulses among the rural population.

4. Resurgence of Millets- Millets are a hardy crop that is comparatively more drought and pest resistant compared to other cereals like rice and wheat. Moreover they are labelled as “Nutri-Cereals” due to the abundance of vitamins and minerals. Recognizing their importance UN recognized 2023 as “International Year of Millets”.

The GoI has promoted millets as “Shree Anna” and formulated NFSM-Nutri Cereals to encourage cultivation. Also, lifestyle modifications has led

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to consumers adopting millets as part of their diet.

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Millets also provide an alternative source of income due to their resilience making them an attractive option. So MSP policies coupled with other schemes have led to the resurgence of millets as part of Indian diet in the revcent years.

In Q4 f 2024, rural demand for FMCG goods outpaced urban demand. This shows an increase in capacity of the rural population to partake in the market. This change can be attributed to shifts in agriculture based on production and consumption demans. Continuous government support has also aided in this growth. To keep this momentum, e-technologies can play an important role in further growth of the agriculture sector.

10. What are the direct and indirect subsidies provided to farm sector in India? Discuss the issues raised by the World Trade Organization (WTO) in relation to agricultural subsidies.

(15 marks; 250 words)

Ans. Agriculture Subsidies refer to government financial support paid to farmers and agribusinesses to promote agricultural development and self-sufficiency by lowering the investment barriers required to increase crop productivity..

Agricultural subsidies are given at two levels- centre and state and based on mode of payment it can be classified as follows:

1. Direct subsidy: These include direct payment to farmers, mostly in the form of cash. Schemes under this category include:

- Direct Cash Transfer: PM-KISAN, PM-PKVY for organic farming
- Distribution of agri-inputs: SMAM distributed tractors, generators, etc for farm mechanisation, Namo Drone Didi scheme supplied drones to SHGs

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for rentals by farmers.

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2. Indirect Subsidy: These include subsidies through price reduction and welfare measures without DBT. Ex:

- Fertiliser subsidy (NBS scheme), Power subsidy (PM-Surya Ghar Muft Bijli)
- MSP for 23 crops
- Credit Subsidy through interest subvention
- Farm loan waivers

The WTO has classified support provided by countries in agriculture into **three categories based on their trade-distorting capacity:** Green Box, Amber Box and Blue Box.

Intending to reduce trade distortion through Amber Box measures, the WTO has raised the following issues in the context of agricultural subsidies:

1. **Amber Box Subsidies and outdated reference prices:** Under the Agreement on Agriculture, developing countries are to cap at 10 % and developed countries at 5 % on Aggregate Measurement Support.

However, India has argued that **1986-88 reference prices** and large AMS base of developed countries underplay their support, while developing countries are sought to be penalised under the 10% cap, which is comparatively meagre when compared to production by developed countries.

2. Export and Domestic Support: Despite not having explicit subsidies, India faces scrutiny from Brazil & Australia for agricultural marketing and transport support for the export of sugar as trade-distorting.

Public schemes like MSP for crops are seen as trade-distorting as they artificially inflate rice and wheat prices in international markets, thereby

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distorting trade.

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3. Public Stockholding: Developing countries often procure food to ensure food security and to supplement PDS schemes. India's procurement by the government through FCI has come under scrutiny.

Despite the temporary 'peace clause', developed countries accuse that it possesses the potential to distort the international trade of wheat and rice by releasing surplus from public stockholding.

Nevertheless, India asserts that these subsidies are crucial for ensuring food security, alleviating poverty, and supporting rural livelihoods, advocating for a reform of WTO rules that is more equitable and focused on development.

Multistakeholder consultations in WTO and domestic interventions like **Universal Basic Income** over subsidies, R&D investment in agriculture and development of food storage and processing infrastructure over trade-distorting tax subsidies can be the way forward to arrive at a consensus with developed countries at the WTO.

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PART C Disaster Management

Part C – Disaster Management

Section - I **Disaster Management: A Complete Overview**

Chapter - 1 Disasters in India

Chapter - 2 Institutional Framework

Chapter - 3 Prevention and Mitigation

Chapter - 4 Preparedness and Response

Chapter - 5 Recovery, Reconstruction and Rehabilitation

Chapter - 6 Capacity Development

Chapter - 7 International Cooperation

Chapter - 8 Types of Disaster – Their Causes, Vulnerability and Mitigation Measures

Section - II **Glossary**

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Section - III UPSC Mains Model Answers: Disaster Management (2025–2023)

(2013–2022 onwards Disaster Management: Mains Model Answers available on McGraw Hill Edge platform)

Upcoming Disaster Management: Mains 2026 -2027 Model Answers

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SECTION I

Disaster Management: A Complete Overview

Disasters in India 1

IN THIS CHAPTER...

1.1 Etymology
1.2 Definition
1.3 Hazard
1.4 Vulnerability
1.5 Risk
1.6 Types of Disaster
1.7 Disaster Management and its stages
1.8 Phases of Disaster Management
- 1.8.1 Disaster Preparedness
- 1.8.2 Disaster Response
- 1.8.3 Disaster Recovery
- 1.8.4 Disaster Mitigation

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1.9 Database
1.10 NDEM
1.11 India's Deadliest disasters
1.12 Indian Scenario
1.13 Vulnerability Profile of India
1.14 Hazard Profile of India
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1.14.3 Landslides
1.14.4 Industrial, Chemical and Biological Disasters
1.14.5 Tsunami
1.14.6 Nuclear Emergencies
1.14.7 Epidemics in India
1.15 Cascading effect of Disaster – Earthquake, Tsunami, fire and nuclear disaster in Japan

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1.1 ETYMOLOGY

The term 'disaster' originates from the French word **Desastre** and Italian word **Disastro**. The word is a combination of two words, 'des' meaning bad and 'aster' meaning star refers to '**Bad or Evil Star**' and resultantly termed as the calamity due to position of a planet or a star.

1.2 DEFINITION

Disaster is defined as serious interruption in the operation of society causing huge material, economic, social or environmental losses, which surpass the capability of affected society to cope using its own resources. It is an outcome of the hazard, vulnerability, and insufficient capacity

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combination to reduce potential chances of risk. Disaster occurs when hazard occurs on vulnerable population causing huge damage, casualties and disruption.

The **Disaster Management Act, 2005** defines disaster as “a catastrophe, **mishap**, calamity or grave occurrence in any area, arising from natural or manmade causes, or by accident or negligence which results in substantial loss of life or human suffering or damage to, and destruction of, property, or damage to, or degradation of, environment, and is of such a nature or magnitude as to be beyond the coping capacity of the community of the affected area”.

The UNISDR (2016) defines `disaster' as "A serious disruption of the functioning of a community or a society at any scale due to hazardous events interacting with conditions of exposure, vulnerability and capacity, leading to one or more of the following: Humans, material, economic and environmental losses and its impact."

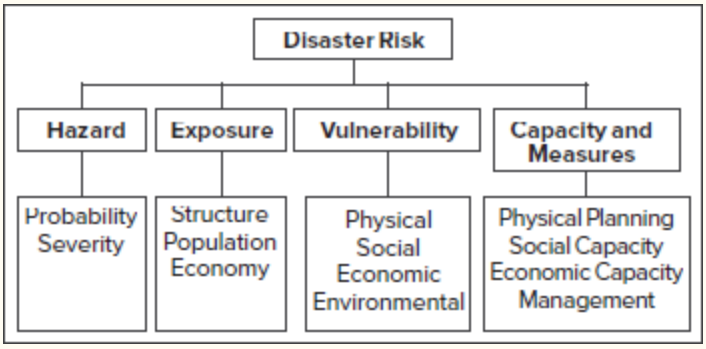

Figure 1.1 Types of Disaster Risk

1.3 HAZARD

Hazard may be defined as “a phenomena, dangerous condition or event that has a threat or the potential to cause damage to life or property or the environment.” Hazards are classified into two broad categories one is natural and the other one is manmade.

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"Hazard is a natural event while disaster is its consequence". A hazard is perceived as a natural event that threatens both life and property. A hazard consisting of risks, vulnerabilities and capacities may lead to a disaster. Hazards may be inevitable but disasters can be prevented or controlled. According to John Whittow, "**A disaster is a realisation of this hazard**."

Various types of Hazards are categorised on the basis of their genesis as follows:

Table 1.1 Various Types of Hazards

Types	Hazards
Geological	Earthquake, Tsunami, Volcanic eruption, Landslide, Dam burst, Mine fire
Water & Climatic	Tropical Cyclone, Tornado and Hurricane, Floods, Drought, Hailstorm, Cloud-burst, Landslide, Heat & Cold wave, Snow, Avalanche, Sea erosion
Environmental	Environmental pollutions, Desertification, Deforestation, Pest Infection
Biological	Human / Animal Epidemics, Food poisoning, Pest attacks, Weapons of Mass Destruction
Chemical, Industrial and Nuclear	Chemical disasters, Oil spills/Fires, Nuclear, Industrial disasters

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Types	Hazards
Accidents, Accident Related	Boat / Road / Train accidents / air crash/ Rural or Urban fires / Bomb /serial bomb blasts, Building collapse, Electric Accidents, Festival related disasters, Mine flooding, Forest fires

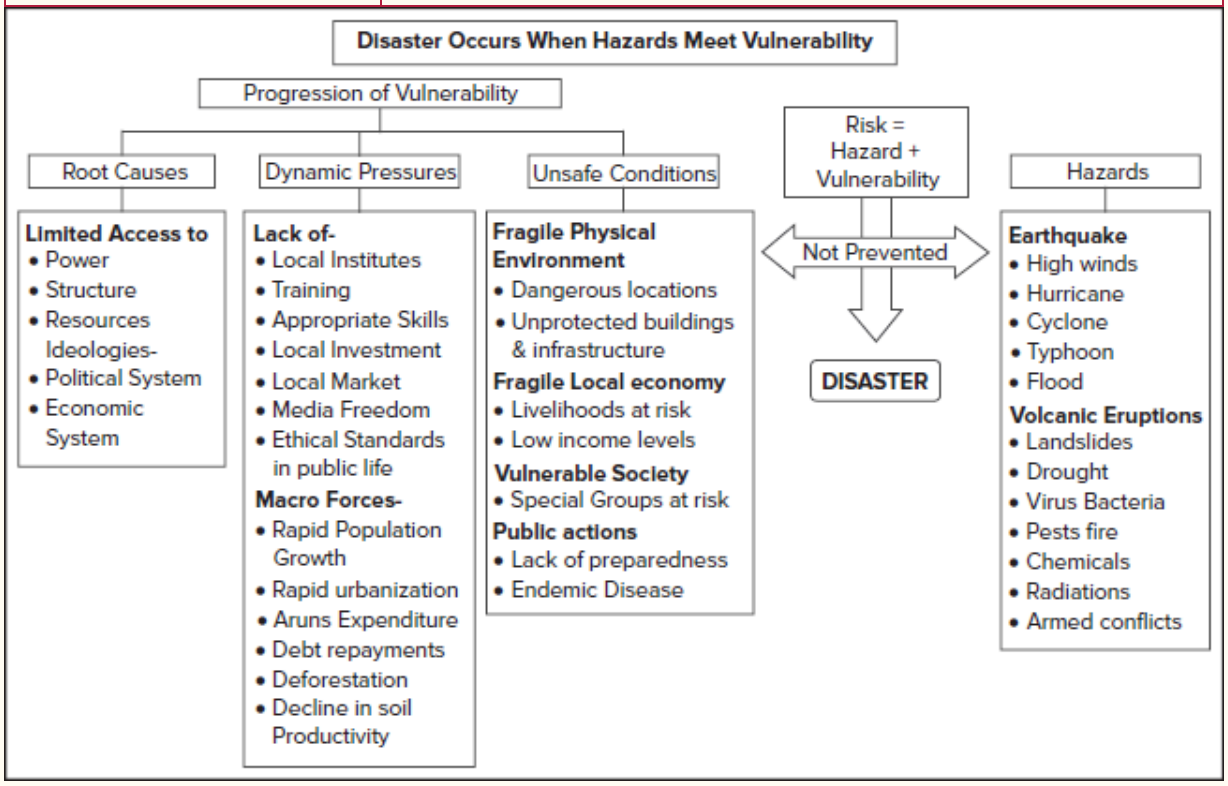

Figure 1.2 Interactions among Disaster, Vulnerability and Hazards

1.4 VULNERABILITY

As per the fourth assessment report (AR4) of IPCC (2007), vulnerability is defined as "the degree to which geophysical, biological, and socio-economic systems are susceptible to, and unable to cope with, adverse impacts of climate change, including climate variability and extremes". In a

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simplified way, vulnerability is the propensity of things to be damaged by the hazard.

Vulnerability to disasters can be characterised by identifying the factors and conditions that make people, communities, or systems susceptible to harm when exposed to hazards. AR4 characterises vulnerability as a function of three factors: exposure, sensitivity, and adaptive capacity.

1. **Exposure:** The degree to which a system experiences climate-related stresses, like the duration of a drought or the scale of changes in rainfall patterns.
2. **Sensitivity:** How much a system is affected by these climate-related stresses or extreme events.
3. **Adaptive Capacity:** The system's ability to cope with, withstand, or bounce back from the damage caused by extreme events.

Ultimately, a system's vulnerability is determined by its adaptive capacity, which influences both its exposure to stress and how sensitive it is to those impacts.

Vulnerabilities can be classified based on the reasons of disaster risk, such as:

Physical Vulnerability: Refers to the susceptibility of physical structures and people to damage. Example: Poorly built houses in seismic zones or settlements in low-lying floodplains.

Social Vulnerability: Determined by demographic and social factors such as age, gender, education, and health. Marginalised groups, children, the elderly, and persons with disabilities are more vulnerable during disasters.

Economic Vulnerability: Reflects exposure due to poverty, low income, and dependence on fragile livelihoods. Example: Small-scale farmers or fishers in coastal areas may lose their only source of income after floods or cyclones.

Organisational Vulnerability: Involves weak governance, poor disaster management systems, and inadequate policies. Example: Lack of

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enforcement of building codes or absence of early warning systems increases risk.

Environmental Vulnerability: Occurs when ecosystems are degraded or natural buffers are absent, amplifying disaster impacts. Example: Deforestation increases landslide risk or mangrove loss, exposing coasts to storm surges.

Cultural and Psychological Vulnerability: Relates to beliefs, practices, and awareness that affect disaster response. Communities unaware of hazards or resistant to evacuation are more at risk.

1.5 RISK

We come to know about risk, a process, by analysing the potential hazards and evaluating existing vulnerability conditions that together could harm people, infrastructure, services, livelihoods and the environment on which they depend. It not only evaluates the immensity and probability of potential losses but also provides a full understanding of the causes and impact of those losses. It is a measure of the potential losses due to a hazard occurring in a given area over a specific time period and a function of the probability of particular hazardous event and the losses each would cause. The level of risk depends upon:

- Nature of the hazard
- Vulnerability of the elements which are affected
- Economic value of those elements

Thus, disaster risk is a function of hazard exposure and societal vulnerability, determining the scale and severity of impacts.

1.6 TYPES OF DISASTER

Disasters can be natural or man-made. Earthquake, cyclone, hailstorm, cloud-burst, landslide, soil erosion, snow avalanche, flood etc. are the examples of natural disasters while fire, epidemics, road, air, rail accidents

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and leakages of chemicals/ nuclear installations etc. fall under the category of human-made disasters.

The **High Power Committee** on Disaster Management, constituted in 1999, by department of Agriculture and Cooperation, Ministry of Agriculture, The Government of India has identified 31 various disasters categorised into five major sub-groups.

Types (Subgroups)		Disasters
1.	**Water and Climate related**	1. Floods and drainage management 2. Cyclones 3. Tornadoes and hurricanes 4. Hailstorm 5. Cloud burst 6. Heat wave and cold wave 7. Snow avalanches 8. Droughts 9. Sea erosion 10. Thunder and lightening 11. Tsunami
2.	**Geology related**	12. Landslides and mudflows

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Types (Subgroups)		Disasters
		13. Earthquakes 14. Dam failures/ Dam bursts 15. Minor fires
3.	**Chemical, industrial and nuclear related**	16. Chemical and industrial disasters 17. Nuclear disasters
4.	**Accident related**	18. Forest fires 19. Urban fires 20. Mine flooding 21. Oil spills 22. Major building collapse 23. Serial bomb blasts 24. Festival related disasters 25. Electrical disasters and fires 26. Air, road and rail accidents 27. Boat capsizing

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Types (Subgroups)		Disasters
		28. Village fire
5.	**Biology related**	29. Biological disasters and epidemics 30. Pest attacks 31. Cattle epidemics and Food poisoning

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	Types (Subgroups)	Disasters
1.	Water and Climate related	1. Floods and drainage management 2. Cyclones 3. Tornadoes and hurricanes 4. Hailstorm 5. Cloud burst 6. Heat wave and cold wave 7. Snow avalanches 8. Droughts 9. Sea erosion 10. Thunder and lightening 11. Tsunami
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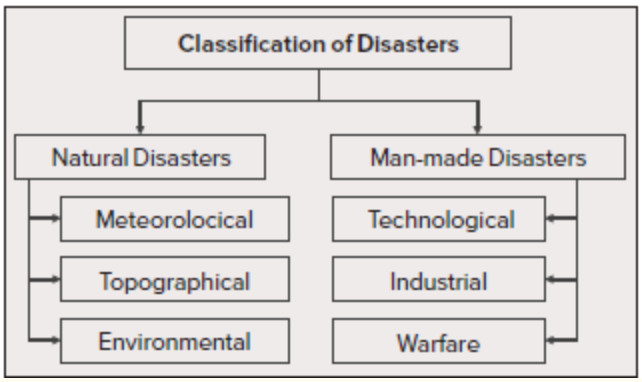

Figure 1.3 Classification of Disasters

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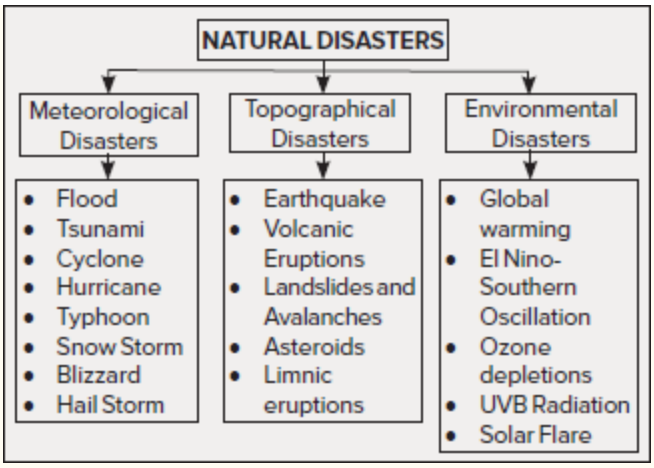

Figure 1.4 Types of Natural Disasters

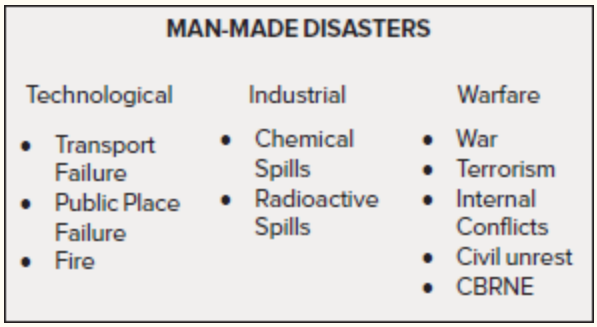

Figure 1.5 Types of Anthropogenic Disaster

1.7 DISASTER MANAGEMENT AND ITS STAGES

Disaster management comprises all the tasks, activities, schemes and measures which are taken up before, during and after a disaster to avoid it, reduce its impact and recover from its damages. As per the Disaster Management (Amendment) Act, 2025, disaster management is inclusive of Disaster Risk reduction, which refers to a systematic effort to analyse and manage the causal factors of disaster through –

- Reduced exposure and vulnerability
- Improved preparedness, resilience and management capacity

As per the Sendai Framework on Disaster Risk Reduction (2015-30), disaster management can be grouped into three stages based on two approaches:

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- **Proactive Approach:** It involves taking steps to prevent or reduce the impact of a disaster through actions that improve the reliance of vulnerable entities to a disaster.
  - Pre-Disaster
- **Reactive Approach:** It deals with the consequences of a disaster during and post the event to minimise the amount of damage to the vulnerable entities.
  - During Disaster
  - Post-Disaster

Pre-disaster and post-disaster can be divided into various sub parts based on operations and administrations.

Pre-Disaster Management

Pre-Disaster management is another form of risk disaster management consisting of the following activities

- Prediction
- Mitigation
- Preparedness

Post Disaster Management

Post-Disaster management is another form of crisis disaster management consisting of the following activities

- Response
- Recovery
- Reconstruction
- Rehabilitation (as per DM (Amendment) Act, 2025)

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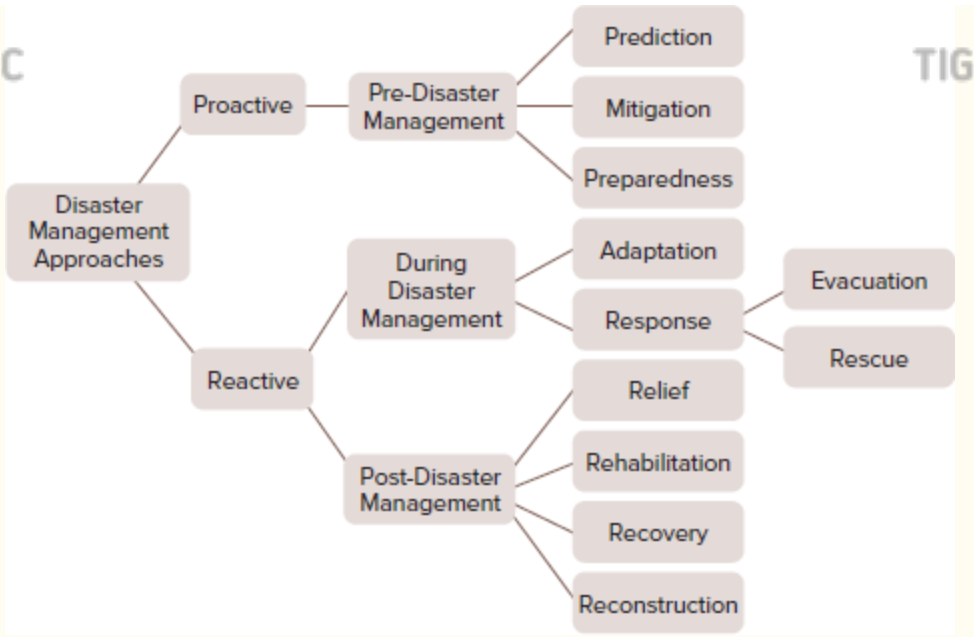

Figure 1.6 Disaster Management Approaches

1.8 PHASES OF DISASTER MANAGEMENT

There are four phases of disaster management

1. Preparedness
2. Response
3. Recovery
4. Mitigation

1.8.1 Disaster Preparedness

Disaster Preparedness helps to face disaster. It performs by the coordination of a variety of sectors

- Evaluation of risk factors
- Adopting standards and regulations
- Organising communication and response mechanism
- Ensuring all resources ready and easily mobilized
- Develop public education programme
- Co-ordinate information with news media
- Disaster simulation exercise

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1.8.2 Disaster Response

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It indicates immediate reaction to disaster as the disaster is anticipated, or soon after it begins to assess the needs, reduce suffering, limit the spread and consequences of the disaster, and open up the way to rehabilitation.

by-

- Mass evacuation
- Search and rescue
- Emergency medical services
- Securing food and water
- Maintenance of Law & Order

1.8.3 Disaster Recovery

There are different modes of disaster recovery: **REPATRIATION:** After the emergency is over, displaced people return to their place of origin.

REHABILITATION: Restoration of basic social functions.

- Providing temporary shelters
- Stress debriefing for responders and victims
- Economic Rehabilitation
- Psycho-social Rehabilitation
- Scientific Damage Assessment

Elements of Recovery

- Community recovery (including psychological)
- Infrastructure recovery (services and lifelines)
- Economy recovery (financial, Political)
- Environmental Recovery

1.8.4 Disaster Mitigation

Disaster mitigation is the effort to reduce loss of life and property (economic assets) under the impact of disaster. There are two types of mitigation:

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- **Primary Mitigation:** Reducing hazard and mitigation
- **Secondary Mitigation:** Reducing effect of hazard.

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Components of Disaster Mitigation

- **Reducing hazard:** Protection against threat by removing the cause of threat.
- **Reducing Vulnerability:** Reducing the effect of threat.

1.9 DATABASE

National Institute for Disaster Management (NIDM), under the Disaster Management Act, 2005 has been assigned with research work and it is expected that it would develop a mechanism to capture such data on the disasters including its impact on socio-economic life of the nation and GDP growth.

At the **global level**, the **three main sources**, which have been **collecting the data**, they are **EM-DAT**, **Natcat** and **Sigma**. EM-DAT is the acronym for data that the Centre for Research on the Epidemiology on Disasters (CRED) has been collecting from the countries since 1987 (going back to 1900). Munich Reinsurance Company maintains, Natcat and Swiss Reinsurance Company maintains, Sigma.

Lack of Central Statistical database as a record of past disasters is a major constraint for risk assessment and compilation of disaster history in the country. Different sources of data have different figures of casualties and impacts, thereby hindering an objective assessment.

In India, **Ministry of Statistics and Programme Implementation** have taken initiatives for collection of data in respect of different disasters. A Technical Committee was constituted by the **Central Statistics Office** with a view to prepare a framework for disaster statistics for developing a database on disaster and related aspects.

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Five formats have been developed for capturing the information about

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(i) Statistics on disaster at district level,
(ii) Statistics on relief, rehabilitation and reconstruction at district level,
(iii) District-wise compilation of statistics on disaster,
(iv) District- wise compilation of statistics on relief, rehabilitation and reconstruction
(v) Aggregation of damage and relief data at state level.

The state Government and District level institutions would be required to furnish this information for preparing the database at centralised level.

In a latest development, the Disaster Management (Amendment) Act, 2025, has sought to formalise the concept of a disaster database by defining it as a collection of data based on a policy determined by the Central Government. It should contain the following data:

- Disaster Assessment
- Fund allocation details and expenditure
- Preparedness and mitigation plan
- Risk register based on risk type and severity

1.10 NDEM

ISRO developed spaced-based comprehensive disaster management system (DMS) named **National Database for Emergency Management System (NDEM)** for tackling disasters and has associated the same with the concerned ministries. NDEM is a GIS based data repository to support the total cycle of disaster/emergency management for the whole country on real time basis. It is envisaged to have core, hazard-specific,
and dynamic data in spatial as well as textual form. The Disaster specific database available with ISRO, Central and State Departments covering Flood/Cyclone, Forest Fires, Tsunami, Drought, Landslides, Earthquakes etc. are organised into NDEM. NDEM services have been operationalised since 2013 providing timely information along with disaster specific products for effective decision making.

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1.11 INDIA'S DEADLIEST DISASTERS

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Some of India's major disasters are elaborated in the following table.

Table 1.2 Major Natural Disasters in India (2000–Present)

2025	**Flash Flood**	Uttarkashi Cloudburst	Uttarakhand
2024	**Landslide**	Wayanad Landslide	Kerala
2024	**Cyclone**	Cyclone Remal	West Bengal
2023	**Landslides**	Himachal Pradesh Landslides	Himachal Pradesh
2022	**Floods**	Assam & Northeast Floods	Assam, Meghalaya
2021	**Cyclone**	Cyclone Tauktae	West Coast
2021	**Cyclone**	Cyclone Yaas	Odisha, WB
2020	**Cyclone**	Cyclone Amphan	West Bengal
2019	**Cyclone**	Cyclone Fani	Odisha
2018	**Floods**	Kerala Floods	Kerala
2017	**Floods**	Assam Floods	Assam
2016	**Cyclone**	Cyclone Vardah	Tamil Nadu

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2015	**Heatwave**	Severe Heatwave	Andhra Pradesh, Telangana
2014	**Floods**	Jammu & Kashmir Floods	J&K
2013	**Flash Floods**	Uttarakhand Disaster (Kedarnath)	Uttarakhand
2011	**Cyclone**	Cyclone Thane	Tamil Nadu
2010	**Cloudburst**	Leh Cloudburst	Ladakh
2008	**Floods**	Kosi River Floods	Bihar
2005	**Floods**	Mumbai Floods	Maharashtra
2004	**Tsunami**	Indian Ocean Tsunami	Tamil Nadu, Andaman & Nicobar Islands
2001	**Earthquake**	Bhuj Earthquake	Gujarat

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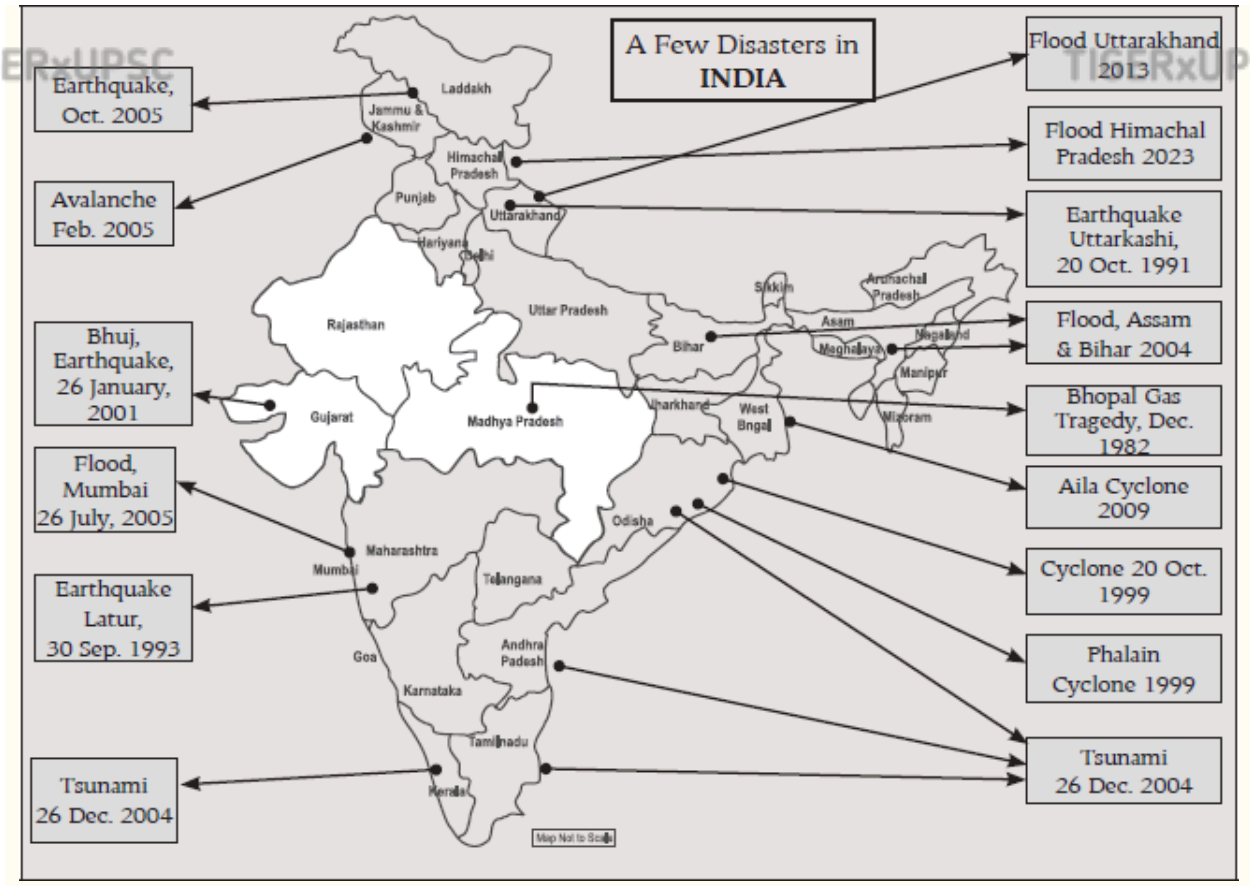

Figure 1.7 Different types of Disaster in Various Locations of India

1.12 INDIAN SCENARIO

India is prone to various disasters due to its geo-climatic and socio-economic condition. In the past fourty years, the country has witnessed **431 major disasters** resulting into huge loss of life and property. As per **Prevention Web statistics**, various disasters in the country has resulted into the killings of 1,43,039 people and affecting another 150 crore in three decades.

In India, the 25th November 1839 cyclone had resulted into death toll of three lakh people. The after effects of **Bhuj earthquake** in 2001 and the **Super Cyclone** of Odisha on 29th October, 1999 are still fresh in our mind. The natural **cloud burst** in 2010 caused heavy flash floods and mudflow in Leh and surrounding areas, causing deaths of 196 persons, missing of 65

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persons, damaging of 3661 houses and finally affecting 27,350 hectares of crop area hence, a huge damage to human lives as well as property. In 2013, India witnessed the Kedarnath disaster, a combination of heavy rains leading to overflow of Mandakini river as well as melting of Chorabari glacier causing a GLOF which led to the death toll of over 6000. More recently in 2025, the village of Dharali in the Uttarkashi was swept away by a cloudburst over the Kheer Ganga river, with more than 60 people still missing.

1.13 VULNERABILITY PROFILE OF INDIA

India is vulnerable to large number of disasters. **27 out of 29 states** and 7 union territories in the country are **disaster prone**. Almost 58.6% of the landmass is prone to moderate to very high intensity earthquakes. More than 40 million hectares (12% of land) are prone to floods and river erosion and nearly 5700 km out of the 7,516 km coastline is prone to cyclones and tsunamis. Around **68% of the cultivable area is drought prone** and its hilly areas are vulnerable to landslides and avalanches. Besides this, India is also prone to **chemical, biological, radiological and nuclear (CBRN)** disaster emergencies. Resultantly, disaster in India really threatens its economy, population and sustainable development.

1.14 HAZARD PROFILE OF INDIA

- **India is one of the ten most disaster vulnerable countries of the world**. The reason for the country's high vulnerability to natural disasters is its unique geographical and geological condition. There are specific problems in the country regarding the country's disaster vulnerability to four distinctive regions of the country i.e. **Himalayan region**, the **alluvial plains**, the **hilly part of the peninsula** and the **coastal zone**. On one side, there are frequent earthquakes and landslides in the Himalayan region, whereas on the other side flood affect the plains almost every year. The

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country's deserts are prone to droughts and famine while the coastal zones are vulnerable to cyclones and storms.

- The Himalayan region and its adjacent alluvial plains have geo-tectonic features that makes them vulnerable to earthquakes, landslides, water erosion etc.
- The plains of Uttar Pradesh and Bihar face frequent flood because of river channel siltation resulted due to huge quantity of sediments brought by various major river systems from the Himalaya.
- The western part of the country, including Rajasthan, Gujarat and some parts of Maharashtra are frequently affected by drought. Moreover, the coastal areas are affected by cyclone due to the disturbance in the pressure condition over the oceans and on an average; the country is exposed to five to six tropical cyclones both from the Arabian Sea and the Bay of Bengal, annually.
- The extreme weather condition, huge quantity of ice and snow stored in the glaciers etc. are other natural factors which make the country prone to various forms of disasters.
- Apart from natural factors, various **human activities** like increasing demographic pressure, deteriorating environmental conditions, deforestation, unscientific development, faulty agricultural practices and grazing, unplanned urbanisation, construction of large dams on river channels etc. are also responsible for faster impact and increase in frequency of disasters in the country.

1.14.1 Climate Profile

India is place to an amazing variety of climatic regions, extending from temperate and alpine in the north Himalayan to tropical in the south. The country's climate is strongly impacted by the Himalayas and the Thar deserts. The probability of drought occurrence is found maximum in Rajasthan (25%), Saurastra & Kutch (23%), followed by Jammu & Kashmir (21%) and Gujarat (21%) region in the period of 124 years.

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Rainfall in India

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The country is affected by two seasons of rains, followed by seasonal reversal of winds from January to July. The summer monsoon are caused due to the intense heat of summer months as the north Indian landmass in the summer becomes hot and draws moist winds over the oceans causing a reversal of the winds over the region and termed as south west monsoon. This climate feature is the most important feature regulating the Indian climate as nearly 75% of the annual rainfall is received during a short span of four months i.e. June to September. The average annual rainfall over the western Rajasthan is less than 13 cm where as the same over Mawsynram in the Meghalaya is as high as 1141 cm. The major reason for drought in our country is **El Nino-Southern Oscillation (ENSO)** pattern and its connection with Indian food grain production.

- **South West Monsoon:** Nearly, 74.2% of the annual rainfall (1182.8 mm) in the country is due to the mean south-west monsoon (June, July, August and September) rainfall (877.2 mm).
- **North East Monsoon:** Post-monsoon or Northeast monsoon or Retreating SW Monsoon Season (October, November and December) is a transition phase related to the formation of the north-easterly wind regime over the Indian sub-continent. Indian states such as coastal Andhra Pradesh, Tamil Nadu, Kerala and South Interior Karnataka receive good amount (nearly 35%) of annual rainfall in these months. Many parts of these states are affected due to forming of cyclones and storms in the Bay of Bengal and results in large scale destruction of life and property due to the occurrence of heavy rainfall and strong winds in the coastal region.

The table demonstrates the examples of extreme weather events:

Table 1.3 Primary and Secondary Climatic Events

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Primary Climatic Events	Secondary Events (May be climate-driven)
• Cold wave, fog, snow storms and avalanches	• Incidence of epidemics or diseases
• Hailstorm, thunderstorm and dust storms	• Urban and rural water shortage
• Extreme temperature	• Crop plantation failure or harvest failure
• Tropical cyclone and tidal wave	• Malnutrition or under nutrition and hunger
• Flood, heavy rain	• Landslides, saline water intrusion and mudflows
• Drought (hydrological, meteorological and agricultural etc.)	

■ *Drought*

The rainfall deficiency and the timing, distribution and intensity of this deficiency particularly in linkage to the existing reserves are the primary cause of drought. A widely recognised climate anomaly known as drought is a period of relatively prolonged dry weather. In this weather condition, water supplies get dry, crops fail to grow, animals die, malnutrition and ill health become wide spread which can be devastating in nature. The drought effects on environment include the salinisation of soil and groundwater decline, increased pollution of freshwater ecosystems and regional extinction of animal species.

In India, around 68% of the area is prone to drought in varying degrees. Nearly, 35% of the total area receives rain between 750 mm and 1125 mm

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which is considered as drought prone while 33% receives rainfall less than 750 mm which is considered to be persistently drought prone.

Flood

India is one of the **worst flood prone** countries in the world. The major reasons for flood lie in the very essence of ecological system in this country such as the monsoon, the highly silted river system and the steep and highly erodible mountains, specifically those of the Himalayan Mountains. The average rainfall in India is 1150 mm with notable variation across the country. The annual rainfall amount to more than 2500 mm along the western coast and Western Ghats, Khasi hills and over most of the Brahmaputra valley. Floods usually occur during the monsoon and are linked with tropical storms or depressions, active monsoon conditions and break monsoon situations.

Simultaneously, in flood, there is an apprehension for out breaking of serious epidemics, such as malaria and cholera and the rising of water scarcity condition. It has a serious and drastic effect on agricultural produce as the water remains standing over large areas for long span of time hampering the Rabi crops. Moreover, 40 million hectares of land, nearly **one-eighth of the country's geographical area, is vulnerable to floods**. The **National Flood Control Program** was launched in the country in 1954 and till 1976, nearly one third of the flood prone area had been provided adequate protection and considerable amount of experience has been gained in planning, implementation and performance of flood warning, protection and control measures.

Almost all river basins in India experiences flood. The main causes of floods are heavy rainfall, inadequate capacity of the rivers to carry high flood water discharge, insufficient drainage system capacity to carry away the rain water quickly to streams/ rivers. The other causes are torrential rain due to typhoons and cyclones and ice jams or landslides in hilly areas

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blocking river water streams. High flow of water during rain and soil's poor permeability is the main cause for the occurrence of Flash floods. Hardpan areas just below the soil surface are more prone to flood as water fails to seep down to the deeper soil layers.

■ *Tropical Cyclones*

The **major natural disaster affecting India's coastal regions** is **cyclone** and the same is exposed to nearly 10% of the world's tropical cyclones. Nearly, 71% of the cyclone prone area in India is in **ten states** (Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, Puducherry, Andhra Pradesh, Odisha and West Bengal) and the Andaman, Nicobar and Lakshadweep islands. On an average, approximately five to six tropical cyclones originated in the Bay of Bengal and the Arabian Sea hit the country's coastline every year and two to three out of them are severe causing huge destruction to life and property.

When a cyclone is in the way of hitting a coast, there is always risk of causing serious loss or damage to life and property due to severe winds, heavy rainfall, storm surges and flood. The cyclonic storm hitting Tamil Nadu and Andhra Pradesh coast, with wind speed between 65 Km/h and 117 Km/h and severe cyclonic storm with wind speed between 119 Km/h and 164 Km/h, is high during the north east monsoon i.e. October – December, whereas there is also occurrence of the highest number of annual storms, severe storms along the coast of Odisha - West Bengal in the same months. For example, a severe super cyclonic storm with winds of nearly 250 km/hour hit the Odisha coast on October 29, 1999 which was the country's one of worst cyclones and was responsible for causing nearly 10,000 deaths, rendering millions homeless and extensive damage to property and environment.

■ *Thunderstorm, Hailstorm and Dust Storm*

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On the transformation of winter season into spring, initially there is rising of temperature in the southern parts of India, making a way for thunderstorms and squally weather which are hazardous in nature. As such there are no dust storms and hailstorms in the southern most part of the country, whereas such hazardous weather is prevalent in the central, northeastern, north and northwestern parts of the country. The frequencies of hailstorm are highest in the Assam valley, followed by hills of Uttarakhand, Jharkhand and Vidarbha Maharashtra. However, there is also occurrence of thunderstorms in Kolkata, Delhi, Jaipur and Ahmedabad. Tornadoes are rare in India but some of them are quite devastating.

■ *Urban Floods*

Urban flooding phenomena are very much significant these days not only in major urban cities of the country but also prevalent globally due to intense and heavy rainfall which engulfs the capacity of the drainage system resulting in flash floods. **According to the UN report, India's average annual economic loss due to disasters is estimated to be $ 9.8 billion, out of which $7 billion loss is due to flood**. Flood in an urban area is a challenging problem. Urban cities are densely populated and flood in these areas not only affects the people's livelihood and endangers their lives but also results in the damage of infrastructure amounting heavy loss to life, property, and economy. The flood volume is nearly six times and flood peak is nearly two to eight times in urban areas as compared to rural areas. Urban flooding is not prevalent in India rather it is a global phenomenon.

1.14.2 Earthquakes

Earthquakes on a global scale result in the loss of about 50,000 lives annually. Earthquakes above 5.5 magnitudes on the Richter scale are progressively damaging to property and human life. There is occurrence of massive earthquakes near the junction of two tectonic plates, e.g., along the Himalayan range, where the Indian plate goes below Eurasian plate. The Indian sub- continent situated on the boundaries of two continental plates is

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very prone to earthquakes. According to the **Bureau of Indian Standard (BIS)** latest seismic zone map, the country's **65% area is prone to earthquake of intensity Modified Mercalli Intensity Scale (MSK) VII or more**.

India has been divided into **four seismic zones (Zone II, III, IV and V)** according to the maximum intensity of earthquake. Out of these, the most active zone is zone V which consists of whole of Northeast India, the northern portion of Uttarakhand, J&K, Himachal Pradesh, Bihar, Gujarat and Andaman & Nicobar Islands. Recent scientific journals have warned regarding the likely occurrence of severe earthquakes any time in the Himalayan Region, which could adversely affect the lives of several million people in India.

1.14.3 Landslides

The Himalayan region and the Western Ghats of India are mainly affected by the landslides and the same are also prevalent in the Nilgiri range. There is occurrence of nearly 30% of the world's landslides in the Himalayas. The Himalayas have a landslides history unlike no other mountain range in the world due to its unique nature. Landslides are also common in Western Ghats and Nilgiris. Due to unprecedented rains in the year 1978 alone, Nilgiris had witnessed nearly one hundred landslides which caused severe damage to communication lines, tea gardens and other cultivated crops. Moreover, a **valley in Nilgiris** is termed as **Avalanches Valley**. Scientist's observation in the Himalayas, north Sikkim and Garhwal regions clearly reveal an average of two landslides per sq. km. Landslide is a major natural hazard in our country, which accounts for huge loss of life and damage to communication routes, human settlements, agricultural fields and forest lands.

1.14.4 Industrial, Chemical and Biological Disasters

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'Chemical Disasters' and **'Industrial Disasters'** are often used interchangeably but actually they are a sub-category of the other. Natural or man-made both sources are the major reason for the occurrence of a chemical disaster. However, the pre-disaster prevention and mitigation of chemical (industrial) disasters is a serious concern in view of growing chemical usage and industrial development worldwide.

- **Industrial disaster:** Industrial disasters are the disasters caused due to an accident in an industrial plant which takes place by mechanical, chemical,civil, electrical or other process failures and further may spill over to the areas outside the plant or within, ultimately causing damage to life, property and environment.
- **Chemical disaster:** Chemical disasters occur due to fire, emission, or explosion occurrence involving one or more hazardous chemicals in the course of industrial activity (handling), transportation or storage due to natural events leading to serious effects inside or outside the installation which can likely cause loss of life and property including adverse effects on the environment. "Chemical accident or emergency can cause extensive damage to the environment along with considerable human and economic costs. The nodal Ministry for the management of Chemical Disasters is the **Ministry of Environment and Forest** and **NDMA** has prepared the guidelines to direct ministries, department and state authorities for the preparation of Disaster management plan.

Do You Know?

There are various reasons for the rising of Chemical and industrial emergencies such as -

- Explosion in a plant
- Accidents in storage facilities of chemicals
- Accidents during the transportation of chemicals, misuse of chemicals
- Improper waste management
- Technological system failures
- Failures of plant safety design
- Human error

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- Accidents in treatment plants

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- **Biological Disasters:** Biological disasters refer to an enormous spread of a disease, virus or infestations of plant, animal or insect life causing a catastrophic effect on an epidemic or pandemic level. The out breaks of Cholera and Influenza H1N1 (Swine flu) are the examples of biological disasters. Huge numbers of people within a given community or area get affected in an epidemic-level biological disaster, whereas in a pandemic-level biological disaster, a bigger region covering entire continents or the globe get affected. The example of cholera is an epidemic level biological disaster where as the swine flu is a pandemic level disaster.

1.14.5 Biological Hazards

Biological hazards, or bio hazards, refer to biological substances or organic matters including the samples of medical waste or microorganism produced by parasites, viruses, bacteria, fungi and further pose a threat to living organism's health, mainly that of human. The harmful effects posed by the Biological hazards to human health are of three types–infections, allergy and poisoning.

The nodal ministry for handling epidemics, decision making, advisory body and emergency medical relief provider is the Ministry of Health and Family Welfare. As per the constitution, health is a state subject and the primary responsibility of dealing with biological disasters rests with the state government. The **National Institute of Communicable Disease (NICD)** is nodal agency for investing outbreaks and NICD/ICMR (Indian Council of Medical Research) also provide teaching/training, research and laboratory support.

1.14.6 Biological Warfare (BW) or Germ Warfare

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It is termed as the usage of biological toxins or infectious agents' such as bacteria, viruses, and fungi as an act of war with the purpose to kill or incapacitate humans, animals or plants. **Biological Weapons** are the living organism or replicating entities that reproduce within their host victims. Moreover, the entomological (insect) warfare is also considered as type of biological warfare. **NBC** is the military acronym for **Nuclear, Biological and Chemical warfare** using weapons of mass destruction. **Ministry of Home Affairs (MHA)** is the nodal ministry for BW and partners with Ministry of Health and Family Welfare in its management. MHA is responsible for assessing threats perception, setting up of deterrent mechanism and providing intelligence inputs.

1.14.7 Tsunami

A tsunami (in Japanese **'tsu' means harbor** and **'nami' means wave**) is a series of water waves caused by the displacement of water body of very large volume, usually an ocean. In Tamil language it is known as **Aazhi Peralai**. Seismicity generated tsunamis are caused due to vertical displacement of the overlying water as a result of abrupt deformation of sea floor. The water above the reformed area is displaced from its equilibrium position on the occurrence of earthquakes beneath the sea level. The tsunami waves have small amplitude but a very long wavelength (often hundreds of kilometer long) and are produced on the release of energy and may be caused by non-seismic event also such as a landslide or impact of a meteor. Tsunami waves in the deep ocean may have very long wavelength of hundreds of kilometer and travels at about 800 km per hour, but an amplitude of only about 1 km.

- **Tsunami Risk Assessment:** The Tsunamis generated mainly by subduction zone related earthquakes from the two potential source regions, viz., the Andaman-Nicobar-Sumatra Island Arc and the Makran subduction zone, north of Arabian Sea are likely to affect the east and west coasts of India and the island regions.

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Tsunami in Indian Ocean: The Tsunami of 26th December 2004 caused huge damage to life and property in the states of Tamil Nadu, Andhra Pradesh, Kerala, UTs of Puducherry and Andaman & Nicobar Islands (A & NI). The fishermen community was badly affected by the Tsunami disaster who not only lost their family members but also lost their means of livelihood. Nearly, 26.63 lakhs population in 1396 villages of five states and UTs were affected by this disaster. Almost 9395 people lost their lives and 3964 people were reported missing and feared dead. Most of the missing persons were from Andaman & Nicobar Islands.

1.14.8 Nuclear Emergencies

Nuclear emergency/Disaster is caused due to a tremendous release of radioactive material or radiation either in the nuclear reactors operation or other nuclear events like **Radiological Dispersal Device (RDD)** or nuclear weapon explosion. The disaster is accompanied with sudden release of harmful radiations or radioactive materials or both together into the environment.

- **Nuclear emergency may be encountered in the following situations:** Intentional use of nuclear weapons in the event of war. Nuclear weapon attacks are extremely destructive and powerful which are enough to destroy an entire city. With the recent advancement of scientific research in the world, acquiring of technology to produce nuclear weapons by several countries can be more destructive and harmful.
- **Accidents in nuclear power project:** The nuclear Power Plants take care of safety by incorporating safety features by design and redundancy in safety systems to avoid any miss happening and further to bring the system to a safe shut down in case of any abnormalities.
- **Accidents in handling radiation sources:** Other reasons for accidental radiation exposure could be handling of radioactive material during transportation or wrong/faulty practices.

  Release/leakage of Nuclear Radiation due to any above reason can be devastating and the example is **Chernobyl Nuclear Disaster**.

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1.14.9 Epidemics in India

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Highly infectious diseases are major public health issue in India. As such many infectious diseases like tuberculosis and malaria are endemic in nature; whereas some of them occasionally attain epidemic proportion. An epidemic refers to increase in sudden number of infected in a community clearly excess of what is normally expected in that population. It includes viral infections (meningitis, measles, dengue, polio, typhoid fever etc.) and bacterial infections (cholera, diarrhea, etc.) The main causes for epidemic are non-availability of hygienic and clean drinking water, contamination of drinking water sources, lack of sanitation awareness, unhygienic food, overcrowding, biological conditions in addition to ecological factors.

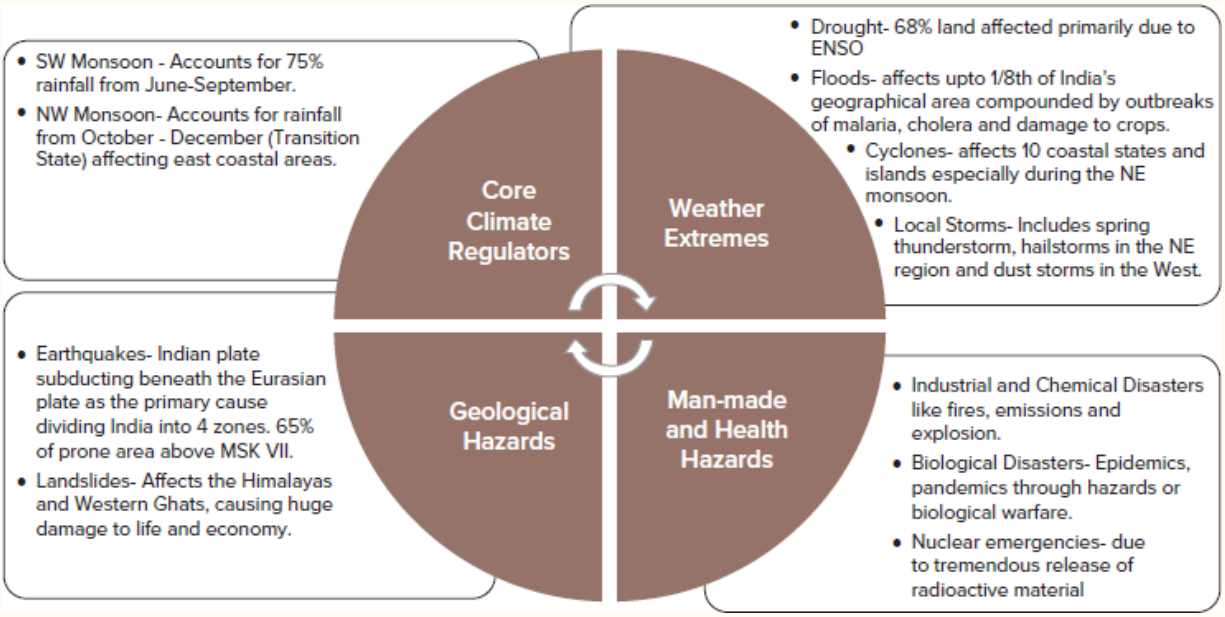

Figure 1.8 India's Disaster Profile

1.15 CASCADING EFFECT OF DISASTER – EARTHQUAKE, TSUNAMI, FIRE AND NUCLEAR DISASTER IN JAPAN

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Event of one kind of disaster sometimes lead to a cascading effect and lead to other disasters. Recently, the experience of Japan earthquake leading to

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Tsunami, fire and then nuclear emergency is an eye opener to those under estimating the power and fury of nature.

1. **Earthquake:** On 11 March 2011, at 14:46 Japan's northeastern part was struck by a major earthquake of magnitude 9 with an epicenter at 130 km off the pacific coast of Tohoku region and depth of 24 km. According to Japan Meteorological Agency (JMA), the seismic intensity of 7 [in the Japanese scale of 1 to 7] was recorded in Kurihara city of Miyagi prefecture. Moreover, 28 cities and towns in Miyagi, Fukushima, Ibaraki, and Tochigi prefecture including Wakuya town, Tome city and Osaki city recorded the 6+ seismic intensity waves.
2. **Tsunami:** The earthquake was followed by a major Tsunami, with maximum height of 11.87 meter recorded near the coast of Akamae at 15:31 JST. Major tsunami of more than 3 m or more arrived in northern part of Japan, eastern, central, western part of the pacific coast of Hokkaido, Pacific coast of Aomori, Iwate, Miyagi, Fukushima, Ibaraki, Chiba prefecture, Kujukuri, & Sotoboare and Izu islands. The earthquake happened in 14:46, and the tsunami warning was issued at 14:49, and it was on the JMA website on 14:49.
3. **Fire Events:** On 11 March, the largest and main island of Japan near the east coast of Honshu witnessed several fires due to the triggering of powerful earthquake. The major fire broke out at Cosmo Oil Company refinery in Ichihara city and continued till 14 March. The number of events increased from 44 cases after the first day of earthquake to 325 cases on 19 March.
4. **Nuclear Emergency:** The same day around 20:30 CET (04:30 JST), at the Fukushima Daiichi (number one) nuclear power plant, a nuclear emergency situation was declared. As a result of the intense earthquake all the three operating (out of six) reactors automatically began the process of shutting down.

- The back-up diesel generators for cooling down the fuel rods through the supply of water inside the reactors were damaged due to tsunamis. The water available began to evaporate and around 15:30 JST on March 12, in reactor number one, a hydrogen explosion took place which ultimately

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forced the Japanese government to evacuate people within a 20 km radius around the Fukushima Daiichi power plant. On March 14, around 11:01 JST, in reactor 3 another explosion took place at the nuclear power plant and on March 15, a further explosion took place in the number 2 reactor at nuclear power plant. Then a fire broke out in the number 4 reactor where spent fuel rods were stored to undergo cooling process and there after soon the fire was extinguished. These incidents led to increase of radiation emission levels. Afterwards, the Japanese Government established a 30km radius **No Fly Zone** around the Fukushima Daiichi nuclear power plant. Further, on March 17, the water spraying supported through helicopter and heavy water cannon trucks continued for many days by units of the Self Defense Force (SDF) who shot water from the ground. High radiation levels were reported in spinach, milk and tap water outside the demarcated 30 km evacuation zone and farmers were advised to voluntarily avoid selling contaminated food in the markets.

- **Lessons:** This disaster points towards the need to have a multi-hazard approach and plan for handling unexpected disasters. In case of disasters, Japan as a country is one of the best prepared. Still it had to face the cascading impact of disasters initiating from the severe under sea earthquake, thereafter the Tsunami, then fire and nuclear energy related emergencies apart from disaster related loss of lives, livelihood assets and infrastructure.

NOTE: JST - Japan Standard Time & CST - Central Standard Time.

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Institutional Framework

2

IN THIS CHAPTER...

2.1 Emergence of Institutional Arrangement in India
2.2 Organisation and Structure of Disaster Management
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2.1 EMERGENCE OF INSTITUTIONAL ARRANGEMENT IN INDIA

Formation of disaster management cell under the aegis of Ministry of Agriculture in the decade of 1990s led to the beginning of a permanent and institutionalised arrangement in the country for disaster management. The following series of disasters—Latur Earthquake (1993), Malpa Landslide (1994), Odisha Super Cyclone (1999) and Bhuj Earthquake (2001), led to the formation of a high powered Committee under the Chairmanship of Mr. J.C. Pant, Secretary, Ministry of Agriculture for drawing up a systematic, comprehensive and holistic approach towards disasters. Afterwards, there is a need to address disaster management by shifting of policy from a relief approach to holistic approach through financial aid and consequently, the shifting of disaster management took place under the Ministry of Home Affairs (MHA) in 2002. National disaster Management Authority (NDMA) was established on 28 September 2005. Provision of disaster management is under the seventh schedule of our Constitution. On 23 December 2005, Disaster Management Act came in force.

2.2 ORGANISATION AND STRUCTURE OF DISASTER MANAGEMENT

The Disaster Management Division in MHA is headed by Joint Secretary (DM), who is assisted by three Directors, Under Secretaries, Section Officers, Technical Officer, Senior Economic Investigator, consultants and other supporting staff.

Present Structure for Disaster Management in India

The **National Disaster Management Authority** has been established at the center, and the SDMA at state and district authorities at district level are gradually being formalised.

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Disaster Management Act, 2005

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This Act provides for the effective disaster management and further provides the drawing up and monitoring the implementation of the disaster management through institutional mechanisms. The Act under the Chairmanship of the Prime Minister provides the setting up of a National Disaster Management Authority (NDMA), State Disaster Management Authorities (SDMAs) under the Chairmanship of the Chief Ministers, District Disaster Management Authorities (DDMAs) under the Chairmanship of Collectors/District Magistrates/ Deputy Commissioners. The Act further contains the provisions for financial mechanisms such as creation of funds for response, National Disaster Mitigation Fund and similar funds at the state and district levels for the purpose of disaster management. The Act also provides specific roles in disaster management to local bodies and also envisages specific roles to be played by the local bodies in disaster management.

Institutional Framework

Disaster Management Structure

- **NDMA:** Apex Body with Prime Minister as Chairperson
- **National Excutive Committee:** Secretaries of 14 Ministries and Chief of Integrated Defence Staff

Centre Level

- Central Ministries; National Disaster Management Authority
- National Institute of Disaster management
- National Disasterw Response Force (NDRF)

State Level

- SDMA headed by Chief Minister
- State Executive Committee (SEC)

District Level

- DDMA headed by District Magistrate
- Interface between govt. and public

2.2.1 Disaster Management (Amendment) Act, 2025

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The amendment brought about several key modifications, focusing on enhancing preparedness, efficiency, and governance in the realm of disaster management.

Expansion of the role of states:

As per the Disaster Management (Amendment) Act 2025, state governments are empowered to create UDMA (Urban District Management Authority) for state capitals and large cities with a municipal corporation.

Moreover, it formalised the creation of SDRF by states for specialised response operations, defining their functions and terms of service.

Statutory status:

The Act grants statutory status to two existing high-level bodies, enhancing their authority and role clarity:

National Crisis Management Committee (NCMC): Nodal body for major disasters with national ramifications (chaired by the Cabinet Secretary).

High Level Committee (HLC): Responsible for financial assistance to state governments from disaster mitigation funds.

- **Enhancing Preparedness:**
  - **Disaster Database:** A new provision mandates the establishment of a **'Disaster Database'** at the national and state levels. This database will compile crucial information, including:
    - Disaster assessment and risk registers.
    - Fund allocation and expenditure details.
    - Preparedness and mitigation plans.
    - Type and severity of disaster risks.
  - **Expanded Role of NDMA/SDMA:** The National Disaster Management Authority (NDMA) and State Disaster Management Authorities (SDMAs) are now explicitly tasked with:
    - Taking periodic stock of **emerging risks** (including those from extreme climate events).

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- Preparing the main disaster management plans (instead of the Executive Committees).
- Undertaking **post-disaster audits** of preparedness and response.

- **Inclusion of Disaster Risk Reduction:** The definition of "Disaster Management" is expanded to explicitly include efforts towards "disaster risk reduction" and strengthening resilience.

2.2.2 National Level Institutions

National Disaster Management Authority(NDMA): The National Disaster Management Authority (NDMA) was formed on May 30, 2005 as an executive order under the Chairmanship of Prime Minister.

The details of NDMA responsibilities are given as under

(a) To lay down policies on disaster management

(b) To approve the National Plan

(c) To approve plans in accordance with the National Plan; prepared by the Ministries or Departments of the Government of India

(d) To lay down guidelines for drawing up the State Plan, to be followed by the State Authorities

(e) To lay down projects guidelines for integrating the measures for prevention of disaster or its mitigation effects in their development plans which has to be followed by the different Ministries or Departments of the Government of India

(f) To coordinate the implementation and enforcement of the disaster management policy and plan

(g) To recommend funds provisioning for the disaster mitigation purpose.

(h) To provide support to other countries affected by major disasters as determined by the Central Government

(i) To take steps as considered necessary for the disaster prevention, or mitigation, preparedness and capacity building inorder to deal with the serious disaster situation or disaster

(j) To lay down broad guidelines and policies for the National Institute of Disaster Management functioning.

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- **Composition of NDMA:** The NDMA organisational structure consists of a Secretary and five Joint Secretaries including one Financial Advisor apart from the nine members nominated by the Prime Minister who is the Chairperson of the Authority. There are 10 posts of Joint Advisors and Directors, 14 Assistant Advisors, Under Secretaries and Assistant Financial Advisor and Duty Officer along with supporting staff.
- **National Executive Committee (NEC):** A National Executive Committee is constituted under Section 8 of DM Act, 2005 to assist the National Authority in its functional performance.

2.2.3 State Level Institutions

Similarly, at the state level institutions, the State Disaster Management Authority (SDMA) and the State Executive Committee (SEC) work, but at the district level District Disaster Management Authority (DDMA) works. The 2025 amendment has also provided for the creation of the Urban District Management Authority (UDMA) for a specific focus on urban areas with municipal corporations.

2.2.4 Hierarchical Structure of Authority and Committee

National Disaster Management Authority is the authority for formulation of policy and guidelines for all disaster management related work in the country. The state authorities further lay down the guidelines for state department and the districts falling in their respective jurisdictions. Similarly, district authorities direct the civil administration, departments and local authorities such as the municipalities, police department and civil administration. The State Executive Committees are responsible for execution of the tasks envisaged by the authorities.

2.3 NATIONAL INSTITUTE OF DISASTER MANAGEMENT (NIDM)

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Against the backdrop of the **International decade for Natural Disaster Reduction (IDNDR)**, a National Centre for Disaster Management was established at the Indian Institute for Public Administration (IIPA) in 1995. On 16th October 2003, the Centre was upgraded and designated as the **National Institute of Disaster management (NIDM)** and the same has now achieved the status of a statutory organisation under the Disaster Management Act, 2005. There are numerous responsibilities assigned to the institute as per section 42 of Chapter VII of the Disaster Management Act, 2005 such as developing of training modules, undertaking research and documentation in disaster management, organising training programmes, undertaking and organising study courses, conferences, lectures and seminars in order to promote and institutionalise disaster management, undertaking the publication of journals, research papers and books.

- **Organisational Structure:** The President of NIDM is Union Home Minister. The institute was constituted on 23 Feb 2007 and has a 42 members general body consisting of secretaries of various ministries, Union Government departments, national level scientific research and organisation heads.

2.4 NATIONAL DISASTER RESPONSE FORCE (NDRF)

NDRF has been constituted *vide* Section 44 of the DM Act, 2005 by an up-gradation/conversion of eight standard battalions of Central Para Military Forces i.e. two battalions each from Border Security Force (**BSF**), Indo-Tibetan Border Police (**ITBP**), Central Industrial Security Force (**CISF**) and Central Reserve Police Force (**CRPF**) to build them as a specialist force for responding to disaster or disaster like situations. Moreover, two more battalions from the SSB were added in NDRF in 2015. Presently, NDRF is having strength of 12 battalions each with an authorised strength of 1149 personnel.

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2.5 STATE DISASTER RESPONSE FORCE (SDRF)

The states/UTs have also been advised for setting up their own Specialist Response Force for disasters response, on the lines of National Disaster Response Force *vide* Ministry of Home Affairs letter dated 26 July 2007 and 8 March 2011. The Union Government is also providing assistance to state governments for trainers training and further also been advised to utilise 10 percent of their State Disaster Response Fund and Capacity Building Grant for the procurement of search and rescue equipment and for training purposes of the Response Force.

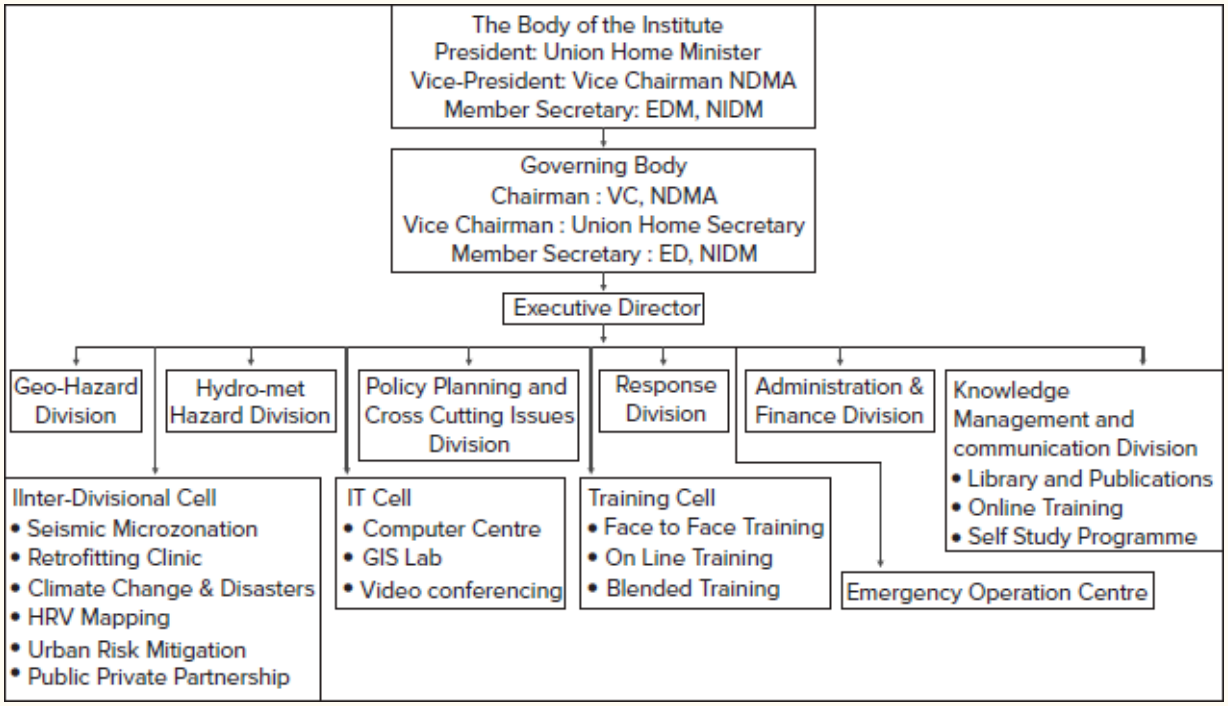

Figure 2.1 Institutional Bodies Working for Disaster Management

2.6 CIVIL DEFENCE

2.6.1 Aims and Objectives of Civil Defence Act

The legislation on Civil Defence (CD) known as Civil Defence Act was enacted in 1968 that is in force throughout the country. The Act has since

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been amended in the year 2010 to cater the need of disaster management so as to utilise the services of Civil Defence volunteers effectively for enhancement of public participation in disaster management related activities in the country.

2.6.2 Role of Civil Defence

The CD organisation has the important role of guarding the hinterland, supporting the armed forces, minimising damage, mobilising the citizens and helping civil administration for saving life and property, maintaining continuity in production centers and raising public morale at the times of war and emergencies.

2.7 NATIONAL CIVIL DEFENCE COLLEGE (NCDC), NAGPUR

The country's first disaster management training institute was founded on 29 April 1957 at Nagpur as the Central Emergency Relief Training Institute (CERTI) to support the Emergency Relief Organisation of the Indian Government. This Institute organised specialist and advanced training for revenue officials responsible for Disaster Relief Operations against any natural or manmade disaster.

2.8 NATIONAL FIRE SERVICE COLLEGE (NFSC), NAGPUR

The National Fire Service College was established in 1956 as a subordinate office of Ministry of Home Affairs with the aim of providing training to the country's fire officers regarding advanced fire-fighting and rescue techniques,and creating uniformity across the country in fire service organisations and their management.

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2.9 NATIONAL POLICY ON DISASTER MANAGEMENT (NPDM)

The National Policy on Disaster Management (NPDM) has been approved on October 22, 2009 by the union government and circulated to all concerned nodal ministries/department for management of disasters. The policy being exhaustive, covers all the disaster management aspects such as institutional and legal arrangements, financial arrangements, disaster prevention, mitigation and preparedness, response, relief and rehabilitation, reconstruction and recovery, capacity development, techno-legal regime, knowledge management, research and development.

2.10 CONVENER - NODAL MINISTRY/DEPARTMENT FOR MANAGEMENT/MITIGATION OF DIFFERENT DISASTERS

Table 2.1 Nodal Ministries Related to Disasters

Types of Disasters	Nodal Ministries
• **Natural-Flood, Tsunami, Cyclone, Earthquake, Manmade-civil Strife**	• Home Affairs
• **Drought**	• Agriculture
• **Biological, Epidemics**	• Health & Family Walfare
• **Chemical, Forest related**	• Environment & Forest
• **Nuclear**	• Atomic Energy

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Types of Disasters	Nodal Ministries
• **Air Accidents**	• Civil Avaition
• **Railway Accidents**	• Railway
• **Industial Accidents**	• Labour

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Prevention and Mitigation

3

IN THIS CHAPTER...

3.1 National Disaster Mitigation Fund
3.2 Measures Taken for Prevention and Mitigation of Hazards
3.3 Earthquakes
- 3.3.1 National Earthquake Risk Mitigation Project (NERMP)
- 3.3.2 National Building Code (NBC)
- 3.3.3 Earthquake and Seismic Zones

3.4 Cyclones
- 3.4.1 National Cyclone Risk Mitigation Project (NCRMP) Initiative
- 3.4.2 Integrated Coastal Zone Management Project (ICZMP)

3.5 Floods
- 3.5.1 National Flood Risk Mitigation Project (NFRMP)
- 3.5.2 Flood Management Programme

3.6 Landslides
3.7 Droughts
3.8 Epidemics
3.9 Civil Aviation

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3.1 NATIONAL DISASTER MITIGATION FUND

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The Disaster Management Act 2005, section 47 caters to the constitution of National Disaster Mitigation Fund. The Act provisions are as under:

- The Central Government constitute a fund to be called the National Disaster Mitigation Fund (NDMF) exclusively for disaster mitigation projects.
- The National Disaster Mitigation fund shall be applied by the National Disaster Management Authority.

The Ministry of Home Affairs (MHA) have discussed modalities of NDMF constitution with the Ministry of Finance, Planning Commission and NDMA from time to time and a reference was made to 13th Finance Commission. As per the recommendations of the 13th Finance Commission, activities of mitigation and reconstruction should be kept out of the schemes funded through Finance Commission grants and met out of the center and state overall development plan funds. The issue is under Ministry of Home Affairs consideration with other concerned Ministries.

3.2 MEASURES TAKEN FOR PREVENTION AND MITIGATION OF HAZARDS

There are two main types of categorisation for mitigation activities, one is structural and the other is non-structural. **Structural mitigation** refers to construction of any physical structure, such as engineering measures and hazard-resistant construction of protective structures and infrastructure to lessen or avoid the possible hazards impacts.

Non-structural mitigation refers to policies, measures for reducing risk related impacts such as policies, awareness, public commitment, knowledge development, and operating practices, including participatory mechanisms and the provision of information.

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To reduce the disaster risk, government has adopted several mitigation measures and the same are being implemented by the concerned ministries.

3.3 EARTHQUAKES

3.3.1 National Earthquake Risk Mitigation Project (NERMP)

The Union Government launched NERMP project for strengthening the structural and non-structural earthquake mitigation efforts and further reducing the hazard in the high risk districts vulnerable to earthquakes. The proposed project components include techno-legal regime, institutional strengthening, capacity building and public awareness etc.

3.3.2 National Building Code (NBC)

The National Building Code of India (NBC) is a national instrument providing guidelines for regulating the building construction activities across the country. It is a comprehensive building code, and was first published in 1970 at the instance of Planning Commission and was revised in 1983 and thereafter three major amendments were issued, two in 1987 and one in 1997. The comprehensive project for revision of NBC was taken by the National Building Code committee and then the revised NBC was brought out as National Building Code of India in 2005 (NBC 2005). The salient features of the revised NBC contains the recent meeting of challenges as posed by natural disaster and reflecting the contemporary and state-of-the-art international applicable practices.

3.3.3 Earthquake and Seismic Zones

The entire Indian landmass, susceptible to different earthquakes levels, covered from Zone II to V as per the Seismic Zoning Map of India.

The general basis of the zones is as follows:

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- **Zone V:** This is the most severe seismic zone and is referred as Very High Damage Risk Zone.
- **Zone IV:** This is referred as High Damage Risk Zone.
- **Zone III:** This is termed as moderate Damage Risk Zone.
- **Zone II:** This zone is referred as Low Damage Risk Zone.

At the time of seismic Zone Map revision, earlier mentioned in the Vulnerability Atlas of India 1997, seismic zone I has now been merged into Seismic Zone II and renamed as Zone II, Zone III has been extended to cover more areas in Maharashtra, Andhra Pradesh and Tamil Nadu. Zones IV and V have remained unchanged.

Exam Pointer

India's Seismic Zonation Revised (November 2025)

India's seismic map underwent a major revision in late November 2025. A new **highest-risk Zone VI** was created, placing the entire **Himalayan arc from Jammu & Kashmir to Arunachal Pradesh** under maximum hazard classification. This change reflects ongoing tectonic stress from the convergence of the Indian and Eurasian plates.

Key Updates

Zone VI introduced: This replaces Zone V as the highest-risk category.

Uniform Himalayan reclassification: The entire Himalayan belt now falls in Zone VI. This corrects an earlier oversight in which some areas were categorised as **Zones IV and V**.

Rising vulnerability:

61% of India's land area is now in moderate-to-high risk zones, an increase from 59%.

Nearly 75% of the population lives in seismically active regions.

Updated safety standards: The BIS Earthquake Code IS 1893:2025 imposes stricter construction rules across the Himalayan states. It also limits development in areas that are highly vulnerable.

New tectonic evidence (Apr 2025): Recent studies suggest that the Indian plate may be splitting beneath the Tibetan Plateau. This changes the understanding of how strain builds up and the behaviour of earthquakes.

Revised Seismic Zones (2025)

Zone VI: Highest risk — Entire Himalayan Arc

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Zone V: Very high risk — Northeast India, Andaman and Nicobar Islands, and Rann of Kutch

Zone IV: High risk — Indo-Gangetic plains, parts of Maharashtra and Gujarat

Zone III: Moderate risk — Large parts of Peninsular India

Zone II: Low risk — Stable shield regions

This revision strengthens earlier assessments. It confirms that around 65% of vulnerable areas experience intensities above MSK VII. The upgrade of the Himalayan zones significantly increases the high-intensity risk areas.

3.4 CYCLONES

3.4.1 National Cyclone Risk Mitigation Project (NCRMP) Initiative

NCRMP is launched by MHA in cyclone vulnerable coastal states and union to reduce the loss of lives and property arising from the cyclone disaster. NDMA is given the task of implementing the scheme and the same is regularly monitored by NDMA and MHA. The scheme aims to construct major infrastructures including multi-purpose cyclone shelters and embankments and further aims to upgrade cyclone forecasting, tracking and warning systems, and building capacity in multi-hazard risk management.

3.4.2 Integrated Coastal Zone Management Project (ICZMP)

The Government of India has launched ICZMP under the support of Ministry of Environment and Forest (MoEF). The objective of the project is to assist union government in preparing the national capacity for the implementation of a comprehensive coastal management approach in the country and piloting the approach of integrated coastal zone management in states of Gujarat, Odisha and West Bengal. The project has four components:

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- Capacity Building
- Piloting ICZM approaches in Gujarat
- Piloting ICZM approaches in Odisha
- Piloting ICZM approaches in West Bengal

3.5 FLOODS

3.5.1 National Flood Risk Mitigation Project (NFRMP)

NFRMP has been envisaged to mitigate or reduce the risk, severity or consequences of floods. NDMA has been given responsibility for the preparation of a detailed project report on Flood Risk Mitigation project.

3.5.2 Flood Management Programme

The Government of India, at a total cost of 8000 crores under the support of Ministry of Water Resources launched the "**Flood Management Programme** (FMP)" for the 11th Plan period (2007–2012).

- **Aim:** The scheme provides financial assistance for undertaking flood management works in critical areas to the state governments during the 11th Plan period (2007–2012).
- **Study of Land Contour by GSI:** Geological Survey of India (GSI) studied the land shape and material getting inundated and generates data on area, shape, slope, infiltration and soil permeability of the basin, drainage pattern, landform and longitudinal and cross profiles of the channels. Based on these studies, GSI creates flood hazard maps pointing prohibitive, cautionary, restricted and flood free zones.

3.6 LANDSLIDES

National Landslide Risk Mitigation Project (NLRMP), aims at strengthening the structural and non-structural mitigation efforts of landslides, reducing the landslide risk and vulnerability of landslide in the

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hilly areas. NDMA has been given the responsibility for preparation of detailed project report on landslide mitigation project.

3.7 DROUGHTS

National Institute of Agriculture and Extension (MANAGE), Hyderabad has been entrusted to launch a National Project for Integrated Drought Monitoring & Management. In this regard, a proposal has been submitted by MANAGES for implementation of this national project that is under consideration. Another proposal to set up a National Institute of Drought Management is also under the consideration by the Ministry of Agriculture & Cooperation.

The **Drought Prone Areas Programme (DPAP)** and **Desert Development Programme (DDP)** are being implemented by the Union Government since 1973–74 and 1977–78 respectively. These programmes aim at drought reduction and minimizing desertification of fragile areas often affected by severe drought conditions in the arid, semi-arid and dry-sub humid regions.

3.8 EPIDEMICS

The Ministry of Health & Family Welfare is the nodal agency in the implementation of various health programmes in the areas of prevention and control of major communicable diseases on a national level and further promotion of conventional and indigenous systems of medicines. The Ministry is actively involved in diagnosis of disease at the time of epidemics and outbreaks, operational research, advisory role and other activities responsible towards prevention and control of any epidemic prone disease.

3.9 CIVIL AVIATION

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The Directorate General Civil Aviation (DGCA) has the regulatory responsibility for aviation safety. Its responsibility is to ensure the highest safety level in Indian Aviation System through the implementation of International Civil Aviation Organization standards and recommended practices.

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Preparedness and Response

4

IN THIS CHAPTER...

4.1 Disaster Preparedness
4.2 Disaster Wise Nodal Agencies for Forecast
4.3 India Meteorological Department (IMD)
4.4 Tsunami Warning – Indian National Centre for Oceanic Information System (INCOIS)
4.5 Warning about Landslide Hazard – Geological Survey of India (GSI)
4.6 Radiological and Nuclear Emergencies
4.7 Epidemic
4.8 Control Rooms
4.8.1 National Emergency Operation Centre (NEOC)
4.8.2 National Disaster Response Force (NDRF)

4.1 DISASTER PREPAREDNESS

Disaster preparedness means getting prepared to respond to disasters effectively as and when it strikes so that lives can be saved and human sufferings reduced to the maximum extent possible through measures like evacuation, search and rescue and humanitarian assistance like shelter and relief. It further envisages the policies, strategies and resources in place for

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'building back better' livelihoods, houses and infrastructures devastated during disasters.

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Table 4.1 Guidelines on Preparedness and Response

• Vulnerability assessment	• Community participation
• Dissemination of warning	• Evacuation Plan
• Emergency Response activities	• Activating EOCs
• Coordination	• Search & Rescue Teams
• Rapid Damage Assessment	• Communication
• Maintenance of essential services	• Identification of Nodal Officer
• Stocking of essential commodities	• Status of SDRF
• Medicines	• Preparedness Drill
• Drinking water	• Shelter/Camps

4.2 DISASTER WISE NODAL AGENCIES FOR FORECAST

Government of India has designated the following offices as the nodal agencies in the under mentioned table for early warning of different natural hazards.

Table 4.2 Disaster-Wise Nodal Agencies for Forecast

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Disasters	Agencies
Cyclone	Indian Meteorological Department
Tsunami	Indian Nation Centre for Oceanic Information Services
Floods	Central Water Commission
Landslides	Geological Survey of India
Avalanches	Snow and Avalanches Study Establishment
Heat & Cold waves	Indian Meteorological Department

4.3 INDIA METEOROLOGICAL DEPARTMENT (IMD)

India's first meteorological observatory was set up at Madras (now Chennai) in 1793. In the year 1875, IMD was formally established with a network of about 90 weather observatories for systematic observation and research.

Atmospheric Monitoring

IMD coordinates and operates 25 types of atmospheric monitoring network which include climatological, meteorological, air pollution, environmental and atmospheric trace constituents specialized observation.

IMD maintains 559 meteorological surface observatories, nearly 35 radio-stations and 64 pilot balloon stations for monitoring the upper atmosphere. At 45 stations, radiation parameters are monitored and at 219 agro meteorological purposes stations, specialized observations are made.

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There are around 70 observatories that monitor current aviation weather conditions.

4.4 TSUNAMI WARNING – INDIAN NATIONAL CENTRE FOR OCEANIC INFORMATION SYSTEM (INCOIS)

The Ministry of Earth Sciences established the Indian National Tsunami Warning System, post tsunami at Indian National Centre for Ocean Information Services (INCOIS), Hyderabad and was made operational on 15 October 2007. The agency has developed a protocol for issue of Tsunami watch, Alert and Warnings. The Centre issues bulletins and provides information to all responders about the origin, time and location of the epicenter, magnitude and depth of an earthquake inside the ocean.

4.5 WARNING ABOUT LANDSLIDE HAZARD – GEOLOGICAL SURVEY OF INDIA (GSI)

Geological Survey of India (GSI), is a government organization being one of the world's oldest organization, established in 1851 under the Union Ministry of Mines with the Headquarter at New Delhi for carrying out geological surveys and studies.

The GSI was designated as nodal agency on 29th January, 2004 for coordinating and conducting geological studies, monitoring landslides, landslides hazard zonation, avalanches, studying the responsible factors and suggesting precautionary and preventive measures. The GSI input as nodal agency helps the concerned Central Government as well as the State Government Ministries and Departments for taking appropriate preventive and mitigation measures in order to reduce landslide hazard and respond in the mountainous and hill areas in the Himalayan and sub Himalayan regions. GSI act as a nodal agency responsible for early warning related to the landslides in the country.

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4.6 RADIOLOGICAL AND NUCLEAR EMERGENCIES

The nodal agency for responding to any nuclear or radiological emergency is Department of Atomic Energy (DAE) for providing the necessary technical inputs to the National or local authorities. In the event of any nuclear or radiological disaster, the Ministry of Home Affairs (MHA) is the nodal ministry for coordinating with the various response agencies in the public domain. Since 1987, a Crisis Management Group (CMG) has been functioning at DAE for this purpose and the same is chaired by the Additional Secretary, DAE, and has expert members from different units of DAE and Atomic Energy Regulatory Board (AERB). Radioactive Detection Equipment (RDE's) are being installed in the country at 29 entry and exit points (12 seaports, 14 airports and 3 international borders) as identified by BARC as a part of preparedness measure.

4.7 EPIDEMIC

The following three core capacities are essentially required to deal with any epidemics. These are

(i) Establishment/strengthening of a laboratory based disease surveillance system on infectious diseases for collection of any baseline data, monitoring the disease trends and further detecting the epidemics in early rising phase.

(ii) Development of clinical, entomological, epidemiological and laboratory capacities for investigating the epidemics to understand the transmission dynamics and to characterize the cases in terms of time, person and place.

(iii) Development of response capacities to prevent/control the epidemics in order to minimize the morbidity and mortality.

The district or state Rapid Response teams generally investigate any outbreaks/epidemics. Various central/regional institutes in the country such as National Centre for Disease Control (formerly National Institute of

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Communicable Diseases), Delhi; National Institute of Virology, Pune; National Institute of Cholera and Enteric Diseases, Kolkata; Vector Control Research Centre, Puducherry and other ICMR institutes provide epidemiological, laboratory and entomological support to the states for investigation and control of any widespread outbreaks/epidemics. There is also increasing collaboration with Department of Animal Husbandry in managing emerging zoonotic diseases such as avian influenza and Crimean Congo Hemorrhagic fever.

4.8 CONTROL ROOMS

4.8.1 National Emergency Operation Centre (NEOC)

The National Emergency Operation Centre (NEOC) functions 24×7 to monitor the disaster or disaster like situation in the Ministry of Home Affairs. The advisories to the concerned States/UTs are issued from time to time for keeping watch on the developing situation and take mitigation measures such as vulnerable persons evacuation, relief camps operation, prepositioning of essential commodities etc. based on the feedback received from National Forecasting Agencies viz Indian Meteorological Department, Central Water Commission, Snow & Avalanche Study Establishment etc.

At the time of south west monsoon, daily situation reports are sent to all concerned based on the feedback received from the affected States and concerned Central Ministries and organizations. During the severe nature calamities, special situation reports are also prepared and issued to all concerned. NEOC also issue SMS alerts to all concerned and help set up state Level and district level Control Room.

4.8.2 National Disaster Response Force (NDRF)

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NDRF has been constituted under Section 44 of the DM Act, 2005 by an up-gradation/conversion of eight standard battalions of Central Para Military Forces i.e. two battalions each from Border Security Force (BSF), Indo-Tibetan Border Police (ITBP), Central Industrial Security Force (CISF) and Central Reserve Police Force (CRPF) in order to build them as a specialist force for responding to disaster or disaster like situations. Moreover, two more battalions from the SSB were added in NDRF in the year 2015. The total strength of each battalion is 1149. All 15 battalions have been equipped and trained to respond to natural as well as man-made disasters. Battalions are also trained and equipped for a response during chemical, biological, radiological, and nuclear (CBRN) emergencies.

Task and Role of NDRF

NDRF is serving the needs of all classes of people and the force is working with great enthusiasm and courage in different types of disaster related fields such as CBRN (Chemical, Biological, Radiological and Nuclear Defense), floods, earthquakes, tsunamis and major accidents etc. The main task of NDRF is to provide disasters specialist response which is as follows:

- Nuclear, Biological, and chemical weapon disasters (Decontamination of the area and personnel)
- Debris removal
- Victims extrication - live or dead
- First medical response to victims
- Moral support extension to victims
- Assistance to civil authorities in relief material distribution.
- Co-ordination with sister agencies
- Capacity building
- Providing assistance to foreign countries
- Other activities of NDRF: Besides search and rescue operation undertaken during emergency situation, NDRF is engaged in the following activities
- Acquainting personnel with their area of responsibility in vulnerability to different disasters through the conduction of familiarization exercise.

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- To conduct mock drill/exercises in coordination with other stake holders for improving coordination response during disasters.
- To conduct community awareness program for capacity building.
- To organise demonstrations and exhibitions as part of community awareness.
- To increase the skill and expertise of NDRF personnel by conducting different kinds of trainings.
- Trainings of State Disaster Response Force (SDRF), community and NGO's for disaster management.

NDRF has played an important role in the country's disaster management. The emergency response and community capacity building programme conducted by NDRF has inculcated a sense of confidence in the people towards the responsibility of Government in providing quick response at the time of disaster. The NDRF's present strength is not enough to provide an immediate response across the country and hence, after analysing the vulnerability profile of the country, there exist a need to raise more battalions.

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Recovery, Reconstruction, and Rehabilitation

5

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5.1 Disaster Recovery
5.2 Guiding Principles for Post-Recovery
5.3 Role of Effective communication in Disaster Recovery

5.1 DISASTER RECOVERY

The International Strategy for Disaster Reduction (ISDR) defines recovery as the "decisions and actions taken after a disaster with a view to restore or improve the pre-disaster living conditions of the stricken community, while encouraging and facilitating necessary adjustments to reduce disaster risk".

There are different modes of disaster recovery: **Repatriation** and **Rehabilitation**

- **Repatriation:** After the emergency is over, displaced people return to their place of origin.
- **Rehabilitation:** It is a restoration of basic social functions-
  (i) Community recovery (including psychological)
  (ii) Infrastructure recovery (services and lifelines)
  (iii) Economy recovery (financial, political)
  (iv) Environment recovery

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5.2 GUIDING PRINCIPLES FOR POST-RECOVERY

- Inclusion of disaster risk reduction in mainstream recovery and development process.
- Coordination improvement.
- Decentralising recovery planning and programming and promotion of participatory approaches.
- Enhancement of safety standards and integration of disaster risk reduction in reconstruction and development activities.
- Improvement of the affected communities and sectors living conditions.
- Building local and national capacities for increased resilience, risk management and sustainable development.
- Taking advantage of ongoing initiatives
- Gender sensibility
- Demonstrative effects
- Monitoring, evaluation and learning

5.3 ROLE OF EFFECTIVE COMMUNICATION IN DISASTER RECOVERY

Effective communication is an important component of Disaster Recovery and Response during and immediately after the disaster situation as it connects the affected people, families and communities with first responders, support systems and other family members. As,the communication technology has a vital role in all the four phases of disaster management such as mitigation, preparedness, response and recovery, mostly it is used in Disaster Recovery and response phases. Now a days, the interoperability of various communication systems such as internet, mobile-phones, fax, e-mail, radio and television is increasingly becoming useful in Disasters mitigation and prevention. Telecommunication is used as a channel for disseminating information about the coming danger in pre-disaster phase and thus making it possible for people to take necessary

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precautions to mitigate the impact of the disaster hazards.Various telecommunication applications such as remote sensing and GPS also have critical role in tracking up coming hazards, awaking the authorities and giving warning. The best example of telecommunication can be taken at the time of super cyclone in Odisha, when all the telephone lines and mobile towers were destroyed and the Andhra Government made the use of modern technology phones which proved to be very effective in the cyclone.

NOTE: The critical phases of Disaster Management Cycle include recovery consisting of early recovery, long-term recovery, lasting recovery, reconstruction and rehabilitation processes. They offer an opportunity in risk reduction from future hazards and the 'Build back better' approach to be followed after recent disasters give us beneficial lessons to approach in the recovery processes. Hence, it is considered that the experiences mentioned would help the stakeholders in planning effective, efficient and time-bound action plan in the event of disasters.

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Capacity Development

6

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6.2 Sendai Framework and Capacity Development
6.3 Capacity Development Levels and Modes
6.4 Components of capacity development
6.5 National Policy and Plan (NPP) for Capacity Development
6.6 Institutional Arrangements for Capacity Development
6.7 National Institute of Disaster Management (NIDM)
6.7.1 Training Programmes of NIDM
6.8 National Disaster Management Authority (NDMA)
6.9 Financial Arrangements

6.1 CAPACITY BUILDING

Trained man power is required in effective disaster management to deal with complex situations abruptly and effectively to reduce the disaster impact on human life and property. Capacity building needs to include all resources available within a community, society and organisation to reduce the level of risk or the effects of a disaster. It is an important component of **Disaster Risk Reduction (DRR)**. The focus is not only on developing human resources, but also on developing the necessary infrastructure and institutional capacity to have an effective disaster risk reduction. Capacity

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building alone is not the responsibility of the state but rather it requires the involvement of communities, professionals, experts, private sector, NGOs and non-state actors.

The **United Nation Development Programme (UNDP)** describes 'capacity building' as the creation of an enabling environment with appropriate policy and legal frameworks, institutional development, including community participation (of women in particular), human resource development and strengthening of managerial systems.

The **United Nations International Strategy for Disaster Reduction (UNISDR)** define 'Capacity Development' for Disaster Risk Reduction as "the process by which people, organisations and society systematically stimulate and develop their capability over time to achieve social and economic goals, including improvement of knowledge, skills, systems, and institutions. It is important component of disaster risk reduction.

6.2 SENDAI FRAMEWORK AND CAPACITY DEVELOPMENT

Sendai Framework emphasises the necessity for enhancing the technical, financial, and administrative capabilities of institutions, governments and communities to deal with the disaster risks at different levels. It also focused the need for capacity building of women and further aim for social inclusion by developing the capacity of vulnerable population in disaster management. Capacity building to reduce the disaster risk is also necessary for achieving **Sustainable Development Goals (SDGs)**.

6.3 CAPACITY DEVELOPMENT LEVELS AND MODES

There are three levels of capacity development–

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- **Individual level** pertains to people's skills and knowledge including individuals, communities, groups and teams.
- **Institutional level** pertains to internal policies, systems, strategies, arrangements, procedures and framework that allow an organisation to deliver on its mandate by enabling individual capacities to achieve its goal
- **Enabling Environment** refers to policies, legislation, institutional arrangements, leadership, political processes, power relations and social norms.

Modes

There are two modes of enhancing capacity development:

- **Functional capacities** that are needed to formulate, implement, review policies, strategies, programmes and projects.
- **Technical capacities** are linked with particular area of needs and sector requirements or themes.

6.4 COMPONENTS OF CAPACITY DEVELOPMENT

Following are considered as vital components of the capacity development:

- **Training:** It involves the knowledge acquisition, skill sharpening, concepts, rules, or change of attitude and behaviour to enhance the performance of individuals associated with different departments and institutions.
- **Education:** The education sector, is planning for more contextual, practical and application oriented curriculum for students at different levels of schooling.
- **Research:** An organised and systematic way of answers finding to questions is research.

Table 6.1 Various Research in India

Involvement of Various Universities and Research Institutes to Carry out research for Disaster Management
• Case Studies and lesson learnt exercises by NIDM

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- Preparation of Digital Maps
- Development of GIS Platform for Vulnerability Analysis and Risk Assessment
- Seismic Microzonation
- Improved Modelling for Advanced forecasting Capability
- National Disaster Management Information System (NDMIS)

4. **Awareness:** Awareness is defined as the knowledge created through interaction between an agent and its environment. This concept of awareness involves state of knowledge as well as dynamic processes of perception and action.

6.5 NATIONAL POLICY AND PLAN (NPP) FOR CAPACITY DEVELOPMENT

The NPP envisaged the need of capacity building for all stake holders in disaster risk reduction. The National policy emphasised the need of putting in place the appropriate institutional framework, management systems and allocation of resources for efficient prevention and handling of disasters. The theme of capacity building under three broad categories are defined as prevention and mitigation of risk reduction, effective preparedness, response and recovery. In order to have the capacity building for Disaster Risk reduction (DRR), there is need to engage with stakeholders, particularly the communities to understand the existing capacity, needs and the gaps.

6.6 INSTITUTIONAL ARRANGEMENTS FOR CAPACITY DEVELOPMENT

Government of India has setup several institutions for undertaking wide range of activities for capacity development to support different state governments. The Ministry of Home Affairs (MHA) is the overall coordinator of disaster management. The Cabinet Committee on Security

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and National Crisis Management Committee are the key committees involved in top level decision making in regard to disaster management.

6.7 NATIONAL INSTITUTE OF DISASTER MANAGEMENT (NIDM)

An act of Parliament created the **National Institute of Disaster Management (NIDM)** with an aim to play the role of a major institute in India for capacity development. NIDM has been assigned nodal responsibilities for human resource development, capacity building, training, research, documentation and policy advocacy in the field of disaster management which comes under the Disaster Management Act 2005. The vision to create a Disaster Resilient India at all levels, for disaster prevention and preparedness, is given the utmost priority under NIDM.

6.7.1 Training Programmes of NIDM

NIDM undertakes the training along with research, documentation and development of a national level information base. It collaborates with other knowledge based organisation and arranges training for trainers, Disaster Management officials and other stakeholders. NIDM strives to emerge as a 'Centre of excellence' in the field of training. It plays an important role in implementing the National Training Schedule for Disaster Management.

The training imparted by NIDM in different formats includes:

(i) Face-to-face training
(ii) Web-based training
(iii) Satellite based training
(iv) Self study courses
(v) India Disaster Management Congress
(vi) Capacity Building Programme for Engineers and Architects in Earthquake Risk Management.

Table 6.2 **Disaster Management Training**

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- NIDM Provides web Based Training Course **www.onlinenidm.gov.in**
- Comprehensive Disaster Risk Management Framework 6 weeks course; Fee 1500/-
- Damage and Reconstruction Needs Assessment
- Risk Sensitive land use planning
- Climate Change & Disaster Risks
- Financial Strategies for Managing the Economic Impacts of Nature disaster
- Earthquake Risk Reduction
- Safe Cities
- Gender Aspects of Disaster Recovery & Reconstruction.

6.8 NATIONAL DISASTER MANAGEMENT AUTHORITY (NDMA)

NDMA, **chaired by the Prime Minister**, was set up in 2005 as the nodal agency for undertaking disaster management in India. It undertakes the following initiatives in the process of capacity development.

(i) Lays down policies, plan and guidelines for disaster management to ensure timely and effective response to disaster and long term disaster risk reduction.

(ii) Conducts Mock drills and stimulation exercises at district, state and regional level to improve the response capacity of state governments.

(iii) Conducts mock exercises on School Safety through National School Safety Programme to build the capacity of students and teachers for better preparedness.

(iv) Undertakes training of Panchayati Raj Institutions (PRIs) and Urban Local bodies (ULBs)

(v) Awareness Campaigns: NDMA has launched a number of public awareness campaigns through electronic and print media. The focus is

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on building an appropriate environment for DM and creating a high level of impact on the target audience.

- Audio Visual Campaigns
- Earthquake Awareness Campaign
- Cyclone Awareness Campaign
- Flood Awareness Campaign
- Information, Education and Communication (IEC) activities

The NDMA is working on two major aspects of capacity building, one is creating an enabling environment and the other is building organisational capacity through the preparation of various hazard specific guidelines and reports and further identification of several hazard specific building measures. The example of undertaking the biggest capacity building exercise in the country by the NDMA is the National Cyclone Risk Mitigation Project to empower the government, communities and especially the women for capacity building of Cyclone mitigation and risk reduction.

To deal with complex situation effectively and to mitigate the impact of disaster on human life and property, disaster management requires trained manpower. Capacity Development is an integral process of knowledge and skill building and it may include physical, institutional, social or economic means as well as skilled personal or collective attributes such as leadership and management. Capacity Development is much bigger then training and it includes human resource development, skills and access to information, equipping individual with understanding, imparting knowledge and training that enables the individual to perform effectively. Capacity building is not a onetime activity rather it is a continuous process. The programmes, projects and trainings for capacity building should be constantly evaluated to make them relevant and suitable. National Policy on Disaster Management (NPDM) outlines its approach to capacity development. NPDM focused on institutionalised capacity development, training of communities, technical

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education, education on disaster management in school, artisan and other group training and national priorities. NPDM capacity development programme emphasised on training of the DM officials, functionaries, trainers, elected representatives and community representatives. Several institutions like, NDMA, NIDM, NCDC, NFSC and ministries have developed the capacity development programme for disaster management.

6.9 FINANCIAL ARRANGEMENTS

Post disaster, relief and reconstruction is resource intensive and the Government has ensured that an established financial mechanisms be made available both at the state and central level to address the resource requirements for such eventualities. The present scheme of State Disaster Response Fund (SDRF) and National Disaster Response Fund (NDRF) are based on the 13th Finance Commission recommendation operated from 1st April 2010 to 31st March 2015. In case of disasters requiring central support above the SDRF, the processing of request of the state government for support from the union Government is done by Ministry of Home Affairs (DM Division), while the budgetary provisions for the relief funds is dealt by the Ministry of Finance. Moreover, the funds have also been allocated for strengthening of Disaster Management Institutions, stakeholder's capacity development, prevention & mitigation and the response mechanism. Efforts are also underway to ensure that the disaster mitigation and prevention are given adequate importance by utilizing these funds to minimise the impact of disasters.

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International Cooperation

7

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7.2 United Nations Disaster Assessment and Coordination
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- 7.3.1 Three Strategic Goals
- 7.3.2 Five Priority Action Areas

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7.1 OVERVIEW

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As far as global initiatives on disaster management are concerned, India has played an important role. India is playing a lead role in strengthening regional cooperation among South Asian countries for reducing disasters with its expertise and multi-dimensional initiatives. Further, India has hosted the SAARC Disaster Management Centre in New Delhi with a clear mandate for enhancing regional cooperation. India is also a signatory to the **Hyogo Framework of Action** and is committed to achieve the goals set under it through systematic and sustained efforts under the five priority action areas mentioned in the framework.

International Co-operation

- UN office for coordination of Humanitarian Affairs (UNOCHA) for all international disaster response.
- United National Development Programme (UNDP), for mitigation and prevention aspects.
- UN Disaster Assessment and Coordination (UNDAC) System, streamlining Institutional Arrangements for Disaster Response.
- The Asian Disaster Reduction Centre in Kobe (1998) to enhance disaster resilience for the 30 member countries, to build safe communities, to create a society where sustainable development is possible.

Disaster Reduction Day

- NIDM observe Disaster Reduction Day on the **second Wednesday of October**.
- UN General Assembly in 2009, designated October 13 as International Day for Disaster Reduction.
- 2013 Theme: Living with disability and Disasters.
- Rallies and lectures for Awareness of disaster reduction amongst youth, children and general people.

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7.2 UNITED NATIONS DISASTER ASSESSMENT AND COORDINATION

The **United Nations Disaster Assessment and Coordination (UNDAC)**, being a part of the international emergency response system for managing sudden-onset emergencies was formed in the year 1993. It is created to help the United Nations and the governments of disaster-affected countries at the time of sudden-onset emergency and further assists in the coordination of incoming international relief at national level and at the emergency site. The **Office for the Coordination of Humanitarian Affairs** (OCHA), on request of disaster affected government, deploys the UNDAC team to the affected country within short notice (12–48 hours), anywhere in the world. UNDAC team provides technical services such as damage, assessment, on site coordination and information management. It consists of more than 70 members and participating countries along with the OCHA staff and 16 other international and regional organisation including UN agencies.

OCHA has established a structure of clusters so that can work in a coordinated manner with non-UN agencies like NGOs in order to deliver humanitarian assistance to the disaster affected countries. There are a total of eleven different clusters and each one is focusing on a set of specific tasks or functions. Each cluster is headed by one or two UN agencies. To tackle the emergency situation, both UNDAC team and the OCHA clusters coordinate their efforts in the affected country with the UN Humanitarian Coordinator and the UN Resident Coordinator.

7.3 HYOGO FRAMEWORK OF ACTION

In 2005, United Nation convened second world conference on disaster reduction in Kobe, Hyogo, Japan. The representatives from 168 countries attended the conference and adopted the present Hyogo framework for action 2005–15; building the resilience of Nations and Communities to

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disaster. The **Hyogo Framework of Action (HFA)** 2005–2015 was negotiated and adopted by 168 countries to work globally towards sustainable reduction of losses in lives and in the social, economic and environmental assets of communities and countries due to disasters. **The framework set three strategic goals and five priority action areas** given as under:

7.3.1 Three Strategic Goals

The goals as agreed under the HFA are-

(i) The more effective integration of disaster risk reduction into sustainable development policies, planning and programming at all levels, with a special emphasis on disaster prevention, mitigation, preparedness and vulnerability reduction.

(ii) The development and strengthening of institutions, mechanisms and capacities at all levels, in particular at the community level that can systematically contribute to building resilience to hazards.

(iii) The systematic incorporation of risk reduction approaches into the design and implementation of emergency preparedness, response and recovery programmes in the reconstruction of the affected communities.

7.3.2 Five Priority Action Areas

Under HFA, the following priority areas have been identified to concentrate, for making the countries disaster resilient–

(i) Ensuring that disaster risk reduction is a national and a local priority with a strong institutional basis for implementation.

(ii) Identifying, assessing and monitoring risks of the disasters and enhancing early warning.

(iii) Using knowledge, innovation and education to build a culture of safety and resilience at all levels.

(iv) Reducing the underlying risk factors.

(v) Strengthening disaster preparedness for effective response at all levels.

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7.4 ENGAGEMENT OF UN AGENCIES IN DRR

There are various agencies working under the United Nations System, engaged in promoting the policies and programme in Disaster Risk Reduction (DRR).The table depicting the information of such agencies based on the publication of United Nations International Strategy for Disaster Reduction (UNISDR), Geneva is as follows:

Table 7.1 Engagement of UN Agencies in Disaster Risk Reduction (DRR)

Name of Agency	Headquarter	Disaster Reduction Goal
Food and Agriculture Organisation (FAO)	Rome, Italy	FAO addresses DRR, aiming to reduce the vulnerability of people before, during and after disasters.
International Labour Organisation (ILO)	Geneva, Switzerland	ILO aims to reduce the impact of disasters on employment and income by promoting risk reduction measures for productive livelihood and effective and efficient post-disaster recovery efforts.
International Telecommunication	Geneva, Switzerland	ITU's goal in disaster reduction is to strengthen

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Name of Agency	Headquarter	Disaster Reduction Goal
Union (ITU)		emergency communications for disaster prevention and mitigation.
Office of the United Nation High Commissioner for Human Rights (OHCHR)	Geneva, Switzerland	OHCHR emphasises the need for a human right-based approach to be integrated into DRR and programmes, HFA and post HFA activities.
United Nations Centres for Regional Development (UNCRD)	Japan	UNCRD focuses on various disaster management initiatives through multi-lateral collaboration at an international level through its disaster management planning office, HFA office, Japan.
The United Nations Environment Programme (UNEP)	Nairobi, Kenya	Through its disaster risk reduction programme, UNEP aims to reduce the impact of Natural hazards on vulnerable

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Name of Agency	Headquarter	Disaster Reduction Goal
		communities and countries through sound environmental management.
United Nations Educational, Scientific and Cultural Organisation (UNESCO)	Paris, France	UNESCO assists member states in building capacities to with stand and cope with natural events and human induced disasters, including those of technological nature.
The United Nations Framework Convention on Climate Change Secretariat (UNFCCC Secretariat)	Bonn, Germany	The Cancun Adaptation Framework includes specific provisions for Disaster Risk Reduction (DRR) and risk management as part of enhanced action on adaptation to climate change.

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Name of Agency	Headquarter	Disaster Reduction Goal
United Nation Population Fund (UNFPA)	New York	To ensure adequate emergency preparedness and contingency planning at country level and improve environmental protection.
United Nations Human Settlement Programmes (UN-HABITAT)	Nairobi	Increasing investment in disaster risk reduction, addressing root causes of disasters, incorporating vulnerability reduction in development plan and building a culture of prevention.
United Nations Development Programme	New York, USA	UNDP supports disaster-prone countries to develop-comprehensive disaster risk reduction programmes, strengthen institutional and legislative systems, implement community

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Name of Agency	Headquarter	Disaster Reduction Goal
		level disaster preparedness activities including contingency planning and early warning, establish coordination mechanisms to ensure the integration of risk reduction into human development as well as developing national capacities for recovery planning.
United Nations Children's Fund (UNICEF)	New York, USA	UNICEF has developed a global framework for disaster reduction. This focuses on protection and risk reduction for those who are the most vulnerable to disasters: Children and Women.
United Nation Institute for Training and Research	Geneva, Switzerland	Develop capacities of Member States and their local authorities to

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Name of Agency	Headquarter	Disaster Reduction Goal
		reduce risks related to disaster. Develop applied research in the use of remote sensing and satellite based vulnerability and risk mapping.
Office of the High Representative for the least Developed Countries, Landlocked Developing Countries and Small Island Developing States (UNOHRLLS)	United State of America	To strengthen disaster mitigation and preparedness and promote the sustainable development of the Least Developed Countries (LDCs), Land locked developing Counties (LLDCs) and Small Island Developing States (SIDS)
United Nation Office for Outer Space Affairs (UNOOSA)	Vienna	Ensure that all countries and international and regional organisations have access to and develop the capacity to use all types of space-

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Name of Agency	Headquarter	Disaster Reduction Goal
		based information and solutions to support the full disaster management cycle
United Nations University (UNU)	Tokyo	Knowledge generation, capacity developments, knowledge transfer, networking are the four corner stones guiding philosophy and work at UNU.
United Nations Volunteers (UNV)	New York	To mobilise volunteers and strengthen volunteerism initiatives in support of disaster risk reduction and management, especially to strengthen the community capacity to respond to and prevent disasters.
World Food Programme (WFP)	Nanaimo	WFP Disaster Risk Reduction policy aims at strengthening WFP's

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Name of Agency	Headquarter	Disaster Reduction Goal
		compliance with the HFA and mainstreaming DRR at all level within the organisation.
World Health Organisation (WHO)	Geneva, Switzerland	To support countries in building capacity in risk reduction and emergency preparedness.
World Meteorological Organisation (WMO)	Geneva, Switzerland	To enhance the contributions of National Meteorological and Hydrological services, in a more cost-effective, systematic and sustainable manner, towards the protection of lives, livelihoods and property.
The World Bank	Washington DC	The overarching objective is to maintain disaster reduction and climate change adaptation in

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Name of Agency	Headquarter	Disaster Reduction Goal
		development strategies of the countries.

7.5 SENDAI FRAMEWORK ON DISASTER RISK REDUCTION

The third United Nations (UN) **World Conference on Disaster Risk Reduction (WCDRR)** held in Sendai, Japan adopted the **Sendai Framework for Disaster Risk Reduction 2015–2030 (SFDRR)** on 14–18 March 2015. It is the first major agreement post-2015 development agenda, with **seven targets and four priorities for action**. The Sendai Framework is the successor agreement to the **Hyogo Framework for Action (HFA) 2005–2015**. The Sendai Framework's primary focus on risk reduction and resilience is a common element highlighted in all the 2030 development agendas adopted by all member states of the United Nations. The Sendai Framework introduces seven global targets to assess global progress toward the strengthening of disaster risk reduction to lessen the losses of life and property occurring world wide due to disasters.

7.5.1 Global Targets

- Substantially reduce global disaster mortality by 2030, aiming to lower average per 100,000 global mortality rates in the decade 2020–2030 compared to the period 2005–2015.
- Substantially reduce the number of affected people globally by 2030, aiming to lower average global figure per 100,000 in the decade 2020–2030 compared to the period 2005–2015.
- Reduce direct disaster economic loss in relation to global gross domestic product (GDP) by 2030.

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- Substantially reduce the damage related to disasters with the help of critical infrastructure and disruption of basic services, especially health and education facilities and also develop resilience by 2030.
- Substantially increase the number of countries with national and local disaster risk reduction strategies by 2020.
- Substantially enhance international cooperation to developing countries through adequate and sustainable support to complement their national actions for implementation of this Framework by 2030.
- Substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information and assessments to the people by 2030.

7.5.2 Priorities Areas for Action

SFDRR defined four areas where action is required:

(i) Understanding disaster risk.

(ii) Strengthening disaster risk governance to manage disaster risk.

(iii) Investing in disaster risk reduction for resilience.

(iv) Enhancing disaster preparedness for effective response and to **Build Back Better** in recovery, rehabilitation and reconstruction

7.6 POST SENDAI GOVERNMENT INITIATIVE AND WAY FORWARD

In accordance with its commitment to Sendai Framework, Government has taken several important initiatives after Sendai Declaration. Some of them are as follows;

1. Hosting of the **Asian Ministerial Conference on Disaster Risk Reduction (AMCDRR)** and adopting 'New Delhi Declaration' and 'Regional Action Plan in order to implement Sendai Framework'. Moreover, the first World Tsunami Awareness Day was also commemorated at the conference to spread tsunami awareness.

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2. The AMCDRR also provided an opportunity for multiple stake holders to come together on a common platform and further make specific commitments and guidance for the implementation of Sendai Framework in Asia and Pacific. At AMCDRR, the Prime Minister figured ten point agenda for the implementation of Disaster Risk Reduction efforts. The PM ten point agenda is as follows:

(i) Ensure that all the development projects –airport, roads, canals, hospitals, schools, and bridges-are built to appropriate disaster resilient standards and contribute to the resilience of the communities they seek to serve. Build a coalition to support disaster resilient infrastructure.

(ii) Work towards risk coverage for all –starting from poor households to small and medium enterprises to multi-national corporations to nation states.

(iii) Encourage greater involvement and leadership of woman in disaster risk management.

(iv) Invest in risk mapping globally for all hazards.

(v) Leverage technology to enhance the efficiency of our disaster risk management efforts.

6. Develop a network of universities to work on disaster issues.

7. Utilise the opportunities provided by social media and mobile technologies.

8. Build on local capacity and initiative.

9. Ensure that opportunity to learn from a disaster is not wasted. Establish a facility for technical support to post-disaster reconstruction of houses.

10. Bring about greater cohesion in international response to disaster.

(c) Government of India has issued set of priority actions to all the State Government on the basis of Sendai Framework (2015–2030) Goals, Targets and priorities. The government has also extended US $1 million grant to UNSIDR for the effective implementation of Sendai Framework in Asian Region.

(d) Government has strengthened the NDRF both in terms of training and equipment so as to empower it as a professional disaster response force.

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(e) Government has approved the creation of National Disaster Response Reserve (NDRR) through a revolving fund of Rs. 250 crore to be operated by the NDRF. This fund would enable the NDRF to maintain a ready inventory of emergency goods and services comprising of essential items required at the time of disaster.

(f) The government is keen in sharing India's expertise and help other countries in disaster response as done during Japan Earthquake in 2011 and Nepal earthquake of 2015. The government is making consistent efforts by hosting the SAARC Disaster Management Centre to reduce disaster risks and promote knowledge sharing to enhance regional cooperation among SAARC member countries. As such the INCOIS, Hyderabad provides early warning not only to India but also to 28 countries in the Indian Ocean Rim.

(g) In order to expand the capacity building, NIDM has signed a MoU with Jawaharlal Nehru University (JNU) for financial assistance and academic cooperation for the establishment of a Centre for Excellence in Disaster Research and Resilience Building at JNU. Besides this, the government has formed the NDRF Academy at Nagpur for training of personnel involved in disaster management and allocated 250 crores for strengthening and up-gradation of National Fire Safety College to make it state of the art facility.

(h) The Government has implemented the recommendations of 14th Finance Commission and allocated funds of nearly Rs 61,220 crores in State Disaster Response Fund (SDRF) for all the states w.e.f. 2015–16 to 2019–20.

(i) The Sendai framework also stressed upon other stakeholders such as the private sector other than the Government for involvement in disaster preparedness and mitigation planning as well as relief and recovery phase.

7.7 ASIAN MINISTERIAL CONFERENCE ON DISASTER RISK REDUCTION 2016

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The conference was held in New Delhi in November 2016 with the aim of providing a common platform to member countries for sharing best practices in the field of **Disaster Risk Reduction**. Nearly, 4000 participants from the Asian region participated in the conference. It was the first Asian conference on Disaster Risk Reduction after Sendai Framework (2015–2030). The conference aims to identify targets and priority action areas towards the reduction of disaster risk by '**reducing the damage caused by natural hazards like earthquakes, floods, droughts and cyclones, through an ethic of prevention**'. The AMCDRR, formed in 2005, was a biennial conference organised jointly by different Asian countries and the United Nations Office for Disaster Risk Reduction (UNISDR). Two important documents –The New Delhi Declaration and Asian Regional Plan for the implementation of Sendai Framework were adopted in the conference.

7.8 NEW DELHI DECLARATION

The 'New Delhi Declaration' is a political statement spelling out the commitment of participating governments towards preventing and reducing disaster risk, and strengthening the resilience of communities, nations and the Asian region. Recognising the need to accelerate the implementation of global frameworks, it commits to a people-centred and whole-of-society approach towards DRR. It also emphasises the need to enhance the capacity of communities and ensure participation of all stakeholder groups towards achieving resilience.

■ *Particularly, the declaration urges stakeholders to:*

- Shift from **response to risk reduction**, ensuring that planning prioritises prevention, preparedness, and resilience rather than post-disaster relief.
- **Strengthen risk-informed governance** by improving data collection and sharing, aligning policies across sectors, and enabling stakeholder participation at national and local levels.

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- **Increase investment** in resilience, including capacity-building, science and technology, resilient infrastructure, and multi-hazard early warning systems with inclusive contingency planning.
- **Promote inclusive and community-centred DRR**, ensuring leadership roles and meaningful participation of women, youth, and persons with disabilities, while leveraging local knowledge and initiatives.
- Integrate risk reduction into recovery and reconstruction, adopting "build back better" principles to reduce future vulnerabilities.
- **Enhance cooperation and partnerships**, reinforcing global, regional, and cross-sectoral collaboration, including support through international assistance and shared regional frameworks.
- Use **awareness platforms and adopt monitoring** frameworks, such as global DRR awareness days and Sendai Framework indicators aligned with national priorities.

In 2023, the G20 New Delhi Declaration underscored that DRR is part of the G20 Architecture, which reflects that DRR is not a standalone issue but an integral part of addressing the interconnected challenges facing the world today.

7.9 ASIAN REGIONAL PLAN

The 'Asian Regional Plan for Implementation of the Sendai Framework' focuses on 'How to' reduce disaster risk at national and local levels. It has arrived a longer road map of cooperation and collaboration, spanning the 15-year horizon of the Sendai Framework, as well as a two-year action plan to further disaster risk reduction with specific, actionable activities.

7.10 DISASTER RISK MANAGEMENT IN TERMS OF SUSTAINABLE DEVELOPMENT

Disaster Management has progressed a long way from managing disaster events to managing disaster risk. Disaster risk management refers to

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scientific assessment of underlying natural or anthropogenic hazard and vulnerabilities and undertaking necessary measures to prevent the creation of disaster risks initiation. In disaster risk management, the existing risks are reduced through a combination of various structural and non-structural measures. Disasters are closely linked with the development as per the following three points-

1. Disasters eat away the development hard earned gains over the years.
2. The vulnerable communities are exposed to the disaster risks due to lack of development.
3. Ironically, development creates new disaster risks such as houses and infrastructure development without the compliance of zoning and building regulations are vulnerable; mining and industries in ecologically sensitive zones may destroy the natural buffer to disasters, while fossil fuel based production and consumption enhance risks of climate related disasters.

The below figure shows the process transformation from disaster management to disaster risk management.

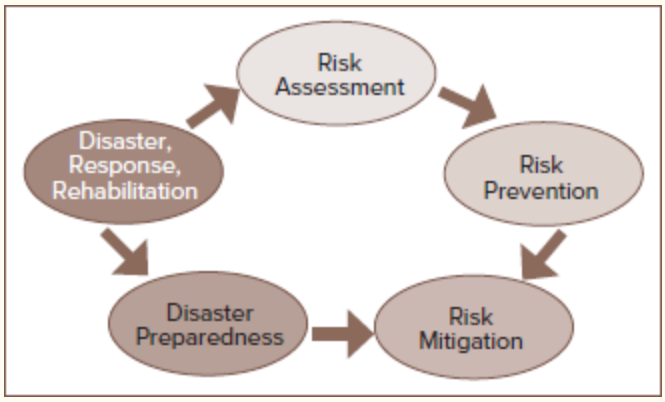

Figure 7.1 Disaster Risk Management

The following three independent processes, in 2015, provided a new direction and momentum to the disaster resilient sustainable development.

- Sendai Framework for Disaster risk Reduction (2015–2030) adopted in Sendai, Japan in March 2015.
- The 2030 agenda for sustainable development adopted by the UN General Assembly in Sept. 2015 inserted disaster risk management in 8 out of 17

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Sustainable Development Goals.

- The Paris agreement on climate change signed in Dec. 2015 defined eight specific action areas for enhancing 'understanding, action and support for disaster risk reduction.

In the finalisation of all three global agreements of 2015, India has played an important role. The Sendai Framework implementation in combination with the Sustainable Development Goals and Paris Climate Agreement provide opportunities to address this so far neglected but challenging task of disaster risk management in India.

7.11 MUTUAL REINFORCEMENT OF DRR, SDGS, AND CLIMATE CHANGE

A comprehensive approach to disaster risk reduction requires the need for coherence between the Sendai Framework, the Sustainable Development Goals, and the commitments under the Paris Agreement. Instead of treating these as separate domains, Disaster risk reduction, sustainable development, and climate change adaptation and mitigation are mutually reinforcing processes that must evolve in an integrated manner.

The connection between the three frameworks follows the process of integrating risk reduction into the development agenda. Disaster risk reduction must not only involve response mechanisms, but should also integrate planning of development and livelihood strategies. Since the impact of climate change acts as risk multiplier, therefore integrating adaptation becomes essential to secure the development gains and prevent new risk creation. Similarly, sustainable development should take into account hazard exposure, vulnerability reduction, and the creation of resilient infrastructure to embed resilience structurally within the framework of disaster risk reduction.

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Operational Convergence among Sendai Framework, SDGs, and Paris Agreement

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- Alignment of Targets and Priorities: The Sendai global targets correspond closely with key SDGs and the Paris Agreement's adaptation goals. Efforts to reduce disaster mortality and the number of affected populations align with:
  - SDG 1: Poverty Reduction
  - SDG 11: Resilient and Sustainable Cities
  - SDG 13: Climate Action
- **Shared Focus on Reducing Vulnerability:** The Paris Agreement prioritises enhancing adaptive capacity and reducing vulnerability. This reinforces Sendai's emphasis on protecting critical infrastructure and essential services such as health, education, and transport systems.

National Initiatives Demonstrating Convergence: India's overarching strategies reflect this integration, such as:

- PM's Ten Point Agenda for DRR
- National Missions under the National Action Plan on Climate Change (NAPCC)
- National Initiative on Climate Resilient Agriculture (NICRA)

Together, these show how development planning, preparedness, and climate adaptation can be advanced through shared strategies and coordinated investment.

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Types of Disaster – Their Causes, Vulnerability and Mitigation Measures

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8.1 EARTHQUAKES

8.1.1 Definition

Earthquake is the shaking of Earth's surface due to sudden release of energy in the earth's lithosphere that creates seismic waves. Earthquakes can vary in size, from weak to violent enough to destroy the cities.

8.1.2 Causes

Mostly, the earthquakes are caused by natural factors due to rupture of geological faults and tectonics plate movement and some of them are also caused by manmade factors such as mining and nuclear tests.

8.1.3 Major Earthquakes in India

The India's major earthquakes are 1897 Shillong, 1905 Kangra , 1934 Bihar-Nepal 1950 Assam-Tibet, Latur Maharashtra in 1993, Chamoli-Uttarakhand in 1999, Bhuj-Gujarat in 2001, Andaman Island in 2002 and

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2009, Kashmir in 2005 and 2013, Sikkim in 2006 and 2011, India-Nepal in 2015, North-east India 2016.

8.1.4 Earthquakes Vulnerability Area

According to the current seismic code, the country is divided into four seismic zones (zone 2, 3, 4, 5) and zone 5 expects the highest level of seismicity whereas zone 2 is associated with the lowest level of seismicity. Zone 5 is known as Very High Damage Risk Zone. The Western and Central Himalayas, North and Middle Bihar, the North-East Indian region, the Rann of Kutch, the Kashmir region and the Andaman and Nicobar group of islands fall under this zone. The country's 10.90% of the area falls under Zone 5.Zone 4 is termed as the High Damage Risk Zone. The region of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, the parts of Indo-Gangetic plains (North Punjab, Chandigarh, Western Uttar Pradesh, Terai Region, North Bengal, Sundarbans) and the capital of the country, Delhi fall under this zone. Moreover, the Patan area in Maharashtra and the northern part of Bihar such as Raxaul near Nepal and India border also falls under this zone. The country's 17.30% of the area falls under Zone 4. Zone 3 is referred as Moderate Damage Risk Zone and Mumbai, some parts of Kashmir and Andaman and Nicobar Islands fall under Zone 3. The country's 30.40% of the area falls under Zone 3. Zone 2 and Zone 1 region is classified as the Low Damage Risk Zone. The rest of country falls under this Zone. The country's 41.40% of the area falls under Zone 1 & 2.

8.1.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate earthquakes–

- In most of the earthquakes, the widespread damage to civil buildings results in huge loss of life and property. In spite of having a mandatory **National building codes (NBC)** for all structures in earthquake prone area to be earthquake-resistant, the regulations are not being strictly followed. To have the better implementation of such building civil codes

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regulations, all structures should be made to comply with building rules and revised town planning bye-laws. All central ministries, state government/SDMA and ULBs need to facilitate the implementation and enforcement of seismically safe design and construction of buildings, bridges, flyover, ports and harbours etc. and should organize training programmes among civil professionals and masons for designing and construction of building as per earthquake resistant safety codes.

- Seismic retrofitting of priority and lifeline buildings are very much required in high risk areas which are prone to earthquake such as in seismic zones III, IV and V and **Building Materials and Technology Promotion Council** (BMTPC) is undertaking various buildings/projects under this category. The state government in consultation with Hazard safety cells reviews the existing buildings and prepares the list priority wise for seismic strengthening and retrofitting of structures in order to protect them from collapse at the time of earthquakes. Buildings of national importance such as Rashtrapati Bhawan, Parliament house, Supreme Court of India, Central and State Secretariats, historical monuments, museums, heritage buildings, strategic assets and vital installations such as power plants, and water works should be included in a priority list of seismic strengthening and retrofit.
- The state government/SDMA in collaboration with nodal agencies and other key stakeholders should increase public awareness for incorporating seismic safety standards in the construction of residential buildings with the help of brouchers, manuals, booklets, action plans, videos and demonstration kits and trainings. A handbook of earthquake safety manual consisting of vulnerability risk information must be prepared for general public highlighting the safety of persons, buildings, and structures.
- Various organizations such as NGOs, SHGs, youth's organization, women's groups, volunteers agencies, civil defence etc. in collaboration with SDMA and DDMA will coordinate in performing various earthquake response activities.
- The government has taken a national initiative in launching a **National Earthquake Risk Mitigation Project (NERMP)** in order to reduce the

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vulnerability in the high risk earthquakes prone district and further strengthen the structural and non-structural earthquake mitigation measures. NDMA has prepared a detailed project report and it is under with various stake holders.

8.2 TSUNAMI

8.2.1 Definition

A tsunami (in Japanese 'tsu' means harbour and 'nami' means wave) is a series of water waves caused by the displacement of a water body of very large volume, usually an ocean. Tsunami waves in the deep ocean may have very long wavelength of hundreds of kilometre and travels at about 800 km per hour, but an amplitude of only about 1 km.

8.2.2 Causes

Seismicity generated tsunamis are caused due to vertical displacement of the overlying water as a result of abrupt deformation of sea floor. The water above the reformed area is displaced from its equilibrium position on the occurrence of earthquakes beneath the sea level. The tsunami waves have small amplitude but a very long wavelength (often hundreds of kilometre long) and are produced on the release of energy and may be caused by non-seismic event also such as a landslide or impact of a meteor.

8.2.3 Major Tsunami in India

On 26th Dec 2004, one of the most destructive tsunamis, the Indian Ocean Tsunami has hit the India coast and 13 other countries in Indian Ocean Region. These tsunamis resulted in the huge devastation and destruction of property, assets and infrastructure in coastal areas including the combined toll of 238,000 casualties (including 51,500 people missing) and nearly more than 1.5 million people displaced in fourteen countries. Around

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10,749 people lost their lives and 5,640 people were missing in the Tsunami affected areas in India.

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8.2.4 Tsunami Vulnerability Area

The areas to be affected in India by tsunami are the country's east and west coasts and the island regions which are prone to subduction zone earthquakes originating from the two high risk regions, named as the Andaman-Nicobar-Sumatra Island Arc and the Makran subduction zone, north of Arabian Sea.

8.2.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate earthquakes

- Setting up a network of land based seismic station for early earthquake detection and estimating the source parameters affecting Indian Ocean region in the major tsunamigenic zones (Java-Sumatra-Andaman-Myanmar belt and the North Arabian Sea belt) to communicate the same to tsunami early warning system.
- Installation of a network of 10–12 bottom pressure recorders (BPRs) (10 in Bay of Bengal and 2 in Arabian Sea) for the detection of tsunami generation around the major tsunamigenic zones.
- Installation of 50 real time tide gauges in order to monitor the progress of tsunami and storm surges. In this regard, 17 station real time Seismic Monitoring Network is to be established by IMD and the data from these 17 stations will be transmitted to IMD, Delhi and INCOIS, Hyderabad for interpretation.
- Setting up of a **National Early Warning Centre** (NEWC) by Ministry of Earth Sciences at INCOIS, Hyderabad for tsunami and storm surges monitoring in India on 24x7 basis to generate Tsunami watch, advisory and alert bulletin to local, state, national and international entities, media and public. The NEWC will be connected with the **National Emergency**

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Operation Centre (NEOC) for all emergency communication and decisions.

- Undertaking the capacity building, training and education of all stakeholders on the utilization of maps, warnings and advisories in the tsunami affected coastal areas.
- Continuous R&D efforts need to be undertaken for the development of a high resolution model for tsunami wave propagation in Indian Ocean.
- MoES with the help of its nodal institutes will prepare and distribute the manuals and tsunami hazard zonation maps to the public through SDMAs and concerned Ministries & Departments of the Government of India in order to create awareness on tsunami risk and vulnerability among the coastal communities, state administrative authorities and other stakeholders in the coastal districts.
- Regular comprehensive public awareness campaigns will be developed and launched by SDMAs/DDMAs at the national, state and district levels, especially in high risk areas for familiarization with the tsunami warning dissemination mechanism and responsibilities of various stakeholder groups. These awareness campaigns will be undertaken through workshops, drills and exercises, screening of video films, distribution of information resources,
posters etc.

8.3 LANDSLIDES

8.3.1 Definition

It is a form of mass movement that includes a wide range of ground movements, such as **rockfalls**, deep failure of slopes, shallow debris flows and gravity is the major driving force for the occurrence of landslide. Landslides also takes place underwater, termed as submarine landslide.

8.3.2 Causes of Landslides

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The major reason for landslides is the changing of slope from stable to an unstable condition and this change in slope stability is caused by both the natural as well as man-made disaster factors. The natural factors are groundwater pressure, absence of vertical vegetation, soil nutrients, soil structure, erosion of the slope toe by rivers or sea waves, slope weakening by snow, glaciers melting, earthquakes and volcanic eruptions. The man-made factors are deforestation, cultivation, construction which destabilizes slopes, blasting, earthwork, removal of deep-rooted vegetation, and construction, agricultural or forestry activities.

8.3.3 Major Landslides in India

The major landslides in India are the Kedarnath landslide in Uttarakhand, Malin landslide and Mumbai landslide in Maharashtra, Amboori landslide in Kerala, Varnavat landslide in Uttarkashi District, Malpha landslide of Pithoragarh district, Okhimath landslide in Chamoli district, Uttarakhand, Wayanad landslides in Kerala, Katra (Vaishno Devi) landslide in Jammu and Kashmir and Paglajhora in Darjeeling district in West Bengal and Guwahati landslide in Assam.

8.3.4 Landslide Vulnerable Area

The major landslide vulnerable areas in India are the Western Himalayan region in Uttar Pradesh, Uttarakhand, Himachal Pradesh and Jammu & Kashmir, the Eastern & North-eastern Himalayan region in West Bengal, Sikkim and Arunachal Pradesh states, the Naga-Arakkan Mountain belt in states of Nagaland, Manipur, Mizoram and Tripura, the Western Ghats region including Nilgiris in states of Maharashtra, Goa, Karnataka, Kerala & Tamil Nadu, the Plateau margins of the Peninsular India and Meghalaya plateau in North-east India.

8.3.5 Mitigation Measures

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The following measures are required to be implemented to mitigate earthquakes

- Preparation of Disaster Management Plans both at National, State and District level for the management of landslide hazard. The plan is to be prepared by the NEC (National Executive Committee) at the National Level, State Government at the state level and District Administration at District level in accordance with the guidelines laid down by the NDMA and SDMAs.
- The NDMA has proposed to launch a **National landslide Mitigation Project** (NLMP) to manage and reduce the risk of landslide occurrence.
- The nodal ministry i.e., Ministry of Mines MoM-GSI Geological Survey of India in collaboration with the Technical Advisory Committee (TAC) and in collaboration with the DST, NDMAs, SDMAs, CSIO, IITs, universities, and other academic institutions will undertake the development of landslide early warning and forecasting system.
- The nodal ministry i.e., Ministry of Mines MoM-GSI in consultation with the TAC and in collaboration with the DST, IITs, universities, and other academic institutions will undertake the efforts in research and development of scientific and innovative designs of surface and sub-surface drainage systems being the most important component of landslide stabilization measures.
- The GSI in consultation with the TAC and in collaboration with the NIDM, IITs, universities, and other academic institutions will provide training to geologists, geo-technical engineers, and professionals from other disciplines involved in landslide hazard investigation and management who need to be kept updated with the latest global developments in this field.
- The State governments in collaboration with the SDMAs, DDMAs, state DGMs Directorates of Geology and Mines, will provide training to all the officers of landslide affected states, such as engineers, geologists, geophysicists, and hydrologists from the SDGMs and other departments involved in developmental activities in hilly regions, especially in the ULBs and PRIs of such states.

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- The SDMAs in collaboration with the DDMAs and NGOs will focus on the community education through the promotion of disaster education, public awareness, community leadership development, and disaster education of unemployed youth, physically challenged, elderly, women, and school children.
- The state governments/SDMAs, district administration of landslide affected areas, (CoA) Council of Architecture and in consultation with the NDMA will establish the necessary techno-legal and techno-financial mechanisms to address the problem of landslide hazards in their respective states in order to ensure that all stakeholders like builders, architects, engineers, and government departments responsible for regulation and enforcement adopt landslide safe land use practices and provide for safety norms.
- The SDMAs/state governments of landslide affected areas in collaboration with the nodal agency such as the MoM-GSI, NIDM, CDMM, IITs, universities and other academic institutions, PTI, DAVP, Prasar Bharati, private broadcasters, NGOs through the use of electronic and print media will make special efforts to organise communities to carry out the landslide mitigation efforts and prepare material for awareness generation campaigns.
- The MoM will establish the Centre for Landslide Research, Studies and Management (CLRSM) as a premier geo-hazard institute with state-of-the art facilities, to carry out various research tasks in the field of landslide management plans and its reduction.
- The NDMA, GSI, NIDM and state governments/SDMAs will set up websites and portals to disseminate information related to landslide safety and will assist in the preparation of films, manuals, and other material in order to create greater public awareness about landslide hazard and importance of land use zoning practices.

8.4 TROPICAL CYCLONES

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8.4.1 Definition

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A tropical cyclone is described as a low-level closed atmospheric circulation of strong winds characterized by a low-pressure centre and is having a spiral arrangement of thunder storms that result in heavy rain. The cyclones occurring between Tropics of Cancer and Capricorn are termed as **Tropical Cyclones**. The tropical cyclone is also named as **hurricane, typhoon, tropical storm, cyclonic storm, tropical depression and cyclone,** according to its location and strength. They are termed as Tropical cyclone in Indian Ocean and are generally have diameter ranging between 100 and 2,000 km.

8.4.2 Causes

Tropical cyclones receive their energy through the water evaporation from the ocean surface and ocean's heat and further form over large warm ocean water bodies which condenses into clouds and results in heavy rain. The frequency of cyclones in the **North Indian Ocean** (NIO) basin is bi-modal, specific to this region and it generally occur in the months of May–June and October–November, with the November is its primary peak and May is its secondary peak period.

8.4.3 Major Cyclones in India

The major cyclones hitting the different states of Indian Coast are the Odisha Super Cyclone (1999), Cyclone Nisha (2008),Cyclone Phyan (2009), Cyclone Jal (2010), Cyclone Laila (2010), **Cyclone Thane (2011), Cyclone Nilam (2012), Cyclone Phailin (2013),** Cyclone Hudhud (2014), Cyclone Vardah
(2016), Cyclone Nada (2016), Cyclone Roanu (2016), Cyclone Maarutha (2017), Severe Cyclonic Strom Mora (2017), Cyclone Ockhi (2017), Cyclone Titli (2018), Cyclone Gaja (2018), Cyclone Phethai (2018), Cyclone Fani (2019), Cyclone Kyarr (2019), Cyclone Maha (2019), Cyclone Vayu (2019), Cyclone Amphan (2020) and Cyclone Tauktae

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(2021), Cyclone Biparjoy (2023), Cyclone Michaung (2023), Cyclone Fengal (2024), Cyclone Asna (2024), and Cyclone Montha (2025).

8.4.4 Cyclone Vulnerable Area in India

The Indian sub-continent exposed to nearly 7% of world's Topical Cyclone is having around 11098 kms coastline and is the worst affected region in the world as far as cyclones are concerned. The country's 13 coastal states/UTs consisting of 84 Coastal districts are affected by the tropical cyclones. Moreover, four states Tamil Nadu, Andhra Pradesh, Odisha and West Bengal and one UT Puducherry on the east coast and one state (Gujarat) on the west coast are highly vulnerable to Tropical Cyclones.

8.4.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate earthquakes

- The Ministry of Earth Science (MoES), MoWR, Coastal State governments as well as some central government departments such as Indian Air Force, Indian Navy, Department of space (DoS), CWC, Indian Council of Agricultural Research will carry out the continuous improvement of the existing meteorological observational and communication network with additional augmentation such as rain gauge network tide gauges, etc. in their respective domain on priority basis for the mitigation of cyclone.
- The MoES, Indian Meteorological Department (IMD), IAF, Indian Navy, Department of space, National remote Sensing Agency through the commissioning of Aircraft Probing of Cyclone (APC) facility along with a combination of manned aircraft and high altitude Unmanned Aerial Vehicles (UAV) fill the critical observational data gaps in case of cyclones over the Bay of Bengal and the Arabian Sea, thereby reducing errors in cyclones track and intensity prediction.
- The NDMA, SDMA, DDMA along with other stakeholders such as Department of Space, Survey of India is commissioning the Nation

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Disaster Communication Infrastructure (NDCI) in order to have state of the art operation centre OC for effective coordination of Disaster Response Actions at the state and local levels and for fusion of information and data pertaining to cyclones early warning system.

- NDMA along with the financial assistance from the World Bank, has launched the **National Cyclone Risk Mitigation Project (NCRMP)**, in order to strengthen cyclone risk mitigation measure and to improve the early cyclone warning dissemination system.
- The coastal state governments along with Ministry of Environment and Forest (MoEF) and coastal Area Development Authority will carry out the developing of **Integrated Coastal Zone Management (ICZM)** frameworks for addressing the sustainability and optimal utilization of coastal resources as an effective cyclone risk reduction measure.
- The SDMAs in coastal states/UT will specify the roles and responsibilities of the State Disaster Management Departments in institutionalizing the Cyclone Risk Mitigation with Developmental Planning and further launch the **Community Based Disaster Management (CBDM)** activities similar to the DRM Project initiatives of MHA in all villages of the 84 districts vulnerable to cyclones.
- The NDMA should conceptualize the establishing of **National Cyclone Disaster Management Institute (NCDMI)** in one of the coastal states to address all the issue related to cyclone risk management and will focus on preparedness, mitigation, response, rehabilitation and recovery process related to cyclone disaster.
- The SDMAs of Coastal states and UTs will establish a comprehensive **Cyclone Disaster Management Information System (CDMIS)** in order to cover all the phases of DM to provide on-line services to the state departments of Disaster Management.
- In order to have speedy completion of digital spatial data generation for covering 84 coastal districts that are vulnerable to cyclones and to evolve the holistic cyclone risk reduction strategies, the integration for ongoing efforts of the Survey of India, Department of Space (DoS) under National

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Spatial Data Infrastructure, National Database for Emergency Management and MoEF will be carried out on priority basis.

8.5 FLOOD

8.5.1 Definition

Flood is overflow of water from water bodies like river, lake that results in the submergence of dry land. From time immemorial, flood is a frequent phenomenon in the country. Every year, floods of varying intensity affects the different parts of country. Due to different climates and rainfall pattern, some parts of the country face flood and some experiences drought like situation. The country receives maximum rainfall (around 80%) due to south west monsoon season.

8.5.2 Causes

The various reason which causes flood are riverbeds silting, reduction in the capacity of river channels, erosion of river beds and banks, obstruction to river flow due to landslide, inadequate capacity of the rivers, poor drainage facilities.

8.5.3 Flood Vulnerability Regions

India is broadly divided into four major flood prone regions. The first one is the Brahmaputra River region consisting of the rivers Brahmaputra and Barak and their tributaries, and the states of Assam, Arunachal Pradesh, Meghalaya, Mizoram, Manipur, Tripura, Nagaland, Sikkim and the northern parts of West Bengal. The Second is the Ganga River Region having many tributaries, including the Yamuna, Sone, Ghaghra, Raphti, Gandak, Burhi Gandak, Bagmati, Kamla Balan, Adhwara group of rivers, Kosi and the Mahananda and covering the states of Uttarakhand, Uttar Pradesh, Jharkand, Bihar, south and central parts of West Bengal, Punjab, parts of Haryana, Himachal Pradesh, Rajasthan, Madhya Pradesh and Delhi. Thirdly

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is the North-west River Region including the main rivers in this region are the Indus, Sutlej, Beas, Ravi, Chenab and Jhelum and covering the states of Jammu and Kashmir, Punjab and parts of Himachal Pradesh, Haryana and Rajasthan. Fourth is the Central India and Deccan Region having the important rivers like Narmada, Tapi, Mahanadi, Godavari, Krishna and Cauvery and covering the states of Andhra Pradesh, Karnataka, Tamil Nadu, Kerala, Odisha, Maharashtra, Gujarat and parts of Madhya Pradesh.

8.5.4 Major Floods in India

Bihar Floods (1987), Assam Floods (2012), Maharashtra Floods (2005), Gujarat Floods (2005), Ladakh Floods (2010), Uttarakhand Floods (2013), Jammu and Kashmir Floods (2014) are the major and deadliest floods in the history of India.

8.5.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate floods.

- The NDMA has launched a National Flood Mitigation Project in the eleventh five year plan to mitigate the risk, severity and consequences of floods and to have effective preparedness of floods and will prepare the detailed project report along with the help of other stakeholders such as central ministries and various other central and state government departments.
- The SDMAs will carry out the studies for the design and construction of flood embankments/ flood walls/flood levees along with other works such as reservoirs, channel improvement works, drainage improvement and construction structures as short term and long term solution to flood problems.
- The **Ministry of Water Resources (MOWR)**, **Central Water Commission (CWC)** and the state governments/SDMAs will solve the problem of rise in riverbeds with the help of science and technology, academic institutions and reputed corporate sector firms and explore the

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techno-economic viability of desilting/dredging riverbeds in order to mitigate the effects of water level rise in the riverbeds.

- The state Governments/SDMA will review the capacity of existing sluices and drainage channel in the regions facing drainage congestion and further construct the new channels and increase their capacity to ensure excess rainwater flow and further bye-pass the water to the new channels to prevent floods.
- The state governments/SDMAs will undertake appropriate watershed management measures including afforestation, check dams, detention basins etc. in the rivers catchment to prevent soil erosion, enhance water conservation and minimize water and sediment runoff. The coastal states/port authorities will plan and execute the construction of sea walls/coastal protection works in order to minimise the flood risk in coastal areas.
- The state governments/SDMAs will provide adequate number of raised platforms/flood shelters for the people affected by flood at suitable locations along with the basic amenities such as drinking water, sanitation, medical treatment, cooking, tents, lantern etc.
- The state governments/SDMAs will carry out the inspection of all structural measures such as dams, flood embankments, levees before and after the monsoon in order to ensure the identification, restoration/strengthening of vulnerable spots and take immediate measures to strengthen them. Moreover, the State governments/SDMAs with CWC and other states will implement the IWRM system for all the river basins and sub-basins.
- The CWC, IMD and the state governments will enhance the density of the basin-wise network of rain gauge and river gauge stations and further establish basin-wise system of Flood Forecast and early warning including data collection, data transmission using modern automatic telemetry data transmission techniques e.g. satellite, VSAT, Internet/e-mail, mobile phones etc., Flood Forecast dissemination using computer networks and satellite, flood hazard mitigation model, damage assessment and

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quantification models, advisories for flood relief routes in order to mitigate the effect of flood hazard.

- The State governments/SDMAs will carry out the study for the creation of natural depressions, swamps and lakes near rivers and utilise them for temporary storage of flood waters. The state governments/SDMAs in consultation with MOWR will form and enforce appropriate laws for implementing flood plain zoning regulations.
- The ISRO and NRSA will create the formulation of digital, thematic and cartographic data base for hazard zonation and risk assessment of Flood vulnerable areas in the country.
- The MOWR will take steps for strengthening of the **Ganga Flood Control Commission (GFCC)**/Brahmaputra Board and in close collaboration with the NDMA, will establish a National Flood Management Institute (NFMI) as a centre of excellence and having state-of-the-art equipment in one of the flood prone states.
- The State Disaster Management Authority (SDMA) laid down the policies and plans for DM in the state and will collaborate with the NFMI in organising the training of personnel in the field of FM. Similarly, the DDMA and PRIs and ULBs at district and panchayat level respectively will carry out relief, rehabilitation and reconstruction activities in the affected areas and will prepare DM plans as per the guidelines of the NDMA, SDMAs and DDMAs.
- The NDRF, SDRF and Civil Defence coordinated by state governments/SDMAs/DDMAs, will impart training to public in the field of flood preparedness, mitigation and response.
- The NEC with the approval of NDMA will prepare the National Disaster Management Plan (DMP) consisting of the Flood Management (FM) aspects and similarly, the state governments/SDMAs will prepare their FMPs in accordance with these guidelines These FM plans will be disseminated widely among various stakeholders for creating greater public awareness and will incorporate all the features of the **Emergency Operation centre (EOCs)** including their establishment and operations.

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8.6 URBAN FLOODS

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8.6.1 Definition

Urban floods are the type of floods caused due to lack of drainage in an urban area. Urban flooding can take place due to flash floods, coastal floods or river floods. Urban cities are densely populated and flooding in these areas not only affect the people's livelihood and endanger their life but also results in the damage of infrastructure amounting heavy loss of life, property and economy.

8.6.2 Causes

The urban flooding is caused mainly by two factors- meteorological/natural and human factors. The natural/meteorological factors are heavy rainfall, cyclonic storms and flash floods due to climate change, shortage and extinction of lakes, urban heat island effect which has increased the rainfall in cities, heavy silting in rivers and reduction in the catchment area of rivers. The human factors responsible for urban flooding are increasing population, deforestation, unplanned urban development, unauthorized colonies, unscientific drainage system, lack of proper solid waste management system and illegal encroachment on flood plains and hindrance to flood flow.

8.6.3 Major Urban Floods

In the past decade, the country has witnessed major urban floods such as Mumbai Floods (2005) (2007) (2015), Chennai Floods (2004) (2015), Kolkata (2007) (2013), Srinagar (2014) (2015), Bangalore (2009) (2013), Surat (2006) (2013).

8.6.4 Mitigation Measures

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The following measures are required to be implemented in order to mitigate urban floods.

- Improving the existing drainage system in the cities by ending all the encroachment and enlarging the existing drainage cover to cater the increased flow of water during heavy rains.
- Creation of small storage ponds and underground water tunnels to cater the water storage during rains may be formed at various places in the cities and the same can be utilized for rain water harvesting.
- Creation of lakes, land use planning in urban cities and restoration of flood plains in the cities to cater the increased amount of water during heavy rain.
- Urban planning and development of cities in a scientific manner.
- Modification of river system including increasing the flow area, protect the banks from the erosion, increase the heights of the banks and desilting of drains.
- District administration should form emergency operation centres to get prepared for urban flash floods in coordination with other departments of state government.
- The administration should strive to work in order to increase the cover of wetland and plain areas in the cities for the flow of water during heavy rains.
- The administration should also increase the awareness and provide training to masses in close collaboration with NGO and different civil communities for plantation of trees and cleaning of drains in the cities.
- Improvement of municipal solid waste management system to improve the flow of water in the drains.
- Strict enforcement of laws in respect of construction and encroachment activities.
- The administration should prepare action plan for pre-monsoon and post-monsoon checks and activities to prevent the flash floods.

8.7 CLOUD BURSTS

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8.7.1 Definition

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A cloudburst is short-term extreme precipitation that takes place over a small area resulting in the release of huge amount of water. Mostly, the cloud bursts occur in hilly and mountainous regions such as Himachal Pradesh, Leh & Ladakh and Uttarakhand regions.

8.7.2 Causes

The main causes for increasing of cloudburst are global warming, climate change, deforestation, reduction of flood plains and river banks and increased construction on hill plains and seismic zones.

8.7.3 Major Cloud Burst in India

The major cloudbursts are Ladakh Cloudburst (2010), Uttarakhand Cloudburst(2013), Mandi (HP) Cloudburst (2015), Kashmir valley cloudburst (2015), Lahaul-Spiti cloudburst (2017), and Kishtwar (J&K) Cloudburst (2025).

8.7.4 Mitigation Measures

The following measures are required to be implemented to mitigate cloudburst effect.

- The State Government facing the cloudburst disaster needs to implement the provisions of flood plain zoning act in order to regulate the construction within river flood plain.
- Hydropower projects in the ecological sensitive zone of hilly area must have comprehensive disaster impact assessment mandatory for new hydro project clearance apart from environment impact assessment (EIA).
- Alternative sources of energy like solar, geothermal, biogas need to be explored apart from existing sources of hydro power in the mountains slopes.

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- There is need to identify landslide risk zone in mountains slopes along with the enforcement of proper guidelines, regulation, and zones for assessing landslide risk for mitigation of floods through an integrated approach involving state and local administration.
- Various scientific techniques can be used for slopes stabilization at landslide prone locations.
- **Hazard, vulnerability, capacity and risk assessment (HVCRA)** must be exercised to compute risk for entire state which is essential for the development of sound disaster management plan.
- There is a need to build new roads and widening of existing roads in consultation with GSI (Geological survey of India) and other scientific agencies along with the construction of separate evacuation roads.
- Encroachment in riverbeds must be strictly prohibited and all past encroachments have to be removed.
- Blasting for developmental activities must be avoided as the same results in destabilizing of rocks resulting in frequent floods and landslides.
- Illegal mining in riverbeds needs to be stopped immediately as the progressive depletion of riverbed results in dramatic readjustments of river course.
- Disaster management plan needs to be regularly reviewed and updated by the state government in order to enhance functional capabilities and coordination among various agencies for preparedness.
- Advance Doppler radar system and automated rain gauge stations must be deployed in the inaccessible area throughout the regions.
- Exhaustive training is to be regularly provided for capacity building required on various aspects of disaster management for all stakeholders such as police, locals, home guards, volunteers, government servants, NGO’s etc.
- Updating **Indian disaster Resource Management** (IDRN) for enhancing preparedness on disasters by providing all essential information about resources, equipment,critical supplies available at the district level with inline department and agencies.

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- Availability of essential commodities in remote disaster prone areas should be ensured through animal based transport at block/panchayat level.
- Strict laws need to be prepared for avoiding construction on slopes and ensure efficient land use planning in order to prevent future floods on the basis of hazard, risk and vulnerability studies.
- To tackle the disaster of cloudburst in future, NDMA guidelines should be more focused on community based disaster risk management so as to build people's capacity for coping with disaster risks and further reduce their vulnerability thereby developing safer and more resilient communities.

8.8 DROUGHT

8.8.1 Definition

Drought is a condition of below average precipitation in a given region resulting in the shortage of water supply and can last for months or years amounting big loss to agriculture, ecosystem and economy of that region.

8.8.2 Causes

The major causes of drought in India are delay in the onset of monsoon/failure of monsoon, variability and deficiency of monsoon rainfall, long break in monsoon, changes in climate, El Nino and La Nina climate phenomena,and human activities such as excessive irrigation, deforestation, and erosion activities.

8.8.3 Major Drought in India

Agra drought (1837–38), Odisha famine of 1866, Maharashtra Drought (2013, 2015), Rajasthan, Gujarat, Odisha, Andhra Pradesh and Madhya Pradesh Drought (2000), Rajasthan, Gujarat, Bihar, Odisha and MP drought (1992), Whole of eastern and North-western India drought (1987).

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8.8.4 Drought Vulnerable Region

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Around 68% to 70% of geographical area of the country is vulnerable to drought. The major drought prone states in the country are southern and eastern Maharashtra including Marathwada and Vidarbha region, northern Karnataka such as Chitradurga district, Andhra Pradesh, Mayurbhanj-Balasore region of Odisha, Telangana, Bikaner and Jaisalmer district of Rajasthan, Saurashtra-Kutch region of Gujarat, Chhattisgarh, North-west Madhya Pradesh, Western Uttar Pradesh, Jharkhand, West Bengal and Tamil Nadu.

8.8.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate drought effect.

- State Drought Monitoring Cells (DMC) under the control of State Disaster Management Authorities (SDMA's) and in collaboration with National remote sensing Centre, IMD, ICAR will undertake the preparation of drought vulnerability maps for their respective states.
- The project named as **National Agricultural Drought Assessment and Monitoring System (NADAMS)**, sponsored by the Department of Agriculture and Cooperation and the Department of Space (DoS) was taken up by the National Remote Sensing Agency in collaboration with the Indian Meteorological Department (IMD), Central Water Commission (CWC) and concerned State Government agencies in order to strengthen the shortcomings of drought monitoring.
- Automatic weather stations and rain gauges are going to set up by different agencies, including IMD consisting of moisture sensors for obtaining information about the soil moisture levels and will have a dedicated web page on drought monitoring and forecast for micro level analysis and forecasting.
- Large scale research will be conducted by state agricultural universities in collaboration with central research institute of dry land agriculture

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(CRIDA) and National rain fed area authority (NRAA) to develop new drought resistant crop varieties.

- The watershed development approach such as **Integrated Watershed Development authority scheme (IMWP)** through various center schemes is an important measure of drought management taken up by the state government in collaboration with GoI, **Ministry of Rural Development (MoRD)**, and **Disaster Assistance Center (DAC)**.
- Cloud seeding will be taken as a measure for drought management by IITM (Indian Institute of Tropical Meteorology) in close collaboration with SAUs and CRIDA. Cloud seeding policy needs to be formulated at national and state level as development of drought mitigation measure.
- Administrative Training Institutes (ATIs), NGOs, PRIs, and ULBs should be encouraged to take up awareness programmes on drought management consisting of sensitizing the farmers, communities and other stake holders at house hold level, community level in rural and urban areas and organizational/ institutional level on a large scale through films, manuals, and other training materials. Farmers must be sensitized to make judicial use of water in drought prone areas through the use of water saving agricultural practices such as drip and sprinkler practices, suitable arid areas farming techniques and on farm reservoir technology used at Chattishgarh, Odisha and Jharkhand area of the country.
- Community based disaster management initiatives should be taken in states with the help of PRIs, ULBs, NGOs and (CBOs). Community based organization at district, mandal and village level to increase the awareness among farmers to practice drought management techniques in their area.
- The NIDM and ATIs must formulate and implement a comprehensive training plan covering all aspects of Drought Management training and should organize training for the MPs, MLAs, and other elected representatives of PRIs and ULBs as a drought mitigation measure.

8.9 HEAT WAVE

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8.9.1 Definition

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Heat Wave is a phenomenon of abnormally high temperatures, more than the normal maximum temperature which occurs in the summer season between the month of March and June and sometimes extended till July. In Heat Wave, the maximum temperature at a grid point consecutively for three days or more is 3 degree Celsius more than the normal temperature. According to **WMO**, the heat waves are defined as the daily maximum temperature which is more than the average maximum temperature by five degree Celsius for five or more consecutive days. The extreme temperatures and resultant atmospheric conditions cause physiological stress to the people living in these regions and sometimes lead to death. In case the maximum temperature of a station reaches at least 40°C or more for plains, 37°C or more for coastal stations and at least 30°C or more for hilly regions, then the condition of Heat Wave is considered in India. Heat Wave is often termed as silent disaster. The following criteria for the declaration of Heat Wave is:

Heat wave is considered if maximum temperature of a station reaches at least 40°C or more for Plains and at least 30°C or more for Hilly regions.

(a) **Based on Departure from Normal**

Heat Wave:	Departure from normal is 4.5°C to 6.4°C
Severe Heat Wave:	Departure from normal is >6.4°C

(b) **Based on Actual Maximum Temperature**

Heat Wave:	When Actual Maximum Temperature >= 45°C
Severe Heat Wave:	When Actual Maximum Temperature >=47°C

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If above criteria met at least in 2 stations in a Meteorological sub-division for at least two consecutive days and it declared on the second day.

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8.9.2 Causes

The heat waves occurs when high atmospheric pressure moves into an area and air from upper levels of atmosphere is pulled toward the ground in such a high-pressure system, where it gets compressed and increases in temperature. This high pressure system makes it difficult for other weather system into the area resulting in the lasting of heat waves for several days or weeks. The more it stays in the system, the hotter the area becomes. The various factors responsible for high temperature are lesser pre-monsoon showers, the El-Nino effect, the dry wind termed as LOO originating in north-western part of the country and the global climate change effect.

8.9.3 Vulnerable Heat Waves Area in India

In 2016, the following states got affected from heat waves are Bihar, Jharkhand, Gangetic West Bengal, Odisha, Punjab, Haryana, Chandigarh, Delhi, Rajasthan, Maharashtra, West Madhya Pradesh, and Gujarat.

8.9.4 Major Areas Affecting from Heat Waves

The major areas affecting from heat waves in 2016 are Phalodi (Rajasthan) 51°C, Titlagarh (Odisha) 48.5°C, Churu (Rajasthan) 48.5°C, Sheopur (MP) 48.0°C, Hisar (Haryana) 47.8°C, Banda (UP) 47.2°C, Ramagundem (Telangana) 46.4°C and Ganganagar (Rajasthan) 46.0°C, Jamshedpur (Jharkhand) 45.5°C.

Do You Know?

2024, globally recognised as the hottest year, recorded the longest heatwave in India since 2010, exacerbated by climate change and an urban heat island effect.

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Temperatures reached as high as 50.5°C in Churu, Rajasthan; Delhi neared 50°C and recorded its warmest night at 35.2°C.

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8.9.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate Heat Wave effect.

- The State Government /SDMA/DDMA/Municipal Corporation/Local Bodies should prepare and implement the heat wave action plan on the NDMA guidelines which aims to provide a framework for implementation, coordination and evaluation of extreme heat mitigation activities in cities and to help early warning agencies to protect people from the after effects of extreme heat.
- The state should identify vulnerable areas and populations in order to establish priorities for heat alert and should appoint a nodal officer at the State or district level, and depute an agency to supervise the Heat Action Plan. It should also build the capacity of key officials and agencies to recognise their roles in the State Heat Action Plan.
- The state Government Nodal officer in coordination with local IMD office should start receiving summer seasons forecast annually and establish early warning system and inter-agency coordination in order to disseminate the information to the public regarding the heat wave alerts. It also provides weather prediction and real time data of maximum temperature, heat wave warning, and heat alerts for vulnerable cities/rural areas.
- The state government in coordination with various central ministries should call for preparedness at local level for health system in mitigating heat wave related illness. The state Government should construct shelters, sheds, bus stands, identification of vulnerable places and provisioning of drinking water at those places, following the alerts/warnings, rescheduling of school timings and disseminate do’s and don’ts for public.
- The state department of health on the directives of Ministry of Health and Family Welfare should implement heat action plan, stockpiling of ORS

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and medicines and further creation of medical post at various places of mass gathering.

- Capacity building in respect of Heath system by the State Health Department in coordination with Ministry of Health and Family Welfare will be undertaken as a monitoring and response activity so as to minimize the effect of Heat Wave. The various capacity building task include deployment of rapid medical response scheme, specific health care for vulnerable groups and hospital preparedness in case of Heat Wave.
- Government leaders should ensure that State officials and agencies are well prepared for the heat season, key officials are well-trained and have information regarding pre, during and post heat season activities. Develop a clearly defined interagency emergency response plan with roles and information flows clearly marked out.
- Comprehensive media campaigns and IEC (Information, education and communication) activities should be carried out to create awareness among general public and vulnerable groups such as elderly, young children, outdoor workers and slum residents through print, electronic and social media regarding heat wave mitigation activities.
- States should adopt various long term measures to reduce the impact of extreme heat, such as increasing the green cover and forest area so as to reduce urban heat island effect, promoting cool roofs to provide comfort as well as reduce the impact of increased urbanization.

8.10 SNOW AVALANCHES

8.10.1 Definition

It is the rapid movement or slide of snow, ice and associated debris such as rocks and vegetation down the mountain slope and mostly occurs in snow covered mountains areas. There are two basic types of avalanche one is slab avalanche and the other is loose snow avalanche and further sub-divided into categories on the basis of nature of snow such as dry, damp or wet, origination of snowslide in a surface layer or involves the whole snow

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cover (slides to the ground), and its motion on the ground, in the air, or mixed.

8.10.2 Causes

Snow Avalanche generally initiates in a zone where the forces on the snow exceed its strength and results in a mechanical failure in the snowpack (slab avalanche). Avalanches are formed due to the increased load on the snowpack due to its weakening or more load due to precipitation. They are initiated by triggered mechanism such as internal metamorphism or stress development or external triggers such as ice fall, falling cornices, earthquakes, rock falls, thermal changes, blizzards and loud sounds such as shouts, machine noise, and sonic booms. Then, the avalanches accelerate rapidly and continuously grow in mass and volume due to the entrainment of more snow. Avalanches are also caused due to other loading conditions such as human or biological related activities. Large Avalanches also entrain rocks, trees, mud and other surficial material apart from snow and ice. Due to the rapid movement of the avalanche, some of the snow mixes with the air results in the formation of a gravity current known as powder snow avalanche. The probability of avalanche increases in case of occurrence of heavy snowfall for short duration and can reach upto the speed of 200 km an hour. They are the most serious and destructive natural hazard in the mountainous region.

8.10.3 Vulnerable Areas

The avalanches mostly occur in India in the Himalayan Region of Jammu and Kashmir, Himachal and Uttarakhand states.

8.10.4 Major Avalanches in India

The list of avalanches are Lahul and Spiti Avalanche (1979), Ladakh Avalanche (1984), Himachal Pradesh avalanche (1991,1995), Siachen Avalanche (2016), Jammu and Kashmir avalanche (2017).

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8.10.5 Mitigation Measures

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The following measures are required to be implemented to mitigate Snow Avalanche effect.

- The Central and state governments in association with the Snow and Avalanche Study Establishment (SASE) and Border Road Organization (BRO) will be responsible for identification and monitoring of snow avalanches and further implementation of clearance and control strategies against identified snow avalanches. The SASE will undertake the zonation of avalanche prone areas and their forecasting in order to mitigate its effect.
- Construction of supporting engineering structures in the starting zone on the upper path of the avalanche zone will be undertaken so as to prevent the initiation of avalanches and retarding the movement before its gain momentum. The structures example are massive earth or stone walls and terraces, rigid structures made from wood, steel, aluminium, pre-stressed concrete or a combination of these materials; and flexible supporting structures called ‘snow nets’ constructed of steel cables or nylon straps and held up by steel poles, snow fence and increase in the forest area coverage for the prevention of avalanche disaster.
- Construction of structures for the deflection or retardation of avalanche in the run-out zone will be undertaken so as to deflect the direction of moving snow or to keep it away from the structures in critical locations. These are massive structures such as usually made of earth, rock, or concrete located in or near the avalanche track or run-out zone.
- Construction of direct protection structures will be undertaken in adjacent to the object to be protected such as in case of building avalanche sheds or shelters which are merely roofs over roads or railroads that allow avalanches to cross the road/railroad without any interruption of the traffic.
- The SDMAs in collaboration with the district administration (DDMAs) and the NDRF will identify organisations/ institutions which can take up programmes in order to educate the communities living in avalanche

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prone areas and prepare them with the latest techniques of self-survival, and further equip them with simple and essential tools. Similarly, the organizations at high altitudes which are engaged in the development and strategic tasks will be educated on initial search and rescue operations and the use of basic equipment necessary for these operations. The **Quick Response Team (QRTs)** consisting of trained personnel from the local administration and the NDRF will be organised and equipped with the latest rescue equipment like snow clearing tools, probes, communication capability, and medical emergency aids so as to reach the affected sites within the shortest possible time. These teams will have the capability to be mobilized at very short notice.

- The occurrence of large avalanches can be prevented by the planned release of small snow slides through explosive techniques. The explosive technique have been used over the years for the deliberate release of small and controlled avalanches and has proved very effective in areas with easy access to avalanche starting zones and that can tolerate many small snow slides without causing damage. However, explosive control may be unfeasible in areas with human habitation.
- The State Government/SDMAs in collaboration with other academic institutes and central government departments will formulate the guidelines and further will organize the workshops and training programmes for different target groups in order to establish the early warning system and training for professionals on the projects such as the installation of pulsed radar, high-resolution camera and motion sensors for the detection of snow avalanches hazard.
- The SASE, in collaboration with knowledge institutions and the NDMA, will envisage, design, and implement R&D programmes for snow avalanche studies and will promote the R&D on the some of the active fields such as simulation, snow cover modelling, avalanche forecasting, snow cover parameter monitoring using optical imagery and microwave data, and the development of virtual reality based snow cover variability visualization.

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- The NDMA, state governments/SDMAs along with mountaineering institutes will set up websites and portals to disseminate information related to avalanche safety and will assist in the preparation of films, manuals, and other material in order to create greater public awareness about avalanche hazard and importance of standard avalanche equipment usage for self-rescue such as avalanche cords, beacons, probes, shovels and other devices like Emergency Position-
  Indicating Radio Beacons (EPIRB) containing the Global Positioning System (GPS), and mobile phones apart from first aid kit and equipment which is further useful for assisting survivors who may have cuts, broken bones, or other injuries, in addition to hypothermia.
- Research programmes and study on global warming and climate change encouraged to be undertaken by DST, IMD, IITs and academic institutions to mitigate the effect of snow avalanches.

8.11 CHEMICAL DISASTER

8.11.1 Definition

Chemical disasters is the release of one or more hazardous chemicals which poses threat to human, health and environment. It occurs due to fire, emission, or explosion occurrence involving one or more hazardous chemicals in the course of industrial activity such as handling, commissioning, manufacturing, maintenance, disposal and transportation or storage inside or outside the installation which can likely cause loss of life and property including adverse effects on the environment. “Chemical accident or emergency can cause extensive damage to the environment along with considerable human and economic costs. The nodal Ministry for the management of Chemical Disasters is the Ministry of Environment and Forest and NDMA has prepared the guidelines to direct ministries, department and state authorities for the preparation of Disaster management plan.

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8.11.2 Causes

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The various causes for the occurrence of chemical disaster are explosion or fire or toxic release in a plant due to number of reasons such as temperature and pressure deviations from set limits (process deviation), runaway reactions, mixing of incompatible materials, catastrophic rupture of reactors, storage vessels, pipelines, leaks, failure of hardware systems, inadequate mixing or agitation failures, improper design of reactor vents, inadequate process hazard analysis etc., loss of containment of non-flammable gases, liquids and dusts, technological system failures, accidents in storage facilities of chemicals, failures of plant safety design due to defects in design, accidents during the transportation of chemicals, misuse of chemicals, improper waste management of hazardous materials, accidents in treatment plants, Human error, organization error, improper maintenance of equipment and due to induced effect of natural calamities.

8.11.3 Major Chemical Disaster in India

The Bhopal Gas tragedy was the most devastating natural disaster in 1984, in which thousands of people died due to the release of toxic Methyl Isocyanate gas. Secondly, a fire in an oil well in Andhra Pradesh (2003), a vapour cloud explosion in the Hindustan Petroleum Corporation Limited Refinery (HPCL), Vishakhapatnam (1997), and third is an explosion in the Indian Petrochemicals Corporation Limited (IPCL) Gas Cracker Complex, Nagothane, Maharashtra.

There have been increasing instances of industrial chemical explosions as observed by 3 factory explosions in 2024 in Virudhunagar, Thane, and Vishakhapatnam.

Chennai Petroleum Corporation Limited (CPCL) in 2023 faced a 10,000 litres oil spill in Nagapattinam due to leakage in pipelines carrying crude oil.

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In 2025, the Liberian-flagged vessel MSC Elsa 3, carrying hazardous materials like calcium carbide capsized in the Arabian Sea leading to an oil spill across the Kerala coast as detected by ISROs EOS-4 SAR satellite and confirmed by the Indian Coast Guard.

8.11.4 Vulnerable Areas

Around 1861 Major Accident Hazard (MAH) units are there in the countries which are spread across 301 districts and 25 states & 3 Union Territories. In addition to this, there are thousands of registered and hazardous factories (below MAH criteria) and un-organized sectors dealing with numerous range of hazardous material posing serious threat.

8.11.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate the chemical disaster effect.

- Identification of hazardous activities should be carried out in order to evaluate the hazard and risk generation from daily activities in the industries operations through a dedicated and knowledgeable team of qualified professionals and through the use of appropriate hazard identification tools such as checklists analysis, safety audit, HAZOP, FTA/ETA, FMEA and LOPA. Moreover, a sound knowledge and study on the safety aspects of the industrial operations is essential and on its basis, the development of safety controls methods such as elimination of hazardous activity, substitution with safer chemicals, isolating the activity that could pose hazard and providing engineering controls would be undertaken to further minimize the chemical hazard risk.
- Preventive Maintenance of all the equipment and machinery in the industrial establishment should be carried out at regular intervals and regular inspection regarding the safety and health of the industrial site need to be carried out to mitigate the effect of chemical hazard risk.

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- In order to detect the leaks at micro level and at an early stage, installation of vapour/gas detection system along with the alarms should be carried out.
- Strict compliance of the rules and regulations already legislated by the state and centre government such as Explosives Act 1884, Petroleum act 1934, Factories Act 1948, Insecticides Act 1968, Environment Protection Act 1986, Motor Vehicles Act 1988, Public Liability Insurance Act 1991 and Disaster Management Act 2005 should be enforced so as to contain the effect of chemical disaster risk.
- The Union Government has enforced the legal framework on Chemical Safety and management of chemical accidents through the enactment of new rules and furthers their continuous amendments such as MSIHC Rules, EPPR Rules, SMPV Rules, CMV Rules, Gas Cylinder Rules, Hazardous Waste Rules, and Dock Workers Rules.
- The NDMA has prepared the guidelines on Chemical Disaster Management (CDM) in order to provide the directions to ministries, departments and state authorities for the preparation of their detailed chemical disaster management plans and further calls for a proactive, participatory, multi-disciplinary and multi-sectorial approach at various levels for chemical disaster preparedness and response.
- In order to strengthen the aspects of chemical safety in India, the NDMA is working on the restoration of Chief Inspectorate of Factories (CIFs) in India. Besides, the National Action Plan on Chemical Industrial Disaster Management (NAP-CIDM) is in the finalization state which will act as the roadmap for chemical disaster management in India.
- Development of best human resource management practices must be carried out consisting of the top most officials so as to improve the safety systems in the chemical industry. Further, a congenial environment has to be created in the industry for the smooth interaction between the top management and the employees so that workers at the bottom level should report substandard practices or hazards in the plant.
- In order to have an emergency preparedness in the chemical industry, there should be formation of a good response team that can react swiftly

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and mock drills should also be carried out frequently so as to equip the workers in case of emergency to mitigate the chemical disaster initiation and propagation.

- Setting up of an accident investigation board on the lines of chemical safety board of USA in order to investigate the chemical disasters and outlines the guidelines based on the lessons learnt in each accident to avoid its reoccurrence by the state and central government. Similarly, initiatives are required to be undertaken in formation of an accident reporting system and an exclusive chemical disaster database through an online portal that would help industries in tackling a wide range of experts concern around the world in prevention of any untoward incident in plant operation.
- Undertaking of awareness campaign by the government in order to provide the awareness to the workers as well as general public regarding the hazards arising out of a chemical disaster and further will help in reducing unsafe acts as well as in tackling the occurrence of chemical disaster.
- The government needs to undertake the R&D efforts for the development of new technologies in the field of chemical industries which can reduce the toxicity of the by-products from the industry and will substantially reduce the adverse effect of the chemical disaster occurrence.
- In order to minimise the effect of major accident in MAH located zones affecting the people and environment and to ensure the discharging of duty by the local administration, an offset emergency planning need to be carried out as per the extant rules and regulations.
- In order to mitigate the risk of disaster during the transportation of Hazardous chemicals and to rescue the general public, there should be a rapid and timely availability of emergency disaster response programme. Moreover, the Indian Chemical Council (ICC) has initiated a program called ‘Nicerglobe’ (nicerglobe.in) which provide GPRS tracking of trucks right form its origin to the place of destination and the programme platform is well linked to the emergency response providers. All chemical

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companies are being advised and encouraged to join this initiative of ICC so as to protect the hazard in the goods transportation.

8.12 BIOLOGICAL DISASTER

8.12.1 Definition

Biological disasters refer to an enormous spread of a disease, virus or infestations of plant, animal or insect life causing a catastrophic effect on an epidemic or pandemic level. Huge numbers of people within a given community or area get affected in an epidemic-level biological disasters, whereas in a pandemic-level biological disasters, a bigger region covering entire continents or the globe got affected. The example of cholera is an epidemic level biological disaster whereas the swine flu is a pandemic level disaster.

8.12.2 Causes

The disaster caused due to the spreading of micro-
organism or toxins in food, water or insect vector or by aerosol which could potentially harm human population, food crops and livestock as a result of any natural epidemics or international use of biological agents such as viruses or bacteria etc. The **United States Centre for Disease Control** classified biological hazards into four **Bio-Safety level** or BSL1-4 that are the potential causes of a Biological disaster. **BSL1** level consist of bacteria and viruses types such as *Bacillussubtilis*, canine hepatitis, *Escherichiacoli*, varicella (chicken pox), some cell cultures and non-infectious bacteria and precautions at this level are minimal, involving gloves and some sort of facial protection. **BSL 2** level includes bacteria and viruses that cause only mild disease to humans, such as salmonella, mumps, measles, scrapie, dengue fever,hepatitis A, B, and C, some influenza A strains, Lyme disease, HIV and at this level more extreme precautions for safety purposes, including the use of autoclaves for sterilizing and biological safety cabinets are taken. **BSL 3** level includes bacteria and viruses that can cause severe to

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fatal disease in humans such as anthrax, West Nile virus, yellow fever, malaria, Venezuelanequine encephalitis, SARS virus, MERS coronavirus, hantaviruses tuberculosis, typhus, rift valley fever, rocky mountain spotted fever and parasites under this level are *Plasmodium falciparum*, which causes malaria, and *Trypanosoma cruzi*, which causes trypanosomiasis. It requires much more stringent safety protocols including the use of respirators to prevent airborne infection and hazards in this group generally have known vaccines or treatments. **BSL 4** consists of viruses that are potentially harmful to humans for which there is no known treatment or vaccine, such as Marburg virus, Ebola virus, Lassa fever virus, Crimean–Congo hemorrhagic fever, and other hemorrhagic diseases and at this level the protective measures include the use of a positive pressure personnel suit, with a segregated air supply. Presently, there are no bacteria classified at this level.

8.12.3 Major Biological Disaster in India

The outbreak of Smallpox Epidemic in Bihar, Odisha and West Bengal (1974), Plague Epidemic in Surat and Beed (1994) and in Shimla, Himachal Pradesh (2002), Swine Flu (H1N1) Epidemic (2009) (2015), Hepatitis B Virus in Gujarat (2009), Hepatitis A and E virus in Odisha (2014), H5N1 Virus (2006) and Chikungunya Virus (2006).

One of the world's deadliest pandemics, COVID-19 spread worldwide, caused by the coronavirus SARS-CoV-2. It was declared as a public health emergency of international concern (PHEIC) on 30 January 2020, and was assessed it as having become a pandemic on 11 March. It lasted for over 3 years, with WHO declaring its end in May 2023. India has the second-highest number of cases after the United States and third-highest deaths after the USA and Brazil. As of 2021, the disease was considered to have an endemic stage instead of complete disappearance.

8.12.4 Vulnerable Area

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Some of the vulnerable areas responsible for epidemic outbreak are poor sanitation practices such as indiscriminate dumping, polluting of drainage channels and sewers, consumption of contaminated water or food etc., improper waste disposal from hospitals and medical centres, practicing unsafe sex or living in poor hygiene condition, limited medical infrastructure and healthcare facilities, limited access to vaccines.

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8.12.5 Mitigation Measures

The following measures are required to be implemented to mitigate the chemical disaster effect.

- The nodal Ministry **(Ministry of Health and Family Welfare)** along with other ministries and state and district department of health in coordination with NDMA, SDMA, DDMA, NEC (National Executive Committee), NCMC (National Management Crisis Committee) will develop DM plans on the basis of guidelines issued by NDMA and establish legislative framework for the BDM (Biological Disaster Management). The legislative framework includes the establishment of a legal, institutional and operational framework which clearly defines the policy, programmes, plans, SOPs and institutional and operational framework and its role in implementation of IHR (International Health Regulations) 2005 and other legal mechanism for the management biosafety and biosecurity aspects and the upgradation of existing infrastructure.
- The SDMA/DDMA in coordination with NDMA and state and central government will undertake the upgradation and strengthening of IDSP (Integrated Disease Surveillance Programme) for the capacity development of various biological disasters. The existing IDSP programme will be expanded and amenities will be developed to cover all issues of environmental management like water supply, personal hygiene, vector control, burial/disposal of the dead and the risk of occurrence of zoonotic disorders and personnel in state/district IDSP units will be trained for creating the awareness programme among public in respect of

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personal hygiene and dissemination of appropriate information to the public health authorities about any epidemic.

- The state/district authorities will develop tools in order to monitor the status of available pharmaceutical interventions including antibiotics, chemotherapeutics and anti-virals, including the listing of essential drugs and undertaking various immunization and vaccination programmes along with the development of on-site contingency plan for the management of biological disaster. Further, mechanisms to employ various non-pharmaceutical interventions like social distancing measures, and isolation and quarantine techniques will be adopted along with the development of a database of various laboratories handling hazardous microorganisms in order to ensure the implementation of various biosafety and biosecurity measures.
- The NDMA/NEC/SDMA/DDMAs will coordinate the various functions of medical response at their respective levels for the management of any epidemic. The CMO is the main coordinator along with the district collector as commander for the management of biological emergencies at the district level.
- Capacity development of human resource training will be developed/standardised and implemented at each level by district/state authorities and nodal/line ministries through various training modules. Further, community awareness about services delivery in various civic amenities will be strengthened so that appropriate knowledge is developed and provided to the stakeholders in order to enhance the participation of the community in all phases of the DM cycle and be resilient enough to tackle biological emergencies with the help of various practices and training schedules along with mock exercises followed by documentation and evaluation of lessons learnt to improve the existing system.
- The SDMA/DDMA along with NDMA will identify the development of critical infrastructure such as the development of a laboratory network including national/state level referral laboratories, district level diagnostic laboratories with medical colleges to confirm diagnosis under a single integrated framework, chain of public health laboratories, development of

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biomonitoring techniques and further upgradation of the existing emergency communication network, health network, including IAN (Integrated Ambulance Network) and mobile tele-health, print and electronic media channels, networking of NGOs and international organizations for the management of biological emergencies in an efficient way.

- The MoH&FW as nodal ministry, other line ministries having health care facilities and departments of health at the state/district levels will provide an integrated framework to manage public health emergencies which includes an effective inbuilt mechanism to keep a check on the early warning signs of an outbreak, make available safe food, water, personal hygiene facilities and also have the capacity to provide psycho-social care. Moreover, different activities like hospital disaster management planning, upgradation of earmarked hospitals, development of mobile hospitals and mobile medical teams supported by adequate medical logistics including essential medicines, antibiotics, vaccines, PPEs, etc., will be undertaken on priority basis at each level for the BDM.
- The MoH&FW, MoD and MHA in collaboration with ICMR, CSIR, DRDO and other research organizations will undertake various research and development activities in order to lay the foundation for long-term research interventions so as to mitigate the impact of biological emergencies. These R&D activities will cater for biodefence and operational research with models to develop checks on various public health consequences, thereby evaluating various mitigation strategies after testing them at numerous stages. NDMA will act as a facilitator, and advisory and monitoring body for such types of activities.

8.13 NUCLEAR DISASTER

8.13.1 Definition

Nuclear emergency/Disaster is caused due to tremendous release of radioactive material or radiation either in the nuclear reactors operation or

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other nuclear events like Radiological Dispersal Device (RDD) or nuclear weapon explosion. The disaster is accompanied with sudden release of harmful radiations or radioactive materials or both together in to the environment.

8.13.2 Causes

The major reasons for the occurrence of nuclear disaster are accident in nuclear power plant due to faulty design, lack of safety features, nuclear meltdown and uncontrolled nuclear chain reaction, accidents in handling of nuclear radiation source due to faulty practices and at the time of transportation of radioactive material, malevolent use of radioactive material as RDD (Radiological Dispersal Device) or dirty bomb by terrorist group for large scale destruction or nuclear weapon attack. There are various secondary factors also such as human error, system failure, sabotage, earthquake, cyclone, flood, tsunami etc.

8.13.3 Major Nuclear Disasters in the World

The Nuclear reactor accident in **Three Mile Island (TMI)** in USA (1979) and **Chernobyl** in erstwhile USSR (1986), **Fukushima** in Japan (2011) and nuclear weapon attack in **Hiroshima** and **Nagasaki** (1945).

8.13.4 Vulnerable Areas

The major vulnerable areas/sources that can lead to contamination of radioactive radiations in the environment are Nuclear Power plants, research reactors, naval fuel facilities, uranium enrichment plants, fuel fabrication plants, and uranium mines.

8.13.5 Mitigation Measures

The following measures are required to be implemented in order to mitigate the nuclear disaster effect.

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- Preparation of DM Plans by all the central ministries/departments and all the states/UT on the guidelines of NDMA along with technical support from DAE will be undertaken for the management of Nuclear Disaster management. Further, induction of nuclear science and technology specialists at all levels of administration in the formulation of the DM plans and their effective monitoring will be carried out.
- Establishment of a dedicated and reliable communication system through the National Disaster Communication Network of NDMA in coordination with the nuclear plant/facility operators and the SDMAs/ DDMAs concerned will be undertaken to ensure proper and adequate last-mile connectivity during all the phases of a nuclear emergency to the disaster affected site.
- Ensuring the inherent safety philosophy in design features, adoption of best available manufacturing/fabrication procedures, best operating practices and stringent regulatory measures in the plant and further carrying out the periodic review of these nuclear reactors safety features by the AERB for the prevention of any unlikely accidents in nuclear power plants. Moreover, the nuclear plant will be designed for Operating Basis Earthquake (OBE) and Safe Shutdown Earthquake (SSE) as per the standard design codes and guides of the AERB in order to make it resistant from any type of disaster. In order to prevent the leakage of radioactive materials during its transportation, the regulatory guidelines have already been framed which specify the design of the container as well as the manner in which they should be handled and transported.
- Before the issuance of any license for the operation of any nuclear plant, the Atomic Energy Regulatory Board (AERB) should ensure that the nuclear facility has the emergency Response manuals and plans in order to cope up with the three main types of emergencies such as plant, on-site and off-site emergency.
- Specialised disaster response NDRF teams, specially trained by NDMA with the assistance from DAE/DRDO/Army for nuclear/radiological emergency/disaster and fully equipped at the state (by the SDMAs/SECs/DDMAs) as well as at the central levels (by MHA) will be

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raised in order to provide first response emergency at the time of any nuclear emergency.

- Large number of ERCs (Emergency Response Centers) equipped with appropriate radiation detection instruments, PPG (Personal Protective Gear) and a mobile monitoring van will be established by DAE as well as the state government covering all the metros and the other vulnerable areas in the country so as to provide the specialized services during nuclear/radiological emergencies.
- The state government in consultation with the DAE and MHA will install IERMON system (Indian Environment Radiation Monitoring Network) for radiation monitoring in various metros and vulnerable areas in order to strengthen the real time surveillance and early warning systems at the time of nuclear emergency.
- The nuclear/radiological facility operators, NIDM and SDMAs/DDMAs with assistance from MHRD, MHA, DAE, and DRDO will conduct the community education and awareness generation programmes about the nuclear energy in order to prepare them in case of any nuclear emergency.
- The MoH&FW will in close coordination with NDMA, SDMA, MHA, DAE, and MoD will outline the medical preparedness plan for nuclear/radiological emergencies such as the establishment/upgradation of primary, secondary, and tertiary care hospitals to handle sufficient number of people affected during a nuclear emergency, creation of medical professionals database and training them in the management of radiation injuries, creation of Radiological Safety Officer (RSOs) database and development of a model Chemical, Biological, Radiological and Nuclear (CBRN) research and treatment centre to take care of any type of nuclear emergency/disaster.

The CBRN trained NDRF trainers and NIDM with assistance from DAE, DRDO and NDMA will impart training at regular intervals to the various first responders and the administrative personnel involved in DM at various levels of administration and further, the SDMA/DDMA will conduct regular mock-drills and emergency preparedness exercises in order

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to cope up with any type of nuclear emergencies. The MoH&FW will undertake training programmes in respect of medical professionals for radiation injuries treatment, formation of Quick Reaction Medical Teams (QRMTs)/Medical First Responders (MFRs) along with maintaining sufficient stock of essential medicines and database for experienced medical professionals on priority basis in order to deal with any nuclear disaster crisis.

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SECTION II

Glossary

Adaptation

The adjustment in natural or human systems in response to actual or expected climatic stimuli or their effects, which moderates harm or exploits beneficial opportunities.

Building Code

A set of ordinances or regulations and associated standards intended to control aspects of the design, construction, materials, alteration and occupancy of structures that are necessary to ensure human safety and welfare, including resistance to collapse and damage.

Capacity

The combination of all the strengths, attributes and resources available within a community, society or organization that can be used to achieve agreed goals.

Capacity Development

The process by which people, organizations and society systematically stimulate and develop their capacities over time to achieve social and economic goals, including through improvement of knowledge, skills, systems, and institutions.

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Disaster

A serious disruption of the functioning of a community or a society involving widespread human, material, economic or environmental losses and impacts, which exceeds the ability of the affected community or society to cope using its own resources.

Disaster Risk

The potential disaster losses, in lives, health status, livelihoods, assets and services, which could occur to a particular community or a society over some specified future time period.

Disaster Risk Management

The systematic process of using administrative directives, organizations, and operational skills and capacities to implement strategies, policies and improved coping capacities in order to lessen the adverse impacts of hazards and the possibility of disaster.

Disaster Risk Reduction

The concept and practice of reducing disaster risks through systematic efforts to analyze and manage the causal factors of disasters, including through reduced exposure to hazards, lessened vulnerability of people and property, wise management of land and the environment, and improved preparedness for adverse events.

Disaster Risk Reduction Plan

A document prepared by an authority, sector, organization or enterprise that sets out goals and specific objectives for reducing disaster risks together with related actions to accomplish these objectives.

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El Nino-Southern Oscillation Phenomenon

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A complex interaction of the tropical Pacific Ocean and the global atmosphere that results in irregularly occurring episodes of changed ocean and weather patterns in many parts of the world, often with significant impacts over many months, such as altered marine habitats, rainfall changes, floods, droughts, and changes in storm patterns.

Environmental Impact Assessment

Process by which the environmental consequences of a proposed project orprogramme are evaluated and undertaken as an integral part of planning and decision-making processes with a view to limiting or reducing the adverse impacts of the project or programme.

Forecast

Definite statement or statistical estimate of the likely occurrence of a future event or conditions for a specific area.

La Nina

La Nina is the reverse of El Nino. At the end of December, westbound winds get stronger than usual, pushing warm water further west than normal. As a result, there is enhanced upwelling of cold sea water. This action allows hurricanes crossing the Atlantic to move farther west and to become more powerful than usual. However, La Nina is generally good for the Indian summer monsoon.

Mitigation

The lessening or limitation of the adverse impacts of hazards and related disasters.

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Knowledge and capacities developed by governments, professional response and recovery organizations, communities and individuals to effectively anticipate, respond to, and recover from, the impacts of likely, imminent or current hazard events or conditions.

Residual Risk

The risk that remains in unmanaged form, even when effective disaster risk reduction measures are in place, and for which emergency response and recovery capacities must be maintained.

Vulnerability

The characteristics and circumstances of a community, system or asset that make it susceptible to the damaging effects of a hazard.

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SECTION III

UPSC Mains Model Answers: Disaster Management (2025 – 2023)

Mains Model Answers (2023 onwards)

UPSC MAINS 2025

1. Seawater intrusion in the coastal aquifers is a major concern in India. What are the causes of seawater intrusion and the remedial measures to combat this hazard? (10 marks; 150 words)

Ans. The Minjur block in Tiruvallur district of Tamil Nadu has experienced severe contamination of coastal aquifers due to seawater intrusion, leading to heavy dependence on municipal and private water supply for daily needs.

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In this context, seawater intrusion refers to the landward movement of saline water into freshwater aquifers in coastal regions, leading to salinisation of groundwater.

India, with a coastline of ~11000 km, has several vulnerable areas such as Saurashtra and Kutch (Gujarat), Chennai–Minjur Belt (Tamil Nadu), Coastal Kerala, Visakhapatnam Coast (Andhra Pradesh), Konkan (Maharashtra), and Lakshadweep Islands.

The main causes of seawater intrusion include:

1. **Natural Causes:** Coastal regions with highly permeable alluvial and limestone formations allow faster saline ingress.
2. **Anthropogenic Causes:**
    a. **Over-extraction of Groundwater** causes a lowering of the water table below sea level, reversing the natural hydraulic gradient. Eg: North Chennai and Kollam Alapuzzha Belt.
    b. **Urbanisation** leads to the replacement of soil with paved surfaces, reducing groundwater recharge.
    c. **Reduction in river recharge of seas** due to dams, barrages and canal diversions, which reduces the pressure on seawater to maintain the natural gradient.
3. **Climate causes:** El Niño leads to reduced monsoons, which reduce fresh groundwater levels, enabling seawater intrusion.

Seawater intrusion affects the quality of fresh water, making it unfit for consumption. It can also lead to increased salinity in agricultural soil, corrosion of infrastructure and resultant internal migration due to lack of water security.

Remedial Measures to combat this hazard include:

1. **Regulatory measures:**
    a. Groundwater Pumping Regulation in coastal blocks notified as over-exploited

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b. Mandatory permits for new borewells, use of digital flow meters to track extraction

c. Promote surface water and treated wastewater reuse to reduce dependence on groundwater (e.g., in Chennai Industrial Corridors)

2. Structural measures:

a. Impermeable sub-surface cut-off walls to prevent the inland movement of saline water. Example: Minjur–Chennai Coastal Aquifer Barrier Projects.

b. Freshwater Injection Wells (Induced Recharge) to maintain hydraulic head and push back the saline wedge.

c. Rainwater Harvesting Systems in high rainfall areas like Tamil Nadu and percolation wells to improve groundwater recharge.

3. Aquifer Management Measures:

a. Managed Aquifer Recharge (MAR)using filtered stormwater or wastewater

b. Balanced use of surface water and groundwater to avoid excessive drawdown.

c. Regular observation of the freshwater–saltwater interface using EC/TDS surveys and remote sensing. (EC-electrical conductivity, TDS-total dissolved salts)

4. Ecological measures:

a. Mangrove & Coastal Wetland Restoration to enhance soil moisture retention, and buffer sea-wave energy.

b. Protection of Sand Dunes & Coastal Vegetation

c. Regulation of Coastal Development to avoid or minimize aquifer exposure during construction of ports, resorts, and SEZs

5. Technological Measures:

a. Desalination Plants to reduce dependence on coastal groundwater. E.g. Mijur and Nemmeli plants in Tamil Nadu

b. Use of Low-Salinity Irrigation Techniques like drip system to reduce demand of freshwater.

c. Salt-Resilient Crop Varieties to sustain livelihoods.

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Seawater intrusion is a result of human-induced aquifer stress, compounded by climate change and geological vulnerability. Addressing it requires sustainable groundwater governance i.e. a combination of regulatory enforcement, community-based water management, artificial recharge infrastructure, and long-term coastal ecosystem protection.

UPSC MAINS 2024

2. What is disaster resilience? How is it determined? Describe various elements of a resilience framework. Also mention the global targets of the Sendai Framework for Disaster Risk Reduction (2015-2030). (15 marks; 250 words)

Ans. Disaster Resilience, as per UNISDR, refers to the ability of a community exposed to a hazard to resist, absorb, accommodate, adapt to, transform and recover from the effects of a hazard in a timely and efficient manner.

Determinants of Disaster Resilience:

1. **Degree of Exposure** of the people, nature and property located in hazard-prone areas. For example, coastal areas are more prone to cyclonic storm surges, while hilly areas are prone to flood-aided landslides and GLOF events.
2. **Capacity to manage** disaster situations in terms of both reactive and proactive approaches. For example, in India, NDMA laid down its 2010 guidelines addressing a wide range of natural and man-made disasters.
3. **Vulnerability-** Social, economic, and environmental- makes people susceptible to the damaging effects of a disaster. For example: Densely populated urban areas, encroached water bodies and marginalised social sections.
4. **Adaptive ability,** including restoration of critical infrastructure and basic services through risk management.

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Various elements of the disaster resilience framework:

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As per the Sendai Framework for DRR (2015-30), disaster resilience can be grouped into three stages based on two approaches:

1. **Proactive Approach:**
   - **a. Pre-Disaster Management:**
     - Disaster Mitigation through risk assessment, forecasting, and early warning systems.
     - Preparedness planning through community planning, mock exercises and disaster management plans at the local levels.
2. **Reactive Approach:**
   - **b. During Disaster Management**:
     - Adaptation through climate-resilient infrastructure and regulated public movement.
     - Rescue Operations through trained responders like NDRF and timely evacuation.
   - **c. Post-Disaster Management:**
     - Relief and rehabilitation of affected people by providing temporary shelter, medical services and food supplies.
     - Recovery and Reconstruction through state efforts like compensation, restoration of services, and **"Building Back Better".**

7 Global targets of Sendai Framework: The Sendai Framework is a successor to the Hyogo Framework of 2005 for reducing disaster risks and building community resilience. Its targets are grouped into two categories:

1. **Reduce:**
   a. Mortality, substantially by 2030 compared to the decade 2005-15.
   b. Affected people through proactive measures like hazard zonation mapping
   c. Economic losses through climate resilient infrastructure and public services, and faster recovery. E.g.: PM Natural Relief Fund
   d. Damage to critical infrastructure and disruption of basic services through reconstruction and recovery efforts.

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2. Increase

a. Number of countries with DRR strategies. For example: India's National Plan for DRR and PM's 10-point agenda on DRR.

b. International support and cooperation to developing countries to complement their national plans by 2030.

c. Availability and accessibility of Early Warning Systems and Disaster Risk Information, and assessment technologies.

The 7 global targets of the Sendai Framework reflect a shift from disaster response to risk reduction and resilience-building.

For India, aligning national frameworks such as the Disaster Management Act (2005), NDMA guidelines, and state-level DRR strategies with these global targets ensures that disaster management is not an isolated exercise but an integral part of sustainable development and climate adaptation.

3. Flooding in urban areas is an emerging climate-induced disaster. Discuss the causes of this disaster. Mention the features of two such major floods in the last two decades in India. Describe the policies and frameworks in India that aim at tackling such floods. (15 marks; 250 words)

Ans. As per NDMP 2019, urban floods are a result of a combination of meteorological, hydrological and human factors. Aided by human use of land, urban flooding happens very rapidly with large flows.

Flooding in urban areas has become one of the most pressing climate-induced disasters, reflecting the combined effects of rapid urbanisation and global climate change.

Causes of Urban Flooding as a disaster:

1. **Climate change:** As per the IPCC, climate change is driving both the intensity and frequency of extreme weather events.
   a. Altered precipitation patterns, like increased cloudbursts and flash floods, cause a sudden rise in river levels and landslides. E.g.: 2025

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Uttarkashi and Chamoli cloudburst events.

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   b. Sea-level rise leads to coastal cities facing increased storm surges and tidal flooding
   c. Erratic monsoon and shifting storm tracks have increased vulnerability to urban flood disasters due to unpredictability.
2. **Urban Vulnerabilities:** In 2025, India received 30% increased rainfall across many areas, leading to increased urban flooding events.
   a. Impermeable surfaces like concrete roads and pavements reduce infiltration, increasing runoff.
   b. Outdated or clogged stormwater drains cannot cope with sudden rainfall. E.g.: Regular floods in Bengaluru and Guwahati
   c. Encroachment on wetlands and floodplains has led to a reduction in the carrying capacity of natural buffers.
   d. High population density has led to increased vulnerability to urban flood disasters.

Features of major Urban Floods of the last two decades:

1. **Chennai Floods 2015:** The floods, triggered by the annual north-east monsoons, were characterised by:
   a. Chennai received the highest rainfall of about 400 mm in one day, a record of 100 years, affecting more than 4 million people.
   b. The cause behind the rainfall was a depression over the region due to El Niño effects, which exacerbates the winter monsoons.
   c. Overflowing water bodies, drains, and surface runoff led to the opening of sluice gates, which further flooded the Adyar and Cooum rivers.
   d. The floods subsided within a week but affected all sectors, inflicting heavy losses on real estate, communication and street trade.
2. **Delhi floods 2023:** The floods are attributed to the surge of the Yamuna River in July 2023, affecting the catchment areas of the river basin.
   a. The high runoff from the river caused breaching of the Highest Flood level (HFL) of the Old Delhi Railway Bridge.

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b. The rainfall also increased inflow at Hathnikund Barrage, leading to a record peak release.
c. The floods impacted the city with disruption of basic services like communication, transportation and electricity.
d. The city, however, saw an unprecedented impact, unlike as predicted by CWC due to river siltation, which marred forecast data.

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Since the challenges associated with urban floods tend to be considerably distinct, he policy and institutional frameworks that address this problem span **national guidelines, mission-mode programs, and local urban governance measures**.

1. NDMA issued **separate guidelines** for Urban Flood Disaster Management (UFDM) in 2010, the first such recognition. It provided for:
   - Preparation of city-specific flood management plans.
   - Stormwater drainage master plans in all cities.
   - Encroachment removal from natural buffers
   - Integration of flood risk with urban planning under the town planning authorities
   - Use of technology like flood modelling, GIS mapping, and early warning systems.
2. **National Mission on Sustainable Habitat (NMSH)** – under NAPCC (2008): It promotes climate-resilient urban planning through:
   - Rainwater Harvesting
   - Sustainable Drainage Systems
   - Urban Water Bod conservation
3. **Hazard Risk Vulnerability and Capacity Assessment** (HRVCA) by NIDM with technical support issued by NDMA at the central level.
4. **Smart Cities Mission (2015):** It aims to build flood resilience through proactive approaches like:
   - **Technology** integration of GIS and early warning systems.
   - **Integrated Command and Control Centres (ICCCs)** for disaster management

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- Improve drainage networks and install stormwater drains as part of the mission's **alignment with initiatives like AMRUT**.

5. International Alignment:

- Alignment with **Sendai Framework for Disaster Risk Reduction** (2015–30).
- MoHUA and NIUA promote **blue-green infrastructure** for urban resilience.

Disaster resilience is a **multidimensional capacity**—determined by exposure, vulnerability, capacity, and adaptive ability. For India, strengthening resilience means aligning development planning with **risk reduction and climate adaptation**, ensuring that disasters do not reverse decades of progress.

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4. Dam failures are always catastrophic, especially on the downstream side, resulting in a colossal loss of life and property. Analyse the various causes of dam failures. Give two examples of large dam failures. (10 marks; 150 words)

Ans. As per NIDM, a dam **failure** is defined simply as an uncontrolled release of water from a reservoir through a dam as a result of structural failures or construction deficiencies in the dam

While the events are considered **low probability**, they are classified as **high-impact** events, where the consequences are disproportionately large compared to the frequency.

The **sudden onset leaves little time for evacuation**, leading to severe casualties, large-scale displacement, and massive economic losses. For example, the Machhu dam failure of 1979 was caused by excessive rain and massive flooding, leading to the disintegration of the earthen walls.

Causes of Dam Failure:

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1. Structural causes:

a. **Design deficiencies** – incorrect estimation of flood levels, inadequate spillway capacity.

b. **Poor construction quality** – use of substandard materials, weak foundations.

c. **Ageing infrastructure** – wear and tear, cracks, seepage through the dam body.

Example: Machhu Dam-II (Gujarat, 1979) failed partly due to design and structural inadequacies.

2. Geographic causes:

a. **Seismic activity** – earthquakes causing cracks or structural collapse

b. **Landslides** in the catchment area leading to overtopping (blocking, then sudden release).

c. **Unusual rainfall patterns** due to climate change exceed the designed capacity.

d. **Weak geology** – permeable or unstable foundation rock leading to seepage or sliding.

Example: Vajont Dam disaster (Italy, 1963) caused by a massive landslide into the reservoir.

3. Industrial causes:

a. **Mining or blasting nearby** – weakens the dam foundation.

b. **Construction of heavy infrastructure projects** in the vicinity, altering hydrology or geology.

c. **Pollution or chemical seepage** weakening embankments.

Example: Mining-induced tremors around reservoirs have raised concerns in Chhattisgarh and Jharkhand dams.

4. Operational causes

a. **Improper reservoir regulation** – overfilling without timely water release.

b. **Sudden release of water** – leading to downstream floods and embankment breaches.

c. **Malfunctioning of gates/sluices** – jammed gates during peak inflow.

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d. **Inadequate monitoring** – ignoring warning signs like seepage, cracks, and unusual vibrations.

e. **Human error** – negligence in flood forecasting or communication failure.

Example: Chennai 2015 floods aggravated by the late release of water from the Chembarambakkam reservoir.

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Some major dam failures include:

1. **Banqiao Dam failure (China), 1975:** The failure killed more than 1 lakh people in the Henan province of China and triggered a chain reaction of failure of over 60 other reservoirs. The cause of failure was attributed to one year worth of rainfall flooding the dam in a single day, a fact not considered during the construction of the reservoir.
2. **Machchu (Gujarat) dam failure, 1979:** The dam, built to serve irrigation purposes, failed due to excessive rain and massive flooding leading to the disintegration of the earthen walls**,** killing over 25000 people downstream**.**

SDG Goal 13 (climate action) mentions dam resilience to extreme weather. Preventing dam failures needs strict safety audits, continuous real-time monitoring, timely maintenance, robust Emergency Action Plans, climate-resilient design, and community preparedness.

India has strengthened this through the **Dam Safety Act (2021)**, the Dam Rehabilitation and Improvement Project **(DRIP)**, and CWC-led inspections, aiming for climate-resilient, well-governed, and safer dam management systems.

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